The Spacetime–Electromagnetic Framework

Mathematical Structures and Obstructions in Split-Signature Clifford Geometry

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Abstract

We investigate the real split-signature Clifford algebra $\mathrm{Cl}(3,3) \cong M_8(\mathbb{R})$ and its graded extensions $\mathrm{Cl}(3n,3n)$ as a common setting for spacetime, electromagnetic and matter constructions. The aim is to distinguish consequences of specified algebraic data from additional assumptions, physical interpretations and compatibility obstructions. For a chosen Lorentz embedding and stated commutation requirements, the complex structure is fixed up to sign, and the associated real Dirac formulation is equivalent to ordinary complex Dirac theory. We construct distinct displacement and field-strength representations on $\mathbb{R}^{3,3}$, with no nonzero Lorentz-equivariant linear intertwiner between the specified modules. We analyse factorised interior dynamics and a Lie–Poisson closure within a restricted ansatz, including bounded motions in a neutral-signature model whose Hamiltonian is unbounded below. In $\mathrm{Cl}(9,9)$, a chosen rank-64 projector realises one chiral Pati–Salam family with a right-handed neutrino; its Lorentz-invariant mass kernels form a 272-dimensional real space. Vacuum-selection results are established under explicit assumptions for specified potentials and sectors, while global claims for the full radiative model remain conjectural. Electric-charge identification depends on selected vacua, and the normal-plane charge is anomalous on the chosen matter block. Curvature identities recover the Einstein tensor of a prescribed four-metric. We also analyse obstructions to causal selection within an invariant adjoint-scalar ansatz and formulate surplus-time constraints at the test-worldline level. The resulting constructions identify mathematical relationships and limits of compatibility. A common action, a general interacting causal reduction and empirical predictions remain open.

Keywords: split-signature Clifford algebras; Lorentz representations; electromagnetic duality; chiral projectors; Pati–Salam; vacuum selection; causal reduction; computer-assisted algebra.

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Introduction and scope

I.1 Introduction

I.1.1 The question

Can the mathematical structures underlying spacetime, electromagnetism and matter be understood as different manifestations of a common algebraic foundation, rather than introduced as independent structures from the outset?

The present work investigates this question through the real split-signature Clifford algebra

$$\mathrm{Cl}(3,3) \cong M_8(\mathbb{R})$$

and its graded extensions $\mathrm{Cl}(3n, 3n)$. The objective is to determine which structures follow from specified algebraic data, which require additional assumptions, which are mutually compatible and which are obstructed. Representing a known theory within the algebra is distinguished from deriving a restriction on its physical content.

The algebraic prototype and its temporal interpretation. We begin with a six-dimensional real vector space

$$V = V^+ \oplus V^-, \qquad \eta = \mathrm{diag}(+,+,+,-,-,-),$$

where V^+ and V^- each have dimension three. A prescribed pairing identifies each positive basis direction with a corresponding negative direction. We assign a temporal interpretation to the positive triple and a spatial interpretation to the negative triple as part of the proposed spacetime reading.

One motivation for this choice is the six-dimensional space of electromagnetic field strengths in four-dimensional spacetime, organised relative to an observer into three electric and three magnetic components. This motivates a comparison of representations; it does not establish the existence of three physical time coordinates. The displacement and field-strength readings use distinct Lorentz actions, and their compatibility must be examined explicitly (Chapter 3).

A related construction uses the three commuting pole operators

$$\omega_i = e_i e_{i+3}, \qquad \omega_i^2 = \mathbf{1}.$$

Their joint eigenspaces are eight one-dimensional subspaces of the real spinor module. Following the principal convention of Section 3.5, the field components are assigned to six of these octants:

component	pole eigenvalues	component	antipodal pole eigenvalues
E_x	$(-,+,+)$	B_x	$(+,-,-)$
E_y	$(+,-,+)$	B_y	$(-,+,-)$
E_z	$(+,+,-)$	B_z	$(-,-,+)$

The electric components occupy the three octants of Fock weight one and the magnetic components their antipodal octants of Fock weight two. The sign triples label joint eigenspaces of

the pole operators, not metric signatures. The assignment is a constructed interpretation rather than a unique consequence of the algebra, and its electromagnetic transformation law is distinct from the interior Dirac spinor action on the same underlying real space.

For the observational spacetime construction, selecting a unit clock $\hat{n} \in V^+$ gives the Lorentzian four-plane

$$E_{\hat{n}} = \mathbb{R}\hat{n} \oplus V^-.$$

Its orthogonal complement is a positive two-plane. The existence of this splitting is algebraically straightforward; its physical selection, the treatment of the two surplus directions and causal consistency require further analysis. The signature $(3, 3)$ is therefore adopted as a balanced prototype with three prescribed pairs. Whether additional requirements characterise this choice uniquely remains a mathematical question.

From a single prototype to constituent multiplicity. The algebra $\mathrm{Cl}(3, 3)$ supplies the single-constituent starting point. With the specified clock and representation data it supports a Lorentzian sector and a real Dirac module, and these constructions can be defined before a second constituent is introduced. We investigate whether repeated prototypes provide a useful mathematical setting for composite systems, multiple-particle descriptions and alternative collective states. The factors are isomorphic copies, and their initial labels are not assigned an intrinsic physical identity; distinctions may enter through states, relations, selected structures or interactions, and whether factor permutations are physical symmetries depends on those additional data. The extensions are formed through orthogonal direct sums and graded tensor products,

$$\mathrm{Cl}(3n, 3n) \cong \widehat{\bigotimes}_{k=1}^{n} \mathrm{Cl}(3, 3).$$

This classical identity supplies a systematic way to enlarge the representation space while retaining copies of the prototype. The integer n counts algebraic constituents. An n-particle interpretation additionally requires physical state spaces, kinematics, dynamics and exchange properties; the tower does not supply particle number or statistics by definition.

Constructive method and compatibility. We begin with the algebraic data, construct the geometric and physical readings they admit, and examine their behaviour under extension. Alternative readings are compared through their representations, invariant structures, normalisation conditions and possible dynamics, and a construction in one sector transfers to another only when the necessary map and compatibility conditions are supplied. Known physical theories provide benchmarks: familiar identities and representations are identified as inherited results, and explicit realisations, conditional classifications and obstructions are stated with their assumptions. The research objective is to determine whether compatible structures can be assembled into a common formulation with fewer independent physical inputs.

The role of a common action. Dynamics is introduced through sector-specific actions whose assumptions and scope can be stated explicitly. These actions provide candidate models and consistency tests; their separate existence does not establish a common variational origin. The construction of a universal parent action is deferred while the available representations, kinematic identities and compatibility constraints are investigated. A nearer objective is a common action for a specified compatible subset of sectors, with controlled constraints, causal evolution and physical degrees of freedom. Whether such an action can incorporate all the proposed readings remains open. A satisfactory common formulation would need to address the selection of the observational spacetime sector, the matter and gauge content, the dynamics of the metric and the stability of the resulting physical states. No universal action is claimed in this manuscript.

The governing criterion. Throughout the investigation, the assumptions entering a result are distinguished from its consequences, and mathematical constructions are distinguished from their physical interpretations. Both successful compatibility results and demonstrated obstructions are part of the outcome. The central question is whether the common algebraic setting reduces the number of independent physical assumptions while preserving mathematical consistency. Convergence toward such a formulation is the objective; identifying precisely where proposed combinations fail is equally part of the research.

Related work. The classification of real Clifford algebras and their gradings is classical [85, 88, 125, 103]. Several programmes extract Standard-Model structure from division and Clifford algebras [45, 58, 25], and the representation-theoretic limits of such embeddings are discussed in [34]. Krasnov realises one generation on the forms of a doubled split space [211]. Chapter 14 compares the constructions of this manuscript with these and other approaches, and the comparisons that concern a single statement are made where the statement occurs.

I.1.2 A deflationary principle

One theorem, proved in Section 2.3, governs everything else. Take the four interior gamma matrices $\gamma^\mu = (e_1, e_4, e_5, e_6)$ and the operator $J = e_2e_3$, which commutes with all four and squares to $-\mathbf{1}$. Then the algebra generated over $\mathbb{C}$, with J playing the role of i, by the four gamma matrices is the whole of $M_4(\mathbb{C})$. The real Dirac theory built on $\mathrm{Cl}(3,3)$ is therefore ordinary complex Dirac theory written in real variables, exactly and not in a limit. A change of variables predicts nothing, so a physical claim about the arena carries information only if it could have come out otherwise, that is, only if it adds a constraint beyond a change of notation or a freely chosen embedding. This is a requirement on physical claims, not a definition of mathematical content, and we apply it throughout. Facts that are true of Dirac theory in any notation, such as $g = 2$ at tree level, current conservation or the Ward identity, are *inherited*; they are reproduced here only where they are needed to state something else. The statements we report are of a different kind: which bilinears are admissible, which generators may be gauged, how large the internal centraliser is at each rung, what the faithful gauge algebra of a projected block is, which potentials can select a vacuum, and which constructions are impossible under stated hypotheses.

Every claim in this paper belongs to one of four classes, and we say in the text which one. A result is *fixed by the algebra* when it is an identity or a theorem proved from explicitly stated hypotheses. It is *constructed* when the algebra admits it and we exhibit it, without anything selecting it among alternatives. It is *obstructed* when it is impossible under hypotheses that we state, and those hypotheses are part of the result. It is *open* when it is neither established nor refuted. Three further labels say where a statement comes from or how far it reaches: *inherited* marks a known result of the literature, used with its hypotheses; *restated* marks a result proved earlier in this programme and re-verified here; *interpretive* marks a physical reading of a fixed statement, which the mathematics does not decide. No statement is a prediction of a specified physical model unless it says so. Chapter 13 applies the same classes to the links between results, with five statuses: *established* (fixed, or inherited and used within its hypotheses), *constructed*, *conditional* (holding under a stated assumption, an adopted action or a chosen datum), *obstructed* and *open*.

Every blade of the realisation used here is a signed permutation matrix, so products, commutators, traces and ranks of blades and of their integer combinations are integer computations. Statements with irrational coefficients, such as those of the symmetric clock $(e_1 + e_2 + e_3)/\sqrt{3}$, parameter-dependent spectra, minimisations and numerical ranks are not, and the numerical checks are identified as such (Chapter 15). Most sections have a derivation script and a separately written referee script; two sections were refereed by reading only, and the photon scripts `p1` and `p2` have no referee script. Every script prints one `PASS` or `FAIL` line per check and exits

with status zero only if every check passes. How the statements were verified is described in Section 15.2.

I.1.3 Summary of results

The arena (Sections 1.1, 2.1, 2.3 and 2.5). A bivector $e_a e_b$ squares to $-\mathbf{1}$ when its two directions have the same sign and to $+\mathbf{1}$ when they have opposite signs. The first kind generates a compact $U(1)$, which can be gauged once a connection, a kinetic term and a matter coupling are supplied; the second generates a boost, which is not unitary for a positive form on the module. As operators on the spinor module the first are complex structures and the second have real ± 1 eigenspaces, while conjugation by either kind is an involution of the algebra. The three *pole operators* $e_i e_{i+3}$ are commuting involutions grading the algebra by $\mathbb{Z}_2^3$, and they span a maximally split Cartan subalgebra of $\mathfrak{so}(3,3)$. The pairing has two square roots: a para-complex structure $\mathcal{K}$ with $\mathcal{K}^2 = +\mathbf{1}$, which grades, and a complex structure G with $G^2 = -\mathbf{1}$, which is self-adjoint for η and, as we show later, binds (gives bounded classical orbits of) the interior dynamics and acts as a Hodge star. A clock is a unit vector of the positive triple; the symmetric clock $\widehat{W} = (e_1 + e_2 + e_3)/\sqrt{3}$ is the unique permutation-invariant point and the entropy maximum of the clock simplex under the Fisher–Rao metric, once clocks are described by probability weights; that description is a choice, and every unit clock is related to $\widehat{W}$ by a rotation. Conversion along any single clock is invisible in the equations of motion, so this does not single out $\widehat{W}$. The imaginary unit is selected rather than adjoined: inside the joint commutant of the interior Lorentz action and the internal bivectors, the only square roots of $-\mathbf{1}$ that commute with the gamma matrices are $\pm J$. Charge conjugation is the surplus plane $\langle e_2, e_3\rangle$ and coincides with the Majorana structure; time reversal is forced to square to $-\mathbf{1}$; the kinetic forms alone fix the internal symmetry algebra $\mathfrak{u}(1) \oplus \mathfrak{su}(2)$ of the free massless action, which becomes a gauge algebra only once a connection is introduced and which the mass form cuts to the $\mathfrak{u}(1)$ of J; and for the real representation of the canonical anticommutation relations and the observables constructed in Chapter 2, the quantum state space must be formed by complexifying from outside.

The tower and its matter (Sections 5.1, 5.2 and 6.1 to 6.3). The bivectors commuting with the interior Lorentz action of $\mathrm{Cl}(3n, 3n)$ form $\mathfrak{so}(3n-1, 3n-3)$, nested along the tower, of dimensions $1, 28, 91, 190, \ldots$. At one constituent this centraliser is the single bivector J, the only compact $U(1)$ candidate of the interior structure; identifying it with the photon would need a connection, a kinetic term, a matter coupling and physical degrees of freedom, and the electromagnetic generator is constructed only in Chapter 8. At three constituents we construct an orthogonal projector

$$P_* = \tfrac{1}{8}\,(\mathbf{1} + \Gamma\chi_W)(\mathbf{1} - J\chi_C)(\mathbf{1} - Jq_A)$$

of rank 64 which commutes with the Lorentz action and with J, is axial in the sense that $P_*\gamma^\mu P_* = 0$, keeps the whole of $\mathfrak{su}(4)$ acting faithfully, and carries $(\mathbf{4}, \mathbf{2}, \mathbf{1})_L \oplus (\mathbf{4}, \mathbf{1}, \mathbf{2})_R$: one Pati–Salam family with a right-handed neutrino. Its two central abelian factors cannot be gauged on this matter. Charges built from three commuting involutions are conjugately paired whenever they are anomaly free, so hypercharge cannot be read off from anomaly cancellation inside that class. When the Lorentz action is instead taken diagonally from all three constituents, the internal algebra becomes the entirely compact $\mathfrak{so}(6) \oplus \mathfrak{so}(3)$; the copy algebra and the Lorentz algebra form a reductive dual pair satisfying $C_{\mathrm{cop}} + C_\Delta = \frac{15}{2}$, both inherited [224, 296], and the module is vector-like. A single theorem gives the candidate gauge algebra of every Lorentz action on the tower, and with it a rung-by-rung census of the vector bosons the tower can carry.

Gauge fields and the electromagnetic reading (Sections 3.2, 3.7 to 3.9, 7.1 and 8.1). A varying projector induces a connection whose available curvature algebra is far too large,

so a parent action would have to cut the projector's freedom rather than build a gauge field. The centraliser of the complex structure G in $\mathfrak{so}(3,3)$ is a Lorentz algebra, under which $\mathbb{R}^{3,3}$ *is* $\Lambda^2\mathbb{R}^{1,3}$: boosts act by $\delta\mathbf{E} = \mathbf{v} \times \mathbf{B}$, $\delta\mathbf{B} = -\mathbf{v} \times \mathbf{E}$, G is the Hodge star, and the two invariants $|\mathbf{B}|^2 - |\mathbf{E}|^2$ and $\mathbf{E} \cdot \mathbf{B}$ are the real and imaginary parts of one holomorphic quadratic form. This reading excludes the interior four-plane on which the matter is built. Covariance fixes the form of Maxwell's equations, because the vector module occurs in $\Lambda^2 \otimes V$ with multiplicity exactly two, and fixes nothing else. The photon's spin one appears in two places, related by the Klein correspondence $V \cong \Lambda^2(S^+)^*$, and at the second rung both can be coupled to a Dirac block in a single gauge theory. The orthogonality of $\mathbf{E}$ and $\mathbf{B}$ is the reality of the invariant $q = z \cdot z$; its failure breaks parity and time reversal, is never spontaneous for the Maxwell energy, and does not generate mass. The energy of a field point is one undivided, duality-invariant quantity $E_6 = \frac{1}{2}|z|^2$; its rest part $E_0 = \frac{1}{2}|q|$ is the infimum over boosts, attained for non-null fields only, and $E_6^2 = E_0^2 + |\mathbf{S}|^2$ is the field-point form of $E^2 = p^2c^2 + m^2c^4$, an identity of a field point and not a particle mass; c enters only as the conversion between the two triples. $\mathrm{Spin}(3,3)$ keeps the Dirac mass form, so a mass breaks no spacetime symmetry, and particle and antiparticle are the two senses of one rest rotation.

Interior dynamics (Sections 4.2 to 4.5 and 9.1). The Lagrangian of a single shifted operator $\partial_\tau + \kappa g$ factorises exactly when the shift is symmetric, and then it describes an inverted oscillator; for every constant shift some solution is unbounded, and with a positive form no solution with nonzero momentum recurs. In the split form the same factorisation admits bound motion, and in the factorised shift model it does so exactly in neutral signature, with the complex structure G as the shift. The force law is a dual transport which is a Lie–Poisson equation only on an extension of the frame; within the closure ansatz that extension is unique, its rotor angle is the duality angle of the momentum, and its free motion is a pendulum that drifts the frame except on null data. The rest energy sets the amplitude of that pendulum; a constant rate and a bounded frame coexist only on null data or on the balance surface of several constituents, and identifying the rest rotation of the Dirac field with the rate of the closure, $mc^2 = \hbar\kappa$, is an added hypothesis. The antisymmetric part of a shift is a connection: the part commuting with the binding root is removed by a moving frame, and no connection binds on its own.

The vacuum (Sections 7.3 to 7.7). At one constituent we classify the vacuum of every coordinate three-plane in four senses. At three constituents the space of Lorentz-invariant mass kernels of the chiral block has dimension 272 and, with the complex structure $\widetilde{\mathcal{C}}$, decomposes as $(\overline{\mathbf{10}}, \mathbf{3}, \mathbf{1}) \oplus (\mathbf{10}, \mathbf{1}, \mathbf{3}) \oplus (\mathbf{15}, \mathbf{2}, \mathbf{2}) \oplus 2(\mathbf{6}, \mathbf{1}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{2}, \mathbf{2})$; the Standard-Model invariant kernels form one complex line, the right-handed neutrino Majorana kernel. We give the complete renormalizable potential of the $(\mathbf{10}, \mathbf{1}, \mathbf{3})$ component, including a holomorphic quartic that violates baryon number by two units, and sufficient conditions, together with separate necessary conditions, under which the Standard-Model orbit is its global minimum (the two do not coincide, and numerics support, but do not locate exactly, the transition between them); within the adopted leading-logarithm model we compute the radiatively generated spectrum of all 271 angular modes, the Goldstone content and the discrete symmetries of the vacuum, and, for a given bidoublet value in the kernel model, an electroweak stage with the Georgi–Jarlskog factor and a seesaw.

Curvature and the photon (Sections 8.5 and 11.1 to 11.3). In a curved arena the pairing becomes an almost para-Hermitian structure. An explicit six-dimensional metric with a normal $U(1)$ has a closed-form Einstein tensor whose tangential block on the four-dimensional slice is the Einstein tensor of the slice, while relative to the ambient Einstein–Hilbert action and the stated ansatz the normal field strength gravitates with negative energy, as the signature forces. In the field-strength reading the curvature operator of a four-metric lives on the frame,

the first Bianchi identity is the single condition $\mathrm{Tr}(G\mathcal{R}) = 0$, and the duality rotation of the curvature ends at the double dual, which contracts to the Einstein tensor. These are identities of given metrics; no gravitational action is derived. The projector current sources a massive normal vector, and a massless photon requires a vacuum that commutes with the charge; such a vacuum exists at the second rung and not at the first. At the third rung the normal charge is half the fermion number on the chiral block and anomalous, and the electric charge is the photon candidate. In this projector model the selected vacuum is electroweak symmetric and electroweak breaking is not obtained; the electroweak stage of the kernel model belongs to a different model, and a common realisation of the two is not established.

Causal structure (Sections 12.1 to 12.4). For unrestricted evolution, a field propagating freely in six dimensions of signature $(3, 3)$ is ill-posed and has no ground state, and its time-like cone has no past and future; constrained ultrahyperbolic problems can be well posed [27]. Choosing a Lorentzian four-plane inside $\mathbb{R}^{3,3}$ is the same as choosing its definite orthogonal two-plane, represented by a normalised simple Clifford bivector whose two signs encode an orientation, and that bivector is a complex structure, and at one constituent the light cone of that plane is the set of chiral spinors squared with the help of J. For an unconstrained adjoint scalar, no bounded-below invariant polynomial potential of degree below eight isolates that choice, and every invariant bilinear kinetic term has ghost directions. The surplus-time constraints follow from a gauge symmetry of the worldline that is chosen, not derived, and is shown for a test worldline only.

I.1.4 Conventions

Spacetime signature is $(+, -, -, -)$, units are Heaviside–Lorentz with $\hbar = c = 1$ unless stated. A *constituent* is a tensor factor $\mathrm{Cl}(3, 3)$ of the tower; the word never refers to a particle, and three constituents do not describe three particles. Anomaly coefficients are chiral traces per Weyl component with every field written left-handed. In the field-strength reading of $\mathbb{R}^{3,3}$ the vector $X = \sum_i (B_i e_i + E_i e_{i+3})$ is identified with the two-form F of components $F_{0i} = E_i$, $F_{jk} = \epsilon_{ijk} B_i$; with this orientation $\mathbf{B}$ is minus the textbook magnetic field, which we write $\mathbf{B}_{\mathrm{t}} = -\mathbf{B}$, while $\mathbf{E}$ is the textbook electric field. Formulas taken from the literature are written in textbook variables, and where a sign matters the variables are named. Compact Lie algebras are realised by skew operators and the corresponding charges by the symmetric operators $\hat{t} = -Jt$.

I.1.5 Scope

The manuscript is a mathematical study of one algebraic arena and of the constructions it admits. It contains: the representation theory of $\mathrm{Cl}(3n, 3n)$ relevant to Lorentz structures, complex units, centralisers and projected blocks; two readings of $\mathbb{R}^{3,3}$, as displacements and as field strengths, with their Lagrangians and energies; an interior dynamics and its rigid-body form; the chiral block, its charges and its mass kernels, with a conditional vacuum selection; the identification of a photon candidate; dynamics between constituents; classical and observational consistency checks of stated readings; curvature identities; and the causal structure of the arena. It does not contain a derivation of gravitational dynamics, a derived action for any sector, a spectral-triple formulation, quantised interacting fields, or predictions of a specified physical model. Where a statement is a reading rather than a theorem, it is marked interpretive. The colour and Standard-Model representation results are kept; no work on a gluon identification is included.

symbol	meaning	chapter
$V = \mathbb{R}^{3,3}$, η, $V^{\pm}$	the arena, $\eta = \mathrm{diag}(1,1,1,-1,-1,-1)$, temporal and spatial triples	1
$e_1, \dots, e_6$; $\mathrm{Cl}(3n,3n)$, S_n	generators; tower algebra on $S_n = (\mathbb{R}^8)^{\otimes n}$, n the number of constituents	1, 5
$\omega_i = e_i e_{i+3}$, ϖ	pole operators, their sum	1
$\mathcal{K}$, G	grading root ($\mathcal{K}^2 = \mathbf{1}$); binding root ($G^2 = -\mathbf{1}$, η-self-adjoint)	1, 3
$T = e_1 e_2 e_3$	temporal volume, the Dirac mass form	2
J, $J(\hat{n})$; $\Gamma = J\Omega$	complex structure (imaginary unit); chirality	2, 6
$\hat{n}$, $\widehat{W}$, $E_{\hat{n}}$	clock, symmetric clock, observable four-plane $\mathbb{R}\hat{n} \oplus V^-$	2
$\mathfrak{l}_{\mathrm{int}}$, $\mathfrak{l}_G$	interior Lorentz algebra; Lorentz algebra centralising G	2, 3
$\mathbf{E}$, $\mathbf{B}$, z, q; E_6, E_0	fields ($\mathbf{B} = -\mathbf{B}_{\mathrm{t}}$), $z = \mathbf{B} + i\mathbf{E}$, $q = z \cdot z$; undivided and rest energy	3
$\Theta^{\mu\nu}$	stress–energy tensor	3
$\kappa = c/\ell$, τ	rate and parameter of the interior dynamics	4, 12
M_6, $m_4(\hat{n})$, m	six-dimensional mass, mass read by a clock, Dirac mass (an input)	3, 10
P_*, V_*; P_{fam}	chiral projector and matter block; family projector	6
$\mathcal{M}$, $\widetilde{\mathcal{C}}$, Φ	kernel space of mass kernels, its complex structure, the kernel field	7
Π, $\Pi_{\pm A}$, $\Pi_{\pm W}$; B_{em}, Q_A	vacuum projectors and aligned points; electric and normal charges	8
$\mathcal{R}$, $\mathcal{G}$	curvature operator on the frame, Einstein tensor	11

Table 1: Symbols with one meaning throughout the manuscript.

I.1.6 Notation

Table 1 lists the symbols used throughout. A symbol in the table has one meaning in the whole manuscript; symbols local to a chapter are defined there, and those of the matter, vacuum and photon chapters are collected in Table B.2. Units of the two readings are in Table B.1.

I.1.7 Organisation and dependencies

Chapters 1 to 3 develop one constituent: the algebra, the Lorentz structures and complex units with the clock reduction, and the two readings of $\mathbb{R}^{3,3}$. Chapter 4 treats the interior dynamics and its rigid-body realisation. Chapters 5 to 8 climb the tower: duplication, the chiral block, the mass kernels and the vacuum, and the photon. Chapters 9 and 10 treat the dynamics between constituents and the classical and observational readings. Chapters 11 and 12 pass to curvature and causal structure. Chapter 13 holds the sectors against one another, Chapter 14 compares with related approaches, Chapter 15 describes the AI-assisted workflow and the verification, and Chapter 16 concludes. Appendix A.1 gives the explicit matrices, so that every finite-dimensional claim can be checked by hand; the appendices also hold the dictionaries, the branching tables and the verification manifest. Figure 1 shows which chapters each chapter relies on.

I.2 Historical motivation: the rest-energy hypothesis

Part of the programme began with a proposed reading of the rest-energy relation, quoted below. What survived the tests is stated first; the assessment is in Sections 13.5 and 13.6. In the field-strength reading the undivided energy of a field point is the duality-invariant E_6; its rest part $E_0 = \frac{1}{2}|q|$ is the infimum of E_6 over boosts, and $E_6^2 = E_0^2 + |\mathbf{S}|^2$ is an inherited identity of a field point, not a particle mass. Schur's lemma fixes the isotropic form of the conversion

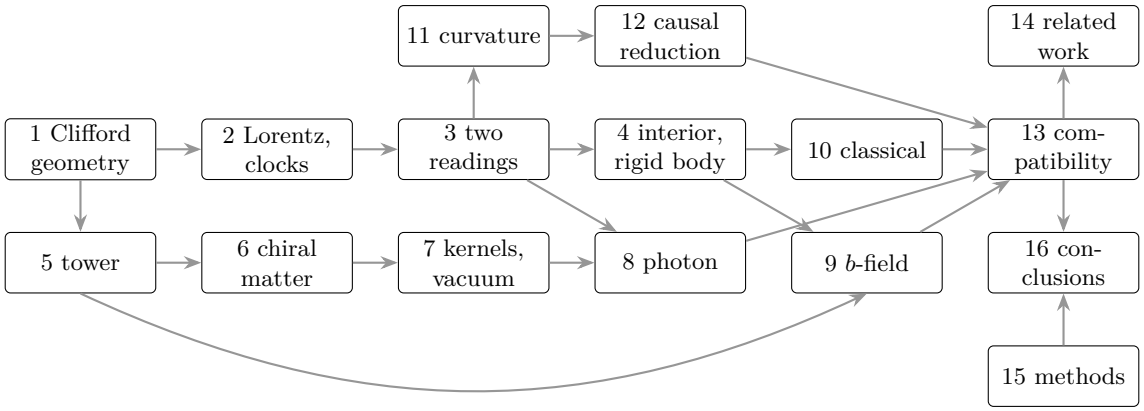


Figure 1: Dependencies between chapters: an arrow from one chapter to another means that the second uses results of the first. Arrows implied by two others are omitted, so Chapter 11, for instance, also uses Chapter 2. Chapter 13 draws on all earlier chapters, and Chapter 15 describes the evidence for every chapter.

between the two triples, not the value of c, and c is a conversion, not a symmetry. A mass breaks duality and chirality but no spacetime symmetry. The b-field splits or destabilises the binding but never creates a rate or a mass. None of this derives a mass, and the dictionary below consists of readings under test, not results.

I.2.1 The hypothesis in its author's words

One strand of the present research programme begins with a proposed interpretation of the rest-energy relation:

> *The mass–energy relation $E = mc^2$ is to be regarded as a unified expression comprising two complementary components: c^2, representing the symmetric component, and m, representing the symmetry-breaking component, hypothesized to arise through the b-particle associated with the Yang–Mills b-field.*

Three clarifications fix the intended reading. First, the energy is to be taken as *one undivided entity*, and the mass is to emerge through an *internal differentiation* of that unity, before any external interaction is applied. Second, c^2 corresponds to the variable X of the interior dynamics of Chapter 4, where the shift operators are $\alpha_\pm = \partial_\tau \pm \kappa G$ with $\kappa = c/\ell$. Third, the energy is six-dimensional. The b-field is the non-abelian gauge potential of Yang and Mills [172], whose quanta carry isotopic spin; Yang and Mills left open whether those quanta are massive.

Several points are settled at once. If E and c^2 are invariant under a group then so is $m = E/c^2$, so a mass cannot *be* the breaking of a symmetry that E and c^2 share. Positive four-dimensional mass cannot be obtained from momentum along the surplus times, since $m_4^2 = \mu_6^2 - (p_u^2 + p_v^2)/c^2$ (Proposition 3.31), and non-orthogonality of $\mathbf{E}$ and $\mathbf{B}$ generates no mass (Section 3.8). The assessment follows the order the hypothesis asks for: the undivided energy, its differentiation, the dynamics of the differentiation, and the b-field last.

I.2.2 The second question: duplication, and the b-field between manifolds

After Sections 3.9, 4.5, 7.1 and 9.1 had established that, on one manifold, the b-field can split or destabilise the binding but never generate it, a further proposal of the author defines the second half of this strand: *the b-particle, induced by the b-field through the interaction of two or more manifolds, is proposed as the mechanism enabling the energy cascade across successive Clifford-algebraic structures, from* $\mathrm{Cl}(3,3)$ *to* $\mathrm{Cl}(6,6)$ *and beyond.* This is a hypothesis, not a

result. We read it as two questions: whether a b-field acting *between* copies of the arena can do what it cannot do on one copy, and which constructions and processes take the undivided energy of one copy to two, and what the duplicate can do that a single copy cannot. Section 5.3, Section 5.6, Section 9.3 answer them and Section 13.6 collects the answers.

I.2.3 The dictionary under test

All constructions are made inside the arena of Chapter 1: $V = \mathbb{R}^{3,3}$ with $\eta = \mathrm{diag}(1,1,1,-1,-1,-1)$, the realisation of $\mathrm{Cl}(3,3)$ on $\mathbb{R}^8$, the binding root G ($Ge_i = e_{i+3}$, $Ge_{i+3} = -e_i$), which squares to $-\mathbf{1}$, is self-adjoint for η and acts as the Hodge star in the field-strength reading $X = \sum_i (B_i e_i + E_i e_{i+3})$, $z = \mathbf{B} + i\mathbf{E}$, $q(z) = z \cdot z$ (Theorem 3.3). The working dictionary, whose entries are conjectural readings, is

hypothesis	candidate object in the arena
E, undivided, six-dimensional	the positive, duality-invariant energy $E_6 = \frac{1}{2}\lvert z\rvert^2 = \frac{1}{2}(\lvert\mathbf{B}\rvert^2 + \lvert\mathbf{E}\rvert^2)$ of a frame vector; in the closure of the force law, the Casimir $\lvert p\rvert^2$
c^2, the symmetric part	the isotropic form $C = c\,\mathbf{1}$ of the conversion between the temporal and the spatial triple, fixed by Schur's lemma, which does not fix the value of c (Theorem 2.2), and the η-self-adjoint, factorising part κG of the shift, $(\kappa G)^2 = -c^2/\ell^2$
m, the differentiation	the Lorentz-class invariant $E_0 = \frac{1}{2}\lvert q\rvert$, the rest-frame energy density of the field; for Dirac matter the mass form $T = e_1e_2e_3$
b, the Yang–Mills field	the η-antisymmetric (transport) part of the shift, $\mathrm{End}(V) = \mathfrak{so}(3,3) \oplus \{X = X^\dagger\}$ (Theorem 4.18); at three constituents the binding-compatible algebra $\mathfrak{so}(4)_\Delta \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$ (Theorem 4.36)

Every entry is an interpretive reading to be tested, not a result; Tables 13.4 and 13.5 give the verdicts. The claims of this strand are classified as in Section I.1.2. The relation $E = mc^2$ itself is inherited; what can be new is only how the arena realises its three ingredients, and which of the realisations are forced.

I.3 Use of AI assistance

This work was developed by the author over roughly three years in an iterative process with AI systems. The questions and the programme are the author's. Claude models assisted extensively with candidate theorems, exploratory proofs, alternative constructions, code, calculations and drafting; ChatGPT assisted with discussion, methodology and referee-style criticism. The author examined, objected to and corrected the results throughout. No particular model version was in use over the whole period. AI-assisted review is not peer review, and agreement between models is not proof: each statement of this manuscript carries its own evidence, a proof, an exact computation, a numerical check with its tolerance, or a citation, as described in Chapter 15 and listed in Appendix E. The author is responsible for the manuscript and its conclusions.

Chapter 1

Split-Signature Clifford Geometry

This chapter fixes the algebra and makes no physical reading, with two forward references: Proposition 1.10 uses the Fisher–Rao clock of Chapter 2, and (1.6) writes a vector in the electric and magnetic components of Chapter 3. We work with the real Clifford algebra $\mathrm{Cl}(3,3) \cong M_8(\mathbb{R})$ of $\mathbb{R}^{3,3}$ in an explicit realisation, with the blade rules and the signature rule, which separates compact bivectors from boosts. The trace form on an orbit splits by the Cartan involution (Theorem 1.13); for actions built on this trace metric, this decides the sign of the kinetic terms later on. The spinor module is then described as the forms on a null three-plane, with its octants and Fock grading, and the tower $\mathrm{Cl}(3n,3n)$ is defined. Every statement is algebra, proved in the text and checked exactly (Appendix E). Chapter 2 introduces the Lorentz structures and the clock; Chapter 3 the two readings of a vector of the arena.

1.1 The split space and its Clifford algebra

1.1.1 A fixed realisation

Let $V = \mathbb{R}^{3,3}$ with orthonormal basis $e_1, \dots, e_6$, $e_a^2 = \eta_{aa}$ and $\eta = \mathrm{diag}(1,1,1,-1,-1,-1)$. We call $V^+ = \langle e_1, e_2, e_3\rangle$ the temporal triple and $V^- = \langle e_4, e_5, e_6\rangle$ the spatial triple. The Clifford algebra $\mathrm{Cl}(3,3)$ is generated by the e_a subject to $e_a e_b + e_b e_a = 2\eta_{ab}\mathbf{1}$; it is isomorphic to $M_8(\mathbb{R})$, and we fix once and for all its realisation on $S = \mathbb{R}^8 = (\mathbb{R}^2)^{\otimes 3}$,

$$\begin{aligned} e_1 &= \sigma_x \otimes \mathbf{1} \otimes \mathbf{1}, \quad & e_2 &= \sigma_z \otimes \sigma_x \otimes \mathbf{1}, \quad & e_3 &= \sigma_z \otimes \sigma_z \otimes \sigma_x, \\ e_4 &= \varepsilon \otimes \mathbf{1} \otimes \mathbf{1}, \quad & e_5 &= \sigma_z \otimes \varepsilon \otimes \mathbf{1}, \quad & e_6 &= \sigma_z \otimes \sigma_z \otimes \varepsilon, \end{aligned} \tag{1.1}$$

with $\sigma_x = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $\sigma_z = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$ and $\varepsilon = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. All six matrices are signed permutation matrices. For a subset $I = \{a_1 < \cdots < a_r\}$ we write $e_I = e_{a_1} \cdots e_{a_r}$ for the corresponding coordinate blade of grade r; the 64 blades form a basis of $M_8(\mathbb{R})$.

The tower is built by graded tensor products. At n constituents the spinor module is $S_n = (\mathbb{R}^8)^{\otimes n}$ and the generators of the k-th constituent are

$$e_a^{(k)} = \underbrace{\omega \otimes \cdots \otimes \omega}_{k-1} \otimes e_a \otimes \mathbf{1} \otimes \cdots \otimes \mathbf{1}, \qquad \omega := e_1 e_2 \cdots e_6. \tag{1.2}$$

Since $\omega^2 = \mathbf{1}$ and ω anticommutes with every e_a, generators of different constituents anticommute, and the $6n$ matrices $e_a^{(k)}$ generate $\mathrm{Cl}(3n,3n) \cong M_{8^n}(\mathbb{R})$. When we need a single index we write $E_{6(k-1)+a} = e_a^{(k)}$, so that at three constituents the eighteen generators are $E_1, \dots, E_{18}$ and the temporal ones are E_A with $A \in \{1,2,3,7,8,9,13,14,15\}$.

Two elementary rules carry most of the finite-dimensional arguments of this paper.

Lemma 1.1 (Blade rules)**.** *For coordinate blades e_I, e_J of grades $|I|, |J|$,*

$$e_I e_J = (-1)^{|I||J| - |I\cap J|}\, e_J e_I, \tag{1.3}$$

and for a coordinate blade B of grade r

$$B^\top = (-1)^{r(r-1)/2}\Big(\prod_{a\in I}\eta_{aa}\Big)B = B^2B. \tag{1.4}$$

In particular a coordinate blade is antisymmetric if and only if it squares to $-\mathbf{1}$, and symmetric if and only if it squares to $+\mathbf{1}$.

Proof. For (1.3), move each generator of e_J past the generators of e_I: it anticommutes with the $|I|$ of them it differs from and commutes with itself when it occurs in I, which gives the exponent. For (1.4), the matrices σ_x, σ_z are symmetric and ε is antisymmetric, so each e_a satisfies $e_a^\top = \eta_{aa}e_a$; transposing a product reverses it, and restoring the order costs $(-1)^{r(r-1)/2}$. The same two factors give $B^2 = (-1)^{r(r-1)/2}\prod_{a\in I}\eta_{aa}\,\mathbf{1}$, so $B^\top = B^2B$. □

Example 1.2. The bivector e_2e_3 has $(e_2e_3)^2 = -e_2^2e_3^2 = -\mathbf{1}$, so it is antisymmetric and $\exp(\theta e_2e_3) = \cos\theta\,\mathbf{1} + \sin\theta\,e_2e_3$ is a rotation of period 2π. The bivector e_1e_4 has $(e_1e_4)^2 = -e_1^2e_4^2 = +\mathbf{1}$, so it is symmetric and $\exp(\theta e_1e_4) = \cosh\theta\,\mathbf{1} + \sinh\theta\,e_1e_4$ is a boost, unbounded in θ. These two bivectors share no index, $|I||J| - |I\cap J| = 4-0 = 4$ is even, and they commute; so do e_2e_3 and e_5e_6. The boost e_1e_4 and the rotation e_1e_2 share the index 1, $|I||J| - |I\cap J| = 4-1 = 3$ is odd, and they anticommute.

1.1.2 The signature rule

Theorem 1.3 (The signature rule for coordinate bivectors)**.** *For $a \neq b$, $(e_ae_b)^2 = -e_a^2e_b^2$. A bivector therefore squares to $-\mathbf{1}$ when its two directions have the same sign and to $+\mathbf{1}$ when they have opposite signs, and by Lemma 1.1 the first kind are exactly the antisymmetric bivectors. The three time–time and three space–space bivectors span the maximal compact subalgebra $\mathfrak{so}(3)\oplus\mathfrak{so}(3) \subset \mathfrak{so}(3,3)$, and the nine mixed bivectors are boosts.*

Corollary 1.4 (A mixed bivector generates no compact $U(1)$)**.** *For a mixed bivector B, $\exp(\theta B) = \cosh\theta\,\mathbf{1} + \sinh\theta\,B$ generates $\mathbb{R}$ and not $U(1)$: it carries no phase and defines no compact charge.*

The obstruction to a boost as a gauge generator lies in its action on the module, not in the kinetic metric of the gauge field. For an abelian algebra that metric decides nothing: on $\mathbb{R}B$ every scalar product is invariant, and $A_\mu = a_\mu B$ has the ordinary, healthy abelian kinetic term. What fails is unitarity. No positive-definite H on S satisfies $B^\top H + HB = 0$, since an eigenvector $Bv = \pm v$ would give $v^\top Hv = 0$. A boost is therefore not unitary for any fixed positive form on the module, and the gauge form $-\operatorname{Tr}$ has the wrong sign on it. Relative to the chosen positive matter metric H, we call an internal generator X *gauge-admissible* when $X^\top H + HX = 0$; in the orthonormal realisation used here, $H = \mathbf{1}$, this is $X^\top = -X$, that is, X is compact. This is the sense in which the corollary is used later (Theorem 5.39, Chapters 9 and 13), and the conclusions drawn there hold under it. Allowing an indefinite invariant matter form removes this positive-metric obstruction. Whether physical ghosts then occur depends on the chosen action and its constraint reduction (an invariant constraint can remove the negative direction); an unconstrained quadratic kinetic term with propagating negative-sign directions is unacceptable. The present model therefore imposes a fixed positive matter metric and restricts its internal gauge generators accordingly.

The criterion that matters for a gauge generator is antisymmetry with respect to the positive form on the module that a kinetic term supplies, equivalently membership of the $+1$ eigenspace of the Cartan involution $\theta(X) = -X^\top$ [80, 68]. For a single blade this coincides with $X^2 = -\mathbf{1}$ by (1.4), which is why the rule is decisive for bivectors. For general elements it is not: $\operatorname{diag}(J_2, 2J_2)$, with J_2 the standard 2×2 rotation generator, generates a group of period 2π although its square is $\operatorname{diag}(-\mathbf{1}, -4\mathbf{1})$.

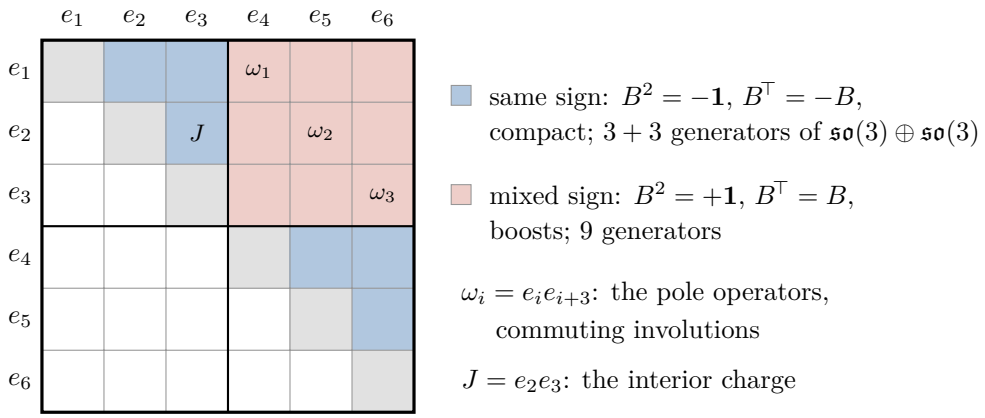


Figure 1.1: The fifteen bivectors $e_a e_b$, $a < b$, of $\mathrm{Cl}(3,3)$. Blue cells pair two directions of the same sign and generate compact rotations; red cells pair a temporal with a spatial direction and generate boosts (Theorem 1.3). The diagonal of the red block holds the three pole operators, which grade the algebra (Proposition 1.5); the cell $(2,3)$ holds the bivector J, the compact rotation that commutes with the interior Lorentz algebra (Proposition 1.7) and is used as the imaginary unit in Chapter 2. It is not identified with electromagnetism: the electromagnetic generator is constructed in Chapter 8.

1.1.3 The pole operators and a multigrading

The mixed bivectors are not idle. The three *pole operators*

$$\omega_i := e_i e_{i+3} \qquad (i = 1, 2, 3)$$

are boosts, so $\omega_i^2 = +\mathbf{1}$, and they commute pairwise by (1.3), since any two share no index. As operators on S the two kinds of bivector differ: one of square $-\mathbf{1}$ is a complex structure, while each ω_i has real ± 1 eigenspaces, which grade the module. By conjugation on the algebra either kind induces an involution: conjugation by $J = e_2 e_3$, of square $-\mathbf{1}$, has ± 1 eigenspaces of dimension 32 each.

Proposition 1.5 ($\mathrm{A}\ \mathbb{Z}_2^3$ grading of $\mathrm{Cl}(3,3)$)**.** *For a blade e_I put $\chi_i(e_I) = (-1)^{|I \cap \{i, i+3\}|}$. Then $\omega_i e_I \omega_i^{-1} = \chi_i(e_I)\, e_I$, and $\mathrm{Cl}(3,3) = \bigoplus_{\sigma \in \{\pm\}^3} \mathrm{Cl}_\sigma$ is a multiplicative grading with eight sectors of dimension eight.*

(i) The joint spectrum of $(\omega_1, \omega_2, \omega_3)$ on S consists of all eight sign triples, each on a line, so the pole operators cut $\mathbb{R}^8$ into eight one-dimensional eigenlines.

(ii) The vectors occupy the three weights with exactly one minus sign, two each (e_i and e_{i+3}); the bivectors occupy the four even weights with multiplicities $3, 4, 4, 4$; the all-minus weight is spanned by eight trivectors, namely $e_1 e_2 e_3$, $e_4 e_5 e_6$ and six mixed ones.

(iii) Correspondingly $\mathfrak{so}(3,3) = \mathfrak{g}_0 \oplus \mathfrak{g}_1 \oplus \mathfrak{g}_2 \oplus \mathfrak{g}_3$ is graded by the Klein four-group, of dimensions $3 + 4 + 4 + 4$, with $\mathfrak{g}_0 = \mathrm{span}\{\omega_1, \omega_2, \omega_3\}$ abelian, self-normalising and consisting of semisimple elements with real eigenvalues: a maximally split Cartan subalgebra of $\mathfrak{so}(3,3) \cong \mathfrak{sl}(4, \mathbb{R})$.

Proof. The conjugation sign is (1.3) with $|J| = 2$: $(-1)^{2|I| - |I \cap \{i, i+3\}|}$. The weight is multiplicative because $|I \triangle I'|$ and $|I| + |I'|$ have the same parity in each pair. The census in (ii) is a count of how many generators of each pair a blade contains. For (iii), a bivector inside one pair is a pole operator, of weight $(+,+,+)$; a bivector with one generator in each of two pairs has two minus signs, and each of the three such weights holds $2 \times 2 = 4$ bivectors. The pole operators commute, are symmetric with eigenvalues ± 1, and the centraliser of $\mathfrak{g}_0$ in $\mathfrak{so}(3,3)$ is $\mathfrak{g}_0$ itself,

because every other bivector has a nontrivial weight; its dimension equals the real rank three of $\mathfrak{so}(3,3)$. □

The three characters are grading weights, not metric signatures: e_1 and e_4 carry the same weight and opposite squares, so nothing in the grading tells a temporal from a spatial direction. The decomposition in (iii) is the root decomposition of $\mathfrak{sl}(4,\mathbb{R})$ with respect to a split Cartan subalgebra, grouped by the coordinate parity of the roots, $Q(D_3)/2\mathbb{Z}^3$, a group of order four (not the root lattice modulo twice itself, $Q(D_3)/2Q(D_3)$, which has order eight); gradings of Clifford and classical Lie algebras by $\mathbb{Z}_2^n$ are well known [103], and we use this one because the pole operators will reappear in the field-strength reading as the three paired planes (e_i, e_{i+3}), $i = 1, 2, 3$, that carry one complex component each.

Example 1.6 (The paired-mode projector)**.** Let $\varpi = \omega_1 + \omega_2 + \omega_3$ be the sum of the pole operators and $\omega = e_1e_2\cdots e_6$ the volume element of (1.2). On the joint eigenline of signs $(\varepsilon_1, \varepsilon_2, \varepsilon_3)$, $\varpi = \varepsilon_1 + \varepsilon_2 + \varepsilon_3 \in \{3, 1, -1, -3\}$ with multiplicities $1, 3, 3, 1$. Reordering $\omega_1\omega_2\omega_3 = e_1e_4e_2e_5e_3e_6$ into $e_1\cdots e_6$ costs three transpositions, so $\omega = -\omega_1\omega_2\omega_3 = -\varepsilon_1\varepsilon_2\varepsilon_3$ on that line. Hence $\frac{1}{2}(\varpi + \omega)$ takes the values $1, 1, -1, -1$ on the four strata: it is the sign of ϖ, and equals the polynomial $(13\varpi - \varpi^3)/12$. The operator

$$\Pi := \tfrac{1}{2}\mathbf{1} - \tfrac{1}{4}(\varpi + \omega) \tag{1.5}$$

is therefore the orthogonal projector of rank four onto the strata with $\varpi < 0$. We call it the paired-mode projector; it is a natural candidate for an observation projector at one constituent and we test it against the photon in Section 8.5. It is not Lorentz invariant: of the six interior Lorentz generators $e_1e_4, e_1e_5, e_1e_6, e_4e_5, e_4e_6, e_5e_6$ only e_1e_4 commutes with Π (the commutators have ranks $0, 4, 4, 4, 4, 4$), and $[\Pi, J]$ has rank four. Its stabiliser in $\mathfrak{so}(3,3)$ is nine-dimensional, a count of linear combinations, not of commuting basis bivectors.

1.1.4 The interior four-plane and the surplus plane

The physical reading of the arena singles out a Lorentzian four-plane inside V. We take

$$E = \langle e_1, e_4, e_5, e_6\rangle, \qquad \gamma^0 = e_1, \quad \gamma^a = e_{a+3}\ (a = 1, 2, 3),$$

of signature $(1, 3)$, and call the orthogonal complement $U = \langle e_2, e_3\rangle$ the surplus plane. The interior Lorentz algebra $\mathfrak{l}_{\text{int}}$ is spanned by the six bivectors of E, with rotations $S_1 = \frac{1}{2}\gamma^2\gamma^3$ (cyclically) and boosts $T_a = \frac{1}{2}\gamma^0\gamma^a$. Its spin Casimir on S is $-\sum_a S_a^2 = \frac{3}{4}\mathbf{1}$, and a rotation through 2π is $-\mathbf{1}$ on S while it is $+\mathbf{1}$ on vectors (Section 3.7): the same rotors give spin one-half to spinors and spin one to vectors.

Proposition 1.7 (The interior charge is unique)**.** *As $\mathfrak{l}_{\text{int}}$-modules, $\Lambda^2 V = \Lambda^2 E \oplus (E \otimes U) \oplus \Lambda^2 U$, of dimensions $6 + 8 + 1$, is the adjoint representation, two copies of the vector representation, and the trivial representation. The centraliser of $\mathfrak{l}_{\text{int}}$ in $\mathfrak{so}(3,3)$ is therefore the line spanned by*

$$J := e_2e_3,$$

which satisfies $J^2 = -\mathbf{1}$, $J^\top = -J$, and commutes with each of the four gamma matrices.

Proof. The adjoint and vector representations of $\mathfrak{so}(1,3)$ contain no invariant, so only $\Lambda^2 U$ survives. That J commutes with e_1, e_4, e_5, e_6 is (1.3) with disjoint index sets and $|I||J| = 2$. □

Proposition 1.7 is the first instance of a pattern that recurs throughout: a structure is fixed once the right ambient algebra and the right conditions are named, and it ceases to be fixed when either is relaxed. Here the ambient space is the span of the bivectors and the condition is commutation with $\mathfrak{l}_{\text{int}}$. In Section 2.3 we enlarge the ambient algebra to all of $\mathrm{End}_{\mathbb{R}}(S)$ and find that a different condition, commutation with the internal bivectors, is what keeps the answer unique.

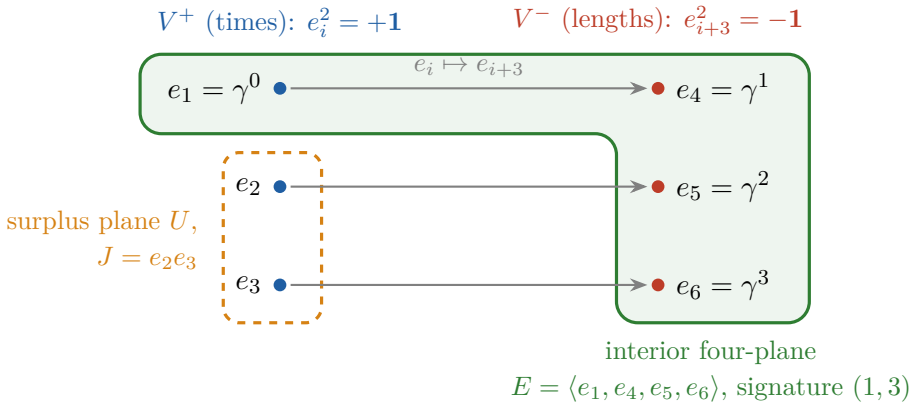


Figure 1.2: The arena. The positive triple V^+ (times) and the negative triple V^- (lengths) are paired by $e_i \mapsto e_{i+3}$; the pairing has two square roots, the grading $\mathcal{K}$ and the binding operator G (Proposition 2.3). The interior four-plane E, on which the Dirac sector is built, contains the clock e_1 and the whole spatial triple; its complement is the surplus plane $U = \langle e_2, e_3 \rangle$, whose rotation generator J is the only bivector commuting with the interior Lorentz algebra (Proposition 1.7).

1.2 The spinor module as forms

The octants and the octant group

Proposition 1.8 (Octants and Fock grading). *(i) The pole operators are $\omega_i|b\rangle = (-1)^{b_i}|b\rangle$ on the basis $|b_1 b_2 b_3\rangle$ of S; their joint eigenlines, the* octants, *are the eight basis lines, labelled by the sign triples $\sigma = (\sigma_1, \sigma_2, \sigma_3)$, $\sigma_i = (-1)^{b_i}$.*

(ii) $a_i^{\pm} := \frac{1}{2}(e_i \pm e_{i+3})$ satisfy $\{a_i^+, a_j^+\} = \{a_i^-, a_j^-\} = 0$, $\{a_i^+, a_j^-\} = \delta_{ij}$, and $\omega_i = \mathbf{1} - 2a_i^+ a_i^-$. Hence $N := \sum_i a_i^+ a_i^-$ is diagonal and counts the minus signs of σ.

(iii) $|\mathrm{vac}\rangle := |000\rangle$, of signs $(+,+,+)$, spans the common kernel of the a_i^- (the Fock vacuum, $N = 0$), and $a_{i_1}^+ \cdots a_{i_r}^+ |\mathrm{vac}\rangle = |b_I\rangle$ for $i_1 < \cdots < i_r$, where b_I has ones exactly at I. The filled pole is $|\mathrm{full}\rangle := |111\rangle$, of signs $(-,-,-)$ and $N = 3$.

(iv) The octant σ is the pure spinor line of the maximal isotropic plane $N_\sigma := \mathrm{span}\{e_i - \sigma_i e_{i+3}\}$: the vectors $v \in V$ with $v|b\rangle = 0$ are exactly those of N_σ.

(v) $\Gamma = \omega = -\omega_1\omega_2\omega_3$, so $(-1)^N = \omega_1\omega_2\omega_3 = -\Gamma$; the element Γ is, up to the sign fixed by the choice $i = \pm J$, the chirality γ_5 of Section 2.5. On the octant with $w = 3 - N$ plus signs, $\Gamma = (-1)^w$.

Proof. (i) is in Proposition 1.5 and Appendix A.1: $e_i|b\rangle = (-1)^{b_1+\cdots+b_{i-1}}|b \oplus \delta_i\rangle$ and $e_{i+3}|b\rangle = (-1)^{b_1+\cdots+b_i}|b \oplus \delta_i\rangle$. (ii) The Clifford relations give $\{e_i \pm e_{i+3}, e_j \pm e_{j+3}\} = 2(\delta_{ij} - \delta_{ij}) = 0$ and $\{e_i + e_{i+3}, e_j - e_{j+3}\} = 4\delta_{ij}$; then $\omega_i = (a_i^+ + a_i^-)(a_i^+ - a_i^-) = a_i^- a_i^+ - a_i^+ a_i^- = \mathbf{1} - 2a_i^+ a_i^-$. (iii)/(iv) By the two formulas of (i), $(e_i - s\, e_{i+3})|b\rangle = (-1)^{b_1+\cdots+b_{i-1}}(1 - s(-1)^{b_i})|b \oplus \delta_i\rangle$, which vanishes iff $s = \sigma_i$. So N_σ annihilates the octant σ; an annihilator is isotropic, since $2\eta(u,v)\phi = (uv + vu)\phi = 0$, hence of dimension at most three, and it equals N_σ. For $\sigma = (+,+,+)$ this is the statement $a_i^-|000\rangle = 0$. The sign of $a_{i_1}^+ \cdots a_{i_r}^+|\mathrm{vac}\rangle$ is a product of factors $(-1)^{b_1+\cdots+b_{i-1}}$ taken while all bits before i are still zero. (v) Reordering $e_1 e_4 e_2 e_5 e_3 e_6$ into $e_1 \cdots e_6$ costs three transpositions (Example 1.6). □

All five items are restated from Section 1.1 or fixed here (r3-1, r3-2, r3-3). Item (v) gives $\Gamma = -\omega_1\omega_2\omega_3$ with eigenvalue $(-1)^w$, where the temporal weight $w = 3 - N$ counts plus signs.

The element $\omega = e_1 \cdots e_6$ of (1.2), the Γ of (v) and of Theorem 2.17, and the volume element used in Example 1.6 are one and the same.

Proposition 1.9 (The octant group)**.** *For each axis a, the automorphisms of* $\mathrm{Cl}(3,3)$ *extending the reflection $R_a : e_a \mapsto -e_a$, $e_b \mapsto e_b$ $(b \neq a)$ are the conjugations by the multiples of $g_a := e_a\omega$. Each g_a is a signed permutation matrix; it conjugates ω_i to $-\omega_i$ for $a \in \{i, i+3\}$ and fixes the other two, so it carries the octant σ to the octant with σ_i reversed. The reflections of a temporal axis and of its spatial partner act alike on the octants, and $g_i g_{i+3} = \pm\omega_i$ fixes every octant. The lifts g_1, g_2, g_3 satisfy $g_i^2 = -\mathbf{1}$, anticommute pairwise and generate a group of order* 16 *whose centre $\{\pm\mathbf{1}, \pm g_1g_2g_3\}$ contains $g_1g_2g_3 = e_4e_5e_6$, the element mapping each octant to its antipode; the kernel of its action on the eight octants is $\{\pm\mathbf{1}\}$, and it acts through $\mathbb{Z}_2^3$, simply transitively.*

Proof. $\mathrm{Cl}(3,3) \cong M_8(\mathbb{R})$ is central simple, so every automorphism is inner and its implementing element is unique up to a real factor (Skolem–Noether). Since ω anticommutes with every generator, $g_a e_b g_a^{-1} = -e_a e_b e_a^{-1}$, which is $-e_b$ for $b = a$ and e_b otherwise. Hence $g_a \omega_i g_a^{-1} = R_a(e_i)R_a(e_{i+3})$. Further $g_i^2 = -e_i^2\omega^2 = -\mathbf{1}$, and $g_ig_j = -e_ie_j$ anticommutes with $g_jg_i = -e_je_i$. Moving g_i past the other two lifts costs two signs, so $g_1g_2g_3 = -e_1e_2e_3\,\omega = e_4e_5e_6$ commutes with every g_i; it is not $\pm\mathbf{1}$, and it reverses all three signs. The images of $|\mathrm{vac}\rangle$ under the eight sign classes are the eight octants (r3-4, r3-5). □

Fixed. One may also define the octant group as the reflections of the temporal $\mathbb{R}^3$ acting on its eight *sign sectors* $\{x \in V^+ : \sigma_i x_i > 0\}$, and identify them with the spinor octants. Both sets are $\mathbb{Z}_2^3$-torsors, so equivariant bijections between them exist, eight of them, one for each choice of the octant assigned to the positive sector. One of them is singled out by an operator of the clock analysis (Section 2.1):

Proposition 1.10 (Clock octant and vacuum)**.** *For a dimensionless $x \in V^+$ the conversion operator of Theorem 2.7 with $C = \mathrm{diag}(x)$ is $B_x = \sum_i x_i e_i e_{i+3} = \sum_i x_i \omega_i$. It has eigenvalue $\sum_i \sigma_i x_i$ on the octant σ; if no x_i vanishes, its unique top eigenline is the octant* $\mathrm{sign}(x)$*; and $g_i B_x g_i^{-1} = B_{R_i x}$. The Fisher–Rao octant of clocks of Section 2.1.3, the closed positive octant $x_i = 2\sqrt{w_i} \geq 0$ of the sphere of radius two, therefore corresponds to the Fock vacuum: for x in its interior the top eigenline of B_x is $|\mathrm{vac}\rangle$. At the symmetric point the radius-two coordinates $x_i = 2/\sqrt{3}$ give $B_x = 2\varpi/\sqrt{3}$, twice the unit clock $B_{\widehat{W}} = \varpi/\sqrt{3}$, whose top eigenvalue $\sqrt{3}$ lies on $|\mathrm{vac}\rangle$; the top eigenspaces agree. The weights $w_i = x_i^2/4$ do not see the signs of the amplitudes, and the clock simplex is the quotient of the clock sphere by the octant group. The quotient has boundary stabilisers: if k coordinates of x vanish, the top eigenspace of B_x has dimension 2^k (*8 *at $x = 0$), and a point on such a face of the simplex has 2^{3-k} preimages. The map is not an eight-sheeted covering.*

Proof. B_x is diagonal by Proposition 1.8(i), and $\sum_i \sigma_i x_i < \sum_i |x_i|$ unless $\sigma = \mathrm{sign}(x)$. Equivariance follows from Proposition 1.9 (r3-6, with twenty random rational x). □

The spectral statements are fixed. The identification of the clock octant with the vacuum pole is constructed: it uses the order in $\omega_i = e_ie_{i+3}$ and the choice of the top rather than the bottom eigenline, and the opposite choice assigns the filled pole. The two objects are different in kind, a two-dimensional region of the sphere of clock directions in V^+ and a line of the spinor module; read physically, the correspondence is interpretive.

The exterior-algebra structure

Proposition 1.11 (The spinor module as an exterior algebra)**.** $L := \mathrm{span}\{e_i + e_{i+3}\}$ *and* $L' := \mathrm{span}\{e_i - e_{i+3}\}$ *are maximal isotropic, transverse, and dual to each other under η. The map $\Lambda^r(L) \to S$, $\ell_1 \wedge \cdots \wedge \ell_r \mapsto 2^{-r}\ell_1 \cdots \ell_r|\mathrm{vac}\rangle$, is an isomorphism $\Lambda^*(L) \cong S$ carrying the degree to N, with $\dim \Lambda^r = 1, 3, 3, 1$ for $r = N = 0, 1, 2, 3$: the weights of the octants. Writing*

$X = X_L + X_{L'}$, a vector acts as exterior multiplication by $2X_L$ plus contraction with the form $\eta(X_{L'}, \cdot)$ on L,

$$X = \sum_i \big[(B_i + E_i)\, a_i^+ + (B_i - E_i)\, a_i^-\big]. \tag{1.6}$$

Moreover $a \mapsto a|\mathrm{vac}\rangle$ is an isomorphism from the temporal Clifford algebra $\mathrm{Cl}(V^+) = \langle e_1, e_2, e_3\rangle \cong \mathrm{Cl}(3,0)$ onto S, and $e_I|\mathrm{vac}\rangle = a_I^+|\mathrm{vac}\rangle$: the octant weight N is the Clifford grade of the temporal blade e_I.

Proof. $\eta(e_i \pm e_{i+3}, e_j \pm e_{j+3}) = \delta_{ij} - \delta_{ij} = 0$, while $\eta(e_i + e_{i+3}, e_j - e_{j+3}) = 2\delta_{ij}$. Elements of L anticommute, so the map is well defined, and it is bijective by Proposition 1.8(iii). Under it a_i^+ is $\ell_i\wedge$ with $\ell_i = e_i + e_{i+3} = 2a_i^+$, and $\ell_i' = e_i - e_{i+3} = 2a_i^-$ is the contraction with $\eta(\ell_i', \cdot)$, since $\eta(\ell_i', \ell_j) = 2\delta_{ij}$. (1.6) is $e_i = a_i^+ + a_i^-$, $e_{i+3} = a_i^+ - a_i^-$. Finally $e_j = a_j^+ + a_j^-$ and a_j^- annihilates $a_I^+|\mathrm{vac}\rangle$ for $j \notin I$, which gives $e_I|\mathrm{vac}\rangle = a_I^+|\mathrm{vac}\rangle$ by induction (r3-7). □

Inherited: this is Chevalley's realisation of the spinors of a space $L \oplus L'$ with $L' \cong L^*$ on $\Lambda^*(L)$ [201, 85], and the pure spinor lines of Proposition 1.8(iv) are the lines of Cartan and Chevalley (see Section 2 of [203]). The last statement is the classical identification of the spinors of $\mathrm{Cl}(3,3) \cong M_8(\mathbb{R})$ restricted to $\mathrm{Cl}(3,0)$ with the Pauli algebra itself. The same realisation of the spinors of $\mathrm{SO}(n,n)$ by the forms on a null n-plane is the starting point of Krasnov [211], who reads the forms as fermion fields; for $\mathrm{SO}(3,3)$ he finds the commutant $\mathrm{SO}(2)$ of the Lorentz group and a single charged two-component Weyl fermion. Here the forms carry the field strength, $\Phi_{\mathrm{oct}}(X) = \mathbf{E}\cdot f + \mathbf{B}\cdot g$ (Theorem 3.44), and Theorem 3.48 shows that this field reading is not the spin representation. The module $\Lambda^*(L)$ is shared; the physical identification is not.

Proposition 1.12 (The commutant of the grading root)**.** *The elements of $\mathfrak{so}(3,3)$ commuting with the grading root $\mathcal{K} = \frac{1}{2}\operatorname{ad}_\varpi|_V$ (Proposition 2.3) form a subalgebra of dimension nine, the adjoint image of the bivectors $E_{ij} := a_i^+ a_j^- - \frac{1}{2}\delta_{ij}\mathbf{1}$ (so $E_{ij} = a_i^+ a_j^-$ for $i \neq j$ and $E_{ii} = -\frac{1}{2}\omega_i$). They commute with N; $\operatorname{ad} E_{ij}$ acts on L by the matrix unit e_{ij} and on L' by $-e_{ji}$, so the subalgebra is $\mathfrak{gl}(L) \cong \mathfrak{gl}(3,\mathbb{R})$; and the spin lift of $A \in \mathfrak{gl}(3)$ acts on $\Lambda^r(L)$ as $d\Lambda^r(A) - \frac{1}{2}\operatorname{tr} A$, where $d\Lambda^r(A)$ is the derived action, extending A to $\Lambda^r(L)$ as a derivation (for $A = \mathbf{1}$ the eigenvalue on degree r is $r - \frac{3}{2}$). On $GL^+(L)$ the integrated action is $\rho(g) = (\det g)^{-1/2}\Lambda^*(g)$. As a $\mathfrak{gl}(3)$-module, $S \cong \Lambda^*(L) \otimes (\det L)^{-1/2}$.*

Proof. $\mathcal{K}$ has the eigenspaces L (eigenvalue -1) and L' $(+1)$, so an element commuting with it preserves both; η-skewness then forces $A|_{L'} = -(A|_L)^*$ under $L' \cong L^*$, and every $A|_L \in \mathfrak{gl}(L)$ occurs: dimension nine. Since $\mathcal{K} = \frac{1}{2}\operatorname{ad}_\varpi$ and $\varpi = 3\,\mathbf{1} - 2N$, an element $\operatorname{ad}\beta$ commutes with $\mathcal{K}$ iff $[N, \beta] = 0$. A bivector is quadratic in the $a^\pm$; the number-conserving ones are the $a_i^+ a_j^-$ up to scalars, and E_{ij} removes the scalar part. By the CAR $[a_i^+ a_j^-, a_k^+] = \delta_{jk} a_i^+$ and $[a_i^+ a_j^-, a_k^-] = -\delta_{ik} a_j^-$, and $a_i^+ a_j^-$ acts on $a_K^+|\mathrm{vac}\rangle$ as the derivation replacing a_j^+ by a_i^+ (r3-8). □

The statement is standard (Example 2.12 of [203] gives $\Lambda^* V^* \otimes (\det V)^{1/2}$ with V^* in the role of L); it is fixed here in the realisation. It shows that $\mathfrak{gl}(3)$ does not act on Λ^r by the bare derived action: the density factor $-\frac{1}{2}\operatorname{tr} A$, the same on all degrees, is part of the spin representation.

1.2.1 The trace form splits by the Cartan involution

Theorem 1.13 (The trace form splits by the Cartan involution)**.** *Let $P = P^\top = P^2$, let B be a bivector and $\Delta = [B, P]$. Then $\Delta^\top = PB^\top - B^\top P$, and*

(i) if B is compact, $B^\top = -B$, then Δ is symmetric and $\operatorname{Tr}(\Delta^2) = \operatorname{Tr}(\Delta^\top\Delta) > 0$ unless $\Delta = 0$;

(ii) if B is a boost, $B^\top = B$, then Δ is antisymmetric and $\operatorname{Tr}(\Delta^2) = -\operatorname{Tr}(\Delta^\top\Delta) < 0$ unless $\Delta = 0$.

Symmetric and antisymmetric matrices are trace-orthogonal, so on an orbit the trace form is an orthogonal sum of a positive definite piece on the directions generated by compact bivectors and a negative definite piece on those generated by boosts. Only the former are tangent to the manifold of orthogonal projectors.

Proof. $\Delta^\top = (BP - PB)^\top = PB^\top - B^\top P$, which is Δ when $B^\top = -B$ and $-\Delta$ when $B^\top = B$. The trace of the product of a symmetric and an antisymmetric matrix vanishes. A curve of orthogonal projectors has symmetric velocity, so the antisymmetric directions leave that manifold. □

Theorem 1.3 and Theorem 1.13 classify bivectors relative to the fixed Cartan decomposition, compact meaning antisymmetric for the chosen positive form. A general element $X = X_a + X_s$ of $\mathfrak{so}(3,3)$, split into antisymmetric and symmetric parts, gives a tangent to the manifold of orthogonal projectors exactly when $[X_s, P] = 0$, so generators are to be counted modulo the stabiliser of P. The theorem fixes the sign of the trace form on an orbit; it fixes the sign of a kinetic term only for actions built on this trace metric, not that of an arbitrary action, spacetime contraction, constraint or propagator residue.

Chapter 2

Lorentz Structures, Complex Units and Observational Spacetime

The question of this chapter is which Lorentz algebra, which clock and which complex unit the algebra selects, and under which hypotheses. A conversion constant ties the temporal triple to the spatial one; a clock is a unit vector of the temporal triple, and all clocks are equivalent under rotations (Remark 2.6). A clock selects an observable four-plane, and the restriction of physical states to a single clock is an assumption, stated as Assumption 2.16 and analysed in Chapter 12. The joint commutant of the interior Lorentz algebra and the internal bivectors fixes the imaginary unit up to sign (Theorem 2.17), and the real Dirac theory built on it is complex Dirac theory. Discrete symmetries and the state space close the chapter. Uniqueness statements hold under the hypotheses they state; global covariance under a change of clock is not a local gauge symmetry.

2.1 Time, conversion, and the two square roots of the pairing

2.1.1 One conversion constant

The split form is not dimensionally innocent. If V^+ carries times and V^- lengths, then $\eta(v,v) = \sum_i (t^i)^2 - \sum_a (x^a)^2$ adds squared times to squared lengths, and a conversion is needed, a priori one constant for each temporal axis.

Proposition 2.1 (A pairing alone does not give isotropy)**.** *A map exchanging V^+ and V^- cannot carry a single dimension; it converts times to lengths one way and lengths to times the other. In block form*

$$\mathcal{K}_C = \begin{pmatrix} 0 & C^{-1} \\ C & 0 \end{pmatrix}, \qquad C : V^+ \to V^-, \qquad [C] = LT^{-1}, \tag{2.1}$$

one has $\mathcal{K}_C^2 = \mathbf{1}$ for every invertible C, including $C = \mathrm{diag}(c_1, c_2, c_3)$ with distinct entries. An involutive pairing therefore does not force $c_1 = c_2 = c_3$.

Theorem 2.2 (One conversion constant)**.** *If a common $SO(3)$ acts on V^+ and on V^- by the vector representation and C intertwines the two actions, then $C = c\,\mathbf{1}$ with $[c] = LT^{-1}$.*

Proof. The vector representation of $SO(3)$ is irreducible of real type, so its commutant is $\mathbb{R}$ and Schur's lemma places C in $\mathbb{R}\,\mathbf{1}$ [57]. □

2.1.2 The two square roots of the pairing

Dimensional analysis fixes $[C]$ and leaves the sign of the lower-left block free. The two choices are different operators, and the rest of the paper turns on the difference.

Proposition 2.3 (The grading root and the binding root)**.** *In units with $c = 1$, and in block form on $V^+ \oplus V^-$, put*

$$\mathcal{K} = \begin{pmatrix} 0 & -\mathbf{1} \\ -\mathbf{1} & 0 \end{pmatrix}, \qquad G = \begin{pmatrix} 0 & -\mathbf{1} \\ \mathbf{1} & 0 \end{pmatrix},$$

that is $\mathcal{K}e_i = -e_{i+3}$, $\mathcal{K}e_{i+3} = -e_i$, and $Ge_i = e_{i+3}$, $Ge_{i+3} = -e_i$. Then:

	square	η-adjoint	$\eta(X\cdot, X\cdot)$	role
$\mathcal{K}$	$+\mathbf{1}$	$\mathcal{K}^\dagger = -\mathcal{K}$	$-\eta$	an element of $\mathfrak{so}(3,3)$; grades
G	$-\mathbf{1}$	$G^\dagger = +G$	$-\eta$	a complex structure outside $\mathfrak{so}(3,3)$; binds

with $X^\dagger := \eta^{-1}X^\top\eta$. Both are anti-isometries. The eigenspaces of $\mathcal{K}$ are the two totally null three-planes spanned by $e_i \mp e_{i+3}$.

Both roots can be built inside the algebra, and they are related by a single trivector.

Lemma 2.4 (Both roots from Clifford data)**.** *Let $\varpi = \sum_i e_i e_{i+3}$ and $T = e_1e_2e_3$, and define $\mathcal{K}(v) = \frac{1}{2}[\varpi, v]$ and $S(v) = TvT^{-1}$ on vectors. Then $\mathcal{K}(e_i) = -e_{i+3}$, $\mathcal{K}(e_{i+3}) = -e_i$, so $\mathcal{K}^2 = \mathbf{1}$; S fixes the temporal and negates the spatial generators, so $S = \eta$ as a map of V; and*

$$G = -\mathcal{K} \circ S.$$

Proof. $[e_ie_{i+3}, e_i] = e_ie_{i+3}e_i - e_i^2e_{i+3} = -2e_{i+3}$ and $[e_ie_{i+3}, e_{i+3}] = e_ie_{i+3}^2 - e_{i+3}e_ie_{i+3} = -2e_i$, while the other pole operators commute with both by (1.3). T is a product of the three temporal generators, so it commutes with each of them up to the sign $(-1)^{3-1} = +1$ and anticommutes with each spatial generator. Finally $-\mathcal{K}(Se_i) = -\mathcal{K}(e_i) = e_{i+3} = Ge_i$ and $-\mathcal{K}(Se_{i+3}) = \mathcal{K}(e_{i+3}) = -e_i = Ge_{i+3}$. □

So the root that grades is $\frac{1}{2}\operatorname{ad}_\varpi$, the adjoint action of the sum of the pole operators, and the root that binds is the same map composed with the orthogonal involution $S = \operatorname{Ad}_{e_1e_2e_3}$ of V, which induces the Cartan involution $A \mapsto SAS^{-1}$ of $\mathfrak{so}(3,3)$. That the two roots are distinguished exactly by the Cartan involution is the cleanest single statement the flat and the curved analyses below jointly support.

Shift operators. A clock coordinate λ has dimension T, the operator G is dimensionless, and their sum requires a scale. With a length ℓ,

$$\kappa = \frac{c}{\ell}, \qquad \alpha_\pm = \partial_\lambda \pm \kappa G. \tag{2.2}$$

The pointwise η-adjoint of κG is κG, while the formal adjoint of the differential operator, after an integration by parts with compatible boundary conditions, is $(\partial_\lambda + \kappa G)^* = -\partial_\lambda + \kappa G^\dagger = -\alpha_-$. The two adjoints must be kept apart: the factorisations below are insensitive to the overall sign, positivity statements are not.

2.1.3 The clock

A clock is a unit vector of V^+. Which one? Let $w = (w_1, w_2, w_3)$ be normalised non-negative weights on the three temporal axes, so that w ranges over the two-simplex Δ^2. The map $x_i = 2\sqrt{w_i}$ sends Δ^2 onto the part of the sphere $\sum_i x_i^2 = 4$ in the closed positive octant, and since $dx_i = dw_i/\sqrt{w_i}$ the Euclidean metric pulls back to

$$\sum_i dx_i^2 = \sum_i \frac{dw_i^2}{w_i},$$

the Fisher–Rao metric. Probability weights alone do not fix it: a permutation-invariant metric such as $(1+\varepsilon\sum_i w_i^2)\sum_i dw_i^2/w_i$, $\varepsilon>0$, is not a multiple of it. If the clock metric is required to belong to a family compatible with congruent Markov embeddings, between simplices of every size, Čencov's characterisation fixes it up to scale to Fisher–Rao [23, 3, 323]; we adopt this compatibility requirement. It fails for arbitrary stochastic maps. It is a smooth metric on the open simplex, and the closed octant is its metric completion. That description is itself a modelling choice, not a consequence of the algebra. It restricts the clocks to the closed positive octant of a round sphere of radius two, and the octant is preserved by the permutations of the axes but not by the rotations of V^+, under which every unit vector of V^+ is equivalent to every other (Remark 2.6).

Proposition 2.5 (The symmetric clock)**.** *The barycentre $w=(\frac{1}{3},\frac{1}{3},\frac{1}{3})$ is the unique point of Δ^2 fixed by the permutations of the three axes, and the unique maximum of the entropy $-\sum_i w_i\log w_i$, with value $\log 3$ and Hessian eigenvalues -9 and -3 in the chart (w_1,w_2). Its amplitude direction is the unit vector*

$$\widehat{W}=\frac{e_1+e_2+e_3}{\sqrt{3}},\qquad \widehat{W}^2=\mathbf{1},\qquad \lambda:=\eta(\widehat{W},v)=\frac{t^1+t^2+t^3}{\sqrt{3}}=\sqrt{3}\,\bar{t}.$$

The permutation argument carries the weight here; the entropy argument is Jaynes' principle and is a hypothesis, recorded as such. What the proposition says is modest: if a clock is to treat the three temporal axes alike, it is $\widehat{W}$, and the physical time it reads is $\sqrt{3}$ times the mean of the three constituent times. It does not say that λ is a proper time, that it is invariant under a group, or that the directions orthogonal to $\widehat{W}$ are unphysical. Those directions, $\widehat{W}^\perp\cap V^+=\{t^1+t^2+t^3=0\}$, spanned by e_1-e_2 and $e_1+e_2-2e_3$, are the *relative times*.

Remark 2.6 (Clocks: covariance, not selection)**.** Five objects must be kept apart: a unit clock $\hat{n}\in V^+$ ($\hat{n}^2=\mathbf{1}$), the unnormalised sum $e_1+e_2+e_3$ (square $3\,\mathbf{1}$), the temporal trivector $T=e_1e_2e_3$ ($T^2=-\mathbf{1}$, not a vector), the discrete permutations of the axes, and the continuous rotations $SO(3)$ of V^+. The rotor $R=(\mathbf{1}+\widehat{W}e_1)/|\mathbf{1}+\widehat{W}e_1|$ of $\mathrm{Spin}(V^+)$ carries the axial clock e_1 to $\widehat{W}$; it commutes with T and carries $J(e_1)=Te_1$ to $J(\widehat{W})=T\widehat{W}$ and the free Dirac Hamiltonian $H_{e_1}(\mathbf{p})=me_1+\sum_a p_a e_1e_{a+3}$ to $H_{\widehat{W}}(\mathbf{p})$. For every unit clock $J(\hat{n})=T\hat{n}$, $J(\hat{n})^2=-\mathbf{1}$ and $\hat{n}J(\hat{n})=T$. So the axial and the symmetric clock lie on one $SO(3)$ orbit, although not on one orbit of the permutations, under which an odd permutation reverses T and with it $J(\hat{n})$. Three cautions follow. (a) This is a *global* covariance of the free data: it shows that physically equivalent descriptions exist, not that a change of clock is a gauge redundancy, which would need an action, a redundancy of its solutions and the invariance of the observables. A position-dependent rotor $R(x)$ adds the term $R^{-1}\partial_\mu R\in\mathfrak{so}(3)_+$ to the Dirac operator, so a local clock symmetry requires a connection with its transformation law, which is not constructed here. (b) Structures that are *transported* with the rotor keep their algebraic properties; structures held *fixed* need not be invariant. The binding root G is preserved by a rotation acting alike on the temporal and the spatial triple and is changed by a rotation of the temporal triple alone; a matter projector chosen once, such as P_* of Section 6.2, is not invariant under clock rotations unless it is transported. (c) Neither the permutation symmetry nor the entropy maximum selects a clock dynamically; they single out $\widehat{W}$ only within the chosen octant description. Global covariance establishes neither a local clock gauge symmetry, nor a dynamical clock selection, nor a well-posed interacting theory with three times (Section 12.1).

2.1.4 The spectrum of a conversion operator

An operator that converts times into lengths is a mixed bivector $B_C=\sum_{i,a}C_{ia}\,e_ie_{a+3}$, symmetric by Theorem 1.3. Its square controls the factorised interior dynamics of Section 4.2, where the equation of motion becomes $\ddot{X}=\kappa^2B_C^2X$ for $X\in S\cong\mathbb{R}^8$, so its spectrum is worth knowing exactly.

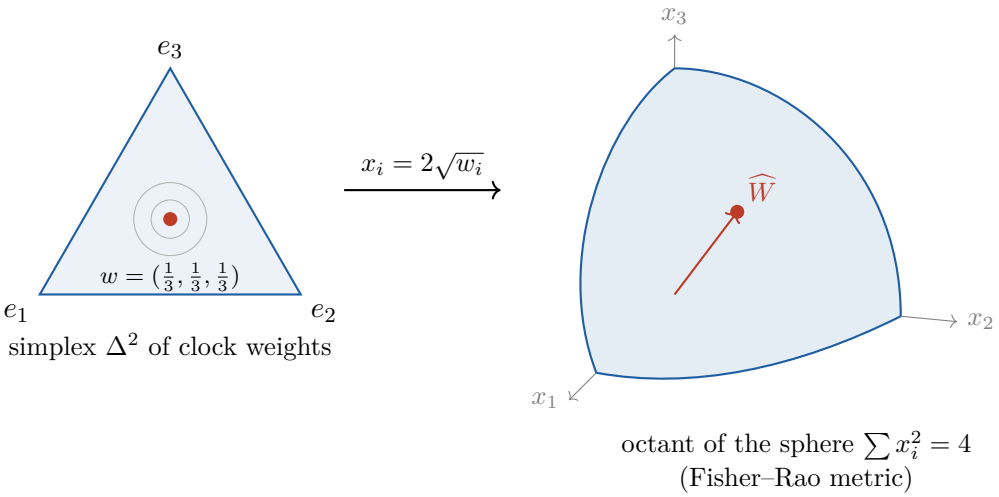


Figure 2.1: The space of clocks. Normalised weights on the three temporal axes form a simplex; the amplitude map carries it onto an octant of a round sphere, whose round metric is the Fisher–Rao metric. The barycentre, the only permutation-invariant point and the entropy maximum, corresponds to the symmetric clock $\widehat{W} = (e_1 + e_2 + e_3)/\sqrt{3}$ (Proposition 2.5).

Theorem 2.7 (Spectrum of a conversion operator)**.** *Write $C = \sum_k \sigma_k u_k v_k^\top$ in singular-value form and put $U_k = \sum_i u_k^i e_i$, $V_k = \sum_a v_k^a e_{a+3}$ and $B_k = U_k V_k$. Then $B_k^2 = \mathbf{1}$, the B_k commute, and*

$$B_C^2 = \|C\|_F^2\, \mathbf{1} + 2 \sum_{k<l} \sigma_k \sigma_l\, \Omega_{kl}, \qquad \Omega_{kl} = B_k B_l, \qquad \Omega_{12}\Omega_{13} = \Omega_{23}.$$

Consequently

$$\operatorname{spec}(B_C^2) = \{(\eta_1\sigma_1 + \eta_2\sigma_2 + \eta_3\sigma_3)^2 : \ \eta \in \{\pm 1\}^3\},$$

and the characteristic polynomial of B_C^2 on $\mathbb{R}^8$ is $\prod_{\epsilon_2,\epsilon_3=\pm1} [t - (\sigma_1 + \epsilon_2\sigma_2 + \epsilon_3\sigma_3)^2]^2$. When the four values are distinct each has multiplicity two; when values coincide their multiplicities add (for $C = \mathbf{1}_3$ the spectrum is 9 twice and 1 six times).

Proof. U_k and V_k are orthogonal unit vectors of square $+\mathbf{1}$ and $-\mathbf{1}$, so $B_k^2 = -U_k^2 V_k^2 = \mathbf{1}$; for $k \neq l$ the four vectors U_k, V_k, U_l, V_l are mutually orthogonal and B_k, B_l commute by (1.3). Expanding the square gives the formula. The seven nontrivial products of the commuting involutions B_k are, in a rotated orthonormal basis of V, blades of grade two, four or six, hence traceless; the joint eigenspace with signs η therefore has dimension $\frac{1}{8} \operatorname{Tr} \prod_k (\mathbf{1} + \eta_k B_k) = 1$, and on it $B_C = \eta \cdot \sigma$. □

Corollary 2.8 (Rank counts clocks)**.** *B_C^2 is a multiple of $\mathbf{1}$ if and only if $\operatorname{rank} C \leq 1$, and some eigenvalue of B_C^2 vanishes if and only if one singular value equals the sum of the other two. The rank of C is the number of independent temporal directions the conversion uses.*

Example 2.9 (Conversion along one clock is dynamically blind)**.** For any unit clock $n \in V^+$ the operators $G_a(n) := n\, e_{a+3}$, which convert along n, have $C = n\, \hat{e}_a^\top$ of rank one and singular values $(1, 0, 0)$. Hence $G_a(n)^2 = \mathbf{1}$, the factorised equation of motion is $\ddot{X} = +\kappa^2 X$, whose solutions grow or decay exponentially and do not oscillate, and the conversion operator does not appear in it at all. This holds for every clock, the axial e_1 and the symmetric $\widehat{W}$ alike, so it does not single out $\widehat{W}$: the blindness is a property of converting along a single temporal direction, forced by the rank. A conversion visible in the equations of motion needs at least two independent temporal directions in C, that is, a relative time with respect to the clock used. Raising the rank does not produce oscillation either: the eigenvalues $(\eta \cdot \sigma)^2$ are real and non-negative at every rank, so the factorised equation has growing or affine modes, never bounded oscillations.

Higher rank makes the instability anisotropic, with up to four rates $\kappa|\eta\cdot\sigma|$ instead of one, and can create flat directions, along which the motion is affine. Bounded motion has to come from elsewhere, and Section 4.3 shows where.

2.2 The observational four-plane of a clock

This section fixes a clock and the four-plane it selects. That physical states and observables use a single clock is Assumption 2.16, stated at the end of the section; it is an assumption, analysed in Chapter 12.

Configuration, clock and the selected four-plane

Definition 2.10 (Configuration and clock)**.** The classical configuration is $Y = \sum_i c\,t^i e_i + \sum_a x^a s_a$, three converted times on V^+ and three lengths on V^-, with $\eta(Y,Y) = c^2|t|^2 - |x|^2$ in m^2. A *clock* is a unit vector $\hat{n} \in V^+$. It defines

$$E_{\hat{n}} = \mathbb{R}\hat{n} \oplus V^-, \qquad U_{\hat{n}} = \hat{n}^\perp \cap V^+, \qquad V = E_{\hat{n}} \oplus U_{\hat{n}},$$

an oriented orthonormal frame $(\hat{n}, u_1, u_2)$ of V^+, the time $t_{\hat{n}} := \eta(\hat{n}, Y)/c = \hat{n}\cdot t$ read by the clock and the surplus coordinates $u^a := \eta(u_a, Y)$, in m. With $\pi_\pm$ the projectors onto $V^\pm$ put $\mathrm{pr}_{\hat{n}} = \hat{n}\,\eta(\hat{n},\cdot) + \pi_-$ and $\mathrm{pr}_{\hat{n}}^\perp = \pi_+ - \hat{n}\,\eta(\hat{n},\cdot)$. The clocks of non-negative weight form the Fisher–Rao octant $\hat{n}_i = \sqrt{w_i}$, $w \in \Delta^2$, which is the octant of radius two of Section 2.1.3 scaled by one half; the symmetric clock is $\widehat{W} = (e_1+e_2+e_3)/\sqrt{3}$ (Proposition 2.5).

This is constructed: a reading rule. The splitting $V^+ \oplus V^-$ is background structure, not invariant under $SO(3,3)$.

Proposition 2.11 (The four-planes that contain the three lengths)**.** *(i)* $\mathrm{pr}_{\hat{n}}$ *and* $\mathrm{pr}_{\hat{n}}^\perp$ *are complementary,* η*-self-adjoint,* η*-orthogonal projectors onto* $E_{\hat{n}}$ *and* $U_{\hat{n}}$*;* $\eta|_{E_{\hat{n}}}$ *has signature* $(1,3)$ *and* $\eta|_{U_{\hat{n}}}$ *is positive definite.*

(ii) Every four-plane $E \subset V$ *that contains* V^- *is* $E_{\hat{n}}$ *for exactly one pair* $\pm\hat{n}$*. Hence the four-planes containing the three lengths are parametrised by the clock sphere modulo* $\pm$*, a two-dimensional family inside the eight-dimensional space of Lorentzian four-planes of Theorem 12.4(iii).*

Proof. (i) $\mathrm{pr}_{\hat{n}} + \mathrm{pr}_{\hat{n}}^\perp = \pi_+ + \pi_- = \mathbf{1}$, $\mathrm{pr}_{\hat{n}}\, n = \hat{n}$, $\mathrm{pr}_{\hat{n}}\, s_a = s_a$, $\mathrm{pr}_{\hat{n}}^\perp\, u_a = u_a$, and the Gram matrices are $\mathrm{diag}(1,-1,-1,-1)$ on $(\hat{n}, s_1, s_2, s_3)$ and $\mathbf{1}$ on (u_1, u_2). (ii) Since V^- is nondegenerate, $E = V^- \oplus (E \cap V^+)$, and $E \cap V^+$ is a line on which η is positive, spanned by a unit $\hat{n}$ determined up to sign. □

Fixed (r2-1a, symbolic for a general frame in the quaternion parametrisation of $SO(3)$; r2-1b). Taking X "in the three directions $(-,-,-)$" therefore leaves exactly one choice, that of a clock, and every observer of the three lengths is an $E_{\hat{n}}$.

The six-momentum. The covector $K = K_A\, dY^A$ conjugate to Y has the raised form

$$K^\sharp = \eta^{-1}K = \frac{E}{c}n + \sum_a p_a s_a + \sum_a q_a u_a, \tag{2.3}$$

so that $E/c = \eta(\hat{n}, K^\sharp)$, $p_a = -\eta(s_a, K^\sharp)$ and $q_a = \eta(u_a, K^\sharp)$. With the phase-space form $K_A\, dY^A = E\, dt_{\hat{n}} - \sum_a p_a\, dx^a + \sum_a q_a\, du^a$ and $\{Y^A, K_B\} = \delta^A_B$ one finds

$$\{ct_{\hat{n}}, E/c\} = 1, \qquad \{x^a, p_b\} = -\delta_{ab}, \qquad \{u^a, q_b\} = \delta_{ab},$$

and all other brackets among these twelve functions vanish (fixed; r2-5a, for the clocks e_1, $(2,3,6)/7$ and $(-1,4,8)/9$). So $\mathbf{p}$ is conjugate to the three lengths, E/c to the converted time $ct_{\hat{n}}$ of the clock and q_a to the surplus coordinates; the relative sign between the first two is the Lorentzian lowering $K_{a+3} = -p_a$. This is the convention of Theorem 12.12; the action $-Mc\int ds$ has the opposite overall sign, which exchanges which pair carries the minus sign and changes no equation of motion.

Proposition 2.12 (Covariance of the selected projection)**.** *(i) An element of $O(V,\eta)$ commutes with $\mathrm{pr}_{\hat{n}}$ iff it preserves $E_{\hat{n}}$ and $U_{\hat{n}}$; the commutant of $\mathrm{pr}_{\hat{n}}$ in $\mathfrak{so}(3,3)$ is $\mathfrak{l}_{\mathrm{int}}(\hat{n}) \oplus \mathfrak{so}(U_{\hat{n}})$, of dimension 7, where $\mathfrak{l}_{\mathrm{int}}(\hat{n})$ is spanned by the six bivectors of $E_{\hat{n}}$.*

(ii) For $\Lambda \in SO^+(E_{\hat{n}})$ extended by the identity on $U_{\hat{n}}$, $\mathrm{pr}_{\hat{n}}(\Lambda K^\sharp) = \Lambda\,\mathrm{pr}_{\hat{n}}\,K^\sharp$ and $q_a(\Lambda K^\sharp) = q_a(K^\sharp)$: the projection is a four-vector and the q_a are Lorentz scalars.

(iii) The rotations $R_\theta = \exp(\theta\, u_1 \wedge u_2)$ of $U_{\hat{n}}$ commute with $\mathrm{pr}_{\hat{n}}$ and turn (u^1,u^2) and (q_1,q_2) through the same angle; their cotangent lift preserves $K_A\,dY^A$.

(iv) The eight generators of $E_{\hat{n}} \wedge U_{\hat{n}}$ do not commute with $\mathrm{pr}_{\hat{n}}$, and their eight commutators are linearly independent. For instance at $\hat{n} = e_1$ the rotation $\exp(\varphi\, e_1 \wedge e_2)$ carries the rest momentum $K^\sharp = (E/c)\,e_1$ to one with $|q_1| = (E/c)|\sin\varphi|$.

Proof. (i) A map commuting with a projector preserves its image and kernel; an isometry preserving $E_{\hat{n}}$ preserves $E_{\hat{n}}^\perp = U_{\hat{n}}$. On Lie algebras this is $\mathfrak{so}(E_{\hat{n}}) \oplus \mathfrak{so}(U_{\hat{n}})$, of dimension $6+1$. (ii) Λ commutes with $\mathrm{pr}_{\hat{n}}$ by (i), and $\eta(u_a, \Lambda K^\sharp) = \eta(\Lambda^{-1}u_a, K^\sharp) = \eta(u_a, K^\sharp)$. (iii) η is positive on $U_{\hat{n}}$, so contravariant and covariant components of $U_{\hat{n}}$ rotate alike; for an isometry R the contragredient on K_A is $R^{-\top} = \eta R \eta^{-1}$, so $K^\sharp$ moves as Y and $K_A\,dY^A$ is invariant. (iv) A generator of $E_{\hat{n}} \wedge U_{\hat{n}}$ maps $U_{\hat{n}}$ into $E_{\hat{n}}$; the example is a direct computation. □

Nothing here gives covariance under the whole of $SO(3,3)$ (r2-2a for seven clocks including $\widehat{W}$, the dimension count for six rational clocks; r2-2b for a finite boost and the $SO(2)$ at $\hat{n} = (2,3,6)/7$; r2-2c). Part (i) is the case $C = J(\hat{n})$ of Theorem 12.4(iii).

Proposition 2.13 (The clock, the surplus plane and $J(\hat{n})$)**.** *Let $I_T = e_1e_2e_3$.*

(i) $I_T^2 = -\mathbf{1}$; I_T commutes with e_1, e_2, e_3 and anticommutes with e_4, e_5, e_6.

(ii) $J(\hat{n}) := I_T\hat{n} = \hat{n}_1e_2e_3 + \hat{n}_2e_3e_1 + \hat{n}_3e_1e_2$ satisfies $J(\hat{n})^2 = -|\hat{n}|^2\mathbf{1}$ and commutes with $\hat{n}$ and with s_1, s_2, s_3, hence with all of $\mathfrak{l}_{\mathrm{int}}(\hat{n})$.

(iii) For every oriented orthonormal frame $(\hat{n}, u_1, u_2)$ of V^+, $nu_1u_2 = I_T$; hence $J(\hat{n}) = u_1u_2$ is the oriented unit bivector of $U_{\hat{n}}$, and $J(e_1) = e_2e_3 = J$.

(iv) Under the clock reversal $\hat{n} \mapsto -\hat{n}$ at fixed six-momentum, a reversal of the clock's orientation distinct from the physical time reversal of Theorem 2.27: $\mathrm{pr}_{-\hat{n}} = \mathrm{pr}_{\hat{n}}$, $\mathrm{pr}_{-\hat{n}}^\perp = \mathrm{pr}_{\hat{n}}^\perp$, $E(-\hat{n}) = -E(\hat{n})$ for the energy $E(\hat{n}) := c\,\eta(\hat{n}, K^\sharp)$ read by the clock, $J(-\hat{n}) = -J(\hat{n})$ and $\Omega_{-\hat{n}} = -\Omega_{\hat{n}}$ for $\Omega_{\hat{n}} := ns_1s_2s_3$, so that $\Gamma_{\hat{n}} = J(\hat{n})\Omega_{\hat{n}}$ is unchanged.

Proof. (i) is the blade rule (Lemma 1.1): $|I||J| - |I \cap J|$ equals 2 for e_i and 3 for s_a. (ii) I_T and $\hat{n}$ commute, so $J(\hat{n})^2 = I_T^2\hat{n}^2 = -|\hat{n}|^2$; I_T and $\hat{n}$ both commute with $\hat{n}$ and both anticommute with each s_a. (iii) nu_1u_2 is the volume element of an oriented orthonormal frame of V^+, which is $\det(\hat{n}, u_1, u_2)\,e_1e_2e_3 = I_T$; then $u_1u_2 = \hat{n}^{-1}I_T = \hat{n}I_T = I_T\hat{n}$. (iv) $\mathrm{pr}_{\hat{n}}$ is even in $\hat{n}$, J and Ω are odd. □

Checks: r2-3a and r2-3c for symbolic $\hat{n}$, r2-3b for a symbolic frame, r2-3d). The formula $J(\hat{n}) = I_T\hat{n}$ is the identity $J = \gamma^0 T$ of Remark 2.18 with $\gamma^0 = \hat{n}$. Item (iv) says that time orientation and the sign of the complex structure are tracked together.

Theorem 2.14 (The selected complex structure at every clock)**.** *Let* $\gamma^0 = \hat{n}$, $\gamma^a = s_a$ *and* $\Omega_{\hat{n}} = ns_1s_2s_3$.

(i) The joint commutant of $\mathfrak{l}_{\mathrm{int}}(\hat{n})$ *and of the surplus bivector* $J(\hat{n})$ *in* $\mathrm{End}_{\mathbb{R}}(S) = M_8(\mathbb{R})$ *is* $\mathrm{span}\{\mathbf{1}, \Omega_{\hat{n}}, J(\hat{n}), \Omega_{\hat{n}}J(\hat{n})\} \cong \mathbb{C} \oplus \mathbb{C}$. *Its solutions of* $x^2 = -\mathbf{1}$ *are* $\pm\Omega_{\hat{n}}, \pm J(\hat{n})$, *and only* $\pm J(\hat{n})$ *commute with the four* γ^μ.

(ii) The commutant of the four γ^μ *alone is* $\mathrm{span}\{\mathbf{1}, J(\hat{n}), u_1\Omega_{\hat{n}}, u_2\Omega_{\hat{n}}\} \cong \mathbb{H}$, *and the elements of it that commute with* I_T *are* $\mathrm{span}\{\mathbf{1}, J(\hat{n})\}$.

(iii) The mass form $\gamma^0 J(\hat{n}) = \hat{n}I_T\hat{n} = I_T$ *is the same for every clock; the kinetic kernels are* $I_T J(\hat{n})\gamma^\mu = -\hat{n}\gamma^\mu$; *and* $H_{\hat{n}}(\mathbf{p}) = c\, ns_a p_a + mc^2\hat{n}$ *is symmetric, commutes with* $J(\hat{n})$ *and satisfies* $H_{\hat{n}}(\mathbf{p})^2 = (c^2|\mathbf{p}|^2 + m^2c^4)\mathbf{1}$.

Proof. A rotor R of $\mathrm{Spin}(V^+)$ with $R\hat{n}R^{-1}$ prescribed exists for every unit $\hat{n}$, and since temporal bivectors commute with each s_a by the blade rule, Ad_R fixes s_1, s_2, s_3 and I_T. It therefore carries (e_1, e_2, e_3) to an oriented frame $(\hat{n}, u_1, u_2)$, $\mathfrak{l}_{\mathrm{int}}(e_1)$ to $\mathfrak{l}_{\mathrm{int}}(\hat{n})$, e_2e_3 to $u_1u_2 = J(\hat{n})$ and Ω to $\Omega_{\hat{n}}$. Conjugation is an automorphism of $M_8(\mathbb{R})$, so (i) and (ii) are Theorem 2.17 and Remark 2.18 transported. For (ii), I_T commutes with u_a and anticommutes with $\Omega_{\hat{n}}$. (iii) $\hat{n}$ commutes with I_T and $\hat{n}^2 = \mathbf{1}$; the square of $H_{\hat{n}}$ is that of equation (2.4) with $\gamma^0 = \hat{n}$. □

Fixed. Each part is checked directly, not only through the transport: r2-4a,b compute the commutants and solve $x^2 = -\mathbf{1}$ exactly at $\hat{n} = e_1, e_2, (1,2,2)/3, (2,3,6)/7, (-1,4,8)/9$; r2-4c is symbolic in a general frame; r2-4d checks the transport by the rotor $\cos\frac{\theta}{2} - \sin\frac{\theta}{2}\, e_1e_2$. So the Dirac spinor read by the clock $\hat{n}$ has imaginary unit $J(\hat{n})$, its sign fixed by the orientation of V^+ and of $\hat{n}$, and its energy for momentum $\mathbf{p}$ is the E of the four-momentum below. The commutant conditions therefore hold at every clock $\hat{n}$. That the mass form does not depend on the clock is consistent with Corollary 7.10, where $T = e_1e_2e_3$ is the mass kernel invariant under all of $\mathfrak{so}(3,3)$.

Remark 2.15 (Three readings of $(-,-,-)$)**.** The sign triple has three meanings in the arena, and only the first bears on the four-momentum. (a) The values of η on V^-, the reading used here (interpretive: a reading of the question). (b) The sign sector $(-,-,-)$ of the temporal $\mathbb{R}^3$, which was proposed, in an earlier and superseded version of the programme, as the carrier of the Lorentzian interior: by Proposition 2.13(iv) the clocks of that sector give the same four-planes and surplus planes as those of the Fisher–Rao octant $(+,+,+)$, with the opposite time orientation, the opposite energy and the opposite complex structure, so the sector carries nothing the Fisher–Rao octant does not (fixed). (c) In the spinor module the joint eigenline of the pole operators with signs $(-,-,-)$ is the filled Fock line $N = 3$ (r2-0b), a spinor and not a direction of V. The octant readings are the subject of Section 3.5.

2.2.1 The clock reduction is an assumption

The observable four-plane $E_{\hat{n}}$ of a clock and the four-momentum it reads are used throughout the following chapters. They rest on one step that the algebra does not supply, and it is recorded here once.

Assumption 2.16 (Clock reduction)**.** Physical states and observables are those of the reduced shell of a single clock $\hat{n}$: the temporal part of every six-momentum is parallel to $\hat{n}$, so that momenta lie in $E_{\hat{n}} = \mathbb{R}\hat{n} \oplus V^-$, of signature $(1,3)$, and the surplus temporal momenta vanish.

The assumption is imposed, not derived. Section 12.4 shows where the constraints that implement it can come from, a worldline gauge symmetry chosen rather than derived (Theorem 12.12); that analysis establishes constraint preservation for a worldline in a background field only, and the constraints remove every surplus-momentum mode, the stable ones included. Statements below that use the four-momentum of a clock, the selected four-plane or the absence of uphill decays are statements under Assumption 2.16, and they cite it.

2.3 The imaginary unit and the Dirac structure

2.3.1 The complex structure is selected

At n constituents write $\Omega = \gamma^0\gamma^1\gamma^2\gamma^3 = e_1^{(1)}e_4^{(1)}e_5^{(1)}e_6^{(1)}$ for the volume element of the interior four-plane and ω_I for the product, in increasing order, of the remaining $6n-4$ generators, the *internal* directions. At $n=1$ these are e_2, e_3 and $\omega_I = e_2e_3 = J$.

Theorem 2.17 (Selection of the imaginary unit)**.** *Inside* $\mathrm{End}_{\mathbb{R}}(S_n)$ *the joint commutant of the interior Lorentz algebra and of all internal bivectors is* $\mathrm{span}\{\mathbf{1}, \Omega, \omega_I, \Omega\omega_I\} \cong \mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \oplus \mathbb{C}$*, at every rung. Its solutions of* $x^2 = -\mathbf{1}$ *are the four isolated points* $\pm\Omega$, $\pm\omega_I$*. Of these only* $\pm\omega_I$ *commute with the interior gamma matrices, from which the kinetic operator is built. Hence the imaginary unit is* $i = \pm\omega_I$*, and the chirality is* $\gamma_5 = \mp\Omega\omega_I = \mp\Gamma$ *with*

$$\Gamma := \omega_I\,\Omega, \qquad \Gamma^2 = \mathbf{1}.$$

Proof. By (1.3), commuting with the six interior bivectors and with every internal bivector is a parity condition on each blade, and the blades that satisfy all of them are exactly the four listed, at every rung. A four-dimensional commutative real algebra with two commuting square roots of $-\mathbf{1}$ and no nilpotents is $\mathbb{C} \oplus \mathbb{C}$, whose square roots of -1 are those four points. Finally Ω is the product of the four interior generators, so it anticommutes with each of them, while ω_I contains no interior index and commutes with each. □

From now on J denotes ω_I; at one constituent this is e_2e_3, the generator of Proposition 1.7. The selection depends on its conditions, and it is worth saying exactly how.

Remark 2.18 (The ambient algebra is part of the statement)**.** If commutation with the internal bivectors is dropped, then at $n=1$ the commutant of the four gamma matrices in $M_8(\mathbb{R})$ is the quaternion algebra

$$\mathbb{H} = \mathrm{span}\{\mathbf{1},\ J,\ e_2\Omega,\ e_3\Omega\},$$

and its square roots of $-\mathbf{1}$ form a two-sphere. One further datum removes that freedom. The volume element of the temporal triple, $T := e_1e_2e_3$, depends only on the splitting $V = V^+ \oplus V^-$ and on an orientation of V^+; it commutes with J and anticommutes with $e_2\Omega$ and $e_3\Omega$ by (1.3), so

$$\{X \in \mathbb{H} :\ [X,T] = 0\} = \mathbb{R}\,\mathbf{1} \oplus \mathbb{R}\,J,$$

and $J = \gamma^0 T$ is recovered from the clock and the temporal orientation, up to the sign of that orientation. Higher up the tower the commutant of the gamma matrices alone acquires a matrix multiplicity algebra; at $n=3$ the Lorentz commutant is $M_{128}(\mathbb{C})$, whose own complex structure is Ω and not ω_I (Proposition 8.1). Selection tracks the conditions imposed, and every uniqueness statement in this paper is made with them.

That the imaginary unit is not adjoined but read off the algebra is the cleanest structural result at one constituent. It has an independent counterpart: on a real Hilbert space, for an elementary relativistic system described by a locally faithful, irreducible, continuous unitary representation of the Poincaré group with nonnegative squared mass, the complex structure is forced [102]. That is an analogy between different selection results under different hypotheses, not a proof of the finite-dimensional one.

2.3.2 Real Dirac theory is complex Dirac theory

Theorem 2.19 (Exactness of the realification)**.** *The* $\mathbb{C}$*-linear algebra generated by the interior gamma matrices on* $(S, J) \cong \mathbb{C}^4$ *is all of* $M_4(\mathbb{C})$*. Consequently, with the standard Maxwell–Dirac action and its field structures transported through this identification, the real classical Maxwell–Dirac system on* $\mathrm{Cl}(3,3)$ *is ordinary complex Maxwell–Dirac theory written in real variables.*

Proof. J commutes with the four γ^μ. The sixteen monomials in the γ^μ and their J-multiples are thirty-two linearly independent real matrices commuting with J, and the commutant of J in $M_8(\mathbb{R})$ is $\mathrm{End}_{\mathbb{C}}(\mathbb{C}^4) = M_4(\mathbb{C})$, of real dimension thirty-two. □

Corollary 2.20. *Anything derivable from the real formulation alone was derivable from the complex one, and conversely, and no prediction can follow from the rewriting. Transporting the standard quantisation yields a real-variable presentation of quantum electrodynamics; the algebra identification itself does not construct that quantisation. Real geometric formulations of Dirac theory have a long precedent [71].*

This is the deflationary principle of the introduction. It does not make the arena uninteresting; it says where its content can lie. Everything below that depends only on the four gamma matrices and J is inherited. What is not inherited is what the arena adds: the selection of J, the surplus plane and its uses, the pairing, and the tower.

2.3.3 Admissible bilinears

A field ψ with values in S is real. If its components anticommute, a term $\psi^\top A\psi$ vanishes when A is symmetric, and $\psi^\top A\,\partial\psi = \frac{1}{2}\partial(\psi^\top A\psi)$ is a total derivative when A is antisymmetric. A mass term therefore needs an antisymmetric kernel, and a first-order kinetic term a symmetric one. With the clock $\gamma^0 = e_1$:

Proposition 2.21 (Kernels of fixed symmetry)**.** *The mass form $\gamma^0 J$ is antisymmetric. The kinetic kernels*

$$K^\mu := (\gamma^0 J)(J\gamma^\mu) = -\gamma^0\gamma^\mu, \qquad K^0 = -\mathbf{1},$$

are symmetric, while $(\gamma^0 J)\gamma^\mu$ is antisymmetric. The kinetic form must therefore carry the factor J: its J-free analogue defines a total derivative and constrains nothing a field equation sees.

The mass form $\gamma^0 J = e_1e_2e_3 = T$ is the temporal volume element of Remark 2.18. It is not the only admissible mass.

Proposition 2.22 (Scalar and pseudoscalar mass)**.** *The antisymmetric elements of $M_8(\mathbb{R})$ that commute with J and whose bilinear form is invariant under the interior Lorentz algebra ($L^\top A + AL = 0$ for every $L \in \mathfrak{l}_{\mathrm{int}}$) form the two-dimensional space* $\mathrm{span}\{T,\ T\Omega\}$*. Used as the kernel of the Dirac bilinear, the second contributes nothing to the kinetic term, because $T\Omega J\gamma^\mu$ is antisymmetric for every μ; the two differ only in the mass term. T is a scalar mass and $T\Omega$ a pseudoscalar one, since $\Omega^2 = -\mathbf{1}$ and Ω anticommutes with every γ^μ, so that Ω is the real form of $i\gamma_5$. The family $T(\cos\vartheta + \sin\vartheta\,\Omega) = Te^{\vartheta\Omega}$ is the chiral mass angle.*

So T is singled out among admissible masses by the requirement that the mass be scalar, and that requirement is an input. The central element Ω will reappear in Section 2.5 as the axial $U(1)$.

The Hamiltonian. The free Dirac Hamiltonian at momentum p is

$$H(p) = \gamma^0\gamma^a p_a + m\gamma^0, \qquad H(p)^2 = E^2\mathbf{1}, \qquad E^2 = |\mathbf{p}|^2 + m^2, \tag{2.4}$$

with spectral projectors $\Lambda_\pm(p) = \frac{1}{2}(\mathbf{1} \pm H(p)/E)$, defined for $E > 0$ (at $E = 0$ they are undefined). Both $\gamma^0 = e_1$ and $\gamma^0\gamma^a = e_1e_{a+3}$ are symmetric, so $H(p)$ is a symmetric matrix that commutes with J; its eigenvalues are $\pm E$, each with real multiplicity four.

Example 2.23. At $\mathbf{p} = (0,0,3)$ and $m = 4$, $H = 3e_1e_6 + 4e_1$ satisfies $H^2 = (9+16)\mathbf{1} = 25\,\mathbf{1}$, because e_1e_6 and e_1 anticommute and $(e_1e_6)^2 = +\mathbf{1}$. The eigenvalues are ± 5; each eigenspace is a complex plane for J, and the helicity $\Sigma_3 = J\,\frac{1}{2}e_4e_5$ commutes with H and splits each of them into two real planes.

2.3.4 A first-order real form of free evolution

For a commuting real field the same data give a first-order symplectic system. Take

$$\mathcal{L}_1 = \tfrac{1}{2}\Psi^\top J\dot\Psi - \tfrac{1}{2}\Psi^\top H(p)\Psi.$$

Its Euler–Lagrange equation is $\dot\Psi = -JH(p)\Psi$. Since JH is antisymmetric with $(JH)^2 = -E^2\mathbf{1}$, the flow lies in $SO(8)$ and

$$\Psi(\lambda) = \cos(E\lambda)\,\Psi_0 - \frac{\sin(E\lambda)}{E}\,JH\Psi_0.$$

Reading J as i turns this into $i\dot\psi = h\psi$ with $h = \alpha\cdot\mathbf{p} + \beta m$, the free Dirac Hamiltonian at fixed momentum: an exact embedding of free Dirac evolution in a real symplectic form, with eight second-class constraints reducing a sixteen-dimensional phase space to eight. Three things norm conservation does not buy. The energy $\frac{1}{2}\Psi^\top H\Psi$ is unbounded below, since $\operatorname{Tr} H = 0$ and $H^2 = E^2\mathbf{1}$ force the spectrum $\{\pm E\}$ with equal multiplicity. For anticommuting components the action collapses to a boundary term, $\Psi^\top H\Psi = 0$ because H is symmetric and $\Psi^\top J\dot\Psi = \frac{1}{2}\frac{d}{d\lambda}(\Psi^\top J\Psi)$ because J is antisymmetric, which is why a fermionic Lagrangian must use the kernels of Proposition 2.21. And the bilinear $\Psi^\top J\dot\Psi$ is not preserved by the boosts. The real form of Dirac theory in which a bivector commuting with the gamma matrices plays the role of i is old [71]; what is specific here is that the bivector is the surplus rotation J, selected by Theorem 2.17.

2.4 The real Dirac Lagrangian

The Dirac Lagrangian

Section 2.3 fixes the kernels and the conditions they must meet, not a normalised Lagrangian: the mass form $T = \gamma^0 J$ is antisymmetric, the kinetic kernels $K^\mu = (\gamma^0 J)(J\gamma^\mu) = -\gamma^0\gamma^\mu$ are symmetric with $K^0 = -\mathbf{1}$, and $(\gamma^0 J)\gamma^\mu$ is antisymmetric (Proposition 2.21; r1-14, which also checks $T^2 = -\mathbf{1}$, $[K^\mu, J] = [T, J] = 0$ and that $K^\mu J$ is antisymmetric). The normalised one-constituent Lagrangian is that of Proposition 7.2(iii). For a real Grassmann field $\psi : \mathbb{R}^{1,3} \to S\otimes\Lambda$, in Heaviside–Lorentz units with $\hbar = c = 1$,

$$L_D = \tfrac{1}{2}\psi^\top(K^\mu D_\mu - mT)\psi = \tfrac{1}{2}\psi^\top T(J\gamma^\mu D_\mu - m)\psi, \qquad D_\mu = \partial_\mu + qA_\mu J, \tag{2.5}$$

the real form of $\bar\psi(i\gamma^\mu D_\mu - m)\psi$ with J in the place of i [71], the charge q being dimensionless in these units ($D_\mu = \partial_\mu + \frac{q}{\hbar}A_\mu J$ with $[q] = \mathrm{C}$ in SI); in SI units the density is $\frac{1}{2}\hbar c\,\psi^\top(K^\mu\partial_\mu - \frac{mc}{\hbar}T)\psi$ with $x^0 = ct$ and $[\psi] = \mathrm{m}^{-3/2}$ (r1-36).

Proposition 2.24 (The real Dirac field)**.** *(i) For arbitrary real 8×8 kernels, the Euler–Lagrange expression of $\frac{1}{2}\psi^\top A^\mu\partial_\mu\psi - \frac{1}{2}\psi^\top M\psi$ is $A^\mu_{\mathrm{sym}}\partial_\mu\psi - M_{\mathrm{anti}}\psi$. A symmetric mass kernel gives a vanishing bilinear and an antisymmetric kinetic kernel a total derivative.*

(ii) The field equation of (2.5) at $q = 0$ is $K^\mu\partial_\mu\psi = mT\psi$, that is $T(J\gamma^\mu\partial_\mu - m)\psi = 0$, equivalently $J\gamma^\mu\partial_\mu\psi = m\psi$ or $\gamma^\mu\partial_\mu\psi = -mJ\psi$; since $(\gamma^\mu\partial_\mu - mJ)(\gamma^\nu\partial_\nu + mJ) = \Box + m^2$, each component obeys $(\Box + m^2)\psi = 0$.

(iii) L_D is invariant under $\psi \mapsto e^{\theta J}\psi$; D_μ is covariant under $\psi \mapsto e^{-q\lambda J}\psi$, $A \mapsto A + d\lambda$, since $e^{-\theta J}$ preserves every K^μ and T. The Noether current $j^\mu = \frac{1}{2}\psi^\top K^\mu J\psi$ satisfies $\partial_\mu j^\mu = \psi^\top J\,\mathcal{E}_D$ identically, with $\mathcal{E}_D = K^\mu\partial_\mu\psi - mT\psi$; $j^0 = -\frac{1}{2}\psi^\top J\psi$.

(iv) The canonical tensor $\Theta^\mu{}_\nu = \frac{1}{2}\psi^\top K^\mu\partial_\nu\psi - \delta^\mu_\nu L_D$ satisfies $\partial_\mu\Theta^\mu{}_\nu = -(\partial_\nu\psi)^\top\mathcal{E}_D$ identically, and its energy density is $\Theta^0{}_0 = \frac{1}{2}\psi^\top(\gamma^0\gamma^a\partial_a + mT)\psi$; on a mode with $\partial_a\psi = Jp_a\psi$ it is $\frac{1}{2}\psi^\top JH(p)\psi$, with $H(p) = \gamma^0\gamma^a p_a + m\gamma^0$ the free Hamiltonian of equation (2.4).

(v) *On shell* $\Theta^{\mu\nu} - \Theta^{\nu\mu} = \partial_\rho S^{\rho\mu\nu}$ *with the totally antisymmetric spin tensor* $S^{\rho\mu\nu} = \frac{1}{4}\psi^\top\gamma^0\gamma^{[\rho}\gamma^\mu\gamma^{\nu]}\psi$*, so the Belinfante tensor* $T_B^{\mu\nu} = \Theta^{\mu\nu} - \frac{1}{2}\partial_\rho S^{\rho\mu\nu} = \frac{1}{4}\psi^\top(K^\mu\partial^\nu + K^\nu\partial^\mu)\psi - \eta^{\mu\nu}L_D$ *is symmetric, conserved on shell, and has* $T_B^{00} = \Theta^{00}$*.*

Proof. For a Grassmann ψ, $\psi^\top M\delta\psi = \delta\psi^\top M\psi$ when M is antisymmetric and $\psi^\top A\partial\delta\psi = \partial(\psi^\top A\delta\psi) + \delta\psi^\top A^\top\partial\psi$; collecting the left variations gives (i), which the jet algebra confirms for random integer kernels (r1-15). (ii) $K^\mu = TJ\gamma^\mu$ and T is invertible (r1-16). (iii) $(K^\mu J)^\top = -K^\mu J$, and $\partial_\mu j^\mu = \psi^\top K^\mu J\partial_\mu\psi = \psi^\top J(\mathcal{E}_D + mT\psi)$ with $\psi^\top JT\psi = 0$ because JT is symmetric (r1-17). (iv) Exchanging Grassmann factors and using the symmetry of K^μ; $\gamma^0\gamma^a J = J\gamma^0\gamma^a$ and $mT = Jm\gamma^0$ give the mode form (r1-18). (v) The coefficient $\frac{1}{4}$ is solved for, and the identity holds for all six pairs (μ,ν) after the time derivatives are eliminated with the field equation (r1-19). □

Item (i) is fixed and turns the admissibility conditions of Proposition 2.21 into a theorem of the variation: a Grassmann field sees only the symmetric part of a first-order kinetic kernel and only the antisymmetric part of a mass kernel. Items (ii)–(iii) are restated (Section 2.5, where the Dirac operator is written $J\gamma^\mu\partial_\mu - m$); the sign of the mass term in $\gamma^\mu\partial_\mu\psi = -mJ\psi$ follows from $K^0 = -\mathbf{1}$ and $T = \gamma^0 J$, and $m \mapsto -m$ passes to the opposite sign. Items (iv)–(v) are inherited [198, 160]. Equation (7.4), on the rank-64 block of P_* in the three-constituent module, uses the kinetic kernels $P_*\gamma^0\gamma^\mu P_*$, which are minus the compressed K^μ, with $K^0 = +\mathbf{1}$ on its block, and the Yukawa term $+\frac{y}{2}\psi^\top\Phi\psi$ in place of a mass term: with $y\Phi$ in the place of mT its fermion Lagrangian is minus (2.5), and its field equation $P_*\gamma^0\gamma^\mu P_* D_\mu\psi + y\Phi\psi = 0$ has the same form (r1-16). For Grassmann fields ${\Theta^0}_0$ is not a number, and the sign of the energy is decided only after quantisation. On the state space that Theorems 2.30 to 2.31 construct from outside the real module, the one-momentum Hamiltonian has the spectrum $\{0, \pm E, \pm 2E\}$ with multiplicities $6, 4, 4, 1, 1$, symmetric about zero and bounded below (r1-37); a positive energy requires measuring it from the lowest state, the usual choice of vacuum.

2.5 Discrete symmetries, the massless gauge algebra, and the states

2.5.1 Charge conjugation, parity and time reversal

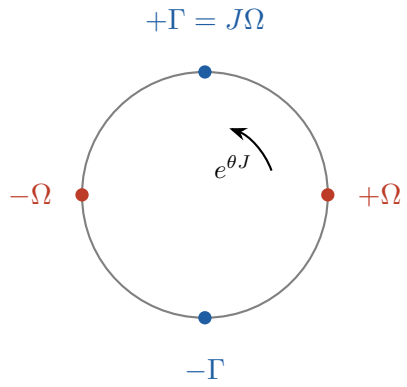
At one constituent write $U = \langle e_2, e_3\rangle$ for the surplus plane, $\Omega = \gamma^0\gamma^1\gamma^2\gamma^3$, and $\Gamma = J\Omega$. An operator X on S is *linear* when $XJ = JX$ and *antilinear* when $XJ = -JX$. It intertwines the free Dirac operator $J\gamma^\mu\partial_\mu - m$ when it commutes with every $J\gamma^\mu$. Parity is required to be linear and time reversal antilinear, and each must satisfy $XH(p)X^{-1} = H(-p)$ for every p and m.

Theorem 2.25 (Charge conjugation is the surplus plane)**.** *The antilinear operators that intertwine the Dirac operator form exactly the two-dimensional space* U*. Every unit vector* $C \in U$*, normalised by* $C^2 = \mathbf{1}$*, satisfies*

$$CH(p)C^{-1} = -H(-p), \qquad CJC^{-1} = -J, \qquad C\Gamma C^{-1} = -\Gamma, \qquad C\Lambda_+(p)C^{-1} = \Lambda_-(-p),$$

commutes with every interior Lorentz generator, and preserves the mass form and every kinetic kernel K^μ*. Hence* C *is unique up to a gauge rotation* $e^{\theta J}$*, and the operator known as the Majorana real structure is charge conjugation.*

Proof. If $XJ = -JX$, then $X(J\gamma^\mu) = (J\gamma^\mu)X$ is equivalent to $X\gamma^\mu = -\gamma^\mu X$. The conditions are linear, and conjugation by a blade sends every blade to plus or minus itself, so the solution space is spanned by the blades that anticommute with J and with e_1, e_4, e_5, e_6. By (1.3) a blade e_I anticommutes with a vector e_a exactly when $|I| - [a \in I]$ is odd. For odd $|I|$ this



$P = \gamma^0$: $CP\mathsf{T} = -\Omega$, $(CP\mathsf{T})^2 = -\mathbf{1}$

$P = \gamma^0 J$: $CP\mathsf{T} = +\Gamma$, $(CP\mathsf{T})^2 = +\mathbf{1}$

both lie on the orbit of Ω under left multiplication by $U(1)_J$

Figure 2.2: The CPT circle at one constituent (Theorem 2.28). The product of the three discrete operators is the spacetime pseudoscalar or the chirality according to the parity chosen; the two lie on one orbit of left multiplication by the rotation $e^{\theta J}$, a choice of phase.

forces $I \subset \{2,3\}$, hence $I = \{2\}$ or $\{3\}$, and both anticommute with J. For even $|I|$ it forces $\{1,4,5,6\} \subset I$, hence $e_I = \Omega$ or $e_1 \cdots e_6$, and both commute with J. The solutions are therefore e_2 and e_3. The listed identities follow from the same rule, since e_2, e_3 anticommute with every γ^μ, with J and with Γ, and commute with every interior bivector. □

Corollary 2.26 (Equal mass, equal spin, opposite charge)**.** *C commutes with every generator of the interior Lorentz algebra, so it preserves the Casimir $-\sum_a S_a^2 = \frac{3}{4}\mathbf{1}$, and $(CH(p)C^{-1})^2 = E^2\mathbf{1}$. The Majorana condition $\Psi = C\Psi$ makes the current vanish: a particle that is its own antiparticle is neutral.*

Theorem 2.27 (Parity and time reversal)**.** *The admissible parities form the space $\langle \gamma^0, \gamma^0 J\rangle$. Normalised by $P^2 = \pm\mathbf{1}$, they are $P = \pm\gamma^0$, with $P^2 = +\mathbf{1}$, and $P = \pm\gamma^0 J$, with $P^2 = -\mathbf{1}$. The admissible time reversals form $U \cdot \gamma^1\gamma^2\gamma^3$, so $\mathsf{T} = C\gamma^1\gamma^2\gamma^3$ with $C \in U$; normalised by $\|C\| = 1$,*

$$\mathsf{T}^2 = -\mathbf{1}.$$

The sign is not a convention: $B = \gamma^1\gamma^2\gamma^3$ has $B^2 = +\mathbf{1}$ and anticommutes with every vector C of the surplus plane U, so $\mathsf{T}^2 = CBCB = -C^2B^2 = -\|C\|^2\mathbf{1}$ for every choice of clock in the positive triple.

The time-reversal operator is written T because the letter T denotes the temporal trivector $e_1e_2e_3$ (Remark 2.18), and the two meet here: the parity $P = \gamma^0 J$ is that trivector.

Theorem 2.28 (CPT lies on a circle)**.** *The product of the three operators depends on the parity chosen:*

$$P = \gamma^0: \quad CP\mathsf{T} = -\Omega,\ (CP\mathsf{T})^2 = -\mathbf{1}; \qquad P = \gamma^0 J: \quad CP\mathsf{T} = +\Gamma,\ (CP\mathsf{T})^2 = +\mathbf{1}.$$

What is convention-free is the orbit of Ω under left multiplication by $U(1)_J$, $\theta \mapsto e^{\theta J}\Omega = \cos\theta\,\Omega + \sin\theta\,J\Omega$, a choice of phase (conjugation by $e^{\theta J}$ fixes Ω, since $[J,\Omega] = 0$). Its four distinguished points $\pm\Omega, \pm\Gamma$ are the spacetime pseudoscalar and the chirality.

These are statements about explicitly constructed operators of the free theory; no CPT theorem is proved, and the interacting theory is not constructed.

2.5.2 The symmetry algebra of the massless kinetic forms

A gauge generator must in particular leave the kinetic form invariant. Solving that condition over all of $\mathrm{End}_{\mathbb{R}}(S)$, with no ansatz, gives the following.

Proposition 2.29 (Symmetry algebra of the kinetic forms)**.** *The solutions of* $X^\top K^\mu + K^\mu X = 0$ *for* $\mu = 0, \dots, 3$ *form*

$$\mathfrak{g}_{\text{massless}} = \mathbb{R}\,\Omega \;\oplus\; \operatorname{span}\{J,\ e_2\Omega,\ e_3\Omega\} \;\cong\; \mathfrak{u}(1)_\Omega \oplus \mathfrak{su}(2).$$

The condition at $\mu = 0$ *is* $X^\top = -X$*, and the conditions at* $\mu = a$ *then say that* X *commutes with the three boosts* $\gamma^0\gamma^a$ *and hence with the whole interior Lorentz algebra. The trace form is* $-8\,\mathbf{1}$ *on this basis, so the algebra is compact. Adding invariance of the mass form,* $X^\top(\gamma^0 J) + (\gamma^0 J)X = 0$*, leaves exactly* $\mathbb{R}J$*. The algebra meets the span of the pole operators in zero.*

The algebra computed is the internal symmetry algebra of the free kinetic action. It becomes a gauge algebra only once a connection is introduced and quantum consistency is checked; the next paragraph does the second for its factors.

The $\mathfrak{su}(2)$ factor contains the fermion-number direction J and two generators that anticommute with J; it therefore mixes a field with its conjugate and is the Pauli–Gürsey algebra of a massless Dirac field [117, 61]. It is a symmetry of the free massless theory, not a gauge candidate on this matter: its two Weyl components form a single doublet, and gauging $SU(2)$ on an odd number of doublets is forbidden by the global anomaly [169]. The central factor is the axial one. Since $\Gamma = J\Omega$, $\Omega = -J\Gamma$, which with $i = J$ and the chirality $\gamma_5 = -\Gamma$ of Theorem 2.17 is the real form of $i\gamma_5$, as in Proposition 2.22; $U(1)_\Omega$ is broken by a Dirac mass and is anomalous on this matter content [15, 2]. The $\mathfrak{u}(1)$ of J, the charge of the Dirac reading, is therefore singled out twice at one constituent: as the whole internal centraliser (Proposition 1.7) and as what survives of the massless symmetry algebra once a mass is admitted. Whether it is the electromagnetic $\mathfrak{u}(1)$ is a separate question, which needs a connection, a kinetic term, a matter coupling and physical degrees of freedom (Chapter 8).

2.5.3 The state space

Theorem 2.30 (The real module is not a state space)**.** *Eight* 16×16 *integer matrices* θ_i*, antisymmetric with* $\theta_i^2 = -\mathbf{1}$ *and pairwise anticommuting, realise the canonical anticommutation relations of the eight real field components exactly. Their commutant in* $M_{16}(\mathbb{R})$ *is* $\mathbb{R}\,\mathbf{1}$*, because the* 256 *Clifford monomials span the whole matrix algebra. Hence no internal complex structure commutes with all eight field operators, and the quadratic observables built from them are real antisymmetric matrices with imaginary spectra. The quantum state space has to be formed from outside, as* $\mathcal{F} = \mathbb{C} \otimes_{\mathbb{R}} \mathbb{R}^{16}$*, with charge* $Q = -i\rho(J)$ *and Hamiltonian* $H = -i\rho(JH(p))$*, where* $\rho(A) = -\frac{1}{4}\sum_{ij} A_{ij}\theta_i\theta_j$ *lifts an antisymmetric* 8×8 *matrix to the module.*

The absence of a commuting complex structure is a statement about the chosen real representation of the field algebra and the observables built in it. The convention for the anticommutation relations is that of the $\chi_i := i\theta_i$, which satisfy $\{\chi_i, \chi_j\} = 2\delta_{ij}$.

It is the complex structure, not the reality, that is missing, which is the lesson of Remark 2.18 arriving one level up.

Theorem 2.31 (The free spectrum of one momentum)**.** *Let* $E > 0$ *(at* $E = 0$ *all sixteen energies coincide). The lifted generators have characteristic polynomials*

$$\chi_{2\rho(J)}(x) = x^6(x^2+4)^4(x^2+16), \qquad \chi_{2\rho(JH)}(x) = x^6(x^2+4E^2)^4(x^2+16E^2),$$

so $\operatorname{spec} Q = \{0, \pm 1, \pm 2\}$ *and* $\operatorname{spec} H = \{0, \pm E, \pm 2E\}$*, with multiplicities* $6, 4, 4, 1, 1$*. The joint eigenspaces of* (H, Q) *are*

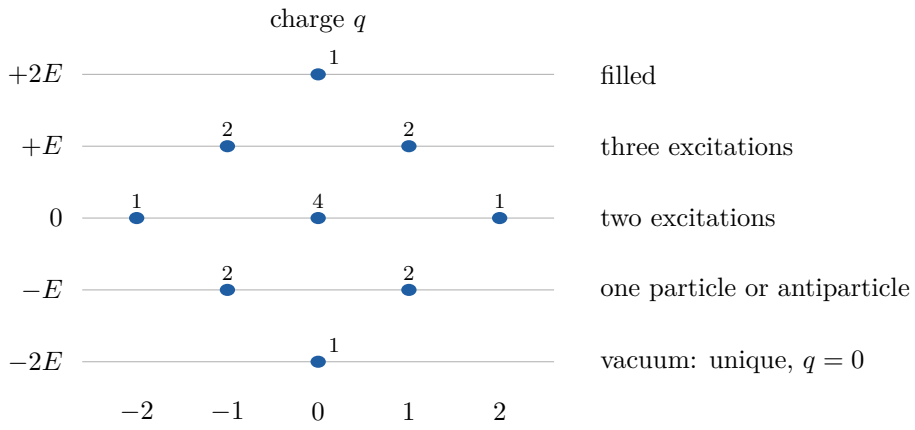


Figure 2.3: The Fock space of one momentum (Theorem 2.31). Each dot is a joint eigenspace of energy and charge, labelled by its dimension. The multiplicities $1, 4, 6, 4, 1$ are those of four fermionic modes: two particle and two antiparticle states of one Dirac flavour, read off rather than inferred.

H	dimension	charges
$-2E$	1	$q = 0$
$-E$	4	$q = \pm 1$, twice each
0	6	$q = \pm 2$ once each, $q = 0$ four times
$+E$	4	$q = \pm 1$, twice each
$+2E$	1	$q = 0$

so $H + 2E$ has spectrum $0, E, 2E, 3E, 4E$ with multiplicities $1, 4, 6, 4, 1$.

The multiplicities $1, 4, 6, 4, 1$ are the binomial coefficients of a four-mode Fock space, as a single Dirac flavour with two particle and two antiparticle states must give. The ground state at $-2E$ is the zero-point energy $-\frac{1}{2}\sum E$ of the four modes; summed over momenta it is the familiar fermion term of the one-loop effective potential. A projector built from $\rho(JH)^2$ could not have produced the table: squaring loses the sign of the energy and conflates the vacuum with the filled state.

Proposition 2.32 (Conjugation on states)**.** *$C \in SO(8)$ lifts to* Spin(8)*; for the representative $C = e_2$ the lift is a product of four integral reflections, and for a general $C = \cos\theta\, e_2 + \sin\theta\, e_3$ four orthogonal reflections suffice. The lift is unitary on $\mathcal{F}$, anticommutes with Q and carries $H(p)$ to $H(-p)$, so $\langle\psi|Q|\psi\rangle = 0$ for every C-eigenstate. The stronger statement $Q|\psi\rangle = 0$ is false: the $+1$ eigenspace of C has dimension eight and Q annihilates only a three-dimensional part of it (real dimensions on $\mathbb{R}^{16}$, complex dimensions on $\mathcal{F}$).*

Remark 2.33 (Many-body chirality is a group lift)**.** ρ is a homomorphism of Lie algebras only, so the product $\rho(J)\rho(\Omega)$, of rank eight, is not a Spin(8) element and is not the chirality of the module. The correct object is $\Gamma_{\mathcal{F}} = \exp[\frac{\pi}{2}(\rho(J) + \rho(\Omega))]$, an involution of $\mathcal{F}$ whose eigenvalue on a joint eigenvector is $i^{q_J + q_\Omega}$; the sectors are labelled by the parity of $\frac{1}{2}(q_J + q_\Omega)$.

Chapter 3

Classical and Electromagnetic Readings of X

The question of this chapter is what the two readings of one vector $X \in \mathbb{R}^{3,3}$—a displacement and a field strength—share, and what keeps them apart. Both carry the metric η and the Clifford action; they differ in units, in their natural Lorentz algebras, in their canonical conjugates and in their observables, and no equivariant linear map joins them (Theorem 3.15). The chapter derives the Lagrangians of both readings, the four-momentum read by a clock (under Assumption 2.16), the field-octant map and Maxwell's equations as one first-order equation on the octants, the bridge $T = \hat{n}J(\hat{n})$ between the Dirac structure and the octants, spin one in both readings, the orthogonality of $\mathbf{E}$ and $\mathbf{B}$, and the undivided energy E_6 with its rest part E_0. The rest energy E_0 is a pointwise invariant, not a generated mass; the even-Clifford obstruction is specific to the octant representation; the coupling classification holds in its stated class.

3.1 Introduction

The letter X is used in this arena for two different things. In the momentum–force construction it is a displacement, a vector of $V = \mathbb{R}^{3,3}$ with the dimension of a length; in the field-strength reading of Section 3.2 the same vector space carries the electric and magnetic fields, $X = \sum_i (B_i e_i + E_i e_{i+3})$. One may keep the two readings strictly apart, X_{cl} for the displacement and $X_{\mathrm{EM}} = (\mathbf{B}, \mathbf{E})$ for the six components of the field two-form, and to build the four-momentum of each separately. This chapter answers four questions put on that basis:

(1) construct the Lagrangians for X, for the Dirac field and for the energies;

(2) differentiate between the two X's, the first a length and the second the pair $(\mathbf{B}, \mathbf{E})$;

(3) construct the four-momentum, the length-valued X being taken along the three directions of the spatial triple V^-, of signature $(-,-,-)$;

(4) place $X_{\mathrm{EM}} = (\mathbf{B}, \mathbf{E})$ on the octants, each of the six components $E_x, E_y, E_z, B_x, B_y, B_z$ on an octant.

Question (4) is read as the pair: X_{EM} is not a sum of two vectors but the six independent components of one field strength. Questions (1) and (2) are answered in Section 3.3, question (3) in Section 3.4 and question (4) in Section 3.5. Section 3.6 takes up the element that the answers share: the temporal volume $T = e_1 e_2 e_3$, which is the Dirac mass form of the classical reading for every clock $\hat{n}$, through $T = \hat{n}J(\hat{n})$ with $J(\hat{n})$ the Dirac complex structure of that clock, and the duality operator of the field reading on the octants. Section 3.10 collects the answers.

What is established, in brief. The two readings share a vector space and a Clifford action, but their units, their natural Lorentz algebras, their canonical conjugates and their observables differ, and no equivariant linear map relates them (Theorem 3.15). The Lagrangians are the six-dimensional worldline and the interior shifted Lagrangian for X_{cl}, the real Grassmann Lagrangian for the Dirac field, and the Maxwell Lagrangian $-\frac{1}{2}\operatorname{Re} q$ for the field, whose Hamiltonian is the undivided energy E_6; the rest energy $E_0 = \frac{1}{2}|q|$ is a Lorentz-invariant and conformal Lagrangian, but not a duality-invariant one and not an admissible Hamiltonian (Theorem 3.21 and Proposition 3.22). With a clock $\hat{n}$ on the temporal triple, the four-momentum $(E/c, \mathbf{p})$ has $\mathbf{p}$ conjugate to the three lengths of V^- and E/c conjugate to the time read by $\hat{n}$, and the four-dimensional mass read by $\hat{n}$ never exceeds the six-dimensional one (Propositions 3.25 and 3.31); the field's four-momentum comes from its stress tensor and is a four-vector only for an isolated whole (Section 3.4.3). On the octants, the joint eigenlines of the pole operators, $\mathbf{E}$ sits on the three octants of Fock weight one and $\mathbf{B}$ on the three of weight two; Maxwell's four equations become one first-order equation with the two Gauss laws on the two poles; and the infinitesimal generators of the boosts and of the duality rotation of the field act on the octants through odd operators (their finite transformations generally have mixed Clifford parity), so that the octant representation of the field is not the spin representation in which a Dirac field transforms (Theorem 3.48 and Proposition 3.49).

Conventions and classes. The conventions are those of Section I.1.4: $\eta = \operatorname{diag}(1,1,1,-1,-1,-1)$, the temporal triple $V^+ = \langle e_1, e_2, e_3\rangle$ carries the converted times of the classical reading and $\mathbf{B}$ of the field reading, the spatial triple $V^- = \langle e_4, e_5, e_6\rangle$ the lengths and $\mathbf{E}$; the paper's $\mathbf{B}$ is minus the textbook magnetic field. Each section states its units. Every claim carries a class stated in words: *fixed* (an identity or theorem that could have failed), *constructed* (admitted and exhibited, not selected), *obstructed* (impossible under stated hypotheses), *open*, *inherited* (true of ordinary physics in any notation, cited), *interpretive* (a reading), or *restated* (established in another chapter and re-verified here). Every finite-dimensional and symbolic claim is checked by a script; the checks are described in Section 15.2.

3.2 The split space as a space of field strengths

The arena was introduced with an electromagnetic picture in mind: six directions, three of one sign and three of the other, paired one to one like the components of a magnetic and an electric field. This section turns the picture into a theorem and measures its price. The binding root G of Proposition 2.3 is the key, and its role in the interior dynamics, where it is the operator that makes bound motion possible, is established in Section 4.3. Here it is used only as a linear map with $G^2 = -\mathbf{1}$ and $G = G^\dagger$.

3.2.1 Arithmetic first

The space $\Lambda^2\mathbb{R}^{1,3}$ with the induced metric $\langle \theta^a \wedge \theta^b, \theta^a \wedge \theta^b\rangle = \eta^{aa}\eta^{bb}$ is negative on the three planes $\theta^0 \wedge \theta^i$ and positive on the three planes $\theta^j \wedge \theta^k$. It therefore has signature $(3,3)$, the Lorentz group acts on it by the adjoint representation, and the Hodge star satisfies $\star^2 = -\mathbf{1}$. This is counting, and it must be said before anything is built on it: every six-dimensional space of split signature is isometric to a Lorentzian field-strength space, so exhibiting one isometry predicts nothing. What is not arithmetic is which subalgebra $\mathfrak{so}(1,3) \subset \mathfrak{so}(3,3)$ realises the action, and whether the operator playing the Hodge star is one the arena already had.

3.2.2 The Lorentz algebra of the binding root

Theorem 3.1 (The centraliser of G)**.** *The centraliser* $\mathfrak{l}_G$ *of* G *in* $\mathfrak{so}(3,3)$ *is isomorphic to* $\mathfrak{so}(3,\mathbb{C}) \cong \mathfrak{sl}(2,\mathbb{C}) \cong \mathfrak{so}(1,3)$*. Its maximal compact subalgebra is the* $\mathfrak{so}(3)$ *of* diagonal *rotations,*

each acting identically on a temporal plane (e_i, e_j) and on its spatial partner (e_{i+3}, e_{j+3}).

Proof. Since $G^2 = -\mathbf{1}$, the identification $z_j = x_j + ix_{j+3}$ makes $\mathbb{R}^6$ a complex space $\mathbb{C}^3$ on which G acts as i, and commuting with G is $\mathbb{C}$-linearity. Under this identification

$$\eta(X, X) = \sum_j x_j^2 - \sum_j x_{j+3}^2 = \operatorname{Re} q(z), \qquad q(z) := \sum_j z_j^2,$$

with q holomorphic. A $\mathbb{C}$-linear map A preserving $\operatorname{Re} q$ preserves it on every phase rotation $e^{i\vartheta}z$, and $q(e^{i\vartheta}z) = e^{2i\vartheta}q(z)$, so $\operatorname{Re}(e^{2i\vartheta}[q(Az) - q(z)]) = 0$ for all ϑ forces $q(Az) = q(z)$. The group is $O(3, \mathbb{C})$ and its algebra $\mathfrak{so}(3, \mathbb{C})$, whose underlying real Lie algebra is $\mathfrak{sl}(2, \mathbb{C}) \cong \mathfrak{so}(1, 3)$ with maximal compact $\mathfrak{so}(3)$. A rotation of (e_i, e_j) alone does not commute with G, because G carries e_i to e_{i+3}; only the rotation performed identically on the partner plane does. □

Proposition 3.2 (Two Lorentz algebras, not conjugate)**.** *The interior algebra $\mathfrak{l}_{\text{int}}$ and $\mathfrak{l}_G$ are isomorphic as Lie algebras but not as subalgebras of $\mathfrak{so}(3, 3)$. The interior algebra fixes the two-dimensional surplus plane pointwise, while $\mathfrak{l}_G$ contains simultaneous rotations of both triples and has no nonzero common fixed vector. Conjugation preserves the dimension of the common fixed subspace, so the two are not conjugate in $GL(6, \mathbb{R})$, let alone in $O(3, 3)$.*

Write L_i for the diagonal rotations and put, for cyclic (i, j, k), $K_i = \frac{1}{2}(e_k e_{j+3} - e_j e_{k+3})$. The six operators L_i, K_i span $\mathfrak{l}_G$.

Theorem 3.3 (The field-strength reading)**.** *Write a vector of $\mathbb{R}^{3,3}$ as $X = \sum_i (B_i e_i + E_i e_{i+3})$. Then*

$$[L_i, L_j] = \epsilon_{ijk} L_k, \qquad [L_i, K_j] = \epsilon_{ijk} K_k, \qquad [K_i, K_j] = -\epsilon_{ijk} L_k,$$

both triples $\mathbf{B}$ and $\mathbf{E}$ transform as vectors under the L_i, and the K_i act by

$$\delta\mathbf{E} = \mathbf{v} \times \mathbf{B}, \qquad \delta\mathbf{B} = -\mathbf{v} \times \mathbf{E}. \tag{3.1}$$

Identify X with the two-form F_X of components $(F_X)_{0i} = E_i$ and $(F_X)_{jk} = \epsilon_{ijk} B_i$. Then G is the Hodge star, $F_{GX} = \star F_X$ with $(\star F)_{ab} = \frac{1}{2}\varepsilon_{abcd} F^{cd}$ and $\varepsilon_{0123} = +1$, and

$$\eta(X, X) = |\mathbf{B}|^2 - |\mathbf{E}|^2, \qquad \eta(X, GX) = -2\,\mathbf{E} \cdot \mathbf{B}$$

are the only invariant quadratic forms. With $\mathfrak{l}_G$ as its Lorentz algebra, $\mathbb{R}^{3,3}$ is $\Lambda^2 \mathbb{R}^{1,3}$.

The orientation matters once, and we fix it here. With $F_{jk} = \epsilon_{ijk} B_i$ the vector $\mathbf{B}$ is minus the textbook magnetic field of the convention $F_{jk} = -\epsilon_{ijk} B_i$ [76], so we write $\mathbf{B}_{\text{t}} = -\mathbf{B}$, as announced in Section I.1.4; the electric field is the textbook one. Passing to a frame moving with velocity $\mathbf{u}$ changes the frame variables by $\delta\mathbf{B} = \mathbf{u} \times \mathbf{E}$ and $\delta\mathbf{E} = -\mathbf{u} \times \mathbf{B}$, which is (3.1) with $\mathbf{v} = -\mathbf{u}$; in textbook variables this is $\mathbf{E}' = \mathbf{E} + \mathbf{u} \times \mathbf{B}_{\text{t}}$ and $\mathbf{B}'_{\text{t}} = \mathbf{B}_{\text{t}} - \mathbf{u} \times \mathbf{E}$, the familiar first-order transformation.

Example 3.4 (A boost along the third axis)**.** Take the pure magnetic field $X = e_1$, that is $\mathbf{B} = (1, 0, 0)$ and $\mathbf{E} = 0$, and the boost generator $K_3 = \frac{1}{2}(e_2 e_4 - e_1 e_5)$, acting on vectors by the commutator. Since e_1 anticommutes with e_2 and with e_4 it commutes with $e_2 e_4$, while $[e_1 e_5, e_1] = -2e_1^2 e_5 = -2e_5$; hence $[K_3, e_1] = e_5$. Under $\delta X = \delta\,[K_3, X]$ the field acquires $\mathbf{E} = (0, \delta, 0)$, in accordance with $\delta\mathbf{E} = \mathbf{v} \times \mathbf{B}$ for $\mathbf{v} = (0, 0, \delta)$. In the same way $[K_3, e_5] = e_1$, so a pure electric field $\mathbf{E} = (0, 1, 0)$ acquires $\mathbf{B} = (\delta, 0, 0) = -\mathbf{v} \times \mathbf{E}$. The invariant $\eta(X, X) = |\mathbf{B}|^2 - |\mathbf{E}|^2$ changes only at second order, as it must.

Corollary 3.5 (The invariant complex structure)**.** *$\mathbb{R}^6$ is irreducible under $\mathfrak{l}_G$, and the commutant of $\mathfrak{l}_G$ in $\operatorname{End}(\mathbb{R}^6)$ is $\operatorname{span}\{\mathbf{1}, G\} \cong \mathbb{C}$. Hence, once $\mathfrak{l}_G$ is taken as the Lorentz algebra, its invariant complex structure is $\pm G$ and nothing else. The interior algebra, by contrast, has a commutant of dimension five, $\mathbb{R}$ on the four-plane and all of $M_2(\mathbb{R})$ on the surplus plane.*

The logic of this corollary runs from G to $\mathfrak{l}_G$ and back to $\pm G$, so the choice of G is recovered, not removed. It is a consistency statement, a uniqueness conditional on taking $\mathfrak{l}_G$ as the Lorentz algebra. Removing the choice would require a characterisation of $\mathfrak{l}_G$ that does not mention G, and we know none.

3.2.3 The two readings exclude one another

The price of the field-strength reading is not small.

Proposition 3.6 (Exclusion)**.** *Under* $\mathfrak{l}_{\text{int}}$ *the six directions decompose as* $V \oplus 2\cdot\mathbf{1}$*, a four-vector and two scalars; under* $\mathfrak{l}_G$ *they are irreducible, with no fixed vector. The two representations are not similar, so a single theory cannot use both algebras as the Lorentz action on the same six directions. If the vector space* $\mathbb{R}^{3,3}$ *is read as the electromagnetic field strength, those six directions are not a four-plane plus a surplus pair, and the Dirac structure, the projector census,* P_* *and the diagonal sector all lose the arena on which they are built.*

What is excluded is reading one six-dimensional space in two ways, not a theory that contains both kinds of module. A theory may contain spacetime vectors and field strengths as different representations of one Lorentz algebra, which is the ordinary arrangement, and $\mathrm{Cl}(3,3)$ realises it with the vectors at grade one and the field strength at grade two.

Proposition 3.7 (The field strength at grade two)**.** *Let* $\Lambda^2 E$ *be the span of the six bivectors of the interior four-plane. It is closed under the adjoint action of* $\mathfrak{l}_{\text{int}}$*, on which it is the adjoint representation; the trace form* $\frac{1}{8}\operatorname{Tr}(XY)$ *on it has signature* $(3,3)$*; and the interior volume* $\Omega = \gamma^0\gamma^1\gamma^2\gamma^3$*, with* $\Omega^2 = -\mathbf{1}$*, commutes with* $\mathfrak{l}_{\text{int}}$*, preserves* $\Lambda^2 E$ *by left multiplication, exchanges* $\gamma^0\gamma^i$ *with the spatial bivectors and acts on* $\Lambda^2 E$ *as* minus *the Hodge star of Theorem 3.3 (*$\varepsilon_{0123} = +1$*, the bivector* $\frac{1}{2}F_{ab}\gamma^a\gamma^b$ *standing for the two-form* F*). Equivalently, the space-time-algebra pseudoscalar* $I_4 := \gamma_0\gamma_1\gamma_2\gamma_3 = -\Omega$*, written with the lower-index generators* $\gamma_\mu = \eta_{\mu\nu}\gamma^\nu$*, acts as the Hodge star.*

Grade three behaves differently under the two algebras, and the difference sharpens the exclusion. Under $\mathfrak{l}_G$ one has $(\Lambda^3\mathbb{R}^6)_{\mathbb{C}} = 2\,(0,0) \oplus 2\,(1,1)$, of dimension $2 + 18$, so no field-strength module $(1,0)\oplus(0,1)$ occurs at grade three at all: trivectors cannot carry the field for this action. Under $\mathfrak{l}_{\text{int}}$, $\mathbb{R}^6 = E \oplus \mathbb{R}^2$ gives $\Lambda^3\mathbb{R}^6 \cong 2E \oplus 2\Lambda^2 E$, which contains two field-strength modules. The quadratic Casimir distinguishes the cases cleanly, with spectrum $\{0^{(2)}, 8^{(18)}\}$ on $\Lambda^3\mathbb{R}^6$ under $\mathfrak{l}_G$ and $\{3^{(8)}, 4^{(12)}\}$ under $\mathfrak{l}_{\text{int}}$, the value 4 being that of $(1,0)\oplus(0,1)$ and 3 that of a four-vector. An explicit attempt confirms it: the six trivectors $C_i = \sum_{j,k}\epsilon_{ijk}e_j N_k$, $N_j = e_{k+3}e_{l+3}$ cyclically, together with $D_i = \mathcal{V}C_i$, transform as two vectors under the L_i, but all eighteen brackets $[K_i, C_j]$, $[K_i, D_j]$ leave their span, for instance $\frac{1}{2}[K_1, C_1] = e_2e_3e_4 + e_4e_5e_6$, and the Lorentz closure of the span has dimension 18.

3.2.4 How much of Maxwell's theory is forced

Theorem 3.3 supplies a field with the right transformation law and nothing that the field satisfies. How much of the dynamics does the transformation law fix by itself? The answer is: the form of the equations completely, and nothing else.

Theorem 3.8 (Exactly two first-order covariant operators)**.** *Consider linear first-order differential operators with constant coefficients on Minkowski space, acting on a field* F *valued in* $\Lambda^2\mathbb{R}^{1,3}$ *and taking values in the vector representation. The derivative* ∂F *lies in* $\Lambda^2 \otimes V$*, of dimension* 24*, and under* $\mathfrak{so}(1,3)$

$$\Lambda^2 \otimes V = \underbrace{(\tfrac{3}{2},\tfrac{1}{2}) \oplus (\tfrac{1}{2},\tfrac{3}{2})}_{16} \;\oplus\; \underbrace{2\,(\tfrac{1}{2},\tfrac{1}{2})}_{8},$$

so the vector module occurs with multiplicity exactly two, and no other module of dimension four occurs. The two copies are realised by

$$(dF)_{\mu\nu\rho} = 3\,\partial_{[\mu}F_{\nu\rho]} \quad (\textit{through } \Lambda^3 \cong V), \qquad (\delta F)_\nu = \partial^\mu F_{\mu\nu},$$

and every covariant operator of the class is $\mathcal{D}F = a\,\delta F + b\,\delta(\star F)$*, since* $\delta\star = \pm\star d$*.*

The Casimir operator confirms the count: on $\Lambda^2 \otimes V$ its spectrum is $\{3^{(8)}, 9^{(16)}\}$, and the value 3 is the vector one. Nothing in the theorem constrains operators of higher order, operators with variable coefficients, or operators valued in the remaining $(\frac{3}{2},\frac{1}{2}) \oplus (\frac{1}{2},\frac{3}{2})$.

Corollary 3.9 (The form of Maxwell's equations)**.** *Within this class the two available equations are $dF = \mathcal{J}_m$ and $d \star F = \mathcal{J}_e$. Covariance does not rule out a magnetic current, which is a covariant field exactly as an electric current is; it identifies the two operators and leaves open which equations are imposed and which sources are admitted. Every source-free choice needs no external data; the equation $\delta F = 0$ alone qualifies as well. What duality selects is the pair: among the nonzero real subspaces of the two operator equations, only the whole space is invariant under the continuous duality rotation, because the Hodge star G exchanges the two operators. Within this linear class the source-free system $dF = \delta F = 0$ is therefore the only duality-invariant choice. Finally, on a contractible region $dF = 0$ gives $F = dA$ by the Poincaré lemma, so the gauge potential is a consequence of the first equation rather than an assumption.*

In the variables of the frame the system takes a compact form, which we use repeatedly below.

Proposition 3.10 (Maxwell's equations on the frame)**.** *Read $\mathbb{R}^{3,3}$ as $\mathbb{C}^3$ through G, $z_j = B_j + iE_j$. Then $dF = 0$ and $d \star F = 0$ for the two-form F_X of Theorem 3.3 are*

$$\partial_t X = -G\,(\nabla \times X), \qquad \nabla \cdot X = 0, \tag{3.2}$$

that is $\partial_t z = -i\,\nabla \times z$ and $\nabla \cdot z = 0$, with the curl built from the rotation generators L_i. Splitting into the two triples gives $\partial_t \mathbf{B} = \nabla \times \mathbf{E}$, $\partial_t \mathbf{E} = -\nabla \times \mathbf{B}$ and $\nabla \cdot \mathbf{E} = \nabla \cdot \mathbf{B} = 0$; in textbook variables these are Faraday's law $\partial_t \mathbf{B}_{\mathrm{t}} = -\nabla \times \mathbf{E}$ and the Ampère–Maxwell law $\partial_t \mathbf{E} = \nabla \times \mathbf{B}_{\mathrm{t}}$ in vacuum. With a source, $\partial_\mu F^{\mu\nu} = (\rho, \mathbf{j})$ reads

$$\partial_t z = -i\,\nabla \times z - i\,\mathbf{j}, \qquad \nabla \cdot z = i\rho, \tag{3.3}$$

and taking the divergence of the first equation gives $\partial_t \rho + \nabla \cdot \mathbf{j} = 0$: continuity is a consequence of the system and the compatibility condition that any proposed current must satisfy.

Up to a factor i, z is the Riemann–Silberstein vector $\mathbf{E} + i\mathbf{B}_{\mathrm{t}}$, which obeys $i\partial_t(\mathbf{E} + i\mathbf{B}_{\mathrm{t}}) = \nabla \times (\mathbf{E} + i\mathbf{B}_{\mathrm{t}})$ [142, 13], and the holomorphic form of the proof of Theorem 3.1 has

$$\operatorname{Re} q(z) = |\mathbf{B}|^2 - |\mathbf{E}|^2, \qquad \operatorname{Im} q(z) = 2\,\mathbf{E} \cdot \mathbf{B},$$

so the two Maxwell invariants are the real and imaginary parts of one holomorphic quadratic form. Three classical facts fall into place. The duality rotation is the phase rotation $z \mapsto e^{i\vartheta} z$ that the proof of Theorem 3.1 already used, and it turns the pair of invariants by 2ϑ. The two quadratic forms that will compete in the interior dynamics of Sections 4.2 and 4.3 reappear with physical names: the η-form $\operatorname{Re} q$ is the Lorentz invariant, while the Hermitian form $|z|^2 = |\mathbf{B}|^2 + |\mathbf{E}|^2$ is twice the energy density and depends on the frame. And the remaining sesquilinear covariant is the Poynting vector, $\frac{i}{2}\,\bar{z} \times z = \mathbf{E} \times \mathbf{B}$, so that $\mathbf{E} \times \mathbf{B}_{\mathrm{t}} = -\frac{i}{2}\,\bar{z} \times z$.

Example 3.11 (A circularly polarised wave on the frame)**.** The field $z = (1, i, 0)\,e^{ik(x_3 - t)}$ solves (3.2): $\nabla \cdot z = 0$ because z has no third component, and $\nabla \times z = ik\,(-i, 1, 0)e^{ik(x_3-t)} = k\,(1, i, 0)e^{ik(x_3-t)}$, so $-i\nabla \times z = -ik\,z = \partial_t z$. Its invariant is $q(z) = (1 + i^2)e^{2ik(x_3-t)} = 0$, so $|\mathbf{B}| = |\mathbf{E}|$ and $\mathbf{E} \cdot \mathbf{B} = 0$ at every point: the wave is null. Its Poynting vector $\frac{i}{2}\bar{z} \times z = \frac{i}{2}(1, -i, 0) \times (1, i, 0) = \frac{i}{2}(0, 0, 2i) = (0, 0, -1)$ is $\mathbf{E} \times \mathbf{B}$, so $\mathbf{E} \times \mathbf{B}_{\mathrm{t}} = (0, 0, 1)$ points along the direction of propagation, as it should.

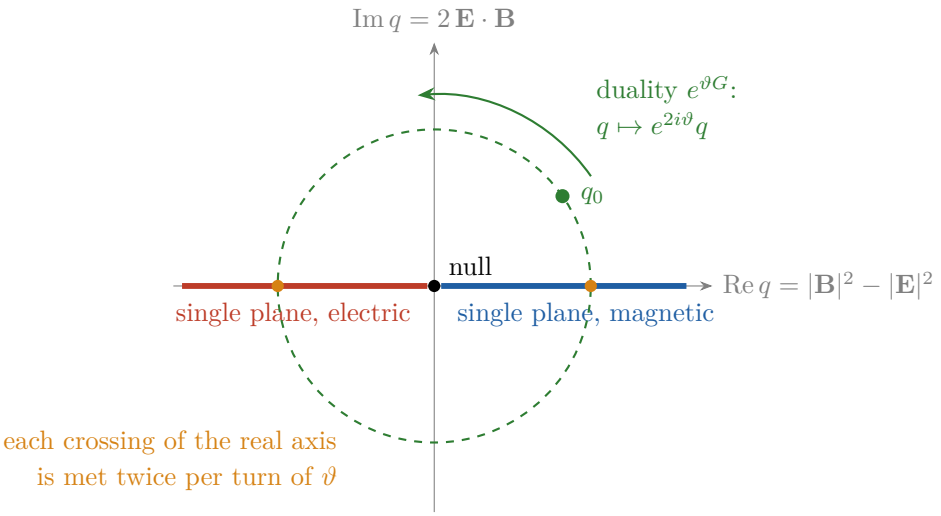


Figure 3.1: The plane of the complex invariant $q = z \cdot z$ of a frame vector (Theorem 3.3 and Proposition 3.12). The real axis is the locus $\mathbf{E}\cdot\mathbf{B} = 0$ on which the spacetime two-form is a single plane, magnetic on the right and electric on the left; the origin is the null cone. A duality rotation $e^{\vartheta G}$ preserves $|q|$ and turns q at twice the rate, so a non-null field becomes a single plane four times as ϑ runs once around.

3.2.5 Frame vectors as spacetime planes

Each pairing plane (e_i, e_{i+3}) carries one complex component $z_i = B_i + iE_i$, and a frame vector is at the same time a bivector of spacetime. The two descriptions meet in the following way.

Proposition 3.12 (Frame vectors as planes)**.** *Let $X = \sum_i (B_i e_i + E_i e_{i+3})$ and F_X as above. Then* $\operatorname{Pf} F_X = \mathbf{E}\cdot\mathbf{B}$ *and* $\det F_X = (\mathbf{E}\cdot\mathbf{B})^2$*. A nonzero F_X is a single plane $u\wedge v$ exactly when $\mathbf{E}\cdot\mathbf{B} = 0$, that is when* $\operatorname{Im} q = 0$*. Otherwise there is a Lorentz frame in which $F_X = a\,\theta^0\wedge\theta^1 + b\,\theta^2\wedge\theta^3$ is a pair of orthogonal planes, and the duality rotation $e^{-\vartheta G}$ with $\tan\vartheta = a/b$ carries it to the single plane $\sqrt{a^2+b^2}\,\theta^2\wedge\theta^3$, the complexion of Rainich and of Misner and Wheeler [129, 98].*

Figure 3.1 shows the content of this proposition in the plane of the complex invariant. The real axis is the locus of single planes, the positive half of magnetic type and the negative half of electric type, the origin is the null cone, and the duality rotation moves a field around a circle at twice its own angle. In Section 4.3 the bound orbits of the interior dynamics will turn out to be exactly such rotations.

Everything in this section is a statement about a representation and its invariants. Theorem 3.8 is a multiplicity count that fixes the form of the equations and nothing about their content: it does not produce a source, does not say that a source is the current of anything, supplies no coupling constant and no Lagrangian. Nor is any of it special to this arena, since the count uses only $\mathfrak{so}(1,3)$ and Λ^2. What the arena adds is that the objects the count needs, the module, the duality and the invariants, were already present, so that the system can be written in the arena's own variables (3.2). The deflationary test of Section I.1.2 applies to it in full.

3.3 Two readings of X and their Lagrangians

The question asked in the introduction contains two requests that this section answers together: that the two uses of the letter X be told apart, the first being a length and the second the pair $(\mathbf{B}, \mathbf{E})$ of three-vectors, and that Lagrangians be written for X, for the Dirac field and for the energies. The section types the two readings (Section 3.3.1), writes the Lagrangians of the classical reading (Section 3.3.2), of the Dirac field (Section 2.4) and of the energies (Section 3.3.3),

assembles them (Section 3.3.4) and states what is fixed and what is not (Section 3.3.5). Identifiers r1-$\hat{n}$ refer to the lines `PASS r1-`$\hat{n}$ printed by `r1_lagrangians.py`, which checks every finite-dimensional and symbolic claim in exact arithmetic; Euler–Lagrange equations, Noether currents and stress tensors are derived there on general symbolic fields, and the Grassmann-valued Dirac field is handled by an exact Grassmann jet algebra.

Conventions are those of Section I.1.4: $V = \mathbb{R}^{3,3}$ with $\eta = \mathrm{diag}(1,1,1,-1,-1,-1)$, the temporal triple $V^+ = \langle e_1, e_2, e_3\rangle$ and the spatial triple $V^- = \langle e_4, e_5, e_6\rangle$, the realisation of Appendix A.1, the binding root $Ge_i = e_{i+3}$, $Ge_{i+3} = -e_i$, which satisfies $G^2 = -\mathbf{1}$, $\eta G = (\eta G)^\top$ and $G^\top \eta G = -\eta$ (r1-1), the interior Lorentz algebra $\mathfrak{l}_{\mathrm{int}}$ on $E = \langle e_1, e_4, e_5, e_6\rangle$ with $\gamma^\mu = (e_1, e_4, e_5, e_6)$, $J = e_2e_3$ and $T = e_1e_2e_3 = \gamma^0 J$. Spacetime signature is $(+,-,-,-)$. The paper's $\mathbf{B}$ is minus the textbook magnetic field, $\mathbf{B}_{\mathrm{t}} = -\mathbf{B}$ (Section I.1.4); every Maxwell formula below names which of the two it uses. Each claim carries a class in words: *fixed*, *constructed*, *obstructed*, *open*, *inherited*, *interpretive* or *restated*.

3.3.1 The two readings, typed

Definition 3.13 (The classical reading)**.** A classical displacement in the arena is

$$X_{\mathrm{cl}} = Y = \sum_{i=1}^{3} c\,t^i\, e_i + \sum_{a=1}^{3} x^a\, e_{a+3}, \qquad \eta(Y,Y) = \sum_i (ct^i)^2 - \sum_a (x^a)^2, \tag{3.4}$$

with $[t^i] = \mathrm{s}$, $[x^a] = \mathrm{m}$, $[c\,t^i] = \mathrm{m}$ and $[\eta(Y,Y)] = \mathrm{m}^2$. Its length-valued part $\sum_a x^a e_{a+3}$ lies on the spatial triple: this is the precise content of the phrase that X is taken in the three directions of sign $(-,-,-)$. The temporal components are times, converted to lengths by the constant c.

The conversion is not a choice. A map $C : V^+ \to V^-$ that commutes with the common rotations of the two triples is $C = c\,\mathbf{1}$ with $[c] = \mathrm{m\,s^{-1}}$ (Theorem 2.2; Proposition 3.98), and the binding root acts on the unconverted coordinates $(t;x)$ as

$$G_C = \begin{pmatrix} 0 & -C^{-1} \\ C & 0 \end{pmatrix} = \Phi^{-1} G\, \Phi, \qquad \Phi = \mathrm{diag}(c,c,c,1,1,1), \qquad \Phi(t;x) = Y. \tag{3.5}$$

Hence G is dimensionless on Y, while G_C carries entries of dimension $\mathrm{m\,s^{-1}}$ and $\mathrm{s\,m^{-1}}$: G maps lengths to lengths only after the conversion, and on Y it sends the length x^a to the converted time slot e_a (r1-2; restated from Section 2.1, the commutation criterion being fixed there).

Definition 3.14 (The field reading)**.** The field-strength reading of Theorem 3.3 is, with SI units and c explicit,

$$X_{\mathrm{EM}} = \sum_{i=1}^{3} (cB_i\, e_i + E_i\, e_{i+3}), \qquad z_c = c\mathbf{B} + i\mathbf{E}, \qquad \eta(X_{\mathrm{EM}}, X_{\mathrm{EM}}) = c^2|\mathbf{B}|^2 - |\mathbf{E}|^2 = \mathrm{Re}\,(z_c \cdot z_c), \tag{3.6}$$

with $[E_i] = [cB_i] = \mathrm{V\,m^{-1}}$, so $[\eta(X_{\mathrm{EM}}, X_{\mathrm{EM}})] = \mathrm{V^2\,m^{-2}}$. X_{EM} is the pair $(c\mathbf{B}; \mathbf{E})$ of three-vectors, the coordinates of one vector of $\mathbb{R}^{3,3}$ in the basis $e_1, \dots, e_6$, and not a sum of $\mathbf{E}$ and $\mathbf{B}$: its six entries label the six components of the field two-form, $F_{0i} = E_i$ and $F_{jk} = \epsilon_{ijk} B_i$ in Heaviside–Lorentz units with $c = 1$. In these units $X = \sum_i (B_i e_i + E_i e_{i+3})$, $z = \mathbf{B} + i\mathbf{E}$, $q = z \cdot z$, $E_6 = \frac{1}{2}|z|^2$, $E_0 = \frac{1}{2}|q|$, and $E_6^2 = E_0^2 + |\mathbf{S}|^2$ with $\mathbf{S} = \mathbf{E} \times \mathbf{B}$ (Proposition 3.93). The same matrix Φ of (3.5) converts $(\mathbf{B}; \mathbf{E})$ to $(c\mathbf{B}; \mathbf{E})$.

Theorem 3.15 (The type rule)**.** *The two readings share the vector space $\mathbb{R}^{3,3}$, its form η and the Clifford action on $S = \mathbb{R}^8$, and differ in the following four respects.*

(i) Units. *A linear map $X_{\mathrm{EM}} \mapsto X_{\mathrm{cl}}$ has entries of dimension $\mathrm{m^2\,V^{-1}}$. No monomial $c^a \ell^b \hbar^d$, with real exponents, has this dimension, where ℓ is the length of the shift operator (equation (2.2)); with the vacuum permittivity admitted, the only such monomial is $\ell^3(\varepsilon_0/\hbar c)^{1/2}$.*

(ii) Lorentz action. *Once the clock e_1 is chosen, $\mathfrak{l}_{\text{int}}$ acts on Y as $E \oplus 2\cdot\mathbf{1}$: the four-vector $(ct^1; x^1, x^2, x^3)$, which complexifies to $(\frac{1}{2}, \frac{1}{2})$, and the two scalars ct^2, ct^3. The algebra that acts naturally on X_{EM} is $\mathfrak{l}_G$, the centraliser of G in $\mathfrak{so}(3,3)$, under which V is irreducible and complexifies to $(1,0)\oplus(0,1)$ (Section 3.2). The two subalgebras of $\mathfrak{so}(3,3)$ meet in 0.*

(iii) No equivariant map. *For every Lie algebra isomorphism $\phi : \mathfrak{l}_{\text{int}} \to \mathfrak{l}_G$ let V_ϕ be V with $\mathfrak{l}_{\text{int}}$ acting through ϕ. Then $\operatorname{Hom}_{\mathfrak{l}_{\text{int}}}(E, V_\phi) = \operatorname{Hom}_{\mathfrak{l}_{\text{int}}}(V_\phi, E) = 0$ and $\operatorname{Hom}_{\mathfrak{l}_{\text{int}}}(E \oplus 2\cdot\mathbf{1}, V_\phi) = \operatorname{Hom}_{\mathfrak{l}_{\text{int}}}(V_\phi, E\oplus 2\cdot\mathbf{1}) = 0$. By contrast $\operatorname{Hom}_{\mathfrak{l}_{\text{int}}}(\Lambda^2 E, V_\phi) = \{(a+bG)M_0\}$ is two-dimensional: the field strength of the interior four-plane, at grade two (Proposition 3.7), is carried to X_{EM} uniquely up to a scale and a duality angle.*

(iv) Conjugates and observables. *The six components of Y are configuration coordinates that Poisson-commute, with conjugate covector K_A of dimension $\mathrm{kg\,m\,s^{-1}}$; its observables are a worldline, a four-momentum and a mass. In Maxwell's theory the configuration variable is the potential $\mathbf{a}$ with $\mathbf{B} = \nabla\times\mathbf{a}$, and $\mathbf{E}$ is (up to ε_0) the momentum conjugate to $\mathbf{a}$, so X_{EM} is a point of phase space, not of configuration space: $\{\int \mathbf{f}\cdot\mathbf{E}, \int \mathbf{g}\cdot\mathbf{B}\} = -\int \mathbf{f}\cdot\nabla\times\mathbf{g}$ (Heaviside–Lorentz) for test fields $\mathbf{f}, \mathbf{g}$, while the $\mathbf{E}$–$\mathbf{E}$ and $\mathbf{B}$–$\mathbf{B}$ brackets vanish. Its observables are E_6, $\mathbf{S}$, the invariants q and the stress tensor.*

Proof. (i) Over the base $(\mathrm{m}, \mathrm{kg}, \mathrm{s}, \mathrm{A})$ the volt is $\mathrm{kg\,m^2\,s^{-3}\,A^{-1}}$; none of c, ℓ, $\hbar$ contains the ampere, so the ampere exponent of $\mathrm{m^2V^{-1}}$ cannot be matched, and with ε_0 the linear system for the exponents has the unique solution stated (r1-3). (ii) $\mathfrak{l}_{\text{int}}$ is spanned by $R_i = \frac{1}{2}e_{j+3}e_{k+3}$ and $B_i = \frac{1}{2}e_1e_{i+3}$ ((i,j,k) cyclic), acting on vectors by the commutator; they annihilate e_2, e_3 and preserve E. The centraliser of G in $\mathfrak{so}(3,3)$ has dimension 6 and is spanned by $L_i = \frac{1}{2}(e_{j+3}e_{k+3} - e_je_k)$ and $K_i = \frac{1}{2}(e_ke_{j+3} - e_je_{k+3})$; its commutant on V is $\operatorname{span}\{\mathbf{1}, G\} \cong \mathbb{C}$, a division algebra, and V is irreducible (Corollary 3.5). The spans of $\{R_i, B_i\}$ and $\{L_i, K_i\}$ inside $\mathfrak{so}(3,3)$ have trivial intersection (r1-4). (iii) Both triples satisfy $[R_i, R_j] = \epsilon_{ijk}R_k$, $[R_i, B_j] = \epsilon_{ijk}B_k$, $[B_i, B_j] = -\epsilon_{ijk}R_k$ and the same with (L, K), so $R_i \mapsto L_i$, $B_i \mapsto K_i$ is an isomorphism (r1-4). For any isomorphism ϕ, V_ϕ is irreducible of dimension 6 and E irreducible of dimension 4. The image of an equivariant $M : E \to V_\phi$ is a submodule of dimension at most 4, hence 0; the kernel of $M : V_\phi \to E$ is a submodule of dimension at least 2, hence all of V_ϕ. On the trivial summand an equivariant map lands in the fixed vectors of V_ϕ, of which there are none, and a map out of V_ϕ into it is zero by irreducibility. An intertwiner $\Lambda^2E \to V_\phi$ exists for every ϕ: the six bivectors of E are $2R_i$ and $2B_i$, so Λ^2E is the adjoint module of $\mathfrak{l}_{\text{int}}$ (Proposition 3.7), ϕ intertwines it with the adjoint module of $\mathfrak{l}_G$ pulled back by ϕ, and that module is isomorphic to V (r1-5). Schur's lemma [57] on the irreducible $\Lambda^2E \cong V_\phi$ then gives an intertwiner space equal to the commutant $\operatorname{span}\{\mathbf{1}, G\}$. The exact solution of the intertwining equations confirms all these dimensions for ϕ and for its parity twist $B_i \mapsto -K_i$ (r1-5), and the Casimirs separate the modules: $\sum R_i^2 - \sum B_i^2 = -3$, $\sum R_iB_i = 0$ on E, against -4 and $-2G$ (eigenvalues $\pm 2i$) on V (r1-6). (iv) The bracket follows from $\{a_i(\mathbf{x}), E_j(\mathbf{y})\} = \delta_{ij}\delta^3(\mathbf{x}-\mathbf{y})$, read off the first-order action of Proposition 3.23(ii), and from $\mathbf{g}\cdot\nabla\times\mathbf{a} - \mathbf{a}\cdot\nabla\times\mathbf{g} = \nabla\cdot(\mathbf{a}\times\mathbf{g})$ (r1-31). □

Parts (i) and (iii) are fixed: (iii) could have failed for a different pair of subalgebras, and (i) obstructs only conversions built from c, ℓ and $\hbar$ in SI units. Since $(\varepsilon_0/\hbar c)^{1/2}\mathbf{E}$ has dimension $\mathrm{m^{-2}}$, which is the field of Heaviside–Lorentz units with $\hbar = c = 1$, in those units ℓ^3 alone supplies the dimension (r1-3); what obstructs a dictionary between the two readings is (iii), not (i). Part (ii) is restated from Proposition 3.2 and Proposition 3.6 except its last sentence, which is fixed (r1-4), and (iv) is inherited electrodynamics [69]. The identity map of $\mathbb{R}^{3,3}$ is linear but not equivariant between the two actions, so sharing the vector space identifies nothing.

Remark 3.16 (Coordinates and modules of the two readings)**.** The surplus coordinates u^a are the converted times ct^2, ct^3 of the temporal triple, length-valued only through $C = c\,\mathbf{1}$. The distinction between the cotangent momentum K and the momentum π of the interior

Lagrangian is a distinction between two Lagrangians for the same coordinates, as Section 3.3.2 shows: applied to Y, the interior Lagrangian has canonical momentum $p = \eta\pi$. As a statement about $\mathfrak{l}_{\mathrm{int}}$-modules, $V_{\mathrm{EM}} \cong \Lambda^2 E^*_{\hat{n}}$ holds through an isomorphism ϕ, unique up to the factor $a+bG$, but $\Lambda^2 E$ is not a subspace of V, and $\mathfrak{l}_G$, under which V is irreducible (Corollary 3.5), preserves no proper subspace of V, while G itself preserves no Lorentzian four-plane (Proposition 3.40). Parts (i) and (iii) give the rule of interpretation: no linear equivariant map exists in either direction, for any isomorphism of the two Lorentz algebras, and in SI units no monomial in c, ℓ, $\hbar$ supplies its dimension; in particular there is no linear equivariant map $\Lambda^2 E^*_{\hat{n}} \to E_{\hat{n}}$, nor one in the opposite direction.

3.3.2 Lagrangians for the classical reading

Proposition 3.17 (The free worldline in the arena)**.** *Let $Y(\lambda)$ be a curve with $\eta(Y', Y') > 0$, λ an arbitrary parameter, $[M] = \mathrm{kg}$, and*

$$L_{\mathrm{free}} = -Mc\sqrt{\eta(Y', Y')}, \qquad S = \int L_{\mathrm{free}}\, d\lambda, \quad [S] = \mathrm{J\,s}. \tag{3.7}$$

(i) $K_A = \partial L/\partial Y'^A = -Mc\,\eta_{AB}Y'^B/\sqrt{\eta(Y', Y')}$, with $\eta^{AB}K_AK_B = M^2c^2$ and $K_AY'^A - L = 0$: the canonical Hamiltonian vanishes, L is homogeneous of degree one in Y', and the dynamics is the constraint $H = \frac{N}{2}(\eta^{AB}K_AK_B - M^2c^2)$. The kinetic six-momentum is $P^A = -\eta^{AB}K_B = McY'^A/\sqrt{\eta(Y', Y')}$.

(ii) The Euler–Lagrange equations are $dK_A/d\lambda = 0$; their solutions are the straight lines, in every parametrisation.

(iii) The timelike set $\{\eta > 0\}$ and the shell $\{\eta(K, K) = M^2c^2\}$ are connected: the curve $(\cos s, \sin s, 0; 0, 0, 0)$ joins e_1 to $-e_1$ with $\eta = 1$, so there is no past and no future. With the clock e_1 and $P = (\mathsf{E}/c, q_1, q_2; \mathbf{p})$, the shell reads $\mathsf{E}^2 = c^2|\mathbf{p}|^2 + M^2c^4 - c^2|\mathbf{q}|^2$; for $M = c = 1$ and $\mathbf{q} = (2, 0)$ the four-dimensional mass squared is -3.

Proof. (i)–(ii) by differentiation (r1-7). A straight line $Y_0 + Vf(\lambda)$ with $f' > 0$ solves the equations, while the timelike curve $(2\cos\lambda, 2\sin\lambda, 0; \lambda, 0, 0)$, with $\eta(Y', Y') = 3$ and Y'' not parallel to Y', does not (r1-8). (iii) Write $Y = (u; v)$ with $u \in V^+$, $v \in V^-$. Along $(u; (1-s)v)$, $0 \le s \le 1$, η stays positive, so $\{\eta > 0\}$ retracts onto $\{(u; 0) : u \neq 0\}$ and hence onto the unit sphere of V^+, which is connected; the shell is the image of $S^2 \times \mathbb{R}^3$ under $(\omega, v) \mapsto (\sqrt{M^2c^2 + |v|^2}\,\omega; v)$, with inverse $(u; v) \mapsto (u/|u|, v)$. Finally $\eta(P, P) = \mathsf{E}^2/c^2 + |\mathbf{q}|^2 - |\mathbf{p}|^2$ (r1-9). □

The form (3.7) is inherited [82]; (iii) is restated from Propositions 12.1 to 12.2 and Example 12.3, where the surplus momenta $\mathbf{q}$ produce the unstable band and the first-class constraints $q_a \approx 0$ (Proposition 11.4, Theorem 12.12) remove it. This constraint is Option II of Proposition 3.28.

The interior Lagrangian of Chapter 4 is the split-form Lagrangian (4.11). Applied to the length-valued Y it reads

$$L_\kappa = \frac{m}{2}\,\eta(Y' + \kappa GY,\, Y' + \kappa GY), \qquad \kappa = \frac{c}{\ell}, \quad [\tau] = \mathrm{s}, \quad [m] = \mathrm{kg}, \quad [L_\kappa] = \mathrm{J}, \tag{3.8}$$

with $Y' = dY/d\tau$; the sum $Y' + \kappa GY$ is homogeneous in dimension only because G acts on the converted coordinates ((3.5), r1-2, r1-36).

Proposition 3.18 (The interior Lagrangian on lengths)**.** *(i) $p = \partial L_\kappa/\partial Y' = m\eta(Y' + \kappa GY)$, $\pi := \eta^{-1}p = m(Y' + \kappa GY)$, and the Euler–Lagrange equation is $Y'' = -\kappa^2 Y$ on all six components; on shell $p' = \kappa G^\top p$ and $\pi' = \kappa G\pi$.*

(ii) $L_\kappa = \frac{m}{2}(\eta(Y',Y') - \kappa^2\eta(Y,Y)) + \frac{d}{d\tau}\Lambda$ *with* $\Lambda = \frac{m\kappa}{2}\eta(Y,GY) = -m\kappa\sum_a (ct^a)\,x^a$, *and* $p - m\eta Y' = \partial\Lambda/\partial Y$.

(iii) The energy $\mathcal{E} = p\cdot Y' - L_\kappa = \frac{m}{2}(\eta(Y',Y') + \kappa^2\eta(Y,Y))$ *is conserved and equals* $H(Y,p) = \frac{1}{2m}\eta^{-1}(p,p) - \kappa\, p^\top GY$, *which generates the flow. Explicitly* $\mathcal{E} = \frac{m}{2}\sum_i((c\dot{t}^i)^2 + \kappa^2(ct^i)^2) - \frac{m}{2}\sum_a((\dot{x}^a)^2 + \kappa^2(x^a)^2)$*: the three converted-time oscillators carry positive and the three length oscillators negative energy, and both* $\mathcal{E}$ *on* (Y,Y') *and* H *on* (Y,p) *have signature* $(6,6)$.

Proof. (i) Differentiation and $G^\dagger = G$ (r1-10). (ii) $\eta(GY,GY) = -\eta(Y,Y)$, $\frac{d}{d\tau}\eta(Y,GY) = 2\eta(Y',GY)$ because ηG is symmetric, and $\eta(Y,GY) = -2\sum_i y_i y_{i+3}$ (r1-11). (iii) Substitution and Hamilton's equations; the inertia of the two 12×12 Hessians is computed exactly (r1-12). □

Bounded solutions do not make the energy bounded below. Every solution oscillates with the single frequency κ because the system is linear and its two signs of energy are decoupled; $\mathcal{E}$ and H have signature $(6,6)$, so once the length oscillators are coupled to a system of positive energy, or quantised with a positive norm, they are ghosts whose excitation lowers the energy. The surplus constraints of Section 3.4 concern the free worldline and do not remove these interior modes. Which constraints of the coupled interior system, if any, remove the negative-energy oscillators, and which modes are physical, is not derived here.

Item (i) is restated (equations (4.11) and (4.12), and Theorem 4.8); in particular the raised momentum obeys $\pi' = \kappa G\pi$ for constant G. Item (ii) is Theorem 4.12 read on lengths: the boundary term that is $-m\kappa\,\mathbf{E}\cdot\mathbf{B}$ in the field reading is $-m\kappa\sum_a(ct^a)x^a$, the correlation of converted time and length along each paired axis. Item (iii) is fixed; the indefinite sign is that of Theorem 9.12.

Proposition 3.19 (The spatial triple is not invariant)**.** *Every solution is* $Y(\tau) = e^{-\kappa G\tau}Y_0 + \frac{\sin\kappa\tau}{\kappa m}\pi_0$, $\pi(\tau) = e^{\kappa G\tau}\pi_0$, *with* $e^{sG} = \cos s\,\mathbf{1} + \sin s\,G$.

(i) Phase-space data of lengths and spatial momenta, $Y_0 = \sum_a x_0^a e_{a+3}$ *and* $\pi_0 \in V^-$, *acquire the temporal part* $\sin(\kappa\tau)\sum_a x_0^a e_a$*: lengths are rotated into converted times at once.*

(ii) Configuration data $(Y_0, Y_0') \in V^-\times V^-$ *stay in* V^-, *since* $Y'' = -\kappa^2 Y$ *is diagonal, but their canonical momentum has the temporal part* $\pi(0)^+ = -m\kappa\sum_a x_0^a e_a$, *nonzero for* $Y_0 \neq 0$, *and* $\pi' = \kappa G\pi$ *maps* V^- *into* V^+*; for example* $\pi(0) = e_5$ *gives* $\pi(\tau) = \cos(\kappa\tau)e_5 - \sin(\kappa\tau)e_2$.

(iii) The zero-momentum orbits $e^{-\kappa G\tau}Y_0$ *are duality rotations;* $e^{-\kappa G\tau}e_4 = \cos(\kappa\tau)e_4 + \sin(\kappa\tau)e_1$.

Proof. The formula solves $Y'' = -\kappa^2 Y$ and $m(Y' + \kappa GY) = \pi$ with $Y(0) = Y_0$; the components are read off (r1-13). □

This is fixed. Restricting X_{cl} to the spatial triple is therefore not preserved by the interior dynamics as a statement about phase space, while it is preserved as a statement about configurations. Proposition 3.41 extends the rank of the leakage of the example in (ii) to every clock. Its reading needs one correction. By (ii) of Proposition 3.18 the rotation of π out of a subspace is carried by the gradient $m\kappa\eta GY$ of a boundary term, which a canonical transformation removes (Theorem 4.12); the kinetic momentum mY' obeys $mY'' = -m\kappa^2 Y$ and preserves V^- and every four-plane $E_{\hat{n}}$. The obstruction of Proposition 3.40 and of the leakage is therefore an obstruction to taking the canonical π as the observable momentum, not a property of the motion. Neither π nor mY' is conserved, so neither is a four-momentum.

3.3.3 Lagrangians for the energies

In this subsection units are Heaviside–Lorentz with $c = 1$, A_μ is covariant, $\mathbf{a} = (A_1, A_2, A_3)$, and $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ with $F_{0i} = E_i$ and $F_{jk} = \epsilon_{ijk}B_i$ (Theorem 3.3). In SI every density below is multiplied by ε_0 with $\mathbf{B}\to c\mathbf{B}$, which makes it a density in $\mathrm{J\,m^{-3}}$ (r1-36).

Proposition 3.20 (Maxwell's Lagrangian in the field reading)**.** *(i)* $\mathbf{E} = \partial_t \mathbf{a} - \nabla A_0$, $\mathbf{B} = \nabla \times \mathbf{a}$, *and* $L_M = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} = \frac{1}{2}(|\mathbf{E}|^2 - |\mathbf{B}|^2) = -\frac{1}{2}\eta(X_{\mathrm{EM}}, X_{\mathrm{EM}}) = -\frac{1}{2}\operatorname{Re} q$*; in SI,* $L_M = \frac{\varepsilon_0}{2}(|\mathbf{E}|^2 - c^2|\mathbf{B}|^2)$.

(ii) The Euler–Lagrange equations of $L_M - A_\mu j^\mu$ *are* $\partial_\mu F^{\mu\nu} = j^\nu$*:* $\nabla \cdot \mathbf{E} = \rho$ *and* $\partial_t \mathbf{E} = -\nabla \times \mathbf{B} - \mathbf{j}$ *(in textbook variables* $\partial_t \mathbf{E} = \nabla \times \mathbf{B}_{\mathrm{t}} - \mathbf{j}$*), together with the identities* $\partial_t \mathbf{B} = \nabla \times \mathbf{E}$, $\nabla \cdot \mathbf{B} = 0$.

(iii) The symmetric tensor $\Theta^{\mu\nu} = F^{\mu\lambda} F_\lambda{}^\nu + \frac{1}{4}\eta^{\mu\nu} F_{\alpha\beta} F^{\alpha\beta}$ *is traceless, with* $\Theta^{00} = E_6$ *and* $\Theta^{0i} = -(\mathbf{E} \times \mathbf{B})_i = (\mathbf{E} \times \mathbf{B}_{\mathrm{t}})_i$*: the momentum density is* $-\mathbf{S}$ *in the paper's variables. It obeys* $\partial_\mu \Theta^{\mu\nu} = -F^\nu{}_\lambda \, \partial_\mu F^{\mu\lambda}$ *identically, and* $\Theta^\mu{}_\lambda T^\lambda{}_\nu = E_0^2 \delta^\mu_\nu$*, so that* $E_0 = \frac{1}{2}\sqrt{T_{\mu\nu}\Theta^{\mu\nu}}$.

(iv) The second invariant $\mathbf{E} \cdot \mathbf{B} = \frac{1}{2} \operatorname{Im} q = -\mathbf{E} \cdot \mathbf{B}_{\mathrm{t}}$ *is the divergence* $\mathbf{E} \cdot \mathbf{B} = -\partial_\mu K^\mu$ *of the Chern–Simons current* $K^\mu = \frac{1}{2}\varepsilon^{\mu\nu\rho\sigma} A_\nu \partial_\rho A_\sigma$ *(*$\varepsilon_{0123} = +1$*), with* $K^0 = -\frac{1}{2}\mathbf{a} \cdot \nabla \times \mathbf{a}$*; its Euler–Lagrange expression vanishes identically, so* $\theta\, \mathbf{E} \cdot \mathbf{B}$ *changes no field equation.*

Proof. Symbolic differentiation on general $A_\mu(t, \mathbf{x})$ and $j^\mu(t, \mathbf{x})$ (r1-20, r1-21, r1-23); (iii) on general $(\mathbf{E}, \mathbf{B})$ (r1-22). □

All of this is inherited [76, 82, 129]; the Rainich identity and $E_6^2 = E_0^2 + |\mathbf{S}|^2$ are restated from Propositions 3.93 to 3.94. Two signs are fixed by the paper's orientation and are easy to lose: the second invariant is $+\frac{1}{2}\operatorname{Im} q$ in the paper's variables and $-\frac{1}{2}\operatorname{Im} q$ in textbook ones, and the momentum density Θ^{0i} is $-\mathbf{S}$, not $\mathbf{S}$. Maxwell's Lagrangian is the η-form of X_{EM}, a member of the Lorentz class of Theorem 4.26.

The rest energy as a Lagrangian. Write $S = \frac{1}{2}(|\mathbf{E}|^2 - |\mathbf{B}_{\mathrm{t}}|^2)$ and $P = \mathbf{E} \cdot \mathbf{B}_{\mathrm{t}}$, so that $E_0 = \frac{1}{2}|q| = \sqrt{S^2 + P^2}$, and set $L_0 := E_0$. For a Lagrangian $L(S, P)$ put $\mathbf{D} = \partial L / \partial \mathbf{E}$ and $\mathbf{H} = -\partial L / \partial \mathbf{B}_{\mathrm{t}}$.

Theorem 3.21 (The rest-energy Lagrangian)**.** *(i)* L_0 *is Lorentz invariant (annihilated by* $\mathfrak{l}_G$*), homogeneous of degree two, and invariant pointwise under the duality rotation* $z \mapsto e^{i\theta} z$*, which does not leave* $L_M = -\frac{1}{2}\operatorname{Re} q$ *invariant. It is not differentiable at nonzero null fields (at* $F = 0$ *its derivative vanishes): along* $\mathbf{B}_{\mathrm{t}} = (1, 0, 0)$, $\mathbf{E} = (0, e, 0)$ *it equals* $\frac{1}{2}|1 - e^2|$*, with one-sided derivatives* -1 *and* $+1$ *at* $e = 1$.

(ii) $\mathbf{D} = (S\mathbf{E} + P\mathbf{B}_{\mathrm{t}})/E_0$, $\mathbf{H} = (S\mathbf{B}_{\mathrm{t}} - P\mathbf{E})/E_0$*, and the Hamiltonian density is* $\mathbf{D} \cdot \mathbf{E} - L_0 = \frac{S}{E_0} E_6 = -\cos(\arg q)\, E_6$*, which ranges over* $[-E_6, E_6]$ *and equals* $-\frac{1}{2}$ *on the magnetic field* $\mathbf{B}_{\mathrm{t}} = (1, 0, 0)$, $\mathbf{E} = 0$*, where* L_0 *is not convex in* $\mathbf{E}$.

(iii) For every $L(S, P)$ *the gauge-invariant tensor* $\Theta^\mu{}_\nu = \frac{\partial L}{\partial(\partial_\mu A_\lambda)} F_{\nu\lambda} - \delta^\mu_\nu L$ *equals* $L_S T_M{}^\mu{}_\nu + \delta^\mu_\nu (S L_S + P L_P - L)$ *and obeys* $\partial_\mu \Theta^\mu{}_\nu = -\mathcal{E}^\lambda F_{\nu\lambda}$*, with* $\mathcal{E}^\lambda$ *the Euler–Lagrange expression of* A_λ*. Since* L_0 *is homogeneous of degree one in* (S, P)*, its tensor* $T_0 = (S/E_0) T_M$ *is traceless:* L_0 *is conformal.*

(iv) For any $L(S, P)$, $\mathbf{D} \cdot \mathbf{H} - \mathbf{E} \cdot \mathbf{B}_{\mathrm{t}} = P(L_S^2 - L_P^2 - 1) - 2 S L_S L_P$*. The defect is* $-2P$ *for* L_0 *and* $-(1 + \lambda^2)P$ *for* λL_0*: no multiple of* E_0 *is a duality-invariant Lagrangian in the sense of Gaillard and Zumino.*

(v) The ModMax Lagrangians $L_\gamma = \cosh\gamma\, S + \sinh\gamma\, E_0 = \cosh\gamma\, L_M + \sinh\gamma\, L_0$ *have vanishing defect for every* γ*, are conformal, and for* $\gamma \geq 0$ *have* $L_S \geq e^{-\gamma} > 0$ *and* $\partial_\gamma L_\gamma = \frac{1}{2}\sqrt{T^{(\gamma)}_{\mu\nu} T^{(\gamma)\mu\nu}}$*; in particular* $L_0 = \partial_\gamma L_\gamma|_{\gamma=0} = \frac{1}{2}\sqrt{T^M_{\mu\nu} T^{M\mu\nu}}$*. As* $\gamma \to \infty$, $L_\gamma / \sinh\gamma \to S + E_0 = L_M + L_0$*, whose defect is* $-P$.

Proof. (i) Under L_i, K_i the variations of S and P vanish; $|q|$ is invariant under $q \mapsto e^{2i\theta}q$ (r1-24). (ii) Differentiation and $\mathbf{D}\cdot\mathbf{E} - L_0 = (SE^2 + P^2 - S^2 - P^2)/E_0$ with $E^2 - S = E_6$; the Hessian in $\mathbf{E}$ at the stated point has a negative eigenvalue (r1-25). (iii) With $G^{\mu\lambda} = \partial L/\partial(\partial_\mu A_\lambda) = L_S \partial S/\partial(\partial_\mu A_\lambda) + L_P \partial P/\partial(\partial_\mu A_\lambda)$ the identity is polynomial in $(\partial A, L_S, L_P, L)$; conservation uses $G^{\mu\lambda}\partial_\mu F_{\nu\lambda} = \frac{1}{2}G^{\mu\lambda}\partial_\nu F_{\mu\lambda}$, which holds for antisymmetric G by the Bianchi identity (r1-27). (iv) $\mathbf{D} = L_S\mathbf{E} + L_P\mathbf{B}_\mathrm{t}$, $\mathbf{H} = L_S\mathbf{B}_\mathrm{t} - L_P\mathbf{E}$ (r1-25, r1-26). (v) $L_S = \cosh\gamma + \sinh\gamma\,(S/E_0)$ with $S/E_0 \in [-1, 1]$; by (iii) and Proposition 3.20(iii), $\frac{1}{2}\sqrt{T^{(\gamma)}\cdot T^{(\gamma)}} = L_S E_0 = \partial_\gamma L_\gamma$ (r1-26, r1-28). □

The general defect formula in (iv) is restated from Section 3.8; every other computation in the theorem is fixed. The duality invariance and conformality of ModMax are inherited [197, 210], as is the Gaillard–Zumino criterion [46, 53] (Section 3.8) and the generation of ModMax by the root-$T\bar{T}$ flow [196]; the scripts re-derive them. Calling L_0 "the rest-energy Lagrangian" is interpretive, and the theorem says what the name costs. E_0 is a pointwise invariant, the energy density in the local rest frame of a non-null field; the invariant mass of a localised configuration is the Lorentz length of the integrated $T^{0\mu}$, which is at least $\int E_0$, with equality only when all points share one rest frame. The mass m of the Dirac Lagrangian below is an input, not generated. L_0 is an admissible Lagrangian, Lorentz invariant and conformal, but it is not duality invariant although E_0 is pointwise duality invariant; its energy density is indefinite; and it is singular exactly on the null fields that carry radiation. A positive-energy field theory with Lagrangian λE_0 is obstructed. What the rest energy is, in the Lagrangian formalism, is the operator that deforms Maxwell's theory into ModMax: $L_0 = \partial_\gamma L_\gamma|_0$, equal to $\frac{1}{2}\sqrt{T_{\mu\nu}\Theta^{\mu\nu}}$ by Rainich's identity. The limit $\gamma \to \infty$ does not isolate it but gives $L_M + L_0$.

Hamiltonians. In Hamiltonian variables $(\mathbf{D}, \mathbf{B}_\mathrm{t})$ put $w = \mathbf{B}_\mathrm{t} + i\mathbf{D}$, $s = E_6(w) = \frac{1}{2}(|\mathbf{D}|^2 + |\mathbf{B}_\mathrm{t}|^2)$, $p = |\mathbf{D}\times\mathbf{B}_\mathrm{t}| = |\mathbf{S}(w)|$ and $E_0(w) = \frac{1}{2}|w\cdot w| = \sqrt{s^2 - p^2}$, the last by the identity of Proposition 3.93. For $\mathcal{H}(\mathbf{D}, \mathbf{B}_\mathrm{t})$ the field equations are $\dot{\mathbf{D}} = \nabla\times\mathbf{H}$, $\dot{\mathbf{B}}_\mathrm{t} = -\nabla\times\mathbf{E}$ with $\mathbf{E} = \partial\mathcal{H}/\partial\mathbf{D}$, $\mathbf{H} = \partial\mathcal{H}/\partial\mathbf{B}_\mathrm{t}$; the energy flux is $\mathbf{E}\times\mathbf{H}$, the momentum density $\mathbf{D}\times\mathbf{B}_\mathrm{t}$, and Lorentz invariance requires $\mathbf{E}\times\mathbf{H} = \mathbf{D}\times\mathbf{B}_\mathrm{t}$; duality invariance is $\mathbf{E}\cdot\mathbf{B}_\mathrm{t} = \mathbf{D}\cdot\mathbf{H}$ and conformal invariance $\mathbf{D}\cdot\mathbf{E} + \mathbf{B}_\mathrm{t}\cdot\mathbf{H} = 2\mathcal{H}$ [199, 197].

Proposition 3.22 (Which energies are Hamiltonians)**.** *For $\mathcal{H} = a\,s + b\,E_0(w)$ one has $\mathbf{E}\times\mathbf{H} = (a^2 - b^2)\,\mathbf{D}\times\mathbf{B}_\mathrm{t}$, and $\mathcal{H}$ is duality invariant and conformal for all (a, b). Hence $\mathcal{H}$ is Lorentz invariant exactly when $a^2 - b^2 = 1$; with $a > 0$ these are $\cosh\gamma\,E_6 - \sinh\gamma\,E_0$, the Maxwell Hamiltonian $s = E_6$ at $\gamma = 0$ and the ModMax Hamiltonians for $\gamma > 0$. The rest energy alone, $\mathcal{H} = E_0$, gives $\mathbf{E}\times\mathbf{H} = -\mathbf{D}\times\mathbf{B}_\mathrm{t}$ and is not Lorentz invariant. The Bialynicki-Birula Hamiltonian $\mathcal{H} = p = |\mathbf{S}(w)| = \sqrt{E_6^2 - E_0^2}$ satisfies all three conditions.*

Proof. With $\xi = \frac{1}{2}(|\mathbf{D}|^2 - |\mathbf{B}_\mathrm{t}|^2)$, $\eta_w = \mathbf{D}\cdot\mathbf{B}_\mathrm{t}$ and $E_0 = \sqrt{\xi^2 + \eta_w^2}$, $\mathbf{E} = a\mathbf{D} + b(\xi\mathbf{D} + \eta_w\mathbf{B}_\mathrm{t})/E_0$ and $\mathbf{H} = a\mathbf{B}_\mathrm{t} + b(\eta_w\mathbf{D} - \xi\mathbf{B}_\mathrm{t})/E_0$; the cross terms cancel and the b^2 term is $-(\xi^2 + \eta_w^2)\mathbf{D}\times\mathbf{B}_\mathrm{t}/E_0^2$ (r1-29). □

The classification of conformal duality-invariant Hamiltonians, ModMax and Bialynicki-Birula's, is inherited [197, 199]. What is fixed here is its form in the variables of Section 3.9: the differentiation $E_6^2 = E_0^2 + |\mathbf{S}|^2$ of the undivided energy supplies all three duality invariants, the admissible Hamiltonians are $\cosh\gamma\,E_6 - \sinh\gamma\,E_0$ and $|\mathbf{S}|$, and the rest energy is admissible only together with E_6, never alone.

Proposition 3.23 (The undivided energy is a Hamiltonian, not a Lagrangian)**.** *(i) Used as a Lagrangian of the potential, $E_6 = \frac{1}{2}(|\mathbf{E}|^2 + |\mathbf{B}|^2)$ with $\mathbf{E} = \partial_t\mathbf{a} - \nabla A_0$, $\mathbf{B} = \nabla\times\mathbf{a}$ gives $\nabla\cdot\mathbf{E} = 0$ and $\partial_t\mathbf{E} = +\nabla\times\mathbf{B}$; a transverse plane wave then needs $\omega^2 + k^2 = 0$. In gauge-fixed (transverse) form the equations are elliptic.*

(ii) *The first-order action* $L_H = \mathbf{E}\cdot\partial_t\mathbf{a} - E_6 + A_0\nabla\cdot\mathbf{E}$, *with* $\mathbf{E}$ *independent, has Euler–Lagrange equations* $\mathbf{E} = \partial_t\mathbf{a} - \nabla A_0$, $\partial_t\mathbf{E} = -\nabla\times\mathbf{B}$, $\nabla\cdot\mathbf{E} = 0$, *and on the first of them* $L_H = L_M + \nabla\cdot(A_0\mathbf{E})$*:* E_6 *is the Legendre transform of* L_M*, with* $\mathbf{E}$ *conjugate to* $\mathbf{a}$.

(iii) *With two potentials,* $\mathbf{B} = \nabla\times\mathbf{A}$ *and* $\mathbf{E} = \nabla\times\mathbf{C}$, $X_A = (\mathbf{A};\mathbf{C})$*, the Schwarz–Sen Lagrangian* $L_{SS} = \frac{1}{2}(\nabla\times X_A)\cdot G\dot{X}_A - \frac{1}{2}|\nabla\times X_A|^2 = \frac{1}{2}(\mathbf{E}\cdot\dot{\mathbf{A}} - \mathbf{B}\cdot\dot{\mathbf{C}}) - E_6$ *is first order in time with Hamiltonian* E_6*, is exactly invariant under* $X_A \mapsto e^{\theta G}X_A$*, and the duality charge density is* $-\frac{1}{2}X_A\cdot\nabla\times X_A$*, minus the optical helicity density.* $L_{SS} - (\mathbf{E}\cdot\dot{\mathbf{A}} - E_6)$ *has identically vanishing Euler–Lagrange expression.*

Proof. Symbolic Euler–Lagrange equations on general fields, with the plane wave $\mathbf{a} = (0, e^{i(kx-\omega t)}, 0)$ and $A_0 = 0$ in (i) (r1-30, r1-31, r1-32). □

Item (i) is fixed, (ii) inherited [69], and (iii) restated from equation (3.15) and Theorem 3.81, the duality-symmetric form of Deser and Teitelboim and of Schwarz and Sen [31, 137]. In (iii) duality is a symmetry off shell because both terms are G-invariant: $E_6 = \frac{1}{2}|z|^2$ has duality weight zero, and G is orthogonal for the Euclidean product of the symplectic term. This is the field form of the dichotomy of Theorem 4.26 and Theorem 3.90: the Lagrangian $-\frac{1}{2}\eta(X,X)$ lies in the Lorentz class, the Hamiltonian E_6 in the duality class, and the two classes meet in 0, so no quadratic form is both. E_0 enters neither L_M nor L_H nor L_{SS}; it enters as the deformation of Theorem 3.21(v) and of Proposition 3.22.

3.3.4 The total Lagrangian

With $\hbar = c = 1$ the total Lagrangian of the field theory and of a classical charged test particle is

$$\begin{aligned} \mathcal{L}_{\mathrm{tot}} &= -\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} + \tfrac{1}{2}\psi^{\top}(K^{\mu}D_{\mu} - mT)\psi, \\ L_{\mathrm{pp}} &= -M\sqrt{\eta(Y',Y')} - e\,A_{\mu}(x)\,x'^{\mu}, \qquad x = (Y^1;Y^4,Y^5,Y^6). \end{aligned} \tag{3.9}$$

Proposition 3.24 (Coupled equations and conservation)**.** *(i)* $\partial\mathcal{L}_{\mathrm{tot}}/\partial A_{\nu} = q\,j^{\nu}$*, so that Maxwell's equations read* $\partial_{\mu}F^{\mu\nu} = J_e^{\nu} := -q\,j^{\nu}$*; the Dirac equation is* $K^{\mu}D_{\mu}\psi = mT\psi$.

(ii) *The gauge-covariant tensor* $\Theta_D{}^{\mu}{}_{\nu} = \frac{1}{2}\psi^{\top}K^{\mu}D_{\nu}\psi - \delta^{\mu}_{\nu}L_D$ *satisfies* $\partial_{\mu}\Theta_D{}^{\mu}{}_{\nu} = qF_{\lambda\nu}j^{\lambda}$ *on the Dirac shell, and* $\partial_{\mu}T_M{}^{\mu}{}_{\nu} = -F_{\nu\lambda}\partial_{\mu}F^{\mu\lambda}$*, which equals* $qF_{\nu\lambda}j^{\lambda}$ *on the Maxwell shell. Hence* $\partial_{\mu}(T_M + \Theta_D)^{\mu}{}_{\nu} = 0$ *on the coupled shell.*

(iii) *For* L_{pp} *the surplus momenta* K_2, K_3 *are conserved, and the interior equations are the Lorentz force* $\frac{d}{d\lambda}(Mx'_{\mu}/\sqrt{\eta(Y',Y')}) = eF_{\mu\nu}x'^{\nu}$.

Proof. (i)–(ii) in the Grassmann jet algebra with coefficients depending on general $A_{\mu}(t,\mathbf{x})$, the time derivatives of ψ being eliminated with the Dirac equation (r1-33, r1-34). (iii) Symbolic Euler–Lagrange equations for general $A_{\mu}(x)$ (r1-35). □

The coupled statement is inherited [160]; in the arena it is the one-constituent version of Theorem 5.77, with the Grassmann kernels of (2.5) in place of the commuting ones there, and the checks make it fixed for this realisation. The four-momentum of the field is $P^{\mu} = \int T^{0\mu}_{\mathrm{tot}}\,d^3x$, developed in Section 3.4.3.

Each reading enters $\mathcal{L}_{\mathrm{tot}}$ in its own place. The classical reading is the argument: the coordinates $x^{\mu} = (ct^1;x^a)$ on which A_{μ} and ψ depend, and the worldline $Y(\lambda)$, whose surplus components ct^2, ct^3 couple to nothing but the six-dimensional mass shell. The field reading is a value: X_{EM} is the field strength $F = dA$ of the interior potential, entering through $L_M = -\frac{1}{2}\eta(X_{\mathrm{EM}}, X_{\mathrm{EM}})$, through $\Theta^{\mu\nu}$ and through the Lorentz force, and it becomes a vector of $\mathbb{R}^{3,3}$ with the action of $\mathfrak{l}_G$ only through the two-parameter dictionary of Theorem 3.15(iii). The two meet only in the derivative ∂_{μ}, of dimension m^{-1} in the classical reading, which carries the potential to the field strength.

symbol	reading	SI unit	natural Lorentz action	Lagrangian	canonical conjugate	Noether charges
$Y = X_{\mathrm{cl}}$	displacement; lengths on V^-	m	$\mathfrak{l}_{\mathrm{int}}$ on $(ct^1; x)$ after the clock; ct^2, ct^3 scalars	$-Mc\sqrt{\eta(Y', Y')}$	K_A, $\mathrm{kg\,m\,s^{-1}}$	K_A (translations), $\mathfrak{so}(3,3)$ charges $Y_A K_B - Y_B K_A$
Y, interior	configuration	m	strict fixed-G invariance supplies none; the oscillator action is $O(3,3)$-invariant up to a boundary term	$\frac{m}{2}\eta(Y' + \kappa GY, Y' + \kappa GY)$	$p = m\eta(Y' + \kappa GY)$	energy, signature $(6,6)$
$A_\mu = (A_0; \mathbf{a})$	potential	$\mathrm{V\,s\,m^{-1}}$ ($A_0 = \phi/c$)	$\mathfrak{l}_{\mathrm{int}}$, $(\frac{1}{2}, \frac{1}{2})$	L_M through $F = dA$	$\varepsilon_0 \mathbf{E}$ conjugate to $\mathbf{a}$	$P^\mu = \int T^{0\mu} d^3x$
$X_{\mathrm{EM}} = (c\mathbf{B}; \mathbf{E})$	field strength	$\mathrm{V\,m^{-1}}$	$\mathfrak{l}_G$, $(1,0) \oplus (0,1)$	$L_M = -\frac{\varepsilon_0}{2}\eta(X_{\mathrm{EM}}, X_{\mathrm{EM}})$	none: a phase-space point	observables E_6, $\mathbf{S}$, q, $\Theta^{\mu\nu}$
$X_A = (\mathbf{A}; \mathbf{C})$	dual potentials	$\mathbf{A}$: $\mathrm{V\,s\,m^{-1}}$, $\mathbf{C}$: V	boosts only on shell; duality $e^{\theta G}$	L_{SS}	$\frac{1}{2}G^\top \nabla \times X_A$ (second class)	energy $\int E_6$, helicity
ψ	Dirac field	$\mathrm{m^{-3/2}}$	$\mathfrak{l}_{\mathrm{int}}$, $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$	$\frac{1}{2}\psi^\top (K^\mu D_\mu - mT)\psi$	$\frac{1}{2}\psi^\top$ (second class)	$\int j^0$, P^μ
E_6	undivided energy	$\mathrm{J\,m^{-3}}$	Θ^{00}, a tensor component	Hamiltonian of L_M, L_{SS}; not a Lagrangian	–	generates time translation
E_0	rest energy	$\mathrm{J\,m^{-3}}$	scalar, duality invariant	L_0 (interpretive); ModMax deformation	–	–

Table 3.1: The objects of Section 3.3: reading, unit, Lorentz action, Lagrangian, conjugate and charges.

3.3.5 Answer

The Lagrangians asked for in the introduction, and the status of each statement, are the following.

(1) *The two X's.* $X_{\mathrm{cl}} = Y$ is a displacement in metres whose length-valued part lies on the spatial triple, the temporal part being times converted by the unique c; $X_{\mathrm{EM}} = (c\mathbf{B}; \mathbf{E})$ is a field strength in volts per metre. They share $\mathbb{R}^{3,3}$, η and the Clifford action, and nothing else: no nonzero linear equivariant map joins them for any identification of their Lorentz algebras, and in SI units no monomial in c, ℓ, $\hbar$ supplies the unit $\mathrm{m^2 V^{-1}}$ such a map would need (Theorem 3.15; fixed, the first statement obstructing a dictionary; the second holds only in SI units, since in Heaviside–Lorentz units with $\hbar = c = 1$ the monomial ℓ^3 supplies the conversion). They are kept apart by treating X_{cl} as the argument and X_{EM} as a value of the fields (Section 3.3.4; interpretive).

(2) *Lagrangians for X_{cl}.* The free worldline (3.7) (inherited form; its split-signature consequences restated) and the interior Lagrangian (3.8) on lengths, with $\kappa = c/\ell$, Euler–Lagrange equation $Y'' = -\kappa^2 Y$, $\pi' = \kappa G\pi$ (restated) and an energy of signature $(6,6)$ (fixed). The interior flow does not preserve the spatial triple in phase space (fixed); the obstruction concerns the canonical momentum, which differs from the kinetic one by a boundary-term gradient (fixed).

(3) *The Dirac Lagrangian.* (2.5), with the Grassmann variation rule (fixed), the field equation $J\gamma^\mu\partial_\mu\psi = m\psi$ (restated), the current $\frac{1}{2}\psi^\top K^\mu J\psi$ and the canonical and Belinfante tensors (inherited, checked).

(4) *Lagrangians for the energies.* Maxwell's $L_M = -\frac{1}{2}\eta(X_{\mathrm{EM}}, X_{\mathrm{EM}})$ (inherited); E_6 is its Hamiltonian, in the first-order and the duality-symmetric forms (inherited, restated), and as a Lagrangian it gives elliptic equations (fixed); E_0 is a Lorentz-invariant conformal Lagrangian with indefinite energy, pointwise duality invariant but not duality invariant in the sense of Gaillard and Zumino (fixed), and it is the generator of ModMax and the partner of E_6 in the ModMax Hamiltonian (fixed; the ModMax facts inherited). Reading L_0 as "the rest-energy Lagrangian" is interpretive, and as the Lagrangian of a positive-energy theory it is obstructed.

(5) *The total Lagrangian.* (3.9), with total conservation on the coupled shell (inherited, checked) and Table 3.1.

(6) *Open.* A dynamical relation between the two readings, which would have to be a new, dimensionful, covariant map, and the quantisation of either reading in the arena's own terms, remain open.

3.4 The four-momentum with X on the spatial triple

The third question asked in the introduction is how the four-momentum is constructed when X is taken in the three directions $(-,-,-)$. We read the sign triple as the values $\eta(e_{a+3}, e_{a+3}) = -1$ of the metric on the spatial triple $V^- = \langle e_4, e_5, e_6\rangle$: the classical configuration X_{cl} has its three lengths there, and its three remaining components are times converted by c on the temporal triple V^+. In brief: (1) a clock $\hat{n} \in V^+$ selects the four-plane $E_{\hat{n}} = \mathbb{R}\hat{n} \oplus V^-$, and these are exactly the Lorentzian four-planes that contain the three lengths; Theorem 2.17 holds at every clock, with complex structure $J(\hat{n}) = e_1e_2e_3\,\hat{n}$ (Section 2.2); (2) the four-momentum of a particle is the projection $P_{\hat{n}} = (E/c, \mathbf{p})$ of the six-momentum, conjugate to $(ct_{\hat{n}}, x^a)$ and conserved by every dynamics that is translation invariant along $E_{\hat{n}}$; of the three mass shells the arena's metric fixes the sign of the ambient one, and the gauge principle of Section 12.4 selects the reduced one (Section 3.4.1); (3) rotating the clock reads a fixed six-momentum as a four-dimensional mass with $m_4(\hat{n})^2 \le \mu^2$, with equality exactly at the clocks parallel to the temporal momentum (Section 3.4.2); (4) the field X_{EM} has no linear, and no quadratic, covariant map to a four-vector: its four-momentum is the integrated first row of the stress tensor, a four-vector only for an isolated whole (Section 3.4.3); (5) the no-go for the binding root concerns G on the vector module $E_{\hat{n}}$, where it leaks two of the three spatial directions into the surplus plane at every clock; it does not concern $G = \star$ on the field module (Section 3.4.4). Section 3.4.5 collects the verdicts.

Conventions and units. As in Section 1.1, Section 2.1 and Section 3.2: $V = \mathbb{R}^{3,3}$, $\eta = \mathrm{diag}(1,1,1,-1,-1,-1)$, the realisation of equation (1.1), $s_a := e_{a+3}$, the binding root $Ge_i = e_{i+3}$, $Ge_{i+3} = -e_i$ ($G^2 = -\mathbf{1}$, η-self-adjoint, an anti-isometry) and the para-complex $\mathsf{K}e_i = e_{i+3}$, $\mathsf{K}e_{i+3} = e_i$ ($\mathsf{K}^2 = \mathbf{1}$, η-skew; this is $-\mathcal{K}$ for the grading root $\mathcal{K}$ of Proposition 2.3). A bivector $a \wedge b$ acts on V by $v \mapsto a\,\eta(b,v) - b\,\eta(a,v) = \frac{1}{2}[ab, v]$. In the particle reading units are SI with c explicit: times t^i in s, lengths x^a and the coordinates Y^A in m, $[c] = \mathrm{m\,s^{-1}}$, momenta p_a, q_a and K in $\mathrm{kg\,m\,s^{-1}}$, energies in J, masses M_6, m_4, μ in kg; clocks $\hat{n}$ and frames are dimensionless. In the field reading the computations use Heaviside–Lorentz units with $c = 1$, with $\mathbf{B}$ measured as $c\mathbf{B}$; the SI form is restored through $z_c = c\mathbf{B} + i\mathbf{E}$, $[\mathbf{E}] = [c\mathbf{B}] = \mathrm{V\,m^{-1}}$ (Proposition 3.99). The paper's $\mathbf{B}$ is minus the textbook magnetic field, $\mathbf{B}_{\mathrm{t}} = -\mathbf{B}$ (Section I.1.4); the frame Poynting vector is $\mathbf{S} = \mathbf{E}\times\mathbf{B} = -\mathbf{S}_{\mathrm{t}}$, with $\mathbf{S}_{\mathrm{t}} = \mathbf{E}\times\mathbf{B}_{\mathrm{t}}$ the textbook one (equation (3.17)). Identifiers r2-$\hat{n}$

refer to `r2_fourmomentum.py`; r2-0a–c check these conventions, the Fock modes $a_i^{\pm} = \frac{1}{2}(e_i \pm e_{i+3})$ (CAR, N counting the minus signs of the pole operators $\omega_i = e_i e_{i+3}$, which are diagonal) and the generators L_i, K_i of $\mathfrak{l}_G$.

3.4.1 The four-momentum of a particle

The four-momentum read by the clock $\hat{n}$ is the projection

$$P_{\hat{n}} := \mathrm{pr}_{\hat{n}}\, K^\sharp = \frac{E}{c} n + \sum_a p_a s_a, \qquad P^\mu_{\hat{n}} = (E/c, \mathbf{p}), \qquad P_{n,\mu} = (E/c, -\mathbf{p}), \tag{3.10}$$

whose three spatial components are the conjugates of the three lengths and whose time component is the conjugate of the clock's time.

Proposition 3.25 (Four-vector and conservation)**.** *(i) Under the boost* $\exp(\chi\, s_1 \wedge \hat{n})$*, extended by the identity on* $U_{\hat{n}}$*, the components* $(E/c, \mathbf{p})$ *transform by the* 4×4 *boost of rapidity* χ *along the first spatial axis, an element of* $SO^+(1,3)$*.*

(ii) The Hamiltonian flows of E/c *and of* $-p_a$ *translate* $ct_{\hat{n}}$ *and* x^a *by one unit and leave every other coordinate fixed.*

(iii) For every Hamiltonian $H(K_A, u^1, u^2)$ *that does not depend on* $(ct_{\hat{n}}, x^a)$*,* $\{E/c, H\} = \{p_a, H\} = 0$*. Adding a term* $V(ct_{\hat{n}})$ *destroys only the conservation of* E*;* q_a *is conserved iff* H *does not depend on* u^a*.*

Proof. (i) is Proposition 2.12(ii) in the basis $(\hat{n}, s_a)$. (ii) is the bracket table above. (iii) $\{P_\mu, H\} = -\partial H/\partial x^\mu$ in the adapted canonical coordinates. □

Fixed (r2-5c at $\hat{n} = (1,2,2)/3$; r2-5b for a symbolic frame and an arbitrary function H). The axiom alone does not make E/c a Noether energy, since the axiom fixes no dynamics; (iii) supplies what is missing: E/c and $\mathbf{p}$ are the generators of the translations of $E_{\hat{n}}$, hence the Noether charges of every dynamics invariant under them.

Three mass shells. The six-momentum is free until a dynamics is chosen. Propositions 3.26, 3.28 and 3.29 treat three options.

Proposition 3.26 (Option I: the ambient shell)**.** *For every clock,* $\eta(K^\sharp, K^\sharp) = E^2/c^2 - |\mathbf{p}|^2 + |\mathbf{q}|^2$*. On the shell* $\eta(K^\sharp, K^\sharp) = M_6^2 c^2$*,*

$$E^2 = c^2|\mathbf{p}|^2 + M_6^2 c^4 - c^2|\mathbf{q}|^2, \qquad m_{\mathrm{eff}}^2 = M_6^2 - |\mathbf{q}|^2/c^2,$$

so the four-dimensional mass vanishes at $|\mathbf{q}| = M_6 c$ *and is imaginary (tachyonic) beyond.*

Proof. $(\hat{n}, s_a, u_a)$ is η-orthonormal with norms $(+1, -1, +1)$. □

Fixed (r2-6, symbolic frame). The sign of $|\mathbf{q}|^2$ is not a choice: the normal plane of a four-plane of signature $(1,3)$ in $\mathbb{R}^{3,3}$ has the sign of time (Theorem 12.4(ii)). The option is Proposition 3.101 in another notation.

Proposition 3.27 (Option I is the momentum reading of the undivided energy)**.** *Let* $K^\sharp = \sum_i (\mathsf{E}_i/c) e_i + \sum_a p_a s_a$ *with* $\eta(K^\sharp, K^\sharp) = \mu^2 c^2$*, as in Proposition 3.101, and write* $\mathbf{E} = (\mathsf{E}_1, \mathsf{E}_2, \mathsf{E}_3)$*, in* J*. For every clock* $\hat{n}$*,*

$$E = \hat{n} \cdot \mathbf{E}, \qquad c\, q_a = u_a \cdot \mathbf{E}, \qquad c^2|\mathbf{q}|^2 = |\mathbf{E}|^2 - (\hat{n} \cdot \mathbf{E})^2 = |\hat{n} \times \mathbf{E}|^2,$$

so Option I holds with $M_6 = \mu$ *and* m_{eff} *equal to the mass* $m_4(\hat{n})$ *read by the clock. At* $\hat{n} = \widehat{W}$*,* $c\,|\mathbf{q}| = |\mathbf{E}_\perp|$ *and* $m_4^2 c^4 = \mu^2 c^4 - |\mathbf{E}_\perp|^2$*, which is the statement of Proposition 3.101 that splitting the temporal energy lowers the mass. For* $\mathbf{E} = (3,4,0)$*,* $\mathbf{p} = (5,0,0)$*,* $c = 1$*:* $\mu = 0$*,* $m_4^2 = -16$ *at* e_1 *and* $m_4^2 = -\frac{26}{3}$ *at* $\widehat{W}$*.*

Proof. $E = c\,\eta(\hat{n}, K^\sharp)$, $q_a = \eta(u_a, K^\sharp)$, and $(\hat{n}, u_1, u_2)$ is orthonormal; the last identity is Lagrange's. □

Restated from Proposition 3.101, re-verified for a symbolic clock (r2-7). The surplus momenta q_a are the relative-time energies of Section 3.9 divided by c, and the tachyonic shell is realised by the example of Section 3.9.8. The obstruction stated there, that in the momentum reading the differentiation of the temporal energy subtracts from the mass, is the same obstruction as the sign in Proposition 3.26.

Proposition 3.28 (Option II: the first-class reduction)**.** *In adapted canonical coordinates take the action*

$$S_{\mathrm{red}} = \int d\tau \Big(P_\mu \dot{x}^\mu + q_a \dot{u}^a - \tfrac{N}{2}\big(P_\mu P^\mu - M_6^2 c^2 + |\mathbf{q}|^2\big) - \lambda^a q_a \Big),$$

with $[N] = \mathrm{kg}^{-1}$ *and* $[\lambda^a] = \mathrm{m\,s}^{-1}$ *for* τ *in seconds. Then* $\{q_a, q_b\} = 0$ *and* $\{q_a, H\} = 0$*, so the constraints* $q_a \approx 0$ *are first class;* $\varepsilon^a q_a$ *generates* $\delta u^a = \varepsilon^a$ *and nothing else;* S_{red} *is exactly invariant under* $\delta u^a = \varepsilon^a(\tau)$*,* $\delta\lambda^a = \dot{\varepsilon}^a$*; the reduced phase space is* $T^*E_{\hat{n}}$*, of dimension* $12 - 2 \times 2 = 8$*, with the shell* $P_\mu P^\mu = M_6^2 c^2$*. With a normal connection* $A_\mu(x)$*,* $k_\mu = P_\mu - A_\mu \ell$ *and* $\ell = v p_u - u p_v$ *at* $\hat{n} = e_1$*, where* $\mathbf{q} = (p_u, p_v)$*, the brackets become* $\{p_u, H\} = -C p_v$*,* $\{p_v, H\} = C p_u$ *with* $C = A_\mu h^{\mu\nu} k_\nu$*, and the constraints stay first class.*

Proof. H does not depend on u; the variation of the surplus part of the Lagrangian is $q_a \dot{\varepsilon}^a - \dot{\varepsilon}^a q_a = 0$. For the last statement, H depends on (u, v) only through ℓ, with $\partial_u \ell = -p_v$ and $\partial_v \ell = p_u$, so $\{p_u, H\} = -\partial_u H = -C p_v$ and $\{p_v, H\} = -\partial_v H = C p_u$, while $\{p_u, p_v\} = 0$. The same computation for a curved four-metric is Proposition 11.4 in Chapter 11. □

At $A = 0$ the reduction is that of Theorem 12.12 (Chapter 12) and its brackets are those of Proposition 11.4 (Chapter 11); it is the implementation of Assumption 2.16 with the clock e_1 replaced by $\hat{n}$ (r2-8a; r2-8b re-verifies the brackets with $A \neq 0$). Option II is the first-class surplus-momentum construction of Section 12.4. Two consequences are restated from Section 12.4: gauging only the rotation of $U_{\hat{n}}$ leaves the unstable band (Proposition 12.13), and in Dirac quantisation the physical states are functions on $E_{\hat{n}}$ obeying the Klein–Gordon equation (Theorem 12.12 and Corollary 12.14). By Proposition 3.27, $q \approx 0$ means $\mathbf{E} \parallel \hat{n}$: the reduction does not select a clock in V^+, it ties the temporal momentum to whichever clock is chosen.

Proposition 3.29 (Option III: a separate internal mass)**.** *For the action* $S_{\hat{n}} = \int d\tau\, [P_\mu \dot{x}^\mu + q_a \dot{u}^a - \tfrac{N}{2}(P_\mu P^\mu - c^2 \mathcal{M}^2(u, q))]$ *with an arbitrary function* $\mathcal{M}^2$ *(dimension* kg^2*):*

(i) $\dot{P}_\mu = 0$*,* $\frac{d}{d\tau}\mathcal{M}^2 = 0$*, and the constraint is* $E^2 = c^2|\mathbf{p}|^2 + c^4 \mathcal{M}^2$*;*

(ii) for the toy $\mathcal{M}^2 = m_0^2 + |\mathbf{q}|^2/c^2$ *the surplus velocity is* $\dot{u}^a = -N q_a$*, against* $+N q_a$ *on the ambient shell, and the constraint form is* $P_\mu P^\mu - |\mathbf{q}|^2 = \eta(R_{\hat{n}} K^\sharp, K^\sharp)$ *with* $R_{\hat{n}} = \mathrm{pr}_{\hat{n}} - \mathrm{pr}_{\hat{n}}^\perp = \exp(\pi\, u_1 \wedge u_2) \in SO(3,3)$*, a quadratic form of signature* $(1, 5)$*.*

Proof. (i) H has no x-dependence, and $\{\mathcal{M}^2, H\} = -\frac{N c^2}{2}\{\mathcal{M}^2, \mathcal{M}^2\} = 0$. (ii) $\partial H/\partial q_a$; $R_{\hat{n}}$ is the identity on $E_{\hat{n}}$ and $-\mathbf{1}$ on $U_{\hat{n}}$, the half-turn of $U_{\hat{n}}$. □

Option III is constructed: $\mathcal{M}^2$ is supplied from outside the arena (r2-9a for an arbitrary function; r2-9b at $\hat{n} = e_1, (2, 3, 6)/7$). The positive sign of the toy is opposite to the ambient one, and (ii) makes this precise: the toy is the free particle of the twisted form $\eta(R_{\hat{n}}\cdot, \cdot)$, which has one time and five space directions, so its sign is that of surplus *lengths*, the Kaluza–Klein sign, and not that of the arena's surplus times.

Remark 3.30 (Which option the arena prefers)**.** The arena's metric fixes the sign of Option I. For fields the same sign is the ill-posed, unbounded-below baseline of Proposition 12.1, whose unstable band Option II removes together with every other nonzero surplus momentum (Figure 12.1 and Theorem 12.12; r2-8a). Option III needs a functional whose surplus sign is opposite

to that of the metric; the clock supplies one, the twisted form $\eta(R_{\hat n}\cdot,\cdot)$ of Proposition 3.29(ii), but nothing in the arena selects it in place of η. So the arena prefers Option II, as the only one of the three that has the arena's sign and no instability; the preference is conditional, because the gauge principle of the normal translations is chosen and not derived (Section 12.4). This is interpretive; whether a principle makes the normal translations gauge is open.

3.4.2 Rotating the clock

Fix a six-momentum $K^\sharp = \sum_i(\mathsf{E}_i/c)e_i + \sum_a p_a s_a$ with temporal energy vector $\mathbf{E}$ and six-dimensional mass $\mu^2c^4 = |\mathbf{E}|^2 - |\mathbf{p}|^2c^2$, and let the clock vary. The energy and the four-dimensional mass read by $\hat n$ are

$$E(\hat n) := c\,\eta(K^\sharp,\hat n), \qquad m_4(\hat n)^2c^4 := E(\hat n)^2 - |\mathbf{p}|^2c^2.$$

Proposition 3.31 (The mass read by a rotating clock)**.** *(i) On the clock sphere $E(\hat n) = \hat n\cdot\mathbf{E}$ and*

$$m_4(\hat n)^2c^4 = \mu^2c^4 - |\hat n\times\mathbf{E}|^2 = \mu^2c^4 - |\mathbf{E}|^2\sin^2\angle(\hat n,\mathbf{E}).$$

Hence $m_4(\hat n)^2 \le \mu^2$. For $\mathbf{E}\neq 0$, equality holds iff $\hat n \parallel \mathbf{E}$ (in either orientation), and $E(\hat n)\le|\mathbf{E}|$, with equality iff $\hat n = \mathbf{E}/|\mathbf{E}|$: the clock that reads the largest energy is the direction of the temporal part of K, and there $m_4=\mu$. For $\mathbf{E}=0$ every clock reads zero energy and the same $m_4^2=\mu^2$, and no clock is distinguished.

(ii) On the Fisher–Rao octant, $E(\hat n) = \sum_i\sqrt{w_i}\,\mathsf{E}_i$ and $m_4^2c^4 = \big(\sum_i\sqrt{w_i}\,\mathsf{E}_i\big)^2 - |\mathbf{p}|^2c^2$. Let $\mathbf{E}^+$ be the componentwise positive part. If $\mathbf{E}^+\neq 0$, $\max E(\hat n) = |\mathbf{E}^+|$, attained at $w_i = (\mathsf{E}_i^+)^2/|\mathbf{E}^+|^2$; if $\mathbf{E}^+=0$, the maximum is $\max_i\mathsf{E}_i\le 0$, at a vertex, and no clock of the octant reads a positive energy. If $\mathbf{E}$ has no negative component the maximising clock of (i) lies in the octant; otherwise, if $\mathbf{E}^+\neq 0$, the largest mass read by a future-reading clock of the octant is $m_4^2c^4 = \mu^2c^4 - |\mathbf{E}-\mathbf{E}^+|^2 < \mu^2c^4$.

(iii) At $\widehat W$, $E(\widehat W) = (\mathsf{E}_1+\mathsf{E}_2+\mathsf{E}_3)/\sqrt3$ and $m_4^2c^4 = \mu^2c^4 - |\mathbf{E}_\perp|^2$, the E_W of Section 3.9.8.

Proof. (i) $|\hat n|=1$ and Lagrange's identity $|\mathbf{E}|^2 = (\hat n\cdot\mathbf{E})^2 + |\hat n\times\mathbf{E}|^2$. (ii) For $\hat n_i\ge 0$, $\hat n\cdot\mathbf{E}\le\hat n\cdot\mathbf{E}^+\le|\mathbf{E}^+|$, with equality iff $\hat n = \mathbf{E}^+/|\mathbf{E}^+|$ when $\mathbf{E}^+\neq 0$; if $\mathbf{E}^+=0$, then $\sum_i n_i\ge|\hat n|=1$ gives $\hat n\cdot\mathbf{E}\le\max_i\mathsf{E}_i\sum_i n_i\le\max_i\mathsf{E}_i$, with equality at the vertex of the largest component; m_4^2 increases with $E(\hat n)\ge 0$, and $|\mathbf{E}^+|^2 = |\mathbf{E}|^2 - |\mathbf{E}-\mathbf{E}^+|^2$. (iii) is (i) at $\hat n=\widehat W$. □

Fixed (r2-10a symbolic; r2-10c a random test of (ii), 200 vectors $\mathbf{E}$ against 4000 octant clocks each, in floating point with tolerance 10^{-12}; r2-10d the case $\mathbf{E}^+=0$, exactly at $\mathbf{E}=(-1,-2,-3)$ and by a random test). An exact example (r2-10b): $\mathbf{E}=(2,3,6)$, $\mathbf{p}=(0,0,3)$, $c=1$ give $\mu^2=40$; the aligned clock $(2,3,6)/7$ lies in the Fisher–Rao octant with weights $(4,9,36)/49$ and reads $m_4^2=40$, as does the opposite clock, which reads the energy -7; the clock e_1 reads $m_4^2=-5$, and $\widehat W$ reads $\frac{94}{3} = 40-|\mathbf{E}_\perp|^2$.

Remark 3.32 (The mass as a projection)**.** Interpretive. In the six-momentum the temporal energy is one vector $\mathbf{E}$, undivided; a clock reads its component $E(\hat n)$, and what the reading leaves out, $c\,\mathrm{pr}^\perp_{\hat n}K^\sharp$, is the part orthogonal to the clock. The energy read by the clock is the projection of the undivided temporal energy onto it; the four-dimensional mass computed from it is maximal, and equal to the six-dimensional μ, only for the clocks parallel to $\mathbf{E}$, and is lowered by every misalignment, which is Proposition 3.27 read on the clock sphere. Nothing in this reading prefers a direction of V^+: the aligned clock is fixed relative to the momentum, and for several particles read by one clock the reduction of Proposition 3.28 requires all their temporal momenta to be parallel to it.

3.4.3 The four-momentum of the field

In the field reading the field is a function on spacetime $E_{\hat{n}}$, with coordinates $x^\mu = (ct_{\hat{n}}, x^a)$, valued in a *second* copy V_{EM} of $\mathbb{R}^{3,3}$ on which $X_{\mathrm{EM}} = \sum_i (B_i e_i + E_i e_{i+3})$ is read as a two-form through $\Phi : X \mapsto F_X$, $(F_X)_{0i} = E_i$, $(F_X)_{jk} = \epsilon_{ijk} B_i$, with $\mathfrak{l}_G$ as Lorentz algebra (Theorem 3.3). Both copies are needed: by Proposition 3.6 one six-dimensional space cannot carry $\mathfrak{l}_{\mathrm{int}}(\hat{n})$ and $\mathfrak{l}_G$ as its Lorentz action at once.

Proposition 3.33 (The dictionary is equivariant)**.** *Φ intertwines $\mathfrak{l}_G$ on V_{EM} with $\mathfrak{l}_{\mathrm{int}}(\hat{n})$ on $\Lambda^2 E^*_{\hat{n}}$ through a unique isomorphism $\phi : \mathfrak{l}_G \to \mathfrak{l}_{\mathrm{int}}(\hat{n})$, namely $L_i \mapsto s_j \wedge s_k$ and $K_i \mapsto s_i \wedge \hat{n}$ for cyclic (i,j,k). Moreover $F_{GX} = \star F_X$ with $\varepsilon_{0123} = +1$; on $\Lambda^2 E^*_{\hat{n}}$ the Hodge star satisfies $\star^2 = -\mathbf{1}$, commutes with $\mathfrak{l}_{\mathrm{int}}(\hat{n})$, and is self-adjoint and anti-isometric for $\langle F, F\rangle = \frac{1}{2} F_{ab} F^{ab} = |\mathbf{B}|^2 - |\mathbf{E}|^2 = \eta(X,X)$, a form of signature $(3,3)$.*

Restated from Theorem 3.3 and Proposition 3.7 (r2-12a, which also finds ϕ; r2-12b). With ϕ, $E_{\hat{n}}$ becomes a four-vector module of $\mathfrak{l}_G$, and the question of a covariant map from the field to a four-vector becomes a question about $\mathfrak{l}_{\mathrm{int}}(\hat{n})$-modules.

Proposition 3.34 (No covariant map from the field to a four-vector)**.** *(i) For $\hat{n} = e_1$ and $\hat{n} = (2,3,6)/7$ the space $\mathrm{Hom}_{\mathfrak{l}_{\mathrm{int}}(\hat{n})}(\Lambda^2 E^*_{\hat{n}}, E_{\hat{n}})$ is zero, and so is $\mathrm{Hom}_{\mathfrak{l}_G}(V_{\mathrm{EM}}, E_{\hat{n}})$ for the module structure given by ϕ.*

(ii) For quadratic maps, $\mathrm{Hom}_{\mathfrak{l}_G}(\mathrm{Sym}^2 V_{\mathrm{EM}}, E_{\hat{n}}) = 0$, $\dim \mathrm{Hom}_{\mathfrak{l}_G}(\mathrm{Sym}^2 V_{\mathrm{EM}}, \mathbb{R}) = 2$ (the invariants $|\mathbf{B}|^2 - |\mathbf{E}|^2$ and $\mathbf{E}\cdot\mathbf{B}$), $\dim \mathrm{Hom}_{\mathfrak{l}_G}(\mathrm{Sym}^2 V_{\mathrm{EM}}, \mathrm{Sym}^2 E_{\hat{n}}) = 3$, and the traceless part of the last space is one-dimensional.

Proof. Complexified, $V_{\mathrm{EM}} \cong (1,0) \oplus (0,1)$ and $E_{\hat{n}} \cong (\frac{1}{2}, \frac{1}{2})$, and $\mathrm{Sym}^2 V_{\mathrm{EM}} \cong (2,0) \oplus (0,2) \oplus (1,1) \oplus 2(0,0)$ contains no $(\frac{1}{2}, \frac{1}{2})$; Schur's lemma [57] gives the dimensions, and the exact computation of the intertwiner spaces confirms them. □

Part (i) is inherited representation theory, computed (r2-11a, with two controls: the endomorphisms of $\Lambda^2 E^*$ commuting with $\mathfrak{l}_{\mathrm{int}}$ are spanned by the identity and the Hodge star, and $\dim \mathrm{Hom}_{\mathfrak{l}_{\mathrm{int}}}(V, E) = 1$). Part (ii) is inherited as well, computed (r2-11b), and shows that a field momentum must be quadratic in the field and integrated over space: both halves are needed. Quadratic alone is not enough: the only traceless symmetric quadratic covariant is the stress tensor, up to scale, and a four-vector arises only after it is contracted with the unit normal of a spacelike hyperplane, which is a clock, and integrated.

Proposition 3.35 (Pointwise energy–momentum is a tensor row)**.** *For the stress tensor $\Theta^{\mu\nu} = \eta^{\mu\alpha} F_{\alpha\lambda} F^{\lambda\nu} + \frac{1}{4} \eta^{\mu\nu} F_{\alpha\beta} F^{\alpha\beta}$ of F_X [76]:*

(i) $\Theta^{00} = E_6$ and $\Theta^{0a} = (\mathbf{S}_{\mathrm{t}})_a = -S_a$; Θ is symmetric and traceless, $\Theta^\mu{}_\lambda \Theta^\lambda{}_\nu = E_0^2 \delta^\mu{}_\nu$ and $E_6^2 = E_0^2 + |\mathbf{S}|^2$.

(ii) The rest field $X = e_1$ ($E_6 = E_0 = \frac{1}{2}$, $\mathbf{S} = 0$) boosted by $\exp(\chi \,\mathrm{ad}_{K_3})$ has $\mathbf{B} = (\cosh\chi, 0, 0)$, $\mathbf{E} = (0, \sinh\chi, 0)$, $E_6 = \frac{1}{2}\cosh 2\chi$ and $\mathbf{S} = -\frac{1}{2}\sinh 2\chi\, e_3$, while a four-vector with rest components $(\frac{1}{2}, 0)$ would have time component $\frac{1}{2}\cosh\chi$.

(iii) For every X, $\Theta[\exp(\chi \,\mathrm{ad}_{K_3})X] = L\,\Theta[X]\,L^\top$ with L the boost of rapidity χ in the $(0,3)$ plane, the passage to the frame moving with velocity $-\tanh\chi\, e_3$.

Restated from Proposition 3.93 and Proposition 3.94 and Theorem 3.95, re-verified symbolically (r2-13, r2-14a,b); the identities are inherited [129, 82]. In the paper's variables the momentum density is $-\mathbf{S} = -\mathbf{E} \times \mathbf{B}$, the textbook $\mathbf{E} \times \mathbf{B}_{\mathrm{t}}$, that is minus the frame Poynting vector. So $(E_6, -\mathbf{S})$ at a point is the first row of a tensor, quadratic in the boost, and not a four-vector, as Section 3.9 states; the density seen by the clock $\hat{n}$ is $\Theta(\cdot, \hat{n})$, different for each clock and of Lorentz length E_0 (Proposition 3.94).

Proposition 3.36 (The four-momentum of an isolated field)**.** *(i) If $\partial_\mu \Theta^{\mu\nu} = 0$ and the Cauchy data have compact support, $P^\mu = (U/c, \mathbf{P})$ with $U = \int u\, d^3x$ and $\mathbf{P} = \int \mathbf{g}\, d^3x$, $\mathbf{g} = \varepsilon_0 \mathbf{E} \times \mathbf{B}_{\mathrm{t}}$ in SI, is independent of the spacelike hyperplane and is a future causal four-vector, with $M^2c^4 = U^2 - c^2|\mathbf{P}|^2 \geq 0$.*

(ii) For two free solutions the interference part $\Theta[X_1 + X_2] - \Theta[X_1] - \Theta[X_2]$ is divergence free. Hence $P[X_1 + X_2] = P[X_1] + P[X_2]$ whenever the two fields have disjoint supports at some time, and two null wholes with momenta at angle ϑ have $M^2c^4 = 2U_1U_2(1 - \cos\vartheta)$, timelike unless $\vartheta = 0$, that is unless the two four-momenta are proportional.

(iii) In Heaviside–Lorentz units with $c = 1$ and $f(s) = (1 - s^2)^2$ on $|s| \leq 1$, zero elsewhere, the field $E_x = af(z - t) + bf(z + t)$, $B_{\mathrm{t}y} = af(z - t) - bf(z + t)$ has, per unit transverse area, $U = \frac{256}{315}(a^2 + b^2)$ and $P_z = \frac{256}{315}(a^2 - b^2)$ at every t, so $M = \frac{512}{315}|ab|$. Boosting the fields along z with rapidity χ gives exactly $(U \cosh\chi - P_z \sinh\chi,\ P_z \cosh\chi - U \sinh\chi)$. For $|t| \geq 1$ the pulses are disjoint, $q \equiv 0$ and $\int E_0 = 0$, while $M > 0$.

Proof. (i) Integrate the conservation law over the region between two spacelike hyperplanes; the flux through the timelike boundary vanishes by finite propagation speed [84, 82]. Each $\Theta(\cdot, \hat{n})$ is future causal (the dominant energy condition [65]), hence so is the integral. (ii) Θ is quadratic and divergence free on every solution, so its polarisation is divergence free on pairs of solutions; the interference charge is conserved and vanishes when the supports are disjoint. The mass formula is that of two future null vectors. (iii) The cross terms of u and of $S_{\mathrm{t}z}$ cancel pointwise; a right-moving pulse is multiplied by $e^{-\chi}$ and stretched by e^{χ} under the boost, and a left-moving one the reverse. □

Item (i) is inherited (von Laue [84]) and is the mass of a whole of Proposition 3.100, restated; items (ii) and (iii) are fixed by computation (r2-16: the interference part for two Maxwell plane waves with wave vectors $(0, 0, 3)$ and $(4, 0, 0)$, the null sum $(3; 0, 0, 3) + (4; 4, 0, 0)$ of mass squared 24 and the antiparallel sum $(3; 0, 0, 3) + (4; 0, 0, -4)$ of mass squared 48; r2-15, by a genuine Lorentz transformation of the fields and exact integration). With $a = b = 1$, (iii) is item (4) of Proposition 3.100. The mass of the whole comes from the non-collinearity of the two parts' four-momenta (their three-momenta are antiparallel), not from their interference, which is absent at $|t| \geq 1$. The four-momentum of the field is the integrated row $\int \Theta^{0\mu} d^3x$, which uses one hyperplane, that is one clock; the result does not depend on it.

Remark 3.37 (The photon)**.** Inherited, and open in the arena. Quantising the free field gives translation generators with $\widehat{P}^\mu|\mathbf{k}, \lambda\rangle = \hbar k^\mu |\mathbf{k}, \lambda\rangle$, k null and $\lambda = \pm 1$ [162, 160]; this quantisation is not derived in the arena. The arena supplies the classical counterparts: a null field such as $z = (1, i, 0)e^{ik(x_3 - t)}$ of Example 3.11 has $E_6 = |\mathbf{S}|$ and momentum density along its propagation (r2-17), and on positive-frequency modes the eigenvalue of G is i times the helicity (Example 3.82), where the quantum of $\hbar$ per photon is put in by hand. Nor are the arena's own Fock modes photons: $a_i^{\pm}$ obey the anticommutation relations on $\mathbb{R}^8$, and no finite matrices obey $[a, a^\dagger] = \mathbf{1}$, since the trace of a commutator vanishes (r2-0b, r2-17).

Remark 3.38 (Base and fibre)**.** The field's four-momentum is a vector of the base $E_{\hat{n}}$, the spacetime of the classical reading; the six components of X_{EM} are values in the fibre $V_{\mathrm{EM}} \cong \Lambda^2 E_{\hat{n}}^*$, with units of field strength, and its temporal triple carries $\mathbf{B}$, not time. So X_{EM} is not a displacement and supplies no extra time components of momentum, (constructed: the field is a function on the base $E_{\hat{n}}$, so its translation charges carry indices of $E_{\hat{n}}$; covariance alone would allow more, since the invariants $|\mathbf{B}|^2 - |\mathbf{E}|^2$ and $\mathbf{E} \cdot \mathbf{B}$ map covariantly to the trivial module $U_{\hat{n}}$, r2-11b). A field that depended on the surplus coordinates would carry surplus translation charges, the field analogue of q_a; these are what the gauge principle of Proposition 3.28 removes (Corollary 12.14). The two readings of X thus meet in one place, which is a reading (interpretive): the classical reading supplies the base $E_{\hat{n}}$, and the field reading the fibre over it.

3.4.4 The no-go in its scope

Lemma 3.39 (Anti-isometry)**.** *If $G^\dagger = G$ and $G^2 = -\mathbf{1}$ then $\eta(Gv, Gw) = -\eta(v, w)$.*

Proof. $\eta(Gv, Gw) = \eta(v, G^\dagger G w) = \eta(v, G^2 w)$. □

Restated (Proposition 2.3 and Theorem 4.8; r2-0a, r2-18).

Proposition 3.40 (No Lorentzian plane is G-invariant)**.** *Every nondegenerate G-invariant subspace of V has neutral signature (k, k). In particular no four-plane of signature $(1, 3)$, and no $E_{\hat{n}}$, is invariant under an η-self-adjoint G with $G^2 = -\mathbf{1}$.*

Proof. If $GW \subset W$, then $G|_W$ is invertible and by Lemma 3.39 makes $\eta|_W$ congruent to $-\eta|_W$; by Sylvester's law of inertia [111] a signature (p, q) then equals (q, p), so it is (k, k). For the binding root the Gram matrix of $\operatorname{span}(v, Gv)$ is $\left(\begin{smallmatrix} a & b \\ b & -a \end{smallmatrix}\right)$, of determinant $-(a^2 + b^2) \le 0$. □

Restated (r2-18: the symbolic identities, and 300 random G-invariant four-planes of signature $(2, 2)$, in floating point with tolerance 10^{-9}).

Proposition 3.41 (Leakage at every clock)**.** *For the interior Lagrangian $L_{\text{int}} = \frac{m}{2}\,\eta(\dot{X} + \kappa G X, \dot{X} + \kappa G X)$ of equation (4.11), with X length-valued and $\kappa = c/\ell$ as in (2.2):*

(i) $p = m\,\eta(\dot{X} + \kappa G X)$, the Euler–Lagrange equation is $\dot{p} = \kappa G^\top p$, and $\pi = \eta^{-1} p$ obeys $\dot{\pi} = \kappa G \pi$, so $\pi(\lambda) = (\cos\kappa\lambda + \sin\kappa\lambda\, G)\pi(0)$. At $\hat{n} = e_1$, $\pi(0) = e_5$ gives $\pi(\lambda) = \cos(\kappa\lambda) e_5 - \sin(\kappa\lambda) e_2$ and $\operatorname{pr}_{e_1} \pi = \cos(\kappa\lambda) e_5$; along every solution $\eta(\pi, \pi)$ becomes $\cos 2\kappa\lambda\, \eta(\pi, \pi) + \sin 2\kappa\lambda\, \eta(\pi, G\pi)$.

(ii) For every clock, $\operatorname{pr}_{\hat{n}}^{\perp} G \hat{n} = 0$ and $\operatorname{pr}_{\hat{n}}^{\perp} G \operatorname{pr}_{\hat{n}} = -(\mathbf{1} - nn^\top) \circ \sigma$, where $\sigma : s_a \mapsto e_a$; hence $\operatorname{rank} \operatorname{pr}_{\hat{n}}^{\perp} G \operatorname{pr}_{\hat{n}} = 2$, with kernel $\operatorname{span}(\hat{n}, G\hat{n})$ in $E_{\hat{n}}$.

(iii) A vector $v \in E_{\hat{n}}$ stays in $E_{\hat{n}}$ under $\exp(\theta G)$ for all θ iff $v \in \operatorname{span}(\hat{n}, G\hat{n})$, a plane of signature $(1, 1)$; at $\hat{n} = e_1$ it is $\langle e_1, e_4 \rangle$.

Proof. (i) Direct variation; $\eta^{-1} G^\top \eta = G$, and $e^{\theta G} = \cos\theta + \sin\theta\, G$ since $G^2 = -\mathbf{1}$; G is an anti-isometry, so $e^{\theta G}$ is not an isometry. (ii) G maps V^+ into $V^- \subset E_{\hat{n}}$, so $\operatorname{pr}_{\hat{n}}^{\perp} G\hat{n} = 0$; on V^-, $G = -\sigma$ lands in V^+, of which $\operatorname{pr}_{\hat{n}}^{\perp}$ keeps the part orthogonal to $\hat{n}$: an idempotent of trace two composed with an isomorphism. (iii) If $e^{\theta G} v \in E_{\hat{n}}$ for all θ, the derivative at 0 gives $\operatorname{pr}_{\hat{n}}^{\perp} G v = 0$, so $v \in \operatorname{span}(\hat{n}, G\hat{n})$; conversely this plane is G-invariant, and $\eta(G\hat{n}, G\hat{n}) = -1$. □

Fixed (r2-19a with sympy Euler–Lagrange equations; r2-19b symbolic in a general frame, and exact ranks at six rational clocks; r2-19c). At $\hat{n} = e_1$ the rank is two and only $\langle e_1, e_4 \rangle$ stays; (ii) extends the rank to every clock, and the mechanism is that G maps the three spatial directions into V^+, and of V^+ only the $\hat{n}$-component returns to $E_{\hat{n}}$. If the interior momentum π is identified with $K^\sharp$ and $q \approx 0$ is imposed on its flow, the momentum is confined to the $(1, 1)$ plane $\operatorname{span}(\hat{n}, G\hat{n})$ and two of the three spatial momenta are lost: the identification is obstructed. Allowing $q \neq 0$ does not help, since neither $\operatorname{pr}_{\hat{n}} \pi$ nor even $\eta(\pi, \pi)$ is conserved.

Proposition 3.42 (The two Lorentz algebras and the clock)**.** *$\mathfrak{l}_{\text{int}}(\hat{n})$ fixes $U_{\hat{n}}$ pointwise and $\mathfrak{l}_G$ fixes no nonzero vector; $\mathfrak{l}_G \cap \mathfrak{l}_{\text{int}}(\hat{n}) = 0$; and $\mathfrak{l}_G$ meets the stabiliser $\mathfrak{so}(E_{\hat{n}}) \oplus \mathfrak{so}(U_{\hat{n}})$ of the clock exactly in the line of the diagonal rotation $\sum_i n_i L_i$. G commutes with $\mathfrak{l}_G$ and not with $\mathfrak{l}_{\text{int}}(\hat{n})$.*

The first two statements restate Proposition 3.2; the rest is fixed (r2-20 at $\hat{n} = e_1, (1, 2, 2)/3$, and $\widehat{W}$ for the fixed planes). Proposition 3.41 is the vector-module face of the exclusion of Proposition 3.6: the operator that is the Hodge star for $\mathfrak{l}_G$ moves every four-plane of $\mathfrak{l}_{\text{int}}(\hat{n})$, and the only symmetry the two structures share is the rotation about the clock, whose axis plane $\operatorname{span}(\hat{n}, G\hat{n})$ is the one that survives in Proposition 3.41(iii).

The no-go does not obstruct G as the Hodge star. On the field module $\Lambda^2 E^*_{\hat{n}}$ the star acts on the whole module, which is neutral of signature $(3,3)$ (Proposition 3.33); Proposition 3.40 forbids only invariant subspaces that are not neutral, and Theorem 4.8 requires exactly $P = Q$, which $\Lambda^2 E^*_{\hat{n}}$ satisfies. So $G_{\mathrm{EM}} = \star$ is electric–magnetic duality on field strengths, commuting with the Lorentz action, while G_{cl} on the vector module $E_{\hat{n}}$ is obstructed. This is restated (Theorem 3.3 and Theorem 4.8; r2-12b with Proposition 3.40) .

3.4.5 Answer

For the particle reading the four-momentum is constructed as follows. X is taken with its lengths on the three directions of V^-; a clock $\hat{n} \in V^+$ selects the four-plane $E_{\hat{n}} = \mathbb{R}\hat{n} \oplus V^-$, and every four-plane containing the lengths is of this form. The six-momentum conjugate to Y splits into the four-momentum $P_{\hat{n}} = (E/c, \mathbf{p})$, conjugate to $(ct_{\hat{n}}, x^a)$, and two surplus momenta q_a, Lorentz scalars. $P_{\hat{n}}$ is a four-vector and the Noether charge of the translations of $E_{\hat{n}}$. What it needs beyond the arena is a dynamics: the arena's metric gives the ambient shell, on which q lowers the mass; the gauge principle of Section 12.4 gives the reduced shell $q \approx 0$, on which the temporal momentum is parallel to the clock and $m_4 = \mu$; a positive internal mass needs a functional of the opposite sign, that of extra space dimensions. For the field reading the four-momentum is the integrated first row of the stress tensor on the base $E_{\hat{n}}$, a four-vector for an isolated whole, with the mass of a whole from parts whose four-momenta are not collinear; no linear or quadratic covariant map produces it pointwise, and X_{EM} supplies no time components. The two readings share the clock: it fixes $E_{\hat{n}}$, the complex structure $J(\hat{n})$ of the Dirac spinor and the hyperplane on which the field's momentum is integrated.

statement	class	checks
four-planes containing V^- are the $E_{\hat{n}}$ (Proposition 2.11)	fixed	r2-1a,b
$\mathrm{pr}_{\hat{n}}$ covariant under $SO^+(1,3) \times SO(2)$ only (Proposition 2.12)	restated	r2-2a–c
$J(\hat{n}) = I_T\hat{n}$, $J(\hat{n}) = u_1u_2$, $J(-\hat{n}) = -J(\hat{n})$ (Proposition 2.13)	restated	r2-3a–d
Theorem 2.17 at every clock; mass form I_T clock-independent (Theorem 2.14)	fixed	r2-4a–d
$P_{\hat{n}}$ four-vector, Noether charge of translations (Proposition 3.25)	fixed	r2-5a–c
ambient shell, tachyonic for $\lvert\mathbf{q}\rvert > M_6c$ (Proposition 3.26)	fixed	r2-6
Option I is the momentum reading of the undivided energy (Proposition 3.27)	restated	r2-7
first-class reduction = Theorem 12.12 at $A = 0$ (Proposition 3.28)	restated	r2-8a,b
internal mass functional; toy has signature $(1,5)$ (Proposition 3.29)	constructed; sign fixed	r2-9a,b
the arena prefers Option II (Remark 3.30)	interpretive	–
$m_4(\hat{n})^2 \le \mu^2$, equality iff $\hat{n} \parallel \mathbf{E}$; octant maximum (Proposition 3.31)	fixed	r2-10a–d
mass as projection onto the clock (Remark 3.32)	interpretive	–
dictionary equivariant, $\star$ on a neutral module (Proposition 3.33)	restated	r2-12a,b
no linear map to a four-vector (Proposition 3.34(i))	inherited	r2-11a,b
no quadratic map to a four-vector (Proposition 3.34(ii))	inherited	r2-11b
pointwise $(E_6, -\mathbf{S})$ is a tensor row (Proposition 3.35)	restated	r2-13, r2-14a,b
von Laue four-vector, mass of a whole (Proposition 3.36(i))	inherited	–

statement	class	checks
interference conserved; exact packets (Proposition 3.36(ii),(iii))	fixed	r2-15, r2-16
photon $P = \hbar k$, two helicities (Remark 3.37)	inherited; open in the arena	r2-17
X_{EM} supplies no time components (Remark 3.38)	constructed	r2-11b
no G-invariant $(1,3)$ plane (Proposition 3.40)	restated	r2-18
$\operatorname{rank} \mathrm{pr}_{\hat{n}}^{\perp} G \mathrm{pr}_{\hat{n}} = 2$ at every clock; $\pi \equiv K^{\sharp}$ with $q \approx 0$ (Proposition 3.41)	fixed; obstructed	r2-19a–c
$\mathfrak{l}_G \cap (\mathfrak{so}(E_{\hat{n}}) \oplus \mathfrak{so}(U_{\hat{n}})) = \mathbb{R} \sum_i n_i L_i$ (Proposition 3.42)	fixed	r2-20
$G = \star$ on $\Lambda^2 E_{\hat{n}}^*$ not obstructed	restated	r2-12b
a principle making the normal translations gauge	open	–

What remains open, located precisely, is a principle that makes the surplus translations gauge rather than chosen, a dynamics that selects the clock rather than aligning it with the momentum, and a derivation of the photon's Hilbert space from the arena. Nothing in this section identifies X_{cl} with X_{EM}; they share the clock, and through it the base on which both four-momenta live.

3.5 The field-strength sector $X_{\mathrm{EM}} = (\mathbf{B}; \mathbf{E})$ on the Clifford octants

The question asked in the introduction about the octants concerns the field reading $X_{\mathrm{EM}} = (\mathbf{B}; \mathbf{E})$, where each of the six components $E_x, E_y, E_z, B_x, B_y, B_z$ is to sit on an octant. As a $\mathfrak{l}_{\mathrm{int}}$-module ($V_{\mathrm{EM}} \cong \Lambda^2 E_{\hat{n}}^*$, Section 3.3.1), X_{EM} is not a sum of two vectors and not a displacement: the pair $(\mathbf{B}; \mathbf{E})$ labels the six components of the field two-form F, and the question is how these six components, as six independent components of one field strength, sit on the eight octants of the spinor module. Everything below is a classical field representation: placing $(\mathbf{B}; \mathbf{E})$ on six lines of S quantises nothing. A construction of this kind was proposed in an earlier, superseded version of the programme: the eight octants are the joint eigenspaces of the pole operators ω_i; the six mixed octants are field octants, those with one plus sign (temporal weight one there, Fock weight $N = 2$ here) carrying the electric components (as boosts) and those with two plus signs ($N = 1$) the magnetic components (as rotations); the two poles carry "mass" and "statistics"; and Maxwell's equations are "pooled" across the octants. This section establishes in the conventions of Section I.1.4 what of this is true. In brief: the octants are the eight lines of a Fock basis of the spinor module $S = \mathbb{R}^8$, which is the exterior algebra of a null three-plane; there is a field-octant map Φ placing the electric member $\mathbf{E}$ of the pair on the Fock weight $N = 1$ and the magnetic member $\mathbf{B}$ on $N = 2$, unique up to a scale and a quarter turn of duality once rotations and Maxwell's equations are required, the quarter turn giving the earlier labelling; it carries the rotations, a parity and the premetric $\mathfrak{gl}(3)$, but no even element of $\mathrm{Cl}(3,3)$ carries the boosts or the duality rotation; the $\mathrm{Spin}(3,3)$-invariant mass form pairs each octant with its antipode and $\mathbf{E}$ with $\mathbf{B}$; and Maxwell's four equations are one first-order equation on S, with the two Gauss laws on the poles. In textbook variables the whole construction is Hestenes' Pauli-algebra electrodynamics, written with the three temporal generators acting on the vacuum pole.

Conventions. $V = \mathbb{R}^{3,3}$, $\eta = \mathrm{diag}(1,1,1,-1,-1,-1)$, the generators $e_1, \ldots, e_6$ are those of eq. (1.1), $\omega := e_1 \cdots e_6 = \Gamma$, $T := e_1 e_2 e_3$ (the mass form, Corollary 7.10), and $\varpi = \sum_i \omega_i$, $\omega_i := e_i e_{i+3}$. The field point is $X = \sum_i (B_i e_i + E_i e_{i+3})$ (Theorem 3.3), with $\mathbf{B}$ minus the textbook magnetic field, $\mathbf{B}_{\mathrm{t}} = -\mathbf{B}$; $z = \mathbf{B} + i\mathbf{E}$, $q = z \cdot z$, $E_6 = \frac{1}{2}|z|^2$, $E_0 = \frac{1}{2}|q|$ (eq. (3.17)).

The binding root is $Ge_i = e_{i+3}$, $Ge_{i+3} = -e_i$; the grading root is $\mathcal{K} = \frac{1}{2}\operatorname{ad}_\varpi$, $\mathcal{K}e_i = -e_{i+3}$, $\mathcal{K}e_{i+3} = -e_i$ (Proposition 2.3 and Lemma 2.4). On V we use coordinates $(\mathbf{B};\mathbf{E}) \in \mathbb{R}^6$. Units are Heaviside–Lorentz with $c = 1$ unless SI is stated. Dimensions: in SI $[\mathbf{E}] = \mathrm{V\,m^{-1}}$, $[\mathbf{B}] = \mathrm{T}$, so that $\mathbf{E}$ and $c\mathbf{B}$ share a unit; $[t] = \mathrm{s}$, $[x^a] = \mathrm{m}$, $[\partial_a] = \mathrm{m^{-1}}$; $[\rho] = \mathrm{C\,m^{-3}}$, $[\mathbf{j}] = \mathrm{A\,m^{-2}}$, $[\varepsilon_0] = \mathrm{F\,m^{-1}}$; the generators, the Fock operators, the octant basis vectors and T are dimensionless, and $\psi = \Phi_{\mathrm{oct}}(X)$ carries the dimension of a field strength. Identifiers r3-n refer to the script `r3_octants.py`, which checks every finite-dimensional and symbolic claim below exactly.

3.5.1 The field-octant map

The map should intertwine the rotations of the field. Which rotations act on both triples as the field's rotations is fixed first.

Lemma 3.43 (The diagonal rotations)**.** *For cyclic (i,j,k) the bivector $L_i = \frac{1}{2}(e_{j+3}e_{k+3} - e_je_k)$ of Section 3.9 acts on V by $\delta\mathbf{B} = e_i \times \mathbf{B}$, $\delta\mathbf{E} = e_i \times \mathbf{E}$ and commutes with G and $\mathcal{K}$. The paired bivector $e_je_k + e_{j+3}e_{k+3}$ rotates the two triples in opposite senses and commutes with neither. The diagonal rotations $\mathfrak{so}(3)_\Delta = \operatorname{span}\{L_i\}$ are the compact part of $\mathfrak{l}_G$ and the antisymmetric part of $\mathfrak{gl}(3)$, and $\mathfrak{so}(3)_\Delta = \mathfrak{l}_G \cap \mathfrak{gl}(3)$. In Fock form*

$$L_i = a_k^+ a_j^- - a_j^+ a_k^-, \qquad K_i = \tfrac{1}{2}(e_ke_{j+3} - e_je_{k+3}) = a_j^- a_k^- - a_j^+ a_k^+.$$

The spin lift of L_i (left multiplication) satisfies $L_i^3 = -L_i$, so $\exp(2\pi L_i) = \mathbf{1}$ on S. Each L_i annihilates the four-dimensional space $\operatorname{span}\{|\mathrm{vac}\rangle, |\mathrm{full}\rangle, f_i, \mathrm{g}_i\}$, the joint kernel of L_1, L_2, L_3 is exactly $\operatorname{span}\{|\mathrm{vac}\rangle, |\mathrm{full}\rangle\}$, and L_i acts as $L_i f_m = \epsilon_{imn} f_n$, $L_i \mathrm{g}_m = \epsilon_{imn}\mathrm{g}_n$ on

$$f_k := a_k^+|\mathrm{vac}\rangle, \qquad \mathrm{g}_m := \tfrac{1}{2}\epsilon_{mjk}\, a_j^+ a_k^+|\mathrm{vac}\rangle.$$

Proof. $[e_je_k, e_k] = 2e_j$ and $[e_je_k, e_j] = -2e_k$, while $[e_{j+3}e_{k+3}, e_{k+3}] = -2e_{j+3}$ because $e_{k+3}^2 = -\mathbf{1}$: the relative minus sign makes the triples turn alike. G and $\mathcal{K}$ map e_i to $\pm e_{i+3}$, so they commute with $R \oplus R'$ iff $R = R'$. The Fock forms follow from $e = a^+ + a^-$, $e' = a^+ - a^-$ and the CAR, and the action on f, g from $[L_i, a_m^+] = \epsilon_{imn}a_n^+$ and $L_i|\mathrm{vac}\rangle = 0$. The intersection is computed in r3-9, which also checks $K_i = GL_i$ on V (Proposition 3.77) and the brackets of Theorem 3.3. □

Restated and fixed (r3-9). The paired combination $e_je_k + e_{j+3}e_{k+3}$, a natural first guess, is the wrong one; with the minus sign it is the diagonal rotation of Theorem 3.1, whose spin lift has integer spin on S, as Proposition 3.83 states.

Theorem 3.44 (The field-octant map)**.** *(i) The linear maps $V \to S$ intertwining $\mathfrak{so}(3)_\Delta$ form a four-dimensional space. Those sending $\mathbf{E}$ into weight one and $\mathbf{B}$ into weight two form the two-parameter family*

$$\Phi_{\mathrm{oct},\alpha,\beta}(X) = \alpha \sum_k E_k f_k + \beta \sum_m B_m g_m,$$

and $\Phi := \Phi_{1,1}$ is an isometry of $\mathbb{R}^6$ onto the six field octants.

(ii) Under $a_I^+|\mathrm{vac}\rangle \leftrightarrow dx^I$, $\Phi_{\mathrm{oct}}(X) = E_k\, dx^k + \frac{1}{2}\epsilon_{mjk}B_m\, dx^j \wedge dx^k$: $\mathbf{E}$ is a one-form and $\mathbf{B}$ a two-form on $\mathbb{R}^3$.

(iii) The spin lift of $\mathfrak{gl}(3)$ preserves the image of Φ and induces $\mathbf{E} \mapsto (A - \frac{1}{2}\operatorname{tr}A)\mathbf{E}$ and $\mathbf{B} \mapsto (\frac{1}{2}\operatorname{tr}A - A^\top)\mathbf{B}$: $\mathbf{E}$ transforms in $L \otimes (\det L)^{-1/2}$ and $\mathbf{B}$ in $L^* \otimes (\det L)^{1/2}$, contragredient up to the determinant.

(iv) The same $\mathfrak{gl}(3) \subset \mathfrak{so}(3,3)$ acts on the field point through the light-cone components, $\mathbf{B}+\mathbf{E} \mapsto A(\mathbf{B}+\mathbf{E})$, $\mathbf{B}-\mathbf{E} \mapsto -A^\top(\mathbf{B}-\mathbf{E})$. For symmetric A this is $\mathbf{B} \mapsto A\mathbf{E}$, $\mathbf{E} \mapsto A\mathbf{B}$, a set of boosts, and Φ intertwines the vector action of $\mathfrak{gl}(3)$ with its spin lift only on $\mathfrak{so}(3)_\Delta$.

Proof. (i) As $\mathfrak{so}(3)_\Delta$-modules $V = 2\cdot\mathbb{R}^3$ and, by Lemma 3.43, $S = 2\cdot\mathbb{R}^3 \oplus 2\cdot\mathbb{R}$ (the f's, the g's and the poles). The vector representation is absolutely irreducible, so by Schur's lemma [57] $\mathrm{Hom}(\mathbb{R}^3,\mathbb{R}^3) = \mathbb{R}$ and $\mathrm{Hom}(\mathbb{R}^3,\mathbb{R}) = 0$; the weight condition leaves one scale on each copy. (ii) a_k^+ acts as $dx^k\wedge$, and $g_1 = a_2^+ a_3^+|\mathrm{vac}\rangle \leftrightarrow dx^2\wedge dx^3$ cyclically. (iii) follows from Proposition 1.12: on Λ^1 the action is $A - \frac{1}{2}\operatorname{tr} A$, and on $\Lambda^2(L) \cong L^*\otimes\det L$, in the basis $\mathfrak{g}_m$, it is $\operatorname{tr} A - A^\top$, from which $\frac{1}{2}\operatorname{tr} A$ is subtracted. (iv) is Proposition 1.12 on V with (1.6) (r3-10, r3-11). □

Items (i), (iii), (iv) are fixed; Φ itself is constructed. Item (ii) is the premetric statement that the electric field strength is a one-form and the magnetic field strength a two-form on space, $F = \mathbf{E}\wedge d\sigma + \mathbf{B}$ in the notation of Hehl and Obukhov [204] (inherited). Item (iv) makes precise the statement that under $\mathfrak{gl}(3,\mathbb{R})$ $\mathbf{E}$ is a vector and $\mathbf{B}$ a two-form: it is true of the spin lift on the octants and false of the vector action on the field point, where the symmetric part of $\mathfrak{gl}(3)$ mixes $\mathbf{E}$ with $\mathbf{B}$. The two scales of (i) are tied together by Maxwell's equations in Proposition 3.54.

Proposition 3.45 (Parity on the octants)**.** *(i)* $\mathrm{Ad}_T|_V = \eta$ *is the frame parity, fixing* $\mathbf{B}$ *and reversing* $\mathbf{E}$ *(Proposition 7.5), and* T *is, up to a factor, the only element of* $\mathrm{Cl}(3,3)$ *implementing it.* T *carries each octant to its antipode,*

$$T|\mathrm{vac}\rangle = |\mathrm{full}\rangle, \quad Tf_k = \mathfrak{g}_k, \quad Tg_k = -f_k, \quad T|\mathrm{full}\rangle = -|\mathrm{vac}\rangle,$$

and $T\Phi = -\Phi\circ G$*: on the field octants the spinor lift of parity acts as minus the Hodge star, exchanging electric and magnetic octants.*

(ii) $(-1)^N\Phi = \Phi\circ\eta$*: the weight parity, which is the chirality* $\gamma_5 = -\Gamma$*, acts as* -1 *on the image of* $\mathbf{E}$ *and* $+1$ *on the image of* $\mathbf{B}$*; but* $\mathrm{Ad}_{(-1)^N} = -\mathbf{1}$ *on* V*.*

(iii) The Dirac parity $P = \gamma^0 = e_1$ *of Theorem 2.27 acts on the frame field by* $(\mathbf{B};\mathbf{E}) \mapsto (B_1, -B_2, -B_3; -\mathbf{E})$*, and* $e_1 f_1 = |\mathrm{vac}\rangle$*: it leaves the field octants.*

Proof. (i) T commutes with e_1, e_2, e_3 and anticommutes with e_4, e_5, e_6; uniqueness is Skolem–Noether. $T = \prod_i (a_i^+ + a_i^-)$ flips all three signs, and the displayed values are r3-12. With them $T\Phi_{\mathrm{oct}}(X) = \sum_k (E_k g_k - B_k f_k)$, while GX has $\mathbf{B}' = -\mathbf{E}$, $\mathbf{E}' = \mathbf{B}$. (ii) $(-1)^N$ is -1 on f and $+1$ on g; Γ anticommutes with every vector. (iii) e_1 commutes with itself and anticommutes with the other generators, and it flips the first sign (r3-12). □

Fixed. The polar and axial characters of $\mathbf{E}$ and $\mathbf{B}$ are present on the octants, but they are carried by the weight parity $(-1)^N = -\Gamma$, whose Clifford action on V is the total reflection $X \mapsto -X$, and not by the spinor lift T of the parity, which acts on the field octants as the duality rotation by $-\pi/2$. This is possible because Φ intertwines the rotations only, not the Clifford action of $\mathrm{Pin}(3,3)$. The statement that parity acts as -1 on the electric and $+1$ on the magnetic octants is therefore false if parity means the spinor lift of the spatial inversion, and true if it means the weight parity.

Remark 3.46 (The earlier field-octant bijection)**.** An alternative labelling calls the octants with one plus sign electric and those with two plus signs magnetic, the octant with its single plus sign in pair k carrying E_k and the octant with its single minus sign in pair m carrying B_m. Its electric octants are the lines of the $\mathfrak{g}_k$, of weight $N = 2$, and its magnetic ones those of the f_m, of weight $N = 1$: up to signs, which it does not fix, its labelling is that of $\Phi\circ G$ (Proposition 3.54), not of Φ. In both labellings E_k and B_k sit on antipodal octants, which the mass form pairs (Proposition 3.51). The earlier bijection mapped octants to a basis of interior bivectors, a bijection of labels and not a linear map; it is admissible as a labelling and equivalent to Φ by a duality rotation of a quarter turn, and nothing in the octant structure prefers it. Its derivation of $E = cB$ from a count of "temporal kernels" is not reproduced: in the present construction the ratio of the two scales of $\Phi_{\mathrm{oct},\alpha,\beta}$ is c in SI units, the conversion constant of Proposition 3.98, fixed by Faraday's law (Proposition 3.54 and Theorem 3.53(iii)).

3.5.2 Clifford multiplication does not give the pair reading

The field point is also a Clifford vector, and one may ask what it does to the octants by left multiplication. The answer shows why the pair reading $X_{\mathrm{EM}} = (\mathbf{B};\mathbf{E})$ needs the map Φ and is not obtained by multiplying.

Proposition 3.47 (Light-cone amplitudes)**.** *For every* $X = \sum_i (B_i e_i + E_i e_{i+3})$

$$X|\mathrm{vac}\rangle = \sum_i (B_i + E_i)\, f_i, \qquad X|\mathrm{full}\rangle = \sum_i (B_i - E_i)\, g_i = \sum_i (-1)^{i-1}(B_i - E_i)\, |111 \oplus \delta_i\rangle, \tag{3.11}$$

and $X^2 = \eta(X,X)\mathbf{1} = \sum_i (B_i + E_i)(B_i - E_i)\,\mathbf{1}$*. The amplitudes* $B_i \pm E_i$ *are the coordinates of* X *on* L *and* L'*, the eigen-coordinates of* $\mathcal{K}$*, with* $\mathbf{B} + \mathbf{E} = \sqrt{2}\,\mathrm{Re}(e^{-i\pi/4} z)$ *and* $\mathbf{B} - \mathbf{E} = -\sqrt{2}\,\mathrm{Im}(e^{-i\pi/4} z)$*. Further*

$$X\big(|\mathrm{vac}\rangle + |\mathrm{full}\rangle\big) = \sqrt{2}\,\Phi\big(e^{\pi G/4} X\big). \tag{3.12}$$

No spinor χ *satisfies* $X\chi = \Phi_{\mathrm{oct}}(X)$ *for all* X*; the Clifford multiplications* $X \mapsto X\chi$ *that intertwine the rotations are those with* $\chi \in \mathrm{span}\{|\mathrm{vac}\rangle, |\mathrm{full}\rangle\}$*, and none of them places* $\mathbf{E}$ *and* $\mathbf{B}$ *on separate weights.*

Proof. (3.11) follows from (1.6) with $a_i^-|\mathrm{vac}\rangle = 0$, $a_i^+|\mathrm{full}\rangle = 0$ and $a_i^-|\mathrm{full}\rangle = g_i$; X^2 follows from the CAR. The duality rotation $e^{\pi G/4}$ gives $\mathbf{E}' = (\mathbf{E} + \mathbf{B})/\sqrt{2}$, $\mathbf{B}' = (\mathbf{B} - \mathbf{E})/\sqrt{2}$, which is (3.12). If $X\chi = \Phi_{\mathrm{oct}}(X)$, then $e_1\chi = g_1$ and $e_4\chi = f_1$, i.e. $\chi = e_1|011\rangle = |111\rangle$ and $\chi = -e_4|100\rangle = |000\rangle$, a contradiction. $L_i(X\chi) = [L_i, X]\chi + XL_i\chi$, so equivariance requires $L_i\chi = 0$, and then $X(a|\mathrm{vac}\rangle + b|\mathrm{full}\rangle)$ puts $a(\mathbf{B} + \mathbf{E})$ on weight one (r3-13). □

Fixed; the reading is interpretive. Read as a Clifford vector acting on the vacuum pole, the field point does not keep the two members of the pair apart: it places on the i-th electric octant the sum $E_i + B_i$, which in textbook variables is $E_i - B_{\mathrm{t}i}$, and on the filled pole it places $B_i - E_i = -(E_i + B_{\mathrm{t}i})$ on the i-th magnetic octant. These sums are the coordinates of X on the null planes L and L', the light-cone components of the para-complex structure; they mix the electric and the magnetic member and are not the field strength. A literal reading "$X = \mathbf{E} + \mathbf{B}$" would therefore be wrong: the pair $(\mathbf{B};\mathbf{E})$ is placed on the octants by Φ, which is not Clifford multiplication by any spinor, and the Clifford action differs from Φ by a duality rotation of an eighth of a turn, (3.12). (An eighth of a turn also separates the Klein correspondence from the frame reading in Theorem 3.84; the maps are different and we draw no conclusion from the coincidence.)

3.5.3 Boosts and duality cannot act by even elements

Theorem 3.48 (The even obstruction)**.** *Every even element of* $\mathrm{Cl}(3,3)$ *commutes with* $(-1)^N$ *and maps an octant of weight* w *into the weights* w *and* $w \pm 2$*; this includes* $\mathfrak{so}(3,3)$*,* $\mathrm{Spin}(3,3)$ *and the spin lifts of* $\mathfrak{l}_G$ *and* $\mathfrak{l}_{\mathrm{int}}$*. No even* Q *satisfies* $Q\Phi = \Phi K_k$ *for a boost* K_k *of* $\mathfrak{l}_G$*, or* $Q\Phi = \Phi G$*. At the level of groups, no element of* $\mathrm{Cl}(3,3)$ *of definite parity intertwines* e^{sK_k} $(s \neq 0)$ *with* Φ*, nor* $e^{\vartheta G}$ *unless* $\vartheta \in \frac{\pi}{2}\mathbb{Z}$*.*

Proof. $(-1)^N = -\omega$ commutes with even and anticommutes with odd elements, and weights of equal parity differ by 0 or 2. By Example 3.4, $K_3 e_1 = e_5$; so $\Phi_{\mathrm{oct}}(e_1) = g_1$ has weight two and $\Phi_{\mathrm{oct}}(K_3 e_1) = f_2$ weight one, and an even Q cannot map the first to the second. Likewise $\Phi_{\mathrm{oct}}(e_1) = g_1$ and $\Phi_{\mathrm{oct}}(Ge_1) = \Phi_{\mathrm{oct}}(e_4) = f_1$. For the groups, $e^{sK_3}e_1 = \cosh s\, e_1 + \sinh s\, e_5$, so $\Phi_{\mathrm{oct}}(e^{sK_3}e_1)$ has components of both parities when $s \neq 0$, while an element of definite parity maps g_1 to a vector of definite parity; the same holds for $e^{\vartheta G}e_1 = \cos\vartheta\, e_1 + \sin\vartheta\, e_4$ unless $\cos\vartheta \sin\vartheta = 0$ (r3-14, r3-15). □

Obstructed, under the hypothesis that the operator has definite parity; the hypothesis covers every element of the spin group and of its Lie algebra. The scope is the realisation chosen here: the theorem says that the spin group of $\mathrm{Cl}(3,3)$, acting on S by left multiplication, does not implement the Lorentz action of $\mathfrak{l}_G$ on the field values placed in the octants by the map Φ. It does not say that electromagnetic fields fail to be Lorentz covariant: they transform covariantly as two-forms, through $\mathfrak{l}_G$ acting on V, with no reference to spinors. Nor does a shared operator identify the two modules: that the same temporal volume element enters both the spin-$\frac{1}{2}$ module S and the octant representation of the spin-one field strengths does not make these modules equivalent, and they are not (Proposition 3.49 and r4-13). The intertwiners that do exist are odd:

Proposition 3.49 (Odd intertwiners and the octant representation)**.** *(i)* $-T\Phi = \Phi G$ *and* $Q_k\Phi = \Phi K_k$ *for*

$$Q_k := -TL_k = -\tfrac{1}{2}e_k(\mathbf{1} - \omega_j\omega_l) = -\tfrac{1}{2}(e_k + e_{k+3}\omega), \qquad \{k,j,l\} = \{1,2,3\},$$

an odd element with parts of grades one and five. The solutions of $Q\Phi = \Phi K_k$ *form an affine space of dimension* 16*, of dimension* 8 *among odd elements, and contain no Clifford vector; no Clifford vector implements* G *on the octants.*

(ii) On $S = \mathrm{Cl}(V^+)|\mathrm{vac}\rangle$ *(Proposition 1.11),* $Q_k(a|\mathrm{vac}\rangle) = -\frac{1}{2}[e_k, a]|\mathrm{vac}\rangle$ *and* $-T$ *is left multiplication by the central element* $-e_1e_2e_3$ *of* $\mathrm{Cl}(V^+)$*.*

(iii) $\rho_{\mathrm{oct}}(L_k) = L_k$*,* $\rho_{\mathrm{oct}}(K_k) = Q_k$ *is a representation of* $\mathfrak{l}_G$ *on* S*. Its Casimir* $\sum_k \rho_{\mathrm{oct}}(K_k)^2 - \rho_{\mathrm{oct}}(L_k)^2$ *is* 4 *on the field octants, the field-strength value, and* 0 *on the poles, and its intertwiners* $V \to S$ *are* $\mathrm{span}\{\Phi, \Phi G\}$*.*

(iv) The spin lift of $\mathfrak{l}_G$ *has Casimir* 3 *on all of* S *and admits no nonzero intertwiner* $V \to S$*. Its boosts carry* f_k *to* $-|\mathrm{full}\rangle$ *and* $|\mathrm{vac}\rangle$ *to* $-\mathfrak{g}_k$*: under the genuine Lorentz action the electric octants are the spatial components of the four-vector* S^+ *whose time component is the filled pole, and the magnetic octants those of the four-vector* S^- *whose time component is the vacuum.*

Proof. (i) T commutes with the same-sign bivectors, hence with L_k, so $-TL_k\Phi = -T\Phi L_k = \Phi G L_k = \Phi K_k$ by Proposition 3.45(i) and $K_k = GL_k$. The closed forms use $T(e_je_l) = -e_k$ for cyclic (k,j,l) and $e_{k+3}\omega = -e_k\omega_j\omega_l$. The values of a solution on the two poles are free, eight components each, four of them of the parity opposite to the pole (r3-15). (ii) $e_{k+3}\omega$ commutes with e_1, e_2, e_3 and sends $|\mathrm{vac}\rangle$ to $-e_k|\mathrm{vac}\rangle$, so it acts on $a|\mathrm{vac}\rangle$ as $-a\,e_k$. (iii) T commutes with $\rho_{\mathrm{oct}}(L)$ and $T^2 = -\mathbf{1}$, so the brackets of $\mathfrak{l}_G$ follow from those of $\mathfrak{so}(3)$; $\rho_{\mathrm{oct}}(K)^2 = -L^2$ gives the Casimir, and an intertwiner is a T-linear map $\mathbb{C}^3 \to \mathbb{C}^3$ commuting with $\mathfrak{so}(3)$, hence a complex multiple of Φ (r3-16). (iv) The Casimir is Proposition 3.83; $S = 2(\frac{1}{2}, \frac{1}{2})$ contains no $(1,0) \oplus (0,1)$; the boost formulas are $K_k = a_j^- a_l^- - a_j^+ a_l^+$ on f_k and $|\mathrm{vac}\rangle$ (r3-16). □

Constructed: the operators Q_k and the representation ρ_{oct}; fixed: the solution counts of (i) and the Casimir values and the intertwiner space of (iii); restated and fixed: (iv). Item (ii) identifies the construction: in textbook variables (Theorem 3.53) the field octants carry Hestenes' Pauli-algebra field $F = \mathbf{E} + I\mathbf{B}_{\mathrm{t}}$ [70], written with $\sigma_k = e_k$ and $I = T$, and the boosts and the duality rotation act as they do in the Pauli algebra, through the commutator with σ_k and through multiplication by I. The commutator is a left multiplication combined with a right multiplication, and on S the right multiplication is the odd element $e_{k+3}\omega$: this is the reason the boosts are odd. Which symmetries of the field the octant picture carries is now settled.

Corollary 3.50 (What the octants carry)**.** *By even elements of* $\mathrm{Cl}(3,3)$ *the field octants carry the rotations* $\mathfrak{so}(3)_\Delta$ *and the premetric* $\mathfrak{gl}(3)$ *of Theorem 3.44(iii), through their spin lifts, and the parity, through* $(-1)^N = -\omega$*. They do not carry the boosts or the duality rotations, which*

act only through the odd operators Q_k and $-T$ of the representation ρ_{oct}, the Pauli algebra of the temporal triple. The generators are odd; the finite transformations they generate are in general of mixed parity, for example $e^{-\theta T} = \cos\theta\,\mathbf{1} - \sin\theta\,T$. The spin-lifted Lorentz algebra $\mathfrak{l}_G$ makes the field octants parts of two four-vectors, not of a field strength.

There is no nonzero linear Lorentz-equivariant map from the field-strength module to the four-vectors. Item (iv) gives the same fact for the spinor module, whose modules under $\mathfrak{l}_G$ are four-vectors (for E_n itself the statement is Proposition 3.34); on the octants the Hodge star of the field is represented, but by the odd element $-T$, with $T = I_T = e_1e_2e_3$, not by an element of $\mathrm{Spin}(3,3)$.

3.5.4 Energies on the octants

Proposition 3.51 (Energies and the Chevalley–Mukai pairing)**.** *Let $\psi = \Phi_{\mathrm{oct}}(X) = \psi_{\mathbf{E}} + \psi_{\mathbf{B}}$ with $\psi_{\mathbf{E}} = \sum_k E_k f_k$, $\psi_{\mathbf{B}} = \sum_m B_m g_m$.*

(i) $|\psi|^2 = |\mathbf{E}|^2 + |\mathbf{B}|^2 = 2E_6$ *and* $\psi^{\top}(-1)^N\psi = |\mathbf{B}|^2 - |\mathbf{E}|^2 = \mathrm{Re}\,q$.

(ii) $\psi_{\mathbf{E}}^{\top}T\psi_{\mathbf{B}} = -\mathbf{E}\cdot\mathbf{B} = -\frac{1}{2}\,\mathrm{Im}\,q$, *while* $\psi^{\top}T\psi = 0$ *and* $\psi^{\top}T\Gamma\psi = 2\,\mathbf{E}\cdot\mathbf{B} = \mathrm{Im}\,q$. *Hence*

$$E_0 = \tfrac{1}{2}|q| = \tfrac{1}{2}\Big[\big(\psi^{\top}(-1)^N\psi\big)^2 + 4\big(\psi_{\mathbf{E}}^{\top}T\psi_{\mathbf{B}}\big)^2\Big]^{1/2}.$$

(iii) *The* $\mathrm{Spin}(3,3)$*-invariant bilinear forms on S are* $\mathrm{span}\{T, T\Gamma\}$, *$T$ antisymmetric and $T\Gamma$ symmetric. Under $\Lambda^*(L) \cong S$ the Chevalley–Mukai pairing $(\varphi,\chi) = [\tau(\varphi)\wedge\chi]_{\mathrm{top}}$, with τ the reversal, equals $-\varphi^{\top}T\chi$. Two octants pair to a nonzero value iff their null planes N_σ, N_τ are transverse, iff they are antipodal.*

(iv) $(-1)^N$ *is invariant under the six compact bivectors and under ρ_{oct}, and under none of the nine boosts. The Euclidean norm is invariant under the compact bivectors and the $\rho_{\mathrm{oct}}(L_k)$, but not under the boosts $\rho_{\mathrm{oct}}(K_k) = Q_k$, which are symmetric: $|\psi|^2 = 2E_6$ changes under boosts, $|\Phi_{\mathrm{oct}}(e^{sK_3}e_1)|^2 = \cosh 2s$.*

(v) *In SI units, with $\psi = \Phi_{\mathrm{oct}}(c\mathbf{B};\mathbf{E})$ $[\mathrm{V\,m^{-1}}]$, $u = \frac{\varepsilon_0}{2}|\psi|^2$ is the energy density and $\psi^{\top}(-1)^N\psi = \mathrm{Re}\,q_c$, $q_c = z_c\cdot z_c$, $z_c = c\mathbf{B} + i\mathbf{E}$ (Proposition 3.99).*

Proof. The f_k, g_m are orthonormal, $(-1)^N = \mp 1$ on them, $Tg_m = -f_m$ and $\Gamma = -(-1)^N$, which gives (i), (ii) and the formula for E_0 by $|q|^2 = (\mathrm{Re}\,q)^2 + (\mathrm{Im}\,q)^2$. (iii) A form M is invariant iff $\beta^{\top}M + M\beta = 0$ for the fifteen bivectors, a linear system with a two-dimensional solution space (r3-18, the count of Theorem 3.84); T commutes with the antisymmetric (same-sign) and anticommutes with the symmetric (mixed) bivectors (Proposition 7.5), and Γ, which anticommutes with every vector, commutes with all of them. The Mukai identity is checked on all 64 pairs of basis forms, and $\dim(N_\sigma\cap N_\tau)$ is the number of agreeing signs. (iv) e_k and $e_{k+3}\omega$ are symmetric matrices, so $Q_k^{\top} + Q_k = 2Q_k \neq 0$; (iv), (v): r3-17, r3-18. □

Items (i), (ii), (iv) are fixed, (iii) is inherited in substance and verified here, (v) restates Section 3.9. By (i), the Maxwell Lagrangian $L_M = -\frac{1}{2}\,\mathrm{Re}\,q$ of Proposition 3.20 is $-\frac{1}{2}\psi^{\top}(-1)^N\psi = \frac{1}{2}(|\psi_{\mathbf{E}}|^2 - |\psi_{\mathbf{B}}|^2)$ on the octants, half the squared electric minus the squared magnetic octant amplitudes. The Chevalley form on the spinors of $L\oplus L'$, antisymmetric for $\dim L = 3$ since $(-1)^{3\cdot 2/2} = -1$, and its non-vanishing on pure spinors exactly for transverse null planes are Chevalley's [201]; Gualtieri identifies the form with the Mukai pairing (Section 2.4 of [203], whose Proposition 2.21 restates the transversality criterion). So the $\mathrm{Spin}(3,3)$-invariant mass form T is, on the octants, the Chevalley–Mukai pairing of each octant with its antipode: of the vacuum with the filled pole and of f_k with g_k, that is, of E_k with B_k, with the sign fixed by (ii): $\psi_{\mathbf{E}}^{\top}T\psi_{\mathbf{B}} = -\mathbf{E}\cdot\mathbf{B} = -\frac{1}{2}\,\mathrm{Im}\,q$. The Lorentz invariant $\mathrm{Re}\,q$ appears on the octants through the

form $(-1)^N$, which is not Spin(3, 3)-invariant, while the invariant symmetric form $T\Gamma$ gives $\operatorname{Im} q$; both facts are consistent with Theorem 3.48 and with the representation ρ_{oct}, which preserves $(-1)^N$.

3.5.5 Maxwell's equations on the octants

Proposition 3.52 (Exterior derivative and codifferential)**.** *On S-valued fields put $d := \sum_i a_i^+ \partial_i$ and $\delta := -\sum_i a_i^- \partial_i$. Then $d^2 = \delta^2 = 0$,*

$$d - \delta = \sum_i e_i \partial_i =: D, \qquad d + \delta = \sum_i e_{i+3} \partial_i =: D', \qquad D^2 = \Delta, \qquad D'^2 = -\Delta,$$

and under $f_k \leftrightarrow dx^k$, $\mathrm{g}_m \leftrightarrow \star dx^m$, $|\mathrm{vac}\rangle \leftrightarrow 1$, $|\mathrm{full}\rangle \leftrightarrow dx^1 \wedge dx^2 \wedge dx^3$, d and δ are the exterior derivative and the codifferential of the Euclidean $\mathbb{R}^3$: $d(\mathbf{E}\cdot f) = (\nabla \times \mathbf{E})\cdot g$, $d(\mathbf{B}\cdot g) = (\nabla\cdot\mathbf{B})|\mathrm{full}\rangle$, $\delta(\mathbf{E}\cdot f) = -(\nabla\cdot\mathbf{E})|\mathrm{vac}\rangle$, $\delta(\mathbf{B}\cdot g) = (\nabla\times\mathbf{B})\cdot f$.

Proof. $d^2 = \sum_{ij} a_i^+ a_j^+ \partial_i \partial_j = 0$ because $a_i^+ a_j^+$ is antisymmetric, and likewise for δ; $a_i^+ \pm a_i^-$ are e_i and e_{i+3}, and $D^2 = \sum_i e_i^2 \partial_i^2$, $D'^2 = \sum_i e_{i+3}^2 \partial_i^2$. The four formulas use $a_i^- g_m = \epsilon_{mik} f_k$ and $a_m^+ g_m = |\mathrm{full}\rangle$ (r3-19). □

Inherited, and verified here: $D = d - \delta$ is Kähler's operator on inhomogeneous forms [205], the forms read as spinors [202] and the Dirac–Kähler equation of Becher and Joos [200], whose Minkowski solutions Benn and Tucker read as a fourfold-degenerate Dirac field [209]. Here the operator acts on space, and the forms carry the field strength, not a fermion.

Theorem 3.53 (Maxwell's equations as one equation on the octants)**.** *For every smooth field $X(t, \mathbf{x})$,*

$$(\partial_t - D)\,\Phi_{\mathrm{oct}}(X) = -(\nabla\cdot\mathbf{E})\,|\mathrm{vac}\rangle + (\partial_t\mathbf{E} + \nabla\times\mathbf{B})\cdot f + (\partial_t\mathbf{B} - \nabla\times\mathbf{E})\cdot g - (\nabla\cdot\mathbf{B})\,|\mathrm{full}\rangle. \quad (3.13)$$

Consequently:

(i) Maxwell's equations with sources in the frame variables, $\partial_t z = -i\nabla\times z - i\mathbf{j}$ and $\nabla\cdot z = i\rho$ (eq. (3.3)), are equivalent to

$$(\partial_t - D)\,\Phi_{\mathrm{oct}}(X) = -\big(\rho\,|\mathrm{vac}\rangle + \mathbf{j}\cdot f\big)\,:$$

the vacuum component is Gauss's law $\nabla\cdot\mathbf{E} = \rho$, the filled component the absence of magnetic charge, the electric octants carry the Ampère–Maxwell law $\partial_t\mathbf{E} = -\nabla\times\mathbf{B} - \mathbf{j}$ and the magnetic octants Faraday's law $\partial_t\mathbf{B} = \nabla\times\mathbf{E}$.

(ii) $(\partial_t + D)(\partial_t - D) = \partial_t^2 - \Delta$ on all eight components, while $(\partial_t + D')(\partial_t - D') = \partial_t^2 + \Delta$. In $D = \sum_i e_i\partial_i$ the derivatives ∂_i are taken along the three physical spatial coordinates x^i, the lengths of the classical reading, which sit on V^-; the Clifford generators multiplying them are the temporal e_1, e_2, e_3, of square $+\mathbf{1}$, the generators that carry $\mathbf{B}$ in the field reading, and not the spatial ones.

(iii) In SI units $\big(\frac{1}{c}\partial_t - D\big)\Phi_{\mathrm{oct}}(c\mathbf{B}; \mathbf{E}) = -(c\rho\,|\mathrm{vac}\rangle + \mathbf{j}\cdot f)/(\varepsilon_0 c)$.

(iv) In textbook variables $\Phi_{\mathrm{t}}(X) := \mathbf{E}\cdot f + \mathbf{B}_{\mathrm{t}}\cdot g = -(-1)^N\Phi_{\mathrm{oct}}(X)$ satisfies $T\Phi_{\mathrm{t}} = \Phi_{\mathrm{t}}\circ G$ and

$$(\partial_t + D)\,\Phi_{\mathrm{t}}(X) = \rho\,|\mathrm{vac}\rangle - \mathbf{j}\cdot f, \qquad \Phi_{\mathrm{t}}(X) = \Big(\sum_k E_k e_k + \sum_k B_{\mathrm{t}k}\,T e_k\Big)|\mathrm{vac}\rangle,$$

which, under $S \cong \mathrm{Cl}(V^+)$ with $\sigma_k = e_k$ and $I = T$, is the equation $(\partial_t + \nabla)(\mathbf{E} + I\mathbf{B}_{\mathrm{t}}) = \rho - \mathbf{j}$ of the Pauli algebra.

(v) *With pole components* $\psi = \Phi_{\mathrm{oct}}(X) + \phi_0|\mathrm{vac}\rangle + \phi_3|\mathrm{full}\rangle$, $(\partial_t - D)\psi = 0$ *reads* $\partial_t\phi_0 = \nabla\cdot\mathbf{E}$, $\partial_t\phi_3 = \nabla\cdot\mathbf{B}$, $\partial_t\mathbf{E} = -\nabla\times\mathbf{B} + \nabla\phi_0$, $\partial_t\mathbf{B} = \nabla\times\mathbf{E} + \nabla\phi_3$; *every component of a solution obeys the wave equation, so the Gauss constraints hold for all times if* ϕ_0, ϕ_3 *and the two divergences vanish initially.*

Proof. (3.13) is $\partial_t\Phi_{\mathrm{oct}}(X) - (d-\delta)\Phi_{\mathrm{oct}}(X)$ with the four formulas of Proposition 3.52; it is checked on general symbolic fields $E_k(t,\mathbf{x})$, $B_k(t,\mathbf{x})$ (r3-20), and the real and imaginary parts of the residuals of eq. (3.3) are exactly the components of $(\partial_t - D)\Phi_{\mathrm{oct}}(X) + \rho|\mathrm{vac}\rangle + \mathbf{j}\cdot f$. (ii) $(\partial_t + D)(\partial_t - D) = \partial_t^2 - D^2$ (r3-21). (iii) is (i) with $\Phi_{\mathrm{oct}}(\mathbf{E}, c\mathbf{B})$ and $\partial_t\mathbf{E} = -c^2\nabla\times\mathbf{B} - \mathbf{j}/\varepsilon_0$, $\nabla\cdot\mathbf{E} = \rho/\varepsilon_0$ (r3-24, on first-order jets). (iv) $\Phi_{\mathrm{t}}(X) = \mathbf{E}\cdot f - \mathbf{B}\cdot g$; $(-1)^N$ anticommutes with D; $Te_k = e_j e_l$ cyclically gives the second form; under $a|\mathrm{vac}\rangle \leftrightarrow a$ left multiplication by e_k is that by σ_k (r3-24). (v) is (3.13) together with $D(\phi_0|\mathrm{vac}\rangle) = \nabla\phi_0\cdot f$ and $D(\phi_3|\mathrm{full}\rangle) = \nabla\phi_3\cdot g$ (r3-21); the wave equation follows from (ii), and ϕ_0 with $\phi_0 = \partial_t\phi_0 = 0$ at $t = 0$ vanishes identically. □

Fixed in the realisation and inherited in substance. Item (ii) is a statement about the representation and not about spacetime: the octant form of Maxwell's equations uses the Pauli triple e_1, e_2, e_3 because it is the triple of square $+\mathbf{1}$ that makes the operator hyperbolic, and this choice is constructed. It does not show that spatial differentiation follows from three temporal directions, and the spatial coordinates x^i remain the lengths of V^-; the Dirac operator of the particle reading couples the same derivatives to the spatial generators through $\gamma^0\gamma^a$ instead (Section 3.6). Item (iv) is the Pauli-algebra form, obtained by multiplying with γ_0, of the spacetime-algebra equation $\nabla F = J$ of Hestenes [70], which Theorem 3.74 realises on the interior four-plane; here it is realised on the temporal triple acting on the vacuum. The element T that plays the pseudoscalar I is I_T, with $I_T^2 = -\mathbf{1}$ and $[I_T, e_i] = 0$ for the temporal e_i (r3-24). The circularly polarised wave of Example 3.11 solves $(\partial_t - D)\Phi_{\mathrm{oct}}(X) = 0$, and the same wave with $\mathbf{B}$ reversed does not (r3-23). The textbook signs, $\partial_t\mathbf{E} = \nabla\times\mathbf{B}_{\mathrm{t}} - \mathbf{j}$ and $\partial_t\mathbf{B}_{\mathrm{t}} = -\nabla\times\mathbf{E}$, become those of (i) in the frame variable $\mathbf{B} = -\mathbf{B}_{\mathrm{t}}$.

Proposition 3.54 (Maxwell fixes the scales and leaves a duality angle)**.** *Let* $\Phi_M(X) = (a\mathbf{E} + b\mathbf{B})\cdot f + (c\mathbf{E} + d\mathbf{B})\cdot g$ *be the general rotation-equivariant map into the field octants. Then* $(\partial_t - D)\Phi_M(X)$ *vanishes for every source-free Maxwell field iff* $c = -b$ *and* $d = a$*, that is iff* $\Phi_M = r\,\Phi\circ e^{\vartheta G}$ *for some* $r \geq 0$ *and* ϑ*. The map puts* $\mathbf{E}$ *and* $\mathbf{B}$ *on single weights iff* $\vartheta \in \frac{\pi}{2}\mathbb{Z}$*:* Φ *for* $\vartheta = 0$ *and* $\Phi\circ G$*, with* $\mathbf{E}$ *on weight two, for* $\vartheta = \frac{\pi}{2}$*.*

Proof. Substituting $\partial_t\mathbf{E} = -\nabla\times\mathbf{B}$, $\partial_t\mathbf{B} = \nabla\times\mathbf{E}$ and the vanishing divergences, the residual has f-component $(d-a)\nabla\times\mathbf{B} + (b+c)\nabla\times\mathbf{E}$, g-component $(d-a)\nabla\times\mathbf{E} - (b+c)\nabla\times\mathbf{B}$ and no pole component; it vanishes on all first-order Maxwell jets iff $d = a$, $c = -b$ (r3-22). Conversely $\Phi\circ e^{\vartheta G}$ maps solutions to solutions, since $e^{\vartheta G}$, the duality rotation $z \mapsto e^{i\vartheta}z$, preserves the source-free system. □

Fixed. The two scales of Theorem 3.44(i) are tied by Faraday's and Ampère's laws, $\beta = \alpha$ in Heaviside–Lorentz units and $\beta = c\,\alpha$ in SI, and the energy normalisation of Proposition 3.51(i) fixes α up to sign. What remains is the duality angle. With electric sources, Φ places the charge density on the vacuum pole and the current on the electric octants, and $\Phi\circ G$ places them on the filled pole and on the weight-two octants; the convention that all charges are electric (Section 3.7, [76]) fixes which octants carry the current once the labelling is chosen, not the labelling.

Remark 3.55 (Octant pooling)**.** Each e_i maps the octant σ to the octant with σ_i reversed, the image of σ under the octant generator R_i, and $e_{i+3} = e_i\omega_i$ acts on the same pair of octants (r3-25). Hence the σ-component of $D\psi$ collects ∂_i of the three neighbouring octants only: the Gauss law on the vacuum collects $\sum_i \partial_i E_i$ from the three adjacent electric octants, and each evolution equation collects a curl from two octants of the other weight. This is the exact content of the proposed "pooling" of Maxwell's equations across octants, and it holds as $D = \sum_i e_i\partial_i$.

The proposed mechanism, a per-pair exchange of e_i with e_{i+3} converging on the pole $(-,-,-)$, is not confirmed: the exchange replaces D by D' and the wave operator by the four-dimensional Laplacian (Theorem 3.53(ii)), and nothing converges on one pole, the two poles carrying one Gauss law each.

3.5.6 The answer: $X_{\mathrm{EM}} = (\mathbf{B}; \mathbf{E})$ with each component on an octant

The pair reading has one exact realisation. The field-octant map (Theorem 3.44 and Proposition 3.54), $\Phi_{\mathrm{oct}}(X) = \sum_k (E_k + B_k T) e_k |\mathrm{vac}\rangle$, puts each E_k on the octant with a single minus sign in pair k and each B_k on its antipode; it is unique up to a scale once rotation equivariance, Maxwell's equations and the separation of $\mathbf{E}$ from $\mathbf{B}$ are required, apart from the alternative $\Phi \circ G$ that the earlier labelling uses; in textbook variables the same octants carry, through $\Phi_{\mathrm{t}} = -(-1)^N \Phi$ (Theorem 3.53(iv)), Hestenes' $\mathbf{E} + I\mathbf{B}_{\mathrm{t}}$ on the temporal Pauli algebra, with $I = T$. Clifford multiplication by the field point does not realise the pair: it places the light-cone sums $\mathbf{B} \pm \mathbf{E}$ on the octants (Proposition 3.47), and differs from Φ by an eighth of a turn of duality. The construction is a representation of the classical field strength; it is not a quantisation of the field.

The main obstruction. On the octants the rotations act through the spin representation, but the boosts and the duality rotation of the field have the odd generators Q_k and $-T$ of ρ_{oct}, their finite transformations being of mixed parity in general (Theorem 3.48 and Proposition 3.49), whereas a Dirac field transforms under the spin lift of the interior Lorentz algebra. A covariant theory containing both the Dirac field and the field octants must say how ρ_{oct}, the spin lift acting on Dirac spinors and the map Φ coexist. This is the principal question the octant picture leaves, and the element T, which is at once the mass form of the Dirac reading and the duality operator on the octants, is where the two meet (Section 3.6).

What the poles carry. With Φ, the vacuum pole carries Gauss's law and the charge density, the filled pole the absence of magnetic charge (Theorem 3.53); they are the Fock vacuum and the filled state, the rotation scalars, the eigenlines of $\mathfrak{gl}(3)$ of weights Λ^0 and Λ^3, and, under the spin-lifted $\mathfrak{l}_G$, the time components of the two four-vectors $S^{\mp}$ (Proposition 3.49(iv)). The mass form pairs them with each other as it pairs E_k with B_k (Proposition 3.51). Of the earlier labels, "statistics" survives as the constructed correspondence between the Fisher–Rao octant of clocks and the vacuum pole $(+,+,+)$ (Proposition 1.10). "Mass" does not survive as a property of the pole $(-,-,-)$: the mass form treats both poles alike, and the interior Lorentz algebra distinguishes no octant, since the stabiliser in $\mathfrak{l}_{\mathrm{int}}$ of the octant σ is the three-dimensional span of $e_1 e_4 = \omega_1$ and the null rotations $(e_1 - \sigma_1 e_4)e_5$, $(e_1 - \sigma_1 e_4)e_6$, depending on σ_1 only, while g_2, g_3 commute with $\mathfrak{l}_{\mathrm{int}}$ and g_1 normalises it (r3-26). The octant Fock structure is the canonical anticommutation algebra of one constituent; it is not a photon Fock space, and the quantisation of the field stays open.

Open. Whether a dynamics, rather than a convention, selects Φ or $\Phi \circ G$; what the two scalar fields ϕ_0, ϕ_3 of the extended system of Theorem 3.53(v) mean, if anything; and how ρ_{oct}, which is not part of the spin representation, enters a covariant dynamics with the Dirac sector of Chapter 2; its covariant couplings linear in the field are determined in Theorem 3.64.

Table 3.3 lists every claim of the section with its check and its class.

Table 3.3: The claims of this section, with their checks and classes.

id	claim	class
r3-1	octants = basis lines, sign triples, CAR, $N = \#$ minus signs, vacuum $(+,+,+)$	restated (Proposition 1.5), fixed
r3-2	octant σ is the pure spinor of $N_\sigma = \mathrm{span}\{e_i - \sigma_i e_{i+3}\}$	fixed; inherited (Cartan–Chevalley)
r3-3	$\Gamma = -\omega_1\omega_2\omega_3$, $(-1)^N = -\Gamma = \gamma_5$; the earlier $I = -\omega_1\omega_2\omega_3$ confirmed	restated, fixed
r3-4, 5	reflections lift to $e_a\omega$, unique; octant group simply transitive; lifted group of order 16, centre $\{\pm\mathbf{1}, \pm e_4e_5e_6\}$	fixed
r3-6	top eigenline of $B_{\mathrm{diag}\,x}$ is the octant sign x; clock octant $\leftrightarrow$ vacuum	fixed; identification constructed
r3-7	$S \cong \Lambda^*(L) \cong \mathrm{Cl}(V^+)\lvert\mathrm{vac}\rangle$, weights $1,3,3,1$; X acts by $2X_L\wedge$ and contraction with $\eta(X_{L'},\cdot)$	inherited (Chevalley), verified
r3-8	commutant of $\mathcal{K} = \mathfrak{gl}(3)$, acting as $\Lambda^w(A) - \frac{1}{2}\,\mathrm{tr}\,A$	inherited (Gualtieri), fixed here
r3-9	diagonal rotations $L_i = \frac{1}{2}(e_{j+3}e_{k+3} - e_je_k)$, Fock forms of L_i, K_i	restated, fixed
r3-10	equivariant maps: dimension 4; $\Phi_{\mathrm{oct},\alpha,\beta}$	fixed; Φ constructed
r3-11	premetric $\mathfrak{gl}(3)$ on the image; vector action mixes $\mathbf{E}, \mathbf{B}$	fixed; premetric reading inherited
r3-12	T lifts parity and acts as $-G$ on the octants; $(-1)^N$ acts as parity	fixed
r3-13	$X\lvert\mathrm{vac}\rangle$, $X\lvert\mathrm{full}\rangle$ light-cone amplitudes; (3.12); Φ is no Clifford map	fixed; reading interpretive
r3-14	no even intertwiner of boosts or of G	obstructed
r3-15, 16	odd intertwiners $Q_k, -T$; ρ_{oct}; spin lift: four-vectors, no intertwiner	Q_k, ρ_{oct} constructed; counts fixed; restated (iv)
r3-17, 18	energies; invariant forms; T = Chevalley–Mukai pairing of antipodes; $\lvert\psi\rvert^2$ not ρ_{oct}-invariant	fixed; pairing inherited
r3-19	d, δ on the octants, $d \mp \delta = D, D'$	inherited (Kähler), verified
r3-20, 21, 23, 24	Maxwell $\Leftrightarrow$ $(\partial_t - D)\Phi_{\mathrm{oct}}(X) = -(\rho\lvert\mathrm{vac}\rangle + \mathbf{j}\cdot f)$; wave operator; SI; Hestenes form	fixed; inherited in substance
r3-22	Maxwell-compatible equivariant maps $= r\Phi \circ e^{\vartheta G}$	fixed
r3-25	pooling = nearest-neighbour structure of D	fixed; mechanism refuted
r3-26	$\mathfrak{l}_{\mathrm{int}}$ distinguishes no octant	fixed; "mass pole" refuted

3.6 The bridge $T = \hat{n}J(\hat{n})$ between the two readings

Two sections of this chapter meet in one element of $\mathrm{Cl}(3,3)$. The clock $\hat{n} \in V^+$ of Section 3.4 determines the complex structure $J(\hat{n}) = T\hat{n}$ of the Dirac reading (Proposition 2.13 and Theorem 2.14), while the octant representation of the field in Section 3.5 uses $T = e_1e_2e_3$ itself, through $-T$, as the duality operator (Propositions 3.45 and 3.49). Since $\hat{n}$ commutes with T and $\hat{n}^2 = \mathbf{1}$, $T = \hat{n}J(\hat{n}) = J(\hat{n})\hat{n}$ for every clock. This section asks whether this identity can be built into both readings at once while their distinct Lorentz representations are kept apart, also when the clock varies or the temporal axes are permuted, and it answers the points left open in Section 3.5: the remark on the spatial derivatives of the octant Maxwell operator after Theorem 3.53, the paragraph "The main obstruction" of Section 3.5.6, and the question of its paragraph "Open" whether ρ_{oct} can enter a coupling to the Dirac sector. Three expectations are tested on the way: (E1) that an odd permutation of the temporal axes reverses the mass form and the duality operator together; (E2) that the couplings of the Dirac field to the field,

which contain T, change sign with it under such a permutation; (E3) that such a coupling can be written through T and $J(\hat{n})$. In brief: (1) T is invariant and $J(\hat{n})$ covariant under every rotation of the temporal triple and under the diagonal rotations; an odd permutation of the temporal axes reverses T and $J(\hat{n})$ together, hence the mass form; paired with the same permutation of the spatial axes, the only form that respects G, it also reverses the duality operator and turns the octant map into its textbook-orientation form; reversing the clock is a half-turn, which fixes T, and the charge conjugation of Section 2.5 is a half-turn composed with T (Section 3.6.1); (2) T is at once the Dirac mass form, the Dirac parity and the lift of the frame parity, and the duality on the octants; $J(\hat{n})$ acts on the octants as a rotation of the transverse field about $\hat{n}$ and maps the longitudinal field to the two poles that carry the Gauss laws (Section 3.6.2); (3) the spin lift of $\mathfrak{l}_{\mathrm{int}}(\hat{n})$ and the octant representation ρ_{oct} are inequivalent on the same module, their joint commutant is $\mathbb{R}\mathbf{1}$, no covariant bilinear coupling exists, and the covariant trilinear couplings of a Grassmann Dirac field to the field octants form a two-dimensional space, the Pauli term and its dual, written through T and $J(\hat{n})$ (Section 3.6.3); (4) the two first-order operators use conjugate Pauli triples, related only by elements of mixed parity, and neither derives spatial differentiation from temporal directions (Section 3.6.4); (5) the field reading does not depend on the clock and the particle reading does, and T is, up to scale, the only element that is the clock times a Dirac complex structure for every clock (Section 3.6.5). Section 3.6.6 collects the verdicts.

Conventions and units. Those of Sections 3.4 and 3.5: $V = \mathbb{R}^{3,3}$, $\eta = \mathrm{diag}(1,1,1,-1,-1,-1)$, the realisation of eq. (1.1), $s_a := e_{a+3}$, $T := e_1e_2e_3$, $\Gamma := \omega = e_1 \cdots e_6$, the binding root $Ge_i = e_{i+3}$, $Ge_{i+3} = -e_i$, the diagonal rotations $L_i = \frac{1}{2}(e_{j+3}e_{k+3} - e_je_k)$ and boosts $K_i = \frac{1}{2}(e_ke_{j+3} - e_je_{k+3})$ of $\mathfrak{l}_G$ for cyclic (i,j,k), acting on V by the commutator, and the octant notation $|\mathrm{vac}\rangle$, $|\mathrm{full}\rangle$, f_k, $\mathtt{g}_m$, $\Phi_{\mathrm{oct}}(X) = \mathbf{E}\cdot f + \mathbf{B}\cdot g$ of Proposition 1.8, Lemma 3.43, and Theorem 3.44; for $v \in \mathbb{R}^3$ put $f(v) = \sum_k v_k f_k$ and $\mathtt{g}(v) = \sum_m v_m g_m$. The field point is $X = \sum_i (B_i e_i + E_i e_{i+3})$ with $\mathbf{B}$ minus the textbook magnetic field, $\mathbf{B}_{\mathrm{t}} = -\mathbf{B}$. A *clock* is a unit $\hat{n} \in V^+$ with an oriented orthonormal frame $(\hat{n}, u_1, u_2)$ of V^+, so $nu_1u_2 = T$ (Proposition 2.13); $\gamma^\mu_{\hat{n}} = (\hat{n}, s_1, s_2, s_3)$, $\mathfrak{l}_{\mathrm{int}}(\hat{n})$ is spanned by the bivectors of $E_{\hat{n}} = \mathbb{R}\hat{n} \oplus V^-$, $\Omega_{\hat{n}} := ns_1s_2s_3$, and $\phi_{\hat{n}} : \mathfrak{l}_G \to \mathfrak{l}_{\mathrm{int}}(\hat{n})$ is the dictionary of Proposition 3.33, $L_i \mapsto s_j \wedge s_k$, $K_i \mapsto s_i \wedge \hat{n}$. A bivector $a \wedge b$ of orthogonal vectors is the Clifford element $\frac{1}{2}ab$, and its spin lift is left multiplication by that element, so the Dirac reading's Lorentz representation of $\mathfrak{l}_G$ on S is

$$\rho_{\hat{n}} := \mathrm{spin} \circ \phi_{\hat{n}}, \qquad \rho_{\hat{n}}(L_i) = \tfrac{1}{2}s_js_k, \qquad \rho_{\hat{n}}(K_i) = \tfrac{1}{2}s_in,$$

and the field reading's is $\rho_{\mathrm{oct}}(L_i) = L_i$, $\rho_{\mathrm{oct}}(K_i) = Q_i = -TL_i$ (Proposition 3.49). Further $t_i := \frac{1}{2}e_je_k$ spans the rotations $\mathfrak{so}(3)_+$ of the temporal triple alone, and $A_i := \frac{1}{2}(e_je_k + e_{j+3}e_{k+3})$ rotates the two triples in opposite senses (Lemma 3.43); $L(\hat{n}) := \sum_i n_iL_i$, $A(\hat{n}) := \sum_i n_iA_i$. Dimensions: $\hat{n}$, $J(\hat{n})$, T, Γ, all Clifford elements and all intertwiners are dimensionless; $[x^a] = [x^0] = [ct] = \mathrm{m}$ and $[\partial_\mu] = \mathrm{m}^{-1}$; in SI units a Dirac field has $[\psi] = \mathrm{m}^{-3/2}$ and a Lagrangian density has the unit $\mathrm{J\,m}^{-3}$, as in (2.5); the field enters as $X_{\mathrm{EM}} = (c\mathbf{B}; \mathbf{E})$ with $[E_i] = [cB_i] = \mathrm{V\,m}^{-1}$; a magnetic moment μ has $[\mu] = \mathrm{J\,T}^{-1}$ and an electric dipole moment d has $[d] = \mathrm{C\,m}$. Computations use Heaviside–Lorentz units with $\hbar = c = 1$. Identifiers r4-$\hat{n}$ refer to `r4_bridge.py`, which checks every finite-dimensional and symbolic claim below in exact arithmetic (r4-1 re-verifies the conventions).

3.6.1 The identity and its covariance

Proposition 3.56 (The identity)**.** *For every $\hat{n} \in V^+$: $\hat{n}^2 = |\hat{n}|^2\mathbf{1}$, $J(\hat{n}) := T\hat{n} = \hat{n}T = \hat{n}_1e_2e_3 + \hat{n}_2e_3e_1 + \hat{n}_3e_1e_2$, $J(\hat{n})^2 = -|\hat{n}|^2\mathbf{1}$, and $\hat{n}J(\hat{n}) = J(\hat{n})\hat{n} = |\hat{n}|^2\,T$. Hence for every clock*

$$T = \hat{n}J(\hat{n}) = J(\hat{n})\hat{n}, \qquad J(\hat{n}) = T\hat{n} = \hat{n}T, \qquad \hat{n}^2 = \mathbf{1}, \qquad J(\hat{n})^2 = T^2 = -\mathbf{1}.$$

$J(\hat{n})$ commutes with $\hat{n}, s_1, s_2, s_3$, and $\hat{n} \mapsto J(\hat{n})$ is linear and injective.

Proof. By the blade rule (Lemma 1.1) T commutes with each e_i and anticommutes with each s_a. So $J(\hat{n})^2 = T\hat{n}T\hat{n} = T^2\hat{n}^2$ and $\hat{n}J(\hat{n}) = nnT = |\hat{n}|^2 T$; T and $\hat{n}$ each anticommute with s_a, so $J(\hat{n})$ commutes with s_a (r4-2, symbolic in $\hat{n}$ and on the rational parametrisation $\hat{n} = (2a, 2b, 1-a^2-b^2)/(1+a^2+b^2)$ of the clock sphere). □

Restated from Proposition 2.13 and from Remark 2.18 ($J = \gamma^0 T$), re-verified (r4-2).

Proposition 3.57 (Rotations)**.** *(i) For $Y \in \mathfrak{so}(3)_+ \oplus \mathfrak{so}(3)_\Delta$, $[Y, T] = 0$ and $[Y, J(\hat{n})] = J([Y, \hat{n}])$, with $[Y, \hat{n}] \in V^+$. Hence for the spin lift R of a rotation of the temporal triple alone, or of a diagonal rotation, $RTR^{-1} = T$ and $RJ(\hat{n})R^{-1} = J(R\hat{n})$, where $R\hat{n} := R\hat{n}R^{-1}$.*

(ii) The rotations $\mathfrak{so}(3)_+$ do not commute with G; the diagonal rotations do, and are symmetries of the field reading: $\rho_{\mathrm{oct}}(L_i) = L_i$ and Φ is equivariant (Theorem 3.44).

(iii) A rotor $q \in \mathrm{Spin}(V^+)$ with $qe_1q^{-1} = \hat{n}$ fixes T, the s_a and $e_4e_5e_6$, and it carries $J(e_1)$, Ω_{e_1}, $\mathfrak{l}_{\mathrm{int}}(e_1)$, ϕ_{e_1} to $J(\hat{n})$, $\Omega_{\hat{n}}$, $\mathfrak{l}_{\mathrm{int}}(\hat{n})$, $\phi_{\hat{n}}$.

Proof. T commutes with every same-sign bivector, hence with t_i and L_i, and $[Y, T\hat{n}] = T[Y, \hat{n}]$. The spatial part of L_i commutes with $\hat{n}$, so $[L_i, \hat{n}] = -[t_i, \hat{n}] \in V^+$. A temporal bivector commutes with every s_a and with $e_4e_5e_6$; with $\phi_{\hat{n}}(L_i) = s_j \wedge s_k$, $\phi_{\hat{n}}(K_i) = s_i \wedge \hat{n}$ this gives (iii). (ii) is Lemma 3.43. The finite statements are checked with the rational rotors $q = u'(e_1 - \hat{n})$, $u' \perp \hat{n}$, at $\hat{n} = (2,3,6)/7, (1,2,2)/3, (-1,4,8)/9$ and with an exact Cayley transform of L_3 (r4-3, r4-4). □

Fixed. Rotations of the temporal triple move the clock and with it the whole Dirac structure, but they are not symmetries of the field reading; the diagonal rotations move the clock as well and are symmetries of both.

Proposition 3.58 (Permutations of the temporal axes and the reversal of the clock)**.** *For $\sigma \in S_3$ let A_σ be the automorphism of $\mathrm{Cl}(3,3)$ with $e_i \mapsto e_{\sigma(i)}$, $s_a \mapsto s_a$, and A'_σ the one with $e_i \mapsto e_{\sigma(i)}$, $s_a \mapsto s_{\sigma(a)}$; write $\sigma\hat{n} = \sum_i n_i e_{\sigma(i)}$.*

(i) Both are inner, $A = \mathrm{Ad}_U$ with U unique up to scale and orthogonal up to scale, and

$$A(T) = \mathrm{sgn}(\sigma)\, T, \qquad A(J(\hat{n})) = \mathrm{sgn}(\sigma)\, J(\sigma\hat{n}), \qquad A(\hat{n}) = \sigma\hat{n}.$$

(ii) A'_σ commutes with G, permutes the pole operators, $A'(\omega_i) = \omega_{\sigma(i)}$, and with $U|\mathrm{vac}\rangle = |\mathrm{vac}\rangle$ it acts on the octants by $U\Phi_{\mathrm{oct}}(X) = \Phi_{\mathrm{oct}}(\tilde{X})$, $\tilde{X} = (\mathrm{sgn}(\sigma)P_\sigma\mathbf{B}; P_\sigma\mathbf{E})$, where P_σ permutes the components: $\mathbf{B}$ is transformed as an axial vector. The transported octant map $\Phi' := U\Phi_{\mathrm{oct}}(P_\sigma \oplus P_\sigma)^{-1}$ is Φ for even σ and $\Phi \circ (-\eta) = \Phi_{\mathrm{t}} = -(-1)^N\Phi$, the textbook-orientation map of Theorem 3.53(iv), for odd σ, and its duality operator is $A'(-T)$: $A'(-T)\Phi' = \Phi' G$.

(iii) No nontrivial A_σ commutes with G.

(iv) Let $R = nu_2$, the rotor of the half-turn about u_1. Then $R\hat{n}R^{-1} = -\hat{n}$, $RTR^{-1} = T$, $RJ(\hat{n})R^{-1} = J(-\hat{n}) = -J(\hat{n})$, and $RT = u_1$: every unit vector C of the surplus plane, the charge conjugation of Theorem 2.25 at the clock $\hat{n}$, is the half-turn about C, which reverses the clock, composed with T.

Proof. (i) $\mathrm{Cl}(3,3) \cong M_8(\mathbb{R})$ is central simple (Skolem–Noether); $e_{\sigma(1)}e_{\sigma(2)}e_{\sigma(3)} = \mathrm{sgn}(\sigma)T$ by reordering anticommuting factors, and $A(J(\hat{n})) = A(T)A(\hat{n})$. (ii) $P_\sigma \oplus P_\sigma$ commutes with the block matrix of G. $A'(a_k^+) = a_{\sigma(k)}^+$, so $Uf_k = f_{\sigma(k)}$ and $Ug_m = \frac{1}{2}\epsilon_{mjk}a_{\sigma(j)}^+ a_{\sigma(k)}^+|\mathrm{vac}\rangle = \mathrm{sgn}(\sigma)g_{\sigma(m)}$, since $\epsilon_{\sigma(m)\sigma(j)\sigma(k)} = \mathrm{sgn}(\sigma)\epsilon_{mjk}$. Hence $\Phi'(Y) = \Phi_{\mathrm{oct}}(\mathrm{sgn}(\sigma)\mathbf{B}_Y; \mathbf{E}_Y)$, and $\Phi \circ (-\eta) = -(-1)^N\Phi$ because $(-1)^N$ is -1 on the f's and $+1$ on the g's; the duality relation is

operation	T; $J(\hat{n})$	Dirac reading: rest flow $e^{-m\lambda T}$, senses	field reading: $-T$, G, Φ
rotation R of V^+ alone	T; $J(R\hat{n})$	flow invariant; eigenspaces $\hat{n}=\pm1$ go to $R\hat{n}=\pm1$, senses kept	not a symmetry: $[\mathfrak{so}(3)_+, G]\neq 0$
diagonal rotation	T; $J(R\hat{n})$	as above, with V^- rotated alike	symmetry: $\rho_{\mathrm{oct}}(L)=L$, $-T$, G, Φ preserved
clock reversal (half-turn)	T; $-J(\hat{n})$	flow invariant; the eigenspaces $\hat{n}=\pm1$ are exchanged, so the particle and antiparticle labels relative to $\hat{n}$ are exchanged	the octant structure does not involve $\hat{n}$; the half-turn itself is not a field symmetry
odd σ, temporal alone	$-T$; $-J(\sigma\hat{n})$	$e^{-m\lambda T}\mapsto e^{+m\lambda T}$: sense reversed on each eigenspace; $(m,q)\mapsto(-m,-q)$	not a symmetry: breaks G
odd σ, paired	$-T$; $-J(\sigma\hat{n})$	as for σ alone, spatial axes permuted	G kept, octants permuted, $\Phi\mapsto\Phi_{\mathrm{t}}$, duality $-T\mapsto+T$
$C=RT$, $C\in U_{\hat{n}}$	T; $-J(\hat{n})$	same flow; particle $\leftrightarrow$ antiparticle (Proposition 7.3)	not a field symmetry (C is a vector)

Table 3.4: What the operations of Propositions 3.57 and 3.58 do to T, $J(\hat{n})$ and the two readings (r4-3 to r4-8).

the transport of $-T\Phi=\Phi G$. (iii) A temporal permutation alone moves e_i and fixes its partner e_{i+3}. (iv) R is an even element of $\mathrm{Cl}(V^+)$, so it commutes with T; $nu_2nu_1u_2=-u_2u_1u_2=u_1$. All of this is checked for the six permutations, alone and paired, and for three clocks (r4-5, r4-6, r4-7). □

Fixed. Expectation (E1) is confirmed in the form (i): an odd permutation reverses T and $J(\hat{n})$ together, and the paired one, which is the only one that respects G and the octants, reverses the mass form T and the duality operator $-T$ together. The reason is (ii): an odd permutation of the axes is an improper rotation, under which $\mathbf{B}$ is axial; on the octants it changes the orientation convention $\mathbf{B}_{\mathrm{t}}=-\mathbf{B}$ of Section I.1.4 into the textbook one, for which the duality operator is $+T$ (Theorem 3.53(iv)). Reversing $\hat{n}$ reverses $J(\hat{n})$, and (iv) sharpens this: the reversal is a rotation in $\mathrm{Spin}(V^+)$, which leaves T, the product of the two reversed objects, fixed.

Corollary 3.59 (The two senses at every clock)**.** *On the eigenspaces $\hat{n}=\pm1$ of any clock, $T=\pm J(\hat{n})$, so the rest flow $e^{-m\lambda T}$ of Proposition 4.46 is $e^{\mp m\lambda J(\hat{n})}$, the two senses of the charge rotation that are particle and antiparticle there (Proposition 7.3). An odd temporal permutation carries the Dirac data $(K^\mu_{\hat{n}}, T, J(\hat{n}))$, with $K^\mu_{\hat{n}}=-\hat{n}\gamma^\mu_{\hat{n}}$, to $(K^\mu_{\sigma\hat{n}}, -T, -J(\sigma\hat{n}))$: the transported Lagrangian (2.5) has mass $-m$ and charge $-q$. The axial rotation by $\Omega_{\sigma\hat{n}}$, which preserves every K^μ and J and reverses T (Proposition 7.2(iv)), restores the mass, leaving $q\mapsto-q$, the effect of C. This is a classical change of variables; in a quantum theory the axial rotation must also account for the measure, that is for the axial anomaly and possible topological terms.*

Proof. With $\Pi_\pm=\frac{1}{2}(\mathbf{1}\pm\hat{n})$, $T\Pi_\pm=\frac{1}{2}(T\pm J(\hat{n}))=\pm J(\hat{n})\Pi_\pm$ because $J(\hat{n})\hat{n}=T$. The rest follows from Proposition 3.58(i) and from $\Omega^\top T\Omega=-T$, $\Omega^\top K^\mu\Omega=K^\mu$, $[\Omega,J]=0$ (r4-8). □

Proposition 4.46 is restated at every clock (fixed); the effect of the permutations on the Dirac reading is fixed; that a Dirac mass sign is not observable alone is inherited [206]. In the decomposition $C=RT$ of Proposition 3.58(iv), T preserves the flow, the eigenspaces of $\hat{n}$ and $J(\hat{n})$ (it is a parity, Theorem 3.60), so the reversal of the sense by C in Proposition 7.3 comes entirely from the half-turn that reverses the clock (fixed; the reading of particle and antiparticle as the two clock orientations is interpretive).

3.6.2 The three roles of T, and $J(\hat{n})$ on the octants

Theorem 3.60 (The three roles of T)**.** *(a)* Mass form. $\gamma^0_{\hat{n}} J(\hat{n}) = T$ *for every clock: the mass kernel of the Dirac reading is the same at every clock, while the kinetic kernels* $K^\mu_{\hat{n}} = TJ(\hat{n})\gamma^\mu_{\hat{n}} = -\hat{n}\gamma^\mu_{\hat{n}}$ *move with it.*

(b) Parity. $\mathrm{Ad}_T|_V = \eta$, *the frame parity fixing* $\mathbf{B}$ *and reversing* $\mathbf{E}$, *and* T *is up to scale the only element implementing it. At every clock* T *commutes with* $J(\hat{n})$ *and* $TH_{\hat{n}}(\mathbf{p})T^{-1} = H_{\hat{n}}(-\mathbf{p})$ *for* $H_{\hat{n}}(\mathbf{p}) = \sum_a n s_a p_a + mn$*:* $T = \gamma^0 J$ *is the admissible Dirac parity with* $P^2 = -\mathbf{1}$ *of Theorem 2.27, at every clock.*

(c) Duality. $-T\Phi = \Phi G$, $Tf_k = \mathrm{g}_k$, $Tg_k = -f_k$, $T|\mathrm{vac}\rangle = |\mathrm{full}\rangle$*: on the field octants* $-T$ *is the Hodge star.*

The complex structure $J(\hat{n})$ *is, for the clock* $\hat{n}$*, the selected imaginary unit and the charge generator of the Dirac reading (Theorem 2.17 at every clock, Theorem 2.14); in (2.5) it enters as* $D_\mu = \partial_\mu + qA_\mu J(\hat{n})$.

Proof. (a) is Proposition 3.56 with $\gamma^0_{\hat{n}} = \hat{n}$, and $TJ(\hat{n}) = -\hat{n}$. (b) T commutes with $\hat{n}$ and anticommutes with s_a, so it commutes with $J(\hat{n})$ and reverses the momentum term of $H_{\hat{n}}$; uniqueness: the solutions of $Ye_a = \eta_{aa} e_a Y$ form a line (Skolem–Noether). (c) is Proposition 3.45(i) (r4-9). □

Restated and fixed: (a) is Remark 2.18 and Proposition 2.21 and Theorem 2.14(iii) at every clock, (b) is Proposition 7.5 and Proposition 3.45 together with Theorem 2.27 at every clock, (c) is Proposition 3.49(i); proved together here (r4-9). The three are compatible: (b) and (a) say that the mass is parity-even, (b) and (c) that the parity lift acts on the field octants as minus the Hodge star, which is Proposition 3.45. Role (a) belongs to the particle reading and uses T as the product $\hat{n}J(\hat{n})$; (c) belongs to the field reading and uses T alone; (b) belongs to both, since $\mathrm{Ad}_T|_V = \eta$ is the parity of the frame $(\mathbf{B};\mathbf{E})$ and of the three lengths, and T is the Dirac parity.

Proposition 3.61 ($J(\hat{n})$ relative to the field structure)**.** *With* $L(\hat{n})$ *and* $A(\hat{n})$ *as in the conventions:*

(i) $J(\hat{n}) = -L(\hat{n}) + A(\hat{n})$, *with* $L(\hat{n})A(\hat{n}) = A(\hat{n})L(\hat{n}) = 0$*; for a clock,* $-L(\hat{n})^2$ *and* $-A(\hat{n})^2$ *are complementary projectors of rank four,* $L(\hat{n})^3 = -L(\hat{n})$ *and* $A(\hat{n})^3 = -A(\hat{n})$.

(ii) $L(\hat{n}) = \rho_{\hat{n}}(L(\hat{n})) - \frac{1}{2}J(\hat{n})$, *in which* $\rho_{\hat{n}}(L(\hat{n})) = \frac{1}{2}\sum_i n_i s_j s_k$ *is the Dirac spin rotation of* V^- *about the spatial axis* $\sum_i n_i s_i$ *paired with* $\hat{n}$ *(it acts trivially on* V^+*); and* $\rho_{\mathrm{oct}}(L_i) = \rho_{\hat{n}}(L_i) - \frac{1}{2}J(e_i)$ *for every* i *and every clock. Hence* $e^{2\pi L(\hat{n})} = e^{2\pi\rho_{\hat{n}}(L(\hat{n}))}e^{-\pi J(\hat{n})} = (-\mathbf{1})(-\mathbf{1}) = \mathbf{1}$.

(iii) For every $\hat{n}$,

$$J(\hat{n})\Phi_{\mathrm{oct}}(X) = -\Phi_{\mathrm{oct}}(\hat{n}\times\mathbf{B};\,\hat{n}\times\mathbf{E}) + (\mathbf{E}\cdot\hat{n})\,|\mathrm{full}\rangle - (\mathbf{B}\cdot\hat{n})\,|\mathrm{vac}\rangle,$$

and $J(\hat{n})|\mathrm{vac}\rangle = g(\hat{n})$, $J(\hat{n})|\mathrm{full}\rangle = -f(\hat{n})$. *The kernel of* $L(\hat{n})$ *is spanned by the two poles and the longitudinal octants* $f(\hat{n})$, $g(\hat{n})$*; on it* $J(\hat{n}) = A(\hat{n})$. *On the transverse octants* $f(u), g(u)$, $u \perp \hat{n}$, $J(\hat{n}) = -L(\hat{n})$, *the rotation about* $\hat{n}$*:* $L(\hat{n})f(u) = f(\hat{n}\times u)$, $L(\hat{n})g(u) = g(\hat{n}\times u)$.

(iv) $J(\hat{n})$ *does not commute with* ρ_{oct} *and does not preserve the field octants* $\Phi_{\mathrm{oct}}(V)$.

Proof. (i) With $t(\hat{n}) = \sum_i n_i e_j e_k = J(\hat{n})$ and $s(\hat{n}) = \sum_i n_i s_j s_k$: $L(\hat{n}) = \frac{1}{2}(s(\hat{n}) - t(\hat{n}))$, $A(\hat{n}) = \frac{1}{2}(s(\hat{n}) + t(\hat{n}))$, $s(\hat{n})$ and $t(\hat{n})$ commute and $s(\hat{n})^2 = t(\hat{n})^2 = -|\hat{n}|^2\mathbf{1}$, so $L(\hat{n})A(\hat{n}) = \frac{1}{4}(s^2 - t^2) = 0$ and $-L(\hat{n})^2 = \frac{1}{2}(\mathbf{1} + st)$, $-A(\hat{n})^2 = \frac{1}{2}(\mathbf{1} - st)$ with $(st)^2 = \mathbf{1}$. (ii) $\rho_{\hat{n}}(L(\hat{n})) = \frac{1}{2}s(\hat{n})$, and $L_i = \frac{1}{2}s_j s_k - \frac{1}{2}e_j e_k$ with $e_j e_k = J(e_i)$; $\frac{1}{2}s(\hat{n})$ and $\frac{1}{2}J(\hat{n})$ commute and square to $-\frac{1}{4}\mathbf{1}$. (iii)

$J(\hat{n}) = \sum_i n_i e_j e_k$ and $e_I|\mathrm{vac}\rangle = a_I^+|\mathrm{vac}\rangle$ (Proposition 1.11) give $J(\hat{n})|\mathrm{vac}\rangle = g(\hat{n})$; $L(\hat{n})$ acts on the f's and g's as the vector rotation (Lemma 3.43) and kills the poles; $A(\hat{n})f(u) = (\hat{n}\cdot u)|\mathrm{full}\rangle$ and $A(\hat{n})g(u) = -(\hat{n}\cdot u)|\mathrm{vac}\rangle$. (iv) follows from (iii), since ρ_{oct} preserves $\Phi_{\mathrm{oct}}(V)$ (r4-10, r4-11, symbolic in $\hat{n}$, $\mathbf{E}$, $\mathbf{B}$, u). □

Fixed. What this means is the following, and the reading is interpretive. The Dirac charge rotation $e^{\theta J(\hat{n})}$, read on the octant module, is not a transformation of the field: on transverse fields it is the rotation by $-\theta$ about the axis paired with the clock, and on the four-dimensional complement it exchanges the longitudinal components with the pole components, $\mathbf{E}\cdot\hat{n}$ with the slot of $\nabla\cdot\mathbf{B}$ and $\mathbf{B}\cdot\hat{n}$ with the slot of $\nabla\cdot\mathbf{E}$ (Theorem 3.53). So the complex structure of the particle reading mixes the field with the Gauss constraints, and it is not in the commutant of the field reading's Lorentz representation. Item (ii) shows where integer and half-integer spin part on S: the field rotation $L(\hat{n})$, which turns $\mathbf{B}$ about $\hat{n}$ and $\mathbf{E}$ about $\sum_i n_i s_i$, is the Dirac rotation about $\sum_i n_i s_i$ combined with half a turn of the charge, and the two signs -1 cancel. $J(\hat{n})$ must be kept apart from $G_{\mathrm{EM}} = \star$: on the octants $\star$ is $-T$ (Proposition 3.49), $J(\hat{n}) = \hat{n}T$ differs from it by the clock, and $J(\hat{n})$ commutes with $\mathfrak{l}_{\mathrm{int}}(\hat{n})$ while G commutes with $\mathfrak{l}_G$.

3.6.3 Two Lorentz representations on one module, and their couplings

Proposition 3.62 (Commutants)**.** *(i) The commutant of $\rho_{\hat{n}}(\mathfrak{l}_G)$, the spin lift of $\mathfrak{l}_{\mathrm{int}}(\hat{n})$, in $\mathrm{End}_{\mathbb{R}}(S)$ is $\mathrm{span}\{\mathbf{1}, u_1, u_2, J(\hat{n})\} \otimes \mathrm{span}\{\mathbf{1}, \Omega_{\hat{n}}\} \cong M_2(\mathbb{C})$, of dimension 8.*

(ii) The commutant of $\rho_{\mathrm{oct}}(\mathfrak{l}_G)$ is $\mathrm{span}\{\Pi_F, T\Pi_F\} \oplus \mathrm{End}(\mathrm{span}\{|\mathrm{vac}\rangle, |\mathrm{full}\rangle\})$, of dimension 6, with Π_F the projector onto the field octants.

(iii) The joint commutant of $\rho_{\hat{n}}(\mathfrak{l}_G)$ and $\rho_{\mathrm{oct}}(\mathfrak{l}_G)$ is $\mathbb{R}\mathbf{1}$, for every clock.

Proof. (i) u_a anticommutes with each vector of $E_{\hat{n}}$ and so commutes with its bivectors; $\Omega_{\hat{n}}$ commutes with the even elements of $\mathrm{Cl}(E_{\hat{n}})$; the eight products are independent, and the commutant has dimension 8 because S complexifies to $2(\frac{1}{2},0)\oplus 2(0,\frac{1}{2})$, whose commutant is $M_2(\mathbb{C})\oplus M_2(\mathbb{C})$ [57]. (ii) $\Phi_{\mathrm{oct}}(V)$ is irreducible of complex type with commutant $\mathrm{span}\{\mathbf{1}, T\}$ (Proposition 3.49(iii)), the poles are two trivial summands, and there is no map between them. (iii) Exact computation at $\hat{n} = e_1, (2,3,6)/7, (1,2,2)/3$ (r4-12); structurally, $\rho_{\mathrm{oct}}(L_i) - \rho_{\hat{n}}(L_i) = -\frac{1}{2}J(e_i)$ spans $\mathfrak{so}(3)_+$, which with $\mathfrak{l}_{\mathrm{int}}(\hat{n})$ generates $\mathfrak{so}(3,3)$, whose commutant $\mathrm{span}\{\mathbf{1}, \Gamma\}$ meets that of the odd Q_i in $\mathbb{R}\mathbf{1}$. □

Fixed; (i) contains the commutant $\mathbb{H}$ of the four $\gamma^\mu_{\hat{n}}$ of Theorem 2.14(ii). No complex structure, and no operator other than a scalar, commutes with both Lorentz actions.

Proposition 3.63 (Three inequivalent representations on S)**.** *On the same module S the Lie algebra $\mathfrak{l}_G \cong \mathfrak{so}(1,3)$ acts in three ways: through $\rho_{\hat{n}}$ (complexification $2(\frac{1}{2},0)\oplus 2(0,\frac{1}{2})$, two Dirac spinors), through ρ_{oct} ($(1,0)\oplus(0,1)\oplus 2(0,0)$, the field strength and two singlets) and through the spin lift of $\mathfrak{l}_G$ itself ($2(\frac{1}{2},\frac{1}{2})$). The Casimirs $\sum_k(\rho(K_k)^2 - \rho(L_k)^2)$ and $(\sum_k \rho(K_k)\rho(L_k))^2$ are $\frac{3}{2}$ and $-\frac{9}{16}$ for $\rho_{\hat{n}}$; 4 and -4 on the field octants and 0 on the poles for ρ_{oct}; 3 and 0 for the spin lift of $\mathfrak{l}_G$. In $\rho_{\hat{n}}$, $\rho_{\hat{n}}(L_3)^2 = -\frac{1}{4}\mathbf{1}$ and $e^{2\pi\rho_{\hat{n}}(L_3)} = -\mathbf{1}$; in the other two $\rho(L_3)^3 = -\rho(L_3)$ and $e^{2\pi\rho(L_3)} = \mathbf{1}$. The three are pairwise inequivalent, and no invertible U satisfies $U\rho_{\hat{n}}(Y)U^{-1} = \rho_{\mathrm{oct}}(Y)$ for all Y.*

Proof. Equivalent representations have conjugate Casimirs; the values are computed exactly (r4-13). The decompositions are Proposition 3.83 and Proposition 3.49. □

Restated (the decompositions) and fixed (the Casimirs and the inequivalence). That ρ_{oct} does not commute with the spin lift of $\mathfrak{l}_{\mathrm{int}}(e_1)$ (r4-13) is the weakest form of this; the proposition says that the two are not even conjugate. They therefore cannot both be the Lorentz action of one field on S. A theory containing a Dirac field and the field octants must treat them as

two fields, ψ in a copy of S with $\rho_{\hat n}$ and $\Phi_{\rm oct}(X)$ in another with $\rho_{\rm oct}$, and the joint action of $Y \in \mathfrak{l}_{\rm int}(\hat n)$ is its spin lift $\rho_{\hat n}(\phi_{\hat n}^{-1}Y)$ on ψ and $\rho_{\rm oct}(\phi_{\hat n}^{-1}Y)$ on the octants; equivalently $Y' \in \mathfrak{l}_G$ acts by $\rho_{\hat n}(Y')$ and by $\rho_{\rm oct}(Y')$. Since Φ intertwines $\mathfrak{l}_G$ on V with $\rho_{\rm oct}$ (Proposition 3.49(iii)), a coupling to $\Phi_{\rm oct}(X)$ is a coupling to X with its own action $Y \cdot X = [Y, X]$, and conversely.

For $X \in V$ let $\ell(X) := \sum_i (B_i L_i + E_i K_i) \in \mathfrak{l}_G$, an equivariant isomorphism $V \to \mathfrak{l}_G$ by the brackets and eq. (3.1) of Theorem 3.3, and

$$S_{\hat n}(X) := \rho_{\hat n}(\ell(X)) = \tfrac{1}{2}\sum_i (B_i\, s_j s_k + E_i\, s_i n),$$

the field strength as a bivector of $E_{\hat n}$; under the dictionary of Proposition 3.33 it is the Clifford image of the bivector $\sum_{\mu<\nu}(F_X)_{\mu\nu}\,\gamma_\mu \wedge \gamma_\nu = \sum_i (E_i\, s_i \wedge \hat n + B_i\, s_j \wedge s_k)$, $\gamma_\mu := \eta_{\mu\nu}\gamma^\nu_{\hat n}$, while the same sum with $F_X^{\mu\nu}$ in place of $(F_X)_{\mu\nu}$ is $S_{\hat n}(\eta X)$ (r4-16).

Theorem 3.64 (Covariant couplings of the Dirac field to the field octants: the derivative-free, field-strength-linear class)**.** *Let $Y \in \mathfrak{l}_G$ act by $\rho_{\hat n}(Y)$ on ψ and by $\rho_{\rm oct}(Y)$ on $\Phi_{\rm oct}(X)$, that is, by $Y \cdot X = [Y, X]$ on X. The class classified is that of couplings linear in the field strength X, bilinear in ψ, without derivatives and invariant under $\mathfrak{l}_G$; minimal coupling to a potential A_μ through a covariant derivative, couplings with derivatives and couplings nonlinear in X lie outside it.*

(i) No nonzero bilinear form $\psi^\top M \Phi_{\rm oct}(X)$ is invariant, nor one on all of S with $\rho_{\rm oct}$.

(ii) The invariant trilinear couplings $\psi^\top M(X)\psi$, $M : V \to \mathrm{End}(S)$ linear with $\rho_{\hat n}(Y)^\top M(X) + M(X)\rho_{\hat n}(Y) + M(Y \cdot X) = 0$, are exactly $M(X) = \hat n C\, S_{\hat n}(X)$ with C in the commutant of Proposition 3.62(i): an eight-dimensional space. For a real Grassmann ψ only antisymmetric kernels contribute, and these form the two-dimensional space spanned by

$$M_{\hat n}(X) = T\, J(\hat n)\, S_{\hat n}(X) = -\hat n\, S_{\hat n}(X), \qquad M^\star_{\hat n}(X) = T\,\Gamma\, S_{\hat n}(X) = M_{\hat n}(GX),$$

and $T\Gamma S_{\hat n}(X) = TJ(\hat n)\Omega_{\hat n} S_{\hat n}(X)$. For a commuting ψ the symmetric kernels form the six-dimensional space given by $C \in \mathrm{span}\{u_1, u_2, J(\hat n)\} \otimes \mathrm{span}\{\mathbf{1}, \Omega_{\hat n}\}$. These dimensions hold at every clock.

(iii) Explicitly $M_{\hat n}(X) = \frac{1}{2}\sum_a E_a s_a - \frac{1}{2}\hat n \sum_i B_i s_j s_k$ and $M^\star_{\hat n}(X) = \frac{1}{2}\sum_a B_a s_a + \frac{1}{2}\hat n \sum_i E_i s_j s_k$. Both commute with $J(\hat n)$ (they carry no charge); under the parity $\psi \mapsto \hat n\psi$ (or $\psi \mapsto T\psi$), $X \mapsto \eta X$, $M_{\hat n}$ is even, like the mass form T, and $M^\star_{\hat n}$ is odd.

Proof. (i) The rotation $e^{2\pi\rho_{\hat n}(L_3)}$ is $-\mathbf{1}$ on ψ and $e^{2\pi\rho_{\rm oct}(L_3)} = \mathbf{1}$ on the octants (Proposition 3.63), so an invariant form satisfies $\beta(-\psi, \chi) = \beta(\psi, \chi)$; the exact solution spaces are zero at $\hat n = e_1, (2,3,6)/7$ (r4-14). (ii) For every Y, $\rho_{\hat n}(Y)^\top = -\hat n \rho_{\hat n}(Y) \hat n$: the rotations $\frac{1}{2}s_j s_k$ are antisymmetric and commute with $\hat n$, the boosts $\frac{1}{2}s_i n$ are symmetric and anticommute with it. Hence, for C commuting with $\rho_{\hat n}$,

$$\rho_{\hat n}(Y)^\top \hat n C S_{\hat n}(X) + \hat n C S_{\hat n}(X)\rho_{\hat n}(Y) = \hat n C\big(S_{\hat n}(X)\rho_{\hat n}(Y) - \rho_{\hat n}(Y) S_{\hat n}(X)\big) = -\hat n C\, S_{\hat n}(Y \cdot X),$$

using $[\rho_{\hat n}(Y), S_{\hat n}(X)] = \rho_{\hat n}([Y, \ell(X)]) = \rho_{\hat n}(\ell(Y \cdot X))$. The map $C \mapsto \hat n C S_{\hat n}$ is injective, and the space of invariant maps $V \to S^* \otimes S^*$ has dimension 8: complexified, $V = (1,0) \oplus (0,1)$, $\Lambda^2 S = (1,0) \oplus (0,1) \oplus 6(0,0) \oplus 4(\frac{1}{2}, \frac{1}{2})$ and $\mathrm{Sym}^2 S = 3(1,0) \oplus 3(0,1) \oplus 2(0,0) \oplus 4(\frac{1}{2}, \frac{1}{2})$, so the dimensions are 2 and 6 [57]. The symmetry of $\hat n C S$ follows from the transpose rule and $C^\top$: $(\hat n C S)^\top = -\hat n S \hat n C^\top \hat n$, which is $-\hat n C S$ for $C = \mathbf{1}, \Omega_{\hat n}$ and $+\hat n C S$ for $C = u_a, J(\hat n)$ and their products with $\Omega_{\hat n}$. Finally $TJ(\hat n) = -\hat n$, $TJ(\hat n)\Omega_{\hat n} = T\Gamma$, $T\Gamma = -e_4 e_5 e_6$, and $M_{\hat n}(GX) = M^\star_{\hat n}(X)$ by (iii). The exact row reduction of the 2304×168 and 2304×216 linear systems at $\hat n = e_1$ and $\hat n = (2,3,6)/7$ gives 2 and 6, and the displayed kernels span them (r4-15, r4-16). By Proposition 3.65(i) the dimensions are the same at every clock. (iii) $\hat n S_{\hat n}(X) \hat n$ reverses the electric part; $\hat n\, e_4 e_5 e_6\, \hat n = -e_4 e_5 e_6$ (r4-17). □

Fixed: the dimensions, the explicit kernels and their properties. The textbook comparison is inherited. In complex Dirac theory the Lorentz- and gauge-invariant couplings of ψ to the field strength that are bilinear in ψ and contain no derivative are the Pauli term $\bar\psi\sigma^{\mu\nu}\psi F_{\mu\nu}$, the coupling of an anomalous magnetic moment introduced by Pauli [207], and the electric-dipole term $i\bar\psi\sigma^{\mu\nu}\gamma_5\psi F_{\mu\nu}$, odd under parity and time reversal [208, 206]. With $J(\hat n)$ in the place of i, $\bar\psi\cdots\psi \leftrightarrow \frac12\psi^\top T\cdots\psi$ as in (2.5), and the real form $\Omega_{\hat n}$ of $i\gamma_5$ (Proposition 2.22), $\sigma^{\mu\nu}F_{\mu\nu}$ becomes a multiple of $J(\hat n)S_{\hat n}(X)$: $M_{\hat n}$ is the real form of the Pauli term and $M^\star_{\hat n}$ that of the electric-dipole term, up to normalisation and the sign convention for γ^a; their parities in (iii) agree with the textbook ones. The Lagrangian density

$$\mathcal{L}_{\mathrm P}=\tfrac12\,\psi^\top\Big(\tfrac{\mu}{c}\,M_{\hat n}(X_{\mathrm{EM}})+d\,M^\star_{\hat n}(X_{\mathrm{EM}})\Big)\psi,\qquad X_{\mathrm{EM}}=(c\mathbf B;\mathbf E),\tag{3.14}$$

has dimension $\mathrm{J\,m^{-3}}$ in SI units for $[\mu]=\mathrm{J\,T^{-1}}$ and $[d]=\mathrm{C\,m}$, and its two couplings have mass dimension -1 in natural units: they are not renormalisable [208]. The arena fixes neither constant (open). The kernel $M_{\hat n}$ is written through T and $J(\hat n)$, as (E3) proposed: T is the adjoint form, $J(\hat n)$ the imaginary unit of $\sigma^{\mu\nu}$, and $S_{\hat n}(X)$ the field bivector; the dual uses the other $\mathrm{Spin}(3,3)$-invariant form $T\Gamma$ (Proposition 3.51(iii)) and no $J(\hat n)$. With these couplings the Dirac sector reads, at every clock,

$$\mathcal{L}_D+\mathcal{L}_{\mathrm P}=\tfrac12\,\psi^\top T\Big[J(\hat n)\Big(\gamma^\mu_{\hat n}D_\mu+\tfrac{\mu}{c}S_{\hat n}(X_{\mathrm{EM}})+d\,\Omega_{\hat n}S_{\hat n}(X_{\mathrm{EM}})\Big)-m\Big]\psi,$$

in the units of (2.5): T is the common left factor of the kinetic, mass and Pauli kernels.

Proposition 3.65 (The couplings when the clock moves or the axes are permuted)**.** *The couplings of Theorem 3.64 transform as follows.*

(i) Let $q\in\mathrm{Spin}(V^+)$ be a rotor with $qe_1q^{-1}=\hat n$. Then, with X fixed,

$$M_{\hat n}(X)=qM_{e_1}(X)q^{-1},\qquad M^\star_{\hat n}(X)=qM^\star_{e_1}(X)q^{-1},\qquad qTq^{-1}=T,\qquad qT\Gamma q^{-1}=T\Gamma:$$

under $\psi\mapsto q\psi$ the couplings at e_1 become those at $\hat n$. The factors $J(\hat n)$, $\Omega_{\hat n}$, $\gamma^\mu_{\hat n}$, $S_{\hat n}$ are conjugated, and T, the clock-independent part of every kernel in the display above, is fixed. The couplings themselves depend on the clock: $M_{e_1}\neq M_{e_2}$.

(ii) For a diagonal rotation R, $M_{R\hat n}(RX)=RM_{\hat n}(X)R^{-1}$ and likewise for $M^\star$: clock, spinor and field turn together.

(iii) For an odd temporal permutation alone, $A_\sigma(M_{\hat n}(X))=M_{\sigma\hat n}(X)$ and $A_\sigma(M^\star_{\hat n}(X))=M^\star_{\sigma\hat n}(X)$, while $A_\sigma(T)=-T$. For the paired permutation, $A'_\sigma(M_{\hat n}(X))=M_{\sigma\hat n}(\tilde X)$ and $A'_\sigma(M^\star_{\hat n}(X))=\mathrm{sgn}(\sigma)M^\star_{\sigma\hat n}(\tilde X)$, with $\tilde X$ of Proposition 3.58(ii).

Proof. (i) q is orthogonal and conjugates ρ_{e_1} to $\rho_{\hat n}$ (Proposition 3.57(iii)); $S_{\hat n}$, $J(\hat n)$, $\Omega_{\hat n}$ follow, and T, Γ are fixed. (ii) The same with the spin lift of a diagonal rotation, which also rotates s_a and X. (iii) $A(TJ(\hat n))=TJ(\sigma\hat n)$ because the two signs cancel; A fixes s_a and maps $\hat n$ to $\sigma\hat n$, so $A(S_{\hat n}(X))=S_{\sigma\hat n}(X)$; $A(\Gamma)=\mathrm{sgn}(\sigma)\Gamma$ for A_σ and Γ for A'_σ, and $A'(S_{\hat n}(X))=S_{\sigma\hat n}(\tilde X)$ because $s_{\sigma(j)}s_{\sigma(k)}$ is $\mathrm{sgn}(\sigma)$ times the cyclically ordered pair (r4-18, r4-19). □

Fixed. Expectation (E2), that the couplings change sign with T under an odd permutation, is refuted for the Pauli term under both kinds of permutation and for the dual term under the temporal permutation alone; it holds only for the dual term under the paired permutation: $M_{\hat n}=TJ(\hat n)S_{\hat n}$ contains the product $TJ(\hat n)=-\hat n$, in which the two reversals cancel. What changes sign is the Pauli term *relative to the mass term*, since T reverses and $M_{\hat n}$ does not; after the axial rotation that restores the mass (Corollary 3.59), $M_{\hat n}$ is reversed. Physically the paired odd permutation is an improper rotation, and the electric-dipole coupling is the parity-odd one (Theorem 3.64(iii)).

3.6.4 The two first-order operators and the spatial derivatives

The Dirac operator of the clock $\hat{n}$ is $\gamma^\mu_{\hat{n}}\partial_\mu = \hat{n}\partial_0 + \sum_a s_a\partial_a$, and $\gamma^0\gamma^\mu\partial_\mu = \partial_0 + \sum_a \alpha_a(\hat{n})\partial_a$ with $\alpha_a(\hat{n}) := ns_a$; the octant Maxwell operator is $\partial_t - D$ with $D = \sum_a e_a\partial_a$ (Theorem 3.53). In both, $\partial_a = \partial/\partial x^a$ differentiates along the three spatial coordinates, the lengths on V^-, in m^{-1}.

Proposition 3.66 (Two Pauli triples)**.** *(i) $\alpha_1(\hat{n}), \alpha_2(\hat{n}), \alpha_3(\hat{n})$ are symmetric, square to $\mathbf{1}$ and anticommute; so do e_1, e_2, e_3. For any such triple π_a, $(\sum_a k_a\pi_a)^2 = |k|^2\mathbf{1}$, so both $\partial_0 \pm \sum_a \pi_a\partial_a$ factor the wave operator.*

(ii) $J(\hat{n})\alpha_a(\hat{n}) = Ts_a$ and $e_a = -TJ(e_a)$: the Dirac triple is one clock times the spatial generators, $\alpha_a(\hat{n}) = -TJ(\hat{n})s_a$; the octant triple consists of the three coordinate clocks themselves, $e_a = -TJ(e_a)$, one for each spatial axis.

(iii) No fixed element Z gives $\alpha_a(\hat{n}) = Ze_a$ for all a, nor $\alpha_a(\hat{n}) = e_aZ$; in particular none of T, Γ, ω_i, $J(\hat{n})$ does.

Proof. (i) $\hat{n}$ and s_a are orthogonal matrices, $\hat{n}$ symmetric, s_a antisymmetric, anticommuting: $(ns_a)^2 = -\hat{n}^2s_a^2 = \mathbf{1}$ and $ns_ans_b = -s_as_b$. (ii) $J(\hat{n})\hat{n} = T$, and $TJ(e_a) = T^2e_a$. (iii) $\alpha_a(\hat{n}) = Ze_a$ for all a would give $Z = ns_ae_a$ for $a = 1, 2$, hence $s_1e_1 = s_2e_2$, which is false; $\alpha_a(\hat{n}) = e_aZ$ would give $e_1ns_1 = e_2ns_2$, whose grade-one part $\hat{n}_1s_1 = \hat{n}_2s_2$ forces $\hat{n} = \pm e_3$, and $e_1e_3s_1 \neq e_2e_3s_2$ (r4-20, symbolic in $\hat{n}$). □

Theorem 3.67 (The triples are conjugate only by elements of mixed parity)**.** *For each clock the $U \in M_8(\mathbb{R})$ with $U\alpha_a(\hat{n}) = e_aU$ $(a = 1, 2, 3)$ form an eight-dimensional space, $\mathcal{C}\,U_0$ for any invertible solution U_0, where $\mathcal{C} = \mathrm{span}\{\mathbf{1}, T\} \otimes \mathrm{Cl}^0(V^-) \cong M_2(\mathbb{C})$ is the commutant of the e_a; its invertible elements are dense. No nonzero solution is even and none is odd, so no element of the Clifford group, of* $\mathrm{Spin}(3,3)$ *or of* $\mathfrak{so}(3,3)$ *relates the two triples. Every invertible solution carries the Dirac representation to the left-multiplication representation of the temporal Pauli algebra,*

$$U\rho_{\hat{n}}(L_i)U^{-1} = -\tfrac{1}{2}e_je_k = -\tfrac{1}{2}J(e_i), \qquad U\rho_{\hat{n}}(K_i)U^{-1} = -\tfrac{1}{2}e_i,$$

whereas $\rho_{\mathrm{oct}}(L_i) = -\frac{1}{2}e_je_k + \frac{1}{2}s_js_k$ and $\rho_{\mathrm{oct}}(K_i) = -\frac{1}{2}e_i - \frac{1}{2}s_i\omega$; by Proposition 3.63 no U relates $\rho_{\hat{n}}$ to ρ_{oct}.

Proof. Both triples generate copies of $\mathrm{Cl}(3,0) \cong M_2(\mathbb{C})$, which has a unique simple module, $\mathbb{R}^4$; S is twice it for each, so intertwiners exist and form a torsor of the commutant (inherited: Wedderburn's theorem). If U had definite parity, $U\alpha_a$ would have the parity of U and e_aU the opposite one, so both would vanish, and $U = 0$ since e_a is invertible. The exact solution spaces, their even and odd parts, invertibility of a generic solution and the conjugated generators are computed at $\hat{n} = e_1$ and $(2, 3, 6)/7$ (r4-21); $\rho_{\hat{n}}(L_i) = -\frac{1}{2}\alpha_j\alpha_k$ and $\rho_{\hat{n}}(K_i) = -\frac{1}{2}\alpha_i$. □

Proposition 3.68 (Left and right in the Fock picture)**.** *On $S = \mathrm{Cl}(V^+)|\mathrm{vac}\rangle$ (Proposition 1.11) the temporal generators act by left multiplication, $e_a(x|\mathrm{vac}\rangle) = (e_ax)|\mathrm{vac}\rangle$, and the spatial ones by twisted right multiplication, $s_a(x|\mathrm{vac}\rangle) = (\hat{x}e_a)|\mathrm{vac}\rangle$, $\hat{x} = (-1)^{|x|}x$. Hence $\alpha_a(\hat{n})(x|\mathrm{vac}\rangle) = (\hat{n}\hat{x}e_a)|\mathrm{vac}\rangle$.*

Proof. On the eight blades $x = e_I$, $I \subset \{1, 2, 3\}$ (r4-22). □

Restated: the right action is that of Proposition 3.49(ii), where $e_{a+3}\omega$ acts as $x \mapsto -xe_a$, on the Fock model of Chevalley [201] (Proposition 1.11); verified here on all blades. Inherited: in the Pauli and space-time algebras a Dirac spinor transforms by one-sided and the field bivector by two-sided multiplication [70, 71], which is the difference between $\rho_{\hat{n}}$ and ρ_{oct} in Theorem 3.67.

These results answer the remark after Theorem 3.53. The Dirac reading couples ∂_a to the spatial generators, through $\gamma^0\gamma^a = \alpha_a(\hat{n}) = ns_a$; in the Fock picture this is the clock $\hat{n}$ on the left and the temporal Pauli generator e_a on the right. The octant field reading couples ∂_a to the

depends on the clock $\hat{n}$	independent of the clock
$\gamma^0_{\hat{n}} = \hat{n}$, $J(\hat{n})$, $\Omega_{\hat{n}}$; $\mathfrak{l}_{\text{int}}(\hat{n})$, $\phi_{\hat{n}}$, the boosts $\rho_{\hat{n}}(K_i)$; $\alpha_a(\hat{n})$; $K^\mu_{\hat{n}}$, $H_{\hat{n}}(\mathbf{p})$; the Dirac complex structure $\pm J(\hat{n})$ and charge; the eigenspaces $\hat{n} = \pm 1$ that carry the two senses; the couplings $M_{\hat{n}}$, $M^\star_{\hat{n}}$; the base time $t_{\hat{n}}$	T, $T\Gamma = -e_4e_5e_6 = -\hat{n}\Omega_{\hat{n}}$, $\Gamma = J(\hat{n})\Omega_{\hat{n}}$; the rest flow $e^{-m\lambda T}$; the rotations $\rho_{\hat{n}}(L_i) = \frac{1}{2}s_j s_k$; ρ_{oct}, Φ, G, $\mathfrak{l}_G$, the octants and pole operators; $D = \sum_a e_a\partial_a$ and the coefficients of the octant Maxwell equation (3.13)

Table 3.5: Clock dependence of the objects of the two readings (r4-2, r4-3, r4-18, r4-23).

temporal generators e_a on the left. In both the index a of the derivative is matched to the index a of a generator by the pairing $a \leftrightarrow a+3$ of the arena, which is background structure, and the derivative is along the length x^a on V^- in both. The two triples are conjugate (fixed), by elements of mixed parity only, which are no symmetry of the arena (fixed), and any Pauli triple would make the operator hyperbolic (fixed). So neither operator shows that spatial differentiation follows from temporal directions: the octant operator uses the three coordinate clocks e_a as spatial Pauli matrices, one per axis, not one clock, and the statement that space is differentiated "through time" is a reading of the choice of representation (interpretive), not a consequence of either equation; as an inference it is refuted, since conjugating by U rewrites Maxwell's equations with the Dirac triple for the octant map $U^{-1}\Phi$.

3.6.5 Varying and permuting the clock

Proposition 3.69 (The field reading does not see the clock; the particle reading does)**.** *The objects of the right column of Table 3.5 are defined without $\hat{n}$; the octant Maxwell equation at clock $\hat{n}$ is $(\partial_{t_{\hat{n}}} - D)\Phi_{\text{oct}}(X) = -(\rho|\text{vac}\rangle + \mathbf{j}\cdot f)$, with the same matrices for every clock, the clock entering only through the time coordinate of the base $E_{\hat{n}}$. The objects of the left column depend on $\hat{n}$: $\hat{n} \mapsto J(\hat{n})$ is injective, $\mathfrak{l}_{\text{int}}(\hat{n}) = \mathfrak{l}_{\text{int}}(\hat{n}')$ only for $\hat{n}' = \pm\hat{n}$, and $M_{e_1} \neq M_{e_2}$. All of them are carried from one clock to another by the rotors of* $\text{Spin}(V^+)$*, which are not symmetries of the field reading, and, together with the field, by the diagonal rotations, which are.*

Proof. By inspection of the definitions in Section 3.5; Propositions 3.56, 3.57 and 3.65; $\mathfrak{l}_{\text{int}}(\hat{n})$ determines $E_{\hat{n}}$ and hence $\pm\hat{n}$ (Proposition 2.11). □

Fixed. This answers the question whether different clocks are physically different sectors, gauge-equivalent choices or an order parameter: the diagonal rotations map the coupled data at $(\hat{n}, \psi, X)$ to those at $(R\hat{n}, R\psi, RX)$ (Proposition 3.65(ii)), so for theories built from the structures of this chapter a change of clock is a change of frame (fixed as a statement of equivalence; its reading as "no sector" is interpretive); whether a dynamics singles out a clock is open, as in Remark 3.32. Equivalence under a global symmetry is not gauge redundancy: symmetry-related configurations can be distinct physical states, and a gauge reading would need an action, a redundancy of its solutions and gauge-invariant observables. A position-dependent clock rotor adds $R^{-1}\partial_\mu R$ to the Dirac operator and so needs a connection that is not constructed here.

Theorem 3.70 (Uniqueness of T)**.** *Let $Y \in \text{Cl}(3,3)$.*

(i) If $\hat{n}^{-1}Y$ commutes with $\gamma^0_{\hat{n}}, \ldots, \gamma^3_{\hat{n}}$ for three linearly independent clocks, then $Y \in \mathbb{R}T$. Three clocks of the Fisher–Rao octant suffice, e.g. e_1, $(1,2,2)/3$, $(2,3,6)/7$. For two linearly independent clocks $\hat{n}, \hat{n}'$ the solutions are $\text{span}\{T, (\hat{n}\times\hat{n}')\,e_4e_5e_6\}$*; for $\hat{n}' = \pm\hat{n}$ the conditions are those of one clock.*

(ii) If $\hat{n}^{-1}Y$ lies, for two independent clocks, in the joint commutant $\text{span}\{\mathbf{1}, \Omega_{\hat{n}}, J(\hat{n}), \Omega_{\hat{n}}J(\hat{n})\}$ *of $\mathfrak{l}_{\text{int}}(\hat{n})$ and $J(\hat{n})$ in which Theorem 2.17 selects the imaginary unit, then $Y \in$* $\text{span}\{T, e_4e_5e_6\} = \text{span}\{T, T\Gamma\}$*, the two* $\text{Spin}(3,3)$*-invariant forms. Indeed $\hat{n}J(\hat{n}) = T$ and $\hat{n}\Omega_{\hat{n}} = e_4e_5e_6$ for every clock.*

(iii) Consequently T is, up to scale, the only element of $\mathrm{Cl}(3,3)$ *that is the product of the clock with a Dirac complex structure (a square root of* $-\mathbf{1}$ *in the commutant of the* $\gamma^\mu_{\hat n}$*) for every clock and that does not depend on the clock: the products are* $\pm\hat n J(\hat n) = \pm T$.

Proof. (i) The commutant of the $\gamma^\mu_{\hat n}$ is $\mathrm{span}\{\mathbf{1}, J(\hat n), u_1\Omega_{\hat n}, u_2\Omega_{\hat n}\}$ (Theorem 2.14(ii)), so $Y \in \hat n \cdot (\dots) = \mathrm{span}\{n, T, u_1 e_4e_5e_6, u_2 e_4e_5e_6\}$, using $n u_a \Omega_{\hat n} = -u_a e_4e_5e_6$. The four have grades $1,3,4,4$, so for Y fixed the grade-one part lies in $\bigcap \mathbb{R}\hat n = 0$, the grade-four part in $(\bigcap U_{\hat n})\, e_4e_5e_6$, which is 0 for three independent clocks and the line of $\hat n \times \hat n'$ for two. (ii) $\hat n \cdot \mathrm{span}\{\mathbf{1}, \Omega_{\hat n}, J(\hat n), \Omega_{\hat n} J(\hat n)\} = \mathrm{span}\{n, e_4e_5e_6, T, \hat n\Gamma\}$, of grades $1,3,3,5$; the grades one and five vanish for two independent clocks; $T\Gamma = -e_4e_5e_6$ (r4-1) and $\mathrm{span}\{T, T\Gamma\}$ is the space of invariant forms (Proposition 3.51(iii)). (iii) If $\hat n^{-1}Y$ is a square root of $-\mathbf{1}$ commuting with the $\gamma^\mu_{\hat n}$ for every clock, then $Y \in \mathbb{R}T$ by (i), so $\hat n^{-1}Y = aJ(\hat n)$, and $(aJ(\hat n))^2 = -a^2\mathbf{1}$ gives $a = \pm 1$; with the selection of Theorem 2.17 at every clock (Theorem 2.14) the Dirac complex structures are $\pm J(\hat n)$ and the products are $\pm T$ directly. For two clocks the elements $\cos\vartheta\, T + \sin\vartheta\, \hat w\, e_4e_5e_6$, $\hat w$ the unit vector along $\hat n \times \hat n'$, meet the conditions at both, so (iii), like (i), needs three clocks. Exact solutions in r4-23. □

Fixed; by the two-clock exception in (i) two clocks do not single out T, and three are needed. At a single fixed clock the theorem gives no uniqueness. The commutant $\mathrm{span}\{\mathbf{1}, J(\hat n), u_1\Omega_{\hat n}, u_2\Omega_{\hat n}\}$ is a quaternion algebra, and its square roots of $-\mathbf{1}$ form the two-sphere $aJ(\hat n) + b\, u_1\Omega_{\hat n} + c\, u_2\Omega_{\hat n}$, $a^2+b^2+c^2 = 1$; for instance $e_2\Omega_{e_1}$ is a Dirac complex structure at $\hat n = e_1$ other than $\pm J(e_1)$ (z1-7c). So "a square root of $-\mathbf{1}$ in the commutant" selects $\pm J(\hat n)$ only with an extra datum: pointwise, the joint commutant with $\mathfrak{l}_{\mathrm{int}}(\hat n)$ and $J(\hat n)$ of Theorem 2.17 (or commutation with T); globally, the two hypotheses of (iii), that the product with the clock be the same element for three independent clocks. The second element of (ii), $e_4e_5e_6$, is the clock times the other complex structure $\Omega_{\hat n}$, which Theorem 2.17 excludes because it anticommutes with the $\gamma^\mu_{\hat n}$.

Remark 3.71 (The "covariant mass" $e_4e_5e_6$)**.** A possible assignment is that, of the two trivectors anticommuting with the three Dirac matrices α_a, the spatial volume $e_4e_5e_6$ is the scalar mass and $e_1e_2e_3$ its pseudoscalar partner. In the present conventions the first half is confirmed at every clock: the trivectors anticommuting with $\alpha_1(\hat n), \alpha_2(\hat n), \alpha_3(\hat n)$ are $\mathrm{span}\{T, e_4e_5e_6\}$, the clock-independent space of Theorem 3.70(ii) (r4-24, at four clocks; at all clocks by Proposition 3.57(iii), since conjugation by a rotor preserves the grade). The assignment is corrected. As the matrix β of $H = \sum_a \alpha_a p_a + m\beta$, $e_4e_5e_6 = \hat n\Omega_{\hat n}$ is admissible (symmetric, square $\mathbf{1}$, commuting with $J(\hat n)$, $H^2 = (|\mathbf{p}|^2 + m^2)\mathbf{1}$), but relative to the kinetic kernels with $\gamma^0 = \hat n$ it gives the pseudoscalar mass of Proposition 2.22, whose Grassmann kernel $J(\hat n)e_4e_5e_6 = T\Omega_{\hat n}$ depends on the clock; the scalar mass has $\beta = \hat n$ and the clock-independent kernel $J(\hat n)\hat n = T$. The element $e_1e_2e_3$ is not a β at all (it is antisymmetric with square $-\mathbf{1}$); it is the kernel.

Interpretation. In the particle reading the clock and the complex structure both move, and their product T does not; in the field reading T is the duality on the octants and, as the Chevalley–Mukai pairing, the form pairing E_k with B_k (Proposition 3.51). The mass form is the part of the particle reading that the field reading shares. This is interpretive; what is fixed is Theorems 3.60 and 3.70.

3.6.6 Answer

(1) *Consistency.* $T = \hat n J(\hat n)$ holds at every clock and is compatible with all operations considered: T is invariant under all rotations of the temporal triple and the diagonal rotations, reversed together with $J(\hat n)$ by odd permutations, and with the duality operator by the paired ones, and fixed by the reversal of the clock (Propositions 3.57 and 3.58; fixed). Its three roles, mass form, parity and duality, are compatible (Theorem 3.60; restated and fixed). It incorporates consistently into both readings.

(2) *The Lorentz representations stay apart.* On S the Dirac representation $\rho_{\hat{n}}$ and the octant representation ρ_{oct} are inequivalent, their joint commutant is $\mathbb{R}\mathbf{1}$, and no operator, T included, relates them (Propositions 3.62 and 3.63 and Theorem 3.67; fixed). A single module carrying both as the Lorentz action of one field is obstructed; the Dirac field and the field octants are two fields.

(3) *The bridge as a coupling.* The answer to "The main obstruction" of Section 3.5.6: ρ_{oct}, $\rho_{\hat{n}}$ and Φ coexist as the actions on two different fields, and the covariant local couplings without derivatives that are linear in the field strength are trilinear, quadratic in ψ: the Pauli kernel $TJ(\hat{n})S_{\hat{n}}(X)$ and the electric-dipole kernel $T\Gamma S_{\hat{n}}(X)$, a two-dimensional space at every clock, bilinear couplings being excluded by the 2π rotation (Theorem 3.64; fixed, the textbook terms inherited). Couplings of higher order in X exist, for instance $(\psi^{\top}T\psi)\,\eta(X,X)$, since T is $\rho_{\hat{n}}$-invariant and $\eta(X,X)$ is $\mathfrak{l}_G$-invariant (r4-25). Since Φ is an equivalence onto its image, the octants add no coupling that X itself does not have. T is the clock-independent factor of every Dirac kernel; the couplings are conjugated by the rotation that moves the clock, and an odd permutation paired with the spatial one reverses the electric-dipole term with T but not the Pauli term, and the temporal one alone reverses neither (Proposition 3.65; fixed, (E2) refuted for the Pauli term).

(4) *$J(\hat{n})$ on the field.* $J(\hat{n}) = -L(\hat{n}) + A(\hat{n})$ rotates the transverse octants about the axis with the components of $\hat{n}$ and maps the longitudinal octants to the Gauss-law poles; this field rotation $L(\hat{n})$ is the Dirac rotation about the spatial axis $\sum_i n_i s_i$ combined with half a turn of the charge (Proposition 3.61; fixed; the reading interpretive).

(5) *Spatial derivatives.* The remark after Theorem 3.53 is answered by Propositions 3.66 and 3.68 and Theorem 3.67: both operators differentiate along the lengths of V^-, the Dirac reading through the spatial generators and one clock, the octant reading through the three temporal generators; the triples are conjugate only by mixed-parity elements, and nothing derives spatial differentiation from temporal directions (fixed; the inference refuted).

(6) *The clock.* The field reading is clock-independent and the particle reading is not; T is the unique clock-independent product of the clock with a Dirac complex structure, and with the second complex structure $\Omega_{\hat{n}}$ the space becomes $\mathrm{span}\{T, T\Gamma\}$ (Proposition 3.69 and Theorem 3.70; fixed). Clocks are related by the diagonal rotations, a symmetry of both readings and of the couplings, so a change of clock is a change of frame (fixed as an equivalence); whether a dynamics selects a clock is open.

(7) *What remains for a covariant Dirac–Maxwell theory in the arena.* The minimal coupling through the potential and $J(\hat{n})$, $D_\mu = \partial_\mu + qA_\mu J(\hat{n})$ (Proposition 3.24, with the Lagrangian (3.9); inherited, restated), which uses A_μ on the base and not the octants; the Pauli and electric-dipole couplings (3.14), with two constants the arena does not fix (open); the absence of a common Lorentz representation on one S (obstructed at one constituent; the two readings can be housed together at the mixed second rung of Section 5.8, at the price stated there, restated); and the quantisation of either reading (open).

Open items. Reversing $\hat{n}$ reverses $J(\hat{n})$ (Proposition 3.58(iv)); the temporal permutations and rotations are Propositions 3.57 and 3.58; $J(\hat{n})$ is not the duality, a common Lorentz representation on S does not exist, and the cross-sector coupling has the units, covariance and parity stated in Theorem 3.64 and (3.14) (Proposition 3.61, Section 3.6.3); the clock dependence is Proposition 3.69. The projector vacuum, the mismatch between the mixed and the spectator embeddings, and the symplectic and quantum consistency of a cross-sector coupling are not treated here (open).

Table 3.6: The claims of this section, with their checks and classes.

id	claim	class
r4-1	conventions: Clifford relations, octants, T, Γ, $T\Gamma = -e_4e_5e_6$	restated
r4-2	$T = \hat{n}J(\hat{n}) = J(\hat{n})\hat{n}$, $J(\hat{n})^2 = -\mathbf{1}$ at every clock (Proposition 3.56)	restated (Proposition 2.13)
r4-3, 4	T invariant, $J(\hat{n})$ covariant under $\mathfrak{so}(3)_+$ and $\mathfrak{so}(3)_\Delta$; only $\mathfrak{so}(3)_\Delta$ commutes with G (Proposition 3.57)	fixed
r4-5, 6	odd permutations: $T \mapsto -T$, $J(\hat{n}) \mapsto -J(\sigma\hat{n})$; paired: $\Phi \mapsto \Phi_{\mathrm{t}}$, duality $-T \mapsto T$ (Proposition 3.58)	fixed
r4-7	clock reversal is a half-turn fixing T; $C = RT$ (Proposition 3.58(iv))	fixed
r4-8	$T = \pm J(\hat{n})$ on $\hat{n} = \pm 1$; odd permutation $\Rightarrow (m, q) \mapsto (-m, -q)$ (Corollary 3.59)	restated (Proposition 7.3); fixed
r4-9	three roles of T; T is the Dirac parity at every clock (Theorem 3.60)	restated, fixed
r4-10	$J(\hat{n}) = -L(\hat{n}) + A(\hat{n})$, $LA = 0$; $L(\hat{n}) = \rho_{\hat{n}}(L(\hat{n})) - \frac{1}{2}J(\hat{n})$ (Proposition 3.61)	fixed
r4-11	$J(\hat{n})$ rotates transverse octants, maps longitudinal ones to the poles (Proposition 3.61)	fixed; reading interpretive
r4-12	commutants $M_2(\mathbb{C})$, $\mathbb{C} \oplus M_2(\mathbb{R})$, joint $\mathbb{R}\mathbf{1}$ (Proposition 3.62)	fixed
r4-13	three inequivalent Lorentz representations on S (Proposition 3.63)	restated (decompositions), fixed
r4-14	no invariant bilinear coupling $\psi^\top M\Phi_{\mathrm{oct}}(X)$ (Theorem 3.64(i))	obstructed
r4-15, 16	trilinear couplings: dimension 2 (Grassmann), 6 (commuting); Pauli $TJ(\hat{n})S_{\hat{n}}$ and dual $T\Gamma S_{\hat{n}}$	fixed; textbook terms inherited
r4-17	Pauli term parity-even, dual parity-odd (Theorem 3.64(iii))	fixed; inherited in substance
r4-18	couplings conjugated by the clock rotor, T fixed; diagonal covariance (Proposition 3.65)	fixed
r4-19	odd permutation: Pauli term does not change sign, dual does when paired (Proposition 3.65(iii))	fixed; (E2) refuted for the Pauli term
r4-20	two Pauli triples, $J(\hat{n})\alpha_a = Ts_a$, $e_a = -TJ(e_a)$, no one-sided relation (Proposition 3.66)	fixed
r4-21	intertwiners: dimension 8, none of definite parity; Dirac $\mapsto$ left Pauli representation (Theorem 3.67)	fixed; existence inherited
r4-22	temporal = left, spatial = twisted right multiplication (Proposition 3.68)	restated (Proposition 3.49), verified; one-/two-sided inherited
r4-23	uniqueness of T; two-clock exception; span$\{T, T\Gamma\}$ (Theorem 3.70)	fixed
r4-24	trivectors anticommuting with $\alpha_a(\hat{n})$; $e_4e_5e_6$ as β is the pseudoscalar mass (Remark 3.71)	fixed; earlier assignment corrected
r4-25	a covariant quartic coupling $(\psi^\top T\psi)\,\eta(X, X)$: the trilinear couplings are the only ones linear in X, not the only ones (Section 3.6.6)	fixed
–	spatial derivatives "from temporal directions" (Section 3.6.4)	interpretive; inference refuted
–	a change of clock is a change of frame (Proposition 3.69)	fixed (equivalence); interpretive
–	mass form = the shared part of the two readings	interpretive
–	values of μ, d; a dynamics selecting the clock; quantisation	open

3.7 Spin one in both readings

A photon is a massless particle of spin one with two helicities. Its covariant description is a field strength, a two-form, and its potential is a four-vector defined up to gauge. The arena offers two places for it. In the *interior reading* the potential is a grade-one element of the four-plane E on which the Dirac sector is built and the field strength is the grade-two module of Proposition 3.7. In the *frame reading* the field strength occupies the six directions themselves, under the algebra $\mathfrak{l}_G$ of Theorem 3.1. This section shows that each reading produces a free field with the spin content of the photon, that the two are related by the Klein correspondence, and that at the second rung both can be housed under one Lorentz action and coupled to a Dirac block. Figure 3.2 summarises the comparison.

3.7.1 The univectorial photon

Proposition 3.72 (Grade one under the interior action)**.** *Under* $\mathfrak{l}_{\mathrm{int}} \oplus \mathfrak{u}(1)_J$*, with* $J = e_2e_3$ *acting by* $\mathrm{ad}(J/2)$*,*

$$V = \left(\tfrac{1}{2},\tfrac{1}{2}\right)_0 \oplus (0,0)_{+1} \oplus (0,0)_{-1}.$$

The Lorentz Casimir is $\mathrm{diag}(3,0,0,3,3,3)$ *on* $(e_1,\dots,e_6)$*, and* $\mathrm{ad}(J/2)$ *vanishes on the four-plane and acts on* (e_2,e_3) *by* $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$*, with eigenvalues* $\pm i$*.*

The potential $A = A^\mu\gamma_\mu$ is therefore neutral under the charge, as a photon must be, while the surplus pair is a Lorentz scalar of charge ± 1. The photon potential is the four-plane part of grade one, not the whole of it.

Proposition 3.73 (Spin one and spin one-half from the same rotors)**.** *The interior rotations* S_k *satisfy* $[S_i,S_j] = \epsilon_{ijk}S_k$*. On grade one* $\operatorname{ad} S_k$ *acts on* (e_4,e_5,e_6) *as the standard rotation generator* r_k *and annihilates* e_1,e_2,e_3*, so that* $-\sum_k(\operatorname{ad} S_k)^2 = \mathrm{diag}(0,0,0,2,2,2)$ *and* $\operatorname{spec}\operatorname{ad} S_3 = \{\pm i,0,0,0,0\}$*. On* $S = \mathbb{R}^8$*, by left multiplication,* $-\sum_k S_k^2 = \frac{3}{4}\mathbf{1}$ *and* S_3 *has eigenvalues* $\pm\frac{i}{2}$*. Consequently* $\exp(2\pi \operatorname{ad} S_3) = \mathbf{1}$ *on grade one while* $\exp(2\pi S_3) = -\mathbf{1}$ *on* S*.*

A vector transforms by conjugation, $A \mapsto RAR^{-1}$ with $R = e^{\theta S_3}$, and a spinor by left multiplication, $\psi \mapsto R\psi$. At $\theta = 2\pi$ the rotor is $-\mathbf{1}$; the sign cancels in the first case and survives in the second. This is the space-time-algebra account of the difference between vectors and spinors [70], realised here by the rotors of the Dirac sector: the same S_k give spin one to the potential and spin one-half to matter.

Theorem 3.74 (Maxwell's equations as one Clifford equation)**.** *Put* $\nabla = \gamma^\mu\partial_\mu$ *and* $\sigma_k = \gamma_k\gamma_0$*.*

(i) For $A = A^\mu\gamma_\mu$*,* $\nabla A = \partial\cdot A + \nabla\wedge A$*, and for* $F := \nabla\wedge A$ *the grade-one part of* ∇F *is* $(\Box A^\nu - \partial^\nu\partial\cdot A)\gamma_\nu$*, while its grade-three part vanishes identically.*

(ii) $F = \mathbf{E} + I_4\,\mathbf{B}$ *with* $I_4 = \gamma_0\gamma_1\gamma_2\gamma_3 = -\Omega$ *(Proposition 3.7),* $\mathbf{E} = \sum_k E_k\sigma_k$*,* $E_k = -\partial_k A^0 - \partial_t A^k$ *and* $\mathbf{B} = \nabla\times\mathbf{A}$*; in terms of* Ω*,* $F = \mathbf{E} - \Omega\,\mathbf{B}$*.*

(iii) For every $F = \sum_k(E_k\sigma_k + B_kI_4\sigma_k)$*,*

$$\nabla F = (\nabla\cdot\mathbf{E})\gamma_0 + (\nabla\times\mathbf{B} - \partial_t\mathbf{E})^k\gamma_k + I_4\big[-(\nabla\cdot\mathbf{B})\gamma_0 + (\partial_t\mathbf{B} + \nabla\times\mathbf{E})^k\gamma_k\big],$$

with no other grades.

Hence $\nabla F = \mathcal{J}$*, with* $\mathcal{J} = \rho\gamma_0 + \mathcal{J}^k\gamma_k$*, is Gauss's law, the Ampère–Maxwell law, the absence of magnetic charge and Faraday's law, and for* $F = \nabla\wedge A$ *the last two hold identically.*

This is Hestenes' single Maxwell equation [70], here on the four-plane of the Dirac sector inside the 8×8 matrices of $\mathrm{Cl}(3,3)$. Plane waves show how the gauge freedom removes the unphysical polarisations and how helicity appears.

Example 3.75 (A plane wave of the univectorial photon)**.** Let $k = \gamma_0 + \gamma_3$, so $k^2 = 0$, and $\varepsilon = \varepsilon^\mu \gamma_\mu$ with $k \cdot \varepsilon = 0$, that is $\varepsilon^0 = \varepsilon^3$. Then

$$F(\varepsilon) = k\varepsilon = \varepsilon^1 F_1 + \varepsilon^2 F_2, \qquad F_j = k\gamma_j,$$

because the component of ε along k drops out, $k^2 = 0$. The condition $k \cdot \varepsilon = 0$ has removed the component along the other null direction $\gamma_0 - \gamma_3$, so only the two transverse polarisations remain, and the gauge freedom $\varepsilon \mapsto \varepsilon + ck$ is exactly the helicity-zero direction. Moreover $F_j^2 = 0$ and $F_1F_2 + F_2F_1 = 0$, so every $F(\varepsilon)$ squares to zero: in the space-time algebra $F^2 = \mathbf{E}^2 - \mathbf{B}^2 + 2I_4\, \mathbf{E}\cdot\mathbf{B}$, so this is the null condition $|\mathbf{E}| = |\mathbf{B}|$, $\mathbf{E} \perp \mathbf{B}$. On $\operatorname{span}\{F_1, F_2\}$,

$$\operatorname{ad} S_3 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad I_4 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = -\operatorname{ad} S_3 = -\Omega,$$

both with eigenvalues $\pm i$. For positive-frequency waves the eigenvectors of $\operatorname{ad} S_3$ with eigenvalues $\mp i$ carry helicity ± 1, and on them the Hodge star is i times the helicity: circular polarisation is self-duality or anti-self-duality of F.

Proposition 3.76 (Every field-strength module under the interior action)**.** *The Lorentz Casimir on the grades of* $\mathrm{Cl}(3,3)$*, the number of field-strength modules (Casimir value* 4*) and their charges under* $\operatorname{ad}(J/2)$ *are*

grade	spectrum of the Casimir (multiplicity)	field-strength modules	charges
$0, 6$	$0\,(1)$	0	—
$1, 5$	$3\,(4),\ 0\,(2)$	0	—
$2, 4$	$4\,(6),\ 3\,(8),\ 0\,(1)$	1	0
3	$4\,(12),\ 3\,(8)$	2	± 1

and on S the Casimir is $\frac{3}{2}$ throughout.

Because J commutes with the four-plane, the grade-four module is J times the grade-two one, and J is the complex structure at one constituent (Theorem 2.17). Up to that complex structure the neutral field strength is unique and sits at grade two; the two grade-three copies carry charge ± 1 and are not candidates for the photon.

3.7.2 The frame photon

In the frame reading the field strength is the vector X itself, and the boosts of $\mathfrak{l}_G$ turn out to be the spin matrices of the photon.

Proposition 3.77 (Spin matrices on the frame)**.** *$K_k = GL_k$ for $k = 1, 2, 3$, both families commute with G and preserve η, and the Casimir $\sum K_k^2 - \sum L_k^2$ equals $4\,\mathbf{1}$ on V, the field-strength value. On $z = \mathbf{B} + i\mathbf{E}$ the rotation L_k acts as r_k and the boost K_k as*

$$s_k = i\, r_k, \qquad (s_k)_{ab} = -i\epsilon_{kab},$$

which are Hermitian with $[s_i, s_j] = i\epsilon_{ijk} s_k$, $\sum_k s_k^2 = 2\,\mathbf{1}$ and $\operatorname{spec} s_3 = \{-1, 0, 1\}$. With $p = -i\nabla$ one has $(s \cdot p)z = \nabla \times z$ identically, so the frame Maxwell system (3.2) is the photon wave equation

$$i\partial_t z = (s \cdot p)\, z, \qquad p \cdot z = 0,$$

and the kernel of s_3 is spanned by $\hat{e}_3$: transversality removes exactly the helicity-zero component of a wave moving along $\hat{e}_3$.

These are the spin-one matrices that Oppenheimer and Bialynicki-Birula use to write the photon wave function [112, 14]. The rotations act on z as r_k, and their Hermitian generators $s_k = ir_k$ are the action of the boosts, because on this module a boost is G times a rotation.

Proposition 3.78 (Orthogonality on the frame)**.** *For* $X = \sum_i (B_i e_i + E_i e_{i+3})$,

$$z \cdot z = \eta(X, X) - i\,\eta(X, GX).$$

Hence $\mathbf{E} \perp \mathbf{B}$ *exactly when* $X \perp_\eta GX$, $|\mathbf{E}| = |\mathbf{B}|$ *exactly when* X *is* η*-null, and both hold exactly when the plane* $\operatorname{span}\{X, GX\}$ *is totally* η*-null.*

Theorem 3.79 (The axis of a null field)**.** *For real* $a, b \in \mathbb{R}^3$,

$$(s \cdot (a \times b))(a + ib) = |b|^2 a + i|a|^2 b - (a \cdot b)(b + ia).$$

Consequently a nonzero $z = \mathbf{B} + i\mathbf{E}$ *satisfies* $z \cdot z = 0$ *if and only if* $(s \cdot n)z = z$ *for some real unit vector* n*, and then* $n = \mathbf{B} \times \mathbf{E}/|\mathbf{B} \times \mathbf{E}|$ *uniquely.*

Proof. If $z \cdot z = 0$ then $\mathbf{B} \cdot \mathbf{E} = 0$ and $|\mathbf{B}| = |\mathbf{E}| \neq 0$; the identity gives $(s \cdot (\mathbf{B} \times \mathbf{E}))z = |\mathbf{B}|^2 z$ and $|\mathbf{B} \times \mathbf{E}| = |\mathbf{B}|^2$. Conversely $n \cdot ((s \cdot n)z) = 0$ and $z \cdot ((s \cdot n)z) = 0$ identically, and the characteristic polynomial of $s \cdot n$ is $-\lambda(\lambda^2 - |n|^2)$, so the eigenvalue 1 forces $n \cdot z = 0$ and $z \cdot z = 0$. For uniqueness, $(s \cdot n)z = in \times z = z$ splits into $\mathbf{B} = -n \times \mathbf{E}$ and $\mathbf{E} = n \times \mathbf{B}$, whence $n \perp \mathbf{B}$ and $\mathbf{B} \times \mathbf{E} = |\mathbf{B}|^2 n$. □

Since $\mathbf{B} \times \mathbf{E} = \mathbf{E} \times \mathbf{B}_{\mathrm{t}}$ is the Poynting vector, a null field is the $m = +1$ state of the photon spin along its own energy flux, the frame form of the classical fact that a null field determines a unique direction of propagation [148, 131]. Orthogonality with equal magnitudes therefore singles out the null fields and their axis. It does not single out a helicity.

Example 3.80 (Three null waves with one axis)**.** The fields

$$z_\pm = (1, i, 0)\, e^{\pm ik(x_3 - t)}, \qquad z_{\mathrm{lin}} = (1, i, 0) \cos k(x_3 - t)$$

all solve (3.2). In each, $\mathbf{B} \cdot \mathbf{E} = 0$, $|\mathbf{B}| = |\mathbf{E}|$ and $\mathbf{B} \times \mathbf{E}$ points along $+\hat{e}_3$ wherever $z \neq 0$, so each is an $m = +1$ state along $\hat{e}_3$. At $x_3 = 0$ the electric field turns according to $(\mathbf{E} \times \dot{\mathbf{E}}) \cdot \hat{e}_3 = +k$, $-k$, 0: helicity $+1$, helicity -1, and a linear superposition. The difference is the sign of the frequency, $z_+ \propto e^{-ikt}$ against $z_- \propto e^{+ikt}$, which is the Bialynicki-Birula assignment for the Riemann–Silberstein vector [14]. Not every radiation field is null either: $z = (1, i, 0) \cos k(x_3 - t) + (0, 1, i) \cos k(x_1 - t)$ solves the system and has $|\mathbf{B}| = |\mathbf{E}|$, but $\mathbf{B} \cdot \mathbf{E} = \cos k(x_1 - t) \cos k(x_3 - t)$.

Orthogonality alone is weaker still. The field $X = (\mathbf{B}; \mathbf{E}) = (1, 0, 0;\ 0, 2, 0)$ has $\eta(X, GX) = 0$, so $\mathbf{E} \perp \mathbf{B}$, but $\eta(X, X) = 1 - 4 = -3$; the boost $e^{\theta K_3}$ with $\theta = -\frac{1}{2} \log 3$ carries it to $(0, 0, 0;\ 0, \sqrt{3}, 0)$, a purely electric field with the same invariant. A field with $\mathbf{E} \perp \mathbf{B}$ and $|\mathbf{E}| \neq |\mathbf{B}|$ is purely electric or purely magnetic in a suitable frame [82]. On the null line $\mathbb{C}(1, i, 0)$, by contrast, $e^{\theta K_3}$ acts by multiplication by e^θ; with $\theta = -\operatorname{artanh} \beta$ for a frame moving with velocity $\beta \hat{e}_3$ this is the Doppler factor $\sqrt{(1 - \beta)/(1 + \beta)}$. The stabiliser of the axis acts on a null field by the helicity phase and by the Doppler factor.

3.7.3 Potentials on the frame, and helicity as a charge

Place a vector potential on each triple, $\mathbf{A}$ on e_1, e_2, e_3 and $\mathbf{C}$ on e_4, e_5, e_6, write $X_A = (\mathbf{A}; \mathbf{C})$ and $X_F = \nabla \times X_A$, the curl acting on each triple, and take

$$\mathcal{L} = \tfrac{1}{2} (\nabla \times X_A) \cdot G \dot{X}_A - \tfrac{1}{2} |\nabla \times X_A|^2 \tag{3.15}$$

with the Euclidean dot product of $\mathbb{R}^6$, for which G is orthogonal.

Theorem 3.81 (Duality-symmetric Maxwell theory on the frame)**.** *(i) The Euler–Lagrange expression of* (3.15) *is* $G\nabla \times (\dot{X}_A + G\nabla \times X_A)$*. Every solution of the first-order equation* $\dot{X}_A = -G\nabla \times X_A$ *solves it, and on a contractible region every solution satisfies the first-order equation up to a time-dependent gauge transformation.*

(ii) *On solutions, X_F obeys the frame Maxwell system (3.2) with $\mathbf{B} = \nabla \times \mathbf{A}$ and $\mathbf{E} = \nabla \times \mathbf{C}$.*

(iii) *Under $X_A \mapsto X_A + (\nabla\lambda_1; \nabla\lambda_2)$, X_F is unchanged and $\mathcal{L}$ changes by a divergence.*

(iv) *The Hamiltonian is $\frac{1}{2}|\nabla \times X_A|^2 = \frac{1}{2}(|\mathbf{B}|^2 + |\mathbf{E}|^2)$, the energy density.*

(v) *$\mathcal{L}(e^{\theta G} X_A) = \mathcal{L}(X_A)$ exactly, and the Noether current of $\delta X_A = G X_A$ has density $j^0 = -\frac{1}{2} X_A \cdot \nabla \times X_A$. For fields that decay or are periodic the conserved charge is $-\mathcal{K}$, where*

$$\mathcal{K} = \tfrac{1}{2} \int X_A \cdot \nabla \times X_A \, d^3x = \tfrac{1}{2} \int (\mathbf{A} \cdot \mathbf{B} + \mathbf{C} \cdot \mathbf{E}) \, d^3x$$

is the optical helicity.

This is the duality-symmetric first-order formulation of Schwarz and Sen [137], with two vector potentials in the temporal gauge and with the binding root G in the place of their $SO(2)$ generator; the conserved charge of the duality rotation is, up to sign, the optical helicity [18, 31, 19].

Example 3.82 (The G-charge is the helicity)**.** For waves $e^{i(kx_3 - \omega t)}$ the first-order equation reads $\omega a = G(k \times a)$, a 6×6 matrix with eigenvalues $\pm k$, each twice, and 0 twice (the longitudinal, pure-gauge pair). The eigenspace $\omega = +k$ is spanned by

$$\begin{aligned} a_1 &= (1, -i) \otimes (1, i, 0), \quad Ga_1 = ia_1, \qquad s_3 a_1 = a_1, \\ a_2 &= (1, i) \otimes (1, -i, 0), \quad Ga_2 = -ia_2, \quad s_3 a_2 = -a_2, \end{aligned}$$

with s_3 acting on each triple. For the real waves $X_A = \mathrm{Re}(a\, e^{ik(x_3 - t)})$ the helicity density $\frac{1}{2} X_A \cdot \nabla \times X_A$ and the energy density are (k, k^2) for a_1, $(-k, k^2)$ for a_2, and $(0, 4k^2 \sin^2 k(x_3 - t))$ for $a_1 + a_2$. On positive-frequency modes the eigenvalue of G is i times the helicity. The ratio of helicity to energy is $\pm 1/\omega$, which for a quantised field of energy $N\hbar\omega$ is $\pm N\hbar$: one unit of $\hbar$ per photon, the spin-one value.

The frame potentials are not components of a Lorentz tensor. Neither the constant boost $\delta X_A = K_3 X_A$ nor the boost of the coordinates is a symmetry of (3.15); the boost that is one is the Schwarz–Sen transformation $\delta X_A = t\, \partial_3 X_A - x_3\, G \nabla \times X_A$, whose curl is, identically,

$$\nabla \times (t\, \partial_3 X_A - x_3\, G \nabla \times X_A) = t\, \partial_3 X_F - x_3\, G \nabla \times X_F - K_3 X_F,$$

which on solutions is the boost of the field strength along $\hat{e}_3$ with its orbital part and its spin part K_3. Boosts reach the potentials only through their curl and only on shell [137]. Duality rotations, for their part, are not Lorentz transformations: $e^{\theta G}$ preserves the energy density and rotates $\eta(X, X)$ into $\eta(X, GX)$.

3.7.4 The bridge between the readings

Proposition 3.83 (Half-spinors under the two algebras)**.** *Let $S^{\pm}$ be the ± 1 eigenspaces of the volume element $e_1 e_2 \cdots e_6$, whose square is $\mathbf{1}$. Under $\mathfrak{l}_{\mathrm{int}}$ the Casimir on $S^{\pm}$ is $\frac{3}{2}$, and S^+ carries a two-parameter family of invariant antisymmetric forms and no invariant symmetric one: it is a Weyl module. Under $\mathfrak{l}_G$ the Casimir on all of S is 3, the invariant symmetric forms on S^+ form a one-parameter family of Lorentzian forms, and there is no invariant antisymmetric form: S^+ is a Minkowski four-vector, and $\exp(2\pi L_3) = +\mathbf{1}$ on S. On $\mathbb{C}^8$ the two algebras give $2\,(\frac{1}{2}, 0) \oplus 2\,(0, \frac{1}{2})$ and $2\,(\frac{1}{2}, \frac{1}{2})$ respectively, and they meet in zero.*

So under the binding algebra every spinor of $\mathrm{Cl}(3,3)$ has integer spin, in accordance with $\mathrm{Spin}(3,3) \cong SL(4, \mathbb{R})$, whose half-spinor modules are $\mathbb{R}^4$ and its dual [125]. This sharpens Proposition 3.6: with $\mathfrak{l}_G$ as the Lorentz algebra one constituent carries no spin one-half at all.

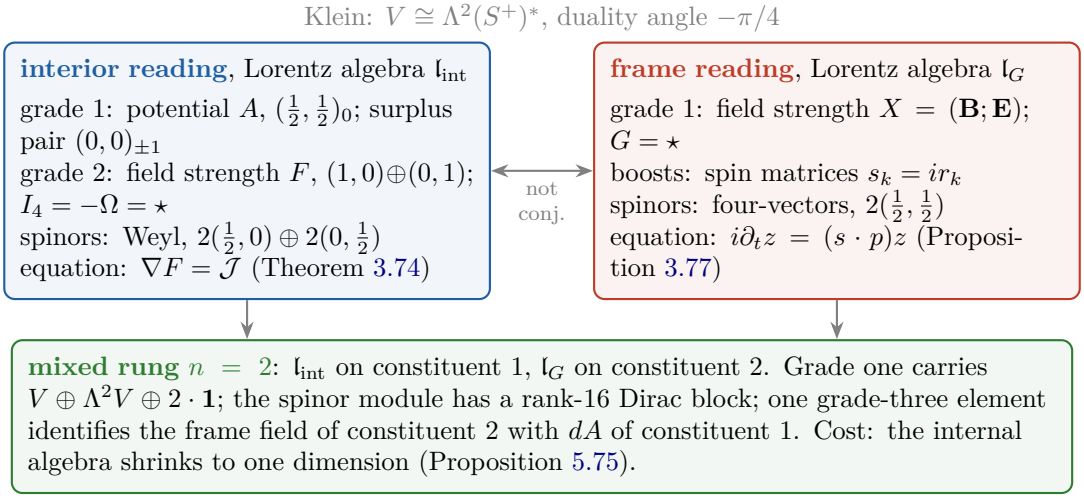


Figure 3.2: Two places for the photon in the arena (Section 3.7). Each reading gives a free field with the spin content of the photon; on one constituent they use non-conjugate Lorentz algebras and do not merge, and at the second rung they can be housed together at the price of the internal algebra.

Theorem 3.84 (The Klein correspondence)**.** *The* $\mathrm{Spin}(3,3)$*-invariant bilinear forms on* S *form a two-dimensional space. For one of them,* β*, the map* $W(X)(\psi,\chi) = \beta(\psi, X\chi)$ *on* S^+ *is antisymmetric and equivariant, and*

$$\operatorname{Pf} W(X) = \eta(X,X).$$

Hence $V \cong \Lambda^2(S^+)^*$ *as* $\mathrm{Spin}(3,3)$*-modules, and the null cone of* η *is the set of decomposable two-forms.*

This is the Klein correspondence between the points of the Klein quadric and the lines of projective three-space [157]. It can be compared with the frame reading. Let $T : \mathbb{R}^{1,3} \to S^+$ be the $\mathfrak{l}_G$-intertwiner, unique up to scale. Then

$$T^*W(X) = -F_X + \star F_X = \sqrt{2}\, F_{e^{3\pi G/4}X} \qquad \text{for all } X,$$

so the canonical identification of V with two-forms on the half-spinor and the frame reading of the field strength differ by a duality rotation of an eighth of a turn, $-\pi/4$ modulo π. In the Klein frame $\eta(X,X)$ is the Pfaffian, the invariant of $\mathbf{E}\cdot\mathbf{B}$ type, and $\eta(X,GX)$ the Hodge pairing; in the frame reading the roles are the other way round. For free fields the two readings are equivalent, since duality rotations map solutions to solutions; a statement that singles out one invariant, such as which of them is the Lagrangian, holds in one reading and not in the other. A plane wave $\kappa\wedge\alpha$ built on S^+, with κ the lowered null momentum $(1,0,0,1)$ and α the polarisation $(0,1,0,0)$, lands on the doubly null vector $X = (-\frac{1}{2},-\frac{1}{2},0;\,\frac{1}{2},-\frac{1}{2},0)$, whose Poynting axis $\mathbf{B}\times\mathbf{E} = (0,0,\frac{1}{2})$ is the spatial momentum; the Klein planes of X and of GX meet in the null line $[\kappa]$, as two lines of projective space joined by a line of the quadric must.

3.8 The orthogonality of the electric and magnetic fields

In the frame reading the orthogonality of $\mathbf{E}$ and $\mathbf{B}$ is the reality of the complex invariant, $\operatorname{Im} q = 2\,\mathbf{E}\cdot\mathbf{B} = 0$, and by Proposition 3.12 it is also the condition for the spacetime two-form to be a single plane. Since the arena puts $\mathbf{E}$ and $\mathbf{B}$ on two orthogonal triples, it is natural to ask what orthogonality means physically: what breaks it, whether its failure is a symmetry breaking, and whether it can produce mass. The answers are classical once the right questions are asked, and they come out sharply negative on the last point.

3.8.1 What orthogonality is

The field strength $F^{\mu}{}_{\nu}$ has eigenvalues $\pm\epsilon_E$ and $\pm i\epsilon_M$ with

$$\epsilon_E = |\operatorname{Im}\sqrt{q}|, \qquad \epsilon_M = |\operatorname{Re}\sqrt{q}|,$$

and $\mathbf{E}\cdot\mathbf{B} = \operatorname{Pf} F = 0$ exactly when F is one plane: a pure boost, a pure rotation or a null rotation. Otherwise F generates a screw, a boost and a rotation about a common axis. Two simple facts locate orthogonality among all fields.

Proposition 3.85 (Single modes are orthogonal)**.** *Let $F = k \wedge b$ with k, b real four-vectors. Then $\operatorname{Pf} F = 0$ and $q = (k\cdot k)(b\cdot b) - (k\cdot b)^2$ is real. The field of a mode $A_\mu = \operatorname{Re}(a_\mu e^{-ik\cdot x})$ with real k and $k\cdot a = 0$ is $k \wedge b(x)$ with $k\cdot b = 0$, for every polarisation. A Maxwell mode, $k\cdot k = 0$, is therefore null at every point, and a Proca mode, $k\cdot k = m^2$, has $q = m^2(b\cdot b) \le 0$: it is orthogonal and of electric type everywhere.*

A mass does not show in the angle between the fields of a single mode; it shows in the other invariant, which it makes nonzero. The null cone $q = 0$ is the massless locus, and orthogonality is a much larger set, containing every single mode of every mass.

Proposition 3.86 (Superposition breaks it)**.** $\operatorname{Pf}(k_1 \wedge b_1 + k_2 \wedge b_2) = \det(k_1^\flat, b_1^\flat, k_2^\flat, b_2^\flat)$. *At a point, two modes give $\mathbf{E}\cdot\mathbf{B} \neq 0$ exactly when their two planes together span spacetime. Waves propagating along one null direction ℓ never do: the total field is $\ell \wedge \sum_j \omega_j b_j$, a single null plane.*

Example 3.87 (Counter-propagating waves)**.** In textbook variables, with $\omega = k$, let a wave along $+z$ polarised along x meet a wave along $-z$. With crossed polarisations, $\mathbf{E} = (\cos(kz-kt), \cos(kz+kt), 0)$ and $\mathbf{B}_{\mathrm{t}} = (\cos(kz+kt), \cos(kz-kt), 0)$; with parallel ones, $\mathbf{E} = (\cos(kz-kt) + \cos(kz+kt), 0, 0)$ and $\mathbf{B}_{\mathrm{t}} = (0, \cos(kz-kt) - \cos(kz+kt), 0)$. Both solve Maxwell's equations, and

$$\begin{aligned} &\text{crossed:} && \mathbf{E}\cdot\mathbf{B}_{\mathrm{t}} = \cos 2kz + \cos 2kt, && |\mathbf{B}_{\mathrm{t}}|^2 - |\mathbf{E}|^2 = 0, \\ &\text{parallel:} && \mathbf{E}\cdot\mathbf{B}_{\mathrm{t}} = 0, && |\mathbf{B}_{\mathrm{t}}|^2 - |\mathbf{E}|^2 = -2(\cos 2kz + \cos 2kt). \end{aligned}$$

The two configurations have the same energy density and the same Poynting vector at every point. The reason is structural: the crossed configuration is the parallel one turned by $\pi/4$ about the z axis and then duality-rotated by $\pi/4$; the rotation preserves the energy density and the Poynting vector along z, the duality rotation preserves the entire energy–momentum tensor, and only the phase of q moves.

Nothing exotic is needed to break orthogonality. Near the ground the fair-weather electric field of the atmosphere, about 100 V/m and vertical [40], meets the geomagnetic field, which is inclined to the horizontal everywhere except on the magnetic equator.

3.8.2 How it changes in time

Proposition 3.88 (The local law)**.** *With $\partial_t\mathbf{B} = \nabla\times\mathbf{E}$ and $\partial_t\mathbf{E} = -\nabla\times\mathbf{B} - \mathbf{j}$,*

$$\partial_t q = -2i\, z\cdot(\nabla\times z) - 2i\, z\cdot\mathbf{j}, \qquad \partial_t(\mathbf{E}\cdot\mathbf{B}) = \mathbf{E}\cdot\nabla\times\mathbf{E} - \mathbf{B}\cdot\nabla\times\mathbf{B} - \mathbf{j}\cdot\mathbf{B},$$

in textbook variables $\partial_t(\mathbf{E}\cdot\mathbf{B}_{\mathrm{t}}) = \mathbf{B}_{\mathrm{t}}\cdot\nabla\times\mathbf{B}_{\mathrm{t}} - \mathbf{E}\cdot\nabla\times\mathbf{E} - \mathbf{j}\cdot\mathbf{B}_{\mathrm{t}}$. With potentials, $\mathbf{B}_{\mathrm{t}} = \nabla\times\mathbf{A}$, $\mathbf{E} = -\partial_t\mathbf{A} - \nabla\phi$, and in a source-free region $\mathbf{E} = -\nabla\times\mathbf{C}$, $\mathbf{B}_{\mathrm{t}} = -\partial_t\mathbf{C} + \nabla\chi$,

$$\partial_t(\mathbf{A}\cdot\mathbf{B}_{\mathrm{t}}) + \nabla\cdot(\phi\mathbf{B}_{\mathrm{t}} + \mathbf{E}\times\mathbf{A}) = -2\,\mathbf{E}\cdot\mathbf{B}_{\mathrm{t}}, \qquad \partial_t(\mathbf{C}\cdot\mathbf{E}) - \nabla\cdot(\mathbf{B}_{\mathrm{t}}\times\mathbf{C} + \chi\mathbf{E}) = -2\,\mathbf{E}\cdot\mathbf{B}_{\mathrm{t}}.$$

In vacuum, $\mathbf{E}\cdot\mathbf{B}$ changes at a point only through $\mathbf{E}\cdot\nabla\times\mathbf{E}$ and $\mathbf{B}\cdot\nabla\times\mathbf{B}$, which measure how much each field twists about itself. Integrated over a region with vanishing boundary fluxes,

$\int \mathbf{E}\cdot\mathbf{B}_{\mathrm{t}}$ is minus half the rate of change of the magnetic helicity $\int \mathbf{A}\cdot\mathbf{B}_{\mathrm{t}}$ [170, 11], and equally of the electric helicity, so that their difference, the optical helicity of Theorem 3.81, is conserved in free space [18, 19]. Non-orthogonality is the agent that changes the helicity of each field while their combination is kept. In matter, charges moving along the magnetic field change $\mathbf{E}\cdot\mathbf{B}$ through $-\mathbf{j}\cdot\mathbf{B}$.

Conductors screen it. For the generalised Ohm law $\mathbf{E}+\mathbf{v}\times\mathbf{B}_{\mathrm{t}} = \eta_{\mathrm{R}}\mathbf{j} + (\mathbf{j}\times\mathbf{B}_{\mathrm{t}})/(ne) - (\nabla\cdot\mathsf{P}_e)/(ne) + (m_e/ne^2)\,d\mathbf{j}/dt$ [127],

$$\mathbf{E}\cdot\mathbf{B}_{\mathrm{t}} = \eta_{\mathrm{R}}\,\mathbf{j}\cdot\mathbf{B}_{\mathrm{t}} - \frac{\mathbf{B}_{\mathrm{t}}\cdot(\nabla\cdot\mathsf{P}_e)}{ne} + \frac{m_e}{ne^2}\,\mathbf{B}_{\mathrm{t}}\cdot\frac{d\mathbf{j}}{dt}\,:$$

the flow term and the Hall term drop out, ideal magnetohydrodynamics enforces orthogonality everywhere, and only resistivity, electron pressure and electron inertia allow $\mathbf{E}\cdot\mathbf{B}_{\mathrm{t}}\neq 0$. This is the criterion of Schindler, Hesse and Birn for magnetic reconnection, a nonvanishing integral of $E_{\parallel}$ along field lines [135]. Where charges are scarce the screening fails: a rotating magnetised neutron star in vacuum induces $\mathbf{E}\cdot\mathbf{B}_{\mathrm{t}}\neq 0$ at its surface, charges drawn from the surface fill the magnetosphere until $\mathbf{E}\cdot\mathbf{B}_{\mathrm{t}}=0$ [56], and the regions left without enough charge keep $\mathbf{E}\cdot\mathbf{B}_{\mathrm{t}}\neq 0$ and accelerate particles [133].

3.8.3 Non-orthogonality breaks parity and time reversal, and is never spontaneous

Theorem 3.89 (Orthogonality is achirality)**.** *For a field strength F at a point the following are equivalent: (a) $\mathbf{E}\cdot\mathbf{B}=0$; (b) the mirror image PF equals ΛF for some proper orthochronous Lorentz transformation Λ; (c) the time-reversed field TF equals ΛF for some such Λ.*

Proof. If $\mathbf{E}\cdot\mathbf{B}=0$ and $\mathbf{B}\neq 0$, the half-turn about $\mathbf{B}$ fixes $\mathbf{B}$ and reverses every vector perpendicular to it, sending $(\mathbf{B};\mathbf{E})$ to $(\mathbf{B};-\mathbf{E}) = P(\mathbf{B};\mathbf{E})$; if $\mathbf{B}=0$ a half-turn about an axis perpendicular to $\mathbf{E}$ does the same. Conversely the Pfaffian is invariant under proper Lorentz transformations and changes sign under P, so $\operatorname{Pf} F = -\operatorname{Pf} F$. For (c), $T=CP$ on fields and $CF=-F$ has the same invariant q; since $\mathfrak{l}_G\cong\mathfrak{so}(3,\mathbb{C})$ acts transitively on each level set of q with the origin removed, $-F$ lies in the proper Lorentz orbit of F, and (c) is equivalent to (b). □

A non-orthogonal field is therefore chiral in Kelvin's sense, a screw of definite handedness, and the sign of $\mathbf{E}\cdot\mathbf{B}$ is that handedness; it is odd under P and T and even under C, PT and CPT. This is a statement about a field at a point. A configuration can be orthogonal everywhere and still chiral as a whole: a static Beltrami magnetic field with $\mathbf{E}=0$ carries a nonzero helicity density. The global chirality of a magnetic configuration is its helicity, P-odd and T-even, and $\mathbf{E}\cdot\mathbf{B}_{\mathrm{t}}$, P-odd and T-odd, is minus half its rate of change.

For the Maxwell energy the breaking is never spontaneous. Since $u^2-|\mathbf{S}|^2 = \frac{1}{4}|q|^2$ for the energy density $u=\frac{1}{2}(|\mathbf{E}|^2+|\mathbf{B}|^2)$ and the Poynting vector $\mathbf{S}$,

$$u \;\geq\; \tfrac{1}{2}|q| \;\geq\; |\mathbf{E}\cdot\mathbf{B}|, \tag{3.16}$$

and the minimum of the energy is the zero field; there is no degenerate family of lowest-energy configurations among which $\mathbf{E}\cdot\mathbf{B}$ could choose a sign. In the quantum theory the corresponding statement is the theorem of Vafa and Witten: in a vector-like gauge theory at $\theta=0$, zero temperature and zero density, parity is not broken spontaneously through a Bose order parameter such as $F\tilde{F}$ [153]. When $\mathbf{E}\cdot\mathbf{B}_{\mathrm{t}}$ is imposed from outside the vacuum responds, for instance by rotating the chiral condensate of quantum chromodynamics through the electromagnetic triangle anomaly [21]; that is the response of a vacuum to an explicit breaking, not a breaking of its own. Both statements have hypotheses that are part of them. The classical one uses the quadratic Maxwell energy, whose minimum is the zero field; it does not cover modified actions

whose energy has minima at nonzero fields. The quantum one needs a vector-like theory with a positive Euclidean measure ($\theta = 0$, zero density); it says nothing about chiral sectors such as the electroweak block, or about actions with an explicit $\mathbf{E}\cdot\mathbf{B}$ term.

Duality behaves differently. The rotation $X \mapsto e^{\vartheta G}X$ leaves the energy–momentum tensor invariant and multiplies q by $e^{2i\vartheta}$, so for $q \neq 0$ only $\vartheta \in \pi\mathbb{Z}$ fixes the invariants, while for $q = 0$ every rotation does. The fields whose invariants are unchanged by duality are the null ones, not the orthogonal ones, and orthogonality is not a symmetric point of duality. On $\mathbb{R}^{3,3}$, moreover, $(e^{\vartheta G})^{\top}\eta\, e^{\vartheta G} = \cos 2\vartheta\,\eta + \sin 2\vartheta\,\eta G$, so $e^{\vartheta G}$ is an isometry of η only for $\vartheta \in \pi\mathbb{Z}$. Duality is broken explicitly by electric sources, as Section 5.8 showed, and by the vacuum of a theory with charged particles. For a Lagrangian $L(\mathcal{F},\mathcal{G})$ of the invariants $\mathcal{F} = \frac{1}{2}(|\mathbf{E}|^2 - |\mathbf{B}_{\mathrm{t}}|^2)$ and $\mathcal{G} = \mathbf{E}\cdot\mathbf{B}_{\mathrm{t}}$, with $\mathbf{D} = \partial L/\partial\mathbf{E}$ and $\mathbf{H} = -\partial L/\partial\mathbf{B}_{\mathrm{t}}$,

$$\mathbf{D}\cdot\mathbf{H} - \mathbf{E}\cdot\mathbf{B}_{\mathrm{t}} = \mathcal{G}(L_{\mathcal{F}}^2 - L_{\mathcal{G}}^2 - 1) - 2\mathcal{F}L_{\mathcal{F}}L_{\mathcal{G}},$$

which vanishes for the Maxwell and Born–Infeld Lagrangians, the duality-invariant ones [46, 53]. For the weak-field Euler–Heisenberg Lagrangian [66, 37] it is $-\frac{8\alpha^2}{15m^4}\mathcal{F}\mathcal{G} + O(\alpha^3)$: the duality defect of the vacuum of quantum electrodynamics is proportional, at leading order, to $\mathbf{E}\cdot\mathbf{B}_{\mathrm{t}}$. For every parity-even Lagrangian $L = h(\mathcal{F},\mathcal{G}^2)$ it carries the explicit factor $\mathcal{G}$, so it vanishes on every orthogonal field.

3.8.4 Non-orthogonality does not generate mass

Mass can enter in four ways: as the rest energy of the field itself, as a photon mass, as a mass of charged matter in the background, and through an $\mathbf{E}\cdot\mathbf{B}$-type term in an action. None of them is produced by non-orthogonality.

Rest energy. For $q \neq 0$ the field has a local rest frame, moving with velocity $\mathbf{S}/(u + \frac{1}{2}|q|)$, in which the Poynting vector vanishes and the energy density is $\frac{1}{2}|q| = u\sqrt{1 - |\mathbf{v}_E|^2}$, $\mathbf{v}_E = \mathbf{S}/u$; null fields transport their energy at the speed of light and have no rest frame. The rest-frame energy density is the modulus of q, orthogonality its phase, and duality turns the phase at fixed energy, momentum and stress. For a finite configuration with total energy U and momentum $\mathbf{P}$, $M^2 = U^2 - |\mathbf{P}|^2$ is positive whenever $q \neq 0$ on a set of positive measure, but M does not detect orthogonality: the Cauchy data $(\mathbf{E},\mathbf{B}_{\mathrm{t}}) = (0,\mathbf{b})$ and $(\mathbf{b}\sin\vartheta, \mathbf{b}\cos\vartheta)$ for a compactly supported divergence-free $\mathbf{b}$ have the same energy density and zero Poynting vector at every point, hence the same M, while $\mathbf{E}\cdot\mathbf{B}_{\mathrm{t}}$ is 0 for the first and $\frac{1}{2}\sin 2\vartheta\,|\mathbf{b}|^2$ for the second.

Photon mass. Every Lorentz-covariant linear map from two-forms to vectors is zero; more strongly, a continuous covariant vector-valued function of the value of F at a point vanishes, because a non-null field is fixed by a rotation and a boost about a common axis that have no common fixed vector, and every tensor power of $(1,0)\oplus(0,1)$ contains only representations (j_1,j_2) with integer j_1, j_2. A mass term in $\partial_\mu F^{\mu\nu} = j^\nu$ must therefore involve the potential, other fields or derivatives of F; no function of the two invariants can serve. The Podolsky-type higher-derivative extension introduces an additional massive pole with opposite residue, a ghost [324]. A constant nonlinear background moves the light cone without opening a gap, $\det W(s\omega, s\mathbf{k}) = s^6 \det W(\omega,\mathbf{k})$. For $L = \mathcal{F} + c_1\mathcal{F}^2 + c_2\mathcal{G}^2$ and a normalised propagation direction k, the two phase velocities are to first order $1 - c_i\Theta^{\mu\nu}k_\mu k_\nu$. This Lagrangian is not duality invariant in general: its duality defect is $4(c_1 - c_2)\mathcal{F}\mathcal{G} + 4c_1(c_1 - 2c_2)\mathcal{F}^2\mathcal{G} - 4c_2^2\mathcal{G}^3$, which vanishes to first order only if $c_1 = c_2$. Written into the field equations, the binding rotation of Section 4.3 gives the frequencies $\kappa \pm |\mathbf{k}|$ to the two circular polarisations, a frequency offset that the time-dependent duality rotation $z = e^{-i\kappa t}w$ removes.

Charged matter. The invariant ϵ_E vanishes exactly when $\mathbf{E}\cdot\mathbf{B} = 0$ and $|\mathbf{B}| \geq |\mathbf{E}|$, so every non-orthogonal field has $\epsilon_E > 0$. In a constant background, or one controlled as locally constant, this makes the vacuum of spinor electrodynamics decay into pairs [138], converting field energy into particles whose mass is a given parameter; for general backgrounds the spatial extent, the time dependence and the available energy also matter. For massless fermions the same field

Maxwell field	Dirac field
duality $e^{\vartheta G}$, turning q by 2ϑ	chiral rotation $e^{i\alpha\gamma_5}$, turning $\sigma + i\omega$ by 2α
energy–momentum tensor invariant	vector current J^μ invariant
$u^2 - \lvert\mathbf{S}\rvert^2 = \frac{1}{4}\lvert q\rvert^2$	$J^\mu J_\mu = \sigma^2 + \omega^2$
$\lvert\mathbf{B}\rvert^2 - \lvert\mathbf{E}\rvert^2$ and $2\,\mathbf{E}\cdot\mathbf{B}$	σ and ω
null field, $q = 0$	Weyl spinor, $\sigma = \omega = 0$
one mode: q real	one plane wave: $\omega = 0$
two modes whose planes span spacetime: $\operatorname{Im} q \neq 0$	both signs of frequency: $\omega \neq 0$
$\arg q$, twice the Rainich complexion	$\arg(\sigma + i\omega)$, the Yvon–Takabayasi angle

Table 3.7: The invariants of a field strength and the bilinears of a Dirac spinor. Each line is an identity or a worked example; the parallel is exact at the level of these identities and is not an identification of the two fields.

pumps chirality, $dN_5/dt = (e^2/2\pi^2)\int \mathbf{E}\cdot\mathbf{B}_\mathrm{t}$, the axial anomaly [1, 10, 108]; a fermion mass relaxes it, and none is produced. The background that does catalyse a fermion mass is a pure magnetic field, which is orthogonal [62].

Topological terms. In four dimensions the abelian term $\theta\,\mathbf{E}\cdot\mathbf{B}_\mathrm{t}$ is a total derivative and invisible to the field equations; a varying θ makes it a source, as in axion electrodynamics [164], and a constant gradient gives birefringence [22]. For a timelike gradient both branches remain gapless; for a spacelike gradient p^μ the dispersion relation $(k\cdot k)^2 + (k\cdot k)(p\cdot p) - (k\cdot p)^2 = 0$ gives, at zero wave vector, $\omega = 0$ on one branch and $\omega = |\mathbf{p}|$ on the other. That gap is a Lorentz-violating dispersion fixed by an external pseudoscalar, not a Proca mass, and it does not arise without the varying θ. In every known mechanism in which a term of this type makes mass, something is essential that a single abelian field on a topologically trivial space lacks: the Chern–Simons term in three dimensions [35], the anomaly in the two-dimensional Schwinger model [139], the topological susceptibility of a non-abelian theory with light fermions for the η' [168, 156], or a pseudoscalar partner for the axion [159, 163]. The frame field of this arena is a single abelian field strength with no fermions and no pseudoscalar partner, so no route from $\mathbf{E}\cdot\mathbf{B}$ to mass is open within what the arena has built.

The Dirac field supplies an exact counterpart, collected in Table 3.7. Its scalar and pseudoscalar densities $\sigma = \bar\psi\psi$ and $\omega = \bar\psi i\gamma_5\psi$ turn under a chiral rotation $e^{i\alpha\gamma_5}$ by the angle 2α, as $(|\mathbf{B}|^2 - |\mathbf{E}|^2, 2\mathbf{E}\cdot\mathbf{B})$ turn under duality; a plane wave has $\omega = 0$ as a single mode has $\operatorname{Im} q = 0$; and a pseudoscalar mass enters the mass shell only through $\sqrt{m_s^2 + m_p^2}$. The pseudoscalar density is a phase, not a mass, and $\mathbf{E}\cdot\mathbf{B}$ has exactly its place.

In the interior dynamics of the next part, orthogonality is broken by the bound motion itself. On an orbit without momentum the flow is the duality rotation $z \mapsto e^{-i\kappa\tau}z$, so $\mathbf{E}\cdot\mathbf{B}(\tau) = \frac{1}{2}|q_0|\sin(\phi_0 - 2\kappa\tau)$, which vanishes four times per period, and the rest-frame field turns from magnetic through screws to electric and back at the fixed rest-frame energy density $\frac{1}{2}|q_0|$. That is the content of Figure 3.1 read along a trajectory, and Section 4.3 derives it. It is a rotation of the phase of q at fixed modulus, and by everything in this section it moves no mass.

3.9 The undivided energy and its differentiation

The energy density of a field point, $\frac{1}{2}(|\mathbf{B}|^2 + |\mathbf{E}|^2)$, is a single quantity: it treats the magnetic and the electric triple alike and it is unchanged by duality. A field point also has a rest energy, the energy density seen in a frame in which its Poynting vector vanishes. This section asks how the second is contained in the first: which quadratic energy the arena singles out and what its symmetry is, how that energy differentiates into a Lorentz-invariant rest part and a part carried by motion without any interaction or second field, where the conversion constant c enters, and which of the arena's own structures can carry the name of a mass. The relation

$E = mc^2$ [38] is inherited; what is at stake is only how the arena realises its ingredients. Most of the answers are inherited electrodynamics written in the arena's variables, and are said to be so. In particular, the local invariant $E_0 = \frac{1}{2}|z \cdot z|$ read as a rest energy and an inertia, the energy-flow velocity $|\mathbf{S}|/E_6$ and the standing-wave examples below appear in Kaiser's analysis of electromagnetic inertia and reactive energy [298], which this section restates and compares in the arena's variables. The arena adds which of its forms plays which role. Throughout, c enters only as the conversion between the two triples of Theorem 2.2, never as a symmetry.

Notation. We use the field-strength reading of Theorem 3.3: $X = \sum_i (B_i e_i + E_i e_{i+3})$, $z = \mathbf{B} + i\mathbf{E} \in \mathbb{C}^3$, the binding root G acts as multiplication by i, and $q(z) = z \cdot z$ has $\operatorname{Re} q = \eta(X, X) = |\mathbf{B}|^2 - |\mathbf{E}|^2$ and $\operatorname{Im} q = 2\,\mathbf{E} \cdot \mathbf{B} = -\eta(X, GX)$. With the Hermitian product $h(z, w) = \sum_j \bar{z}_j w_j$ we put

$$\begin{aligned} E_6 &:= \tfrac{1}{2}|z|^2 = \tfrac{1}{2}(|\mathbf{B}|^2 + |\mathbf{E}|^2), & E_0 &:= \tfrac{1}{2}|q(z)|, \\ \mathbf{S} &:= \mathbf{E} \times \mathbf{B} = \tfrac{i}{2}\,\bar{z} \times z = -\mathbf{S}_{\mathrm{t}}, & \mathbf{S}_{\mathrm{t}} &:= \mathbf{E} \times \mathbf{B}_{\mathrm{t}}. \end{aligned} \tag{3.17}$$

E_6 is the energy density u of Section 3.8, $\mathbf{S}$ the Poynting vector of the frame (Example 3.11) and $\mathbf{S}_{\mathrm{t}}$ the textbook one. Applied to a frame field X these are the frame field's energies; for the momentum field $\pi = \eta^{-1} p$ of the interior dynamics studied below, (4.11) and Section 4.4 we write $E_6(\pi)$, $E_0(\pi)$ and $\mathbf{S}(\pi)$. Units are Heaviside–Lorentz with $c = 1$ until Section 3.9.6. The generators L_i, K_i of $\mathfrak{l}_G$ (Theorem 3.3) act by $\delta\mathbf{B} = e_i \times \mathbf{B}$, $\delta\mathbf{E} = e_i \times \mathbf{E}$ and by $\delta\mathbf{E} = e_i \times \mathbf{B}$, $\delta\mathbf{B} = -e_i \times \mathbf{E}$ respectively, so that $\boldsymbol{\omega} \cdot L + \mathbf{v} \cdot K$ acts on z as $z \mapsto (\boldsymbol{\omega} + i\mathbf{v}) \times z$. We write $\mathfrak{p}_G$ for the part of $\mathfrak{so}(3,3)$ anticommuting with G, so that $\mathfrak{so}(3,3) = \mathfrak{l}_G \oplus \mathfrak{p}_G$, and the Lorentz-class inertia of Theorem 4.26 as $N = (\nu_1 \eta^{-1} + \nu_2 G \eta^{-1})/m$, its coefficients renamed so that the letter b is free for a gauge generator in Section 9.2.

3.9.1 One undivided energy, with pointwise symmetry $U(3)$

Theorem 3.90 (Uniqueness of the undivided energy)**.** *(i) The symmetric forms on V invariant under the diagonal rotations are spanned by $|\mathbf{B}|^2$, $|\mathbf{E}|^2$ and $\mathbf{B} \cdot \mathbf{E}$, and they split as $\mathbb{R}\,\mathbf{1} \oplus \operatorname{span}\{\eta, \eta G\}$, the first summand commuting and the second anticommuting with G.*

(ii) A quadratic energy invariant under the diagonal rotations and under the duality rotations $e^{\vartheta G}$ is a multiple of E_6, which is positive definite.

(iii) The forms invariant under all of $\mathfrak{l}_G$ are $\operatorname{span}\{\eta, \eta G\}$, that is $|\mathbf{B}|^2 - |\mathbf{E}|^2$ and $\mathbf{E} \cdot \mathbf{B}$; none of them is definite.

Proof. The diagonal rotations act on $V = V^+ \oplus V^-$ as $\mathbb{R}^2 \otimes \mathbb{R}^3$ with $SO(3)$ on the second factor, which is absolutely irreducible, so by Schur's lemma [57] the invariant forms are $N = \left(\begin{smallmatrix} \alpha & \gamma \\ \gamma & \beta \end{smallmatrix}\right) \otimes \mathbf{1}$. With $G = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right) \otimes \mathbf{1}$ one finds $\frac{1}{2}(N - GNG) = \frac{1}{2}(\alpha + \beta)\mathbf{1}$ and $\frac{1}{2}(N + GNG) = \frac{1}{2}(\alpha - \beta)\eta - \gamma\,\eta G$. Since $G^{\top} G = \mathbf{1}$ and $G^{\top} = -G$, N is invariant under $e^{\vartheta G}$ iff $G^{\top} N + NG = 0$ iff $[N, G] = 0$, which gives (ii). For (iii), the boost law gives $\delta(\alpha|\mathbf{B}|^2 + \beta|\mathbf{E}|^2 + 2\gamma\,\mathbf{B} \cdot \mathbf{E}) = -2(\alpha + \beta)\,\mathbf{v} \cdot (\mathbf{E} \times \mathbf{B})$, the γ term being unchanged, so $\beta = -\alpha$. □

The theorem restates Theorem 4.26(iii) and Proposition 4.19 on V rather than on V^*, and it is inherited: for every conjugate root $G' = gGg^{-1}$, $g \in SO(3,3)$, the forms invariant under the compact part of $\mathfrak{l}_{G'}$ and under G' are the multiples of the positive form $g^{-\top} g^{-1}$, so the statement could not have failed. It is the uniqueness of the energy density among rotation- and duality-invariant quadratic forms. What the arena adds is that this form is the Euclidean form of its own basis: G is η-self-adjoint and anticommutes with $\operatorname{Ad}_{e_1 e_2 e_3}|_V = \eta$ (Lemma 2.4), so $G^{\top} G = \eta G \eta G = \mathbf{1}$. E_6 has duality weight zero, while $e^{\vartheta G}$ turns $(\operatorname{Re} q, \operatorname{Im} q)$ through 2ϑ. In this sense E_6 is undivided: it distinguishes neither the magnetic triple from the electric one nor, as Section 3.9.3 shows, rest from motion.

Proposition 3.91 (The pointwise symmetry of the undivided energy)**.** *(i) The infinitesimal maps of V that preserve E_6 and commute with G form the Lie algebra $\mathfrak{u}(3)$, of dimension 9, of the group $U(3)$: $A = \left(\begin{smallmatrix} P & Q \\ -Q & P \end{smallmatrix}\right)$ acts on z by the anti-Hermitian matrix $P - iQ$.*

(ii) The G-linear maps preserving q up to a phase form the group $O(3,\mathbb{C}) \cdot U(1)_G$, whose two factors meet in $\{\pm\mathbf{1}\}$; its Lie algebra is $\mathfrak{l}_G \oplus \mathbb{R}G$, of dimension 7.

(iii) The intersection of the two is $\mathfrak{so}(3) \oplus \mathbb{R}G$, of dimension 4, the maximal compact subalgebra of $\mathfrak{l}_G \oplus \mathbb{R}G$; and $\mathfrak{u}(3) \cap \mathfrak{l}_G = \mathfrak{so}(3)$.

(iv) The stabiliser of the function E_0 in $U(3)$ is $U(1) \cdot O(3) = \{e^{i\varphi} O\}$. For instance the unitary map

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 & 0 \\ i & -i & 0 \\ 0 & 0 & \sqrt{2} \end{pmatrix}$$

carries the field $(1,0,0)$, at rest with $E_0 = E_6 = \frac{1}{2}$, to the null field $\frac{1}{\sqrt{2}}(1,i,0)$ at the same E_6.

Proof. (i) Commuting with G forces the block form, which is multiplication by $P - iQ$ on z; $A^\top = -A$ means $P^\top = -P$ and $Q^\top = Q$, so $P - iQ$ is anti-Hermitian, and $3 + 6 = 9$. (ii) $q(e^{tM} z) = e^{i\varphi t} q(z)$ gives $M^\top + M = i\varphi\,\mathbf{1}$, so $M = \frac{i\varphi}{2} + M_0$ with $M_0 \in \mathfrak{so}(3,\mathbb{C}) = \mathfrak{l}_G$ (Theorem 3.1), and $i\,\mathbf{1}$ is G. (iii) A complex antisymmetric anti-Hermitian matrix is real. (iv) If $U \in U(3)$ preserves $|q|$, the form $z^\top U^\top U z$ vanishes on the cone $q = 0$; q is irreducible, so $U^\top U = e^{i\varphi}\mathbf{1}$, and $O = e^{-i\varphi/2} U$ is unitary and complex orthogonal, hence real. □

All four parts are inherited Hermitian linear algebra, which makes the word *unitary* literally true of the pointwise energy; part (iv) is the stabiliser of $|z \cdot z|$ among unitary maps of the Riemann–Silberstein vector [142, 13]. Read as a statement about the rest energy, (iv) says exactly in which sense E_0 reduces a symmetry: $U(3)$ is reduced to $U(1) \cdot O(3)$, and the five-dimensional orbit $U(3)/U(1) \cdot O(3)$ parametrises the real structures $z \mapsto \bar{z}$ up to phase, of which the arena has chosen one (Proposition 7.5). But $U(3)$ is a symmetry of the pointwise form E_6 only. It is not a symmetry of Maxwell's equations on the frame, $\partial_t z = -i\nabla \times z$ (Proposition 3.10): the map U of (iv) carries the wave $(1,i,0)e^{ik(x_3 - t)}$ of Example 3.11 to $\frac{1+i}{\sqrt{2}}(1,1,0)e^{ik(x_3-t)}$, which does not solve them. Nor is it a symmetry of the split-form interior, since $G = i\,\mathbf{1}$ lies in $U(3)$ and reverses η, and $U(3) \cap O(\eta) = O(3)$. Of the closures of Section 4.4 it is a symmetry only for the inertia $N = \mathbf{1}/m$, because the $U(3)$-invariant symmetric forms are the multiples of $\mathbf{1}$.

3.9.2 The rest energy is the complete invariant of Lorentz transformations and duality

Theorem 3.92 (Orbits)**.** *(i) Each level set $\{|q| = r^2\}$, $r > 0$, is one orbit of $O(3,\mathbb{C}) \times U(1)_G$, of real dimension 5, with two-dimensional stabilisers; every non-null z is carried to $(r,0,0)$ with $r = \sqrt{2E_0}$.*

(ii) The nonzero null fields form one orbit of $SO(3,\mathbb{C})$ alone, of dimension 4, and at a null point Gz is tangent to the Lorentz orbit.

(iii) Hence E_0 is a complete invariant of Lorentz transformations and duality on non-null fields, and E_6 is the complete invariant of $U(3)$, whose orbits are the spheres.

Proof. (i) Write $q(z) = r^2 e^{i\varphi}$ and $w = e^{-i\varphi/2} z/r$, a duality rotation, so that $q(w) = 1$. Since q is nondegenerate on $w^\perp = \{f : f \cdot w = 0\}$, some $f \in w^\perp$ has $f \cdot f = 1$; with $f_3 = w \times f$, the identity $(a \times b) \cdot (a \times b) = (a \cdot a)(b \cdot b) - (a \cdot b)^2$ gives $f_3 \cdot f_3 = 1$, so $A = (w\ f\ f_3)^\top \in SO(3,\mathbb{C})$ and $Ae^{-i\varphi/2} z = re_1$. An element $M_0 + \frac{i\varphi}{2}$ fixes e_1 iff $\varphi = 0$ and M_0 is a complex rotation about e_1;

the stabiliser is $\mathfrak{so}(2,\mathbb{C})$ and the orbit has dimension $7-2=5$. (ii) A rotation carries a null z to $\lambda(1,i,0)$ with $\lambda>0$. K_3 acts by $\delta z = i\,e_3\times z$, and $i\,e_3\times(1,i,0)=(1,i,0)$, so the boost $e^{\chi K_3}$ with $\chi=-\log\lambda$ finishes. The Lorentz tangent space at a null z is its q-orthogonal complement, which contains iz; at $(1,i,0)$, $iz=-e_3\times z=-L_3 z$ is the action of a rotation about e_3. (iii) follows from (i) and (ii). □

This is inherited, the classification of a field strength at a point [148, 82]. Without duality the complete invariant on nonzero fields is q itself (the transitivity used in the proof of Theorem 3.89), and duality removes its phase, which is twice the complexion of Proposition 3.12. At the null point $(1,i,0)$ the tangent $Gz=-L_3 z$ is generated by the rotation about the propagation axis, the identity of the G-charge and the helicity of Example 3.82.

3.9.3 The differentiation identity

The relation $u^2-|\mathbf{S}|^2=\frac{1}{4}|q|^2$ was used in Section 3.8 for the bound (3.16). Its structure is the following.

Proposition 3.93 (Differentiation of the undivided energy)**.** *For every $z\in\mathbb{C}^3$*

$$|z|^4=|z\cdot z|^2+|\bar z\times z|^2,\qquad \textit{equivalently}\qquad E_6^2=E_0^2+|\mathbf{S}|^2. \tag{3.18}$$

The rank-one Hermitian matrix $H=zz^\dagger$ carries all three terms: $\operatorname{tr}H=2E_6$, $(\operatorname{Im}H)_{jk}=\epsilon_{jkl}S_l$, $\operatorname{tr}(H\bar H)=|q|^2=4E_0^2$ *and* $H^2=(\operatorname{tr}H)H$. *Hence $E_0\le E_6$, with equality iff $\mathbf{S}=0$, iff $\bar z\parallel z$, iff $\mathbf{B}\parallel\mathbf{E}$ or one of them vanishes.*

Proof. $\bar z\times z=-2i\,\mathbf{E}\times\mathbf{B}$, and $(a\times b)\cdot(c\times d)=(a\cdot c)(b\cdot d)-(a\cdot d)(b\cdot c)$ at $(\bar z,z,z,\bar z)$ gives $|\bar z\times z|^2=|z|^4-|q|^2$. Next $H_{jk}=B_jB_k+E_jE_k+i\epsilon_{jkl}(\mathbf{E}\times\mathbf{B})_l$ and $\operatorname{tr}(H\bar H)=|\sum_j z_j^2|^2$. The equality case is the equality case of the Cauchy–Schwarz inequality $|h(\bar z,z)|\le|z|^2$. □

The identity is inherited, a complex Lagrange identity. It is the exact form of the differentiation asked about: the undivided E_6 is the hypotenuse of a right triangle whose legs are a Lorentz-invariant rest part E_0 and a motion part $|\mathbf{S}|$, and the split needs no interaction and no second field. In Minkowski terms E_0 is the Lorentz length of $(E_6,\mathbf{S})$, although $(E_6,\mathbf{S}_{\mathrm{t}})$ is not a four-vector, as the next proposition makes precise. Three quantities must be kept apart. E_0 is a pointwise invariant: for a non-null field it is the energy density in the local frame where $\mathbf{S}=0$, and it vanishes on null fields. It is not the invariant mass of a localised field configuration, $M=\big((\int E_6)^2-|\int\mathbf{S}|^2\big)^{1/2}$ with $\Theta^{0\mu}$ integrated over space. The vectors $(E_6,\mathbf{S})(x)$ are causal, so by the reverse triangle inequality $M\ge\int E_0\,d^3x$, with equality only when all points share one rest frame. And neither is a generated particle mass: the mass m of the Dirac form of Section 2.3 is an input of the Lagrangian. To avoid a clash with the trivector $e_1e_2e_3$ we write $\Theta^{\mu\nu}$ for the stress–energy tensor.

Proposition 3.94 (Rainich identity)**.** *For the stress–energy tensor of $(\mathbf{E},\mathbf{B}_{\mathrm{t}})$, $\Theta^{00}=\frac{1}{2}(|\mathbf{E}|^2+|\mathbf{B}_{\mathrm{t}}|^2)=E_6$, $\Theta^{0i}=(\mathbf{S}_{\mathrm{t}})_i$ and $\Theta^{ij}=\Theta^{00}\delta_{ij}-E_iE_j-B_{\mathrm{t}i}B_{\mathrm{t}j}$, one has $\Theta^\mu{}_\mu=0$ and*

$$\Theta^\mu{}_\lambda\Theta^\lambda{}_\nu=E_0^2\,\delta^\mu{}_\nu,\qquad \Theta_{\mu\nu}\Theta^{\mu\nu}=4E_0^2. \tag{3.19}$$

$\Theta^{\mu\nu}$ is invariant under duality, and for every unit timelike observer t the density $j_t^\mu=\Theta^\mu{}_\nu t^\nu$ is future causal with $\eta(j_t,j_t)=E_0^2$.

Proof. Duality invariance is a polynomial identity. Both sides of (3.19) are covariant, so by Theorem 3.92 it suffices to take $z=(r,0,0)$, where $\Theta^\mu{}_\nu=\frac{r^2}{2}\operatorname{diag}(1,1,-1,-1)$; null fields follow by continuity. That j_t is causal and future directed is the dominant energy condition of the Maxwell field [65]. □

This is inherited [129, 148]. The rest energy of a field point is the Lorentz scalar of the tensor whose 00 component is the undivided energy; but each observer t sees a different density j_t, all of Lorentz length E_0, and no single four-vector carries the energy and momentum of a field point.

3.9.4 The rest energy as a minimum, and two boost factors

Theorem 3.95 (Variational form of the rest energy)**.** *(i) For non-null z, E_0 is the minimum of E_6 over the Lorentz orbit, and over the orbit of Lorentz transformations and duality, attained exactly where $\mathbf{S}=0$; for null $z\neq 0$ the infimum 0 is not attained. Moreover $E_6 = E_0/\sqrt{1-v_E^2}$ with $v_E := |\mathbf{S}|/E_6 < 1$.*

(ii) If $\mathbf{B} \parallel \mathbf{E} \parallel \mathbf{n}$ and the boost $e^{\chi K}$ of rapidity χ has axis orthogonal to $\mathbf{n}$, then $E_6 = E_0\cosh 2\chi$, $|\mathbf{S}| = E_0|\sinh 2\chi|$ and $v_E = |\tanh 2\chi| = 2|\beta|/(1+\beta^2)$ with $\beta=\tanh\chi$ (for $\chi\geq 0$ the absolute values can be dropped). A boost along $\mathbf{n}$ changes neither E_6 nor $\mathbf{S}=0$.

(iii) The observer moving with velocity $\mathbf{v}_{\text{rest}} = \mathbf{S}_{\text{t}}/(E_6+E_0)$ sees $\mathbf{S}=0$ and $E_6=E_0$, and $|\mathbf{v}_{\text{rest}}| = |\mathbf{S}|/(E_6+E_0)$, which is $|\beta|$ in the setting of (ii). For a nonzero null field the formula gives unit speed, and there is no timelike rest observer.

Proof. (i) By (3.18), $E_6\geq E_0$ with equality iff $\mathbf{S}=0$, and Theorem 3.92 supplies points with $\mathbf{S}=0$ in both orbits; for null z the boost of Theorem 3.92(ii) drives E_6 to 0. Dividing (3.18) by E_6^2 gives the last formula. (ii) Take $\mathbf{n}=e_1$ and $X=\sigma_1 e_1+\sigma_2 e_4$. By Example 3.4 and the same computation for the other basis vectors, $[K_3,e_1]=e_5$, $[K_3,e_5]=e_1$, $[K_3,e_4]=-e_2$ and $[K_3,e_2]=-e_4$, so $\operatorname{ad}_{K_3}^3=\operatorname{ad}_{K_3}$ and $e^{\chi\operatorname{ad}_{K_3}} = \mathbf{1}+\sinh\chi\,\operatorname{ad}_{K_3}+(\cosh\chi-1)\operatorname{ad}_{K_3}^2$. Hence $\mathbf{B}=(\sigma_1\cosh\chi,-\sigma_2\sinh\chi,0)$, $\mathbf{E}=(\sigma_2\cosh\chi,\sigma_1\sinh\chi,0)$ and $\mathbf{S}=-E_0\sinh 2\chi\, e_3$; likewise for K_2, while K_1 fixes e_1 and e_4. (iii) $|\mathbf{S}|/(E_6+E_0) = |\sinh 2\chi|/(\cosh 2\chi+1) = |\tanh\chi|$; for a null field $E_0=0$ and $|\mathbf{S}|=E_6$. For the direction, $e^{\chi K_3}$ is the passage to a frame moving with velocity $-\tanh\chi\, e_3$ (after Theorem 3.3), in which the rest frame moves along $+e_3$, and so does $\mathbf{S}_{\text{t}}=-\mathbf{S}$. □

The theorem is inherited [82]. Its velocity is the rest-frame velocity of Section 3.8, where the Poynting vector is the textbook $\mathbf{S}_{\text{t}}$, as Section I.1.4 stipulates for formulas from the literature; with the frame vector $\mathbf{S}=\mathbf{E}\times\mathbf{B}$ the direction would be reversed. A field point carries two boost factors, $E_6=E_0\cosh 2\chi$ against $E=mc^2\cosh\chi$ for a particle, because the energy density is a component of a second-order tensor and is quadratic in the boost: the rest frame moves with β, the energy with the relativistic sum $2\beta/(1+\beta^2)$ of β with itself, and an observer riding with the energy velocity sees the same E_6 and the reversed Poynting vector. E_0 is the rest-energy density of a field point, not the rest energy of a particle carried by the flux.

3.9.5 Duality on the invariants, and the stabilisers of field points

The duality rotation $D_\vartheta = e^{\vartheta G} = \cos\vartheta\,\mathbf{1}+\sin\vartheta\, G$ acts as $z\mapsto e^{i\vartheta}z$: it preserves E_6 and $\Theta^{\mu\nu}$, multiplies q by $e^{2i\vartheta}$, preserves the positive form for every ϑ and η only for $\vartheta\in\pi\mathbb{Z}$ (Section 3.8). For $q\neq 0$ exactly the angles $\vartheta\in\pi\mathbb{Z}$ fix q, and for $q=0$ every angle does. If $q\neq 0$, Proposition 3.12 supplies Lorentz coordinates with $z=(f_m+if_e)\hat{e}_1$, f_m, f_e real; then $q=(f_m^2+f_e^2)e^{2i\varphi}$ with $\varphi=\arg(f_m+if_e)$, and $D_{-\varphi}$ makes the field purely magnetic, $z=|f_m+if_e|\,\hat{e}_1$. The angle must be the argument and not the principal value of $\arctan(f_e/f_m)$, which gives $-|f_m+if_e|\,\hat{e}_1$ when $f_m<0$. All of this is the inherited duality rotation of the Maxwell invariants [129, 18, 31].

Proposition 3.96 (The rest energy is the doubly invariant modulus)**.** *Of the four functions E_6, $\operatorname{Re} q$, $\operatorname{Im} q$ and E_0, the first is invariant under $U(1)_G$ but not under $\mathfrak{l}_G$, the next two are invariant under $\mathfrak{l}_G$ but not under $U(1)_G$, and $E_0=\frac{1}{2}|q|$ is invariant under both. On the space*

of Lorentz orbits of nonzero vectors, labelled by $q \in \mathbb{C}$, duality rotates the label through 2ϑ; the stabiliser of a nonzero label is $\{D_0, D_\pi\} = \{\pm\mathbf{1}\}$.

Proof. The six generators of $\mathfrak{l}_G$ commute with G and lie in $\mathfrak{so}(3,3)$, so $\eta(X, AX) = \eta(X, GAX) = 0$; a boost K_i is symmetric for the positive form and $X^\top K_i X \neq 0$ for generic X. The labelling is the transitivity of $\mathfrak{l}_G$ on the level sets of q with the origin removed (proof of Theorem 3.89), and D_ϑ acts on the label by $q \mapsto e^{2i\vartheta} q$. □

The proposition is inherited. With $\mathbf{g} = \mathbf{S}_\mathrm{t}/c^2$ and units restored, (3.18) reads $u^2 - c^2|\mathbf{g}|^2 = E_0^2$, the field form of $E^2 - p^2c^2 = m^2c^4$ (Proposition 3.99). A nonzero rest energy thus breaks $U(1)_G$ to $\mathbb{Z}_2$ on the space of Lorentz orbits, and $\arg q$ is the angle along the broken direction. That is a reading of the proposition, and it comes with two qualifications. The breaking is a property of a configuration, not of a vacuum: the energy is minimised by the zero field and no potential selects $|q| \neq 0$ (Section 3.8). And the broken symmetry is not one of the interior Lagrangian, which duality rotates into its ηG partner.

Theorem 3.97 (Stabilisers of field points)**.** *Let $z \neq 0$.*

(i) $\mathrm{Stab}_{\mathfrak{l}_G}(z) = \{(\zeta z)\times\cdot : \zeta \in \mathbb{C}\}$ *is abelian of real dimension two. If $q \neq 0$ and $\hat{z} = z/\sqrt{q}$, then $R = \hat{z}\times\cdot$ satisfies $R^3 = -R$ (a rotation) and $(GR)^3 = GR$ (a boost) about the common complex axis $\hat{z}$: the stabiliser is $\mathfrak{so}(2) \oplus \mathfrak{so}(1,1)$. If $q = 0$, every nonzero element N of the stabiliser has $N^3 = 0 \neq N^2$: null rotations, $\mathbb{R}^2$.*

(ii) In $\mathfrak{l}_G \oplus \mathbb{R}G$ a non-null z has the same stabiliser, with no G component. A null $z = \mathbf{n}_1 + i\mathbf{n}_2$ has a three-dimensional stabiliser $\cong \mathfrak{e}(2)$, spanned by the null rotations and $L_\mathbf{n} + G$, with $\mathbf{n} = \mathbf{n}_1 \times \mathbf{n}_2/|\mathbf{n}_1|^2 = \mathbf{B}\times\mathbf{E}/|\mathbf{B}\times\mathbf{E}|$ the propagation axis, which acts on the null rotations with eigenvalues $\pm i$. For $z = (1, i, 0)$, $(L_3 + G)z = 0$ and $e^{\vartheta G} z = e^{-\vartheta L_3} z$.

(iii) In $\mathfrak{l}_\mathrm{int}$, on the interior four-plane $\langle e_1, e_4, e_5, e_6\rangle$, $\mathrm{Stab}(e_1) = \mathfrak{so}(3)$ *and* $\mathrm{Stab}(e_1 + e_6) = \mathfrak{e}(2)$*, both three-dimensional.*

Proof. (i) $w \times z = 0$ iff $w \in \mathbb{C}z$. From $(z\times)^2 w = z(z\cdot w) - qw$ one has $(z\times)^3 = -q\,(z\times)$; with $\hat{z}\cdot\hat{z} = 1$ this is $R^3 = -R$, and $G \leftrightarrow i$ gives $(GR)^3 = GR$; for $q = 0$, $(z\times)^3 = 0 \neq (z\times)^2$. (ii) The condition is $w \times z + i\vartheta z = 0$, and dotting with z gives $i\vartheta q = 0$. For null z, $\mathbf{n}_1 \cdot \mathbf{n}_2 = 0$ and $|\mathbf{n}_1| = |\mathbf{n}_2|$ give $\mathbf{n}\times\mathbf{n}_1 = \mathbf{n}_2$ and $\mathbf{n}\times\mathbf{n}_2 = -\mathbf{n}_1$, so $\mathbf{n}\times z = -iz$, $(L_\mathbf{n} + G)z = 0$ and $[L_\mathbf{n}, z\times\cdot] = (\mathbf{n}\times z)\times\cdot = -i(z\times\cdot)$. (iii) is the classical computation of the little groups of a timelike and a null vector. □

The theorem is inherited: (iii) is Wigner's classification [162, 160], and (i)–(ii) are the classical stabilisers of a bivector and the identity of helicity and duality on null fields [121, 18], which on the frame are Theorem 3.79 (the axis of a null field) and Example 3.82 (the G-charge is the helicity). It separates two notions that the word *mass* runs together. The little group of a momentum is three-dimensional with or without mass, and the stabiliser of a field point is two-dimensional with or without q. A mass changes the type of a Lorentz stabiliser, never its dimension, and breaks no Lorentz generator, since q and the square of a momentum are invariants. What a nonzero q removes is the duality generator: a null field point is invariant under duality up to a rotation about its axis, so the group $\mathfrak{e}(2)$ of Wigner reappears once duality is admitted, with the helicity as its compact generator, while for a non-null point duality is broken to $\{\pm\mathbf{1}\}$.

3.9.6 Where c enters

Adding the dimensionless G to a derivative along a clock coordinate needs a length ℓ and gives the rate $\kappa = c/\ell$ of (2.2). On the frame, c enters as follows.

Proposition 3.98 (c^2 is an isotropic weight)**.** *Let the temporal triple carry* $\mathbf{B}$ *and the spatial triple* $\mathbf{E}$*, let* $C : V^+ \to V^-$ *be the conversion and* $G_C = \left(\begin{smallmatrix} 0 & -C^{-1} \\ C & 0 \end{smallmatrix}\right)$*, so that* $G_C^2 = -\mathbf{1}$*.*

(i) G_C *commutes with the diagonal rotations iff* $C = c\,\mathbf{1}$*.*

(ii) For $C = c\,\mathbf{1}$*, writing* G_c *for* G_C*, the forms invariant under the diagonal rotations and under* $e^{\vartheta G_c}$ *are the multiples of* $c^2|\mathbf{B}|^2 + |\mathbf{E}|^2$*.*

(iii) $(\kappa G)^2 = -c^2/\ell^2\,\mathbf{1}$*, and* ℓ *is not fixed.*

Proof. (i) The lower-left block of $[G_C, R \oplus R]$ is $CR - RC$, and Schur's lemma gives $C = c\,\mathbf{1}$; this is Theorem 2.2. (ii) For $N = \left(\begin{smallmatrix} \alpha & \gamma \\ \gamma & \beta \end{smallmatrix}\right) \otimes \mathbf{1}$, $G_c^{\top} N + N G_c = \left(\begin{smallmatrix} 2c\gamma & c\beta - \alpha/c \\ c\beta - \alpha/c & -2\gamma/c \end{smallmatrix}\right) \otimes \mathbf{1}$, which vanishes iff $\gamma = 0$ and $\alpha = c^2\beta$. (iii) $G^2 = -\mathbf{1}$. □

Part (i) is Theorem 2.2, fixed there, and it fails for $C = \operatorname{diag}(c_1, c_2, c_3)$ with distinct entries; part (ii) is inherited, and part (iii) is constructed, a definition, since $G^2 = -\mathbf{1}$ is adopted in Theorem 4.8 and ℓ is chosen. Even the weight is a unit convention: $\Phi = \operatorname{diag}(c, c, c, 1, 1, 1)$ conjugates G_c to G and turns $c^2|\mathbf{B}|^2 + |\mathbf{E}|^2$ into $|\Phi X|^2$, so with $\mathbf{B}$ measured as $c\mathbf{B}$, as in Heaviside–Lorentz units, c disappears from E_6 and appears only in the rest-mass density ρ_0 below.

Proposition 3.99 (SI form)**.** *With* $\varepsilon_0\mu_0 c^2 = 1$*,* $z_c := c\mathbf{B} + i\mathbf{E}$*,* $u = \frac{\varepsilon_0}{2}|z_c|^2$*,* $\mathbf{S}_{\mathrm{t}} = \mathbf{E} \times \mathbf{B}_{\mathrm{t}}/\mu_0$*,* $\mathbf{g} = \mathbf{S}_{\mathrm{t}}/c^2$*,* $q_c := z_c \cdot z_c = c^2|\mathbf{B}_{\mathrm{t}}|^2 - |\mathbf{E}|^2 - 2ic\,\mathbf{B}_{\mathrm{t}} \cdot \mathbf{E}$ *and* $u_0 := \frac{\varepsilon_0}{2}|q_c|$*, Proposition 3.93 applied to* z_c *gives*

$$u^2 = c^2|\mathbf{g}|^2 + u_0^2, \qquad \rho_0 := u_0/c^2, \qquad [u] = [c\,\mathbf{g}] = [u_0] = \mathrm{J\,m^{-3}}, \quad [\rho_0] = \mathrm{kg\,m^{-3}}. \tag{3.20}$$

This is inherited, and calling ρ_0 a rest-mass density is a reading, since $(u, c\,\mathbf{g})$ is not a four-vector (Proposition 3.94). It is the exact content of the clause, taken up in Section 13.5, that c^2 is the symmetric part of $E = mc^2$, and it is less than the statement suggests. The constant c is the unique conversion between the triples that commutes with their common rotations, which are the residual symmetry of both E_6 and E_0; in SI units it enters E_6 as the isotropic weight c^2 of $|\mathbf{B}|^2$, which a choice of units removes. It is a conversion, not a symmetry, and nothing breaks it: u_0 uses the same c, and $\rho_0 = u_0/c^2$ only converts units.

3.9.7 The mass of a whole

For a free field with compactly supported Cauchy data, $(U, \mathbf{P}) = \int (u, \mathbf{S}_{\mathrm{t}})\, d^3x$ is a four-vector and its Lorentz length M is positive whenever $q \neq 0$ on a set of positive measure, while M does not detect orthogonality (Section 3.8). The following bounds and examples locate M relative to the rest energies of the field points.

Proposition 3.100 (Mass of a whole)**.** *(i)* $M^2 = U^2 - |\mathbf{P}|^2 \geq U \int (u - |\mathbf{S}|)\, d^3x \geq 0$*, and in every frame* $M \geq \int E_0\, d^3x$*, with equality iff the densities* $(u, \mathbf{S}_{\mathrm{t}})(x)$ *are all proportional.*

(ii) An infinite monochromatic plane wave has no finite integrated U*,* $\mathbf{P}$ *or* M*; its energy-momentum density is null, and in a finite mode volume the integrated tuple is null,* $M = 0$ *(a finite wave packet with angular spread can have timelike total four-momentum). In SI units the waves* $E_x = a\cos(kz \mp \omega t)$*,* $cB_{\mathrm{t}y} = \pm a\cos(kz \mp \omega t)$ *superpose to* $E_x = 2a\cos kz\cos\omega t$*,* $cB_{\mathrm{t}y} = 2a\sin kz\sin\omega t$*. Per cell (unit area, length* $2\pi/k$*) one has* $U = 2\pi\varepsilon_0 a^2/k$ *and* $\mathbf{P} = 0$*, so* $Mc^2 = U$*, although each constituent has* $q_c = 0$ *and* $U_{\pm} = |\mathbf{P}_{\pm}|c$*. The total has* $q_c = -2a^2(\cos 2kz + \cos 2\omega t)$*,* $\int_{\mathrm{cell}} \Theta^{zz} = U$ *and* $\int_{\mathrm{cell}} \Theta^{xx} = -U\cos 2\omega t$*, and the integrated rest energy* $\int_{\mathrm{cell}} u_0$ *oscillates between* $\frac{2}{\pi}U$ *and* U*.*

(iii) With $c = 1$ and $f(s) = (1 - s^2)^2$ for $|s| \le 1$, $f(s) = 0$ otherwise, the field $E_x = f(z - t) + f(z + t)$, $B_{ty} = f(z - t) - f(z + t)$ has, per unit transverse area, $U = \frac{512}{315}$ and $\mathbf{P} = 0$ at all times, so $M = U$, with (U, P_z) a two-vector under boosts along z only. For $|t| \ge 1$ the two pulses are disjoint, $q \equiv 0$ and $\int E_0 \, d^3x = 0$.

Proof. (i) Conservation and finite propagation speed make $\int_\Sigma \Theta^{\mu\nu} d\Sigma_\nu$ independent of the spacelike hyperplane Σ [84]. Since $u - |\mathbf{S}| = E_0^2/(u + |\mathbf{S}|) \ge 0$ and $M^2 = (U - |\mathbf{P}|)(U + |\mathbf{P}|)$, the first bound follows. Each $(u, \mathbf{S}_t)(x)$ is future causal of Lorentz length $E_0(x)$ (Proposition 3.94), and future causal vectors obey the reversed triangle inequality $|P_1 + P_2| \ge |P_1| + |P_2|$, with equality iff they are proportional. (ii) The cell integrals are elementary. For the last statement, with $s = \sin^2 \omega t$ one has $q_c = 4a^2(s - \cos^2 kz)$, so $u_0 = 2\varepsilon_0 a^2 |s - \cos^2 kz|$; its cell mean is least at the median $s = \frac{1}{2}$ of the arcsine distribution of $\cos^2 kz$, where it is $\frac{2}{\pi}U$, and equals U at $s = 0$ and $s = 1$. (iii) $u = f(z - t)^2 + f(z + t)^2$ and $S_{tz} = f(z - t)^2 - f(z + t)^2$, the cross terms cancelling, and $q = -4f(z - t)f(z + t)$; finally $2\int_{-1}^{1}(1 - s^2)^4 ds = \frac{512}{315}$. □

The first bound and item (ii) are inherited. The cell in (ii) is not isolated: $\int \Theta^{zz} = U$ is the pressure on whatever confines it, so its $(U, \mathbf{P})$ is a four-vector only with the walls included, as for radiation in a box [84]. The inequality $M \ge \int E_0$ and item (iii) obstruct the reading that the mass of a whole emerges from the interference that makes $q \ne 0$: interference suffices for $M > 0$, by (i) and the example (ii), but it is not necessary, by (iii). A whole gets its mass from the non-collinearity of the energy–momenta of its parts, as a pair of photons does.

3.9.8 Splitting the temporal energy lowers the mass

There is a second reading of the phrase that the energy is six-dimensional: the three temporal directions of V^+ carry three energies, and a mass would arise by differentiating them. On the arena this reading fails.

Proposition 3.101 (Splitting the temporal energy lowers the mass)**.** *Let $P = \sum_i (\mathsf{E}_i/c) e_i + \sum_a p_a e_{a+3}$ with $\eta(P, P) = \mu^2 c^2$, let $\widehat{W}$ be the symmetric clock of Proposition 2.5, $\mathsf{E}_W := (\mathsf{E}_1 + \mathsf{E}_2 + \mathsf{E}_3)/\sqrt{3}$, and let $\mathsf{E}_\perp$ be the part of $(\mathsf{E}_1, \mathsf{E}_2, \mathsf{E}_3)$ orthogonal to $(1, 1, 1)$, the energies of the relative times. The mass read by the clock, $m_4^2 c^4 := \mathsf{E}_W^2 - |\mathbf{p}|^2 c^2$, is*

$$m_4^2 c^4 = \mu^2 c^4 - |\mathsf{E}_\perp|^2 \le \mu^2 c^4.$$

For the clock e_1 instead, $m_4^2 c^4 = \mu^2 c^4 - \mathsf{E}_2^2 - \mathsf{E}_3^2$; for example $\mathsf{E} = (3, 4, 0)$, $\mathbf{p} = (5, 0, 0)$ and $c = 1$ give $\mu = 0$ and $m_4^2 = -16$.

Proof. $|(\mathsf{E}_1, \mathsf{E}_2, \mathsf{E}_3)|^2 = \mathsf{E}_W^2 + |\mathsf{E}_\perp|^2$ and $\mu^2 c^4 = c^2 \eta(P, P) = |(\mathsf{E}_1, \mathsf{E}_2, \mathsf{E}_3)|^2 - |\mathbf{p}|^2 c^2$. □

This is an obstruction, and it restates Proposition 12.1 and Example 12.3 in a form covariant under the choice of clock. In the momentum reading the differentiation of the temporal energy subtracts from the mass and can make it tachyonic, and under the surplus constraints of Theorem 12.12, that is under Assumption 2.16, $\mathsf{E}_\perp = 0$, nothing is left to differentiate. In the field-strength reading the motion part adds instead, $E_6 \ge E_0$. Only the latter carries the structure of an undivided energy with an internal differentiation.

3.10 Synthesis

The four questions of the introduction have the following answers; each is stated with its class, and the sections hold the proofs.

(1) Lagrangians. For the classical reading the free worldline of the arena gives the six-momentum and the shell $E^2 = c^2|\mathbf{p}|^2 + M^2c^4 - c^2|\mathbf{q}|^2$, and the interior shifted Lagrangian gives $\ddot{Y} = -\kappa^2 Y$ with canonical momentum obeying $\dot{\pi} = \kappa G\pi$ (restated, Propositions 3.17 and 3.18). The Dirac Lagrangian $\frac{1}{2}\psi^\top(K^\mu\partial_\mu - mT)\psi$ gives $\gamma^\mu\partial_\mu\psi = -mJ\psi$, its current and its stress tensors (restated and inherited, Proposition 2.24). The energies are governed by the Maxwell Lagrangian $-\frac{1}{2}\operatorname{Re} q$, whose Hamiltonian is the undivided energy E_6 and whose first-order form makes duality an off-shell symmetry (inherited, Propositions 3.20 and 3.23). The rest energy $E_0 = \frac{1}{2}|q|$, taken as a Lagrangian, is Lorentz invariant and conformal but not duality invariant in the sense of Gaillard and Zumino, has an indefinite energy, and is singular on radiation; its natural place is the first-order term of the ModMax family. Neither E_0 alone nor any Hamiltonian $aE_6 + bE_0$ with $a^2 - b^2 \neq 1$ is Lorentz invariant (fixed, Theorem 3.21 and Proposition 3.22).

(2) The two X's. X_{cl} is a displacement, with converted times on V^+ and lengths on V^-; $X_{\mathrm{EM}} = (\mathbf{B}, \mathbf{E})$ is the pair labelling the six components of the field two-form. They share the vector space and the Clifford action and differ in units, natural Lorentz algebra ($\mathfrak{l}_{\mathrm{int}}$ on a four-plane against $\mathfrak{l}_G$ on the two-form), canonical conjugates and observables. No linear map equivariant for the interior Lorentz algebra relates them in either direction, whatever dictionary is chosen (fixed, Theorem 3.15). The two readings must therefore be interpreted separately.

(3) The four-momentum. With a clock $\hat{n}$ on the temporal triple, $E_{\hat{n}} = \mathbb{R}\hat{n} \oplus V^-$ has signature $(1,3)$; the projection $P_{\hat{n}} = (E/c, \mathbf{p})$ of the six-momentum is a four-vector of $\mathrm{SO}^+(1,3)_{\hat{n}}$, with $\mathbf{p}$ conjugate to the three lengths and E/c to the time read by $\hat{n}$, and it is conserved for every dynamics independent of $(ct_{\hat{n}}, \mathbf{x})$ (fixed, Propositions 2.12 and 3.25). Of three mass shells, the ambient one is the statement of Section 3.9 that splitting the temporal energy lowers the mass, the first-class reduction is the surplus-time constraint of Section 12.4 with the clock $\hat{n}$, and the third is constructed (restated and constructed, Propositions 3.26, 3.28 and 3.29). The four-dimensional mass read by a clock satisfies $m_4(\hat{n})^2c^4 = \mu^2c^4 - |\hat{n} \times \mathbf{E}|^2$, never above the six-dimensional value (fixed, Proposition 3.31). The field has no linear covariant map to a four-vector; its four-momentum is the integral of its stress tensor, a four-vector only for an isolated whole, and the photon's $P = \hbar k$ is not derived in the arena (fixed, inherited and open, Section 3.4.3). The anti-isometry no-go holds for every clock and obstructs the identification of the interior momentum with the six-momentum, not the duality of the field (fixed, Propositions 3.40 and 3.41).

(4) The pair on the octants. The octants are the joint eigenlines of the pole operators, the Fock basis of $S \cong \Lambda^*(L)$. The field-octant map places the electric member of the pair on the three octants of weight one and the magnetic member on the three of weight two, uniquely up to a scale and a quarter turn of duality once rotations and Maxwell's equations are imposed; $\mathbf{E}$ is then a one-form and $\mathbf{B}$ a two-form (fixed and constructed, Theorem 3.44 and Proposition 3.54). Maxwell's four equations are one first-order equation on S, with the two Gauss laws on the two poles and the evolution laws on the field octants (fixed, inherited in substance, Theorem 3.53); the spatial derivatives in it multiply the temporal generators, a property of the representation and not a derivation of space from time. The mass form pairs each electric octant with its magnetic antipode and gives $-\mathbf{E} \cdot \mathbf{B}$ (fixed, Proposition 3.51). Clifford multiplication by the field point mixes the two members of the pair and does not realise it (Proposition 3.47). The infinitesimal generators of the boosts and of the duality rotation are odd on the octants, their finite transformations generally of mixed parity (obstructed for even operators, Theorem 3.48 and Proposition 3.49). The construction represents the classical field strength; it quantises nothing.

The bridge. The element shared by the two readings is $T = e_1e_2e_3 = \hat{n}J(\hat{n})$: for every clock it is the Dirac mass form of the particle reading, it lifts the frame parity, and on the octants $-T$ is the duality of the field reading (fixed, Theorem 3.60); it is, up to scale, the only element that is the clock times a Dirac complex structure for every clock (fixed, Theorem 3.70). The field reading does not see the clock and the particle reading does (fixed, Proposition 3.69). The Dirac action, the octant representation of the field and the spin lift of $\mathfrak{l}_G$ are three inequivalent Lorentz representations on one module, whose joint commutant is the scalars; no bilinear coupling of a Dirac field to the field octants is covariant, while the trilinear couplings are exactly the Pauli term and its dual, at every clock (fixed and inherited, Proposition 3.63 and Theorem 3.64). The answer to the question whether $T = \hat{n}J(\hat{n})$ incorporates consistently into both readings is therefore yes at the level of the module: the two readings keep their Lorentz representations, and they meet in T and in the Pauli coupling. The classical minimally coupled Maxwell–Dirac sector, through the potential and the charge $J(\hat{n})$, is written down (Proposition 3.24); what remains open is its selection from the arena rather than its adoption, its extension to the further sectors, a common parent dynamics, a dynamics that selects the clock, and a quantisation derived from the construction.

Chapter 4

Classical Rigid-Body Realization

The question of this chapter is whether the interior dynamics of the arena is a rigid-body system, and what an identification with one preserves. Section 4.1 recalls conventional rigid-body mechanics, against which the identification is tested. Sections 4.2 to 4.5 contain the arena's interior dynamics: the shifted Lagrangian in a positive form, binding in neutral signature, the closure of the force law into a unique rotor, and the rest energy in the closure. Section 4.6 tests the identification against equations, brackets, conserved quantities and degrees of freedom, and Section 4.7 compares the interior with planetary orbital dynamics. The closure is a Lie–Poisson system of rigid-body type, on a group of one rotation and six translations, and its reduced motion is a pendulum (Theorem 4.53). Its rotation, however, is an internal phase with no axis, its Lorentz-invariant inertia is indefinite, and the bounded orbits of the interior are Hooke's, not Newton's (Proposition 4.56). The algebra is proved and checked; every physical reading is marked as one. Bounded orbits do not make the Hamiltonian bounded below: the interior energy has signature $(6,6)$, and the bound orbits are not bound states (Corollary 4.15). The gravitational potential of Section 4.7 is assumed.

4.1 Conventional rigid-body mechanics

This section recalls the mechanics of a rigid body in the form in which it will be compared with the arena: the inertia tensor, angular velocity and momentum, Euler's equations as an Euler–Arnold equation, the Lie–Poisson structure and its conservation laws, and the planar body that rotates and translates. Everything here is inherited [175, 5, 176, 177, 93]; the checks rb1-1 to rb1-8 only fix the signs and conventions used later.

4.1.1 Configuration, angular velocity and inertia

A rigid body with one point held fixed has the configuration space $SO(3)$: a material point with body coordinates $\mathbf{X}$ is at $\mathbf{x}(t) = R(t)\mathbf{X}$, $R \in SO(3)$. The body angular velocity $\boldsymbol{\Omega}$ is defined by $\dot{R} = R\,\widehat{\boldsymbol{\Omega}}$, where $\widehat{\boldsymbol{\Omega}}\mathbf{v} = \boldsymbol{\Omega} \times \mathbf{v}$, and the spatial one is $\boldsymbol{\omega} = R\boldsymbol{\Omega}$. For a mass density ρ the inertia tensor is

$$\mathbb{I} = \int \rho(\mathbf{X})\,(|\mathbf{X}|^2\,\mathbf{1} - \mathbf{X}\mathbf{X}^\top)\,d^3X, \tag{4.1}$$

symmetric and positive definite unless the body lies on a line. In principal axes its moments satisfy $I_j + I_k - I_i = 2\int \rho X_i^2\,d^3X \geq 0$, the triangle inequalities (rb1-4). The kinetic energy is $T = \frac{1}{2}\boldsymbol{\Omega} \cdot \mathbb{I}\boldsymbol{\Omega}$, the body angular momentum is $\mathbf{M} = \mathbb{I}\boldsymbol{\Omega}$ and the spatial one is $\mathbf{m} = R\mathbf{M}$. The units are $[\mathbb{I}] = \mathsf{M}\mathsf{L}^2$, $[\boldsymbol{\Omega}] = \mathsf{T}^{-1}$ and $[\mathbf{M}] = \mathsf{M}\mathsf{L}^2\mathsf{T}^{-1}$; the evolution parameter is the Newtonian time t, and the action of the free body is $\int \frac{1}{2}\boldsymbol{\Omega} \cdot \mathbb{I}\boldsymbol{\Omega}\,dt$.

4.1.2 Euler's equations as an Euler–Arnold equation

The Lagrangian $\frac{1}{2}\boldsymbol{\Omega}\cdot\mathbb{I}\boldsymbol{\Omega}$ is invariant under $R \mapsto QR$, $Q \in SO(3)$, and depends on R only through $\boldsymbol{\Omega} = (R^{-1}\dot{R})^{\vee}$. Varying R with $\widehat{\boldsymbol{\Sigma}} = R^{-1}\delta R$ gives $\delta\boldsymbol{\Omega} = \dot{\boldsymbol{\Sigma}} + \boldsymbol{\Omega}\times\boldsymbol{\Sigma}$, and Hamilton's principle becomes Euler's equations

$$\dot{\mathbf{M}} = \mathbf{M}\times\boldsymbol{\Omega}, \qquad \text{that is} \qquad I_1\dot{\Omega}_1 = (I_2 - I_3)\,\Omega_2\Omega_3 \quad \text{and cyclically} \tag{4.2}$$

(rb1-2). This is the case $G = SO(3)$ of Arnold's description of a left-invariant Lagrangian $\frac{1}{2}\langle\xi, A\xi\rangle$ on a Lie group: with $\mu = A\xi$ the motion obeys $\dot{\mu} = \mathrm{ad}_{\xi}^{\top}\mu$, and the solutions are the geodesics of the left-invariant metric A [5, 6]. The variations $\delta\xi = \dot{\Sigma} + [\xi, \Sigma]$ are those of Euler–Poincaré reduction [93, 72]. The steady rotations about the axes of largest and smallest moment are stable, and those about the intermediate axis are unstable: for $(I_1, I_2, I_3) = (1, 2, 3)$ the linearisations have characteristic polynomials $\lambda(\lambda^2 + \frac{1}{3}\Omega^2)$, $\lambda(\lambda^2 - \frac{1}{3}\Omega^2)$ and $\lambda(\lambda^2 + \Omega^2)$ (rb1-5).

4.1.3 Lie–Poisson structure and conservation laws

On $\mathfrak{so}(3)^* \cong \mathbb{R}^3$ the Lie–Poisson bracket and the Hamiltonian are

$$\{F, K\}(\mathbf{M}) = -\mathbf{M}\cdot(\nabla F\times\nabla K), \qquad H = \tfrac{1}{2}\,\mathbf{M}\cdot\mathbb{I}^{-1}\mathbf{M}, \tag{4.3}$$

and $\dot{\mathbf{M}} = \{\mathbf{M}, H\}$ is (4.2) (rb1-1, rb1-2). Three conservation laws follow, of three different kinds (rb1-3). The energy H is conserved because it generates the flow. The Casimir $|\mathbf{M}|^2$ Poisson-commutes with every function, whatever the Hamiltonian; its level sets, the spheres, are the symplectic leaves. The spatial angular momentum $\mathbf{m} = R\mathbf{M}$ is the Noether charge of the left action $R \mapsto QR$. Reduction thus takes the six-dimensional $T^*SO(3)$ to the three-dimensional $\mathfrak{so}(3)^*$ and then to two-dimensional leaves: one degree of freedom, and the orbits are the intersections of the spheres with the energy ellipsoids.

4.1.4 Rotation with translation: the planar body

A body that also moves has the group of rigid motions as its configuration space. In the plane this is $SE(2) = SO(2)\ltimes\mathbb{R}^2$, with the product $(\theta, \mathbf{x})(\theta', \mathbf{x}') = (\theta + \theta', \mathbf{x} + e^{\theta J}\mathbf{x}')$, $J = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, and the Lie algebra $\mathbb{R}\hat{g}\ltimes_J\mathbb{R}^2$ with $[\hat{g}, X] = JX$. For the Lagrangian $\frac{1}{2}I\dot{\theta}^2 + \frac{1}{2}\,\mathbf{u}^{\top}N^{-1}\mathbf{u}$ in the body velocity $\mathbf{u} = e^{-\theta J}\dot{\mathbf{x}}$, with $I > 0$ and N^{-1} a positive translational inertia, the coordinates (h, p) of $\mathfrak{se}(2)^*$ have the brackets $\{h, p_1\} = -p_2$, $\{h, p_2\} = p_1$, $\{p_1, p_2\} = 0$ and the Casimir $|p|^2$ (rb1-6). With $H = h^2/(2I) + \frac{1}{2}p^{\top}Np$,

$$\dot{p} = \frac{h}{I}\,J^{\top}p, \qquad \dot{h} = -\,p^{\top}JNp, \qquad \dot{\theta} = \frac{h}{I}, \qquad \dot{\mathbf{x}} = e^{\theta J}Np. \tag{4.4}$$

On a leaf $p = e^{-\theta J}p_0$. With $N = \mathrm{diag}(a, b)$ and $p_0 = (|p_0|, 0)$ the rotation obeys $I\ddot{\theta} = \frac{1}{2}(a - b)|p_0|^2\sin 2\theta$, a pendulum in the doubled angle; with $a = b$ the body rotates uniformly and translates uniformly, the two motions decoupled (rb1-7). The spatial momentum $p_s = e^{\theta J}p$ and the quantity $J_s = h + p_s^{\top}J\mathbf{x}$, the total angular momentum about the origin, are conserved for every N (rb1-8). For a body symmetric about two perpendicular axes this is Kirchhoff's description of planar motion through an ideal fluid at rest at infinity, with the mass of the body and the added masses of the fluid in N^{-1} [178, 179]. It is the benchmark that Section 4.6 needs: the closure of Section 4.4 has the same form, with three planes in place of one.

In this chapter a system is *of rigid-body type* when it is the Lie–Poisson system of a left-invariant Lagrangian on a group of rotations, or of rotations and translations. A *physical* rigid body has a positive-definite inertia and a rotation group acting on the directions of space.

4.2 The shifted Lagrangian in a positive form

Everything in the arena so far has been kinematics. The arena supplies a candidate first-order shift operator, built from the conversion between times and lengths (Section 2.1), and this part adopts a shifted quadratic action for it and asks what motion it generates. For the adopted shifted quadratic action, the comparison concerns the choice of the quadratic form, positive definite or the split form η of the arena itself, and the two answers are opposite. This section treats the positive form, where the verdict is negative and sharp; Section 4.3 treats the split form, where bound motion appears exactly in neutral signature.

4.2.1 Momentum and force

Let $X(\tau) \in \mathbb{R}^N$ be an interior configuration, g a constant real $N \times N$ matrix, and

$$\alpha_\pm := \partial_\tau \pm \kappa g, \qquad \mathcal{L} = \frac{m}{2} \left|\alpha_+ X\right|^2 = \frac{m}{2} \left|\dot{X} + \kappa g X\right|^2. \tag{4.5}$$

Only the product κg appears: $\kappa = c/\ell$ carries the conversion of (2.2) and g its direction. The Lagrangian is the square of a single first-order operator, which is what makes it the natural object to ask Clifford questions of.

Proposition 4.1 (Momentum and force)**.** *The canonical momentum and the Euler–Lagrange equation are*

$$p = \frac{\partial \mathcal{L}}{\partial \dot{X}} = m\, \alpha_+ X, \qquad \dot{p} = \kappa\, g^\top p, \tag{4.6}$$

and in second-order form, with $a := \frac{1}{2}(g - g^\top)$*,*

$$\mathcal{E}_g := m(\ddot{X} + 2\kappa a \dot{X} - \kappa^2 g^\top g X) = 0. \tag{4.7}$$

Proof. $\partial\mathcal{L}/\partial\dot{X} = m(\dot{X}+\kappa g X)$ and $\partial\mathcal{L}/\partial X = \kappa g^\top p$. Substituting $p = m(\dot{X}+\kappa g X)$ into $\dot{p} = \kappa g^\top p$ gives $m(\ddot{X} + \kappa g \dot{X}) = \kappa m g^\top \dot{X} + \kappa^2 m g^\top g X$, which rearranges to (4.7). □

The force law is $\dot{p} = \kappa g^\top p$. It takes the symmetric-looking form $\dot{p} = \kappa g p$ exactly when g is symmetric, and the next two results show that this is not a cosmetic condition.

Theorem 4.2 (Factorisation)**.** $\mathcal{E}_g = m\, \alpha_- \alpha_+ X$ *identically if and only if* $a = 0$*.*

Proof. For constant g, $\alpha_-\alpha_+ X = \ddot{X} - \kappa^2 g^2 X$, so $\mathcal{E}_g - m\alpha_-\alpha_+ X = m(2\kappa a \dot{X} + \kappa^2(g^2 - g^\top g)X)$. The two terms involve $\dot{X}$ and X independently, and both vanish for all trajectories if and only if $a = 0$, which makes g symmetric and $g^2 = g^\top g$. □

Proposition 4.3 (The factorised branch is an inverted oscillator)**.** *The Noether energy*

$$E = \langle p, \dot{X}\rangle - \mathcal{L} = \frac{m}{2}|\dot{X}|^2 - \frac{m\kappa^2}{2}\,|gX|^2 \tag{4.8}$$

is conserved for every g*. When* $a = 0$ *the motion is* $m\ddot{X} = -\nabla V$ *with* $V = -\frac{1}{2}m\kappa^2|gX|^2 \le 0$*, and the initial data whose solutions stay bounded for* $\tau \ge 0$ *form a linear subspace of dimension exactly* N *in the* $2N$*-dimensional phase space; for all real* τ *only the zero-frequency sector, of dimension* $\dim\ker g$*, stays bounded.*

Proof. Writing $u = \alpha_+ X$, $E = \frac{m}{2}|u|^2 - m\kappa\langle u, gX\rangle$, and expanding u gives (4.8). On shell $\dot{E} = \frac{\kappa}{m}\langle p, (g^\top - g)p\rangle = -\frac{2\kappa}{m}\langle p, ap\rangle = 0$ because a is antisymmetric. With $a = 0$ the equation is $\ddot{X} = \kappa^2 g^2 X$, and on the eigenplane of an eigenvalue $\lambda \neq 0$ of g the solutions are $Ae^{\kappa|\lambda|\tau} + Be^{-\kappa|\lambda|\tau}$, bounded for $\tau \ge 0$ iff $A = 0$; on $\lambda = 0$ they are $x_0 + v_0\tau$, bounded iff $v_0 = 0$. In each plane one condition is imposed on two dimensions. □

The factorisation condition and the inverted-potential condition are the same condition. Since $g^\top g \succeq 0$, the potential is never a well, and the bounded set has measure zero: no open set of initial data stays bounded. Only the velocity-coupled term $2\kappa a\dot{X}$ could bind, and it is exactly the term that obstructs factorisation.

Example 4.4 (The conversion operator as the shift)**.** Take $g = B_C$, the mixed bivector of Theorem 2.7 acting on S, which is symmetric. The equation of motion is $\ddot{X} = \kappa^2 B_C^2 X$, and by Theorem 2.7 the growth rates are $\kappa|\eta_1\sigma_1 + \eta_2\sigma_2 + \eta_3\sigma_3|$ for the four sign patterns η. For the conversion along a single clock, the symmetric one or any other, C has singular values $(1, 0, 0)$, every rate equals κ, and the dynamics cannot see the Clifford structure (Example 2.9). For C of rank two with $\sigma = (2, 1, 0)$ the rates are 3κ and κ, each on a four-dimensional subspace; for $\sigma = (1, 1, 0)$ they are 2κ and 0, and the second is a flat direction. Raising the rank makes the instability anisotropic and can create flat directions, never an oscillation.

4.2.2 No recurrence at nonzero momentum

It is tempting to stop here and blame the tension between factorisation and binding. That would be too generous to the unfactorised branch. A single identity settles the question for every g.

Theorem 4.5 (Virial identity)**.** *With the positive form of (4.5), along every solution and for every constant g,*

$$\frac{d}{d\tau}\langle X, p\rangle = \frac{|p|^2}{m} \;\geq\; 0. \tag{4.9}$$

Consequently: (i) no solution with $p(0) \neq 0$ is periodic or recurrent; (ii) purely imaginary characteristic exponents may occur, but never semisimply, so the motion they carry is secular, and the Jordan blocks need not have size two; (iii) a trajectory bounded in phase space has $p \in L^2$, so either $p \equiv 0$ or p decays exponentially and the orbit is not recurrent.

Proof. $\frac{d}{d\tau}\langle X, p\rangle = \langle \dot{X}, p\rangle + \langle X, \kappa g^\top p\rangle = \langle \dot{X} + \kappa g X, p\rangle = |p|^2/m$. A recurrent orbit would return $\langle X, p\rangle$ to an earlier value, which (4.9) forbids unless $p \equiv 0$, and $p = e^{\kappa g^\top \tau} p(0)$ vanishes identically only if $p(0) = 0$. A semisimple imaginary exponent would supply a periodic solution with $p \neq 0$. For (iii), boundedness makes $\langle X, p\rangle$ bounded and monotone, hence convergent, so $\int_0^\infty |p|^2 < \infty$, and a bounded square-integrable solution of a constant-coefficient linear system is zero or exponentially decaying. □

Example 4.6 (A Jordan block of size four)**.** Let J_2 be the standard 2×2 rotation generator and $g = \left(\begin{smallmatrix} J_2 & \mathbf{1}_2 \\ 0 & J_2 \end{smallmatrix}\right)$ on $\mathbb{R}^4$, with $\kappa = m = 1$. The eight-dimensional phase-space generator M has its spectrum on $\pm i$, and $\operatorname{rank}\big((M^2 + \mathbf{1})^k\big) = 6, 4, 2, 0$ for $k = 1, 2, 3, 4$. The blocks therefore have size four and the generic solution grows like τ^3, although every exponent is purely imaginary.

The zero-momentum sector is not empty, and it does oscillate. The general solution is

$$p(\tau) = e^{\kappa g^\top \tau} p_0, \qquad X(\tau) = e^{-\kappa g \tau}\Big[X_0 + \frac{1}{m}\int_0^\tau e^{\kappa g s} e^{\kappa g^\top s} p_0 \, ds\Big], \tag{4.10}$$

and the kernel of α_+, $X(\tau) = e^{-\kappa g \tau} X_0$, has $p \equiv 0$. The integral collapses to τp_0 only when $g^\top = -g$. For $g = J$ antisymmetric with $J^2 = -\mathbf{1}$, the zero-momentum solutions are bounded rotations of period $2\pi/|\kappa|$, and any nonzero momentum adds the secular term $\tau\, e^{-\kappa J\tau} p_0/m$: the orbit becomes an outward spiral (Figure 4.1, left). In that marginal case $g^\top g = \mathbf{1}$, $a = J$, the equation reads $\ddot{X} + 2\kappa J\dot{X} - \kappa^2 X = 0$, and in the eigenbasis of J the characteristic polynomial is $(\lambda + i\kappa)^2$, a double root on the imaginary axis; the corresponding Jordan block is exactly the instability just described.

So with a positive form, oscillation exists only on a measure-zero sector and is destroyed by any momentum. What is missing is neither curvature nor a dependence of g on τ. It is the positive form itself, and the next section removes it.

4.2.3 Towards a stress–energy tensor

The worldline statement lifts to a curved metric in two ways, which together indicate where a stress–energy tensor would come from. First, replacing ∂_τ by $D/D\tau$ along a worldline in (M, g) and promoting g to a section G of $\mathrm{End}(TM)$, the variation of $\int \frac{m}{2} g_{AB}(\alpha_+X)^A(\alpha_+X)^B\, d\tau$ gives $Dp_B/D\tau = \kappa G^A{}_B p_A$, and the second-order form acquires one new term,

$$\frac{D^2X}{D\tau^2} + 2\kappa a \frac{DX}{D\tau} + \kappa(D_\tau G)X - \kappa^2 G^\top G X = 0;$$

the factorisation computation goes through verbatim, since the extra term appears on both sides, so $a = 0$ remains the factorisation condition. Second, for a field X in a bundle with fibre metric h and $D_M = \nabla_M + \kappa G_M$, the action $\int \sqrt{|g|}\, \frac{m}{2} h_{AB} g^{MN} (D_M X)^A (D_N X)^B$ has, when g is varied with the fibre metric h and the bundle connection held fixed (a connection that depends on g, such as the Levi-Civita connection on TM, needs further terms),

$$T_{MN} = m\Big[h_{AB}(D_M X)^A (D_N X)^B - \tfrac{1}{2} g_{MN}\, h_{AB}\, g^{PQ} (D_P X)^A (D_Q X)^B\Big],$$

whose trace $m(1 - \frac{D}{2})|DX|^2$ vanishes identically for arbitrary fields only in $D = 2$, the same dimension in which the projector sigma model of Section 8.1.5 is renormalisable. This is the stress tensor of the shifted kinetic term only, with G a background, and Theorem 4.5 is a warning about it: a matter sector whose only oscillations sit at zero momentum, with an energy unbounded below, is not yet a source one can couple to gravity.

4.3 The split form and binding in neutral signature

The norm in (4.5) is positive definite, and that is what makes $g^\top g \succeq 0$ in Proposition 4.3 and $|p|^2 \geq 0$ in Theorem 4.5. But the arena does not hand us a positive form; it hands us η, of signature $(3, 3)$. Redoing the derivation with η changes one inequality, and with it the verdict.

4.3.1 The split-form Lagrangian

Write $G^\dagger := \eta^{-1} G^\top \eta$ for the η-adjoint and take

$$\mathcal{L} = \frac{m}{2}\, \eta(\alpha_+X, \alpha_+X), \qquad p = m\, \eta\, (\alpha_+X), \qquad \pi = \eta^{-1} p, \qquad \dot{p} = \kappa\, G^\top p, \tag{4.11}$$

with second-order form and energy

$$\ddot{X} + 2\kappa a \dot{X} - \kappa^2 G^\dagger G X = 0, \qquad a = \tfrac{1}{2}(G - G^\dagger), \qquad E = \frac{m}{2}\eta(\dot{X}, \dot{X}) - \frac{m\kappa^2}{2}\eta(GX, GX). \tag{4.12}$$

Everything in Section 4.2 survives with the transpose replaced by the η-adjoint: factorisation holds exactly when $a = 0$, and E is conserved because a is η-antisymmetric. What does not survive is the sign of the virial identity.

Theorem 4.7 (The split virial identity)**.** *Let X be a section along a path in a manifold with metric of signature $(3, 3)$ and let $D/D\tau$ come from any metric-compatible connection, torsion included. Then for every G,*

$$\frac{d}{d\tau}\, \eta(X, \pi) = \frac{\eta(\pi, \pi)}{m}. \tag{4.13}$$

Consequently every closed orbit satisfies $\oint \eta(\pi, \pi)\, d\tau = 0$: a bound orbit has null momentum on average.

Proof. Note that $\eta(X, \pi) = \langle p, X\rangle$. Metric compatibility gives $\frac{d}{d\tau}\eta(X, \pi) = \eta(DX/D\tau, \pi) + \eta(X, D\pi/D\tau)$, and torsion does not enter this Leibniz rule. Substituting $DX/D\tau = \pi/m - \kappa GX$ and $D\pi/D\tau = \kappa G^\dagger \pi$ gives $\eta(\pi, \pi)/m - \kappa\, \eta(GX, \pi) + \kappa\, \eta(X, G^\dagger \pi)$, and the last two terms cancel by the definition of the adjoint. □

Neither curvature nor torsion appears in (4.13); the identity sees only η. In definite signature $\eta(\pi,\pi)$ has one sign and the identity forbids recurrence, which is Theorem 4.5. In split signature it does not, and the question becomes whether solutions with null momentum on average exist. They do, and the operator that produces them is one we already have.

4.3.2 Binding exactly in neutral signature

Theorem 4.8 (Factorisation and binding coexist exactly in neutral signature)**.** *Let η have signature (P,Q) on $\mathbb{R}^N$. There is a linear map G with*

(i) $G = G^\dagger$, so that the factorisation of Theorem 4.2 holds, and

(ii) $G^2 = -\mathbf{1}$, so that (4.12) reads $\ddot{X} = -\kappa^2 X$ and every solution is periodic of period $2\pi/\kappa$,

if and only if $P = Q$. Such a G is an anti-isometry, $\eta(GX,GY) = -\eta(X,Y)$, and on $\mathbb{R}^{3,3}$ one of them, maximal in that it uses every paired direction, is the binding root of Proposition 2.3:

$$G:\ e_i \longmapsto e_{i+3}, \qquad e_{i+3} \longmapsto -e_i \qquad (i = 1,2,3). \tag{4.14}$$

Proof. Given (i) and (ii), $\eta(GX,GY) = \eta(X,G^\dagger GY) = \eta(X,G^2Y) = -\eta(X,Y)$, so G carries a maximal positive subspace isometrically onto a negative one and back, which forces $P = Q$. Conversely (4.14) satisfies $G^2 = -\mathbf{1}$, and $\eta(Ge_i,e_{i+3}) = \eta(e_{i+3},e_{i+3}) = -1 = \eta(e_i,-e_i) = \eta(e_i,Ge_{i+3})$, so $G = G^\dagger$. With $G^\dagger G = G^2 = -\mathbf{1}$, (4.12) has $a = 0$ and becomes $\ddot{X} = -\kappa^2 X$. □

The linear algebra is inherited: a complex structure that is anti-isometric for η is a Norden, or anti-Kähler, structure, and its existence forces neutral signature [301]. What the arena adds is the reading of (4.14) as the binding root of the pairing.

This reverses the verdict of Section 4.2 at its root. In definite signature the factorised branch is an inverted oscillator and nothing binds; in neutral signature the same factorisation condition $G = G^\dagger$ admits operators of purely imaginary spectrum, and the maximal one makes the whole interior a harmonic well of one frequency. The obstruction of Theorem 4.5 was a theorem about the positive form, not about the arena. Two qualifications keep the statement honest. The normalisation $G^2 = -\mathbf{1}$ imposes one common frequency; any η-self-adjoint shift built from independent paired blocks $\mu_i J_2$ with distinct $\mu_i > 0$ also factorises and binds, at unequal frequencies, so the construction is a family and $G^2 = -\mathbf{1}$ picks a member. And no equation of motion for G has been written: G is adopted, and calling its consequences a selection would be premature.

Proposition 4.9 (How much can bind)**.** *Let $G = G^\dagger$ be diagonalisable over $\mathbb{C}$ and $\kappa \neq 0$. An eigenvector v with non-real eigenvalue satisfies $\eta(v,\bar{v}) = 0$, and eigenvectors of different eigenvalues are η-orthogonal. The real span of the eigenvectors with non-real eigenvalues is therefore the η-orthogonal complement of the real eigenspaces; it is nondegenerate of signature (k',k'). Let V_{osc} be its part belonging to the nonzero purely imaginary eigenvalues. V_{osc} is nondegenerate of signature (k,k) with $k \le \min(P,Q)$, η-orthogonal to the remaining spectral subspaces, and every solution of $\ddot{X} = \kappa^2 G^2 X$ with position and velocity in V_{osc} is bounded (quasi-periodic; periodic when the frequencies are commensurate). Zero eigenvalues give constant solutions and linear drift, and eigenvalues with nonzero real part give no nonzero solution bounded for all times. On Minkowski space $\mathbb{R}^{1,3}$ at most a single two-plane can bind; on $\mathbb{R}^{3,3}$ all six directions can, simultaneously and at one frequency.*

A non-real spectrum is not enough. For $\eta = \mathrm{diag}(1,-1)$ the operator $G = \left(\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right)$ is η-self-adjoint and diagonalisable with eigenvalues $1 \pm i$, and $X(\tau) = e^{\kappa\tau}(\cos\kappa\tau, \sin\kappa\tau)$ solves $\ddot{X} = \kappa^2 G^2 X$: the oscillation is exponentially modulated, and no nonzero solution of this block is bounded on $\mathbb{R}$. Hence the restriction to the purely imaginary spectrum, $\operatorname{spec} G \subset \mathbb{R} \cup i\mathbb{R}$ being the case in which the two subspaces agree (a3-11, exact).

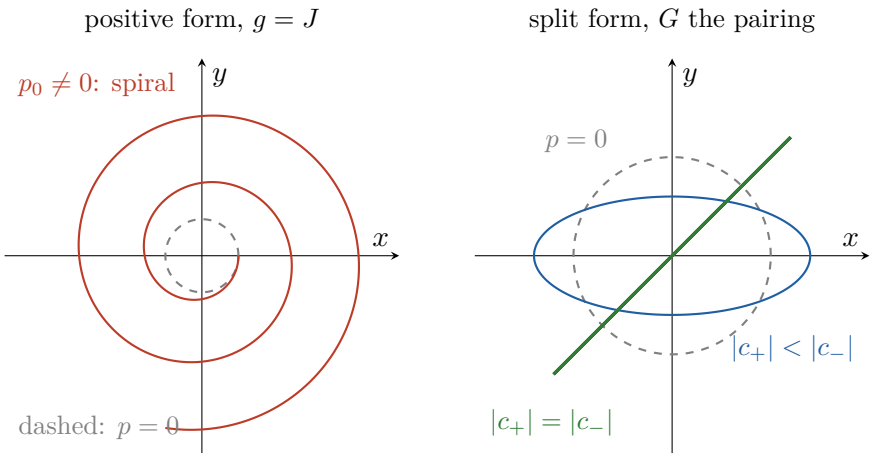


Figure 4.1: Interior motion in one pairing plane (Example 4.10). Left: with the positive form and an antisymmetric shift the zero-momentum orbits are circles, and any momentum turns them into outward spirals, as Theorem 4.5 requires. Right: with the split form and the binding root G every orbit is closed; the zero-momentum orbits are circles, the momentum-carrying ones ellipses, degenerating to a segment when the two rotating parts have equal size.

The statement concerns the whole subspace V_{osc}. Let $\eta = \left(\begin{smallmatrix} 0 & K \\ K & 0 \end{smallmatrix}\right)$ on $\mathbb{R}^4$, with $K = \mathrm{diag}(1,-1)$. For the diagonalisable $G = \mathrm{diag}(J_2, J_2)$, which is η-self-adjoint, the plane of one conjugate pair is totally null, although the whole of V_{osc} has signature $(2,2)$. For the non-diagonalisable $G = \left(\begin{smallmatrix} J_2 & \mathbf{1} \\ 0 & J_2 \end{smallmatrix}\right)$ even the subspace statement fails: G is η-self-adjoint and its bounded motions lie in a plane on which η vanishes. The conclusions used later concern $G^2 = -\mathbf{1}$, which is diagonalisable, and stand.

Example 4.10 (One pairing plane)**.** In the plane (e_1, e_4) write $X = x\,e_1 + y\,e_4$, so that G acts as $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ and $z = x + iy$. The general solution of $\ddot{X} = -\kappa^2 X$ is $z(\tau) = c_- e^{-i\kappa\tau} + c_+ e^{i\kappa\tau}$, an ellipse with semi-axes $|c_-| + |c_+|$ and $||c_-| - |c_+||$. The momentum $\alpha_+ X = \dot{X} + \kappa G X$ corresponds to $\dot{z} + i\kappa z = 2i\kappa\, c_+ e^{i\kappa\tau}$, so it is carried entirely by the counter-rotating part, and the zero-momentum orbits are the circles $z = c_- e^{-i\kappa\tau}$, traversed at angular velocity $-\kappa$. The virial quantity $\eta(\pi, \pi) = m^2 \eta(\alpha_+ X, \alpha_+ X) = m^2\,\mathrm{Re}((2i\kappa c_+ e^{i\kappa\tau})^2)$ oscillates with mean zero, as Theorem 4.7 requires of a closed orbit.

4.3.3 The orbits as duality rotations

Read through the field-strength dictionary of Theorem 3.3, an interior configuration X is a two-form and the binding root is the Hodge star. The bound motion then has a transparent meaning.

Proposition 4.11 (Bound orbits in field-strength language)**.** *Let G be the pairing (4.14) and $z = \mathbf{B} + i\mathbf{E}$.*

(i) The zero-momentum orbits are $X(\tau) = e^{-\kappa G\tau} X_0$, that is $z(\tau) = e^{-i\kappa\tau} z_0$: continuous duality rotations, along which $q(z(\tau)) = e^{-2i\kappa\tau} q(z_0)$. A non-null orbit is a single spacetime plane at exactly four instants per period, alternately of magnetic and electric type, and a null orbit is a single plane at every instant.

(ii) Every solution is $z(\tau) = c_- e^{-i\kappa\tau} + c_+ e^{i\kappa\tau}$ with $c_\pm \in \mathbb{C}^3$, $\alpha_+ X$ corresponds to $2i\kappa c_+ e^{i\kappa\tau}$ (the momentum field $\pi = m\alpha_+ X$ to m times it), and the period averages are

$$\langle q \rangle = 2\, c_- \cdot c_+, \qquad \langle \mathbf{E} \cdot \mathbf{B} \rangle = \mathrm{Im}(c_- \cdot c_+), \qquad \langle |\mathbf{B}|^2 - |\mathbf{E}|^2 \rangle = 2\,\mathrm{Re}(c_- \cdot c_+).$$

(iii) The energy is $\mathcal{E} = 2m\kappa^2 \operatorname{Re}(c_- \cdot c_+)$*, and the complex quantity* $\frac{m}{2}(q(\dot z)+\kappa^2 q(z)) = 2m\kappa^2\, c_- \cdot c_+$ *is conserved; its imaginary part is a second constant of motion.*

(iv) On the orbits with $c_+ = \bar c_-$ *the vector* z *is real at every instant:* $\mathbf{E} = 0$ *and* $\mathbf{E}\cdot\mathbf{B} = 0$ *throughout, while the momentum is not zero.*

On a zero-momentum orbit $\mathbf{E}\cdot\mathbf{B}(\tau) = \frac{1}{2}|q_0|\sin(\phi_0 - 2\kappa\tau)$ with $q_0 = |q_0|e^{i\phi_0}$, and the rest-frame field of Section 3.8 turns from purely magnetic through screws to purely electric and back, at the fixed rest-frame energy density $\frac{1}{2}|q_0|$. So the binding flow is what breaks orthogonality on every non-null orbit without momentum, while orbits that carry momentum can stay orthogonal throughout. The mean non-orthogonality and the energy of a general orbit are the imaginary and real parts of one conserved complex number $c_- \cdot c_+$, and neither determines the other.

Theorem 4.12 (The parity-odd term is a total derivative)**.** *For the split-form Lagrangian with the pairing,*

$$\mathcal{L}_\kappa = \frac{m}{2}\Big(\eta(\dot X,\dot X) - \kappa^2\eta(X,X)\Big) - m\kappa\,\frac{d}{d\tau}(\mathbf{E}\cdot\mathbf{B}).$$

Consequently the cross term has identically vanishing Euler–Lagrange expression, the equation $\ddot X = -\kappa^2 X$ *comes from the first bracket alone, and the momenta of* $\mathcal{L}_\kappa$ *and* $\mathcal{L}_{-\kappa}$ *differ by* $2m\kappa\,\eta GX$*. Parity maps* $\mathcal{L}_\kappa$ *to* $\mathcal{L}_{-\kappa}$ *and exchanges the orbits without momentum with those carried entirely by the counter-rotating part, while* C *and* T *leave* $\mathcal{L}_\kappa$ *invariant.*

Proof. G is η-self-adjoint, so $\frac{d}{d\tau}\eta(X,GX) = 2\eta(\dot X, GX)$, and $\eta(X,GX) = -2\,\mathbf{E}\cdot\mathbf{B}$ by Theorem 3.3; $\eta(GX,GX) = -\eta(X,X)$ because G is an anti-isometry. Expanding $\eta(\dot X + \kappa GX, \dot X + \kappa GX)$ gives the formula. Parity reverses $\mathbf{E}$ and fixes $\mathbf{B}$, so $PGP^{-1} = -G$ and P acts on z as complex conjugation. □

The binding term carries the one parity-odd ingredient of the interior, the sign of κ, and it sits entirely in a total derivative. Since $\mathbf{E}\cdot\mathbf{B}$ is a single-valued function on the configuration space, the canonical transformation $p \mapsto p - m\kappa\,\eta GX$ removes it, classically and as a phase of wave functions; the shift by $2m\kappa\,\eta GX$ of Theorem 4.12 relates the two gauges $\pm\kappa$. The structure is that of the θ-term of electrodynamics, a parity-odd coefficient times a total derivative, with the difference that here it has no topological sector in which to act. In the positive form the same cross term is gyroscopic, with Euler–Lagrange expression $2m\kappa G\dot X$, and the sign of κ enters the equations of motion; that is another way of seeing why the two forms behave so differently.

4.3.4 Frame field and momentum field

Two fields can be read along an interior solution: the frame field X itself and the momentum field $\pi = \eta^{-1}p = m\,\alpha_+ X$. For a field point Y with $z(Y) = \mathbf{B} + i\mathbf{E}$, $E_6(Y)$, $E_0(Y)$ and $\mathbf{S}(Y)$ are its undivided energy, its rest energy and its Poynting vector, and $h(z,w) = \sum_j \bar z_j w_j$ ((3.17)). By Proposition 4.11, every solution of $\ddot X = -\kappa^2 X$ is $z(\tau) = c_- e^{-i\kappa\tau} + c_+ e^{i\kappa\tau}$, $c_\pm \in \mathbb{C}^3$, the momentum field $\pi = m\alpha_+ X$ corresponds to $2im\kappa c_+ e^{i\kappa\tau}$, the Noether energy is $\mathcal{E} = 2m\kappa^2 \operatorname{Re}(c_-\cdot c_+)$, and the complex quantity $\mathcal{E}_\mathbb{C} := \frac{m}{2}(q(\dot z)+\kappa^2 q(z)) = 2m\kappa^2\, c_-\cdot c_+$ is conserved. The pointwise energies of the frame field and of the momentum field behave differently along these solutions.

Proposition 4.13 (Frame field and momentum field in the interior)**.** *Along every solution of the split-form interior with the binding root:*

(i) The momentum field is $z(\pi) = 2im\kappa\, c_+ e^{i\kappa\tau}$*, that is* $\pi(\tau) = e^{\kappa\tau G}\pi_0$*: it moves by pure duality, and* $E_6(\pi) = \frac{1}{2}|p|^2 = 2m^2\kappa^2|c_+|^2$*,* $E_0(\pi) = 2m^2\kappa^2|c_+\cdot c_+|$ *and* $\mathbf{S}(\pi)$ *are constant.*

(ii) The frame field has

$$|z|^2 = |c_-|^2 + |c_+|^2 + 2\operatorname{Re}\big(h(c_-,c_+)e^{2i\kappa\tau}\big), \qquad q(z) = e^{-2i\kappa\tau}c_-\cdot c_- + 2c_-\cdot c_+ + e^{2i\kappa\tau}c_+\cdot c_+,$$

so its E_6 is constant iff $h(c_-, c_+) = 0$, and its E_0 oscillates in general: $c_- = e_1$, $c_+ = e_2$ give $\mathcal{E} = 0$, $E_6 = 1$ and $E_0 = |\cos 2\kappa\tau|$.

(iii) Both pure senses, $z = e^{-i\kappa\tau}c_-$ and $z = e^{i\kappa\tau}c_+$, are duality rotations of the frame field, with $\mathcal{E}_{\mathbb{C}} = 0$ and with constant E_6 and E_0 of the frame field. On the first, the orbits without momentum, $p = 0$ while $E_6 = \frac{1}{2}|c_-|^2 > 0$; on the second $|p|^2 = 8m^2\kappa^2 E_6$.

(iv) The parameter m is an overall factor of the Lagrangian and does not enter $\ddot{X} = -\kappa^2 X$.

Proof. $\dot{z} + i\kappa z = 2i\kappa c_+ e^{i\kappa\tau}$ is $z(\alpha_+ X)$, since G acts as i, and $\pi = m\alpha_+ X$; with $|\pi| = |p|$ (η is orthogonal) this gives (i), and $\overline{z(\pi)} \times z(\pi)$, $|z(\pi)|$ and $|z(\pi) \cdot z(\pi)|$ are invariant under multiplication of $z(\pi)$ by a phase. (ii) is the expansion of $\bar{z} \cdot z$ and $z \cdot z$; in the example $|z|^2 = 2$ and $q = 2\cos 2\kappa\tau$. (iii) is (i) and (ii) at $c_+ = 0$ or $c_- = 0$, with $|p|^2 = |z(\pi)|^2 = 4m^2\kappa^2|c_+|^2$. (iv) is read off (4.11) and Theorem 4.8. □

Item (i) and the first half of (iii) make explicit what Proposition 4.11 states about the momentum and about the orbits without momentum; item (ii) and the second half of (iii) are fixed by the algebra and complete the picture, in which parity exchanges the two senses (Theorem 4.12). The quantities the interior conserves belong to the momentum field: $E_6(\pi)$, $E_0(\pi)$ and $\mathbf{S}(\pi)$ are constant already in the clamped interior, because π moves by a duality rotation. The same functions of the frame field are not conserved, and E_6 of the frame field is not a function of $|p|^2$: on the orbits without momentum $|p|^2 = 0$ while $E_6 > 0$, on the duality rotations of the other sense $|p|^2 = 8m^2\kappa^2 E_6$. Duality acts along both pure senses, so duality orbits are not the orbits without momentum; that sense is singled out by the momentum, not by duality. And the interior's conserved energy is not undivided: it is the real part of a constant of duality charge two, carried by neither sense alone.

4.3.5 What the binding root selects, and what tears it

The binding root is not merely available; it carries structure. By Theorem 3.1 its centraliser is a Lorentz algebra $\mathfrak{l}_G$, not conjugate to the interior one (Proposition 3.2). The same operator limits what transport can do to the bound sector.

Proposition 4.14 (Transport and a homogeneous threshold)**.** *Let a connection along the path take values ω in $\mathfrak{so}(3,3)$. If ω lies in the compact part of $\mathfrak{l}_G$, transport and the shift commute and the orbit is carried rigidly. If ω has a boost component of strength c in a pairing plane (e_a, e_{a+3}), the homogeneous 2×2 block there has exponents $\pm\sqrt{c^2 - \kappa^2}$: oscillatory for $c < \kappa$ and real for $c > \kappa$.*

Proof. In the basis (e_a, e_{a+3}) the boost generator is $\sigma = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and G is $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, so $-c\sigma + \kappa G = \left(\begin{smallmatrix} 0 & -c-\kappa \\ -c+\kappa & 0 \end{smallmatrix}\right)$, whose square is $(c^2 - \kappa^2)\mathbf{1}$. □

The threshold classifies exponents of the homogeneous block, not boundedness of the full flow, and the difference is real. With the momentum included, the phase-space generator $M = \left(\begin{smallmatrix} -c\sigma-\kappa G & \mathbf{1}/m \\ 0 & -c\sigma+\kappa G \end{smallmatrix}\right)$ satisfies $M^2 - (c^2 - \kappa^2)\mathbf{1} = \left(\begin{smallmatrix} 0 & -2c\sigma/m \\ 0 & 0 \end{smallmatrix}\right)$, nilpotent and nonzero for $c \neq 0$. At $\kappa = m = 1$ and $c = \frac{1}{2}$ one finds $\det(\lambda\mathbf{1} - M) = (\lambda^2 + \frac{3}{4})^2$ with $\operatorname{rank}(M^2 + \frac{3}{4}\mathbf{1}) = 2$: Jordan blocks, and generic secular growth strictly below the threshold. Along an open path, moreover, a connection coefficient can be changed by a frame rotation, so growth on one worldline proves nothing; the invariant is the holonomy around a closed path. Section 11.1 computes it in an explicit curved model and finds that normal curvature converts zero-momentum orbits into momentum-carrying ones, through exactly the failure of the normal rotation to commute with G.

What this part delivers is a bound sector, a frequency, a centralising Lorentz algebra and an exponent threshold, all from an adopted binding operator. It does not deliver a spectrum: κ is

not fixed, the single frequency is degenerate across the three pairs, and the energy is indefinite, so these are bound orbits and not bound states in any quantum sense. Nor does it settle the relation between the two square roots of the pairing beyond Lemma 2.4: $\mathcal{K}$ is η-antisymmetric and grades, G is η-self-adjoint and binds, and they differ by the Cartan involution $\mathrm{Ad}_{e_1e_2e_3}$.

4.3.6 Bound orbits are not bound states

The orbits of Theorem 4.8 are classical. Quantised in the most direct way, the same oscillators have no state of lowest energy.

Corollary 4.15 (No ground state)**.** *Quantise the interior oscillators of the full frame canonically, mode by mode, with positive norm. Then the spectrum of the energy* $\mathcal{E} = \frac{m}{2}(\eta(\dot{X},\dot{X}) + \kappa^2\eta(X,X))$ *is* $\hbar\kappa\mathbb{Z}$*, every level is infinitely degenerate, and every state has one of lower energy. Making the spatial energies positive instead requires negative norm.*

Proof. Since G is an anti-isometry, $\mathcal{E}$ is the energy of (4.12), and $\mathcal{E} = \sum_a \eta_{aa}\,\frac{m}{2}(\dot{x}_a^2 + \kappa^2 x_a^2)$. The cross term of $\mathcal{L}_\kappa$ is a total derivative (Theorem 4.12); it shifts the momenta by a canonical transformation and leaves the spectrum unchanged. A spatial mode x has Lagrangian $-\frac{m}{2}(\dot{x}^2 - \kappa^2x^2)$, momentum $p_x = -m\dot{x}$ and Hamiltonian $-(p_x^2/2m + m\kappa^2x^2/2)$. With $x = \sqrt{\hbar/2m\kappa}\,(\hat{a}+\hat{a}^\dagger)$ and $p_x = i\sqrt{\hbar m\kappa/2}\,(\hat{a}^\dagger - \hat{a})$, $p_x^2/2m + m\kappa^2x^2/2 = \frac{\hbar\kappa}{2}(\hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a})$, so a temporal mode contributes $\hbar\kappa(j+\frac{1}{2})$ and a spatial one $-\hbar\kappa(j+\frac{1}{2})$, $j = 0, 1, \ldots$. On $\mathbb{R}^{3,3}$ the half-integers cancel and a state of occupations $(j_1, j_2, j_3; j_4, j_5, j_6)$ has energy $\hbar\kappa(j_1 + j_2 + j_3 - j_4 - j_5 - j_6)$. Every integer occurs; adding one temporal and one spatial quantum keeps the energy, and adding a spatial quantum lowers it by $\hbar\kappa$. A positive spatial energy needs $[\hat{a}, \hat{a}^\dagger] = -1$ on the spatial modes, a negative norm. □

The per-mode quantisation is the inherited indefinite oscillator [114], and the corollary is an obstruction under the hypothesis of positive norm. The half-integers cancel because the signature is neutral, which is the condition of Theorem 4.8 under which factorisation and binding coexist: the signature that closes every classical orbit leaves the quantised oscillators without a ground state. The zero-point values on coordinate subspaces are those of Theorem 7.13.

4.4 Closing the force law

The force law $\dot{p} = \kappa G^\top p$ carries its rate κ as a constant put in by hand. A force law of this shape is familiar from rigid bodies and ideal fluids, where the equation of motion is a geodesic equation on a Lie group and the rate is itself dynamical. This section asks whether the interior law can be closed in that sense: whether there is a Lie–Poisson system whose bracket and inertia are built from the same data that build G, and whose solutions contain the force law with κ as a dynamical variable. A bracket or an inertia is called *derived from the Clifford structure* when it is equivariant under the algebra $\mathfrak{l}_G$ of Theorem 3.1 and built from η, G and the bivectors. The answers, in brief: no closure lives on the frame alone; the closure on one extra direction is unique; its inertia is constrained by two requirements that exclude each other; and the dynamics it generates is a pendulum in the duality angle that drifts the frame except on null data.

Throughout, $V = \mathbb{R}^{3,3}$, a covector p is a column with $\langle p, X\rangle = p^\top X$, $\pi = \eta^{-1}p$, and $z(X) = x + iy$ for $X = \sum_j (x_j e_j + y_j e_{j+3})$, so that G acts as i. For a covector put

$$C(p) = p^\top\eta^{-1}p, \qquad D(p) = p^\top G\eta^{-1}p, \qquad w(p) := q(z(\pi)) = C(p) - iD(p). \tag{4.15}$$

In the field-strength reading, with $(\mathbf{B};\mathbf{E}) = \pi$, $w = |\mathbf{B}|^2 - |\mathbf{E}|^2 + 2i\,\mathbf{B}\cdot\mathbf{E}$ is the complex invariant of the field carried by the momentum.

4.4.1 The force law as a dual transport

Euler–Arnold equations are geodesic equations on a Lie group with a one-sided invariant metric [5, 6]; for a left-invariant metric they read $\dot{\mu} = \mathrm{ad}_w^{\top}\mu$ with $\mu = Aw$ for an inertia A. The rigid body is the case of $SO(3)$, where this is $\dot{\mathbf{M}} = \mathbf{M} \times \boldsymbol{\Omega}$ (Section 4.1.2), and incompressible Euler flow the case of the volume-preserving diffeomorphisms.

Proposition 4.16 (The dictionary)**.** *(i) Since $\langle G^{\top}p, v\rangle = \langle p, Gv\rangle$, the law $\dot{p} = \kappa G^{\top}p$ transports the covector p by the transpose of κG. It is the Lie–Poisson equation $\dot{p} = \mathrm{ad}_w^{\top}p$ of a Lie bracket on the frame exactly when $\kappa G = \mathrm{ad}_w$ is an inner derivation.*

(ii) For a bracket with trivial centre on the $6n$-dimensional frame the inner derivations fill a $6n$-dimensional subspace of $\mathfrak{so}(3n, 3n)$, of dimension $3n(6n-1)$, when η is invariant: a fraction $2/(6n-1)$.

(iii) When $\kappa G = \mathrm{ad}_u$ and $p = Au$, the shift operators $\alpha_{\pm} = \partial_{\tau} \pm \kappa G$ are the constrained variations $\delta u = \dot{\xi} \pm [u, \xi]$ of Euler–Poincaré reduction [93, 72], in the body and in the spatial frame.

Proposition 4.17 (The binding root is inner for no bracket)**.** *For every Lie bracket and every w, $\mathrm{ad}_w(w) = 0$, so every inner derivation is singular. The binding root satisfies $G^2 = -\mathbf{1}$ and is invertible, so κG is inner for no Lie bracket on $\mathbb{R}^{3n,3n}$, invariant or not, and the force law with that G is never a Lie–Poisson equation on the frame itself. It is one on the extension $\mathbb{R}\hat{g} \ltimes \mathbb{R}^{3n,3n}$, the frame taken abelian and $\hat{g}$ acting on it by G, where it is the Lie–Poisson equation of the fixed element $\kappa\hat{g}$ restricted to the ideal.*

Theorem 4.18 (Transport and factorisation are complementary)**.** $\mathrm{End}(\mathbb{R}^{3n,3n}) = \mathfrak{so}(3n, 3n) \oplus \{G : G = G^{\dagger}\}$, *of dimensions $3n(6n-1) + 3n(6n+1) = 36n^2$. Along $\dot{p} = \kappa G^{\top}p$ the Casimir $C = \eta^{-1}(p, p)$ changes at the rate $dC/d\tau = 2\kappa\, p^{\top}G\eta^{-1}p$, which vanishes for every p exactly when G lies in the first summand, while factorisation in the split form requires the second. The binding root lies in the second summand and moves the Casimir at the rate $dC/d\tau = -4\kappa\, \mathbf{B}\cdot\mathbf{E}$ in the variables $(\mathbf{B}; \mathbf{E}) = \eta p$.*

At the first rung the frame itself is a Lie algebra. With $L_i = e_i$ and $K_i = e_{i+3}$ obeying the brackets of Theorem 3.3, $\mathbb{R}^{3,3}$ becomes $\mathfrak{so}(1, 3)$ with Killing form -4η; in the variable $z = \mathbf{B} + i\mathbf{E}$ the bracket is the complex cross product, G is multiplication by i and lies in the centroid, $[Gx, y] = G[x, y]$, and $z \cdot z$ is the form q. The inner derivations fill six of the fifteen dimensions of $\mathfrak{so}(3, 3)$. Closing the force law into an Euler–Arnold equation on the frame would need a bracket for which the shift is inner, and by Proposition 4.17 there is none. A closure must leave the frame.

4.4.2 No closure on the frame; the rotor is unique

Proposition 4.19 (Equivariant maps and forms)**.** *For $\mathfrak{l}_G = \mathrm{span}\{L_i, K_i\}$: $\mathrm{Hom}_{\mathfrak{l}_G}(V, V) = \mathrm{span}\{\mathbf{1}, G\}$ and $\mathrm{Hom}_{\mathfrak{l}_G}(V, \mathbb{R}) = 0$; $\mathrm{Hom}_{\mathfrak{l}_G}(\Lambda^2 V, V) = \mathrm{span}\{\beta_0, G\beta_0\}$, where $\beta_0(X, Y)$ is the complex cross product $z(X) \times z(Y)$ returned to V, and $\mathrm{Hom}_{\mathfrak{l}_G}(\Lambda^2 V, \mathbb{R}) = 0$; and the $\mathfrak{l}_G$-invariant symmetric forms on V^* are $\mathrm{span}\{\eta^{-1}, G\eta^{-1}\}$.*

Theorem 4.20 (The brackets on the frame)**.** *The $\mathfrak{l}_G$-equivariant Lie brackets on V are $\beta_L = L\beta_0$ with $L = a + bG$. The bracket β_0 makes $(V, \beta_0) \cong \mathfrak{l}_G$, its derivations are the six inner ones, its centroid is $\mathrm{span}\{\mathbf{1}, G\} \cong \mathbb{C}$, so G is not a derivation, and β_L has Killing form $-4\big[(a^2 - b^2)\eta + 2ab\,\eta G\big]$.*

Proof. By Proposition 4.19 every equivariant antisymmetric bilinear map is $(c_1 + c_2 G)\beta_0$. Since L lies in the centroid, $\beta_L(\beta_L(X, Y), Z) = L^2\beta_0(\beta_0(X, Y), Z)$, and β_L inherits the Jacobi identity. □

Theorem 4.21 (No closure on the frame alone)**.** *Let $L = a + bG \neq 0$. For every symmetric N, with $\mathfrak{g}^*$ identified with $\mathfrak{g}$ through η, the Euler–Arnold field $\beta_0(M, NM)$ conserves $\eta(M, M)$ and $\eta(M, GM)$, whereas along the force law, in the form $\dot{M} = \kappa GM$, $\frac{d}{d\tau}\eta(M, M) = 2\kappa\,\eta(M, GM)$ and $\frac{d}{d\tau}\eta(M, GM) = -2\kappa\,\eta(M, M)$. No Hamiltonian flow on $(V, \beta_L)^*$ coincides with a nontrivial force law ($\kappa \neq 0$) on an open set U, and this remains true with a continuous state-dependent rate that does not vanish identically on U.*

Proof. The two quadratic forms are the Casimirs of $(V, \beta_L)^*$ and are conserved by every Hamiltonian flow, while the force law rotates them into each other. Equality on an open set would make both vanish there, that is $z \cdot z = 0$ on an open set, which no nonzero polynomial does. For a state-dependent rate f not identically zero on U, continuity gives an open $W \subset U$ on which $f \neq 0$, and the same argument applies on W. (On the null cone, where both Casimirs vanish, the equations $\beta_0(M_s, NM_s) = f_s\, GM_s$ at twelve rational null vectors force every $f_s = 0$ and $N \in \operatorname{span}\{\mathbf{1}, G\}$, for which $\beta_0(M, NM) \equiv 0$; this sampled computation is a check, not part of the proof.) □

A closure therefore needs a direction beyond the frame. Adjoin one generator $\hat{g}$, the *rotor*, on which $\mathfrak{l}_G$ acts trivially; the rate κ is a Lorentz scalar, since G commutes with $\mathfrak{l}_G$.

Theorem 4.22 (Uniqueness of the closure)**.** *Let $\mathfrak{s} = \mathbb{R}\hat{g} \oplus V$ carry an $\mathfrak{l}_G$-equivariant Lie bracket for which $\mathrm{ad}_{\hat{g}}$ does not vanish on V. Then $[\hat{g}, X] = (a + bG)X$ and $[X, Y] = 0$ for $X, Y \in V$, with $(a, b) \neq (0, 0)$. If moreover the rotor rotates the frame covector without dilating it, as the force law does, then $a = 0$ and, after rescaling $\hat{g}$, $\mathfrak{s} \cong \mathbb{R} \ltimes_G V$: $[\hat{g}, X] = GX$, $[X, Y] = 0$.*

Proof. Equivariance gives $[\hat{g}, X] = AX + \alpha(X)\hat{g}$ with $A \in \operatorname{span}\{\mathbf{1}, G\}$ and $\alpha = 0$, and $[X, Y] = (c_1 + c_2 G)\beta_0(X, Y)$, the $\hat{g}$-component vanishing by Proposition 4.19. The Jacobi identity for $(\hat{g}, X, Y)$ reads $(a + bG)(c_1 + c_2 G)\beta_0(X, Y) = 0$, a linear system in (c_1, c_2) with determinant $a^2 + b^2 \neq 0$, so $c_1 = c_2 = 0$. On $\mathfrak{s}^*$ the rotor acts on p through $(a + bG)^\top$, and $\frac{d}{d\tau}|p|^2 = 2a\,(h/I_0)|p|^2$ because G is antisymmetric; a pure rotation needs $a = 0$. □

The extension of Proposition 4.17 is therefore not one choice among several: at one constituent it is the only equivariant closure. The frame is abelian, all the dynamics sits in the rotor, and the rotor acts by the binding root itself. With several constituents the answer depends on the symmetry imposed. If each constituent carries its own copy of $\mathfrak{l}_G$, so that the symmetry is $\mathfrak{l}_G^{(1)} \oplus \cdots \oplus \mathfrak{l}_G^{(n)}$ and no constituents mix, equivariant maps between distinct constituents vanish, the argument runs factor by factor, and a common rotation rate gives $\mathfrak{s}_n = \mathbb{R} \ltimes_{G_n} V_n$ with $G_n = \bigoplus_k G$. Under a common, diagonal Lorentz action this fails: equivalent constituents admit cross-constituent intertwiners, $\mathrm{End}_{\mathfrak{l}_G}(V_n) = \{A \otimes \mathbf{1} + B \otimes G\} \cong M_n(\mathbb{C})$, so the rotor may act by any element of $\mathfrak{u}(n)$ that rotates without dilating. When its rates are resonant the frame then admits non-abelian equivariant brackets, already at $n = 2$ for rates in the ratio $1 : 2$. Under the diagonal action $\mathbb{R} \ltimes_{G_n} V_n$ with an abelian frame is therefore the selected subclass of a common rate and no mixing, not the only closure.

4.4.3 The Lie–Poisson dynamics, and the two interiors as clamped reductions

Proposition 4.23 (Lie–Poisson equations)**.** *On $\mathfrak{s}^*$ with coordinates (h, p) let $H = h^2/(2I_0) + \frac{1}{2}p^\top N p$, with $I_0 > 0$ the rotor inertia and N a symmetric form on V^*, the inverse frame inertia. The Lie–Poisson equations are*

$$\dot{p} = \frac{h}{I_0}\, G^\top p, \qquad \dot{h} = -\,p^\top G N p, \tag{4.16}$$

so the force law holds with the dynamical rate $\kappa = h/I_0$, and $|p|^2$ is a Casimir. For $p \neq 0$ the coadjoint orbit through (h, p_0) is the cylinder $\mathbb{R} \times \{e^{\theta G} p_0\} \cong T^*S^1$, on which (θ, h) are canonical for $h^2/(2I_0) + U(\theta)$ with $U(\theta) = \frac{1}{2}p(\theta)^\top N p(\theta)$ and $p(\theta) = e^{-\theta G} p_0$: every reduced motion is a motion of the rotor angle alone.

Proposition 4.24 (The two interiors are clamped reductions)**.** *On the group S with product $(\theta,x)(\theta',x')=(\theta+\theta',x+e^{\theta G}x')$ let $L=\frac{1}{2}I_0\dot\theta^2+\frac{m}{2}\eta(u,u)$ with body velocity $u=e^{-\theta G}\dot x$, and put $X=e^{-\theta G}x$. Then $u=\dot X+\dot\theta\,GX$, the Euler–Lagrange equations are (4.16) with $N=\eta^{-1}/m$, and clamping $\theta=\kappa\tau$ gives*

$$L=\tfrac{1}{2}I_0\kappa^2+\tfrac{m}{2}\,\eta(\dot X+\kappa GX,\ \dot X+\kappa GX),$$

the split-form interior (4.11). With $\frac{m}{2}|u|^2$ in place of $\frac{m}{2}\eta(u,u)$ the clamped Lagrangian is the positive-form interior (4.5) with $g=G$.

So the two interiors of Sections 4.2 and 4.3 are the clamped reductions of one closure, with inverse inertias $N=\eta^{-1}/m$ and $N=\mathbf{1}/m$. Which inertias the Clifford data permit is the next question.

4.4.4 Two requirements on the inertia

Definition 4.25 (The two requirements)**.** A symmetric N on V^* satisfies the *Lorentz requirement* if it is $\mathfrak{l}_G$-invariant, $rN+Nr^\top=0$ for all $r\in\mathfrak{l}_G$, and the *duality requirement* if the rate h/I_0 is conserved for all data. Write $N=N_c+N_a$ with $N_c=\frac{1}{2}(N-GNG)$ commuting and $N_a=\frac{1}{2}(N+GNG)$ anticommuting with G.

Theorem 4.26 (The two requirements exclude each other)**.** *(i) N_c and N_a are symmetric; the commuting forms have dimension 9 and the anticommuting ones 12.*

(ii) The Lorentz requirement gives $N=(a\eta^{-1}+bG\eta^{-1})/m$, a two-dimensional family inside the anticommuting forms, with frame energy $\frac{1}{2}p^\top Np=\mathrm{Re}\big((a+ib)w(p)\big)/(2m)$.

(iii) The duality requirement holds exactly when $[N,G]=0$. Among the forms invariant under the compact part of $\mathfrak{l}_G$ the only ones commuting with G are the multiples of $\mathbf{1}$: the positive form.

(iv) The two requirements meet in 0, and they do not exhaust the symmetric forms: $N=(2\,\mathbf{1}+\eta^{-1})/m$ is positive definite and satisfies neither.

In the field-strength reading the dichotomy is the classical one. The Lorentz class consists of the two invariants $|\mathbf{B}|^2-|\mathbf{E}|^2$ and $\mathbf{B}\cdot\mathbf{E}$; the duality class contains the energy density $|\mathbf{B}|^2+|\mathbf{E}|^2$; and a duality rotation preserves the energy density while it rotates the Lagrangian density [18, 31, 46]. The split form lies in the Lorentz class and the positive form in the duality class.

Theorem 4.27 (The frame under the closure)**.** *Reconstruct the group curve from the body velocity,*

$$(\theta,x)^{-1}\tfrac{d}{d\tau}(\theta,x)=(h/I_0,\ Np),\qquad \theta(0)=0.$$

Then $p=e^{-\theta G}p_0$ and

$$\dot x=(N_c+e^{2\theta G}N_a)p_0,\qquad \dot h=-\,p_0^\top Ge^{2\theta G}N_ap_0.$$

With the rotor clamped at $\theta=\kappa\tau$, $\kappa\neq0$,

$$x(\tau)=x_0+\tau\,N_cp_0+(e^{2\kappa\tau G}-\mathbf{1})(2\kappa G)^{-1}N_ap_0,$$

so the frame stays bounded for all data exactly when $N_c=0$ and drifts at the velocity N_cp_0 otherwise, while the rate is constant for all data exactly when $N_a=0$.

A clamped rate that is also the dynamical one, and a bounded clamped frame, are therefore exclusive requirements: each is met by one summand of $\mathrm{Sym}(V^*)=\mathrm{Sym}_c\oplus\mathrm{Sym}_a$. For $N=\mathbf{1}/m$ the rate is conserved and the frame drifts at p_0/m, which is the non-recurrence of Theorem 4.5, $X=e^{-\kappa\tau G}(X_0+\tau p_0/m)$; for $N=\eta^{-1}/m$ the clamped frame is bounded, as in Theorem 4.8, but the rate is conserved only on null data.

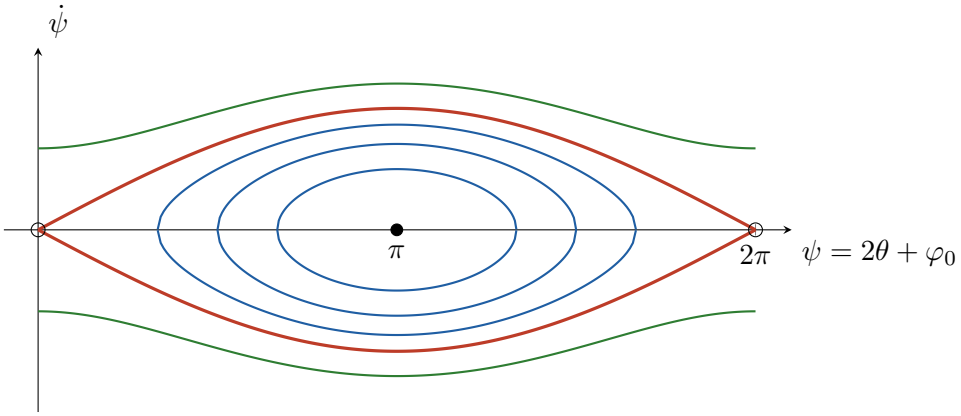


Figure 4.2: Phase portrait of the duality pendulum (Theorem 4.28), in units with $2R_0/(mI_0) = 1$; the filled dot is the stable point $\psi = \pi$ and the open dots the unstable one. Blue: librations, in which the rate $\kappa = \dot\theta$ changes sign periodically and the duality rotation reverses. Red: the separatrix. Green: rotations, in which the rotor turns for all time with a modulated rate. The constant-rate law of Section 4.3 is the limit of large rotor energy, far above the separatrix (Corollary 4.30).

4.4.5 The free rotor: a pendulum in the duality angle

Theorem 4.28 (The duality pendulum)**.** *Let $N = (a\eta^{-1} + bG\eta^{-1})/m$ and write $(a+ib)w(p_0) = R_0 e^{i\varphi_0}$, $R_0 \geq 0$. Along the flow $z(\pi)$ turns as $e^{i\theta}$ and $w(p) = e^{2i\theta}w(p_0)$, the frame energy is $U(\theta) = \frac{R_0}{2m}\cos(2\theta + \varphi_0)$, and*

$$I_0\ddot\theta = \frac{R_0}{m}\sin(2\theta + \varphi_0), \qquad E_\theta = \tfrac{1}{2}I_0\dot\theta^2 + \frac{R_0}{2m}\cos(2\theta + \varphi_0) = H.$$

With $\psi := 2\theta + \varphi_0$ this is the pendulum $\ddot\psi = (2R_0/mI_0)\sin\psi$, stable at $\psi = \pi$. The rotor angle is the duality angle of the momentum field, and the potential is its Lorentz-invariant frame energy.

Corollary 4.29 (When the rate is constant)**.** *With Lorentz-class inertia the rate is constant and nonzero along the flow exactly when $R_0 = 0$ and $h(0) \neq 0$; $R_0 = 0$ means $(a, b) = (0, 0)$ or $w(p_0) = 0$, that is $|\mathbf{B}| = |\mathbf{E}|$ and $\mathbf{B}\cdot\mathbf{E} = 0$ for the field of π_0: a null field, the class that contains plane waves [148, 131].*

Corollary 4.30 (The multiscale criterion)**.** *Let $R_0 > 0$, $\kappa_0 := \dot\theta(0) \neq 0$, $\psi_0 = 2\theta(0) + \varphi_0$ and $\varepsilon := R_0/(mI_0\kappa_0^2)$. Then $\kappa^2 = \kappa_0^2[1 + \varepsilon(\cos\psi_0 - \cos\psi)]$. The rotor turns for all time exactly when $\varepsilon(1 - \cos\psi_0) < 1$, and $\varepsilon < \frac{1}{2}$ suffices for every phase, in which case $|\kappa^2/\kappa_0^2 - 1| \leq 2\varepsilon$. At $\varepsilon(1 - \cos\psi_0) = 1$ the motion is the separatrix, and beyond it the rotor librates and the duality rotation reverses. The constant-rate law is the limit $\varepsilon \to 0$, and the multiscale requirement is*

$$I_0\,\kappa_0^2 \;\gg\; R_0/m. \tag{4.17}$$

A clamped rotor is therefore an idealisation of a heavy one. It is not a harmless idealisation, because the frame feels the modulation.

Theorem 4.31 (The free rotor drifts the frame)**.** *Let N be of Lorentz class with $R_0 > 0$ and $\kappa_0 \neq 0$, in the rotating regime. During one period of the rate, while θ changes by π, the frame is displaced by*

$$\Delta x = \lambda\, e^{-\varphi_0 G}Np_0, \qquad \lambda = \tfrac{1}{2}\int_0^{2\pi} \frac{\cos\psi\, d\psi}{|\kappa_0|\sqrt{1 + \varepsilon(\cos\psi_0 - \cos\psi)}} > 0,$$

and its mean velocity is $\frac{\varepsilon}{4}e^{-\varphi_0 G}Np_0 + O(\varepsilon^2)$. Every rotating solution with $R_0 \neq 0$ therefore has an unbounded frame.

Proof. For Lorentz-class N, $N = N_a$ and $\dot{x} = e^{2\theta G} N p_0$. In the rotating regime $\dot{\theta}$ never vanishes, so $d\tau = |d\theta|/|\dot{\theta}|$, and with $e^{2\theta G} = e^{-\varphi_0 G} e^{\psi G}$ and $e^{\psi G} = \cos\psi + G\sin\psi$ the $\sin\psi$ part integrates to zero because $1/|\dot{\theta}|$ depends on $\cos\psi$ only. Since $1/|\dot{\theta}|$ is a strictly increasing function of $\cos\psi$ when $R_0 > 0$, the integrand of λ exceeds its value at $\cos\psi = 0$ wherever $\cos\psi > 0$ and falls short of it where $\cos\psi < 0$, so $\lambda > 0$. Expanding to first order gives $\lambda = \pi\varepsilon/(4|\kappa_0|) + O(\varepsilon^2)$ over a period $\pi/|\kappa_0| + O(\varepsilon)$. □

Example 4.32 (A drift in closed form)**.** Take $p_0 = e^1$ and $m = I_0 = \kappa_0 = 1$, so that $w = 1$, $\varphi_0 = 0$, $\varepsilon = 1$ and $\dot{\theta}^2 = 1 + 2\sin^2\theta$. Writing $(1-2s)/\sqrt{1+2s} = 2/\sqrt{1+2s} - \sqrt{1+2s}$ with $s = \sin^2\theta$, the displacement per full turn of the rotor, $\theta \mapsto \theta + 2\pi$, which is two periods of the rate in Theorem 4.31, is

$$\Delta x = \big(8K(-2) - 4E(-2)\big)e_1 \approx 0.6336081022\, e_1,$$

with $K(m) = \int_0^{\pi/2}(1 - m\sin^2 t)^{-1/2}dt$ and $E(m) = \int_0^{\pi/2}(1 - m\sin^2 t)^{1/2}dt$ the complete elliptic integrals in the parameter m [36]. A numerical integration of the full system, frame included, reproduces this value to ten digits and returns h and p to their initial values after each turn.

The clamped frame of Theorem 4.27 moves on a circle of diameter $|Np_0|/|\kappa_0|$, while the free rotor adds $\frac{\pi\varepsilon}{2|\kappa_0|}|Np_0|$ per turn to first order, so the constant-rate picture tracks the bounded orbit for about $2/(\pi\varepsilon)$ turns. With Lorentz-class inertia and a free rotor, a constant rate and a bounded frame coexist exactly on null data.

Proposition 4.33 (Several constituents and one rotor)**.** *Let one rotor act on n constituents with $N_k = (a_k\eta^{-1} + b_k G\eta^{-1})/m_k$ and $w_k = w(p^{(k)})$. Then $\dot{w}_k = 2i(h/I_0)w_k$ for every k, all forces carry the common phase $e^{2i\theta}$, and with $R_{\mathrm{eff}}e^{i\varphi_{\mathrm{eff}}} := \sum_k (a_k + ib_k)w_k(0)/m_k$ the rotor obeys $I_0\ddot{\theta} = R_{\mathrm{eff}}\sin(2\theta + \varphi_{\mathrm{eff}})$. The* balance surface $R_{\mathrm{eff}} = 0$ *is preserved by the flow; on it the rate is exactly constant although no w_k need vanish, and if $h(0) \neq 0$ every frame moves on a bounded orbit; with $h(0) = 0$ the frames drift linearly. With equal inertias, $p^{(2)} = -Gp^{(1)}$ gives $w_2 = -w_1$.*

The Clifford data η, G, $G\eta^{-1}$ and β_0 are integer matrices. The rescaling $(I_0, m) \mapsto (sI_0, tm)$ preserves admissibility and, at fixed initial rate κ_0, multiplies ε by $1/(st)$ (at fixed $h(0)$ it multiplies it by s/t), so no value of ε, and no ratio among I_0, m and $|a + ib|$, is determined by the Clifford structure: the form of the inertia is derived, and (4.17) is an input.

4.4.6 Three constituents: the closure symmetry and the two projectors

At $n = 3$ write $e_A = e_i^{(k)}$ with $A = 6(k-1)+i$; the temporal indices are $\{1, 2, 3, 7, 8, 9, 13, 14, 15\}$, each with its partner $i' = i + 3$, and for temporal $i < j$ put

$$\mathrm{rot}(i,j) := e_{i'}e_{j'} - e_ie_j, \qquad \mathrm{bst}(i,j) := e_je_{i'} - e_ie_{j'}.$$

For one constituent these span $\mathfrak{l}_G$, with $\mathrm{rot}(2,3) = 2L_1$ and $\mathrm{bst}(2,3) = 2K_1$ and cyclically up to sign. In this numbering the weak directions of P_* are $W = \{7, 13, 14, 15\}$, the colour directions $\{10, 11, 12, 16, 17, 18\}$, and E denotes the span of the temporal directions $\{7, 8, 9, 13, 14, 15\}$ of constituents two and three; $G_3 = G \oplus G \oplus G$.

Theorem 4.34 (The closure symmetry is $\mathfrak{so}(9,\mathbb{C})$)**.** *$G_3$ is an anti-isometry of η_3 with $G_3^2 = -\mathbf{1}$, and its centraliser in $\mathfrak{so}(9,9)$ is* 72*-dimensional, spanned by the* 36 *paired rotations and the* 36 *paired boosts. In the basis of the nine temporal directions followed by their partners every element is $\left(\begin{smallmatrix} R & S \\ -S & R \end{smallmatrix}\right)$ with R, S antisymmetric, and $R - iS$ identifies the centraliser with $\mathfrak{so}(9,\mathbb{C})$, whose compact part consists of the paired rotations.*

In the field-strength reading $V_3 \cong \mathbb{C}^9$ carries three field strengths, and $\mathfrak{so}(9,\mathbb{C})$ consists of the complex orthogonal transformations of $\sum_k z^{(k)} \cdot z^{(k)}$. The identification of the centraliser of an anti-isometric complex structure with a complex orthogonal algebra is the inherited linear algebra of Norden geometry [301].

Theorem 4.35 (No common Lorentz algebra)**.** *The centraliser of G_3 meets the interior $\mathfrak{so}(1,3)$ of the first constituent in 0; the elements commuting with both G_3 and that algebra form the 30-dimensional $\mathfrak{so}(6,\mathbb{C})$ of paired elements of constituents two and three. Under $\mathfrak{l}_G^{(1)}$ the complexified spinor module is $128 \cdot (\frac{1}{2},\frac{1}{2})$, with no Weyl spinor.*

No nonzero Lorentz generator leaves both the closure and the spinor structure of matter invariant: the two Lorentz algebras are different subalgebras of $\mathfrak{so}(9,9)$ with zero intersection, which is Proposition 3.2 lifted to three constituents. A Pati–Salam generator acts on the closure variables only through its grade-two part, since ad_Z maps vectors to vectors exactly when Z has grades zero and two. The intersection with the Pati–Salam algebra can then be computed in general.

Theorem 4.36 (The Pati–Salam algebra and the binding root)**.** *Let U be any four-dimensional subspace of the span of the eight internal temporal directions, and $\mathfrak{ps}_U = \mathfrak{so}(U) \oplus \mathfrak{so}(6)_C$. Then neither summand meets the centraliser of G_3, and*

$$\mathfrak{ps}_U \cap \mathrm{Cent}(G_3) = \{A + G_3 A G_3^{-1} :\ A \in \mathfrak{so}(U \cap E)\} \cong \mathfrak{so}(U \cap E),$$

of dimension $\binom{d}{2} \in \{1,3,6\}$ with $d = \dim(U \cap E)$. For the Pati–Salam algebra of P_*, $U = \operatorname{span} W \subset E$, and the binding-compatible part is the diagonal $\mathfrak{so}(4)_\Delta = \operatorname{span}\{\mathrm{rot}(i,j) : i < j \in W\} \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$, compact and non-abelian.

Proof. Let $A \in \mathfrak{so}(U)$ and $B \in \mathfrak{so}(6)_C$ with $[A + B, G_3] = 0$, and split the internal temporal span as $E \oplus F$ with $F = \operatorname{span}\{e_2, e_3\}$. G_3 maps F into the interior plane $\operatorname{span}\{e_5, e_6\}$, where A and B vanish, so $Av = 0$ for $v \in F$. For $v \in E$, $G_3 v$ is a colour direction and $Bv = 0$, so $G_3 A v = B G_3 v$ lies in the colour span and $Av \in E$. Hence $A \in \mathfrak{so}(U \cap E)$ and $B = G_3 A G_3^{-1}$ on the colour span, and conversely. Finally $d \geq 4 + 6 - 8 = 2$. □

Neither Pati–Salam factor contains an element commuting with the binding root; the compatible part pairs every rotated temporal direction with its spatial partner, which the colour $\mathfrak{su}(4)$, acting on spatial directions alone, never does.

Theorem 4.37 (What survives each projector)**.** *(i) The part of $\mathrm{Cent}(G_3)$ commuting with P_{fam} is $\operatorname{span}\{\mathrm{rot}(2,3), \mathrm{rot}(8,9), \mathrm{rot}(14,15), \mathrm{bst}(14,15)\}$, abelian of dimension four. The kinetic condition holds for the two internal rotations and fails for $\mathrm{bst}(14,15)$, so the gauge-admissible part is $\mathfrak{u}(1)^2 = \operatorname{span}\{\mathrm{rot}(8,9), \mathrm{rot}(14,15)\}$.*

(ii) The part commuting with P_* is $\operatorname{span}\{\mathrm{rot}(2,3), \mathrm{rot}(8,9)\} \oplus \mathfrak{so}(4)_\Delta$, compact of dimension eight, and its internal part $\mathfrak{so}(4)_\Delta \oplus \mathbb{R}\,\mathrm{rot}(8,9)$ acts faithfully and satisfies the kinetic condition.

(iii) Under either projector the frame Lorentz algebra $\mathfrak{l}_G^{(1)}$ keeps only $\mathrm{rot}(2,3) = e_5 e_6 - e_2 e_3$, a rotation of the interior plane (e_5, e_6) locked to the rotation of the surplus pair.

The rotor acts through G_3 and commutes with all of $\mathrm{Cent}(G_3)$, so a one-rotor closure is compatible with either projector. What differs is the symmetry each leaves: an abelian one for P_{fam}, and for P_* the non-abelian diagonal of weak isospin and a colour $\mathfrak{so}(4)$.

Theorem 4.38 (Chirality and anomalies of what survives)**.** *With anomaly coefficients $\mathcal{A}(f) = \frac{1}{4}[\mathrm{Tr}(fP_L) - \mathrm{Tr}(fP_R)]$ and charges $-JB$:*

(q_1, q_2)	$(-1,-1)$	$(-1,0)$	$(-1,1)$	$(0,-1)$	$(0,0)$	$(0,1)$	$(1,-1)$	$(1,0)$	$(1,1)$
left, $\Gamma=-1$	1	0	1	1	2	1	0	2	0
right, $\Gamma=+1$	0	2	0	1	2	1	1	0	1

Table 4.1: Weyl content of the block of P_{fam} under the two surviving charges $Q_1 = -\frac{1}{2}J\,\mathrm{rot}(8,9)$ and $Q_2 = -\frac{1}{2}J\,\mathrm{rot}(14,15)$: the number of Weyl components with charges (q_1, q_2) on each chirality half.

(i) On P_{fam} the content is Table 4.1; $\mathcal{A}((\alpha Q_1 + \beta Q_2)^3) = -12\alpha\beta^2$ and $\mathcal{A}(\alpha Q_1 + \beta Q_2) = 0$, and the left and right charge multisets coincide exactly when $\alpha\beta = 0$. The anomaly-free subalgebras of $\mathfrak{u}(1)^2$ are 0, $\mathbb{R}Q_1$ and $\mathbb{R}Q_2$, each vector-like.

(ii) On P_*, with $\mathfrak{so}(4)_\Delta = \mathfrak{su}(2)_+ \oplus \mathfrak{su}(2)_-$ and $Q = -\frac{1}{2}J\,\mathrm{rot}(8,9)$, the halves are $L = (0,0)_0 \oplus (\frac{1}{2},\frac{1}{2})_{+1} \oplus (1,0)_0$ and $R = (0,0)_0 \oplus (\frac{1}{2},\frac{1}{2})_{-1} \oplus (0,1)_0$. For $H = \alpha H_3^+ + \beta H_3^- + \gamma Q$, $\mathcal{A}(H^3) = 2\gamma(3\alpha^2 + 3\beta^2 + 4\gamma^2)$ and $\mathcal{A}(H) = 8\gamma$, so the anomaly-free subalgebras are exactly those of $\mathfrak{so}(4)_\Delta$, which has neither a perturbative nor a global anomaly.

(iii) The two halves of P_* carry inequivalent representations of $\mathfrak{so}(4)_\Delta$, so it acts chirally; but every summand is a real representation of $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$, so the left-handed content is real and gauge-invariant Majorana masses exist [147]. The chirality of $\mathfrak{so}(4)_\Delta$ on P_* forbids no mass.

The rotor has no continuous lift to spinors: $(e^{\theta G})^\top \eta\, e^{\theta G} = \cos 2\theta\,\eta + \sin 2\theta\,\eta G$, so $e^{\theta G}$ is an isometry of η only for $\theta \in \pi\mathbb{Z}$, and at $\theta = \pi$, $e^{\pi G_3} = -\mathbf{1}$ is implemented by Γ. This excludes a coupling of the rotor through the spin representation and says nothing about other couplings.

Theorem 4.39 (Inertia freedom at three constituents)**.** *The real dimensions of the symmetric forms on V_3^* in the Lorentz class, $\{N, G_3\} = 0$, and in the duality class, $[N, G_3] = 0$, that are invariant under a given symmetry are:*

symmetry	Lorentz class	duality class
$\mathrm{Cent}(G_3) = \mathfrak{so}(9,\mathbb{C})$	2	0
$\bigoplus_k \mathfrak{l}_G^{(k)}$	6	0
the P_{fam}-surviving algebra	18	13
the P_*-surviving algebra	8	6
none	90	81

The Lorentz-class inertia is fixed up to one complex scale under the full closure symmetry and up to three under the single-constituent Lorentz algebras, and imposing only what a projector leaves enlarges the freedom to nine complex scales for P_{fam} and four for P_*: neither projector selects the inertia.

4.4.7 Brackets supplied by hand

The closure above is derived; one may also supply a bracket on the frame by hand and ask what the arena's own inertias do with it. A simple quadratic Lie algebra whose invariant form is η must have a maximal compact subalgebra of exactly half its dimension. For algebras that are not simple this fails: $\mathfrak{so}(4)$ with invariant forms of opposite sign on its two factors is compact and has signature $(3,3)$. Such *balanced* simple real forms are rare: a complex simple algebra regarded as real, $\mathfrak{su}(p,q)$ with $|p-q| = 1$, or $\mathfrak{so}(p,q)$ with $(p,q) = (\frac{k(k+1)}{2}, \frac{k(k-1)}{2})$, and no exceptional form. At the first rung $\mathbb{R}^{3,3}$ admits exactly three semisimple quadratic structures, $\mathfrak{so}(1,3)$, $\mathfrak{so}(4)$ and $\mathfrak{so}(2,2)$, and only $\mathfrak{so}(1,3)$ if η is to be proportional to the Killing form. For $n \le 20$ a simple closure exists exactly at $n = 1, 4, 5, 6, 7, 8, 12, 15, 16, 20$.

Two observations about the arena's inertias hold across these closures. With the split-form inertia $A = m\eta$, which is an invariant form, every solution of the Euler–Arnold equation is a one-parameter subgroup and $\dot{p} \equiv 0$. With the positive form $A = m\,\mathbf{1}$ on $\mathfrak{so}(1,3)$ the system is $\dot{\mathbf{B}} = 0$, $\dot{\mathbf{E}} = \frac{2}{m}\mathbf{E} \times \mathbf{B}$: the magnetic half is frozen and the electric half precesses about it. The inertias with $G^{\top}A^{-1}G = -A^{-1}$ form a twelve-dimensional family of holomorphic Euler tops $\dot{Z} = Z \times MZ$ with M complex symmetric; they are Liouville integrable and never positive definite.

At the rungs $n = (N^2 - 1)/3$ with $3 \nmid N$, that is $(N, n) = (2, 1), (4, 5), (5, 8), \ldots$, the compact half of the balanced form $\mathfrak{sl}(N, \mathbb{C})_{\mathbb{R}}$ is $\mathfrak{su}(N)$, which is Zeitlin's finite truncation of two-dimensional incompressible Euler flow [173]. Giving the 't Hooft clock–shift modes the discrete-Laplacian inertia, numerical integration at $N = 4$ and $N = 5$ conserves energy and enstrophy to 10^{-12} and moves up to 37% of the energy and 54% of the enstrophy between Laplacian shells: transfer between scales inside the frame. With either of the arena's own inertias the same data are exact equilibria, because both $\mathbf{1}$ and η restrict to multiples of the invariant form on $\mathfrak{su}(N)$. What the frame lacks for a fluid is the inertia. Finally, no closure dissipates: every Euler–Arnold flow stays on a coadjoint orbit and preserves every Casimir, which is exactly what viscosity destroys, so no closure of the force law can produce $\nu\Delta u$ on its own; and a constant generator that uniformly decreases the split form C has, by the inertia theorem of Ostrowski and Schneider [113], exactly $3n$ exponentially unstable directions.

4.5 Rest energy in the closure

The closure of Section 4.4 is the one dynamics that the arena closes by itself: the rate κ of the shift operators $\alpha_{\pm} = \partial_{\tau} \pm \kappa G$ becomes the angular velocity of a rotor. This section asks what that dynamics does to the energies of a field point, the undivided energy $\frac{1}{2}(|\mathbf{B}|^2 + |\mathbf{E}|^2)$ and the rest energy, and whether it relates the rest energy to the rate. Two fields are kept apart throughout: the *momentum field* $\pi = \eta^{-1}p$, which is $m\alpha_{+}X$ in the clamped split-form interior, and the *frame field* X. The answers, in brief, are these. The closure freezes the undivided energy, the rest energy and the Poynting vector of the momentum field, for every Hamiltonian; it changes those of the frame field only by an oscillation or an unbounded drift, and selects no value. The rest energy of the momentum field is exactly the amplitude of the potential of the duality angle, and the conserved total splits into a rate part and a Lorentz-invariant frame part, whose exchange of energy is fixed by that rest energy alone. The constant rate of the interior is the limit of a heavy rotor. The rest flow of Dirac matter and the binding rotation of the frame are two rotations at rest that the spin homomorphism does not relate; m and κ are independent inputs, and $mc^2 = \hbar\kappa$ is an added identification, and nothing in the algebra fixes the length ℓ in $\kappa = c/\ell$. A quantised rotor gives a rate ladder only under stated assumptions.

Notation. As in Section 4.4, $V = \mathbb{R}^{3,3}$, G is the binding root, $G^2 = -\mathbf{1}$, $G^{\top} = -G$, $\eta G = -G\eta$, a covector p is a column, and $z(Y) = \mathbf{B} + i\mathbf{E}$ for $Y \in V$ with components $(\mathbf{B}; \mathbf{E})$ along $(e_1, e_2, e_3; e_4, e_5, e_6)$, so that G acts as i (Theorem 3.3). With $q(z) = z \cdot z$, the undivided energy $E_6(Y)$, the rest energy $E_0(Y)$ and the Poynting vector $\mathbf{S}(Y) = \mathbf{E} \times \mathbf{B}$ of Y, minus the textbook one, are those of (3.17); only $|\mathbf{S}|$ enters below. For the momentum field, $E_6(\pi) = \frac{1}{2}|p|^2$ because η is orthogonal, and $E_0(\pi) = \frac{1}{2}|w(p)|$ with w as in (4.15). The Lorentz-class inertia of Theorem 4.26(ii) is written $N = (\nu_1\eta^{-1} + \nu_2 G\eta^{-1})/m$, $\nu := \nu_1 + i\nu_2$, its coefficients being the (a, b) of that theorem. Units are $\hbar = c = 1$ unless a symbol is shown, τ is a time, and $\mathrm{K}(\mu)$, $\mathrm{E}(\mu)$ are the complete elliptic integrals of Example 4.32.

4.5.1 What the closure conserves

On $\mathfrak{s}^*$, the dual of the unique closure $\mathfrak{s} = \mathbb{R}\hat{g} \ltimes_G V$ of Theorem 4.22, with coordinates (h, p), $F_h = \partial F/\partial h$ and $\nabla F = \partial F/\partial p$, the Lie–Poisson bracket and the equations of an arbitrary Hamiltonian H are

$$\{F_1, F_2\} = p^\top (F_{2,h}\, G\nabla F_1 - F_{1,h}\, G\nabla F_2), \qquad \dot{p} = H_h\, G^\top p, \qquad \dot{h} = -\, p^\top G\, \nabla H. \tag{4.18}$$

For $H = h^2/(2I_0) + \frac{1}{2} p^\top N p$ these are the equations (4.16) of Proposition 4.23.

Theorem 4.40 (The momentum field is frozen up to its duality phase)**.** *(i) A function $f(p)$ is a Casimir of $\mathfrak{s}^*$ if and only if it is invariant under $p \mapsto e^{tG}p$, that is under the duality rotations $z(\pi) \mapsto e^{-it} z(\pi)$ of the momentum field. Hence $E_6(\pi)$, $E_0(\pi)^2$ and the vector $\mathbf{S}(\pi)$ are smooth Casimirs, while $\operatorname{Re} w$ and $\operatorname{Im} w$ are not; $E_0 = |w|/2$ is conserved as well but is not smooth at $w = 0$.*

(ii) At $p \neq 0$ the symplectic leaves are two-dimensional, so there are five independent local Casimirs. The real and imaginary parts of the products $\bar{z}_j z_k$, $z = z(\pi)$, provide five, and $E_6(\pi)$, $E_0(\pi)^2$, $\mathbf{S}(\pi)$ account generically for four of them, subject to $E_6(\pi)^2 = E_0(\pi)^2 + |\mathbf{S}(\pi)|^2$.

(iii) Along the flow of every Hamiltonian, $p(\tau) = e^{-\theta(\tau)G} p_0$ with $\dot{\theta} = H_h$ and $\theta(0) = 0$; so $z(\pi(\tau)) = e^{i\theta(\tau)} z(\pi_0)$ and $w(p(\tau)) = e^{2i\theta(\tau)} w(p_0)$.

Proof. (i) For $f = f(p)$, (4.18) gives $\{f, F\} = F_h\, p^\top G \nabla f = -F_h\, (Gp) \cdot \nabla f$, which vanishes for every F exactly when f is constant along $t \mapsto e^{tG}p$. Since $\eta G = -G\eta$, $\eta^{-1} e^{tG} p = e^{-tG}\pi$, on which G acts as i. The quantities $|z|^2$, $|z \cdot z|$ and $\bar{z} \times z$ are invariant under a phase, while $w = z \cdot z$ is multiplied by e^{-2it}. (ii) From (4.18), $\{p_i, p_j\} = 0$ and $\{h, p_j\} = (Gp)_j$, so the Poisson tensor has rank two at $p \neq 0$ and the leaves have codimension five in the seven-dimensional $\mathfrak{s}^*$. The map $z \mapsto z\bar{z}^\top$ from $\mathbb{C}^3 \setminus \{0\}$ onto the rank-one positive Hermitian matrices has the phase circles as its fibres, so its image is five-dimensional and its nine real components contain five independent functions, all phase invariant. The relation is the identity (3.18) of Proposition 3.93. (iii) $\dot{p} = -H_h G p$ integrates to $p = e^{-\theta G} p_0$, so $\pi = e^{\theta G}\pi_0$, and w is quadratic in z. □

The theorem is fixed by the algebra. In substance it is a corollary of Proposition 4.23, whose coadjoint orbits are the cylinders $\mathbb{R} \times \{e^{\theta G} p_0\}$; item (iii) is the phase law of Theorem 4.28, extended to every Hamiltonian, and the relation in (ii) is inherited. That $E_6(\pi)$, $E_0(\pi)$ and $\mathbf{S}(\pi)$ are conserved for every Hamiltonian follows from Proposition 4.23, and that they are constant already in the clamped interior is Proposition 4.13(i). What the theorem adds is that the Casimirs are exactly the duality-invariant functions of p, and their count.

Corollary 4.41 (What the closure can and cannot change)**.** *(i) For every Hamiltonian on $\mathfrak{s}^*$, quadratic or not, of Lorentz class or not, $E_6(\pi)$, $E_0(\pi)$ and $\mathbf{S}(\pi)$ are constant along every solution: a null momentum field stays null, and a momentum field with rest energy $E_0(\pi_0)$ keeps it.*

(ii) The same functions of the frame field are not constants of motion. In the clamped split-form interior, where $z(X) = c_- e^{-i\kappa\tau} + c_+ e^{i\kappa\tau}$, they oscillate in general (Proposition 4.13(ii)); for instance $c_- = (1, i, 0)$, $c_+ = (1, 0, 0)$ give $|q(z(X))|^2 = 9 - 8\sin^2 \kappa\tau$. Under the free-rotor drift of Example 4.32 with $x(0) = 0$, the frame field $X = e^{-\theta G} x$ has $E_0(X) = 0$ at $\tau = 0$ and $E_0(X) = E_6(X) = \frac{1}{2} k^2 |\Delta x|^2$ after k full turns, Δx the displacement per turn of that example; after three turns this is ≈ 1.807. On every rotating solution with Lorentz-class inertia and $R_0 > 0$, in the notation of Theorem 4.28, $E_6(X)$ and $E_0(X)$ are unbounded.

Proof. (i) Casimirs Poisson-commute with every Hamiltonian (Theorem 4.40(i)). (ii) By the expansion of $z \cdot z$ in Proposition 4.13(ii), in the example $c_- \cdot c_- = 0$ and $c_- \cdot c_+ = c_+ \cdot c_+ = 1$,

so $|q|^2 = |2 + e^{2i\kappa\tau}|^2 = 5 + 4\cos 2\kappa\tau$. Since G acts as i, $e^{-\theta G}$ multiplies z by a phase, so $E_6(X) = E_6(x)$ and $E_0(X) = E_0(x)$. In Example 4.32 the motion of (θ, h) and the velocity $\dot{x} = e^{2\theta G} N p_0$ are periodic in θ, so at $\theta = 2\pi k$ the frame is $x = k\,\Delta x$, a real multiple of e_1, and $q(z(x)) = k^2|\Delta x|^2$. In general, by Theorem 4.31 the frame advances by $\Delta x = \lambda e^{-\varphi_0 G} N p_0$, $\lambda > 0$, in each period of the rate, so $x = x_0 + n\Delta x$ after n periods. As $z(Np_0) = \nu\, z(\pi_0)/m$, $q(z(\Delta x)) = \lambda^2 e^{-2i\varphi_0}\nu^2 w(p_0)/m^2$, which is nonzero because $|\nu w(p_0)| = R_0 > 0$; hence $|q(z(x))|$ and $|x|^2$ grow like n^2. □

Item (i) is an obstruction, under the hypothesis that the dynamics is a Hamiltonian flow on $\mathfrak{s}^*$: no such flow produces, removes or changes the rest energy of the momentum field. Item (ii) is fixed: its statement on the clamped interior is Proposition 4.13(ii), its growth follows from Proposition 4.11 and Theorem 4.31, and the two examples are added. The first verdict therefore cuts both ways. For the momentum field, $E_6(\pi)$ and its split $E_6(\pi)^2 = E_0(\pi)^2 + |\mathbf{S}(\pi)|^2$ label the leaf: the rest energy is carried by the dynamics, not produced by it. For the frame field, to which the field-strength reading of E_6 and E_0 refers, the closure does change the rest energy, by an oscillation or by an unbounded drift, but not by any mechanism that selects a value. Nor is $E_6(X)$ a function of $E_6(\pi)$: on the orbits without momentum with $X \neq 0$, $|p|^2 = 0 < E_6(X)$ (Proposition 4.13(iii)). What moves is the duality angle θ and the rate h/I_0.

4.5.2 The duality pendulum

Unless duality-class inertia is named, N is from now on of Lorentz class, $\nu w(p_0) = R_0 e^{i\varphi_0}$ with $R_0 \geq 0$, and $\psi := 2\theta + \varphi_0$. By Theorem 4.26(ii) the frame energy is $\frac{1}{2}p^\top N p = \mathrm{Re}(\nu w(p))/(2m)$, and by Theorem 4.28 the rotor is the pendulum

$$\frac{h^2}{2I_0} + U(\theta), \qquad U(\theta) = \frac{R_0}{2m}\cos\psi, \qquad I_0\ddot{\theta} = \frac{R_0}{m}\sin\psi, \qquad R_0 = |\nu|\,|w(p_0)| = 2|\nu|\,E_0(\pi_0).$$

The amplitude of the potential of the duality angle is therefore the rest energy of the momentum field, times $|\nu|/m$; its undivided energy $E_6(\pi)$ and its Poynting vector $\mathbf{S}(\pi)$ do not enter.

Proposition 4.42 (Small oscillations)**.** *The stable equilibrium is $\psi = \pi$, where $U = -R_0/(2m)$. With $\psi = \pi + \xi$, $\ddot{\xi} = -\Omega^2 \sin\xi$ and*

$$\Omega^2 = \frac{2R_0}{mI_0} = \frac{4|\nu|\,E_0(\pi_0)}{mI_0};$$

θ oscillates at the same frequency with half the amplitude of ξ, and at amplitude $0 < \xi_0 < \pi$ the period is $4\,\mathrm{K}(\sin^2(\xi_0/2))/\Omega$.

Proof. $\ddot{\psi} = 2\ddot{\theta} = (2R_0/mI_0)\sin\psi$ and $\sin(\pi + \xi) = -\sin\xi$. The period follows from $\frac{1}{2}\dot{\xi}^2 = \Omega^2(\cos\xi - \cos\xi_0)$ with the substitution $\sin(\xi/2) = \sin(\xi_0/2)\sin t$. □

The proposition is fixed, as a corollary of Theorem 4.28, which it linearises; the period is the inherited one of the pendulum [36]. By Corollary 4.29, with Lorentz-class inertia the rate is constant and nonzero exactly when $R_0 = 0$, that is when $\nu = 0$ or the momentum field is null, and $h(0) \neq 0$; then $U \equiv 0$ and $\theta = h\tau/I_0$ is an exact flat direction. For duality-class inertia, $[N, G] = 0$, the rate is constant for all data (Theorem 4.26(iii), Theorem 4.27), and the potential $U(\theta) = \frac{1}{2}p_0^\top e^{\theta G} N e^{-\theta G} p_0 = \frac{1}{2}p_0^\top N p_0$ is constant, in general not zero. The remaining case is the following.

Corollary 4.43 (A constant rate with a rest energy)**.** *Let N be of Lorentz class with $\nu \neq 0$, and let $E_0(\pi_0) > 0$. The rate h/I_0 is constant along a solution if and only if $h(0) = 0$ and $\psi(0) \in \pi\mathbb{Z}$. There the rate vanishes and the frame moves at the constant velocity Np_0, with $|Np_0| = |\nu|\,|p_0|/m \neq 0$.*

Proof. $\dot h = (R_0/m)\sin\psi$ with $R_0 > 0$. If h is constant, $\sin\psi \equiv 0$, so θ is constant and $h = I_0\dot\theta = 0$; conversely these points are equilibria. With $\theta \equiv 0$, Theorem 4.27 ($N_c = 0$) gives $\dot x = Np_0$, and $N^2 = |\nu|^2\mathbf{1}/m^2$ because $\eta^{-1}G\eta^{-1} = -G$. □

The corollary is fixed by the algebra. It complements Corollary 4.29, which covers zero rest energy, and adds the uniform motion of the frame at the equilibria.

Remark 4.44 (A pseudo-Goldstone reading, and its limits)**.** On a leaf, θ is the angle of the duality group $U(1)_G$ of the momentum field, which a nonzero rest energy breaks to $\{\pm\mathbf{1}\}$ on the space of Lorentz orbits (Proposition 3.96; Section 7.1.5). In the duality class it is a symmetry of H; in the Lorentz class it is broken on each leaf in proportion to $E_0(\pi)$, and exactly

$$I_0\,\Omega^2 = U''(\theta_{\min}) = \frac{2R_0}{m} = \frac{4|\nu|\,E_0(\pi_0)}{m}.$$

The first equality defines Ω for every oscillator with one degree of freedom; the content specific to the closure is the second, which is Theorem 4.28. With I_0 in the place of the square f_π^2 of the pion decay constant and $|\nu|E_0(\pi_0)/m$ in the place of the explicit breaking, the relation has the shape of the Gell-Mann–Oakes–Renner relation for a pseudo-Goldstone boson [48, 158], as every pendulum does. That is a reading, and its limits are definite. The breaking is explicit, by the inertia, not spontaneous: there is no vacuum manifold, and $E_0(\pi)$ is a label fixed by the data (Corollary 4.41(i)), not an order parameter; and by the bound (3.16) the energy of a field point is least at the zero field, with no degenerate family of lowest-energy configurations among which a phase of q could be chosen. With one degree of freedom, "mass" here means a frequency. What survives the reading is exact: on a null momentum field the angle is a zero mode, and a rest energy of the momentum field gives it the frequency $\Omega \propto \sqrt{E_0(\pi_0)}$.

4.5.3 The energy budget: a rate part and a frame part

Proposition 4.45 (Rate part and frame part)**.** *With Lorentz-class inertia, along every solution:*

(i) $H = K_{\rm rot} + U$ *with* $K_{\rm rot} = \frac12 I_0\kappa^2 \geq 0$, $\kappa = h/I_0$, $U = \frac{R_0}{2m}\cos\psi = \mathrm{Re}(\nu\,w(p))/(2m)$, *and* $\dot K_{\rm rot} = -\dot U = \frac{R_0}{m}\,\kappa\sin\psi$;

(ii) in the rotating regime $K_{\rm rot}$ *runs through exactly* $[H - \frac{R_0}{2m},\, H + \frac{R_0}{2m}]$ *in every period of the rate, so the energy exchanged between the two parts is* $R_0/m = 2|\nu|E_0(\pi_0)/m$; *in the librating regime it runs through* $[0,\, H + \frac{R_0}{2m}]$;

(iii) $K_{\rm rot}$ *and* U *are separately invariant under* $\mathfrak{l}_G$ *acting by* $(h,p) \mapsto (h, \Lambda^{-\top}p)$, $\Lambda = e^r$, $r \in \mathfrak{l}_G$, *and the flow commutes with this action;*

(iv) $(\theta, h)(\tau)$ *depends on* p_0 *only through* $\nu w(p_0)$; $E_6(\pi_0)$ *and* $\mathbf{S}(\pi_0)$ *do not enter the motion.*

Proof. (i) $\dot K_{\rm rot} = h\dot h/I_0$ and $\dot U = -(R_0/2m)\,2\kappa\sin\psi$, with $\dot h = (R_0/m)\sin\psi$. (ii) In rotation $\dot\psi = 2\kappa$ never vanishes and $\cos\psi$ covers $[-1,1]$ in each period; in libration $K_{\rm rot} = 0$ at the turning points and is largest at $\psi = \pi$. (iii) The Lorentz requirement $rN + Nr^\top = 0$ of Definition 4.25 integrates to $\Lambda^{-1}N\Lambda^{-\top} = N$, so $U(\Lambda^{-\top}p) = U(p)$; since $[r,G] = 0$, $\Lambda^{-\top}$ commutes with $G^\top$, and both equations (4.16) are preserved. (iv) $\eta^{-1}\Lambda^{-\top} = \Lambda\eta^{-1}$ because $\Lambda \in O(3,3)$, and $\mathfrak{l}_G \cong \mathfrak{so}(3,\mathbb{C})$ acts on z complex-linearly and preserves q (Theorem 3.1), so $w(\Lambda^{-\top}p) = w(p)$; the equations for (θ, h) contain only R_0 and φ_0. □

Item (i) restates the energy $E_\theta = H$ of Theorem 4.28; items (ii)–(iv) are fixed by the algebra. The conserved total thus differentiates into two Lorentz-invariant parts, a rate part and a frame part, whose exchange is fixed by the rest energy of the momentum field alone; neither part is the undivided energy $E_6(\pi)$, which is inert.

Units, and the factor one half. Take $[\tau] = \mathsf{T}$ and $[\kappa] = \mathsf{T}^{-1}$, as in (2.2), and $[X] = \mathsf{L}$, a convention by Proposition 4.48(i) under which m is a mass. Then $[I_0] = \mathsf{ML}^2$, $[h] = \mathsf{ML}^2\mathsf{T}^{-1}$ is an action, $[p] = \mathsf{MLT}^{-1}$, $[N] = \mathsf{M}^{-1}$ and $[H] = [K_{\mathrm{rot}}] = [U] = \mathsf{ML}^2\mathsf{T}^{-2}$; $E_0(\pi)$ is a squared momentum and $|\nu|E_0(\pi)/m$ an energy. With the clamped value $\kappa = c/\ell$ and $M_{\mathrm{rot}} := I_0/\ell^2$, a mass,

$$K_{\mathrm{rot}} = \tfrac{1}{2} I_0 \kappa^2 = \tfrac{1}{2} M_{\mathrm{rot}} c^2.$$

Reading this as the "c^2 part" of the conserved total is a reading: it is literally the Newtonian kinetic energy of a mass M_{rot} on a circle of radius ℓ at rim speed c, outside the domain of Newtonian kinetics. It is not $E = mc^2$. The factor $\frac{1}{2}$ comes from an energy quadratic in the rate, where Planck's relation is linear; M_{rot} is not tied to m, since the rescaling $(I_0, m) \mapsto (sI_0, tm)$ recalled in Section 4.5.6 changes the two independently; and $\kappa = c/\ell$ holds only for the clamped rotor.

4.5.4 When the rate is constant: the multiscale criterion

Corollary 4.30 answers when the rate is nearly constant. In terms of the two parts of the energy its parameter is

$$\varepsilon = \frac{R_0}{mI_0\kappa_0^2} = \frac{U_{\max}}{K_{\mathrm{rot}}(0)}, \qquad U_{\max} = \frac{R_0}{2m} = \frac{|\nu|\, E_0(\pi_0)}{m},$$

the ratio of the rest-energy amplitude of the momentum field to the initial rotor energy. If c is the fixed conversion of Theorem 2.2, a varying rate $\kappa = c/\ell$ is a varying length $\ell(\tau) = c/\kappa(\tau)$, and the law $\kappa^2 = \kappa_0^2[1 + \varepsilon(\cos\psi_0 - \cos\psi)]$ of Corollary 4.30 gives it the relative spread $\varepsilon + O(\varepsilon^2)$ along a rotating solution. With $\kappa_0 = c/\ell$, the multiscale requirement (4.17) becomes

$$\tfrac{1}{2} M_{\mathrm{rot}} c^2 \;\gg\; \frac{|\nu|\, E_0(\pi_0)}{m} :$$

for "c/ℓ" to be constant, the rotor energy must dominate the rest-energy amplitude of the momentum field. The clamped split-form interior of Section 4.3 is the limit $\varepsilon \to 0$ (Proposition 4.24 and Corollary 4.30); the positive-form interior of Section 4.2 has duality-class inertia and an exactly constant rate (Theorem 4.26(iii)). Read in this way, the shift κG, with $(\kappa G)^2 = -c^2/\ell^2\,\mathbf{1}$, which Section 13.5 reads as the "symmetric part" of $E = mc^2$ (Proposition 3.98(iii), Theorem 4.18), is constant only insofar as the differentiated part is negligible beside the rotor; ε is an input, and the reading is ours.

4.5.5 Two rotations at rest

Two rotations at rest are available. The first belongs to Dirac matter. At $\mathbf{p} = 0$ the Hamiltonian (2.4) of the commuting first-order form $\mathcal{L}_1 = \frac{1}{2}\Psi^\top J\dot\Psi - \frac{1}{2}\Psi^\top H\Psi$ is $H(0) = m\gamma^0 = me_1$.

Proposition 4.46 (The rest flow)**.** *$JH(0) = mT$, so the rest flow is $\Psi(\lambda) = e^{-m\lambda T}\Psi_0 = (\cos m\lambda\,\mathbf{1} - \sin m\lambda\, T)\Psi_0$. On the eigenspaces $e_1 = \pm 1$, each four-dimensional and J-invariant, $T = \pm J$, so $\Psi(\lambda) = e^{\mp m\lambda J}\Psi_0$ with energy $\frac{1}{2}\Psi^\top H\Psi = \pm\frac{1}{2}m|\Psi|^2$. Moreover $\gamma^0 = -TJ$.*

Proof. $Je_1 = e_2e_3e_1 = e_1e_2e_3 = T$. Since $T = e_1J$ with $[e_1, J] = 0$, $T\Pi_\pm = \pm J\Pi_\pm$ for $\Pi_\pm = \frac{1}{2}(\mathbf{1} \pm e_1)$; $T^2 = -\mathbf{1}$ gives the exponential, and $-TJ = -e_1J^2 = e_1$. □

The proposition is inherited: it is the realification of $\psi = e^{\mp im\lambda}\psi_0$ on $\beta = \pm 1$, at angular frequency $mc^2/\hbar$ in restored units, de Broglie's internal clock [30]. The second rotation belongs to the frame: the orbits without momentum of the split-form interior are the duality rotations $z \mapsto e^{-i\kappa\tau}z$ of angular frequency $\kappa = c/\ell$, along which q turns at 2κ, and the duality rotations of the other sense carry all the momentum (Proposition 4.11(i)–(ii), Proposition 4.13(iii)); the sense without momentum is singled out by the momentum, not by duality.

Proposition 4.47 (No operator relates the two rates)**.** *(i)* $\mathrm{Ad}_{e^{\vartheta T}}$ *fixes* e_1, e_2, e_3 *and maps a spatial generator* e_a *to* $\cos 2\vartheta\, e_a + \sin 2\vartheta\, Te_a$*, which is a vector only for* $\vartheta \in \frac{\pi}{2}\mathbb{Z}$*. The Dirac rest flow is therefore not a motion of* V*, and its generator* T*, of grade three, is not in* $\mathfrak{so}(3,3)$*.*

(ii) $(e^{\theta G})^\top \eta\, e^{\theta G} = \cos 2\theta\, \eta + \sin 2\theta\, \eta G$*, so* $e^{\theta G}$ *lies in* $O(3,3)$*, and lifts to* $\mathrm{Pin}(3,3)$*, only for* $\theta \in \pi\mathbb{Z}$*.*

Hence neither one-parameter group is the image of the other under the spin homomorphism, and this route supplies no relation between m *and* κ*.*

Proof. (i) T commutes with the temporal and anticommutes with the spatial generators (Lemma 1.1), so $e^{\vartheta T} e_a e^{-\vartheta T} = e^{2\vartheta T} e_a$, and Te_a has grade four. (ii) Expand with $e^{\theta G} = \cos\theta + \sin\theta\, G$, using that $G^\top \eta = \eta G$ is symmetric and $G^\top \eta G = -\eta$. □

The proposition is an obstruction to one route only, a map of the arena carrying one rate to the other; item (ii) is the remark closing the three-constituent part of Section 4.4, and the corresponding statement in Section 3.8. That m and κ are independent is not a consequence of it: both are free inputs. What remains is an identification, and it is added, not derived. With $E = \hbar\times\text{rate}$, $\hbar\kappa = mc^2$ holds if and only if $\ell = \hbar/(mc)$, the reduced Compton wavelength. The dynamics does not supply it. The interior equation $\ddot{X} = -\kappa^2 X$ does not contain m, which multiplies the Lagrangian; and the rescaling $(\tau, \kappa, I_0, m) \mapsto (s\tau, \kappa/s, sI_0, sm)$ of Proposition 4.48, which maps solutions to solutions and preserves the action, hence any quantisation of h, keeps $m\kappa$ and multiplies m/κ by s^2, so that $m = \hbar\kappa/c^2$ picks one point of each orbit by fiat. The rate is moreover ambiguous by a factor two: q turns at 2κ while z turns at κ, and matching $2\hbar\kappa = mc^2$ instead gives $\ell = 2\hbar/(mc)$. The algebra decides neither.

4.5.6 Scale freedom

The Clifford data η, G, $G\eta^{-1}$, ηG, the operator $\mathcal{K}$ of Lemma 2.4 and the structure constants of the complex cross product β_0 of Proposition 4.19 are integer matrices, and the rescaling $(I_0, m) \mapsto (sI_0, tm)$, $s, t > 0$, preserves admissibility and multiplies ε by $1/(st)$ at fixed initial rate κ_0, by s/t at fixed $h(0)$ (the remark following Proposition 4.33). Two further scalings act on the time.

Proposition 4.48 (Scalings)**.** *(i) The interior law* $\ddot{X} = -\kappa^2 X$ *is invariant under* $X \mapsto \sigma X$*, which multiplies* $\frac{m}{2}\eta(\alpha_+ X, \alpha_+ X)$ *by* σ^2*, absorbed by* $m \mapsto m/\sigma^2$*; and under* $(\tau, \kappa) \mapsto (s\tau, \kappa/s)$*, which together with* $m \mapsto sm$ *preserves the action.*

(ii) The closure is invariant under $(\tau, I_0, m) \mapsto (s\tau, sI_0, sm)$*, with* (θ, h, p) *unchanged as functions of the rescaled time;* $\kappa \mapsto \kappa/s$*, while* $m\kappa$*,* ε *and the action are invariant.*

Proof. (i) Linearity; $Y(\tau') = X(\tau'/s)$ solves $Y'' = -(\kappa/s)^2 Y$, the Lagrangian with sm in place of m is multiplied by s^{-1}, and $d\tau' = s\, d\tau$. (ii) With $d/d\tau' = s^{-1} d/d\tau$, $I_0' = sI_0$ and $N' = N/s$, both equations (4.16) are reproduced, $H' = H/s$, and $\kappa_0' = \kappa_0/s$ gives $\varepsilon' = \varepsilon$; h is unchanged. □

The proposition is fixed by the algebra.

Proposition 4.49 (Nothing in the algebra fixes ℓ)**.** *(i) Schur's lemma [136] fixes only the form* $c\,\mathbf{1}$ *of the conversion operator (Theorem 2.2), and* c *converts units of time into units of length.*

(ii) No monomial $c^{n_1}\hbar^{n_2}$ *is a length, a mass or a time; with a mass* m *adjoined, the only length is* $\hbar/(mc)$*, up to a number.*

(iii) The interior systems with different $\kappa > 0$ *are carried into each other by a rescaling of* τ *that fixes every Clifford datum (Proposition 4.48(i)), so no condition formulated in the Clifford data distinguishes one value of* κ*, or of* $\ell = c/\kappa$*; in the closure none fixes* ε*.*

Proof. (i) The commutant of the vector representation of $\mathfrak{so}(3)$ in $M_3(\mathbb{R})$ is $\mathbb{R}\,\mathbf{1}$. (ii) In the basis $(\mathsf{M}, \mathsf{L}, \mathsf{T})$ the dimensions $[c] = (0, 1, -1)$ and $[\hbar] = (1, 2, -1)$ span a plane containing none of $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$; adjoining $[m] = (1, 0, 0)$ gives an invertible matrix, with $(0, 1, 0) = -[c] + [\hbar] - [m]$. (iii) The data are integer matrices, the rescaling acts on τ and κ only, and ε is moved by the rescaling of (I_0, m). □

Item (i) restates Theorem 2.2, fixed there, and item (ii) is inherited dimensional analysis, already implicit in (2.2), where the sum of ∂_λ and G requires a scale; (iii) is fixed by the algebra. This is the one exact sense available here of the phrase that m breaks the symmetry of c^2, the symmetric part of $E = mc^2$ in the reading of Section 13.5 (compare Section 7.1.5): c carries no scale, the shift κG needs a length ℓ that the algebra cannot supply, and through $\ell = \hbar/(mc)$ that length would be a mass. What it would break is the change of the unit of τ (Proposition 4.48), not a symmetry of a vacuum; that is a reading.

4.5.7 The frame drifts

Example 4.32 computes the drift of the frame in closed form, $\Delta x = (8\mathrm{K}(-2) - 4\mathrm{E}(-2))e_1$ per full turn of the rotor. After half a turn h has returned and $p = -p_0$, since $e^{\pi G} = -\mathbf{1}$; after a full turn both have returned. By Theorem 4.31 every rotating solution with $R_0 > 0$, that is with $\nu \neq 0$ and a nonzero rest energy of the momentum field, has an unbounded frame, and by Corollary 4.41(ii) the rest energy of the frame field grows with it, as $\frac{1}{2}k^2|\Delta x|^2$ after k turns in the example. A constant nonzero rate and a bounded frame coexist, for a free rotor with Lorentz-class inertia, exactly on null data (Section 4.4); at the equilibria of Corollary 4.43 the rate is zero and the frame moves uniformly; and with several constituents they coexist on the balance surface of Proposition 4.33. Read in the terms of this section, and only as a reading, the differentiation of the energy in motion unbinds the frame.

4.5.8 A quantum rotor, conditionally

Proposition 4.50 (The half turn)**.** *(i)* $e^{\pi G} = -\mathbf{1}$, $e^{2\pi G} = \mathbf{1}$, *and for* $p_0 \neq 0$ *the leaf circle* $\{e^{\theta G} p_0\}$ *has period exactly* 2π.

(ii) For every symmetric N, $U(\theta + \pi) = U(\theta)$.

(iii) $\Gamma = e_1 e_2 \cdots e_6$ *anticommutes with every* e_a *and* $\Gamma^2 = \mathbf{1}$, *so* $\mathrm{Ad}_\Gamma = -\mathbf{1}$ *on* V *and* Γ *implements* $e^{\pi G}$ *on spinors. By Proposition 4.47(ii) the multiples of the half turn are the only elements of the rotor group that lift to* $\mathrm{Pin}(3, 3)$.

Proof. (i) $(e^{\theta G} - \mathbf{1})p_0 = (\cos\theta - 1)p_0 + \sin\theta\, G p_0$ with $p_0 \perp G p_0$ and $|G p_0| = |p_0|$. (ii) $p(\theta + \pi) = -p(\theta)$ and U is quadratic. (iii) A product of all six anticommuting generators anticommutes with each of them, and $(e_1 \cdots e_6)^2 = (-1)^{15} \prod_a \eta_{aa} = \mathbf{1}$. □

The proposition restates the remark on the lift of the rotor at the end of the three-constituent part of Section 4.4 and makes its ingredients explicit; it is fixed. The quantum statement it permits is conditional on an assumption.

Assumption 4.51 (A quantised leaf)**.** The leaf T^*S^1 of Proposition 4.23 (Theorem 4.40(ii)) is quantised by wave functions with $\Phi(\theta + 2\pi) = e^{2\pi i\delta}\Phi(\theta)$, $h = -i\hbar\,\partial_\theta$, and the Casimirs of the leaf are treated as numbers.

Proposition 4.52 (A conditional rate ladder)**.** *Under Assumption 4.51:*

(i) h *has spectrum* $\hbar(\mathbb{Z} + \delta)$. *On the leaves on which* U *is constant* ($R_0 = 0$, *or duality-class inertia) the levels are* $E_n^{\mathrm{rot}} = \hbar^2(n + \delta)^2/(2I_0)$, *plus the constant* $\frac{1}{2}p_0^\top N p_0$ *in the duality class, the rates are* $\kappa_n = (n + \delta)\hbar/I_0$, *and* $E_{n+1}^{\mathrm{rot}} - E_n^{\mathrm{rot}} = \frac{\hbar}{2}(\kappa_n + \kappa_{n+1})$.

(ii) *If $R_0 > 0$ the Hamiltonian is a Mathieu operator [94, 36] whose potential couples $e^{i(n+\delta)\theta}$ only to $e^{i(n\pm2+\delta)\theta}$, so for every δ the parity of n is conserved; it labels the eigenvalues $\pm e^{i\pi\delta}$ of the half-turn translation $\theta \mapsto \theta+\pi$. For $q_{\mathrm{M}} := I_0R_0/(2m\hbar^2) \gg 1$ the lowest gap in either parity sector is $\hbar\Omega - \hbar^2/(2I_0) + O(q_{\mathrm{M}}^{-1/2}\hbar^2/I_0)$, and the two sectors are degenerate up to tunnelling.*

(iii) *If moreover $\delta = 0$, and, as a separate assumption, the first level $h = \hbar$ is matched by $\hbar\kappa_1 = mc^2$ with $\ell = \hbar/(mc)$, then $I_0 = \hbar^2/(mc^2)$, $M_{\mathrm{rot}} = m$ and $E_1^{\mathrm{rot}} = \frac{1}{2}mc^2$.*

Proof. (i) Spectrum of $-i\hbar\partial_\theta$ on twisted periodic functions; in the duality class U is the constant found above. (ii) $\cos(2\theta+\varphi_0)$ shifts n by ±2, and $e^{i(n+\delta)(\theta+\pi)} = (-1)^n e^{i\pi\delta}e^{i(n+\delta)\theta}$. Dividing the Schrödinger equation by $-\hbar^2/(2I_0)$ gives $\Phi'' + (a_{\mathrm{M}} - 2q_{\mathrm{M}}\cos(2\theta+\varphi_0))\Phi = 0$ with $a_{\mathrm{M}} = 2I_0E/\hbar^2$, and $a_{\mathrm{M}} \sim -2q_{\mathrm{M}} + 2(2r+1)\sqrt{q_{\mathrm{M}}} - \frac{1}{8}((2r+1)^2+1)$ as $q_{\mathrm{M}} \to \infty$ [36, §28.8]. The levels $r = 0, 1$ lie in the same sector and differ by $4\sqrt{q_{\mathrm{M}}} - 1$, while $\hbar\Omega = (\hbar^2/2I_0)\,4\sqrt{q_{\mathrm{M}}}$. (iii) $\hbar/I_0 = \kappa_1 = c/\ell = mc^2/\hbar$, $M_{\mathrm{rot}} = I_0/\ell^2$ and $E_1^{\mathrm{rot}} = \hbar^2/(2I_0)$. □

Numerically, at $R_0/m = 2000$ and $\hbar = I_0 = 1$ the lowest gap in each sector is 62.7415 for $\delta = 0$ and for $\delta = 0.3$, against $\hbar\Omega - \hbar^2/(2I_0) = 62.7456$, and 62.7416 with the next term of the expansion. The proposition is constructed. The quantisation of the leaf exists: it is the one the orbit method assigns [78], with δ a character of the stabiliser $2\pi\mathbb{Z}$ in the universal cover of the rotor group. Nothing selects δ, I_0 or the matchings, and whether the ladder is physical is open. Missing are a reason for the assumption, that is a quantum theory of the momentum field, which the closure does not give and the split-form frame obstructs, since quantised mode by mode with positive norm it has no ground state (Corollary 4.15). Missing too are a value of I_0; a coupling to matter, since only the $\mathbb{Z}_2$ of the half turn reaches spinors, as Γ, and nothing ties the parity of n to chirality; and any derivation of the two matchings in (iii). The level $E_1^{\mathrm{rot}} = \frac{1}{2}mc^2$ is $K_{\mathrm{rot}} = \frac{1}{2}h\kappa$ at $h = \hbar$: the factor $\frac{1}{2}$ is built in, and mc^2 appears only as $\hbar$ times the rate put in, neither as a level nor as a transition energy.

4.6 The identification tested

The preceding sections speak of a rotor, an inertia, a Lie–Poisson system and Euler–Arnold equations. This section states the identification with a rigid body as a map, and tests whether the map preserves equations, brackets, conserved quantities and degrees of freedom. The checks are rb2-1 to rb2-15.

4.6.1 The map

Table 4.2 sets the closure of Section 4.4 beside the two rigid bodies of Section 4.1. The closure is taken in the group form of Proposition 4.24: its configuration is (θ, x) in the group S with product $(\theta,x)(\theta',x') = (\theta+\theta', x + e^{\theta G}x')$, where θ is the rotor angle and $x \in V = \mathbb{R}^{3,3}$ the frame, with body form $X = e^{-\theta G}x$. Its Lagrangian is $\frac{1}{2}I_0\dot\theta^2 + \frac{m}{2}\,\eta(u,u)$ in the body velocity $u = e^{-\theta G}\dot x$, or $\frac{m}{2}|u|^2$ for the positive form. The evolution parameter is the interior parameter τ, with the dimension of a time; this chapter does not relate it to the time read by a clock (Assumption 2.16). With X a length, the convention of Proposition 4.48(i), the units are those of Table 4.2, and both equations of (4.16) are dimensionally consistent (rb2-12): h is an angular momentum, p a linear momentum and I_0 a moment of inertia.

4.6.2 What the map preserves

Theorem 4.53 (The closure is a rigid body of the planar kind)**.** *(i)* Equations. *The bracket (4.18) is the Lie–Poisson bracket of $\mathfrak{s} = \mathbb{R}\hat g \ltimes_G V$, and for every symmetric N the equations of $H = h^2/(2I_0) + \frac{1}{2}p^{\top}Np$ are (4.16). On each pairing plane (e_i, e_{i+3}) the bracket is that of*

	rigid body	planar body	closure
configuration	$R \in SO(3)$	$(\theta, \mathbf{x}) \in SE(2)$	$(\theta, x) \in S$: rotor angle and frame $x \in \mathbb{R}^{3,3}$
parameter	time t	time t	interior parameter τ, $[\tau] = \mathsf{T}$
velocity	$\boldsymbol{\Omega} = (R^{-1}\dot{R})^\vee$	$(\dot\theta, \mathbf{u})$, $\mathbf{u} = e^{-\theta J}\dot{\mathbf{x}}$	$(\dot\theta, u)$, $u = e^{-\theta G}\dot{x}$
inertia	$\mathbb{I} > 0$, $[\mathbb{I}] = \mathsf{ML}^2$	$I > 0$, $N^{-1} > 0$	$I_0 > 0$, $[I_0] = \mathsf{ML}^2$; frame $m\eta$ (split) or $m\,\mathbf{1}$ (positive), $[m] = \mathsf{M}$
action	$\int \frac{1}{2}\boldsymbol{\Omega}\cdot\mathbb{I}\boldsymbol{\Omega}\,dt$	$\int(\frac{1}{2}I\dot\theta^2 + \frac{1}{2}\mathbf{u}^\top N^{-1}\mathbf{u})\,dt$	$\int(\frac{1}{2}I_0\dot\theta^2 + \frac{m}{2}\eta(u,u))\,d\tau$
momenta	$\mathbf{M}$, $[\mathbf{M}] = \mathsf{ML}^2\mathsf{T}^{-1}$	(h, p)	(h, p), $[h] = \mathsf{ML}^2\mathsf{T}^{-1}$, $[p] = \mathsf{MLT}^{-1}$
equations	(4.2)	(4.4)	(4.16)
bracket	(4.3)	Section 4.1.4	(4.18)

Table 4.2: The identification as a map: the closure of Section 4.4 beside the free rigid body and the planar body of Section 4.1. Units with X a length (Proposition 4.48); checks rb2-12.

$\mathfrak{se}(2)^*$, *with G acting as J. The group S consists of the translations of $\mathbb{R}^6$ and the common rotation $e^{\theta G}$ of the three pairing planes.*

(ii) Brackets. *At $p \neq 0$ the Poisson tensor of $\mathfrak{s}^*$ has rank two, with five independent Casimirs, the duality-invariant functions of p (Theorem 4.40); those of $\mathfrak{so}(3)^*$ and $\mathfrak{se}(2)^*$ have rank two with one Casimir. $\mathfrak{s}$ is solvable and $\mathfrak{so}(3)$ is perfect, so every Lie algebra homomorphism $\mathfrak{so}(3) \to \mathfrak{s}$ is zero, and every homomorphism $\mathfrak{s} \to \mathfrak{so}(3)$ has an image of dimension at most one.*

(iii) Conserved quantities. *The energy H, the Casimirs, the spatial momentum $p_s = e^{\theta G}p$ and*

$$J_s = h + p_s^\top G x$$

are conserved for every symmetric N: the analogues of the energy, $|\mathbf{M}|^2$, and the linear and total angular momenta of a rigid body that rotates and translates.

(iv) Degrees of freedom. *The configuration group, the dual Lie algebra, a generic leaf and the Casimirs have dimensions $(7, 7, 2, 5)$ for the closure, against $(3, 3, 2, 1)$ for $SO(3)$ and for $SE(2)$. In all three the reduced motion has one degree of freedom; in the closure it is the rotor angle, and the six frame coordinates are cyclic.*

(v) Inertia. *For duality-class inertia, $[N, G] = 0$, the rotor and the frame decouple: $\dot h = 0$ for all data. For Lorentz-class inertia, $N = (\nu_1\eta^{-1} + \nu_2 G\eta^{-1})/m$, they exchange angular momentum at the rate $\dot h = \mathrm{Im}(\nu w(p))/m$, and N has the eigenvalues $\pm|\nu|/m$, three of each: no nonzero Lorentz-class inertia is positive definite.*

(vi) Energy. *With nonzero Lorentz-class inertia ($\nu \neq 0$) H is unbounded below on $\mathfrak{s}^*$, and on each leaf it is bounded below by $-|\nu w(p_0)|/(2m)$.*

Proof. (i) For $\mu = (h, p)$ and $\nabla F = (F_h, \nabla_p F) \in \mathfrak{s}$, $[\nabla F_1, \nabla F_2] = F_{1,h}G\nabla_p F_2 - F_{2,h}G\nabla_p F_1$, and $-\langle\mu, [\nabla F_1, \nabla F_2]\rangle$ is (4.18); the equations are Proposition 4.23, for general N. On (e_1, e_4), G is $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ and $\{h, p_1\} = -p_4$, $\{h, p_4\} = p_1$ (rb2-1, rb2-2, rb2-4). $G^\top = -G$, so $e^{\theta G} \in SO(6)$ rotates each pairing plane by θ. (ii) The rank is that of $\{h, p_j\} = (Gp)_j$, $\{p_i, p_j\} = 0$; the five Casimirs are Theorem 4.40(ii) (rb2-3). $[\mathfrak{s}, \mathfrak{s}] = V$ and $[V, V] = 0$. A homomorphism from the simple $\mathfrak{so}(3)$ has kernel 0 or $\mathfrak{so}(3)$, and its image cannot be a copy of $\mathfrak{so}(3)$ inside the solvable $\mathfrak{s}$. The image of $\mathfrak{s}$ in $\mathfrak{so}(3)$ is solvable, and $\mathfrak{so}(3)$ has no two-dimensional subalgebra, since for independent $\mathbf{a}, \mathbf{b}$ the vector $\mathbf{a} \times \mathbf{b}$ is nonzero and orthogonal to both (rb2-14). (iii) $\dot p = -\dot\theta G p$ gives $\frac{d}{d\tau}(e^{\theta G}p) = 0$. With $\dot x = e^{\theta G}Np$ and $p_s^\top G e^{\theta G} = p^\top G$, since G commutes with $e^{\theta G}$ and $(e^{\theta G})^\top = e^{-\theta G}$, one has

$\dot{J}_s = -p^\top GNp + p^\top GNp = 0$ (rb2-5). (iv) Count from (ii) (rb2-13). (v) If $[N, G] = 0$ then GN is antisymmetric and $p^\top GNp = 0$. For the Lorentz class, $-p^\top GNp = \mathrm{Im}(\nu w(p))/m$ and $\frac{1}{2}p^\top Np = \mathrm{Re}(\nu w(p))/(2m)$, and the characteristic polynomial of N is $(\lambda^2 - |\nu|^2/m^2)^3$ (rb2-6, rb2-7). (vi) For $\nu \neq 0$, along an eigenvector of $-|\nu|/m$, $H \to -\infty$ (at $\nu = 0$, $H = h^2/(2I_0)$). On a leaf $w(p) = e^{2i\theta}w(p_0)$, and $\mathrm{Re}(\nu w) \geq -|\nu w(p_0)|$ (rb2-8). □

The theorem is fixed. Items (i) and (iii) restate Propositions 4.23 and 4.24 and Theorem 4.27 in the language of Section 4.1. The conserved J_s is the momentum map of semidirect-product mechanics [72, 93]; it and items (ii), (iv), (v) and (vi) are computed here for this closure. A numerical integration of the full closure, rotor, momentum and frame, with Lorentz-class inertia keeps H, $|p|^2$, $|w(p)|$, p_s and J_s to 2.4×10^{-10} or better over $\tau \in [0, 200]$, while h varies over a range 0.737: the rotor and the frame exchange angular momentum, and their total is conserved (rb2-15; Table 4.3).

The verdict is therefore two-sided. As a structure the identification holds. The closure is a Lie–Poisson system of a left-invariant Lagrangian on a group of rotations and translations, its reduced motion on every leaf is one rotation angle, and on each pairing plane its bracket and its equations are those of the planar body of Section 4.1.4. The pendulum of Theorem 4.28 is the pendulum of (4.4) for an anisotropic translational inertia: on each pairing plane the Lorentz-class N has the principal values $|\nu|/m$ and $-|\nu|/m$, and $|w(p_0)| = |p_0|^2$ for p_0 in that plane. As an identification with a free rigid body it fails: the closure contains no rotation group $SO(3)$, and it maps to one at most through its rotor. As an identification with a physical body it fails as well. A positive inertia that couples the rotor to the frame exists, for instance $N = (2\,\mathbf{1} + \eta^{-1})/m$ (Theorem 4.26(iv)), but it is not Lorentz invariant; the Lorentz-invariant inertia that produces the pendulum is indefinite.

4.6.3 Five distinctions

Five pairs of notions must be kept apart by a rigid-body reading. In the arena they separate as follows.

(1) *An internal rotation and the orientation of a body.* The rotor group $e^{\theta G}$ commutes with the whole of $\mathfrak{gl}(3, \mathbb{C}) = \{A : [A, G] = 0\}$: it is the centre $U(1)$ of $U(3)$ acting on $\mathbb{C}^3 = \mathbb{R}^6$. It fixes no nonzero vector unless $\theta \in 2\pi\mathbb{Z}$, since $\det(e^{\theta G} - \mathbf{1}) = (2 - 2\cos\theta)^3$, and it is an isometry of η only for $\theta \in \pi\mathbb{Z}$ (rb2-9, rb2-10; Proposition 4.47(ii)). In the field-strength reading it is the duality rotation $z \mapsto e^{i\theta}z$ (Theorem 4.28). The orientation of a body is an element of $SO(3)$ acting on the directions of space, and every such element fixes its axis. The rotor angle is a phase: it has no axis, so it can point nowhere and precess about nothing.

(2) *A full dynamical rotor and a rate-clamped reduction.* The clamped interiors of Sections 4.2 and 4.3 are the reductions $\theta = \kappa\tau$ of the closure (Proposition 4.24). A clamped motion solves the closure exactly when $\dot{h}$ vanishes along it: always with duality-class inertia, and with Lorentz-class inertia only when $\nu w(p_0) = 0$, that is for a null momentum field (rb2-11; Corollary 4.29). Otherwise the clamped rate is the limit $\varepsilon \to 0$ of a heavy rotor (Corollary 4.30), and the free rotor drifts the frame (Theorem 4.31).

(3) *Bounded solutions and a Hamiltonian bounded below.* Every solution of the split-form interior is periodic (Theorem 4.8), while its energy has signature $(6, 6)$ (Proposition 3.18(iii)). In the closure the Hamiltonian is unbounded below on $\mathfrak{s}^*$ and bounded below on each leaf (Theorem 4.53(vi)). Boundedness of the orbits is a property of each leaf; a positive energy is not available.

(4) *A classical oscillation and a healthy quantum state.* The bound orbits, quantised mode by mode with positive norm, have no ground state (Corollary 4.15).

(5) *An algebraic gap and a binding energy.* The gap $\hbar\Omega - \hbar^2/(2I_0)$ of Proposition 4.52(ii) separates two eigenvalues of one Mathieu operator on one leaf, under the assumption Assumption 4.51. The closure has no threshold of separated constituents against which it would be a binding energy. On the tower the analogous comparison is Proposition 5.44: at rest the graded level $(m_1^2+m_2^2)^{1/2}$ lies below the free threshold m_1+m_2, and the inequality is not a bound-state condition.

Item (1) fixes how far a rigid-body vocabulary carries in this chapter. "Rotor" and "inertia" name the structure of Theorem 4.53, and "rotation" names the internal phase of item (1); none of them refers to a body in space.

4.7 From Rigid-Body Rotation to Planetary Orbital Dynamics

This section is a bounded extension. Rigid-body rotation and the orbital motion of a centre of mass are different systems. The first lives on $SO(3)$, the second in $\mathbb{R}^3$, and in the Newtonian two-body problem they separate exactly for spherically symmetric bodies and are coupled by a gravitational torque otherwise. A comparison with the arena therefore needs a map, which is stated first. The benchmark is Newtonian, and its potential is assumed. The checks are rb3-1 to rb3-13, and the numerical runs are collected in Table 4.3.

4.7.1 Two maps

Definition 4.54 (The orbit map and the spin map)**.** (i) The *orbit map* reads the interior configuration in one pairing plane, $z = x + iy$ for $X = x\,e_1 + y\,e_4$, as the relative position of two bodies in their orbital plane, and the interior parameter τ as their time.

(ii) The *spin map* reads the rotor angle θ of the closure as the spin angle of a rigid body turning about an axis normal to its orbital plane, measured from the line to the primary, and I_0 as its moment of inertia C about that axis. The rotor momentum $h = I_0\dot\theta$ is then the spin angular momentum relative to the rotating line to the primary; the spin angular momentum itself is $h+Cn$ on a circular orbit of mean motion n, and $h+C\dot f$ on an eccentric one, with f the true anomaly.

Neither map is supplied by the arena. Both are readings, and the propositions below test what they carry.

4.7.2 The Newtonian benchmark

Assumption 4.55 (Newtonian benchmark)**.** Two bodies of masses M_1 and M_2 interact through the potential $-GM_1M_2/r$. A non-spherical body is a rigid body with principal moments $A < B < C$ whose axis of largest moment is normal to the orbital plane, and it moves on a Keplerian orbit about a point mass M that it does not perturb.

The assumption is the comparison system. It is not derived from the arena, which contains no gravitational potential: the curvature identities of Chapter 11 come without a gravitational action. Its consequences are inherited [176, 177, 185, 186].

- The centre of mass moves uniformly. The relative position obeys $\ddot{\mathbf{r}} = -\mu\,\mathbf{r}/r^3$ with $\mu = G(M_1+M_2)$, and conserves the energy $E = \frac{1}{2}|\dot{\mathbf{r}}|^2 - \mu/r$, the angular momentum $\mathbf{L} = \mathbf{r}\times\dot{\mathbf{r}}$ and the Laplace–Runge–Lenz vector $\mathbf{A} = \dot{\mathbf{r}}\times\mathbf{L} - \mu\,\mathbf{r}/r$. The bounded orbits are ellipses with a focus at the centre of force, and their period is $2\pi\sqrt{a^3/\mu}$.
- By Bertrand's theorem the inverse-square law and Hooke's law are the only central forces all of whose bounded orbits close [180, 176]. Because $\mathbf{A}$ is conserved, a Kepler orbit does

not precess. Apsidal precession needs a perturbation, such as an oblate primary, a third body or relativity, and the benchmark contains none.

- Two spherically symmetric bodies attract exactly as point masses and exert no torque on each other, so their spin and orbit separate. A triaxial body feels the torque $-\frac{3}{2}(GM/r^3)(B-A)\sin 2(\theta-f)$, with θ its spin angle and f the true anomaly. On a circular orbit of radius a, the angle $\gamma=\theta-nt$ to the line to the primary obeys

$$C\ddot{\gamma} = -\tfrac{3}{2}\,\frac{GM}{a^3}\,(B-A)\sin 2\gamma, \tag{4.19}$$

  a pendulum in 2γ whose stable point $\gamma=0$, with the long axis towards the primary, is synchronous rotation, the Moon's state. On an eccentric orbit the torque contains resonances at the spin rates jn, $j\in\frac{1}{2}\mathbb{Z}$, of strengths $H_j(e)$, among them Mercury's 3:2; at larger asphericity the rotation can be chaotic [185, 186, 187].

- The planar model has no spin-axis precession, which needs a torque component in the orbital plane.

4.7.3 The orbit map: Hooke, not Kepler

Proposition 4.56 (Interior orbits under the orbit map)**.** *(i) In a pairing plane the split-form interior is Hooke's law, $\ddot{x}=-\kappa^2 x$ and $\ddot{y}=-\kappa^2 y$. Its bounded orbits are ellipses centred on the centre of force, $z(\tau+\pi/\kappa)=-z(\tau)$, all of period $2\pi/\kappa$ whatever their size. A Kepler ellipse has the centre of force at a focus, and its period scales as $a^{3/2}$.*

(ii) The interior conserves $\ell=x\dot{y}-y\dot{x}$, but as the Noether charge $m(\dot{x}y-\dot{y}x)$ of the boost of the mixed plane (e_1,e_4). The Euclidean rotation of that plane carries solutions to solutions but does not leave the Lagrangian invariant. The conserved energy $\frac{m}{2}[(\dot{x}^2+\kappa^2x^2)-(\dot{y}^2+\kappa^2y^2)]$ is indefinite.

(iii) The arena's complex invariant $q(z)=z\cdot z$, which is z^2 in one plane, carries the orbit $z=c_-e^{-i\kappa\tau}+c_+e^{i\kappa\tau}$ onto an ellipse with one focus at the origin, the other at $4c_-c_+$, semi-major axis $a_K=|c_-|^2+|c_+|^2$ and eccentricity $2|c_-||c_+|/a_K$. Away from collisions, $z\neq 0$, and with the new time $dt=\lambda|z|^2d\tau$, $\lambda>0$, the image $Q(t)=z(\tau)^2$ solves

$$\frac{d^2Q}{dt^2}=-\mu_K\frac{Q}{|Q|^3},\qquad \mu_K=\frac{4\kappa^2a_K}{\lambda^2},\qquad E_K=-\frac{2\kappa^2}{\lambda^2}.$$

For $|c_-|=|c_+|$ the orbit passes through the origin, and its image, of eccentricity one, is a regularised collision orbit.

(iv) The coupling μ_K depends on the orbit: at fixed λ the images form a family of Kepler problems of one energy, one coupling per orbit, not one Kepler problem with its family of orbits. Beyond one plane the image of an orbit is an ellipse with foci $2c_-\cdot c_+\pm 2\big((c_-\cdot c_-)(c_+\cdot c_+)\big)^{1/2}$. The origin is a focus only when $(c_-\cdot c_+)^2=(c_-\cdot c_-)(c_+\cdot c_+)$. For $c_-=e_1$ and $c_+=e_2$ the image $q=2\cos 2\kappa\tau$ passes through the origin while $|z|$ stays constant.

Proof. (i) The Euler–Lagrange equations of $\frac{m}{2}[(\dot{x}^2-\dot{y}^2)-\kappa^2(x^2-y^2)]-m\kappa\frac{d}{d\tau}(xy)$, which is (4.11) on the plane by Theorem 4.12 (rb3-1). Periodicity and the half-period identity are read off the general solution, and Kepler's third law scales as stated (rb3-2). (ii) Under $(\delta x,\delta y)=(y,x)$ the Lagrangian changes by $\frac{d}{d\tau}(-m\kappa(x^2+y^2))$, and Noether's theorem gives the charge. Under $(\delta x,\delta y)=(-y,x)$ the change has the nonzero Euler–Lagrange expression $2m(\ddot{y}+\kappa^2y)$ in x, so it is not a total derivative (rb3-1). (iii) $z^2-4c_-c_+=(c_-e^{-i\kappa\tau}-c_+e^{i\kappa\tau})^2$, so $|q|+|q-4c_-c_+|=|z|^2+|c_-e^{-i\kappa\tau}-c_+e^{i\kappa\tau}|^2=2(|c_-|^2+|c_+|^2)$ (rb3-3); this is Bohlin's observation [181]. The equation

for Q is the Levi-Civita transformation [182]. One checks it directly, using $|\dot{z}|^2+\kappa^2|z|^2=2\kappa^2 a_K$ along the orbit (rb3-4). (iv) $\mu_K \propto a_K$ by (iii); for instance $(c_-,c_+)=(1,\frac{2}{5})$ and $(1,\frac{4}{5})$ give $\mu_K=\frac{116}{25}$ and $\frac{164}{25}$ at $\kappa=\lambda=1$ (rb3-5). With $\alpha_-=c_-\cdot c_-$, $\alpha_+=c_+\cdot c_+$, $\alpha_0=2c_-\cdot c_+$ and $u_-=\sqrt{\alpha_-}e^{-i\kappa\tau}$, $u_+=\sqrt{\alpha_+}e^{i\kappa\tau}$, the image is $q=\alpha_0+u_-^2+u_+^2$ with u_-u_+ constant. So $q-\alpha_0\mp 2u_-u_+=(u_-\mp u_+)^2$, and $|q-F_+|+|q-F_-|=2(|\alpha_-|+|\alpha_+|)$ for $F_\pm=\alpha_0\pm 2\sqrt{\alpha_-\alpha_+}$ (rb3-6). □

Items (i), (ii) and (iv) are fixed; (iii) is inherited and applied to the arena's own q [181, 182, 183]. In Bertrand's dichotomy the interior realises the Hooke family, not the Kepler family. The squaring map that carries one to the other is the arena's invariant q, but the inverse-square law does not follow. The time change, with its free scale λ, is put in by hand. The coupling μ_K is a property of each orbit, not of the force. And the map works plane by plane. In three dimensions the Kepler problem is regularised through the Kustaanheimo–Stiefel map $\mathbb{R}^4\to\mathbb{R}^3$ under a bilinear constraint [184]; the arena's $q\colon\mathbb{C}^3\to\mathbb{C}$ is not that map. The numerical test rb3-11 integrates an interior orbit with the new time and, independently, the Kepler orbit of its image. They agree to 2.5×10^{-9} over ten Kepler periods (Table 4.3): the equivalence is exact, and it is an equivalence of orbits, not of laws.

4.7.4 The spin map: the duality pendulum as spin–orbit coupling

Proposition 4.57 (The duality pendulum under the spin map)**.** *(i) With $\theta=\gamma+\frac{1}{2}(\pi-\varphi_0)$, $I_0=C$ and $R_0/m=\frac{3}{2}(GM/a^3)(B-A)$, the duality pendulum of Theorem 4.28 is the circular-orbit equation (4.19). Its stable point $\psi=\pi$ is $\gamma=0$, synchronous rotation, and the frequency Ω of Proposition 4.42 is the libration frequency $\big(3(GM/a^3)(B-A)/C\big)^{1/2}$. The librating solutions are synchronous rotation with a libration, and the rotating ones are non-synchronous spin with a modulated rate.*

(ii) $R_0=0$ and duality-class inertia, under which the rotor decouples from the frame, correspond to $B=A$, under which the spin decouples from the orbit. The closure conserves $J_s=h+p_s^{\top}Gx$ while h varies, as the full two-body problem conserves the sum of the spin angular momentum $h+Cn$ and the orbital angular momentum.

(iii) The correspondence holds only on the circular orbit. On an eccentric orbit the spin–orbit equation is not autonomous: at $e=0.2$ the resonance strengths are $H_{1/2}(e)=-0.0995$, $H_1(e)=0.9013$, $H_{3/2}(e)=0.6397$ and $H_2(e)=0.3101$, against $H_j(0)=\delta_{j,1}$. The duality pendulum is autonomous and has one family of equilibria.

Proof. (i) Substitution, with $\sin(2\gamma+\pi)=-\sin 2\gamma$ and $\Omega^2=2R_0/(mI_0)$ (rb3-7). (ii) The torque is proportional to $B-A$ (rb3-8); for the closure see Theorem 4.53(iii) and (v). (iii) $H_j(e)=\frac{1}{2\pi}\int_0^{2\pi}(a/r)^3\cos(2f-2jnt)\,d(nt)$, with t measured from pericentre, computed by quadrature; at $e=0.05$ it agrees with the series $H_{1/2}(e)=-\frac{e}{2}+\frac{e^3}{16}$, $H_1(e)=1-\frac{5}{2}e^2+\frac{13}{16}e^4$ and $H_{3/2}(e)=\frac{7}{2}e-\frac{123}{16}e^3$ to 1.2×10^{-6} (rb3-9). □

Items (i) and (ii) are fixed; the spin–orbit facts in (iii) are inherited [185, 186]. The equivalence in (i) is one of equations, and its coefficients come from different places: in the spin–orbit problem from the assumed potential and the shape of the body, in the closure from the inertia and the rest energy of the momentum field, $R_0=2|\nu|E_0(\pi_0)$. The arena supplies the form of the pendulum, not the orbit. In the closure the angular momentum h is exchanged with the frame, whose motion is the drift of Theorem 4.31, a translation in $\mathbb{R}^{3,3}$ and not an orbit. Two numerical runs make the boundary visible (Table 4.3, Figure 4.3). On a circular orbit the spin–orbit angle and the rotor of the closure, integrated independently from corresponding data, agree to 5.1×10^{-10} rad over 50 orbits (rb3-12). On an orbit of eccentricity 0.2, a body started at pericentre with spin rate $1.52\,n$ librates in the 3:2 resonance for these initial data,

check	system and initial data	tolerance	invariant drift	result
rb2-15	closure, $\nu = 1 + \frac{i}{2}$; $\theta = 0$, $h = 1.1$, $p = (1, 0.3, -0.2, 0.1, 0.4, -0.5)$, $x = 0$; $\tau \le 200$	10^{-12}	H, $\lvert p\rvert^2$, $\lvert w\rvert$, p_s: $\le 7.4 \times 10^{-11}$; J_s: 2.4×10^{-10}	h varies by 0.737; rotor and frame exchange angular momentum
rb3-10	Kepler, $\mu = a = 1$, $e = 0.5$, pericentre; 100 periods	10^{-12}	E: 4.5×10^{-10}; $\lvert\mathbf{L}\rvert$: 1.6×10^{-10}; $\mathbf{A}$: 1.9×10^{-10}	return to pericentre within 3.3×10^{-7}; no apsidal precession
rb3-11	interior orbit $(c_-, c_+) = (1, \frac{2}{5})$, $\kappa = \lambda = 1$, with $dt = \lambda\lvert z\rvert^2 d\tau$; Kepler orbit, $\mu_K = 4.64$, of its image; 10 periods	10^{-12}	–	images agree to 2.5×10^{-9}; $\tau = 10\pi$ is $t = 10 \times 2\pi(a_K^3/\mu_K)^{1/2}$
rb3-12	spin–orbit, $e = 0$, $(B - A)/C = 0.01$, $\gamma(0) = 0.5$, $\dot\gamma(0) = 0$; closure, $N = 0.015\,\eta^{-1}$, $p_0 = e^1$; 50 orbits	10^{-11}	closure H: 1.1×10^{-12}	angles agree to 5.1×10^{-10} rad (libration)
rb3-13	spin–orbit, $e = 0.2$, $(B - A)/C = 10^{-3}$, $\theta(0) = f(0) = 0$, $\dot\theta(0) = 1.52\,n$; closure from the same data; 1000 orbits	10^{-11}	orbital energy: 4.0×10^{-8}	spin–orbit librates in 3:2, mean rate $1.49988\,n$; closure $1.51855\,n$ (closed form $1.51855\,n$)

Table 4.3: Numerical runs of Sections 4.6 and 4.7: DOP853, relative and absolute tolerance as shown, dimensionless units with $GM = a = 1$ for the orbits and $m = I_0 = 1$ for the closure. Drifts are maximal deviations over the run. Every entry is printed by the check named in the first column.

with mean spin rate $1.49988\,n$ over orbits 200 to 1000; the model has no tidal dissipation, so this is libration, not capture. The duality pendulum with the same initial data keeps its own mean rate $1.51855\,n$, which is the closed-form value for the circular orbit (rb3-13).

4.7.5 Numerical examples

Table 4.3 collects the numerical runs of this chapter: each tests an equivalence or its failure. Units are dimensionless. The orbital runs take $GM = a = 1$, so that the mean motion is $n = 1$ and the orbital period 2π; the closure takes $m = I_0 = 1$. The solver is the explicit Runge–Kutta method DOP853 of order eight [188], as implemented in SciPy [189], with the tolerances shown. The environment is recorded with the run outputs in `reproducibility/selected-results/v1.0.1/` (Python 3.13, NumPy 2.5, SciPy 1.18), and the checks are listed in Appendix E. The commands are `python3 verification/ch04_rigid_body/rb2_es_map.py` and `python3 verification/ch04_rigid_body/rb3_orbital.py --write-data`, which also writes the data of Figure 4.3(b).

4.7.6 What the extension shows

- *Elliptical motion.* The interior has closed elliptical orbits, of Hooke's kind. Kepler ellipses appear only through the squaring map q and a time change put in by hand, with a coupling that varies from orbit to orbit. No inverse-square law is inferred, and no relativistic correction; a familiar orbit can be drawn, which is not a derivation.

- *Conservation of energy and angular momentum.* The interior's energy is indefinite. Its angular momentum ℓ is a boost charge, not the charge of a rotation of space.

(a) an interior orbit and its image under q (b) $e = 0.2$: libration in 3:2 and not

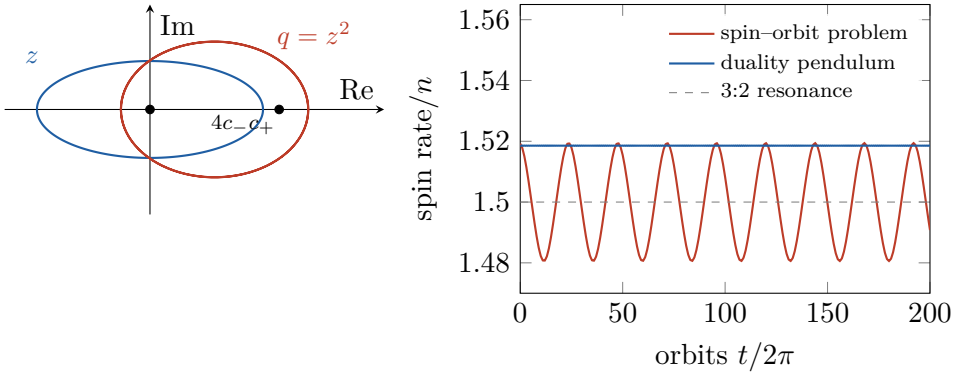


Figure 4.3: (a) Analytical. The interior orbit $z = e^{-i\varphi} + \frac{2}{5}e^{i\varphi}$ in one pairing plane (blue), an ellipse centred on the centre of force, and its image $q = z^2$ (red), an ellipse with one focus at the origin and the other at $4c_-c_+ = \frac{8}{5}$ (Proposition 4.56). (b) Numerical, check rb3-13. The spin rate averaged over each orbit, $(\theta(t + 2\pi) - \theta(t))/2\pi$, of a body with $(B - A)/C = 10^{-3}$ on an orbit of eccentricity 0.2, started at pericentre at $1.52\,n$, over the first 200 orbits: it librates about the 3:2 resonance (red), between $1.481\,n$ and $1.519\,n$, with a period of about 24 orbits. The duality pendulum, which is the circular-orbit problem, keeps the rate $1.5186\,n$ (blue). Data: `manuscript/figures/ch04/spinrate_{so,es}.dat`, written by `python3 verification/ch04_rigid_body/rb3_orbital.py --write-data`; units $GM = a = 1$, DOP853 with tolerance 10^{-11}, 401 samples.

- *Spin–orbit separation, torque and coupling.* Their form is present: duality-class inertia decouples rotor and frame, as a spherical body decouples spin and orbit. Lorentz-class inertia couples them with the duality pendulum, as a triaxial body on a circular orbit is coupled. The Lorentz-class inertia is indefinite, and the coupling constant is the rest energy of the momentum field, not a gravitational quantity.

- *Synchronous rotation.* The stable libration of the duality pendulum has the form of synchronous rotation. The resonances of eccentric orbits, 3:2 among them, have no counterpart.

- *Precession.* There is none. The rotor has no axis to precess about (Section 4.6.3, item (1)). The interior orbits close, and the drift of the frame under the free rotor is a translation, not an apsidal precession.

- *The gravitational potential* is assumed throughout (Assumption 4.55). The arena derives none.

Chapter 5

The Clifford Tower and Constituent Duplication

The question of this chapter is what the tower $\mathrm{Cl}(3n, 3n)$ establishes about the number of particles. It treats the centralisers along the tower, the Lorentz actions on several constituents, the duplication of $\mathrm{Cl}(3, 3)$ into $\mathrm{Cl}(6, 6)$, the second rung (rest energies, complex structures, the Fock reading, exchange), the processes that take one constituent to two, and the ladder with its spin parity and dual pair. The algebra is proved; reading its quanta as particles is conditional. Fock quanta are not particles, $\Lambda^n\mathbb{C}^4 = 0$ for $n > 4$ is a statement about one mode, and graded exchange does not prove a spin–statistics theorem.

5.1 The tower: centralisers, nesting, and carriers

Climbing the tower adds six directions per rung and leaves the local structure of each factor unchanged. The interior four-plane stays in the first factor; the question is what the added directions do to the internal symmetry.

Theorem 5.1 (Internal centraliser)**.** *In* $\mathrm{Cl}(3n, 3n)$ *the bivectors commuting with the interior Lorentz action form*

$$\mathfrak{g}_{\mathrm{int}}^{(n)} \cong \mathfrak{so}(3n-1, 3n-3), \qquad \dim \mathfrak{g}_{\mathrm{int}}^{(n)} = \tfrac{1}{2}(6n-4)(6n-5).$$

Proof. By (1.3) a bivector sharing exactly one index with an interior bivector anticommutes with it, and one sharing none or two commutes. A bivector commuting with all six interior bivectors therefore has either both indices in E, which gives $\mathfrak{l}_{\mathrm{int}}$ itself and is excluded by commutation with the boosts, or both indices in the Lorentz complement

$$W_n = \langle e_2^{(1)}, e_3^{(1)} \rangle \oplus \bigoplus_{k=2}^{n} \langle e_1^{(k)}, \ldots, e_6^{(k)} \rangle,$$

which has $3n-1$ positive and $3n-3$ negative directions. All bivectors of W_n commute with $\mathfrak{l}_{\mathrm{int}}$, and they span $\mathfrak{so}(W_n)$. □

Corollary 5.2 (The photon at one constituent)**.** *At* $n = 1$, $W_1 = U$ *and* $\mathfrak{g}_{\mathrm{int}}^{(1)} = \mathbb{R}J$. *The electromagnetic* $\mathfrak{u}(1)$ *is not a symmetry of the interior structure but the symmetry, by exhaustion within the Lorentz-commuting grade-two connection algebra. In agreement with Proposition 2.29, no other charge direction of that algebra gauges at* $n = 1$. *This does not by itself select a gauging, a Maxwell kinetic term or the electromagnetic identification, and it says nothing about symmetries of a larger endomorphism algebra.*

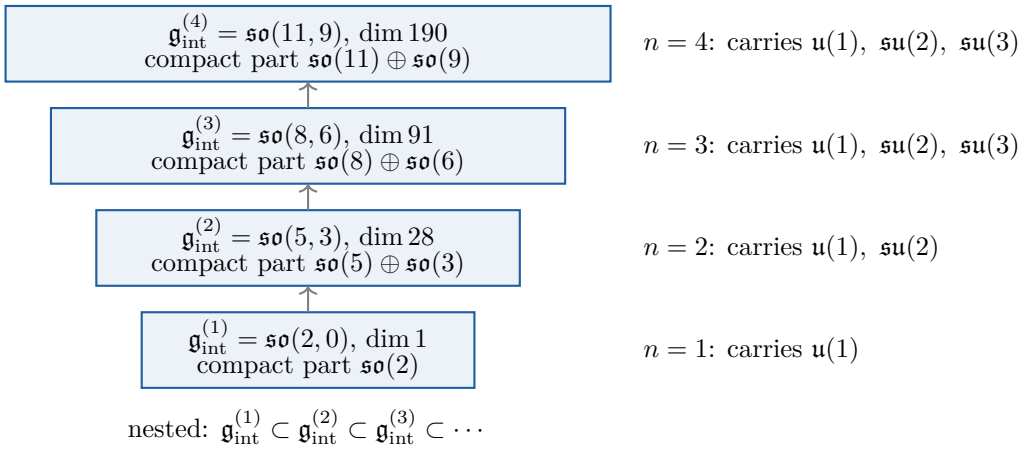


Figure 5.1: The tower of internal centralisers (Theorems 5.1 and 5.4). Each rung contains the previous one; the compact part decides which gauge factors can be carried. The electromagnetic $\mathfrak{u}(1)$ is the whole centraliser only at the first rung, the one rung with no room for a nonabelian factor.

Proposition 5.3 (One frame, and nesting)**.** *The interior four-plane lies in the first tensor factor, so the six interior bivectors have the same index sets at every rung and identical structure constants. The tower is one Lorentz frame with an increasing family of modules over it. Moreover* $W_1 \subset W_2 \subset \cdots$*, hence*

$$\mathfrak{g}_{\mathrm{int}}^{(1)} \subset \mathfrak{g}_{\mathrm{int}}^{(2)} \subset \mathfrak{g}_{\mathrm{int}}^{(3)} \subset \cdots, \qquad \dim = 1,\ 28,\ 91,\ 190,\ 325, \ldots,$$

with direct limit $\mathfrak{so}(\infty,\infty)$*. A connection valued at one rung is a connection at every higher rung with no matching condition. The inclusion runs in the wrong direction to justify a truncation to low rungs, which would be a separate postulate.*

A gauge algebra has to be compact, and every compact subalgebra of a reductive real Lie algebra is conjugate into a maximal compact one [80, 68], here $\mathfrak{so}(3n-1) \oplus \mathfrak{so}(3n-3)$. Since $\mathfrak{su}(2)$ needs a faithful real representation of dimension at least three and $\mathfrak{su}(3)$ one of dimension at least six [57], the rungs can be sorted by what they can carry.

Theorem 5.4 (Carrier census)**.**

n	$\mathfrak{g}_{\mathrm{int}}^{(n)}$	*maximal compact*	dim	*rank*	$\mathfrak{u}(1)$ *alone*	$\mathfrak{su}(2)$	$\mathfrak{su}(3)$
1	$\mathfrak{so}(2,0)$	$\mathfrak{so}(2)$	1	1	*yes*	*no*	*no*
2	$\mathfrak{so}(5,3)$	$\mathfrak{so}(5) \oplus \mathfrak{so}(3)$	13	3	*no*	*yes*	*no*
3	$\mathfrak{so}(8,6)$	$\mathfrak{so}(8) \oplus \mathfrak{so}(6)$	43	7	*no*	*yes*	*yes*
4	$\mathfrak{so}(11,9)$	$\mathfrak{so}(11) \oplus \mathfrak{so}(9)$	91	9	*no*	*yes*	*yes*

At $n = 2$ *the projection of a simple* $\mathfrak{su}(3)$ *to* $\mathfrak{so}(3)$ *must vanish and* $\mathfrak{so}(5)$ *has no proper subalgebra of dimension above six; at* $n = 3$ *colour enters through* $\mathfrak{su}(3) \subset \mathfrak{su}(4) \cong \mathfrak{so}(6)$*.*

Lemma 5.5 (Rank obstruction)**.** $\operatorname{rank}(\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)) = 4$*, while the maximal compact subalgebras above have ranks* $1, 3, 7, 9$*. The full Standard-Model algebra therefore embeds at no rung* $n \leq 2$*. At* $n = 2$ *the maximal compact subalgebra has dimension* $13 \geq 12$*, so dimension counting alone would admit it, and rank is the sharp invariant.*

Corollary 5.6 (The tower's trade-off)**.** *The* $\mathfrak{u}(1)$ *is the whole internal centraliser only at* $n = 1$*, the one rung with no room for either nonabelian carrier. Inside* $\mathfrak{g}_{\mathrm{int}}^{(n)}$ *the centraliser of the*

embedded one-rung charge $e_2^{(1)}e_3^{(1)}$ *has dimensions* $1, 16, 67, 154, 277$ *for* $n = 1, \dots, 5$*, since by* (1.3) *exactly* $12(n-1)$ *internal bivectors fail to commute with it. From* $n = 2$ *onwards the photon is one direction in a space that grows without bound; its uniqueness is lost to no dynamical process, only to the arrival of room.*

At $n \geq 2$ the embedded one-rung charge $e_2^{(1)}e_3^{(1)}$ must not be confused with the complex structure $J = \omega_I$ of Theorem 2.17, which commutes with every internal bivector and has all of $\mathfrak{g}_{\text{int}}^{(n)}$ as centraliser; they agree only at $n = 1$.

Remark 5.7 (The gaugeable fraction does not improve)**.** Of the $\binom{6n}{2} = 3n(6n-1)$ bivectors of $\mathrm{Cl}(3n, 3n)$, $2\binom{3n}{2} = 3n(3n-1)$ pair directions of the same sign: the fractions $6/15$, $30/66$, $72/153$ rise only to $1/2$. Climbing adds boosts at essentially the rate at which it adds rotations. The census permits the chain $\mathfrak{u}(1) \subset \mathfrak{u}(1) \oplus \mathfrak{su}(2) \subset \mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ at $n = 1, 2, 3$; it permits it, and nothing selects those subalgebras among others of the same rank.

5.2 Other Lorentz actions, and the vector bosons of the tower

Everything so far takes the Lorentz action from one constituent and treats the others as passengers. Theorem 5.1 is a statement about that choice, which we call the *spectator* action. The choice is adopted, not forced, and the alternatives are different mathematics with different verdicts.

5.2.1 The diagonal action

At three constituents let $\gamma_\mu^{(k)}$ be the interior four-plane of constituent k, of metric η. We call the two remaining generators $s_1^{(k)}, s_2^{(k)}$ of that constituent its surplus pair; both square to $+\mathbf{1}$. Write $\mathcal{S}$ for the six surplus directions and put

$$\Sigma_{\mu\nu}^{(k)} = \tfrac{1}{2}\gamma_\mu^{(k)}\gamma_\nu^{(k)}, \qquad \Delta_{\mu\nu} = \sum_{k=1}^{3} \Sigma_{\mu\nu}^{(k)}.$$

Bivectors of distinct constituents commute, so Δ satisfies the $\mathfrak{so}(1,3)$ brackets, and as a quadratic space

$$\mathbb{R}^{9,9} = \left(\mathbb{R}^3 \otimes \mathbb{R}^{1,3}\right) \oplus \mathbb{R}^{6,0}, \tag{5.1}$$

the first factor carrying a copy index with δ_{kl} and a Lorentz index with $\eta_{\mu\nu}$. The actions Δ and $\Sigma^{(1)}$ are not conjugate: their centralisers have different dimensions and their Casimirs different spectra.

Definition 5.8. $M_{kl} = \frac{1}{2}\sum_\mu \eta^{\mu\mu}\gamma_\mu^{(k)}\gamma_\mu^{(l)}$ for $k < l$ span the copy algebra $\mathfrak{k}_{\text{cop}}$, and $\mathfrak{k}_{\mathcal{S}}$ is the span of the fifteen surplus bivectors.

Theorem 5.9 (The diagonal centraliser is compact)**.** $\mathfrak{z}(\Delta) = \mathfrak{k}_{\mathcal{S}} \oplus \mathfrak{k}_{\text{cop}} \cong \mathfrak{so}(6) \oplus \mathfrak{so}(3)$*, of dimension* 18*, and every direction of it is compact.*

Proof. Under Δ the vector representation is $3V \oplus 6\cdot\mathbf{1}$ by (5.1), with V irreducible and nontrivial. By Schur an equivariant endomorphism acts as $A \otimes \mathbf{1}_V$ on the first summand and arbitrarily on the second, with no mixing since $\mathrm{Hom}_{\mathfrak{so}(1,3)}(V, \mathbf{1}) = 0$; antisymmetry for $\delta \otimes \eta$ makes $A \in \mathfrak{so}(3)$, and the surplus block gives $\mathfrak{so}(6)$. Both are realised: the surplus bivectors commute with Δ by (1.3), and M_{kl} is the spin lift of the copy rotation $E_{kl} - E_{lk}$ tensored with the identity on $\mathbb{R}^{1,3}$. Every surplus direction squares to $+\mathbf{1}$, and each term of M_{kl} pairs $\gamma_\mu^{(k)}$ with $\gamma_\mu^{(l)}$, two generators of the same square, so every generator is compact by Theorem 1.3. □

For comparison, the centraliser of $\Sigma^{(1)}$ is $\mathfrak{so}(8,6)$, of dimension 91 with 43 compact and 48 boost directions; that of all three $\Sigma^{(k)}$ together is $\mathfrak{k}_{\mathcal{S}}$ alone, of dimension 15; and that of $\Sigma^{(1)}$ and Δ together is $\mathfrak{k}_{\mathcal{S}} \oplus \mathbb{R}M_{23}$, of dimension 16. The diagonal choice exchanges 91 internal directions, only 43 of them compact, for 18 of which all are compact. By Theorem 1.13 below the noncompact directions are exactly the ghost-signed ones, so the diagonal internal algebra is free of ghosts as a whole while the spectator one is not.

5.2.2 A reductive dual pair

Put $C_\Delta = -\sum_{a<b} \eta^{aa}\eta^{bb}\Delta_{ab}^2$, $C_{\mathrm{cop}} = -\sum_{k<l} M_{kl}^2$ and $C_{\mathcal{S}} = -\sum_{s<s'}(\frac{1}{2}ss')^2$. For a single constituent $-\sum_{a<b}\eta^{aa}\eta^{bb}(\Sigma_{ab}^{(k)})^2 = \frac{3}{2}$ identically, the Dirac value, and $C_{\mathcal{S}} = \frac{15}{4}$ identically, so the surplus $\mathfrak{so}(6)$ acts isotypically on S_3 and its Casimir separates nothing.

Theorem 5.10 (Spectrum of the diagonal Casimir)**.** $\operatorname{spec} C_\Delta = \{\frac{3}{2}\,(160),\ \frac{11}{2}\,(288),\ \frac{15}{2}\,(64)\}$, *each eigenprojector being Lorentz invariant. The* $C_\Delta = \frac{3}{2}$ *block, the* Dirac block $\mathcal{D}$*, has rank* 160.

Proof. A quadratic Casimir of $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$ is $\alpha j_L(j_L+1) + \beta j_R(j_R+1)$, and the Dirac value $\frac{3}{2}$ together with the vector value 3 gives $\alpha = \beta = 2$. Since the constituent Casimir is $\frac{3}{2}$ identically, each $\mathbb{R}^8$ is $M \oplus M$ with M the real Majorana module, and $S_3 \cong M^{\otimes 3} \otimes \mathbb{R}^8$. Writing $A = (\frac{1}{2},0)$ and $B = (0,\frac{1}{2})$,

$$AAA = (\tfrac{3}{2},0) \oplus 2(\tfrac{1}{2},0), \quad AAB = (1,\tfrac{1}{2}) \oplus (0,\tfrac{1}{2}),$$
$$ABB = (\tfrac{1}{2},1) \oplus (\tfrac{1}{2},0), \quad BBB = (0,\tfrac{3}{2}) \oplus 2(0,\tfrac{1}{2}),$$

with three words of each mixed type. This gives multiplicities $1,3,5$ for $(\frac{3}{2},0)\oplus(0,\frac{3}{2})$, $(1,\frac{1}{2})\oplus(\frac{1}{2},1)$ and $(\frac{1}{2},0)\oplus(0,\frac{1}{2})$, of complex dimensions $8,12,4$; the totals $8,36,20$ add up to 64, and the multiplicity factor 8 turns them into $64, 288, 160$. □

Theorem 5.11 (A Casimir identity)**.** *As operators on* S_3*,*

$$C_{\mathrm{cop}} + C_\Delta = \tfrac{15}{2}\,\mathbf{1}, \tag{5.2}$$

and for n *constituents with a* d*-dimensional interior,* $C_{\mathrm{cop}} + C_\Delta = \big[nd(d-1)/8 + \binom{n}{2}d/4\big]\mathbf{1}$*; for* $d = 4$ *this is* $\frac{1}{2}n(n+2)\,\mathbf{1}$*.*

Proof. Write $B_{ab}^{kl} = \gamma_a^{(k)}\gamma_b^{(k)}\gamma_a^{(l)}\gamma_b^{(l)}$, symmetric in $k \leftrightarrow l$ and in $a \leftrightarrow b$. Expanding,

$$\Delta_{ab}^2 = \tfrac{1}{4}\Big[-n\,\eta_{aa}\eta_{bb} + 2\sum_{k<l} B_{ab}^{kl}\Big], \qquad M_{kl}^2 = -\tfrac{d}{4} - \tfrac{1}{2}\sum_{\mu<\nu}\eta^{\mu\mu}\eta^{\nu\nu}B_{\mu\nu}^{kl},$$

the first from $\gamma_a\gamma_b\gamma_a\gamma_b = -\eta_{aa}\eta_{bb}$, the second from $\gamma_\mu^{(k)}\gamma_\mu^{(l)}\gamma_\nu^{(k)}\gamma_\nu^{(l)} = -B_{\mu\nu}^{kl}$ for $\mu \neq \nu$. Hence $C_\Delta = nd(d-1)/8 - \frac{1}{2}\mathsf{S}$ and $C_{\mathrm{cop}} = \binom{n}{2}d/4 + \frac{1}{2}\mathsf{S}$ with $\mathsf{S} = \sum_{k<l}\sum_{a<b}\eta^{aa}\eta^{bb}B_{ab}^{kl}$, the only term in which two constituents interact. It cancels in the sum. □

At $n = 3$, $d = 4$ the constant is $3\cdot 4\cdot 3/8 + 3\cdot 4/4 = \frac{9}{2} + 3 = \frac{15}{2}$. The identity is inherited: its general form $nd(d+n-2)/8$ is the case $d_{\mathrm{N}} = n$, $k_{\mathrm{N}} = d/2$ of the relation $C_{\mathfrak{o}(d_{\mathrm{N}})} + C_{\mathfrak{o}(2k_{\mathrm{N}})} = \frac{1}{4}k_{\mathrm{N}}d_{\mathrm{N}}(d_{\mathrm{N}} + 2k_{\mathrm{N}} - 2)$ between the Casimirs of the $\mathfrak{o}(d_{\mathrm{N}})$–$\mathfrak{o}(2k_{\mathrm{N}})$ dual pair on a fermionic Fock space of k_{N} species [296, eq. (73)], an integer k_{N} (2 for $d = 4$, 3 for $d = 6$) at every rung, odd ones included (see also [297, 224]). The proof given here is an independent derivation in the real Clifford realisation.

Corollary 5.12 (A dual pair)**.** $\mathfrak{so}(3)_{\mathrm{cop}}$ *and* $\mathfrak{so}(1,3)_\Delta$ *are mutual centralisers inside* $\mathfrak{so}(3,9)$*, and the joint spectrum of* $(C_{\mathrm{cop}}, C_\Delta)$ *has exactly three points:*

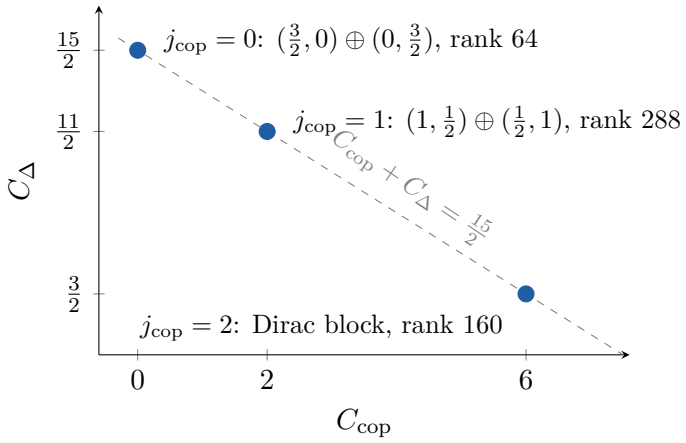


Figure 5.2: The diagonal sector at three constituents (Theorem 5.11 and Corollary 5.12). The copy Casimir and the Lorentz Casimir add up to a constant on the whole module, so copy spin is locked to Lorentz type: the Dirac block is exactly the copy quintet.

j_{cop}	dim	C_{cop}	C_Δ	Lorentz content	rank in S_3
0	1	0	$\frac{15}{2}$	$(\frac{3}{2}, 0) \oplus (0, \frac{3}{2})$	$8 \cdot 1 \cdot 8 = 64$
1	3	2	$\frac{11}{2}$	$(1, \frac{1}{2}) \oplus (\frac{1}{2}, 1)$	$12 \cdot 3 \cdot 8 = 288$
2	5	6	$\frac{3}{2}$	$(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$	$4 \cdot 5 \cdot 8 = 160$

The multiplicities $1, 3, 5$ *of Theorem 5.10 are the copy representations, so* $\mathcal{D} \cong L \otimes U$ *with* $L = (\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$ *and* $U = [j_{\mathrm{cop}} = 2] \otimes \mathbb{R}^8$ *carrying the whole gauge action.*

A projector selected by a rotation condition alone need not be Lorentz invariant. The rotation Casimir $-\sum_{i<j} \Delta_{ij}^2$ has spectrum $\{\frac{3}{4}\,(256), \frac{15}{4}\,(256)\}$, and its rank-256 spin-$\frac{1}{2}$ projector commutes with the rotations and with no boost: inside the irreducible $(1, \frac{1}{2})$ a boost carries the $j = \frac{1}{2}$ piece into the $j = \frac{3}{2}$ piece. The Lorentz-invariant answer is rank 160.

5.2.3 Gamma matrices on the Dirac block

The averaged gammas $V_\mu = 3^{-1/2} \sum_k \gamma_\mu^{(k)}$ satisfy $\{V_\mu, V_\nu\} = 2\eta_{\mu\nu}$ and transform as a Lorentz vector, but they are a copy vector, not a copy singlet: they map $\mathcal{D}$ into itself and into the rank-288 block with equal norms, and their compression to $\mathcal{D}$ degenerates on a rank-32 locus. The block does carry gamma matrices; they come from a copy-singlet cubic.

Theorem 5.13 (A gauge-invariant $\mathrm{Cl}(1,3)$ on the Dirac block)**.** *Set* $\Psi_{abc} = \varepsilon_{klm} \gamma_a^{(k)} \gamma_b^{(l)} \gamma_c^{(m)}$,

$$Q_\mu = \tfrac{1}{2} \eta^{\nu\rho} \Psi_{\mu\nu\rho} = \sum_\nu \eta_{\nu\nu} (\gamma_\mu^{(1)} \gamma_\nu^{(2)} \gamma_\nu^{(3)} + \gamma_\nu^{(1)} \gamma_\mu^{(2)} \gamma_\nu^{(3)} + \gamma_\nu^{(1)} \gamma_\nu^{(2)} \gamma_\mu^{(3)}), \qquad G_\mu = \tfrac{1}{6}\, \omega_{\mathcal{S}} Q_\mu|_{\mathcal{D}},$$

with $\omega_{\mathcal{S}}$ *the product of the six surplus generators. Then* Ψ *is a copy singlet and totally symmetric in* (a, b, c)*; each* Q_μ *is a sum of ten grade-three blades, commutes with all eighteen gauge generators and with* C_Δ*, is a Lorentz vector, and satisfies* $\{Q_\mu, Q_\nu\}|_{\mathcal{D}} = -72\, \eta_{\mu\nu}$*. Consequently, on all* 160 *dimensions,*

$$\{G_\mu, G_\nu\} = 2\eta_{\mu\nu} \mathbf{1}_{\mathcal{D}}, \qquad [\Delta_{ab}, G_c] = \eta_{bc} G_a - \eta_{ac} G_b, \qquad [G_\mu, \mathfrak{z}(\Delta)] = 0, \qquad \tfrac{1}{4}[G_a, G_b] = \Delta_{ab}|_{\mathcal{D}}.$$

Proof. Reordering $\gamma^{(\sigma 1)} \gamma^{(\sigma 2)} \gamma^{(\sigma 3)}$ into copy order costs $\operatorname{sign} \sigma$, because generators of distinct constituents anticommute; against the factor ε this gives $(\operatorname{sign} \sigma)^2 = 1$, so antisymmetry in the copies is symmetry in the Lorentz labels. Block preservation follows because a copy singlet commutes with C_{cop} and hence, by (5.2), with C_Δ. Finally $\omega_{\mathcal{S}}$ commutes with Q_μ and $\omega_{\mathcal{S}}^2 =$

$-\mathbf{1}$, so $\{G_\mu, G_\nu\} = \frac{1}{36}(-\mathbf{1})(-72\eta_{\mu\nu})$. The two restricted identities $\{Q_\mu, Q_\nu\}|_{\mathcal{D}} = -72\,\eta_{\mu\nu}$ and $\frac{1}{4}[G_a, G_b] = \Delta_{ab}|_{\mathcal{D}}$ are not derived by hand here; they are certified by exact integer computation on the 160-dimensional block (a9-1, a9-2). □

The sign flip is not cosmetic. $\mathrm{Cl}(1,3) \cong M_2(\mathbb{H})$ has minimal real module of dimension eight, so the four-dimensional Majorana module carries no real $\mathrm{Cl}(1,3)$; the Q_μ alone generate $\mathrm{Cl}(3,1)$, and $\omega_{\mathcal{S}}$, with square $-\mathbf{1}$, converts them. The operator $\not{D} = G^\mu \nabla_\mu$ is Lorentz covariant and gauge invariant on $\mathcal{D}$; $G_5 = G_0G_1G_2G_3$ has $G_5^2 = -\mathbf{1}$, a complex structure rather than a chirality, as $M_2(\mathbb{H})$ requires.

5.2.4 The diagonal sector is vector-like

Theorem 5.14 (Vector-likeness)**.** *Let $W_\mu = \gamma_\mu^{(1)}\gamma_\mu^{(2)}\gamma_\mu^{(3)}$. Then $W_\mu^2 = -\eta_{\mu\mu}$, so W_μ is invertible; $[W_\mu, \mathfrak{z}(\Delta)] = 0$; W_μ anticommutes with Δ_{ab} when $\mu \in \{a,b\}$ and commutes otherwise, hence commutes with C_Δ and preserves every block; and W_μ anticommutes with the gauge-invariant chirality $\omega = E_1 \cdots E_{18}$. Therefore W_0 is an invertible $\mathfrak{z}(\Delta)$-equivariant map exchanging the two chiral halves of every C_Δ block, and all of S_3 is vector-like for $\mathfrak{z}(\Delta)$.*

Proof. Each property is (1.3) with $|I| = 3$: against a surplus bivector the sign is $(-1)^6$; against M_{kl}, the term with $\nu = \mu$ meets W_μ in two indices and the others in none; against Δ_{ab} it is $(-1)^{6-1}$ when $\mu \in \{a,b\}$; against ω it is $(-1)^{54-3}$. The intertwining conclusion is Schur's lemma. □

By Corollary 5.12 the reason is structural: $\mathcal{D} \cong L \otimes U$ with the gauge algebra acting on U only, so chirality and gauge charge are carried by different tensor factors. Vector-like means that the gauge representation permits a mass for every state, not that any mass is large; nothing here fixes a scale. A Weyl projection of the block is possible, and it keeps the kinetic operators and the gauge algebra, but it is then a choice of field content, and at three constituents it carries a colour anomaly (Proposition 5.17).

5.2.5 The candidate gauge algebra of the constituent-wise Lorentz actions

A Lorentz action on the tower can be taken from each constituent in one of three ways: by the interior algebra $\mathfrak{l}_{\mathrm{int}}$, by the algebra $\mathfrak{l}_G$ that centralises the binding operator (Section 3.2), or not at all. These constituent-wise actions are the class treated here. Embeddings of the Lorentz algebra that mix constituents, for example by non-diagonal pairings of the Lorentz bases of different constituents, are not covered, and the no-go statements below that rely on this classification hold within the tested class (Section 13.1).

Theorem 5.15 (Candidate algebras)**.** *Let the Lorentz algebra act on rung n by $\mathfrak{l}_{\mathrm{int}}$ on k_I constituents, by $\mathfrak{l}_G$ on k_B constituents and trivially on k_S constituents, $k_I + k_B \geq 1$. The grade-two elements of $\mathrm{Cl}(3n, 3n)$ commuting with the action form*

$$\mathfrak{g}(k_I, k_B, k_S) \cong \mathfrak{so}(k_I) \oplus \mathfrak{so}(2k_I + 3k_S,\, 3k_S) \oplus \mathfrak{so}(k_B, \mathbb{C}),$$

a direct sum of ideals. Its maximal compact subalgebra is $\mathfrak{so}(k_I) \oplus \mathfrak{so}(2k_I + 3k_S) \oplus \mathfrak{so}(3k_S) \oplus \mathfrak{so}(k_B)$, and the algebra is compact exactly when $k_S = 0$ and $k_B \leq 1$.

Proof. Grade two is $\mathfrak{so}(V_n)$, and $V_n = U \oplus W \oplus F$ with $U = \mathbb{R}^{k_I} \otimes \mathbb{R}^{1,3}$ the interior four-planes, W the Lorentz-trivial directions (the surplus pairs of the interior constituents and all directions of the spectators, of signature $(2k_I + 3k_S, 3k_S)$), and $F = \mathbb{R}^{k_B} \otimes V_f$ with $V_f = (1,0) \oplus (0,1)$ the binding frames. The modules $(\frac{1}{2}, \frac{1}{2})$, $\mathbf{1}$ and V_f are pairwise inequivalent and irreducible, so the mixed parts of $\Lambda^2 V_n$ contain no invariant. $\Lambda^2 W$ is invariant and gives $\mathfrak{so}(W)$. In $\Lambda^2 U = (\Lambda^2\mathbb{R}^{k_I} \otimes S^2\mathbb{R}^4) \oplus (S^2\mathbb{R}^{k_I} \otimes \Lambda^2\mathbb{R}^4)$ the invariants of $S^2\mathbb{R}^4$ are the multiples of η and those of $\Lambda^2\mathbb{R}^4$ vanish, leaving $\mathfrak{so}(k_I)$. On V_f the invariant symmetric forms are spanned by η and $\eta(\cdot, G\cdot)$ and there is no invariant antisymmetric form, leaving $\Lambda^2\mathbb{R}^{k_B} \otimes \mathrm{span}\{\mathbf{1}, G\} \cong \mathfrak{so}(k_B, \mathbb{C})$. □

Here "compact" means contained in the chosen compact orthogonal subgroup, the one compatible with a positive kinetic form, in the stated embedding; it is not a statement about abstract Lie algebras. A compact candidate centraliser is not by itself a gauge theory.

Proposition 5.16 (Orthogonal against spin actions)**.** *A copy rotation of interior four-planes is a sum of four commuting terms of square $-\frac{1}{4}$, and one of binding frames a sum of six; their eigenvalues on S_n are i times integers, and a rotation through 2π is the identity. A rotation $\frac{1}{2}e_s e_t$ of two Lorentz-trivial directions of equal sign squares to $-\frac{1}{4}$ and its rotation through 2π is $-\mathbf{1}$. The copy algebras act through orthogonal groups, the algebra of the Lorentz-trivial directions through spin groups. Consequently the spin content of S_n is half-integer if k_I is odd and integer if k_I is even: fermions require an odd number of interior constituents.*

5.2.6 The diagonal tower

Under the diagonal interior action the $4n$ four-plane generators and the $2n$ surplus generators $e_2^{(k)}, e_3^{(k)}$ anticommute, so $S_n \otimes \mathbb{C} \cong S_{4n} \otimes S_{2n}$, with the Lorentz and copy algebras acting on the first factor and the surplus $\mathfrak{so}(2n)$ on the second. The connection algebra is the compact $\mathfrak{so}(n) \oplus \mathfrak{so}(2n)$, of dimension $n(5n-3)/2$.

Proposition 5.17 (The diagonal tower)**.** *(i) For $n \leq 7$ the factor S_{4n} decomposes under $\mathfrak{l}_{\text{int}} \oplus \mathfrak{so}(n)$ without multiplicity. Spin is half-integer exactly at odd n, and the Dirac components occur only at odd n, each once, tensored with the copy representation of highest weight $(2, \ldots, 2)$, of dimension $1, 5, 35, 294$ at $n = 1, 3, 5, 7$.*

(ii) At odd n the product $\mathcal{J}$ of the $2n$ surplus generators is a complex structure commuting with the whole connection algebra, and $\Gamma = \mathcal{J}\Omega$ is a chirality. With $\mathcal{J}$ as i the Dirac block is $[(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})] \otimes V_{(2,\ldots,2)} \otimes S^+$, with S^+ a half-spinor of the surplus $\mathfrak{so}(2n)$, and the charge $q = -\mathcal{J}\tilde{J}/2$ of $\tilde{J} = \sum_k J^{(k)}$ takes on S^+ the values

n	$\dim S^+$	values of q (multiplicity)
1	1	$\frac{1}{2}$
3	4	$-\frac{3}{2}$ (1), $\frac{1}{2}$ (3)
5	16	$-\frac{3}{2}$ (5), $\frac{1}{2}$ (10), $\frac{5}{2}$ (1)
7	64	$-\frac{7}{2}$ (1), $-\frac{3}{2}$ (21), $\frac{1}{2}$ (35), $\frac{5}{2}$ (7)

(iii) At $n = 3$ the centraliser of $\tilde{J}$ in the surplus $\mathfrak{so}(6)$ is $\mathfrak{u}(3)$, whose derived algebra is a compact $\mathfrak{su}(3)$; on a Weyl half it acts as a triplet with $q = \frac{1}{2}$ and a singlet with $q = -\frac{3}{2}$, so that with $B-L = \frac{2}{3}q$ the triplet has $B-L = \frac{1}{3}$ and the singlet -1: lepton number as the fourth colour [116]. The eight gluons are neutral, the six remaining generators of $\mathfrak{so}(6)$ carry $q = \pm 2$ and turn the triplet into the singlet, as leptoquark bosons do. The copy $\mathfrak{so}(3)$ acts on the Dirac block with Casimir 6, that is isospin two: it is not weak isospin.

(iv) At $n = 5$, with colour on three surplus planes and weak isospin on two, S^+ is one Standard-Model family with a right-handed neutrino, $(\mathbf{3}, \mathbf{2})_{1/6} \oplus (\bar{\mathbf{3}}, \mathbf{1})_{-2/3} \oplus (\bar{\mathbf{3}}, \mathbf{1})_{1/3} \oplus (\mathbf{1}, \mathbf{2})_{-1/2} \oplus (\mathbf{1}, \mathbf{1})_1 \oplus (\mathbf{1}, \mathbf{1})_0$, the spinor $\mathbf{16}$ of $\mathfrak{so}(10)$ [43, 144], occurring in 35 copies. At $n = 7$, S^+ under $\mathfrak{so}(10) \oplus \mathfrak{so}(4)$ is two families and two mirror families, the pattern found for the spinor of $\mathfrak{so}(18)$ [165].

(v) The Dirac block is vector-like at every odd n. Its Weyl projection admits no gauge-invariant mass, since no weight of S^+ has its negative in S^+; at $n = 3$ it carries a colour anomaly, $\mathrm{Tr}(P_L \mathcal{J} H^3) = 960$ for the colour Cartan element $H = J^{(1)} + J^{(2)} - 2J^{(3)}$, and at $n = 5$ every Standard-Model anomaly sum vanishes and $\mathfrak{so}(10)$ has no cubic anomaly [50].

Lemma 5.18 (No grade-two generator is chiral by itself)**.** *Let Γ be, up to sign, a product of $k \geq 5$ distinct generators with $\Gamma^2 = \mathbf{1}$, and $P_{L,R} = \frac{1}{2}(\mathbf{1} \mp \Gamma)$. A grade-two element X with $XP_R = 0$ is zero.*

Proof. $XP_R = 0$ means $X\Gamma = -X$. For $a \neq b$, $e_a e_b \Gamma$ is a blade of grade $k-2$, k or $k+2$, so $X\Gamma$ has no grade-two component, and $X\Gamma = -X$ forces $X = 0$. □

Under the diagonal and the spectator actions the chirality is, up to sign, the volume element of $\mathrm{Cl}(3n, 3n)$, so a gauge interaction built from connection components can be chiral only through the representation the Weyl block carries.

5.2.7 The compact part of the spectator algebra

Under the spectator action the connection components form $\mathfrak{so}(3n-1, 3n-3)$, noncompact for $n \geq 2$, and the gauge algebra is its compact part $\mathfrak{so}(3n-1) \oplus \mathfrak{so}(3n-3)$, acting through spinors. Let w be the product of the $6n-4$ Lorentz-trivial generators; it commutes with the whole connection algebra and serves as its complex structure, and $\Gamma = w\,\Omega^{(1)}$ is the chirality.

Proposition 5.19 (What the spectator gauge algebra acts on)**.** *At $n = 2$ the Weyl block is $(\frac{1}{2}, 0) \otimes (\mathbf{4}, \mathbf{2})$ under $\mathfrak{sp}(2) \oplus \mathfrak{su}(2) \cong \mathfrak{so}(5) \oplus \mathfrak{so}(3)$: isospin doublets without colour, in a real representation, so the weak interaction it carries is vector-like. At $n = 3$ the Weyl block is*

$$(\tfrac{1}{2}, 0) \otimes \left[(\mathbf{8}_s, \mathbf{4}) \oplus (\mathbf{8}_c, \bar{\mathbf{4}})\right] \quad \text{under } \mathfrak{so}(8) \oplus \mathfrak{so}(6).$$

With colour on three negative planes, weak isospin on two positive planes and a family $\mathfrak{so}(4)$ on the remaining two, its 64 weights are two Standard-Model families with ν^c and two mirror families. The cubic $\mathfrak{su}(4)$ sum and the Standard-Model anomaly sums vanish, there are sixteen weak doublets, the weak roots carry $Q = \pm 1$, the gluons are neutral and the leptoquarks carry $\pm\frac{2}{3}$. The block is chiral and anomaly free under $\mathfrak{so}(8) \oplus \mathfrak{so}(6)$ and vector-like under the Standard-Model algebra.

Executable check: e5-3a, e5-3b, e5-3c for $n = 2$ and a7-6 for $n = 3$ (exact symbolic or integer arithmetic; Section 15.2).

At $n = 3$ the compact spectator algebra therefore already contains the Pati–Salam algebra $\mathfrak{so}(4) \oplus \mathfrak{so}(6) \cong \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R \oplus \mathfrak{su}(4)$. The projector P_* of Section 6.2 is a way of cutting one chiral family out of this content.

5.2.8 Vector bosons rung by rung

n	diagonal connection algebra	bosons	fermions	action on the Dirac block
1	$\mathfrak{so}(2) = \mathfrak{u}(1)_J$	1	yes	one charge, also the complex structure
2	$\mathfrak{so}(2) \oplus \mathfrak{so}(4)$	7	no	integer spin only
3	$\mathfrak{so}(3) \oplus \mathfrak{so}(6) \cong \mathfrak{su}(2) \oplus \mathfrak{su}(4)$	18	yes	$5 \times (\mathbf{3} \oplus \mathbf{1})$; the $\mathfrak{su}(2)$ is the copy algebra
4	$\mathfrak{so}(4) \oplus \mathfrak{so}(8)$	34	no	—
5	$\mathfrak{so}(5) \oplus \mathfrak{so}(10)$	55	yes	$35 \times \mathbf{16}$, one family each
7	$\mathfrak{so}(7) \oplus \mathfrak{so}(14)$	112	yes	$294 \times (2 \cdot \mathbf{16} \oplus 2 \cdot \overline{\mathbf{16}})$
n	spectator gauge algebra	bosons		action on the Weyl block
1	$\mathfrak{so}(2)$	1		one charge
2	$\mathfrak{so}(5) \oplus \mathfrak{so}(3)$	13		$(\mathbf{4}, \mathbf{2})$: doublets, no colour, real
3	$\mathfrak{so}(8) \oplus \mathfrak{so}(6)$	43		two families and two mirror families

The tempting assignment "$n = 1$: photon; $n = 2$: W and Z; $n = 3$: gluons" can now be tested. The photon at $n = 1$ is supported: the single connection component J gauges the Dirac spinor and is also its complex structure. Chiral W and Z at $n = 2$ are supported by no action:

the diagonal action has no fermions there, the action that is interior on one constituent and binding on the other has the abelian algebra $\mathfrak{u}(1)$ only, and the spectator Weyl block is a real representation. Gluons at $n = 3$ are supported, without weak bosons under the diagonal action, where the $\mathfrak{su}(2)$ is the copy algebra, and with them under the spectator action. Rung three is the first at which some gauge algebra contains $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ acting on fermions in Standard-Model representations, there together with mirror families; among the actions with a compact connection algebra the first such rung is $n = 5$, through $\mathfrak{so}(10)$, with one family per copy and thirty-five copies. Nothing computed here reduces the copy multiplicity to three, and the rung index counts tensor factors, so none of this identifies particles with constituents.

5.3 Duplication: from $\mathrm{Cl}(3,3)$ to $\mathrm{Cl}(6,6)$

The author's second remark asks which physical and mathematical process makes a unitary energy "result in a duplication", from $\mathrm{Cl}(3,3)$ to $\mathrm{Cl}(6,6)$. A duplication acts on four levels at once: on the vector space, $V \to V_2 := V \oplus V$; on the algebra, $\mathrm{Cl}(V) \to \mathrm{Cl}(V_2) = \mathrm{Cl}(6,6)$; on the spinor module, $S \to S_2 = S \otimes S$; and on the energy, $E_6 \to E_6 \oplus E_6$. This section takes the four steps in turn, asks which are forced and which are chosen, and records what the duplicate can do that one copy cannot. The answers, with their classes, are these. The step on the algebra is forced: it is the graded tensor product, its Klein factor is the Jordan–Wigner sign, and S_2 is the Fock space of six fermionic modes (inherited; Section 5.3.1). The step on the energy is a choice within a three-sphere of energy-preserving, Lorentz-covariant duplications, which is the algebra of a lossless beam splitter. Only a circle in it, the copy rotation, respects η, and the two notions of rest energy that coincide on one copy separate on two (fixed; Section 5.3.3). The duplicate carries structures one copy cannot carry: an η-isometric complex structure, a compact, Lorentz-invariant copy charge, the copy algebra $\mathfrak{so}(2,\mathbb{C})$, a closure that runs at constant rate with non-null copies ($E_0(\pi^{(k)}) \neq 0$), and a non-rigid, Lorentz-invariant vacuum (fixed, partly restated from earlier chapters; Sections 5.3.4 to 5.3.7). Finally, the copy rotation is the only admissible b-field between two binding frames that commutes with the binding root and the rotations. When the two rates differ it cannot be removed, and it confines a frame that has no rate of its own; it creates no rate where both frames are free (fixed, with the mixing formulas inherited; Section 5.3.8). Every finite-dimensional statement was checked by `d5_duplication_math.py`; identifiers d5-n refer to its `PASS` lines (Table 5.3).

Notation. $V_2 = V \oplus V$ has the basis $e_1, \dots, e_6$ of copy 1 followed by that of copy 2, the metric $\eta_2 = \mathbf{1}_2 \otimes \eta$, and the binding root $G_2 = \mathbf{1}_2 \otimes G$ of both copies (as G_3 in Theorem 4.34). $\varepsilon = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, σ_x, σ_z act on the copy index, and $C_v := \varepsilon \otimes \mathbf{1}_6$ is the copy rotation on vectors. A pair of field strengths is $(z_1, z_2) \in \mathbb{C}^3 \oplus \mathbb{C}^3$, with $q_2 := q(z_1) + q(z_2)$ and duplicated energy $E_6(z_1) + E_6(z_2)$. On $S_2 = \mathbb{R}^8 \otimes \mathbb{R}^8$ the tower of equation (1.2) gives $e_a^{(1)} = e_a \otimes \mathbf{1}$, $e_a^{(2)} = \omega \otimes e_a$, $\omega = e_1 \cdots e_6$, and $E_{6(k-1)+a} = e_a^{(k)}$. Statements about V_2 are pointwise linear algebra and apply to whichever field is placed there. Where the field matters we name it: the frame field X, or the momentum field $\pi = \eta^{-1} p$ of the closure, to which the Casimirs belong (Section 4.4, Section 4.5). The rates κ_k of the binding, the spectrum of the self-adjoint part of the shift, are kept apart from the lines of the motion, the spectrum of the shift itself: a b-field can move the lines but never the rates (Theorem 9.21(iii)). Reading either as a rest energy through $mc^2 = \hbar\kappa$ is an added identification.

5.3.1 The duplicated algebra: graded tensor product and Klein factor

Proposition 5.20 (The duplicated algebra)**.** *(i)* $\omega^2 = \mathbf{1}$, $\omega^\top = \omega$, $\operatorname{Tr}\omega = 0$, *and* ω *anti-commutes with every* e_a. *The twelve matrices* $e_a^{(k)}$ *satisfy the Clifford relations of* $\mathbb{R}^{6,6}$.

Their 4096 *blades are signed permutation matrices, and all except* $\mathbf{1}$ *are traceless, so* $\mathrm{Cl}(6,6) \cong M_{64}(\mathbb{R})$.

(ii) For blades e_I, e_J *of* $\mathrm{Cl}(3,3)$, $e_I^{(1)} e_J^{(2)} = (e_I\, \omega^{|J|}) \otimes e_J$ *and* $e_J^{(2)} e_I^{(1)} = (-1)^{|I||J|} e_I^{(1)} e_J^{(2)}$. *Hence* $a \hat{\otimes} b \mapsto a\, \omega^{|b|} \otimes b$ *is an algebra isomorphism* $\mathrm{Cl}(3,3) \hat{\otimes} \mathrm{Cl}(3,3) \to \mathrm{Cl}(6,6)$, *where* $\hat{\otimes}$ *is the graded tensor product,* $(a \hat{\otimes} b)(c \hat{\otimes} d) = (-1)^{|b||c|} ac \hat{\otimes} bd$.

(iii) The Klein factor is forced. In $M_{64}(\mathbb{R})$ *the elements commuting with every* $e_a \otimes \mathbf{1}$ *form* $\mathbf{1} \otimes M_8(\mathbb{R})$, *and those anticommuting with every* $e_a \otimes \mathbf{1}$ *form* $\omega \otimes M_8(\mathbb{R})$. *Any realisation of* $\mathrm{Cl}(V_2)$ *on* $\mathbb{R}^{64}$ *that extends* $e_a \mapsto e_a \otimes \mathbf{1}$ *is therefore conjugate to the tower by an element of* $\mathbf{1} \otimes GL(8, \mathbb{R})$.

Proof. (i) $\omega^2 = (-1)^{15}(-1)^3 \mathbf{1}$ by the transpose rule (1.4), and ω has even grade in an even dimension. The relations follow from $(\omega \otimes e_a)(\omega \otimes e_b) = \mathbf{1} \otimes e_a e_b$ and $(e_a \otimes \mathbf{1})(\omega \otimes e_b) = -(\omega \otimes e_b)(e_a \otimes \mathbf{1})$. A product of signed permutation matrices is one, and if $I \neq J$ then $e_I^{\top} e_J = \pm e_{I \triangle J}$ is traceless, so the blades are orthogonal for the trace form; $4096 = 64^2$. (ii) $e_J^{(2)} = \omega^{|J|} \otimes e_J$, and moving the $|J|$ copy-2 generators past the $|I|$ copy-1 generators costs $(-1)^{|I||J|}$. (iii) By (1.3) a blade $e_I^{(1)} e_J^{(2)}$ commutes with $e_a^{(1)}$ exactly when $|I| + |J| - [a \in I]$ is even. For all six a this forces either $I = \emptyset$ and $|J|$ even, or $I = \{1, \dots, 6\}$ and $|J|$ odd: 64 blades, each $\pm \mathbf{1} \otimes e_J$. The anticommuting case is the same with the parities reversed. If e'_a anticommutes with all $e_b \otimes \mathbf{1}$, then $e'_a = \omega \otimes f_a$, and $(\omega \otimes f_a)(\omega \otimes f_b) = \mathbf{1} \otimes f_a f_b$, so the f_a satisfy the relations of $\mathrm{Cl}(3,3) \cong M_8(\mathbb{R})$. By the Skolem–Noether theorem $f_a = u e_a u^{-1}$, and conjugation by $\mathbf{1} \otimes u$ fixes $e_a \otimes \mathbf{1}$ and $\omega \otimes \mathbf{1}$. □

Everything here is inherited: it is the exponential law $\mathrm{Cl}(V \oplus W) \cong \mathrm{Cl}(V) \hat{\otimes} \mathrm{Cl}(W)$ [201, 214], and (i) restates equation (1.2). The arena contributes only the fixed realisation (d5-1a to d5-1d). The duplication of the algebra is not a choice; given copy 1, copy 2 is determined up to conjugation.

Proposition 5.21 (Fock space and Jordan–Wigner strings)**.** *Put* $n_i^{\pm} = \frac{1}{2}(e_i \pm e_{i+3})$, $i = 1, 2, 3$. *Then* $\{n_i^{\pm}, n_j^{\pm}\} = 0$ *and* $\{n_i^{+}, n_j^{-}\} = \delta_{ij}$, *so* $\mathrm{Cl}(3,3)$ *is the real algebra of canonical anticommutation relations of three fermionic modes and* $S = \mathbb{R}^8$ *is their Fock space. The occupation numbers are* $N_i = n_i^{+} n_i^{-} = \frac{1}{2}(\mathbf{1} - \omega_i)$, *with* $\omega_i = e_i e_{i+3}$ *the pole operators. The eight eigenlines of Proposition 1.5(i) are therefore the eight Fock states, and* $(-1)^{N_1+N_2+N_3} = -\omega$. *At* $n = 2$ *the tower supplies six modes. The modes of copy* 2 *are* $\omega \otimes n_i^{\pm}$, *each carrying the string* $-(-1)^{N^{(1)}}$ *of copy* 1. *The total number has spectrum* $0, \dots, 6$ *with multiplicities* $1, 6, 15, 20, 15, 6, 1$, *and* $(-1)^N$ *is the volume element of* $\mathrm{Cl}(6,6)$.

Inherited. In split signature the vectors $e_i \pm e_{i+3}$ span two maximal isotropic subspaces, and the spinor module is the exterior algebra of one of them [201]. The string is the Jordan–Wigner transformation [212], and the Klein factor is the device that makes the fields of two subsystems anticommute [213]. On the level of states, the mathematical process behind "$\mathrm{Cl}(3,3) \to \mathrm{Cl}(6,6)$" is the second-quantisation rule $\mathrm{Fock}(\mathcal{H} \oplus \mathcal{H}) = \mathrm{Fock}(\mathcal{H}) \otimes \mathrm{Fock}(\mathcal{H})$ (d5-1e).

5.3.2 Centre-of-mass and relative halves

Proposition 5.22 (Two halves, and the exchange)**.** *Let* $f_a = (e_a^{(1)} + e_a^{(2)})/\sqrt{2}$ *and* $g_a = (e_a^{(1)} - e_a^{(2)})/\sqrt{2}$.

(i) $\{f_a, f_b\} = \{g_a, g_b\} = 2\eta_{ab}$ *and* $\{f_a, g_b\} = 0$ *for all* a, b, *so* $\mathrm{Cl}(6,6) = \mathrm{Cl}(f) \hat{\otimes} \mathrm{Cl}(g)$: *two mutually anticommuting copies of* $\mathrm{Cl}(3,3)$.

(ii) The copy rotation $C_b = \frac{1}{2} \sum_a \eta_{aa} e_a^{(1)} e_a^{(2)}$ *acts on vectors by* $e_a^{(1)} \mapsto -e_a^{(2)}$, $e_a^{(2)} \mapsto e_a^{(1)}$, *that is by* $-C_v$, *so it lifts the copy rotation with the opposite orientation; it is a sum of*

six commuting terms of square $-\frac{1}{4}$, so $\exp(2\pi C_b) = \mathbf{1}$. Conjugation by $\exp(\frac{\pi}{4}C_b)$ carries $e_a^{(1)}$ to g_a and $e_a^{(2)}$ to f_a, and $C_b = -\frac{1}{2}\sum_a \eta_{aa} f_a g_a$. The split into halves is the split into constituents turned by an eighth of a turn of the copy rotation, which rotates one half into the other.

(iii) Let $\omega = (-1)^{\mathsf{p}}$ on the basis of equation (1.1), in which ω is diagonal, let $\tau(x\otimes y) = y\otimes x$, and let $D = (-1)^{\mathsf{p}\otimes\mathsf{p}}$. The Koszul swap $\tau_K = D\tau$ satisfies $\tau_K e_a^{(1)} \tau_K = e_a^{(2)}$, $\tau_K^2 = \mathbf{1}$, $\operatorname{Tr}\tau_K = 0$, and $\tau_K = g_1 g_2 \cdots g_6$. So the exchange commutes with every f_a and anticommutes with every g_a. The naive swap τ does not implement the exchange.

(iv) The summed clock is $e_1^{(1)} + e_1^{(2)} = \sqrt{2} f_1$. Every diagonal bivector satisfies $\sum_k e_a^{(k)} e_b^{(k)} = f_a f_b + g_a g_b$; this covers the diagonal interior and binding Lorentz generators and the summed charge $J^{(1)} + J^{(2)} = f_2 f_3 + g_2 g_3$, which act on both halves alike with no cross term. G_2 preserves both halves. The copy rotations C_b and M_{12} (the copy rotation of the interior four-planes, Section 5.2.1) are pure cross terms and anticommute with τ_K.

Proof. (i) and (iv) are expansions, using $e_a^{(1)} e_a^{(2)} = -f_a g_a$. (ii) By (1.3), $[C_b, e_b^{(1)}] = -e_b^{(2)}$. The six terms $\frac{1}{2}\eta_{aa} e_a^{(1)} e_a^{(2)}$ share no index, so they commute, and each squares to $-\frac{1}{4}$; hence $\exp(\theta C_b) = \prod_a (\cos\frac{\theta}{2} + \sin\frac{\theta}{2}\,\eta_{aa} e_a^{(1)} e_a^{(2)})$, which is $(-1)^6\mathbf{1}$ at $\theta = 2\pi$. On vectors $\exp(\theta \operatorname{ad} C_b)$ turns $e_a^{(1)}$ into $\cos\theta\, e_a^{(1)} - \sin\theta\, e_a^{(2)}$, which is g_a at $\theta = \frac{\pi}{4}$. (iii) The swap of the copies fixes every f_a and negates every g_a: it is the reflection of V_2 in the span of the f_a along the span of the g_a, and the product $g_1 \cdots g_6$ of an even number of unit vectors implements it by conjugation. The Koszul swap implements it too, by the sign rule. The commutant of $\mathrm{Cl}(6,6) = M_{64}(\mathbb{R})$ is $\mathbb{R}$, so the two agree up to a scalar, which is $+1$. $\operatorname{Tr}\tau_K$ counts ± 1 over the diagonal states $x_i \otimes x_i$ with sign $(-1)^{\mathsf{p}_i}$, and ω is traceless. □

Inherited: centre-of-mass and relative coordinates, and the graded swap, which is the exchange operator of two fermionic subsystems. The identity $\tau_K = g_1 \cdots g_6$, with its sign, and the list in (iv) are facts of the realisation (d5-1f, d5-2a to d5-2c). Because τ_K anticommutes with the copy rotation, exchanging the constituents reverses the copy charge, as exchanging two real vector fields W^1, W^2 exchanges $W^{\pm}$ (inherited). The halves are not invariant under the copy rotation. Only the exchange selects them, as its two eigenspaces on vectors.

5.3.3 Duplicating the undivided energy

Proposition 5.23 (Symmetry of the duplicated energy)**.** *(i) The linear maps of V_2 that commute with the diagonal rotations and with G_2 form $\mathfrak{gl}(2)\otimes \operatorname{span}\{\mathbf{1}, G\} \cong \mathfrak{gl}(2,\mathbb{C})$. The infinitesimal symmetries of $E_6 \oplus E_6$ among them form $\mathfrak{u}(2) = \operatorname{span}\{C_v,\ \mathbf{1}\otimes G,\ \sigma_z\otimes G,\ \sigma_x\otimes G\}$. Those in $\mathfrak{so}(6,6)$ form $\mathfrak{so}(2,\mathbb{C}) = \operatorname{span}\{C_v,\ \varepsilon\otimes G\}$, and those in both form $\mathfrak{so}(2) = \mathbb{R}\, C_v$. The overall duality $\mathbf{1}\otimes G = G_2$, the relative duality $\sigma_z \otimes G$ (the binding rotation in opposite senses on the two copies) and $\sigma_x \otimes G$ are η_2-self-adjoint, like G.*

(ii) Inside $\mathfrak{u}(6) = \mathfrak{so}(12) \cap \operatorname{Cent}(G_2)$, of dimension 36, the diagonal $\mathfrak{u}(3)$ and $\mathfrak{u}(2)\otimes\mathbf{1}$ are each other's commutants: $U(3)\times U(2)$ is a dual pair in $U(6)$.

Part (i) restates rows three to five of Proposition 5.72 (fixed there), with the generators named. Row three is stated there for the diagonal $\mathfrak{l}_G$, whose commutant in $\mathfrak{so}(6,6)$ is the same $\mathfrak{so}(2,\mathbb{C})$ (Proposition 5.27(iii)). The four generators are those of a lossless beam splitter on two modes [272]: a real splitter C_v, an overall phase, a relative phase and a symmetric splitter. Their group $U(2)$ is the energy-preserving duality group of two free Maxwell fields [46], as Section 5.8 notes. Part (ii) is inherited from Howe duality for $\mathbb{C}^3 \otimes \mathbb{C}^2$ [224] (d5-3a, d5-3b).

Theorem 5.24 (Duplication maps)**.** *(i) The G-linear, rotation-equivariant maps $V \to V_2$ are $\iota_{\alpha,\beta}(z) = (\alpha z, \beta z)$ with $\alpha, \beta \in \mathbb{C}$. Such a map preserves the energy, $E_6(\alpha z) + E_6(\beta z) =$*

$E_6(z)$, if and only if $|\alpha|^2+|\beta|^2=1$, which gives a three-sphere $U(2)/U(1)$: the first column of a beam splitter. It is an isometry $(V,\eta)\to(V_2,\eta_2)$ if and only if $\alpha^2+\beta^2=1$ over $\mathbb{C}$, a non-compact complex conic, the orbit of $\iota_{1,0}$ under the group of the copy algebra $\mathfrak{so}(2,\mathbb{C})$. An energy-preserving $\iota_{\alpha,\beta}$ is an isometry if and only if α,β are real with $\alpha^2+\beta^2=1$, which gives a circle, the orbit of the copy rotation.

(ii) On the image, $q_2=(\alpha^2+\beta^2)\,q$, $E_0^{(1)}+E_0^{(2)}=(|\alpha|^2+|\beta|^2)E_0$, and the summed Maxwell stress tensor is $T(\alpha z)+T(\beta z)=(|\alpha|^2+|\beta|^2)\,T(z)$. Every energy-preserving duplication therefore preserves the summed stress tensor, the sum of the copies' rest energies, and the pair invariant $E^2_{\text{pair}}:=\frac{1}{4}T_{\text{tot}}{}^{\mu}{}_{\nu}T_{\text{tot}}{}^{\nu}{}_{\mu}$, which equals E_0^2 by Rainich's identity $(T\eta)^2=E_0^2\,\mathbf{1}$ [129]. By contrast $|q_2|=|\alpha^2+\beta^2|\,|q|=(1-4(\operatorname{Im}\alpha\bar\beta)^2)^{1/2}|q|$ sweeps $[0,|q|]$. The complex invariant q_2 is preserved exactly on the circle. Its modulus $|q_2|$ is preserved exactly where $\operatorname{Im}\alpha\bar\beta=0$, that is on the torus $(\alpha,\beta)=e^{i\psi}(\cos t,\sin t)$, the orbit of the copy rotation and the overall duality rotation $e^{\psi G_2}$; for example $\iota_{i,0}$ gives $q_2=-q$. At the equal split with relative duality angle θ ($\beta=e^{i\theta}\alpha$), $|q_2|=|\cos\theta|\,|q|$, which vanishes at $\theta=\pi/2$.

(iii) Without equivariance, consider $X\mapsto(\cos\phi\,X,\ \sin\phi\,RX)$ with $R\in U(3)$ and $0<\phi<\pi/2$. It preserves the summed stress tensor for all X iff $R=e^{\vartheta G}$ is a duality rotation, preserves q_2 iff $R\in O(3)$, and preserves both iff $R=\pm\mathbf{1}$. More generally, a G-linear embedding $W=U+iV:\mathbb{C}^3\to\mathbb{C}^6$ that preserves both $E_6\oplus E_6$ and q_2 is real, since $W^*W-W^\top W=2V^\top V-2iV^\top U$.

Proof. (i) The rotations act on $V=\mathbb{R}^2\otimes\mathbb{R}^3$ through the absolutely irreducible second factor, so by Schur's lemma [136] the equivariant maps are $(A,B)\otimes\mathbf{1}_3$ with $A,B\in M_2(\mathbb{R})$, and G-linearity puts A,B in $\operatorname{span}\{\mathbf{1},G\}\cong\mathbb{C}$. With $G^\top G=\mathbf{1}$, $G^\top\eta G=-\eta$ and $G^\top\eta=\eta G$, the map $L=\iota_{\alpha,\beta}$ has $L^\top L=(|\alpha|^2+|\beta|^2)\mathbf{1}$ and $L^\top\eta_2 L=\operatorname{Re}(\alpha^2+\beta^2)\,\eta+\operatorname{Im}(\alpha^2+\beta^2)\,\eta G$. Since $|\alpha|^2+|\beta|^2-\operatorname{Re}(\alpha^2+\beta^2)=2[(\operatorname{Im}\alpha)^2+(\operatorname{Im}\beta)^2]$, the two conditions together force real coefficients. (ii) q is holomorphic and quadratic. The components of T are $\frac{1}{2}|z|^2$, the textbook Poynting vector $-\mathbf{S}=-\frac{i}{2}\bar z\times z$, and $\frac{1}{2}\delta_{ij}|z|^2-\operatorname{Re}(\bar z_i z_j)$, all sesquilinear, so $T(\alpha z)=|\alpha|^2T(z)$. Rainich's identity is the tensor form of the identity $E_6^2=E_0^2+|\mathbf{S}|^2$ of Section 3.9.3. (iii) T determines the Hermitian matrix zz^*, the real part from T^{ij} and the imaginary part from T^{0i}, and conversely. So $T(Rz)=T(z)$ for all z iff R commutes with every rank-one Hermitian matrix, iff R is a phase. $q(Rz)=q(z)$ for all z iff $R^\top R=\mathbf{1}$, which for unitary R means $\bar R=R$. If $W^*W=W^\top W=\mathbf{1}$ the identity gives $V^\top V=0$. □

All three parts are inherited: they are the linear algebra of a beam splitter acting on $\mathbb{C}^3\otimes\mathbb{C}^2$ (d5-3c to d5-3e). What is not inherited is the reading.

Remark 5.25 (Two rest energies of a duplicate)**.** On one copy $E_0=\frac{1}{2}|q|$ is at once the holomorphic invariant and the stress-energy invariant; this is Rainich's identity, the field-point form of $E^2=(pc)^2+(mc^2)^2$ (Section 3.9.3). On two copies the two separate. The stress-energy content $(T_{\text{tot}},\ \sum_k E_0^{(k)},\ E_{\text{pair}})$ is preserved by the whole three-sphere, the holomorphic invariant q_2 only by the circle (fixed). Reading this as the point where duplication changes the meaning of "rest energy" is interpretive. Which one deserves the name depends on the group. The invariants of a pair under the diagonal $\mathfrak{so}(3,\mathbb{C})$ are the entries of the complex Gram matrix $Q_{kl}=z_k\cdot z_l$; the copy algebra $\mathfrak{so}(2,\mathbb{C})$ preserves $\operatorname{Tr}Q=q_2$ and $\det Q$ but not Q_{11} (d5-3f); and the closure sees $|q_2|$ of the momentum field (Proposition 5.28). Two counter-propagating null fields, $z_1=(1,i,0)$ and $z_2=(1,-i,0)$, have $E_0^{(k)}=0$ and $q_2=0$, yet $E^2_{\text{pair}}=2$ (d5-3i). Their rest energy as a pair is the rest-energy density $M_{\text{pair}}=2$ of the summed stress tensor (Theorem 5.57), not $E_{\text{pair}}=\sqrt{2}$: E_{pair} is positive exactly when M_{pair} is, and equals it only when $\det Q=0$, as on every duplicate of one field (Theorem 5.57). A nonzero M_{pair} is a kinematic analogue, at one field point, of the invariant mass of two photons, which permits $\gamma\gamma\to e^+e^-$ [221]; it is a local rest-energy density, not an invariant mass, and the analogy describes no production process. A single null

field duplicated by any $\iota_{\alpha,\beta}$ stays null in each copy, since $q(\alpha z) = \alpha^2 q(z)$; that is the analogue of the absence of $\gamma \to e^+e^-$ in vacuum (the photon kinematics inherited; both analogies interpretive). No isometric linear map copies the energy, $z \mapsto (z, z)$ has $E_6 \oplus E_6 = 2E_6$, and the genuine duplication of a quantum, $z \mapsto z \otimes z$, is quadratic; that is the no-cloning theorem [277] (inherited; treating the field point z as a quantum state is interpretive). A linear duplication divides the undivided energy; it does not multiply it.

Proposition 5.26 (An isometric complex structure exists only on the duplicate)**.** *(i) No $J \in O(3,3)$ satisfies $J^2 = -\mathbf{1}$.*

(ii) On $\mathbb{R}^{6,6}$, C_v satisfies $C_v^2 = -\mathbf{1}$, $C_v^\top C_v = \mathbf{1}$ and $C_v^\top \eta_2 C_v = \eta_2$, and commutes with G_2 and with the diagonal $\mathfrak{l}_G$. It is a compact charge, invariant under the Lorentz group that acts on both frames alike.

(iii) $\mathfrak{u}(6) \cap \mathfrak{so}(6,6) = \mathfrak{so}(6)$ has dimension 15; it consists of the paired rotations and is the compact part of $\mathrm{Cent}(G_2)$. At $n = 1$ the corresponding algebra is $\mathfrak{u}(3) \cap \mathfrak{so}(3,3) = \mathfrak{so}(3)$. Inside $\mathfrak{so}(6)$ the centraliser of C_v is $\mathfrak{u}(3)' = \{\mathbf{1} \otimes \mathrm{diag}(A, A) + \varepsilon \otimes \mathrm{diag}(B, B) : A^\top = -A,\ B^\top = B\}$, of dimension 9, and its part commuting with the diagonal boosts is $\mathbb{R}\, C_v$.

Proof. (i) If $J^\top \eta J = \eta$ and $J^2 = -\mathbf{1}$, then $h(x, y) = \eta(x, y) + i\,\eta(x, Jy)$ is a Hermitian form on $(\mathbb{R}^6, J) \cong \mathbb{C}^3$ with $\mathrm{Re}\, h = \eta$. A Hermitian form of signature (p, q) has real signature $(2p, 2q)$, never $(3, 3)$. (ii) and (iii) are computed (d5-3g, d5-3h). □

Part (i) is inherited, the signature parity of Hermitian forms. It bites because the arena's triples are odd, and that much is fixed. Read physically, $w = X^{(1)} + iX^{(2)}$ combines two neutral copies into one field charged under C_v, as two real fields form a charged one (inherited). The pointwise unitary symmetry $U(3)$ of E_6 is cut to $O(3)$ by Lorentz compatibility on one copy (Section 3.9). It reappears inside $SO(6,6)$ as $U(3)'$, with the copy index as imaginary unit and not the binding root (interpretive), and $U(3)'$ is not Lorentz invariant.

5.3.4 The duplicated closure symmetry

Proposition 5.27 ($\mathrm{Cent}(G_2) = \mathfrak{so}(6, \mathbb{C})$ and the copy algebra)**.** *(i) The centraliser of $G_n = G \oplus \cdots \oplus G$ in $\mathfrak{so}(3n, 3n)$ has dimension $3n(3n-1)$: 6, 30 and 72 at $n = 1, 2, 3$. It is $\mathfrak{so}(3n, \mathbb{C})$, with compact part $\mathfrak{so}(3n)$ of dimension 3, 15 and 36. At $n = 2$ every element is $\left(\begin{smallmatrix} R & S \\ -S & R \end{smallmatrix}\right)$ in the basis of the six temporal directions followed by their partners, with R, S antisymmetric.*

(ii) $\mathfrak{so}(6, \mathbb{C}) = \mathfrak{l}_G^{(1)} \oplus \mathfrak{l}_G^{(2)} \oplus (\mathbb{C}^3 \otimes \mathbb{C}^3)$, of dimensions $6 + 6 + 18$. Under the diagonal rotations the cross part is $\mathbf{1} \oplus \mathbf{3} \oplus \mathbf{5}$ (complex), with Casimir 0, 2, 6 on 2, 6, 10 real dimensions.

(iii) The commutant of the diagonal $\mathfrak{l}_G$ in $\mathfrak{so}(6,6)$ is the singlet of (ii), the copy algebra $\mathfrak{so}(2, \mathbb{C}) = \mathrm{span}\{C_v, \varepsilon \otimes G\}$. Here $C_v^2 = -\mathbf{1}$ is compact, while $(\varepsilon \otimes G)^2 = +\mathbf{1}$ is a copy boost, acting on (z_1, z_2) by $\left(\begin{smallmatrix} 0 & -i \\ i & 0 \end{smallmatrix}\right)$; it preserves q_2 and not $E_6 \oplus E_6$.

(iv) $\mathrm{Cent}(G_2)$ meets $\mathfrak{l}_{\mathrm{int}}^{(1)}$ in 0, and the elements of $\mathfrak{so}(6,6)$ commuting with both G_2 and $\mathfrak{l}_{\mathrm{int}}^{(1)}$ form $\mathfrak{l}_G^{(2)} \cong \mathfrak{so}(3, \mathbb{C})$.

Part (i) is the $n = 2$ member of Theorem 4.34, with the same proof, and the dimension 72 at $n = 3$ re-verifies that theorem. Part (iii) restates row three of Proposition 5.72 and Theorem 5.15 with $k_B = 2$ (fixed there). Parts (ii) and (iv) are fixed, and (iv) is the $n = 2$ analogue of Theorem 4.35 (d5-4a to d5-4c).

Proposition 5.28 (Duplicated momentum fields in the closure)**.** *Let one rotor act on two constituents with the same inertia $N = (\nu_1 \eta^{-1} + \nu_2 G \eta^{-1})/m$, whose momentum fields are a duplication of one momentum field: $\pi^{(1)} = \alpha\pi$ and $\pi^{(2)} = \beta\pi$, with α, β acting through $\mathrm{span}\{\mathbf{1}, G\}$.*

action	(k_I, k_B, k_S)	algebra	dim	compact	content of $\mathbb{R}^{6,6}$	spin on S_2
spectator	$(1,0,1)$	$\mathfrak{so}(5,3)$	28	13	$V \oplus 8 \cdot \mathbf{1}$	half-integer
binding + spectator	$(0,1,1)$	$\mathfrak{so}(3,3)$	15	6	$\Lambda^2 V \oplus 6 \cdot \mathbf{1}$	integer
diagonal interior	$(2,0,0)$	$\mathfrak{so}(2) \oplus \mathfrak{so}(4)$	7	7	$2V \oplus 4 \cdot \mathbf{1}$	integer
mixed	$(1,1,0)$	$\mathfrak{so}(2)$	1	1	$V \oplus \Lambda^2 V \oplus 2 \cdot \mathbf{1}$	half-integer
binding on both	$(0,2,0)$	$\mathfrak{so}(2,\mathbb{C})$	2	1	$2\,\Lambda^2 V$	integer

Table 5.1: The Lorentz actions at $n = 2$ and the grade-two elements commuting with them (Theorem 5.15), with their compact parts, the content of the vector space (Lorentz Casimir 3 on a four-vector V, 4 on a field strength $\Lambda^2 V$), and the effect of a 2π rotation on S_2. Every entry was computed in the realisation (d5-5a, d5-5b, d5-6b).

Then $w_1 + w_2 = (\alpha^2 + \beta^2)\, w(\pi)$ with $w = q(z(\pi))$ (equation (4.15)). The effective strength of Proposition 4.33 is therefore $R_{\text{eff}} = |\nu_1 + i\nu_2|\, |q_2(\pi)|/m$: the closure sees the modulus of the holomorphic invariant of the duplicated momentum field. For the equal split with a relative quarter turn of duality, $\pi^{(2)} = G\pi^{(1)}$, equivalently $p^{(2)} = -Gp^{(1)}$, one has $w_2 = -w_1$ and $R_{\text{eff}} = 0$. The rate is then exactly constant, and nonzero when $h(0) \neq 0$, while each copy keeps $E_0(\pi^{(k)}) = \frac{1}{2} E_0(\pi) \neq 0$. On one constituent with the same inertia and $(\nu_1, \nu_2) \neq (0,0)$ a constant nonzero rate requires $E_0(\pi_0) = 0$ (Corollary 4.29); for $(\nu_1, \nu_2) = (0,0)$ the rate is constant for all data.

This restates Proposition 4.33 and Corollary 4.29 (fixed there). The expression of R_{eff} through q_2 for duplicates is a one-line consequence (fixed; d5-9f). So, for $(\nu_1, \nu_2) \neq (0,0)$, a duplicate can do what one copy cannot: run at an exactly constant rate while its copies carry nonzero rest energy in the momentum field, in the sense that $E_0(\pi^{(k)}) \neq 0$. Reading $E_0(\pi^{(k)})$ as a rest energy is the interpretive dictionary stated after Remark 7.6. The identification $mc^2 = \hbar\kappa$ concerns the common rate $\kappa = h/I_0$ of both copies, which does not depend on $E_0(\pi^{(k)})$, so it does not make $E_0(\pi^{(k)})$ a mass (interpretive).

5.3.5 The five Lorentz actions at the second rung

What the duplication does depends on how the Lorentz algebra acts on the two copies. Theorem 5.15 lists the choices, and at $n = 2$ there are five (Table 5.1).

Read row by row: under the spectator action copy 2 is a passenger whose six directions join a non-compact internal algebra. With $\mathfrak{l}_G$ on copy 1, copy 2's whole $\mathfrak{so}(3,3)$ becomes internal symmetry and no spin one-half survives. The diagonal interior action gives a compact algebra containing the copy rotation M_{12}, but only integer spin. The mixed action keeps only $J_1 = e_2^{(1)} e_3^{(1)}$. With $\mathfrak{l}_G$ on both copies the frames are two field strengths mixed by $\mathfrak{so}(2,\mathbb{C})$. The table restates Theorem 5.15 and Proposition 5.16, Proposition 5.74 and Proposition 5.75 (fixed there), re-verified. For the mixed rung we also re-verified the Casimir identity $(C_{\text{mix}} - \frac{3}{2})(C_{\text{mix}} - \frac{11}{2}) = 0$ with ranks 16 and 48 (d5-5b). We checked the grade-three element Θ_F of Section 5.8, which identifies the frame field of copy 2 with dA of copy 1. It and Θ_F followed by G are invariant and span the invariants in $\Lambda^2(\text{four-plane}_1) \otimes V^{(2)}$, of dimension two as stated there. Here $\text{four-plane}_1 = \langle e_1^{(1)}, e_4^{(1)}, e_5^{(1)}, e_6^{(1)} \rangle$ is the interior four-plane of copy 1 and $V^{(2)}$ is the span of the $e_a^{(2)}$, the space of copy 2 alone, which Section 5.8 writes V_2. The full space of invariant grade-three elements has dimension four: the other two lie in $\Lambda^3 V^{(2)}$, among the trivectors of copy 2 alone, the two $\mathfrak{l}_G$-invariant trivectors of $(\Lambda^3 \mathbb{R}^6)_{\mathbb{C}} = 2\,(0,0) \oplus 2\,(1,1)$ (Section 3.2.3; d5-5c). This is a precision, not a correction. The mixed rung is the only action at $n = 2$ that houses both readings, at the price of the whole internal algebra.

5.3.6 Spin and statistics of the duplicate

Proposition 5.29 (Spin of the duplicate)**.** *(i) The copy rotations M_{12} (four commuting terms of square $-\frac{1}{4}$) and C_b (six) have eigenvalues $i\{-2,\dots,2\}$ and $i\{-3,\dots,3\}$, and $\exp(2\pi M_{12}) = \exp(2\pi C_b) = \mathbf{1}$. A surplus rotation $\frac{1}{2}e_se_t$ has $\exp(\pi e_se_t) = -\mathbf{1}$.*

(ii) A rotation through 2π is $+\mathbf{1}$ on S_2 under the diagonal interior action and $-\mathbf{1}$ under the spectator action. Under the mixed action it is $-\mathbf{1}$, and under $\mathfrak{l}_G$ on one or both copies it is $+\mathbf{1}$. The spin Casimir is 0 on 16 and 2 on 48 dimensions for the diagonal action ($j = 0, 1$), and $\frac{3}{4}$ on all of S_2 for the spectator action.

(iii) Under the diagonal interior action the Lorentz Casimir is $0, 3, 4$ on $8, 32, 24$ dimensions and $C_{\mathrm{cop}} = -M_{12}^2$ is $0, 1, 4$ on $24, 32, 8$, with $C_{\mathrm{cop}} + C_\Delta = 4$. The exchange τ_K commutes with the diagonal Lorentz action and anticommutes with M_{12}, and its ±1 eigenspaces carry identical Lorentz content, $0, 3, 4$ on $4, 16, 12$ dimensions each.

This restates Proposition 5.16, Theorem 5.11 at $n = 2$ ($\frac{1}{2}n(n+2) = 4$) and the last paragraph of Section 5.8 (fixed there), with the 2π eigenvalues computed (d5-6a, d5-6b). That two spin-$\frac{1}{2}$ constituents make an integer-spin composite is inherited [222]: under the diagonal action the duplicate of a fermion is a boson, and fermions need an odd number of interior constituents. The Lorentz isomorphism of the two exchange sectors in (iii) is fixed (d5-6c). It holds because exchange reverses the copy charge, pairing charge $\pm m$, and splits the neutral sector evenly; there is no "singlet antisymmetric, triplet symmetric" rule once the surplus directions and the Koszul signs are included.

5.3.7 What the second rung adds: the vacuum and the imaginary unit

Proposition 5.30 (The photon vacuum and the imaginary unit, re-verified)**.** *(i) At $n = 1$ the commutant of $\mathfrak{l}_{\mathrm{int}}$ and $Q = \frac{1}{2}e_2e_3$ in $M_8(\mathbb{R})$ has dimension four, and its symmetric part is spanned by $\mathbf{1}$ and $\omega = e_1\cdots e_6$. The rank-four projectors in it are $\frac{1}{2}(\mathbf{1} \pm \omega)$, and every bivector commutes with ω, so they are rigid (Proposition 8.55).*

(ii) (Theorem 8.57 and Theorem 8.58) With $a, b = e_2^{(1)}, e_3^{(1)}$ and $c, d, e = e_1^{(2)}, e_2^{(2)}, e_3^{(2)}$: $H = abcd$ has $H^2 = \mathbf{1}$, $H^\top = H$ and $\operatorname{Tr} H = 0$, and $\Pi_0 = \frac{1}{2}(\mathbf{1}+H)$ commutes with $Q = \frac{1}{2}ab$, with the four interior gammas and with the internal volume. Moreover $\pi = [\frac{1}{2}be, \Pi_0] = -\frac{1}{2}acde$ is symmetric with $\operatorname{Tr}\pi^2 = 16$ and $[Q, \pi] = \frac{1}{2}bcde$. The $Z_i = I_5v_i$ are symmetric of grade four with $\{Z_i, Z_j\} = 2\delta_{ij}$. $\Pi(n)$ is a rank-32 projector commuting with the gammas (checked at $n = (2, 3, 6, 0, 0)/7$). The tangents $T_i = \frac{1}{2}I_5v_i$ have $\operatorname{Tr} T_iT_j = 16\delta_{ij}$, $[Q, T_a] = -T_b$, $[Q, T_b] = T_a$ and $[Q, T_c] = [Q, T_d] = 0$.

(iii) The grade-four blades in the five positive internal generators that commute with Q are exactly abcd, abce and abde. Every such vacuum straddles the two copies.

(iv) (Theorem 2.17 at $n = 2$) The product of the eight internal generators is $J = (e_2e_3) \otimes \omega$, with $J^2 = -\mathbf{1}$. The joint commutant of $\mathfrak{l}_{\mathrm{int}}^{(1)}$ and the 28 internal bivectors in $M_{64}(\mathbb{R})$ is spanned by the blades $\mathbf{1}, \Omega, J, \Omega J$. The solutions of $x^2 = -\mathbf{1}$ in it are exactly $\pm\Omega$ and $\pm J$, and only $\pm J$ commute with the gammas.

Parts (i), (ii) and (iv) restate earlier results (fixed there), re-verified in the realisation of $\mathrm{Cl}(6, 6)$ (d5-7a to d5-7c, d5-8). Part (iii) is fixed but elementary (d5-7d). The vacuum of Theorem 8.57 is built on four positive internal directions, two of them the surplus pair a, b of copy 1. A grade-four blade in the five positive internal generators commutes with Q exactly when it contains both a and b, so it straddles the copies. Part (iii) covers only these positive blades, not blades that also use the negative internal directions of copy 2. At one copy the first rung is rigid by the commutant argument of Proposition 8.55, without reference to blades, so

the vacuum exists because there are two copies. Part (iv) adds one identity: the imaginary unit of the duplicate is the imaginary unit of copy 1 times the volume element of copy 2, which is that copy's fermion parity up to sign (Proposition 5.21). It is neither the embedded one-rung charge $J_1 \otimes \mathbf{1}$ nor $J^{(1)} + J^{(2)}$.

5.3.8 The copy rotation as a b-field between two manifolds

On one copy a b-field commuting with G is removable along a worldline (Proposition 9.5), and a purely anticommuting one tears the binding (Theorem 9.7). An admissible one with both parts keeps every orbit bounded when the two triples rotate at equal rates, outside one exceptional ratio, and it is then not removable (Theorem 9.9 and Proposition 9.8). On two copies the binding term is $K \otimes G$ with $K = \operatorname{diag}(\kappa_1, \kappa_2)$. That is a multiple of G_2 only when the rates agree, and the following theorem measures what changes. The interior Lagrangian of each copy is that of Section 9.1, with $S = K \otimes G + b$; the frame field obeys $\dot{X} = -SX + \pi/m$ and the momentum field $\dot{\pi} = S^{\dagger}\pi$.

Theorem 5.31 (The inter-copy b-field)**.** *Let $\kappa_1, \kappa_2 \geq 0$, $\beta \in \mathbb{R}$, $b = \beta\, C_v$, and $M = \left(\begin{smallmatrix} -S & \mathbf{1}/m \\ 0 & S^{\dagger} \end{smallmatrix}\right)$ on (X, π).*

(i) Up to scale, b is the only element of $\mathfrak{so}(6,6) \cap \mathfrak{so}(12)$ commuting with G_2 and the diagonal rotations. It is compact, and $S^{\dagger} = K \otimes G - b$.

(ii) $\det(\lambda - S) = \big(\lambda^4 + (\kappa_1^2 + \kappa_2^2 + 2\beta^2)\lambda^2 + (\kappa_1\kappa_2 - \beta^2)^2\big)^3$, so $\operatorname{spec} S = \{\pm i\omega_{\pm}\}$ with $\omega_{\pm} = \frac{1}{2}(\kappa_1 + \kappa_2) \pm \sqrt{\frac{1}{4}(\kappa_1 - \kappa_2)^2 + \beta^2}$; equivalently, $\omega_{\pm}$ are the eigenvalues of $\left(\begin{smallmatrix} \kappa_1 & i\beta \\ -i\beta & \kappa_2 \end{smallmatrix}\right)$.

(iii) $[K \otimes G, b] = \beta(\kappa_2 - \kappa_1)\, \sigma_x \otimes G$, so for $\beta \neq 0$, b is removable along the worldline (Larmor [83]) iff $\kappa_1 = \kappa_2$. The centraliser of $K \otimes G$ in $\mathfrak{so}(6,6)$ has dimension 30 if $\kappa_1 = \kappa_2 \neq 0$, 12 if $\kappa_1 \neq \kappa_2$ are both nonzero, and 21 if exactly one vanishes.

(iv) $(M^2 + \omega_+^2)(M^2 + \omega_-^2) = 0$ identically in $\kappa_1, \kappa_2, \beta, m$.

(v) Every orbit is bounded iff either $\kappa_1 = \kappa_2 > 0$, or $\kappa_1 \neq \kappa_2$ and $\kappa_1\kappa_2 \neq \beta^2$. In particular, for $\kappa_1 = 0 < \kappa_2$ the orbits are bounded iff $\beta \neq 0$: the inter-copy field confines a frame with no rate of its own. For $\kappa_1 = \kappa_2 = 0$ they are unbounded for every β.

(vi) For $\kappa_1 = 0 < \kappa_2$, $\omega_+\omega_- = -\beta^2$ and $\omega_+ + \omega_- = \kappa_2$. For $\beta \ll \kappa_2$ the slow line is $|\omega_-| = \beta^2/\kappa_2 - \beta^4/\kappa_2^3 + \cdots$, the seesaw. For $\beta \gg \kappa_2$ both lines approach $\pm\beta$, and the orbit size grows like $2/(m\kappa_2)$.

Proof. (i) is Proposition 5.23(i), and C_v is η_2-antisymmetric while $K \otimes G$ is self-adjoint. (ii) On the sector $G_2 = i$, $S = (iK + \beta\varepsilon) \otimes \mathbf{1}_3$, whose characteristic polynomial is $\lambda^2 - i(\kappa_1 + \kappa_2)\lambda - \kappa_1\kappa_2 + \beta^2$; the sector $G_2 = -i$ is its conjugate. (iii) $K\varepsilon - \varepsilon K = (\kappa_2 - \kappa_1)\sigma_x$. If $\kappa_1 = \kappa_2 = \kappa$, then $X = e^{-b\tau}Y$ and $\pi = e^{-b\tau}\varpi$ turn the system into $b = 0$ with rate κ; if $\beta \neq 0$ and $\kappa_1 \neq \kappa_2$, this substitution makes the binding $e^{b\tau}(K \otimes G)e^{-b\tau}$ time-dependent. The centraliser dimensions are computed. (iv) is a symbolic identity (d5-9c). (v) If $\kappa_1 \neq \kappa_2$ and $\kappa_1\kappa_2 \neq \beta^2$, then $\omega_+ \neq \omega_-$, $\omega_- \neq 0$ and $\omega_+ + \omega_- = \kappa_1 + \kappa_2 > 0$, so the polynomial of (iv) has four simple imaginary roots and M is semisimple with imaginary spectrum. If $\kappa_1 = \kappa_2 = \kappa > 0$, use (iii) and the boundedness of the $b = 0$ system (Theorem 4.8); if $\kappa = 0$ the substituted system is free drift. If $\kappa_1 \neq \kappa_2$ and $\kappa_1\kappa_2 = \beta^2$, then $\omega_- = 0$, and M is semisimple at 0 iff $\ker S^{\dagger} \subset \operatorname{im} S$. Because $S^{\top} = -S$ and $S^{\dagger} = -\eta_2 S \eta_2$, this says that $\ker S$ is η_2-totally isotropic. Complexifying, $\ker S$ is spanned by the real parts of $v \otimes c$ in the $G_2 = i$ sector, with $c \in \mathbb{C}^3$ in the basis $e_j - ie_{j+3}$ and $v = (\beta, i\kappa_1)$ (or $v = (1, 0)$ when $\beta = \kappa_1 = 0$), and $\eta_2(\operatorname{Re}(v \otimes c), \operatorname{Re}(v \otimes c')) = \operatorname{Re}\big[(v \cdot v)(c \cdot c')\big]$. This vanishes for all c, c' iff $v \cdot v = \beta^2 - \kappa_1^2 = 0$, which together with $\kappa_1\kappa_2 = \beta^2$ means $\kappa_1 = \kappa_2$ or $\beta = \kappa_1 = 0$, and in the second case $v = (1, 0)$ is not isotropic; so there is a Jordan block. (vi) Vieta and expansion. The orbit sizes are numerical (d5-9e, d5-9g). □

Parts (ii) and the seesaw in (vi) are inherited: two-level mixing with detuning $\frac{1}{2}(\kappa_1 - \kappa_2)$ and coupling β, the generalised Rabi splitting, and the seesaw formula [96]. Part (i) restates Proposition 5.72 (fixed there). Parts (iii)–(v) are fixed (d5-9a to d5-9d). The exact checks are: bounded at $(\kappa_1, \kappa_2, \beta) = (1, 3, \frac{2}{5}), (0, 5, \frac{1}{2}), (2, 2, \frac{1}{3}), (1, 1, 1), (2, 2, 0)$ and $(0, \frac{1}{100}, 1)$; unbounded at $(0, 0, \frac{1}{2}), (1, 4, 2), (1, 9, 3), (0, 3, 0)$. At $(0, 5, \frac{1}{2})$, $\omega_- = -0.04951$ against $\beta^2/\kappa_2 = 0.05$, and $\sup_{t \le 600} \|e^{Mt}\| = 19.86$ against $1/|\omega_-| = 20.2$. At $(0, \frac{1}{100}, 1)$ the lines are 1.00501 and -0.99501, and the supremum over a beat period is $200.0 = 2/(m\kappa_2)$. So the seesaw orbit is bounded, with amplitude of order $1/(m|\omega_-|)$ for $\beta \ll \kappa_2$ and $2/(m\kappa_2)$ for $\beta \gg \kappa_2$.

Remark 5.32 (The author's first remark at two manifolds)**.** On one manifold a b commuting with G is removable along a worldline, and a constant connection alone gives free drift (Propositions 9.8 and 9.11). An admissible b that does not commute with G can already there be non-removable, change the spectrum and keep every orbit bounded (Theorem 9.9), so these two features are not new at $n = 2$. What two manifolds with different rates add is this. The compact inter-copy field commutes with G_2, so on one manifold a field of that kind would be removable, but it does not commute with the binding term $K \otimes G$. It is therefore not removable, it shifts the lines, and it confines a frame with $\kappa_1 = 0$ by coupling it to a frame with $\kappa_2 > 0$. The no-go of Proposition 9.11 still holds at $n = 2$: for $\kappa_1 = \kappa_2 = 0$ the motion is free drift for every β. In that sense, and only in it, the b-field is a "deal breaker" at $n \ge 2$ (fixed). Two qualifications remain. The field can set the lines, which approach $\pm\beta$ when $\beta \gg \kappa_2$, but not the rates, and the size of the orbit is set by κ_2 and diverges as $\kappa_2 \to 0$. And reading a line or a rate as a rest energy still needs $mc^2 = \hbar\kappa$ (an added identification).

5.3.9 Assessment

What is forced, what is chosen, what is added. *Forced* (inherited): the algebra step, $\mathrm{Cl}(V \oplus V) = \mathrm{Cl}(V)\hat{\otimes}\mathrm{Cl}(V)$, with the Klein factor determined up to conjugation; $S_2 = S \otimes S$ as the Fock space of six modes; and the exchange as the Koszul swap. *Chosen* (constructed): the Lorentz action, one of five; the energy duplication, a point of the three-sphere, or of the circle if η is to be respected, with the angle ϕ free; and the vacuum point on S^4. *Added* at $n = 2$ (fixed; the copy algebra, the copy charge, the constant-rate closure, the vacuum and the mixed rung restate earlier chapters, while the complex-structure statement of Proposition 5.26, the comparison $\mathfrak{so}(6) \supset \mathfrak{u}(3)'$, the form of R_{eff} through q_2 and the inter-copy b-field are fixed here): the copy algebra $\mathfrak{so}(2, \mathbb{C})$ on two binding frames (dimension 0 at $n = 1$); an η-isometric complex structure and a compact, Lorentz-invariant copy charge (none at $n = 1$); a Lorentz-compatible energy symmetry $\mathfrak{so}(6) \supset \mathfrak{u}(3)'$, against $\mathfrak{so}(3)$; a closure with constant rate and non-null copies, $E_0(\pi^{(k)}) \neq 0$; a non-rigid, Lorentz-invariant vacuum; the mixed rung, which houses both readings; and an inter-copy b-field that confines (d5-10 collects the dimensions). *Lost*: spin one-half under the diagonal interior action, and the uniqueness of the photon direction (Corollary 5.6).

Processes. Table 5.2 lists the physical processes that take one copy to two, and what each preserves.

The precise sense of "a unitary energy results in a duplication". Three statements, in increasing order of what the arena adds. First, the undivided energy $(\mathbb{C}^3, E_6, U(3))$ embeds isometrically in $(\mathbb{C}^3 \otimes \mathbb{C}^2, E_6 \oplus E_6, U(6))$. The Lorentz-covariant embeddings form the three-sphere of a beam splitter, and the diagonal $U(3)$ and the copy $U(2)$ are a dual pair (inherited). Second, compatibility with η cuts the copy $U(2)$ to $O(2)$ and $U(6)$ to $O(6)$, since $U(n) \cap O(n, \mathbb{C}) = O(n)$. The copy rotation $SO(2)$, the identity component of $O(2)$, is the one connected family that preserves the energy, the metric, the binding root and the diagonal Lorentz group at once; the other component contains the swap of the copies and $\mathrm{diag}(1, -1)$. Neither exists on one

process	mathematics	preserves	class
splitting one field into two (beam splitter)	$\iota_{\alpha,\beta}$, $S^3 = U(2)/U(1)$; circle if also η-isometric	E_6, T, $\sum E_0^{(k)}$; q_2 only on the circle	inherited; reading fixed
second quantisation	$\mathrm{Cl}(V \oplus V) = \mathrm{Cl}(V)\hat{\otimes}\mathrm{Cl}(V)$; Fock space; Jordan–Wigner strings	Clifford relations	inherited (forced)
two neutral copies into one charged field	C_v as i, $w = X^{(1)} + iX^{(2)}$	η_2, $E_6 \oplus E_6$	inherited; exists only at $n \geq 2$: fixed
two-body composite	diagonal interior action; exchange τ_K	integer spin	inherited; restated
pair creation from null energy	linear covariant duplication keeps null copies; two non-collinear null fields have $E_{\mathrm{pair}} > 0$	T	inherited; pair-creation reading interpretive
cloning, $z \mapsto z \otimes z$	quadratic, not isometric and linear	—	obstructed (inherited)
closure balance	$\pi^{(2)} = G\pi^{(1)}$, $R_{\mathrm{eff}} \propto \lvert q_2(\pi)\rvert = 0$	constant rate, $E_0^{(k)} \neq 0$; bounded frames if $h(0) \neq 0$	restated (Proposition 4.33)
inter-copy b-field	$S = K \otimes G + \beta C_v$	bounded iff Theorem 5.31(v)	fixed; mixing inherited

Table 5.2: Processes that take one copy of the arena to two.

copy (fixed). Third, the duplicate's rest energy splits into a stress-energy part, preserved by every duplication, and a holomorphic part. The complex invariant q_2 is preserved only on the circle of the copy rotation. Its modulus $|q_2|$, which is what the closure sees, is preserved on that circle composed with the overall duality rotation G_2, which multiplies q_2 by a phase (fixed). Read literally as a multiplication of the energy, the phrase fails either way: the classical map $z \mapsto (z, z)$ is linear and doubles the quadratic energy, $(E_6 \oplus E_6)(z, z) = 2E_6(z)$ (Remark 5.25), and its normalised version $z \mapsto (z, z)/\sqrt{2}$ preserves the total energy and divides it equally between the copies. Copying classical field data is well defined; the no-cloning theorem, which concerns the universal copying of an unknown quantum state by a physical quantum operation, is a separate statement (inherited).

5.4 One particle or two: the second rung

The conventions of Section I.1.4 say that a constituent is a tensor factor $\mathrm{Cl}(3, 3)$ of the tower, that the word never refers to a particle, and that three constituents do not describe three particles (Section I.1.4). The duplication $\mathrm{Cl}(3, 3) \to \mathrm{Cl}(6, 6)$ of Section 5.3 is therefore not yet the emergence of two particles from one. This section asks in what precise sense, and under which added structures, the second rung $\mathrm{Cl}(6, 6)$ carries two particles rather than one particle with a larger internal space. Four tests separate the two readings: whether the copies have separate mass shells, whether there is one complex structure under which both copies are complex Dirac spinors, whether some operator counts particles, and whether exchange of the copies sorts states by statistics. The answer has three parts. In the graded tower the two mass terms anticommute, they add in quadrature, and no state has a sharp rest energy in each copy (fixed); read as particle content, the graded rung is one particle (interpretive). A two-particle reading exists, and it is fixed as an algebra: the commutant of the first copy is a second, commuting $\mathrm{Cl}(3, 3)$, whose generators are the graded ones dressed by the chirality of the first copy (fixed). But it needs three further structures, and none of them is selected. They are a common complex structure, which exists only on a 32-dimensional subspace, an additive Hamiltonian that is not a vector of $\mathrm{Cl}(6, 6)$, and an ungraded exchange that is not the Clifford exchange (constructed; the selection of a complex structure is obstructed under the diagonal action). The Fock number of the tower counts quanta, not particles. No gauge-admissible field that commutes with the binding root

statement	class	checks
Proposition 5.20: graded tensor product, $\mathrm{Cl}(6,6) = M_{64}(\mathbb{R})$, Klein factor forced	inherited (restated: (1.2))	d5-1a to d5-1d
Proposition 5.21: Fock space, Jordan–Wigner strings, pole operators $= \mathbf{1} - 2N_i$	inherited (reading of Proposition 1.5)	d5-1e
Proposition 5.22: halves, $\pi/4$ copy rotation, $\tau_K = g_1 \cdots g_6$, what each half carries	inherited; identities fixed	d5-1f, d5-2a to d5-2c
Proposition 5.23: $\mathfrak{u}(2)$, $\mathfrak{so}(2,\mathbb{C})$, $\mathfrak{so}(2)$; dual pair	fixed (restated: Proposition 5.72); inherited	d5-3a, d5-3b
Theorem 5.24: duplication maps, S^3 and circle, q_2 versus T	inherited; reading fixed	d5-3c to d5-3e
Remark 5.25: two rest energies; Gram invariants; pair invariant; no cloning	fixed; inherited; analogies interpretive	d5-3f, d5-3i
Proposition 5.26: isometric complex structure only at $n \geq 2$; $\mathfrak{so}(6) \supset \mathfrak{u}(3)'$	inherited (parity); fixed	d5-3g, d5-3h
Proposition 5.27: $\mathrm{Cent}(G_n) = \mathfrak{so}(3n,\mathbb{C})$; $\mathbf{1} \oplus \mathbf{3} \oplus \mathbf{5}$; copy algebra; analogue of Theorem 4.35	fixed (restated: Theorem 4.34, Proposition 5.72)	d5-4a to d5-4c
Proposition 5.28: closure sees $\lvert q_2(\pi)\rvert$; balance with non-null copies	fixed (restated: Proposition 4.33, Corollary 4.29)	d5-9f
Table 5.1: five Lorentz actions; mixed rung; Θ_F (dimension 2 within $\Lambda^2(\text{four-plane}_1) \otimes V^{(2)}$, 4 in all)	fixed (restated: Theorem 5.15, Propositions 5.74 to 5.75)	d5-5a to d5-5c
Proposition 5.29: spin and statistics, Casimirs, exchange sectors	fixed (restated: Proposition 5.16, Theorem 5.11); inherited	d5-6a to d5-6c
Proposition 5.30: rigid first rung, S^4 vacuum, straddling, $J = J_1 \otimes \omega$	fixed (restated: Proposition 8.55, Theorem 8.57, Theorem 8.58, Theorem 2.17)	d5-7a to d5-7d, d5-8
Theorem 5.31: inter-copy b-field: uniqueness, spectrum, removability, boundedness criterion, seesaw	fixed; mixing and seesaw inherited	d5-9a to d5-9e, d5-9g
Remark 5.32: the deal-breaker reading	fixed; obstructed for $\kappa_1 = \kappa_2 = 0$; $mc^2 = \hbar\kappa$ an identification	—
Section 5.3.9: comparison $n = 1$ against $n = 2$	fixed; mostly restated (Proposition 5.72)	d5-10

Table 5.3: Claims of Section 5.3, their classes, and the checks printed by `d5_duplication_math.py`. "Restated": established in an earlier chapter, restated and re-verified here.

changes it (fixed). It is not invariant under the interior Lorentz algebra, under the boosts of $\mathfrak{l}_G$, or under the charge J (fixed), so as a particle number it is an internal reading (interpretive). Identifiers e1-n refer to the `PASS` lines of `e1_two_from_one.py` (Table 5.6).

Notation. $S = \mathbb{R}^8$ with the realisation of equation (1.1), $\omega = e_1 \cdots e_6$, $J = e_2e_3$, $T = e_1e_2e_3$, $\Omega = e_1e_4e_5e_6$, and $\gamma^\mu = (e_1, e_4, e_5, e_6)$. Appendix A.1 records $\omega = J\Omega = \Gamma$, the chirality. On $S_2 = S \otimes S = \mathbb{R}^{64}$ the graded generators are $E_{(1,a)} = e_a \otimes \mathbf{1}$ and $E_{(2,a)} = \omega \otimes e_a$ (equation (1.2)), and the *ungraded* copy-2 generators are $u_a = \mathbf{1} \otimes e_a$. We write $\Gamma^{(1)} = \omega \otimes \mathbf{1}$ and $\Gamma^{(2)} = \mathbf{1} \otimes \omega$ for the volume elements of the two constituents, $J^{(1)} = J \otimes \mathbf{1}$, $J^{(2)} = \mathbf{1} \otimes J$, $T^{(1)} = T \otimes \mathbf{1}$, $T^{(2)} = \mathbf{1} \otimes T$, and $\omega_S = E_{(1,2)}E_{(1,3)}E_{(2,2)}E_{(2,3)}$ for the volume element of the four surplus directions. On vectors, K_n is the para-complex structure $Ke_i = e_{i+3}$, $Ke_{i+3} = e_i$ on each constituent; it is $-\mathcal{K}$ of Appendix A.1. G_n is the binding root, and $\mathfrak{l}_G$ and $\mathfrak{l}_{\mathrm{int}}$ are the field-strength and interior Lorentz algebras of one constituent. The Fock modes are $a^{\pm}_{k,i} = \frac{1}{2}(E_{(k,i)} \pm E_{(k,i+3)})$, with $N = \sum_{k,i} a^+_{k,i}a^-_{k,i}$ and $N^{(k)}$ the part of constituent k (Proposition 5.21). The rest masses are $m_1, m_2 \geq 0$ and $M = (m_1^2 + m_2^2)^{1/2}$. The diagonal interior Lorentz generators are $\Delta_{\mu\nu} = \frac{1}{2}(E_{(1,\mu)}E_{(1,\nu)} + E_{(2,\mu)}E_{(2,\nu)})$, and $M_{12} = \frac{1}{2}\sum_\mu \eta_{\mu\mu}E_{(1,\mu)}E_{(2,\mu)}$ is the interior copy rotation. The copy rotation is $C_b = \frac{1}{2}\sum_a \eta_{aa}E_{(1,a)}E_{(2,a)}$, with the sign of Proposition 5.22(ii); this sign is used in every section of this chapter. $k_I \in \{1, 2\}$ is the number of constituents on which $\mathfrak{l}_{\mathrm{int}}$ acts (the spectator and the diagonal action; Theorem 5.15).

5.4.1 Two copies of $\mathrm{Cl}(3,3)$ inside $\mathrm{Cl}(6,6)$

Proposition 5.33 (The commuting copy)**.** *(i) The twelve $E_{(k,a)}$ satisfy the Clifford relations of $\mathbb{R}^{6,6}$. The six u_a satisfy those of $\mathbb{R}^{3,3}$ and commute with every $E_{(1,b)}$. Moreover $u_a = \Gamma^{(1)}E_{(2,a)} = E_{(1,1)}\cdots E_{(1,6)}E_{(2,a)}$, a blade of grade 7.*

(ii) The commutant of $\mathrm{Cl}(3,3)\otimes\mathbf{1}$ in $M_{64}(\mathbb{R})$ is spanned by 64 blades of $\mathrm{Cl}(6,6)$, each equal to $\pm\mathbf{1}\otimes e_J$; it is $\mathbf{1}\otimes M_8(\mathbb{R})$, the algebra generated by the u_a. A second copy of $\mathrm{Cl}(3,3)$ commuting with the first is therefore unique up to conjugation by $\mathbf{1}\otimes GL(8,\mathbb{R})$.

(iii) The dressing is the chirality and the Fock parity of the first copy: $\Gamma^{(1)} = (J\Omega)\otimes\mathbf{1} = -(-1)^{N^{(1)}}$.

(iv) Both the ungraded product $\mathrm{Cl}(3,3)\otimes\mathrm{Cl}(3,3)$ and the graded product $\mathrm{Cl}(3,3)\hat{\otimes}\mathrm{Cl}(3,3) = \mathrm{Cl}(6,6)$ are $M_{64}(\mathbb{R})$, with the same 4096 blades as basis. The twelve elements $e_a\otimes\mathbf{1}$, u_a span no quadratic space: $(e_1\otimes\mathbf{1}+u_1)^2 = 2\,\mathbf{1}+2\,e_1\otimes e_1$.

(v) At rung n the ungraded generator of constituent k, $\mathbf{1}^{\otimes(k-1)}\otimes e_a\otimes\mathbf{1}^{\otimes(n-k)}$, equals $\Gamma^{(1)}\cdots\Gamma^{(k-1)}E_{(k,a)}$, a blade of grade $6(k-1)+1$. Ungraded generators of distinct constituents commute and graded ones anticommute (checked at $n=3,4$).

Proof. (i) $(\omega\otimes\mathbf{1})(\omega\otimes e_a) = \omega^2\otimes e_a = \mathbf{1}\otimes e_a$, and $\Gamma^{(1)} = E_{(1,1)}\cdots E_{(1,6)}$. (ii) Conjugation by an invertible blade multiplies every blade by ±1 (equation (1.3)), so the commutant of a set of blades is spanned by the blades it contains. A blade commutes with all $E_{(1,b)}$ exactly when it is $\Gamma^{(1)\varepsilon}$ times a blade of copy 2 with grade parity ε, that is $\pm\mathbf{1}\otimes e_J$ (Proposition 5.20(iii)). The uniqueness is the Skolem–Noether theorem in $\mathbf{1}\otimes M_8(\mathbb{R})\cong M_8(\mathbb{R})$. (iii) $\omega = J\Omega$ by moving e_1 past e_2e_3, and $(-1)^{N_1+N_2+N_3} = -\omega$ is Proposition 5.21. (iv) Distinct blades B, B' have $\mathrm{Tr}(B^\top B') = 0$ because $B^\top B'$ is $\pm$ a blade other than $\mathbf{1}$. The square is a one-line expansion. (v) $\Gamma^{(j)} = \mathbf{1}^{\otimes(j-1)}\otimes\omega\otimes\mathbf{1}^{\otimes(n-j)}$, and the ω's cancel in pairs. □

Parts (i)–(iii) restate Proposition 5.20 and Proposition 5.21 (inherited there) and Appendix A.1; naming $u_a = \Gamma^{(1)}E_{(2,a)}$ as the commuting copy is a restatement in this notation (e1-1a to e1-1d). The Klein–Jordan–Wigner reading is inherited [212, 213], and so is the exponential law [201, 214]. Two remarks organise the rest of the section. First, bivectors do not see the grading: $E_{(2,a)}E_{(2,b)} = \mathbf{1}\otimes e_ae_b$. Hence each constituent's $\mathfrak{l}^{(k)}_{\mathrm{int}}$ and $\mathfrak{l}^{(k)}_{G}$, its charge $J^{(k)} = E_{(k,2)}E_{(k,3)}$ and its number $N^{(k)}$ are the same operators in both readings, and the two Lorentz algebras commute in both. Second, the readings differ only in their odd elements, the gamma matrices and the mass terms. The algebra $M_{64}(\mathbb{R})$ is the same; what changes is which of its elements are taken as the vectors.

5.4.2 Rest energies: quadrature against addition

Put $H_g = m_1E_{(1,1)} + m_2E_{(2,1)}$, the rest Hamiltonian built from the tower's vectors, and $H_u = m_1\,e_1\otimes\mathbf{1} + m_2\,\mathbf{1}\otimes e_1$, built from the commuting copy. On one constituent the rest flow of the real Dirac form is $\dot\Psi = -JH(0)\Psi$ with $JH(0) = mT$, of angular frequency m (Section 2.3.4; Section 4.5.5).

Proposition 5.34 (Graded and ungraded rest energies)**.** *(i) $H_g^2 = M^2\,\mathbf{1}$ and $\mathrm{Tr}\,H_g = 0$, so $\mathrm{spec}\,H_g = \{\pm M\}$, each of multiplicity 32 for $M>0$ (coincident eigenvalues add their multiplicities: at $M=0$, $H_g=0$ with the single eigenvalue 0 on 64 dimensions).*

(ii) With $B = E_{(2,1)}E_{(1,1)}$, $B^2 = -\mathbf{1}$, and $U_\phi = \cos\phi + \sin\phi\,B$, $\tan2\phi = m_2/m_1$: $U_\phi(ME_{(1,1)})U_\phi^{-1} = H_g$. The graded rest Hamiltonian is one Dirac mass along a unit vector of $\mathbb{R}^{6,6}$, turned by a compact element of $\mathrm{Spin}(6,6)$.

(iii) H_u *has eigenvalue* $s_1m_1 + s_2m_2$ *on the* 16*-dimensional joint eigenspace of each sign pair* (s_1, s_2)*, so* $\operatorname{spec} H_u = \{\pm m_1 \pm m_2\}$*; for generic masses each value has multiplicity* 16*, and at* $m_1 = m_2$ *the value* 0 *has multiplicity* 32*.* $H_u = m_1 E_{(1,1)} + m_2 \Gamma^{(1)} E_{(2,1)}$ *has components of grade* 1 *and* 7*. For* $m_1, m_2 \neq 0$ *it is diagonalisable with at least three distinct eigenvalues, so it is not conjugate to any vector of* $\mathbb{R}^{6,6}$*: a vector satisfies* $v^2 = \eta_2(v, v)\mathbf{1}$ *and is either diagonalisable with at most two eigenvalues or nilpotent.*

(iv) Joint mass shells. No nonzero ψ *satisfies* $E_{(1,1)}\psi = s_1\psi$ *and* $E_{(2,1)}\psi = s_2\psi$ *with* $s_1, s_2 = \pm 1$*. Each joint sign* (s_1, s_2) *of* $e_1 \otimes \mathbf{1}$ *and* $\mathbf{1} \otimes e_1$ *occurs on* 16 *dimensions. More generally, the graded Dirac operators* $D_k = \sum_\mu p_\mu^{(k)} E_{(k,\mu)}$ *anticommute, with* $(D_1 + D_2)^2 = (p^{(1)} \cdot p^{(1)} + p^{(2)} \cdot p^{(2)})\mathbf{1}$*, while their ungraded counterparts commute.*

(v) Rest flows. $J^{(1)}$ *and* $J^{(2)}$ *commute with* H_g *and* H_u*. The ungraded generator* $Z_u = m_1 T^{(1)} + m_2 T^{(2)}$ *is a sum of two commuting complex structures, with eigenvalues* $i(\pm m_1 \pm m_2)$*:* $e^{-\lambda Z_u}$ *winds a two-torus with angular frequencies* m_1 *and* m_2*. The graded generator* $Z_g = J^{(1)} H_g$ *has* $Z_g^2 = -M^2\mathbf{1}$*, so* $e^{-\lambda Z_g} = \cos(M\lambda) - \sin(M\lambda) Z_g/M$ *has the single frequency* M *(at* $M = 0$*,* $Z_g = 0$ *and the flow is the identity, the limit of the formula).*

(vi) At rung n *the graded* $E_{(k,1)}$ *anticommute and the ungraded ones commute, with traceless products over every nonempty subset. Hence* $\operatorname{spec}(\sum_k m_k E_{(k,1)}) = \{\pm(\sum_k m_k^2)^{1/2}\}$*, while the ungraded sum has spectrum* $\{\sum_k \pm m_k\}$*, each sign pattern on* 4^n *dimensions (coinciding values add their multiplicities).*

Proof. (i) $E_{(1,1)}$ and $E_{(2,1)}$ are anticommuting traceless involutions. (ii) $U_\phi X U_\phi^{-1} = \cos^2\phi\, X + \sin\phi\cos\phi\,[B, X] - \sin^2\phi\, BXB$, with $[B, E_{(1,1)}] = 2E_{(2,1)}$ and $BE_{(1,1)}B = E_{(1,1)}$. (iii) Commuting involutions with traceless products have all four joint signs with equal multiplicity. (iv) If $E_{(1,1)}\psi = s_1\psi$ and $E_{(2,1)}\psi = s_2\psi$, then $s_1 s_2 \psi = E_{(1,1)}E_{(2,1)}\psi = -E_{(2,1)}E_{(1,1)}\psi = -s_1 s_2 \psi$. Generators of distinct constituents anticommute, so the cross terms of $(D_1 + D_2)^2$ cancel. (v) $T^{(k)} = J^{(k)} \times$ (the mass term of copy k), both $T^{(k)}$ square to $-\mathbf{1}$, and they commute. $J^{(1)}$ commutes with $E_{(1,1)}$ and $E_{(2,1)}$ and squares to $-\mathbf{1}$, so $(J^{(1)}H_g)^2 = -H_g^2$. (vi) As (i) and (iii), with 2^n joint signs. □

Parts (i) and (iii) at $m_1 = m_2$ restate Proposition 5.62(iv). The general masses, the rotation (ii), the joint mass shells (iv), the flows (v) and the ladder (vi) are fixed (e1-2a to e1-2g). The free-particle facts behind (iv) are inherited. A system of two particles has two mass-shell conditions, one per particle, imposed simultaneously, as in the two-body Dirac (Breit) equation [215]. A single particle in a doubled momentum space has one, $p^{(1)} \cdot p^{(1)} + p^{(2)} \cdot p^{(2)} = M^2$. By (ii) and (iv) the graded tower is of the second kind. Its two copies admit no simultaneous nonzero-mass Dirac constraints of the graded form, its rest energy is one value M, and the second copy enters the one particle as further momentum and mass directions. The ungraded operators are of the first kind: the energies add, and the pair threshold is $m_1 + m_2$.

5.4.3 Complex structures: the two-particle space and the selected unit

The paper reads $J = e_2 e_3$ as the imaginary unit of one constituent (Theorem 2.17). A complex two-particle space needs one imaginary unit acting as J on both factors.

Proposition 5.35 (The complex two-particle space)**.** *(i)* $J^{(1)}$ *and* $J^{(2)} = E_{(2,2)}E_{(2,3)}$ *commute and square to* $-\mathbf{1}$*, and* $J^{(1)}J^{(2)} = \omega_S$ *is a symmetric traceless involution. The subspace* $W = \{\psi : J^{(1)}\psi = J^{(2)}\psi\} = \ker(\omega_S + \mathbf{1})$ *has dimension* 32*, and its orthogonal complement is* $\overline{W} = \ker(J^{(1)} + J^{(2)}) = \ker(\omega_S - \mathbf{1}) = \operatorname{im}(J^{(1)} - J^{(2)})$*. With* $J := J^{(1)}|_W = J^{(2)}|_W$ *as imaginary unit,* $W \cong S \otimes_{\mathbb{C}} S$*, and* $\overline{W} \cong S \otimes_{\mathbb{C}} \overline{S}$*.*

(ii) *A blade of* $\mathrm{Cl}(6,6)$ *with an even number of surplus indices preserves* W *and* $\overline{W}$*. A blade with an odd number exchanges them, for instance* $\mathbf{1}\otimes e_2$ *(charge conjugation* $C = e_2$ *of copy 2 alone, Theorem 2.25) and* $E_{(1,1)}E_{(2,2)}$*.*

(iii) $J^{(1)}$ *and* $J^{(2)}$ *commute with* $\mathfrak{l}^{(1)}_{\mathrm{int}}$, $\mathfrak{l}^{(2)}_{\mathrm{int}}$*, the eight graded and the eight ungraded interior gammas,* H_g*,* H_u *and the interior copy rotation* $M_{12} = \frac{1}{2}\sum_\mu \eta_{\mu\mu}E_{(1,\mu)}E_{(2,\mu)}$*. All of these are* $\mathbb{C}$*-linear on* W *and on* $\overline{W}$*.*

(iv) *On* W *the surplus part of the copy rotation vanishes, since* $E_{(1,3)}E_{(2,3)} = E_{(1,2)}E_{(2,2)}\omega_S$*, so the copy rotation* $C_b = \frac{1}{2}\sum_a \eta_{aa}E_{(1,a)}E_{(2,a)}$ *of Proposition 5.22(ii) acts on* W *as* M_{12}*,* $\mathbb{C}$*-linearly. The four cross surplus bivectors* $E_{(1,s)}E_{(2,t)}$ *(*$s,t\in\{2,3\}$*) preserve* W *but anticommute with* $J^{(1)}$ *and* $J^{(2)}$*, so they act on* W *antilinearly.*

Proof. (i) Bivectors of distinct constituents commute; $(J^{(1)}J^{(2)})^2 = (J^{(1)})^2(J^{(2)})^2 = \mathbf{1}$ and $\mathrm{Tr}(J\otimes J) = (\mathrm{Tr}\,J)^2 = 0$. $J^{(1)}\psi = J^{(2)}\psi$ is equivalent to $J^{(1)}J^{(2)}\psi = -\psi$. $J^{(1)} - J^{(2)}$ is antisymmetric, so its kernel is the orthogonal complement of its image, and $\omega_S(J^{(1)} - J^{(2)}) = J^{(1)} - J^{(2)}$. The canonical map $S\otimes_{\mathbb{R}} S \to S\otimes_{\mathbb{C}} S$ has kernel $\mathrm{im}(J^{(1)} - J^{(2)})$, spanned by $Jx\otimes y - x\otimes Jy$, so it restricts to an isomorphism on W; the case of $\overline{W}$ is the same with $-J$ on the second factor. (ii) By equation (1.3) a blade commutes with the surplus four-blade ω_S iff it shares an even number of indices with it, and otherwise anticommutes. (iii) By the blade rule: the graded gammas, the bivectors of $\mathfrak{l}^{(k)}_{\mathrm{int}}$ and the terms of M_{12} contain no surplus index; the ungraded gammas $u_\mu = \Gamma^{(1)}E_{(2,\mu)}$ contain both surplus indices of copy 1, hence meet $J^{(1)}$ evenly and $J^{(2)}$ not at all. Directly, $\mathbf{1}\otimes\gamma$ commutes with $J\otimes\mathbf{1}$ and $\mathbf{1}\otimes J$. (iv) $E_{(1,2)}E_{(2,2)}\omega_S = -E_{(2,2)}E_{(1,3)}E_{(2,2)}E_{(2,3)} = E_{(1,3)}E_{(2,3)}$, by the anticommutation of distinct generators. On W, $\omega_S = -\mathbf{1}$, so the two surplus terms of C_b cancel. A cross surplus bivector shares one index with each $J^{(k)}$. □

The tensor-product linear algebra of (i) is inherited. Parts (ii)–(iv) are fixed (e1-3a, e1-3b, e1-3f). With the external i of Proposition 5.62(i), $J^{(1)} + J^{(2)}$ is $2J$ on W and 0 on $\overline{W}$. So W carries total charge ±2, two particles of equal charge, and $\overline{W}$ the neutral pairs, among them the particle–antiparticle state $\psi\otimes C\phi$ of that proposition (restated; the charge reading interpretive). The question is now whether anything in the algebra selects W and the complex structure on it.

Proposition 5.36 (Which imaginary unit the second rung selects)**.** *(i) Per-particle selection. The commutant of the eight ungraded gammas* $\gamma^\mu\otimes\mathbf{1}$*,* $\mathbf{1}\otimes\gamma^\mu$ *is* $\mathbb{H}\otimes\mathbb{H}\cong M_4(\mathbb{R})$*, spanned by* 16 *blades. Adding the two temporal volume elements* $T^{(1)}, T^{(2)}$ *(the datum of Remark 2.18 for each copy) cuts it to* $\mathrm{span}\{\mathbf{1}, J^{(1)}, J^{(2)}, \omega_S\}\cong\mathbb{C}\otimes_{\mathbb{R}}\mathbb{C}\cong\mathbb{C}\oplus\mathbb{C}$*. Its solutions of* $x^2 = -\mathbf{1}$ *are exactly* $\pm J^{(1)}$ *and* $\pm J^{(2)}$*, four isolated points, none of which acts as* J *on both factors of* S_2*.*

(ii) *The spectator action,* $k_I = 1$*. The selected unit of Theorem 2.17 at* $n = 2$*, the product of the eight internal generators, is* $J_{\mathrm{sel}} = J\otimes\omega = J^{(1)}\Gamma^{(2)}$ *(Proposition 5.30(iv)). It commutes with* $J^{(1)}$ *and* $J^{(2)}$ *and so preserves* W*. It equals* $J^{(1)}$ *on* $\{\Gamma^{(2)} = 1\}$ *and* $-J^{(1)}$ *on* $\{\Gamma^{(2)} = -1\}$*, and* $J^{(2)}$ *on* $\{\Gamma^{(2)}\omega_S = -1\}$ *and* $-J^{(2)}$ *on* $\{\Gamma^{(2)}\omega_S = 1\}$*,* 32 *dimensions each; on* W *it is* J *on* 16 *and* $-J$ *on* 16 *dimensions. It anticommutes with every generator of copy 2, graded or ungraded. So no gamma matrix of copy 2 is* J_{sel}*-linear, and neither* H_g *nor* H_u *is for* $m_2\neq 0$*;* $m_1E_{(1,1)}$ *is.*

(iii) *The diagonal action,* $k_I = 2$*. The commutant of the eight graded interior gammas* $E_{(k,\mu)}$ *is spanned by* 16 *blades (*$M_2(\mathbb{H})$*). Adding the six surplus bivectors cuts it to* $\mathrm{span}\{\mathbf{1},\omega_S\}\cong\mathbb{R}\oplus\mathbb{R}$*, with* $\omega_S^2 = +\mathbf{1}$*. No complex structure commutes with the gammas of both copies and the surplus bivectors. On* W *alone the restrictions of the same operators have commutant* $\mathbb{R}$*.*

(iv) Up the ladder, under the diagonal action of rung n the same commutant is $\operatorname{span}\{\mathbf{1}, \omega_S\}$ *with* $\omega_S = J^{(1)} \cdots J^{(n)}$ *and* $\omega_S^2 = (-1)^n$ *(checked at $n = 2, 3$ for the commutant and $n = 2, 3, 4$ for the square). An imaginary unit is selected exactly at odd n. The common subspace* $J^{(1)} = \cdots = J^{(n)}$*, the complex n-fold product $S \otimes_{\mathbb{C}} \cdots \otimes_{\mathbb{C}} S$, has dimension $2 \cdot 4^n$.*

Proof. (i) By the blade argument of Proposition 5.33(ii), since $\mathbf{1} \otimes \gamma^\mu = \Gamma^{(1)} E_{(2,\mu)}$ and $T^{(2)} = \Gamma^{(1)} E_{(2,1)} E_{(2,2)} E_{(2,3)}$ are blades. Equivalently, the commutant of the gammas on S is the quaternion algebra of Remark 2.18, and that of $\{\gamma^\mu, T\}$ is $\mathbb{R} \oplus \mathbb{R}J$. For $x = a + bJ^{(1)} + cJ^{(2)} + d\,\omega_S$, $x^2 = (a^2 - b^2 - c^2 + d^2) + 2(ab - cd)J^{(1)} + 2(ac - bd)J^{(2)} + 2(ad + bc)\omega_S$, and $x^2 = -\mathbf{1}$ has exactly the four solutions listed. (ii) J_{sel} is a blade of grade 8 containing every copy-2 index, so it anticommutes with each $E_{(2,a)}$; it commutes with $\Gamma^{(1)}$, so it also anticommutes with each $u_a = \Gamma^{(1)} E_{(2,a)}$. $\Gamma^{(2)}$ commutes with $J^{(1)}$ and $J^{(2)}$, and $\operatorname{Tr} \Gamma^{(2)} = 0$, also on W. Finally $J_{\mathrm{sel}} \mp J^{(2)} = J^{(1)}(\Gamma^{(2)} \pm \omega_S)$ with $J^{(1)}$ invertible, and $\Gamma^{(2)}\omega_S$ is a traceless involution. (iii) A blade commuting with all eight interior generators meets them in all or none of them. Meeting all of them requires odd grade, and meeting none requires even grade. Commuting with every surplus bivector forces it to meet the four surplus directions in all or none. Both conditions together leave $\mathbf{1}$ and ω_S. The algebra generated contains the central idempotent $\frac{1}{2}(\mathbf{1} - \omega_S)$, so its commutant on W is the compression of $\operatorname{span}\{\mathbf{1}, \omega_S\}$, which is $\mathbb{R}$. (iv) The same parity count with $4n$ interior and $2n$ surplus generators: a blade containing all $4n$ interior generators has grade $4n$ or $6n$, both even, so it fails. The square follows from equation (1.4), and the dimension is $8^n/2^{n-1}$, because the $n - 1$ involutions $J^{(k)}J^{(k+1)}$ commute and have traceless products. □

Part (i) is fixed, with the quaternion commutant of one copy restated from Remark 2.18. Part (ii) restates Theorem 2.17 and Proposition 5.30(iv) (fixed there), and the comparison with W is fixed. Part (iii) is obstructed under the stated hypotheses (e1-3c to e1-3e, e1-3g). Part (iv) restates the selection at odd n of Proposition 5.17(ii) (fixed there) and adds that nothing is selected at even n (fixed; e1-3h).

Corollary 5.37 (No selected two-particle complex structure)**.** *At the second rung none of the selection principles of Chapter 2 gives a complex structure under which both copies are complex Dirac spinors with $\mathbb{C}$-linear gamma matrices. Per copy they give $\pm J^{(1)}$ and $\pm J^{(2)}$ separately. The spectator action gives J_{sel}, under which copy 2 carries no linear Dirac operator, so that copy 2 is internal. The diagonal action gives none. What the diagonal action does select is the splitting $S_2 = W \oplus \overline{W}$, the eigenspaces of the one nontrivial element ω_S of its commutant. The complex structure $J|_W$ that makes W the two-particle space commutes with every interior gamma matrix, with both Hamiltonians and with the copy rotation, but not with the surplus $\mathfrak{so}(4)$. It is an added datum.*

The corollary is fixed as a statement about these selection principles (Proposition 5.36). That the choice of W, rather than $\overline{W}$, and of $J|_W$ is what turns the second rung into a two-particle space is interpretive. This also bears on the phrase "two particles from one". The paper's own second quantisation of one constituent (Theorem 2.31, Theorem 7.14) already has two-excitation states of one Dirac momentum (four fermionic modes), a level of dimension 6 with charges $q = \pm 2$ once each and $q = 0$ four times. In ordinary physics that is where several particles live (inherited [216]). It does not need the second rung.

5.4.4 The Fock reading: quanta, not particles

The tower has its own number operator. S_n is the Fock space of $3n$ fermionic modes, and N counts them (Proposition 5.21). We ask which elements of $\mathfrak{so}(3n, 3n)$, the bivectors, conserve N.

Proposition 5.38 (Bivectors and the number of quanta)**.** *(i) $N = \frac{3n}{2} - \frac{1}{2}\sum_{k,i} E_{(k,i)}E_{(k,i+3)}$ is diagonal, with spectrum $0, \ldots, 3n$; at $n = 2$ the multiplicities are $1, 6, 15, 20, 15, 6, 1$.*

$N = N^{(1)} + N^{(2)}$, *the same operator in the graded and ungraded readings.* $(-1)^N = \Gamma^{(1)}\Gamma^{(2)}$ *is the volume element of* $\mathrm{Cl}(6,6)$.

(ii) On vectors $[N, v] = K_n v$. *A bivector* X *commutes with* N *iff* $\operatorname{ad} X$ *commutes with* K_n. *These form* $\mathfrak{gl}(L_+) \cong \mathfrak{gl}(3n, \mathbb{R})$, *of dimension* $9n^2$ *(*36 *at* $n = 2$*), the stabiliser of the transverse maximal isotropic subspaces* $L_\pm = \ker(K_n \mp \mathbf{1})$. *The remaining* $3n(3n-1)$ *directions (*30 *at* $n = 2$*) are* $\Lambda^2 L_+ \oplus \Lambda^2 L_-$, *on which* $[N, X] = \pm 2X$*: they create or annihilate a pair of quanta.*

(iii) $(-1)^N$ *commutes with every bivector and anticommutes with every generator. No element of* $\mathfrak{so}(3n, 3n)$ *changes* N *by an odd number.*

Proof. (i) $N_i = \frac{1}{2}(\mathbf{1} - e_i e_{i+3})$, and the pole operators are diagonal (Appendix A.1). (ii) $[N, a^\pm_{k,i}] = \pm a^\pm_{k,i}$, and $a^\pm_{k,i}$ is proportional to $e_i \pm e_{i+3}$ in constituent k, the ± 1 eigenvectors of K. $N - \frac{3n}{2}$ is a bivector, so $[N, X]$ is one, and the spin representation of $\mathfrak{so}(V_n)$ is faithful. Hence $[N, X] = 0$ iff $[\operatorname{ad} N, \operatorname{ad} X] = [K_n, \operatorname{ad} X] = 0$. Under $\operatorname{ad} K_n$, $\mathfrak{so}(V_n) \cong \Lambda^2 V_n = \Lambda^2 L_+ \oplus (L_+ \otimes L_-) \oplus \Lambda^2 L_-$ has eigenvalues $2, 0, -2$, and $L_\pm$ are isotropic because $\eta(e_i \pm e_{i+3}, e_j \pm e_{j+3}) = \delta_{ij} - \delta_{ij}$. (iii) Grade parity. □

Part (i) restates Proposition 5.21. The rest is the standard structure of the spin representation on the exterior algebra of a maximal isotropic subspace [201], inherited, with its realisation in the tower fixed (e1-4a, e1-4b, e1-4g). The question is which of these directions are physical.

Theorem 5.39 (Compact fields conserve the quanta)**.** *Let* $X \in \mathrm{Cent}(G_n) \cap \mathfrak{so}(3n, 3n)$, *the algebra* $\mathfrak{so}(3n, \mathbb{C})$ *of binding-compatible bivectors (the case* $n = 3$ *is Theorem 4.34, the cases* $n \leq 3$ *Proposition 5.27(i); the block form of the proof gives every* n*). Then* $K_n X K_n = -X^\top$, *the Cartan involution. Consequently:*

(i) X *commutes with* N *iff* X *is compact (antisymmetric), that is iff it is a paired rotation* $\left(\begin{smallmatrix} R & 0 \\ 0 & R \end{smallmatrix}\right)$. *These form* $\mathfrak{so}(3n)$, *of dimension* $\frac{3}{2}n(3n-1)$.

(ii) The non-compact part, the paired boosts $\left(\begin{smallmatrix} 0 & S \\ -S & 0 \end{smallmatrix}\right)$, *anticommutes with* K_n. *A nonzero paired boost has nonzero components in both* $\Lambda^2 L_+$ *and* $\Lambda^2 L_-$, *so it changes* N *by* $+2$ *and by* -2*: it has no component commuting with* N, *and* $[N, X] \neq 0$ *for* $X \neq 0$. *It may still conserve partial occupation numbers; for instance* $(e_3 e_5 - e_2 e_6)/2$ *conserves* N_1 *and* $N_2 - N_3$.

(iii) By Corollary 1.4 a boost generates no compact $U(1)$, *so it is not gauge-admissible in the sense fixed after that corollary. Hence no gauge-admissible, binding-compatible field creates or annihilates a pair of quanta, at any rung.*

Proof. On vectors $\eta G_n = -K_n$ and $G_n \eta = -\eta G_n$. For X commuting with G_n, $K_n X K_n = \eta G_n X \eta G_n = -\eta G_n X G_n \eta = -\eta X G_n^2 \eta = \eta X \eta$, and $\eta X \eta = -X^\top$ for $X \in \mathfrak{so}(\eta)$. (i) follows from Proposition 5.38(ii). The block forms are those of Proposition 5.27(i). (ii) X maps $L_\pm$ to $L_\mp$, so its components lie in $\Lambda^2 L_+ \oplus \Lambda^2 L_-$. G_n commutes with X, maps L_+ onto L_- (because $G_n K_n = -K_n G_n$) and normalises $\mathfrak{so}(\eta)$, so conjugation by G_n carries the $\Lambda^2 L_+$ component of X to its $\Lambda^2 L_-$ component. They vanish together. (iii) Gauge admissibility is compactness, the criterion of Section 1.1.2. □

The theorem is fixed. It was checked exactly on vectors at $n = 1, \ldots, 4$, where $\dim \mathrm{Cent}(K_n) \cap \mathfrak{so}(3n, 3n) = 9, 36, 81, 144$, $\dim \mathrm{Cent}(G_n) \cap \mathfrak{so}(3n, 3n) = 6, 30, 72, 132$ and the compact part has half of it. It was checked entry by entry on S_n at $n = 1, \ldots, 4$ for every paired rotation $e_i e_j - e_{i+3} e_{j+3}$ and every paired boost $e_i e_{j+3} - e_j e_{i+3}$, within one constituent and across two (e1-4c, e1-4d). At $n = 4$, S_4 has dimension 4096, and the generators were handled as signed permutations.

Proposition 5.40 (The quanta and the Lorentz algebras)**.** *The number N behaves as follows under the copy algebra and under the two Lorentz algebras of one constituent.*

(i) The copy rotation C_b commutes with the total number N, but it does not commute with the number $N^{(1)}$ of the first constituent. The quarter turn

$$U = e^{\pi C_b/2} = \tfrac{1}{8}\textstyle\prod_a(\mathbf{1} + \eta_{aa}E_{(1,a)}E_{(2,a)})$$

satisfies $UN^{(1)}U^{-1} = N^{(2)}$. The spin lift of the copy boost $\varepsilon \otimes G$ changes N by ± 2 only, with both signs.

(ii) In $\mathfrak{l}_{\text{int}}$ the elements commuting with N form $\mathbb{R}\, e_1e_4$, a boost (the pole operator $\mathbf{1} - 2N_1$). The rotations e_4e_5, e_4e_6, e_5e_6, the boosts e_1e_5, e_1e_6 and the charge $J = e_2e_3$ do not commute with N.

(iii) In $\mathfrak{l}_G$ the elements commuting with N form its compact $\mathfrak{so}(3)$, the paired rotations. No nonzero boost of $\mathfrak{l}_G$ commutes with N.

Proof. (i) C_b is a paired rotation (Theorem 5.39(i)). Its six terms commute and square to $-\frac{1}{4}$, so $e^{\theta C_b} = \prod_a(\cos\frac{\theta}{2} + \sin\frac{\theta}{2}\,\eta_{aa}E_{(1,a)}E_{(2,a)})$, which at $\theta = \frac{\pi}{2}$ is U; it carries $E_{(1,a)}$ to $\pm E_{(2,a)}$ (Proposition 5.22(ii)). The copy boost is a paired boost (Proposition 5.27(iii)); apply Theorem 5.39(ii). (ii) $\mathfrak{l}_{\text{int}} \cap \mathfrak{gl}(3,\mathbb{R})$ is computed: of the six planes in $\langle e_1, e_4, e_5, e_6\rangle$ only $\langle e_1, e_4\rangle$ is K-invariant, and J involves the modes $a_2^+a_3^+$. (iii) is Theorem 5.39 at $n = 1$, with $\mathfrak{l}_G = \mathrm{Cent}(G) \cap \mathfrak{so}(3,3)$ (Theorem 3.1). □

All three parts are fixed (e1-4e, e1-4f), and (i) gives a precise sense to the copy rotation as a hop of quanta between the constituents at fixed total number. Part (ii) corrects an easy expectation. The Fock number is not invariant under the spatial rotations of the interior four-plane, only under one of its boosts, and it does not commute with the charge J. The quanta counted by N are therefore not Lorentz-invariant particles. Within $\mathfrak{l}_{\text{int}}$ they are invariant only under the one boost e_1e_4, the pole operator of the first mode; within $\mathfrak{l}_G$ only under its compact $\mathfrak{so}(3)$; they are not invariant under the charge J or under any boost of $\mathfrak{l}_G$. The bivectors that preserve N form $\mathfrak{gl}(3n,\mathbb{R})$ (Proposition 5.38(ii)), which contains boosts (at $n = 1$ six of its nine dimensions), so N is tied to the split structure K and not to a Lorentz algebra. Reading N as a particle number is therefore an internal reading (interpretive). What Theorem 5.39 does establish is the following. If quanta are counted by N, the only elements of the binding-compatible algebra that create them in pairs are the paired boosts. These are excluded as gauge fields, so within the binding-compatible algebra pair creation of quanta would need a non-compact, non-gauge interaction (fixed, with the identification of quanta interpretive). Outside it, compact gauge-admissible fields that do not commute with G_n, such as the charge J and the time–time rotation e_1e_2, do create and annihilate quanta in pairs: both have components with $\Delta N = +2$, 0 and -2. This agrees with the statement of Section 5.6.5 that duplication in the tower's sense is not pair creation.

5.4.5 Exchange and statistics

Two swaps act on S_2: the ungraded swap $P(x \otimes y) = y \otimes x$, and the Koszul swap $\tau_K = D\,P$ of Proposition 5.22(iii), where $D = (-1)^{\mathsf{p}\otimes\mathsf{p}}$ and $\omega = (-1)^{\mathsf{p}}$ on the basis of equation (1.1).

Proposition 5.41 (Two exchanges)**.** *(i) $P(e_a \otimes \mathbf{1})P = u_a = \Gamma^{(1)}E_{(2,a)}$. So P exchanges the ungraded copies and maps the vectors $E_{(1,a)}$ of $\mathrm{Cl}(6,6)$ to elements of grade 7. It commutes with the diagonal Lorentz generators $\Delta_{\mu\nu}$, with N, with ω_S (so it preserves W) and with H_u at $m_1 = m_2$, but not with H_g.*

(ii) $\tau_K E_{(1,a)}\tau_K = E_{(2,a)}$. τ_K commutes with $\Delta_{\mu\nu}$, N, ω_S and H_g at $m_1 = m_2$, but not with H_u.

sector	on $W = S \otimes_{\mathbb{C}} S$ (32)	on $\overline{W} = S \otimes_{\mathbb{C}} \overline{S}$ (32)	on S_2 (64)
$P = -1$	$0^4\ 3^8$	$0^2\ 3^8\ 4^6$	$0^6\ 3^{16}\ 4^6$
$P = +1$	$3^8\ 4^{12}$	$0^2\ 3^8\ 4^6$	$0^2\ 3^{16}\ 4^{18}$
$\tau_K = -1$	$0^2\ 3^8\ 4^6$	$0^2\ 3^8\ 4^6$	$0^4\ 3^{16}\ 4^{12}$
$\tau_K = +1$	$0^2\ 3^8\ 4^6$	$0^2\ 3^8\ 4^6$	$0^4\ 3^{16}\ 4^{12}$

Table 5.4: Casimir content c^d ($C = c$ on d real dimensions) of the exchange sectors under the diagonal interior Lorentz algebra. Checks e1-5d to e1-5f.

(iii) $D = \frac{1}{2}(\mathbf{1} + \Gamma^{(1)} + \Gamma^{(2)} - \Gamma^{(1)}\Gamma^{(2)})$. *So* $\tau_K = P$ *except on the sector* $\Gamma^{(1)} = \Gamma^{(2)} = -1$, *where* $\tau_K = -P$.

Proof. (i) $P(A \otimes B)P = B \otimes A$. Diagonal bivectors, $N = N^{(1)} + N^{(2)}$, $\omega_S = J \otimes J$ and H_u at equal masses are symmetric under it, while $P(\omega \otimes e_1)P = e_1 \otimes \omega$. (ii) The first identity restates Proposition 5.22(iii). Hence $\tau_K \Gamma^{(1)} \tau_K = \Gamma^{(2)}$ and $\tau_K u_1 \tau_K = e_1 \otimes \omega \neq u_1$, while bivectors $E_{(1,a)}E_{(1,b)}$ go to $E_{(2,a)}E_{(2,b)}$. (iii) D is diagonal with entry -1 exactly where both ω-eigenvalues are -1. □

Each swap is a symmetry of its own reading and not of the other: P of the additive Hamiltonian, τ_K of the graded one (fixed; e1-5a, e1-5b). For the Lorentz content let $C = -\sum_{\mu<\nu} \eta^{\mu\mu}\eta^{\nu\nu}\Delta_{\mu\nu}^2$ be the diagonal Casimir, normalised as in Section 5.2.2. It is $\frac{3}{2}$ on one constituent, it satisfies $C(C-3)(C-4) = 0$ on S_2, and it takes the values 0, 3 and 4 on $(0,0)$, $(\frac{1}{2},\frac{1}{2})$ and $(1,0) \oplus (0,1)$. On one constituent each boost is $J\Gamma$ times its dual rotation, $e_1e_4 = J\Gamma\, e_5e_6$ and cyclically, so $\Gamma = \pm 1$ are the two Weyl halves (e1-5c).

Proposition 5.42 (Statistics sectors)**.** *The dimensions of the eigenspaces of* C *in the sectors of* P *and* τ_K *are those of Table 5.4. In particular:*

(i) On W, P *is* $\mathbb{C}$*-linear. Its* -1 *eigenspace,* $\Lambda^2_{\mathbb{C}}S$, *carries* $2(0,0) \oplus (\frac{1}{2},\frac{1}{2})$, *and its* $+1$ *eigenspace,* $\mathrm{Sym}^2_{\mathbb{C}}S$, *carries* $(1,0) \oplus (0,1) \oplus (\frac{1}{2},\frac{1}{2})$. *In the chirality sectors,* $\Gamma^{(1)} = \Gamma^{(2)} = \pm 1$ *gives spin one in* $P = +1$ *and spin zero in* $P = -1$, *while the mixed sector (*16 *dimensions,* $C = 3$*) splits* $8 + 8$.

(ii) On W, τ_K *is* $\mathbb{C}$*-linear, and its two eigenspaces have the same Casimir content,* $C = 0, 3, 4$ *on* $2, 8, 6$ *dimensions. The spin-one part of* $\tau_K = +1$ *lies in* $\Gamma^{(1)} = \Gamma^{(2)} = +1$, *and that of* $\tau_K = -1$ *in* $\Gamma^{(1)} = \Gamma^{(2)} = -1$*: the two sectors are mirror images.*

(iii) On $\overline{W}$, P *is antilinear,* $PJ^{(1)} = -J^{(1)}P$, *and* $J^{(1)}$ *exchanges its two eigenspaces. They carry the same content.*

Proof. All entries are traces of products of commuting integer matrices: the projectors $\frac{1}{2}(\mathbf{1} \pm X)$ for $X \in \{\omega_S, P, \tau_K, \Gamma^{(1)}, \Gamma^{(2)}\}$ and the spectral projectors of C (e1-5d to e1-5f). (i) $PJ^{(1)}P = J^{(2)}$, which equals $J^{(1)}$ on W. (ii) Use Proposition 5.41(iii). (iii) $J^{(2)} = -J^{(1)}$ on $\overline{W}$, and $J^{(1)}$ commutes with C. □

The decompositions of Λ^2 and Sym^2 of a Dirac spinor are inherited [160]; their appearance on W, and only on W, is fixed. The Koszul column restates Proposition 5.29(iii) on S_2 (fixed there) and refines it on W. The two τ_K sectors are isomorphic as real representations, since the realification of $(1,0)$ is that of $(0,1)$, but they are mirror images once J is the imaginary unit (fixed). So the ungraded swap sorts the complex two-particle space by statistics in the textbook way. Identical fermions are antisymmetric under exchange [217], and the antisymmetric square of a Dirac spinor has only scalar and vector content at a single momentum mode (inherited [160]). The Clifford exchange τ_K does not separate symmetric from antisymmetric content. Its two sectors have the same real Lorentz content and differ only in the chirality of their spin-one

structure	what it does at $n = 2$	class
graded vectors $E_{(2,a)}$	one Dirac clock of mass M; no simultaneous nonzero-mass graded Dirac constraints; τ_K as exchange	fixed (they define $\mathrm{Cl}(6,6)$)
commuting copy $u_a = \Gamma^{(1)} E_{(2,a)}$	two mass shells, energies $\pm m_1 \pm m_2$	fixed as an algebra (unique up to $\mathbf{1} \otimes GL(8,\mathbb{R})$); use as vectors constructed
common complex structure $J\|_W$	$W = S \otimes_{\mathbb{C}} S$, 32 dimensions, total charge ± 2	$W \oplus \overline{W}$ fixed by ω_S; $J\|_W$ constructed; selection obstructed ($k_I = 2$), copy 2 internal ($k_I = 1$)
additive Hamiltonian H_u	threshold $m_1 + m_2$; two-torus rest flow	constructed; not a vector (grades 1 and 7)
ungraded exchange P	$\Lambda^2_{\mathbb{C}} S = 2(0,0) \oplus (\frac{1}{2}, \frac{1}{2})$ on W	constructed; τ_K fixed but statistics-blind
Fock number N	counts quanta; conserved by paired rotations, changed by ± 2 by paired boosts	fixed; as a particle number interpretive (not Lorentz- or J-invariant)

Table 5.5: Structures that bear on a two-particle reading of the second rung.

part, so its eigenvalue is a chirality label, not a statistics label. Fermi statistics for two copies is therefore a constructed requirement: antisymmetry under P on W. The algebra supplies τ_K, the exchange that preserves the vectors of $\mathrm{Cl}(6,6)$, and τ_K is blind to it.

5.4.6 Answer: what makes the second rung carry two particles

Table 5.5 collects the structures. Read together they answer the question in four sentences; the particle language is an interpretation under the stated assumptions, as Section 5.9 says. (1) As given by the tower, and read in the Dirac picture, the second rung carries one particle. Its vectors $E_{(k,a)}$ give one Dirac clock along a unit vector of $\mathbb{R}^{6,6}$, of mass $M = (m_1^2 + m_2^2)^{1/2}$, with no simultaneous nonzero-mass Dirac constraint of the graded form on the copies. Under the selected unit J_{sel} copy 2 carries no linear Dirac operator (fixed; Propositions 5.34 and 5.36). Read as particle content, the rung is one particle with a larger internal space (interpretive). (2) The algebra also contains, uniquely up to conjugation, a second commuting copy, the generators dressed by the chirality of the first. With it the copies have separate mass shells and additive energies (fixed; Propositions 5.33 and 5.34). (3) Three further structures make that a two-particle system in the sense of ordinary quantum mechanics: the subspace W with the complex structure $J|_W$, the additive Hamiltonian H_u, and the ungraded exchange P with antisymmetry. None of them is selected. W is singled out only together with $\overline{W}$, and the diagonal action selects no imaginary unit at even rungs (constructed; obstructed as a selection; Corollary 5.37 and Proposition 5.42). (4) The tower's own counting operator N counts quanta of the split structure, not Lorentz-invariant particles. No gauge-admissible, binding-compatible field changes it (fixed; Theorem 5.39 and Proposition 5.40).

Which process takes one to two. Three candidates have been tested. The step from rung 1 to rung 2 is a change of arena, not a dynamical process. In the graded reading it enlarges the particle and does not add one. In the Fock reading the Lie algebra $\mathfrak{so}(3n, 3n)$ changes N only by 0 or ± 2, and among binding-compatible fields only the paired boosts do it, which are not gauge fields. A step from one quantum to two would need an odd element, which is a field operator and not a symmetry (fixed; Proposition 5.38 and Theorem 5.39). In the two-particle reading the step is the tensor embedding $\psi \mapsto \psi \otimes \phi_0$, which intertwines $A \mapsto A \otimes \mathbf{1}$ but needs a state ϕ_0 of the second particle that nothing selects (constructed). In ordinary physics the process is creation from a field, and the paper's second quantisation of one constituent already contains

statement	class	checks
Proposition 5.33: commuting copy, commutant $\mathbf{1} \otimes M_8(\mathbb{R})$, grade-7 generators, chirality dressing, rung n	restated (Proposition 5.20, Proposition 5.21, Appendix A.1); commutant dimension and uniqueness fixed; exponential law inherited	e1-1a to e1-1d
Proposition 5.34: quadrature against addition, rotated clock, joint mass shells, rest flows, rung n	fixed; restated at $m_1 = m_2$ (Proposition 5.62(iv)); two-body kinematics inherited	e1-2a to e1-2g
Proposition 5.35: $W = S \otimes_{\mathbb{C}} S$, ω_S, what preserves W, $C_b\|_W = M_{12}\|_W$	fixed; tensor algebra inherited	e1-3a, e1-3b, e1-3f
Proposition 5.36: per-particle units $\pm J^{(k)}$; $J_{\mathrm{sel}} = J^{(1)}\Gamma^{(2)}$; no unit at $k_I = 2$; parity of n	fixed; restated (Theorem 2.17, Proposition 5.17; Proposition 5.30); obstructed	e1-3c to e1-3e, e1-3g, e1-3h
Corollary 5.37: no selected two-particle complex structure	fixed (as stated); reading interpretive	—
Proposition 5.38: $\mathrm{ad}\, N = K_n$, $\mathfrak{gl}(3n, \mathbb{R})$, pairs, parity	restated (Proposition 5.21); inherited; realisation fixed	e1-4a, e1-4b, e1-4g
Theorem 5.39: compact $\Leftrightarrow$ conserves N; boosts change it by ± 2; no gauge field creates pairs	fixed	e1-4c, e1-4d
Proposition 5.40: copy rotation hops quanta; N against $\mathfrak{l}_{\mathrm{int}}$, J, $\mathfrak{l}_G$	fixed; particle reading interpretive	e1-4e, e1-4f
Proposition 5.41: P against τ_K	fixed; τ_K restated (Proposition 5.22)	e1-5a, e1-5b
Proposition 5.42, Table 5.4: statistics sectors	fixed; Dirac$\otimes$Dirac inherited; τ_K on S_2 restated (Proposition 5.29)	e1-5c to e1-5f
Section 5.4.6: which process; up the ladder; dynamical selection of W	constructed; fixed; open	see text

Table 5.6: Claims of Section 5.4 and their classes. The checks are the `PASS` lines printed by the script `e1_two_from_one.py`.

it (Theorem 2.31). Whether some mechanism of the arena selects W and $J|_W$ dynamically is open.

Up the ladder. What has been proved at every rung is this. The commuting copies exist, dressed by $\Gamma^{(1)} \cdots \Gamma^{(k-1)}$. The graded masses add in quadrature and the ungraded ones additively. The rule of Theorem 5.39 holds. The common complex subspace $J^{(1)} = \cdots = J^{(n)}$ has dimension $2 \cdot 4^n$. The diagonal action selects an imaginary unit, $\omega_S = J^{(1)} \cdots J^{(n)}$, exactly at odd n. On the common subspace it acts as $(-1)^{(n-1)/2} J$, so at odd rungs the selected unit and the common complex structure agree up to sign there, while at even rungs there is no selected unit to compare (fixed; e1-1d, e1-2f, e1-3h, e1-4c, e1-4d). The physics of the odd rungs, whose diagonal sector is the vector-like matter of Section 5.2, is not examined here.

5.5 How one becomes two: processes

Section 5.4 asked which structures make a rung carry several particles. This section asks which processes take one to two. The word "particle" has four precise senses in the arena, and each has its own process. A particle can be a spin module of a Lorentz algebra, and then the process is a vertex, an equivariant map. It can be a mass shell, and then the process is a decay above a threshold. It can be a field, as in the field-strength reading (Chapter 3), and then the process is a duplication of fields. And it can be a quantum of the Fock structure of the tower, and then the process is a change of the number N. The answers, in order, are these. (1) Under the common interior Lorentz action no equivariant map takes a spin-$\frac{1}{2}$ module into the space of

two constituents, in either direction. Maps from a fermion to a fermion and a boson exist, and Clifford multiplication provides them (the selection rule inherited, the counts in the realisation fixed; Section 5.5.1). (2) At equal momentum labels the top level E_+ of the graded composite of two masses lies strictly below the free-pair energy $E_1 + E_2$ iff $m_1 m_2 \neq 0$; at rest the gap is $m_1 + m_2 - M$, with $(m_1 + m_2)^2 - M^2 = 2m_1 m_2$ (fixed). This compares two operators on one space, not a bound state with its decay products: at fixed total momentum the graded levels with large relative momentum lie above the free-pair threshold, so "binding" is not established (Section 5.5.2). (3) One null field never becomes a massive pair covariantly, and two non-collinear null fields always form one (restated from Section 3.9; Section 5.5.3). (4) Boosts create pairs of Fock quanta, with the amplitudes of a fermionic Bogoliubov transformation, and so do the compact generators that do not commute with the binding root, among them the spatial rotations of the interior four-plane and the charge J. The compact generators that commute with the binding root never do (fixed; Section 5.5.4). Under the common binding action a Lorentz-invariant number of quanta exists exactly at even rungs, and its quanta are shared equally by two constituents (fixed; Section 5.5.5). (5) The copy rotation moves one quantum from one constituent to another and never doubles it; at unequal rates it does so by Rabi mixing (fixed; Section 5.5.6). Section 5.5.7 collects the processes with their classes. Identifiers e2-n refer to the `PASS` lines of `e2_processes.py` (Table 5.8).

Notation. As in Section 5.4: $S = \mathbb{R}^8$ with the realisation of equation (1.1), the tower generators $E_{(k,a)} = \omega^{\otimes(k-1)} \otimes e_a \otimes \mathbf{1}^{\otimes(n-k)}$ on $S_n = S^{\otimes n}$ (equation (1.2)), $V = \mathbb{R}^{3,3}$, $V_n = \mathbb{R}^n \otimes V$, $\eta_n = \mathbf{1}_n \otimes \eta$, $G_n = \mathbf{1}_n \otimes G$ and $K_n = \mathbf{1}_n \otimes K$ with $Ke_i = e_{i+3}$, $Ke_{i+3} = e_i$. The spin lift of $A \in \mathfrak{so}(\eta_n)$ is $\sigma(A) = \frac{1}{4}\sum_{a,b} A_{ab}\eta_{bb}E_aE_b$, so that $[\sigma(A), E_c] = \sum_a A_{ac}E_a$. The interior Lorentz algebra $\mathfrak{l}_{\text{int}}$ has the rotations $S_1 = \frac{1}{2}e_5e_6$, $S_2 = \frac{1}{2}e_6e_4$, $S_3 = \frac{1}{2}e_4e_5$ and the boosts $T_a = \frac{1}{2}e_1e_{a+3}$; it acts on the interior four-plane $E = \langle e_1, e_4, e_5, e_6\rangle$ with $\gamma^\mu = (e_1, e_4, e_5, e_6)$, and trivially on the surplus pair $\langle e_2, e_3\rangle$ (Section 2.3). The field-strength algebra $\mathfrak{l}_G$, the commutant of G in $\mathfrak{so}(3,3)$ (Theorem 3.1), has the boosts $K_i = \frac{1}{2}(e_ke_{j+3} - e_je_{k+3})$ and the rotations $L_k = -[K_i, K_j]$, for (i,j,k) cyclic. $J = e_2e_3$. The Fock modes are $a^{\pm}_{k,i} = \frac{1}{2}(E_{(k,i)} \pm E_{(k,i+3)})$, $N^{(k)} = \sum_i a^+_{k,i}a^-_{k,i}$ and $N = \sum_k N^{(k)}$ (Proposition 5.21); $a^-_{k,i}$ is the transpose of $a^+_{k,i}$. On S the basis vectors of the realisation are the Fock states, and $|n_1n_2n_3\rangle$ denotes the one with $N_i = n_i$; the Fock vacuum $|000\rangle$ is the first basis vector, and $|0\rangle$ is the first basis vector of S_n (e2-1). $\varepsilon = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ acts on the copy index. The copy rotation on vectors is $\varepsilon \otimes \mathbf{1}_6$ and the copy boost is $\varepsilon \otimes G$ (Section 5.3). We write

$$\hat{C} := \sigma(\varepsilon \otimes \mathbf{1}_6) = -\tfrac{1}{2}\textstyle\sum_a \eta_{aa}E_{(1,a)}E_{(2,a)},$$

the negative of C_b of Proposition 5.22(ii), which lifts the copy rotation with the opposite orientation. Dimensions of spaces of equivariant maps are real dimensions.

5.5.1 Vertices

A vertex that takes one particle into two is an equivariant linear map from the module of the one into the module of the two. The ungraded and graded copies of $\mathrm{Cl}(3,3)$ in $\mathrm{Cl}(6,6)$ have the same bivectors, $E_{(2,a)}E_{(2,b)} = \mathbf{1} \otimes e_ae_b$ (e2-1; Proposition 5.33), so every Lorentz action built from bivectors is the same operator on $S_2 = S \otimes S$ in both readings, and the statements below hold for both.

Proposition 5.43 (Vertices)**.** *Let $\mathfrak{l}_{\text{int}}$ act on S_n diagonally, on every constituent, unless stated otherwise.*

(i) $\dim \mathrm{Hom}_{\mathfrak{l}_{\text{int}}}(S, S_2) = \dim \mathrm{Hom}_{\mathfrak{l}_{\text{int}}}(S_2, S) = 0$. *The rotation $e^{2\pi S_3}$ is $-\mathbf{1}$ on S and $(-1)^n\mathbf{1}$ on S_n, so $\mathrm{Hom}_{\mathfrak{l}_{\text{int}}}(S, S_n) = 0$ for every even n. At $n = 3$, $\dim \mathrm{Hom}_{\mathfrak{l}_{\text{int}}}(S, S_3) = 160$.*

(ii) The spaces of $\mathfrak{l}_{\text{int}}$-equivariant maps have the dimensions

$$\dim \mathrm{End}_{\mathfrak{l}_{\text{int}}}(S) = 8, \quad \dim \mathrm{Hom}_{\mathfrak{l}_{\text{int}}}(S, S \otimes E) = 8, \quad \dim \mathrm{Hom}_{\mathfrak{l}_{\text{int}}}(S, S \otimes V) = 24,$$

$$\dim \mathrm{Hom}_{\mathfrak{l}_{\mathrm{int}}}(S, S\otimes\Lambda^2 E) = 8, \qquad \dim \mathrm{Hom}_{\mathfrak{l}_{\mathrm{int}}}(S, S\otimes\Lambda^2 V) = 32.$$

(iii) The Clifford multiplications $c_1(\psi) = \sum_\mu \eta_{\mu\mu}\gamma_\mu\psi\otimes e_\mu$ *and* $c_2(\psi) = \sum_{\mu<\nu}\eta_{\mu\mu}\eta_{\nu\nu}\gamma_\mu\gamma_\nu\psi\otimes e_\mu\wedge e_\nu$ *are* $\mathfrak{l}_{\mathrm{int}}$*-equivariant, and* $\mathrm{Hom}_{\mathfrak{l}_{\mathrm{int}}}(S, S\otimes E) = c_1\circ\mathrm{End}_{\mathfrak{l}_{\mathrm{int}}}(S)$, $\mathrm{Hom}_{\mathfrak{l}_{\mathrm{int}}}(S, S\otimes\Lambda^2 E) = c_2\circ\mathrm{End}_{\mathfrak{l}_{\mathrm{int}}}(S)$. *Without the factors* $\eta_{\mu\mu}$, c_1 *is not equivariant.*

(iv) Under the spectator action, $\mathfrak{l}_{\mathrm{int}}$ *on constituent* 1 *alone,* $\dim\mathrm{Hom}(S, S_2) = 64$. *Under the mixed action,* $\mathfrak{l}_{\mathrm{int}}$ *on constituent* 1 *and* $\mathfrak{l}_G$ *on constituent* 2, $\dim\mathrm{Hom}(S, S_2) = 16$ *and* $\dim\mathrm{Hom}(S, S\otimes V^{(2)}) = 8$, *where* $V^{(2)}$ *is the vector space of constituent* 2. *On* S *the* $\mathfrak{l}_G$ *Casimir* $\sum_i K_i^2 - \sum_k L_k^2$ *equals* $3\,\mathbf{1}$, *and* $e^{2\pi L_3} = \mathbf{1}$.

Proof. (i) $e^{2\pi S_3} = e^{\pi e_4 e_5} = \cos\pi + \sin\pi\, e_4e_5 = -\mathbf{1}$ since $(e_4e_5)^2 = -\mathbf{1}$. The diagonal lift is a product of n commuting copies, one per constituent. An equivariant T intertwines the group elements, so $T(-\psi) = (-1)^n T\psi$, and $T = 0$ for even n; the same holds for $\mathrm{Hom}(S_n, S)$. For $n = 3$, $S\otimes\mathbb{C} = 2(\frac12,0)\oplus 2(0,\frac12)$ by (ii), and the joint weights of the commuting pair (S_3, T_3) on $S_3\otimes\mathbb{C}$ contain $(\frac12,0)$ and $(0,\frac12)$ forty times each, so $\dim_{\mathbb{C}}\mathrm{Hom} = 2\cdot 40 + 2\cdot 40$. (ii) Over $\mathbb{C}$, $\mathfrak{l}_{\mathrm{int}}\cong\mathfrak{sl}(2,\mathbb{C})$ and $S\otimes\mathbb{C} = 2(\frac12,0)\oplus 2(0,\frac12)$, $E\otimes\mathbb{C} = (\frac12,\frac12)$, $\Lambda^2E\otimes\mathbb{C} = (1,0)\oplus(0,1)$, while $V = E\oplus 2\cdot\mathbf{1}$ and $\Lambda^2V = \Lambda^2E\oplus 2E\oplus\mathbf{1}$. The Clebsch–Gordan rule gives $2^2 + 2^2 = 8$ for each of $\mathrm{End}(S)$, $S\to S\otimes E$ and $S\to S\otimes\Lambda^2E$, hence $8 + 2\cdot 8$ and $8 + 2\cdot 8 + 8$. (iii) $\sum_\mu\eta_{\mu\mu}\gamma_\mu\otimes e_\mu$ is the invariant element of $\mathrm{Cl}(E)\otimes E$ defined by η; the equivariance, its failure without η, and the ranks 8 of the composites are computed exactly. (iv) Constituent 2 is a trivial module under the spectator action, so $S_2\cong 8S$ and the count is $8\cdot 8$. On S the $\mathfrak{l}_G$ Casimir is $3\,\mathbf{1}$, the value on a four-vector, so $S\otimes\mathbb{C}$ is two copies of $(\frac12,\frac12)$, and $\dim\mathrm{Hom}(S, S\otimes 2E) = 2\cdot 8$. V is the field-strength module of $\mathfrak{l}_G$ (Theorem 3.3), $V\otimes\mathbb{C} = (1,0)\oplus(0,1)$, so $\mathrm{Hom}(S, S\otimes V^{(2)})\cong\mathrm{Hom}(S, S\otimes\Lambda^2E)$. □

Part (i) is the angular-momentum selection rule: a spin-$\frac12$ state cannot become a state of integer spin, and two spin-$\frac12$ constituents have integer spin [222, 160]. It restates Proposition 5.16 and Proposition 5.62(v) as a statement about vertices (inherited; verified in the realisation, e2-2, e2-6). Parts (ii) and (iii) are inherited Clebsch–Gordan counts, fixed in the realisation (e2-3, e2-4). The vertex c_1 is the structure of the coupling $\bar\psi\gamma^\mu\psi A_\mu$, and c_2 that of the Pauli coupling $\bar\psi\gamma^\mu\gamma^\nu\psi F_{\mu\nu}$. In both the boson is an added tensor factor E or Λ^2E, not a second constituent. Part (iv) is fixed (e2-5). Under the mixed action the second constituent itself is the boson. A fermion of constituent 1 has an equivariant vertex into $S\otimes V^{(2)}$, where $V^{(2)}$ is read through $\mathfrak{l}_G$ as a field strength, the frame field of constituent 2 in the field-strength reading. The mixed action is one of the five Lorentz actions at $n = 2$ (Theorem 5.15; Section 5.3.5), and nothing selects it, so this vertex is constructed. Under the diagonal action a fermion has vertices into the odd rungs only, and the first is $n = 3$.

5.5.2 Thresholds

The free Dirac Hamiltonian of one constituent is $H(\mathbf{p}) = me_1 + \sum_a p_a e_1e_{a+3}$, with $H(\mathbf{p})^2 = E^2\mathbf{1}$, $E = (m^2 + |\mathbf{p}|^2)^{1/2}$ (equation (2.4)). For two constituents with masses $m_1, m_2\ge 0$ and momenta $\mathbf{p}, \mathbf{q}$ put $E_1 = (m_1^2 + |\mathbf{p}|^2)^{1/2}$, $E_2 = (m_2^2 + |\mathbf{q}|^2)^{1/2}$,

$$a = E_1^2 + E_2^2, \qquad b = E_1^2E_2^2 - m_1^2m_2^2 = m_1^2|\mathbf{q}|^2 + m_2^2|\mathbf{p}|^2 + |\mathbf{p}|^2|\mathbf{q}|^2.$$

The ungraded two-particle operator is $H_u = H(\mathbf{p})\otimes\mathbf{1} + \mathbf{1}\otimes H(\mathbf{q})$. The graded composite is $H_g = H^{(1)}(\mathbf{p}) + H^{(2)}(\mathbf{q})$, with $H^{(k)}(\mathbf{p}) = m_kE_{(k,1)} + \sum_a p_aE_{(k,1)}E_{(k,a+3)}$ built from the tower's vectors.

Proposition 5.44 (Thresholds). *(i)* $\mathrm{spec}\,H_u = \{\pm E_1\pm E_2\}$, *the four sign choices on* 16 *dimensions each, and* $E_1 + E_2\ge m_1 + m_2$, *with equality iff* $\mathbf{p} = \mathbf{q} = 0$.

(ii) $H_g^2 = a\,\mathbf{1} + X$ *with* X *traceless and* $X^2 = 4b\,\mathbf{1}$*, and* $E_{(1,1)}E_{(2,1)}$ *anticommutes with* H_g*. Hence* $\operatorname{spec} H_g = \{\pm E_+, \pm E_-\}$ *with*

$$E_\pm^2 = E_1^2 + E_2^2 \pm 2\big(E_1^2E_2^2 - m_1^2m_2^2\big)^{1/2},$$

each on 16 *dimensions when the four values are distinct; when values coincide the multiplicities add, and* $E_- = 0$ *exactly when* $m_1m_2 = 0$ *and* $E_1 = E_2$ *(for* $m_1 = m_2 = 0$ *and* $\mathbf{p} = \mathbf{q} = (1,0,0)$ *the spectrum is* -2*,* 0*,* 2 *on* 16*,* 32*,* 16 *dimensions).* E_+ *is the largest eigenvalue. At rest* $\operatorname{spec} H_g = \{\pm M\}$*,* $M = (m_1^2 + m_2^2)^{1/2}$*, each on* 32 *dimensions.*

(iii) $\max(E_1, E_2) \le E_+ \le E_1 + E_2$*, and* $(E_1 + E_2)^2 - E_+^2 = 2\big((b + m_1^2m_2^2)^{1/2} - b^{1/2}\big)$*, which is positive iff* $m_1m_2 \neq 0$*. If* $m_1m_2 = 0$*, then* $E_\pm = |E_1 \pm E_2|$*: the graded composite has the additive spectrum.*

Proof. (i) $H(\mathbf{p}) \otimes \mathbf{1}$ and $\mathbf{1} \otimes H(\mathbf{q})$ commute, are diagonalisable with eigenvalues $\pm E_1$ and $\pm E_2$ on four dimensions each, and the joint eigenspaces have dimension $4 \cdot 4$. $E_k \ge m_k$. (ii) Write $H^{(1)} = m_1A + P$ and $H^{(2)} = m_2B + Q$ with $A = E_{(1,1)}$, $B = E_{(2,1)}$, and P, Q the momentum terms. Each $H^{(k)}$ squares to E_k^2, as on one constituent. A anticommutes with P and B, and commutes with Q; B anticommutes with Q and commutes with P; P and Q commute. Hence $X = \{H^{(1)}, H^{(2)}\} = 2(m_1AQ + m_2PB + PQ)$. The squares of the three terms are $m_1^2|\mathbf{q}|^2$, $m_2^2|\mathbf{p}|^2$ and $|\mathbf{p}|^2|\mathbf{q}|^2$, and the three mutual anticommutators vanish by the same rules, so $X^2 = 4b$. Each term is a product of distinct blades, so X is traceless, and H_g^2 has the eigenvalues $a \pm 2b^{1/2}$ on 32 dimensions each. $W = E_{(1,1)}E_{(2,1)}$ anticommutes with A, B and every term of P and Q by the blade rule (equation (1.3)), and $W^2 = -\mathbf{1}$, so $WH_gW^{-1} = -H_g$; W commutes with X, so each eigenspace of H_g^2 carries $\pm E_\pm$ on equal halves. (iii) $E_+^2 - E_1^2 = E_2^2 + 2b^{1/2} \ge 0$, and $(E_1 + E_2)^2 = a + 2E_1E_2 = a + 2(b + m_1^2m_2^2)^{1/2}$. □

Part (i) is the threshold of two free particles, inherited: a particle of mass M at rest decays into two of masses m_1, m_2 only if $M \ge m_1 + m_2$, since the final energy at zero total momentum is $E_1(\mathbf{p}) + E_2(-\mathbf{p}) \ge m_1 + m_2$ (e2-7). Part (ii) extends Proposition 5.62(iv) and Proposition 5.34(i) from rest to all momenta. The closed form and (iii) are fixed and elementary (e2-8: $(H_g^2 - a)^2 = 4b$ exactly at four integer data, the inequalities in 200 random samples to 10^{-9}). What follows, and what does not. At rest the graded composite has the single level $M = (m_1^2 + m_2^2)^{1/2}$, below the free-pair threshold $m_1 + m_2$; the gap $m_1 + m_2 - M$ is positive iff $m_1m_2 \neq 0$, and it is not proportional to m_1m_2 (at $m_1 = m_2 = m$ it is $(2-\sqrt{2})\,m$, linear in m); what is proportional to the product is $(m_1+m_2)^2 - M^2 = 2m_1m_2$. At equal momentum labels, $E_+(\mathbf{p}, \mathbf{q}) < E_1(\mathbf{p}) + E_2(\mathbf{q})$ iff $m_1m_2 \neq 0$, by (iii). It does not follow that the graded composite lies below the threshold of the free pair at the same total momentum. At $m_1 = m_2 = m$ and $\mathbf{p} = -\mathbf{q}$, the top level E_+ exceeds $m_1 + m_2 = 2m$ as soon as $|\mathbf{p}| > m/2$ (equality at $|\mathbf{p}| = m/2$, exactly), and at $\mathbf{p} = -\mathbf{q} = (0,0,2)$, $m = 1$, it is $E_+ \approx 4.45$ (checks y2-8 and z1-2a); at each total momentum only the lowest graded levels lie below $((m_1 + m_2)^2 + |\mathbf{P}|^2)^{1/2}$. So the inequality is not a bound-state condition, and the graded composite is not shown to be stable against dissociation. Both operators act on the same space S_2, nothing in the tower turns one into the other, and neither is derived from a common interacting theory with asymptotic states; "binding" and "dissociation" therefore compare two readings and describe no dynamical process. Reading $m_1 + m_2 - M$ as a binding energy is not supported.

5.5.3 Fields: one null field and two

At the level of fields three results of the energy analysis (Section 3.9) answer the question directly. We restate them as statements about one and two.

Corollary 5.45 (One field into two)**.** *(i) One null field does not become a massive pair. Every map commuting with the diagonal* $SO(3, \mathbb{C})$ *sends* $(z, 0)$*,* z *null, to* $(\alpha z, \beta z)$*, whose*

Gram matrix Q vanishes, so $M_{\text{pair}} = \frac{1}{2}(\sigma_1 + \sigma_2) = 0$ (Proposition 5.56(iii) and Theorem 5.57(ii)).

(ii) Two null fields that are not collinear do: $M_{\text{pair}} = |z_1 \cdot z_2| > 0$, and the copy rotation can move the whole of it into rest energy of the two constituents, in kinematic analogy with $\gamma\gamma \to e^+e^-$ (Theorem 5.57(iv),(v) and Example 5.58; [221]); M_{pair} is a rest-energy density, not an invariant mass.

(iii) The closure moves nothing between constituents: for every Hamiltonian the per-constituent invariants of the momentum fields are constant, so no closure dynamics fills an empty constituent or makes a null one massive (Corollary 5.54).

All three are restated (fixed there; re-verified for the Gram matrices, e2-9). Read as one and two: at the field level "one becomes two" covariantly only as "two become two". A single null field is shared between the constituents as literal replicas, and a massive pair needs a second, non-collinear field or a non-covariant third body that takes the momentum (Example 5.59).

5.5.4 Quanta: pairs emerge under boosts

The tower carries a counting operator of its own, the Fock number N. We now ask which Lie-algebra elements change it, and with which amplitudes. Throughout, "pair" means two Fock quanta of N on the finite space S_n, and "creates a pair" means that a Lie-algebra element has matrix elements with $\Delta N = \pm 2$, or that its exponential gives the Fock vacuum a component with $N = 2$. These are statements about a fixed finite-dimensional space at one momentum label. They are not statements about observable particle production: that would need a physical vacuum, a positive inner product preserved by the evolution, a particle-number observable defined from a Poincaré-invariant state space, and a dynamics that generates the transformation. None of these is specified here, and boosts are not unitary in the realisation. The first lemma refines Proposition 5.38 and Theorem 5.39.

Lemma 5.46 (The number and the split structure)**.** *(i) $N = \frac{3n}{2}\mathbf{1} + \sigma(K_n)$, and $\operatorname{ad} N = K_n$ on V_n. $(-1)^N = -\omega$ at $n = 1$ and is the volume element of $\mathrm{Cl}(6,6)$ at $n = 2$; it commutes with every bivector.*

(ii) $\mathfrak{so}(3n,3n) = \mathrm{Cent}(K_n) \oplus \mathrm{Anti}(K_n)$, the parts commuting and anticommuting with K_n, of dimensions $9n^2$ and $9n^2-3n$. The lift of an element of $\mathrm{Cent}(K_n)$ commutes with N. The lift of an element of $\mathrm{Anti}(K_n)$ maps each eigenspace $\{N = k\}$ into $\{N = k-2\} \oplus \{N = k+2\}$: it changes N by exactly ± 2.

(iii) An element of the maximal compact subalgebra $\mathfrak{so}(3n) \oplus \mathfrak{so}(3n)$ of $\mathfrak{so}(3n,3n)$ (an antisymmetric matrix in the realisation; called compact *below) commutes with N iff it commutes with G_n, iff it is a paired rotation $\left(\begin{smallmatrix} R & 0 \\ 0 & R \end{smallmatrix}\right)$, $R \in \mathfrak{so}(3n)$, in the basis of the temporal directions followed by their partners. The restriction to $\mathfrak{so}(3n) \oplus \mathfrak{so}(3n)$ is needed: the element $\left(\begin{smallmatrix} A & B \\ B & A \end{smallmatrix}\right)$ with A the rotation of (e_1, e_2) and $B = \mathrm{diag}(\frac{1}{2}, -\frac{1}{2}, 0)$ lies in $\mathfrak{so}(3,3)$, generates a periodic one-parameter group and commutes with K and N, but not with G (y2-13).*

(iv) Let $X \in \mathfrak{so}(3n,3n)$ be compact in this sense or Euclidean-symmetric (a pure boost); for other X the conclusion fails (y2-14). If $X \notin \mathrm{Cent}(K_n)$, then $\sigma(X)|0\rangle$ has a nonzero component with $N = 2$: $e^{t\sigma(X)}|0\rangle$ contains a pair at first order in t.

Proof. (i) and (ii) restate Proposition 5.38: $N_i = \frac{1}{2}(\mathbf{1} - e_i e_{i+3})$ and $\sigma(K) = -\frac{1}{2}\sum_i e_i e_{i+3}$, K is an η-antisymmetric involution, and $\operatorname{ad} K$ has eigenvalues $0, \pm 2$ on $\Lambda^2 V_n$. (iii) A compact element is $X = \mathrm{diag}(A, D)$ with $A, D \in \mathfrak{so}(3n)$. In the same basis $K_n = \left(\begin{smallmatrix} 0 & \mathbf{1} \\ \mathbf{1} & 0 \end{smallmatrix}\right)$ and $G_n = \left(\begin{smallmatrix} 0 & -\mathbf{1} \\ \mathbf{1} & 0 \end{smallmatrix}\right)$, and both $K_n X K_n^{-1}$ and $G_n X G_n^{-1}$ equal $\mathrm{diag}(D, A)$. So each commutation is $A = D$. By (ii) commuting with K_n is commuting with N, because the spin representation is faithful on bivectors. (iv) A

compact X is a combination of blades $e_a e_b$ with $\eta_{aa} = \eta_{bb}$, which are antisymmetric matrices, and a Euclidean-symmetric one of blades with $\eta_{aa} = -\eta_{bb}$, which are symmetric. So $\sigma(X)^\top = \mp\sigma(X)$. Write $\sigma(X) = R + Z + L$ with $[N, R] = 2R$, $[N, Z] = 0$, $[N, L] = -2L$, by (ii). Transposition exchanges raising and lowering parts ($a^- = (a^+)^\top$), so $L = \mp R^\top$, and $X \notin \mathrm{Cent}(K_n)$ gives $R \neq 0$. $R = \sum_{i<j} c_{ij} a_i^+ a_j^+$ in the $3n$ modes, and the states $a_i^+ a_j^+ |0\rangle$, $i < j$, are linearly independent, so $R|0\rangle \neq 0$; it is the $N = 2$ component of $\sigma(X)|0\rangle$. □

(i) and (ii) are inherited, the spin representation on the exterior algebra of a maximal isotropic subspace [201], and fixed in the realisation at $n = 1, 2, 3$ (e2-10, e2-11). (iii) and (iv) are fixed (e2-12, e2-26). Part (iii) sharpens Theorem 5.39, which works inside $\mathrm{Cent}(G_n)$: among all compact elements, the number-conserving ones are exactly the compact binding-compatible ones. By (iv) a compact element that does not commute with G_n creates a pair from the vacuum. Proposition 5.47 gives the examples, and e2-13 records which generators of $\mathfrak{l}_{\mathrm{int}}$, $\mathfrak{l}_G$ and the copy algebras lie where: $\mathfrak{l}_{\mathrm{int}} \cap \mathrm{Cent}(K) = \mathbb{R}\, T_1$, the pole boost; every other generator of $\mathfrak{l}_{\mathrm{int}}$, and J, has a pair part; $\mathfrak{l}_G \cap \mathrm{Cent}(K)$ is spanned by the L_k, and the K_i lie in $\mathrm{Anti}(K)$; the copy rotation $\varepsilon \otimes \mathbf{1}_6$ lies in $\mathrm{Cent}(K_2)$ and the copy boost $\varepsilon \otimes G$ in $\mathrm{Anti}(K_2)$.

Proposition 5.47 (Pair amplitudes on one constituent)**.** *(i) Interior boost.* $\mathrm{ad}\, T_2$ *has eigenvalues* ± 1 *and* 0*, so* $U_\zeta = e^{\zeta T_2}$ *is the boost of rapidity* ζ *in the plane* (e_1, e_5)*.* $U_\zeta|000\rangle = \cosh\frac{\zeta}{2}\,|000\rangle + \sinh\frac{\zeta}{2}\,|110\rangle$ *and* $\|U_\zeta|000\rangle\|^2 = \cosh\zeta$*.* U_ζ *is symmetric and not orthogonal.*

(ii) Field-strength boost. $\mathrm{ad}\, K_1$ *has eigenvalues* ± 1 *(twice) and* 0 *(twice), and* $e^{\zeta K_1}|000\rangle = \cosh\zeta\,|000\rangle - \sinh\zeta\,|011\rangle$*. The rotations* L_k *commute with* N*.*

(iii) Compact generators outside $\mathrm{Cent}(G)$*.* $e^{\theta S_1}|000\rangle = \cos\frac{\theta}{2}\,|000\rangle + \sin\frac{\theta}{2}\,|011\rangle$ *and* $e^{\theta J}|000\rangle = \cos\theta\,|000\rangle + \sin\theta\,|011\rangle$*. A spatial rotation through* π *about the axis of* S_1 *carries the vacuum to a state of two quanta.*

Proof. Each generator is a sum of commuting blades B with $B^2 = \pm\mathbf{1}$, so its exponential is a product of factors $\cos t + \sin t\, B$ or $\cosh t + \sinh t\, B$. In the realisation, with $\varepsilon = \sigma_x\sigma_z$ the matrix used on the copy index, $e_1e_5 = \varepsilon\otimes\varepsilon\otimes\mathbf{1}$, $e_5e_6 = \mathbf{1}\otimes\sigma_x\otimes\varepsilon$ and $e_2e_3 = \mathbf{1}\otimes\varepsilon\otimes\sigma_x$. Each maps $|000\rangle$ to the basis vector with two occupation numbers flipped, $|110\rangle$, $|011\rangle$ and $|011\rangle$, and $(e_1e_5)^2 = \mathbf{1}$, $(e_5e_6)^2 = (e_2e_3)^2 = -\mathbf{1}$. The terms $\frac{1}{2}e_3e_5$ and $-\frac{1}{2}e_2e_6$ of K_1 commute, square to $\frac{1}{4}\mathbf{1}$ and both flip modes 2 and 3, so $e^{\zeta K_1}|000\rangle = (\cosh^2\frac{\zeta}{2} + \sinh^2\frac{\zeta}{2}\, s)|000\rangle + \cosh\frac{\zeta}{2}\sinh\frac{\zeta}{2}\,(e_3e_5 - e_2e_6)|000\rangle$ with $s = -e_3e_5e_2e_6$ on $|000\rangle$; the signs, $s = 1$ and $(e_3e_5 - e_2e_6)|000\rangle = -2|011\rangle$, are computed (e2-14 to e2-16). □

All three parts are fixed (e2-14 to e2-16, at $\zeta = 0.3, 1, 2.5$ and $\theta = 0.4, 1.3, \pi$, to 10^{-11}). The amplitudes have the form of a fermionic Bogoliubov transformation [218, 219], inherited. A boost of rapidity ζ turns the vacuum into $\cos\vartheta\,|0\rangle + \sin\vartheta\,|\mathrm{pair}\rangle$ after normalisation, with $\tan\vartheta = \tanh\frac{\zeta}{2}$ for T_2 and $\tan\vartheta = \tanh\zeta$ for K_1. The normalised persistence $\cos^2\vartheta$ is $\cosh^2\frac{\zeta}{2}/\cosh\zeta$ and $\cosh^2\zeta/\cosh 2\zeta$, and it decreases to $\frac{1}{2}$ as $\zeta \to \infty$. So for these boosts the normalised pair probability $\tanh^2/(1 + \tanh^2)$ stays below $\frac{1}{2}$, which is its large-rapidity limit, where cosh and sinh become equal. This is not a Pauli bound: under the unitary rotation $e^{\pi S_1}$ of (iii) the pair probability is 1, and Pauli exclusion bounds the occupation of each mode pair by one (y2-17). The normalisation is needed because a boost does not preserve the Euclidean inner product of the realisation, which is the Fock inner product ($a_i^- = (a_i^+)^\top$): in the split realisation boosts are symmetric and rotations antisymmetric, and the Lorentz group, being non-compact, has no finite-dimensional unitary representation other than the trivial one [160] (inherited). In ordinary field theory the free-particle number is Poincaré invariant, and a vacuum turns into pairs only for an observer whose modes are not related to the inertial ones by a Poincaré transformation, as in the Unruh effect [220]. Here the inertial boosts of the interior four-plane already mix the modes, because N is built from the split structure K and not from the Lorentz algebra. The parallel with the Unruh effect is interpretive.

Part (iii) corrects an easy expectation. Compactness does not protect the number of quanta. A spatial rotation of the interior four-plane and the charge J, both compact, create pairs unitarily, and at $\theta = \pi$, resp. $\theta = \frac{\pi}{2}$, the vacuum is turned wholly into a pair. What protects N is commutation with the binding root (Lemma 5.46(iii)).

Proposition 5.48 (Pairs across constituents)**.** *At $n = 2$, let $U_\zeta = e^{\zeta\sigma(\varepsilon\otimes G)}$, the copy boost of rapidity ζ.*

(i) $\sigma(\varepsilon \otimes G) = -\sum_{i=1}^{3}(a^+_{1,i}a^+_{2,i} + a^-_{2,i}a^-_{1,i})$. *Hence* $U_\zeta|0\rangle = \prod_i(\cosh\zeta - \sinh\zeta\, a^+_{1,i}a^+_{2,i})|0\rangle$.

(ii) $\langle 0|U_\zeta|0\rangle = \cosh^3\zeta$; *the weight of* $U_\zeta|0\rangle$ *in* $\{N = 2k\}$ *is* $\binom{3}{k}\cosh^{2(3-k)}\zeta\,\sinh^{2k}\zeta$, *of total* $\cosh^3 2\zeta$, *and there is none in odd* N.

(iii) $N^{(1)} = N^{(2)}$ *on* $U_\zeta|0\rangle$: *every pair that emerges has one quantum in each constituent.*

Proof. (i) $\varepsilon\otimes G$ anticommutes with K_2, so its lift lies in $\Lambda^2 L_+ \oplus \Lambda^2 L_-$ (Proposition 5.38(ii)), and it pairs the modes $(1,i)$ and $(2,i)$ because $\varepsilon \otimes G$ maps the plane $(e^{(1)}_i, e^{(1)}_{i+3})$ to $(e^{(2)}_i, e^{(2)}_{i+3})$. The coefficients are computed exactly, with $\sigma(\varepsilon \otimes G)$ symmetric. The three terms commute. On the two-dimensional span of $|0\rangle$ and $a^+_{1,i}a^+_{2,i}|0\rangle$ the i-th term acts as $-\sigma_x$ and it annihilates the states with one quantum in the pair $(1,i),(2,i)$. (ii) and (iii) follow by expanding the product. □

Fixed (e2-17, exact for (i), and for (ii), (iii) at $\zeta = 0.3, 1, 2$ to relative 10^{-9}); the lift of the copy boost is checked against $\varepsilon \otimes G$ on all twelve generators. This is pair emergence across the constituents. The copy boost is a metric connection between two binding frames but not a gauge field (Proposition 9.22; Corollary 1.4). Each emerging pair has one quantum in each constituent, with normalised persistence $\cosh^6\zeta/\cosh^3 2\zeta = (1+\tanh^2\zeta)^{-3}$.

Proposition 5.49 (Lorentz-invariant states)**.** *(i) At $n = 1$ no nonzero state is $\mathfrak{l}_{\text{int}}$-invariant. The two projectors $\frac{1}{2}(\mathbf{1} \pm \omega)$ of Proposition 8.55 are the projectors onto odd and even N. The paired-mode projector of Example 1.6 is the projector onto $N \geq 2$.*

(ii) At $n = 2$ the states invariant under the diagonal $\mathfrak{l}_{\text{int}}$ form a space of dimension 8. Its weights in the sectors $N = 0, \ldots, 6$ (the traces of the invariant projector restricted to them) are $0, \frac{1}{2}, 2, 3, 2, \frac{1}{2}, 0$. It contains no state with $N = 0$ or $N = 6$, and its states of definite N form a space of dimension 2, at $N = 3$, on which $N^{(1)}_i + N^{(2)}_i = 1$ for $i = 1, 2, 3$.

(iii) Under the spectator action no nonzero state of S_2 is invariant. The projector $\Pi_0 = \frac{1}{2}(\mathbf{1}+H)$ of Theorem 8.57 commutes with $(-1)^N$ and not with N, and its compression to every sector $\{N = k\}$ has trace equal to half the dimension of that sector.

Proof. (i) $S \otimes \mathbb{C}$ contains only spin $\frac{1}{2}$. ω is $-(-1)^N$ (Lemma 5.46(i)). With $\varpi = \sum_i e_i e_{i+3} = \sum_i(1 - 2N_i)$ and the volume element ω, the projector $\frac{1}{2}\mathbf{1} - \frac{1}{4}(\varpi + \omega)$ of Example 1.6 vanishes exactly where at most one mode is occupied, by the table of signs given there. (ii) and (iii) are exact kernel computations; the weights are traces of the orthogonal projector onto the kernel. $H = E_{(1,2)}E_{(1,3)}E_{(2,1)}E_{(2,2)}$ is a blade of even grade, so it commutes with the volume element $(-1)^N$. It shares one index with each of the pole operators of the modes $(1,2)$, $(1,3)$, $(2,1)$, $(2,2)$ and none with the other two, so it anticommutes with those four and is a signed permutation of the Fock basis that flips exactly those four occupation numbers. Hence $H|0\rangle$ has $N = 4$, so H and Π_0 do not commute with N, and H has zero diagonal, so the compression of Π_0 to every sector has trace half its dimension. □

Fixed (e2-18). At one constituent the number of quanta is not Lorentz invariant (Proposition 5.40), while its parity is: the Lorentz-invariant projectors of Proposition 8.55 are the parity projectors, and the paired-mode projector, a function of N that is not a function of the parity, is not Lorentz invariant. The Fock vacuum is not Lorentz invariant at either rung. At the

second rung the Lorentz scalars of the diagonal action are spread over all numbers from 1 to 5; the two scalars of definite number carry one quantum in each mode pair i, shared between the constituents. The Lorentz-invariant vacuum of Theorem 8.57 is not a function of N. It selects half of every number sector, and so it is not a Fock vacuum.

5.5.5 A Lorentz-invariant number at even rungs

The Fock number is the lift of the involution K_n (Lemma 5.46(i)), and K_n commutes with no Lorentz algebra of the tower except through the pole boost. Any other η_n-antisymmetric involution defines a number in the same way. The question is whether one commutes with a Lorentz action.

Theorem 5.50 (An invariant number of quanta)**.** *(i) Let $X \in \mathfrak{so}(3n, 3n)$ with $X^2 = \mathbf{1}$. Its eigenspaces $V_\pm$ are totally null of dimension $3n$, and $N_X := \frac{3n}{2}\mathbf{1} + \sigma(X)$ has spectrum $0, \dots, 3n$ with multiplicities $\binom{3n}{k}$. If X commutes with a Lie algebra $\mathfrak{l} \subset \mathfrak{so}(3n, 3n)$, then N_X commutes with $\sigma(\mathfrak{l})$.*

(ii) Under the common binding action (the diagonal $\mathfrak{l}_G$, $(k_I, k_B, k_S) = (0, n, 0)$ in Theorem 5.15) such an X exists iff n is even. At $n = 2$ the only ones are $\pm\varepsilon \otimes G$, with $V_\pm = \{X^{(2)} = \pm GX^{(1)}\}$, the null subspaces of Proposition 5.55(ii). At $n = 4$, $(\varepsilon \oplus \varepsilon) \otimes G$ is one; it is not unique, since each of the three pairings of the four constituents gives one.

(iii) At $n = 2$ none exists under the other four actions: the commutants, of dimensions 7, 1, 28 and 15 for $(2, 0, 0)$, $(1, 1, 0)$, $(1, 0, 1)$ and $(0, 1, 1)$, consist of compact elements for the first two and of elements vanishing on the Lorentz-carrying subspace of constituent 1 for the last two. At $n = 3$ the action $(0, 2, 1)$ admits $(\varepsilon \otimes G) \oplus K$.

(iv) At $n = 2$, with $N_K := N_{\varepsilon\otimes G}$: the diagonal-$\mathfrak{l}_G$-invariant states form a space of dimension 4, with weights $1, 2, 1$ at $N_K = 0, 3, 6$. The vacuum $|0_K\rangle$ of N_K is $\mathfrak{l}_G$-invariant, has weights $\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8}$ at $N = 0, 2, 4, 6$, and equals the limit of $U_\zeta|0\rangle/\|U_\zeta|0\rangle\|$ as $\zeta \to -\infty$. Every vector of $V_\pm$ has components of equal length in the two constituents.

Proof. (i) For $v, w \in V_+$, $\eta(v, w) = \eta(Xv, w) = -\eta(v, Xw) = -\eta(v, w)$, so V_+ is totally null, and likewise V_-; as $V_n = V_+ \oplus V_-$ and null subspaces of $\mathbb{R}^{3n,3n}$ have dimension at most $3n$, both have dimension $3n$. Two pairs of complementary maximal null subspaces are conjugate under $O(\eta_n)$ (Witt's theorem), so $X = gK_ng^{-1}$, and a lift $\hat{g}$ of g to the Pin group gives $\sigma(X) = \hat{g}\sigma(K_n)\hat{g}^{-1}$. The spectrum of $\sigma(K_n) = N - \frac{3n}{2}$ is that of Lemma 5.46. Finally σ is a Lie algebra homomorphism. (ii) The commutant of the diagonal $\mathfrak{l}_G$ in $\mathfrak{so}(3n, 3n)$ is $\{A \otimes \mathbf{1}_6 + B \otimes G : A, B \in \mathfrak{so}(n)\}$, which acts on $z \in \mathbb{C}^n \otimes \mathbb{C}^3$ as $(A + iB) \otimes \mathbf{1}_3$ (Proposition 9.19(i), and Proposition 5.27(iii) at $n = 2$; commuting with the diagonal $\mathfrak{l}_G$ already forces commuting with G_n, since $\mathrm{End}_{\mathfrak{l}_G}(V) = \mathrm{span}\{\mathbf{1}, G\}$). $X^2 = \mathbf{1}$ requires $(A+iB)^2 = \mathbf{1}$, so $A+iB$ is invertible; but a complex antisymmetric matrix of odd size has $\det(A+iB) = \det(A+iB)^\top = -\det(A+iB) = 0$. For even n, $A+iB = i\varepsilon \oplus \cdots \oplus i\varepsilon$ squares to $\mathbf{1}$. At $n = 2$ the solutions of $(a\,\varepsilon \otimes \mathbf{1}_6 + b\,\varepsilon \otimes G)^2 = \mathbf{1}$ are $a = 0$, $b = \pm 1$. (iii) and (iv) are computed. For the limit, $|0_K\rangle$ is the eigenvector of $\sigma(\varepsilon \otimes G)$ for its lowest eigenvalue -3, and $\langle 0|0_K\rangle \neq 0$. On $V_\pm$, $|X^{(2)}| = |GX^{(1)}| = |X^{(1)}|$. □

Part (i) is inherited, the correspondence between polarisations and Fock structures [201, 219]; the spectrum and the commutation are fixed in the realisation at $n = 2$ (e2-19) and on S_4, of dimension 4096, where $\sigma((\varepsilon \oplus \varepsilon) \otimes G)$ was checked to lift its vector and to commute with the six lifted diagonal $\mathfrak{l}_G$ generators in sparse integer arithmetic (e2-22). Part (ii) is fixed; the parity argument is general, and the exact checks cover $n = 2, 3, 4$ (e2-19, e2-21). Part (iii) is fixed and re-verifies the dimensions of Theorem 5.15 (e2-21). Part (iv) is fixed (e2-20; the limit to 10^{-9} at $\zeta = -12$).

So a Lorentz-invariant number of quanta exists, under the common binding action, exactly at even rungs. At the second rung it is unique up to $N_K \mapsto 6 - N_K$. Its quanta are not quanta of one constituent: $V_\pm$ is the graph of $\pm G$ from constituent 1 to constituent 2, and N_K does not commute with $N^{(1)}$, since $\sigma(\varepsilon \otimes G)$ changes $N^{(1)}$ and $N^{(2)}$ by one each (Proposition 5.48). Its vacuum, read in the Fock structure of the constituents, is a condensate of pairs with one quantum in each constituent, the infinite copy boost of the Fock vacuum (fixed). Read as a particle number, N_K counts quanta that are Lorentz invariant under the binding action and belong to two constituents at once (interpretive). The parity pattern can be set beside Proposition 5.36(iv): under the diagonal interior action an imaginary unit is selected exactly at odd rungs, and under the diagonal binding action an invariant number exists exactly at even rungs. The two statements concern different Lorentz actions, and the juxtaposition is a reading.

process	sense	result	class
$S \to S_n$, common interior action	module	none for even n; 160 maps at $n = 3$	inherited (selection rule); fixed
$S \to S \otimes E$, $S \otimes \Lambda^2 E$; $S \to S \otimes V^{(2)}$ (mixed)	module	Clifford multiplication, 8 maps each	inherited counts; fixed; mixed action constructed
free particle of mass $M \to$ two	mass shell	only if $M \geq m_1 + m_2$	inherited
graded composite $\to$ free pair	mass shell	$E_+ < E_1 + E_2$ iff $m_1 m_2 \neq 0$ at equal labels; above threshold at large relative momentum	fixed; binding not supported
one null field $\to$ massive pair	field	no covariant map; closure moves nothing	obstructed (restated)
two null fields $\to$ pair with $M_{\text{pair}} > 0$	field	$M_{\text{pair}} = \lvert z_1 \cdot z_2 \rvert$, a rest-energy density	inherited (restated)
boosts ($\mathfrak{l}_{\text{int}}$, $\mathfrak{l}_G$, copy)	quanta	pairs, fermionic Bogoliubov amplitudes	fixed; form inherited
compact, outside Cent(G_n) (S_k, J)	quanta	pairs, unitarily	fixed
compact, inside Cent(G_n) (paired rotations)	quanta	N conserved	fixed
copy rotation	quanta	one quantum hops; Rabi at unequal rates	fixed; two-level inherited; i constructed
N_X under $(0, n, 0)$	quanta	exists iff n even; vacuum a pair condensate	fixed; particle reading interpretive
dynamical pair-creating agent	—	—	open

Table 5.7: Processes that take one to two, by sense of "particle".

5.5.6 The copy rotation moves one quantum

Proposition 5.51 (Hopping and Rabi transfer). *(i)* $\hat{C} = \sigma(\varepsilon \otimes \mathbf{1}_6) = \sum_i (a^+_{2,i} a^-_{1,i} - a^+_{1,i} a^-_{2,i})$. *It commutes with* N *and not with* $N^{(1)}$; $\hat{C} a^+_{1,i}|0\rangle = a^+_{2,i}|0\rangle$ *and* $\hat{C} a^+_{2,i}|0\rangle = -a^+_{1,i}|0\rangle$, *and* $e^{\theta \hat{C}} a^+_{1,1}|0\rangle = \cos\theta\, a^+_{1,1}|0\rangle + \sin\theta\, a^+_{2,1}|0\rangle$. $e^{\theta\hat{C}}$ *has no matrix element between different values of* N, *and* $e^{2\pi\hat{C}} = \mathbf{1}$.

(ii) *At* $n = 3$ *the copy rotations commute with* K_3, *and their lifts with* N; *the copy boosts anticommute with* K_3.

(iii) *With rates* κ_1, κ_2 *and coupling* β, *let* $\mathsf{H} = \kappa_1 N^{(1)} + \kappa_2 N^{(2)} + i\beta\hat{C}$, *with an external* i. H *is Hermitian and commutes with* N. *On the sector* $N = 1$, *in the basis* $a^+_{1,i}|0\rangle$, $a^+_{2,i}|0\rangle$, *it is* $\left(\begin{smallmatrix} \kappa_1 & -i\beta \\ i\beta & \kappa_2 \end{smallmatrix}\right) \otimes \mathbf{1}_3$, *whose eigenvalues are the lines* $\omega_\pm$ *of Theorem 5.31(ii). From* $a^+_{1,1}|0\rangle$ *the probability of finding the quantum in constituent 2 is* $s \sin^2(\Omega t/2)$, *with* $s = 4\beta^2/((\kappa_2 - \kappa_1)^2 + 4\beta^2)$ *and* $\Omega = ((\kappa_2 - \kappa_1)^2 + 4\beta^2)^{1/2}$.

statement	class	checks
conventions: tower, ungraded copy, common bivectors, CAR, Fock vacuum	restated (Proposition 5.20, Proposition 5.21)	e2-1
Proposition 5.43(i): no vertex $S \leftrightarrow S_2$; $(-1)^n$; 160 at $n = 3$	inherited; fixed in realisation	e2-2, e2-6
Proposition 5.43(ii), (iii): counts $8, 8, 24, 8, 32$; Clifford multiplication	inherited counts; fixed	e2-3, e2-4
Proposition 5.43(iv): spectator 64; mixed 16 and 8; $\mathfrak{l}_G$ Casimir 3	fixed; mixed action constructed	e2-5
Proposition 5.44(i): free-pair threshold	inherited	e2-7
Proposition 5.44(ii), (iii): $(H_g^2 - a)^2 = 4b$; $E_+ < E_1 + E_2$ iff $m_1 m_2 \neq 0$; crossover $\lvert\mathbf{p}\rvert = m/2$	fixed; binding not supported	e2-8, y2-8, z1-2a
Corollary 5.45: one null field, two null fields, closure	restated (Proposition 5.56, Theorem 5.57, Corollary 5.54)	e2-9
Lemma 5.46(i), (ii): $N = \frac{3n}{2} + \sigma(K_n)$; $\mathrm{Cent}(K_n) \oplus \mathrm{Anti}(K_n)$	restated (Proposition 5.38); inherited	e2-10, e2-11
Lemma 5.46(iii), (iv): compact and number-conserving = paired rotations; outside $\mathrm{Cent}(K_n)$ the vacuum acquires a pair	fixed	e2-12, e2-13, e2-26
Proposition 5.47: Bogoliubov amplitudes; pairs from S_1 and J	fixed; form inherited; Unruh analogy interpretive	e2-14 to e2-16
Proposition 5.48: copy boost, pairs across constituents	fixed	e2-17
Proposition 5.49: invariant states at $n = 1, 2$; Proposition 8.55, Theorem 8.57	fixed (compared with these)	e2-18
Theorem 5.50: invariant number iff n even; N_K at $n = 2$	fixed; (i) inherited; reading interpretive	e2-19 to e2-22
Proposition 5.51: hopping, ladder, Rabi transfer	fixed; Rabi inherited; external i constructed	e2-23 to e2-25
Section 5.5.7: dynamical agent of pair creation	open	—

Table 5.8: Claims of Section 5.5, their classes and the checks printed by `e2_processes.py`.

Proof. (i) $\varepsilon \otimes \mathbf{1}_6$ commutes with K_2 and maps the vector of each mode of constituent 1 to the same mode of constituent 2, so its lift lies in $L_+ \otimes L_-$ and is a hopping term; the coefficients are computed. The rest follows from $[\hat{C}, a^{\pm}_{1,i}] = a^{\pm}_{2,i}$ and $[\hat{C}, a^{\pm}_{2,i}] = -a^{\pm}_{1,i}$, and from $\hat{C}$ being a sum of six commuting terms of square $-\frac{1}{4}$ (Proposition 5.22(ii)). (ii) $GK = -KG$. (iii) $\hat{C}$ is real antisymmetric, so $i\beta\hat{C}$ is Hermitian. On the sector $N = 1$, $\hat{C}$ acts as $\varepsilon \otimes \mathbf{1}_3$, and the two-level formula follows. □

Fixed (e2-23 to e2-25; the transfer probability at $(\kappa_1, \kappa_2, \beta) = (1, 3, 0.4)$ to 2×10^{-16}, with weight below 10^{-24} outside $N = 1$, and complete transfer at equal rates at $t = \pi/(2\beta)$). The two-level formula is inherited [223]. The external i is constructed, as in Proposition 5.62(i); on fields the same matrix acts with G_2 as the imaginary unit and gives the per-copy rest energies of Section 5.6.4. Read as one and two: one quantum can be shared between constituents, in any proportion at equal rates and up to the fraction s at unequal rates, but it is never doubled, because $\hat{C}$ commutes with N. At equal rates the coupling is removable (Theorem 5.31(iii)), so which constituent carries the quantum is gauge dependent if the copy rotation is gauged, as the per-copy rest energy is (Proposition 5.60). At unequal rates the partial transfer cannot be removed while the rates are kept fixed. Up the ladder the copy rotations $\mathfrak{so}(n)$ hop quanta among n constituents and the copy boosts create pairs across them (e2-24).

5.5.7 Answer: which processes take one to two

Table 5.7 lists the processes. Read together they say the following. In the sense of Lorentz modules, one fermion never becomes two under the common interior action. It becomes a fermion and a boson, by Clifford multiplication, and under the mixed action the boson can be the second constituent read as a field strength (fixed; the mixed action constructed). In the sense of mass shells, the graded composite at rest lies below the free-pair threshold iff $m_1 m_2 \neq 0$, and at equal momentum labels its top level lies below the free-pair energy; at fixed total momentum its levels with large relative momentum lie above the threshold, so no bound state and no binding energy are established (fixed; binding not supported). In the sense of fields, one null field becomes a massive pair only together with a second field (restated). In the sense of quanta, no element of $\mathfrak{so}(3n, 3n)$ changes N by an odd number, so one quantum never becomes two. The vacuum becomes partly a pair under every compact or Euclidean-symmetric element outside $\mathrm{Cent}(K_n)$: the boosts of $\mathfrak{l}_{\mathrm{int}}$ other than the pole boost, the boosts of $\mathfrak{l}_G$, the copy boost, and the compact generators outside $\mathrm{Cent}(G_n)$, including the spatial rotations of the interior four-plane and the charge J (fixed). The compact binding-compatible generators, and the copy rotation among them, never create quanta; the copy rotation moves one (fixed). The number of Fock quanta is therefore not Lorentz invariant under any action of the tower, since every action contains, on some constituent, the boosts of $\mathfrak{l}_{\mathrm{int}}$ or of $\mathfrak{l}_G$. Under the common binding action an invariant number operator of the specified polarisation type exists exactly at even rungs, where its quanta belong to two constituents at once (fixed); other actions with spectator constituents can also admit one, as Theorem 5.50(iii) shows. Whether a dynamical mechanism of the arena supplies the non-binding-compatible or non-compact generator that creates a pair is open.

Up the ladder. Rung by rung the following changes, each fixed. Under the common interior action $e^{2\pi S_3} = (-1)^n$, so a fermion has vertices into odd rungs only. $\mathfrak{so}(3n, 3n)$ splits into $9n^2$ number-conserving and $9n^2 - 3n$ pair-creating directions, and the compact number-conserving ones are the $\frac{3}{2}n(3n-1)$ paired rotations. The copy rotations $\mathfrak{so}(n)$ hop quanta among the constituents and the copy boosts create pairs across them. An invariant number under the common binding action exists exactly at even n, uniquely up to $N_K \mapsto 6 - N_K$ at $n = 2$ and in several ways at $n = 4$.

5.6 Duplication in motion: the balanced pair

Two further remarks are tested here: that the b-particle of Section 9.1 becomes decisive once two or more manifolds are present, and that one should find the process by which a unitary (undivided) energy results in a duplication, from $\mathrm{Cl}(3,3)$ to $\mathrm{Cl}(6,6)$. Duplication is taken literally: two copies of the arena whose tangent spaces add, $V_2 = V \oplus V \cong \mathbb{R}^{6,6}$, which is grade one of $\mathrm{Cl}(6,6) \cong \mathrm{Cl}(3,3)\hat{\otimes}\mathrm{Cl}(3,3)$ (equation (1.2); the graded tensor product is inherited). It is followed in motion, inside the closure of Section 4.4, and at a single field point. In brief: (1) with one constituent and Lorentz-class inertia a constant nonzero rate forces a null momentum field, while with two or more a codimension-two balance surface carries massive constituents at an exactly constant rate with bounded frames (Theorem 5.52, restating Proposition 4.33 with the dimension count added). This is the precise sense in which a duplicate does what one copy cannot, and it needs no b-field. (2) The closure never creates a pair (Corollary 5.54). (3) The admissible b-field between the manifolds, the copy rotation, duplicates a field literally, and no Lorentz-covariant operation makes a massive pair out of a single null field (Proposition 5.56); a natural candidate that rotates a single direction between the copies is a different, non-covariant element (Example 5.59). (4) The rest-energy density of a duplicated field point is $M_{\mathrm{pair}} = \frac{1}{2}(\sigma_1 + \sigma_2)$, and the copy rotation can move all of it into the two copies (Theorem 5.57 and Example 5.58); the kinematics of $\gamma\gamma \to e^+e^-$ is an analogy only. (5) The rest energy of each copy separately is not invariant under the copy rotation, hence gauge dependent if that rotation is gauged, and with equal inertias the closure's own symmetry maps every balanced pair to one null constituent (Proposition 5.60): pair creation is physical only where the copies are distinguished. (6) On the Dirac side the graded duplication adds Dirac masses in quadrature (Proposition 5.62). (7) The zero-point energy stays zero, but an inter-manifold b at unequal rates can bring one line of the motion to zero exactly, and there the motion is unbounded (Proposition 5.63).

Conventions. As in Section 4.5 for one constituent. For two, $V_2 = V \oplus V$ with coordinates $(X^{(1)}; X^{(2)})$, $\eta_2 = \mathbf{1}_2 \otimes \eta$, $G_2 = \mathbf{1}_2 \otimes G$; in the tower $e_a^{(1)} = e_a \otimes \mathbf{1}$, $e_a^{(2)} = \omega \otimes e_a$, $\omega = e_1 \cdots e_6$, and V_2 is spanned by the twelve $e_a^{(k)}$. $Z = (z_1, z_2) \in \mathbb{C}^2 \otimes \mathbb{C}^3$, $z_k = z(X^{(k)})$, so G_2 acts as i; per copy $E_6^{(k)}, E_0^{(k)}, \mathbf{S}^{(k)}$ and $q_k = z_k \cdot z_k$ are those of (3.17), and $q_{\mathrm{tot}} = q_1 + q_2$. The Gram matrix $Q_{kl} = z_k \cdot z_l$ (complex bilinear, symmetric) has singular values $\sigma_1 \geq \sigma_2 \geq 0$; $H_{\mathrm{sum}} = \sum_k z_k z_k^\dagger$. The *copy rotation* is generated by $E := \epsilon \otimes \mathbf{1}_6$, $\epsilon = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$: $R_\varphi = e^{\varphi E}$ maps $(z_1, z_2) \mapsto (\cos\varphi\, z_1 - \sin\varphi\, z_2,\ \sin\varphi\, z_1 + \cos\varphi\, z_2)$, identically in all six directions; $E^2 = -\mathbf{1}$, $[E, G_2] = 0$, E is antisymmetric for η_2 and for the Euclidean form. $T(z)$ is the stress tensor of Proposition 3.94, $P(t) = T(z)t$ the density seen by a unit future timelike observer t, $P_{\mathrm{sum}}(t) = \sum_k T(z_k)t$, and

$$M_{\mathrm{pair}} := \inf_t |P_{\mathrm{sum}}(t)|,$$

the rest-energy density of the duplicated field point: a local invariant of the summed stress tensor, not the invariant mass of two photons. In the closure the per-copy quantities are those of the momentum fields $\pi^{(k)} = \eta^{-1}p^{(k)}$, $w_k = q(z(\pi^{(k)}))$; inertias are of Lorentz class, $N_k = (\nu_{k,1}\eta^{-1} + \nu_{k,2}G\eta^{-1})/m_k$, with $c_k := \nu_k/m_k \in \mathbb{C}$ and $R_{\mathrm{eff}}e^{i\varphi_{\mathrm{eff}}} := \sum_k c_k w_k(0)$ (Proposition 4.33). Two Lorentz actions occur: the per-copy algebras $\mathfrak{l}_G^{(1)} \oplus \mathfrak{l}_G^{(2)}$, and the diagonal $\mathfrak{l}_{G,\Delta}$ ($k_B = 2$ in Theorem 5.15). Each $E_0^{(k)}$ is invariant under both; M_{pair}, defined with one observer for both copies, is invariant under the diagonal action but not under the per-copy one, since a boost of one copy alone changes it. A constituent is called *massive* when its momentum field is not null, $E_0(\pi^{(k)}) \neq 0$: massive in the sense of the dictionary of Section I.2, an interpretive reading that does not involve $mc^2 = \hbar\kappa$. Rates are rates: $mc^2 = \hbar\kappa$ is an added identification (Section 4.5). Identifiers d6-n refer to `d6_pair_balance.py` (integrations DOP853, relative tolerance 10^{-12}; d6-0 checks the conventions).

5.6.1 Mass with a constant rate needs a second constituent

Theorem 5.52 (Constant rate with massive constituents)**.** *Let one rotor act on n constituents with Lorentz-class inertias and all $c_k \neq 0$.*

(i) Along every solution $w_k(\tau) = e^{2i\theta(\tau)}w_k(0)$, so R_{eff} is constant. The rate h/I_0 is constant and nonzero iff $R_{\text{eff}} = 0$ and $h(0) \neq 0$; then $\theta = \kappa\tau$ and every frame moves on the circle $x_k(\tau) = x_k(0) + (e^{2\kappa\tau G} - \mathbf{1})(2\kappa G)^{-1}N_k p^{(k)}(0)$, of diameter $|N_k p^{(k)}(0)|/|\kappa|$.

(ii) For $n = 1$, $R_{\text{eff}} = 0$ iff $E_0(\pi) = 0$: no datum with a constant nonzero rate has a massive momentum field.

(iii) For $n \geq 2$ the balance set $\mathcal{B} = \{\sum_k c_k w_k = 0\} \subset \mathbb{R}^{6n}$ is, away from 0, a smooth submanifold of codimension two; its part on which every $E_0(\pi^{(k)})$ vanishes has dimension $4n < 6n - 2$, so the data with a constant nonzero rate and some $E_0(\pi^{(k)}) > 0$ are open and dense in $\mathcal{B} \times \{h \neq 0\}$. For $n = 2$ both constituents are then massive, with $|c_1|E_0(\pi^{(1)}) = |c_2|E_0(\pi^{(2)})$.

Proof. (i) With a common rotor the momenta evolve as $p^{(k)} = e^{-\theta G}p^{(k)}(0)$ (Proposition 4.23 and the closing remark of Theorem 4.22, $\mathfrak{s}_n = \mathbb{R} \ltimes_{G_n} V_n$), so $w_k = e^{2i\theta}w_k(0)$ (Theorem 4.40) and, by Theorem 4.26(ii) per constituent, $U(\theta) = \sum_k \frac{1}{2}p^{(k)\top}N_k p^{(k)} = \frac{1}{2}\operatorname{Re}\big(e^{2i\theta}\sum_k c_k w_k(0)\big)$, whence $I_0\ddot{\theta} = R_{\text{eff}}\sin(2\theta + \varphi_{\text{eff}})$. If $R_{\text{eff}} = 0$, $\ddot{\theta} = 0$; if $R_{\text{eff}} > 0$ and h is constant, then $\sin(2\theta + \varphi_{\text{eff}}) \equiv 0$, so θ is constant and $h = 0$. Lorentz-class N_k anticommutes with G, so $\dot{x}_k = e^{\theta G}N_k e^{-\theta G}p^{(k)}(0) = e^{2\theta G}N_k p^{(k)}(0)$ (Theorem 4.27), which integrates to the circle. (ii) $R_{\text{eff}} = |c|\,|w(p)| = 2|c|E_0(\pi)$. (iii) $F(p) = \sum_k c_k w(p^{(k)})$ has $dF_p(\delta) = 2\sum_k c_k\, z(\pi^{(k)}) \cdot z(\delta\pi^{(k)})$. At $p \neq 0$ some $z(\pi^{(k)}) \neq 0$, and $u \mapsto 2c_k z(\pi^{(k)}) \cdot u$ maps $\mathbb{C}^3$ onto $\mathbb{C}$, so 0 is a regular value of F on $\mathbb{R}^{6n} \setminus 0$. The null set $\{w = 0\} \subset \mathbb{R}^6$ is the cone $|\mathbf{B}| = |\mathbf{E}|$, $\mathbf{B} \cdot \mathbf{E} = 0$, of dimension four, and the all-null part of $\mathcal{B}$ is its n-fold product. Points with $w(p^{(1)}) = 1$, $w(p^{(2)}) = -c_1/c_2$ show that the massive part is not empty. □

Items (i) and (ii) restate Proposition 4.33 and Corollary 4.29 (fixed there); (iii) is fixed, the dimension count being added. Checks: $\frac{1}{2}p^\top Np = \operatorname{Re}(\nu w(p))/(2m)$ exactly, and the rank of dF at random balanced points for $n = 1, 2, 3$ (d6-1a,b); integrations of the full two-constituent closure, rotor, momenta and frames (d6-2a–d; $I_0 = 1$, momenta drawn at random from the script's seeded generator, so the values quoted are those of that sample). With equal inertias and $p^{(2)} = -Gp^{(1)}$ ($E_0(\pi^{(k)}) = 4.95$) the rate stays constant to 1.6×10^{-14} over $\tau \in [0, 60]$ and the frames lie on the circles of (i) to 4×10^{-12}; a generic balanced pair, and unequal inertias ($m = 1, 2.5$; $\nu = 1 + 0.3i$, $0.4 - 0.7i$), give the same to 10^{-13}. Off the surface (the unequal-inertia pair with $p^{(2)}$ perturbed at random, $R_{\text{eff}} = 0.0752$, rotating) the rotor follows $I_0\ddot{\theta} = R_{\text{eff}}\sin(2\theta + \varphi_{\text{eff}})$ to 2×10^{-10} and frame 1 drifts by 0.1795 per half turn, exactly additively (Theorem 4.31 for several constituents).

Remark 5.53 (The Lorentz requirement is what empties (ii))**.** $N = \operatorname{diag}(1, -1, 0, -1, 1, 0)/m$ anticommutes with G but is not $\mathfrak{l}_G$-invariant. For $p_0 = e^1 + e^2$, $z(\pi_0) = (1, 1, 0)$ and $E_0(\pi_0) = 1$, while $U(\theta) = \frac{1}{2}p_0^\top e^{2\theta G}Np_0 \equiv 0$: the rate is exactly constant and the frame moves on a circle (d6-1c). One constituent can thus combine a constant rate, a massive momentum field and a bounded, moving frame only through a cancellation between directions that breaks Lorentz invariance: a bounded clamped frame needs $N_c p_0 = 0$, and a nonzero Lorentz-class N_a then keeps U constant only on null data. The balance surface is the Lorentz-invariant form of the same cancellation, between constituents. Without the bounded frame no cancellation is needed: a duality-class inertia, $[N, G] = 0$, gives a constant rate for every datum, massive ones included, with $U = \frac{1}{2}p_0^\top Np_0$ constant, and the frame drifts at the velocity Np_0 (Corollary 4.43(iii) and Theorem 4.27; restated). Fixed, as a counterexample to (ii) without the hypothesis.

Corollary 5.54 (The closure creates no pair)**.** *For every Hamiltonian on $\mathfrak{s}_n^*$, including any coupling of the constituents through H, $p^{(k)}(\tau) = e^{-\theta(\tau)G}p^{(k)}(0)$ with one angle θ. Hence E_6, E_0*

and $\mathbf{S}$ *of each* $\pi^{(k)}$*, the Hermitian Gram matrix* $\bar{z}_k \cdot z_l$ *and the moduli* $|Q_{kl}|$ *of the momentum fields are constants of motion, and* $Q \mapsto e^{2i\theta}Q$*. No closure Hamiltonian turns a null constituent into a massive one, fills an empty copy, or moves rest energy between constituents.*

Proof. The Lie–Poisson equations of $\mathfrak{s}_n$ are (4.18) with $G_n = \bigoplus_k G$; the first integrates to $p = e^{-\theta G_n} p_0$, $\dot{\theta} = H_h$. □

Obstructed, for the momentum field and every H; it extends Theorem 4.40 to n constituents. With a random Hamiltonian that couples the two copies (quartic in h, trigonometric in p) these quantities stay constant to 2×10^{-13} and Q follows $e^{2i\theta}Q$ to the same accuracy (d6-3). Inside the closure a pair is therefore a label of the initial data, carried and not generated, and the balance surface is where such labels are compatible with a constant rate.

5.6.2 The balanced pairs

Proposition 5.55 (Balanced pairs)**.** *Let* $z_k = z(\pi^{(k)})$.

(i) Equal inertias. With $a = \frac{1}{2}(z_1 - iz_2)$ *and* $b = \frac{1}{2}(z_1 + iz_2)$*,* $q_1 + q_2 = 4\, a \cdot b$ *and* $q_1 = a \cdot a + 2a \cdot b + b \cdot b$*. The balance surface is the quadric* $a \cdot b = 0$*, and on it* $q_2 = -q_1$ *and* $E_0^{(1)} = E_0^{(2)} = \frac{1}{2}|a \cdot a + b \cdot b|$.

(ii) $EG_2 = \epsilon \otimes G$ *is an* η_2*-antisymmetric involution, the noncompact generator of* $\mathfrak{g}(0,2,0) \cong \mathfrak{so}(2,\mathbb{C})$*. Its eigenspaces* $V_\pm = \{X^{(2)} = \pm GX^{(1)}\}$ *are six-dimensional and totally* η_2*-null, and* $E = \mp G_2$ *on* $V_\pm$*. A momentum field lies in* V_+ *iff* $p^{(2)} = -Gp^{(1)}$ *iff* $b = 0$*, and in* V_- *iff* $p^{(2)} = Gp^{(1)}$ *iff* $a = 0$*; so* $V_\pm$ *lie in the balance surface, and on them the copy rotation* R_φ *is the duality rotation* $e^{\mp\varphi G_2}$ *of both copies.*

(iii) On V_+*,* $z_2 = iz_1$*:* $w_2 = -w_1$*, and the two momentum fields have equal* E_6*,* $\mathbf{S}$ *and* E_0*; when* $q_1 > 0$ *the first is of magnetic and the second of electric type.* $T(z_2) = T(z_1)$*, so the summed stress tensor is* $2T(z_1)$ *and* $M_{\text{pair}} = 2E_0^{(1)}$*, although* $q_{\text{tot}} = 0$.

(iv) Unequal inertias. If $q_1 \neq 0$ *the balanced pairs are* $z_2 = \rho\Lambda z_1$ *with* $\Lambda \in SO(3,\mathbb{C})$ *and* $\rho^2 = -c_1/c_2$*, and* $|c_1|E_0^{(1)} = |c_2|E_0^{(2)}$.

Proof. (i) Expand. (ii) $(\epsilon \otimes G)^2 = \epsilon^2 \otimes G^2 = \mathbf{1}$ and $(\epsilon \otimes G)^\dagger = \epsilon^\top \otimes G^\dagger = -\epsilon \otimes G$. The eigenvalue $+1$ means $-GX^{(2)} = X^{(1)}$, i.e. $X^{(2)} = GX^{(1)}$, and then $\eta_2(X,Y) = \eta(X^{(1)}, Y^{(1)}) + \eta(GX^{(1)}, GY^{(1)}) = 0$ since G is an anti-isometry; on V_+, $E(X^{(1)}, X^{(2)}) = (-GX^{(1)}, X^{(1)}) = -G_2(X^{(1)}, X^{(2)})$. Finally $\eta^{-1}p^{(2)} = G\eta^{-1}p^{(1)}$ iff $p^{(2)} = -Gp^{(1)}$, as $\eta G = -G\eta$, and $z_2 = iz_1$ iff $b = 0$. (iii) T is duality invariant (Proposition 3.94), and Proposition 3.94 gives $|2T(z_1)t| = 2E_0^{(1)}$ for every t. (iv) $c_1 q_1 + c_2\rho^2 q(\Lambda z_1) = c_1(q_1 - q_1) = 0$; conversely $q(z_2/\rho) = q_1 \neq 0$, and $SO(3,\mathbb{C})$ is transitive on each complex sphere $\{q = q_1\}$ (Theorem 3.92). □

The first sentence of (iii) restates the last sentence of Proposition 4.33 (fixed there); (i), (ii) and (iv) are fixed (that $V_\pm$ are null, and are exactly where the copy rotation is a duality rotation, could have failed); "magnetic and electric pair" and "duality-balanced" are readings. Checked exactly (d6-4a–c) and numerically (d6-4d). So the balanced pair of Proposition 4.33 is a magnetic and an electric constituent a quarter duality turn apart, with equal rest energies, whose duality phases the rotor turns together. Its total q vanishes, but q_{tot} is not the rest energy of the pair: its summed density has rest energy $2E_0^{(1)}$ for every observer. Thus $\frac{1}{2}|q_{\text{tot}}|$ is what the rotor sees ($R_{\text{eff}} = |\nu|\,|q_{\text{tot}}|/m$ for equal inertias) and $M_{\text{pair}} \geq \frac{1}{2}|q_{\text{tot}}|$ is what an observer of the two copies sees (Theorem 5.57). The block-diagonal inertia is the per-copy Lorentz requirement: the invariant symmetric forms on V_2^* number four under $\mathfrak{l}_G^{(1)} \oplus \mathfrak{l}_G^{(2)}$, six under $\mathfrak{l}_{G,\Delta}$ alone (cross terms, $U = \frac{1}{2}\operatorname{Re}\operatorname{tr}(CQ(\pi))$, C complex symmetric, balance $\operatorname{tr}(CQ(\pi)) = 0$, again of codimension two) and two under $\operatorname{Cent}(G_2) \cap \mathfrak{so}(6,6)$, all of Lorentz class (d6-4e; the $n = 2$ analogue of Theorem 4.39).

5.6.3 What a copy rotation can and cannot duplicate

Proposition 5.56 (Lorentz-covariant duplication)**.** *(i) The real linear maps of V_2 commuting with $\mathfrak{l}_{G,\Delta}$ are the $A \otimes \mathbf{1}_3$, $A \in M_2(\mathbb{C})$, acting on Z with $G_2 = i$. Inside $\mathfrak{so}(6,6)$ they form* $\mathrm{span}\{E, EG_2\} \cong \mathfrak{so}(2,\mathbb{C})$*, which is $\mathfrak{g}(0,2,0)$ of Theorem 5.15; its compact part is $\{R_\varphi\}$.*

(ii) $A \otimes \mathbf{1}_3$ maps $Q \mapsto AQA^\top$ and $Z \mapsto ZA^\top$ (Z the 3×2 matrix of columns z_k). Hence R_φ preserves H_{sum}, so every summed density $P_{\mathrm{sum}}(t)$, and q_{tot}, $\det Q$, σ_1, σ_2. The unitary A, the group $U(2)$ of energy-preserving duality rotations of two free Maxwell fields (end of Section 5.8; [46]), preserve H_{sum}, σ_1 and σ_2 but not q_{tot}, and $\mathfrak{u}(2) \cap \mathfrak{so}(6,6) = \mathbb{R}E$.

(iii) Every map $F : \mathbb{C}^6 \to \mathbb{C}^6$, linear or not, commuting with the diagonal $SO(3,\mathbb{C})$ sends $(z,0)$ with $q(z) = 0 \neq z$ to $(\alpha z, \beta z)$: two null constituents. In particular $R_\varphi(z,0) = (\cos\varphi\, z, \sin\varphi\, z)$.

Proof. (i) V is an irreducible $\mathfrak{l}_G$-module with commutant $\mathrm{span}\{\mathbf{1}, G\} \cong \mathbb{C}$ (Proposition 4.19), so the commutant of $\mathfrak{l}_{G,\Delta}$ on $\mathbb{R}^2 \otimes V$ is $M_2(\mathbb{R}) \otimes \mathrm{span}\{\mathbf{1}, G\} = M_2(\mathbb{C})$; antisymmetry for η_2 leaves $A \in \mathbb{C}\epsilon$, which is Proposition 5.72 (row "diagonal $\mathfrak{l}_G$", dimension 2). (ii) $Q = Z^\top Z$ and $H_{\mathrm{sum}} = ZZ^\dagger$, and T is linear in H (Proposition 3.93). (iii) For $\Lambda \in SO(3,\mathbb{C})$ fixing z, $F(z,0) = F(\Lambda z, 0) = \Lambda F(z,0)$. In the action $u \mapsto w \times u$ of $\mathfrak{so}(3,\mathbb{C})$ ((9.1)) the stabiliser algebra of a null z is $\{w : w \times z = 0\} = \mathbb{C}z$, whose common fixed vectors satisfy $z \times v = 0$, i.e. $v \in \mathbb{C}z$. □

Item (i) restates Proposition 5.72 and Theorem 5.15 (fixed there), (ii) is inherited linear algebra and restates the remark on $U(n)$ at the end of Section 5.8, and (iii) is an obstruction; all checked exactly (d6-5a–c). This settles in the negative the reading of the copy rotation as a pair-creating operation. The copy rotation acting identically in all six directions is the admissible b-field between the two manifolds, and it duplicates a field literally: the undivided energy and the rest energy of $(z,0)$ are shared in the same ratio $\cos^2\varphi : \sin^2\varphi$, a null field becomes two null fields and a massive one two massive replicas. It cannot produce a balanced massive pair from a single null field, and no other Lorentz-covariant operation can.

Theorem 5.57 (The rest-energy density of a duplicated field point)**.** *Let $z_1, z_2 \in \mathbb{C}^3$, with Gram matrix Q of singular values $\sigma_1 \geq \sigma_2$.*

(i) $T(z_1) \cdot T(z_2) = |z_1 \cdot z_2|^2$, and as mixed tensors $\big(T(z_1) + T(z_2)\big)^2 = \frac{1}{4}\|Q\|_F^2\,\mathbf{1} + T(z_1 \times z_2)$, with $(z_1 \times z_2) \cdot (z_1 \times z_2) = \det Q$.

(ii) $|P_{\mathrm{sum}}(t)|^2 = \frac{1}{4}\|Q\|_F^2 + u_t(z_1 \times z_2)$ for every observer t, $u_t(w)$ being the energy density of w seen by t. Hence $M_{\mathrm{pair}} = \frac{1}{2}(\sigma_1 + \sigma_2)$, attained exactly in the rest frames of $z_1 \times z_2$ when it is not null.

(iii) In every frame, $E_0^{(1)} + E_0^{(2)} \leq M_{\mathrm{pair}} \leq |P_{\mathrm{sum}}(t)| \leq E_6^{(1)} + E_6^{(2)}$, and $M_{\mathrm{pair}} \geq \frac{1}{2}|q_{\mathrm{tot}}|$.

(iv) $\max_\varphi (E_0^{(1)} + E_0^{(2)})(R_\varphi Z) = M_{\mathrm{pair}}$.

(v) For two null fields $M_{\mathrm{pair}} = |z_1 \cdot z_2| = (E_6^{(1)} E_6^{(2)})^{1/2}(1 - \cos\vartheta_{12})$, ϑ_{12} the angle between their Poynting vectors; for $(z,0)$ with z null, $M_{\mathrm{pair}} = 0$.

Proof. (i) Polynomial identities, verified symbolically in twelve real variables (d6-6a); in spinor form [121], $T = \sum_k \phi_k \bar{\phi}_k$ and $\phi^k_{AB}\phi^{l\,B}{}_C = \frac{1}{2}(\phi^k \cdot \phi^l)\epsilon_{AC} + \chi^{kl}_{AC}$ with χ^{kl} antisymmetric in (k,l), so the cross terms of T^2 pair the symmetric Q with the antisymmetric χ and drop out. The last identity is the complex Lagrange identity. (ii) T is g-self-adjoint, so $|P(t)|^2 = t \cdot T^2 t$; insert (i). By Theorem 3.95, $u_t(w) \geq E_0(w) = \frac{1}{2}|\det Q|$, with equality iff t is a rest frame of w, and $\frac{1}{4}\|Q\|_F^2 + \frac{1}{2}|\det Q| = \frac{1}{4}(\sigma_1 + \sigma_2)^2$. For null $w \neq 0$ the infimum is approached by Doppler boosts; for $w = 0$, $|P(t)| = \frac{1}{2}\sigma_1$ for all t. (iii) Each $P_k(t)$ is future causal of length

$E_0^{(k)}$ (Proposition 3.94), future causal vectors obey $|P_1+P_2|\geq|P_1|+|P_2|$ and $|P|\leq P^0$, and $\sigma_1+\sigma_2\geq|\operatorname{tr}Q|$. (iv) Write $q_{1,2}(\varphi)=t\pm d(\varphi)$ and $Q_{12}(\varphi)=r(\varphi)$ with $t=\frac{1}{2}q_{\mathrm{tot}}$. The pair (d,r) turns by 2φ, $d^2+r^2=t^2-\det Q$ is fixed, and $(|q_1|+|q_2|)^2=2|t|^2+2|d|^2+2|t^2-d^2|\leq 2|t|^2+2|d|^2+2|r|^2+2|\det Q|=(\sigma_1+\sigma_2)^2$, by $|t^2-d^2|=|\det Q+r^2|\leq|\det Q|+|r|^2$. Equality holds exactly when $\overline{\det Q}\,r(\varphi)^2\geq 0$, hence for every φ when $\det Q=0$; as φ varies, $r(\varphi)$ runs over an ellipse centred at 0 or a segment through 0, so such φ exists. (v) $Q=\left(\begin{smallmatrix}0&r\\r&0\end{smallmatrix}\right)$, $r=z_1\cdot z_2$; and $T(z_k)=k_k\otimes k_k$ with k_k future null, $k_k^0=(E_6^{(k)})^{1/2}$, so $|r|=k_1\cdot k_2$ by (i), and $|\mathbf{S}^{(k)}|=E_6^{(k)}$. □

Items (i), (iii) and (v) are inherited: ordinary electrodynamics of two field strengths, with the reverse triangle inequality for future causal momentum densities. The closed form of M_{pair} in (ii) and its saturation by the copy rotation in (iv) are fixed; their priority is not established (compare Kaiser's local invariants [298] and Thompson's results on complex symmetric matrices [299]). Checked symbolically (i), at the rest frames of $z_1\times z_2$ to 10^{-10} against 300 random observers (ii), by maximising over φ (relative deviation 5×10^{-16}) (iv), and on random and null pairs (iii), (v) (d6-6a–d). Since $T(z_1\times z_2)$ is traceless, the trace of (i) gives the pair invariant of Remark 5.25, $E_{\mathrm{pair}}=(\frac{1}{4}\operatorname{tr}T_{\mathrm{tot}}^2)^{1/2}=\frac{1}{2}\|Q\|_F$, and (ii) gives $M_{\mathrm{pair}}^2=E_{\mathrm{pair}}^2+\frac{1}{2}|\det Q|$. The two agree exactly when $\det Q=0$, which holds on every duplicate $(\alpha z,\beta z)$ of a single field (Theorem 5.24); for the counter-propagating null pair of Example 5.58, $E_{\mathrm{pair}}=\sqrt{2}$ while $M_{\mathrm{pair}}=2$. M_{pair} is the local rest-energy density of the summed stress tensor at a point, not the invariant mass of two photons. For two null fields it equals $(u_1u_2)^{1/2}(1-\cos\vartheta_{12})$ by (v), whereas two photons of energies E_1,E_2 have invariant mass $\big(2E_1E_2(1-\cos\vartheta_{12})\big)^{1/2}$; for $z_1=(1,i,0)$ and $z_2=(0,1,i)$ the values are 1 and $\sqrt{2}$, and the counter-propagating example of Example 5.58 agrees only by coincidence.

Example 5.58 (Two counter-propagating null fields)**.** $z_1=(1,i,0)$ and $z_2=(1,-i,0)$ are null with $\mathbf{S}^{(1)}=-e_3=-\mathbf{S}^{(2)}$, so $\mathbf{S}_{\mathrm{sum}}=0$, $E_6^{\mathrm{tot}}=2$, $Q=\left(\begin{smallmatrix}0&2\\2&0\end{smallmatrix}\right)$ and $M_{\mathrm{pair}}=2$. The rotation $R_{\pi/4}$ gives $z_1'=(0,\sqrt{2}\,i,0)$ and $z_2'=(\sqrt{2},0,0)$: a pure electric and a pure magnetic field, both at rest, with $q=-2,+2$ and $E_0=1$ each. In this frame, the centre-of-momentum frame, the chain of Theorem 5.57(iii) holds with equality for the rotated pair, $E_0^{(1)}+E_0^{(2)}=M_{\mathrm{pair}}=|P_{\mathrm{sum}}|=E_6^{\mathrm{tot}}=2$, while $M_{\mathrm{pair}}=2>\frac{1}{2}|q_{\mathrm{tot}}|=0$: the rotated pair is at rest in the centre-of-momentum frame and takes the whole of M_{pair} (d6-7). The kinematics of $\gamma\gamma\to e^+e^-$ [221] is an analogy, not a production process. Theorem 5.57 adds the two stress tensors, $T(z_1)+T(z_2)$, which differs in general from the tensor $T(z_1+z_2)$ of the coherent superposition (here $\operatorname{diag}(2,0,0,2)$ against $\operatorname{diag}(2,-2,2,2)$); the coherent superposition is, by analogy, the standing wave of Proposition 3.100(3).

Example 5.59 (The rotation of one direction)**.** Let $R_\varphi^{(1)}$ rotate the planes $(e_1^{(1)},e_1^{(2)})$ and $(e_4^{(1)},e_4^{(2)})$ through φ. It is compact, commutes with G_2 and lies in $\mathrm{Cent}(G_2)\cap\mathfrak{so}(6,6)\cong\mathfrak{so}(6,\mathbb{C})$, the $n=2$ analogue of Theorem 4.34. On $((1,i,0),0)$ it gives $z_1'=(\cos\varphi,i,0)$, $z_2'=(\sin\varphi,0,0)$: $q_1'=-\sin^2\varphi$, $q_2'=\sin^2\varphi$, a balanced pair with $E_0=\frac{1}{2}\sin^2\varphi$ each. $E_6^{\mathrm{tot}}=1$ and $q_{\mathrm{tot}}=0$ are conserved, $\mathbf{S}_{\mathrm{sum}}$ goes from $-e_3$ to $-\cos\varphi\,e_3$ and M_{pair} from 0 to $|\sin\varphi|$, so $E_6^{\mathrm{tot}}=1\geq M_{\mathrm{pair}}=|\sin\varphi|\geq\sum_k E_0^{(k)}=\sin^2\varphi$, with equality throughout at $\varphi=\pi/2$ (a magnetic field along e_1 and an electric one along e_2, at rest). $R^{(1)}$ does not commute with $\mathfrak{l}_{G,\Delta}$, and lies neither in $\mathfrak{g}(0,2,0)$ nor in $\mathfrak{g}(0,1,1)=\mathfrak{so}(3,3)$ (Lorentz algebra on copy 1, copy 2 a spectator): it is a candidate gauge transformation under no Lorentz action of Theorem 5.15 with $k_I=0$ (d6-8), and under none with $k_I>0$ either: it lies outside $\mathfrak{g}(2,0,0)$ and $\mathfrak{g}(1,0,1)=\mathfrak{so}(5,3)$ (x6-8a), and it has components from the interior four-plane of one copy into the binding frame of the other, which the proof of Theorem 5.15 excludes from $\mathfrak{g}(1,1,0)$. It is therefore non-covariant under all five Lorentz actions at $n=2$. Read kinematically it is an infinitely heavy third body, absorbing momentum and no energy, and $E_6\geq E_0^{(1)}+E_0^{(2)}$ is the field-point form of the threshold $E_\gamma\geq 2mc^2$ for pair production in the field of a nucleus [271]. The numbers and the non-covariance are fixed

(d6-8; x6-8a,b), the third-body reading is interpretive, the threshold kinematics is inherited, and the agent is constructed: it exists, and nothing in the arena supplies it.

A tempting reading of pair creation applies to $((1,i,0),0)$ a rotation said to mix the copies "identically in all six directions". The rotation that produces the balanced pair above is $R_\varphi^{(1)}$, which rotates two directions only, not the copy rotation R_φ. For $R_\varphi^{(1)}$ the conservation of E_6^{tot} and q_{tot} and the non-conservation of $\mathbf{S}_{\mathrm{sum}}$ hold, while the copy rotation, which does act identically in all six directions, conserves $\mathbf{S}_{\mathrm{sum}}$ and the whole summed stress tensor.

5.6.4 Is the rest energy of one copy observable?

Proposition 5.60 (Gauge status of the per-copy rest energy)**.** *(i) R_φ is admissible for the diagonal action: compact, commuting with G_2 and $\mathfrak{l}_{G,\Delta}$, and the compact part of $\mathfrak{g}(0,2,0)$. On S_2 it is generated by $M = \frac{1}{2}\sum_a \eta_{aa} e_a^{(1)} e_a^{(2)}$, with $\operatorname{ad} M = -E$ on V_2 and eigenvalues $i\{-3,\dots,3\}$; $U_\varphi = \prod_a (\cos\frac{\varphi}{2} + \sin\frac{\varphi}{2}\,\eta_{aa} e_a^{(1)} e_a^{(2)})$ has $U_{2\pi} = \mathbf{1}$, so the copy rotation acts through $SO(2)$; and $U_{\pi/2} J^{(1)} U_{\pi/2}^{-1} = J^{(2)}$, $J^{(k)} = e_2^{(k)} e_3^{(k)}$.*

(ii) R_φ preserves H_{sum}, q_{tot}, $\det Q$, $\sigma_{1,2}$, M_{pair} and the total charge $J^{(1)} + J^{(2)}$; it does not preserve $E_0^{(k)}$, q_k or the per-copy charges (Example 5.58, (i)). On $V_\pm$ it is a duality rotation of both copies, and there the $E_0^{(k)}$ are invariant.

(iii) Equal inertias. The two-constituent closure is invariant under $O(\eta_2)\cap\mathrm{Cent}(G_2) \cong O(6,\mathbb{C})$, whose compact part, the real orthogonal group of $\mathbb{C}^6$, acts transitively on $\{q_{\mathrm{tot}} = 0,\ |Z| = r\}$: every balanced pair is carried to one null constituent and an empty copy at the same E_6^{tot}; for instance $(e_1, ie_1) \mapsto ((1,i,0),0)$, with M_{pair} going from 1 to 0.

(iv) If $c_1 \neq c_2$, no nonzero element of $\mathfrak{g}(0,2,0)$ preserves the inertia.

Proof. (i) Each term pairs two generators of the same square, hence is compact (Theorem 1.3), squares to $-\frac{1}{4}$, and the six terms commute; exponentiate term by term. (ii) Proposition 5.56(ii) and Proposition 5.55(ii). (iii) Every Λ commuting with G_2 preserves the Lie–Poisson structure, and for $\Lambda \in O(\eta_2)$, $\Lambda\eta_2^{-1}\Lambda^\top = \eta_2^{-1}$ and $\Lambda G_2 \eta_2^{-1}\Lambda^\top = G_2\eta_2^{-1}$, so the equal Lorentz-class inertia is invariant. A nonzero Z with $q_{\mathrm{tot}}(Z) = 0$ is $u + iv$ with $u, v \in \mathbb{R}^6$, $|u| = |v|$, $u \perp v$, and $SO(6)$ is transitive on such pairs. (iv) $sE + tEG_2$ acts on the copy index as $(s+it)\epsilon$, and $\epsilon^\top \operatorname{diag}(c_1,c_2) + \operatorname{diag}(c_1,c_2)\,\epsilon = (c_2 - c_1)\left(\begin{smallmatrix}0 & 1\\ 1 & 0\end{smallmatrix}\right)$, which must vanish for the form $\sum_k c_k\, z_k \cdot z_k$ to be invariant. □

Item (i) restates Theorem 5.15 and Proposition 5.16 (fixed there), with the lift computed; (ii)–(iv) are fixed; checked in integer arithmetic (d6-9a, d6-9b, d6-9c, d6-10). This decides whether "mass creation in pairs" is physical. *If the copy rotation is gauged*, the rest energy of each copy is not an observable: Example 5.58 is a change of gauge, not a process, and two counter-propagating null fields and an electric–magnetic pair at rest are one physical state, whose gauge-invariant content $M_{\mathrm{pair}} = 2$ was there before. *In the equal-inertia closure alone*, its own compact symmetry maps every balanced pair to one null constituent, changing M_{pair} as it does so (by (iii)), and the rotor sees only $\frac{1}{2}|q_{\mathrm{tot}}| = 0$; the rest-energy density M_{pair} of the pair is invisible to the rate. M_{pair} becomes an invariant only through the diagonal Lorentz action, and the per-copy rest energies only through a per-copy one, neither of which the closure selects (the Lorentz action at several constituents is left open, Table 13.1; at $n = 3$ the closure symmetry meets the interior Lorentz algebra of the first constituent in 0, Theorem 4.35, and Proposition 5.27(iv) is the $n = 2$ analogue). *If the copies are distinguished* — by unequal inertias, which by (iv) break $\mathfrak{g}(0,2,0)$, by unequal binding rates, by a projector, or by a Lorentz action on one copy only, whose algebra $\mathfrak{g}(0,1,1)$ contains no copy mixing — the per-copy rest energies are observables, but then the closure creates no pair (Corollary 5.54). In the interior

with $S = \mathrm{diag}(\kappa_1, \kappa_2) \otimes G + \beta E$ the momentum field obeys $\dot\pi = S^\dagger \pi$, i.e. $z(\pi)(\tau) = e^{ih\tau} z(\pi)(0)$ with $h = \left(\begin{smallmatrix} \kappa_1 & -i\beta \\ i\beta & \kappa_2 \end{smallmatrix}\right)$ on the copy index. For $\kappa_1 = \kappa_2$ this is $e^{i\kappa\tau} R_{-\beta\tau}$, and the per-copy rest energies change only by the gauge rotation (Larmor [83]; Section 9.1). For $\kappa_1 \neq \kappa_2$ they change physically: for the pair of Example 5.58, $E_0(\pi^{(1)})(\tau) = |z_1 \cdot z_2|\,|U_{11}U_{12}|$, $U = e^{ih\tau}$, which for $\kappa = (1, 3)$, $\beta = 0.4$ peaks at 0.6897, the Rabi value $2\sqrt{s(1-s)}$ with $s = 4\beta^2/((\kappa_2 - \kappa_1)^2 + 4\beta^2)$, at fixed E_6^{tot}; from $(z, 0)$ with z null both copies stay null for all τ (d6-11). This is two-level mixing [223], inherited. It changes the per-copy rest energies but not M_{pair}, so their sum stays at most M_{pair} (Theorem 5.57(iii)) without being conserved in general: for the pair of Example 5.58 both copies gain the same rest energy, so none passes from one copy to the other, and from two null fields rest energy appears only where the second is non-collinear and supplies a nonzero M_{pair}.

Remark 5.61 (A copy rotor)**.** The uniqueness argument of Theorem 4.22 uses that equivariant maps between distinct constituents vanish, which holds for $\mathfrak{l}_G^{(1)} \oplus \mathfrak{l}_G^{(2)}$ but not for $\mathfrak{l}_{G,\Delta}$: E is $\mathfrak{l}_{G,\Delta}$-equivariant, commutes with G_2 and rotates p without dilating it, so a second rotor $\hat c$ with $[\hat c, X] = EX$, $[\hat c, \hat g] = 0$ gives an equivariant closure $\mathbb{R}^2 \ltimes V_2$, with Lie–Poisson equations $\dot p = (H_h G_2^\top + H_{h_c} E^\top)p$, $\dot h = -p^\top G_2 \nabla H$, $\dot h_c = -p^\top E \nabla H$. With equal Lorentz-class inertias $\dot h_c = 0$, the copy angle turns uniformly, and the per-copy $E_0(\pi^{(k)})$ oscillate while the summed momentum tensor, $M_{\mathrm{pair}}(\pi)$ and $|q_{\mathrm{tot}}(\pi)|$ stay constant; $((1, i, 0), 0)$ stays null–null (d6-12). The inter-manifold b-field made dynamical changes the rest energies of the two manifolds at fixed pair invariants. Their sum is not conserved in general (for the data of Example 5.58 the copy rotation takes it from 0 to 2) and never exceeds M_{pair} (Theorem 5.57(iii)), so the rotor creates per-copy rest energy but no pair invariant. Constructed.

5.6.5 The Dirac side

Proposition 5.62 (Pairs of Dirac states in the tower)**.** *On $S_2 = \mathbb{R}^8 \otimes \mathbb{R}^8$ with the tower generators:*

(i) $J^{(1)} = J \otimes \mathbf{1}$ and $J^{(2)} = \mathbf{1} \otimes J$ (the Klein factors cancel on bivectors); they commute with each other and with the eight interior gammas $e_a^{(k)}$, $a \in \{1, 4, 5, 6\}$. With an external i, $Q_{\mathrm{tot}} = -i(J^{(1)} + J^{(2)})$ has spectrum $\{-2, 0, 2\}$ with multiplicities $16, 32, 16$.

(ii) Let $\psi, \phi \in S \otimes \mathbb{C}$ be particles at rest, $e_1\psi = \psi$, $J\psi = i\psi$ ($H(0) = me_1$, equation (2.4)), and $C = e_2$ (Theorem 2.25). Then $\Psi = \psi \otimes C\phi$ has charges $(1, -1)$, $Q_{\mathrm{tot}}\Psi = 0$, and $-i\big(JH(0) \otimes \mathbf{1} + \mathbf{1} \otimes JH(0)\big)\Psi = 2m\Psi$.

(iii) $C^{(1)}C^{(2)} = e_2^{(1)}e_2^{(2)}$ anticommutes with $J^{(1)}$ and $J^{(2)}$, but $(C^{(1)}C^{(2)})^2 = -\mathbf{1}$: no state is mapped to $\pm$ itself. The Klein-corrected $C \otimes C = -\omega^{(1)}C^{(1)}C^{(2)}$, $\omega^{(1)} = \omega \otimes \mathbf{1}$, squares to $\mathbf{1}$, and $\Psi_\pm = \psi \otimes C\psi \pm C\psi \otimes \psi$ satisfy $(C \otimes C)\Psi_\pm = \pm\Psi_\pm$ and $Q_{\mathrm{tot}}\Psi_\pm = 0$.

(iv) The tower's mass terms anticommute: $(e_1^{(1)} + e_1^{(2)})^2 = 2\,\mathbf{1}$, so $m(e_1^{(1)} + e_1^{(2)})$ has spectrum $\{\pm\sqrt{2}\,m\}$, whereas the two-particle operator $m(e_1 \otimes \mathbf{1} + \mathbf{1} \otimes e_1) = m(e_1^{(1)} + \omega^{(1)}e_1^{(2)})$ has spectrum $\{-2m, 0, 2m\}$, multiplicities $16, 32, 16$.

(v) Under the diagonal interior action ($k_I = 2$) a rotation through 2π is $+\mathbf{1}$ on S_2.

Proof. (i) $(\omega \otimes e_2)(\omega \otimes e_3) = \mathbf{1} \otimes J$ as $\omega^2 = \mathbf{1}$; the commutations are the blade rule (equation (1.3)); $(J^{(1)} + J^{(2)})^2 = -2(\mathbf{1} - J^{(1)}J^{(2)})$, where $\frac{1}{2}(\mathbf{1} - J^{(1)}J^{(2)})$ is a projector of trace 32. (ii) C anticommutes with e_1 and J, and J commutes with $H(0)$. (iii) Generators of distinct constituents anticommute and $e_2^2 = \mathbf{1}$; $\mathbf{1} \otimes e_2 = \omega^{(1)}e_2^{(2)}$. (iv) The same anticommutation; $e_1 \otimes \mathbf{1}$ and $\mathbf{1} \otimes e_1$ are commuting traceless involutions with traceless product. (v) Each copy's rotation through 2π is $-\mathbf{1}$, and the two commute. □

Inherited: Klein–Jordan–Wigner signs [212], the charge-conjugation parity of a particle–antiparticle pair [160], and integer spin for two spin-$\frac{1}{2}$ constituents, which is Proposition 5.16 at $k_I = 2$ (restated, fixed there); checked exactly (d6-13a–e). Two points are specific to the tower. The total charge is invariant under the copy rotation while $U_{\pi/2}$ exchanges $J^{(1)}$ and $J^{(2)}$ (Proposition 5.60(i)), so which copy carries the particle is not invariant under the copy rotation, hence gauge dependent if that rotation is gauged, exactly as the per-copy rest energy is. And by (iv) the graded duplication $\mathrm{Cl}(3,3)\hat{\otimes}\mathrm{Cl}(3,3) = \mathrm{Cl}(6,6)$ adds Dirac masses in quadrature, like two anticommuting Clifford directions — one particle of mass $\sqrt{2}\,m$ — while the pair threshold $2m$ appears only for Klein-corrected, ungraded operators: duplication in the tower's sense is not pair creation.

5.6.6 Zero-point energy of the duplicated frame

Proposition 5.63 (Zero-point energy)**.** *(i) For* $\mathcal{L} = \sum_k \frac{m_k}{2}\eta(\dot{X}^{(k)} + \kappa_k G X^{(k)}, \dot{X}^{(k)} + \kappa_k G X^{(k)})$ *each rate* κ_k *carries Krein signature* $(3,3)$*; with positive-norm quantisation the zero-point energy is* $\sum_k \frac{\hbar\kappa_k}{2}(3-3) = 0$ *for all* $\kappa_1, \kappa_2 > 0$ *and* $m_1, m_2 \neq 0$*. At* $\kappa_k = 0$ *the copy is free, its phase-space generator is not semisimple, and no zero-point energy is defined.*

(ii) With an inter-manifold $b = \beta E$ *(*$m_1 = m_2$*,* $\kappa_k \geq 0$ *not both zero),* $S = \mathrm{diag}(\kappa_1, \kappa_2)\otimes G + \beta E$ *has spectrum* $\pm i\omega_\pm$*,* $\omega_\pm = \frac{1}{2}(\kappa_1+\kappa_2) \pm \left(\frac{1}{4}(\kappa_1-\kappa_2)^2+\beta^2\right)^{1/2}$*, each three times; these are the lines of Theorem 5.31(ii). Every solution is bounded unless* $\beta^2 = \kappa_1\kappa_2$ *and* $\kappa_1 \neq \kappa_2$*; when bounded, each nonzero-frequency line (coincident lines combined) has Krein signature* $(3,3)$ *and their zero-point energy vanishes. At* $\kappa_1 = \kappa_2 = |\beta| > 0$ *the zero line is semisimple and the motion bounded, but it is not an oscillator line. At* $\kappa_1 = 0$*,* $|\omega_-| \approx \beta^2/\kappa_2$*.*

(iii) At $\beta^2 = \kappa_1\kappa_2$*,* $\kappa_1 \neq \kappa_2$*,* $\omega_- = 0$ *and the phase-space generator is not semisimple: some solutions grow linearly in* τ*, and on that line there is no oscillator vacuum.*

The cancellation in (i) and (ii) concerns the nonzero-frequency lines of a Hamiltonian that is unbounded below; no vacuum-energy conclusion follows from it (compare Theorem 4.12).

Proof. (i) Corollary 4.15 per copy, or Theorem 9.12. (ii) On the $G_2 = i$ sector S acts on the copy index as $i\,\mathrm{diag}(\kappa_1,\kappa_2) + \beta\epsilon = i(K - i\beta\epsilon)$ with $K - i\beta\epsilon$ Hermitian. $S^\dagger = K\otimes G - \beta E$ commutes with G_2, so for $\mu \neq 0$ its eigenspaces lie in the $G_2 = \pm i$ sectors, which are totally isotropic for h (a vector with $GY = iY$ has $\mathbf{E} = -i\mathbf{B}$); Theorem 9.4(ii) then gives boundedness except possibly at $\omega_- = 0$, that is at $\beta^2 = \kappa_1\kappa_2$; there the kernel of $S^\dagger$ computed in (iii) is η_2-null exactly when $\kappa_1 = \kappa_2$, so the case $\kappa_1 = \kappa_2 = |\beta|$ is bounded as well. Theorem 9.12 gives the signatures. (iii) At $\beta^2 = \kappa_1\kappa_2$ the kernel of $S^\dagger$ consists of the $(v_1 c, v_2 c)$, $c \in \mathbb{C}^3$, with (v_1, v_2) a nonzero multiple of $(i\kappa_2, \beta)$ if $\kappa_2 > 0$ and of $(0,1)$ if $\kappa_2 = 0$ (then $\beta = 0$ and copy 2 is free), on which $\eta_2 = \mathrm{Re}((v_1^2+v_2^2)\,c\cdot c)$ and $v_1^2+v_2^2$ is a nonzero multiple of $\kappa_2(\kappa_1-\kappa_2)$, respectively of 1, hence $\neq 0$; so the kernel is not η_2-null and Theorem 9.4(i) fails. □

Item (i) restates Theorem 7.13 (fixed there) for each copy: duplication preserves the neutral signature, and the cancellation survives unequal rates and masses. Since $E = C_v$, (ii) and (iii) specialise Theorem 5.31 to $m_1 = m_2$. The lines in (ii) are inherited two-level mixing and restate Theorem 5.31(ii); the boundedness criterion restates Theorem 5.31(v); the seesaw value 0.0495097568 at $(\kappa_1, \kappa_2, \beta) = (0, 5, \frac{1}{2})$ against $\beta^2/\kappa_2 = 0.05$ restates Theorem 5.31(vi). What is new here is the Krein signature $(3,3)$ of each line and the vanishing zero-point energy at $n = 2$, fixed by Theorem 9.12. (iii) restates the Jordan block in the proof of Theorem 5.31(v) (fixed there); it is the inter-manifold analogue of the exceptional ratio of Theorem 9.9. Checked numerically for the signatures (d6-14a,b) and with rational data for semisimplicity: not semisimple at $(1,4,2)$, semisimple at $(1,4,1)$, $(2,2,2)$ and $(1,3,\frac{2}{5})$ (d6-14c). Read with the added identification $mc^2 = \hbar\kappa$ applied to the lines: an inter-manifold b can give an unbound manifold the line

β^2/κ_2 borrowed from the binding of the other, or bring a line to zero, at unequal rates exactly at the price of unbounded motion; it never changes a rate, the spectrum of the self-adjoint part of the shift (Theorem 9.21(iii)), and never produces a line without an existing binding, since at $\kappa_1 = \kappa_2 = 0$ the lines $\pm\beta$ are removed by $X \mapsto e^{-\beta E\tau}X$.

5.6.7 Assessment

On the b-particle with two manifolds. The remark is partly borne out. With two manifolds there is an admissible b-field between them, the copy rotation, the compact part of $\mathfrak{g}(0,2,0)$; it commutes with the binding root G_2, and with the binding $K \otimes G$ only at equal rates (Theorem 5.31(iii)), keeps every orbit bounded except when $\beta^2 = \kappa_1\kappa_2$ with $\kappa_1 \neq \kappa_2$, or when $\kappa_1 = \kappa_2 = 0$ and nothing binds (Theorem 5.31(v)), and changes the rest energy of each manifold, their sum bounded by M_{pair} but not conserved in general, which on one manifold only the anti-diagonal part of b can do, and only by tearing or by screening (Proposition 9.18 and Theorem 9.9); with unequal rates it mixes them, lending a line to an unbound manifold or bringing one to zero, while the rates themselves never change (Theorem 9.21(iii)). It does not generate an invariant mass: the copy rotation preserves M_{pair}, the summed stress tensor and q_{tot}, and the interior flow $e^{ih\tau}$, unitary on the copy index, preserves M_{pair}, the summed stress tensor and E_6^{tot} at any rates but $|q_{\text{tot}}|$ only at equal rates (for the pair of Example 5.58 at $\kappa = (1,3)$, $\beta = 0.4$ it leaves 0); with equal rates the copy rotation is removable along a worldline (Theorem 5.31(iii)); the rest energy of each copy is not invariant under the copy rotation, hence gauge dependent if that rotation is gauged; and the one thing two constituents do in the closure that one cannot, carrying rest energy at a constant rate, needs no b-field at all: it is the balance surface.

On how a unitary energy results in a duplication. The mathematical process is the graded tensor product (inherited), under which the undivided energy becomes $E_6^{\text{tot}} = \frac{1}{2}|Z|^2$; the Lorentz-covariant, energy–momentum-preserving duplications form $U(2)\otimes\mathbf{1}_3$, of which the metric ones form $O(2)\otimes\mathbf{1}_3$ (the real orthogonal A); its identity component, the copy rotation $SO(2)$, is the part generated by an admissible b-field, and the other component contains the swap of the copies and $\mathrm{diag}(1,-1)$, which are metric but not copy rotations (Proposition 5.56). The physical process it describes shares the energy of one copy between two as literal replicas, preserving the type. From one null field, a balanced magnetic–electric pair requires either a second, non-collinear null field, with the inherited kinematics of $\gamma\gamma \to e^+e^-$ ($\sum_k E_0^{(k)} \leq M_{\text{pair}}$, saturated), or a non-covariant agent that takes the momentum, with the kinematics of pair production on a nucleus ($\sum_k E_0^{(k)} \leq E_6^{\text{tot}}$). From one massive field z the copy rotation never balances $(z,0)$, since it preserves $q_{\text{tot}} = q(z) \neq 0$; the non-metric equal split with a relative quarter turn of duality, $z \mapsto (z, iz)/\sqrt{2}$ (Theorem 5.24(ii) at $\theta = \pi/2$), is covariant and energy-preserving, lands on V_+ and gives the balanced pair of Proposition 5.55(iii), with $E_0^{(k)} = \frac{1}{2}E_0(z)$ each and $M_{\text{pair}} = E_0(z)$ unchanged (Proposition 5.28). In motion the closure conserves the content of each copy and carries the pair, on the balance surface, at a constant rate. The decisive caveat is Proposition 5.60: every statement that a mass has been created in one copy needs a structure that distinguishes the copies, and the symmetric closure and the gauged copy rotation provide none. Whether such a structure is physical is open.

5.7 Up the ladder

Sections 5.4 and 5.5 treat the second rung. This section treats the rung $\mathrm{Cl}(3n,3n)$ for every n and the step from n to $n+1$. The question is the same as at $n = 2$: in what precise sense, and under which added structures, does rung n carry n particles rather than one particle with a larger internal space, and what changes from one rung to the next. The answers, with their classes, are these. (1) $\mathrm{Cl}(3n,3n) = M_{8^n}(\mathbb{R})$ at every rung. The step embeds rung n in rung $n+1$ with multiplicity eight and adds three Fock modes, one frame and one copy direction; the

commuting copy of the new constituent is its generators dressed by the volume element of rung n (fixed; Section 5.7.1). (2) For n equal masses the graded rest Hamiltonian is one clock of mass $\sqrt{n}\,m$, and the ungraded one is n clocks of mass m, with spectrum $m(n-2j)$ of multiplicity $\binom{n}{j}4^n$ (fixed, partly restated from Proposition 5.34; Section 5.7.2). (3) A rotation through 2π of the diagonal interior Lorentz action is $(-1)^n$ on S_n, and one of the diagonal field-strength action is $+\mathbf{1}$. The interior composite has half-integer spin exactly at odd n, a fermion under the spin–statistics reading, and the field-strength composite has integer spin at every rung (restated from Proposition 5.16, with the second half fixed; Section 5.7.3). (4) On vectors the copy rotations of all six directions span the commutant of the diagonal $\mathfrak{so}(3,3)$, and the copy group $O(n)$ has the diagonal $\mathfrak{so}(3,3)$ as its commutant. On S_n the Casimirs satisfy $C_{\rm cop}+C_{\rm diag}=\frac{3}{4}n(n+4)$. This is Howe's skew duality $(O(n),\mathfrak{so}(6))$ on the spinor module of $\mathfrak{so}(6n)$, and at $n=2,3$ the joint decomposition has the predicted multiplicity-free form (fixed; the duality inherited; Section 5.7.4). (5) The Koszul swaps realise the permutation matrices of that $O(n)$, twisted by the sign character. Their symmetric and antisymmetric sectors are mirror images, of equal dimension and never empty. The plain permutations of the common complex space give $\Lambda^n_{\mathbb{C}}$ of the Dirac module, which vanishes for $n\geq 5$; this concerns one momentum label of the finite construction, with antisymmetry imposed, and is not a bound on fermions in physical space (fixed; the exterior powers inherited; Section 5.7.5). (6) The algebra singles out no rung, and its invariant forms single out only the parity of n, through a period of four. Odd rungs are singled out by the interior Lorentz action, even rungs by the binding action, and $n=3$ by the projector P_* (restated and fixed; Section 9.7.1). (7) Going up the ladder enlarges one particle; it becomes the addition of a particle only through the constructed structures of Section 5.4, now needed at every rung (interpretive, drawn from the fixed statements; Section 5.7.6). Identifiers e3-n refer to the `PASS` lines of `e3_ladder.py` (Table 5.11).

Notation. As in Section 5.4: $S=\mathbb{R}^8$, $S_n=S^{\otimes n}$, $E_{(k,a)}=\omega^{\otimes(k-1)}\otimes e_a\otimes\mathbf{1}^{\otimes(n-k)}$, and $V_n=\mathbb{R}^n\otimes V=\mathbb{R}^{3n,3n}$ with $\eta_n=\mathbf{1}_n\otimes\eta$ and $G_n=\mathbf{1}_n\otimes G$. $\Gamma^{(k)}=E_{(k,1)}\cdots E_{(k,6)}=\mathbf{1}\otimes\cdots\otimes\omega\otimes\cdots\otimes\mathbf{1}$ is the volume element of constituent k, and $\Gamma_n=\Gamma^{(1)}\cdots\Gamma^{(n)}$. The ungraded generators are $u_{(k,a)}=\mathbf{1}^{\otimes(k-1)}\otimes e_a\otimes\mathbf{1}^{\otimes(n-k)}$, $J^{(k)}=E_{(k,2)}E_{(k,3)}$, and $a^{\pm}_{k,i}=\frac{1}{2}(E_{(k,i)}\pm E_{(k,i+3)})$, $N=\sum_{k,i}a^+_{k,i}a^-_{k,i}$ (Proposition 5.21); $|0\rangle$ is the first basis vector of S_n, the Fock vacuum. The diagonal generators are $\Delta_{ab}=\frac{1}{2}\sum_k E_{(k,a)}E_{(k,b)}$. All fifteen span the *diagonal* $\mathfrak{so}(3,3)$; those with $a,b\in\{1,4,5,6\}$ span the diagonal $\mathfrak{l}_{\rm int}$; and the paired rotations $\Delta_{ij}-\Delta_{i+3,j+3}$ and paired boosts $\Delta_{i,j+3}-\Delta_{j,i+3}$ span the diagonal $\mathfrak{l}_G$ (Theorem 3.1). The copy rotations are $C^{kl}=\frac{1}{2}\sum_a\eta_{aa}E_{(k,a)}E_{(l,a)}$ for $k<l$, with $C^{12}=C_b$ of Proposition 5.22(ii), and M_{kl} are their interior parts (Section 5.2.1). The Casimirs are $C_{\rm diag}=-\sum_{a<b}\eta^{aa}\eta^{bb}\Delta^2_{ab}$ over all six directions, C_Δ the same over the interior four (Section 5.2.2), $C_G=-\sum_i L_i^2+\sum_i K_i^2$ for the paired rotations L_i and boosts K_i of $\mathfrak{l}_G$, $C_{\rm cop}=-\sum_{k<l}(C^{kl})^2$ and $C^{\rm int}_{\rm cop}=-\sum_{k<l}M^2_{kl}$. A Lorentz Casimir takes the value $2[j_L(j_L+1)+j_R(j_R+1)]$ on (j_L,j_R). We write $\mathfrak{S}_n$ for the symmetric group, to keep S_n for the module.

5.7.1 The rung and the step

Proposition 5.64 (The rung and the step)**.** *(i) Every blade of* $\mathrm{Cl}(3n,3n)$ *other than* $\pm\mathbf{1}$ *is traceless on* S_n*. Hence the* 64^n *blades are linearly independent and* $\mathrm{Cl}(3n,3n)\cong M_{8^n}(\mathbb{R})$*, with* S_n *its unique irreducible module. The volume element is* $E_{(1,1)}\cdots E_{(n,6)}=\Gamma_n=\omega^{\otimes n}$*, and* $(-1)^N=(-1)^n\Gamma_n$*.*

(ii) $E^{(n+1)}_{(k,a)}=E^{(n)}_{(k,a)}\otimes\mathbf{1}$ *for* $k\leq n$ *and* $E_{(n+1,a)}=\Gamma_n\otimes e_a$*. So* $\mathrm{Cl}(3n,3n)\otimes\mathbf{1}\subset\mathrm{Cl}(3n+3,3n+3)$*, and* S_{n+1} *restricted to* $\mathrm{Cl}(3n,3n)$ *is eight copies of* S_n*. The commutant of* $\mathrm{Cl}(3n,3n)\otimes\mathbf{1}$ *is* $\mathbf{1}\otimes M_8(\mathbb{R})$*, generated by the ungraded generators* $\mathbf{1}\otimes e_a=(\Gamma_n\otimes\mathbf{1})E_{(n+1,a)}$ *of the new constituent.*

(iii) $N_{n+1} = N_n \otimes \mathbf{1} + \mathbf{1} \otimes N_1$, $\operatorname{spec} N_n = \{0, \dots, 3n\}$ *with multiplicities* $\binom{3n}{j}$, *and* $[N, E_{(k,i)}] = E_{(k,i+3)}$, $[N, E_{(k,i+3)}] = E_{(k,i)}$. *Each step adds three modes.*

(iv) Every diagonal bivector satisfies $\sum_{k \le n+1} E_{(k,a)} E_{(k,b)} = \big(\sum_{k \le n} E_{(k,a)} E_{(k,b)}\big) \otimes \mathbf{1} + \mathbf{1} \otimes e_a e_b$. *Hence* $S_{n+1} \cong S_n \otimes S$ *and* $S_n \cong S^{\otimes n}$ *as modules of the diagonal* $\mathfrak{so}(3,3)$, *and so of the diagonal* $\mathfrak{l}_{\mathrm{int}}$ *and* $\mathfrak{l}_G$.

Proof. (i) Moving the generators of each constituent to their own factor gives $e_I = \pm \bigotimes_j e_{I_j} \omega^{p_j}$ with $p_j = |I_{j+1}| + \cdots + |I_n| \bmod 2$, and the trace is multiplicative. On S, $\operatorname{tr}(e_J \omega^p) \neq 0$ only for $(J,p) = (\emptyset, 0)$ and for J the full index set with $p = 1$. The last factor has $p_n = 0$, so a nonzero trace needs $I_n = \emptyset$; then $p_{n-1} = 0$, and so on, down to $I = \emptyset$. Trace orthogonality gives independence, and $64^n = (8^n)^2$. The volume element is $\prod_k \Gamma^{(k)}$ because $\prod_a E_{(k,a)} = (\omega^{\otimes(k-1)})^6 \otimes \omega \otimes \mathbf{1}$. On one constituent $(-1)^N = -\omega$ (Proposition 5.21), and $N = \sum_k N^{(k)}$. (ii) The formulas are equation (1.2) at rungs n and $n+1$. The commutant of $M_{8^n}(\mathbb{R}) \otimes \mathbf{1}$ in $M_{8^{n+1}}(\mathbb{R})$ is $\mathbf{1} \otimes M_8(\mathbb{R})$, and $\Gamma_n^2 = \mathbf{1}$. (iii) $N = \sum_{k,i} \frac{1}{2}(\mathbf{1} - E_{(k,i)} E_{(k,i+3)})$, the terms are commuting diagonal involutions with traceless products, and the commutators are those of Proposition 5.21. (iv) $E_{(n+1,a)} E_{(n+1,b)} = \Gamma_n^2 \otimes e_a e_b = \mathbf{1} \otimes e_a e_b$. □

Part (i) is fixed in the realisation (e3-1a, exhaustive over all 64^n blades at $n \le 4$). That $\mathrm{Cl}(p,q) \cong M_{2^{(p+q)/2}}(\mathbb{R})$ for $p - q \equiv 0 \pmod 8$ is inherited [214]: the real structure of the algebra is the same at every rung, so it singles out none. Parts (ii) and (iii) restate Proposition 5.20 and Proposition 5.21 and Proposition 5.33(ii),(v) at the last factor (e3-1b, with the commutant exhaustive over blades at $n = 1, 2$; e3-1c, spectrum also at $n = 5$). Part (iv) is fixed (e3-6c, $n \le 3$). Under (iv) every statement about the diagonal action is a statement about tensor powers of the one-constituent module, and the step is tensoring with S.

5.7.2 Mass up the ladder: one clock or n clocks

Put $H_g = m \sum_k E_{(k,1)}$, built from the vectors of the rung, and $H_u = m \sum_k u_{(k,1)}$, built from the commuting copies.

Proposition 5.65 (Mass up the ladder)**.** *(i) The* $E_{(k,1)}$ *are pairwise anticommuting traceless involutions. So* $H_g^2 = nm^2 \mathbf{1}$ *and* $\operatorname{spec} H_g = \{\pm\sqrt{n}\, m\}$, *each of multiplicity* $8^n/2$. *Every* $J^{(k)}$ *commutes with* H_g, *and* $Z_g = J^{(1)} H_g$ *satisfies* $Z_g^2 = -nm^2 \mathbf{1}$: *the rest flow* $e^{-\lambda Z_g}$ *is one clock of angular frequency* $\sqrt{n}\, m$.

(ii) The ungraded generators are $u_{(k,a)} = \Gamma^{(1)} \cdots \Gamma^{(k-1)} E_{(k,a)}$, *with no further sign. The* $u_{(k,1)}$ *commute, square to* $\mathbf{1}$, *and every product over a nonempty subset is traceless. So* $\operatorname{spec} H_u = \{m(n-2j) : j = 0, \dots, n\}$, *the value* $m(n-2j)$ *with multiplicity* $\binom{n}{j} 4^n$.

(iii) $T_u^{(k)} = J^{(k)} u_{(k,1)} = \mathbf{1} \otimes \cdots \otimes T \otimes \cdots \otimes \mathbf{1}$ *are* n *commuting complex structures:* n *clocks of angular frequency* m.

(iv) $H_g^{(n+1)} = H_g^{(n)} \otimes \mathbf{1} + \Gamma_n \otimes m e_1$ *with anticommuting summands, and* $H_u^{(n+1)} = H_u^{(n)} \otimes \mathbf{1} + \mathbf{1} \otimes m e_1$ *with commuting summands. The graded mass grows in quadrature,* $\sqrt{n}\, m \to \sqrt{n+1}\, m$, *and the ungraded energies add.*

Proof. (i) Graded generators of distinct constituents anticommute, and each $E_{(k,1)}$ is traceless because $\operatorname{tr} e_1 = 0$. $J^{(k)}$ is a bivector sharing no index with any $E_{(l,1)}$, so it commutes with each (equation (1.3)), and $(J^{(k)})^2 = -\mathbf{1}$. (ii) $\Gamma^{(1)} \cdots \Gamma^{(k-1)} = \omega^{\otimes(k-1)} \otimes \mathbf{1}$ and $\omega^2 = \mathbf{1}$, so the product is $\mathbf{1}^{\otimes(k-1)} \otimes e_a \otimes \mathbf{1}$. Products of the $u_{(k,1)}$ are tensor products of factors e_1 and $\mathbf{1}$, traceless unless all factors are $\mathbf{1}$. Commuting involutions whose nonempty products are traceless have all 2^n joint signs with equal multiplicity $8^n/2^n = 4^n$, and j minus signs give $m(n-2j)$. (iii) $J e_1 = e_2 e_3 e_1 = T$ on the kth factor. (iv) Proposition 5.64(ii); Γ_n anticommutes with every generator of $\mathrm{Cl}(3n, 3n)$. □

At general masses (i) and (ii) are Proposition 5.34(vi), restated; the multiplicities, the clocks (iii) and the step (iv) are fixed (e3-2a and e3-2b, exact over signed permutations at $n \le 5$, eigenvalue counts in floating point at $n \le 3$; e3-6c). At $n = 2$, (i) is the Dirac duplication of Proposition 5.62(iv). So rung n read through its vectors carries one Dirac mass, $\sqrt{n}\,m$, along the unit vector $n^{-1/2}\sum_k E_{(k,1)}$ of $\mathbb{R}^{3n,3n}$, with one clock; read through its commuting copies it carries n masses m with n clocks and threshold nm. The quadrature is the statement that the copies of the graded tower are directions of one momentum space (Section 5.4.2).

5.7.3 Spin parity

Let $R_n = \frac{1}{2}\sum_k E_{(k,4)}E_{(k,5)}$, the rotation about the third axis of the diagonal $\mathfrak{l}_{\text{int}}$, and $L_n = \frac{1}{2}\sum_k (E_{(k,1)}E_{(k,2)} - E_{(k,4)}E_{(k,5)})$, the paired rotation of the $(1,2)$ and $(4,5)$ planes in the diagonal $\mathfrak{l}_G$. Both generate rotations of period 2π on vectors.

Proposition 5.66 (Spin parity up the ladder)**.** *(i) R_n is a sum of n commuting terms $\frac{1}{2}b_k$ with $b_k^2 = -\mathbf{1}$. Hence $e^{2\pi R_n} = (-1)^n\mathbf{1}$, and the eigenvalues of $2R_n$ are i times integers congruent to n modulo 2. Under the diagonal $\mathfrak{l}_{\text{int}}$ the whole of S_n has half-integer spin for odd n and integer spin for even n. The Casimir spectra are*

$$\begin{aligned}
&n=1: && C_\Delta = \tfrac{3}{2}\,(8),\\
&n=2: && 0\,(8),\ 3\,(32),\ 4\,(24),\\
&n=3: && \tfrac{3}{2}\,(160),\ \tfrac{11}{2}\,(288),\ \tfrac{15}{2}\,(64),\\
&n=4: && 0\,(160),\ 3\,(1024),\ 4\,(864),\ 8\,(864),\ 9\,(1024),\ 12\,(160),
\end{aligned}$$

where the values $0, \frac{3}{2}, 3, 4, \frac{11}{2}, 8, \frac{15}{2}, 9, 12$ are those of $(0,0)$, $(\frac{1}{2},0)\oplus(0,\frac{1}{2})$, $(\frac{1}{2},\frac{1}{2})$, $(1,0)\oplus(0,1)$, $(1,\frac{1}{2})\oplus(\frac{1}{2},1)$, $(1,1)$, $(\frac{3}{2},0)\oplus(0,\frac{3}{2})$, $(\frac{3}{2},\frac{1}{2})\oplus(\frac{1}{2},\frac{3}{2})$ and $(2,0)\oplus(0,2)$.

(ii) L_n commutes with G_n on vectors and is a sum of $2n$ commuting terms $\pm\frac{1}{2}b$ with $b^2 = -\mathbf{1}$. Hence $e^{2\pi L_n} = +\mathbf{1}$ at every rung. The diagonal $\mathfrak{l}_G$ has Casimir $C_G = 3$ on all of S; $0, 4, 8$ on $4, 24, 36$ dimensions at $n = 2$; and $3, 9, 15$ on $128, 256, 128$ at $n = 3$: the content of $(\frac{1}{2},\frac{1}{2})^{\otimes n}$, integer spin at every rung.

(iii) The central element $-\mathbf{1}$ of $\mathrm{Spin}(3,3)$, which is $e^{\pi e_4 e_5} = e^{\pi e_1 e_2}$ on S, acts on S_n through the diagonal $\mathrm{Spin}(3,3)$ as $(-1)^n$. The rotation through 2π of $\mathfrak{l}_{\text{int}}$ is this element, and that of $\mathfrak{l}_G$ is not.

(iv) For the actions of Theorem 5.15, with $\mathfrak{l}_{\text{int}}$ on k_I constituents, $\mathfrak{l}_G$ on k_B and trivially on the rest, the rotation through 2π is $(-1)^{k_I}$.

Proof. (i) $(E_{(k,4)}E_{(k,5)})^2 = -E_{(k,4)}^2E_{(k,5)}^2 = -\mathbf{1}$, and bivectors of distinct constituents share no index and commute. So $e^{2\pi R_n} = \prod_k(\cos\pi + b_k\sin\pi)$. Each listed value of $2[j_L(j_L+1) + j_R(j_R+1)]$ is attained by exactly one unordered pair (j_L, j_R), which identifies the modules. (ii) $e_ie_j - e_{i+3}e_{j+3}$ commutes with G (Proposition 5.27(i)). The $2n$ terms share no index, and $e^{2\pi L_n} = \prod_k(-\mathbf{1})(-\mathbf{1})$. By Proposition 5.64(iv), $S_n \otimes \mathbb{C} \cong (S\otimes\mathbb{C})^{\otimes n} = 2^n(\frac{1}{2},\frac{1}{2})^{\otimes n}$ under the diagonal $\mathfrak{l}_G$, since $S\otimes\mathbb{C} = 2(\frac{1}{2},\frac{1}{2})$ (Proposition 3.83); the Clebsch–Gordan series gives the listed values. (iii) The diagonal $\mathrm{Spin}(3,3)$ acts on $S^{\otimes n}$ by $g \mapsto g^{\otimes n}$ (Proposition 5.64(iv)). $e^{2\pi\cdot\frac{1}{2}e_4e_5}$ is $-\mathbf{1}$, while the double rotation $e^{2\pi\cdot\frac{1}{2}(e_1e_2-e_4e_5)} = e^{\pi e_1e_2}e^{-\pi e_4e_5}$ is $+\mathbf{1}$. (iv) Products of the factors of (i) and (ii) over the respective constituents. □

Part (i) restates Proposition 5.16 and Proposition 5.17(i) (fixed there), with the element and the spectra re-verified (e3-3a, exact at $n \le 5$; e3-3c, floating-point spectra at $n \le 4$). Part (ii) extends Proposition 3.83 from $n = 1$ to every rung and is fixed (e3-3b, e3-3c). Part (iii) is fixed (e3-3b), and (iv) restates Proposition 5.16. The parity of (i) is the Ehrenfest–Oppenheimer rule, a composite of n spin-$\frac{1}{2}$ particles is a fermion iff n is odd (inherited [222]). The rule holds equally

for one particle whose internal space is the n-fold tensor product of spinors. So the spin parity does not decide between one particle and n. Under the field-strength reading it does not arise: every rung carries integer spin, $\mathrm{Spin}(3,3) \cong SL(4,\mathbb{R})$ restricted to $\mathfrak{l}_G$ integrates to $SO^+(1,3)$ on every S_n, and the diagonal $\mathfrak{l}_G$ composite is a boson at every rung.

Remark 5.67 (Comparison with Theorem 5.73)**.** Theorem 5.73 says that a copy-invariant four-vector in $\Lambda^g(\mathbb{R}^n \otimes \mathbb{R}^{1,3})$ needs n odd and g odd, $n \le g \le 3n$. The statement concerns the algebra and (i) concerns the module, and they are logically independent. Both are products over the n constituents: in (i) each constituent contributes a factor -1 to the 2π rotation, and in the theorem the copy invariant of odd grade needs the Levi-Civita symbol of $SO(n)$ with its n indices. They meet on the Dirac block. At odd n the module contains spin $\frac{1}{2}$, and at $n = 3$ the copy-singlet four-vector Q_μ of grade three builds a gauge-invariant $\mathrm{Cl}(1,3)$ on that block (Theorem 5.13). The check e3-6b re-verifies that Q_μ commutes with the interior copy rotations M_{kl} and satisfies $[\Delta_{ab}, Q_c] = \eta_{bc}Q_a - \eta_{ac}Q_b$, and that the $n = 2$ analogue $\sum_\nu \eta_{\nu\nu}(\gamma^{(1)}_\mu\gamma^{(2)}_\nu + \gamma^{(1)}_\nu\gamma^{(2)}_\mu)$ commutes with M_{12} but is not a vector. At even n there is neither a fermion to carry gamma matrices nor a copy-invariant four-vector to build them. Under the diagonal $\mathfrak{l}_G$ neither statement has an analogue: the spin is integral at every rung, and grade one is $n[(1,0)\oplus(0,1)]$, n field strengths and no four-vector (Section 5.8). The juxtaposition is interpretive; the facts are restated and fixed.

5.7.4 A dual pair up the ladder

Theorem 5.68 (The copy algebra and the diagonal $\mathfrak{so}(3,3)$)**.** *(i) On vectors, the commutant of the diagonal $\mathfrak{so}(3,3)$ in $\mathfrak{so}(3n,3n)$ is $\mathfrak{so}(n)\otimes\mathbf{1}_6$, of dimension $\binom{n}{2}$, spanned by the vector actions of the C^{kl}. The commutant of $\mathfrak{so}(n)\otimes\mathbf{1}_6$ together with the copy permutations is $\mathbf{1}_n \otimes \mathfrak{so}(3,3)$ for $n \ge 2$; that of $\mathfrak{so}(n)\otimes\mathbf{1}_6$ alone has dimension 36 at $n = 2$ and 15 for $n \ge 3$. The weights η_{aa} are forced: $\sum_a E_{(k,a)}E_{(l,a)}$ does not commute with the diagonal $\mathfrak{so}(3,3)$.*

(ii) On S_n, C^{kl} is the spin lift of $(e_k e_l^\top - e_l e_k^\top)\otimes\mathbf{1}_6$: $[C^{kl}, E_{(l,a)}] = E_{(k,a)}$ and $[C^{kl}, E_{(k,a)}] = -E_{(l,a)}$. The C^{kl} commute with the fifteen Δ_{ab} and close on $\mathfrak{so}(n)$, with $[C^{kl}, C^{lm}] = C^{km}$. Each is a sum of six commuting terms of square $-\frac{1}{4}$, so $e^{2\pi C^{kl}} = \mathbf{1}$. In Fock form

$$C^{kl} = \sum_{i=1}^{3}(a^+_{k,i}a^-_{l,i} - a^+_{l,i}a^-_{k,i}), \qquad C^{kl}a^+_{l,i}|0\rangle = a^+_{k,i}|0\rangle,$$

so the copy algebra conserves N and moves quanta between constituents.

(iii) As operators on S_n,

$$C_{\mathrm{cop}} + C_{\mathrm{diag}} = \tfrac{3}{4}\,n(n+4)\,\mathbf{1},$$

that is $\frac{15}{4}, 9, \frac{63}{4}, 24, \frac{135}{4}, \frac{81}{2}$ at $n = 1,\dots,6$. The interior identity of Theorem 5.11, $C^{\mathrm{int}}_{\mathrm{cop}} + C_\Delta = \frac{1}{2}n(n+2)\,\mathbf{1}$, holds at every n. Both constants are $nd(d+n-2)/8$, at $d = 6$ and $d = 4$.

(iv) At $n = 2$ the commutant of the diagonal $\mathfrak{so}(3,3)$ in $M_{64}(\mathbb{R})$ has dimension 14 and is spanned by the words in C^{12} and the Koszul swap τ_K, with $C^{12}\tau_K = -\tau_K C^{12}$: it is the image of the group algebra of $O(2)$. The words in C^{12} alone span 7 dimensions.

Proof. (i) $V = \mathbb{R}^{3,3}$ is absolutely irreducible under $\mathfrak{so}(3,3)$, so by Schur's lemma the commutant of $\mathbf{1}_n \otimes \mathfrak{so}(3,3)$ in $\mathrm{End}(\mathbb{R}^n \otimes V)$ is $\mathrm{End}(\mathbb{R}^n)\otimes\mathbf{1}_6$. An element $A\otimes\mathbf{1}_6$ is η_n-antisymmetric iff $A^\top = -A$. Conversely, $\mathbb{R}^n$ is absolutely irreducible under the group generated by $SO(n)$ and the permutation matrices for $n \ge 2$, and under $\mathfrak{so}(n)$ alone for $n \ge 3$; at $n = 2$ the commutant of $\mathfrak{so}(2)$ is $\mathrm{span}\{\mathbf{1}, \varepsilon\}$, which adds $\varepsilon \otimes \mathrm{Sym}_\eta(V)$, of dimension 21. The unweighted

sum lifts $(e_k e_l^\top - e_l e_k^\top) \otimes \eta$, and η does not commute with the boosts. (ii) The spin lift $\sigma(X) = \frac{1}{4}\sum_{A,B} X_{AB}\eta_{BB}E_A E_B$ satisfies $[\sigma(X), E_C] = \sum_A X_{AC}E_A$ and is a Lie algebra homomorphism; the vector matrices commute by (i). The six terms $\frac{1}{2}\eta_{aa}E_{(k,a)}E_{(l,a)}$ share no index, and $(E_{(k,a)}E_{(l,a)})^2 = -\eta_{aa}^2\mathbf{1}$. For the Fock form substitute $E_{(k,i)} = a^+_{k,i}+a^-_{k,i}$, $E_{(k,i+3)} = a^+_{k,i}-a^-_{k,i}$: $C^{kl} = \sum_i (a^+_{k,i}a^-_{l,i} + a^-_{k,i}a^+_{l,i})$, and $a^-_{k,i}a^+_{l,i} = -a^+_{l,i}a^-_{k,i}$ for $k \neq l$ by the CAR of Proposition 5.21. (iii) The proof of Theorem 5.11 uses neither $d = 4$ nor the signature of the interior: with $B^{kl}_{ab} = E_{(k,a)}E_{(k,b)}E_{(l,a)}E_{(l,b)}$ it gives $C_{\mathrm{diag}} = \frac{nd(d-1)}{8} - \frac{1}{2}\mathsf{S}$ and $C_{\mathrm{cop}} = \binom{n}{2}\frac{d}{4} + \frac{1}{2}\mathsf{S}$ with the same cross term $\mathsf{S} = \sum_{k<l}\sum_{a<b}\eta^{aa}\eta^{bb}B^{kl}_{ab}$. At $d = 6$ the constant is $\frac{15n}{4} + \frac{3n(n-1)}{2} = \frac{3}{4}n(n+4)$. (iv) Computed: the nullity of $\sum_X \mathrm{ad}_X^\top \mathrm{ad}_X$ over the fifteen generators, and the rank of the words. □

Part (i) is fixed (e3-4a, exact over $\mathbb{Q}$ at $n \leq 5$); at $n = 3$ its interior analogue is Theorem 5.9. Part (ii) restates Proposition 5.22(ii) at $n = 2$ and Proposition 9.24(i), and is fixed at $n \leq 4$ (e3-4b, e3-4f); the Fock form at $n = 2$ is Proposition 5.51(i) with the opposite orientation. Part (iii) is inherited: the constant $\frac{3}{4}n(n+4)$ is the case $d_{\mathrm{N}} = n$, $k_{\mathrm{N}} = 3$ of Neergård's relation $C_{\mathfrak{o}(d_{\mathrm{N}})} + C_{\mathfrak{o}(2k_{\mathrm{N}})} = \frac{1}{4}k_{\mathrm{N}}d_{\mathrm{N}}(d_{\mathrm{N}} + 2k_{\mathrm{N}} - 2)$ [296, eq. (73)], and the Clifford-algebra proof is an independent derivation in the real realisation. The identity was checked exactly as matrices on S_n at $n \leq 4$ and blade by blade in the abstract Clifford algebra at $n \leq 6$; the interior identity is restated and checked in the same two ways (e3-4c). Part (iv) is fixed (e3-4e; nullity in floating point with gap > 1, rank exact). The η-weighting is where the signature enters: a copy rotation pairs two generators of the same square, so it is compact (Theorem 5.9), and $C_{\mathrm{cop}} \geq 0$.

Howe's skew duality. By Proposition 5.21, $S_n \otimes \mathbb{C}$ is the exterior algebra on the $3n$ creation operators $a^+_{k,i}$, that is $\Lambda(\mathbb{C}^n \otimes \mathbb{C}^3)$ with k the copy index and i the mode index. The complexified diagonal $\mathfrak{so}(3,3)$ is spanned by the quadratics $\sum_k x_k y_k$ with $x, y \in \{a^\pm_{\cdot,i}\}$, the $O(n)$-invariant contractions over the copy index, which form, up to multiples of $\mathbf{1}$ and by the substitution used in (ii), $\Lambda^2\mathbb{C}^3 \oplus \mathfrak{gl}(3) \oplus \Lambda^2\mathbb{C}^3 \cong \mathfrak{so}(6,\mathbb{C})$. By (ii) the copy rotations are the $\mathfrak{so}(n) \subset \mathfrak{gl}(n)$ acting on the copy index. This is the setting of Howe's skew $(O(n), \mathfrak{so}(2m))$ duality on $\Lambda(\mathbb{C}^n \otimes \mathbb{C}^m)$, here with $m = 3$ [224]: the two actions generate mutual commutants, and the module decomposes without multiplicity as $\bigoplus_\lambda \rho_\lambda \otimes \pi_{\lambda'}$, with ρ_λ an irreducible representation of $O(n)$ and $\pi_{\lambda'}$ one of $\mathfrak{so}(6)$ determined by it. The theorem, and the linear relation between the two Casimirs, are inherited; (iii) is its constant and (iv) its commutant at $n = 2$ in the tower. The $O(n)$ contains more than $SO(n)$: at $n = 2$ the reflection is τ_K, and Section 5.7.5 shows that the Koszul swaps realise the permutation matrices at every rung.

Proposition 5.69 (Joint decomposition)**.** *The joint spectrum of $(C_{\mathrm{cop}}, C_{\mathrm{diag}})$ on S_2 and S_3 is that of Table 5.10. Every rank is the dimension of the copy module times the sum of the dimensions of the listed $\mathfrak{so}(6)$ modules, and every $\mathfrak{so}(6)$ Casimir equals $\frac{3}{4}n(n+4) - C_{\mathrm{cop}}$. At $n = 4$, $C_{\mathrm{cop}} = 0, 3, 4, 8, 9, 12, 15, 16, 19, 24$ on $70, 560, 504, 810, 1024, 200, 320, 450, 144, 14$ dimensions, the values $2[j_+(j_+ + 1) + j_-(j_- + 1)]$ of the $\mathfrak{so}(4) = \mathfrak{su}(2) \oplus \mathfrak{su}(2)$ modules $(0,0)$, $(\frac{1}{2},\frac{1}{2})$, $(1,0) \oplus (0,1)$, $(1,1)$, $(\frac{3}{2},\frac{1}{2}) \oplus (\frac{1}{2},\frac{3}{2})$, $(2,0) \oplus (0,2)$, $(\frac{3}{2},\frac{3}{2})$, $(2,1) \oplus (1,2)$, $(\frac{5}{2},\frac{1}{2}) \oplus (\frac{1}{2},\frac{5}{2})$, $(3,0) \oplus (0,3)$, each paired with $\mathfrak{so}(6)$ modules of Casimir $24 - C_{\mathrm{cop}}$ whose dimensions reproduce the ranks.*

Proof. At $n = 2, 3$ the minimal polynomial of C_{cop} is checked exactly and the ranks are traces of its Lagrange projectors, in integer arithmetic. The $\mathfrak{so}(6)$ dimensions are the Weyl dimension formula of D_3. At $n = 4$, C_{cop} commutes with N (by (ii)) and is diagonalised on each of the thirteen number sectors. □

This is fixed (e3-4d), and the $n = 3$ copy spectrum restates Proposition 9.24(ii). The joint spectra lie on the line of Theorem 5.68(iii). Copy type and $\mathfrak{so}(3,3)$ type determine each other, as j_{cop} and Lorentz type do in Corollary 5.12. Two features recur up the ladder. The sector of largest copy Casimir carries the smallest $\mathfrak{so}(3,3)$ module: the trivial module at $n = 2, 4$ and the half-spinors $\mathbf{4}, \bar{\mathbf{4}}$ of one constituent at $n = 3$. And the copy isospin is integral at $n = 2, 3$,

while at $n = 4$ the modules of $\mathfrak{so}(4) = \mathfrak{su}(2) \oplus \mathfrak{su}(2)$ include doublets of each factor, always with $j_+ + j_-$ integral, as single-valuedness requires (Theorem 5.68(ii)).

5.7.5 The symmetric group

Let $\tau_k = \mathbf{1}^{\otimes(k-1)} \otimes \tau_K \otimes \mathbf{1}^{\otimes(n-k-1)}$, with $\tau_K = D\,P$ the Koszul swap of Proposition 5.22(iii). For a surplus direction $s \in \{2,3\}$ let $\Theta_s = E_{(1,s)}E_{(2,s)}\cdots E_{(n,s)}$, and let $C_3 = \sum_{a<b,\,c<d,\,e<f} \varepsilon_{abcdef} L^{ab}L^{cd}L^{ef}$ be the cubic invariant of the diagonal $\mathfrak{so}(3,3)$, with $L^{ab} = \eta^{aa}\eta^{bb}\Delta_{ab}$.

Proposition 5.70 (The Koszul symmetric group)**.** *(i) $\tau_k^2 = \mathbf{1}$, $(\tau_k\tau_{k+1})^3 = \mathbf{1}$ and $\tau_k\tau_l = \tau_l\tau_k$ for $|k-l| > 1$, so the τ_k define an action of $\mathfrak{S}_n$ on S_n. Moreover $\tau_k E_{(j,a)}\tau_k = E_{(s_k j,a)}$ and $\tau_k = \frac{1}{8}\prod_a (E_{(k,a)} - E_{(k+1,a)})$: τ_k is the lift to $\mathrm{Pin}(3n,3n)$ of the transposition matrix of copies $k, k+1$, a product of six reflections. It commutes with the diagonal $\mathfrak{so}(3,3)$, with N, with H_g and with C_{cop}, and conjugates C^{kl} as the permutation does. On the vacuum $\tau_k|0\rangle = -|0\rangle$.*

(ii) At $n = 3$ the $\mathfrak{S}_3$-isotypic parts have the content

	$C_\Delta = \frac{3}{2}$	$\frac{11}{2}$	$\frac{15}{2}$	$C_{\mathrm{cop}} = 0$	2	6	12
trivial	24	48	16	20	36	20	12
sign	24	48	16	20	36	20	12
standard	112	192	32	0	144	160	32

(real dimensions; the standard rows are twice the multiplicity of the two-dimensional representation). The Dirac block, $C_\Delta = \frac{3}{2}$, contains 24 totally symmetric and 24 totally antisymmetric dimensions.

(iii) Θ_s is invertible, anticommutes with every τ_k, commutes with the diagonal $\mathfrak{l}_{\mathrm{int}}$ and with every C^{kl}, and anticommutes with C_3. It maps the Koszul-symmetric part isomorphically onto the Koszul-antisymmetric part, as modules of $\mathfrak{l}_{\mathrm{int}} \oplus \mathfrak{so}(n)$, and exchanges the two chiralities of $\mathfrak{so}(6)$. At $n = 3$ the eigenvalues of $8C_3$ are $-630, -210, -126$ on the symmetric part of the copy sectors $j = 0, 1, 2$ and $+630, +210, +126$ on the antisymmetric part.

(iv) The Koszul-symmetric part and the Koszul-antisymmetric part of S_n have the same dimension, $\sum_k \binom{n-k+3}{3}\binom{4}{k}$, which is $8, 32, 88, 192, 360$ at $n = 1, \ldots, 5$. Neither is ever zero.

Proof. (i) The relations are checked. $(E_{(k,a)} - E_{(k+1,a)})^2 = 2\eta_{aa}$, and the reflection in this vector exchanges $E_{(k,a)}$ and $E_{(k+1,a)}$; the product over the six directions acts on vectors as the transposition matrix P, because the sign $(-1)^6$ of the twisted adjoint action is $+1$. Hence $\tau_k\sigma(X)\tau_k^{-1} = \sigma(PXP^{-1})$ for every $X \in \mathfrak{so}(\eta_n)$ and $\tau_k v \tau_k^{-1} = Pv$ for every vector. P commutes with $\mathbf{1}_n \otimes \mathfrak{so}(3,3)$ and with K_n, and fixes $\sum_k E_{(k,1)}$, so τ_k commutes with the Δ_{ab}, with $N = \frac{3n}{2} + \sigma(K_n)$ (Lemma 5.46(i)) and with H_g; it permutes the C^{kl} up to sign and so fixes C_{cop}. (ii) Traces of the isotypic projectors $\frac{\dim\rho}{6}\sum_\sigma \chi_\rho(\sigma)\tau_\sigma$ times the spectral projectors of C_Δ and C_{cop}, in exact integer arithmetic. (iii) Θ_s is a blade. Against $E_{(k,a)} - E_{(k+1,a)}$ it gives the sign $(-1)^n$ for $a \neq s$ and $(-1)^{n-1}$ for $a = s$, so against τ_k the sign is $(-1)^{6n-1} = -1$. The interior bivectors share no index with Θ_s; the terms of C^{kl} share none or two; a diagonal bivector Δ_{sb} shares one, so $\mathrm{Ad}\,\Theta_s$ acts on the diagonal $\mathfrak{so}(3,3)$ as the reflection of the direction s, an outer automorphism, which reverses C_3. (iv) τ_σ permutes the basis tensors of $S^{\otimes n}$ with the Koszul sign of ω-parity, and ω has four even and four odd basis vectors. The image of the symmetriser is spanned by orbit sums, and an orbit contributes iff no odd basis vector repeats, so its dimension is the number of monomials of degree n in four even and four odd variables, $\sum_k \binom{n-k+3}{3}\binom{4}{k}$; for the antisymmetriser exchange the roles of even and odd, which gives the same number. □

Part (i) restates Proposition 5.22(iii) at $n = 2$ and is fixed at every rung (e3-5a at $n = 3, 4$; e3-5e, relations and vacuum at $n \leq 5$). Parts (ii)–(iv) are fixed (e3-5b; e3-5c, exact at $n \leq 4$, the C_3 eigenvalues in floating point to 10^{-6}; e3-5e). Since τ_k and the natural action $\Lambda(s_k)$ of the transposition on $\Lambda(\mathbb{C}^n \otimes \mathbb{C}^3)$ implement the same automorphism of $\mathrm{Cl}(3n, 3n)$, they differ by a scalar, and the vacuum fixes it: $\tau_\sigma = \mathrm{sgn}(\sigma)\Lambda(\sigma)$. The Koszul swaps are therefore the permutation matrices of Howe's $O(n)$, twisted by the sign character. Part (iii) generalises the mirror statement of Proposition 5.42(ii) to every rung: the Koszul sign does not sort the module by Lorentz type or by copy type, and its two sectors carry the same content up to the reflection of one direction.

Proposition 5.71 (The ungraded n-particle space)**.** *Let P_σ permute the tensor factors of S_n without signs, and let $W_n = \{J^{(1)} = \cdots = J^{(n)}\}$, of dimension $2 \cdot 4^n$, with $J := J^{(1)}|_{W_n}$ as imaginary unit, so that $W_n \cong S \otimes_{\mathbb{C}} \cdots \otimes_{\mathbb{C}} S = (\mathbb{C}^4)^{\otimes_{\mathbb{C}} n}$ (Proposition 5.36(iv)).*

(i) $P_\sigma J^{(k)} P_\sigma^{-1} = J^{(\sigma k)}$, so P_σ preserves W_n and is $\mathbb{C}$-linear there.

(ii) The totally antisymmetric part of W_n under P_σ is $\Lambda^n_{\mathbb{C}}\mathbb{C}^4$, of real dimension $2\binom{4}{n} = 8, 12, 8, 2, 0$ at $n = 1, \ldots, 5$, and the totally symmetric part is $\mathrm{Sym}^n_{\mathbb{C}}\mathbb{C}^4$, of real dimension $2\binom{n+3}{3} = 8, 20, 40, 70, 112$.

(iii) Under the diagonal $\mathfrak{l}_{\mathrm{int}}$, $\Lambda^2_{\mathbb{C}}\mathbb{C}^4$ has $C_\Delta = 0$ on 4 and 3 on 8 dimensions, $2(0,0) \oplus (\frac{1}{2}, \frac{1}{2})$; $\Lambda^3_{\mathbb{C}}\mathbb{C}^4$ has $C_\Delta = \frac{3}{2}$ on all 8, a Dirac spinor; and $\Lambda^4_{\mathbb{C}}\mathbb{C}^4$ has $C_\Delta = 0$, a scalar.

Proof. (i) $P_\sigma(A_1 \otimes \cdots \otimes A_n)P_\sigma^{-1}$ permutes the factors. (ii) Exact traces of $\frac{1}{n!}\sum_\sigma \mathrm{sgn}(\sigma)P_\sigma$ and of $\frac{1}{n!}\sum_\sigma P_\sigma$, composed with the projector $2^{1-n}\prod_k(\mathbf{1} - J^{(k)}J^{(k+1)})$ onto W_n, as signed permutations. (iii) The projector onto the antisymmetric part, as an exact integer matrix, satisfies the stated polynomial identities with C_Δ. Since $\mathbb{C}^4 = (\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$, $\Lambda^2 = \Lambda^2(\frac{1}{2}, 0) \oplus (\frac{1}{2}, 0) \otimes (0, \frac{1}{2}) \oplus \Lambda^2(0, \frac{1}{2})$, $\Lambda^3\mathbb{C}^4 \cong (\mathbb{C}^4)^* \otimes \Lambda^4\mathbb{C}^4$ and $\Lambda^4\mathbb{C}^4 = \det$. □

The representation theory is inherited [160]; its realisation on W_n is fixed (e3-5d). With the plain exchange and antisymmetry imposed, rung n on W_n has the content of n identical Dirac fermions at one momentum label, and exclusion holds: at most four, since a Dirac spinor has four states, so there is no antisymmetric state at $n \geq 5$ (exclusion inherited [217]). The scope is narrow. The vanishing of $\Lambda^n_{\mathbb{C}}\mathbb{C}^4$ is a statement about one mode of a finite construction; it is not a limit on the number of fermions in physical space, which would need a continuum of momenta. The antisymmetry is imposed, not derived: neither exchange of the tower proves a spin–statistics connection, which needs a local relativistic field theory with positive energy. The two exchanges therefore differ at every rung, not only at $n = 2$ (Proposition 5.41). The Koszul exchange is the one the algebra supplies, since it permutes the vectors of $\mathrm{Cl}(3n, 3n)$. Its antisymmetric sector has dimension 88 at $n = 3$ and contains all three Lorentz types, and at $n = 5$ it has dimension 360 where the plain one is empty. It is not a statistics label. The plain exchange is a statistics label, but only on the constructed space W_n, with the constructed generators $u_{(k,a)}$ as gamma matrices.

5.7.6 The step from n to $n + 1$, and what going up the ladder means

What the step preserves. By Proposition 5.64(ii) the step keeps rung n as the subalgebra $\mathrm{Cl}(3n, 3n) \otimes \mathbf{1}$, with all of its operators. Under the spectator action the interior four-plane stays in the first factor, so the tower is one Lorentz frame with nested internal centralisers (Proposition 5.3). The diagonal module is tensored with S (Proposition 5.64(iv)), and the dual pair $(O(n), \mathfrak{so}(3,3))$ sits inside $(O(n+1), \mathfrak{so}(3,3))$.

What the step adds. Three Fock modes, $N_{n+1} = N_n \otimes \mathbf{1} + \mathbf{1} \otimes N_1$ (Proposition 5.64(iii)). One frame: $V_{n+1} = V_n \oplus V$, which under the diagonal $\mathfrak{l}_G$ is one more field strength $(1,0) \oplus (0,1)$ and under the diagonal $\mathfrak{l}_{\text{int}}$ one more potential $(\frac{1}{2}, \frac{1}{2})$ with two scalars (Section 5.8). One copy direction: the copy space $\mathbb{R}^n$ becomes $\mathbb{R}^{n+1}$, the copy algebra gains the n rotations $C^{k,n+1}$, and the Casimir constant grows by $\frac{3}{4}(2n+5)$ (Theorem 5.68(iii); e3-6c). The new constituent has its own commuting copy, $(\Gamma_n \otimes \mathbf{1})E_{(n+1,a)}$.

What the step changes. The graded mass grows in quadrature and the ungraded energies add (Proposition 5.65(iv)). The spin parity under the diagonal $\mathfrak{l}_{\text{int}}$ flips, and that under the diagonal $\mathfrak{l}_G$ does not (Proposition 5.66). Every parity-dependent selection of Section 9.7.1 switches on or off.

One particle or several. Read together, the statements of this section answer the question in four sentences. (1) As given by its vectors, every rung carries one Dirac mass $\sqrt{n}\,m$ along a unit vector of $\mathbb{R}^{3n,3n}$ with one clock, and copies that are the components of a vector of the copy group $O(n)$ (fixed); read as particle content, this is one particle (interpretive). None of the tests used here, mass shells, complex structures, exchanges or counting operators, is a particle count by itself: a particle interpretation needs a state space with a positive inner product, a positive-energy representation of the Poincaré group and asymptotic states, and none is constructed here. That group is the Howe dual of the diagonal $\mathfrak{so}(3,3)$, its type is locked to the $\mathfrak{so}(3,3)$ type by $C_{\text{cop}} + C_{\text{diag}} = \frac{3}{4}n(n+4)$, and its Lie algebra moves quanta between constituents without creating any. So the copy index behaves as a flavour index of one particle, and going up the ladder adds a flavour direction and three modes to that particle (fixed statements, read interpretively; Proposition 5.65 and Theorem 5.68). (2) The Lorentz content does not decide. The parity $(-1)^n$ under the interior action is the parity of n spin-$\frac{1}{2}$ particles and equally that of one particle in an n-fold tensor product, and under the field-strength action every rung is bosonic (fixed; inherited rule; Proposition 5.66). (3) A rung carries n particles in the sense of ordinary quantum mechanics only with the structures of Section 5.4, now needed at every rung: the commuting copies $u_{(k,a)} = \Gamma^{(1)} \cdots \Gamma^{(k-1)} E_{(k,a)}$ (fixed as an algebra), the additive Hamiltonian H_u with its n clocks, the common complex space W_n with $J|_{W_n}$, and the plain exchange with antisymmetry (constructed; the unit selected up to sign at odd n only). With them the rung has the content of n identical Dirac fermions at one momentum label, with exclusion $n \leq 4$ at that label (fixed; inherited; antisymmetry imposed; Proposition 5.71). (4) The exchange the algebra supplies is the Koszul $\mathfrak{S}_n$, the permutation matrices of $O(n)$ up to sign, and it is not a statistics label: its two sectors are mirror images and never empty (fixed; Proposition 5.70).

Which process goes up the ladder. Within a rung, the copy algebra hops quanta among the n constituents and conserves N (Theorem 5.68(ii); Proposition 5.51). Between rungs there is no process in the algebra: rung $n+1$ is not an element or a state of rung n, and the embedding $\mathrm{Cl}(3n,3n) \otimes \mathbf{1}$ makes rung n the part of rung $n+1$ that does not see the last constituent. In the two-particle reading the step is the tensor embedding $\psi \mapsto \psi \otimes \phi_0$, which needs a state ϕ_0 that nothing selects (constructed, as at $n = 2$). Whether a dynamical mechanism selects W_n, the additive Hamiltonian and a rung is open.

5.8 Spin one up the tower

The tower

On the tower the two readings multiply in different ways.

Proposition 5.72 (Commutants along the tower)**.** *On $V_n = \mathbb{R}^{3n,3n}$ the commutants have the following dimensions:*

commutant of	$n=1$	2	3	4	5
spectator $\mathfrak{l}_{\text{int}}$ in $\mathfrak{so}(3n,3n)$	1	28	91	190	325
diagonal $\mathfrak{l}_{\text{int}}$ in $\mathfrak{so}(3n,3n)$	1	7	18	34	55
diagonal $\mathfrak{l}_G$ in $\mathfrak{so}(3n,3n)$	0	2	6	12	20
diagonal rotations and G_n in $\mathfrak{so}(6n)$	1	4	9	16	25
the same inside $\mathfrak{so}(3n,3n)$	0	1	3	6	10

These are the dimensions of $\mathfrak{so}(3n-1,3n-3)$, $\mathfrak{so}(n)\oplus\mathfrak{so}(2n)$, $\mathfrak{so}(n,\mathbb{C})$, $\mathfrak{u}(n)$ *and* $\mathfrak{so}(n)$.

Under the diagonal interior action $V_n = n\,(\frac{1}{2},\frac{1}{2})\oplus 2n\,(0,0)$, so there are n potentials forming the vector of the copy algebra $\mathfrak{so}(n)$, and none of them is copy-invariant at grade one for $n\geq 2$. A single univectorial potential in the diagonal sector therefore needs a choice of direction in copy space, and any such choice breaks $\mathfrak{so}(n)$ to $\mathfrak{so}(n-1)$. The obstruction belongs to grade one only.

Theorem 5.73 (Copy-invariant four-vectors)**.** *A copy-invariant four-vector in* $\Lambda^g(\mathbb{R}^n\otimes\mathbb{R}^{1,3})$ *requires* n *odd and* g *odd with* $n\leq g\leq 3n$*, and for every odd* n *there is one at* $g=n$*,*

$$Q_b^{(n)} = \epsilon_{k_1\cdots k_n}\,\eta^{a_1a_2}\cdots\eta^{a_{n-2}a_{n-1}}\,\gamma_{a_1}^{(k_1)}\wedge\cdots\wedge\gamma_{a_{n-1}}^{(k_{n-1})}\wedge\gamma_b^{(k_n)}.$$

At $n=3$ *there is exactly one at each of the grades* $3,5,7,9$*, and at* $n=2$ *there is none at any grade.*

Proof. An invariant element of $\Lambda^g(\mathbb{R}^n\otimes\mathbb{R}^{1,3})\otimes\mathbb{R}^{1,3}$ is an invariant tensor of $SO(n)\times SO^+(1,3)$ in $(\mathbb{R}^n)^{\otimes g}\otimes(\mathbb{R}^{1,3})^{\otimes(g+1)}$. By the first fundamental theorem for the special orthogonal groups [161], such tensors are spanned by products of metrics and at most one Levi-Civita symbol. On the Lorentz side every factor has an even number of indices, so $g+1$ is even. On the copy side $g=2d+ne$ with $e\in\{0,1\}$; g odd forces $e=1$ and n odd, so $g\geq n$. The Hodge star of $\mathbb{R}^n\otimes\mathbb{R}^{1,3}$ is an equivariant isomorphism $\Lambda^g\cong\Lambda^{4n-g}$, which gives $g\leq 3n$. The displayed $Q_b^{(n)}$ is invariant because ϵ is, and nonzero because the terms of $Q_0^{(n)}$ with all Lorentz indices equal to 0 add up to $n!\,\gamma_0^{(1)}\wedge\cdots\wedge\gamma_0^{(n)}$. □

At $n=3$ the invariant four-vector of grade three is, up to a factor, the copy-singlet cubic Q_μ that builds the gamma matrices of the Dirac block in Theorem 5.13; the parity theorem explains why such an object exists at three constituents and could not exist at two. Under the diagonal $\mathfrak{l}_G$, finally, $V_n = n[(1,0)\oplus(0,1)]$: n field strengths mixed by the complex orthogonal flavour algebra $\mathfrak{so}(n,\mathbb{C})$. The energy-preserving duality group of n free Maxwell fields is $U(n)$, the maximal compact subgroup of the Gaillard–Zumino group $Sp(2n,\mathbb{R})$ [46]; its Lie algebra meets $\mathfrak{so}(3n,3n)$ only in $\mathfrak{so}(n)$, and its centre, generated by G_n and carrying the total helicity, lies outside $\mathfrak{so}(3n,3n)$, as G does at $n=1$. A single photon would be one direction in this flavour space, and nothing selects one.

The mixed rung

Proposition 3.6 forbids reading one six-dimensional space in two ways. At $n\geq 2$ there is more than one constituent, and the obvious repair is to let one constituent carry the interior reading and another the frame reading, under one Lorentz action.

Proposition 5.74 (The mixed embedding)**.** *In* $\mathrm{Cl}(6,6)$ *let* $\mathfrak{l}_{\text{mix}}$ *be spanned by* $S_k\otimes 1+1\otimes L_k$ *and* $T_k\otimes 1+1\otimes K_k$*: the interior algebra on constituent* 1 *and* $\mathfrak{l}_G$ *on constituent* 2*, paired generator by generator. It satisfies the* $\mathfrak{so}(1,3)$ *brackets, and*

$$\mathbb{R}^{6,6} = \underbrace{V}_{C=3}\oplus\underbrace{\Lambda^2 V}_{C=4}\oplus 2\cdot\mathbf{1},$$

so one Lorentz action carries both a four-vector and a field strength at grade one. On $S_2 = \mathbb{R}^{64}$ *the Casimir satisfies* $(C_{\mathrm{mix}} - \frac{3}{2})(C_{\mathrm{mix}} - \frac{11}{2}) = 0$*, with the Dirac value* $\frac{3}{2}$ *on sixteen dimensions and* $\frac{11}{2}$ *on* 48*.*

Proposition 5.75 (The cost)**.** *The centraliser of* $\mathfrak{l}_{\mathrm{mix}}$ *in* $\mathfrak{so}(6,6)$ *is one-dimensional, spanned by the rotation of the surplus pair of constituent* 1*. At the same rung:*

Lorentz action at $n = 2$	centraliser	content of $\mathbb{R}^{6,6}$
interior on one constituent	28	$V \oplus 8 \cdot \mathbf{1}$
$\mathfrak{l}_G$ on one constituent	15	$\Lambda^2 V \oplus 6 \cdot \mathbf{1}$
mixed, across both	1	$V \oplus \Lambda^2 V \oplus 2 \cdot \mathbf{1}$

The mixed embedding buys the field-strength module at the price of the whole internal algebra: $\mathfrak{so}(5,3)$ *collapses to one* $\mathfrak{u}(1)$*.*

The count is for this pairing of the two Lorentz bases at $n = 2$; another pairing or a higher rung could do better and has not been examined. Within its scope, however, the mixed rung is an honest place to test whether the two readings can be coupled, and it turns out that they can.

Theorem 5.76 (The Dirac block of the mixed rung)**.** *Let* $P = \frac{1}{8}(11\,\mathbf{1} - 2C_{\mathrm{mix}})$ *and* $\hat{\gamma}_\mu = 2P\gamma_\mu^{(1)}P$*, with* $J_1 = e_2e_3 \otimes 1$*.*

(i) P *is the orthogonal projector of rank* 16 *onto the block with the Dirac value, and commutes with the mixed generators.*

(ii) $\{\hat{\gamma}_\mu, \hat{\gamma}_\nu\} = 2\eta_{\mu\nu}P$*,* $[\hat{\gamma}_\mu, J_1] = 0$*,* $J_1^2 = -\mathbf{1}$*, and* $P\mathcal{S}_kP = \frac{1}{2}\hat{\gamma}_i\hat{\gamma}_j$ *(cyclic),* $P\mathcal{T}_kP = \frac{1}{2}\hat{\gamma}_0\hat{\gamma}_k$*: the compressed gammas generate the mixed Lorentz action on the block.*

(iii) With J_1 *as the imaginary unit the block is two complex Dirac modules, labelled by the half-spinors of constituent* 2*, with flavour algebra* $\mathfrak{u}(2)$*;* $\mathcal{H}(p) = \hat{\gamma}^0\hat{\gamma}^a p_a + m\hat{\gamma}^0$ *is symmetric and* $\mathcal{H}(p)^2 = (|p|^2 + m^2)P$*.*

(iv) The block is the γ*-trace part* $\sum_a \gamma_a\chi \otimes \tau e_a$ *of the vector-spinors (Dirac spinor of constituent* 1*)* $\otimes$*(four-vector of constituent* 2*), and the rank-*48 *block is their* γ*-traceless* $(1, \frac{1}{2}) \oplus (\frac{1}{2}, 1)$ *part.*

Item (iv) explains the factor 2 in $\hat{\gamma}_\mu$, since $P\gamma_\mu^{(1)}$ acts on the γ-trace part as $-\frac{1}{2}\gamma_\mu$, and it identifies the decomposition with the classical splitting of a vector-spinor into its trace and its traceless part [130].

A covariant identification of the field strength of constituent 1 with the frame field of constituent 2 is an invariant element of $\Lambda^2(\text{four-plane}_1) \otimes V_2$, that is, a grade-three element of the algebra commuting with the mixed generators. The invariant space is two-dimensional, and one invariant is

$$\Theta_F = \gamma_2^{(1)}\gamma_3^{(1)}e_1' + \gamma_3^{(1)}\gamma_1^{(1)}e_2' + \gamma_1^{(1)}\gamma_2^{(1)}e_3' - \gamma_0^{(1)}\gamma_1^{(1)}e_4' - \gamma_0^{(1)}\gamma_2^{(1)}e_5' - \gamma_0^{(1)}\gamma_3^{(1)}e_6',$$

with e_a' the generators of constituent 2; it pairs the spatial bivectors of constituent 1 with the magnetic directions of constituent 2 and the time–space bivectors with the electric ones. The other invariant is Θ_F followed by G, so up to scale the identifications form the family $X = e^{\theta G}X_F$.

Theorem 5.77 (A coupled model)**.** *On the block, in flavour coordinates* $\mathbb{R}^8 \otimes \mathbb{R}^2$ *with* $\hat{\gamma}_\mu = -\gamma_\mu \otimes 1$*,* $J_1 = J \otimes 1$ *and* $U^\mu = \hat{\gamma}^0\hat{\gamma}^\mu$*, take for commuting real fields*

$$\mathcal{L} = -\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} + \psi^\top J_1 U^\mu D_\mu\psi - m\,\psi^\top\hat{\gamma}^0\psi, \qquad D_\mu = \partial_\mu + qA_\mu J_1, \tag{5.3}$$

the real form of $-\frac{1}{4}F^2 + \bar{\psi}(i\gamma^\mu D_\mu - m)\psi$ *with* J_1 *in the place of* i*. Then the field equations are Maxwell's equations* $\partial_\mu F^{\mu\nu} = q\, j^\nu$ *with* $j^\nu = \psi^\top U^\nu \psi$ *and* $j^0 = |\psi|^2$*, and the Dirac equation* $J_1\hat{\gamma}^\mu D_\mu \psi = m\psi$*; the identity* $\partial_\mu j^\mu + \psi^\top J_1 \mathcal{E}_\psi = 0$ *holds with* $\mathcal{E}_\psi$ *the Euler–Lagrange derivative of the action,* $\mathcal{E}_\psi = 2\hat{\gamma}^0(J_1\hat{\gamma}^\mu D_\mu \psi - m\psi)$*, and not with the Dirac expression itself; the Lagrangian is exactly invariant under* $\psi \mapsto e^{-q\lambda J_1}\psi$*,* $A \mapsto A + \partial\lambda$*; and its flavour symmetry is the* $\mathfrak{u}(2)$ *spanned by* $J \otimes 1$*,* $1 \otimes \epsilon$*,* $J \otimes \sigma_x$*,* $J \otimes \sigma_z$*. The potential carries two polarisations of helicity* ± 1 *and no massive mode, and each momentum carries two flavours, two helicities and two signs of frequency.*

In the frame reading the source is electric. For the frame field of an arbitrary potential one has, identically, $\partial_t z + i\nabla \times z + i\boldsymbol{\mathcal{J}} = 0$ and $\nabla \cdot z = i\rho$ with $(\rho, \boldsymbol{\mathcal{J}}) = \partial_\mu F^{\mu\nu}$, and the rotated field $e^{\theta G} X_F$ satisfies the same system with electric source $\cos\theta$ times the current and magnetic source $\sin\theta$ times it. Since the frame field takes no part in the dynamics of (5.3), which member of the family is called the frame field is a choice; requiring it to carry electric sources only fixes $\theta \equiv 0$ modulo π, as in the textbook treatment of duality with sources, where the angle is fixed by the convention that all charges are electric [76]. The duality rotation, a symmetry of the free system and of (3.15), is not a symmetry once electric sources are present.

The coupled model is classical two-flavour spinor electrodynamics with equal charges, written in the algebra of the second rung. It is a construction, not a selection, and it has a definite limitation that we record now and use later: the second-rung vacuum of Section 8.5 is defined for the spectator action, and under the mixed action it is preserved only by the rotations about one axis, so the coupled model lies outside that vacuum sector.

At even rungs, finally, the diagonal interior action places integer-spin modules inside the spinor module itself: at $n = 2$ the Casimir takes the values $0, 3, 4$ on $8, 32, 24$ dimensions, so S_2 contains scalars of copy charge ± 2, four-vectors of copy charge ± 1 and copy-neutral field strengths $(1,0) \oplus (0,1)$, and the copy and Lorentz Casimirs add up to 4, the $n = 2$ case of (5.2). These are modules of one spinor space, not bound states, and we do not propose them as the photon.

5.9 Synthesis: what the tower establishes about particle multiplicity

The question of this section was whether, and in what sense, a rung $\mathrm{Cl}(3n, 3n)$ of the tower carries several particles rather than one particle with a larger internal space. Particle multiplicity was tested in five senses, each with its own operational meaning: separate mass shells, a common complex structure for the copies, a counting operator, an exchange that sorts by statistics, and a process that takes one to two. This section states the outcome claim by claim, in four classes: *proved algebra* (theorems, exact computations and restatements of fixed results), *conditional or constructed structures*, *obstructions within a stated class*, and *missing dynamics*. Table 9.9 gives the evidence for each claim.

5.9.1 The answer

The tower supplies the algebraic room for n-fold multiplicity, but it does not establish particle multiplicity.

(1) *What is proved.* Every rung is $M_{8^n}(\mathbb{R})$, and the step embeds rung n in rung $n+1$ with multiplicity eight. A second, commuting copy of each constituent exists and is unique up to conjugation; its generators are the graded ones dressed by the volume elements of the earlier constituents. The graded masses add in quadrature to one Dirac mass $(\sum_k m_k^2)^{1/2}$, with no simultaneous nonzero-mass Dirac constraints of the graded form, while the commuting copies have additive spectra $\{\sum_k \pm m_k\}$. A common complex structure for all copies exists

only on the subspace $W_n = \{J^{(1)} = \cdots = J^{(n)}\}$, of real dimension $2 \cdot 4^n$. The diagonal interior action selects an imaginary unit exactly at odd n, and none at even n. The Fock number N is conserved exactly by $\mathfrak{gl}(3n, \mathbb{R})$. Within the maximal compact subalgebra, it is conserved exactly by the paired rotations, the compact binding-compatible fields. No element of $\mathfrak{so}(3n, 3n)$ changes it by an odd number. N is invariant under no Lorentz action of the tower, and a Lorentz-invariant number of quanta exists, under the common binding action, exactly at even n. On vectors, the copy algebra with the copy permutations adjoined and the diagonal $\mathfrak{so}(3, 3)$ are mutual centralisers (Theorem 5.68(i)); on S_n the Casimirs satisfy $C_{\text{cop}} + C_{\text{diag}} = \frac{3}{4}n(n+4)$, and the dual-pair statement there concerns the full $O(n)$ action, reflections included [297]. The compact field between constituents is a one-body operator: it moves quanta and never creates one.

(2) *What is constructed, not selected.* A reading with n particles in the sense of ordinary quantum mechanics needs three structures together: the subspace W_n with the complex structure $J|_{W_n}$, the additive Hamiltonian H_u built from the commuting copies, and the plain exchange P with antisymmetry imposed. Each exists. None is selected by the algebra or by a Lorentz action of the tower. With them, rung n has the content of n identical Dirac fermions at one momentum label, with exclusion $n \leq 4$ at that label.

(3) *What is obstructed.* At $n = 2$ no selection principle of the paper gives a complex structure under which both copies are complex Dirac spinors. Under the diagonal action none exists, and under the spectator action copy 2 carries no linear Dirac operator. No gauge-admissible, binding-compatible field changes N. Within the binding-compatible algebra the number-changing directions are the paired boosts, which are not gauge fields; outside it, compact generators and non-compact ones, the interior boosts of Proposition 5.47 among them, also mix sectors differing by two quanta. The Clifford exchange τ_K is not a statistics label. At $n = 2$ none of the compact parts of the five candidate gauge algebras contains the Pati–Salam algebra.

(4) *What is missing.* No state space with a positive inner product carrying a positive-energy representation of the Poincaré group is constructed. There are no asymptotic states, no derived quantum theory of tower-changing processes (Theorem 5.77 is a classical interacting model), no dynamics that selects W_n, H_u, an exchange or a rung, and no proof of a spin–statistics connection. Without these, "one particle with a larger internal space" and "n particles" are two readings of the same algebra, and the tower does not decide between them.

Three readings that earlier drafts suggested are not supported. (a) The gap $m_1 + m_2 - (m_1^2 + m_2^2)^{1/2}$ is not a binding energy. It compares two operators on one space at equal momentum labels, and at fixed total momentum the graded levels with large relative momentum lie above the free-pair threshold (Proposition 5.44; z1-2a). (b) Fock mixing by a boost, a non-unitary operator on a finite space at one momentum label, is not observable pair production. (c) The time-odd triangle signs of the copy field are holonomies in copy space, not physical CP violation.

claim	statement	class	check
Thm. 5.52(i)	rate constant and nonzero iff $R_{\mathrm{eff}} = 0$ and $h(0) \neq 0$; frames then on circles	fixed (restated, Proposition 4.33)	d6-2a–d
Thm. 5.52(ii)	$n = 1$: a constant nonzero rate with nonzero Lorentz-class inertia requires $E_0(\pi) = 0$	fixed (restated, Corollary 4.29)	d6-1a
Thm. 5.52(iii)	$n \geq 2$: codimension-two balance set, massive data dense	fixed	d6-1b
Rem. 5.53	without the Lorentz requirement $n = 1$ suffices	fixed (counterexample)	d6-1c
Cor. 5.54	no closure Hamiltonian creates or moves a pair	obstructed	d6-3
Prop. 5.55	$a \cdot b = 0$; $V_{\pm}$ null, $E = \mp G_2$; magnetic–electric pair, $M_{\mathrm{pair}} = 2E_0$, $q_{\mathrm{tot}} = 0$	fixed (part restated); readings interpretive	d6-4a–e
Prop. 5.56(i,ii)	commutant $M_2(\mathbb{C})$, $\mathfrak{g}(0, 2, 0)$, invariants of R_φ and $U(2)$	restated / inherited	d6-5a,b
Prop. 5.56(iii)	no covariant map makes a pair from one null field	obstructed	d6-5c
Thm. 5.57	$M_{\mathrm{pair}} = \frac{1}{2}(\sigma_1 + \sigma_2)$, the chain, $\max_\varphi$	(i), (iii), (v) inherited; (ii), (iv) derived here, priority unestablished	d6-6a–d
Ex. 5.58	two null fields → electric and magnetic at rest	inherited	d6-7
Ex. 5.59	rotation of one direction: balanced pair from a null field, covariant under none of the five Lorentz actions at $n = 2$; third body	fixed (numbers, non-covariance); agent constructed; reading interpretive	d6-8; x6-8a,b
Prop. 5.60	R_φ admissible; per-copy E_0 not invariant; $O(6, \mathbb{C})$ maps pairs to null fields	fixed (part restated)	d6-9a–c, 10
interior	distinguished copies: per-copy rest energy changed by two-level mixing, sum $\leq M_{\mathrm{pair}}$	inherited	d6-11
Rem. 5.61	a dynamical copy rotor changes $E_0^{(k)}$ and their sum at fixed pair invariants	constructed	d6-12
Prop. 5.62	Klein signs, C-parity, masses in quadrature	inherited	d6-13a–e
Prop. 5.63	formal cancellation of the zero-point energy of the nonzero lines of the indefinite Hamiltonian; exceptional ratio $\beta^2 = \kappa_1\kappa_2$	fixed ((i) restated; lines, boundedness and (iii) restate Theorem 5.31)	d6-14a–c
answers	duplication as copy rotation, pair creation and balance	interpretive	—

Table 5.9: The claims of Section 5.6, their classes and the checks of `d6_pair_balance.py` (d6-0 verifies the conventions; x6-8a,b are printed by `x6_pair_balance_referee.py`).

n	copy label	C_{cop}	C_{diag}	$\mathfrak{so}(6)$ modules $(\lambda_1,\lambda_2,\lambda_3)$	dimensions	rank in S_n
2	$\lvert j\rvert=0$	0	9	$(1,1,1)$, $(1,1,-1)$	$10+10$	$1\cdot 20=20$
	$\lvert j\rvert=1$	1	8	$(1,1,0)$	15	$2\cdot 15=30$
	$\lvert j\rvert=2$	4	5	$(1,0,0)$	6	$2\cdot 6=12$
	$\lvert j\rvert=3$	9	0	$(0,0,0)$	1	$2\cdot 1=2$
3	$j=0$	0	$\frac{63}{4}$	$(\frac{3}{2},\frac{3}{2},\pm\frac{3}{2})$	$20+20$	$1\cdot 40=40$
	$j=1$	2	$\frac{55}{4}$	$(\frac{3}{2},\frac{3}{2},\pm\frac{1}{2})$	$36+36$	$3\cdot 72=216$
	$j=2$	6	$\frac{39}{4}$	$(\frac{3}{2},\frac{1}{2},\pm\frac{1}{2})$	$20+20$	$5\cdot 40=200$
	$j=3$	12	$\frac{15}{4}$	$(\frac{1}{2},\frac{1}{2},\pm\frac{1}{2})$	$4+4$	$7\cdot 8=56$

Table 5.10: Joint decomposition of S_n under the copy algebra and the diagonal $\mathfrak{so}(3,3)$ (Theorem 5.68). At $n=2$, j is the $\mathfrak{so}(2)$ weight; at $n=3$, j is the copy isospin. $C_{\mathrm{diag}}=\langle\lambda,\lambda+2\rho\rangle$ with $\rho=(2,1,0)$, so that the spinor has $\frac{15}{4}$ and the vector 5. Ranks are real dimensions of S_n, equal to complex dimensions of $S_n\otimes\mathbb{C}$. Check e3-4d.

statement	class	checks
Proposition 5.64(i): $\mathrm{Cl}(3n,3n) = M_{8^n}(\mathbb{R})$, volume $\omega^{\otimes n}$, $(-1)^N = (-1)^n\Gamma_n$	fixed in the realisation; classification inherited	e3-1a
Proposition 5.64(ii), (iii): step formulas, eight copies, commutant $\mathbf{1} \otimes M_8(\mathbb{R})$; Fock modes	restated (Proposition 5.20, Proposition 5.21; Proposition 5.33)	e3-1b, e3-1c
Proposition 5.64(iv): diagonal bivectors additive, $S_{n+1} = S_n \otimes S$	fixed	e3-6c
Proposition 5.65(i), (ii): $\pm\sqrt{n}\,m$; $m(n-2j)$ with $\binom{n}{j}4^n$; exact dressing	restated (Proposition 5.34(vi), Proposition 5.62(iv)); multiplicities fixed	e3-2a, e3-2b
Proposition 5.65(iii), (iv): one clock against n clocks; the step	fixed	e3-2a, e3-2b, e3-6c
Proposition 5.66(i), (iv): $e^{2\pi R_n} = (-1)^n$; $(-1)^{k_I}$; C_Δ spectra	restated (Proposition 5.16, Proposition 5.17(i)); spectra fixed	e3-3a, e3-3c
Proposition 5.66(ii), (iii): $e^{2\pi L_n} = +\mathbf{1}$; $(\frac{1}{2},\frac{1}{2})^{\otimes n}$; central element	fixed (extends Proposition 3.83)	e3-3b, e3-3c
Remark 5.67: spin parity against copy-invariant four-vectors	restated (Theorem 5.13, Theorem 5.73); comparison interpretive	e3-6b
Theorem 5.68(i): mutual commutants on vectors; η-weighting forced	fixed	e3-4a
Theorem 5.68(ii): spin lifts, $\mathfrak{so}(n)$, single-valued, Fock form	fixed; restated at $n = 2, 3$ (Proposition 5.22, Proposition 9.24)	e3-4b, e3-4f
Theorem 5.68(iii): $C_{\mathrm{cop}} + C_{\mathrm{diag}} = \frac{3}{4}n(n+4)$; interior identity	inherited (Neergård, eq. (73)); independent proof; interior restated (Theorem 5.11)	e3-4c, z1-3a
Theorem 5.68(iv): commutant at $n = 2$ is the image of $\mathbb{R}[O(2)]$	fixed	e3-4e
Howe's skew duality $(O(n), \mathfrak{so}(6))$ on $\Lambda(\mathbb{C}^n \otimes \mathbb{C}^3)$	inherited [224]	e3-4f
Proposition 5.69, Table 5.10: joint decomposition at $n = 2, 3, 4$	fixed; $n = 3$ copy spectrum restated (Proposition 9.24)	e3-4d
Proposition 5.70(i): Koszul $\mathfrak{S}_n$, Pin lift, commutation, vacuum sign	fixed; restated at $n = 2$ (Proposition 5.22(iii))	e3-5a, e3-5e
Proposition 5.70(ii)–(iv): $\mathfrak{S}_3$ content; mirror Θ_s; sector dimensions	fixed	e3-5b, e3-5c, e3-5e
Proposition 5.71: W_n, $\Lambda^n_{\mathbb{C}}\mathbb{C}^4$, $\mathrm{Sym}^n_{\mathbb{C}}\mathbb{C}^4$, Lorentz content	fixed in the realisation; inherited (exterior powers, exclusion); one momentum label; antisymmetry imposed	e3-5d, z1-3c
Table 9.8: the ladder $n = 1, \ldots, 5$	fixed; restated (Theorem 5.1, Theorem 6.10, Theorem 5.15, Theorem 5.73, Proposition 5.72; Proposition 9.24)	e3-6a, e3-6b, as listed
Section 9.7.1: no rung selected by the algebra; odd, even, third rung	interpretive (drawn from fixed statements); period four inherited	—
Section 5.7.6: step; one particle or several; process	fixed parts as cited; reading interpretive; tensor embedding constructed; mechanism open	e3-1b, e3-1c, e3-6c

Table 5.11: Claims of Section 5.7, their classes and their checks, which are `PASS` lines of `e3_ladder.py`. "Restated": established in an earlier chapter or in Section 5.4, re-verified here.

Chapter 6

Chiral Matter and Internal Symmetries

The question of this chapter is which chiral content the tower carries at three constituents and which of its symmetries can be gauged. It treats the projectors of rank 64, among them the family projector P_{fam}, which is kept apart from the chiral projector; the chiral Pati–Salam block P_*, which carries one family with a right-handed neutrino; and the charges and anomalies of the block. The content is proved. The choice of P_* among the projectors is constructed, not selected, and chirality is imposed, not derived (Proposition 8.5).

6.1 Chirality, projectors, and colour

6.1.1 Unprojected matter is vector-like

With $i = J$ and $\gamma_5 = \Gamma$ as in Theorem 2.17, consider the part of the Lorentz commutant that preserves chirality. It splits into a *vector class* $\mathrm{Cl}^0(W)$, the even subalgebra of the internal directions, whose elements commute with the interior gamma matrices, and an *axial class* $\Omega\,\mathrm{Cl}^0(W)$, whose elements anticommute with them.

Proposition 6.1 (Unprojected matter is vector-like)**.** *Every grade-two internal algebra lies in the vector class. For a gauge algebra in the vector class, γ_μ is an invertible, complex-linear, gauge-equivariant map between the two chirality sectors, so the unprojected matter is vector-like at every rung.*

Proposition 6.2. *The two classes form a symmetric pair, $[\mathrm{a},\mathrm{a}] \subseteq \mathrm{v}$, so the axial class is a Lie triple system and no nonabelian gauge algebra lies entirely inside it.*

This does not exclude generators with both parts. If $[\Gamma, t_a] = 0$ and $P_L = \frac{1}{2}(\mathbf{1} - \Gamma)$, then

$$[P_L t_a, P_L t_b] = P_L^2 [t_a, t_b] = \epsilon_{abc} P_L t_c,$$

a nonabelian algebra, half vector and half axial. Such *dressing* is available to nonabelian carriers exactly as to abelian ones; only the purely axial summand is excluded. For scalar dressings $f(\Gamma)t_a$ with $f = a + b\Gamma$ and unchanged structure constants, closure forces $f^2 = f$, that is $a^2 + b^2 = a$ and $2ab = b$, so $f \in \{0, \mathbf{1}, P_L, P_R\}$.

6.1.2 A family of projectors at three constituents

For the undressed internal bivector action of Propositions 6.1 and 6.2, chiral matter requires a projection; with dressing and spectator states, Theorem 6.9 obtains a chiral interaction otherwise. At $n = 3$ write $J^{(k)} = e_2^{(k)} e_3^{(k)}$ for the surplus bivector and $\omega^{(k)} = e_1^{(k)} \cdots e_6^{(k)}$ for the

volume of constituent k, and $K_{14}^{(3)} = e_1^{(3)} e_4^{(3)}$ for the pole bivector joining the first temporal and the first spatial direction of constituent three. The three operators

$$X_1 = J^{(1)} J^{(2)}, \qquad X_2 = \omega^{(3)}, \qquad X_3 = \omega^{(1)} K_{14}^{(3)}$$

are commuting symmetric traceless involutions, and they generate the orthogonal projectors $P_1 = \frac{1}{2}(\mathbf{1} + X_1)$ of rank 256, $P_{ij} = \frac{1}{4}(\mathbf{1} + X_i)(\mathbf{1} + X_j)$ of rank 128, and $P_{\text{fam}} = \frac{1}{8}\prod_i (\mathbf{1} + X_i)$ of rank 64.

Definition 6.3 (Ambient and faithful algebra)**.** For an orthogonal projector P commuting with the Lorentz action let $\mathfrak{g}_{\text{phys}}(P)$ be the internal bivectors commuting with P, $\mathfrak{k}(P)$ its compact part (the $+1$ eigenspace of $\theta(X) = -X^\top$), and

$$\mathfrak{k}_{\text{eff}}(P) = \mathfrak{k}(P) / \{X \in \mathfrak{k}(P) : XP = 0\}.$$

The quotient is not a technicality: an element can preserve the block and annihilate every state in it.

Example 6.4 (An annihilator)**.** With $X_1 = J^{(1)} J^{(2)}$, $(J^{(1)})^2 = (J^{(2)})^2 = -\mathbf{1}$ and $[J^{(1)}, J^{(2)}] = 0$,

$$(J^{(1)} + J^{(2)})(\mathbf{1} + X_1) = J^{(1)} + J^{(2)} + (J^{(1)})^2 J^{(2)} + J^{(2)} J^{(1)} J^{(2)} = J^{(1)} + J^{(2)} - J^{(2)} - J^{(1)} = 0.$$

So $J^{(1)} + J^{(2)}$ is a compact element of $\mathfrak{k}(P_{\text{fam}})$ that annihilates the block of P_{fam}, although it does not annihilate the block of P_{23}. A table of gauge algebras that omits the quotient overcounts, here by three dimensions.

Theorem 6.5 (The projector family)**.** *Computed on the* 512*-dimensional module as the exact nullspace of* $X \mapsto [X, P]$ *over the* $\binom{14}{2} = 91$ *internal bivectors, with* $\mathfrak{k}_{\text{eff}}$ *decomposed on the image (*$\dim \mathfrak{g}_{\text{phys}}$*,* $\dim \mathfrak{k}$ *and* $\dim \mathfrak{p}$ *count the commuting algebra before the quotient by its kernel on the block, so* $\dim \mathfrak{k}_{\text{eff}} = \dim \mathfrak{k} - \dim \ker$*):*

P	rank	$\dim \mathfrak{g}_{\text{phys}}$	$\dim \mathfrak{k}$	$\dim \mathfrak{p}$	$\dim \ker$	$\mathfrak{k}_{\text{eff}}$	axial
P_{fam}	64	17	9	8	3	$\mathfrak{u}(1)^3 \oplus \mathfrak{su}(2)$	yes
P_{23}	128	23	9	14	0	$\mathfrak{u}(1)^3 \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(2)$	yes
P_{13}	128	33	17	16	3	$\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{so}(5)$	yes
P_{12}	128	27	15	12	3	$\mathfrak{su}(2)^{\oplus 4}$	no
P_1	256	51	27	24	3	$\mathfrak{so}(6) \oplus \mathfrak{su}(2)^{\oplus 3}$	no

Here a projector is axial when $P\gamma^\mu P = 0$*. Factors are identified by dimension and rank:* $(3,1) = \mathfrak{su}(2)$*,* $(10,2) = \mathfrak{so}(5) \cong \mathfrak{sp}(2)$*,* $(15,3) = \mathfrak{so}(6) \cong \mathfrak{su}(4)$*; a six-dimensional semisimple compact factor is necessarily* $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$*. The centres have dimensions* $3, 3, 1, 0, 0$ *and the derived algebras* $3, 6, 13, 12, 24$*.*

Executable check: a4-4 and a10-2 (exact; Section 15.2).

Corollary 6.6 (No colour on the axial members)**.** *The largest simple factor among the rank-*64 *and rank-*128 *blocks is* $\mathfrak{so}(5)$*, every proper subalgebra of which has dimension at most six. Since* $\mathfrak{su}(3)$ *is simple of dimension eight, it embeds in none of them. Dimension eight is necessary and not sufficient:* P_{13} *carries an* $\mathfrak{so}(5)$ *of dimension ten and still contains no* $\mathfrak{su}(3)$*, so the test is an embedding test.*

6.1.3 Colour at rank 256

Theorem 6.7 (Colour on P_1)**.** *Let* $f_1, \ldots, f_6$ *be the six spatial directions of the second and third constituents. The fifteen bivectors* $f_a f_b$ *square to* $-\mathbf{1}$*, are antisymmetric and close on* $\mathfrak{so}(6)$*;*

each commutes with X_1, because the index sets are disjoint and $2 \cdot 4$ is even in (1.3), so all fifteen preserve P_1 and act faithfully on its image. Let $\Omega_6 = f_1 f_4 + f_2 f_5 + f_3 f_6$ generate the pairing $f_k \mapsto f_{k+3}$. Its centraliser in $\mathfrak{so}(6)$ has dimension nine and is $\mathfrak{u}(3)$; the derived algebra has dimension eight and rank two, is compact, closed and simple, hence $\mathfrak{su}(3)$, and its eight generators act faithfully on the block.

Theorem 6.8 (P_1 is vector-like undressed)**.** *X_1 is a product of four generators, none of them interior, so $|I||J| - |I \cap J| = 4$ is even for every interior generator and $[X_1, \gamma_\mu] = 0$. Hence P_1 is not axial and its matter is vector-like by Proposition 6.1.*

The obstruction is to the undressed action only.

Theorem 6.9 (Colour with a chiral doublet on P_1)**.** *Keep the colour generators in the six negative directions of factors two and three, and in the three positive directions of factor three set*

$$t_1 = \tfrac{1}{2} e_1^{(3)} e_2^{(3)}, \qquad t_2 = \tfrac{1}{2} e_1^{(3)} e_3^{(3)}, \qquad t_3 = -\tfrac{1}{2} e_2^{(3)} e_3^{(3)}.$$

With $P_{1,L} = P_1 \frac{1}{2}(\mathbf{1} - \Gamma)$ and $L_i = P_{1,L} t_i$,

$$[L_i, L_j] = \epsilon_{ij}{}^k L_k, \qquad [C_A, L_i] = 0, \qquad -\sum_i L_i^2 = \tfrac{3}{4} P_{1,L}, \qquad L_i P_{1,R} = 0$$

for every colour generator C_A. Vector colour and a commuting, purely left-handed doublet action therefore coexist on P_1.

Proof. The t_i are half time–time bivectors of factor three, so each squares to $-\frac{1}{4}\mathbf{1}$ and they close on $\mathfrak{su}(2)$. The colour generators carry only negative indices of factors two and three, disjoint from $\{e_1^{(3)}, e_2^{(3)}, e_3^{(3)}\}$, and two even blades with disjoint supports commute. $P_{1,L}$ is an idempotent of rank 128 commuting with each t_i, so $L_i = P_{1,L} t_i$ inherits the bracket and $-\sum L_i^2 = P_{1,L}(-\sum t_i^2) = \frac{3}{4} P_{1,L}$, the Casimir of a doublet. $P_{1,L} P_{1,R} = 0$ gives the last identity. □

Ordering the family by $\dim \mathfrak{k}_{\text{eff}}$ and recording whether the projector is axial,

$$P_{\text{fam}}\,(6) \text{ axial}, \quad P_{23}\,(9) \text{ axial}, \quad P_{12}\,(12) \text{ not}, \quad P_{13}\,(14) \text{ axial}, \quad P_1\,(24) \text{ not},$$

the two orderings are not reverses of one another, and what survives is a statement about these five projectors only: no axial one carries $\mathfrak{su}(3)$ in its effective bivector algebra, and $\mathfrak{su}(3)$ appears on exactly one of them, the vector-like P_1. The pattern is not a law of the arena. The next section exhibits a projector outside the family on which colour and chirality coexist.

6.2 A chiral Pati–Salam block

The five projectors of Section 6.1 are one family among many. Enlarging the class to all commuting symmetric blade involutions produces a projector on which colour and chirality coexist. The idea is to correlate spacetime chirality with an *internal* four-dimensional chirality, and to constrain the colour *volume* rather than any colour direction.

6.2.1 Construction

At three constituents take the interior gamma matrices $\gamma^\mu = (e_1^{(1)}, e_4^{(1)}, e_5^{(1)}, e_6^{(1)})$ and split the remaining fourteen directions into three groups:

group	directions	signs	generators E_A
weak W	$e_1^{(2)},\ e_1^{(3)},\ e_2^{(3)},\ e_3^{(3)}$	$+\,+\,+\,+$	$7, 13, 14, 15$
colour C	$e_4^{(2)}, e_5^{(2)}, e_6^{(2)},\ e_4^{(3)}, e_5^{(3)}, e_6^{(3)}$	$-\,-\,-\,-\,-\,-$	$10, 11, 12, 16, 17, 18$
auxiliary A	$e_2^{(1)}, e_3^{(1)},\ e_2^{(2)}, e_3^{(2)}$	$+\,+\,+\,+$	$2, 3, 8, 9$

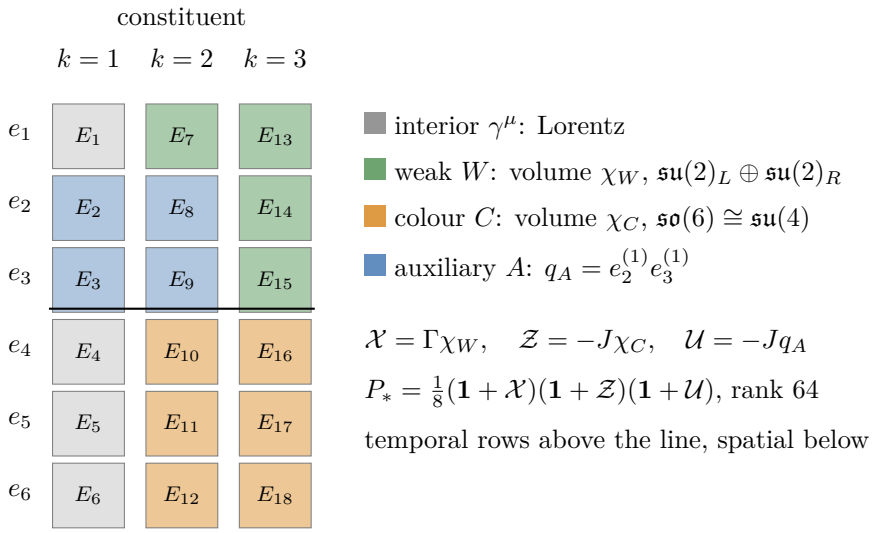


Figure 6.1: The eighteen generators of $\mathrm{Cl}(9,9)$ arranged by constituent, and the grouping that defines the Pati–Salam block. The interior four-plane carries the Lorentz action. The weak and colour groups enter P_* only through their volumes, so every rotation inside each of them survives the projection. The auxiliary group enters through the two-plane volume $q_A = e_2^{(1)} e_3^{(1)}$, not through the volume of its four directions, and of its rotations only E_2E_3 and E_8E_9 survive (E_2E_8 anticommutes with $-Jq_A$, and $P_*E_2E_8P_* = 0$). The surviving bivectors, 15 colour, 6 weak, E_2E_3 and E_8E_9, span the 23-dimensional algebra of Theorem 6.11 (Theorem 6.10).

With $J = \omega_I$ the ordered product of the fourteen internal generators and $\Gamma = J\gamma^0\gamma^1\gamma^2\gamma^3$, define the three volumes

$$\chi_W = e_1^{(2)} e_1^{(3)} e_2^{(3)} e_3^{(3)}, \qquad \chi_C = e_4^{(2)} e_5^{(2)} e_6^{(2)} e_4^{(3)} e_5^{(3)} e_6^{(3)}, \qquad q_A = e_2^{(1)} e_3^{(1)}.$$

Theorem 6.10 (A chiral projector carrying colour)**.** *$\mathcal{X} = \Gamma\chi_W$, $\mathcal{Z} = -J\chi_C$ and $\mathcal{U} = -Jq_A$ are commuting symmetric traceless involutions, and all seven nonidentity products $\mathcal{X}$, $\mathcal{Z}$, $\mathcal{U}$, $\mathcal{X}\mathcal{Z}$, $\mathcal{X}\mathcal{U}$, $\mathcal{Z}\mathcal{U}$, $\mathcal{X}\mathcal{Z}\mathcal{U}$ are traceless. Hence*

$$P_* = \tfrac{1}{8}(\mathbf{1} + \Gamma\chi_W)(\mathbf{1} - J\chi_C)(\mathbf{1} - Jq_A) \tag{6.1}$$

is an orthogonal projector with $\operatorname{Tr} P_* = 512/8 = 64$. It satisfies $[P_*, \Sigma^{\mu\nu}] = [P_*, J] = 0$ for the six Lorentz generators $\Sigma^{\mu\nu} = \frac{1}{2}\gamma^\mu\gamma^\nu$, and $P_*\gamma^\mu P_* = 0$, so it is axial; its two Weyl blocks have real rank 32 each, and $\chi_W P_* = \Gamma P_*$.

Rank is not automatic from the three factors being individually traceless; it is the products that decide, and here all of them are traceless. The statement $\chi_W P_* = \Gamma P_*$ is the correlation of spacetime chirality with the internal chirality of the weak directions.

6.2.2 The effective algebra

Theorem 6.11 (The effective algebra and its Pati–Salam part)**.** *All fifteen colour bivectors e_ae_b, $a, b \in C$, commute with $\mathcal{X}$, $\mathcal{Z}$ and $\mathcal{U}$, so the whole compact $\mathfrak{so}(6)_C \cong \mathfrak{su}(4)_C$ survives and acts faithfully on the block; the colour volume χ_C is invariant under those rotations, so constraining it costs no colour generator. With $w_1, \dots, w_4$ the weak directions, $\tau_i = -\frac{1}{2}w_jw_k$ for cyclic (i,j,k), $t_i^{L,R} = \frac{1}{2}(\mathbf{1} \mp \chi_W)\tau_i$ and $T_i^{L,R} = P_* t_i^{L,R} P_*$,*

$$[T_i^L, T_j^L] = \epsilon_{ijk} T_k^L, \qquad T_i^L P_R = 0, \qquad -\sum_i (T_i^L)^2 = \tfrac{3}{4} P_L,$$

and the mirror identities hold on the right. The $t_i^{L,R}$ are genuine bivectors, for example $t_3^L = -\frac{1}{4}(w_1w_2 + w_3w_4)$. The full effective compact bivector algebra is

$$\mathfrak{k}_{\text{eff}}(P_*) = \mathfrak{su}(4) \oplus \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R \oplus \mathfrak{u}(1)^2,$$

of dimension 23, entirely compact, with a derived algebra of dimension 21 and a two-dimensional centre. All 21 simple generators preserve the kinetic kernels $K^\mu = P_*\gamma^0\gamma^\mu P_*$, and $K^0 = P_*$. (On the block of P_* these are minus the compressed one-constituent kernels of Proposition 2.21; the overall sign of the fermion Lagrangian is a convention, which changes neither the field equation, nor the invariance condition $X^\top K^\mu + K^\mu X = 0$, nor the one-momentum spectrum, see the discussion after Proposition 2.24.)

Proof. Each colour bivector has even overlap with each of the three involution masks, so all fifteen commute with $\mathcal{X}, \mathcal{Z}, \mathcal{U}$ and descend faithfully. For the weak part, $\chi_W^2 = \mathbf{1}$ and $[\chi_W, \tau_i] = 0$ make $\frac{1}{2}(\mathbf{1} \mp \chi_W)$ idempotents commuting with the τ_i; the τ_i close on $\mathfrak{su}(2)$ with $\tau_i^2 = -\frac{1}{4}\mathbf{1}$, so $-\sum_i (T_i^L)^2 = \frac{3}{4}P_L$, and $\chi_W P_* = \Gamma P_*$ places the two ideals on opposite chiralities. The dimension 23 is the full solution of $[B, P_*] = 0$ over the 91-dimensional space of internal bivectors: the nullspace is 23-dimensional, entirely compact, has trivial annihilator on the block, and has a derived algebra of dimension 21. □

Executable check: a10-1 (exact; Section 15.2).

Corollary 6.12 (One Pati–Salam family)**.** *The colour volume is not an involution: χ_C is a grade-six blade on six negative directions, so $\chi_C^2 = -\mathbf{1}$. The involution is*

$$\Gamma_C := J\chi_C, \qquad \Gamma_C^2 = \mathbf{1}, \qquad \Gamma_C P_* = -P_*, \quad \textit{equivalently} \quad \chi_C P_* = JP_*.$$

On the module with $J = +i$ each Weyl block then carries exactly four $\mathfrak{su}(4)$ weights, all of colour chirality $\Gamma_C = -1$, each with multiplicity $4 = 2 \times 2$, a weak doublet times a Weyl spinor, and the two blocks carry the same colour weights. The content is

$$(\mathbf{4}, \mathbf{2}, \mathbf{1})_L \oplus (\mathbf{4}, \mathbf{1}, \mathbf{2})_R,$$

one Pati–Salam family including a right-handed neutrino [116]. Which of the two conjugate four-dimensional representations is called $\mathbf{4}$ is a convention.

The decisive steps of Theorems 6.5 and 6.11 and of this corollary are statements that nullspaces have the stated dimensions. They are certified exactly, in integer arithmetic: the 23-generator basis of Theorem 6.11 together with the completeness argument that every blade commutes or anticommutes with each of the three involutions (a10-1); the centre and derived-algebra dimensions of every row of Theorem 6.5 (a10-2); and the sign of the colour volume, rather than the quadratic Casimir, as what tells $\mathbf{4}$ from $\bar{\mathbf{4}}$ (a10-3).

The content is prior art. Embeddings of one chiral family together with the Lorentz group in a single orthogonal algebra are due to Maraner [192], Nesti and Percacci [191] and Krasnov [211, §§6.4–6.7], and the same family is the fermion content of the spectral Pati–Salam model [292]. What this realisation adds is the projector P_* built from three commuting involutions, with the normal charge on an extra positive four-plane, and the computations inside V_*; the comparison is made in Chapter 14.

6.2.3 Charges

The compact algebra is realised by skew operators; charges are the corresponding symmetric operators $\hat{t} = -Jt$. On the block the centre of $\mathfrak{k}_{\text{eff}}(P_*)$ is spanned by the skew operators JP_* and $J\Gamma P_*$, whose charges are P_* and ΓP_*. Concretely $q_A|_{V_*} = J$ and $E_8E_9|_{V_*} = J\Gamma$ on the image $V_* = \operatorname{im} P_*$.

Proposition 6.13 (Colour intertwiner, $B-L$, and the charge table)**.** *$R = P_*\gamma^0 e_1^{(2)} e_2^{(2)}$ satisfies $R^2 = -P_*$, exchanges the two Weyl blocks and commutes with every colour generator, so the colour representations of the two chiralities are equivalent and the content is $(\mathbf{4},\mathbf{2},\mathbf{1})_L \oplus (\mathbf{4},\mathbf{1},\mathbf{2})_R$ rather than a $\mathbf{4}/\bar{\mathbf{4}}$ pairing. With*

$$H = -(e_4^{(2)} e_4^{(3)} + e_5^{(2)} e_5^{(3)} + e_6^{(2)} e_6^{(3)}), \qquad Q_c = -JH,$$

the derived centraliser of H inside colour $\mathfrak{so}(6)$ is $\mathfrak{su}(3)$, and $(Q_c - \mathbf{1})(Q_c + 3\mathbf{1})P_* = 0$: Q_c has eigenvalues $+1$ and -3 on the block, with real multiplicities 48 and 16, the triplet and the singlet. Setting $B-L = Q_c/3$ and using the weak charges $-JT^3_{L,R}$,

$$Y = -J\,T_R^3 + \tfrac{1}{2}(B-L), \qquad Q = -J\,T_L^3 + Y,$$

gives one family:

field	$SU(3) \times SU(2)_L$	Y	internal Weyl components
Q_L	$(\mathbf{3},\mathbf{2})$	$\frac{1}{6}$	6
L_L	$(\mathbf{1},\mathbf{2})$	$-\frac{1}{2}$	2
u_R	$(\mathbf{3},\mathbf{1})$	$\frac{2}{3}$	3
d_R	$(\mathbf{3},\mathbf{1})$	$-\frac{1}{3}$	3
e_R	$(\mathbf{1},\mathbf{1})$	-1	1
ν_R	$(\mathbf{1},\mathbf{1})$	0	1

Four real spinor coordinates make one internal Weyl component, so rank 64 is sixteen internal Weyl components, eight left and eight right, and not sixty-four particles.

The $\mathfrak{su}(3)$ embedding, the right-handed Cartan direction and the hypercharge combination are chosen; what is established is their mutual compatibility. In the complex structure J every one of the sixteen Weyl fermions has charge one: J is fermion number, $F = 3B + L$, and the baryon number appears as $4B = F + (B-L)$.

Proposition 6.14 (The two central factors cannot be gauged)**.** *For a central charge $a + b\Gamma$, corresponding to the skew element $aJ + bJ\Gamma$, the anomaly coefficients on the block, per Weyl fermion with every field written left-handed, are*

$$SU(2)_L^2 :\ 2(a-b), \qquad SU(2)_R^2 :\ -2(a+b), \qquad SU(4)^2 :\ -2b,$$
$$\textit{cubic} :\ -16(3a^2b + b^3), \qquad \textit{gravitational} :\ -16b,$$

and the mixed anomalies with two central and one Pati–Salam generator vanish. All vanish only for $a = b = 0$. The Pati–Salam algebra itself is anomaly free on the block: the $SU(4)^3$ anomaly cancels between two fundamentals and two antifundamentals in the all-left convention, and each weak $SU(2)$ sees four doublets, so the global anomaly is absent [169]. Hence, with the block as the matter content and within the 23-dimensional effective algebra of Theorem 6.11, $\mathfrak{su}(4) \oplus \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ is the largest anomaly-free gauge algebra containing Pati–Salam, and the two central directions remain global symmetries. This is not a classification of all algebras acting on the block.

The obstruction is stated for this matter content; additional matter or another cancellation mechanism is not excluded. The two central directions, fermion number J and the chiral charge $J\Gamma$, are classical global symmetries with mixed anomalies, not exact quantum symmetries. The fermion-number current has a triangle anomaly with $SU(2)_L$ and $SU(2)_R$ already in perturbation theory [1]; what holds to all orders of perturbation theory is the corresponding charge selection rule in the trivial topological sector, which instantons violate [151].

Proposition 6.15 (Two rank-64 projectors, two orbits)**.** *Under the compact group of the connection sector, whose Lie algebra is the maximal compact* $\mathfrak{so}(8)\oplus\mathfrak{so}(6)$ *of dimension* 43*, the orbit through* P_{fam} *has dimension* $43-9=34$ *and the orbit through* P_* *has dimension* $43-23=20$*. Equal rank does not put two projectors on one orbit.*

Remark 6.16 (What the construction settles)**.** It settles an existence question: colour on a chiral block is not obstructed, and no counting argument based on the five-projector family can be a theorem. It does not select P_*; it does not derive the hypercharge embedding, which is compatible with anomaly cancellation but chosen; it gives one family and not three; and by Corollary 2.20 exhibiting a known representation content is a consistency result, not evidence. One further fact will matter in Section 7.6: γ^0 commutes with all 91 internal bivectors and with J and the rotations, and anticommutes with Γ and the boosts. It maps V_* onto an orthogonal rank-64 subspace on which the same internal algebra acts with Γ reversed. The chirality of the block is put in by the choice of P_*.

6.3 Charges and anomalies

The weak and colour algebras of a block are fixed by the projector; the abelian charges are not. This section asks how much the algebra says about them, first inside the arena and then in a layer that uses nothing from the arena at all.

6.3.1 A pairing obstruction

The family block of P_{fam} carries three abelian charges. The centre of $\mathfrak{k}_{\text{eff}}(P_{\text{fam}})$ is spanned by the block images of $X_a \in \{\frac{1}{2}J^{(1)}, \frac{1}{2}J^{(3)}, \frac{1}{2}e_5^{(3)}e_6^{(3)}\}$, and their charges $q_a := -J(2X_a)P_{\text{fam}}$ are commuting symmetric involutions of the block, with eigenvalues ±1 and $\operatorname{Tr} q_a = 0$. The cubic anomaly of $u = aq_1 + bq_2 + cq_3$ is the chiral trace

$$\mathcal{A}(u^3) = \tfrac{1}{4}\big[\operatorname{Tr}(u^3P_L) - \operatorname{Tr}(u^3P_R)\big], \qquad P_{L,R} = P_{\text{fam}}\tfrac{1}{2}(\mathbf{1}\mp\Gamma),$$

per Weyl component, right-handed components entering with conjugated charge when every field is written left-handed. Since $q_a^2 = P_{\text{fam}}$ and the linear chiral traces vanish, only the mixed term survives:

$$\mathcal{A}(u^3) = 6abc\,\mathcal{A}(q_1q_2q_3) = 96\,abc,$$

while the plain trace $\operatorname{Tr}(u^3)$ vanishes for every u and carries no information about anomalies. The anomaly-free locus is the union of the three coordinate planes $abc = 0$, and on it there is nothing to select.

Theorem 6.17 (Conjugate pairing)**.** *Let a module carry three commuting involutions* q_a *with eigenvalues* ±1 *and* $\operatorname{Tr} q_a = 0$*. For every* $u = aq_1 + bq_2 + cq_3$ *with* $abc = 0$ *the weight multiset of* u *is symmetric,* $\operatorname{mult}(w) = \operatorname{mult}(-w)$*.*

Proof. Take $c = 0$; only the joint multiplicities $M_{s_1s_2}$ of (q_1, q_2) matter. The trace conditions give $M_{++}+M_{+-}-M_{-+}-M_{--} = 0$ and $M_{++}-M_{+-}+M_{-+}-M_{--} = 0$; adding and subtracting, $M_{++} = M_{--}$ and $M_{+-} = M_{-+}$. The weight of (s_1, s_2) is $as_1 + bs_2$, and $(s_1, s_2) \mapsto (-s_1, -s_2)$ negates it while preserving multiplicities. □

The proof uses only the three linear traces, with no reference to rank, rung or realisation, so within its hypotheses no enlargement of the block and no ascent of the tower repairs it. Charge pairing is all that is concluded: calling a full matter representation vector-like would also require the chirality convention and the other gauge factors. And the hypotheses are not decorative.

Example 6.18 (Vanishing traces do not imply pairing)**.** The multiset $\{(-3)^{\times 5}, 1^{\times 10}, 5^{\times 1}\}$ has $\sum q = -15+10+5 = 0$ and $\sum q^3 = -135+10+125 = 0$, yet it is not symmetric under $q \mapsto -q$. It is realised by three commuting involutions with joint signs and multiplicities $(+,+,+)\colon 1$, $(-,-,+)\colon 10$, $(+,-,-)\colon 5$: these have traces $(\operatorname{Tr} q_1, \operatorname{Tr} q_2, \operatorname{Tr} q_3) = (-4,-14,6)$, so the hypotheses of Theorem 6.17 fail, while $Y = q_1 + q_2 + 3q_3$ has spectrum $\{5^{\times 1}, 1^{\times 10}, (-3)^{\times 5}\}$ with $\operatorname{Tr} Y = \operatorname{Tr} Y^3 = 0$. Leaving the hypotheses therefore need not violate the gravitational condition, which constrains $\operatorname{Tr} Y$ for the gauged combination and not $\operatorname{Tr} q_a$ for each basis charge [2, 15].

6.3.2 Dressing, and a benchmark

The escape from Theorem 6.17 is to leave its hypotheses. On the family block let

$$t^1 = -J \cdot \tfrac{1}{2} e_5^{(2)} e_6^{(2)}, \qquad t^2 = -J \cdot \tfrac{1}{2} e_6^{(2)} e_4^{(2)}, \qquad t^3 = -J \cdot \tfrac{1}{2} e_4^{(2)} e_5^{(2)}$$

be the weak charges, whose skew counterparts, the rotations of the spatial directions of constituent two, span the derived $\mathfrak{su}(2)$ of $\mathfrak{k}_{\mathrm{eff}}(P_{\mathrm{fam}})$; let $\varepsilon = \pm 1$ be an orientation sign, and put $t^a_{L,R} = P_{L,R} t^a$,

$$B_0 = \frac{\varepsilon}{6} \Gamma(q_1 + q_2 + q_3), \qquad Y = B_0 + t^3_R, \qquad Q_{\mathrm{em}} = t^3_L + Y.$$

For the displayed charge table we choose $\varepsilon = +1$ in the stated orientation; reversing ε reverses the B_0 charges, and the conventions must then be changed before the Standard-Model identification below is reused. The cell sums $\varepsilon \sum_a q_a$ are then $\{-1,-1,-1,3\}$ on the $\Gamma = -1$ half and $\{-3,1,1,1\}$ on the other, which gives $B_0|_{\Gamma=-1} = \{\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, -\frac{1}{2}\}$ and, after adding $t^3_R = \pm\frac{1}{2}$ on the right-handed cells, $Y \in \{0, -1, \frac{2}{3}, -\frac{1}{3}\}$ there. Over the sixteen left-handed Weyl components $\operatorname{Tr} Y = \operatorname{Tr} Y^3 = \operatorname{Tr} T_3^2 Y = 0$ with four doublets, so all four anomaly conditions hold, and $\operatorname{Tr} T_3^2 / \operatorname{Tr} Q^2 = 3/8$. This hypercharge is adopted, not derived.

Proposition 6.19 (The escape)**.** *Theorem 6.17 constrains undressed elements of* $\mathfrak{u}(1)^3$*, which lie in the vector class. The charge* B_0 *carries a factor* Γ *and is axial, not an element of* $\mathfrak{u}(1)^3$*, so the cubic-trace formula does not apply to it. Once chirality-dressed endomorphisms are admitted, which abelian charges can be gauged changes, and no-go results proved in the undressed grade-two setting do not carry across; a well-posed search must say which Clifford grades are permitted alongside* $Y^\dagger = Y$*,* $[Y, t_{La}] = 0$ *and* $[Y, C_A] = 0$*.*

Since $P_L + P_R = \mathbf{1}$, the chirality projectors on the weak axis cancel in $Q_{\mathrm{em}} = t^3 + B_0$ when the same axis is used in both terms. The dressing of $B_0 = \frac{\varepsilon}{6}\Gamma(q_1+q_2+q_3)$ does not cancel, so Q_{em} still contains Γ. The value $3/8$ is arithmetic on a charge table; it becomes $\sin^2\theta_W$ only after the equality of the normalised couplings is imposed at a matching scale, as a boundary condition, which this chapter does not supply (the adopted action of Chapter 7 supplies it by choice, Theorem 7.34). Five choices enter the benchmark: the projector, the dressing, which abelian direction is Y, which right-handed axis, and the overall sign.

6.3.3 A model-independent layer

The following uses nothing from the arena. It calibrates how much of the hypercharge problem is generic.

Theorem 6.20 (The perturbative hypercharge-anomaly variety of one family)**.** *For one family with* $N_c \geq 2$ *colours and a right-handed neutrino (for* $N_c = 1$ *there is no colour mixed condition), with left-handed hypercharges* q, l *and right-handed* y_u, y_d, y_ν, y_e*, the three linear conditions give* $l = -N_c q$*,* $y_d = 2q - y_u$*,* $y_e = 2l - y_\nu$*. With* $h_q = y_u - q$ *and* $h_\ell = y_\nu - l$ *the cubic condition factorises,*

$$\mathcal{A}_{Y^3} = 6 N_c q \, (h_\ell^2 - h_q^2),$$

using $2q^3-y_u^3-(2q-y_u)^3=-6qh_q^2$. *The solution set is the union of three planes:* Π_A *(*$h_\ell=h_q$*),* Π_B *(*$h_\ell=-h_q$*) and* Π_C *(*$q=l=0$*). On* Π_A *the four Yukawa relations* $y_u=q+h$, $y_d=q-h$, $y_\nu=l+h$, $y_e=l-h$ *hold, so they are consequences of anomaly cancellation rather than assumptions;* Π_B *is the image of* Π_A *under the exchange* $\nu_R\leftrightarrow e_R$. *At* $N_c=3$ *and* $h=\frac{1}{2}$, Π_A *is the line* $Y=Y_{\mathrm{SM}}+\lambda(B-L)$.

Anomaly cancellation does not supply the scalar, its hypercharge, or nonvanishing couplings. Quark Yukawa couplings alone do not exclude Π_C either: at $N_c=3$ the point $(q,l,y_u,y_d,y_\nu,y_e)=(0,0,\frac{1}{2},-\frac{1}{2},1,-1)$ with scalar hypercharge $\frac{1}{2}$ cancels all four anomalies and permits both quark Yukawas, yet $h_q=\frac{1}{2}\neq h_\ell=1$. What excludes it is integrality: with $q=0$ the left-handed quarks carry $\pm\frac{1}{2}$ and a three-quark singlet has half-integral charge.

Theorem 6.21 (Without ν_R hypercharge is quantised on the generic branch)**.** *Deleting* ν_R *gives* $y_e=2l$ *and* $\mathcal{A}_{Y^3}=6N_cq(N_c^2q^2-h_q^2)$, *so on the generic branch* $h_q=\pm N_cq$ *is locked and hypercharge is fixed up to scale and relabelling: at* $N_c=3$, $q=\frac{1}{6}$ *gives the Standard-Model values with no residual* $B-L$ *direction.*

This is the classical charge-quantisation result [41, 49]. Its bearing here is that the difficulty of selecting hypercharge on a family block exists because the block carries sixteen Weyl components rather than fifteen: the sterile neutrino is both the block's virtue and the source of the ambiguity.

Proposition 6.22 (Integrality, and what it does not fix)**.** *On* Π_A *with* $Q_{\mathrm{em}}=T_L^3+Y$, *the leptons carry* $Q_\nu=-\lambda$ *and* $Q_e=-1-\lambda$, *so integrality of the charged lepton forces* $\lambda\in\mathbb{Z}$, *and colour triality then gives integral charge to every colour singlet. It does not fix the fractional part of the quark charges, since* $Q_u=\frac{2}{3}+\lambda/3$, *and neutrality does not close the argument for composites:* $Q(ddd)=\lambda-1$ *and* $Q(uuu)=\lambda+2$, *so neutral colour-singlet operators of odd fermion parity exist at* $\lambda=1$ *and* $\lambda=-2$. *A neutral* elementary *colour-singlet fermion exists if and only if* $\lambda\in\{0,-1\}$. *These give* $(Q_u,Q_d,Q_\nu,Q_e)=(\frac{2}{3},-\frac{1}{3},0,-1)$ *and* $(\frac{1}{3},-\frac{2}{3},1,0)$*; exchanging* ν *and* e *alone does not map one to the other, but charge conjugation together with* $u\leftrightarrow d$ *and* $\nu\leftrightarrow e$ *does.*

Three other routes fail to single out the Standard Model. The kernel of the representation is not a selection principle: its order depends on λ through the integer normalisation of the weights. Nor is a $\mathbb{Z}_6$ kernel evidence for three colours: with a neutral ν_R the kernel is $\mathbb{Z}_{2N_c}$ for odd N_c and $\mathbb{Z}_{N_c}$ for even N_c. And integrality does not select three either: on the standard branch every N_c-quark colour singlet has integral charge exactly for odd N_c, and the number N_c+1 of weak doublets per family, which the global $SU(2)$ anomaly requires to be even [169], is even exactly for odd N_c. All three facts hold at every odd N_c. With N_g generations the parameter count depends on how the family is specified. If anomalies cancel only after summing over generations, the perturbative anomaly variety of unnormalised hypercharges has local dimension $6N_g-4$ at the Standard-Model point (14 for three generations), before an overall normalisation is fixed. A $2N_g$-dimensional family is obtained with one common Yukawa difference h, $y_u=q_i+h$, $y_d=q_i-h$, $y_\nu=l_i+h$, $y_e=l_i-h$ with $\sum_i(N_cq_i+l_i)=0$, and it contains L_e-L_μ; imposing the one-family branch generation by generation also gives $2N_g$ parameters but excludes L_e-L_μ. The anomaly-free directions include L_e-L_μ and $L_\mu-L_\tau$ [67], so family universality is an input that grows with N_g. The global structure of the gauge group is a separate datum [152, 47].

6.3.4 Charge at one constituent

Complexifying, the spinor module of one constituent is $\Lambda^\bullet\mathbb{C}^3$, and its even half is spanned by $|0\rangle,|12\rangle,|13\rangle,|23\rangle$. The diagonal Fock charge N_1-N_2 has spectrum $\{0,0,+1,-1\}$ there, the charge skeleton of $(\nu,\bar\nu,e^+,e^-)$.

Theorem 6.23 (No unique charge at one constituent)**.** *The condition* $\operatorname{spec}Q=\{0,0,+1,-1\}$ *on the even half, for* $Q=\sum_i q_iN_i+c$, *has exactly twelve solutions, forming two orbits of* $S_3\times\{\pm1\}$*:*

the six differences $Q = N_i - N_j$, with $c = 0$, and the six signed sums $Q = \pm(N_i + N_j - 1)$, with $c = \mp 1$. Hodge conjugation C maps exterior degree k to $3 - k$ and so exchanges the even and odd halves; the relation $CQC^{-1} = -Q$, equivalent to $\sum_i q_i + 2c = 0$, holds on the whole exterior algebra for both orbits, not as an operation inside the even four-state sector, and does not separate them.

Proof. Summing the four values gives $4c + 2\sum_i q_i = 0$. If the neutral pair is $\{|0\rangle, |ij\rangle\}$ then $c = 0$ and $q_i + q_j = 0$, which forces the third coefficient to vanish; if it is $\{|ij\rangle, |ik\rangle\}$ then $q_j = q_k$, $c = -q_j$ and $q_i = 0$. □

The charge skeleton alone does not identify neutrinos and electrons. The further step, $Q|0\rangle = 0$ and hence $c = 0$, is the separate assumption that the state identified with the neutrino is the Fock vacuum. Choosing between the orbits is an identification, not a derivation, and it has the same shape as the hypercharge problem two rungs up, differing in size and not in kind.

Remark 6.24 (Relation to Krasnov's construction)**.** Krasnov describes one generation, with the right-handed neutrino, as a real Weyl spinor of $\mathrm{SO}(7,7)$ realised on the even forms on $\mathbb{R}^7$; the Pati–Salam group appears as the compact part of the commutant $\mathrm{SO}(4,6)$ of the Lorentz group [211]. The block V_* has the same real dimension, 64, but it is not a module of $\mathfrak{so}(4,6)$: of the 45 bivectors of $\mathfrak{so}(4,6)$ on the weak and colour directions, only the 21 compact ones preserve P_*, and each of the 24 non-compact ones maps the block to its complement (check z3-9). As representations of the common algebra $\mathfrak{so}(1,3) \oplus \mathfrak{so}(4) \oplus \mathfrak{so}(6)$ the two modules are nevertheless equivalent: both are the underlying real representation (the realification) of $(\mathbf{2},\mathbf{1}) \otimes [(\mathbf{4},\mathbf{2},\mathbf{1}) \oplus (\bar{\mathbf{4}},\mathbf{1},\mathbf{2})]$ (Corollary 6.12), so an intertwiner exists by complete reducibility. It has not been written down, and whether it carries the kinetic structures is open; the non-compact generators of $\mathfrak{so}(4,6)$ obstruct only its extension to the ambient action. The comparison is made in Chapter 14.

Chapter 7

Mass Kernels and Conditional Vacuum Selection

The question of this chapter is which Lorentz-invariant mass terms the chiral block admits and what selects a vacuum. It treats mass at one constituent, the one-constituent vacua and the trace potential (Lemma 7.11 and Theorem 7.12), the 272-dimensional space $\mathcal{M}$ of mass kernels with its complex structure (Theorem 7.17), the selection of the Standard-Model orbit by an adopted potential at leading logarithm, matter in the selected vacuum, and the Goldstone bosons, antiparticles, baryon number and topology of the vacuum orbit. The classification is proved. The selection is conditional on the adopted potential and scheme, global minimality beyond the local result is a conjecture supported by numerical searches, curvature masses are not pole masses, and the electroweak stage is missing. The normalisation of couplings and masses is fixed in Appendix B.

7.1 What a rest energy breaks, and the two senses of rotation

The arena carries a mass in two readings. In the field-strength reading the rest energy of a field point is a function of the frame vector, studied in Section 3.9; the duality rotation $e^{\vartheta G}$ generated by the binding root multiplies the complex invariant q by $e^{2i\vartheta}$, so for $q \neq 0$ only $\vartheta \in \pi\mathbb{Z}$ fixes it (Section 3.8). In the Dirac reading the mass is the kernel $T = e_1e_2e_3 = \gamma^0 J$ of the Grassmann bilinear (Proposition 2.21 and Corollary 7.10). This section asks what the second breaks and compares the rest motion it generates with the binding rotation of the frame. In brief: the Dirac mass breaks no generator of $\mathfrak{so}(3,3)$, and what reduces the six-dimensional symmetry to the interior one is the kinetic term; it breaks the chiral and Pauli–Gürsey symmetries of the massless kinetic term down to $\mathfrak{u}(1)_J$, along three explicit directions; and the two senses of the Dirac rest rotation are exchanged by charge conjugation, those of the frame's binding rotation by parity. Units are $\hbar = c = 1$ unless a symbol is shown.

7.1.1 The Dirac mass: invariant in six dimensions, breaking the kinetic algebra

With $\gamma^0 = e_1$ and $\gamma^a = e_{a+3}$ the kinetic kernels are $K^\mu = -\gamma^0\gamma^\mu$ (Proposition 2.21). A Grassmann bilinear $\psi^\top M\psi$ is invariant under $\delta\psi = Y\psi$ exactly when $Y^\top M + MY = 0$, and by Corollary 7.10 the antisymmetric kernels with this property for all fifteen bivectors $Y = e_ae_b$ are the multiples of T. Write $\mathrm{Spin}(3,3)$ for the group generated by the exponentials of the bivectors.

Proposition 7.1 ($\mathrm{Spin}(3,3)$ keeps the mass form, not the kinetic forms)**.** *On the spinor module* $\mathbb{R}^8$*:*

(i) For every bivector $Y = e_ae_b$ *and every real* s, $(e^{sY})^\top T\, e^{sY} = T$.

(ii) An element $Y \in \mathfrak{so}(3,3)$ satisfies $Y^\top K^\mu + K^\mu Y = \Lambda^\mu{}_\nu K^\nu$ for some real 4×4 matrix Λ if and only if $Y \in \mathfrak{l}_{\text{int}} \oplus \mathbb{R}J$, a space of dimension seven.

Proof. (i) By Lemma 1.1 the trivector T commutes with a bivector that shares two or no indices with $\{1,2,3\}$ and anticommutes with a mixed one, and by Theorem 1.3 the former are the antisymmetric bivectors and the latter the symmetric ones. For antisymmetric Y, $Y^2 = -\mathbf{1}$ and $e^{sY} = \cos s\,\mathbf{1} + \sin s\,Y$ is orthogonal and commutes with T. For symmetric Y, $Y^2 = \mathbf{1}$, $e^{sY} = \cosh s\,\mathbf{1} + \sinh s\,Y$ is symmetric, and $(e^{sY})^\top T = e^{sY}T = Te^{-sY}$. (ii) The conditions are 256 linear equations in the fifteen bivector coefficients of Y and the sixteen entries of Λ. Their solution space is seven-dimensional and projects injectively onto the bivectors. It contains $\mathfrak{l}_{\text{int}}$, under which the K^μ transform covariantly, and J, which is antisymmetric and commutes with every $\gamma^0\gamma^\mu$, so that $J^\top K^\mu + K^\mu J = [K^\mu, J] = 0$; these span seven dimensions. □

Item (i) is Corollary 7.10 carried from the Lie algebra to the group; item (ii) is fixed by the algebra. Within Spin(3, 3) the Dirac mass breaks nothing. What reduces the six-dimensional symmetry to $\mathfrak{l}_{\text{int}} \oplus \mathbb{R}J$ is the kinetic term, in which the clock γ^0 is paired with the spatial triple. The breaking that a Dirac mass does describe lives in the internal symmetries of the kinetic term. Let $\mathfrak{g}_{\text{massless}} = \mathbb{R}\,\Omega \oplus \operatorname{span}\{J, e_2\Omega, e_3\Omega\} \cong \mathfrak{u}(1)_\Omega \oplus \mathfrak{su}(2)$ be the algebra of Proposition 2.29, of which invariance of the mass form leaves $\mathbb{R}J$.

Proposition 7.2 (What the Dirac mass breaks)**.** *(i) On $\mathfrak{g}_{\text{massless}}$ the map $Y \mapsto Y^\top T + TY$ has rank three and image $\operatorname{span}\{T\Omega, Te_2\Omega, Te_3\Omega\}$, contained in the six-dimensional kernel space of the interior Lorentz algebra (Theorem 7.9). The kernel $T\Omega$ commutes with J and is the pseudoscalar mass of Proposition 2.22; $Te_2\Omega$ and $Te_3\Omega$ anticommute with J and are Majorana kernels.*

(ii) For $M = T(m_s + m_p\Omega)$ the field equation $K^\mu\partial_\mu\Psi = M\Psi$ of $\frac{1}{2}\Psi^\top(K^\mu\partial_\mu - M)\Psi$, with $\partial_a\Psi = Jp_a\Psi$, is $\dot\Psi = -JH\Psi$, where

$$H = \gamma^0\gamma^a p_a + \gamma^0(m_s + m_p\Omega), \qquad H^\top = H, \qquad [H,J] = 0, \qquad H^2 = (|\mathbf{p}|^2 + m_s^2 + m_p^2)\mathbf{1}. \tag{7.1}$$

(iii) The chiral rotation $e^{\xi\Omega}$ preserves every K^μ and sends M to $(e^{\xi\Omega})^\top M e^{\xi\Omega} = Me^{2\xi\Omega}$; the stabiliser of $M \neq 0$ in $U(1)_\Omega$ is $\{\pm\mathbf{1}\}$.

Proof. (i) Every element of $\mathfrak{g}_{\text{massless}}$ is antisymmetric, so the map is $Y \mapsto [T,Y]$. By (1.3) T anticommutes with Ω, $e_2\Omega$ and $e_3\Omega$ and commutes with J, which gives the images $2T\Omega$, $2Te_2\Omega$, $2Te_3\Omega$ and 0. They are Lorentz-invariant kernels because T is one and $\mathfrak{g}_{\text{massless}}$ commutes with $\mathfrak{l}_{\text{int}}$. T and Ω commute with J, while e_2 and e_3 anticommute with it. (ii) Varying gives $K^\mu\partial_\mu\Psi = M\Psi$, since K^μ is symmetric and M antisymmetric. With $K^0 = -\mathbf{1}$, and J commuting with $\gamma^0\gamma^a$, γ^0 and Ω, $\dot\Psi = (K^aJp_a - M)\Psi = -JH\Psi$. The five matrices $\gamma^0\gamma^a$, γ^0 and $\gamma^0\Omega = e_4e_5e_6$ are symmetric, square to $\mathbf{1}$ and anticommute pairwise, which gives H^2. (iii) $\Omega \in \mathfrak{g}_{\text{massless}}$ is antisymmetric and anticommutes with T, so $e^{-\xi\Omega}Te^{\xi\Omega} = Te^{2\xi\Omega}$; and since $M^\top M = (m_s^2+m_p^2)\mathbf{1}$, a nonzero M is invertible, so $Me^{2\xi\Omega} = M$ exactly when $e^{2\xi\Omega} = \mathbf{1}$, that is for $\xi \in \pi\mathbb{Z}$, where $e^{\xi\Omega} = \pm\mathbf{1}$. □

The count of three in (i) is that of Theorem 7.12(a); the explicit broken directions are fixed by the algebra. Their physical content, that a Dirac mass breaks the axial $U(1)$ and the Pauli–Gürsey $SU(2)$ of a massless Dirac field down to fermion number [117, 61], is inherited through Theorem 2.19. Items (ii) and (iii) are inherited: they are the realification of $h = \boldsymbol{\alpha}\cdot\mathbf{p} + \beta(m_s + i\gamma_5 m_p)$ and of the chiral rotation, here derived from the Grassmann kernels; the commuting first-order form of Section 2.3 with the same H has the same flow. The two readings share a shape, visible already in the bilinears of Table 3.7. In both, the mass is the modulus of an order parameter of charge two under a $U(1)$ generated by a complex structure commuting

with a Lorentz algebra, G for $\mathfrak{l}_G$ and Ω for $\mathfrak{l}_{\text{int}}$, and a nonzero modulus leaves only $\{\pm\mathbf{1}\}$. They differ in kind: the Dirac mass is a coupling, or a value selected by a potential (Theorem 7.12), while the rest energy of a frame vector is a property of a configuration that no potential selects, the minimum of the energy density being the zero field (3.16).

7.1.2 Particle and antiparticle: the two senses of one rotation

At $\mathbf{p} = 0$ Dirac matter moves by the rest flow $\Psi(\lambda) = e^{-m\lambda T}\Psi_0$ of Proposition 4.46, which on the eigenspaces $e_1 = \pm 1$ is $e^{\mp m\lambda J}\Psi_0$, with energy $\pm\frac{1}{2}m|\Psi|^2$; it is the realification of $\psi = e^{\mp im\lambda}\psi_0$ on $\beta = \pm 1$. The arena adds a name. The rest motion is the one-parameter group of the temporal volume element T, the unique $\mathrm{Spin}(3,3)$-invariant mass form (Corollary 7.10 and Proposition 7.1), and the clock $\gamma^0 = -TJ$ is the quotient of that rotation by the charge rotation. Read in this way, the single rotation $e^{-m\lambda T}$ splits into the two senses of the charge rotation according to the eigenvalue of the clock.

The frame has a rotation with two senses as well. With the split-form Lagrangian $\mathcal{L}_\kappa = \frac{m}{2}\eta(\alpha_+X, \alpha_+X)$ of (4.11), every solution is $z(\tau) = c_-e^{-i\kappa\tau} + c_+e^{i\kappa\tau}$ with $c_\pm \in \mathbb{C}^3$ (Proposition 4.11). Below C, ϖ, T name the operations. On spinors they are realised by the operators of Theorems 2.25 and 2.27; on the frame by $C_f = -\mathbf{1}$, $P_f = \eta$ ($\mathbf{E} \mapsto -\mathbf{E}$) and $T_f\colon X(\tau) \mapsto -\eta X(-\tau)$, which combines the reversal of τ with the field map $(\mathbf{B}, \mathbf{E}) \mapsto (-\mathbf{B}, \mathbf{E})$ of C_fP_f, as $T = CP$ on fields in Theorem 3.89.

Proposition 7.3 (Which operation reverses the sense)**.** *(i) Every unit vector C of the surplus plane U satisfies $C^2 = \mathbf{1}$, $[C, T] = 0$, $\{C, J\} = \{C, e_1\} = 0$ and $C^\top H(0)C = -H(0)$. It maps the solution $e^{-m\lambda J}\Psi_0$ on $e_1 = +1$ to the solution $e^{+m\lambda J}C\Psi_0$ on $e_1 = -1$ of the same flow. The parities γ^0 and $\gamma^0 J$ and the time reversal $T_{\text{rev}} = e_2e_4e_5e_6$ of Theorem 2.27 keep the sign of the rest energy.*

(ii) On the frame $P_fGP_f^{-1} = -G$, and on $z = c_-e^{-i\kappa\tau} + c_+e^{i\kappa\tau}$

$$P_f\colon (c_-, c_+) \mapsto (\bar{c}_+, \bar{c}_-), \qquad C_f\colon (c_-, c_+) \mapsto (-c_-, -c_+), \qquad T_f\colon (c_-, c_+) \mapsto (-\bar{c}_-, -\bar{c}_+).$$

(iii) The sense of the Dirac rest rotation is reversed by C, CP, CT *and* CPT *and kept by* ϖ, T *and* PT*. The sense of the frame's binding rotation is reversed by* ϖ, CP, PT *and* CPT *and kept by* C, T *and* CT*.*

Proof. (i) Write $C = \cos\sigma\, e_2 + \sin\sigma\, e_3$. By (1.3) it anticommutes with e_1, with J and with the γ^a, and commutes with T; it is symmetric with $C^2 = \mathbf{1}$. Hence $C^\top e_1 C = -e_1$, C maps the eigenspace $e_1 = +1$ onto $e_1 = -1$, and C commutes with $e^{-m\lambda T}$. The operators $\gamma^0 = e_1$ and $\gamma^0 J = T$ commute with e_1 and with T, so they preserve each eigenspace and the flow. T_{rev} commutes with e_1 and anticommutes with J and T, so $T_{\text{rev}}\Psi(-\lambda) = e^{-m\lambda T}T_{\text{rev}}\Psi_0$ solves the same flow on the same eigenspace. (ii) $\eta G\eta = G^\top = -G$. On z, P_f acts as $z \mapsto \bar{z}$, C_f as $z \mapsto -z$, and T_f as $z(\tau) \mapsto -\overline{z(-\tau)}$; substituting the general solution gives the three maps. (iii) Each operation either exchanges the two senses or keeps both, so the signs multiply under composition; the single operations are (i), (ii) and Theorem 2.27. □

Item (i) is inherited, realised by the arena's charge conjugation and time reversal (Theorems 2.25 and 2.27). Items (ii) and (iii) are fixed by the algebra; (ii) makes explicit, on the two senses, the action of the operations of Theorem 4.12, by which P_f maps $\mathcal{L}_\kappa$ to $\mathcal{L}_{-\kappa} = \mathcal{L}_\kappa + 2m\kappa\,\frac{d}{d\tau}(\mathbf{E}\cdot\mathbf{B})$, a symmetry up to a total derivative, while C_f and T_f leave $\mathcal{L}_\kappa$ invariant.

For Dirac matter the two senses are particle and antiparticle. They are solutions of one Lagrangian, of angular frequency m, each with energy of definite sign, and they are exchanged by C, a symmetry of the Grassmann Lagrangian, since it preserves the mass form and every K^μ (Theorem 2.25); on the commuting first-order form $\mathcal{L}_1$ it reverses the sign of the action, because $CJC^{-1} = -J$ and $CH(\mathbf{p})C^{-1} = -H(-\mathbf{p})$. For the frame the two senses are the kernels of the

stage	order parameter	kept	broken directions	source
$n = 1$, Dirac	$T = \gamma^0 J$	$\mathfrak{u}(1)_J$	3, global: $T\Omega$, $Te_2\Omega$, $Te_3\Omega$	Theorem 7.12(a), Proposition 7.2
$n = 1$, Majorana	kernel on $\Gamma = +1$	$\mathbb{R}(J - J\Gamma)$	3, global	Theorem 7.12(b)
$n = 3$, Pati–Salam	Φ_0 ($\nu_R\nu_R$)	14 of 23	9, eaten by Z', $W_R^{\pm}$ and six X	Theorem 7.41
$n = 3$, electroweak	Φ_0 and bidoublet	10 of 23	12 eaten, 1 axion-like $(1 - \Gamma)$	Theorem 7.42

Table 7.1: Broken directions along the tower.

two factors of $\alpha_-\alpha_+ = \partial_\tau^2 + \kappa^2$. One carries no momentum and the other all of it, and each alone carries zero energy, since the energy is $2m\kappa^2 \operatorname{Re}(c_- \cdot c_+)$ (Proposition 4.11). They are exchanged by parity, which is a symmetry only up to a total derivative, and their frequency $\kappa = c/\ell$ (2.2) is a conversion rate, not a mass. In the interior m and κ are independent inputs: m multiplies the Lagrangian and κ sets the period. Reading the frame's $\hbar\kappa$ as mc^2 would therefore be an added identification, not a result. That the frame's binding is not a particle–antiparticle pair in the Dirac sense is our reading of these facts. In both cases the reversal is CPT-odd.

7.1.3 Two vacua

For $E > 0$ the Fock vacuum of one Dirac momentum is unique and neutral, with energy $-2E$, and charge conjugation lifts to a unitary operator on the Fock space that anticommutes with the charge and carries $H(\mathbf{p})$ to $H(-\mathbf{p})$ (Theorem 2.31 and Proposition 2.32). At rest its energy, $-2m$ ($-2mc^2$ in restored units), is minus half the sum of the rest energies of the four modes. Only in that sense does the vacuum carry the rest energy with the opposite sign, and only this vacuum does.

The frame has a different vacuum. On a coordinate subspace with k temporal and l spatial directions the zero-point energy of the interior oscillators is $\frac{1}{2}\hbar\kappa(k - l)$, and it vanishes on $\mathbb{R}^{3,3}$ (Theorem 7.13). The full frame has no state of lowest energy for this to be the energy of.

Quantised mode by mode with positive norm, the energy of the full frame has the spectrum $\hbar\kappa\mathbb{Z}$, and every state has one of lower energy (Corollary 4.15).

The zero-point values restate Theorem 7.13, and Corollary 4.15 makes explicit for the full space what Section 7.3 records for the mixed planes. The two vacua differ in kind: the Dirac vacuum has energy $-2E$ per momentum and is bounded below, while the frame's zero-point energy cancels between positive temporal and negative spatial oscillators, and there is no ground state for it to be the energy of.

7.1.4 Goldstone directions across the tower

Table 7.1 collects the broken directions at the stages where the paper computes them, at n constituents, counted for the specified generators: at one constituent the four of $\mathfrak{g}_{\text{massless}}$, at three constituents the 21 Pati–Salam generators and the central J and $J\Gamma$ (Proposition 7.40). One element runs through all of them.

Lemma 7.4 (One charge across the scales)**.** *Let J be the complex structure and Γ the chirality, at one constituent or at three, so that $[J, \Gamma] = 0$, and let $J - J\Gamma$ be the generator of charge $1 - \Gamma$.*

(i) If a kernel $M \neq 0$ commutes with J and anticommutes with Γ (Dirac type), then $[J - J\Gamma, M] = -2J\Gamma M \neq 0$.

symmetry	broken by a mass?	where
interior Lorentz $\mathfrak{l}_{\text{int}}$	no: P^2 and T are invariant; the little group of a momentum is three-dimensional with or without mass	inherited [160]; Section 3.9
frame Lorentz $\mathfrak{l}_G$	no: q is invariant; the stabiliser of a nonzero field point is two-dimensional with or without q	inherited [121]; Section 3.9
$\mathrm{Spin}(3,3)$	no: T is invariant; the kinetic term keeps seven dimensions	fixed; Corollary 7.10, Proposition 7.1
duality $U(1)_G$	yes, to $\{\pm\mathbf{1}\}$ on Lorentz orbits, exactly when $q \neq 0$; for a configuration, not a vacuum	inherited; Sections 3.8 and 3.9
axial $U(1)_\Omega$	yes, to $\{\pm\mathbf{1}\}$	inherited; Propositions 2.22 and 2.29, Proposition 7.2
Pauli–Gürsey $\mathfrak{su}(2)$	yes, to $\mathfrak{u}(1)_J$	inherited; Proposition 2.29 and Theorem 7.12
$U(1)_{1-\Gamma}$	by every Dirac-type mass; not by a Majorana mass on $\Gamma = +1$	inherited; Theorems 7.12 and 7.42, Lemma 7.4

Table 7.2: What a mass breaks in the arena.

(ii) If M lives on $\Gamma = +1$, that is $\Gamma M = M\Gamma = M$, and anticommutes with J, then $[J - J\Gamma, M] = 0$.

(iii) At one constituent $J - J\Gamma = J + \Omega$.

Proof. J commutes with Γ. (i) $[J, M] = 0$ and $MJ\Gamma = -J\Gamma M$, so $[J\Gamma, M] = 2J\Gamma M$, which is nonzero because $J\Gamma$ is invertible. (ii) $[J, M] = 2JM$, and $[J\Gamma, M] = JM - MJ\Gamma = JM + JM\Gamma = 2JM$. (iii) $\Gamma = J\Omega$, so $J\Gamma = J^2\Omega = -\Omega$. □

The lemma is inherited: a charge that vanishes on one chirality is kept by a Majorana mass of that chirality and broken by every mass that pairs opposite chiralities. What the arena adds is the identification across scales: one element of the algebra, $J - J\Gamma$, is kept by a Majorana mass on $\Gamma = +1$ and broken by every Dirac-type mass, both at one constituent and at three. At one constituent it is $J + \Omega$; the Majorana phase keeps it, and the Dirac mass T, which commutes with J and anticommutes with Γ, breaks it (Theorem 7.12). At three constituents Φ_0 lives on $\Gamma = +1$, since $\Phi_0^\top \Phi_0 = P_\nu$ and P_ν projects into the eigenspace $\Gamma = +1$, and anticommutes with J (Theorem 7.17); item (ii) is therefore the one-line reason for the equality $[J, \Phi_0] = [J\Gamma, \Phi_0]$ in Theorem 7.41. The bidoublet kernels are of Dirac type (Theorem 7.17), so every nonzero bidoublet value breaks $J - J\Gamma$ by item (i); at the generic electroweak vacuum, with both bidoublet values nonzero, its direction escapes the gauge orbit (Theorem 7.42).

7.1.5 What a mass breaks

Table 7.2 collects the answer. The statement that a mass is what breaks a symmetry is true in the arena for duality in the frame, in the orbit-space sense and for a configuration rather than a vacuum, and for the chiral and Pauli–Gürsey symmetries of Dirac matter. It is false for every spacetime symmetry, the six-dimensional one included: a mass changes the type of a Lorentz stabiliser, never its dimension, and at a field point what a nonzero q removes is the duality generator (Section 3.9).

Which term counts as the symmetric part, in the sense of the clause on c^2 discussed at the end of Section 3.9, and which as the breaking depends on which symmetry is meant. For $\mathrm{Spin}(3,3)$ the mass form is invariant and the kinetic term breaks the symmetry (Proposition 7.1); for the massless kinetic algebra $\mathfrak{u}(1)_\Omega \oplus \mathfrak{su}(2)$ it is the mass that cuts it to $\mathfrak{u}(1)_J$ (Proposition 7.2); and

in the frame the generator of duality is the binding root, which enters the interior through the shift κG but is not an infinitesimal isometry of the split form on which the interior Lagrangian is built (Sections 3.8 and 3.9). In the closure of the force law the breaking of duality is explicit: with Lorentz-class inertia the duality angle of the momentum field moves in a potential whose amplitude vanishes exactly when that field is null, that is when its rest energy vanishes (Theorem 4.28 and Corollary 4.29). This restates Section 3.9; the breaking is explicit, carried by the inertia and not by a vacuum, and the reading of the angle as a pseudo-Goldstone boson of duality, with its limits, is taken up in Section 4.5.

7.2 Candidates for a mass at one constituent

Three candidates for m in the arena

The arena has three structures that might be called a mass: the temporal volume element $T = e_1e_2e_3$, which is the Dirac mass form (Corollary 7.10); the parameter m and the conserved energy of the interior dynamics of Section 4.3; and the rest energy of the momentum field in the closure of Section 4.4. We take them in turn.

Proposition 7.5 (The mass form acts on the frame as parity)**.** *(i)* $\mathrm{Ad}_T|_V = \eta$ *fixes* $\mathbf{B}$ *and reverses* $\mathbf{E}$*: on the frame it is parity. It commutes with the* L_i *and anticommutes with* G *and the* K_i*, so it normalises* $\mathfrak{l}_G$ *without being* $\mathfrak{l}_G$*-invariant.*

(ii) ηX *corresponds to* $\bar{z}$ *and* $q(z) = h(\bar{z}, z)$*, so* $E_6 = \frac{1}{2}h(z,z)$ *and* $E_0 = \frac{1}{2}|h(\bar{z},z)| = \frac{1}{2}\max_\vartheta \eta(e^{\vartheta G}X, e^{\vartheta G}X)$*.*

(iii) As a Grassmann mass kernel on spinors, T *is invariant under all of* $\mathfrak{so}(3,3)$*,* $\mathfrak{l}_G$ *included, in the sense of Definition 7.8, and breaks nothing; only its sign changes, under conjugation by a spatial generator.*

Proof. (i) $\mathrm{Ad}_T|_V = \eta$ is Lemma 2.4; $\eta G \eta = -G$ blockwise, the L_i preserve each triple and the K_i exchange them. (ii) $z(\eta X) = \mathbf{B} - i\mathbf{E}$, $h(\bar{z}, z) = \sum_j z_j^2$, and $\eta(e^{\vartheta G}X, e^{\vartheta G}X) = \mathrm{Re}(e^{2i\vartheta}q)$. (iii) is Corollary 7.10. □

The identities are inherited or restate the paper; their consequence is an obstruction by type. The identification of m with T fails on frame vectors, where the adjoint action of T is parity, while as a kernel on spinors T is invariant under $\mathfrak{l}_G$. The Dirac mass lives on the interior reading of the six directions, which Proposition 3.6 excludes on the same six directions, and even there it breaks nothing. What survives is a reading: the rest energy is the overlap $\frac{1}{2}|h(\bar{z},z)|$ of a field with its mirror image, and Ad_T, the arena's splitting into two triples, is the real structure whose stabiliser in $U(3)$ is $U(1)\cdot O(3)$ (Proposition 3.91(iv)).

The second candidate is the interior. Its frame field and its momentum field are compared in Proposition 4.13: the momentum field moves by pure duality and keeps $E_6(\pi)$, $E_0(\pi)$ and $\mathbf{S}(\pi)$, while the rest energy of the frame field oscillates in general, and the parameter m is an overall factor of the Lagrangian.

As an identification of a rest energy, the interior is obstructed. In it m is an overall factor and an input independent of κ, the pure duality motion of either sense carries zero energy, and, with X a length and $\kappa = c/\ell$, the energy $\mathcal{E} = mc^2 \cdot 2\,\mathrm{Re}(c_- \cdot c_+)/\ell^2$ has the form mc^2 times a dimensionless amplitude; but the amplitude is indefinite, vanishes on the pure duality orbits and is selected by nothing, since ℓ is free. That κG lies in the η-self-adjoint summand of Theorem 4.18 is fixed; nothing in the interior turns κ^2 into a rest energy, and reading the rate as a mass, $mc^2 = \hbar\kappa$, would be an added identification, not a result.

The third candidate is the closure.

Remark 7.6 (The momentum field in the closure)**.** The closure of Section 4.4, whose bearing on the rest energy is the subject of Section 4.5, settles the third candidate by results fixed there. For every Hamiltonian, $E_6(\pi) = \frac{1}{2}|p|^2$, $E_0(\pi) = \frac{1}{2}|w(p)|$, with w as in (4.15), and $\mathbf{S}(\pi)$ are Casimirs (Proposition 4.23 and Theorem 4.40), while the same functions of the frame field oscillate in the clamped interior (Proposition 4.13) and, with a free rotor in the rotating regime, Lorentz-class inertia and $E_0(\pi_0) > 0$, grow without bound (Theorem 4.31 and Corollary 4.41). For Lorentz-class $N = (\nu_1\eta^{-1} + \nu_2 G\eta^{-1})/m$ the potential of the rotor angle, which is the duality angle of the momentum field, has the amplitude $|\nu_1 + i\nu_2|\, E_0(\pi_0)/m$ (Theorem 4.28); for $(\nu_1, \nu_2) \neq (0, 0)$ the rate is constant and nonzero exactly when $E_0(\pi_0) = 0$ and the initial rate is nonzero (Corollary 4.29), and if $E_0(\pi_0) > 0$ it is constant only at the equilibria, where it vanishes (Corollary 4.43). For the duality-class $N = \mathbf{1}/m$ the frame energy $\frac{1}{2}p^\top Np$ is $E_6(\pi)/m$, the undivided energy of the momentum field, and the rate is conserved for all data (Theorem 4.26(iii)).

Read through a dictionary that is not a theorem, in which the clamped rate $\kappa = c/\ell$, with ℓ free, stands for the conversion and $E_0(\pi)$, which has the dimension of a squared momentum, stands for a rest energy, these results say the following. The momentum field's rate keeps its symmetry, constancy and invariance under the duality translation of the rotor, when the momentum field is null; its rest part $E_0(\pi)$ is the amplitude that breaks it; and $E_6(\pi)$ is a Casimir and breaks nothing. But the breaking is explicit, by an inertia that Theorem 4.26 forbids from also conserving the rate; what is broken is the constancy of the rotor rate, a property of duality, not Lorentz invariance and nothing carried by c; and with the undivided energy as Hamiltonian, $N = \mathbf{1}/m$, the rate is constant at the price of the drift of the frame in Theorem 4.27.

7.3 The vacuum at one constituent

The word *vacuum* can mean several things, and the arena gives each of them a precise form. It can mean the structure an algebra selects on its module; the value of an order parameter chosen by a potential; the ground state of a classical field; or the Fock vacuum of a quantised one. At one constituent all four readings can be computed for every coordinate three-plane of $\mathbb{R}^{3,3}$ and for the full algebra. The exercise is worth doing for two reasons: it shows which vacuum properties belong to a part of the arena and which need all of it, and it contains, in miniature, the vacuum that will be selected at three constituents.

7.3.1 The subfields

A three-plane $W \subset \mathbb{R}^{3,3}$ generates a subalgebra $\mathrm{Cl}(W) \subset \mathrm{Cl}(3,3)$. Up to $O(3,3)$ only its signature matters, $(3,0)$, $(0,3)$, $(2,1)$ or $(1,2)$, but relative to the structures already built (the clock e_1, the surplus plane U, the interior four-plane E, and the pairing $e_i \leftrightarrow e_{i+3}$) the coordinate planes fall into the classes of Table 7.3.

7.3.2 The algebraic vacuum

Let ω_W be the ordered product of the three generators of W.

Theorem 7.7 (Complex structure or projector)**.** *For every coordinate three-plane W of signature (p, q), ω_W commutes with the generators in W and anticommutes with those outside it, and*

$$\omega_W^2 = -(-1)^q\,\mathbf{1}, \qquad \omega_W^\top = -(-1)^q\,\omega_W.$$

So ω_W is a complex structure for T and for the $(1,2)$ planes, and a symmetric involution, hence a rank-four orthogonal projector $\Pi_W = \frac{1}{2}(\mathbf{1} + \omega_W)$, for S and the $(2,1)$ planes. Three of these volume elements are operators met before:

$$\omega_T = e_1e_2e_3 = \gamma^0 J, \qquad \omega_S = \gamma^1\gamma^2\gamma^3, \qquad \omega_T\,\omega_S = \Gamma,$$

name	plane	signature	position
T	$\langle e_1, e_2, e_3\rangle$	$(3,0)$	temporal triple: clock and surplus plane
S	$\langle e_4, e_5, e_6\rangle$	$(0,3)$	spatial triple, inside the interior four-plane
M_{21a}	$\langle e_1, e_2, e_4\rangle$	$(2,1)$	clock, one surplus direction, one space; one partner pair
M_{21b}	$\langle e_2, e_3, e_4\rangle$	$(2,1)$	surplus plane and one space; no partner pair
M_{12a}	$\langle e_1, e_4, e_5\rangle$	$(1,2)$	a 2+1 slice of the interior spacetime; one partner pair
M_{12b}	$\langle e_2, e_4, e_5\rangle$	$(1,2)$	a surplus direction and two spaces; one partner pair
M_{12c}	$\langle e_1, e_5, e_6\rangle$	$(1,2)$	a 2+1 slice of the interior spacetime; no partner pair

Table 7.3: A selection of representatives of the coordinate three-planes of $\mathbb{R}^{3,3}$, by signature and by their position relative to the clock, the surplus plane, the interior four-plane and the pairing; the list of positional classes is not exhaustive (for instance $\langle e_1, e_2, e_6\rangle$, of signature $(2,1)$, contains the clock and no partner pair).

the Dirac mass form, the time-reversal factor of Theorem 2.27, and the chirality, which is also the volume element of the full algebra.

Proof. A generator inside W shares one index with ω_W, so by (1.3) the sign is $(-1)^{3-1} = +1$; one outside shares none, and the sign is $(-1)^3 = -1$. The square and the transpose follow from (1.4) with $r = 3$: $(-1)^3 \prod \eta_{aa} = -(-1)^q$. □

The restriction to coordinate planes matters for the transpose, not for the square. For any nondegenerate three-plane W with an orthonormal basis, ω_W commutes with the vectors of W, anticommutes with those of $W^\perp$, and $\omega_W^2 = -(-1)^q\mathbf{1}$. The transpose rule, however, refers to the positive inner product of S, which is adapted to the splitting $V = V^+ \oplus V^-$: it holds for every plane spanned by vectors of V^+ and vectors of V^-, since rotations inside the two triples act on S by orthogonal matrices, and it fails for tilted planes. For the plane $W_t = \langle \cosh t\, e_4 + \sinh t\, e_1,\ e_5,\ e_6\rangle$, of signature $(0,3)$, $\omega_{W_t} = \cosh t\, e_4e_5e_6 + \sinh t\, e_1e_5e_6$ is the sum of a symmetric and an antisymmetric blade, so it is not symmetric for $t \neq 0$, although $\omega_{W_t}^2 = \mathbf{1}$. This is the mechanism behind the statement below that boost-generated variations leave the orthogonal projectors.

Under $\mathfrak{so}(3,3)$ the volume elements ω_T and ω_S have six-dimensional stabilisers that contain all six compact rotations, so no compact bivector moves them; only the nine boosts do, and by Theorem 1.13 boost-generated variations leave the orthogonal projectors. Each mixed volume element, by contrast, has a two-dimensional compact stabiliser and a four-dimensional compact orbit: the mixed planes are the only subfields whose vacuum projector or complex structure admits admissible fluctuations. Charge conjugation $C = e_2$ exchanges Π_S and Π_Γ with their complements; on a coordinate three-plane W it fixes Π_W when $e_2 \in W$ and reverses it when $e_2 \notin W$ (so it fixes the two $(2,1)$ representatives of Table 7.3, which contain e_2, but not, for instance, Π of $\langle e_1, e_3, e_4\rangle$), and the full volume element Γ commutes with all of $\mathfrak{so}(3,3)$: its projector is the rigid, Lorentz-invariant vacuum $\frac{1}{2}(\mathbf{1} \pm \Gamma)$ that will reappear in Section 8.5.

7.3.3 The order-parameter vacuum

Let $\mathfrak{so}(W)$ play the role of a Lorentz algebra, its antisymmetric bivectors acting as rotations and its symmetric ones as boosts.

Definition 7.8 (Mass kernels of a subfield)**.** The kernel space $\mathcal{M}_W$ is the space of antisymmetric $M \in \mathrm{End}(S)$ with $[M, \text{rotation}] = 0$ and $\{M, \text{boost}\} = 0$ for the generators of $\mathfrak{so}(W)$, and the internal algebra $\mathfrak{g}_W$ is the space of antisymmetric operators commuting with $\mathfrak{so}(W)$.

Antisymmetry makes $\psi^\top M\psi$ a nonzero Grassmann bilinear, and the two conditions make it Lorentz invariant, since a rotation is antisymmetric and a boost symmetric. Both conditions are diagonal on blades, so both spaces are spanned by blades.

Theorem 7.9 (Kernel spaces and invariant kernels)**.** *The kernel spaces, the internal algebras and the invariant kernels are:*

Lorentz algebra	$\dim \mathcal{M}_W$	$\dim \mathfrak{g}_W$	$\mathcal{M}_W$ under $\mathfrak{g}_W$	invariant kernel
$\mathfrak{so}(T)$, $\mathfrak{so}(S)$	10	10	adjoint of $\mathfrak{sp}(2)$, irreducible	none
each mixed $\mathfrak{so}(W)$	10	6	$9 \oplus 1$	the plane's own 2+1 mass form
full $\mathfrak{so}(3,3)$	1	0	—	$\gamma^0 J = e_1e_2e_3$
interior $\mathfrak{so}(1,3)$	6	4	irreducible	none

The invariant kernel of each mixed plane is $\pm\gamma^0_W\omega_W$, with γ^0_W the generator of minority signature. For the interior algebra $\mathfrak{g}_I = \langle J, \Omega, e_2\Omega, e_3\Omega\rangle = \mathfrak{u}(1)_\Omega \oplus \mathfrak{su}(2)$ is the massless gauge algebra of Proposition 2.29.

Every entry is a finite computation on the 64 blades of $\mathrm{Cl}(3,3)$: the kernel conditions are diagonal on blades, $\mathfrak{g}_W$ is the commutant of $\mathfrak{so}(W)$ in the kernel-preserving algebra, and the decomposition and the invariant kernels follow from the commutant of $\mathfrak{g}_W$ on $\mathcal{M}_W$. Executable check: a6-1 (exact symbolic or integer arithmetic; Section 15.2).

Corollary 7.10 (The full algebra has one mass term)**.** *The only Grassmann mass kernel invariant under all of $\mathfrak{so}(3,3)$ is $\gamma^0 J = \omega_T$, and no antisymmetric operator commutes with $\mathfrak{so}(3,3)$. The order parameter of the full algebra is a single real number r in $r\gamma^0 J$; it breaks nothing and has no Goldstone direction, and its two signs are exchanged by conjugation with any spatial generator.*

The temporal triple therefore supplies more than a clock: its volume element is the Dirac mass form, the unique Lorentz-invariant mass term of the whole algebra. To see what a potential selects, take on $\mathcal{M}_W$ the trace potential

$$V = -\mu^2 N + \alpha N^2 + \beta Q, \qquad N = \operatorname{Tr} M^\top M, \qquad Q = \operatorname{Tr}(M^\top M)^2. \tag{7.2}$$

Whether this potential has a minimum at all depends on its coefficients; the next lemma gives the complete answer on the six-dimensional kernel space $\mathcal{M}$ of the interior $\mathfrak{so}(1,3)$.

Lemma 7.11 (The trace potential on $\mathcal{M}$)**.** *Let $\mathcal{M}$ be the kernel space of the interior $\mathfrak{so}(1,3)$, spanned by the blades e_1e_2, e_1e_3, $e_1e_2e_3$, $e_2e_4e_5e_6$, $e_3e_4e_5e_6$ and $e_2e_3e_4e_5e_6$. For $M \in \mathcal{M}$, $M^\top M = \frac{N}{8}\mathbf{1} + D$ with D traceless and $D^2 = d^2\mathbf{1}$, so $Q = \frac{1}{8}N^2 + 8d^2$ and, for $M \neq 0$,*

$$q := Q/N^2 \in [\tfrac{1}{8}, \tfrac{1}{4}],$$

with $q = \frac{1}{8}$ exactly when $M^\top M \propto \mathbf{1}$ (for instance on the blades, and on $\gamma^0 J$) and $q = \frac{1}{4}$ exactly when M has rank four. In the coordinates $u = (c_1, c_2, c_6)$, $v = (c_4, c_5, c_3)$ of the six blades in the order listed, $N = 8(|u|^2 + |v|^2)$ and $d = 2|u \times v|$, so $\mathcal{M} \cong \mathbb{C}^3$ with $z = u + iv$: $q = \frac{1}{8}$ iff $u \parallel v$, and $q = \frac{1}{4}$ iff $z \cdot z = 0$, that is $u \perp v$ and $|u| = |v|$. At fixed N each of these two sets is connected and of real dimension three. Write $a(q) = \alpha + \beta q$, so that $V = -\mu^2 N + a(q)N^2$.

(i) V is bounded below on $\mathcal{M}$ iff $a(\frac{1}{8}) > 0$ and $a(\frac{1}{4}) > 0$, or $\min\{a(\frac{1}{8}), a(\frac{1}{4})\} = 0$ and $\mu^2 \leq 0$.

(ii) If $\mu^2 \leq 0$ and both $a(\frac{1}{8}), a(\frac{1}{4}) > 0$, the unique minimum is $M = 0$: no symmetry is broken.

(iii) If $\mu^2 > 0$ and both $a(\frac{1}{8}), a(\frac{1}{4}) > 0$, the minimum on each ray of angular type q is at $N = \mu^2/(2a(q))$ with value $-\mu^4/(4a(q))$, and the global minima are the kernels with $q = q_*$ minimising a: $q_* = \frac{1}{8}$ for $\beta > 0$, $q_* = \frac{1}{4}$ for $\beta < 0$, at $N_* = \mu^2/(2(\alpha + \beta q_*))$. For $\beta = 0$ every direction of $\mathcal{M}$ is a minimum direction.

(iv) On the boundary $\min\{a(\frac{1}{8}), a(\frac{1}{4})\} = 0$: for $\mu^2 < 0$ the unique minimum is $M = 0$ (for instance $\mu^2 = -1$, $\alpha = \beta = 0$ gives $V = N$), for $\mu^2 = 0$ the potential is flat along the rays with $a(q) = 0$, and for $\mu^2 > 0$ it is unbounded below. If $\min\{a(\frac{1}{8}), a(\frac{1}{4})\} < 0$, V is unbounded below; for example, at $\mu^2 = 1$, $\alpha = -1$, $\beta = 1$ one has $a < 0$ on the whole range, and on the ray $M = t\,\gamma^0 J$, $V = -8t^2 - 56t^4$.

Proof. The six blades are the antisymmetric blades that commute with the interior rotations and anticommute with the interior boosts (Definition 7.8); $M^\top M = \frac{N}{8}\mathbf{1} + D$ with $D^2 = d^2\mathbf{1}$ is a polynomial identity in the six coefficients. Since $M^\top M \geq 0$ has the eigenvalues $\frac{N}{8} \pm d$, each on four dimensions, $|d| \leq \frac{N}{8}$, which gives the range of q and its endpoints. For fixed q, V is a quadratic polynomial in $N \geq 0$, and (i)–(iv) follow by minimising it; on $t\,\gamma^0 J$, $N = 8t^2$ and $Q = 8t^4$. □

So the Dirac and Majorana phases below need $\mu^2 > 0$ and $a > 0$ at both ends of the range; outside that region the potential keeps $M = 0$, is not bounded below, or, on the zero-quartic boundary at $\mu^2 = 0$, is flat along the rays with $a(q) = 0$ (Lemma 7.11(iv)), and selects no phase. Inside it, the minimum set at N_* is the three-dimensional set of the lemma, and the kernels named in the theorem are representatives of it: in the Majorana phase the minima are all null z, not only the complex line supported on $\Gamma = +1$, and the other minima are obtained from that line by the elements $e_2\Omega$, $e_3\Omega$ of $\mathfrak{g}_I$, which anticommute with Γ.

Theorem 7.12 (Two phases of the interior Lorentz algebra)**.** *For the interior $\mathfrak{so}(1,3)$, with $\mathcal{M}$ the irreducible six-dimensional module of $\mathfrak{u}(1)_\Omega \oplus \mathfrak{su}(2)$, let $\mu^2 > 0$, $\alpha + \frac{1}{8}\beta > 0$ and $\alpha + \frac{1}{4}\beta > 0$, the bounded case of Lemma 7.11(iii):*

(a) Dirac phase *($\beta > 0$). The mass form $\gamma^0 J$ is a minimum; its stabiliser in $\mathfrak{g}_I$ is $\langle J \rangle$, so $\mathfrak{u}(1)_\Omega \oplus \mathfrak{su}(2)$ breaks to $\mathfrak{u}(1)_J$ with three Goldstone directions. Fermion number is conserved and particle and antiparticle are distinct.*

(b) Majorana phase *($\beta < 0$). The kernels supported on the $\Gamma = +1$ half form a complex line; each anticommutes with J, has rank four, $Q/N^2 = \frac{1}{4}$ and $M^\top M \propto \frac{1}{2}(\mathbf{1} + \Gamma)$, and its stabiliser in $\mathfrak{g}_I$ is exactly $\langle J - J\Gamma \rangle$, whose charge $1 - \Gamma$ is twice the left-handed fermion number, again with three Goldstone directions.*

The Majorana phase is the one-constituent seed of the vacuum selected in Section 7.5. There, too, a concentrated phase of the trace potential selects a Majorana mass on one right-handed line, and the unbroken global charges include $1 - \Gamma$. At one constituent there is no gauge group and the three Goldstone directions are physical, classically; at three constituents the corresponding directions are gauge directions and are eaten.

7.3.4 The field-strength vacuum

Read through Theorem 3.3, a vector X restricted to W is a field strength of restricted type, and the interior dynamics of Section 4.3 restricts to it.

Theorem 7.13 (Field strengths on the subfields)**.** *For the coordinate three-planes and the full space:*

	Lorentz types	$\mathbf{E} \cdot \mathbf{B} \equiv 0$	partner pairs	Lorentz algebra kept	zero-point energy
T	magnetic only	yes	0	$\mathfrak{so}(3)$	$+\frac{3}{2}\hbar\kappa$
S	electric only	yes	0	$\mathfrak{so}(3)$	$-\frac{3}{2}\hbar\kappa$
M_{21b}	magnetic, electric, null	yes	0	$\mathfrak{so}(1,2)$	$+\frac{1}{2}\hbar\kappa$
M_{12c}	magnetic, electric, null	yes	0	$\mathfrak{so}(1,2)$	$-\frac{1}{2}\hbar\kappa$
M_{21a}	all four	no	1	none	$+\frac{1}{2}\hbar\kappa$
M_{12a}, M_{12b}	all four	no	1	none	$-\frac{1}{2}\hbar\kappa$
full	all four	no	3	$\mathfrak{so}(1,3)$	0

On every subfield the cross term of the split-form Lagrangian is a total derivative, the equation of motion is $\ddot{X} = -\kappa^2 X$, and the energy is $\frac{m}{2}(\eta(\dot{X},\dot{X}) + \kappa^2\eta(X,X))$. The zero-momentum flow $\dot{X} = -\kappa G X$ stays in W exactly on $W \cap G^{-1}W$, of dimension 0 or 2 for the subfields and 6 for the full space.

On T the energy is positive definite and $X = 0$ is a true ground state: a pure magnetic field oscillates about zero with a positive quantum zero-point energy. On S the energy is negative definite: $X = 0$ is a maximum, dynamically stable, and its quantum oscillators are ghost-like, of negative energy or, if the energy is made positive, negative norm. On the mixed planes the energy is indefinite and there is no ground state; the zero-point energy is $\frac{1}{2}\hbar\kappa(p-q)$, and for the full $\mathbb{R}^{3,3}$ it vanishes. Neutral signature cancels the zero-point energy exactly. The mixed planes without a partner pair are the field strengths of electrodynamics in 2+1 dimensions, M_{12c} carrying one magnetic and two electric components and M_{21b} its dual, closed under an $\mathfrak{so}(1,2)$ and orthogonal by construction.

7.3.5 The Fock vacuum

A subfield with three anticommuting generators can carry a first-order Hamiltonian, and quantising it with Majorana operators as in Theorem 2.30 gives a Fock space.

Theorem 7.14 (A universal Fock vacuum)**.** *Let A be an antisymmetric first-order generator built from three anticommuting generators of a subfield, with $A^2 = -E^2$ for some $E > 0$ (at $E = 0$ all sixteen states have energy zero), and let Q be a commuting charge, antisymmetric with $Q^2 = -\mathbf{1}$ and balanced against A, $\operatorname{Tr}(AQ) = 0$. On the sixteen-dimensional Fock space the joint spectrum of energy and charge is*

$$(-2E, 0)^1,\ (-E, \pm 1)^{2+2},\ (0, 0)^4,\ (0, \pm 2)^{1+1},\ (E, \pm 1)^{2+2},\ (2E, 0)^1 :$$

a unique charge-zero ground state at $-2E$, the zero-point energy of four modes, and two particles and two antiparticles. The charge is the volume element ω_W for T and the $(1,2)$ planes; for S and the $(2,1)$ planes it must be borrowed from the complementary plane.

Proof. Commuting antisymmetric operators with $A^2 = -E^2\mathbf{1}$ and $Q^2 = -\mathbf{1}$ are simultaneously block diagonal on four orthogonal planes, on the k-th of which $A = Ea_kJ_2$ and $Q = s_kJ_2$ with $a_k, s_k \in \{\pm 1\}$. Each plane is one fermionic mode, and a Fock state is labelled by occupation signs $\epsilon_k = \pm 1$, with energy $\frac{E}{2}\sum_k \epsilon_k a_k$ and charge $\frac{1}{2}\sum_k \epsilon_k s_k$. Since $\operatorname{Tr}(AQ) = -2E\sum_k a_k s_k$, the balance condition says that $a_k s_k = +1$ on two planes and -1 on the other two, and counting the sixteen sign choices gives the table; in particular the ground state $\epsilon_k = -a_k$ has charge $-\frac{1}{2}\sum_k a_k s_k = 0$. □

The balance condition cannot be dropped. $A = Ee_4$ and $Q = e_4$ satisfy every other hypothesis, and then the Fock energy and charge are proportional, so the ground state at $-2E$ carries charge ± 2. The charges named in the theorem are balanced: $\operatorname{Tr}(\omega_T\, p\cdot e\, \omega_T) = -\operatorname{Tr}(p\cdot e) = 0$ on T, and $\operatorname{Tr}(AJ) = 0$ for $A = p\cdot(e_4, e_5, e_6)$ on S, because a product of distinct generators is traceless.

The subfields differ in the Hamiltonians they carry. T carries a massless Weyl-type Hamiltonian $\omega_T\, p\cdot e$ with an intrinsic charge; each $(1,2)$ plane a massive Dirac Hamiltonian in 2+1 dimensions with an intrinsic charge; S a real Majorana Hamiltonian $p\cdot(e_4, e_5, e_6)$ with no intrinsic charge; and the $(2,1)$ planes only a single antisymmetric generator, a mass term without momentum dependence.

Example 7.15 (Charge conjugation needs the whole structure)**.** Call charge conjugation of a charged subfield any operator anticommuting with its charge and commuting with every term of its Dirac generator, the condition of Theorem 2.25. For the interior 3+1 theory with charge J the solutions are exactly the surplus plane $\langle e_2, e_3\rangle$. For T, M_{12a}, M_{12b} and M_{12c} the solution spaces are eight-dimensional, and for S with charge J four-dimensional. For T the space contains e_4, e_5, e_6: the charge conjugation of the temporal Weyl theory lives in the spatial triple, as that of the four-dimensional theory lives in the surplus plane. Only the full four-dimensional Dirac structure fixes C up to the $U(1)_J$ rotation.

reading	temporal T	spatial S	full $\mathrm{Cl}(3,3)$
algebraic	$\omega_T = \gamma^0 J$: complex structure, rigid under rotations	$\omega_S = \gamma^1\gamma^2\gamma^3$: projector, rigid, swapped by C	$\Gamma = \omega_T\omega_S$: projector onto the half-spinors, invariant under $\mathfrak{so}(3,3)$
order parameter	adjoint of $\mathfrak{sp}(2)$; every vacuum breaks it, six Goldstone directions	the same	one invariant kernel $\gamma^0 J$; with the interior $\mathfrak{so}(1,3)$, a Dirac phase keeping J or a Majorana phase keeping $J - J\Gamma$
field strength	magnetic, positive energy, ground state	electric, negative energy	every type, indefinite energy, zero-point energy cancels
Fock	charged Weyl, C in the spatial triple	Majorana, charge only from outside	charged Dirac, $C = \langle e_2, e_3\rangle$ up to $U(1)_J$

Table 7.4: The vacuum at one constituent in four readings.

Table 7.4 summarises the four readings for the two triples and the full algebra. The temporal triple supplies the mass form and an intrinsic charge, the spatial triple the time-reversal factor and, as a field strength, the negative-energy half, and their product is the chirality. Charge conjugation, the alternative between a Dirac and a Majorana phase, and the cancellation of the zero-point energy all require the full algebra; none of them is settled inside a subfield.

7.4 The mass kernels of the chiral block

We now return to three constituents and to the block $V_* = \mathrm{im}\, P_*$ of Section 6.2, which carries one Pati–Salam family $(\mathbf{4},\mathbf{2},\mathbf{1})_{\Gamma=-1} \oplus (\mathbf{4},\mathbf{1},\mathbf{2})_{\Gamma=+1}$ with a right-handed neutrino. A vacuum for this matter is *dynamically selected* if it solves $\delta\Gamma_{\mathrm{eff}}/\delta\Phi = 0$ with a second variation that is non-negative apart from the zero modes of broken symmetries, the equations being those of a stated theory. It is *compatible with matter* if, over it, the fermions carry a well-defined Dirac operator, the unbroken gauge algebra acts consistently on them, anomalies cancel, and the permitted couplings produce acceptable masses. This and the next three sections answer both questions inside the arena. Three ingredients are inputs, and we name them where they enter: the projector P_*, which is the field content; the promotion of the space of mass kernels to a dynamical field; and, for the radiative statements, the form of the tree potential at the matching scale.

7.4.1 Notation on the block

The colour algebra $\mathfrak{so}(6) \cong \mathfrak{su}(4)$ is spanned by the fifteen bivectors of the colour directions $C = \{10, 11, 12, 16, 17, 18\}$; with $w = (E_7, E_{13}, E_{14}, E_{15})$ and $\tau_i = -\frac{1}{2} w_j w_k$ for (i,j,k) a cyclic permutation of $(1,2,3)$, as in Section 6.2, the two weak algebras are $t_i^L = \frac{1}{2}(\mathbf{1} - \chi_W)\tau_i$ and $t_i^R = \frac{1}{2}(\mathbf{1} + \chi_W)\tau_i$. These $15 + 3 + 3 = 21$ generators and the two central elements J and $J\Gamma$ commute with P_*, with the Lorentz action and with J, and on the block $q_A = E_2E_3$ acts as J and E_8E_9 as $J\Gamma$. With $H = -(E_{10}E_{16} + E_{11}E_{17} + E_{12}E_{18})$ the charges are

$$\hat{t} = -Jt, \qquad T_3^L = -Jt_3^L, \quad T_3^R = -Jt_3^R, \qquad B-L = -\tfrac{1}{3}JH, \qquad Y = T_3^R + \tfrac{1}{2}(B-L),$$

and the colour $\mathfrak{su}(3)$ is the part of $\mathfrak{so}(6)$ commuting with H. The charge table is that of Proposition 6.13. Finally γ^0 commutes with all 91 internal bivectors, with J and with the rotations, and anticommutes with Γ and the boosts, so it maps V_* onto an orthogonal rank-64 *mirror* on which the same internal algebra acts with Γ reversed (Remark 6.16).

Several letters of this chapter carry more than one meaning, each fixed by its context: κ is the frame rate c/ℓ in Section 7.1, the coupling of the holomorphic quartic in Section 7.5 and, with indices, the bidoublet values κ_1, κ_{15}, $\kappa_{u,d}$ in Section 7.6; Q is the quartic invariant $\mathrm{Tr}(M^\top M)^2$,

the charge of Theorem 7.14 and the electric charge; H is the colour generator above, and the block Hamiltonian is written $\mathcal{H}(p)$; T is the mass form $e_1e_2e_3$ and, in Section 7.5.5, the function $T(\hat{u})$; and K denotes the kinetic kernels, the bidoublet kernels K_{1u}, K_{1d} and the frame boosts.

7.4.2 The kernel space

Definition 7.16 (Mass kernels)**.** A Lorentz-invariant Grassmann mass kernel on the block is an $M \in \mathrm{End}(V_*)$ with

$$M^\top = -M, \qquad [M, P_*\Sigma^{ab}P_*] = 0, \qquad \{M, P_*\Sigma^{0a}P_*\} = 0,$$

for the interior rotations Σ^{ab} and boosts $\Sigma^{0a} = \frac{1}{2}\gamma^0\gamma^a$. Their real vector space is $\mathcal{M}$, with inner product $\langle M, N\rangle = \mathrm{Tr}\, M^\top N$.

The definition is forced. For a Grassmann-valued $\psi \in V_* \otimes \Lambda$ the form $\psi^\top M\psi$ sees only the antisymmetric part of M, and it is invariant under $\delta\psi = X\psi$ iff $X^\top M + MX = 0$; the rotations are antisymmetric and the boosts symmetric, which gives the two conditions. A kernel is exactly a Lorentz-invariant mass term of the block.

Theorem 7.17 (The kernel space)**.** *(i)* $\dim_{\mathbb{R}} \mathcal{M} = 272$*. A basis is given by the compressions* P_*BP_* *of the blades* $B = \gamma^0 e_I$ *and* $B = \gamma^1\gamma^2\gamma^3 e_I$*,* e_I *an internal blade, that are antisymmetric and commute with the three involutions defining* P_**: there are* 2176 *such blades and* 272 *distinct, mutually orthogonal compressions, and they span all solutions of Definition 7.16.*

(ii) $\mathcal{M}$ *is a complex vector space for the complex structure* $\widetilde{\mathcal{C}}(M) = JM\Gamma$*, defined on every kernel:* $\widetilde{\mathcal{C}}$ *preserves* $\mathcal{M}$*, squares to* $-\mathbf{1}$ *and commutes with the action of the* 23 *generators. (Left multiplication by* J *alone does not preserve* $\mathcal{M}$*: on a Dirac-type kernel* JM *is symmetric.) As a complex representation of* $\mathfrak{su}(4) \oplus \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$*, with the real dimensions of the summands under the braces,*

$$\mathcal{M} \cong \underbrace{(\bar{\mathbf{10}}, \mathbf{3}, \mathbf{1})}_{60} \oplus \underbrace{(\mathbf{10}, \mathbf{1}, \mathbf{3})}_{60} \oplus \underbrace{(\mathbf{15}, \mathbf{2}, \mathbf{2})}_{120} \oplus \underbrace{2(\mathbf{6}, \mathbf{1}, \mathbf{1})}_{24} \oplus \underbrace{(\mathbf{1}, \mathbf{2}, \mathbf{2})}_{8},$$

the conjugate of the Pati–Salam branching of $\mathrm{Sym}^2(\mathbf{16})$ *with* $\mathbf{16} = (\mathbf{4}, \mathbf{2}, \mathbf{1}) \oplus (\bar{\mathbf{4}}, \mathbf{1}, \mathbf{2})$*. The Casimir labels* $(-36, -2, 0)$*,* $(-36, 0, -2)$*,* $(-32, -\frac{3}{4}, -\frac{3}{4})$*,* $(-20, 0, 0)$ *and* $(0, -\frac{3}{4}, -\frac{3}{4})$ *identify the components in this order, and the cubic* $SU(4)$ *trace identifies the conjugation of the two tens, which Casimir labels cannot see: for* $h = \mathrm{diag}(1, 1, 1, -3)$*,* $\mathrm{tr}\, h^3 = -192$ *on* $\mathbf{10}$*, and the cubic trace of* h *over* $(\mathcal{M}, \widetilde{\mathcal{C}})$ *vanishes, as on* $\mathrm{Sym}^2(\mathbf{16})$*. Replacing* $\widetilde{\mathcal{C}}$ *by* $-\widetilde{\mathcal{C}}$ *conjugates every label. As a real representation the two components with* $\mathbf{10}$ *are irreducible of complex type, while* $(\mathbf{15}, \mathbf{2}, \mathbf{2})$*,* $(\mathbf{6}, \mathbf{1}, \mathbf{1})$ *and* $(\mathbf{1}, \mathbf{2}, \mathbf{2})$ *are of real type and occur with multiplicities* 2*,* 4 *and* 2*.*

(iii) Every basis kernel either anticommutes with J *and commutes with* Γ *(*Majorana type*, pairing fermions of equal chirality,* 144 *of them, spanning* $(\bar{\mathbf{10}}, \mathbf{3}, \mathbf{1}) \oplus (\mathbf{10}, \mathbf{1}, \mathbf{3}) \oplus 2(\mathbf{6}, \mathbf{1}, \mathbf{1})$*), or commutes with* J *and anticommutes with* Γ *(*Dirac type*, pairing opposite chiralities,* 128 *of them, spanning* $(\mathbf{15}, \mathbf{2}, \mathbf{2}) \oplus (\mathbf{1}, \mathbf{2}, \mathbf{2})$*).*

(iv) No nonzero kernel is invariant under the 21 *Pati–Salam generators.*

(v) The kernels invariant under the twelve Standard-Model generators form the plane spanned by Φ_0 *and* $J\Phi_0$*, where*

$$\Phi_0 = P_\nu\, P_*\, \gamma^0 E_3 E_9 E_{13} E_{14} E_{16} E_{17} E_{18}\, P_*\, P_\nu, \tag{7.3}$$

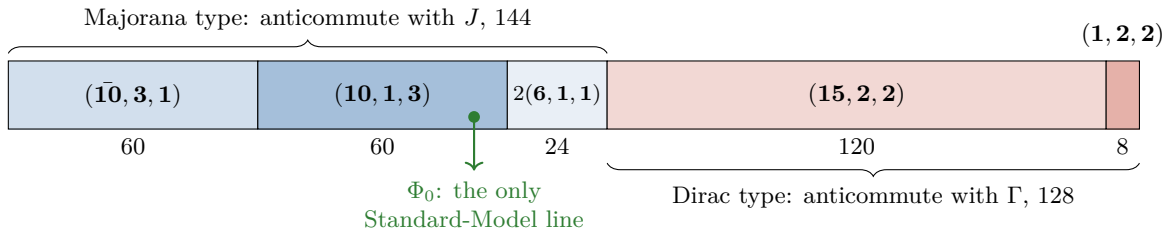


Figure 7.1: The 272-dimensional space of Lorentz-invariant mass kernels of the chiral block (Theorem 7.17), drawn to scale. The Standard-Model invariant kernels form a single complex line, the right-handed neutrino Majorana kernel Φ_0, inside $(\mathbf{10},\mathbf{1},\mathbf{3})$.

P_ν being the projector of the block V_* onto $\Gamma = +1$, $T_3^R = +\frac{1}{2}$, $B-L = -1$, the right-handed neutrino (on all of S_3 the joint eigenspace of these three operators has rank eight). Moreover $\Phi_0^\top \Phi_0 = P_\nu$, of rank four, J anticommutes with Φ_0, and

$$[t_3^R, \Phi_0] = J\Phi_0, \qquad [\tfrac{1}{3}H, \Phi_0] = -2J\Phi_0, \qquad [t_Y, \Phi_0] = 0,$$

so Φ_0 carries $T_3^R = 1$, $B-L = -2$ and $Y = 0$. It lies in the $(\mathbf{10},\mathbf{1},\mathbf{3})$ component.

Proof of the identification in (iii). A kernel commuting with Γ preserves the two chirality halves, which carry $(\mathbf{4},\mathbf{2},\mathbf{1})$ and $(\mathbf{4},\mathbf{1},\mathbf{2})$; one anticommuting with Γ exchanges them. A bilinear in two $(\mathbf{4},\mathbf{2},\mathbf{1})$ is an $\mathfrak{su}(2)_R$ singlet and one in two $(\mathbf{4},\mathbf{1},\mathbf{2})$ an $\mathfrak{su}(2)_L$ singlet, while a bilinear between the halves is a doublet of both. So $(\mathbf{15},\mathbf{2},\mathbf{2})$ and $(\mathbf{1},\mathbf{2},\mathbf{2})$ are of the second type and the others of the first, and $60+60+24 = 144$ and $120+8 = 128$ agree with the counts. The dimension, the components with their Casimir labels and types, and Φ_0 are checked in v2-1 to v2-3 and z3-8, and the complex structure $\widetilde{\mathcal{C}}$ with its cubic trace in a11-1. A complex structure that acts as J on the Majorana-type kernels and as $\widetilde{\mathcal{C}}$ on the Dirac-type ones also preserves $\mathcal{M}$ and commutes with the 23 generators, but it gives both tens the same conjugation and a cubic trace ± 1152, so it does not realise a branching of $\mathrm{Sym}^2(\mathbf{16})$ or of its conjugate. □

The complex dimension of $\mathcal{M}$ is $136 = \dim \mathrm{Sym}^2(\mathbb{C}^{16})$: a Lorentz-invariant mass term of sixteen Weyl fermions is a complex symmetric 16×16 matrix. In the language of $\mathfrak{so}(10)$, $\mathrm{Sym}^2(\mathbf{16}) = \mathbf{10} \oplus \mathbf{126}$ [144], and the decomposition (ii), read with the complex structure $\widetilde{\mathcal{C}}$, is the Pati–Salam branching of $\mathbf{10} \oplus \overline{\mathbf{126}}$, the dual of $\mathrm{Sym}^2(\mathbf{16})$, as befits the coefficients of mass bilinears (for $-\widetilde{\mathcal{C}}$, of $\mathbf{10} \oplus \mathbf{126}$), of complex dimension $136 = \frac{1}{2} \cdot 272$; the $(\mathbf{10},\mathbf{1},\mathbf{3})$ component is the field which, in minimal left–right and Pati–Salam models, gives the right-handed neutrino a Majorana mass [100, 101]. Here this content has been obtained from Definition 7.16, with only the Pati–Salam algebra acting. The only component containing a Standard-Model singlet is $(\mathbf{10},\mathbf{1},\mathbf{3})$: the colour-singlet parts of $(\bar{\mathbf{10}},\mathbf{3},\mathbf{1})$, $(\mathbf{15},\mathbf{2},\mathbf{2})$ and $(\mathbf{1},\mathbf{2},\mathbf{2})$ carry $Y = \pm 1$ or $\pm\frac{1}{2}$, and $(\mathbf{6},\mathbf{1},\mathbf{1})$ has none.

Theorem 7.18 (Symmetric criticality)**.** *Let $V : \mathcal{M} \to \mathbb{R}$ be smooth and invariant under the Pati–Salam group acting by $M \mapsto gMg^{-1}$. Then for every $r \in \mathbb{R}$*

$$\nabla V(r\Phi_0) = \frac{1}{\langle \Phi_0, \Phi_0 \rangle} \frac{d}{dr} V(r\Phi_0)\, \Phi_0,$$

so $r\Phi_0$ is a critical point of V if and only if r is a critical point of $r \mapsto V(r\Phi_0)$.

Proof. The action is orthogonal. If h fixes $x = r\Phi_0$, invariance gives $h\nabla V(x) = \nabla V(x)$, so $\nabla V(x)$ is fixed by the Standard-Model group and lies in $\mathrm{span}\{\Phi_0, J\Phi_0\}$. Since $J\Phi_0 = [t_3^R, \Phi_0]$ is tangent to the orbit, $\langle \nabla V(x), J\Phi_0 \rangle = 0$, and only the radial component remains. □

This is Michel's theorem on critical orbits [95], the principle of symmetric criticality [115]. Its content here is that no invariant coupling on the whole kernel space can generate a tadpole for any other component along Φ_0. It says nothing about the second variation or about other minima, which are the subject of Section 7.5.

7.4.3 The kernel field and its action

Assumption 7.19. The kernels are the values of a scalar field $\Phi(x) \in \mathcal{M}$ with kinetic term $\frac{1}{2}\langle D_\mu\Phi, D^\mu\Phi\rangle$ and a potential $V(\Phi)$ invariant under the gauge algebra, and a single Yukawa coupling y in (7.4). The overall normalisation of the kinetic term is a convention; its use of one trace form on all of $\mathcal{M}$, the single coupling y and the form of V are choices, restricted below only by symmetry.

With a real Grassmann field $\psi(x) \in V_*$, a gauge field A_μ with values in an algebra $\mathfrak{g}$ of antisymmetric operators, the kinetic kernels $K^\mu = P_*\gamma^0\gamma^\mu P_*$ (symmetric, $K^0 = \mathbf{1}$ on V_*; minus the compressed kernels of Proposition 2.21, a sign convention reconciled after Proposition 2.24) and the module metric $\mathsf{G}_{ab} = \mathrm{Tr}(t_a^\top t_b)/512$, take

$$S = \int d^4x \Big[\tfrac{1}{2}\psi^\top K^\mu D_\mu\psi + \tfrac{y}{2}\psi^\top\Phi\psi + \tfrac{1}{2}\langle D_\mu\Phi, D^\mu\Phi\rangle - V(\Phi) - \tfrac{1}{4}\mathsf{G}_{ab}F^a_{\mu\nu}F^{b\mu\nu}\Big], \tag{7.4}$$

with $D_\mu\psi = \partial_\mu\psi + A_\mu\psi$ and $D_\mu\Phi = \partial_\mu\Phi + [A_\mu, \Phi]$. Every term is Lorentz invariant by Definition 7.16, and gauge invariant for any $\mathfrak{g}$ inside the 23-dimensional faithful algebra of the block. The module metric is 1 on the colour bivectors and $\frac{1}{8}$ on t^L and t^R.

Normalisation dictionary. As written, (7.4) has unit gauge coupling for generators that are orthonormal in the module metric; a coupling g_A is restored by $-\frac{1}{4g_A^2}\mathsf{G}_{ab}F^a_{\mu\nu}F^{b\mu\nu}$. The generators of module norm $\frac{1}{8}$, such as t_i^L, t_i^R and the colour generator $\frac{1}{4}(E_{10}E_{16} - E_{11}E_{17})$, are those of the conventional normalisation $\mathrm{Tr}\,T_aT_b = \frac{1}{2}\delta_{ab}$ in the fundamental: their charge operators have eigenvalues $\pm\frac{1}{2}$ on a doublet. The conventionally normalised coupling is therefore

$$g = \sqrt{8}\, g_A,$$

and every vector mass from Section 7.5 on is quoted in g; (7.4) as written is the case $g_A = 1$, $g = 2\sqrt{2}$. The kernel field is normalised by $\frac{1}{2}\langle D\Phi, D\Phi\rangle$ with $\langle M, N\rangle = \mathrm{Tr}\,M^\top N$, and $v := |\Phi|/\sqrt{2}$ (so $v = \sqrt{2}\,|r|$ at $\Phi = r\Phi_0$). Ratios of masses do not depend on these choices.

Proposition 7.20 (Fermion number)**.** *J commutes with the kinetic kernels and with the* 21 *Pati–Salam generators. The transformation $\psi \mapsto e^{\theta J}\psi$, $\Phi \mapsto e^{\theta J}\Phi e^{-\theta J}$, $A \mapsto A$ leaves every term of (7.4) except V invariant; on $\mathcal{M}$ it acts as $\Phi \mapsto e^{2\theta J}\Phi$ on Majorana-type kernels and trivially on Dirac-type ones. In the complex structure J each of the sixteen Weyl fermions has charge* 1*, so this is the fermion number $F = 3B + L$, and $4B = F + (B-L)$.*

By Proposition 6.14 the two central directions J and $J\Gamma$ are anomalous on this matter, so the gauge algebra can be any anomaly-free subalgebra of the 23, and $\mathfrak{su}(4) \oplus \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ is the largest one containing Pati–Salam. A gauge field coupled to an anomalous current makes the quantum theory inconsistent [59, 16], which is why the central directions carry no gauge field; they remain global symmetries. In the vacuum $r\Phi_0$ the stabiliser among the 23 generators is fourteen-dimensional: the twelve Standard-Model generators together with $J + \frac{1}{3}H$ and $J - J\Gamma$, whose charges are $1 + (B-L) = 4B$ and $1 - \Gamma$, baryon number and twice the left-handed fermion number, both global.

Proposition 7.21 (The coupled field equations)**.** *The Euler–Lagrange equations of* (7.4) *are*

$$K^\mu D_\mu\psi + y\Phi\psi = 0, \quad D_\mu D^\mu\Phi + \nabla V(\Phi) - \tfrac{y}{2}\Pi_{\mathcal{M}}(\psi\psi^\top) = 0, \quad \mathsf{G}_{ab}D_\nu F^{b\nu\mu} = -\big(j^\mu_a(\psi) + j^\mu_a(\Phi)\big),$$

with $\Pi_{\mathcal{M}}$ the orthogonal projection onto $\mathcal{M}$. At constant $\Phi = r\Phi_0$, $\psi = 0$, $A = 0$ the first and third hold identically and the second reduces to $\frac{d}{dr}V(r\Phi_0) = 0$ by Theorem 7.18. The same holds with V replaced by the effective potential, which is again Pati–Salam invariant; at one loop

$$V_1 = \frac{1}{64\pi^2}\,\mathrm{STr}\Big[\mathcal{M}^4\Big(\ln\frac{\mathcal{M}^2}{\mu_R^2} - c\Big)\Big], \qquad \mathrm{STr} = \textstyle\sum_{\text{scalars}} + 3\sum_{\text{vectors}} - 2\sum_{\text{Weyl}}, \tag{7.5}$$

the fermion term being the determinant of the Dirac operator of the first equation [26, 75]. By the Nielsen identities the value of the effective potential at a stationary point does not depend on the gauge fixing [107]; the background coordinates and the raw Hessian do.

The fermions, classically zero, act on the vacuum only through the effective potential, and the matter content enters through the Yukawa coupling y. That there is one such coupling is part of Assumption 7.19, not a consequence of calling Φ one field: Φ is reducible under the Pati–Salam algebra, and symmetry alone allows an independent Yukawa coupling and an independent kinetic normalisation for each of its five irreducible components, with a mixing between the two copies of $(\mathbf{6},\mathbf{1},\mathbf{1})$. The same is true of the common gauge coupling of $\mathfrak{su}(4)$ and $\mathfrak{su}(2)_R$ used in Section 7.5. In the vacuum the block Hamiltonian is $\mathcal{H}(p) = p_iK^i + iy\,r\,\Phi_0$; since $\{K^i,K^j\} = 2\delta^{ij}$, $\{K^i,\Phi_0\} = 0$ and $-\Phi_0^2 = P_\nu$,

$$\mathcal{H}(p)^2 = |p|^2 + y^2r^2P_\nu\,: \tag{7.6}$$

one Majorana fermion, the right-handed neutrino, with four real components and mass $|y|r$, and sixty massless real components. Executable check: a6-3 for the three identities used here (exact symbolic or integer arithmetic; Section 15.2). The Euler–Lagrange equations are a direct variation of (7.4), and (7.5) is the inherited Coleman–Weinberg formula; neither is the subject of a check.

Theorem 7.18 settles stationarity for the whole coupled quantum system along Φ_0. Which orbit is the *minimum* is decided by the couplings, and that is the subject of the next section. A non-negative second variation on the $(\mathbf{10},\mathbf{1},\mathbf{3})$ directions alone does not imply it on all of $\mathcal{M}$: fluctuations outside $(\mathbf{10},\mathbf{1},\mathbf{3})$ with the Standard-Model quantum numbers of $(\mathbf{10},\mathbf{1},\mathbf{3})$ fluctuations can mix with them, and Section 7.5.5 exhibits such a mixing.

7.5 Selecting the Standard-Model orbit

The Standard-Model line Φ_0 lies in the $(\mathbf{10},\mathbf{1},\mathbf{3})$ component, which is small enough to be analysed completely. We first classify the renormalizable potential on that component and give sufficient and necessary conditions for Φ_0 to be its global minimum; then we add the gauge and matter fluctuations; and finally we return to the whole of $\mathcal{M}$.

7.5.1 The tensor model

Let $\mathfrak{su}(4)\oplus\mathfrak{su}(2)_R$ act on $\mathbb{C}^8 = \mathbb{C}^4\otimes\mathbb{C}^2$ by Hermitian generators normalised by $\mathrm{Tr}\,T_aT_b = \frac{1}{2}\delta_{ab}$ in the fundamental, and on complex symmetric 8×8 matrices by $S\mapsto USU^{\top}$. The subspace

$$\mathcal{T} = \mathrm{Sym}^2(\mathbb{C}^4)\otimes\mathrm{Sym}^2(\mathbb{C}^2)\subset\mathrm{Sym}^2(\mathbb{C}^8), \qquad \dim_{\mathbb{C}}\mathcal{T} = 30, \qquad |S|^2 = \mathrm{Tr}\,S^{\dagger}S,$$

is invariant and irreducible, isomorphic to $(\mathbf{10},\mathbf{1},\mathbf{3})$. With $e_1,\dots,e_4$ and $e_\pm$ the standard bases of $\mathbb{C}^4$ and $\mathbb{C}^2$, $B-L = \mathrm{diag}(\frac{1}{3},\frac{1}{3},\frac{1}{3},-1)$ and $T_3^R = \mathrm{diag}(\frac{1}{2},-\frac{1}{2})$, the vectors $e_i\otimes e_+$, $e_4\otimes e_+$, $e_i\otimes e_-$, $e_4\otimes e_-$ $(i\le 3)$ carry $Y = \frac{2}{3},0,-\frac{1}{3},-1$: they are u_R, ν_R, d_R, e_R, and a component of S is a pair of them. Put $u_0 = e_4\otimes e_+$ and $S_0 = u_0u_0^{\top}$.

Proposition 7.22 (From kernels to tensors)**.** *The component $\mathcal{M}_{(\mathbf{10},\mathbf{1},\mathbf{3})}$ is J-invariant of real dimension 60, and with J as complex structure it is isomorphic to $\mathcal{T}$ by a Pati–Salam-equivariant*

ι with $\operatorname{Tr}\Phi^\top\Phi = 4|\iota(\Phi)|^2$, unique up to a phase, with $\iota(\Phi_0) = e^{i\varphi}S_0$. For $S = \iota(\Phi)$, $\Phi^\top\Phi$ is supported on the $\Gamma = +1$ half and has the spectrum of $S^\dagger S$, each eigenvalue four times, so

$$N := \operatorname{Tr}\Phi^\top\Phi = 4|S|^2, \qquad Q := \operatorname{Tr}(\Phi^\top\Phi)^2 = 4\operatorname{Tr}(S^\dagger S)^2, \qquad \frac{Q}{N^2} = \frac{1}{4}\frac{\operatorname{Tr}(S^\dagger S)^2}{|S|^4}.$$

Executable check: a8-1 (numerical; Section 15.2). It recomputes that the component is J-invariant of dimension 60, that its Pati–Salam commutant is spanned by $\mathbf{1}$ and J, so that it is irreducible over $\mathbb{C}$ and ι is unique up to a complex scale, which the normalisation fixes up to a phase, that Φ_0 lies in it with $\Phi_0^\top\Phi_0 = P_\nu$, and that $\Phi^\top\Phi$ is supported on $\Gamma = +1$ with every eigenvalue four times. The map ι is explicit and checked exactly in a13-2. On the $\Gamma = +1$ half V_+ of the block, with J as the imaginary unit, an intertwiner U identifies V_+ with $\mathbb{C}^2 \otimes \mathbb{C}^4 \otimes \mathbb{C}^2$ (Lorentz Weyl index, colour, $\mathfrak{su}(2)_R$), as a scaled isometry. In this basis every kernel of the component is the real antilinear map $v \mapsto (\varepsilon \otimes S)\bar{v}$, $\varepsilon = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, with $S \in \mathcal{T}$, and $\iota(\Phi) = S$. Then $\Phi^\top\Phi$ is the realification of $\mathbf{1}_2 \otimes SS^\dagger$, whose eigenvalues are those of $S^\dagger S$, each four times; $\operatorname{Tr}\Phi^\top\Phi = 4|S|^2$ holds exactly on a basis, and $\iota(\Phi_0)$ is of rank one with $B{-}L = -2$, $T_3^R = 1$, the tensor S_0 up to scale (a13-2, exact).

Let $P_1, \dots, P_5$ be the orthogonal projectors of $\operatorname{Sym}^2\mathcal{T}$ onto its five irreducible components

$$(\mathbf{35},\mathbf{5}) \oplus (\mathbf{35},\mathbf{1}) \oplus (\mathbf{20'},\mathbf{5}) \oplus (\mathbf{20'},\mathbf{1}) \oplus (\mathbf{45},\mathbf{3}), \qquad 175 + 35 + 100 + 20 + 135 = 465,$$

and put $I_k(S) = \langle S \otimes S, P_k(S \otimes S)\rangle \geq 0$, so that $\sum_k I_k = |S|^4$.

Theorem 7.23 (The renormalizable potential of $(\mathbf{10},\mathbf{1},\mathbf{3})$)**.** *The most general renormalizable invariant potential on $\mathcal{T}$ is*

$$V(S) = -\mu^2|S|^2 + \sum_{k=1}^{5} \lambda_k I_k(S) + \kappa\,\mathcal{D}(S) + \bar{\kappa}\,\overline{\mathcal{D}(S)}, \qquad \lambda_k \in \mathbb{R},\ \kappa \in \mathbb{C}, \tag{7.7}$$

where the holomorphic quartic is

$$\mathcal{D}(S) = \tfrac{1}{24}\,\epsilon^{abcd}\epsilon^{efgh}P_{ab,ef}(S)P_{cd,gh}(S), \qquad P_{ab,ef}(S) = \varepsilon_{su}\varepsilon_{tv}S_{as,et}S_{bu,fv},$$

with colour indices $a, \dots, h$ and $\mathfrak{su}(2)_R$ indices s, t, u, v. There is no invariant cubic. Moreover:

(i) $\mathcal{D}(e^{i\theta}S) = e^{4i\theta}\mathcal{D}(S)$, every I_k is phase invariant, and $\mathcal{D}$ is not a combination of the I_k.

(ii) $\mathcal{D}$ depends on S only through the $(\mathbf{20'},\mathbf{1})$ component of $S \otimes S$, and $|\mathcal{D}(S)| \leq \frac{3}{8}I_4(S)$, with equality at $S = \mathbf{1}_4 \otimes \mathbf{1}_2$.

(iii) Under fermion number $S \mapsto e^{2i\theta}S$, so the I_k are invariant and $\mathcal{D} \mapsto e^{8i\theta}\mathcal{D}$: (7.7) is invariant under fermion number if and only if $\kappa = 0$. $\mathcal{D}$ has $B{-}L = 0$ and fermion number 8, hence baryon number 2.

The bound in (ii) follows from Schur's lemma. The $(\mathbf{20'},\mathbf{1})$ component carries, up to scale, one invariant complex-bilinear symmetric form β; $\mathcal{D}(S) = c\,\beta(w,w)$ with w the $(\mathbf{20'},\mathbf{1})$ component of $S \otimes S$; and $\beta^\dagger\beta$ commutes with the irreducible action, so all singular values of β are equal and $|\mathcal{D}(S)| \leq |c|\,\sigma\,|w|^2 = |c|\,\sigma\,I_4(S)$. The constant is computed exactly (a13-1): the $(\mathbf{20'},\mathbf{1})$ component is the $\mathfrak{su}(2)_R$ singlet $\varepsilon_{su}\varepsilon_{tv} + \varepsilon_{sv}\varepsilon_{tu}$ times the tensors of $\operatorname{Sym}^2(\operatorname{Sym}^2\mathbb{C}^4)$ with vanishing total symmetrisation, which has an integer basis; on it the matrix Q of the quartic form and the Gram matrix G satisfy $QG^{-1}Q = (\frac{3}{8})^2G$, so $|c|\sigma = \frac{3}{8}$, and $\mathcal{D}(\mathbf{1}_4 \otimes \mathbf{1}_2) = 4$, $I_4(\mathbf{1}_4 \otimes \mathbf{1}_2) = \frac{32}{3}$, so equality holds there (also v1-5, v1-6 in floating point). The invariance and the phase degree of $\mathcal{D}$ are checked in v1-5, the vanishing of $\mathcal{D}$ to second order at S_0 in v1-7.

The trivial representation occurs in $\mathcal{T}\otimes\bar{\mathcal{T}}$, $\operatorname{Sym}^2\mathcal{T}$, $\operatorname{Sym}^3\mathcal{T}$, $\operatorname{Sym}^2\mathcal{T}\otimes\bar{\mathcal{T}}$, $\operatorname{Sym}^4\mathcal{T}$, $\operatorname{Sym}^3\mathcal{T}\otimes\bar{\mathcal{T}}$ and $\operatorname{Sym}^2\mathcal{T}\otimes\operatorname{Sym}^2\bar{\mathcal{T}}$ with multiplicities $1, 0, 0, 0, 1, 0, 5$ (z4-4, exact), which is the content of the

theorem. The phase of κ can be removed by $S \mapsto e^{i\alpha}S$, so only $|\kappa|$ matters. The contraction of four $(\mathbf{10},\mathbf{1},\mathbf{3})$ fields with two $SU(4)$ ϵ tensors is the baryon-number-violating coupling of Pati–Salam models through which diquark components mediate neutron–antineutron oscillation [99, 24].

Two lemmas organise the quartics. The first is the recoupling identity $\mathrm{Tr}(S^\dagger S)^2 = I_1 - \frac{1}{2}I_2 - \frac{1}{2}I_3 + \frac{1}{4}I_4 + 0\cdot I_5$, obtained from Schur–Weyl duality on $(\mathbb{C}^4)^{\otimes 4} \otimes (\mathbb{C}^2)^{\otimes 4}$. The second characterises the orbit of S_0.

Lemma 7.24 (The highest-weight orbit)**.** *For $S \in \mathcal{T}\setminus 0$ the following are equivalent: $S\otimes S$ lies in $(\mathbf{35},\mathbf{5})$; $\mathrm{Tr}(S^\dagger S)^2 = |S|^4$; $\operatorname{rank} S = 1$; $S = (a\otimes b)(a\otimes b)^\top$ with $a\in\mathbb{C}^4$, $b\in\mathbb{C}^2$; S lies on the orbit $\mathcal{O}_{\mathrm{hw}}$ of $\mathbb{C}^\times S_0$.*

This is the Kostant–Lichtenstein description of the highest-weight orbit by quadrics [87] (checked in v1-3). On $\mathcal{O}_{\mathrm{hw}}$ the colour factor of S has rank one, so $\mathcal{D}$ vanishes there; more precisely $\mathcal{D}$ and its first and second derivatives vanish at S_0, and

$$\mathcal{D}(s_0 S_0 + \delta) = s_0\Big(\tfrac{1}{3}\epsilon_{ijk}\epsilon_{lmn}U_{il}D_{jm}D_{kn} - \tfrac{1}{12}\epsilon_{ijk}\epsilon_{lmn}M_{il}M_{jm}D_{kn}\Big) + \mathcal{D}(\delta),$$

with U, M, D the symmetric 3×3 matrices of the colour sextets u_Ru_R, u_Rd_R, d_Rd_R. In the vacuum the holomorphic quartic contributes nothing to the masses and produces the cubic couplings $\kappa v\,\epsilon\epsilon\,\Delta_{uu}\Delta_{dd}\Delta_{dd}$ and $\kappa v\,\epsilon\epsilon\,\Delta_{ud}\Delta_{ud}\Delta_{dd}$ of diquarks of hypercharges $\frac{4}{3},\frac{1}{3},-\frac{2}{3}$, each of baryon number two. With the Yukawa couplings of the diquarks to quark pairs, here the single coupling y of (7.4), the vertex closes into a six-quark $\Delta B = 2$ operator, the Pati–Salam mechanism of neutron–antineutron oscillation; the present limits on the free-neutron oscillation time are 8.6×10^7 s from free neutrons [8] and 4.7×10^8 s inferred from the search with neutrons bound in ^{16}O [146], which is not a lifetime of bound neutrons.

7.5.2 Where the Standard-Model orbit is the global minimum

Number the components $k = 1,\dots,5$ as above, put $f_k(S) = I_k(S)/|S|^4$ (so $f_k \ge 0$, $\sum f_k = 1$), $d(S) = |\mathcal{D}(S)|/|S|^4$, and

$$\Delta_k = \lambda_k - \lambda_1 \ (k\ge 2), \qquad \Lambda(S) = \sum_k \lambda_k f_k(S) = \lambda_1 + \sum_{k\ge 2}\Delta_k f_k(S), \qquad \Lambda_\kappa(S) = \Lambda(S) - 2|\kappa|\,d(S).$$

For a unit vector $\hat S$, $\min_\varphi V(re^{i\varphi}\hat S) = -\mu^2 r^2 + \Lambda_\kappa(\hat S) r^4$, because the I_k do not depend on the phase and $2\,\mathrm{Re}(\kappa e^{4i\varphi}\mathcal{D}(\hat S))$ reaches $-2|\kappa||\mathcal{D}(\hat S)|$. With $\mu^2 > 0$ the potential is bounded below iff $\Lambda_\kappa > 0$ on the unit sphere, the minimum along a ray is $-\mu^4/(4\Lambda_\kappa)$, and the vacuum is the configuration of smallest Λ_κ.

Theorem 7.25 (Selection without the holomorphic quartic)**.** *Let $\kappa = 0$, $\mu^2 > 0$, $\lambda_1 > 0$ and $\Delta_2,\dots,\Delta_5 > 0$. The global minima of (7.7) are exactly the points of $\mathcal{O}_{\mathrm{hw}}$ with $|S|^2 = \mu^2/(2\lambda_1)$, with $V_{\min} = -\mu^4/(4\lambda_1)$. They form one orbit of real dimension nine, and the stabiliser of each point is conjugate to $\mathfrak{su}(3)\oplus\mathfrak{u}(1)_Y$ inside $\mathfrak{su}(4)\oplus\mathfrak{su}(2)_R$ with $Y = T_3^R + \frac{1}{2}(B-L)$; together with $\mathfrak{su}(2)_L$, which acts trivially on $(\mathbf{10},\mathbf{1},\mathbf{3})$, this is the Standard-Model algebra.*

Proof. $\Lambda \ge \lambda_1$, with equality iff $f_2 = \cdots = f_5 = 0$, that is iff $S\in\mathcal{O}_{\mathrm{hw}}$ by Lemma 7.24; the unit-norm part of $\mathcal{O}_{\mathrm{hw}}$ is a single orbit, of dimension $18-9$; the orbit dimension and the stabiliser are checked in v1-4. □

The unbroken $\mathfrak{u}(1)$ is an output, computed as the stabiliser: the weights of the vacuum, not a choice of embedding, fix the hypercharge. This is the vacuum of Pati–Salam and left–right models with a $(\mathbf{10},\mathbf{1},\mathbf{3})$ field [116, 101]; the theorem says for which couplings the dynamics chooses it.

Theorem 7.26 (Selection with the holomorphic quartic)**.** *Let $\mu^2 > 0$ and $\kappa \in \mathbb{C}$.*

(a) If $\lambda_1, \lambda_2, \lambda_3, \lambda_5 > 0$ and $\lambda_4 > \frac{3}{4}|\kappa|$, the potential is bounded below; if it is bounded below then $\lambda_1 > 0$ and $2\lambda_1 + \lambda_2 + 2\lambda_3 + \lambda_4 > \frac{3}{4}|\kappa|$.

(b) If $\lambda_1 > 0$, $\Delta_2, \Delta_3, \Delta_5 > 0$ and $\Delta_4 > \frac{3}{4}|\kappa|$, the global minima are exactly the points of $\mathcal{O}_{\mathrm{hw}}$ with $|S|^2 = \mu^2/(2\lambda_1)$.

(c) If those points are global minima, then $\Delta_2 \geq 0$, $\Delta_3 \geq 0$, $2\Delta_2 + 2\Delta_3 + 4\Delta_4 + 9\Delta_5 \geq 0$ and $\Delta_2 + 2\Delta_3 + \Delta_4 \geq \frac{3}{4}|\kappa|$.

(d) The second variation at the Standard-Model orbit does not depend on κ.

At the benchmark couplings $\lambda = (1, 1.7, 2.3, 3.1, 2.9)$ the bounds read: $\frac{4}{3}\Delta_4 = 2.8$, sufficient for selection; $\frac{4}{3}(\Delta_2 + 2\Delta_3 + \Delta_4) = 7.2$, necessary for selection and equal to the computed threshold; $\frac{4}{3}\lambda_4 = 4.13$, sufficient for boundedness; and $\frac{4}{3}(2\lambda_1+\lambda_2+2\lambda_3+\lambda_4) = 15.2$, necessary for boundedness. Minimisations from random starts are consistent with the transition at $|\kappa| = 7.2$: at $|\kappa| = 6, 7, 7.5, 8$ the runs end on the Standard-Model orbit, which is always a local minimum, or on the orbit of $\mathbf{1}_4 \otimes \mathbf{1}_2$, which is the higher of the two for $|\kappa| \leq 7$ and the lower for $|\kappa| \geq 7.5$, and no run ends below both (v3-1; sixteen archived starts per value, numerical): at the benchmark, $f(\mathbf{1}_4 \otimes \mathbf{1}_2) = (\frac{1}{3}, \frac{1}{6}, \frac{1}{3}, \frac{1}{6}, 0)$ and $d(\mathbf{1}_4 \otimes \mathbf{1}_2) = \frac{1}{16}$, so $\Lambda_\kappa(\mathbf{1}_4 \otimes \mathbf{1}_2) - \Lambda_\kappa(S_0) = 0.9 - \frac{1}{8}|\kappa|$, which changes sign at $|\kappa| = 7.2$.

Theorem 7.27 (Second variation at the Standard-Model orbit)**.** *At $S_v = vS_0$ the sixty real fluctuations split into $SU(3) \times U(1)_Y$ multiplets, each with one mass:*

pair	$SU(3)_Y$	$\dim_{\mathbb{R}}$	m^2/v^2	role
$\nu_R\nu_R$ (modulus)	1_0	1	$4\lambda_1$	radial
$\nu_R\nu_R$ (phase), $\nu_R e_R$, $u_R\nu_R$	1_0, 1_{-1}, $3_{2/3}$	$1+2+6$	0	eaten
$e_R e_R$	1_{-2}	2	$\frac{4}{3}\Delta_2$	Δ_{ee}
$u_R e_R + d_R\nu_R$	$3_{-1/3}$	6	Δ_5	Δ_{ue}
$d_R e_R$	$3_{-4/3}$	6	$\frac{2}{3}\Delta_2 + \Delta_5$	Δ_{de}
$u_R u_R$	$6_{4/3}$	12	$\frac{4}{3}\Delta_3$	Δ_{uu}
$u_R d_R$	$6_{1/3}$	12	$\frac{2}{3}\Delta_3 + \Delta_5$	Δ_{ud}
$d_R d_R$	$6_{-2/3}$	12	$\frac{2}{9}\Delta_2 + \frac{2}{9}\Delta_3 + \frac{4}{9}\Delta_4 + \Delta_5$	Δ_{dd}

Consequently S_v is a non-degenerate local minimum on $\mathcal{T}$ iff $\lambda_1, \Delta_2, \Delta_3, \Delta_5 > 0$ and $2\Delta_2+2\Delta_3+4\Delta_4+9\Delta_5 > 0$, a region that strictly contains the cone of Theorem 7.25; and, independently of the couplings,

$$m^2_{de} = m^2_{ue} + \tfrac{1}{2}m^2_{ee}, \qquad m^2_{ud} = m^2_{ue} + \tfrac{1}{2}m^2_{uu}. \tag{7.8}$$

The table is checked against the full 60×60 mass matrix at the benchmark and at random couplings (v1-9 to v1-11). The $d_R d_R$ entry is $\frac{1}{9}(2\Delta_2+2\Delta_3+4\Delta_4+9\Delta_5)$, whose positivity is the last local-minimum condition stated in the theorem, and with the one-loop gaps of Theorem 7.30 it gives $m^2_{dd} = 120 - 4x$, as in Table 7.5.

Example 7.28 (Checking a sum rule)**.** At the benchmark $\lambda = (1, 1.7, 2.3, 3.1, 2.9)$ one has $\Delta = (0.7, 1.3, 2.1, 1.9)$, so in units of v^2: $m^2_{ee} = \frac{4}{3}\cdot 0.7 = 0.9\overline{3}$, $m^2_{ue} = 1.9$, $m^2_{de} = \frac{2}{3}\cdot 0.7 + 1.9 = 2.3\overline{6}$, and indeed $1.9 + \frac{1}{2}\cdot 0.9\overline{3} = 2.3\overline{6}$. Likewise $m^2_{uu} = \frac{4}{3}\cdot 1.3 = 1.7\overline{3}$ and $m^2_{ud} = \frac{2}{3}\cdot 1.3 + 1.9 = 2.7\overline{6} = 1.9 + \frac{1}{2}\cdot 1.7\overline{3}$. The full 60×60 mass matrix at these couplings and $\mu^2 = 2$ has the predicted spectrum to 4×10^{-15}.

The colour-sextet components $\Delta_{uu}, \Delta_{ud}, \Delta_{dd}$ are the diquarks just met, and Δ_{ee} is the doubly charged scalar of left–right models. The sum rules (7.8) hold for the most general renormalizable potential of a single $(\mathbf{10}, \mathbf{1}, \mathbf{3})$, with or without κ. When one of the bounds $\Delta_2, \Delta_3, \Delta_5 > 0$ fails, gradient flows end on configurations of lower energy that break electric charge or colour: the configuration $(e_4 e_4)(e_+ e_-)$, for instance, keeps colour, but its unbroken abelian factor is T_3^R instead of Y.

7.5.3 The trace potential

The kernel field carries two quartic invariants that do not refer to the Pati–Salam decomposition at all, N^2 and $Q = \mathrm{Tr}(\Phi^\top\Phi)^2$; they are invariant under every orthogonal change of basis of the block that preserves $\mathcal{M}$, in particular under fermion number, so the trace potential has $\kappa = 0$.

Theorem 7.29 (The trace potential on $(\mathbf{10},\mathbf{1},\mathbf{3})$)**.** *Let $V_{\mathrm{tr}} = -\mu^2 N + \alpha N^2 + \beta Q$ with $\mu^2 > 0$. In the variables of Theorem 7.23, $\lambda_k = 16\alpha + 4\beta a_k$ with $a = (1, -\frac{1}{2}, -\frac{1}{2}, \frac{1}{4}, 0)$, so $(\Delta_2, \Delta_3, \Delta_4, \Delta_5) = -4\beta\,(\frac{3}{2}, \frac{3}{2}, \frac{3}{4}, 1)$. If $\beta < 0$ and $\alpha + \frac{1}{4}\beta > 0$, the global minimum is the Standard-Model orbit; if $\beta > 0$ and $\alpha + \frac{1}{32}\beta > 0$, the minima are the S with $S^\dagger S = \frac{1}{8}|S|^2\mathbf{1}_8$, and every one of them breaks colour; if $\beta = 0$ the potential is $O(60)$-symmetric and the orbit is chosen by fluctuations.*

Q/N^2 measures how concentrated the Majorana mass matrix is: it equals $\frac{1}{4}$ when one fermion carries all of it and $\frac{1}{32}$ when the eight right-handed fermions share it equally. A negative β rewards concentration on a single line, and the only line on which the dynamics can concentrate without breaking colour is ν_R. The Standard-Model vacuum is the concentrated phase of the Majorana mass, which is the three-constituent form of the Majorana phase of Theorem 7.12.

7.5.4 The fluctuations decide: back-reaction on $(\mathbf{10},\mathbf{1},\mathbf{3})$

Take the tree potential degenerate on $(\mathbf{10},\mathbf{1},\mathbf{3})$, $\lambda_1 = \cdots = \lambda_5$ and $\kappa = 0$, the case $\beta = 0$. Every unit direction of $\mathcal{T}$ is then a tree-level minimum, and the orbit is chosen by the one-loop potential (7.5). Gauge and Yukawa interactions respect fermion number, so they do not generate κ in perturbation theory. Write $L = \ln(\Lambda_{\mathrm{UV}}^2/v^2)$. The orbit-dependent part of the leading logarithm is

$$\delta V = \frac{L}{64\pi^2}\Big[2\sum_{\mathrm{fam}}\mathrm{Tr}(M_F^\dagger M_F)^2 - 3\,\mathrm{Tr}\,M_V^4 - \mathrm{Tr}\,M_S^4\Big],$$

where for the degenerate tree potential $\mathrm{Tr}\,M_S^4$ depends on $|S|$ only. Here and in Section 7.5.5 the field is normalised canonically. The analysis is to be compared with that of the quantum vacuum of the minimal non-supersymmetric $SO(10)$ models [302, 303], in which one-loop corrections likewise decide the vacuum. All radiative statements of this section share four limitations: they keep the leading logarithm only, with orbit-independent and non-logarithmic terms dropped and their effect not estimated; vectors are counted with three polarisations, as in Landau gauge, and the scheme dependence away from stationary points is not examined; the masses are curvature masses of the effective potential, not pole masses; and the couplings are taken equal where stated, as in Assumption 7.19. By Proposition 7.22, $\mathrm{Tr}\,\Phi^\top\Phi = 4|\iota(\Phi)|^2$, so the kinetic term of Assumption 7.19 is $\mathrm{Tr}(DS)^\dagger(DS)$ for $S := \sqrt{2}\,\iota(\Phi)$, and from here on S denotes this field and $v = |S| = |\Phi|/\sqrt{2} = \sqrt{2}\,r$, the v of Section 7.5.5 and Section 7.6. In these variables the right-handed neutrino mass $|y|r$ of Section 7.4 reads $(|y|/\sqrt{2})\,v$, so the y of $M_F = yS$ below is the coupling of (7.4) divided by $\sqrt{2}$, while the coupling g is the same as in Theorem 7.34. The vector mass matrix is $(M_V^2)_{ab} = 2g^2\,\mathrm{Re}\,\mathrm{Tr}(T_aS + ST_a^\top)^\dagger(T_bS + ST_b^\top)$ with one coupling for $\mathfrak{su}(4)$ and $\mathfrak{su}(2)_R$, and the eight right-handed Weyl fermions get the Majorana matrix $M_F = yS$.

Theorem 7.30 (One-loop selection on $(\mathbf{10},\mathbf{1},\mathbf{3})$)**.** *At leading logarithm, $\mathrm{Tr}\,M_V^4 = g^4\sum_k c_k I_k$ with $c = (33, 3, -3, 39, 5)$ and $\mathrm{Tr}(M_F^\dagger M_F)^2 = y^4\sum_k a_k I_k$, and the fluctuations generate the gaps*

$$\Delta_k = \frac{L}{64\pi^2}\,g^4(G_k - x\,F_k), \qquad G = (90, 108, -18, 84), \quad F = (3, 3, \tfrac{3}{2}, 2), \qquad x = \frac{\sum_{\mathrm{fam}} y^4}{g^4}.$$

Gauge fluctuations favour the Standard-Model orbit in the directions $k = 2, 3, 5$, and matter fluctuations disfavour it in every direction. The Standard-Model orbit is a non-degenerate local minimum on $\mathcal{T}$ iff $x < 30$, the binding conditions being those of Δ_{ee} and Δ_{dd}. The configuration $\mathbf{1}_4 \otimes \mathbf{1}_2$ has excess $\frac{L}{64\pi^2}g^4(48 - \frac{7}{4}x)$ over the Standard-Model orbit, negative for $x > \frac{192}{7} \approx 27.43$.

At $x = 0$ the radiatively generated masses are, in units of $\frac{L}{64\pi^2} g^4 v^2$, $m^2_{ee} = 120$, $m^2_{ue} = 84$, $m^2_{de} = 144$, $m^2_{uu} = 144$, $m^2_{ud} = 156$ and $m^2_{dd} = 120$, which satisfy (7.8) (v1-12, v1-13); the fifty massive scalars are pseudo-Goldstone bosons of the accidental $O(60)$ symmetry of the tree potential [158, 55]. Minimisation on the unit sphere of $\mathcal{T}$ from random starts finds no configuration below the Standard-Model orbit at $x = 0, 20, 27$, and at $x = 28, 29$ finds the orbit of $\mathbf{1}_4 \otimes \mathbf{1}_2$ as the lowest end point (v3-2; sixteen archived starts per value, numerical); its excess $48 - \frac{7}{4}x$ crosses zero at $x = 192/7$ (v1-14); between $192/7$ and 30 the Standard-Model vacuum is metastable on $\mathcal{T}$. We conjecture that on $\mathcal{T}$ the Standard-Model orbit is the global minimum of the one-loop potential exactly for $x < 192/7$.

With $M_N = yv$ and $M_{W_R} = gv$, $x = \sum_{\text{fam}} (M_N/M_{W_R})^4$; in the couplings of (7.4) and Theorem 7.34, $x = \sum_{\text{fam}} y^4/(4g^4)$. Stated in terms of masses, x does not depend on the normalisation. The content of the bound is structural: the Majorana Yukawa coupling, matter's only handle on the vacuum, pushes towards configurations in which the Majorana mass is shared among several fermions, and the gauge fluctuations hold the Standard-Model orbit only while the heavy neutrinos are not much heavier than W_R.

7.5.5 The full kernel space

The general invariant potential on all of $\mathcal{M}$ has too many couplings to be classified usefully: the numbers of linearly independent Pati–Salam-invariant real polynomials of degrees $1, 2, 3, 4$ are $0, 18, 0, 686$, and with fermion number imposed $0, 12, 0, 402$. Two structures do not depend on these couplings.

Proposition 7.31 (A left–right involution)**.** *Let $O = E_1 E_7 E_8$ restricted to V_*. It commutes with P_* and with the rotations, satisfies $O^\top K^0 O = K^0$, $O^\top K^i O = -K^i$, $O\Gamma O^\top = -\Gamma$ and $OJO^\top = J$, and maps t_i^L to $\pm t_i^R$ and each colour bivector to $\pm$ itself. Hence $\psi(t, \mathbf{x}) \mapsto O\psi(t, -\mathbf{x})$, $\Phi \mapsto O\Phi O^\top$, with the corresponding map of the gauge fields, is a symmetry of the kinetic, Yukawa and gauge terms of (7.4), and of the trace invariants. The image $\Phi_0' = O\Phi_0 O^\top$ is a kernel in $(\bar{\mathbf{10}}, \mathbf{3}, \mathbf{1})$ orthogonal to Φ_0, the mirror vacuum of left–right models, and along the great circle $\hat{u}(\theta) = \cos\theta\, \hat{u}_0 + \sin\theta\, \hat{u}_0'$, $Q = \frac{1}{4} - \frac{1}{8}\sin^2 2\theta$ (v2-10).*

The Standard-Model vacuum and its mirror are exchanged by a symmetry of the action, so an O-invariant potential cannot prefer one of them; the choice between breaking $\mathfrak{su}(2)_R$ and breaking $\mathfrak{su}(2)_L$ is spontaneous, as in models of spontaneous parity violation [140].

Now take the tree potential degenerate on all of $\mathcal{M}$, $V_0 = -\mu^2 N + \alpha N^2$, invariant under the accidental $O(272)$. For a unit vector $\hat{u} \in \mathcal{M}$ put

$$Q(\hat{u}) = \operatorname{Tr}(\hat{u}^\top \hat{u})^2, \qquad T(\hat{u}) = \operatorname{Tr}\big(G^{-1} K(\hat{u})\big)^2, \qquad K(\hat{u})_{ab} = \langle [t_a, \hat{u}], [t_b, \hat{u}] \rangle,$$
$$E_x(\hat{u}) = 8x\, Q(\hat{u}) - \tfrac{3}{16} T(\hat{u}),$$

with the 21 Pati–Salam generators t_a and the module metric G. The orientation-dependent part of the leading logarithm is then $\delta V = \frac{L}{64\pi^2} g^4 v^4 E_x(\Phi/|\Phi|)$ with $v = |\Phi|/\sqrt{2}$, and at the Standard-Model point $E_x(\hat{u}_0) = 2x - 99$.

Theorem 7.32 (One-loop second variation on all of $\mathcal{M}$)**.** *Let $H(x)$ be the second variation of E_x at $\hat{u}_0$ on the 271-dimensional tangent space of the unit sphere.*

(i) $H(x) = H_G + xH_F$ with $H_F = -8(\mathbf{1} - \Pi_\nu)$, where Π_ν is the orthogonal projector onto the kernels that pair ν_R with a Weyl fermion of the block, of rank 32 on the 272-dimensional kernel space and of rank 31 on the 271-dimensional tangent space. H_G preserves each Pati–Salam component of $\mathcal{M}$.

(ii) With $m^2 = \frac{1}{2} \times$ (eigenvalue of $H(x)$) in units of $\frac{L}{64\pi^2} g^4 v^2$, the 271 angular modes have the masses of Table 7.5.

fermion pair	multiplet	$\dim_{\mathbb{R}}$	component	m^2
$\nu_R\nu_R$ (phase), $\nu_R e_R$, $\nu_R u_R$	$(1,1)_0, (1,1)_1, (3,1)_{2/3}$	$1+2+6$	$(\mathbf{10},\mathbf{1},\mathbf{3})$	0 (Goldstone)
$e_R e_R$	$(1,1)_2$	2	$(\mathbf{10},\mathbf{1},\mathbf{3})$	$120-4x$
$d_R e_R$	$(3,1)_{4/3}$	6	$(\mathbf{10},\mathbf{1},\mathbf{3})$	$144-4x$
$u_R u_R$	$(6,1)_{4/3}$	12	$(\mathbf{10},\mathbf{1},\mathbf{3})$	$144-4x$
$u_R d_R$	$(6,1)_{1/3}$	12	$(\mathbf{10},\mathbf{1},\mathbf{3})$	$156-4x$
$d_R d_R$	$(6,1)_{2/3}$	12	$(\mathbf{10},\mathbf{1},\mathbf{3})$	$120-4x$
$u_R e_R \pm d_R \nu_R$	$(3,1)_{1/3}$	$6+6$	$(\mathbf{10},\mathbf{1},\mathbf{3})$, $(\mathbf{6},\mathbf{1},\mathbf{1})_R$	eigenvalues of M_3
$u_R d_R$ (antisymmetric)	$(3,1)_{1/3}$	6	$(\mathbf{6},\mathbf{1},\mathbf{1})_R$	$180-4x$
$u_L e_L - d_L \nu_L$, $u_L d_L$	$(3,1)_{1/3}$	$6+6$	$(\mathbf{6},\mathbf{1},\mathbf{1})_L$	$180-4x$
$\ell_L \ell_L$	$(1,3)_1$	6	$(\bar{\mathbf{10}},\mathbf{3},\mathbf{1})$	$108-4x$
$q_L \ell_L$	$(3,3)_{1/3}$	18	$(\bar{\mathbf{10}},\mathbf{3},\mathbf{1})$	$156-4x$
$q_L q_L$	$(6,3)_{1/3}$	36	$(\bar{\mathbf{10}},\mathbf{3},\mathbf{1})$	$180-4x$
q_L–u_R, q_L–d_R (octet)	$(8,2)_{1/2}$	64	$(\mathbf{15},\mathbf{2},\mathbf{2})$	$174-4x$
ℓ_L–u_R, q_L–e_R	$(3,2)_{7/6}$	24	$(\mathbf{15},\mathbf{2},\mathbf{2})$	$162-4x$
ℓ_L–d_R	$(3,2)_{1/6}$	12	$(\mathbf{15},\mathbf{2},\mathbf{2})$	$114-4x$
q_L–ν_R	$(3,2)_{1/6}$	12	$(\mathbf{15},\mathbf{2},\mathbf{2})$	114
q_L–d_R, ℓ_L–e_R (singlets)	$(1,2)_{1/2}$	$4+4$	$(\mathbf{15},\mathbf{2},\mathbf{2})$, $(\mathbf{1},\mathbf{2},\mathbf{2})$	$138-4x$, $186-4x$
q_L–u_R, ℓ_L–ν_R (singlets)	$(1,2)_{1/2}$	$4+4$	$(\mathbf{15},\mathbf{2},\mathbf{2})$, $(\mathbf{1},\mathbf{2},\mathbf{2})$	eigenvalues of M_2

Table 7.5: The radiatively generated masses of the 271 angular modes of the kernel field at the Standard-Model vacuum, in units of $\frac{L}{64\pi^2}g^4v^2$, with $x=\sum_{\text{fam}}(M_N/M_{W_R})^4$ (Theorem 7.32). Pairs written ff' are of Majorana type, pairs written f–f' of Dirac type; multiplets are given under $SU(3)\times SU(2)_L$ with $|Y|$. The two mixing matrices are $M_3=\left(\begin{smallmatrix}84-2x & 2x\\ 2x & 180-2x\end{smallmatrix}\right)$ and $M_2=\left(\begin{smallmatrix}138-x & \sqrt{3}x\\ \sqrt{3}x & 186-3x\end{smallmatrix}\right)$, with eigenvalues $132-2x\mp2\sqrt{x^2+576}$ and $162-2x\mp2\sqrt{x^2-12x+144}$.

(iii) *$H(x)$ is non-negative with kernel exactly the nine Goldstone directions if and only if $x<27$. At $x=27$ the six modes of $\ell_L\ell_L$, the left-handed triplet, become massless, and for $x>27$ they are tachyonic.*

The second variation is computed from the definitions on the 272-dimensional kernel space, and every entry of Table 7.5 is reproduced at several values of x (v2-4 to v2-8); its restriction to $(\mathbf{10},\mathbf{1},\mathbf{3})$ reproduces the tensor-model spectrum of Theorem 7.30, which ties the two normalisations together (v2-9). This is floating-point linear algebra on exactly specified matrices; (iii) follows from the table, in which the left-handed triplet, $108-4x$, gives the first loss of positivity, at $x=27$ (at small x another line is lower, 84 at $x=0$).

Statement (i) is the mechanism by which matter opposes the Standard-Model orbit, now on all of $\mathcal{M}$. At fixed $|\Phi|$ the fermion loop $2\sum m_f^4$ is largest when the whole Majorana mass sits on ν_R; it decreases quadratically along every direction that moves mass onto pairs not containing ν_R and is stationary along the directions that pair ν_R with another fermion. Every mode that does not involve ν_R therefore has slope -4, and the first of them to become massless is the one with the smallest gauge contribution, the left-handed triplet. The mixing in M_3, between the $(\mathbf{10},\mathbf{1},\mathbf{3})$ mode $u_Re_R+d_R\nu_R$ and the $(\mathbf{6},\mathbf{1},\mathbf{1})_R$ mode $u_Re_R-d_R\nu_R$, is an instance of the mixing announced at the end of Section 7.4: it moves the Δ_{ue} threshold from 42 to $315/11$. The thresholds are the zeros of the lowest lines,

$$x=27\ (\ell_L\ell_L),\qquad x=28.5\ (\ell_L\text{–}d_R),\qquad x=\tfrac{315}{11}\approx28.64\ (M_3),\qquad x=30\ (e_Re_R,\ d_Rd_R),$$

and Figure 7.2 shows them. Along the great circle of Proposition 7.31,

$$E_x(\hat{u}(\theta))=2x-99+(27-x)\sin^2 2\theta,$$

so for $x>27$ the Standard-Model vacuum descends towards the left–right symmetric configuration $\theta=\pi/4$, where ν_R and a left-handed lepton carry equal Majorana masses.

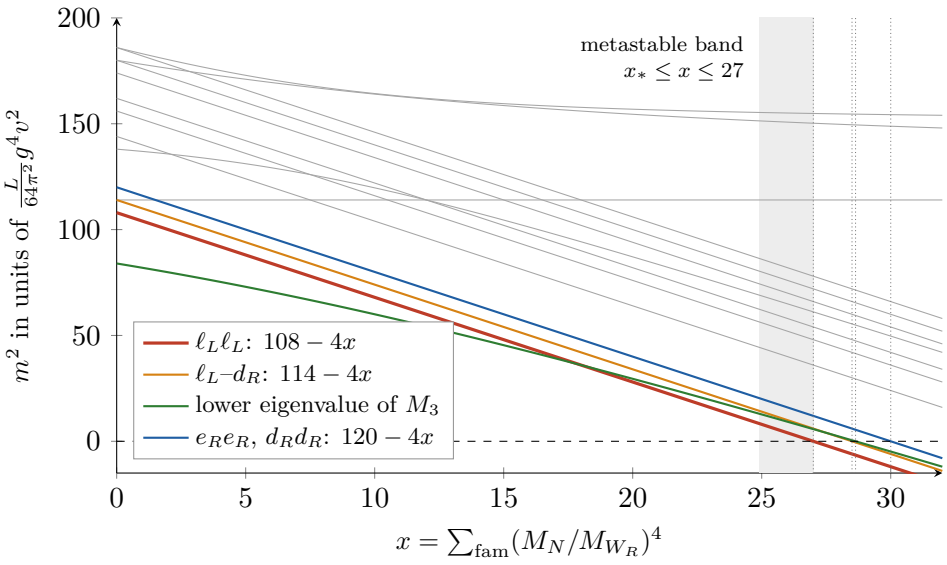


Figure 7.2: Radiatively generated masses of the angular modes of the kernel field at the Standard-Model vacuum (Theorem 7.32, Table 7.5). The coloured lines are the four lowest; they cross zero at $x = 27$, 28.5, 315/11 and 30 (dotted). The Standard-Model orbit is a non-degenerate local minimum for $x < 27$. That it is the global minimum for $x < x_* \approx 24.884$, and metastable in the shaded band, is conjectured from the searches of Proposition 7.33.

Proposition 7.33 (Random-start minimisation)**.** *E_x was minimised on the unit sphere of $\mathcal{M}$ by a quasi-Newton method (L-BFGS) from standard Gaussian starts. For $x = 0, 20, 22$ no run (forty per value) ended below the energy $2x - 99$ of the Standard-Model orbit, and at least* 39 *of the* 40 *ended at that orbit or its mirror; which end points above it are reached depends on the floating-point path (in the regressions of the release candidate and of version 1.0.0, with one BLAS thread per script, one run at $x = 22$ stopped at the higher critical value $E = -51.562388$). At $x = 25$, about one run in five (*22 *or* 23 *of* 100*, according to the floating-point environment) ended below it, at $E = -49.1398704$, in two forms exchanged by the involution of Proposition 7.31. The form with $\mathfrak{su}(2)_L$ unbroken has weight* 0.968 *on* $(\mathbf{10}, \mathbf{1}, \mathbf{3})$ *and* 0.032 *on* $2(\mathbf{6}, \mathbf{1}, \mathbf{1})$*, and its stabiliser among the* 21 *Pati–Salam generators has dimension nine, contains $\mathfrak{su}(2)_L$ and projects onto colour with rank six, whereas any algebra containing $\mathfrak{su}(3)$ projects with rank at least eight: colour is broken. There four right-handed Weyl fermions carry Majorana masses, one of them* 2.04 *times as heavy as the other three. A second optimiser (nonlinear conjugate gradient) reaches the same energy. Followed in x, this minimum crosses the energy of the Standard-Model orbit at $x_* = 24.88405$ (to the stated digits), where it is a local minimum on the sphere.*

The proposition reports numerical experiments; it is not a certificate. They are archived with their seeds, start distribution, optimisers, tolerances and residuals in the third script (v3) of `verification/ch07_mass_vacuum`, checks v3-3 to v3-7: the largest final Riemannian gradient is below 10^{-5}, the Goldstone zero modes are left to the optimiser (the energies of the end points agree to 10^{-9} between runs and optimisers), and x_* is located by Brent's method on warm-started minimisations with tolerance 10^{-10}, giving 24.8840501. The fraction of runs that reaches the competitor depends on the start distribution; with standard Gaussian starts it is 22%. At $x = 24.5$ no run of forty ends below the Standard-Model energy. Only the local statement, Theorem 7.32, is a theorem about the stated approximation; a random search does not establish a global minimum.

We conjecture that, within the model stated below, in the radiative scenario on $\mathcal{M}$ the Standard-Model orbit and its mirror are the global minima exactly for $x < x_*$, and that for $x_* < x < 27$ the Standard-Model vacuum is metastable. The competitor again spreads the

Majorana mass, now over four right-handed fermions instead of one, which is what the fermion loop rewards, at the price of breaking colour, which the gauge loop penalises. In terms of masses the local condition and the conjectured global one read

$$M_N < 27^{1/4} M_{W_R} = 2.280\, M_{W_R} \quad \text{and} \quad M_N < 2.233\, M_{W_R} \quad \text{(one family dominating)},$$
$$M_N < 9^{1/4} M_{W_R} = 1.732\, M_{W_R} \quad \text{and} \quad M_N < 1.697\, M_{W_R} \quad \text{(three degenerate families)}.$$

They hold, like the thresholds $x < 27$ and $x_* \approx 24.884$, only within the stated model: an $O(272)$-symmetric tree potential, degenerate on $\mathcal{M}$ at the matching scale; the trace kinetic normalisation; one loop at leading logarithm; $\kappa = 0$, which fermion number makes automatic; a common gauge coupling for $\mathfrak{su}(4)$ and $\mathfrak{su}(2)_R$; and a single Yukawa coupling for all components of the kernel field (Assumption 7.19). The invariant counts of Section 7.5.5, 18 quadratic and 686 quartic invariants (12 and 402 with fermion number), show that symmetry does not force this small potential: its special coefficients are matching conditions, and their stability under running and finite corrections is not analysed. The masses of Table 7.5 are curvatures of the leading-logarithm effective potential divided by the kinetic norm; pole masses would need the wave-function renormalisation and a gauge-consistent perturbative expansion, which are not attempted. The bounds are therefore statements about this model and approximation, not bounds on heavy-neutrino masses outside it. Within it, the result is that gauge fluctuations make the Standard-Model orientation of the Majorana condensate stable in every direction of the kernel space as long as the heavy neutrinos are not much heavier than W_R; beyond that, the local instability is the condensation of the left-handed triplet, and the configuration that overtakes the Standard-Model orbit globally, at the slightly smaller x_*, breaks colour.

7.6 Matter in the selected vacuum

We return to the vacuum $\Phi = r\Phi_0$, $r > 0$ (for $r < 0$ read $|r|$ in the formulas), with $v = |\Phi|/\sqrt{2} = \sqrt{2}\, r$, and ask whether the matter of the block is compatible with it in the sense stated at the beginning of Section 7.4.

Theorem 7.34 (Compatibility with matter)**.** *(a)* Unbroken algebra. *Among the* 21 *Pati–Salam generators the stabiliser of* Φ_0 *has dimension* 12 *and closes under brackets; its part orthogonal to* $\mathfrak{su}(3) \oplus \mathfrak{su}(2)_L$ *is spanned by* $t_Y = t_3^R + \frac{1}{6}H$*. Among the* 23 *generators it has dimension* 14*; the two further directions* $J + \frac{1}{3}H$ *and* $J - J\Gamma$ *carry the charges* $4B$ *and* $1 - \Gamma$ *and are global.*

(b) Vector bosons. *The matrix* $M_{ab} = \langle [t_a, \Phi_0], [t_b, \Phi_0] \rangle / |\Phi_0|^2$*, taken relative to the module metric, has eigenvalues* 0 *(twelve times),* 4 *(eight times) and* 20 *(once), so the squared vector masses are* 0, g^2v^2 *and* $5g^2v^2$ *in the conventionally normalised coupling* $g = \sqrt{8}\, g_A$ *of Section 7.4, that is* $8g_A^2v^2$ *and* $40g_A^2v^2$ *for the module-normalised fields of* (7.4)*. The eight carry* $Y = \pm\frac{2}{3}$ *(six leptoquarks* X*) and* ± 1 *(the two* $W_R^\pm$*); the one at* $5g^2v^2$ *is neutral, a* Z'*.*

(c) Couplings. *In the module trace* $|t_3^L|^2 = \frac{1}{8}$ *and* $|t_Y|^2 = \frac{5}{24}$*, so*

$$\sin^2\theta_W = \frac{|t_3^L|^2}{|t_3^L|^2 + |t_Y|^2} = \frac{3}{8}$$

at the matching scale, the value of the Pati–Salam and $\mathfrak{so}(10)$ *embeddings [52].*

(d) Dirac operator. *The kinetic kernels are symmetric with* $K^0 = \mathbf{1}$ *and are preserved by the twelve unbroken generators, and the block Hamiltonian* $\mathcal{H}(p)$ *of* (7.6) *satisfies* $\mathcal{H}(p)^2 = |p|^2 + y^2r^2P_\nu$*.*

(e) Chirality. *The massless fermions are* $(3,2)_{1/6} \oplus (1,2)_{-1/2}$ *with* $\Gamma = -1$ *and* $(3,1)_{2/3} \oplus (3,1)_{-1/3} \oplus (1,1)_{-1}$ *with* $\Gamma = +1$*: the unbroken* $\mathfrak{su}(2)_L$ *acts on one Lorentz chirality only.*

(f) Anomalies. *On the block the* Y^3*, mixed gravitational,* $SU(2)^2Y$*,* $SU(3)^2Y$ *and* $SU(3)^3$ *anomalies vanish, with and without* ν_R*, and there are four left-handed* $SU(2)$ *doublets, an even number [169].*

The formula in (c) deserves a line. With one coupling for the whole Pati–Salam algebra, the effective couplings of t_3^L and t_Y are in the ratio of their norms in the module, $g'^2/g^2 = |t_3^L|^2/|t_Y|^2 = \frac{3/24}{5/24} = \frac{3}{5}$, the familiar normalisation of hypercharge in unified models, and $\sin^2\theta_W = g'^2/(g^2+g'^2) = \frac{3}{8}$, at the (unspecified) scale where the single coupling holds; without a matching scale, spectrum, running and thresholds this is a normalisation statement, not a prediction of the low-energy angle. Every requirement of a matter-compatible vacuum is met at this stage except mass generation for the charged fermions, which needs the electroweak stage.

7.6.1 The electroweak stage and the fermion masses

Everything in this subsection assumes, in addition, a vacuum value of the bidoublet kernels; their dynamics and their scale are not derived. In the radiative scenario of Section 7.5.5 the colour-singlet doublets have squared masses $138 - 4x$, $186 - 4x$ and the eigenvalues of M_2, so they sit at the Pati–Salam scale and a light doublet requires a tuning that we do not derive.

Proposition 7.35 (Electroweak stage)**.** *(i) The kernels commuting with* $\mathfrak{su}(3)$ *and with* $Q = T_3^L + Y$ *form a 12-dimensional space: two each in* $(\mathbf{10},\mathbf{1},\mathbf{3})$ *and* $(\bar{\mathbf{10}},\mathbf{3},\mathbf{1})$ *and four each in* $(\mathbf{15},\mathbf{2},\mathbf{2})$ *and* $(\mathbf{1},\mathbf{2},\mathbf{2})$*. In each bidoublet the up-type kernels (no* d,e *entries) and the down-type kernels (no* u,ν *entries) form two-dimensional spaces.*

(ii) On the charge sectors u *(*$Q = \frac{2}{3}$*),* d *(*$-\frac{1}{3}$*),* ν *(*0*) and* e *(*-1*), the* $(\mathbf{1},\mathbf{2},\mathbf{2})$ *kernels give* $m_u = m_\nu^D$ *and* $m_d = m_e$*; the* $(\mathbf{15},\mathbf{2},\mathbf{2})$ *kernels give lepton masses three times the quark masses, with relative sign* -3 *once aligned with a* $(\mathbf{1},\mathbf{2},\mathbf{2})$ *kernel. For* $\Phi = \kappa_1 K_1 + \kappa_{15} K_{15}$ *of each type,*

$$m_{\text{quark}} = |\kappa_1 + \kappa_{15}|, \qquad m_{\text{lepton}} = |\kappa_1 - 3\kappa_{15}|,$$

the Georgi–Jarlskog factor [51] used in fits of fermion masses with $\mathbf{10} \oplus \mathbf{126}$ *[7].*

(iii) With $\Phi = v_R \Phi_0 + \kappa K_{1u}$ *the neutral sector has singular values* $\frac{1}{2}(\sqrt{v_R^2 + 4\kappa^2} \mp v_R)$*, each four times: the seesaw pair [96, 100], with light mass* $\approx \kappa^2/v_R$*.*

(iv) For $\Phi = v_R\Phi_0 + v_u K_{1u} + v_d K_{1d}$ *with* $v_R \neq 0$ *and* $v_u^2 + v_d^2 > 0$ *the stabiliser among the* 21 *generators is nine-dimensional,* $\mathfrak{su}(3) \oplus \mathfrak{u}(1)_Q$*, and* Q *annihilates* Φ *(at* $v_u = v_d = 0$ *it is twelve-dimensional, Theorem 7.42).*

Here K_{1u} and K_{1d} are up- and down-type $(\mathbf{1},\mathbf{2},\mathbf{2})$ kernels of (i), normalised to $|K|^2 = n_K = 32$; each is fixed up to a rotation in its two-dimensional space, which changes none of the statements. Executable check: a8-2, a8-3, a8-4, a8-5 (numerical; Section 15.2).

Example 7.36 (An electroweak spectrum)**.** At $(v_R, v_u, v_d) = (20, 1, \frac{1}{2})$ the squared vector masses, computed from (7.4) as written, that is for $g_A = 1$ and $g^2 = 8$ (divide by 8 for $g = 1$), are 0 (nine times), 79.360 (twice, $W^\pm$), 127.872 (once, Z), 6400 (six times), 6480.64 (twice) and 32032.1 (once). The ratio $m_W^2/m_Z^2 = 0.62062$ is close to $\cos^2\theta_W = \frac{5}{8}$, and at $v_R = 200$ it differs from $\frac{5}{8}$ by -4.4×10^{-5}: $\rho = 1$ at tree level up to corrections of order κ^2/v_R^2. In the same vacuum at $(v_R, v_u, v_d) = (20, 3, 1)$ the fermion mass eigenspaces are a light Majorana neutrino of mass $(\sqrt{436} - 20)/2 \approx 0.440$ and a heavy one of mass $(\sqrt{436} + 20)/2 \approx 20.440$, each of real dimension four, the d quarks and the electron with $m_d = m_e = 1$, and the u quarks with mass 3. The chirality content of the neutral pair is $\langle\Gamma\rangle = \mp\cos 2\vartheta$ with $\tan 2\vartheta = 2v_u/v_R$, which is

their mixing, and the Majorana-type part of the full mass operator is exactly $v_R\Phi_0$: ν_L acquires its Majorana mass only through the mixing.

What is not obtained here is equally definite: the values of v_u and v_d, the potential that would select them together with v_R, and the hierarchy $\kappa \ll v_R$, which in Pati–Salam and left–right models is a tuning of the scalar potential [54].

7.6.2 What the vacuum does not supply

The statements of this subsection are impossibility results within explicitly stated algebras of operators. They say where an input enters, not that no mechanism outside those algebras could replace it.

Proposition 7.37 (Chirality is field content)**.** *Let $\mathcal{A}$ be the algebra of operators on S_3 generated by the 91 internal bivectors and J. Then γ^0 commutes with every element of $\mathcal{A}$ and with the rotations, anticommutes with Γ and the boosts, and maps V_* isometrically onto the orthogonal rank-64 subspace $\gamma^0 V_*$, on which $\mathcal{A}$ acts by the same matrices and Γ has the opposite sign. Hence every eigenspace of an element of $\mathcal{A}$ has a mirror image with the same eigenvalue and the opposite chirality, and no vacuum value of a field with values in $\mathcal{A}$, nor any function of such values, separates V_* from $\gamma^0 V_*$.*

The block is selected by $\mathcal{X} = \Gamma\chi_W$, which contains the spacetime orientation through $\Gamma = J\gamma^0\gamma^1\gamma^2\gamma^3$ and is not in $\mathcal{A}$. The kernel field is defined on V_* and presupposes it, so it does not escape the proposition either. The Standard-Model vacuum of Theorem 7.25 is chiral because the matter on which it acts already is; this is the arena's form of the fact that a vector-like gauge theory does not generate chirality through its vacuum [154].

Proposition 7.38 (Hermitian order parameters keep rank five)**.** *On the spin-up part of the block, the operators commuting with J, Γ and the twelve Standard-Model generators form a twelve-dimensional algebra whose symmetric part is six-dimensional: the commutative algebra of the six multiplet projectors (Q, L, u, d, e, ν), which are pairwise inequivalent. A Hermitian vacuum value in it commutes with T_3^R and B–L, so its stabiliser contains $\mathfrak{su}(3)\oplus\mathfrak{su}(2)_L\oplus\mathfrak{u}(1)_{T_3^R}\oplus\mathfrak{u}(1)_{B-L}$, of rank five; the Standard-Model algebra, of rank four, is not the unbroken algebra of such an order parameter.*

The spin-up part is an eigenspace of Je_4e_5 on the block. Executable check: a8-7 (numerical; Section 15.2), which also finds the six multiplet projectors with multiplicities $12, 4, 6, 6, 2, 2$.

This is the block version of the statement that order parameters of adjoint type do not lower the rank [86]. The kernel field escapes it because a Grassmann bilinear carries charge: $[t_3^R, \Phi_0] = J\Phi_0 \neq 0$.

Proposition 7.39 (Gauge alignment does not see chirality)**.** *Consider rank-64 projectors P commuting with Lorentz and J, with the alignment energy $V(P) = \frac{1}{512}\sum_T \|[T, P]\|^2$ over the 43 compact internal bivectors. For P_*, P_{fam}, the colour-singlet lepton projector P_{lep} and the vector-like $Q_v = P_L + \gamma^0 P_L \gamma^0$ with $P_L = P_*\frac{1}{2}(\mathbf{1} - \Gamma)$:*

	P_*	P_{fam}	P_{lep}	Q_v
$V(P)$	5	9	3	5
stabiliser among the 43	23	9	37	25
axial, $P\gamma^0 P = 0$	yes	—	—	no

Alignment prefers P_{lep} to P_*, and $V(P_*) = V(P_L) + V(P_R)$ with $V(P_L) = V(P_R) = \frac{5}{2}$ while $V(Q_v) = 2V(P_L) = V(P_*)$: the alignment energy does not see chirality.

Vacuum alignment [123, 126] selects among projectors only by the amount of gauge symmetry they keep; it neither selects the Pati–Salam block nor distinguishes a chiral block from a vector-like one of the same rank. Finally, with the single Yukawa coupling of Assumption 7.19, matter opposes the Standard-Model vacuum: on $(\mathbf{10},\mathbf{1},\mathbf{3})$ every F_k of Theorem 7.30 is positive, and on all of $\mathcal{M}$ the Yukawa part of the second variation is $-8x(\mathbf{1}-\Pi_\nu)\leq 0$. The fermion loop lowers the energy along every direction that moves Majorana mass away from ν_R and raises it along none; on its own it would select configurations in which the mass is spread, which break colour. The matter sector is not the agent that selects the Standard-Model orbit. At one loop the gauge fluctuations are, and they succeed for $x<27$ locally (Theorem 7.32), and globally, by conjecture from numerical searches, for $x<x_*$ (Proposition 7.33).

7.7 Goldstone bosons, antiparticles, baryon number and topology

The selected vacuum can be examined from four further angles: which of its zero modes are Goldstone bosons and which are eaten; how the discrete operations that exchange particles and antiparticles act on it; whether the potential protects baryon number; and what topological defects its orbit supports. Each answer depends on counting the two global central generators J and $J\Gamma$ alongside the 21 gauged ones.

7.7.1 Goldstone bosons

Proposition 7.40 (Goldstone directions attached to the symmetries)**.** *J and $J\Gamma$ are antisymmetric, commute with the kinetic kernels and the 21 Pati–Salam generators, span the centre of the 23-dimensional faithful algebra, are anomalous, and carry no gauge fields. If V is invariant under all 23 generators and Φ is stationary, the tangent space of the 23-orbit lies in the kernel of the Hessian; the gauge bosons eat the tangents of the 21-orbit, whose dimension is the rank of the vector mass matrix, and*

$$\#\{\textit{symmetry-generated Goldstone directions not eaten}\}=\dim(\text{23-}\textit{orbit})-\dim(\text{21-}\textit{orbit}).$$

The count is classical, and it counts the Goldstone directions attached to the 23 specified generators, not all massless scalars: a potential with an accidental symmetry has more zero modes, as the degenerate tree potential of Section 7.5.5 has 271 against a gauge orbit of dimension nine. Since J and $J\Gamma$ are anomalous, an uneaten direction loses its Goldstone protection: the anomaly allows a mass, generated by the relevant quantum dynamics through a nonconstant phase-dependent effective potential (in the QCD normalisation $m_a^2 f_a^2=\chi_{\text{top}}$, a dynamical input [325]); the classical count does not compute it.

Theorem 7.41 (No Majoron)**.** *At $r\Phi_0$ the 21-orbit and the 23-orbit both have dimension nine, so the global symmetries add no Goldstone direction beyond those eaten. The reason is*

$$[J,\Phi_0]=[J\Gamma,\Phi_0]=2\,[t_3^R,\Phi_0]:$$

the fermion-number rotation of the $\nu_R\nu_R$ condensate is a gauge rotation, and $J+\frac{1}{3}H$, of charge $4B$, stays unbroken. The nine eaten modes are the $\nu_R\nu_R$ phase (Z'), $\nu_R e_R$ ($W_R^{\pm}$) and $u_R\nu_R$ (six leptoquarks), matching the vector masses of Theorem 7.34.

The would-be Majoron of a spontaneously broken lepton number is eaten by the Z'. At the electroweak stage the situation changes.

Theorem 7.42 (One uneaten direction at the electroweak vacuum)**.** *For $\Phi=v_R\Phi_0+v_uK_{1u}+v_dK_{1d}$:*

vacuum	stab. in 21	stab. in 23	massive vectors	23-orbit	Goldstones
generic, $v_R \neq 0$, $v_u v_d \neq 0$	9	10	12	13	**1**
$v_R \neq 0$, exactly one of v_u, v_d zero	9	11	12	12	0
$v_R \neq 0$, $v_u = v_d = 0$	12	14	9	9	0
$v_R = 0$, $v_u v_d \neq 0$, $\lvert v_u\rvert \neq \lvert v_d\rvert$	16	17	5	6	1
$v_R = 0$, $\lvert v_u\rvert = \lvert v_d\rvert \neq 0$	18	19	3	4	1
$v_R = 0$, one of v_u, v_d zero	16	18	5	5	0
$v_R = v_u = v_d = 0$	21	23	0	0	0

At the generic electroweak vacuum ($v_R \neq 0$, both bidoublet values nonzero) the tangents of J and $J + \frac{1}{3}H$ lie in the gauge orbit and those of $J\Gamma$ and $J - J\Gamma$ do not; $J + \frac{1}{3}H$ annihilates Φ and $J - J\Gamma$, of charge $1 - \Gamma$, does not.

Executable check: d2-24 (exact); the rows with $v_R = 0$, a11-2; the row $v_u = v_d = 0$, a11-3 (numerical; Section 15.2).

The mechanism is simple. $J\Gamma$ acts on every Dirac-type kernel as the complex structure $2J\Gamma$, so it rotates the up-type and the down-type lines by the same phase, while the only Standard-Model generator that moves within these lines, the Z direction, rotates them by opposite phases. When both values are nonzero the $J\Gamma$ direction cannot be absorbed. The broken global charge is $1 - \Gamma$, whose anomalies per Weyl fermion in the all-left convention are

charge	$SU(3)^2$	$SU(2)_L^2$	Y^2	Q^2	gravitational	electroweak vacuum
$1 - \Gamma$	**2**	4	$\frac{4}{3}$	$\frac{16}{3}$	16	broken
$4B = 1 + (B - L)$	0	2	-2	—	0	unbroken

The current of the broken charge has a colour anomaly, so its classical Goldstone direction is a pseudo-Goldstone boson of Peccei–Quinn–Weinberg–Wilczek type [119, 159, 163]: an axion. Since its only order parameter is the bidoublet, its decay constant is of the electroweak scale, and such an axion is excluded experimentally [304, 305, 118]. Within this field content, leaving $U(1)_{1-\Gamma}$ intact is therefore a phenomenological obstruction, and the electroweak-stage potential must break it explicitly. The trace potential and the one-loop gauge and Yukawa potential of Section 7.5 are invariant under every orthogonal conjugation commuting with the Pati–Salam algebra, hence under $\exp \theta(J - J\Gamma)$, and the holomorphic quartic lives on the $\Gamma = +1$ half, where $J - J\Gamma = 0$; none of them lifts the axion.

Proposition 7.43 (Bidoublet mass terms and their CP parity)**.** *On $(\mathbf{1}, \mathbf{2}, \mathbf{2})$ there are three Pati–Salam- and J-invariant symmetric forms, one of them $J\Gamma$-invariant. The CP operator of Theorem 7.45 acts on them with eigenvalues $(+1, +1, -1)$: the invariant form B_{inv} is CP-even, and the two-dimensional breaking space splits into a CP-even B_e and a CP-odd B_o. Along the $J\Gamma$-orbit,*

$$B_e\big(e^{\theta\,\mathrm{ad}\,J\Gamma}\Phi\big) = -v_u v_d \cos 4\theta, \qquad B_o\big(e^{\theta\,\mathrm{ad}\,J\Gamma}\Phi\big) = -v_u v_d \sin 4\theta, \qquad B_{\mathrm{inv}} = \mathit{const},$$

the analogues of Re *and* $\mathrm{Im}\,\mathrm{Tr}\,\tilde{\Phi}^\dagger\Phi$ *in left–right models.*

Executable check: a8-8 (numerical; Section 15.2): the three forms, the CP eigenvalues, and the dependence $A\cos(4\theta + \varphi)$, $A\sin(4\theta + \varphi)$ along the orbit with A proportional to $v_u v_d$ and B_{inv} constant. The amplitude $-v_u v_d$ and $\varphi = \pi$ of the display are a choice of normalisation of B_e, B_o and of the relative phase of K_{1u} and K_{1d}.

Theorem 7.44 (The kinetic norm of the axion direction)**.** *Let $a = J\Phi_0$, $e_u = J\Gamma K_{1u}$, $e_d = J\Gamma K_{1d}$; these are mutually orthogonal with $|a|^2 = 4$ and $|e_u|^2 = |e_d|^2 = n_K = 32$, and $t = [J\Gamma, \Phi] = 2v_R a + 2v_u e_u + 2v_d e_d$. The gauge tangent space splits orthogonally into its part in* $\mathrm{span}\{a, e_u, e_d\}$ *and its part orthogonal to that span; the first is spanned by a and $z = v_u e_u - v_d e_d$*

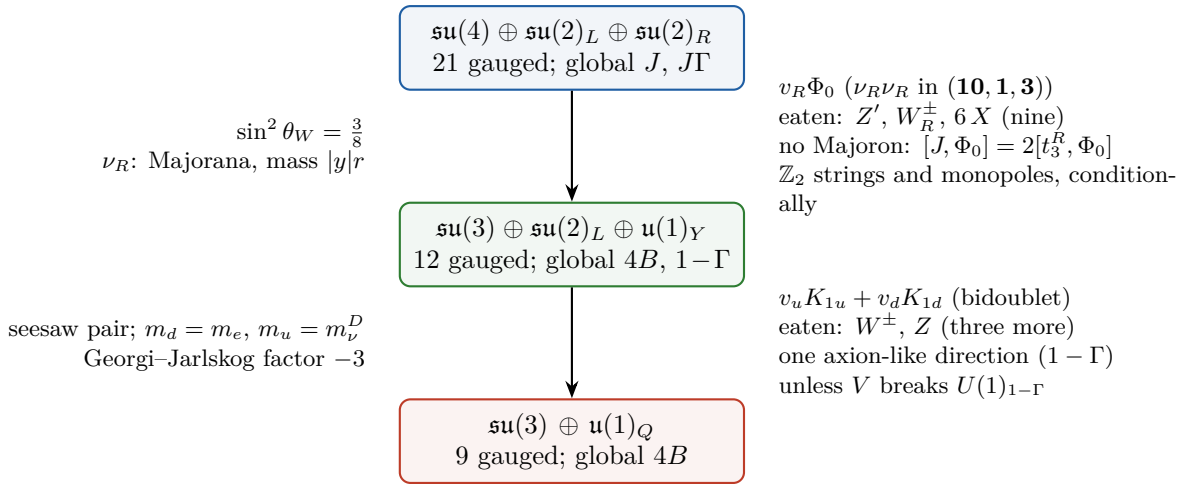


Figure 7.3: The two stages of symmetry breaking in the selected vacuum (Sections 7.6 and 7.7). Each arrow lists the order parameter, the vectors that eat the Goldstone directions, and the uneaten directions attached to the global central charges.

when $v_R \neq 0$, and by z alone when $v_R = 0$, where the gauge tangent space has no Majorana component. For $v_u^2 + v_d^2 > 0$ (the continuous extension to $v_u = v_d = 0$ is 0)

$$|P_\perp t|^2 = |t|^2 - \frac{\langle t,a\rangle^2}{|a|^2} - \frac{\langle t,z\rangle^2}{|z|^2} = \frac{16\, n_K\, v_u^2 v_d^2}{v_u^2 + v_d^2}.$$

Proof. J and $J\Gamma$ are orthogonal, so $|a|^2 = |\Phi_0|^2 = 4$ and $|e_{u,d}|^2 = |K_{1u,1d}|^2 = n_K$. The three vectors are orthogonal because a lies in $(\mathbf{10},\mathbf{1},\mathbf{3})$ and e_u, e_d in $(\mathbf{1},\mathbf{2},\mathbf{2})$, and the up- and down-type kernels have disjoint charge sectors. Φ_0 anticommutes with J and is supported on $\Gamma = +1$, and K_{1u}, K_{1d} anticommute with $J\Gamma$, which gives the expression for t. These facts and the orthogonal splitting of the gauge tangent space, with the part in the span equal to $\mathrm{span}\{a,z\}$ for $v_R \neq 0$, are recomputed in a8-6 (numerical; Section 15.2). Given them, t lies in the span, so its projection onto the gauge tangent space is its projection onto $\mathrm{span}\{a,z\}$, with $a \perp z$; at $v_R = 0$ the span is that of z alone, and since $\langle t,a\rangle = 8v_R = 0$ the same expression results; and $\langle t,a\rangle = 8v_R$, $\langle t,z\rangle = 2n_K(v_u^2 - v_d^2)$, $|z|^2 = n_K(v_u^2 + v_d^2)$, $|t|^2 = 16v_R^2 + 4n_K(v_u^2 + v_d^2)$, so

$$|P_\perp t|^2 = 4n_K(v_u^2 + v_d^2) - \frac{4n_K(v_u^2 - v_d^2)^2}{v_u^2 + v_d^2} = \frac{16\, n_K\, v_u^2 v_d^2}{v_u^2 + v_d^2}. \qquad \square$$

The heavy-condensate component of t is a gauge direction, which is why v_R drops out. If $\mu_e^2 B_e$ is the only term breaking $U(1)_{1-\Gamma}$, with canonical amplitudes $h_{u,d} = \sqrt{32}\,\kappa_{u,d}$ and $m_{12}^2 = \mu_e^2/32$, then to leading order in the explicit breaking

$$m_A^2 = \frac{\partial_\theta^2 V}{|P_\perp[J\Gamma,\Phi]|^2} = \frac{m_{12}^2}{\sin\beta\cos\beta}, \qquad \tan\beta = \frac{v_u}{v_d},$$

the pseudoscalar mass of the two-Higgs-doublet model with a soft m_{12}^2 term [17]. Phase-dependent quartics add to it, and whether this curvature is a mass eigenvalue once mixing with the other neutral scalars is included requires the complete Hessian of a specified potential. With CP imposed ($\mu_o^2 = 0$) every extremum of the breaking mass terms has $\sin 4\theta = 0$, so no quadratic term can break CP spontaneously; that needs terms of charge ± 8 under $J\Gamma$, and the potential has $31 + 31$ of them among its quartics (a6-5), the analogues of the phase-dependent quartics of left–right models [33].

7.7.2 Particles and antiparticles

At one constituent charge conjugation is unique up to a $U(1)_J$ rotation, lies in the surplus plane and is the Majorana real structure (Theorem 2.25), and the Fock space of one momentum has a unique charge-zero ground state of energy $-2E$ (Theorem 2.31). That value is the zero-point energy of two particle and two antiparticle modes; summed over momenta it is the fermion term $-2\sum_{\text{Weyl}}$ of the one-loop potential (7.5), the very term through which matter opposes the Standard-Model vacuum. At three constituents the discrete operations are classified as follows. A map $\psi \mapsto S\psi$ with S orthogonal on V_* exchanges particles and antiparticles when $SJ = -JS$; it keeps or reverses chirality according to $S\Gamma S^{-1} = \pm\Gamma$; it goes with $\mathbf{x} \mapsto -\mathbf{x}$ when $S^{\top} K^i S = -K^i$.

Theorem 7.45 (Four classes of discrete operators)**.** *The blades of* $\mathrm{Cl}(9,9)$ *that commute with the three involutions of* P_* *and with the rotations number* 8192*, and they fall into exactly four classes of* 2048*: internal (*J*-linear,* Γ *kept, boosts kept),* P *(*J*-linear,* Γ *reversed, boosts reversed),* C *(*J*-antilinear,* Γ *reversed, boosts kept) and* CP *(*J*-antilinear,* Γ *kept, boosts reversed). The two missing types are forbidden for every operator, since an antilinear map commuting with the Lorentz action and keeping* Γ *would identify* $(\frac{1}{2}, 0)$ *with its conjugate* $(0, \frac{1}{2})$*. Representatives are*

$$C = E_2E_7E_{10}E_{11}E_{12}, \qquad P = O = E_1E_7E_8, \qquad CP = E_1E_2E_7E_8E_{10}E_{11}E_{12}E_{14},$$

and all three normalise the Pati–Salam algebra and preserve the kinetic, gauge and Yukawa terms.

The factors of C are instructive: E_2 is the surplus generator of the first constituent, where the one-constituent charge conjugation lives; E_7 exchanges left and right; and $E_{10}E_{11}E_{12}$, an odd number of colour generators, induces the outer automorphism of $\mathfrak{su}(4)$ that exchanges $\mathbf{4}$ and $\bar{\mathbf{4}}$. $P = O$ is parity composed with an $SU(2)$ Weyl reflection, and $CP = \pm(C\,O)\,E_7E_{14}$ with E_7E_{14} a gauge element.

Theorem 7.46 (CP fixes the order parameter; C and P move it)**.** $CP\,\Phi_0\,CP^{\top} = \Phi_0$. $C\Phi_0C^{\top}$ *is the* $\nu_L\nu_L$ *kernel, which commutes with* $\mathfrak{su}(3)$*,* $\mathfrak{su}(2)_R$ *and* Q *but not with* $\mathfrak{su}(2)_L$*, and* $P\Phi_0P^{\top}$ *is the* e_Le_L *kernel. Both images lie on the mirror orbit in* $(\bar{\mathbf{10}}, \mathbf{3}, \mathbf{1})$*: the degeneracy of Proposition 7.31 is realised by two distinct discrete symmetries.*

Executable check: a6-2 (exact symbolic or integer arithmetic on the 512-dimensional module, with Φ_0 rebuilt exactly; Section 15.2).

Invariance of the order parameter is not invariance of the theory. The kinetic, gauge and Yukawa terms and the trace invariants are CP-invariant; the general potential on $\mathcal{M}$ is invariant only for couplings satisfying reality conditions; and on $(\mathbf{10}, \mathbf{1}, \mathbf{3})$ the holomorphic quartic needs a generalised CP, composed with a baryon-number rotation $e^{\theta(J+H/3)}$, which is a classical symmetry of the renormalisable theory for every κ but is anomalous, since $4B$ has $SU(2)_L^2$ anomaly 2. Whether a CP survives in the quantum theory is not settled.

Which fermions are their own antiparticles follows from a pairing rule. If M is a mass kernel and t an unbroken generator with $[t, M] = 0$, then a Majorana-type M satisfies $M\hat{t} = -\hat{t}M$, so within one chirality it pairs a representation of the unbroken group with its conjugate, while a Dirac-type M pairs equal representations of opposite chirality. A single irreducible multiplet can therefore carry a Majorana mass only if its representation of the unbroken group is real: a nonzero charge under an unbroken $U(1)$ excludes this, while a nontrivial real representation of a non-abelian group does not. In the block, every multiplet other than ν_R carries a nonzero hypercharge at the Pati–Salam stage, and every multiplet other than ν_L and ν_R is a colour triplet or electrically charged at the electroweak stage. So ν_R is the only Majorana fermion at the first stage, the light ν and heavy N of Example 7.36 are the Majorana fermions at the second, and the massive ν_R is mapped to itself by CP with eigenvalues $\pm i$, since $CP^2 = -\mathbf{1}$ on the block.

7.7.3 Baryon number in the potential

For a Pati–Salam invariant $B{-}L = 0$, so $4B = F$ and $\Delta B = F/4$ in terms of the fermion number F.

Theorem 7.47 (Invariant polynomials by fermion number)**.** *The Pati–Salam-invariant polynomials on $\mathcal{M}$ of degrees two and four, graded by F, number*

degree	$F = 0$	$F = \pm 4$ ($\lvert\Delta B\rvert = 1$)	$F = \pm 8$ ($\lvert\Delta B\rvert = 2$)	total
2	12	$3 + 3$, all on $2(\mathbf{6}, \mathbf{1}, \mathbf{1})$	—	18
4	402	$127 + 127$	$15 + 15$, including $\mathcal{D}$	686

With $U(1)_F$ imposed the couplings number $12 + 402$; with $U(1)_{1-\Gamma}$, $8 + 228$; with both F and $J\Gamma$-charge, $6 + 162$.

Executable check: z4-3, a6-4, a6-5 (exact Molien–Weyl integrals; Section 15.2), with the charges of the components under J and $J\Gamma$ read from the realisation in a8-9 (numerical). $U(1)_{1-\Gamma}$ is generated by $J - J\Gamma$, so its invariants are those with equal charges under J and $J\Gamma$.

Pati–Salam gauge invariance alone does not forbid baryon-number violation. The six $F = \pm 4$ quadratic invariants are holomorphic bilinears in the complex sextets of $2(\mathbf{6}, \mathbf{1}, \mathbf{1})$; under $\mathfrak{su}(3) \oplus \mathfrak{u}(1)_{B-L}$ each sextet is $\bar{\mathbf{3}}_{2/3} \oplus \mathbf{3}_{-2/3}$, diquark components coupling to quark pairs and leptoquark components coupling to quark–lepton pairs, and $\epsilon^{abcd}\Delta_{ab}\Delta'_{cd}$ pairs one with the other. Through the Yukawa coupling such a mass mixing generates tree-level $\Delta B = \Delta L = 1$ four-fermion operators, of the type that mediates proton decay. Two protections are available and they are not equivalent: continuous $U(1)_F$ forbids every $F \neq 0$ term and conserves baryon number perturbatively (it is violated by $SU(2)$ instantons [151]); the $\mathbb{Z}_4$ generated on $\mathcal{M}$ by conjugation with $e^{\pi J/4}$ forbids the $\lvert\Delta B\rvert = 1$ terms but allows the $\lvert\Delta B\rvert = 2$ ones that mediate neutron–antineutron oscillation. $U(1)_F$, which the radiative scenario of Section 7.5 has, is sufficient, not necessary.

7.7.4 Topology of the vacuum orbit

Theorem 7.48 ($\mathbb{Z}_2$ strings)**.** *On the block $H = 3J(B{-}L)$, so $e^{\theta H}$ acts on quarks as $\cos\theta + J\sin\theta$ and on leptons as $\cos 3\theta - J\sin 3\theta$. The colour element $g_* = e^{\pi H/3}$ fixes Φ_0, acts as -1 on the ν_R line and on every lepton, and the path $\theta \mapsto e^{\theta H}\Phi_0 e^{-\theta H} = \cos 6\theta\,\Phi_0 - \sin 6\theta\,J\Phi_0$, $\theta \in [0, \pi/3]$, is a closed loop on which the condensate phase winds once. Every generator of the stabiliser annihilates the ν_R line, so the identity component of the stabiliser acts trivially there and g_* is not in it. Lifting to $SU(4) \times SU(2)_L \times SU(2)_R$, the stabiliser has exactly two components, so the gauge orbit of the vacuum has $\pi_1 = \mathbb{Z}_2$, and $\pi_2 \supseteq \mathbb{Z}$ from $U(1)_Y$.*

If the gauge orbit is the complete set of minima of the chosen potential, as it is on $(\mathbf{10}, \mathbf{1}, \mathbf{3})$ in the region of Theorem 7.25, the vacuum supports topologically stable $\mathbb{Z}_2$ strings around which ν_R changes sign, the Pati–Salam form of the $\mathbb{Z}_2$ strings of $\mathfrak{so}(10)$ broken by a **126** [77], and Pati–Salam monopoles. If the minimum set is larger, as in the degenerate limit with accidental $O(272)$ symmetry, the relevant fundamental group is that of the larger manifold, and a string may unwind through flat directions. The Standard-Model orbit and its mirror lie in different Pati–Salam components, so no gauge transformation connects them; whether they are disconnected components of the full minimum set, which would give domain walls [174], depends on paths through configurations outside both orbits. Along the great circle of Proposition 7.31 the radiative energy has a barrier between them for $x < 27$, but other paths are not controlled.

Chapter 8

Gauge Geometry and Photon Identification

The question of this chapter is which charge is the photon, and what a physical photon still needs. It treats the projector as a field and the connection it induces, the compact families at the third rung, alignment by a gauged subalgebra, the identification of the electric charge B_{em}, and the sources and masslessness of the photon. The charge identification is established; the normal-plane charge Q_A is anomalous with the matter of P_* and is not the photon. The Maxwell term and the selecting potential are adopted, not derived, and electroweak breaking is missing; the adopted potential opposes it.

Several letters carry more than one meaning in this chapter, each fixed by its context: Ω is the interior volume and, in the curved setting, the connection $\Omega_{\mathrm{Lorentz}} + A^a T_a$; Q is the second-rung charge $\frac{1}{2}ab$, the electric charge, and with indices Q_A, Q_c, Q_v; H is the second-rung vacuum blade, H_C a colour generator and $H(p)$ a Hamiltonian; $\mathcal{C}$ is the Lorentz commutant and $\mathcal{C}_\mu$ a current, C the colour group and C an element of the Klein group; D is the duality map and the covariant derivative; and T denotes the tangents T_i, T_{aw} and the temporal trivector. Two invariant forms on the compact algebra are used, $-\operatorname{Tr}(XY)/128$ in Section 8.3 and $-\operatorname{Tr}(XY)/512$ in Section 8.4; the conversion between them is given in the conventions of Section 8.3.

8.1 The projector as a field

Up to this point a projector has been a choice made once and for all. A choice of this kind becomes a field as soon as it is allowed to vary from point to point, and the selection problem then changes character: instead of asking which projector is right we ask which configurations an action prefers. Before any action can be written, three things must be known: the space in which the projector moves, the connection it induces on its own image, and the sign of the natural metric on its variations. All three can be computed exactly, and each of them carries a surprise. A field valued in a homogeneous space of the symmetry group, and its gauging, is the setting of the nonlinear sigma models of Coleman, Wess and Zumino and of Callan, Coleman, Wess and Zumino [306, 307]; what follows computes that space and its metric for the projectors of the arena.

8.1.1 Configuration space

Let $\mathcal{P}_r$ be the set of orthogonal projectors P on $S_3 = \mathbb{R}^{512}$ of rank r that commute with the interior Lorentz action Σ. Differentiating $P^2 = P$ gives $dP\,P + P\,dP = dP$, hence

$$P\,dP\,P = 0, \qquad (\mathbf{1} - P)\,dP\,(\mathbf{1} - P) = 0 :$$

a variation of a projector is purely off-diagonal with respect to the splitting $\operatorname{im} P \oplus \ker P$. The tangent space is therefore

$$T_P\mathcal{P}_r = \{S :\ S^\top = S,\ PSP = 0,\ (\mathbf{1}-P)S(\mathbf{1}-P) = 0,\ [S,\Sigma] = 0\},$$

the tangent space of the real Grassmannian cut down by the Lorentz condition.

Proposition 8.1 (Configuration space)**.** *The Lorentz commutant $\mathcal{C} \subset \mathrm{End}_{\mathbb{R}}(S_3)$ is spanned by the* 32768 *blades that commute with the six interior bivectors. Its centre is $\{\mathbf{1}, \Omega\} \cong \mathbb{C}$ with $\Omega = \gamma^0\gamma^1\gamma^2\gamma^3$, $\Omega^2 = -\mathbf{1}$, and since a semisimple real algebra of dimension $2 \cdot 128^2$ with centre $\mathbb{C}$ is $M_{128}(\mathbb{C})$, we have $\mathcal{C} \cong M_{128}(\mathbb{C})$. Consequently*

$$\mathcal{P}_r \cong \mathrm{Gr}_{\mathbb{C}}(\tfrac{r}{4}, 128), \qquad \dim_{\mathbb{R}} \mathcal{P}_r = 2\,\frac{r}{4}\Big(128 - \frac{r}{4}\Big),$$

which is 3584 *at $r = 64$ and* 6144 *at $r = 128$, against* $\dim \mathrm{Gr}_{\mathbb{R}}(64, 512) = 28672$ *for the unconstrained real Grassmannian. $\mathcal{P}_r$ is non-empty only for ranks r divisible by four.*

Proof. A blade commutes with $e_a e_b$ exactly when it contains both or neither of e_a, e_b, so the blades commuting with the six interior bivectors are those containing all or none of the four interior generators: $2 \cdot 2^{14} = 32768$. They are generated by the fourteen external generators and Ω, and the central ones are $\mathbf{1}$ and Ω; a central element has only central blades in its support, since conjugation by a blade multiplies each blade by ± 1. $\Omega^2 = -\mathbf{1}$ because Ω has one positive and three negative generators. Every e_A satisfies $e_A^\top = \pm e_A$, so $\mathcal{C}$ is closed under transposition, and $\mathrm{Tr}(x^\top x) > 0$ for $x \neq 0$ excludes nilpotent ideals: $\mathcal{C}$ is semisimple, its centre is a field, so it is simple, and a simple $\mathbb{C}$-algebra of complex dimension 128^2 is $M_{128}(\mathbb{C})$. As a module, $\mathbb{R}^{512}$ is two copies of $\mathbb{C}^{128}$, so a projector of real rank r in $\mathcal{C}$ has complex rank $r/4$. Transposition is a positive involution acting on the centre as conjugation ($\Omega^\top = -\Omega$), hence conjugate to the Hermitian adjoint, so orthogonal projectors correspond to Hermitian ones and $\mathcal{P}_r \cong \mathrm{Gr}_{\mathbb{C}}(\frac{r}{4}, 128)$. □

Executable check: a7-1 (exact symbolic or integer arithmetic; Section 15.2).

The complex structure that appears here is the interior volume Ω, not the internal one ω_I of Theorem 2.17. That is not a contradiction. The commutant of the Lorentz action alone contains $e_2^{(1)}$, which commutes with the six interior bivectors but anticommutes with the fourteen-index volume ω_I, so ω_I is not central in $\mathcal{C}$; it becomes central only in the smaller algebra that also commutes with the internal bivectors, which is the algebra of Theorem 2.17. Each structure is central in its own ambient algebra, which is the pattern described after Proposition 1.7.

Remark 8.2 (No blade is tangent)**.** A count of tangent directions by blades fails, for a reason worth recording. Let $P = \prod_i \frac{1}{2}(\mathbf{1} + X_i)$ be one of the projectors of Section 6.1, built from k commuting involutions, and let B be a blade anticommuting with at least one X_i. Then $PBP = 0$, but $(\mathbf{1}-P)B(\mathbf{1}-P) = (\mathbf{1}-P-P_\varepsilon)B$ with $P_\varepsilon = \prod_i \frac{1}{2}(\mathbf{1}+\varepsilon_i X_i)$, which vanishes only when $P + P_\varepsilon = \mathbf{1}$, that is only for $k = 1$. There is also a rank argument: a blade is a signed permutation matrix, invertible of rank 512, while a tangent vector has the block form $\left(\begin{smallmatrix} 0 & A^\top \\ A & 0 \end{smallmatrix}\right)$ and rank at most $2r$. For $k \geq 2$ no single blade is tangent, the tangent space is not spanned by blades, and a blade count overestimates it, by a factor four at $r = 64$ and two at $r = 128$. The dimensions of Proposition 8.1 are obtained by imposing both conditions sector by sector.

8.1.2 The sign of a variation

The trace form on an orbit splits by the Cartan involution (Theorem 1.13): compact directions carry positive and boost directions negative kinetic weight.

The group that the geometry supplies is generated by the internal bivectors, and it moves a projector along an orbit inside $\mathcal{P}_r$. Whether a bivector moves P to another orthogonal projector, and with what sign the natural trace form measures the motion, is decided by the signature rule of Theorem 1.3.

Example 8.3 (The paired-mode projector at one constituent)**.** Take the rank-four projector Π of Example 1.6. Its stabiliser in $\mathfrak{so}(3,3)$ has dimension nine, so its orbit has dimension six. On the bivectors

$$B_I = \tfrac{1}{2}(e_1e_2,\ e_1e_3,\ e_2e_3,\ e_1e_5,\ e_1e_6,\ e_2e_6), \qquad T_I = [B_I, \Pi],$$

an exact 8×8 computation gives $\mathrm{Tr}(T_I T_J) = \mathrm{diag}(1,1,1,-1,-1,-1)$. The first three B_I are time–time rotations, the last three are boosts, exactly as Theorem 1.13 predicts. Only three of the fifteen basis bivectors commute with Π individually, while the stabiliser has dimension nine, so six of its directions are combinations of basis bivectors and no single one of them. Reading a stabiliser off basis elements overestimates the orbit.

Positive norm and tangency are nevertheless different conditions. For unit T_{sym} and T_{anti} of the two kinds, $\mathrm{Tr}((T_{\mathrm{sym}} + \frac{1}{2}T_{\mathrm{anti}})^2) = 1 - \frac{1}{4} > 0$, yet the direction is not symmetric and leaves the orthogonal projectors. What is exact is the orthogonal splitting, not an equivalence between the sign of the norm and tangency. On the manifold of orthogonal projectors itself every variation is symmetric and the trace form is positive definite, so the indefiniteness of the example is an artefact of moving Π along the full $\mathrm{Spin}(3,3)$ orbit. This is a statement about the target of a projector field and not about a field theory over a base: in a base of split signature the kinetic contraction $g^{\mu\nu}\,\mathrm{Tr}(D_\mu P\, D_\nu P)$ is indefinite in the spacetime index whatever the target metric is.

At three constituents the same reasoning counts the physical motions of P_{fam}. Of the 91 internal bivectors 43 are compact, forming the maximal compact $\mathfrak{so}(8) \oplus \mathfrak{so}(6)$, and the compact part of the stabiliser of P_{fam} has dimension nine (Theorem 6.5), so the compact orbit through P_{fam} has dimension

$$43 - 9 = 34,$$

which is also the rank of $B \mapsto [B, P_{\mathrm{fam}}]$ on the compact bivectors. Only seven of the 43 compact basis bivectors commute with P_{fam} individually, which repeats the warning of Example 8.3. Of the 3584 configuration directions at rank 64, thirty-four lie along the only group the geometry supplies. Whether these thirty-four are gauge redundancies or physical modes is a question about an action, which the algebra does not supply.

8.1.3 The induced connection over-generates

A varying projector carries its own connection. If the gauge field is not postulated but induced, $\nabla = P\,d$ on the bundle of images, then its curvature $F_{\mu\nu} = P[\partial_\mu P, \partial_\nu P]P$ is a composite, and the algebra in which it takes values is computed rather than chosen. To leading order in the Ambrose–Singer theorem that algebra is

$$\mathfrak{h}(P) = \mathrm{span}\{P[S, S']P :\ S, S' \in T_P\mathcal{P}_r\}.$$

Theorem 8.4 (Over-generation)**.** *Let $P(x)$ range over all of $\mathcal{P}_{64}$. The curvature values $P[\partial_\mu P, \partial_\nu P]P$ are antisymmetric, hence compact, and lie in the block, and their span on* $\mathrm{im}\, P_{\mathrm{fam}}$ *has dimension*

$$\dim \mathfrak{h}(P_{\mathrm{fam}}) = 256 = \dim \mathfrak{u}(16),$$

the largest compact algebra the block admits at all. It exceeds the six dimensions of $\mathfrak{k}_{\mathrm{eff}}(P_{\mathrm{fam}}) = \mathfrak{u}(1)^3 \oplus \mathfrak{su}(2)$ by a factor of about 43*.*

Proof. Write $P_{\mathrm{fam}} = \prod_i \frac{1}{2}(\mathbf{1} + X_i)$ with the three involutions of Section 6.1. A tangent vector at P_{fam} is $S = Y + Y^\top$ with $Y \in P_{\mathrm{fam}}\mathcal{C}(\mathbf{1} - P_{\mathrm{fam}})$, so $T_{P_{\mathrm{fam}}}\mathcal{P}_{64}$ is spanned by the $S_b = P_{\mathrm{fam}} b(\mathbf{1} - P_{\mathrm{fam}}) + (\mathbf{1} - P_{\mathrm{fam}})b^\top P_{\mathrm{fam}}$ for blades $b \in \mathcal{C}$, and $\mathfrak{h}(P_{\mathrm{fam}})$ by the $P_{\mathrm{fam}}[S_{b_1}, S_{b_2}]P_{\mathrm{fam}} = P_{\mathrm{fam}} b_1(\mathbf{1} - P_{\mathrm{fam}})b_2^\top P_{\mathrm{fam}} - P_{\mathrm{fam}} b_2(\mathbf{1} - P_{\mathrm{fam}})b_1^\top P_{\mathrm{fam}}$. These are antisymmetric, lie in the block and in $\mathcal{C}$, so $\mathfrak{h}(P_{\mathrm{fam}})$ lies in the antisymmetric part of $P_{\mathrm{fam}}\mathcal{C}P_{\mathrm{fam}}$. A blade of $\mathcal{C}$ that anticommutes with some X_i has $P_{\mathrm{fam}} b P_{\mathrm{fam}} = 0$ (Remark 8.2); a blade commuting with all three has $P_{\mathrm{fam}} b P_{\mathrm{fam}} = b P_{\mathrm{fam}} \neq$

0, and b and bX_i give the same element. The 4096 blades of $\mathcal{C}$ commuting with X_1, X_2, X_3 fall into 512 classes modulo $\langle X_1, X_2, X_3\rangle$; distinct classes are trace-orthogonal, since a non-scalar blade is traceless, so they form a basis of $P_{\text{fam}}\mathcal{C}P_{\text{fam}} \cong M_{16}(\mathbb{C})$; the sign in $b^\top = \pm b$ is constant on a class, and 256 classes are antisymmetric. Hence $\dim \mathfrak{h}(P_{\text{fam}}) \le 256$. Conversely, let c be an antisymmetric class, $b_1 = E_2E_7$, which lies in $\mathcal{C}$ and anticommutes with X_1, and $b_2 = b_1c$ up to sign, which then anticommutes with X_1 as well. Then $P_{\text{fam}}b_iP_{\text{fam}} = 0$, so $P_{\text{fam}}[S_{b_1}, S_{b_2}]P_{\text{fam}} = P_{\text{fam}}(b_1b_2^\top - b_2b_1^\top)P_{\text{fam}}$; this is antisymmetric and proportional to cP_{fam}, and the proportionality factor is ± 2 because c is antisymmetric. So every antisymmetric class is reached and $\dim \mathfrak{h}(P_{\text{fam}}) = 256$. The class count is a finite computation on blade parities; check a7-2 performs it and also computes the 256 curvature values from the definition, certifying their span by an exact modular rank of their integer Gram matrix. □

Executable check: a7-2 (exact symbolic or integer arithmetic; Section 15.2).

The induced connection is not flat, every curvature value is compact, and every value lies in the block. The route fails by over-generating and by nothing else. In particular chirality is not among the casualties. The span in the theorem is taken over all projector fields. One field configuration, for instance a solution of a variational model, has curvature values in a subspace that can be much smaller. So the theorem shows that the geometry alone does not confine the induced curvature to $\mathfrak{k}_{\text{eff}}(P_{\text{fam}})$; it does not show that the curvature of every solution leaves $\mathfrak{k}_{\text{eff}}(P_{\text{fam}})$.

Proposition 8.5 (Chirality must be imposed, and costs nothing)**.** *Every projector of the family commutes with* Γ*. If a field* $P(x)$ *is required to satisfy* $[P(x), \Gamma] = 0$ *everywhere, then* $[\partial_\mu P, \Gamma] = 0$ *and*

$$[P[\partial_\mu P, \partial_\nu P]P,\ \Gamma] = 0,$$

so the induced curvature preserves chirality and the holonomy lies in $\mathfrak{u}(8) \oplus \mathfrak{u}(8)$*. That algebra is still far larger than* $\mathfrak{u}(1)^3 \oplus \mathfrak{su}(2)$*.*

The tangent vectors commuting with Γ are those built from even blades, and the argument of the proof of Theorem 8.4, restricted to the 128 even antisymmetric classes, shows that the span is all of $\mathfrak{u}(8) \oplus \mathfrak{u}(8)$. Executable check: a7-2 (exact symbolic or integer arithmetic; Section 15.2).

The comparison of Theorem 8.4 is between what the construction permits and what is wanted, not an impossibility theorem about any particular field. A constant field has zero induced curvature, and a field compatible with Γ has holonomy inside $\mathfrak{u}(8) \oplus \mathfrak{u}(8)$. What the comparison does establish is the direction of the problem. A parent action must *cut* the freedom of the projector, by a constraint that shrinks $T_P\mathcal{P}$ before the curvature is formed or by a potential that freezes all but the intended holonomy directions, rather than build a gauge field up from nothing. This reverses the usual motivation for composite gauge fields. Figure 8.1 collects the numbers.

8.1.4 One explicit variational model

A list of fields does not determine field equations. What can be done without a derived action is to adopt one explicitly, derive its equation, and record what it can and cannot do. On a four-dimensional Lorentzian base, with $D_\mu P = \partial_\mu P + [A_\mu, P]$ and with f and V as inputs, take

$$S_P = \int \sqrt{-g}\left(\tfrac{f^2}{2}\operatorname{Tr}(D_\mu P\, D^\mu P) - V(P)\right). \tag{8.1}$$

Varying along the Grassmannian, $\delta P = [\epsilon, P]$ with ϵ antisymmetric, gives

$$[P,\ f^2 D_\mu D^\mu P + V'(P)] = 0, \tag{8.2}$$

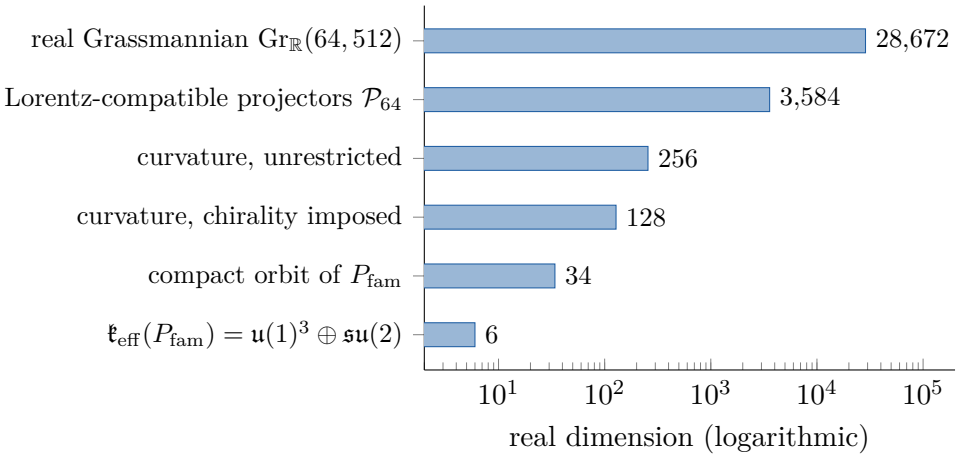


Figure 8.1: How much freedom a projector field has at rank 64, and how little of it is wanted. The configuration space of Lorentz-compatible projectors (Proposition 8.1) is far smaller than the real Grassmannian but still enormous; the group the geometry supplies moves P_{fam} along only 34 directions; and the curvature algebra the induced connection can reach (Theorem 8.4 and Proposition 8.5) exceeds the faithful gauge algebra of the block by more than an order of magnitude.

and for $V = 0$ this is the conservation law $D_\mu \mathcal{C}^\mu = 0$ of the matrix current $\mathcal{C}_\mu = [P, D_\mu P]$. The variation has to be projected onto the admissible tangent space whenever P is restricted further, for instance to commute with Γ.

Three consequences are immediate. First, a potential that depends only on the spectrum of a projector of fixed rank is constant, since $\operatorname{Tr} f(P) = r f(1) + (d-r) f(0)$; such a potential selects nothing, and for the pure kinetic term every constant admissible projector is a critical point. Selection needs couplings that distinguish orbits, and there is more than one orbit to distinguish: at rank 64 the orbits through P_{fam} and P_* have dimensions 34 and 20 (Proposition 6.15). Second, whether varying A_μ gives a Yang–Mills equation depends on how the action depends on A, not on A appearing in a list of fields; a composite A contributes through the chain rule to the equations of its constituents. Third, the pointwise stabiliser condition $[A_\mu, P] = 0$ is not the condition $D_\mu P = 0$ that a moving subbundle be preserved; the two agree only for constant P. Which connection acts on matter, $P(d + A)$, an ambient connection preserving P, or an independent connection in a moving frame of the block, is a further choice, and we make none of them here.

8.1.5 What Yang–Mills theory keeps open

Four conditions on a connection have been established as theorems: commutation with the Lorentz action (Theorem 5.1); commutation with the projector, with a faithful action on its image (Definition 6.3 and Theorem 6.5); antisymmetry, which for bivectors is Theorem 1.3; and preservation of the kinetic kernels K^μ. They are homogeneous and geometric, and together they determine the algebra in which A_μ takes its values, $\mathfrak{u}(1)^3 \oplus \mathfrak{su}(2)$ on the family block, $\mathfrak{so}(6) \oplus \mathfrak{su}(2)^{\oplus 3}$ at rank 256 and the Pati–Salam algebra on P_*, over the single frame of Proposition 5.3. They do not determine the field, since no parent action is derived; nor the couplings, since homogeneous conditions cancel every normalisation; nor which abelian direction is electromagnetic, by Theorem 6.17.

There is no representational obstruction behind this. By the theorem on universal connections [106], every connection on a bundle over a manifold of bounded dimension is the pullback of the canonical connection of a tautological bundle over a Grassmannian of sufficient rank, so the projector representation of a gauge field is always available in principle. The difficulty is

dynamical, and power counting makes it concrete. A projector is dimensionless, so $\mathrm{Tr}(\partial P\,\partial P)$ carries a coupling of mass dimension $D-2$ and $\mathrm{Tr}\,F^2$ one of dimension $D-4$. In two dimensions the sigma model is renormalisable and the composite gauge field acquires a kinetic term at leading order in $1/N$, with linear confinement of the constituents [29, 167]. In four dimensions the sigma model is not power-counting renormalisable and that argument does not transfer.

Confinement is a separate question again. Invariant tensors permit meson-like and baryon-like operators, but they say nothing about masses or about the energy of separated sources, and the diagnostics with dynamical matter are flux-tube and string-breaking correlation matrices [9]. Moreover a block supplies a multiplicity index, not three constituents: three copies of a doublet form a direct sum, whereas three constituents require a tensor product with antisymmetrisation, so even the operator statement needs a Fock construction first.

8.2 The compact families at the third rung

The second-rung vacuum of Definition 8.56 lies on a compact family S^4 of Lorentz-invariant projectors, and a spectrum-only potential is constant on it Section 8.5. A coupling that distinguishes its points has to be found at $n \geq 3$, where colour survives Section 13.1. This section fixes the geometry on which any such coupling acts. The results are as follows. At the third rung the Lorentz-invariant symmetric involutions that are products of four internal generators form three orbits of the compact group K. Only the four-positive orbit F_3, an oriented Grassmannian of dimension 16, commutes with colour, and it commutes with colour at every point (fixed). The second-rung family is one Clifford four-sphere inside F_3 and is not K-invariant (fixed). The vacuum Π_A built on the auxiliary four-plane is the analogue of Definition 8.56. It is neutral under the construction's charge and under electromagnetism, and it is fixed by the Pati–Salam and Standard-Model algebras (fixed). Its sixteen fluctuations are four complex Standard-Model doublets of hypercharge $\frac{1}{2}$ (fixed). Any K-invariant coupling is constant on F_3. The alignment energy of a proper gauged subalgebra vanishes exactly on its fixed set, and these fixed sets are determined here. A K-equivariant duality $P \mapsto P^{\perp}$ preserves every gauge-boson mass matrix, so no alignment energy separates Π_A from Π_W (fixed). On the matter block the construction's charge acts as half the fermion number (fixed), and with $\mathfrak{su}(2)_L$ it is anomalous there (restated). Identifiers p1-n refer to the `PASS` lines of `p1_families.py` (Table 8.1).

Conventions. The realisation is that of Appendix A.1: $E_{6(k-1)+a} = \omega^{\otimes(k-1)} \otimes e_a \otimes \mathbf{1}^{\otimes(n-k)}$ on $\mathbb{R}^{8^n}$, every generator and blade a signed permutation matrix, and $e_a^{\top} = \eta_{aa} e_a$. Products, commutators and traces of blades are therefore integer computations, and blades obey the commutation rule, equation (1.3), and the transpose rule, (1.4).

Definitions. Rung $n = 3$, spectator interior action: interior generators $\gamma^{\mu} = (E_1, E_4, E_5, E_6)$; internal generators the other fourteen. The positive internal span is $V_+ = \mathrm{span}(A \cup W)$, of dimension 8, and the negative internal span is $V_- = \mathrm{span}(C)$, of dimension 6. The groups are those of Section 6.2: auxiliary $A = (E_2, E_3, E_8, E_9)$, weak $W = (E_7, E_{13}, E_{14}, E_{15})$, written $(w_1, \ldots, w_4)$, and colour $C = (E_{10}, E_{11}, E_{12}, E_{16}, E_{17}, E_{18})$. $K = \mathrm{Spin}(8) \times \mathrm{Spin}(6)$ is the group generated by the 43 compact internal bivectors Theorem 1.3, with Lie algebra $\mathfrak{k} = \mathfrak{so}(8) \oplus \mathfrak{so}(6)$. For an oriented four-plane P in V_+ with oriented orthonormal basis (u_1, u_2, u_3, u_4), $\mathrm{vol}_P := u_1u_2u_3u_4$, a symmetric involution commuting with every interior gamma, and $\Pi_P := \frac{1}{2}(\mathbf{1} + \mathrm{vol}_P)$. The compact family $F_3 := \{\Pi_P : P \text{ an oriented four-plane in } V_+\}$ is the K-orbit of Π_A (A the auxiliary four-plane) and is diffeomorphic to the oriented Grassmannian of four-planes in $\mathbb{R}^8$, of dimension 16 (Theorem 8.10). The gauged subalgebras considered are: $\mathfrak{h}_{43} = \mathfrak{k}$, all compact internal bivectors, as in Proposition 7.39; $\mathfrak{h}_{\mathrm{PS}} = \mathfrak{so}(4)_W \oplus \mathfrak{so}(6)_C$, the Pati–Salam algebra $\mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R \oplus \mathfrak{su}(4)$; $\mathfrak{h}_{\mathrm{SM}} = \mathfrak{su}(3)_C \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_Y$, embedded as in Section 6.2; $\mathfrak{u}(1)_{\mathrm{em}}$, the bivector part of $Q = -JT_L^3 + Y$; and $\mathfrak{u}(1)_A = \mathrm{span}(q_A)$, the construction's charge $Q = \frac{1}{2}J_1$

at $n = 3$. The alignment energy of a gauged subalgebra $\mathfrak{h}$ with orthonormal basis (t_a) and couplings g_a is $V_{\mathfrak{h}}(\Pi) := \sum_a g_a^2 \|[t_a, \Pi]\|^2$ (Frobenius norm on $\mathbb{R}^{512}$). With the projector kinetic term $\frac{f^2}{2}\operatorname{Tr}(D\Pi D\Pi)$ of Section 8.5 it is proportional to the trace of the gauge-boson mass matrix at Π.

Notation. J is the ordered product of the internal generators, $\Gamma = J\gamma^0\gamma^1\gamma^2\gamma^3$, $q_A = E_2E_3$, $Q_A = \frac{1}{2}q_A$, $\chi_W = w_1w_2w_3w_4$, χ_C the product of C, and $P_* = \frac{1}{8}(\mathbf{1}+\mathcal{X})(\mathbf{1}+\mathcal{Z})(\mathbf{1}+\mathcal{U})$ with $\mathcal{X} = \Gamma\chi_W$, $\mathcal{Z} = -J\chi_C$, $\mathcal{U} = -Jq_A$ Theorem 6.10; $P_{L,R} = P_*\frac{1}{2}(\mathbf{1} \mp \Gamma)$. The colour Cartan element is $H_C = -(E_{10}E_{16} + E_{11}E_{17} + E_{12}E_{18})$, and $\mathfrak{su}(3)_C$ is its derived centraliser in $\mathfrak{so}(6)_C$ Proposition 6.13. Further $\tau_i = -\frac{1}{2}w_jw_k$ for cyclic (i,j,k), $t_i^{L,R} = \frac{1}{2}(\mathbf{1} \mp \chi_W)\tau_i$ Theorem 6.11, the hypercharge bivector is $t_Y = t_3^R + \frac{1}{6}H_C$, and

$$B_{\mathrm{em}} = \tau_3 + \tfrac{1}{6}H_C = t_3^L + t_Y = -\tfrac{1}{2}E_7E_{13} + \tfrac{1}{6}H_C$$

spans $\mathfrak{u}(1)_{\mathrm{em}}$: on the block $Q = -J(t_L^3 + t_R^3 + \frac{1}{6}H_C)$ Proposition 6.13. The vector representation ρ is defined by $[X,u] = \rho(X)u$ for internal vectors u. Thus $\rho(\frac{1}{2}e_ie_j) = E_{ij} - E_{ji}$ on V_+, with a sign -1 on V_-, and the lift $M \mapsto \sum_{i<j} M_{ij}\frac{1}{2}e_ie_j$ (sign -1 on V_-) is a Lie algebra map inverse to ρ (p1-12). We write $\rho_+(X) \in \mathfrak{so}(8)$ for the V_+ part, p_P for the orthogonal projector of $\mathbb{R}^8$ onto P, θ for $+1$ on V_+ and -1 on V_-, and $\widetilde{\mathrm{Gr}}_k(V)$ for the oriented Grassmannian. On vectors, $\rho(Q_A)$ maps $E_2 \mapsto -E_3$, $E_3 \mapsto E_2$ and kills E_8, E_9, W. $\rho(B_{\mathrm{em}})$ maps $w_1 \mapsto w_2$, $w_2 \mapsto -w_1$ and kills A, w_3, w_4. $J_Y = 2\rho(t_Y)$ is a complex structure on W that kills A.

8.2.1 The second rung, restated

Proposition 8.6 (The second-rung family is a K_2-orbit; restated and fixed)**.** *At $n = 2$ the internal complement of the spectator action has signature $(5,3)$, with $V_+^{(2)} = \operatorname{span}(E_2, E_3, E_7, E_8, E_9)$ and $V_-^{(2)} = \operatorname{span}(E_{10}, E_{11}, E_{12})$. Let $(a,b,c,d,e) = (E_2, E_3, E_7, E_8, E_9)$, $Q = \frac{1}{2}ab$, $H = abcd$, $\Pi_0 = \frac{1}{2}(\mathbf{1}+H)$, $I_5 = abcde$, $Z_i = I_5v_i$.*

(i) (Restated.) *Theorems 8.57 to 8.59 hold: H is a symmetric traceless involution commuting with the interior gammas, Q and the internal volume; $[\frac{1}{2}be, \Pi_0] = -\frac{1}{2}acde$; the Z_i are a symmetric Clifford system, $T_i = [\frac{1}{2}v_ie, \Pi_0] = \frac{1}{2}Z_i$ with $\operatorname{Tr}(T_iT_j) = 16\delta_{ij}$, Q rotates (t_a, t_b) and fixes t_c, t_d; along $\Pi(n) = \frac{1}{2}(\mathbf{1} + I_5n)$, $\operatorname{Tr}(d\Pi^2) = 16|dn|^2$, $-\operatorname{Tr}([Q,\Pi]d\Pi) = 16(n_a dn_b - n_b dn_a)$ and $[\Pi, d\Pi] = \frac{1}{4}[n, dn]$. The stabiliser of H in $\mathfrak{so}(5,3)$ is $\mathfrak{so}(P) \oplus \mathfrak{so}(P^\perp) = \mathfrak{so}(4) \oplus \mathfrak{so}(1,3)$, of dimension 12 with compact part 9, and every other internal bivector anticommutes with H Proposition 8.60.*

(ii) (Fixed.) *The 13 compact internal bivectors preserve $\operatorname{span}\{Z_i\}$: $\mathfrak{so}(5)$ acts on it by the vector representation and $\mathfrak{so}(3)$ trivially. The K_2-orbit of Π_0, $K_2 = \mathrm{Spin}(5) \times \mathrm{Spin}(3)$, is the whole family $S^4 = \{\Pi(n)\}$, and $\exp(\frac{\vartheta}{2}v_ie)$ carries $\Pi(e)$ to $\Pi(\cos\vartheta\, e + \sin\vartheta\, v_i)$. Moreover $I_5n = \mathrm{vol}_{n^\perp}$, so $S^4 = \{\Pi_P : P \in \widetilde{\mathrm{Gr}}_4(V_+^{(2)})\}$.*

(iii) (Fixed.) *$[Q, \Pi(n)] = 0$ exactly on the two-sphere $n_a = n_b = 0$, the planes containing $\operatorname{span}(a,b)$. Every $\Pi(n)$ has spectrum $\{0^{32}, 1^{32}\}$, and $\frac{1}{64}\sum \|[e_ie_j, \Pi(n)]\|^2 = 4$ over the 13 compact blades. A spectrum-only potential and the full alignment energy are constant on S^4.*

Proof. (i) is the computation of Section 8.5, repeated in p1-1 to p1-4 and p1-7. The stabiliser statement is equation (1.3): a bivector e_ie_j and the grade-four blade H commute iff $|\{i,j\} \cap \{a,b,c,d\}|$ is even. (ii) A positive bivector inside $V_+^{(2)}$ shares two indices with I_5 and commutes with it, so $[X, Z_i] = I_5[X, v_i] = I_5\rho(X)v_i$; a negative bivector shares none and commutes with each Z_i. Since $\mathrm{Spin}(5)$ is transitive on S^4, the orbit of $\Pi(e)$ is the family. The rotation formula is $\mathrm{Ad}(\exp\frac{\vartheta}{2}v_ie)e = \cos\vartheta\, e + \sin\vartheta\, v_i$. If $(f_1, \dots, f_4, n)$ is an oriented orthonormal basis of $V_+^{(2)}$,

then $I_5 = f_1f_2f_3f_4n$ and $I_5n = f_1f_2f_3f_4$ (p1-5). (iii) $[Q, Z_i] = I_5\rho(Q)v_i$, and $\rho(Q)$ rotates (a, b) and kills c, d, e (p1-6). Each $\Pi(n)$ is conjugate to Π_0 by an orthogonal matrix, and the alignment sum is K_2-invariant, as in Proposition 8.20 below (p1-4, p1-5). □

At $n = 2$ three descriptions coincide: the Clifford sphere of five positive generators, the K_2-orbit, and the oriented Grassmannian of four-planes in $V_+^{(2)}$. At $n = 3$ they separate. The Grassmannian description is the one that continues.

8.2.2 Four-generator involutions at the third rung

Lemma 8.7 (Blade census; fixed)**.** *At $n = 3$ the eighteen generators satisfy the Clifford relations of $\mathbb{R}^{9,9}$, $V_+ = \mathrm{span}(E_2, E_3, E_7, E_8, E_9, E_{13}, E_{14}, E_{15})$, and the 43 compact internal bivectors square to $-\mathbf{1}$ while the 48 mixed ones square to $+\mathbf{1}$ (p1-8).*

(i) Of the 3060 blades of grade four exactly the 1001 internal ones commute with the four interior gammas. All are traceless. One with q negative factors is a symmetric involution if q is even and an antisymmetric complex structure if q is odd. The counts are $(4, 0) : 70$, $(3, 1) : 336$, $(2, 2) : 420$, $(1, 3) : 160$, $(0, 4) : 15$ (p1-9).

(ii) More generally an internal blade of even grade r with q negative factors is a symmetric involution iff $r(r-1)/2 + q$ is even. The only nontrivial one invariant under K is $\mathrm{vol}_+ = E_2E_3E_7E_8E_9E_{13}E_{14}E_{15}$, and $\mathrm{vol}_+ = -\mathcal{Z} = J\chi_C$ (p1-10).

Proof. For $|I| = 4$, equation (1.3) gives $e_Ie_j = (-1)^{4-|I\cap\{j\}|}e_je_I$, so a generator commutes with e_I iff it is not a factor. By equation (1.4), $e_I^2 = (-1)^{r(r-1)/2+q}\mathbf{1}$ and $e_I^\top = e_I^2e_I$. An even internal blade commutes with e_ie_j iff $|I \cap \{i, j\}|$ is even. For all same-sign pairs this forces $I \cap V_\pm \in \{\emptyset, V_\pm\}$, which leaves vol_+, χ_C ($r = q = 6$, square $-\mathbf{1}$) and $\pm J$ ($r = 14$, square $-\mathbf{1}$). The identity $\mathrm{vol}_+ = J\chi_C$ is checked on the matrices. □

Lemma 8.8 (Transpose and recovery; fixed)**.** *Let P be a nondegenerate internal four-plane with η-orthonormal oriented basis $(u_1, \dots, u_4)$ and $\mathrm{vol}_P = u_1u_2u_3u_4$. Then vol_P commutes with the interior gammas, $\mathrm{vol}_P^2 = (-1)^q\mathbf{1}$, the internal vectors anticommuting with vol_P are exactly those of P, and*

$$\mathrm{vol}_P^\top = \mathrm{vol}_{\theta P}.$$

Hence for q even, Π_P is an orthogonal projector iff $\theta P = P$, that is iff $P = (P \cap V_+) \oplus (P \cap V_-)$. For a positive-definite $P \not\subset V_+$, Π_P is an idempotent but not symmetric (p1-13).

Proof. The interior gammas anticommute with every internal vector, hence commute with a product of four. By equation (1.4), $\mathrm{vol}_P^2 = (-1)^6\prod\eta(u_i)\mathbf{1} = (-1)^q\mathbf{1}$. For an internal vector u, $u^\top = \theta u$. Transposing reverses the product, and reversing four anticommuting vectors costs $(-1)^6 = 1$. So $\mathrm{vol}_P^\top = \theta u_1 \cdots \theta u_4 = \mathrm{vol}_{\theta P}$, and θ is an isometry of η. Each u_i commutes with itself and anticommutes with the other three factors, so every vector of P anticommutes with vol_P. A vector η-orthogonal to P anticommutes with all four factors, so it commutes with vol_P. Since P is nondegenerate, an internal v splits as $v_P + v_\perp$; if v anticommutes with vol_P, so does $v_\perp$, which also commutes, hence $v_\perp\mathrm{vol}_P = 0$ and $v_\perp = 0$. So P is determined by vol_P, and its orientation by the sign. Then vol_P is symmetric iff $\theta P = P$ as oriented planes. □

Lemma 8.9 (Colour acts irreducibly on V_-; fixed)**.** *The centraliser of $\rho(H_C)$ in $\mathfrak{so}(V_-)$ is $\mathfrak{u}(3)$. Its part orthogonal to H_C is $\mathfrak{su}(3)_C$, of dimension 8 and perfect. Lifted to $\mathfrak{k}$, $\mathfrak{su}(3)_C$ commutes with H_C and acts trivially on V_+. Its commutant on $V_- = \mathbb{R}^6$ is $\mathrm{span}(\mathbf{1}, \frac{1}{2}\rho(H_C))$ with $(\frac{1}{2}\rho(H_C))^2 = -\mathbf{1}$, a field, so $\mathfrak{su}(3)_C$ acts irreducibly on V_- (p1-15).*

Proof. The computation is in the six-dimensional vector model and is lifted by the Lie algebra map of p1-12. A proper invariant subspace of an orthogonal representation has an invariant orthogonal complement, and the projector onto it is an idempotent in the commutant other than 0 and $\mathbf{1}$. A field has no such idempotent. □

Theorem 8.10 (The three orbit types and the colour test; fixed)**.** *The Lorentz-invariant symmetric involutions* vol_P*, with* P *a nondegenerate internal four-plane, form three* K*-orbits. Their stabilisers and dimensions are:*

type	orbit	stabiliser in $\mathfrak{so}(8,6)$	dim	compact part	orbit dim
$(4,0)$	$F_3 \cong \widetilde{\mathrm{Gr}}_4(V_+)$	$\mathfrak{so}(4)\oplus\mathfrak{so}(4,6)$	51	27: $\mathfrak{so}(4)_A\oplus\mathfrak{so}(4)_W\oplus\mathfrak{so}(6)_C$	16
$(2,2)$	$\widetilde{\mathrm{Gr}}_2(V_+)\times\widetilde{\mathrm{Gr}}_2(V_-)/\pm$	$\mathfrak{so}(2,2)\oplus\mathfrak{so}(6,4)$	51	23	20
$(0,4)$	$\widetilde{\mathrm{Gr}}_4(V_-)$	$\mathfrak{so}(4)\oplus\mathfrak{so}(8,2)$	51	35	8

Every point has spectrum $\{+1^{256},-1^{256}\}$*, so no spectrum-only potential separates the points or the types. Every point of* F_3 *commutes with all of* $\mathfrak{so}(6)_C\supset\mathfrak{su}(3)_C$*. No point of the* $(2,2)$ *or* $(0,4)$ *orbits commutes with* $\mathfrak{su}(3)_C$*.*

Proof. By Lemma 8.8 the symmetric ones are the $\mathrm{vol}_{P_+}\mathrm{vol}_{P_-}$ with $P_\pm\subset V_\pm$ of dimensions (p,q), q even. For $g\in K$, $\mathrm{Ad}(g)u=\rho(g)u$ on vectors, so $\mathrm{Ad}(g)\mathrm{vol}_P=\mathrm{vol}_{\rho(g)P}$ (p1-14). $SO(8)\times SO(6)$ is transitive on pairs of oriented planes, which gives one orbit per type; reversing both orientations in type $(2,2)$ gives the same element. By the recovery statement of Lemma 8.8, $\exp(tX)$ fixes vol_P iff $\rho(X)$ preserves P, so the stabiliser is $\mathfrak{so}(P)\oplus\mathfrak{so}(P^\perp)$. Bivector by bivector, e_ie_j commutes with vol_P iff both or neither of i,j lie in P, and anticommutes otherwise; the counts are p1-11. The orbit dimension is 43 minus the compact part. A symmetric involution of trace zero has eigenvalues ±1, each 256 times (p1-42). A vector of V_+ anticommutes with both factors of a colour bivector and so commutes with it; hence every colour bivector commutes with every vol_P, $P\subset V_+$. For types $(2,2)$ and $(0,4)$, $\mathfrak{su}(3)_C$ fixes vol_P iff it preserves the plane $P_-\subset V_-$ of dimension 2 or 4, which Lemma 8.9 excludes. p1-16 checks this at the representatives and at random K-translates. □

At $n=3$ the stabiliser of the four-positive type is the row of Proposition 8.60. The same proof covers all even grades: a θ-stable plane of type (p,q) with $r(r-1)/2+q$ even gives a K-orbit of dimension $p(8-p)+q(6-q)$. Among them only $\mathrm{vol}_+=-\mathcal{Z}$ is K-fixed (Lemma 8.7). No compact generator moves $\frac12(\mathbf{1}+\mathrm{vol}_+)$, as for the chirality projectors of Proposition 8.55; only the boosts do. Lemma 8.8 agrees with Theorem 1.13, by which a variation $[B,\Pi]$ stays among the orthogonal projectors exactly when $[B+B^\top,\Pi]=0$, that is, when B is compact modulo the stabiliser: a noncompact generator that does not commute with Π tilts P out of V_+, and Π_P stops being symmetric. At Π_A, for instance, the boost $E_{13}E_{10}/2$ commutes with the projector, so the non-compact $E_2E_7/2+E_{13}E_{10}/2$ gives the same tangent as $E_2E_7/2$; the dimension counts are unaffected.

Proposition 8.11 (Clifford spheres in F_3; fixed)**.** *(i) For an oriented five-plane* $L\subset V_+$ *with oriented orthonormal basis* $(v_1,\dots,v_5)$ *put* $I_L=v_1\cdots v_5$ *and* $Z_i=I_Lv_i$*. The* Z_i *are a symmetric Clifford system, and* $\frac12(\mathbf{1}+I_Ln)=\Pi_{n^\perp\cap L}$*. Hence* $S_L:=\{\frac12(\mathbf{1}+I_Ln):|n|=1\}=\{\Pi_P:P\subset L\}\cong S^4$ *is the* $\mathrm{Spin}(L)$*-orbit of any of its points. Dually, for an oriented three-plane* U *with volume* U_3*,* $S^U:=\{\Pi_P:P\supset U\}=\{\frac12(\mathbf{1}+nU_3):n\in U^\perp,|n|=1\}\cong S^4$*.*

(ii) Through each Π_P *pass the spheres* S_L *with* $L=P\oplus\mathbb{R}u$*,* $u\in P^\perp$ *(a projective space* $\mathbb{R}P^3$ *of them), with tangent* $P\otimes u$*, and the spheres* S^U *with* $U=P\cap a^\perp$*,* $a\in P$*, with tangent* $a\otimes P^\perp$*. These are the two kinds of maximal linear spaces of rank-one elements in* $T_{\Pi_P}F_3\cong P\otimes P^\perp$*. The inner spheres are parametrised by* $\mathrm{Gr}_5(\mathbb{R}^8)$*, of dimension* 15*, and none of them is* K*-invariant.*

(iii) On the inner sphere through $A\oplus\mathbb{R}E_7$ *with* $Q=Q_A$*, and on the outer sphere of planes containing* $\mathrm{span}(E_3,E_8,E_9)$ *with* $Q=B_{\mathrm{em}}$*,* $\mathrm{Tr}(d\Pi^2)=128|dn|^2$ *and* $-\mathrm{Tr}([Q,\Pi]d\Pi)=\pm128(n_a\,dn_b-n_b\,dn_a)$*. On the inner sphere* $[\Pi,d\Pi]=\frac14[n,dn]$*. This is the structure of Theorems 8.58 and 8.59 with* 16 *replaced by* 128 *(p1-19, p1-32).*

(iv) The second-rung family embedded with its own generators, $L_0 = \mathrm{span}(a,b,c,d,e) = \mathrm{span}(E_2,E_3,E_7,E_8,E_9)$*, is the inner sphere* S_{L_0}*. Its point* $n = c$ *is* Π_A*, since* $I_5 c = E_2E_3E_8E_9$*, and its point* $n = e$ *is* $\Pi_0 = \frac{1}{2}(\mathbf{1} + E_2E_3E_7E_8)$*. On* S_{L_0}*,* $Q_A = \frac{1}{2}ab$ *is unbroken on the two-sphere* $n_a = n_b = 0$*. The algebras* $\mathfrak{u}(1)_{\mathrm{em}}$*,* $\mathfrak{h}_{\mathrm{SM}}$ *and* $\mathfrak{h}_{\mathrm{PS}}$ *are each unbroken only at* $n = \pm c$*, that is at* $\Pi_{\pm A}$ *(p1-47).*

Proof. (i) As in Proposition 8.6(ii): $Z_iZ_j + Z_jZ_i = 2\delta_{ij}$ because I_L commutes with each v_i and squares to $+\mathbf{1}$, and $I_L n = f_1f_2f_3f_4$ for an oriented basis $(f_1,\dots,f_4,n)$ of L. For S^U, nU_3 is the volume of $\mathrm{span}(n,U)$; vectors of $U^\perp$ anticommute with U_3, so the nU_3 again form a Clifford system (p1-19). (ii) The orbit of $\mathrm{Spin}(L)$, respectively $\mathrm{Spin}(U^\perp)$, has tangent $[\mathfrak{m}_P \cap \mathfrak{so}(L), \Pi_P]$, respectively $[\mathfrak{m}_P \cap \mathfrak{so}(U^\perp), \Pi_P]$, where $\mathfrak{m}_P = P \wedge P^\perp$ is spanned by the bivectors mixing P and $P^\perp$. These intersections are $P \wedge u$ and $a \wedge P^\perp$. (iii) Π and $d\Pi$ are combinations of the Z_i, and $\mathrm{Tr}(Z_iZ_j) = 512\delta_{ij}$; $[Q, Z_i] = \pm Z_j$ by the action of $\rho(Q)$ on L or on $U^\perp$ (p1-32). (iv) The identities are blade identities. The fixed points follow from Theorem 8.21 below, since $P \subset L_0$ contains or is orthogonal to $\mathrm{span}(E_7, E_{13})$ only for $P = A$, and $E_{13} \notin L_0$; p1-47 computes the kernels of $n \mapsto [X, I_5 n]$ directly. □

Proposition 8.12 (The duality $P \mapsto P^\perp$; fixed)**.** vol_+ *commutes with* $\mathfrak{k}$*, with the interior gammas and with every* vol_P*, and* $\mathrm{vol}_+\mathrm{vol}_A = \mathrm{vol}_W$*. The map* $D(\Pi) = \frac{1}{2}(\mathbf{1} + \mathrm{vol}_+(2\Pi - \mathbf{1}))$ *sends* Π_P *to* $\Pi_{P^\perp}$*, with* $P^\perp \subset V_+$ *suitably oriented. It is a* K*-equivariant involution of* F_3 *without fixed points, and* $D(\Pi_A) = \Pi_W$*. For every* $T, T' \in \mathfrak{k}$*,*

$$\mathrm{Tr}([T, D\Pi][T', D\Pi]) = \mathrm{Tr}([T,\Pi][T',\Pi]), \qquad \mathrm{Tr}([T, \mathbf{1}-\Pi][T', \mathbf{1}-\Pi]) = \mathrm{Tr}([T,\Pi][T',\Pi]).$$

So every gauge-boson mass matrix, and every alignment energy $V_{\mathfrak{h}}$ *with* $\mathfrak{h} \subset \mathfrak{k}$*, takes equal values at* $\Pi_{\pm A}$ *and* $\Pi_{\pm W}$*. On the block,* $\mathrm{vol}_+ P_* = -P_*$*, and* $\mathrm{Tr}(\Pi_A P_L) = 32$*,* $\mathrm{Tr}(\Pi_W P_L) = 0$*,* $\mathrm{Tr}(\Pi_W P_R) = 32$ *(p1-20, p1-46).*

Proof. vol_+ is K-invariant by Lemma 8.7, and it commutes with the interior gammas, with which it shares no index. In an orthonormal basis $(u_1,\dots,u_8)$ of V_+ adapted to $P \oplus P^\perp$, $\mathrm{vol}_+ = \pm u_1 \cdots u_8$, so vol_+ commutes with $\mathrm{vol}_P = u_1u_2u_3u_4$ and $\mathrm{vol}_+\mathrm{vol}_P = \pm u_5u_6u_7u_8 = \pm\mathrm{vol}_{P^\perp}$. D is an involution because $\mathrm{vol}_+^2 = \mathbf{1}$, and it has no fixed point because $P \neq P^\perp$. Since vol_+ commutes with T and Π, $[T, D\Pi] = \mathrm{vol}_+[T,\Pi]$, and $\mathrm{vol}_+^2 = \mathbf{1}$ gives the identity. The second identity is $[T, \mathbf{1}-\Pi] = -[T,\Pi]$. On the block, $\mathrm{vol}_+ P_* = -ZP_* = -P_*$. The traces follow from Proposition 8.16. □

8.2.3 The vacuum Π_A

Definition 8.13 (The third-rung vacuum)**.** $\Pi_A := \frac{1}{2}(\mathbf{1} + E_2E_3E_8E_9)$, built from the surplus pairs of the first and second constituents, and $\Pi_W := \frac{1}{2}(\mathbf{1} + \chi_W)$. Their orientation reversals are $\Pi_{-A} = \mathbf{1} - \Pi_A$ and $\Pi_{-W} = \mathbf{1} - \Pi_W$.

Theorem 8.14 (Lorentz invariant, neutral and not rigid; fixed)**.** Π_A *and* Π_W *are rank-*256 *orthogonal projectors with the following properties.*

(i) Each commutes with the interior gammas, with J*, with* q_A*, with* B_{em}*, with the* 21 *generators of* $\mathfrak{h}_{\mathrm{PS}}$ *and with the* 12 *of* $\mathfrak{h}_{\mathrm{SM}}$ *(p1-21).*

(ii) The compact internal bivectors that move Π_A *are exactly the* 16 *that mix* A *with* W*. The tangents* $T_{aw} = [\frac{1}{2}aw, \Pi_A] = \frac{1}{2}aw\,\mathrm{vol}_A$ *are symmetric with* $\mathrm{Tr}(T_{aw}T_{a'w'}) = 128\,\delta_{aa'}\delta_{ww'}$*; the same holds at* Π_W *(p1-23).*

(iii) The stabiliser of Π_A *in* $\mathfrak{k}$ *is* $\mathfrak{so}(4)_A \oplus \mathfrak{so}(4)_W \oplus \mathfrak{so}(6)_C$*, of dimension* 27*. It contains* $\mathfrak{u}(1)_A$*,* $\mathfrak{u}(1)_{\mathrm{em}}$*,* $\mathfrak{h}_{\mathrm{PS}}$ *and* $\mathfrak{h}_{\mathrm{SM}}$ *(p1-35).*

The literal embedding $\frac{1}{2}(\mathbf{1} + E_2E_3E_7E_8)$ *of the second-rung vacuum is also a point of* F_3 *and commutes with* q_A*. It does not commute with* B_{em} *or with* $\mathfrak{su}(2)_L$*, because* $E_7 \in W$ *(p1-22).*

Proof. All statements are equation (1.3) applied to blades. vol_A shares two indices with q_A, none with E_7E_{13}, the colour bivectors or the weak bivectors, and four with J. A bivector aw shares one index with vol_A and anticommutes with it, so $[\frac{1}{2}aw, \Pi_A] = \frac{1}{4}[aw, \mathrm{vol}_A] = \frac{1}{2}aw\,\mathrm{vol}_A$. Symmetry and the trace follow from $(aw\,\mathrm{vol}_A)^2 = -(aw)^2 = \mathbf{1}$ and the tracelessness of nontrivial blades. B_{em}, $t_i^{L,R}$ and $\mathfrak{su}(3)_C$ are combinations of bivectors inside W or C. □

In Proposition 5.19 the compact spectator algebra carries colour on three negative planes, weak isospin on two positive planes and a family $\mathfrak{so}(4)$ on the remaining two. With the grouping of Section 6.2 the remaining two planes are those of A, so Π_A separates the weak four-plane from the family four-plane and keeps everything else (interpretive). By Proposition 8.11(iv), Π_A is the point $n = c$ of the second-rung family itself, embedded at $n = 3$. The point $n = e$ chosen in Definition 8.56 is not neutral at $n = 3$.

Theorem 8.15 (The fluctuations at Π_A; fixed)**.** *Let* $T_{\Pi_A}F_3 = \mathrm{span}\{T_{aw}\} \cong A \otimes W$*.*

(i) Under $\mathfrak{so}(4)_A \oplus \mathfrak{so}(4)_W \cong \mathfrak{su}(2)^{\oplus 4}$ *it is the real irreducible* $(\mathbf{2},\mathbf{2};\mathbf{2},\mathbf{2})$*, with commutant* $\mathbb{R}$ *and every* $\mathfrak{su}(2)$ *Casimir equal to* $\frac{3}{4}$ *(p1-24).*

(ii) Under $\mathfrak{h}_{\mathrm{SM}}$*,* $\mathfrak{su}(3)_C$ *acts trivially;* $\mathfrak{su}(2)_L$ *has Casimir* $\frac{3}{4}$*, and* $\mathrm{ad}\,t_Y$ *satisfies* $(\mathrm{ad}\,t_Y)^2 = -\frac{1}{4}$ *and commutes with* $\mathfrak{su}(2)_L$*. The space is four complex doublets* $(\mathbf{1},\mathbf{2})_{1/2}$*, one for each direction of* A*, and* $\mathfrak{so}(4)_A$ *rotates the doublet index (p1-25).*

(iii) Under $\mathfrak{u}(1)_{\mathrm{em}}$*,* $\mathrm{ad}\,B_{\mathrm{em}} = \mathrm{ad}(t_L^3 + t_Y)$*. The charged directions are* $A \otimes \mathrm{span}(w_1, w_2)$*, four complex scalars* $\phi_a = x_{a,13} + ix_{a,7}$ *of charge one, the upper components of the doublets. The neutral directions are* $A \otimes \mathrm{span}(w_3, w_4)$*, eight real, the lower components.*

(iv) Under $\mathfrak{u}(1)_A$ *the charged directions are* $\mathrm{span}(E_2, E_3) \otimes W$*, four complex scalars* $\phi_w = x_{2,w} + ix_{3,w}$*. The neutral directions are* $\mathrm{span}(E_8, E_9) \otimes W$*, eight real.*

(v) The same holds at Π_W *with* $\mathrm{Hom}(W, A) \cong W \otimes A$ *(p1-24, p1-25, p1-26).*

Here $x_{a,w}$ *are the coordinates of* $\Pi = \mathrm{Ad}(e^X)\Pi_A$*,* $X = \sum x_{a,w}\frac{1}{2}aw$*. At* $n = 2$ *the same count gives one charged complex scalar and two neutral real scalars Theorem 8.58.*

Proof. For X in the stabiliser, $[X, [Y, \Pi_A]] = [[X, Y], \Pi_A]$. So $T_{\Pi_A}F_3$ is the representation of the stabiliser on $\mathfrak{m}_A = A \wedge W \cong A \otimes W$, on which $\rho(\frac{1}{2}aw)$ is the rotation of the (a, w) plane. All five statements reduce to the action on A and W listed in the conventions. $\mathfrak{su}(2)_L$ acts on W with a quaternionic commutant (p1-37), so W is one complex doublet with complex structure $J_Y = 2\rho(t_Y)$. $\rho(t_L^3 + t_Y) = \rho(\tau_3)$ rotates (w_1, w_2) at unit rate and kills (w_3, w_4). p1-26 gives the explicit pairs, for example $[Q_A, T_{2,w}] = -T_{3,w}$, $[Q_A, T_{3,w}] = T_{2,w}$ and $[B_{\mathrm{em}}, T_{a,7}] = T_{a,13}$, $[B_{\mathrm{em}}, T_{a,13}] = -T_{a,7}$. □

Proposition 8.16 (The vacuum on the matter block; fixed, and restated)**.** *(i)* vol_A *and* χ_W *commute with* $\mathcal{X}, \mathcal{Z}, \mathcal{U}$*, and* $\mathrm{vol}_A P_* = -\Gamma P_*$*,* $\chi_W P_* = \Gamma P_*$*. Hence* $\Pi_A P_* = P_L$ *and* $\Pi_W P_* = P_R$*, the two Weyl blocks of rank* 32 *(p1-27).*

(ii) $q_A P_* = JP_*$*. So* $Q_A = \frac{1}{2}q_A$*, the third-rung form of the construction's charge* $\frac{1}{2}J_1$*, acts on the block as* $\frac{1}{2}J$*: its charge operator* $-JQ_A$ *equals* $\frac{1}{2}$ *on all sixteen Weyl components. It is half the fermion number of Section 6.2, not the electric charge, which takes the values* $-1, -\frac{1}{3}, 0, \frac{2}{3}$ *(p1-18, p1-27).*

(iii) Every T_{aw} *anticommutes with* $\mathcal{X} = \Gamma\chi_W$*, and* $P_*T_{aw}P_* = 0$*: the fluctuations of* Π_A *map the block into its orthogonal complement (p1-27).*

(iv) (Restated, Proposition 6.14 with $a = \frac{1}{2}$, $b = 0$.) *With* $\hat{Q}_A = -JQ_A$ *and* $\hat{T} = -Jt_3^L$, *we have* $\mathrm{Tr}(P_L\hat{Q}_A\hat{T}^2) = 4$ *and* $\mathrm{Tr}(P_R\hat{Q}_A\hat{T}^2) = 0$, *and likewise* 4 *for* $\mathfrak{su}(2)_R$ *on* P_R. *For* $\hat{Q}_{\mathrm{em}} = -JB_{\mathrm{em}}$, $\mathrm{Tr}(P_L\hat{Q}_{\mathrm{em}}\hat{T}^2) = 0$. *The mixed* $SU(2)_L^2\,U(1)_A$ *anomaly does not cancel on the block, while the gravitational one does (p1-28) [16, 50].*

Proof. (i) $\mathrm{vol}_A = \mathrm{vol}_+\chi_W$ by Proposition 8.12. vol_+ commutes with Γ and $\mathrm{vol}_+P_* = -P_*$; with $\chi_W P_* = \Gamma P_*$ Theorem 6.10 this gives $\mathrm{vol}_A P_* = -\Gamma P_*$. (ii) $\mathcal{U}P_* = P_*$ means $-Jq_AP_* = P_*$, and $J^2 = -\mathbf{1}$ gives $q_AP_* = JP_*$; then $-JQ_AP_* = \frac{1}{2}P_*$. (iii) T_{aw} is half a blade of grade four with three factors in A and one in W. It shares one index with χ_W and anticommutes with it, and it commutes with Γ, which is $\pm$ the product of all eighteen generators. Hence it anticommutes with $\mathcal{X}$ (p1-27). Since $\mathcal{X}P_* = P_*$, $P_*TP_* = P_*\frac{1}{2}(\mathbf{1}+\mathcal{X})TP_* = P_*T\frac{1}{2}(\mathbf{1}-\mathcal{X})P_* = 0$. (iv) On P_L, $\hat{Q}_A = \frac{1}{2}$ and $\hat{T}^2 = -(t_3^L)^2 = \frac{1}{4}$, and $\mathrm{Tr}\,P_L = 32$; $t_3^L P_R = 0$. The electric charges of the left doublets sum to zero. □

The charge that the second-rung construction gauges, $Q = \frac{1}{2}J_1$, therefore becomes at $n = 3$ a universal charge on the P_* matter, and with $\mathfrak{su}(2)_L$ it cannot be gauged on that matter. The charge for which Π_A plays the role of Definition 8.56 with electromagnetic content is B_{em}. Both meet the massless criterion of Corollary 8.53 at Π_A (Theorem 8.18). Which of them is the photon is an identification and is not decided by the geometry of F_3 (open).

8.2.4 Kinetic term, current and transport on F_3

Lemma 8.17 (Vector model; fixed)**.** *For* $P \in \widetilde{\mathrm{Gr}}_4(V_+)$ *and* $X, Y \in \mathfrak{k}$,

$$\mathrm{Tr}([X,\Pi_P][Y,\Pi_P]) = 64\,\mathrm{tr}_8([\rho_+X, p_P][\rho_+Y, p_P]).$$

Hence on F_3: $\frac{f^2}{2}\mathrm{Tr}(D\Pi D\Pi) = 32f^2\,\mathrm{tr}_8(Dp\,Dp)$; *for* $Q \in \mathfrak{k}$, $j_\Pi = -f^2\,\mathrm{Tr}([Q,\Pi]D\Pi) = -64f^2\,\mathrm{tr}_8([\rho_+Q, p]Dp)$; *and*

$$V_{\mathfrak{h}}(\Pi_P) = 64\sum_a g_a^2\,\|[\rho_+t_a, p_P]\|^2,$$

in which the $\mathfrak{so}(6)$ *components of the* t_a *drop out (p1-29, p1-30).*

Proof. Both sides are bilinear in (X,Y) and invariant under K, because $\mathrm{Ad}(g)\Pi_P = \Pi_{\rho(g)P}$, $p_{RP} = Rp_PR^\top$ and ρ is equivariant. So it suffices to take $P = A$. Write $X = X_0 + \sum x_{aw}\frac{1}{2}aw$ with X_0 in the stabiliser; then $[X,\Pi_A] = \sum x_{aw}T_{aw}$, and the left side is $128\sum x_{aw}y_{aw}$. On the right, $[\rho(\frac{1}{2}aw), p_A] = -(E_{aw}+E_{wa})$, and ρ_+X_0 commutes with p_A, so the right side is $64\cdot 2\sum x_{aw}y_{aw}$. Every tangent vector at Π_P is $[Y,\Pi_P]$ for some $Y \in \mathfrak{k}$, which gives the kinetic and current forms. □

The factor $64 = 512/8$ is the real trace on $\mathbb{R}^{512}$ against that on $\mathbb{R}^8$. At $n = 2$ the same factor is $8 = 64/8$. With $\mathrm{Tr}(T_iT_j) = 16\delta_{ij}$ there, this reproduces the $8f^2$ and $16f^2$ of Theorem 8.58.

Theorem 8.18 (Kinetic term and current at Π_A; fixed)**.** *With* $\Pi = \mathrm{Ad}(e^X)\Pi_A$, $X = \sum x_{aw}\frac{1}{2}aw$, *and the conventions of Theorem 8.58, to quadratic order in* x:

$$\mathcal{L}_\Pi = 64f^2\sum_{a,w} D_\mu x_{aw} D^\mu x_{aw}, \qquad \begin{aligned} j^\mu_{\Pi,A} &= 128f^2\sum_{w\in W}\mathrm{Im}(\bar\phi_w D^\mu\phi_w),\\ j^\mu_{\Pi,\mathrm{em}} &= 128f^2\sum_{a\in A}\mathrm{Im}(\bar\phi_a D^\mu\phi_a),\end{aligned}$$

with $\phi_w = x_{2,w} + ix_{3,w}$ *and* $\phi_a = x_{a,13} + ix_{a,7}$. *Since* $[Q,\Pi_A] = 0$ *for both charges, no* $A_\mu A^\mu$ *term arises at* $x = 0$, *and the criterion of Corollary 8.53 holds at* Π_A *and* Π_W *for* $\mathfrak{u}(1)_A$ *and for* $\mathfrak{u}(1)_{\mathrm{em}}$. *On the Clifford spheres of Proposition 8.11(iii) the formulas are exact, with* $\mathcal{L}_\Pi = 64f^2\,Dn\cdot Dn$ *(p1-31, p1-32).*

Proof. $D\Pi = [DX, \Pi_A] + O(x^2)$, and j_Π at second order is $-f^2 \operatorname{Tr}([[Q,X],\Pi_A][DX,\Pi_A])$. p1-31 evaluates both bilinear forms with $\operatorname{Tr}(T_{aw}T_{a'w'}) = 128\delta$ and the pairs of Theorem 8.15. The sign pattern is that of Theorem 8.58: $[Q, T_{\mathrm{re}}] = -T_{\mathrm{im}}$ and $[Q, T_{\mathrm{im}}] = T_{\mathrm{re}}$. □

Theorem 8.19 (Curved extension; fixed)**.** *(i) Let $\Omega = \Omega_{\mathrm{Lorentz}} + A^a T_a$ with t_a spanning a subalgebra $\mathfrak{s} \subset \mathfrak{k}$. Then $D\Pi_P = 0$ and $[\mathcal{R}, \Pi_P] = 0$ hold for arbitrary spacetime and gauge curvature iff $\mathfrak{s} \subset \mathfrak{so}(P) \oplus \mathfrak{so}(P^\perp \cap V_+) \oplus \mathfrak{so}(6)_C$. At Π_A and Π_W this includes $\mathfrak{u}(1)_A$, $\mathfrak{u}(1)_{\mathrm{em}}$, $\mathfrak{h}_{\mathrm{PS}}$ and $\mathfrak{h}_{\mathrm{SM}}$ (p1-35).*

(ii) At every point of F_3, if $D_\mu \Pi = [Y_\mu, \Pi]$ with $Y_\mu \in \mathfrak{k}$, the adapted connection of Proposition 11.16 is $\mathcal{C}_\mu = [\Pi, D_\mu \Pi] = -\frac{1}{2}(Y_\mu - \mathrm{vol}_P Y_\mu \mathrm{vol}_P)$. This is a compact bivector of $\Lambda^2 V_+$, so the adapted connection is spin-valued across the whole Grassmannian family (p1-33).

Proof. (i) The six interior bivectors commute with every Π_P and with $\mathfrak{k}$. So $D\Pi_P = A^a[t_a, \Pi_P]$ and $\mathcal{R} = \mathcal{R}_{\mathrm{Lorentz}} + F^a T_a$, and both vanish for arbitrary A and F iff each t_a is in the stabiliser of Theorem 8.10. (ii) With $v = \mathrm{vol}_P$ and $v^2 = \mathbf{1}$, $[\Pi, [Y, \Pi]] = \frac{1}{4}[v, [Y, v]] = \frac{1}{2}(vYv - Y)$. $\mathrm{Ad}(v)$ acts on vectors as -1 on P and $+1$ on $P^\perp \cap V_+$ and V_-. So it maps $\Lambda^2 V_+$ and $\Lambda^2 V_-$ to themselves, and the $\Lambda^2 V_-$ part of Y cancels. □

Proposition 8.20 (Invariant couplings are constant; fixed)**.** *Every K-invariant function on F_3 is constant. In particular the full alignment energy satisfies $\frac{1}{512}\sum \|[e_i e_j, \Pi]\|^2 = 16$ on F_3, with the sum over the 43 compact blades. Every spectrum-only potential is constant on F_3. For a subalgebra $\mathfrak{h} \subsetneq \mathfrak{k}$ and positive couplings, the zero set of $V_\mathfrak{h}$ on F_3 is the fixed set $\mathrm{Fix}(\mathfrak{h}) = \{\Pi \in F_3 : [\mathfrak{h}, \Pi] = 0\}$ (p1-34, p1-42).*

Proof. F_3 is one K-orbit. The 43 blades are orthogonal of equal norm in the trace form, which $\mathrm{Ad}(K)$ preserves, so $\sum_a \|[t_a, \mathrm{Ad}(g)\Pi]\|^2 = \sum_a \|[\mathrm{Ad}(g^{-1})t_a, \Pi]\|^2$ is a sum over another orthonormal basis and has the same value. The value 16 is sixteen movers, each with $\|[e_i e_j, \Pi_A]\|^2 = 512$. $V_\mathfrak{h}$ is a sum of squares with positive weights. □

A coupling that selects a point of F_3 must therefore break K, and the natural candidates are the $V_\mathfrak{h}$ of proper subalgebras [123, 126]. Proposition 8.12 adds that every such $V_\mathfrak{h}$ is also invariant under D and under $\Pi \mapsto \mathbf{1} - \Pi$.

8.2.5 Fixed sets and the colour condition

Write $U_A = \mathrm{span}(E_2, E_3)$, $U_{\mathrm{em}} = \mathrm{span}(w_1, w_2)$ and $W_0 = \mathrm{span}(w_3, w_4)$.

Theorem 8.21 (Fixed sets in F_3; fixed)**.**

algebra	fixed set in F_3	dimensions	contains
$\mathfrak{so}(6)_C$, $\mathfrak{su}(3)_C$	all of F_3	16	all points
$\mathfrak{u}(1)_A$	$\{P \supset U_A\} \sqcup \{P \perp U_A\} \cong \widetilde{\mathrm{Gr}}_2(\mathbb{R}^6) \sqcup \widetilde{\mathrm{Gr}}_4(\mathbb{R}^6)$	8, 8	$\Pi_{\pm A}$; $\Pi_{\pm W}$
$\mathfrak{u}(1)_{\mathrm{em}}$	$\{P \supset U_{\mathrm{em}}\} \sqcup \{P \perp U_{\mathrm{em}}\} \cong \widetilde{\mathrm{Gr}}_2(\mathbb{R}^6) \sqcup \widetilde{\mathrm{Gr}}_4(\mathbb{R}^6)$	8, 8	$\Pi_{\pm W}$; $\Pi_{\pm A}$
$\mathbb{R}t_Y$	$\{\Pi_{\pm A}\} \sqcup \{\Pi_{\pm W}\} \sqcup \{A' \oplus W'\}$	0, 0, 6	
$\mathfrak{su}(2)_L$, $\mathfrak{h}_{\mathrm{SM}}$, $\mathfrak{h}_{\mathrm{PS}}$	$\{\Pi_A, \Pi_{-A}, \Pi_W, \Pi_{-W}\}$	0	four isolated points
$\mathfrak{k}$	$\emptyset$		

In the t_Y row, A' is a two-plane in A and W' a J_Y-complex line in W. That component is $\widetilde{\mathrm{Gr}}_2(A) \times \mathbb{C}P^1$, connected (p1-36 to p1-39).

Proof. For a connected group, Π_P is fixed iff P is invariant under $\rho_+(\mathfrak{h})$ (Lemma 8.8). Let j be a complex structure on a subspace U that kills $U^\perp$. If P is j-invariant and $v = v_U + v_\perp \in P$, then $j^2 v = -v_U \in P$, so $P = (P \cap U) \oplus (P \cap U^\perp)$ with $P \cap U$ itself j-invariant. For $U = U_A$ or U_{em}, $P \cap U \in \{0, U\}$, which gives the two Grassmannians of $U^\perp \cong \mathbb{R}^6$, each connected of dimension 8. For $U = W$ and $j = J_Y$, $P \cap W$ is a complex subspace of dimension 0, 2 or 4. For $\mathfrak{su}(2)_L$: A is trivial and W is irreducible without invariant vectors (p1-37). So $P \cap W \in \{0, W\}$, and if

$P \cap W = 0$ the W-component of every vector of P is invariant, hence zero; thus $P \in \{A, W\}$. $\mathfrak{h}_{\mathrm{SM}}$ and $\mathfrak{h}_{\mathrm{PS}}$ contain $\mathfrak{su}(2)_L$ and fix A and W. $\mathfrak{so}(8)$ is irreducible on V_+. The tangent kernels in p1-36, p1-38 and p1-39 agree with these dimensions, and p1-38 shows that the four points are isolated. □

Proposition 8.22 (Unbroken algebras across F_3; fixed)**.** *The stabiliser of* Π_P *in* $\mathfrak{h}_{\mathrm{PS}}$ *is* $\mathfrak{so}(6)_C \oplus \mathfrak{so}(P \cap W) \oplus \mathfrak{so}(P^\perp \cap W)$*. In* $\mathfrak{h}_{\mathrm{SM}}$ *it is* $\mathfrak{su}(3)_C \oplus (\mathfrak{u}_W \cap (\mathfrak{so}(P \cap W) \oplus \mathfrak{so}(P^\perp \cap W)))$*, with* $\mathfrak{u}_W = \rho(\mathfrak{su}(2)_L \oplus \mathbb{R}t_Y) \cong \mathfrak{u}(2)$*. At random points the dimensions in* $(\mathfrak{h}_{\mathrm{SM}}, \mathfrak{h}_{\mathrm{PS}})$ *are* $(9, 16)$ *on both* $\mathfrak{u}(1)_{\mathrm{em}}$*-fixed components, namely* $\mathfrak{su}(3)_C \oplus \mathfrak{u}(1)_{\mathrm{em}}$ *and* $\mathfrak{so}(6)_C \oplus \mathfrak{u}(1)_{\mathrm{em}}$*. They are* $(8, 16)$ *on both* $\mathfrak{u}(1)_A$*-fixed components,* $(8, 15)$ *at generic points of* F_3 *and* $(12, 21)$ *at* $\Pi_{\pm A}, \Pi_{\pm W}$*. On the six-dimensional* t_Y*-component the stabiliser in* $\mathfrak{h}_{\mathrm{SM}}$ *has dimension* 10 *(p1-39, p1-40).*

Proof. If $X \in \mathfrak{so}(W)$ preserves P and kills A, then $X(P) \subset P \cap W$ and $X(P^\perp) \subset P^\perp \cap W$. So X has image in $(P \cap W) \oplus (P^\perp \cap W)$, and by antisymmetry it lies in $\mathfrak{so}(P \cap W) \oplus \mathfrak{so}(P^\perp \cap W)$; the converse is immediate. On $\{P \perp U_{\mathrm{em}}\}$ one generically has $P \cap W = 0$ and $P^\perp \cap W = U_{\mathrm{em}}$, and $\mathfrak{so}(U_{\mathrm{em}}) = \mathbb{R}\rho(\tau_3) = \mathbb{R}\rho(B_{\mathrm{em}}) \subset \mathfrak{u}_W$. p1-40 computes the dimensions of the intersections and of the stabilisers. □

The $\mathfrak{u}(1)_{\mathrm{em}}$-fixed component through Π_A is $\widetilde{\mathrm{Gr}}_4(A \oplus W_0)$. Its tangent at Π_A is $A \otimes W_0$, the neutral lower components of the four doublets of Theorem 8.15. Along it $\mathfrak{su}(3) \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_Y$ breaks to $\mathfrak{su}(3) \oplus \mathfrak{u}(1)_{\mathrm{em}}$, the pattern of a neutral doublet value (interpretive).

Corollary 8.23 (Critical points and Hessians; fixed)**.** *Every smooth function on* F_3 *invariant under the connected group of* $\mathfrak{h} \in \{\mathfrak{su}(2)_L, \mathfrak{h}_{\mathrm{SM}}, \mathfrak{h}_{\mathrm{PS}}\}$ *has* $\Pi_{\pm A}$ *and* $\Pi_{\pm W}$ *as critical points. Its Hessian there lies in the space of invariant symmetric forms on* $A \otimes W$*. For* $\mathfrak{h}_{\mathrm{PS}}$ *this space has dimension* 10*, the forms* $S \otimes \delta_W$ *with* S *symmetric on* A*. For* $\mathfrak{h}_{\mathrm{SM}}$ *it has dimension* 16*, the forms* $S \otimes \delta_W + N \otimes \omega_Y$ *with* N *antisymmetric and* $\omega_Y(w, w') = \langle w, J_Y w' \rangle$ *(p1-41).*

Proof. At a fixed point the differential is an invariant covector, and by p1-38 there is none other than zero [95, 115]. The Hessian at a critical point is invariant. The counts follow from Schur's lemma, with the commutant $\mathbb{R}$ of $\mathfrak{so}(4)_W$ on W and the commutant $\mathbb{C} = \mathrm{span}(\mathbf{1}, J_Y)$ of $\mathfrak{u}_W$; p1-41 computes them. □

Remark 8.24 (The fourth rung, in dimensions; fixed)**.** At $n = 4$ the internal signature is $(11, 9)$. The stabiliser of $E_2E_3E_8E_9$ in $\mathfrak{so}(11, 9)$ has dimension 126 with compact part 63, the $n = 4$ row of Proposition 8.60. The family $F_4 = \widetilde{\mathrm{Gr}}_4(\mathbb{R}^{11})$ has dimension $91 - 63 = 28$ (p1-43). All 36 bivectors of V_- commute with every point, so any colour algebra in $\mathfrak{so}(V_-)$ passes the colour test everywhere. At Π_A a $\mathfrak{u}(1)$ on a plane inside A fixes a component of dimension $14 = \dim \widetilde{\mathrm{Gr}}_2(\mathbb{R}^9)$, and one on a plane orthogonal to A fixes one of dimension $20 = \dim \widetilde{\mathrm{Gr}}_4(\mathbb{R}^9)$. $\mathfrak{so}(4)_W$ fixes a component of dimension $12 = \dim \widetilde{\mathrm{Gr}}_4(\mathbb{R}^7)$ through Π_A, so at $n = 4$ the Pati–Salam fixed points are no longer isolated (p1-44). In general $\dim F_n = 4(3n - 5)$, and a $\mathfrak{u}(1)$ on one positive plane fixes components of dimensions $2(3n-5)$ and $4(3n-7)$: (2), (8, 8), (14, 20) at $n = 2, 3, 4$ (p1-45).

8.2.6 The answer

(a) *The family.* The $n \geq 3$ analogue of the compact family is $F_n = \{\Pi_P : P \in \widetilde{\mathrm{Gr}}_4(V_+)\}$, the K_n-orbit of Π_A; at $n = 2$ it is the paper's S^4 (fixed). At $n = 3$ it is $\widetilde{\mathrm{Gr}}_4(\mathbb{R}^8)$, of dimension 16, a compact symmetric space [68]. Its metric is 64 times the trace metric of the Grassmannian (Lemma 8.17). The paper's S^4 is one of the inner Clifford spheres in it, and is not K-invariant (fixed).

(b) *Colour.* Every point of F_3 commutes with all of $\mathfrak{so}(6)_C \supset \mathfrak{su}(3)_C$, and so does every point of F_n with $\mathfrak{so}(V_-)$. The other involutive four-blade types never commute with $\mathfrak{su}(3)_C$ (fixed).

statement	class	checks
Proposition 8.6: second-rung data, K_2-orbit $= S^4 = \widetilde{\mathrm{Gr}}_4(\mathbb{R}^5)$, Q-fixed S^2, constant potentials	restated (Theorems 8.57 to 8.59, Proposition 8.60); orbit and fixed set fixed	p1-1 to p1-7
Lemma 8.7: four-blade census, even-grade rule, $\mathrm{vol}_+ = -\mathcal{Z}$	fixed	p1-8 to p1-10
Lemma 8.8: $\mathrm{vol}_P^\top = \mathrm{vol}_{\theta P}$, recovery of P	fixed	p1-13
Lemma 8.9: $\mathfrak{su}(3)_C$ irreducible on V_-	fixed	p1-15
Theorem 8.10: three K-orbits, stabilisers, spectra, colour test	fixed; $n = 3$ row of Proposition 8.60 restated	p1-11, p1-12, p1-14, p1-16, p1-42
Proposition 8.11: Clifford spheres, embedded second-rung family	fixed	p1-19, p1-32, p1-47
Proposition 8.12: duality $P \mapsto P^\perp$, invariance of mass matrices	fixed	p1-20, p1-46
Theorem 8.14: Π_A, Π_W neutral, $\mathfrak{h}_{\mathrm{PS}}$- and $\mathfrak{h}_{\mathrm{SM}}$-fixed, not rigid; literal embedding fails	fixed	p1-21 to p1-23, p1-35
Theorem 8.15: fluctuations, four doublets $(\mathbf{1},\mathbf{2})_{1/2}$, charges	fixed	p1-24 to p1-26
Proposition 8.16: $\Pi_A P_* = P_L$, $q_A = J$ on the block, $P_* T P_* = 0$, anomaly	fixed; (iv) restated (Proposition 6.14); anomaly condition inherited	p1-17, p1-18, p1-27, p1-28
Lemma 8.17: vector model, kinetic term, current, $V_{\mathfrak{h}}$	fixed	p1-29, p1-30
Theorem 8.18: $\mathcal{L}_\Pi$ and j_Π at Π_A and on spheres	fixed	p1-31, p1-32
Theorem 8.19: parallel vacua, compact adapted connection	fixed	p1-33, p1-35
Proposition 8.20: K-invariant couplings constant; zero sets	fixed; selection by K-invariant couplings obstructed	p1-34, p1-42
Theorem 8.21, Proposition 8.22: fixed sets, unbroken algebras	fixed	p1-36 to p1-40
Corollary 8.23: critical points and Hessians	fixed; criticality principle inherited	p1-38, p1-41
Remark 8.24: fourth rung	fixed; $n = 4$ row of Proposition 8.60 restated	p1-43 to p1-45
family reading of $\mathfrak{so}(4)_A$; electroweak pattern on the neutral component	interpretive	—
which charge and coupling select the vacuum	open	—

Table 8.1: Claims of Section 8.2 and their classes. The checks are the `PASS` lines printed by `p1_families.py`.

(c) *The vacuum.* Π_A is the analogue of Definition 8.56. It is Lorentz invariant, neutral under $\mathfrak{u}(1)_A$ and $\mathfrak{u}(1)_{\mathrm{em}}$, fixed by $\mathfrak{h}_{\mathrm{PS}}$ and $\mathfrak{h}_{\mathrm{SM}}$, moved by 16 compact bivectors, and parallel under any connection valued in its stabiliser. It is the point $n = c$ of the embedded second-rung family. The literal point $n = e$ is not neutral at $n = 3$ (fixed).

(d) *Data for a selecting coupling.* A coupling that distinguishes the points of F_3 is not K-invariant (obstructed for K-invariant couplings). Its zero set, if it is an alignment energy, is one of the fixed sets of Theorem 8.21. Its Hessian at the four Pati–Salam points lies in the 10- or 16-dimensional spaces of Corollary 8.23. Its fluctuations there are four doublets $(\mathbf{1},\mathbf{2})_{1/2}$ with a family index (fixed). No alignment energy of a subalgebra of $\mathfrak{k}$ separates Π_A, Π_{-A}, Π_W and Π_{-W}; a coupling to the chiral block does, since on the block D exchanges P_L and P_R (fixed). This is the counterpart on F_3 of Proposition 7.39 (interpretive).

(e) *The charge.* On P_* the construction's charge $\frac{1}{2}J_1$ is $\frac{1}{2}J$, universal, and anomalous together with $\mathfrak{su}(2)_L$ (fixed; restated). The electromagnetic charge B_{em} also leaves Π_A massless and has four charged complex fluctuations. Which charge, which coupling and which scale select the vacuum is not decided here (open).

8.3 Alignment by a gauged subalgebra

Section 8.2 found the compact family F_3 of vacuum projectors at the third rung, and showed that couplings invariant under the whole compact internal algebra $\mathfrak{k}$ are constant on it. This section asks what a gauged subalgebra $\mathfrak{h} \subset \mathfrak{k}$ does. The answers are as follows. The gauge field's mass matrix at a vacuum Π has trace $V_{\mathfrak{h}}(\Pi) = \sum_a g_a^2 \|[t_a, \Pi]\|^2$, its kernel is the unbroken subalgebra (fixed). Its zero set on F_3 is the set of vacua that leave $\mathfrak{h}$ unbroken (fixed). Which point is the vacuum is decided by the sign of the coefficient with which $V_{\mathfrak{h}}$ enters the vacuum energy. That sign is the hypothesis (H3) of Theorem 8.48; the one-loop gauge contribution supplies it only in a cutoff scheme (inherited; scheme dependent). Identifiers p2-n refer to the `PASS` lines of `p2_alignment.py`.

Conventions. As in Sections 8.2 and 8.4. On $\mathfrak{k}$ the invariant form here is $\langle X, Y\rangle = -\operatorname{Tr}(XY)/128$, in which the unit-rate rotations $\frac{1}{2}e_ie_j$ are orthonormal (p2-1). This differs by a factor 4 from the form $-\operatorname{Tr}(XY)/512$ of Section 8.4; norms below are quoted in the form of this section. In it the Standard-Model generators t_i^L and t_Y have squared norms $\frac{1}{2}$ and $\frac{5}{6}$, so the orthonormal basis of $\mathfrak{h}_{\mathrm{SM}}$ used below is $(\sqrt{2}\,t_i^L,\ \sqrt{6/5}\,t_Y)$, with couplings g_2 and g_Y. In the form of Section 8.4 every squared norm is divided by four ($\frac{1}{8}$ and $\frac{5}{24}$, the module metric of (7.4)), the orthonormal generators are doubled, and the entries of M^2 gain a factor four; thresholds expressed through ratios of couplings are unchanged. Physical masses are $M^2_{\mathrm{phys}} = f^2M^2$ with the kinetic scale f, and Hessians are quoted relative to $\operatorname{Tr}(d\Pi^2)$ unless stated otherwise.

Proposition 8.25 (The alignment energy is the trace of the mass matrix; fixed)**.** *For $\mathfrak{h}$ with orthonormal basis t_a and couplings $g_a > 0$, put $M^2_{ab}(\Pi) = g_ag_b \operatorname{Tr}([t_a,\Pi][t_b,\Pi])$. Then M^2 is symmetric and positive semidefinite, $\operatorname{Tr} M^2 = V_{\mathfrak{h}}(\Pi)$, and $\dim\ker M^2$ is the dimension of the stabiliser of Π in $\mathfrak{h}$. If the couplings form an $\mathrm{Ad}(H)$-invariant quadratic form, one coupling on each simple ideal of $\mathfrak{h}$ and any couplings on its abelian centre, $V_{\mathfrak{h}}$ is invariant under the group generated by $\mathfrak{h}$ (p2-2, p2-3). On F_3, $V_{\mathfrak{h}}(\Pi_P) = 64\sum_a g_a^2\|[\rho_+t_a, p_P]\|^2$ is computed on $V_+ = \mathbb{R}^8$, and the colour couplings drop out (p2-4). For $\mathfrak{h} = \mathfrak{u}(1)_A, \mathfrak{u}(1)_{\mathrm{em}}, \mathfrak{h}_{\mathrm{SM}}, \mathfrak{h}_{\mathrm{PS}}, \ldots$ it has the closed forms of p2-13 in the principal angles between P and the planes U_A, U_{em} and W.*

The invariance needs the hypothesis on the couplings. For arbitrary positive g_a it fails: with $\mathfrak{h} = \mathfrak{so}()\langle E_2, E_3, E_8\rangle$, $t = \frac{1}{2}(E_2E_3, E_2E_8, E_3E_8)$ and $g = (1,2,3)$, the two planes $\langle E_2, E_7, E_{13}, E_{14}\rangle$ and $\langle E_3, E_7, E_{13}, E_{14}\rangle$, related by the group, give $V_{\mathfrak{h}} = 640$ and 1280. The Standard-Model and Pati–Salam calculations below use one coupling per simple ideal.

Proposition 8.26 (Gauging all of $\mathfrak{k}$ selects nothing; fixed)**.** *With $\mathfrak{h} = \mathfrak{k}$ and $\mathrm{Ad}(K)$-invariant couplings, the mass matrix has spectrum $\{128g_8^2\ (\times 16),\ 0\ (\times 27)\}$ at every point of F_3, so every spectral function of M^2, the full one-loop potential included, is constant on F_3 (p2-6).*

So the selection of a point needs a proper subalgebra, as Proposition 8.20 already showed for K-invariant couplings in general.

Proposition 8.27 (Zero sets and Goldstone counts; fixed)**.** *For each $\mathfrak{h}$ listed here, whose fixed set on F_3 is nonempty, the global minimum of $V_{\mathfrak{h}} \geq 0$ on F_3 is its zero set, the $\mathfrak{h}$-fixed points of Theorem 8.21: for $\mathfrak{u}(1)_A$ and $\mathfrak{u}(1)_{\mathrm{em}}$ (with or without $\mathfrak{su}(3)_C$) the two components $P \supset U$ and $P \perp U$, for $\mathfrak{h}_{\mathrm{SM}}$, $\mathfrak{h}_{\mathrm{PS}}$ and $\mathfrak{h}_{\mathrm{PS}} \oplus \mathfrak{u}(1)_A$ the four points $\Pi_{\pm A}, \Pi_{\pm W}$ (p2-16: 60 random starts per algebra, random couplings, all reaching $V_{\mathfrak{h}} < 10^{-12}$ relative). At Π_A the Hessian of $V_{\mathfrak{h}}$ is positive semidefinite with kernel $[\mathfrak{c}_{\mathfrak{k}}(\mathfrak{h}), \Pi_A]$: $\mathfrak{u}(1)_A$ and $\mathfrak{u}(1)_{\mathrm{em}}$ leave eight Goldstone directions, and every $\mathfrak{h}$ containing $\mathfrak{su}(2)_L$ or $\mathfrak{so}(4)_W$ leaves none (p2-14, p2-15, p2-18).*

Proposition 8.28 (A second critical set of $\mathfrak{h}_{\mathrm{SM}}$; fixed)**.** *$P_0 = \operatorname{span}(E_2, E_3, E_7, E_{13})$ is a critical point of $V_{\mathfrak{h}_{\mathrm{SM}}}$, on a critical set $\widetilde{\mathrm{Gr}}_2(A) \times \mathbb{C}P^1$. Relative to $\operatorname{Tr}(d\Pi^2)$ its Hessian has eigenvalues 0 ($\times 6$, along the critical set), $\frac{3}{5}g_Y^2 - g_2^2$ ($\times 8$) and $\frac{12}{5}g_Y^2$ ($\times 2$), for the orthonormal basis of*

the conventions above (for the raw generators t_i^L, t_Y they read $\frac{1}{2}(g_Y^2 - g_2^2)$ and $2g_Y^2$). So it is a saddle for $g_Y < \sqrt{5/3}\, g_2$ and a local, non-global minimum for $g_Y > \sqrt{5/3}\, g_2$. There the unbroken algebra is $\mathfrak{su}(3)_C \oplus \mathfrak{u}(1)_{T_L^3} \oplus \mathfrak{u}(1)_Y$, of dimension 10 and containing $B_{\rm em}$, and only the $W^\pm$ are massive (p2-17, p2-20).

This metastable point is not electroweak breaking: the Z stays massless, and it exists only for $g_Y > \sqrt{5/3}\, g_2$, a ratio of couplings that is not physical at low energies and that the equal-coupling convention of Theorem 8.37(d) excludes: equal normalised couplings remain outside the metastable region (interpretive comparison).

Remark 8.29 (The sign hypothesis (H3); inherited, scheme dependent)**.** In the one-loop effective potential with a Euclidean momentum cutoff Λ, a real degree of freedom of mass m contributes $\frac{\Lambda^2 m^2}{32\pi^2} + \frac{m^4}{64\pi^2}(\log\frac{m^2}{\Lambda^2} - \frac{1}{2}) + O(m^6/\Lambda^2)$, and a massive vector in Landau gauge counts three (p2-5, inherited [26]). With the physical masses $M^2_{\rm phys} = f^2 M^2$ of the conventions above, the leading term is $\frac{3\Lambda^2}{32\pi^2}\operatorname{Tr} M^2_{\rm phys} = \frac{3\Lambda^2 f^2}{32\pi^2} V_{\mathfrak{h}}$, so in this scheme $\xi_1 = \frac{3\Lambda^2 f^2}{32\pi^2} > 0$ at order Λ^2, of mass dimension four. The coefficient is regulator dependent (it is absent in dimensional regularisation), the $m^4 \log m^2$ terms are not controlled, and the fermion loops of the matter enter with the opposite sign. The sign is proved for vector-like theories [281]. Here (H3) therefore remains a hypothesis; the curvature masses of M^2 are not pole masses, and no gauge-fixing or scheme dependence of the vacuum energy has been examined.

Remark 8.30 (The second rung; fixed)**.** At $n = 2$, with $\mathfrak{h} = \mathfrak{u}(1)_Q \oplus \mathfrak{so}(3)_{cde}$ on the family S^4, the global minimum is the uncharged two-sphere (photon massless, $\mathfrak{so}(3) \to \mathfrak{so}(2)$) for $g_3^2 < \frac{1}{2} g_Q^2$, and the charged circle (photon massive, $\mathfrak{so}(3)$ unbroken) for $g_3^2 > \frac{1}{2} g_Q^2$; at $g_3^2 = \frac{1}{2} g_Q^2$ the potential is constant (p2-8 to p2-10). Which charge is massless is decided by the ratio of the couplings, not by the algebra.

What this section establishes. Gauging a proper subalgebra that has fixed points on F_3, with a positive alignment coefficient, selects the points that leave it unbroken (fixed). The hypothesis is needed: for a subalgebra without fixed points on F_3 the zero set is empty, and the minimum set is not determined by this argument. The positivity of the coefficient is a hypothesis with one-loop support in a cutoff scheme only (inherited). The vacua selected by $\mathfrak{h}_{\rm SM}$ are the electroweak-symmetric points $\Pi_{\pm A}, \Pi_{\pm W}$. A further critical set of $V_{\mathfrak{h}_{\rm SM}}$ breaks $\mathfrak{su}(2)_L$ to $\mathfrak{u}(1)_{T_L^3}$; it breaks part of the electroweak group but leaves an extra massless neutral vector, so it is not the observed electroweak-breaking pattern (Proposition 8.28). Another critical point, for all positive couplings, is the plane $P_e = \langle E_2, E_3, E_9, E_{14}\rangle$, the $\theta = \pi/2$ member of the neutral deformation of Proposition 8.49: it is critical for the $SU(2)_L$ term and for the hypercharge term separately and leaves $\mathfrak{su}(3)_C \oplus \mathfrak{u}(1)_{\rm em}$ unbroken, with massive W and Z and a massless photon (with unit raw couplings and $f = 1$ its electroweak mass matrix has eigenvalues $0, 32, 32, 64$). It is a saddle: relative to $\operatorname{Tr}(d\Pi^2)$ its Hessian has eigenvalues -2 (once), 0 (nine times) and 1 (six times), and along the deformation $V_{\mathfrak{h}_{\rm SM}}(\Pi_\theta) = 128 c_{\rm SM} \sin^2\theta$, in the conventions of Section 8.4 (with the couplings of Section 8.3 the coefficient is $128(\frac{3}{2}g_2^2 + \frac{3}{10}g_Y^2)$), is maximal there. Random-start minimisation cannot certify the absence of saddles, so no complete list of critical sets is claimed. Electroweak breaking is not produced by alignment as a stable vacuum, in agreement with Proposition 8.49.

8.4 Which charge is the photon, and the residual choice

The second-rung construction gauges the charge $Q = \frac{1}{2} J_1$ and carries it, in the curved extension, by the normal $SO(2)$ connection Theorems 8.58 and 8.59. At the third rung that charge is $Q_A = \frac{1}{2} q_A = \frac{1}{2} E_2 E_3$, and the matter block of Section 6.2 carries a second candidate, the electric charge $B_{\rm em}$. This section decides which of the two is the photon on the family F_3 of Section 8.2,

and what is left to choose once a gauged subalgebra has aligned the vacuum. The results are as follows. The charge operator of Q_A is half the involution $\mathcal{U}$ that cuts out P_*, so on the block Q_A equals $\frac{1}{2}$ on every state: it is half the fermion number, and it is anomalous together with $\mathfrak{su}(2)_L$ (fixed). It is not electromagnetism on that matter, whatever the vacuum (obstructed). The electric charge leaves the four aligned vacua massless, has four charged complex fluctuations with the electron's charge magnitude, and acts on the block with the Standard-Model charges and without anomaly. With B_{em} gauged, the massless vector of the projector current is the candidate for the electromagnetic photon of the block (charge identification fixed; Maxwell term adopted; an electroweak stage, which the potential of Section 8.4.6 disfavours, is still missing), and the normal $U(1)$ is a separate abelian field (fixed). Of the three counts against the normal $U(1)$ in Section 13.1 only the sign count changes status, and a fourth count, the charge on the block, decides. After alignment by $\mathfrak{h}_{\mathrm{SM}}$ four vacua remain. A Klein group of symmetries of the projector sector permutes them simply transitively, so no alignment energy, kinetic term or current separates them (fixed). The overlaps with P_*, with the lepton projector and with the chirality halves do not separate them either (fixed). The overlap with the half P_L of the block on which $\mathfrak{su}(2)_L$ acts selects the pair $\{\Pi_A, \Pi_{-W}\}$, and the two members of the pair coincide on the half of the module that contains the block (fixed). A single point needs a coupling that sees the other half, for instance the overlap with P_{fam} (constructed). Identifiers p3-n refer to the `PASS` lines of `p3_identification.py` (Table 8.5).

Conventions. Definitions and notation are those of Section 8.2; in particular F_3, $\Pi_{\pm A}$, $\Pi_{\pm W}$, Q_A, B_{em}, $t_i^{L,R}$, t_Y, H_C, ρ_+ and vol_+ are as there. For a compact generator X commuting with J the charge operator is $\hat{X} := -JX$. It is symmetric, and $X = J\hat{X}$ acts as $i\hat{X}$ in the complex structure J Section 6.2. The inner product on $\mathfrak{k}$ is $\langle X, Y\rangle = -\operatorname{Tr}(XY)/512$. For it the unit blades e_ie_j have norm one, $|t_3^L|^2 = \frac{1}{8}$ and $|t_Y|^2 = \frac{5}{24}$. Orthonormal bases are $2\sqrt{2}\,t_i^L$, $\sqrt{24/5}\,t_Y$ and an orthonormal basis of $\mathfrak{su}(3)_C$ for $\mathfrak{h}_{\mathrm{SM}}$, with couplings g_2, g_Y, g_3, and the unit blades of $\mathfrak{so}(4)_W \oplus \mathfrak{so}(6)_C$ for $\mathfrak{h}_{\mathrm{PS}}$, with one coupling g. We write $U_A = \operatorname{span}(E_2, E_3)$, $A_0 = \operatorname{span}(E_8, E_9)$, $U_{\mathrm{em}} = \operatorname{span}(E_7, E_{13})$, $W_0 = \operatorname{span}(E_{14}, E_{15})$ and $P_{L,R} = P_*\frac{1}{2}(\mathbf{1} \mp \Gamma)$. For an oriented four-plane $P \subset V_+$ with oriented orthonormal frame $(f_1, \ldots, f_4)$, written as a 4×8 matrix F in the generator basis of V_+, the Plücker coordinates are $p_I(P) = \det F_I$ for the 70 four-subsets I of that basis, and e_I is the blade of I in increasing order. We write p_A and p_W for $I = A$ and $I = W$. The family projector is $P_{\mathrm{fam}} = \frac{1}{8}\prod_i(\mathbf{1} + X_i)$ Section 6.1, and $Q_v = P_L + \gamma^0 P_L \gamma^0$ Proposition 7.39.

Definition 8.31 (The lepton projector)**.** P_{lep} is the spectral projector of the colour charge $Q_c = -JH_C$ for the eigenvalue -3. With $s_k = J\,E_{9+k}E_{15+k}$, $k = 1, 2, 3$, one has $Q_c = s_1+s_2+s_3$ and $P_{\mathrm{lep}} = \frac{1}{8}\prod_k(\mathbf{1} - s_k)$.

On P_* the eigenvalue -3 of Q_c is the colour singlet Proposition 6.13. Proposition 8.51 shows that P_{lep} reproduces the row of Proposition 7.39, where the colour-singlet lepton projector is named but not written out.

8.4.1 Two charges on the matter

Proposition 8.32 (The construction's charge is half of $\mathcal{U}$; fixed, and restated)**.** *(i) $\hat{Q}_A = -JQ_A = \frac{1}{2}\mathcal{U}$ on $\mathbb{R}^{512}$, where $\mathcal{U} = -Jq_A$ is the third involution of P_*. Since $\mathcal{U}P_* = P_*$ and q_A commutes with $\mathcal{X}$, $\mathcal{Z}$ and $\mathcal{U}$,*

$$q_A P_* = JP_*, \qquad \hat{Q}_A P_* = \tfrac{1}{2}P_*.$$

The first identity is the statement $q_A|_{V_*} = J$ of Section 6.2 (p3-1).

(ii) On $V_* = \operatorname{im} P_*$ the operators Γ, $\hat{T}_L^3 = -Jt_3^L$, $\hat{Y} = -Jt_Y$, $\hat{Q}_{\mathrm{em}} = -JB_{\mathrm{em}}$, $\hat{Q}_A$ and $B-L = -JH_C/3$ commute. Their joint spectrum is that of Table 8.2 (p3-2).

field	Γ	$\hat{T}_L^3$	$\hat{Y}$	$\hat{Q}_{\mathrm{em}}$	$\hat{Q}_A$	$B-L$	real dim.	complex dim.
Q_L	-1	$\pm\frac{1}{2}$	$\frac{1}{6}$	$\frac{2}{3},\ -\frac{1}{3}$	$\frac{1}{2}$	$\frac{1}{3}$	$12+12$	$6+6$
L_L	-1	$\pm\frac{1}{2}$	$-\frac{1}{2}$	$0,\ -1$	$\frac{1}{2}$	-1	$4+4$	$2+2$
u_R	$+1$	0	$\frac{2}{3}$	$\frac{2}{3}$	$\frac{1}{2}$	$\frac{1}{3}$	12	6
d_R	$+1$	0	$-\frac{1}{3}$	$-\frac{1}{3}$	$\frac{1}{2}$	$\frac{1}{3}$	12	6
e_R	$+1$	0	-1	-1	$\frac{1}{2}$	-1	4	2
ν_R	$+1$	0	0	0	$\frac{1}{2}$	-1	4	2

Table 8.2: Joint eigenvalues on V_* (p3-2). Complex dimensions refer to the complex structure J; one internal Weyl component has four real, or two complex, dimensions. The columns $\hat{Y}$ and $\hat{Q}_{\mathrm{em}}$ restate Proposition 6.13. The column $\hat{Q}_A$ is constant.

(iii) On the Weyl block $\frac{1}{2}(\mathbf{1}-\Gamma)$ of the spectator action, of rank 256, $\hat{Q}_A$ takes the values $\pm\frac{1}{2}$, each 128 times. There $\hat{Q}_{\mathrm{em}}$ takes the values ±1 (16 times each), $\pm\frac{2}{3}$ (48), $\pm\frac{1}{3}$ (48) and 0 (32), the charges of the two families and two mirror families of Proposition 5.19 (p3-3).

Proof. (i) $\hat{Q}_A = -J\frac{1}{2}q_A = \frac{1}{2}\mathcal{U}$. As index sets, $\mathcal{X} \sim I \cup A \cup C$ with I the interior, $\mathcal{Z} \sim A \cup W$ and $\mathcal{U} \sim \{8,9\} \cup W \cup C$. The bivector q_A shares two indices with $\mathcal{X}$ and with $\mathcal{Z}$ and none with $\mathcal{U}$, so by equation (1.3) it commutes with all three. $\mathcal{X}$, $\mathcal{Z}$, $\mathcal{U}$ commute, so $\mathcal{U}P_* = P_*$, that is $-Jq_AP_* = P_*$; multiplying by J and using $J^2 = -\mathbf{1}$ gives $q_AP_* = JP_*$. (ii) The operators are the charge operators of commuting compact elements, together with Γ, which commutes with every internal bivector; p3-2 diagonalises them jointly on V_*. (iii) $\mathcal{U}$ commutes with Γ and is traceless on the Weyl block; p3-3 computes both spectra there. □

Corollary 8.33 (What the construction's photon couples to)**.** *The items are classed as fixed, restated or obstructed.*

(i) A gauge field coupled through Q_A couples to every Weyl fermion of the block with the same charge $\frac{1}{2}$, on both chiralities. In the complex structure J each of the sixteen Weyl fermions has fermion number one (Section 6.2). The coupling is therefore to $\frac{1}{2}F$, $F = 3B + L$, a universal vector charge (fixed).

(ii) No real c gives $\hat{Q}_AP_* = c\,\hat{Q}_{\mathrm{em}}P_*$, since $\hat{Q}_{\mathrm{em}}$ vanishes on the neutrinos and $\hat{Q}_A$ does not (p3-4). The statement involves only P_*, so the construction's charge is not the electric charge on the block for any vacuum projector (obstructed).

(iii) $\hat{Q}_{\mathrm{em}}$ has the same spectrum on P_L as on P_R, so its cubic, gravitational and mixed colour anomalies cancel between the chiralities, and $\mathrm{Tr}(P_L\hat{Q}_{\mathrm{em}}(\hat{T}_L^3)^2) = 0$ (fixed; the anomaly conditions are inherited [16, 50]). For $\hat{Q}_A$ the same traces vanish except $\mathrm{Tr}(P_L\hat{Q}_A(\hat{T}_L^3)^2) = 4$. This is the mixed $SU(2)_L^2\,U(1)_A$ anomaly of Proposition 6.14 at $a = \frac{1}{2}$, $b = 0$ (restated; p1-28, p3-5). With P_* as the matter and $\mathfrak{su}(2)_L$ gauged, $\mathfrak{u}(1)_A$ cannot be gauged (obstructed under these hypotheses).

Proof. (i) and (ii) read off Table 8.2. (iii) is p3-5. The anomaly of $\hat{Q}_A$ is that of fermion number in the Standard Model, which $SU(2)_L$ instantons violate [151]. □

The universality of Q_A on matter is therefore not a property of the vacuum. It is the factor $\frac{1}{2}(\mathbf{1}+\mathcal{U})$ of P_*: the block is cut out as an eigenspace of the construction's charge (Proposition 8.32(i)), and on the rest of the spectator Weyl block the same charge takes the value $-\frac{1}{2}$ (Proposition 8.32(iii)). At the second rung the coupled model and the vacuum share the charge $\frac{1}{2}J_1$ Section 8.5. At the third rung that shared charge is half the fermion number.

Remark 8.34 (What would make the normal connection admissible; fixed, with options open)**.** The normal $SO(2)$ connection of the curved extension Theorem 8.59 is the component of the

count	normal $U(1)$, generator Q_A	$\mathfrak{u}(1)_{\mathrm{em}}$, generator B_{em}	change
(i) a spinless test particle cannot see it Proposition 11.4 and Theorem 12.12	holds: it concerns the metric charge of the normal plane; the field couples through the spinor sector, at Π_0 to four complex scalars ϕ_w and to the block	not applicable, since B_{em} is not a metric charge; it couples through the spinor sector to four complex scalars ϕ_a and to the block, and a neutral spinless field does not see the ordinary photon either	none in substance (restated; interpretive reading)
(ii) the adapted generator does not commute with the physical generators Section 11.3	removed: $[Q_A, \Pi_0] = 0$, $D\Pi_0 = 0$, and the adapted correction is compact and commutes with the interior gammas Theorem 8.59, Theorem 8.19	removed in the same way	none at $n = 3$; removed already at $n = 2$ (fixed)
(iii) it gravitates with the wrong sign for the ambient Einstein–Hilbert action (Corollary 11.6 and Theorem 12.15)	holds for the Kaluza–Klein vector of the normal plane	not covered: the Maxwell term is adopted, and a positive one exists since $\langle B_{\mathrm{em}}, B_{\mathrm{em}}\rangle > 0$; read as a Kaluza–Klein vector, the plane U_{em} has the sign of time, like the normal plane	from forced to adopted (constructed); the Kaluza–Klein reading is interpretive
(iv) the charge on the block, Corollary 8.33	universal $\frac{1}{2}$, anomalous with $\mathfrak{su}(2)_L$	the Standard-Model charges, anomaly free	decides (obstructed for Q_A, fixed for B_{em})

Table 8.3: The counts against the normal $U(1)$ at the third rung (p3-4, p3-5). In row (iii), E_2, E_3, E_7, E_{13} square to $+\mathbf{1}$, as the time generator $\gamma^0 = E_1$ does, and the colour generators to $-\mathbf{1}$ (p3-4).

connection along Q_A (Theorem 8.37(c)). With P_* as the matter and $\mathfrak{su}(2)_L$ gauged it is anomalous, so before it can be treated as a physical gauge field it must be excluded, cancelled or decoupled. *Cancelled*: on the whole spectator Weyl block $\frac{1}{2}(\mathbf{1} - \Gamma)$ the mixed trace is 0, because the rest of the block, two mirror families with $\hat{Q}_A = -\frac{1}{2}$, contributes -4 (z2-2); cancellation occurs in that larger block, with the mirror matter that P_* removes. *Excluded*: the connection is kept flat or non-dynamical; the curved extension then loses its normal gauge field. *Decoupled*: a Stückelberg mass does not by itself cancel the anomalous Ward identity; decoupling needs a specified Green–Schwarz or Wess–Zumino completion, with its transformations and coefficients [308], which is not constructed. Cancellation by the mirror block is one available completion, not a statement excluding other anomaly-cancelling matter. None of the three is established here, and the normal field is not treated as physical.

8.4.2 The counts against the normal $U(1)$, and the dichotomy

Table 8.3 takes the three counts of Section 13.1 to the third rung and adds the count of Corollary 8.33. The four aligned vacua $\Pi_0 \in \{\Pi_{\pm A}, \Pi_{\pm W}\}$ commute with both Q_A and B_{em} (Theorem 8.14), so both charges can be compared at the same vacuum.

Row (iii) rests on Corollary 11.6 and Theorem 12.15: there a vector obtained from the ambient Einstein–Hilbert action, with the stated connection-metric ansatz, by reduction on a plane of the sign of time has a kinetic term of the wrong sign [74]. That theorem is about a metric origin of the field. An internal gauge field with an adopted Maxwell term is not covered by it. For a compact generator the invariant form is positive (Theorem 1.3, Table 8.3), so a Maxwell term of the healthy sign exists, as in ordinary Yang–Mills theory. If $\mathfrak{u}(1)_{\mathrm{em}}$ were itself obtained by reduction, its weak part would rotate the plane U_{em}, of the sign of time, and the same mechanism would apply to that part and not to the colour part $\frac{1}{6}H_C$ (interpretive). The same holds for Q_A, whose plane U_A also has the sign of time. Count (iii) therefore separates

the two fields only by their origin, and not by their generator.

The dichotomy at the end of Section 11.3 reads: either the normal $U(1)$ is not electromagnetic, or the paired-mode projector is not the observation projector. Section 8.5 resolves it in one direction: if the normal $U(1)$ is to be electromagnetism, the paired-mode projector must go. At the third rung count (iv) decides the first alternative for every projector, because it is a property of P_* through its factor $\mathcal{U}$: on the block matter the normal $U(1)$ is not electromagnetic (obstructed). The replacement of the paired-mode projector is still available, as Π_A or any point of the aligned set, and the charge that uses it is B_{em}. At $n = 3$ both alternatives of the dichotomy hold, and neither is a choice.

8.4.3 Identification at the aligned vacua

Proposition 8.35 (Joint fixed points of the two charges; fixed)**.** *The points of F_3 commuting with both Q_A and B_{em} form six components: the isolated points $\Pi_{\pm(U_A\oplus U_{\mathrm{em}})}$, the isolated points $\Pi_{\pm R}$ with $R = \mathrm{span}(E_8, E_9, E_{14}, E_{15})$, and two copies of $\widetilde{\mathrm{Gr}}_2(\mathbb{R}^4)$, of dimension 4,*

$$\{\Pi_{U_A\oplus P'} : P' \in \widetilde{\mathrm{Gr}}_2(R)\}, \qquad \{\Pi_{U_{\mathrm{em}}\oplus P'} : P' \in \widetilde{\mathrm{Gr}}_2(R)\}.$$

$\Pi_{\pm A}$ lie on the first of these, with tangent $A_0 \otimes W_0$, and $\Pi_{\pm W}$ on the second (p3-6).

Proof. By Theorem 8.21, Π_P commutes with Q_A iff $P \supset U_A$ or $P \perp U_A$, and with B_{em} iff $P \supset U_{\mathrm{em}}$ or $P \perp U_{\mathrm{em}}$. The four combinations give $P = U_A \oplus U_{\mathrm{em}}$; $P = U_A \oplus P'$ with $P' \subset (U_A \oplus U_{\mathrm{em}})^\perp = R$; $P = U_{\mathrm{em}} \oplus P'$; and $P = R$. An orientation of a direct sum is a pair of orientations of the summands up to a common reversal. This gives $\widetilde{\mathrm{Gr}}_2(R)$ in the middle cases and two points in the outer ones. Finally $A = U_A \oplus A_0$ and $W = U_{\mathrm{em}} \oplus W_0$ with $A_0, W_0 \subset R$. p3-6 computes the joint tangent kernels 0, 0, 4, 4 at representatives. □

Proposition 8.36 (Charges of the fluctuations; fixed)**.** *At each of $\Pi_{\pm A}$, $\Pi_{\pm W}$ the sixteen tangent directions decompose under the torus $\mathfrak{u}(1)_A \times \mathfrak{u}(1)_{\mathrm{em}}$ into complex lines of weights*

$$U_A \otimes U_{\mathrm{em}} :\ (1,1) \oplus (1,-1), \qquad U_A \otimes W_0 :\ 2 \times (1,0), \qquad A_0 \otimes U_{\mathrm{em}} :\ 2 \times (0,1),$$

and four real neutral directions $A_0 \otimes W_0$ (p3-7). Eight real directions are charged under each $\mathfrak{u}(1)$, four under both and four under neither, in agreement with Theorem 8.15(iii),(iv).

Proof. $\rho_+(Q_A)$ is the rotation of U_A at unit rate and $\rho_+(B_{\mathrm{em}})$ the rotation of U_{em}; on $A \otimes W$ the first acts on the first factor and the second on the second factor. □

Theorem 8.37 (The photon of the aligned vacua; fixed, with the gauged generator a choice)**.** *Let $\Pi_0 \in \{\Pi_{\pm A}, \Pi_{\pm W}\}$. Let the internal connection carry a field A_μ along a compact generator $Q \in \mathfrak{k}$, acting on spinors by Q and on the projector by $\mathrm{ad}\, Q$, and adopt a Maxwell term.*

(a) For $Q = B_{\mathrm{em}}$ the four requirements on an electromagnetic photon of the block hold. (1) The generator is the electric charge: $\hat{Q}_{\mathrm{em}} = \hat{T}^3_L + \hat{Y}$ on V_*, with the values of Table 8.2. (2) $[B_{\mathrm{em}}, \Pi_0] = 0$, so the kinetic term gives no $A_\mu A^\mu$ term. (3) Charged fluctuations exist: the four complex scalars ϕ_a of Theorem 8.15(iii), with current $j^\mu_{\Pi,\mathrm{em}} = 128 f^2 \sum_a \mathrm{Im}(\bar{\phi}_a D^\mu \phi_a)$ (Theorem 8.18). (4) The matter carries the Standard-Model charges without anomaly (Corollary 8.33(iii)). Both currents have the same unit of charge: $B^2_{\mathrm{em}} = -\hat{Q}^2_{\mathrm{em}}$ on V_*, with value -1 on the electron, and $(\mathrm{ad}\, B_{\mathrm{em}})^2 = -1$ on the charged tangent directions (p3-8). The scalars of the projector current carry the electron's charge magnitude, and no normalisation is free.

(b) For $Q = Q_A$ requirements (2) and (3) hold and (1) and (4) fail (Corollary 8.33). The massless vector couples to $\frac{1}{2}F$ on the block and to four complex scalars ϕ_w of charge 1, twice the fermion charge (p3-8). It is not the electromagnetic photon (obstructed).

(c) *Q_A and B_{em} commute and are orthogonal. Q_A commutes with $\mathfrak{h}_{\mathrm{SM}}$ and $\mathfrak{h}_{\mathrm{PS}}$ and lies in the centraliser $\mathfrak{so}(4)_A$ of $\mathfrak{h}_{\mathrm{PS}}$ in $\mathfrak{k}$ (p3-4). The normal $SO(2)$ connection of Theorem 8.59, whose plane is the surplus pair E_2, E_3 of the first constituent, is the component of the connection along Q_A. It is a second abelian field, not the photon (fixed).*

(d) *All twelve generators of $\mathfrak{h}_{\mathrm{SM}}$ commute with Π_0, so with $\mathfrak{h}_{\mathrm{SM}}$ gauged no gauge boson is massive at Π_0, and $\mathfrak{u}(1)_{\mathrm{em}}$ is singled out only by an electroweak stage. The projector's own neutral directions provide one. At $\Pi_\theta = \mathrm{Ad}(\exp\frac{\theta}{2}E_8E_{14})\Pi_A$, $\theta \notin \pi\mathbb{Z}$, the mass matrix of $\mathfrak{h}_{\mathrm{SM}}$ has the eigenvalue 0 nine times (gluons and photon), m_W^2 twice and m_Z^2 once, with*

$$\frac{m_W^2}{m_Z^2} = \frac{1}{1+\frac{3}{5}\,g_Y^2/g_2^2},$$

and its massless electroweak combination is B_{em} (p3-9). At $g_Y = g_2$ the ratio is $\frac{5}{8} = 1-\frac{3}{8}$, the value $\cos^2\theta_W$ for the $\sin^2\theta_W = \frac{3}{8}$ of Theorem 7.34. That value reflects the normalisation $|t_3^L|^2 = \frac{1}{8}$, $|t_Y|^2 = \frac{5}{24}$ of the invariant form together with the assumption $g_Y = g_2$ for the orthonormalised generators (z2-6). It is the familiar unification-scale value. A physical prediction would need the scale at which the couplings are equal, the particle spectrum, the running down to the electroweak scale and the thresholds, none of which is given.

Proof. (a) (1) is Proposition 8.32(ii). (2) and (3) are Theorems 8.14, 8.15 and 8.18. At $\Pi_{-A} = \mathbf{1}-\Pi_A$ the tangents change sign and nothing else changes, and at $\Pi_{\pm W}$ Theorem 8.15(v) applies. For the unit: on V_*, $B_{\mathrm{em}} = J\hat{Q}_{\mathrm{em}}$ with J commuting and $J^2 = -\mathbf{1}$, so $B_{\mathrm{em}}^2 = -\hat{Q}_{\mathrm{em}}^2$; on the tangent, $[B_{\mathrm{em}}, T_{a,7}] = T_{a,13}$ and $[B_{\mathrm{em}}, T_{a,13}] = -T_{a,7}$ (Theorem 8.15). (b) p3-8 gives $(\mathrm{ad}\,Q_A)^2 = -1$ on $U_A \otimes W$ and $Q_A^2 = -\frac{1}{4}$ on V_*. (c) is p3-4 together with the form $\Omega = \Omega_{\mathrm{Lorentz}} + AQ$, $Q = \frac{1}{2}ab$, of the curved extension Theorem 8.59. (d) $\mathfrak{su}(3)_C$ acts trivially on V_+. In the vector model of Lemma 8.17, $p_\theta = p_{\mathrm{span}(E_2,E_3,E_9)} + uu^\top$ with $u = \cos\theta\,E_8 \pm \sin\theta\,E_{14}$. For $x \in \rho_+(\mathfrak{h}_{\mathrm{SM}})$, which kills A, $[x, uu^\top] = au^\top + ua^\top$ with $a = xu = \pm\sin\theta\,xE_{14}$, so the mass matrix is $128\sin^2\theta$ times the Gram matrix of the vectors $g_a\rho_+(t_a)E_{14}$. The vectors $\rho_+(t_i^L)E_{14}$ are orthogonal of length $\frac{1}{2}$, because the $2\rho_+(t_i^L)$ are anticommuting complex structures, and $\rho_+(t_Y)E_{14} = -\rho_+(t_3^L)E_{14}$, because B_{em} kills E_{14}. In the basis of the conventions the Gram matrix is $2g_2^2$ on the first two generators, and on $(2\sqrt{2}\,t_3^L, \sqrt{24/5}\,t_Y)$ it is $\left(\begin{smallmatrix} 2g_2^2 & -\sqrt{12/5}\,g_2g_Y \\ -\sqrt{12/5}\,g_2g_Y & \frac{6}{5}g_Y^2 \end{smallmatrix}\right)$, with eigenvalues 0 and $2g_2^2 + \frac{6}{5}g_Y^2$. p3-9 computes the full 12×12 matrix on $\mathbb{R}^{512}$. □

So at the third rung the gauge-field candidate for the photon of the aligned vacua is the $\mathfrak{u}(1)_{\mathrm{em}}$ component of the internal connection, and the normal $U(1)$ is a separate abelian field. What is established is a charge identification: a compact generator with the Standard-Model charges, anomaly free, which commutes with the vacuum and couples to charged scalar fluctuations with the electron's unit. What is adopted is the Maxwell term (positive, since the invariant form is positive on compact generators). Gauge redundancy, a conserved current, two physical polarisations, massless spin-one states and quantisation then follow as in ordinary Yang–Mills theory for the adopted action: inherited, not derived from the Clifford data. That the field is the physical photon also requires the electroweak stage, which Proposition 8.49 shows the present potential does not supply. On the block the normal-connection field along Q_A couples to half the fermion number, and with $\mathfrak{su}(2)_L$ gauged it cannot be gauged on that matter (Corollary 8.33). The construction's identification of its photon with the normal connection is therefore not compatible with the matter at $n = 3$ (obstructed). What the second rung established survives with B_{em} in place of Q_A: the massless criterion of Corollary 8.53 at a Lorentz-invariant vacuum, the charged-scalar current, and a curved extension in which the vacuum is parallel (Theorem 8.19). The theorem rests on the choice of the gauged generator, and B_{em} is fixed only once the Standard-Model embedding of Section 6.2 is adopted (constructed). The coupling of

the second abelian field and the values of f and of the gauge couplings are not decided (open). Theorem 8.37(d) shows that the projector can itself carry an electroweak stage: its neutral doublet directions break $\mathfrak{h}_{\mathrm{SM}}$ to $\mathfrak{su}(3)_C \oplus \mathfrak{u}(1)_{\mathrm{em}}$ with the tree-level relation of one doublet [279, 280]. Whether the vacuum moves there is decided by the potential, which Section 8.4.5 examines.

8.4.4 The residual choice

By Proposition 8.20 and Theorem 8.21, with $\mathfrak{h}_{\mathrm{SM}}$ or $\mathfrak{h}_{\mathrm{PS}}$ gauged with positive couplings, the alignment energy vanishes on F_3 exactly at Π_A, Π_{-A}, Π_W and Π_{-W}. This subsection determines what separates these four points.

Lemma 8.38 (Overlaps are Plücker functions; fixed)**.** *For every oriented four-plane $P \subset V_+$, $\mathrm{vol}_P = \sum_I p_I(P)\, e_I$, and for every operator M on $\mathbb{R}^{512}$*

$$\mathrm{Tr}(\Pi_P M) = \tfrac{1}{2}\,\mathrm{Tr}\, M + \tfrac{1}{2}\sum_I p_I(P)\,\mathrm{Tr}(e_I M).$$

The four-blade components $\mathrm{Tr}(e_I M)$ vanish for $M = P_*$, P_{lep} and $\frac{1}{2}(\mathbf{1} \pm \Gamma)$. The nonzero ones are $\mathrm{Tr}(e_A P_{\mathrm{fam}}) = 64$, $\mathrm{Tr}(e_A P_L) = -\,\mathrm{Tr}(e_W P_L) = 32$, their negatives for P_R, $\mathrm{Tr}(e_A Q_v) = -\,\mathrm{Tr}(e_W Q_v) = 64$ and $\mathrm{Tr}(e_A \Pi_A) = 256$ (p3-10).

Proof. The frame vectors are orthonormal, so their Clifford product equals their exterior product, $\sum_I \det F_I\, e_I$. The blades e_I are symmetric involutions with $\mathrm{Tr}(e_I e_J) = 512\,\delta_{IJ}$. For the components it suffices to list index sets. The eight products of $\mathcal{X}$, $\mathcal{Z}$, $\mathcal{U}$ have the index sets $\emptyset$, $I \cup A \cup C$, $A \cup W$, $\{8,9\} \cup W \cup C$ and their symmetric differences, none of which is a four-subset of V_+. With Γ, whose index set is all eighteen generators, the products include $\Gamma\mathcal{X} = \chi_W$ and $\Gamma\mathcal{X}\mathcal{Z} = -\chi_W \mathrm{vol}_+ = -\mathrm{vol}_A$, so $P_L = \frac{1}{16}(\mathbf{1}+\mathcal{X})(\mathbf{1}+\mathcal{Z})(\mathbf{1}+\mathcal{U})(\mathbf{1}-\Gamma)$ contains $\frac{1}{16}(\mathrm{vol}_A - \chi_W)$ and no other four-blade of V_+. The products of the s_k have index sets of sizes 12, 4 inside C, and 8 equal to $A \cup W$. Among the products of the X_i only $X_1 = \mathrm{vol}_A$ is a four-blade of V_+ (Proposition 8.51). γ^0 commutes with every e_I, which doubles the components of P_L in Q_v. □

Proposition 8.39 (The overlaps; fixed)**.** *(i) On all of F_3, $\mathrm{Tr}(\Pi P_*) = \mathrm{Tr}(\Pi P_{\mathrm{lep}}) = 32$ and $\mathrm{Tr}(\Pi \frac{1}{2}(\mathbf{1} \pm \Gamma)) = 128$ are constant, while*

$$\mathrm{Tr}(\Pi_P P_{\mathrm{fam}}) = 32(1+p_A) = \tfrac{1}{4}\,\mathrm{Tr}(\Pi_P \Pi_A), \qquad \mathrm{Tr}(\Pi_P P_L) = 16(1+p_A-p_W) = \tfrac{1}{2}\,\mathrm{Tr}(\Pi_P Q_v),$$

and $\mathrm{Tr}(\Pi_P P_R) = 16(1 - p_A + p_W)$ (p3-12).

(ii) $\mathrm{Tr}(\Pi P_* \Pi P_*) = \|P_* \Pi P_*\|^2 \le \mathrm{Tr}(\Pi P_*) = 32$, with equality iff $[\Pi, P_*] = 0$. The points of F_3 commuting with P_* are $\Pi_{\pm A}$ and $\Pi_{\pm W}$, which are isolated, and the two components $\{\Pi_{U_A \oplus W'}\}$ and $\{\Pi_{A_0 \oplus W'}\}$, $W' \in \widetilde{\mathrm{Gr}}_2(W)$, of dimension 4 (p3-12, p3-13).

(iii) At the four points the linear and quadratic overlaps are those of Table 8.4 (p3-11).

Proof. (i) is Lemma 8.38 with $\mathrm{Tr}\, M$ and the components listed there. (ii) P_* is a contraction, so $\|P_* \Pi P_*\|^2 \le \|\Pi P_*\|^2 = \mathrm{Tr}(P_* \Pi P_*) = 32$, with equality iff $(\mathbf{1} - P_*)\Pi P_* = 0$, that is iff $\Pi P_* = P_* \Pi P_*$ is symmetric, that is iff Π commutes with P_*. Every vol_P commutes with J, Γ and $\mathrm{vol}_+ = -\mathcal{Z}$, so Π_P commutes with P_* iff vol_P commutes with χ_W and with q_A. $\mathrm{Ad}(\chi_W)$ acts on V_+ as the reflection -1 on W, $+1$ on A. It fixes vol_P iff $P = (P \cap A) \oplus (P \cap W)$ with $\dim(P \cap W)$ even. With Theorem 8.21 for q_A this leaves $P = \pm A$, $P = \pm W$, and $P = A' \oplus W'$ with $A' \in \{U_A, A_0\}$ and W' a two-plane in W. p3-13 checks that the four points are isolated and that the components have dimension 4. (iii) is the computation p3-11. □

Lemma 8.40 (A comass inequality; fixed)**.** *For every oriented four-plane $P \subset V_+ = A \oplus W$, $|p_A(P)| + |p_W(P)| \le 1$, with equality iff $P \in \{\pm A, \pm W\}$.*

M	Π_A	Π_{-A}	Π_W	Π_{-W}	$\mathrm{Tr}(\Pi_P M)$ on F_3
P_*	$(32,32)$	$(32,32)$	$(32,32)$	$(32,32)$	32
P_{lep}	$(32,32)$	$(32,32)$	$(32,32)$	$(32,32)$	32
$\frac{1}{2}(\mathbf{1}\pm\Gamma)$	$(128,128)$	$(128,128)$	$(128,128)$	$(128,128)$	128
P_L	$(32,32)$	$(0,0)$	$(0,0)$	$(32,32)$	$16(1+p_A-p_W)$
P_R	$(0,0)$	$(32,32)$	$(32,32)$	$(0,0)$	$16(1-p_A+p_W)$
Q_v	$(64,64)$	$(0,0)$	$(0,0)$	$(64,64)$	$32(1+p_A-p_W)$
P_{fam}	$(64,64)$	$(0,0)$	$(32,16)$	$(32,16)$	$32(1+p_A)$

Table 8.4: Overlaps ($\mathrm{Tr}\,\Pi M$, $\mathrm{Tr}\,\Pi M\Pi M$) at the four aligned points, and the linear overlap on all of F_3 (p3-11, p3-12).

Proof. Let $M_A=(\langle a_i,f_j\rangle)$ and $M_W=(\langle w_i,f_j\rangle)$ for an oriented orthonormal frame (f_j) of P, so that $M_A^\top M_A+M_W^\top M_W=\mathbf{1}$. The two symmetric matrices commute, so in a common eigenbasis they are $\mathrm{diag}(\cos^2\theta_i)$ and $\mathrm{diag}(\sin^2\theta_i)$, with $\theta_i\in[0,\frac{\pi}{2}]$ the principal angles between P and A. Hence $|p_A|=\prod_i\cos\theta_i$ and $|p_W|=\prod_i\sin\theta_i$. For any two indices $i\neq j$,

$$|p_A|+|p_W|\le\cos\theta_i\cos\theta_j+\sin\theta_i\sin\theta_j=\cos(\theta_i-\theta_j)\le 1.$$

Equality for every pair forces all θ_i to equal one θ, and then $\cos^4\theta+\sin^4\theta=1-2\cos^2\theta\sin^2\theta=1$ forces $\theta\in\{0,\frac{\pi}{2}\}$, that is $P=\pm A$ or $P=\pm W$. p3-14 checks the identities at 300 random planes and finds the maximum 1 of p_A-p_W only at A and $-W$. □

Corollary 8.41 (The extremes of the separating overlaps; fixed)**.** *On F_3, $0\le\mathrm{Tr}(\Pi P_L)\le 32$, with the maximum attained exactly at Π_A and Π_{-W} and the minimum exactly at Π_{-A} and Π_W. Likewise $0\le\mathrm{Tr}(\Pi P_{\mathrm{fam}})\le 64$, with the maximum only at Π_A and the minimum only at Π_{-A}.*

Proof. By Proposition 8.39(i) and Lemma 8.40, $|p_A-p_W|\le 1$ with $p_A-p_W=1$ iff $P\in\{A,-W\}$ and $=-1$ iff $P\in\{-A,W\}$. Also $|p_A|\le 1$ with equality iff $P=\pm A$, since $|p_A|=\prod\cos\theta_i$. □

Proposition 8.42 (A Klein group of the projector sector; fixed)**.** *Let $\mathsf{C}(\Pi)=\mathbf{1}-\Pi$, let $\mathsf{D}(\Pi)=\frac{1}{2}(\mathbf{1}+\mathrm{vol}_+(2\Pi-\mathbf{1}))$ be the duality of Proposition 8.12, and let $\mathsf{E}=\mathsf{CD}$, so that $\mathsf{E}(\Pi)=\frac{1}{2}(\mathbf{1}+\mathrm{vol}_+)-\mathrm{vol}_+\Pi$.*

(i) $\{1,\mathsf{C},\mathsf{D},\mathsf{E}\}$ is a Klein four-group of K-equivariant involutions of F_3, with $\mathsf{E}(\Pi_P)=\Pi_{-P^\perp}$. It acts simply transitively on the four points: D exchanges Π_A and Π_W, C exchanges Π_A and Π_{-A}, and E exchanges Π_A and Π_{-W} (p3-16).

(ii) Each element preserves every mass matrix $\mathrm{Tr}([T,\Pi][T',\Pi])$, $T,T'\in\mathfrak{k}$, and, for connections valued in $\mathfrak{k}$ and the interior Lorentz algebra, the kinetic term and the current j_Π of every $Q\in\mathfrak{k}$ (p3-16).

(iii) E is the identity on the half $\mathcal{Z}=+1$, the range of $\frac{1}{2}(\mathbf{1}+\mathcal{Z})$, which contains P_*: $\mathsf{E}(\Pi)\frac{1}{2}(\mathbf{1}+\mathcal{Z})=\Pi\frac{1}{2}(\mathbf{1}+\mathcal{Z})$, and $P_*\mathsf{E}(\Pi)P_*=P_*\Pi P_*$. On that half $\Pi_A=\Pi_{-W}=\frac{1}{2}(\mathbf{1}-\chi_W)$ and $\Pi_W=\Pi_{-A}=\frac{1}{2}(\mathbf{1}+\chi_W)$, and there $t_i^L=\Pi_A\tau_i$ and $t_i^R=\Pi_W\tau_i$. On the half $\mathcal{Z}=-1$, $\Pi_A=\Pi_W$ and $\Pi_{-A}=\Pi_{-W}$ (p3-17).

Hence no alignment energy, kinetic term or current separates any two of the four points. No coupling that depends on Π only through $\Pi\frac{1}{2}(\mathbf{1}+\mathcal{Z})$ separates Π_A from Π_{-W}, or Π_{-A} from Π_W.

Proof. vol_+ is K-invariant, commutes with the interior gammas and with every Π_P, and squares to $\mathbf{1}$ (Lemma 8.7 and Proposition 8.12). C and D commute, since both $\mathsf{CD}(\Pi)$ and $\mathsf{DC}(\Pi)$ equal $\frac{1}{2}(\mathbf{1}-\mathrm{vol}_+(2\Pi-\mathbf{1}))$. (ii) For T in $\mathfrak{k}$ or in the interior Lorentz algebra, $[T,\mathsf{C}\Pi]=-[T,\Pi]$, $[T,\mathsf{D}\Pi]=\mathrm{vol}_+[T,\Pi]$ and $[T,\mathsf{E}\Pi]=-\mathrm{vol}_+[T,\Pi]$. Since vol_+ is orthogonal and commutes with $[T,\Pi]$, the traces are preserved. vol_+ is parallel for such connections, so D_μ transforms in the same way. (iii) On $\mathcal{Z}=+1$, $\mathrm{vol}_+=-\mathbf{1}$, so $\mathsf{E}(\Pi)=\Pi$ there. $\mathrm{vol}_A=\mathrm{vol}_+\chi_W$ equals $-\chi_W$ there and $+\chi_W$ on $\mathcal{Z}=-1$; with $t_i^L=\frac{1}{2}(\mathbf{1}-\chi_W)\tau_i$ Theorem 6.11 the remaining identities follow. □

On the half of the module that contains the matter, the vacuum Π_A is the support projector of $\mathfrak{su}(2)_L$, and Π_W that of $\mathfrak{su}(2)_R$. The residual choice is therefore, on the matter, the choice of which weak $\mathfrak{su}(2)$ acts inside the image of the vacuum. It is not a choice of spacetime chirality: the chirality halves have constant overlap, and the vector-like Q_v separates exactly as the chiral P_L does (Table 8.4).

Proposition 8.43 (Normalisers; fixed)**.** *(i) The fixed vectors of $\rho_+(\mathfrak{h}_{\mathrm{SM}})$ and of $\rho_+(\mathfrak{h}_{\mathrm{PS}})$ in V_+ are exactly A. A reflection of W conjugates $\mathfrak{su}(2)_L$ onto $\mathfrak{su}(2)_R$. Hence every element of K normalising $\mathfrak{h}_{\mathrm{SM}}$ fixes each of Π_A, Π_{-A}, Π_W, Π_{-W} (p3-18).*

(ii) $g = E_9E_{15} = \exp(\frac{\pi}{2}E_9E_{15})$ normalises $\mathfrak{h}_{\mathrm{PS}}$, exchanges $\mathfrak{su}(2)_L$ and $\mathfrak{su}(2)_R$, and sends $\mathrm{vol}_A \mapsto -\mathrm{vol}_A$, $\mathrm{vol}_W \mapsto -\mathrm{vol}_W$ and $\mathcal{X} \mapsto -\mathcal{X}$. So the normaliser of $\mathfrak{h}_{\mathrm{PS}}$ identifies Π_A with Π_{-A} and Π_W with Π_{-W}, by elements that do not preserve P_*. No element of either normaliser maps $\Pi_{\pm A}$ to $\Pi_{\pm W}$ (p3-18).

(iii) $g_{AW} = \prod_i \frac{1}{\sqrt{2}}(\mathbf{1} + a_i w_i)$, with $(a_i) = (E_2, E_3, E_8, E_9)$ and $(w_i) = (E_7, E_{13}, E_{14}, E_{15})$, is an element of $\mathrm{Spin}(8)$ with $\mathrm{Ad}(g_{AW})\mathrm{vol}_A = \mathrm{vol}_W$ and $\mathrm{Ad}(g_{AW})\mathrm{vol}_W = \mathrm{vol}_A$, which normalises neither $\mathfrak{h}_{\mathrm{SM}}$ nor $\mathfrak{h}_{\mathrm{PS}}$. Moreover $\mathrm{vol}_A\mathrm{vol}_W = \mathrm{vol}_+$ (p3-18).

Proof. (i) $\mathrm{Ad}(k)$ acts on F_3 through the rotation $R = \rho(k)$ of V_+ (Theorem 8.10). If $\mathrm{Ad}(k)$ normalises $\mathfrak{h}$, R preserves the fixed space A of $\rho_+(\mathfrak{h})$ and its complement W. For $\mathfrak{h}_{\mathrm{SM}}$, $R|_W$ normalises the derived algebra $\mathfrak{su}(2)_L$ of $\rho_+(\mathfrak{h}_{\mathrm{SM}})$. A rotation of W of determinant -1 is a rotation times a reflection, and it maps $\mathfrak{su}(2)_L$ onto $\mathfrak{su}(2)_R$; so $\det R|_W = 1$, hence $\det R|_A = 1$, and R fixes the oriented planes A and W. (ii) $\mathrm{Ad}(g)$ is the rotation by π in the plane (E_9, E_{15}), of determinant -1 on A and on W; it preserves W and V_-, hence $\mathfrak{h}_{\mathrm{PS}}$. Γ commutes with g and $\mathrm{Ad}(g)\chi_W = -\chi_W$. The last statement follows from (i), which applies to $\mathfrak{h}_{\mathrm{PS}}$ as well. (iii) $\mathrm{Ad}(\exp\frac{\pi}{4}a_iw_i)$ rotates the plane (a_i, w_i) by $\frac{\pi}{2}$; the signs and the failure to normalise are computed in p3-18. □

With $\mathfrak{h}_{\mathrm{SM}}$ gauged the four points are therefore distinct points of the gauged theory, although the projector sector cannot tell them apart. With $\mathfrak{h}_{\mathrm{PS}}$ gauged, Π_A and Π_{-A} are related by a left–right exchange of the Pati–Salam algebra, which the matter block breaks.

8.4.5 Robustness

Lemma 8.44 (Invariant couplings keep fixed critical points; inherited, with proof)**.** *Let $H \subset K$ be connected with Lie algebra $\mathfrak{h}$, let f be a smooth H-invariant function on F_3, and let $\Pi_0 \in F_3$ be fixed by H. Then df_{Π_0} is an H-invariant covector. If $T_{\Pi_0}F_3$ contains no nonzero $\mathfrak{h}$-invariant vector, Π_0 is a critical point of f. In particular Π_0 stays critical for $V_0 + tf$ for every t whenever V_0 is H-invariant too, and it stays a nondegenerate local minimum as long as the Hessian of $V_0 + tf$, an H-invariant symmetric form, stays positive definite.*

Proof. For $h \in H$, $f \circ h = f$ and $h\Pi_0 = \Pi_0$ give $df_{\Pi_0} \circ dh_{\Pi_0} = df_{\Pi_0}$. The metric $\mathrm{Tr}(d\Pi\, d\Pi)$ is K-invariant (Lemma 8.17), so the gradient is an H-invariant tangent vector, and for a connected group the invariant vectors are those killed by $\mathrm{ad}\,\mathfrak{h}$ [95, 115]. The Hessian at a critical point is invariant for the same reason. □

Proposition 8.45 (Invariant tangent vectors; fixed)**.** *The dimensions of the $\mathfrak{h}$-invariant subspaces of the tangent spaces at Π_A and Π_W, and so at Π_{-A} and Π_{-W}, whose tangent spaces are the same up to sign, are: $\mathfrak{h}_{\mathrm{SM}}$, $\mathfrak{h}_{\mathrm{PS}}$, $\mathfrak{su}(2)_L$ and $\mathfrak{u}(1)_Y \oplus \mathfrak{su}(3)_C$: 0; $\mathfrak{u}(1)_{\mathrm{em}} \oplus \mathfrak{su}(3)_C$: 8, the neutral components $A \otimes W_0$; $\mathfrak{u}(1)_A$: 8; $\mathfrak{u}(1)_A \oplus \mathfrak{u}(1)_{\mathrm{em}}$: 4 (p3-19).*

Corollary 8.46 (Matter couplings that respect $\mathfrak{h}$; fixed, with an inherited principle)**.** *Let the vacuum energy on F_3 be the sum of an alignment energy and of couplings to matter that respect*

$\mathfrak{h} \in \{\mathfrak{h}_{\mathrm{SM}}, \mathfrak{h}_{\mathrm{PS}}\}$*: functions of* Π *and of operators fixed by* $\mathfrak{h}$*, such as the overlaps of Proposition 8.39, Yukawa-type kernels commuting with* $\mathfrak{h}$*, or the fermion loop of a Dirac operator with* $\mathfrak{h}$*-invariant couplings Section 7.6. Such an energy is* H*-invariant, since* $f(\mathrm{Ad}(h)\Pi; M) = f(\Pi; \mathrm{Ad}(h^{-1})M)$*, and* $\Pi_{\pm A}$*,* $\Pi_{\pm W}$ *remain critical points of it. All linear overlaps of Table 8.4, including the one with* P_{fam}*, which does not commute with* $\mathfrak{su}(3)_C$*, are* $\mathfrak{h}_{\mathrm{SM}}$*-invariant functions on* F_3*, and their first variations vanish at the four points along all of* $\mathfrak{k}$ *(p3-20). With only* $\mathfrak{u}(1)_{\mathrm{em}} \oplus \mathfrak{su}(3)_C$ *gauged the eight neutral directions are invariant, and an invariant coupling can move the vacuum along the* $\mathfrak{u}(1)_{\mathrm{em}}$*-fixed component* $\widetilde{\mathrm{Gr}}_4(A \oplus W_0)$ *of Theorem 8.21.*

Proof. Lemma 8.44 with Proposition 8.45. The invariance of the overlaps is p3-20; for P_{fam} it also follows from $\mathrm{Tr}(\Pi_P P_{\mathrm{fam}}) = \frac{1}{4}\mathrm{Tr}(\Pi_P \Pi_A)$ and $[\mathfrak{h}_{\mathrm{SM}}, \Pi_A] = 0$. □

What invariant couplings can do is change the Hessian. Write $\Pi(x) = \mathrm{Ad}(e^X)\Pi_0$ with $X = \sum x_{aw} \frac{1}{2} aw$, as in Theorem 8.15, and

$$c_{\mathrm{SM}} = 6g_2^2 + \tfrac{6}{5}g_Y^2, \qquad c_{\mathrm{PS}} = 12g^2.$$

Proposition 8.47 (Hessians at the four points; fixed)**.** *At each of the four points* $V_{\mathfrak{h}_{\mathrm{SM}}}(\Pi(x)) = 128\, c_{\mathrm{SM}}|x|^2 + O(x^3)$ *and* $V_{\mathfrak{h}_{\mathrm{PS}}}(\Pi(x)) = 128\, c_{\mathrm{PS}}|x|^2 + O(x^3)$*. The overlaps expand as*

$$\mathrm{Tr}(\Pi(x)P_L) = \begin{cases} 32 - 8|x|^2 + O(x^3) & \text{at } \Pi_A,\ \Pi_{-W}, \\ 8|x|^2 + O(x^3) & \text{at } \Pi_{-A},\ \Pi_W, \end{cases}$$

$$\mathrm{Tr}(\Pi(x)P_{\mathrm{fam}}) = \begin{cases} 64 - 16|x|^2 + \cdots & \text{at } \Pi_A, \\ 16|x|^2 + \cdots & \text{at } \Pi_{-A}, \\ 32 + O(x^4) & \text{at } \Pi_{\pm W} \end{cases}$$

(p3-21). For $V = \xi_1 V_{\mathfrak{h}_{\mathrm{SM}}} - \xi_2 \mathrm{Tr}(\Pi P_L) - \xi_3 \mathrm{Tr}(\Pi P_{\mathrm{fam}})$*,* $\xi_1, \xi_2, \xi_3 \geq 0$*, every Hessian at the four points is a multiple of the identity: the coefficient of* $|x|^2$ *in* V *is* $128\xi_1 c_{\mathrm{SM}} + h$ *(the coordinate Hessian is twice it, and canonical scalar masses are the Hessian divided by the kinetic metric* $128 f^2$*), where* $h = 8\xi_2 + 16\xi_3$ *at* Π_A*,* $8\xi_2$ *at* Π_{-W}*,* $-8\xi_2$ *at* Π_W *and* $-8\xi_2 - 16\xi_3$ *at* Π_{-A}*. Hence* Π_A *is a nondegenerate local minimum unless* $\xi_1 = \xi_2 = \xi_3 = 0$*, and* Π_{-W} *is one iff* $\xi_1 > 0$ *or* $\xi_2 > 0$*. At* $\xi_1 = \xi_2 = 0 < \xi_3$ *the Hessian at* Π_{-W} *vanishes, and* Π_{-W} *is* not *a local minimum: along the curve that turns each* w_j *towards* a_j *through the same angle* t*, with orientation such that* $p_A = \sin^4 t$*,* $V = -32\xi_3(1+\sin^4 t) < V(\Pi_{-W})$ *(z2-3).* Π_W *is a nondegenerate local minimum iff* $\xi_2 < 16\xi_1 c_{\mathrm{SM}}$*, and* Π_{-A} *iff* $\xi_2 + 2\xi_3 < 16\xi_1 c_{\mathrm{SM}}$*; for the opposite strict inequalities all sixteen directions are unstable at once. On the boundaries the Hessian vanishes, and the quadratic test decides nothing.*

Proof. Since $[t_a, \Pi_0] = 0$, $\|[t_a, \Pi(x)]\|^2 = \|[[t_a, X], \Pi_0]\|^2 + O(x^3)$. $[t_a, X]$ corresponds to $x\,\rho_+(t_a)^\top$ on $A \otimes W$, so the sum is 128 $\mathrm{tr}(x\, C\, x^\top)$ with $C = \sum_a g_a^2 \rho_+(t_a)\rho_+(t_a)^\top$ on W. For $\mathfrak{h}_{\mathrm{SM}}$ the Casimir of $\mathfrak{su}(2)_L$ and $\rho_+(t_Y)^2 = -\frac{1}{4}$ give $C = (6g_2^2 + \frac{6}{5}g_Y^2)\mathbf{1}_W$, and $\mathfrak{su}(3)_C$ gives nothing; for $\mathfrak{h}_{\mathrm{PS}}$ each of the four directions of W lies in three unit blades, so $C = 12g^2\mathbf{1}_W$. The overlaps follow from Proposition 8.39(i) with $p_A(\Pi(x)) = 1 - \frac{1}{2}|x|^2 + O(x^4)$ near A, and $p_W = \det(O(x)) = O(x^4)$ there; near $\pm W$ the roles are exchanged. p3-21 checks the second derivatives $\mathrm{Tr}([X,[X,\Pi_0]]P)$ exactly and by finite differences. The stability conditions compare $128\xi_1 c_{\mathrm{SM}}$ with $8\xi_2$, and with $8\xi_2 + 16\xi_3$ at Π_{-A}. □

The sign of ξ_2 is a choice. For $\xi_2 < 0$ the roles of $\{\Pi_A, \Pi_{-W}\}$ and $\{\Pi_{-A}, \Pi_W\}$ are exchanged. The Hessians found here are the multiples of the identity in the sixteen-dimensional space of $\mathfrak{h}_{\mathrm{SM}}$-invariant forms of Corollary 8.23; a coupling with a component of the form $N \otimes \omega_Y$ would split the doublet directions.

8.4.6 The complete selection

The alignment result used is Proposition 8.20 and Theorem 8.21: for positive couplings $V_{\mathfrak{h}_{\mathrm{SM}}} \geq 0$ on F_3, with zero set exactly $\{\Pi_{\pm A}, \Pi_{\pm W}\}$.

Theorem 8.48 (Selection on F_3)**.** *Assume:*

(H1) the vacuum projector is a point of F_3 (constructed: Section 8.2.6);

(H2) $\mathfrak{h}_{\mathrm{SM}}$ is gauged, embedded as in Section 6.2, with couplings $g_3, g_2, g_Y > 0$ (constructed);

(H3) the sign hypothesis: the gauge fluctuations contribute $\xi_1 V_{\mathfrak{h}_{\mathrm{SM}}}$ to the vacuum energy with $\xi_1 \geq 0$ (inherited mechanism of vacuum alignment [123, 126]; its sign is proved for vector-like theories [281] and assumed here);

(H4) the vacuum couples to the block by $-\xi_2 \operatorname{Tr}(\Pi P_L)$ with $\xi_2 > 0$ (constructed, sign chosen);

(H5) optionally, it couples to the family block by $-\xi_3 \operatorname{Tr}(\Pi P_{\mathrm{fam}})$ with $\xi_3 > 0$ (constructed).

Let $V = \xi_1 V_{\mathfrak{h}_{\mathrm{SM}}} - \xi_2 \operatorname{Tr}(\Pi P_L) - \xi_3 \operatorname{Tr}(\Pi P_{\mathrm{fam}})$, with $\xi_3 = 0$ when (H5) is not assumed. Minima, and global minima, are taken over the sixteen-dimensional family F_3, not over the manifold $\mathcal{P}_{256} \cong \mathrm{Gr}_{\mathbb{C}}(64, 128)$ of all Lorentz-compatible rank-256 projectors, of real dimension 8192.

(a) Under (H1)–(H3) with $\xi_1 > 0$ the minimum set is $\{\Pi_{\pm A}, \Pi_{\pm W}\}$. At $\xi_1 = 0$, $V \equiv 0$ and every point of F_3 is a minimum. Overlaps with P_*, P_{lep} and $\frac{1}{2}(\mathbf{1} \pm \Gamma)$, linear or quadratic, take equal values at these points and do not separate them.

(b) Under (H1)–(H4) the global minimum value is $-32\xi_2$, attained exactly at Π_A and Π_{-W}, for every $\xi_1 \geq 0$. These two points coincide on the half $\mathcal{Z} = +1$ that contains P_*, and they are exchanged by E, which preserves every term of V except the one of (H5). Π_W and Π_{-A} remain critical, and are nondegenerate local minima iff $\xi_2 < 16\xi_1 c_{\mathrm{SM}}$ (with $\xi_3 = 0$; boundary cases as in Proposition 8.47).

(c) Under (H1)–(H5) the global minimum is Π_A alone, with value $-32\xi_2 - 64\xi_3$.

(d) At the selected vacuum all of $\mathfrak{h}_{\mathrm{SM}}$ is unbroken. Π_A is the electroweak-symmetric point, not the observed electroweak-broken vacuum: the generator B_{em} is massless there, but so are the $W^{\pm}$ and the Z, and a photon distinguished from them by mass needs the electroweak stage of Theorem 8.37(d), which the potential V disfavours (Proposition 8.49). The field along Q_A is a second abelian field.

Proof. (a) is Proposition 8.20 and Theorem 8.21 and Table 8.4. (b) $V \geq \xi_1 \cdot 0 - 32\xi_2$ by Corollary 8.41. Equality requires $\operatorname{Tr}(\Pi P_L) = 32$, and, when $\xi_1 > 0$, also $V_{\mathfrak{h}_{\mathrm{SM}}} = 0$; by the overlap maxima of Corollary 8.41 the first condition already fixes $\Pi \in \{\Pi_A, \Pi_{-W}\}$, where $V_{\mathfrak{h}_{\mathrm{SM}}} = 0$ holds as well, so the conclusion is the same at $\xi_1 = 0$. The coincidence and the role of E are Proposition 8.42; the overlap with P_L depends on Π only through $\Pi\frac{1}{2}(\mathbf{1} + \mathcal{Z})$. Stability is Proposition 8.47. (c) With $\xi_3 > 0$ equality requires in addition $\operatorname{Tr}(\Pi P_{\mathrm{fam}}) = 64$, which holds only at Π_A (Corollary 8.41). (d) is Theorem 8.14 and Theorem 8.37. p3-22 minimises V from random starts for several couplings, including $\xi_2 > 16\xi_1 c_{\mathrm{SM}}$ and $\xi_1 = 0$, and finds exactly these minima. □

Two features of the theorem could have come out otherwise. The overlap with P_* itself, which the matter block suggests first, is constant on F_3 and selects nothing, and the selection is made by the weak-chirality half P_L. Once that overlap is present the alignment energy is not needed for the selection; it only stabilises the other two critical points. The pair $\{\Pi_A, \Pi_{-W}\}$ is one projector on the half of the module that carries the matter, so for physics on the block the selected vacuum is unique (interpretive). A unique point of F_3 needs a coupling that sees the

other half. The overlap with P_{fam} is such a coupling, but P_{fam} carries no colour, and adopting it amounts to choosing Π_A, since $\operatorname{Tr}(\Pi P_{\text{fam}}) = \frac{1}{4}\operatorname{Tr}(\Pi\Pi_A)$ on F_3 (constructed). The rank-256 projector Π_A is the projector P_1 of the projector family Theorem 6.5, on which colour and a chiral doublet first coexist Theorems 6.7 and 6.9.

Proposition 8.49 (The potential opposes the electroweak deformation; fixed)**.** *Let* $\Pi_\theta = \operatorname{Ad}(\exp\frac{\theta}{2}E_8E_{14})\Pi_A$*, the neutral deformation of Theorem 8.37(d). Then* $\operatorname{Tr}(\Pi_\theta P_L) = 16(1+\cos\theta)$*,* $\operatorname{Tr}(\Pi_\theta P_{\text{fam}}) = 32(1+\cos\theta)$*,* $V_{\mathfrak{h}_{\text{SM}}}(\Pi_\theta) > 0$ *for* $\theta \notin \pi\mathbb{Z}$*, and*

$$V(\Pi_\theta) - V(\Pi_A) = \xi_1 V_{\mathfrak{h}_{\text{SM}}}(\Pi_\theta) + (16\xi_2 + 32\xi_3)(1-\cos\theta) \;\geq 0.$$

Under the positive coefficients of Theorem 8.48 every term favours the aligned, electroweak-symmetric point (z2-5).

Proof. $\operatorname{Ad}(\exp\frac{\theta}{2}E_8E_{14})$ turns E_8 into $\cos\theta\,E_8 + \sin\theta\,E_{14}$, so $p_A(\Pi_\theta) = \cos\theta$ and $p_W(\Pi_\theta) = 0$; apply Proposition 8.39(i). The mass matrix of Theorem 8.37(d) is nonzero for $\theta \notin \pi\mathbb{Z}$. □

So the electroweak stage that Theorem 8.37(d) exhibits is not reached by this potential. A transition to Π_θ needs a further contribution with the opposite sign along the neutral doublet direction, for instance a negative mass term for the doublets. No such term is derived here, and none is adopted: the mechanism of electroweak breaking is missing.

Remark 8.50 (Status of the selecting potential and of the scalar fluctuations)**.** (a) *Origin.* The alignment term rests on the sign hypothesis (H3). The overlaps with P_L and P_{fam}, and their signs, are adopted. Since $\operatorname{Tr}(\Pi P_{\text{fam}}) = \frac{1}{4}\operatorname{Tr}(\Pi\Pi_A)$ on F_3, the ξ_3-term encodes its target. Theorem 8.48 is therefore a conditional selection given an adopted potential, not a first-principles selection mechanism. (b) *The four doublets.* At each aligned point the sixteen real tangent directions form four complex doublets of $\mathfrak{su}(2)_L$ (Theorem 8.15). That is representation content. If all sixteen modes propagated and an electroweak stage removed three as longitudinal gauge modes, thirteen real scalars would remain. Their masses (with V above, all sixteen share one Hessian coefficient at Π_A), their mixing, their Yukawa couplings and any constraint that removes them are not determined, so no scalar phenomenology is claimed.

Proposition 8.51 (Restated data of the projectors; restated)**.** $X_1 = J^{(1)}J^{(2)} = \text{vol}_A$*, so* $P_1 = \Pi_A$*, with* $\dim\mathfrak{g}_{\text{phys}}(P_1) = 51$*,* $\dim\mathfrak{k}(P_1) = 27$ *and a three-dimensional annihilator, as in Theorem 6.5, and* $P_{\text{fam}} \leq \Pi_A$*. The projector* P_{lep} *of Definition 8.31 has rank* 64*, commutes with the interior gammas and with* J*, and with* P_**,* P_{fam} *and* Q_v *reproduces the alignment energies* 3*,* 5*,* 9*,* 5 *and stabilisers* 37*,* 23*,* 9*,* 25 *of Proposition 7.39 (p3-15).*

Proof. $X_1 = e_2^{(1)}e_3^{(1)}e_2^{(2)}e_3^{(2)} = E_2E_3E_8E_9$. The other statements are the computation p3-15. □

8.4.7 The answer

(a) *Which* $\mathfrak{u}(1)$ *is the photon.* At the third rung the photon of the aligned vacua is the $\mathfrak{u}(1)_{\text{em}}$ gauge field of the internal connection, generated by $B_{\text{em}} = -\frac{1}{2}E_7E_{13} + \frac{1}{6}H_C$. It is massless at $\Pi_{\pm A}$ and $\Pi_{\pm W}$, its projector current is carried by four complex scalars of the electron's charge magnitude, and on the block it couples with the Standard-Model charges without anomaly (fixed, given the adopted embedding, which is constructed).

(b) *The construction's charge.* $Q = \frac{1}{2}J_1$ becomes $Q_A = \frac{1}{2}E_2E_3$, whose charge operator is $\frac{1}{2}\mathcal{U}$. On the block it is half the fermion number, the same on every state, and anomalous with $\mathfrak{su}(2)_L$ (fixed; anomaly restated). It is not electromagnetism (obstructed). The normal $SO(2)$ connection that carries it is a separate abelian field (fixed), and its coupling is not decided (open).

(c) *The counts and the dichotomy.* Of the three counts against the normal $U(1)$, the spinless count is unchanged, the adapted-generator count is removed for both charges, and the sign count changes from forced to adopted; a fourth count, the charge on the block, decides. Both alternatives of the dichotomy of Section 11.3 hold at $n = 3$ (fixed; the Kaluza–Klein reading of the sign count is interpretive).

(d) *Which vacuum.* Alignment by $\mathfrak{h}_{\mathrm{SM}}$ leaves $\Pi_{\pm A}, \Pi_{\pm W}$ (fixed, Theorem 8.21). A Klein group of symmetries of the projector sector permutes them simply transitively; no alignment energy, kinetic term or current separates them (fixed). The weak-chirality overlap $-\xi_2 \operatorname{Tr}(\Pi P_L)$, $\xi_2 > 0$, selects $\{\Pi_A, \Pi_{-W}\}$, which is one projector on the matter half (fixed under constructed hypotheses). A single point requires a coupling off that half, such as $-\xi_3 \operatorname{Tr}(\Pi P_{\mathrm{fam}})$, which selects Π_A (constructed). With only alignment ($\xi_2 = \xi_3 = 0$) the four points need $\xi_1 > 0$, and at $\xi_1 = 0$ every point is a minimum. The selected Π_A preserves all of $\mathfrak{h}_{\mathrm{SM}}$: it is the electroweak-symmetric point, not the observed vacuum, and the adopted potential opposes the neutral electroweak deformation (Proposition 8.49).

(e) *Robustness.* $\mathfrak{h}_{\mathrm{SM}}$- or $\mathfrak{h}_{\mathrm{PS}}$-invariant couplings, including matter couplings, keep the four points critical, because their tangent spaces carry no invariant vector (fixed; principle inherited). They can only change the Hessians, which for the couplings considered are multiples of the identity; Π_W and Π_{-A} lose stability for $\xi_2 > 16\xi_1 c_{\mathrm{SM}}$ and $\xi_2 + 2\xi_3 > 16\xi_1 c_{\mathrm{SM}}$, and Π_{-W} is not a local minimum at $\xi_1 = \xi_2 = 0 < \xi_3$ (fixed; corrected).

(f) *Open.* The dynamical origin and sign of the overlap couplings, the scales f, g, ξ_1, ξ_2, ξ_3, the electroweak stage and its mechanism, the coupling of the second abelian field and how its anomaly is excluded, cancelled or decoupled (Remark 8.34), whether the half $\mathcal{Z} = -1$ is physical, and the origin of the Maxwell term of $\mathfrak{u}(1)_{\mathrm{em}}$ are open.

8.5 Sources, and whether the photon can be massless

Theorem 3.8 fixes the form of Maxwell's equations and leaves the source entirely open. The geometry of the curved arena supplies two candidate currents, and testing them gives a sharp negative result about the photon, together with the criterion that repairs it and a vacuum, at the second rung, that satisfies the criterion.

8.5.1 Two currents

On the slice Σ the oriented normal two-plane bundle of Section 11.1.2 has an $SO(2)$ connection A with curvature $F_N = dA$, an ordinary two-form on spacetime whose six components sit in the module of Theorem 3.3, and which is distinct from the compressed spinor curvature $\mathcal{F}_{\mathrm{obs}}$. Because the structure group is abelian, $dF_N = 0$ identically: the homogeneous half of Maxwell's equations is the Bianchi identity of the normal bundle, not an assumption. Setting $j^{\nu}_{\mathrm{geom}} := \lambda \nabla_\mu F_N^{\mu\nu}$ gives a current conserved identically, by the antisymmetry of F_N. On a flat base the connection $A = Bx\,dz$ has constant curvature and zero current, while $A = a(x)\,dz$ gives $F_N = a'(x)\,dx \wedge dz$ and $j^z_{\mathrm{geom}} = \lambda\, a''(x)$. This current is defined from F_N, so it is the current of a geometry that has been specified, not an independent source; its value is to show that continuity can be met by a geometric object.

The stronger candidate is independent of F. Fix a compact $\mathfrak{u}(1)$ generator Q, let $D_\mu \Pi = \nabla_\mu \Pi + A_\mu[Q, \Pi]$, and add a Maxwell term $-F_{\mu\nu}F^{\mu\nu}/4g_A^2$. Varying the kinetic term $\frac{f^2}{2} \operatorname{Tr}(D_\mu \Pi D^\mu \Pi)$ with respect to A_μ gives

$$j^\mu_\Pi = -f^2 \operatorname{Tr}(Q\,[\Pi, D^\mu \Pi]) = -f^2 \operatorname{Tr}([Q, \Pi]\, D^\mu \Pi), \tag{8.3}$$

the component of projector motion along the chosen charge rotation. It is conserved on the projector equation of motion, $D_\mu \mathcal{C}^\mu = 0$, rather than identically.

Theorem 8.52 (The paired-mode projector sources a massive vector)**.** *For $Q = J_N/2$ and the angle ansatz of Example 11.17, $\mathrm{Tr}([Q,\Pi]^2) = 1$ exactly, so $j^\mu_\Pi = -f^2(\partial^\mu\theta + A^\mu)$, and coupling it to the Maxwell term reproduces the Stückelberg mass $m_A^2 = g_A^2 f^2$. The mass and the current come from the same trace, so no choice of f gives a massless photon with this source.*

Proof. On the ansatz $\Pi = e^{\theta Q}\Pi_0 e^{-\theta Q}$ one has $D_\mu\Pi = b_\mu T$ with $b_\mu = \partial_\mu\theta + A_\mu$ and $T = [Q,\Pi]$, and $\mathrm{Tr}(T^2) = 1$ is check g3-7; it is constant along the orbit. By (8.3), $j^\mu_\Pi = -f^2 b^\mu \mathrm{Tr}([Q,\Pi]T) = -f^2 b^\mu \mathrm{Tr}(T^2) = -f^2 b^\mu$, and the kinetic term is $\frac{f^2}{2}\mathrm{Tr}(T^2)\, b_\mu b^\mu = \frac{f^2}{2} b_\mu b^\mu$, the Stückelberg Lagrangian of Example 11.17. With the Maxwell term, the gauge $\theta = 0$ gives the Proca mass $m_A^2 = g_A^2 f^2$. The current and the mass carry the same factor $f^2\,\mathrm{Tr}(T^2)$, so a nonzero current requires $f \neq 0$ and then gives a mass. □

Corollary 8.53 (The massless criterion)**.** *A massless vector requires a vacuum Π_0 with $[Q,\Pi_0] = 0$, so that the kinetic term generates no mass, and fluctuations π with $[Q,\pi] \neq 0$, so that charged modes still carry the current $-f^2\,\mathrm{Tr}(Q[\pi, D^\mu\pi]) + \cdots$. If $[Q,\Pi(x)] = 0$ is imposed everywhere, the current vanishes identically and nothing is sourced.*

8.5.2 The first rung

Theorem 8.54 (The criterion can be met)**.** *In $\mathrm{Cl}(3,3)$ with $Q = J_N/2$ (J_N of Lemma 11.2, which commutes with the four-plane $\langle \widehat{W}, e_4, e_5, e_6\rangle$ of the symmetric clock) the rank-four orthogonal projectors commuting with Q form $\mathrm{Gr}_{\mathbb{C}}(2,4)$, of real dimension eight, since $2Q$ is a complex structure. For a generic member Π_0,*

$$\dim\{[B,\Pi_0] : B \in \mathfrak{so}(3,3)\} = 14, \qquad \mathrm{rank}(\mathrm{ad}_Q\,|_T) = \dim[Q,T] = 8 \quad (T \text{ that span}),$$

so the vacuum is massless and not rigid; the mechanism is the Jacobi identity $[Q,[B,\Pi_0]] = [[Q,B],\Pi_0]$, with ad_Q of rank eight on $\mathfrak{so}(3,3)$. But only two of the eight charged directions are admissible: by Theorem 1.13 a variation $[B,\Pi_0]$ stays among the orthogonal projectors exactly when $[B + B^\top, \Pi_0] = 0$, that is, when B is compact modulo the stabiliser of Π_0, and of the fourteen orbit directions only five are compact-generated, of the eight charged ones only two.

Proposition 8.55 (Lorentz invariance makes the first rung rigid)**.** *With $\mathfrak{l}_{\mathrm{int}}$ the interior Lorentz algebra of $\langle \widehat{W}, e_4, e_5, e_6\rangle$, the commutant of $\{\mathfrak{l}_{\mathrm{int}}, Q\}$ in $\mathrm{End}(\mathbb{R}^8)$ has dimension four, with symmetric part spanned by $\mathbf{1}$ and ω. So the only rank-four orthogonal projectors commuting with both the interior Lorentz action and Q are $\frac{1}{2}(\mathbf{1} \pm \omega)$, the chirality projectors of Theorem 7.7. Every bivector commutes with ω, so these have a spin-orbit tangent space of dimension zero: they are rigid, and by Corollary 8.53 their current vanishes identically.*

The paired-mode projector fails the same Lorentz condition, since all six bivectors of $\langle \widehat{W}, e_4, e_5, e_6\rangle$ fail to commute with it (with the axial clock e_1 the pole operator e_1e_4 commutes with it, Chapter 11); non-invariance of the observation projector is a standing feature of the one-constituent construction. What Proposition 8.55 adds is that imposing Lorentz invariance together with $[Q,\Pi_0] = 0$ forces rigidity at the first rung. The tower relaxes this: a surplus rotation commuting with the interior generators of every constituent is unique at $n = 1$, while at $n = 2$ there are six, and ad_Q has rank 8 of 15 at $n = 1$ and 20 of 66 at $n = 2$ on the bivectors, so the higher rungs offer a genuine choice of which $\mathfrak{u}(1)$ to gauge.

8.5.3 A Lorentz-invariant charged vacuum at the second rung

Definition 8.56 (The second-rung vacuum)**.** In $\mathrm{Cl}(6,6)$ with the spectator interior action, whose internal complement has signature $(5,3)$, let a, b, c, d, e be orthonormal positive internal

generators: the surplus pair of the first constituent and the three positive directions of the second. Set

$$Q := \tfrac{1}{2}ab, \qquad H := abcd, \qquad \Pi_0 := \tfrac{1}{2}(\mathbf{1} + H).$$

Theorem 8.57 (Lorentz invariant, uncharged and not rigid)**.** *$H^2 = \mathbf{1}$, $H^\top = H$ and $\operatorname{Tr} H = 0$, so Π_0 is an orthogonal projector of rank 32 on $\mathbb{R}^{64}$. Moreover:*

(i) *$[Q, \Pi_0] = 0$, so the kinetic term gives the vacuum no gauge-boson mass;*

(ii) *Π_0 commutes with every interior gamma matrix, hence with the interior Lorentz algebra, and with the internal volume that supplies the complex structure;*

(iii) *it is moved by a compact internal bivector: for $B = \tfrac{1}{2}be$, $\pi = [B, \Pi_0] = -\tfrac{1}{2}acde$ and $[Q, \pi] = \tfrac{1}{2}bcde \neq 0$, with π symmetric and $\operatorname{Tr}(\pi^2) = 16$.*

Proof. H is a product of four positive generators, so $H^2 = (-1)^{4\cdot 3/2}\mathbf{1} = \mathbf{1}$ and $H^\top = H^2 H = H$, and it is a traceless blade. $Q = \tfrac{1}{2}ab$ shares two indices with H and commutes with it by (1.3). Each interior gamma shares no index with H, and $4 \cdot 1$ is even, so it commutes. $B = \tfrac{1}{2}be$ shares one index with H and anticommutes with it, so $[B, \Pi_0] = \tfrac{1}{2}[B, H] = BH = \tfrac{1}{2}\, be\, abcd$; since $b\, e\, a\, b = b^2\, e\, a = e\, a$, this is $\tfrac{1}{2}\, eacd = -\tfrac{1}{2}\, acde$. The blade $acde$ has four positive generators and is symmetric, and $\operatorname{Tr}(\pi^2) = \tfrac{1}{4}\operatorname{Tr}\mathbf{1} = 16$. Finally Q shares one index with π, so $[Q, \pi] = 2Q\pi = -\tfrac{1}{2}\, ab\, acde = \tfrac{1}{2}\, bcde$. □

Theorem 8.58 (The compact family and its current)**.** *Let $I_5 := abcde$ and $Z_i := I_5 v_i$ for $v_i \in \{a, b, c, d, e\}$. The Z_i are symmetric of grade four with $Z_i Z_j + Z_j Z_i = 2\delta_{ij}$, so for every unit $n \in \mathbb{R}^5$*

$$\Pi(n) := \tfrac{1}{2}(\mathbf{1} + I_5 n)$$

is a rank-32 orthogonal projector commuting with every interior gamma matrix. These form a compact family $S^4 = \mathrm{Spin}(5)/\mathrm{Spin}(4)$ with $\Pi_0 = \Pi(e)$. At the vacuum the tangents $T_i = [\tfrac{1}{2}v_i e, \Pi_0] = \tfrac{1}{2}I_5 v_i$ satisfy $\operatorname{Tr}(T_i T_j) = 16\delta_{ij}$ and

$$[Q, T_a] = -T_b, \qquad [Q, T_b] = T_a, \qquad [Q, T_c] = [Q, T_d] = 0 :$$

one charged complex scalar $\phi = n_a + i n_b$ and two neutral real scalars. With $D_\mu\phi = (\partial_\mu - iA_\mu)\phi$ the kinetic term and its current are

$$\mathcal{L}_\Pi = 8f^2\, D_\mu n \cdot D^\mu n, \qquad j^\mu_\Pi = 16 f^2 (n_a D^\mu n_b - n_b D^\mu n_a) = 16 f^2 \operatorname{Im}(\bar\phi D^\mu \phi),$$

the factor 16 being the real trace on the 64-dimensional module. At $n = e$, $\phi = 0$, so no $A_\mu A^\mu$ term is generated, while charged fluctuations carry current.

Executable check: a7-3 (exact symbolic or integer arithmetic on $\mathbb{R}^{64}$; Section 15.2); it also checks Theorem 8.57.

Theorem 8.59 (The curved extension)**.** *Reduce the internal connection so that a, b carry the normal $SO(2)$ connection and c, d, e are parallel, $\Omega = \Omega_{\text{Lorentz}} + AQ$. Both summands commute with Π_0, so*

$$D\Pi_0 = 0, \qquad [\mathcal{R}, \Pi_0] = 0$$

for arbitrary spacetime curvature and arbitrary normal curvature, the two conditions that the transport obstruction of Proposition 11.15 separates, here satisfied together by construction. Throughout the family $\mathcal{C}_\mu = [\Pi, D_\mu\Pi] = \tfrac{1}{4}[n, D_\mu n] = \tfrac{1}{2}\, n \wedge D_\mu n$, a compact internal bivector at every point, so the adapted connection stays spin-valued away from the vacuum as well.

Executable check: a7-4 (exact symbolic arithmetic; Section 15.2).

For a general projector $[\Pi, D\Pi]$ need not be a bivector: in a normal frame with unit clock w and negative s, t, the projector $\Pi = \tfrac{1}{2}(\mathbf{1} + X)$ with $X = (w + ws)/\sqrt{2}$ and $B = st$ gives $[\Pi, [B, \Pi]] = \tfrac{1}{2}(t - st)$, which has a grade-one part. The S^4 family avoids this by construction.

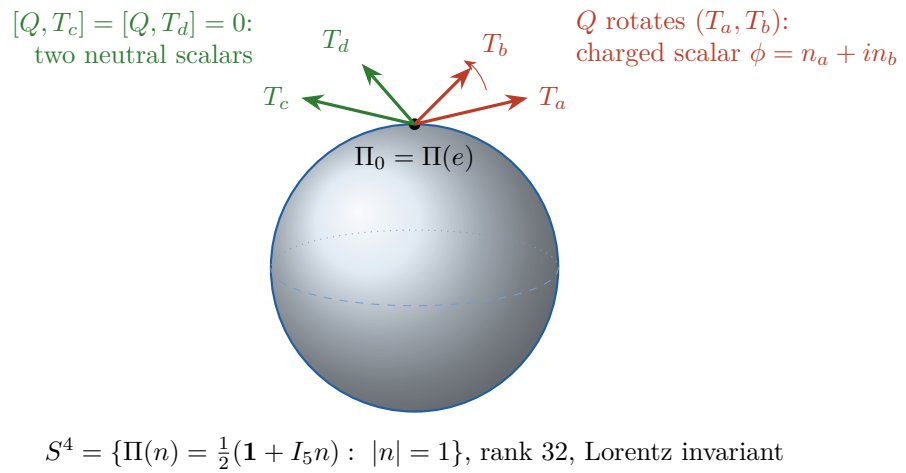


Figure 8.2: The compact family of second-rung vacua (Theorem 8.58). Every point is a Lorentz-invariant projector; at the vacuum Π_0, which commutes with the charge Q, the four tangent directions are generated by compact bivectors, two of them rotated into each other by Q and two neutral.

Proposition 8.60 (The vacuum and the matter)**.** *The stabiliser of* Π_0 *in the internal* $\mathfrak{so}(5,3)$ *is* $\mathfrak{so}(4) \oplus \mathfrak{so}(1,3)$*, of dimension* 12 *with compact part* $\mathfrak{su}(2) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(2)$*, which does not contain* $\mathfrak{su}(3)$*. The construction is therefore incompatible with colour at* $n = 2$*; it is compatible from the third rung on:*

rung	internal algebra	stabiliser of H	compact part
$n = 2$	$\mathfrak{so}(5,3)$, dim 28	$\mathfrak{so}(4) \oplus \mathfrak{so}(1,3)$, dim 12	9, no $\mathfrak{su}(3)$
$n = 3$	$\mathfrak{so}(8,6)$, dim 91	$\mathfrak{so}(4) \oplus \mathfrak{so}(4,6)$, dim 51	27, containing $\mathfrak{so}(6) \supset \mathfrak{su}(3)$
$n = 4$	$\mathfrak{so}(11,9)$, dim 190	$\mathfrak{so}(4) \oplus \mathfrak{so}(7,9)$, dim 126	63

At $n = 3$*, the rung of* P_**, the unbroken compact algebra is* $\mathfrak{so}(4) \oplus \mathfrak{so}(4) \oplus \mathfrak{so}(6)$ *and contains colour.*

A simple $\mathfrak{su}(3)$ has no nonzero homomorphism to $\mathfrak{su}(2)$, so it embeds in no sum of copies of $\mathfrak{su}(2)$. Executable check: a7-5 (exact blade parity; Section 15.2).

The second-rung vacuum settles that a vacuum which is simultaneously Lorentz invariant, uncharged, not rigid and moved by a compact generator exists, so the first-rung rigidity of Proposition 8.55 is a property of that rung and not a law; its fluctuation spectrum is one charged complex scalar and two neutral ones, its current is an ordinary charged-scalar current, and its curved extension has $D\Pi_0 = 0$ with a spin-valued adapted connection across the whole family. It does not settle which point of S^4 is selected, since a spectrum-only potential is constant on the family; whether j_Π is the electric current of a fermion, with which charge normalisation; the values of f and g_A; stability; or a gravitational action. It must be done at $n \geq 3$ to leave room for colour. And it uses the spectator action throughout: under the mixed action of Section 5.8, Π_0 is preserved only by the rotations about one axis, its commutators with the six mixed generators span a five-dimensional space, and the rank-16 Dirac block of the coupled model meets the range of Π_0 only in zero, with $P\Pi_0 P$ having eigenvalues $\frac{1}{4}$ and $\frac{3}{4}$ on it. What the coupled model and the second-rung vacuum share is the charge: $Q = \frac{1}{2}J_1$.

The dichotomy left at the end of Section 11.3 is thereby resolved in one direction. If the normal $U(1)$ is to be electromagnetism, the paired-mode projector must go, and Theorem 8.57 shows what can replace it. At one constituent, the paired-mode projector of Example 1.6 fails the photon test.

Remark 8.61 (The third rung)**.** At $n = 3$ the shared charge becomes $Q_A = \frac{1}{2}E_2E_3$, the generator of the normal $SO(2)$ connection, and on the chiral block of Section 6.2 its charge operator satisfies $-JQ_AP_* = \frac{1}{2}P_*$: it is half the fermion number. Its mixed trace with weak isospin, $\mathrm{Tr}(P_L(-JQ_A)(-Jt_3^L)^2) = 4$, is nonzero, so with $\mathfrak{su}(2)_L$ gauged it cannot be gauged on that matter. Within the block, the trace cancels on the whole spectator Weyl block, against mirror families that P_* removes; other anomaly-cancelling completions are not excluded. The electric charge $B_{\mathrm{em}} = -\frac{1}{2}E_7E_{13} + \frac{1}{6}H_C$ carries the Standard-Model charges without anomaly. It commutes with four aligned vacua $\Pi_{\pm A}, \Pi_{\pm W}$ of the compact family of vacuum projectors at $n = 3$, and it couples to four charged complex scalar fluctuations with the electron's unit of charge. On these counts the normal $U(1)$ is not electromagnetism at $n = 3$, and B_{em} is the charge of the photon candidate (charge identification fixed; the Maxwell term adopted). Vacuum alignment by a gauged $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ with a positive coefficient selects these four points, and adopted overlaps with the weak-chirality half of the block and with the family projector select Π_A. That selection is conditional on the adopted potential and its signs. Π_A leaves all of $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ unbroken, so it is the electroweak-symmetric point. The adopted potential increases along the neutral doublet deformation that would break $\mathfrak{su}(2) \oplus \mathfrak{u}(1)$ to $\mathfrak{u}(1)_{\mathrm{em}}$, so a photon distinguished by mass from the W and Z, and the mechanism of electroweak breaking, are not obtained. The proofs are in Sections 8.2 to 8.4 of this chapter, and the checks are the scripts `p1`–`p3` and `z2` (Section 15.2).

8.6 Synthesis: the photon at the third rung

8.6.1 What is established, adopted and missing

(1) *Charge identification (fixed).* On the chiral block P_*, $\hat{Q}_AP_* = \frac{1}{2}P_*$: the normal-plane charge is half the fermion number and has the mixed anomaly $\mathrm{Tr}(P_L\hat{Q}_A(\hat{T}_L^3)^2) = 4$. The electric charge $B_{\mathrm{em}} = -\frac{1}{2}E_7E_{13} + \frac{1}{6}H_C$ has the Standard-Model values and no anomaly (Proposition 8.32 and Corollary 8.33). The normal $U(1)$ is not electromagnetism on this matter for any vacuum (obstructed). Its anomaly cancels only together with the mirror families of the spectator Weyl block (Remark 8.34). Whether its connection is excluded, cancelled or decoupled is open.

(2) *Vacuum family and alignment (fixed, given hypotheses).* The compact family F_3 of vacuum projectors carries no selection under $\mathfrak{k}$-invariant couplings (Propositions 8.20 and 8.26). A gauged proper subalgebra with fixed points on F_3 and a positive alignment coefficient selects those fixed points (Propositions 8.25 and 8.27). The positivity is the hypothesis (H3), with one-loop support in a cutoff scheme only (Remark 8.29).

(3) *Selection of Π_A (conditional).* With $\mathfrak{h}_{\mathrm{SM}}$ gauged and $\xi_1 > 0$ the minima are $\Pi_{\pm A}, \Pi_{\pm W}$. At $\xi_1 = 0$ nothing is selected. Adopted overlaps $-\xi_2\,\mathrm{Tr}(\Pi P_L)$ and $-\xi_3\,\mathrm{Tr}(\Pi P_{\mathrm{fam}})$ with positive coefficients select $\{\Pi_A, \Pi_{-W}\}$ and then Π_A. Their origin and signs are adopted, and the ξ_3-term encodes its target (Theorem 8.48 and Remark 8.50). Local stability needs the conditions of Proposition 8.47: Π_{-W} is not a local minimum when $\xi_1 = \xi_2 = 0 < \xi_3$.

(4) *Electroweak symmetry (not obtained).* Π_A preserves all of $\mathfrak{h}_{\mathrm{SM}}$, so it is the electroweak-symmetric point. The neutral doublet direction Π_θ would give the one-doublet pattern $m_W^2/m_Z^2 = (1 + \frac{3}{5}g_Y^2/g_2^2)^{-1}$ with B_{em} massless. But the adopted potential increases along it by $\xi_1 V_{\mathfrak{h}_{\mathrm{SM}}} + (16\xi_2 + 32\xi_3)(1 - \cos\theta)$ (Proposition 8.49). Of the other critical sets found, one leaves the Z massless (Proposition 8.28) and the plane P_e, which has the observed pattern, is a saddle (Section 8.3); no complete list of critical sets is claimed. The electroweak mechanism is missing.

(5) *Photon dynamics (adopted, not derived).* With B_{em} gauged and a Maxwell term adopted,

a positive kinetic term exists because the invariant form is positive on compact generators. The field is massless at the aligned vacua and couples to the block and to four charged complex scalars with the electron's unit (Theorem 8.37). Gauge redundancy, a conserved current, two physical polarisations and quantisation are those of the adopted Yang–Mills action. A physical photon distinguished from $W^{\pm}$ and Z requires the missing electroweak stage.

(6) *Scalars and weak angle (representation content only).* The sixteen tangent directions at an aligned vacuum form four complex doublets. After an electroweak stage thirteen real scalars would remain, with masses, mixing and Yukawa couplings undetermined. The value $\sin^2\theta_W = \frac{3}{8}$ follows from the normalisation $|t_3^L|^2 : |t_Y|^2 = 3 : 5$ and the assumption $g_Y = g_2$. A prediction would need a matching scale, the spectrum, running and thresholds.

8.6.2 Two routes to the photon

Two routes to a Maxwell field have been examined, here and in the global manuscript, and their obstructions are kept.

- *The normal-plane vector.* Its connection is the $SO(2)$ of the surplus pair. As a Kaluza–Klein vector of a plane of the sign of time it gravitates with the wrong sign (ghost) relative to the ambient Einstein–Hilbert action, it does not couple to spinless matter (Corollary 11.6 and Theorem 12.15 and Proposition 11.4 and Theorem 12.12), and at $n = 3$ its charge is anomalous half fermion number on the block (obstructed on three counts).

- *The internal* $\mathfrak{u}(1)_{\mathrm{em}}$ *and the field-strength reading.* The internal connection along B_{em} with an adopted Maxwell term is healthy and carries the right charges (constructed, with the charge identification fixed). Separately, the field-strength reading $\mathbb{R}^{3,3} = \Lambda^2\mathbb{R}^{1,3}$ gives the positive Maxwell energy $|\mathbf{B}|^2 + |\mathbf{E}|^2$ for a field strength. That reading is kinematic, and connecting it with the internal $\mathfrak{u}(1)_{\mathrm{em}}$ connection, so that one field is both the gauge field of B_{em} and the field strength of the arena, is not done.

A non-preserving projector, a wrong-sign kinetic term or a curvature term of the wrong sign obstructs each construction where the cited theorems say so; none of these obstructions is removed here.

8.6.3 Discontinued work

The planned track p4, on the gluon if the obstruction persists at $n \geq 3$, is cancelled by the author's decision. No gluon identification, gluon-specific massless-vector, confinement or gluon mass investigation is part of this manuscript. The colour statements that are used, $\mathfrak{su}(3)_C \subset \mathfrak{so}(6)_C$, its trivial action on V_+, and the colour charges of the block, are the established ones of Section 6.2.

statement	class	checks
Definition 8.31, Proposition 8.51: P_{lep}, $P_1 = \Pi_A$, the table of Proposition 7.39	restated (Theorem 6.5, Proposition 7.39)	p3-15
Proposition 8.32: $\hat{Q}_A = \frac{1}{2}\mathcal{U}$, $q_A P_* = J P_*$, charge table, Weyl block	fixed; $q_A P_* = J P_*$ and the $\hat{Y}$, $\hat{Q}_{\mathrm{em}}$ columns restated	p3-1 to p3-3
Corollary 8.33: Q_A is $\frac{1}{2}F$ on the block, not the electric charge, anomalous; $\mathfrak{u}(1)_{\mathrm{em}}$ anomaly free	fixed; obstructed; anomaly restated; anomaly conditions inherited	p3-4, p3-5, z2-1
Remark 8.34: the normal connection must be excluded, cancelled (for instance in the larger spectator block, with mirror matter) or decoupled	fixed (cancellation trace); options open	z2-2
Table 8.3: the four counts and the dichotomy	restated (i); fixed (ii), (iv); constructed and interpretive (iii)	p3-4, p3-5
Proposition 8.35: joint fixed points of Q_A and B_{em}	fixed	p3-6
Proposition 8.36: torus weights of the fluctuations	fixed	p3-7
Theorem 8.37: the photon candidate is $\mathfrak{u}(1)_{\mathrm{em}}$ (charge identification); common charge unit; Q_A field separate; m_W^2/m_Z^2; $\sin^2\theta_W = \frac{3}{8}$ by normalisation and $g_Y = g_2$	fixed; gauged generator constructed; Maxwell term adopted; photon dynamics inherited, not derived; (b) obstructed	p3-4, p3-8, p3-9, z2-6
Lemma 8.38, Proposition 8.39: overlaps as Plücker functions, values, commutant of P_*	fixed	p3-10 to p3-13
Lemma 8.40, Corollary 8.41: comass inequality, extremes of the separating overlaps	fixed	p3-12, p3-14
Proposition 8.42: Klein group, E trivial on the matter half	fixed	p3-16, p3-17
Proposition 8.43: normalisers of $\mathfrak{h}_{\mathrm{SM}}$, $\mathfrak{h}_{\mathrm{PS}}$; g_{AW}	fixed	p3-18
Lemma 8.44: invariant couplings keep fixed critical points	inherited [95, 115], proof given	—
Proposition 8.45, Corollary 8.46: invariant tangent vectors, matter couplings	fixed; invariance of induced energies inherited	p3-19, p3-20
Proposition 8.47: Hessians and stability conditions; Π_{-W} not a minimum at $\xi_1 = \xi_2 = 0 < \xi_3$; boundary cases degenerate	fixed (corrected)	p3-21, z2-3, z2-4
Theorem 8.48: selection of $\{\Pi_A, \Pi_{-W}\}$, and of Π_A with (H5)	fixed under (H1), (H2), (H4), (H5) constructed and (H3) inherited; (a) needs $\xi_1 > 0$; conditional selection, not first-principles	p3-22
Proposition 8.49: the potential opposes the electroweak deformation	fixed; electroweak mechanism missing	z2-5
Remark 8.50: origin of the potential adopted; four doublets as representation content only	constructed; open	—
unique vacuum for physics on the block; Kaluza–Klein reading of the sign count	interpretive	—
origin and sign of the overlap couplings; scales; the second abelian field	open	—

Table 8.5: Claims of Section 8.4 and their classes. The checks are the `PASS` lines printed by `p3_identification.py`.

Chapter 9

Inter-Constituent Dynamics and the b-Field

The question of this chapter is what can act between constituents, and with what rates. It treats the connection part of the shift and the gauge transport of the two energies, the b-field between manifolds, the b-field up the ladder, the closure at two constituents, the b-field on the temporal triple with c on the spatial triple, and, in Section 9.7, the ladder table and what changes up the ladder. The connection part of the shift splits the binding (its part commuting with G) or destabilises it (its anticommuting part), and never creates it. A compact field between constituents is a one-body operator: it moves quanta and keeps their number, so it does not take one particle to two. It is removable between equal rates and, with the prescribed binding tensor kept fixed, non-removable exactly between unequal ones, and it changes no rate. From three constituents it carries time-odd plaquette invariants. Copy isospin is integral at two and three constituents under a common Lorentz action; doublets appear from four. The algebra is proved and checked. Reading a rate as a mass, $mc^2 = \hbar\kappa$, is an identification, and reading the plaquette as CP violation is an analogy; neither is a result.

Several letters carry more than one meaning in this chapter, each fixed by its context: b is an index and the connection; S_n is the spinor module and, in Theorem 9.62, the spatial volume; N is the number operator, the inertias N_k and a nilpotent part; H is the Noether energy, the copy Hamiltonian and the operator H; C is the conversion, the copy rotations $C_\pm$, C^{kl}, $\hat{C}$, C_v, C_b and a complex symmetric matrix; and M is the phase-space generator and a mass.

9.1 The connection part of the shift

A constant shift of the interior Lagrangian has two summands (Theorem 4.18): an η-antisymmetric one, in $\mathfrak{so}(3,3)$, and an η-self-adjoint one. Sections 4.3 to 4.5 were built on the second, the binding term κG, and met the first only as transport along a path (Proposition 4.14) and in Theorem 4.18. This section adds the first, a constant $b \in \mathfrak{so}(3,3)$, and asks what it does to the binding. Only b is a connection, and where it is gauged it is the value of a Yang–Mills potential along the worldline, the "b-field" in the notation of Yang and Mills [172]. In brief, the answers are as follows. For every b the field equation factorises covariantly and the binding term transforms as a tensor, so no gauge transformation creates it (Theorem 9.1). An admissible gauge potential is compact. Its part commuting with G splits the binding frequency into Zeeman lines that a moving frame removes (Proposition 9.5 and Theorem 9.6); a purely anticommuting part tears the binding (Theorem 9.7); a mixed one keeps every orbit bounded exactly when the two triples rotate at equal rates about axes that are not opposite, outside one exceptional ratio (Theorem 9.9). On its own a connection does not bind, and nothing in the interior dynamics ties the mass m to the rate κ (Proposition 9.11). Wherever the motion is bounded the zero-point energy cancels frequency by frequency (Theorem 9.12). At three

constituents the part of the Pati–Salam algebra compatible with the binding splits the frequency by $\mathfrak{su}(2)$ weights (Proposition 9.15). None of this supplies a scale. That the scale of a quantum Yang–Mills theory is the binding length is stated as a conjecture (Conjecture 9.17).

Units are $\hbar = c = 1$ unless a symbol is shown. The parameter τ has the dimension of a time, $\kappa = c/\ell$ of (2.2) and every eigenvalue of b are rates, and $\hbar\kappa$ is a quantum of interior energy.

Notation. $V = \mathbb{R}^{3,3}$, $F^\dagger := \eta^{-1}F^\top\eta$ is the η-adjoint, G is the binding root (4.14) and $\mathcal{K}$ the grading root of Proposition 2.3; on vectors $\mathcal{K} = \frac{1}{2}\operatorname{ad}(\sum_i e_i e_{i+3})$ and $G = -\mathcal{K}\eta$ (Lemma 2.4). For indices $a < b$ (here b is an index, not the connection) the element $R_{ab} \in \mathfrak{so}(3,3)$ is defined by $R_{ab}e_b = \eta_{bb}e_a$, $R_{ab}e_a = -\eta_{aa}e_b$, and zero on the other basis vectors; it is $\frac{1}{2}\operatorname{ad}(e_a e_b)$ on vectors. The generators L_i, K_i of $\mathfrak{l}_G$ (Theorem 3.3) act on $z = \mathbf{B} + i\mathbf{E}$ as $z \mapsto \mathbf{e}_i \times z$ and $z \mapsto i\,\mathbf{e}_i \times z$, so $K_i = GL_i$, the L_i are rotations and the K_i boosts, and

$$b = \textstyle\sum_i (w_i^{\mathrm{r}} L_i + w_i^{\mathrm{b}} K_i) \in \mathfrak{l}_G \quad \text{acts as} \quad z \mapsto w \times z, \qquad w = w^{\mathrm{r}} + i\,w^{\mathrm{b}} \in \mathbb{C}^3. \tag{9.1}$$

In particular $R_{12} - R_{45} = -L_3$. We write σ_j for the boost exchanging e_j and e_{j+3}, $[\mathbf{v}]_\times$ for the matrix of $\mathbf{x} \mapsto \mathbf{v} \times \mathbf{x}$ on $\mathbb{R}^3$, $\mathcal{E}_\mu(T) = \ker(T - \mu)$ for the eigenspaces of an operator on $V \otimes \mathbb{C}$, and $h_\eta(y, y') = \bar{y}^\top \eta\, y'$ for the Hermitian extension of η. The interior Lagrangian with a constant shift is

$$\mathcal{L} = \frac{m}{2}\,\eta(u, u), \qquad u = \dot{X} + SX, \qquad S = b + \kappa G, \qquad b \in \mathfrak{so}(3,3), \tag{9.2}$$

with $p = m\,\eta u$ and $\pi = \eta^{-1}p = mu$. For $b = 0$ it is the split-form Lagrangian (4.11).

9.1.1 Two summands, and which one is a connection

By Theorem 4.18 at $n = 1$, $\operatorname{End}(V) = \mathfrak{so}(3,3) \oplus \{F : F = F^\dagger\}$, of dimensions $15 + 21$, and by Proposition 2.3 the binding root lies in the second summand and the grading root in the first. The η-antisymmetric part of S is therefore b, and its self-adjoint part is κG. The decomposition is linear algebra, valid for any nondegenerate form; the next theorem gives it a dynamical meaning.

Theorem 9.1 (Covariant factorisation)**.** *Let $b \in \mathfrak{so}(3,3)$ be constant and $\nabla := \frac{d}{d\tau} + b$.*

(i) The Euler–Lagrange equations of (9.2) are $(\nabla - \kappa G)\pi = 0$ with $\pi = m(\nabla + \kappa G)X$, that is

$$(\nabla - \kappa G)(\nabla + \kappa G)X = 0 \qquad \Longleftrightarrow \qquad \nabla^2 X + \kappa^2 X + \kappa\,(\nabla_\tau G)\,X = 0, \qquad \nabla_\tau G := [b, G]. \tag{9.3}$$

(ii) Let $X = g(\tau)Y$ with $g(\tau) \in O(3,3)$. Then $(\frac{d}{d\tau} + S)(gY) = g(\frac{d}{d\tau} + S^g)Y$ with $S^g = g^{-1}Sg + g^{-1}\dot{g}$. The $\mathfrak{so}(3,3)$ part $b^g = g^{-1}bg + g^{-1}\dot{g}$ of S^g transforms as a connection, its self-adjoint part $\kappa\, g^{-1}Gg$ as a tensor, and $\frac{d}{d\tau}(g^{-1}Gg) + [b^g, g^{-1}Gg] = g^{-1}(\nabla_\tau G)g$.

Proof. (i) $\partial\mathcal{L}/\partial\dot{X} = p$ and $\partial\mathcal{L}/\partial X = S^\top p$, so $\dot{p} = S^\top p$, that is $\dot{\pi} = S^\dagger\pi = (-b + \kappa G)\pi$, which is $(\nabla - \kappa G)\pi = 0$. Moreover $\pi = m(\nabla + \kappa G)X$, $\nabla(GX) - G\nabla X = [b, G]X$ and $G^2 = -\mathbf{1}$. (ii) Differentiating $g^\top\eta g = \eta$ gives $(g^{-1}\dot{g})^\dagger = -g^{-1}\dot{g}$, and $F \mapsto g^{-1}Fg$ commutes with the η-adjoint, so it preserves both summands; the last identity is a direct differentiation. □

Part (ii) is inherited: it is the transformation law of a connection and of an endomorphism field in any gauge theory. Part (i) is fixed. It is the internal-connection, split-form version of the curved second-order equation at the end of Section 4.2, which, as there, adds to the case $b = 0$ only the term $\kappa(\nabla_\tau G)X$. That the binding part transforms as a tensor restates $G = G^\dagger$: no gauge transformation creates, removes or rescales it. The connection is the transport part of the shift, and no other part of the shift can be one.

Proposition 9.2 (Commuting and anticommuting parts)**.** $\mathrm{Ad}_G : b \mapsto GbG^{-1}$ *is an involution of* $\mathfrak{so}(3,3)$*, so* $\mathfrak{so}(3,3) = \mathfrak{l}_G \oplus \mathfrak{p}_G$*, where* $\mathfrak{l}_G = \{b : [b,G] = 0\}$ *has dimension* 6 *and* $\mathfrak{p}_G = \{b : bG = -Gb\}$ *has dimension* 9*. In the variable* z*,*

$$\mathfrak{l}_G = \{z \mapsto (P + iQ)z\} \cong \mathfrak{so}(3,\mathbb{C}), \qquad \mathfrak{p}_G = \{z \mapsto (C + i\Sigma)\bar{z}\},$$

with P, Q, C *real antisymmetric and* Σ *real symmetric. The Killing form* $4\,\mathrm{Tr}$ *has signature* $(3,3)$ *on* $\mathfrak{l}_G$*. On* $\mathfrak{p}_G$ *it is negative on the three directions* $\Sigma = 0$*, compact counter-rotations of the two triples such as* $R_{12} + R_{45}$*, and positive on the six boosts* $C = 0$*. Both* $\mathcal{K}$*, which acts as* $z \mapsto -i\bar{z}$*, and the pairing boosts* σ_j *lie in* $\mathfrak{p}_G$*.*

Proof. $G^{-1} = -G$ and $G^\dagger = G$ give $(GbG^{-1})^\dagger = -GbG^{-1}$, and $\mathrm{Ad}_G^2 = \mathbf{1}$. In $(\mathbf{B}, \mathbf{E})$ blocks, commuting with G means the form $\left(\begin{smallmatrix} P & -Q \\ Q & P \end{smallmatrix}\right)$ and anticommuting means $\left(\begin{smallmatrix} C & \Sigma \\ \Sigma & -C \end{smallmatrix}\right)$; $b^\dagger = -b$ gives the symmetry types. The Killing form of $\mathfrak{so}(3,3)$ is $4\,\mathrm{Tr}$, and on the two block forms $\mathrm{Tr}\, b^2 = 2\,\mathrm{Tr}(P^2 - Q^2)$, respectively $2\,\mathrm{Tr}(C^2 + \Sigma^2)$, which gives the signatures. □

This is fixed; the description of $\mathfrak{l}_G$ restates Theorem 3.1. The pairing boosts that tear the binding at a threshold in Proposition 4.14 are anticommuting, and so is the grading root.

Gauge admissibility. A gauge-admissible potential is compact in the sense fixed after Corollary 1.4: its generators are unitary for a positive form on the module (Theorem 1.3). It therefore takes values in the compact part $\mathfrak{so}(3) \oplus \mathfrak{so}(3)$ of $\mathfrak{so}(3,3)$, spanned by the R_{ab} with a, b in the same triple. On the two triples,

$$b = \mathrm{diag}(A + D,\, A - D), \qquad A = [\mathbf{a}]_\times, \quad D = [\mathbf{d}]_\times, \tag{9.4}$$

so the temporal triple rotates about $\mathbf{a} + \mathbf{d}$ and the spatial triple about $\mathbf{a} - \mathbf{d}$. The diagonal part (A, A) lies in $\mathfrak{l}_G$, with $w = \mathbf{a}$ real in (9.1); the anti-diagonal part $(D, -D)$ lies in $\mathfrak{p}_G$, with $\Sigma = 0$ (it acts as $z \mapsto \mathbf{d} \times \bar{z}$). The boost-valued connections used below (K_i, σ_j, $\mathcal{K}$, the Σ-part of $\mathfrak{p}_G$) are metric connections in the sense of Proposition 4.14, legitimate in (9.2), but they are not gauge fields. What a compact b does pointwise to the undivided energy and to the rest energy is Proposition 9.18; here it acts through the interior dynamics.

9.1.2 The phase-space generator and a criterion for boundedness

Lemma 9.3 (The phase-space generator)**.** *On* (X, π) *the flow of* (9.2) *is generated by*

$$M = \begin{pmatrix} -S & \mathbf{1}/m \\ 0 & S^\dagger \end{pmatrix}, \qquad S^\dagger = -b + \kappa G,$$

since $\dot{X} = \pi/m - SX$ *and* $\dot{\pi} = S^\dagger \pi$*. It is the Hamiltonian flow of the Noether energy* $H(X,p) = \frac{1}{2m} p^\top \eta^{-1} p - p^\top S X$*: in the canonical variables* (X, p) *the generator is* $\mathbb{J}\mathcal{H}$*, with* $\mathcal{H}$ *the Hessian of* H *and* $\mathbb{J} = \left(\begin{smallmatrix} 0 & \mathbf{1} \\ -\mathbf{1} & 0 \end{smallmatrix}\right)$*, and it is similar to* M*.* H *vanishes on the invariant Lagrangian subspace* $p = 0$*, and* $\det(\lambda - M) = \chi_S(\lambda)\chi_S(-\lambda)$ *with* $\chi_S(\lambda) = \det(\lambda - S)$*, because* $S^\dagger$ *is similar to* $S^\top$*.*

This is Hamiltonian bookkeeping, inherited. The classification below rests on the following criterion.

Theorem 9.4 (When the interior motion stays bounded)**.** *Let* S *be any constant real* 6×6 *matrix.*

(i) M *is semisimple if and only if* S *is semisimple and, for every* $\mu \in \mathbb{C}$*,* $\mathcal{E}_\mu(S^\dagger)$ *and* $\mathcal{E}_{-\mu}(S^\dagger)$ *are orthogonal for the bilinear extension of* η*.*

(ii) Every solution is bounded on $\mathbb{R}$ if and only if, in addition, $\operatorname{spec} S \subset i\mathbb{R}$*. Then (i) says that every eigenspace of $S^\dagger$ is totally isotropic for h_η. Equivalently, every momentum normal mode $\pi = \operatorname{Re}(y\, e^{i\omega\tau})$ with $\omega \neq 0$, superpositions within a degenerate frequency included, has period mean $\langle \eta(\pi,\pi)\rangle = \frac{1}{2}h_\eta(y,y) = 0$, and the static momenta ($\omega = 0$) span a totally η-null space.*

Proof. M is semisimple if and only if $-S$ (its restriction to $\pi = 0$) and $S^\dagger$ (the induced map on the quotient) are semisimple and M is similar to $\operatorname{diag}(-S, S^\dagger)$, which by Roth's theorem [132] (for an invariant proof see [309]) means that $-S\Xi - \Xi S^\dagger = -\mathbf{1}/m$ is solvable. With $-S = F_1\Lambda_1 F_1^{-1}$ and $S^\dagger = F_2\Lambda_2 F_2^{-1}$ diagonal, this holds if and only if $(F_1^{-1}F_2)_{ij} = 0$ whenever $(\Lambda_1)_i = (\Lambda_2)_j =: \mu$. A row $r^\top$ of F_1^{-1} with eigenvalue μ has $S^\top r = -\mu r$, and $S^\top = \eta S^\dagger \eta^{-1}$ puts $\eta^{-1}r$ in $\mathcal{E}_{-\mu}(S^\dagger)$; the condition is $r^\top y = \eta(\eta^{-1}r, y) = 0$ for $y \in \mathcal{E}_\mu(S^\dagger)$. (ii) A linear flow is bounded on $\mathbb{R}$ exactly when its generator is semisimple with imaginary spectrum, and then $\mathcal{E}_{-\mu}(S^\dagger) = \overline{\mathcal{E}_\mu(S^\dagger)}$. □

The theorem is fixed, although it holds for any nondegenerate η and any S. Its necessity half is the split virial identity (Theorem 4.7: a bound orbit has null momentum on average); what it adds is that the averaged virial condition, imposed mode by mode, is also sufficient. For $S = \kappa G$ it is the isotropy of Proposition 4.9.

9.1.3 Connections commuting with G: a Zeeman pattern without a scale

Proposition 9.5 (The commuting part is a moving frame)**.** *Let $b \in \mathfrak{l}_G$. Then $(\frac{d}{d\tau} + b + \kappa G)(e^{-b\tau}Y) = e^{-b\tau}(\dot{Y} + \kappa G Y)$. The exponents are therefore those of the binding, $\pm i\kappa$, shifted by the eigenvalues of $-b$, and with $p_X = e^{b^\top \tau}p_Y$ the energies in the two frames are related by*

$$H_{\text{lab}} = H_{\text{co}} - p_Y^\top b Y.$$

Proof. $e^{-b\tau}$ is an isometry that commutes with b and G; moreover $e^{b\tau}\eta^{-1}e^{b^\top\tau} = \eta^{-1}$ and $e^{b\tau}(b + \kappa G)e^{-b\tau} = b + \kappa G$. □

For compact b this is Larmor's theorem [83], inherited: a rotating frame trades the field for an angular velocity. It is also the first statement of Proposition 4.14. For a boost in $\mathfrak{l}_G$, $e^{-b\tau}$ is a uniformly boosted frame instead. $Q_b = p_Y^\top b Y$ is the Noether charge of the $\mathfrak{l}_G$ generator, so H_{lab} carries a chemical potential for it.

Theorem 9.6 (Connections commuting with G)**.** *Let $b \in \mathfrak{l}_G$ act as in (9.1).*

(i) $\Xi = -G/(2\kappa m)$ solves $-S\Xi - \Xi S^\dagger = -\mathbf{1}/m$, so M is semisimple if and only if S is. The argument uses only $G^2 = -\mathbf{1}$, $G^\dagger = G$ and $[b, G] = 0$.

(ii) $\chi_S(\lambda) = (\lambda^2 + \kappa^2)\big((\lambda - i\kappa)^2 + w\cdot w\big)\big((\lambda + i\kappa)^2 + \overline{w\cdot w}\big)$.

(iii) In the frame in which b is constant, every solution is bounded if and only if $e^{-b\tau}$ is bounded, that is, if and only if $w = 0$ or $w \cdot w > 0$. The frequencies are then κ and $|\kappa \pm s|$ with $s = \sqrt{w\cdot w}$. For $w\cdot w = 0 \neq w$ the zero-momentum flow grows like τ^2; for $w\cdot w \notin [0,\infty)$ it grows exponentially. For a gauge-admissible b, $w = \mathbf{a}$ is real and every solution is bounded.

(iv) For the rotation $b = \beta(R_{12} - R_{45}) = -\beta L_3$ the spectrum of M is $\pm i\kappa$, $\pm i(\kappa+\beta)$, $\pm i(\kappa-\beta)$, each double. For the boost $b = \beta K_3$ the exponents are $\pm\beta \pm i\kappa$, together with $\pm i\kappa$ from the factor $\lambda^2 + \kappa^2$ of (ii): complex rather than real, so the orbits spiral outward at the binding frequency. The growth is the boost factor $e^{-\beta K_3 \tau}$ of Proposition 9.5, and K_3 is not a gauge generator (Corollary 1.4).

Proof. (i) With $[b,G]=0$: $-(b+\kappa G)\Xi-\Xi(-b+\kappa G)=\frac{1}{2\kappa m}(bG-Gb+2\kappa G^2)=-\frac{1}{m}$. (ii) On $(V,G)\cong\mathbb{C}^3$, S is $z\mapsto i\kappa z+w\times z$, and $(w\times)^3=-(w\cdot w)(w\times)$ gives $w\times$ the eigenvalues $0,\pm is$; the real characteristic polynomial of a complex-linear map is its complex one times the conjugate. (iii) By Proposition 9.5, $X=e^{-b\tau}Y$ with Y a solution at $b=0$, bounded by (i) and (ii); conversely the zero-momentum solutions are $e^{-b\tau}e^{-\kappa G\tau}X_0$ with $e^{\kappa G\tau}$ bounded. And $e^{-b\tau}$ is bounded exactly when $w\times$ is semisimple with imaginary eigenvalues: s real, and not $w\cdot w=0\neq w$, where $N=w\times$ has $N^2\bar{w}=|w|^2w\neq 0=N^3$. For real w, $w\times$ is real antisymmetric. (iv) Here $s=\pm\beta$, respectively $\pm i\beta$; in the first case S is semisimple and (i) applies. □

The theorem is fixed, but its content deflates. Criterion (iii) is a property of the frame in which b is constant, not of the binding. In the co-moving frame of Proposition 9.5 the binding survives every commuting connection, as the remark after Proposition 4.14 anticipates (growth on one worldline proves nothing), and for admissible b the motion in the frame in which b is constant is bounded as well. In that frame, not in the co-moving one, a rotation lifts the degeneracy of the single frequency across the three pairs, noted at the end of Section 4.3, into the triplet $\kappa-\beta$, κ, $\kappa+\beta$. Read as a frame vector through the identification of the frame with $\mathfrak{l}_G$ (Theorem 4.20, with $L_i=e_i$ and $K_i=e_{i+3}$), the connection has the invariant $q=w\cdot w$; on this reading the lab-frame motion stays bounded exactly for a connection of magnetic type, $\mathbf{E}_b\cdot\mathbf{B}_b=0$ and $|\mathbf{B}_b|>|\mathbf{E}_b|$, while the binding itself survives every commuting connection, and an admissible one has $\mathbf{E}_b=0$. The Zeeman pattern labels the levels of the binding frequency by their $\mathfrak{l}_G$ charge. On one worldline it has no gauge-invariant content, and it is observable only against a reference that b does not rotate, as the Zeeman effect is observed against the laboratory; that is a reading, but a sharp one. A commuting b supplies a spectrum and no invariant scale, since β/κ is free.

9.1.4 Connections anticommuting with G, and screening

Theorem 9.7 (Connections anticommuting with G)**.** *Let $b\in\mathfrak{p}_G$ with $b\neq 0$.*

(i) $(\pm b+\kappa G)^2=b^2-\kappa^2$, *and* $M^2=\operatorname{diag}(b^2-\kappa^2,b^2-\kappa^2)+N$ *with* $N=\left(\begin{smallmatrix}0&-2b/m\\0&0\end{smallmatrix}\right)\neq 0$. *Hence M is never semisimple, and some solutions grow at least linearly.*

(ii) For the admissible counter-rotation $b=\beta(R_{12}+R_{45})$, which is $(D,-D)$ with $\mathbf{d}=-\beta\mathbf{e}_3$, $\det(\lambda-M)=(\lambda^2+\kappa^2)^2(\lambda^2+\kappa^2+\beta^2)^4$. *The exponents $\pm i\sqrt{\kappa^2+\beta^2}$ have algebraic multiplicity four and geometric multiplicity two, and $\pm i\kappa$ are semisimple. The bounded solutions are exactly those with $\pi_1=\pi_2=\pi_4=\pi_5=0$, that is, with no momentum on the support of b. The zero-momentum orbits are bounded and turn at the rate $\sqrt{\kappa^2+\beta^2}$ on the support.*

(iii) The grading root as a connection satisfies $(\nu\mathcal{K}+\kappa G)^2=(\nu^2-\kappa^2)\mathbf{1}$ on all of V.

Proof. (i) $\{b,G\}=0$ gives both squares, and M^2 has off-diagonal block $(S^\dagger-S)/m=-2b/m$. With $b^2-\kappa^2=Q_s+Q_n$ (Jordan–Chevalley), $M^2=\operatorname{diag}(Q_s,Q_s)+\big(\operatorname{diag}(Q_n,Q_n)+N\big)$ with commuting summands (Q_n is a polynomial in b), so by uniqueness of the decomposition M^2, and hence M, is semisimple only if $\operatorname{diag}(Q_n,Q_n)+N=0$, which is impossible for an off-diagonal $N\neq 0$. A non-semisimple generator of a Hamiltonian flow has a Jordan block at some exponent with non-negative real part. (ii) On the plane $\langle e_3,e_6\rangle$, b vanishes and $S=\kappa G$, semisimple with exponents $\pm i\kappa$, each double, by Theorem 9.6(i). On the support $\langle e_1,e_2,e_4,e_5\rangle$, $b^2=-\beta^2$, so by (i) $S^2=(S^\dagger)^2=-\omega^2$ with $\omega^2=\kappa^2+\beta^2$ and, on the corresponding eight-dimensional block of phase space, $M^2+\omega^2=N$ with $N^2=0$ and $\operatorname{rank}N=\operatorname{rank}b=4$. Hence the exponents there are $\pm i\omega$, each of algebraic multiplicity four, $\ker(M^2+\omega^2)=\ker N$ has dimension four and splits into $\ker(M\mp i\omega)$ of dimension two each, and the bounded data on the block are $\ker N=\{b\pi=0\}$. The zero-momentum flow is $e^{-S\tau}$ with $S^2=-\omega^2$ on the support. (iii) $\mathcal{K}^2=\mathbf{1}$ and $\{\mathcal{K},G\}=0$. □

Part (i) is an obstruction: no purely anticommuting connection, admissible or not, is compatible with bounded normal modes. In particular the Jordan blocks found after Proposition 4.14 for a pairing-plane boost $\nu\sigma_j$ below the threshold occur for every $\nu \neq 0$, below, at and above it. Parts (ii) and (iii) are fixed. The rate changes to $\sqrt{\kappa^2 + |\mathbf{d}|^2}$ for an admissible element and to $\sqrt{\kappa^2 - \nu^2}$ for a pairing boost (a metric connection, not a gauge field), and the momentum sector behaves as under the positive form of Theorem 4.5: exponents carried by Jordan blocks, imaginary for a compact element and for a boost below the threshold.

Proposition 9.8 (What a gauge transformation removes)**.** *Let $b = b_l + b_p \in \mathfrak{l}_G \oplus \mathfrak{p}_G$ be constant. Some $g(\tau) \in O(3,3)$ carries (b, G) to $(0, G')$ with G' constant if and only if $[b, G] = 0$. For $b \in \mathfrak{p}_G$, $e^{b\tau} G e^{-b\tau} = e^{2b\tau} G$. The gauge-covariant remainder is $\nabla_\tau G = [b, G] = 2b_p G$: along a worldline b_l is pure gauge and b_p is not.*

Proof. $b^g = 0$ forces $g = e^{-b\tau} g_0$, and then $G' = g_0^{-1} e^{b\tau} G e^{-b\tau} g_0$ has derivative $g_0^{-1} e^{b\tau} [b, G] e^{-b\tau} g_0$. If b anticommutes with G, then $Gf(b) = f(-b)G$ for every power series f. □

This is fixed. A connection on a line has no curvature, since the non-abelian term $[b_\mu, b_\nu]$ needs two directions (inherited); once the binding root is present, the invariant is $\nabla_\tau G$, which is exactly the anticommuting part. One might expect boundedness to force $b \in \mathfrak{l}_G$. It does not, and the admissible case shows it.

Theorem 9.9 (Compact screening)**.** *Let b be admissible, as in (9.4), and $\kappa > 0$.*

(i) If every solution is bounded, then $\mathbf{a} \cdot \mathbf{d} = 0$: the two triples rotate at equal rates, $|\mathbf{a} + \mathbf{d}| = |\mathbf{a} - \mathbf{d}|$.

(ii) Let $\mathbf{a} \cdot \mathbf{d} = 0$ and $\mathbf{a} \neq 0$. Then $\operatorname{spec} S = \{\pm i\omega_1, \pm i\omega_2, \pm i\omega_3\}$, where the ω_j are the roots of

$$f(\omega) = \omega^3 + \kappa\omega^2 - \left(\kappa^2 + |\mathbf{a}|^2 + |\mathbf{d}|^2\right)\omega - \kappa\left(\kappa^2 + |\mathbf{d}|^2 - |\mathbf{a}|^2\right),$$

and if $|\mathbf{a}|^2 \neq \kappa^2 + |\mathbf{d}|^2$ every solution is bounded, although $[b, G] = 2b_p G \neq 0$ when $\mathbf{d} \neq 0$.

(iii) If $\mathbf{a} \cdot \mathbf{d} = 0$, $\mathbf{d} \neq 0$ and $|\mathbf{a}|^2 = \kappa^2 + |\mathbf{d}|^2$, the vector $X_0 = (\mathbf{n}, (\mathbf{a} + \mathbf{d}) \times \mathbf{n}/\kappa)$ with $\mathbf{n} = \mathbf{a} \times \mathbf{d}$ is static, $SX_0 = 0$, with $\eta(X_0, X_0) = -2|\mathbf{d}|^2|\mathbf{n}|^2/\kappa^2 \neq 0$, and some solutions drift linearly.

For $\mathbf{a} = 0$ and $\mathbf{d} \neq 0$, Theorem 9.7 applies.

Bounded motion is not robust under compact perturbations: the necessary condition $\mathbf{a} \cdot \mathbf{d} = 0$ of (i) has codimension one in the six-dimensional space of compact connections. For $\kappa = 1$, $\mathbf{a} = (0, 0, 2)$ and $\mathbf{d} = (1, 0, \epsilon)$ one has $\operatorname{tr}(\eta(S^\dagger)^2) = -8\,\mathbf{a} \cdot \mathbf{d} = -16\epsilon$, and M is not semisimple at $\epsilon = \frac{1}{10}$ although S is real antisymmetric. A physical model has to explain what restricts the connection to this set.

Proof. $S = b + \kappa G$ is real antisymmetric, since b and G are; so S and $S^\dagger = -\eta S \eta = -b + \kappa G$ are normal with imaginary spectrum, and by Theorem 9.4 the motion is bounded if and only if $P_\mu \eta P_\mu = 0$ for the orthogonal spectral projectors P_μ of $S^\dagger$. (i) Then $\operatorname{Tr}(\eta(S^\dagger)^2) = \sum_\mu \mu^2 \operatorname{Tr}(P_\mu \eta P_\mu) = 0$. But $\{b, G\}$ is block-off-diagonal, $\operatorname{Tr}\eta = 0$ and $\operatorname{Tr}[\mathbf{v}]_\times^2 = -2|\mathbf{v}|^2$, so $\operatorname{Tr}(\eta(S^\dagger)^2) = \operatorname{Tr}(A + D)^2 - \operatorname{Tr}(A - D)^2 = -8\,\mathbf{a} \cdot \mathbf{d}$. (ii) Let ρ be the half-turn about $\mathbf{a}$ and $\hat{\rho} = \operatorname{diag}(\rho, \rho)$. Since $\rho\mathbf{a} = \mathbf{a}$ and $\rho\mathbf{d} = -\mathbf{d}$, $\hat{\rho} b \hat{\rho} = \operatorname{diag}(A - D, A + D) = G^{-1} b G$, so $G' := G\hat{\rho}$ commutes with b and G, hence with S and $S^\dagger$. Moreover $G'^2 = -\mathbf{1}$ and $G'^\dagger = G'$, so, as for G in Proposition 4.9, each eigenspace $\mathcal{V}_\pm$ of G' (eigenvalue $\pm i$) is totally h_η-isotropic: $-i\,h_\eta(y, y') = h_\eta(G'y, y') = h_\eta(y, G'y') = i\,h_\eta(y, y')$ on $\mathcal{V}_+$. In the $(\mathbf{B}, \mathbf{E})$ blocks $\mathcal{V}_+ = \{(\mathbf{x}, -i\rho\mathbf{x})\}$, and S acts there as $\mathbf{x} \mapsto (\mathbf{a} + \mathbf{d}) \times \mathbf{x} + i\kappa\rho\mathbf{x}$. After a common rotation of both triples, which commutes with G and η, to $\mathbf{a} = |\mathbf{a}|\mathbf{e}_3$ and $\mathbf{d} = |\mathbf{d}|\mathbf{e}_1$, one finds $\det(i\omega - [\mathbf{a} + \mathbf{d}]_\times - i\kappa\rho) = -if(\omega)$. So S has the eigenvalues $i\omega_j$ on $\mathcal{V}_+$ and $-i\omega_j$ on $\mathcal{V}_- = \overline{\mathcal{V}_+}$; since η exchanges $\mathcal{V}_\pm$, $S^\dagger$ has the same eigenvalues there. Every eigenspace of $S^\dagger$ splits along $\mathcal{V}_\pm$, and lies in one of them

unless $f(\omega) = f(-\omega) = 0$ for some ω. But $\frac{1}{2}[f(\omega) + f(-\omega)] = \kappa(\omega^2 - \kappa^2 - |\mathbf{d}|^2 + |\mathbf{a}|^2)$ and $\frac{1}{2}[f(\omega) - f(-\omega)] = \omega(\omega^2 - \kappa^2 - |\mathbf{a}|^2 - |\mathbf{d}|^2)$, so a common root forces $\mathbf{a} = 0$, or $\omega = 0$ and $|\mathbf{a}|^2 = \kappa^2 + |\mathbf{d}|^2$. (iii) With $\mathbf{u} = \mathbf{a}+\mathbf{d}$ and $\mathbf{v} = \mathbf{a}-\mathbf{d}$, $SX_0 = \big(\mathbf{u}\times\mathbf{n} - \mathbf{u}\times\mathbf{n},\ \mathbf{v}\times(\mathbf{u}\times\mathbf{n})/\kappa + \kappa\mathbf{n}\big) = 0$, because $\mathbf{v}\cdot\mathbf{n} = 0$ and $\mathbf{u}\cdot\mathbf{v} = |\mathbf{a}|^2 - |\mathbf{d}|^2 = \kappa^2$; and $\eta(X_0, X_0) = |\mathbf{n}|^2 - |\mathbf{u}|^2|\mathbf{n}|^2/\kappa^2$. So $\mathcal{E}_0(S^\dagger) = \eta\,\mathcal{E}_0(S)$ is not η-null, and Theorem 9.4 fails at $\mu = 0$. □

The theorem is fixed, and it is the case that matters for a gauge field. Its mechanism is a second binding root $G' = G\hat{\rho}$ commuting with S. The exceptional ratio $|\mathbf{a}|^2 = \kappa^2 + |\mathbf{d}|^2$ has measure zero. At $\mathbf{d} = 0$ the theorem gives the Zeeman lines of Theorem 9.6. For $\mathbf{d} \neq 0$ the screened frequencies cannot be removed along a worldline (Proposition 9.8), but they are κ times functions of $|\mathbf{a}|/\kappa$ and $|\mathbf{d}|/\kappa$: screening supplies no scale.

Proposition 9.10 (Gyroscopic screening by a boost)**.** *Let $b = \alpha L_3 + b_\Sigma$, where $b_\Sigma \in \mathfrak{p}_G$ is the boost $z \mapsto i\Sigma\bar{z}$ with $\Sigma = \left(\begin{smallmatrix} y_1 & y_2 & 0 \\ y_2 & -y_1 & 0 \\ 0 & 0 & 0 \end{smallmatrix}\right)$, and let $r^2 = y_1^2 + y_2^2 > 0$. Then $[b, G] \neq 0$ and*

$$\chi_S(\lambda) = (\lambda^2 + \kappa^2)(\lambda^2 + (\kappa - \alpha)^2 - r^2)(\lambda^2 + (\kappa + \alpha)^2 - r^2).$$

If $\alpha \neq 0$ and $r < \min(|\kappa - \alpha|, |\kappa + \alpha|)$, every solution is bounded; for $\alpha = 0$ and for $|\kappa - \alpha| < r < |\kappa + \alpha|$ some are not. Numerically, near bounded points at which the five conditions of the proof are independent, the set of $b \in \mathfrak{so}(3,3)$ with all motion bounded appears to be a 10-dimensional submanifold, and at the tested points of $\mathfrak{l}_G$ and of this family the conditions are degenerate. Both statements rest on finite-difference Jacobian ranks, and rank loss of a defining map does not by itself make its zero set singular (as for $x^2 = 0$); they are numerical observations until an exact rank and local-geometry argument is given.

Proof. On $\langle e_1, e_2, e_4, e_5\rangle$, $K_3 = GL_3$ is an η-antisymmetric involution commuting with G, L_3 and b_Σ, so its eigenplanes $\mathcal{U}_\pm$ are totally η-null and invariant. There $L_3 = \mp G$, so $S = (\kappa \mp \alpha)G + b_\Sigma$ and $S^\dagger = (\kappa \pm \alpha)G - b_\Sigma$, each squaring to a multiple of $\mathbf{1}$ ($b_\Sigma^2 = r^2$, $\{b_\Sigma, G\} = 0$), negative under the stated bound on r. For $\alpha \neq 0$ the restrictions of $S^\dagger$ to $\mathcal{U}_+$ and to $\mathcal{U}_-$ have no common eigenvalue, so every eigenspace of $S^\dagger$ lies in $\mathcal{U}_+ \oplus \langle e_3, e_6\rangle$ or in $\mathcal{U}_- \oplus \langle e_3, e_6\rangle$ (complexified). Since $\mathcal{U}_\pm$ are totally η-null and η-orthogonal to the b-free plane $\langle e_3, e_6\rangle$, on which the eigenspaces of κG are isotropic lines, h_η vanishes on every eigenspace, and Theorem 9.4 applies. For $\alpha = 0$, $b \in \mathfrak{p}_G$ and Theorem 9.7 applies; for $|\kappa - \alpha| < r < |\kappa + \alpha|$, χ_S has a real root. For the dimension statement: at simple frequencies Theorem 9.4(ii) imposes five real conditions on b (two for imaginary spectrum after the trace relation, three for isotropy). Their finite-difference Jacobian has rank five at four least-squares solutions and at an exact compact screened point, and rank four, with residuals of second order along the kernel directions, at three points of this family and at two elliptic points of $\mathfrak{l}_G$. □

The family is fixed; the dimension statements are numerical. This is the gyroscopic stabilisation of a saddle [150, 79]: the rotation detunes the two light-like planes of K_3 in which the boost acts, so that $\eta(\pi, \pi) = 2\eta(\pi_+, \pi_-)$, with $\pi_\pm$ the components of π in the null planes $\mathcal{U}_\pm$ of the proof, averages to zero. Since b_Σ is a boost, the family is not a gauge field. Together with Theorem 9.9 it shows that an anticommuting part need not cause secular growth, and that boundedness does not single out $\mathfrak{l}_G$; gauge removability does (Proposition 9.8).

Proposition 9.11 (A connection alone does not bind)**.** *(i) Let $\kappa = 0$ and let $b \in \mathfrak{so}(3,3)$ be any constant connection, $b = \nu\mathcal{K}$ included. Then $X = e^{-b\tau}Y$ gives $\mathcal{L} = \frac{m}{2}\eta(\dot{Y}, \dot{Y})$ and $\ddot{Y} = 0$. The generator $M = \left(\begin{smallmatrix} -b & \mathbf{1}/m \\ 0 & -b \end{smallmatrix}\right)$ has $\det(\lambda - M) = \chi_{-b}(\lambda)^2$ and is never semisimple: the motion is free drift, and there is no binding frequency.*

(ii) The rescaling $(\tau, \kappa, b, m) \mapsto (\vartheta\tau, \kappa/\vartheta, b/\vartheta, \vartheta m)$ maps M to M/ϑ, solutions to solutions and the action $\int \mathcal{L}\, d\tau$ to itself, and leaves η, G, $\mathfrak{l}_G$, $\mathfrak{p}_G$ and $\mathfrak{so}(4)_\Delta$ of Theorem 4.36 unchanged. No value of κ, hence of $\ell = c/\kappa$, is selected, with or without a constant b.

(iii) $M(m) = TM(1)T^{-1}$ *with* $T = \mathrm{diag}(\mathbf{1}, m\mathbf{1})$*: the spectrum of* M *does not involve* m*, and the rescaling of (ii) keeps* $m\kappa$*, not* m/κ*, fixed.*

Proof. (i) $\dot{X} + bX = e^{-b\tau}\dot{Y}$. Write $-\mathrm{diag}(b, b) = S' + N'$ (Jordan–Chevalley). Then $M = S' + (N' + N)$ with $N = \left(\begin{smallmatrix} 0 & \mathbf{1}/m \\ 0 & 0 \end{smallmatrix}\right)$, and $N' + N \neq 0$ because N is off-diagonal and N' is block-diagonal. (ii) and (iii) are direct. □

Part (i) is inherited, since every connection on a line is pure gauge, and its consequence is an obstruction: a Yang–Mills potential on a worldline, with no self-adjoint shift, produces free motion. Part (ii) is dimensional analysis, inherited; it restates that κ is not fixed (end of Section 4.3) and that no ratio among the inertia constants and m is fixed by the Clifford structure (after Proposition 4.33). Part (iii) is fixed and elementary: nothing in the interior dynamics ties m to κ, and $mc^2 = \hbar\kappa$, that is $\ell = \hbar/(mc)$, is a matching added from outside, never a result.

9.1.5 The split virial identity and the vacuum

The split virial identity of Theorem 4.7 holds unchanged with the connection: for every constant S, and so for every $b \in \mathfrak{so}(3,3)$, $\frac{d}{d\tau}\eta(X,\pi) = \eta(\pi/m - SX, \pi) + \eta(X, S^\dagger\pi) = \eta(\pi,\pi)/m$ along every solution. The quantum statement needs the signature of the energy on each frequency.

Theorem 9.12 (Krein signatures and the zero-point energy)**.** *Suppose every solution is bounded, and let* $\omega > 0$ *have multiplicity* k_ω *in* $\mathrm{spec}(-iS)$*. On the eigenspace* $\mathcal{E}_{i\omega}$ *of the generator, the energy form* $v \mapsto \bar{v}^\top \mathcal{H} v$ *has signature* (k_ω, k_ω)*. Hence, with positive-norm quantisation of the normal modes, the zero-point energy* $\sum_j (\pm)\frac{1}{2}\hbar\omega_j$ *cancels frequency by frequency, and every level has a lower one.*

Proof. Work in the canonical variables of Lemma 9.3. On $\mathcal{E}_{i\omega}$, $\mathcal{H}v = -i\omega\mathbb{J}v$, so the energy form equals ω times the Krein form $-i\bar{v}^\top\mathbb{J}v$, which is nondegenerate there because eigenspaces of distinct imaginary eigenvalues are orthogonal for it [171]. The zero-momentum part of $\mathcal{E}_{i\omega}$ is $\mathcal{E}_{i\omega}(-S)$, of dimension k_ω, and H vanishes on it (Lemma 9.3). Since the generator is semisimple, $\dim \mathcal{E}_{i\omega} = 2k_\omega$ by Lemma 9.3. A nondegenerate Hermitian form with a half-dimensional isotropic subspace has signature (k, k), and a mode of energy sign $\pm$ quantises to $\pm\hbar\omega(n + \frac{1}{2})$. □

The theorem is fixed, although structural: it holds for any nondegenerate η. It extends the vanishing of the zero-point energy on $\mathbb{R}^{3,3}$ at $b = 0$ (Theorem 7.13) and the absence of a ground state (Corollary 4.15) to every connection that keeps the motion bounded: the cancellation, forced by the perfect-square form (9.2), survives the splitting line by line, and so does the absence of a ground state. On the coordinate subfields of Theorem 7.13, where the zero-point energy is $\pm\frac{3}{2}\hbar\kappa$ or $\pm\frac{1}{2}\hbar\kappa$, the theorem does not apply, since G does not preserve the subfield and the zero-momentum flow leaves it (Theorem 7.13).

Corollary 9.13 (No Fock vacuum with Jordan blocks)**.** *Suppose* M *is not semisimple at a frequency* $\omega \neq 0$*, as for the counter-rotation of Theorem 9.7(ii) and for a pairing boost below the threshold. Then no linear symplectic change of variables writes the Hamiltonian on that block as a sum of signed oscillators, and there is no Fock vacuum annihilated by normal-mode lowering operators.*

Proof. A sum of signed oscillators generates a semisimple flow, and symplectic changes of variables act on the generator by similarity. □

This is an obstruction. A model block with the same Jordan structure is $H = \omega(q_1 p_2 - q_2 p_1) + \frac{1}{2}(q_1^2 + q_2^2)$, whose two terms Poisson-commute [166]; that its quantisation has continuous spectrum is inherited and is not re-derived here.

9.1.6 Three constituents: the binding-compatible algebra $\mathfrak{so}(4)_\Delta$

At three constituents we use the generators E_A of (1.2), the numbering of Section 4.4 (the temporal directions, their partners $i' = i+3$, the weak directions $W = \{7, 13, 14, 15\}$, the colour directions $\{10, 11, 12, 16, 17, 18\}$ and the span E), $G_3 = G \oplus G \oplus G$, and $\mathrm{rot}(i, j)$ acting by $\frac{1}{2}\,\mathrm{ad}$; R_{AB} is defined as at one constituent. By Theorem 4.36, the part of the Pati–Salam algebra of P_* that commutes with G_3 is $\mathfrak{so}(4)_\Delta = \mathrm{span}\{\mathrm{rot}(i, j) : i < j \in W\} \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$, compact, and neither Pati–Salam factor meets $\mathrm{Cent}(G_3)$. The spectra below use an explicit splitting.

Lemma 9.14 (The two factors of $\mathfrak{so}(4)_\Delta$)**.** *The elements* $T_1^\pm = \frac{1}{2}(\mathrm{rot}(13, 14) \pm \mathrm{rot}(7, 15))$, $T_2^\pm = \frac{1}{2}(\mathrm{rot}(7, 14) \mp \mathrm{rot}(13, 15))$ *and* $T_3^\pm = \frac{1}{2}(\mathrm{rot}(7, 13) \pm \mathrm{rot}(14, 15))$ *satisfy* $[T_i^+, T_j^-] = 0$ *and* $[T_1^\pm, T_2^\pm] = -T_3^\pm$ *cyclically, and the trace form is negative definite on their span, so* $\mathfrak{so}(4)_\Delta = \mathfrak{su}(2)_+ \oplus \mathfrak{su}(2)_-$*. On the eighteen vectors each* $T_3^\pm$ *has the eigenvalues* $\pm\frac{i}{2}$*, four times each, and* 0*, ten times.*

Proof. Each $\mathrm{rot}(i, j)$ with $i, j \in W$ rotates the plane (e_i, e_j) and its partner plane $(e_{i'}, e_{j'})$ alike, so on span W, and identically on the partner span $\{10, 16, 17, 18\}$, the six elements act as the standard generators of $\mathfrak{so}(4)$, and the $T_i^\pm$ are their self-dual and anti-self-dual combinations; the brackets follow by direct computation. Each $T_3^\pm$ is half a sum of two commuting plane rotations on each of the two four-dimensional spans and vanishes on the remaining ten directions. □

The lemma is fixed; it makes the isomorphism $\mathfrak{so}(4)_\Delta \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$ of Theorem 4.36 explicit. Each $\mathfrak{su}(2)$ factor locks a weak rotation of W to the colour rotation of the partners $\{10, 16, 17, 18\}$. Reading $\mathfrak{so}(4)_\Delta$ as Yang and Mills' isospin algebra, doubled, is a reading, as is the description after Theorem 4.37.

Proposition 9.15 (Spectra on the eighteen-dimensional frame)**.** *Let* $\kappa = 5$ *and* $m = 1$*.*

(i) For $b = \beta T_3^+$ *with* $\beta = 2$*,* $\chi_S = (\lambda^2 + 25)^5(\lambda^2 + 16)^2(\lambda^2 + 36)^2$*: the frequency* κ *on the five complex singlet directions, and* $\kappa \mp \beta/2$ *on the two* $\mathfrak{su}(2)_+$ *doublets, of weights* $\pm\frac{1}{2}$*.*

(ii) For $b = \beta_+ T_3^+ + \beta_- T_3^-$ *with* $(\beta_+, \beta_-) = (2, 4)$*, the lines are* $\kappa \pm \frac{1}{2}(\beta_+ \pm \beta_-) = 8, 6, 4, 2$*, together with* κ*.*

(iii) In both cases $\Xi = -G_3/(2\kappa m)$ *solves the Sylvester equation,* M *is semisimple, the Krein signatures are* (k, k)*, and the zero-point energy vanishes.*

(iv) The weak-only rotation $R_{7,13}$ *and the colour-only rotation* $R_{10,16}$ *give a non-semisimple* M*, hence secular growth.*

(v) $b = R_{7,13} \pm R_{10,11} \in \mathfrak{ps}_W$ *has* $[b, G_3] \neq 0$*, and every solution is bounded, for every* $\kappa > 0$*. At* $\kappa = 5$ *the frequencies are* κ *(six pairs) and* $|\omega_j| \approx 4.319, 5.050, 5.731$*, with* ω_j *the roots of* $\omega^3 + 5\omega^2 - 26\omega - 125$*.*

Proof. (i)–(iii) The connection lies in $\mathfrak{so}(4)_\Delta \subset \mathrm{Cent}(G_3)$ and is real antisymmetric, so Theorem 9.6(i), whose argument uses only $G_3^2 = -\mathbf{1}$, $G_3^\dagger = G_3$ and $[b, G_3] = 0$, gives semisimplicity and boundedness, and Theorem 9.12 gives the signatures and the zero-point energy. Since b commutes with G_3, $S = b + \kappa G_3$ is complex linear on $(\mathbb{R}^{18}, G_3) \cong \mathbb{C}^9$ and acts as $i\kappa + b$. By the proof of Lemma 9.14, on the four complex directions of W the commuting T_3^+, T_3^- have the joint eigenvalues $\pm\frac{i}{2}(1, 1)$ and $\pm\frac{i}{2}(1, -1)$, and both vanish on the other five; the characteristic polynomials follow as in the proof of Theorem 9.6(ii). (iv) and (v) G_3 maps the six-plane $\langle e_7, e_8, e_{13}; e_{10}, e_{11}, e_{16}\rangle$ as G maps $\langle e_1, e_2, e_3; e_4, e_5, e_6\rangle$, and each b named preserves that six-plane and vanishes on its η-orthogonal complement, where the motion is the binding at $b = 0$. On the six-plane b is admissible. $R_{7,13}$ rotates the temporal triple alone and $R_{10,16}$ the spatial triple alone, so $\mathbf{a} \cdot \mathbf{d} = \pm|\mathbf{a}|^2 \neq 0$, and Theorem 9.9(i) applies; since S is real antisymmetric, the unbounded solutions grow polynomially. For $R_{7,13} \pm R_{10,11}$ the rotation vectors

of the two triples are the unit vectors $\mathbf{e}_2$ and $\pm\mathbf{e}_3$: equal rates and perpendicular axes, so $\mathbf{a} = \frac{1}{2}(\mathbf{e}_2 \pm \mathbf{e}_3) \perp \mathbf{d} = \frac{1}{2}(\mathbf{e}_2 \mp \mathbf{e}_3)$ and $|\mathbf{a}|^2 = |\mathbf{d}|^2 = \frac{1}{2} \neq \kappa^2 + \frac{1}{2}$. Theorem 9.9(ii) applies, with f as stated at $\kappa = 5$; the three frequencies are its roots, evaluated numerically. □

This is fixed. Every constant $\mathfrak{so}(4)_\Delta$-valued connection is compact and commutes with G_3, so it keeps the interior bounded; the splitting follows the $\mathfrak{su}(2)$ weights, β/κ is free, and by Proposition 9.5 the pattern can be removed along one worldline. The single-factor rotations tear the binding, but boundedness does not single out the locked diagonal: the connection of part (v) is bounded with $[b, G_3] \neq 0$, while the anti-locked $R_{7,13} + R_{10,16}$, which has $\mathbf{a} = 0$ on its six-plane, is not. What singles out $\mathfrak{so}(4)_\Delta$ is commutation with G_3 (Theorem 4.36), that is, removability along a worldline (Proposition 9.8).

9.1.7 Quantum Yang–Mills facts, and the channels of the vacuum

Classically the connection splits, screens or tears the binding but supplies no scale. The only place left for it to generate one is the quantum theory, and the following inherited facts are needed to say what that would involve.

Proposition 9.16 (Inherited facts). *(i) For $\mathfrak{su}(4)$ with $\mathrm{tr}(T^A T^B) = \frac{1}{2}\delta^{AB}$, $\sum_A T^A Z T^A = \frac{1}{2}(\mathrm{tr}(Z)\mathbf{1} - \frac{1}{4}Z)$ for every 4×4 matrix Z: the factor is $\frac{15}{8} = C_F$ on $\mathbf{1}$ and $-\frac{1}{8}$ on traceless Z.*

(ii) The channel factors $T_1 \cdot T_2 = \frac{1}{2}[C_2(pair) - C_2(R_1) - C_2(R_2)]$ are $+\frac{3}{8}, -\frac{5}{8}$ on $\mathbf{10}, \mathbf{6} \subset \mathbf{4}\otimes\mathbf{4}$, $-\frac{15}{8}, +\frac{1}{8}$ on $\mathbf{1}, \mathbf{15} \subset \bar{\mathbf{4}}\otimes\mathbf{4}$, and $+\frac{1}{4}, -\frac{3}{4}$ on $\mathbf{3}, \mathbf{1} \subset \mathbf{2}\otimes\mathbf{2}$ of $\mathfrak{su}(2)$, with $C_2 = \frac{15}{8}, \frac{9}{2}, \frac{5}{2}, 4$ on $\mathbf{4}, \mathbf{10}, \mathbf{6}, \mathbf{15}$. Hence the Dirac bilinear $\overline{(\mathbf{4},\mathbf{2},\mathbf{1})} \otimes (\mathbf{4},\mathbf{1},\mathbf{2}) = (\mathbf{1},\mathbf{2},\mathbf{2}) \oplus (\mathbf{15},\mathbf{2},\mathbf{2})$ has $\mathfrak{su}(4)$ factors $-\frac{15}{8}$ and $+\frac{1}{8}$, and a Lorentz-scalar Majorana bilinear of $(\mathbf{4},\mathbf{1},\mathbf{2})$, symmetric in its internal indices, lies in $\mathrm{Sym}^2(\mathbb{C}^4 \otimes \mathbb{C}^2) = (\mathbf{10},\mathbf{1},\mathbf{3}) \oplus (\mathbf{6},\mathbf{1},\mathbf{1})$, of dimensions $30 + 6$, with factors $(+\frac{3}{8}, +\frac{1}{4})$ and $(-\frac{5}{8}, -\frac{3}{4})$.

(iii) In Euclidean momentum space

$$\int_{k^2 \le k_{\max}^2} \frac{d^4k}{(2\pi)^4} \frac{1}{k^2 + M_{\mathrm{dyn}}^2} = \frac{1}{16\pi^2}\left[k_{\max}^2 - M_{\mathrm{dyn}}^2 \ln(1 + k_{\max}^2/M_{\mathrm{dyn}}^2)\right].$$

The function $f_{\mathrm{gap}}(x) = 1 - x^2 \ln(1 + x^{-2})$ decreases strictly from 1 to 0, so the contact gap equation $1 = \lambda_{\mathrm{c}} f_{\mathrm{gap}}(x)$ has a root $x = M_{\mathrm{dyn}}/k_{\max} > 0$, with M_{dyn} the dynamical mass, $k_{\max}$ the cutoff and λ_{c} the dimensionless contact coupling, if and only if $\lambda_{\mathrm{c}} > 1$.

(iv) The ladder critical coupling $\pi/(3C_F)$ [44, 97] is $8\pi/45 \approx 0.5585$ at $C_F = \frac{15}{8}$. For $SU(2)$ with N_f Dirac doublets the one-loop coefficient is $\beta_0 = (22 - 2N_f)/3$, equal to 6 at $N_f = 2$ [60, 124], and $\Lambda = \mu\, e^{-8\pi^2/(\beta_0 g^2(\mu))}$ is independent of μ. The breaking $SU(4) \to Sp(4)$ of $N_f = 2$ leaves $15 - 10 = 5$ Goldstone directions [123].

All of this is inherited [105, 144]; the strict decrease in (iii) follows from $\ln(1+u) > u/(1+u)$. The use of (ii) is a reading. Under the most-attractive-channel heuristic of one-boson exchange [128], in which a negative factor means attraction, and with a common coupling for $\mathfrak{su}(4)$ and $\mathfrak{su}(2)_R$, which this manuscript adopts only as an assumption (Assumption 7.19 and the remark after Proposition 7.21), the exchange favours the Dirac singlet channel $(\mathbf{1},\mathbf{2},\mathbf{2})$ and the colour-breaking $(\mathbf{6},\mathbf{1},\mathbf{1})$. It disfavours the $(\mathbf{15},\mathbf{2},\mathbf{2})$ that the Georgi–Jarlskog factor -3 needs (Proposition 7.35; [51]), and the $(\mathbf{10},\mathbf{1},\mathbf{3})$ that carries the Standard-Model vacuum of Section 7.5. That vacuum is a minimum of a scalar potential, not a gauge-exchange condensate, so the heuristic addresses a different, dynamical scenario, in which a gauge-generated vacuum would point away from the one selected there. In the selected vacuum the order is moreover the reverse of a mass generated by the gauge field: the Pati–Salam vector bosons outside the Standard Model acquire their masses by eating nine Goldstone directions (Theorem 7.41).

9.1.8 What the connection part supplies

The results of this section combine as follows.

(a) The binding frequency comes from the self-adjoint part of the shift, never from a connection. The rate κ, with $(\kappa G)^2 = -c^2/\ell^2$, sits in the tensor part (Theorem 9.1, fixed), and a connection alone gives free drift (Proposition 9.11, an obstruction).

(b) A commuting connection splits the binding frequency without supplying a scale: the lines κ and $\kappa \pm \sqrt{w \cdot w}$ are bounded for admissible b and removable along any worldline, with β/κ free (Theorem 9.6 and Proposition 9.5, fixed; the Zeeman description is a reading). At three constituents $\mathfrak{so}(4)_\Delta$ gives lines labelled by $\mathfrak{su}(2)$ weights and is not the only bounded part of the Pati–Salam algebra (Proposition 9.15), and the zero-point cancellation survives line by line (Theorem 9.12).

(c) A purely anticommuting connection tears the binding (Theorem 9.7 and Corollary 9.13, obstructions), with rates $\sqrt{\kappa^2 + |\mathbf{d}|^2}$ for an admissible element or $\sqrt{\kappa^2 - \nu^2}$ for a pairing boost, carried by Jordan blocks. A mixed admissible connection keeps every orbit bounded exactly at equal rates of the two triples about axes that are not opposite, outside the ratio $|\mathbf{a}|^2 = \kappa^2 + |\mathbf{d}|^2$ (Theorem 9.9); a boost can be screened gyroscopically (Proposition 9.10); neither gives a scale.

(d) The only role left for the connection in generating a scale is quantum, and it is open. The arena leaves ℓ free (Proposition 9.11, and Section 4.5 for a dynamical rate), and classical four-dimensional Yang–Mills theory has no dimensionful parameter; a scale arises only by dimensional transmutation, $\Lambda_b = \mu\, e^{-8\pi^2/(\beta_0 g^2(\mu))}$ [26, 60], for a Yang–Mills theory whose potential is b.

Conjecture 9.17 (A transmuted binding length)**.** *$\ell = \hbar c/\Lambda_b$ for the scale Λ_b of a Yang–Mills theory whose potential is the connection part of the shift; that is, $\hbar\kappa = \Lambda_b$.*

The conjecture is open: nothing in the arena derives it. Reading the rest energy mc^2 as $\hbar\kappa$ is a separate identification and not part of the conjecture, since the spectrum of M does not involve m and the rescaling of Proposition 9.11 keeps $m\kappa$, not m/κ, fixed. A mechanism would need κ to be dynamical, as the rotor rate of Section 4.4 is, and coupled to a quantum gauge sector with a fixed β_0 and a matching coupling; even then, by Proposition 9.16(ii), the most attractive channels are not those of the vacuum selected in Section 7.5. A classification of the compact connections that keep the binding bounded at three constituents, beyond the commuting ones and the equal-rate family embedded as in Proposition 9.15(v), is also open, as is a description of the bounded set in $\mathfrak{so}(3,3)$ beyond the numerical statement of Proposition 9.10.

Read in the variables of $E = mc^2$, and only through the added identification $mc^2 = \hbar\kappa$, the results of this section complete the reading of Section 13.5, in which the self-adjoint summand κG carries c and alone binds; what follows is a reading of the fixed results above. The Yang–Mills potential is the antisymmetric, transport summand of the same shift, and it is compact. Classically it generates neither m nor κ: when it commutes with G it splits the interior rate κ into charge-labelled lines, a purely anticommuting potential tears the binding, and a mixed one may or may not (Theorem 9.9). Whether its quantum scale is the length ℓ that the arena leaves free is open.

9.2 Gauge transport of the two energies

What transport by a compact gauge field does to the two energies

An admissible Yang–Mills generator is compact: by Theorem 1.3 and Corollary 1.4 gauge generators lie in $\mathfrak{so}(3)_T \oplus \mathfrak{so}(3)_S \subset \mathfrak{so}(3,3)$, the rotations of the two triples. Write such a generator as

$b = (A + D, A - D)$, with $A + D$ acting on the temporal and $A - D$ on the spatial triple. Its diagonal part (A, A) lies in $\mathfrak{l}_G$ and its anti-diagonal part $(D, -D)$ in $\mathfrak{p}_G$. Boost-valued connections are legitimate metric connections (Proposition 4.14) but not admissible gauge potentials.

Proposition 9.18 (What transport preserves)**.** *(i) An element $Y \in \mathfrak{so}(3,3)$ preserves $\mathbf{E} \cdot \mathbf{B}$ for all X, equivalently q, equivalently E_0, iff $[Y, G] = 0$, that is iff $Y \in \mathfrak{l}_G$.*

(ii) A compact $b = (A + D, A - D)$ preserves E_6 and $\operatorname{Re} q = |\mathbf{B}|^2 - |\mathbf{E}|^2$, and it preserves E_0 for all X iff $D = 0$. The anti-diagonal rotation through $\frac{\pi}{4}$ about e_3 carries the null field $(1, i, 0)$ to the field $\frac{1+i}{\sqrt{2}}(1, 1, 0)$ at rest, E_0 going from 0 to 1 at $E_6 = 1$.

(iii) At three constituents the part of the Pati–Salam algebra of P_* commuting with G_3, the algebra $\mathfrak{so}(4)_\Delta \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$ of Theorem 4.36, consists of paired rotations, so it preserves the $\mathfrak{so}(9, \mathbb{C})$ invariants $\frac{1}{2}|Z|^2 = \frac{1}{2}\sum_k |z^{(k)}|^2$ and $q_3(Z) = \sum_k z^{(k)} \cdot z^{(k)}$, hence $\frac{1}{2}|q_3|$.

(iv) It preserves neither the rest energies of the constituents nor that of their summed density: the rotation through $\frac{\pi}{4}$ generated by $\frac{1}{2}\mathrm{rot}(7, 13) \in \mathfrak{so}(4)_\Delta$ maps $(z^{(1)}, z^{(2)}, z^{(3)}) = (0, (1, i, 0), (1, 0, 0))$ to $(0, (0, i, 0), (\sqrt{2}, 0, 0))$, keeping $\frac{1}{2}|Z|^2 = \frac{3}{2}$ and $q_3 = 1$, while $\sum_k E_0^{(k)}$ goes from $\frac{1}{2}$ to $\frac{3}{2}$ and $(\sum_k E_6^{(k)}; \sum_k \mathbf{S}_{\mathrm{t}}^{(k)})$ goes from $(\frac{3}{2}; 0, 0, 1)$ to $(\frac{3}{2}; 0, 0, 0)$, its Lorentz length from $\frac{\sqrt{5}}{2}$ to $\frac{3}{2}$.

Proof. (i) For $Y \in \mathfrak{so}(3,3)$, $Y^\top \eta G + \eta G Y = \eta[G, Y]$, so Y preserves the form ηG, that is $\mathbf{E} \cdot \mathbf{B}$, iff $[Y, G] = 0$. Every $Y \in \mathfrak{so}(3,3)$ preserves $\operatorname{Re} q$; if it also preserves $|q|$, then $\operatorname{Im} q \cdot \delta(\operatorname{Im} q) = 0$, so $\delta(\operatorname{Im} q) = 0$ because polynomials form an integral domain. (ii) b rotates each triple orthogonally, so it preserves $|\mathbf{B}|^2$ and $|\mathbf{E}|^2$; (A, A) commutes with G and $G(D, -D)G^{-1} = (-D, D)$, so by (i) E_0 is preserved for all X iff $D = 0$. In the example $\mathbf{B} = (1, 0, 0) \mapsto \frac{1}{\sqrt{2}}(1, 1, 0)$ and $\mathbf{E} = (0, 1, 0) \mapsto \frac{1}{\sqrt{2}}(1, 1, 0)$. (iii) By Theorem 4.34 the paired rotations are the elements $\left(\begin{smallmatrix} R & 0 \\ 0 & R \end{smallmatrix}\right)$ with R real antisymmetric, which act on $Z \in \mathbb{C}^9$ by a real orthogonal matrix and preserve $|Z|^2$ and q_3; by Theorem 4.36 $\mathfrak{so}(4)_\Delta$ is spanned by such elements. (iv) $\mathrm{rot}(7, 13)$ rotates the first components of $z^{(2)}$ and $z^{(3)}$ into each other by a real rotation, and the stated values follow by direct evaluation, with $\mathbf{S}_{\mathrm{t}} = (0, 0, 1)$ for $(1, i, 0)$. □

Parts (i) and (ii) are inherited linear algebra, and they cut both ways. Pointwise, the anti-diagonal part of an admissible b does turn a null field into one at rest at fixed undivided energy, while the diagonal part, a rotation in $\mathfrak{l}_G$, changes no invariant. But the anti-diagonal part is the part that fails to commute with G: along a worldline it cannot be removed by a rotating frame, unlike the diagonal part (Proposition 9.8; Larmor's theorem [83]), and in the interior it does not commute with the shift κG. There, alone, it produces Jordan blocks and secular growth (Theorem 9.7), and a mixed compact connection keeps every orbit bounded exactly when it rotates the two triples at equal rates about axes that are not opposite, outside one exceptional ratio of the rates to κ at which some solutions drift (Theorem 9.9); these are statements about the interior transport, not about the pointwise invariants, and they are not used here. Nothing in the proposition selects an amount of rest energy. Parts (iii) and (iv) are an obstruction for the $\mathfrak{so}(9, \mathbb{C})$ invariants only: no transport valued in the paired rotations changes $\frac{1}{2}|Z|^2$ or q_3. They are no obstruction for the rest energy of the whole in the diagonal Lorentz reading, which the example changes; $\mathrm{rot}(7, 13)$ does not commute with the diagonal rotations, and which of the two quantities is physical depends on the Lorentz action at three constituents, which the algebra leaves open (Table 13.1; by Theorem 4.35 the closure symmetry meets the interior Lorentz algebra of the first constituent in 0). Moreover $\mathfrak{so}(4)_\Delta$ is the part of the Pati–Salam algebra commuting with G_3, not the only part whose transport keeps orbits bounded (Proposition 9.15(v)). Here b enters only as a constant transport element or a holonomy, with no field strength and no energy of its own, so where a gauge field belongs in the differentiation of the energy is, at this level, a matter of reading.

9.3 The b-field between manifolds

The author's first remark is that the b-particle is a deal breaker when two manifolds or more are in question. Theorem 5.31 tested it for two binding frames and one field, the copy rotation. This section asks the question for any number of manifolds and any field that can act between them, and answers it in five steps. First, which fields exist depends on how the Lorentz algebra acts: a common binding action admits the complex orthogonal copy algebra $\mathfrak{so}(n,\mathbb{C})$, a common interior action only its compact part, the copy rotations $\mathfrak{so}(n)$, and a spectator action decouples the Lorentz-carrying manifold from every binding-compatible field (fixed, partly restated from Section 4.4; Section 9.3.1). Second, the closure itself admits such a field: with a common binding action the rotor may act by any element of $\mathfrak{u}(n)$, which is a copy rotation plus a matrix of rates (constructed; Section 9.3.2). Third, every binding-compatible field turns the frame flow into a Schrödinger flow on $\mathbb{C}^{3n}$, unitary for the undivided energy exactly when the field is compact. The rates are gauge invariants, the field redistributes the frequencies at fixed sum, and it is removable exactly when it preserves the rate eigenspaces (fixed; Sections 9.3.3 and 9.3.4). Fourth, three manifolds carry an invariant that two do not: the sign of $B_{12}B_{23}B_{31}$, odd under interior time reversal, the analogue of Jarlskog's invariant (fixed; Section 9.3.5). Fifth, on two and three manifolds under a common Lorentz action the copy isospin is integral: frames carry isospin one, never one-half. The Yang–Mills doublet acting on two copies of the undivided energy is a connection in one of its three directions and a rate term in the other two, and on three manifolds doublets exist only under the spectator action, where unequal internal rates break the $\mathfrak{so}(4)_\Delta$ of Theorem 4.36 to a diagonal $\mathfrak{so}(3)$ (fixed; Section 9.3.6). The verdict is in Section 9.3.7. Identifiers d7-n refer to the `PASS` lines of `d7_bfield_between.py` (Table 9.1).

Notation. As in Section 5.3, with n copies: $V_n = \mathbb{R}^n \otimes \mathbb{R}^6$ in the copy-major basis, $\eta_n = \mathbf{1}_n \otimes \eta$, $G_n = \mathbf{1}_n \otimes G$, and $z = (z_1, \dots, z_n) \in \mathbb{C}^{3n}$ with $z_k = \mathbf{B}_k + i\mathbf{E}_k$, on which G_n acts as i. E_{kl} are the matrix units on the copy index. The *copy rotation* $C^{kl} = (E_{lk} - E_{kl}) \otimes \mathbf{1}_6$ maps copy k to copy l, and the *copy boost* is $(E_{lk} - E_{kl}) \otimes G$. A *copy field* is $b = B \otimes \mathbf{1}_6$ with $B \in \mathfrak{so}(n)$ real antisymmetric, and $\mathcal{J} := B_{12}B_{23}B_{31}$ at $n = 3$. The binding term with rates $\kappa_k \geq 0$ is $K \otimes G$ with $K = \mathrm{diag}(\kappa_1, \dots, \kappa_n)$, and $S = K \otimes G + b$, $\dot{X} = -SX + \pi/m$, $\dot{\pi} = S^\dagger \pi$, with M as in Theorem 5.31. A field is *binding-compatible* if it commutes with G_n. Rates are rates, and reading one as a rest energy needs $mc^2 = \hbar\kappa$ (Proposition 4.47).

9.3.1 Which fields can act between manifolds

Proposition 9.19 (Inter-manifold fields and the Lorentz action)**.** *(i)* Common binding action *(the diagonal $\mathfrak{l}_G$). The elements of $\mathfrak{so}(3n,3n)$ that commute with it and with G_n are $\{B \otimes \mathbf{1}_6 + B' \otimes G : \ B, B' \in \mathfrak{so}(n)\} \cong \mathfrak{so}(n,\mathbb{C})$, of dimension $0, 2, 6, 12$ for $n = 1, \dots, 4$. The compact part consists of the copy rotations, of dimension $0, 1, 3, 6$; the copy boosts $B' \otimes G$ are noncompact.*

(ii) Common interior action *(the diagonal $\mathfrak{l}_{\mathrm{int}}$, the action Δ of Section 5.2). The elements of $\mathfrak{so}(3n,3n)$ commuting with it and with G_n are the copy rotations alone, of dimension $0, 1, 3$ for $n = 1, 2, 3$; no copy boost commutes with $\mathfrak{l}_{\mathrm{int}}$. At $n = 2$ the copy rotation is $C^{12} = -M_{12} + C_{\mathrm{surp}}$, where M_{12} of Section 5.2.1, the spin lift of $E_{12} - E_{21}$, acts on vectors as $-\varepsilon \otimes P_{\mathrm{int}}$. It is the copy rotation of the interior four-planes, the copy algebra of Theorem 5.9, and $C_{\mathrm{surp}} = \varepsilon \otimes P_{\mathrm{surp}}$ that of the surplus pairs. Both commute with the diagonal $\mathfrak{l}_{\mathrm{int}}$, and neither commutes with G_2.*

(iii) Spectator action *($\mathfrak{l}_G^{(1)}$ or $\mathfrak{l}_{\mathrm{int}}^{(1)}$ on copy 1 alone). The elements of $\mathfrak{so}(3n,3n)$ commuting with it and with G_n form $\mathfrak{so}(3n-3,\mathbb{C})$ of copies $2, \dots, n$, of dimension 6 and 30 at $n = 2, 3$. Every one of them vanishes on copy 1 and has no component into it.*

Part (i) restates the third row of Proposition 5.72 (fixed there) and Proposition 5.27(iii) at $n = 2$. For $\mathfrak{l}_{\text{int}}^{(1)}$, part (iii) restates Theorem 4.35 at $n = 3$ (fixed there) and Proposition 5.27(iv) at $n = 2$. For $\mathfrak{l}_G^{(1)}$ it is fixed here (d7-1d), and its proof is Schur's lemma: V is an irreducible, nontrivial module of $\mathfrak{l}_G$, and the other copies are trivial modules of it. For $\mathfrak{l}_{\text{int}}^{(1)}$ Schur's lemma alone does not suffice. V splits into the interior four-plane $\langle e_1, e_4, e_5, e_6\rangle$, which is irreducible and nontrivial, and the surplus pair $\langle e_2, e_3\rangle$, on which $\mathfrak{l}_{\text{int}}$ acts trivially. The commutant of $\mathfrak{l}_{\text{int}}^{(1)}$ alone, of dimension 28 and 91 at $n = 2, 3$ (the first row of Proposition 5.72), therefore acts on the surplus pair of copy 1. Commutation with G_n removes these elements, because G maps the surplus pair into the interior four-plane, as in the proof of Theorem 4.36. Part (ii) and the decomposition of C^{12} are fixed (d7-1a to d7-1d). Two consequences matter below. A binding-compatible, Lorentz-invariant field can couple the manifold that carries the Lorentz algebra to the others only when that algebra acts on all of them in the same way. Under the spectator action copy 1 is out of reach, and the inter-manifold fields act among the internal manifolds only. And the copy algebra $\mathfrak{k}_{\text{cop}}$ of Chapter 5 does not respect the binding: the binding-compatible copy rotation C^{kl} is $-M_{kl}$ plus the surplus copy rotation.

9.3.2 The closure admits an inter-manifold field

Theorem 4.22 shows that, on one constituent, the only equivariant closure is $\mathbb{R} \ltimes_G V$: the rotor acts by G. After it the paper extends the result to several constituents "factor by factor, since equivariant maps between distinct constituents vanish", with a common rate. That is correct for a Lorentz algebra acting on each constituent separately, once the common rate is imposed. A common action admits more.

Proposition 9.20 (Rotor generators on n binding frames)**.** *The endomorphisms of V_n commuting with the diagonal $\mathfrak{l}_G$ form $\mathfrak{gl}(n, \mathbb{C}) = \mathfrak{gl}(n) \otimes \operatorname{span}\{\mathbf{1}, G\}$, of dimension $2n^2$. Those that rotate the frame covector without dilating it, the condition of Theorem 4.22, form*

$$\mathfrak{u}(n) = \mathfrak{so}(n) \otimes \mathbf{1}_6 \;\oplus\; \operatorname{Sym}(n) \otimes G,$$

of dimension n^2. The first summand is η_n-antisymmetric (copy rotations) and the second is η_n-self-adjoint (rate matrices). The paper's rotor G_n spans the centre. For $\mathfrak{s}_n = \mathbb{R} \ltimes_A V_n$ with $A = K \otimes G + B \otimes \mathbf{1}_6 \in \mathfrak{u}(n)$ and the rotor clamped at $\theta = \Omega\tau$, the substitution $X = e^{-\theta A}x$ of Proposition 4.24 gives $e^{-\theta A}\dot{x} = \dot{X} + \Omega A X$, so the shift is $S = \Omega A$.

At $n = 2$ the algebra is the energy symmetry $\mathfrak{u}(2)$ of Proposition 5.23(i). The algebra $\mathfrak{u}(n)$, its dimension n^2 and its centre G_n restate row four of Proposition 5.72 and the remark after it (fixed there). The commutant $\mathfrak{gl}(n, \mathbb{C})$ and the decomposition into copy rotations and rate matrices are fixed (d7-2a); the reading as rotor generators is new.

The semidirect product $\mathbb{R} \ltimes_A V_n$ is a Lie algebra for every A, so the closure exists. It is not the only closure with rotor A, because the half of Theorem 4.22 that makes the frame abelian does not carry over to a common action. An equivariant bracket on V_n is $[X, Y]_k = \sum_{i,j} \Gamma_{ij}^k \, \beta_0(X_i, Y_j)$ with complex $\Gamma_{ij}^k = \Gamma_{ji}^k$ (G acting as i), a space of real dimension 12 at $n = 2$. In a copy basis in which A acts as $i \operatorname{diag}(\omega_1, \ldots, \omega_n)$, the Jacobi identity with the rotor reads $\Gamma_{ij}^k(\omega_k - \omega_i - \omega_j) = 0$. The frame is therefore abelian for $A = G_n$ and whenever no ω_k equals $\omega_i + \omega_j$. For a resonant A it need not be: for $A = \operatorname{diag}(1, 2) \otimes G$ the bracket $[X, Y] = (0, c\,\beta_0(X_1, Y_1))$, $c \in \mathbb{C}$, satisfies every Jacobi identity (fixed; d7-2b). This chapter uses only the abelian-frame closure, to which the shift $S = \Omega A$ refers.

Which rotor acts is a choice. Theorem 4.22 selects G_n only when the action is taken per constituent and a common rate is imposed. Per constituent alone it admits $K \otimes G$ with any real diagonal K: unequal rates, but no copy field, since equivariant maps between distinct constituents vanish. Under the common action a common rate alone still admits $\kappa G_n + B \otimes \mathbf{1}_6$:

a copy field, but one that commutes with the binding and is removable (Theorem 9.21(v)). The two parts of a non-central A, the η_n-antisymmetric copy field $B \otimes \mathbf{1}_6$ and the η_n-self-adjoint rate matrix $K \otimes G$, are independent; their ratios are fixed by A and their scale by the rotor. The common action is the one place found where the closure admits unequal rates together with a copy field, the combination that Theorem 5.31 needs. That such an A is admissible is constructed: nothing in the algebra selects it. The paper's one-rotor closure, with $A = G_n$, gives every constituent the same rate (Proposition 4.33).

9.3.3 The frame flow is a Schrödinger flow

Theorem 9.21 (Binding-compatible fields between n manifolds)**.** *Let $b \in \mathrm{Cent}(G_n) \cap \mathfrak{so}(3n, 3n)$. By Proposition 5.27(i), b acts on z as $R + iQ$ with $R, Q \in \mathfrak{so}(3n)$ real: R is its paired-rotation part and Q its paired-boost part. Put $\tilde{K} = K \otimes \mathbf{1}_3$ and $H := \tilde{K} - iR + Q$.*

(i) S acts on z as iH, so the free frame flow is $z(\tau) = e^{-iH\tau} z(0)$.

(ii) The flow preserves $E_6 = \frac{1}{2}|z|^2$ for every K and every initial value iff $Q = 0$, that is, iff b is compact; then H is Hermitian. The compact fields, the only ones admissible as Yang–Mills fields (Corollary 1.4), are exactly those whose frame flow is unitary for the undivided energy.

(iii) The rates are gauge invariants. Under any η_n-isometric change of frame $X = U(\tau)Y$, the η_n-self-adjoint part of S is conjugated, so its spectrum $\{\pm i\kappa_k\}$ is unchanged. No field, compact or not, changes a rate.

(iv) For compact b: $\operatorname{tr} H = 3\sum_k \kappa_k$ and $\operatorname{tr} H^2 = 3\sum_k \kappa_k^2 + \|R\|^2$. The field redistributes the frequencies of the frame at fixed sum and widens their spread: their variance grows by $\|R\|^2/(3n)$, the largest frequency does not decrease and the smallest does not increase.

(v) b is removable along the worldline iff $[\tilde{K}, R + iQ] = 0$, that is, iff it preserves every rate eigenspace. At $n = 3$ the compact binding-compatible fields, of dimension 36, have a removable part of dimension 36, 18 and 9 for rates (κ, κ, κ), $(\kappa, \kappa, \kappa')$ and three distinct rates. Within the copy $\mathfrak{so}(3)$ the dimensions are 3, 1 and 0.

(vi) (Boundedness) Let b be compact. Every orbit of (X, π) is bounded iff, for every λ with $\lambda, -\lambda \in \operatorname{spec} H$ (including $\lambda = 0$), the eigenspaces $\mathcal{E}_\lambda(H)$ and $\mathcal{E}_{-\lambda}(H)$ are orthogonal for the bilinear form $v \cdot w = \sum_a v_a w_a$ on $\mathbb{C}^{3n}$.

Proof. (i) On the sector $G_n = i$, $K \otimes G$ acts as $i\tilde{K}$, a paired rotation as R and a paired boost as iQ (Propositions 5.27 and 9.2); $\dot{X} = -SX$ becomes $\dot{z} = -iHz$. (ii) $E_6 = \frac{1}{2}|X|^2$, so it is conserved for all X iff S is Euclidean antisymmetric. $K \otimes G$ and the paired rotations are, and the paired boosts are Euclidean symmetric. (iii) is Theorem 9.1(ii) on V_n. The self-adjoint part transforms as a tensor, $U^{-1}(K \otimes G)U$, while $U^{-1}bU + U^{-1}\dot{U}$ stays in $\mathfrak{so}(3n, 3n)$. (iv) $\operatorname{tr}(\tilde{K}R) = 0$ because $\tilde{K}$ is symmetric and R antisymmetric, and $\operatorname{tr}(-iR)^2 = \|R\|^2$. The variance follows from the two sum rules. For a real unit eigenvector v of $\tilde{K}$, $v^\top R v = 0$, so $\bar{v}^\top H v$ is its rate, and the largest and smallest eigenvalues of H bound it from above and below. (v) is Proposition 9.5: $e^{-b\tau}$ intertwines the systems iff b commutes with $K \otimes G$. (vi) is Theorem 9.4 with $S^\top = -S$ (both summands are Euclidean antisymmetric). Then $\mathcal{E}_\mu(S^\dagger) = \eta_n \mathcal{E}_{-\mu}(S)$, and the condition reads $\eta_n(\mathcal{E}_\nu(S), \mathcal{E}_{-\nu}(S)) = 0$ for every ν. The sectors $G_n = \pm i$ are η_n-orthogonal, and on the sector $G_n = i$, $\eta_n = 2\, v \cdot w$ in the coordinates z. The eigenspaces of S there are those of iH, and on the other sector their conjugates. □

Parts (i), (ii), (iv) and (v) are fixed but elementary: the arena's version of the fact that a compact gauge field acts unitarily (d7-3a, d7-3b, d7-3d, d7-3e). The widening in (iv) concerns the variance and the range, not each gap. At rates $(1, 2, 4)$ with $(B_{12}, B_{23}, B_{31}) = (\frac{1}{2}, \frac{1}{3}, \frac{1}{5})$ the

upper gap falls from 2 to 1.916, so the field does not repel every pair of levels (x7-3d). Part (iii) is Theorem 9.1(ii) restated for n copies, checked exactly on a Cayley curve (d7-3c). Part (vi) is a corollary of Theorem 9.4. On five cases at $n = 3$, exact semisimplicity of M, the criterion and the growth of $\|e^{Mt}\|$ agree (d7-3f). Bounded: rates $(1, 2, 4)$ with $(B_{12}, B_{23}, B_{31}) = (\frac{1}{2}, \frac{1}{3}, \frac{1}{5})$; the seesaw $(0, 3, 5)$ with $(1, \frac{1}{2}, \frac{1}{3})$, a frame without rate confined by two couplings; equal rates $(1, 1, 1)$ with $B_{12} = 1$. Unbounded: $(1, 2, 3)$ with $(1, 1, 1)$, where $\operatorname{spec} H = \{0, 2, 4\}$ and the zero mode has $v \cdot v \neq 0$, a resonance created by the field between manifolds with different rates; and zero rates with $(1, 1, 1)$, where the field produces only drift. Part (vi) generalises Theorem 5.31(v): at $n = 2$, for the field of Theorem 5.31, $\operatorname{spec} H$ is the pair of lines $\omega_\pm$ of Theorem 5.31(ii), with $\omega_+ + \omega_- = \kappa_1 + \kappa_2$, so for $\kappa_1 + \kappa_2 > 0$ the only coincidence is $\omega_- = 0$, at $\beta^2 = \kappa_1\kappa_2$. At $\kappa_1 = \kappa_2 = 0$ and $\beta \neq 0$ it is $\omega_+ = -\omega_- = |\beta|$, with eigenspaces that are not orthogonal for the bilinear form, so at zero rates the orbits are unbounded for every β, as in Theorem 5.31(v).

9.3.4 Copy boosts: metric connections, not gauge fields

Proposition 9.22 (The copy boost between two manifolds)**.** *Let $n = 2$ and $b = \beta'\varepsilon \otimes G$, so that $H = K + \beta'\varepsilon$ on the copy index (Theorem 9.21(i)), real and not symmetric. Its eigenvalues are $\frac{1}{2}(\kappa_1 + \kappa_2) \pm \sqrt{\frac{1}{4}(\kappa_1 - \kappa_2)^2 - \beta'^2}$.*

(i) Below the exceptional point, $|\beta'| < \frac{1}{2}|\kappa_1 - \kappa_2|$, and provided $\kappa_1\kappa_2 + \beta'^2 > 0$, the spectrum is real, M is semisimple with imaginary spectrum, and every orbit is bounded. The flow then preserves a positive form $z^\dagger \mathcal{Q} z$ with $H^\dagger \mathcal{Q} = \mathcal{Q} H$, which for $\beta' \neq 0$ is not E_6. (At $(\kappa_1, \kappa_2, \beta') = (0, 1, 0)$ the first copy is free and M is not semisimple.)

(ii) At $|\beta'| = \frac{1}{2}|\kappa_1 - \kappa_2|$ with $\beta' \neq 0$, M is not semisimple (at $\beta' = 0$ and $\kappa_1 = \kappa_2 > 0$ the copies are uncoupled oscillators and M is semisimple; at $\kappa_1 = \kappa_2 = \beta' = 0$ the motion is free and M has nilpotent Jordan blocks); above it, M has eigenvalues with real part $\pm\sqrt{\beta'^2 - \frac{1}{4}(\kappa_1 - \kappa_2)^2}$.

(iii) At equal rates the threshold is zero: every $\beta' \neq 0$ grows exponentially, although b commutes with $K \otimes G$ and is removable, by a boosted frame $e^{-b\tau}$ that is itself unbounded.

This is fixed, checked exactly at rates $(1, 4)$ with $\beta' = \frac{1}{2}, 1, \frac{3}{2}, 2$, and at $(2, 2)$ with $\beta' = \frac{1}{3}$ (d7-4a, d7-4b); the boundary cases at $\beta' = 0$, rates $(1, 1)$ and $(0, 0)$, are d7-4c. Its threshold is the copy analogue of the homogeneous threshold of the pairing boosts of Proposition 4.14, with the rate difference $\frac{1}{2}|\kappa_1 - \kappa_2|$ in the place of κ. The analogy stops there. A pairing boost anticommutes with G and gives Jordan blocks for every $\nu \neq 0$, below the threshold as well (Theorem 9.7(iii)), whereas the copy boost commutes with G_2 and keeps every orbit bounded below its threshold. At equal rates it is the copy analogue of the commuting boosts K_i of Theorem 9.6(iv), with exponents $\pm\beta' \pm i\kappa$, removed by a boosted frame that is itself unbounded. The spectral pattern is inherited: the non-Hermitian dimer with an exceptional point [273, 270], and the conserved metric of a pseudo-Hermitian Hamiltonian [274]. A copy boost is a metric connection, legitimate in the interior Lagrangian, but not a gauge field (Corollary 1.4). Where it keeps the motion bounded it does so at the price of the undivided energy, which it replaces by another positive form.

9.3.5 Three manifolds: a time-odd invariant

Theorem 9.23 (The copy field between three manifolds)**.** *Let $n = 3$, distinct rates, and $b = B \otimes \mathbf{1}_6$, so $H = (K - iB) \otimes \mathbf{1}_3$.*

(i) $\det(x - K + iB) = x^3 - (\sum \kappa)x^2 + (\sum_{k<l} \kappa_k\kappa_l - \sum B^2)x - (\kappa_1\kappa_2\kappa_3 - \kappa_3 B_{12}^2 - \kappa_1 B_{23}^2 - \kappa_2 B_{31}^2)$: the spectrum depends on the couplings only through their squares.

(ii) *The plaquette* $H_{12}H_{23}H_{31} = iJ$ *is purely imaginary, and* $\det[H,K] = 2iJ(\kappa_1-\kappa_2)(\kappa_1-\kappa_3)(\kappa_2-\kappa_3)$.

(iii) *The endomorphisms commuting with* G_3*, the diagonal* $\mathfrak{l}_G$ *and* $K\otimes G$ *are* $\sum_k (a_k + b_k G)$ *on copy* k*, and the* η_3*-isometries among them are exactly the eight copy reflections* $\mathrm{diag}(\pm1,\pm1,\pm1)\otimes \mathbf{1}_6$*. Under them* $B_{kl}\mapsto s_k s_l B_{kl}$*, so the invariants of the field are* $|B_{12}|, |B_{23}|, |B_{31}|$ *and the sign of* $\mathcal{J}$.

(iv) *With* $P = \mathbf{1}_3\otimes \mathrm{diag}(\mathbf{1}_3, -\mathbf{1}_3)$ *(*$\mathbf{E}\mapsto -\mathbf{E}$*) and* $T = -P$ *(*$\mathbf{B}\mapsto -\mathbf{B}$*), parity maps* (K,B) *to* $(-K,B)$*, and interior time reversal,* $X(\tau)\mapsto TX(-\tau)$*, maps* (K,B) *to* $(K,-B)$*. At* $n=2$ *a copy reflection undoes* $B\mapsto -B$*; at* $n=3$ *with* $\mathcal{J}\neq 0$ *none does, since* $\mathcal{J}\mapsto -\mathcal{J}$*. For general* n*,* (K,B) *and* $(K,-B)$ *are related by a copy reflection iff the graph of the nonzero couplings is bipartite.*

(v) *With energy started in copy* 1*, the copy energies* $P_k(\tau) = |z_k(\tau)|^2/|z(0)|^2$ *obey* $P_k(\tau;-B) = P_k(-\tau;B)$*.* P_1 *is the same for* B *and* $-B$*, and* P_2, P_3 *are not. At rates* $(1,2,4)$ *with* $B_{12}=B_{23}=B_{31}=1$ *and* $\tau=\frac{1}{2}$*,* $P_2 = 0.293$ *for* B *against* 0.116 *for* $-B$*: the sign of* $\mathcal{J}$ *is the sense in which the energy circulates around the three manifolds.*

Proof. (i), (ii) Symbolic expansion (d7-5a, d7-5b). For a 3×3 Hermitian matrix, $\det H$ contains $2\,\mathrm{Re}(H_{12}H_{23}H_{31})$, which vanishes here. (iii) The commutant is exact (d7-5c). $(a+bG)^\top \eta (a+bG) = (a^2-b^2)\eta + 2ab\,\eta G$ because $G^\top\eta = \eta G$ and $G^2 = -\mathbf{1}$, so $ab=0$ and $a^2-b^2=1$. (iv) $PGP^{-1} = -G$, P commutes with $B\otimes\mathbf{1}_6$, and $\dot X = -SX$ goes to $\dot X' = -S'X'$ with $S' = -TST^{-1}$. A copy reflection s maps B to $-B$ iff $s_k s_l = -1$ on every edge, which is a two-colouring (d7-5d, d7-5e). (v) $|(e^{-i\bar H\tau})_{k1}| = |(e^{iH\tau})_{k1}|$, and P_1 is even in τ for Hermitian H (d7-5f). □

The statements are fixed. Their pattern is inherited three times over. With two families there is no rephasing-invariant phase and with three there is one, the Kobayashi–Maskawa phase [226], whose invariant measure is Jarlskog's commutator determinant [225], the form of (ii). A purely imaginary hopping around a triangle is a flux $\pm\pi/2$ through the plaquette, the Peierls phase [275] of a lattice gauge field. And time-reversal-odd hopping makes a quantum walk chiral [278]. What the arena adds is the identification. The copy field is an η-compatible connection on the three-site "lattice" of manifolds in copy space, with $\mathcal{J}$ its holonomy around the triangle, the copy-space analogue of a Wilson loop; it is not the Wilson loop of a spacetime connection, and its sign is not by itself a physical CP phase (Theorem 9.29). Its phase is fixed at $\pm\pi/2$ because the field is real, $H_{kl} = -iB_{kl}$. Like a Wilson loop it is unchanged by per-copy duality rotations, which multiply each H_{kl} by a phase and are in any case not η-isometries (Proposition 4.47(ii)). On one manifold the binding is odd under parity and even under interior time reversal. At two manifolds the copy field is even under both, up to relabelling. From three on, a field whose couplings close a triangle is odd under interior time reversal, and no relabelling removes it.

9.3.6 Isospin one, not one-half

Yang and Mills' b-field acts on the isotopic doublet of nucleons [172]. The question is whether an inter-manifold field can.

Proposition 9.24 (Copy isospin)**.** *(i) On the tower module, the spin lift of the copy rotation of a set* D *of directions is* $-\frac{1}{2}\sum_{a\in D}\eta_{aa}e_a^{(k)}e_a^{(l)}$*, a sum of* $|D|$ *commuting terms of square* $-\frac{1}{4}$*, so* $\exp(2\pi\,\cdot) = (-1)^{|D|}$*. At* $n=2$*: for the six binding directions the eigenvalues are* $i\{-3,\dots,3\}$ *and* $\exp(2\pi C) = +\mathbf{1}$*. For the four interior directions* (M_{12}) *they are* $i\{-2,\dots,2\}$*,* $+\mathbf{1}$*. For the three temporal directions they are* $i\{\pm\frac{1}{2},\pm\frac{3}{2}\}$*,* $-\mathbf{1}$*, and that rotation does not commute with* G_2*. A binding-compatible copy rotation pairs each temporal direction with its partner, so* $|D|$ *is even.*

(ii) *At $n = 3$ the three binding copy rotations close on $\mathfrak{so}(3)$ on the 512-dimensional module, with Casimir $j(j+1) = 0, 2, 6, 12$ on $40, 216, 200, 56$ dimensions ($j = 0, \dots, 3$), and $\exp(2\pi C^{12}) = +\mathbf{1}$. The frames form the vector, $j = 1$.*

(iii) *The doublet $\mathfrak{u}(2)$ acting on (z_1, z_2) preserves $E_6 \oplus E_6$ and commutes with G_2 and the diagonal $\mathfrak{l}_G$. In real form $i\mathbf{1} \mapsto \mathbf{1} \otimes G$, $i\sigma_x \mapsto \sigma_x \otimes G$, $i\sigma_y \mapsto -\varepsilon \otimes \mathbf{1}_6$ and $i\sigma_z \mapsto \sigma_z \otimes G$. Only $i\sigma_y$, the copy rotation, is η_2-antisymmetric. The other three are η_2-self-adjoint, terms of the same type as the binding. For $S = \kappa G_2 + b_{\mathrm{YM}}$, $b_{\mathrm{YM}} = \sum_a \beta_a (i\sigma_a)$, the lines are $\kappa \pm |\boldsymbol{\beta}|$, and the rates, the spectrum of the self-adjoint part, are $\kappa \pm (\beta_x^2 + \beta_z^2)^{1/2}$.*

This is fixed (d7-6a to d7-6c). Part (i) restates Proposition 5.16 ("the copy algebras act through orthogonal groups") and Proposition 5.29(i), and adds that binding-compatibility forces $|D|$ to be even, so every binding-compatible copy rotation lifts single-valuedly. The criterion is the parity of $|D|$, not the binding: M_{12} is not binding-compatible and also lifts single-valuedly. Under a common Lorentz action the inter-manifold field is therefore an $SO(n)$ gauge field. At $n = 3$ it acts with integer isospin only: the frames are triplets, as in the model of Georgi and Glashow [50], the tower module carries $j = 0, \dots, 3$, and it never acts on doublets. Part (iii) restates Proposition 5.23(i) in the Yang–Mills reading and adds the spectra; it is the precise form of "Yang–Mills between two copies". The doublet group exists as a symmetry of the undivided energy (the copy $\mathfrak{u}(2)$), but η_2 cuts it to its circle. The two directions it removes are not connections at all: they belong to the symmetric part, where the binding c^2 lives in the dictionary of Section I.2. A doublet field that splits the rates between two manifolds does so only through those two directions. In the arena that is a modification of the symmetric part, not a b-field.

Proposition 9.25 (The isospin $\mathfrak{so}(4)_\Delta$ between unequal constituents)**.** *At $n = 3$, with the numbering of Section 4.4 and $T_i^\pm$ as in Lemma 9.14, let $K = \mathrm{diag}(\kappa_1, \kappa_2, \kappa_3)$ with $\kappa_2 \neq \kappa_3$.*

(i) *The part of $\mathfrak{so}(4)_\Delta$ commuting with $K \otimes G$ is $\mathrm{span}\{\mathrm{rot}(13,14), \mathrm{rot}(13,15), \mathrm{rot}(14,15)\} = \mathrm{span}\{T_1^+ + T_1^-,\ T_2^+ - T_2^-,\ T_3^+ - T_3^-\}$, the graph of an isomorphism $\mathfrak{su}(2)_+ \to \mathfrak{su}(2)_-$: a diagonal $\mathfrak{so}(3)$. The three directions $\mathrm{rot}(7,13), \mathrm{rot}(7,14), \mathrm{rot}(7,15)$, which join constituent two to constituent three, are not removable.*

(ii) *For $\kappa = (5, 3, 5)$ and $b = 2T_3^+$,*

$$\det(\lambda - S) = (\lambda^2 + 9)^2(\lambda^2 + 16)(\lambda^2 + 25)^3(\lambda^2 + 36)(\lambda^4 + 36\lambda^2 + 196),$$

and M is semisimple. The quartic carries the lines $4 \pm \sqrt{2}$, the two-level mixing of z_7 and z_{13} with rates $3, 5$ and coupling 1 (Theorem 5.31(ii)). At equal rates 5 the same field gives $(\lambda^2 + 25)^5(\lambda^2 + 16)^2(\lambda^2 + 36)^2$, the removable Zeeman pattern of Proposition 9.15(i).

This is fixed (d7-7a, d7-7b). It completes Proposition 9.15, which used equal rates. The paper's binding-compatible non-abelian algebra is itself an inter-manifold field: three of its six directions, $\mathrm{rot}(7,13)$, $\mathrm{rot}(7,14)$ and $\mathrm{rot}(7,15)$, join constituent two to constituent three, and the other three rotate constituent three alone. It carries doublets ($(\frac{1}{2}, \frac{1}{2})$ on W) because the spectator action leaves constituents two and three free of the Lorentz algebra (Proposition 9.19(iii)). With equal rates it is removable along a worldline, as Section 9.1.6 found. With unequal rates it breaks to a diagonal $\mathfrak{so}(3)$, and its three remaining directions become physical. The pattern $\mathfrak{su}(2) \oplus \mathfrak{su}(2) \to \mathfrak{su}(2)_{\mathrm{diag}}$ is that of chiral symmetry breaking and of the custodial symmetry of the Higgs sector [276]. That is an interpretive parallel only: here the breaking is by a constant rate matrix, not by a vacuum.

9.3.7 Assessment

Remark 9.26 (The deal breaker, settled)**.** (a) *True, in this sense* (fixed). A compact binding-compatible field that couples manifolds of different rates cannot be removed:

by Theorem 9.21(v) it is removable exactly when it preserves every rate eigenspace. It shifts and splits the lines, it can confine a frame that has no rate of its own, it can create a resonance, and from three manifolds on it is non-abelian and carries a time-odd invariant. What does not extend from one manifold is the removability of a field that commutes with the binding root (Proposition 9.5). What does extend is the no-go of Proposition 9.11: a connection alone does not bind, and at zero rates it produces only drift (Theorems 5.31, 9.21 and 9.23).

(b) *False, in this sense* (fixed). The field never creates or changes a rate. Rates are the spectrum of the self-adjoint part, a tensor under every change of frame. The field only redistributes the frequencies, the lines of $\operatorname{spec} H$, at fixed sum. With equal rates every binding-compatible field, abelian or the $\mathfrak{so}(4)_\Delta$ of Theorem 4.36, is removable along a worldline. With zero rates it produces only drift (Theorem 9.21(iii)–(vi)).

(c) *Where unequal rates come from.* Not from the closure of Section 4.4: it imposes a common rate, so its single rotor acts through G_n at one rate, and within that closure as it stands the inter-manifold field is removable (fixed for that closure). Under a common binding action the closure admits a rotor acting by a non-central element of $\mathfrak{u}(n)$. The copy field and the rate differences can then both enter, as two independent parts of one generator, and either can occur without the other (constructed: admitted, not selected; Proposition 9.20).

(d) *Isospin one-half* (fixed). On two and three manifolds under a common Lorentz action the copy isospin is integral, and the doublet field on two copies is a connection in one of its three directions only. On three manifolds doublets need the spectator action, where the Lorentz-carrying manifold is decoupled (Propositions 9.19, 9.24 and 9.25). This does not extend to four manifolds. There the binding-compatible copy rotations under a common action form $\mathfrak{so}(4) \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$, the copy index is its vector $(\frac{1}{2}, \frac{1}{2})$, and doublets exist without a spectator (d7-6d).

(e) *The arrow reversed* (interpretive). The non-removable part of the copy field is $[K \otimes G, B \otimes \mathbf{1}_6] = [K, B] \otimes G$, with $[K, B]_{kl} = (\kappa_k - \kappa_l) B_{kl}$ and $\|[K \otimes G, b]\|^2 = 6 \sum_{k \neq l} (\kappa_k - \kappa_l)^2 B_{kl}^2$ (d7-8, fixed). Formally this is the gauge-boson mass term $g^2 \|[T, \langle \Phi \rangle]\|^2$ of the Englert–Brout–Higgs–Kibble mechanism [234, 235, 236], with the rate matrix in the role of an adjoint vacuum value. Read with $mc^2 = \hbar\kappa$, so that a mass is a rate, the arena reverses the arrow of the hypothesis. It is not the b-field that differentiates the rates; the differentiation of the rates, which breaks the copy symmetry, is what gives the b-field an effect. The parallel is not dynamical: here b is a constant connection without kinetic term, and the rates are inputs. Read instead from the lines, the spectrum of H, a mass is shifted and split by the field, and a frame without rate of its own is confined at a nonzero line, the seesaw (Theorem 5.31(vi)). That effect is non-removable only when the rates differ (Theorem 9.21(v)), so in this reading too the field acts through the rate differences and does not create them. In the closure of Proposition 9.20 the copy field and the rate differences are independent parts of one generator, so the order is logical, not causal.

9.4 The b-field up the ladder

Section 9.3 studied the field that acts between the constituents of one rung, the copy field $b = B \otimes \mathbf{1}_6$, and settled what it does to rates and lines on two and three constituents. This section follows that field up the ladder, $n = 2, 3, 4, \ldots$, and reads it in the Fock picture of Sections 5.4.4 and 5.5.4. The field always acts between the constituents of one rung $\mathrm{Cl}(3n, 3n)$; going up the ladder means increasing n.

statement	class	checks
Proposition 9.19(i), (iii): $\mathfrak{so}(n,\mathbb{C})$ under a common binding action; copy 1 decoupled under the spectator action	fixed (restated: Prop. 13.14; Thm 17.20 for $\mathfrak{l}^{(1)}_{\mathrm{int}}$)	d7-1a, d7-1d
Proposition 9.19(ii): $\mathfrak{so}(n)$ under the interior action; $C^{12} = -M_{12} + C_{\mathrm{surp}}$	fixed	d7-1b, d7-1c
Proposition 9.20: rotor generators $\mathfrak{u}(n) = \mathfrak{so}(n) \oplus \mathrm{Sym}(n) \otimes G$; abelian frame only for non-resonant A	fixed (restated: Prop. 13.14); non-central rotor constructed	d7-2a, d7-2b
Theorem 9.21(i), (ii), (iv), (v): Schrödinger form, unitarity iff compact, sum rules, removability	fixed (elementary)	d7-3a, d7-3b, d7-3d, d7-3e
Theorem 9.21(iii): rates are gauge invariants	fixed (restated: Theorem 9.1)	d7-3c
Theorem 9.21(vi): bilinear boundedness criterion; five cases at $n = 3$	fixed (corollary of Theorem 9.4)	d7-3f
Proposition 9.22: copy boost, exceptional point, conserved positive form	fixed; pattern inherited	d7-4a, d7-4b
Theorem 9.23: spectrum, plaquette, Jarlskog determinant, reflections, P and T, bipartite criterion, circulation	fixed; parallels inherited	d7-5a to d7-5f
Proposition 9.24: integral copy isospin at $n = 2, 3$; the doublet is one connection and two rate terms; doublets at $n = 4$	fixed (restated: Prop. 10.9, Proposition 5.23)	d7-6a to d7-6d
Proposition 9.25: $\mathfrak{so}(4)_\Delta \to$ diagonal $\mathfrak{so}(3)$ at unequal rates	fixed; custodial parallel interpretive	d7-7a, d7-7b
Remark 9.26: the deal breaker; the commutator; the arrow reversed	fixed; (c) constructed; (e) interpretive	d7-8

Table 9.1: Claims of Section 9.3, their classes, and the checks printed by `d7_bfield_between.py`.

The answers, with their classes, are these. (1) Every compact binding-compatible field is the second quantisation of a one-quantum Hamiltonian. It conserves the number N of quanta, moves quanta between the constituents, and its many-quantum lines are sums of one-quantum lines. The copy boosts create pairs, one quantum in each of the two constituents they join (fixed; Section 9.4.1). (2) On n constituents with distinct rates and all couplings nonzero, the invariants of the copy field are the $|B_{kl}|$ and the signs of the products of B around the cycles of the complete graph. Exactly $(n-1)(n-2)/2$ of these signs are independent, and all of them can be taken time-odd. The Jarlskog-type determinant $\det[H, K]$ has time parity $(-1)^n$, so it detects time reversal only at odd n (fixed; Section 9.4.2). (3) At $n = 4$ the copy group is $SO(4)$, and each of its two $\mathfrak{su}(2)$ factors has half-integer isospin exactly on the states with an odd number of quanta. The copy inversion acts as $(-1)^N$. Above $n = 4$ the same holds for the $\mathfrak{so}(4)$ of any four constituents, with the quanta in those four. At no rung does a spinor of the copy algebra occur (fixed; Section 9.4.3). (4) For rates of multiplicities $n_1 + \cdots + n_r = n$ the removable part of the copy $\mathfrak{so}(n)$ is $\bigoplus_i \mathfrak{so}(n_i)$, and $\sum_{i<j} n_i n_j$ directions are physical (fixed; the Higgs-like reading is interpretive; Section 9.4.4). (5) For a rotor with rates (κ_k), a frame bracket between copies i, j and k is admitted only if $\kappa_k = \kappa_i + \kappa_j$. A harmonic ladder of rates $\kappa_k = k\kappa_1$ therefore admits non-abelian, nilpotent frames. From $n = 4$ on the Jacobi identity among the frame brackets is a quadratic condition (fixed; the reading is constructed; Section 9.4.5). Section 9.4.6 says what the field can and cannot do for the emergence of particles. Identifiers e4-n refer to the `PASS` lines of `e4_bfield_ladder.py` (Table 9.4).

Notation. As in Section 9.3: $V_n = \mathbb{R}^n \otimes \mathbb{R}^6$, η_n, $G_n = \mathbf{1}_n \otimes G$, $z = (z_1, \ldots, z_n) \in \mathbb{C}^{3n}$ with $z_k = \mathbf{B}_k + i\mathbf{E}_k$, copy rotations $c^{kl} = (E_{lk} - E_{kl}) \otimes \mathbf{1}_6$, copy boosts $(E_{lk} - E_{kl}) \otimes G$, copy field $b = B \otimes \mathbf{1}_6$ with $B \in \mathfrak{so}(n)$, rates $K = \mathrm{diag}(\kappa_1, \ldots, \kappa_n)$, $\tilde{K} = K \otimes \mathbf{1}_3$, and, for a compact binding-compatible b acting on z as the real antisymmetric R, the frame Hamiltonian $H = \tilde{K} - iR$ of Theorem 9.21. σ is the spin lift to $S_n = (\mathbb{R}^8)^{\otimes n}$, with $[\sigma(X), v] = Xv$ for

$v \in V_n$, and $\hat{C}^{kl} = \sigma(c^{kl}) = -C^{kl}$, where $C^{kl} = \frac{1}{2}\sum_a \eta_{aa} E_{(k,a)} E_{(l,a)}$ are the spin-level copy rotations of Section 5.7 and $C^{12} = C_b$. The $3n$ modes are $p = (k,i)$, $k \le n$, $i \le 3$, with $a^{\pm}_{k,i} = \frac{1}{2}(E_{(k,i)} \pm E_{(k,i+3)})$ (Proposition 5.21). $N_{k,i} = a^{+}_{k,i} a^{-}_{k,i}$, $N^{(k)} = \sum_i N_{k,i}$ is the number in constituent k, $N_{[i]} = \sum_k N_{k,i}$ the number in direction i, and $N = \sum_k N^{(k)}$. For a complex $3n \times 3n$ matrix M we write $\mathcal{N}(M) = \sum_{p,q} M_{pq}\, a^{+}_p a^{-}_q$. The first basis vector of the realisation is the Fock vacuum $|0\rangle$ (e4-1a).

9.4.1 Quanta between constituents: transfer, not creation

Theorem 5.39 shows, at every rung, that the compact binding-compatible fields are exactly the binding-compatible fields that conserve N, and Propositions 5.48 and 5.51 give the hop and the pair at $n = 2$. The theorem below adds the second-quantised form, which says how the quanta move, and the bookkeeping by constituent and by direction at $n = 3$.

Theorem 9.27 (The field between constituents, second quantised)**.** *(i) Let X be a paired rotation, acting on z as the real antisymmetric $R \in \mathfrak{so}(3n)$. Then $\sigma(X) = \mathcal{N}(R)$. Hence $\sigma(X)$ commutes with N, and $\sigma(X)\, a^{+}_q|0\rangle = \sum_p R_{pq}\, a^{+}_p|0\rangle$.*

(ii) For the copy field $b = B \otimes \mathbf{1}_6$ and rates K, the operator $\mathsf{H} = \sum_k \kappa_k N^{(k)} - i\sigma(b)$ on $S_n \otimes \mathbb{C}$ equals $\mathcal{N}(H)$, $H = (K - iB) \otimes \mathbf{1}_3$. It is Hermitian and commutes with N. Its block on $\{N = 1\}$, in the basis $a^{+}_p|0\rangle$, is H, and its eigenvalues on $\{N = m\}$ are the sums of m eigenvalues of H with distinct indices.

(iii) At $n = 3$, $\hat{C}^{kl} = \sum_i (a^{+}_{l,i} a^{-}_{k,i} - a^{+}_{k,i} a^{-}_{l,i})$. It commutes with N, with $N^{(k)} + N^{(l)}$, with $N^{(m)}$ for $m \notin \{k,l\}$ and with every $N_{[i]}$, and not with $N^{(k)}$. The three lifts close on $\mathfrak{so}(3)$ as on vectors, $[\hat{C}^{12}, \hat{C}^{13}] = \hat{C}^{23}$.

(iv) At $n = 3$, $\sigma\big((E_{lk} - E_{kl}) \otimes G\big) = -\sum_i (a^{+}_{k,i} a^{+}_{l,i} + a^{-}_{l,i} a^{-}_{k,i})$. It joins only sectors whose N differs by 2, in both directions, and it commutes with $N^{(k)} - N^{(l)}$ and with $N^{(m)}$, $m \notin \{k,l\}$: every pair it creates has one quantum in constituent k and one in constituent l.

(v) At $n = 3$ the quarter turn $U = e^{\pi \hat{C}^{12}/2} = \frac{1}{8}\prod_a (\mathbf{1} + 2t_a)$, with $t_a = -\frac{1}{2}\eta_{aa} E_{(1,a)} E_{(2,a)}$ the six commuting terms of $\hat{C}^{12}$, is orthogonal, commutes with N and $N^{(3)}$, and satisfies $U N^{(1)} U^{-1} = N^{(2)}$. For $\kappa_1 \neq \kappa_2$ it changes the rate operator $\sum_k \kappa_k N^{(k)}$ and keeps its spectrum.

Proof. (i) A paired rotation acts in the same way on the temporal directions and on their partners, so $X(2a^{\pm}_q) = \sum_p R_{pq}\, 2a^{\pm}_p$, and $[\sigma(X), a^{\pm}_q] = \sum_p R_{pq} a^{\pm}_p$. The CAR give $[\mathcal{N}(R), a^{+}_q] = \sum_p R_{pq} a^{+}_p$ and $[\mathcal{N}(R), a^{-}_q] = -\sum_p R_{qp} a^{-}_p$, which is the same because $R^{\top} = -R$. So $\sigma(X) - \mathcal{N}(R)$ commutes with every generator, and S_n is irreducible, so it is a multiple of $\mathbf{1}$. Both operators are traceless: $\sigma(X)$ is a sum of products of two anticommuting generators, and $\mathrm{Tr}(a^{+}_p a^{-}_q) = 0$ for $p \neq q$, while $R_{pp} = 0$. Hence they are equal. Each term $a^{+}_p a^{-}_q$ conserves N, and $a^{-}_q|0\rangle = 0$ gives the action on one quantum. (ii) $N^{(k)} = \mathcal{N}(E_{kk} \otimes \mathbf{1}_3)$, and by (i) $\sigma(b) = \mathcal{N}(B \otimes \mathbf{1}_3)$, so $\mathsf{H} = \mathcal{N}(H)$. $N^{(k)}$ is real diagonal and $\sigma(b)$ real antisymmetric, so H is Hermitian. For $H = W D W^{\dagger}$ with W unitary, $b^{\pm}_r = \sum_p W^{(\pm)}_{pr} a^{\pm}_p$ ($W^{(+)} = W$, $W^{(-)} = \bar{W}$) satisfy the CAR and $\mathsf{H} = \sum_r d_r\, b^{+}_r b^{-}_r$; its eigenvectors are $b^{+}_{r_1} \cdots b^{+}_{r_m}|0\rangle$ with $r_1 < \cdots < r_m$. (iii) is (i) for $R = (E_{lk} - E_{kl}) \otimes \mathbf{1}_3$. Each term moves one quantum between (k,i) and (l,i). (iv) A copy boost anticommutes with K_3, so its lift is a combination of $a^{+}a^{+}$ and $a^{-}a^{-}$ terms (Lemma 5.46(ii)). It maps the plane of mode (k,i) to that of mode (l,i), which fixes the modes paired, and the coefficients are computed. (v) $t_a^2 = -\frac{1}{4}$, so $e^{\theta \hat{C}^{12}} = \prod_a (\cos\frac{\theta}{2} + 2\sin\frac{\theta}{2}\, t_a)$. At $\theta = \frac{\pi}{2}$, $\mathrm{Ad}\, U$ maps $E_{(1,a)}$ to $\pm E_{(2,a)}$ and $E_{(2,a)}$ to $\mp E_{(1,a)}$, and fixes $E_{(3,a)}$. $N^{(k)}$ is a function of the products $E_{(k,i)} E_{(k,i+3)}$, so $U N^{(1)} U^{-1} = N^{(2)}$, and conjugation keeps spectra. □

The theorem is fixed. Part (i) is the standard realisation of $\mathfrak{so}(3n)$ on the exterior algebra by one-body operators [201, 216], inherited, and its realisation in the tower is checked exactly for all 15 and 36 basis elements at $n = 2, 3$ (e4-1b) and for the copy rotations at $n = 4$ (e4-3a). Part (ii) is checked exactly on $\{N = 1\}$ at $n = 3$, rates $(1, 2, 4)$ and $(B_{12}, B_{23}, B_{31}) = (\frac{1}{2}, \frac{1}{3}, \frac{1}{5})$, and the additivity on $\{N = 2\}$ and $\{N = 3\}$ to 10^{-9} (e4-1e). Parts (iii) to (v) are exact (e4-1c, e4-1d, e4-1f); in (iv) a random paired boost of $\mathfrak{so}(9, \mathbb{C})$ also changes N by exactly ± 2. The external i in H is constructed, as in Proposition 5.51(iii). Under $z \mapsto \sum_p z_p a_p^+ |0\rangle$ the one-quantum sector of $e^{-i\mathsf{H}t}$ is the frame flow $z(t) = e^{-iHt} z(0)$ of Theorem 9.21(i).

Quadratic fermion dynamics of this kind is standard [310]. Nonnegative rates do not make H positive: for the bounded seesaw example, rates $(0, 3, 5)$ and couplings $(1, \frac{1}{2}, \frac{1}{3})$, H has the eigenvalue -0.326115, so the empty state is not the lowest state of $\mathsf{H} = \mathcal{N}(H)$, which is bounded below with its negative modes filled. This exterior-algebra model is distinct from the canonical bosonic system of Theorem 9.12 and is not its quantisation; the removability of b refers to removing it while keeping the prescribed constant binding tensor.

Corollary 9.28 (Transfer, not creation)**.** *Let b be a compact element of $\mathrm{Cent}(G_n) \cap \mathfrak{so}(3n, 3n)$, the class of gauge-admissible fields between the constituents (Theorem 9.21(ii); Corollary 1.4).*

(i) $e^{t\sigma(b)}$ conserves N and acts on $\{N = m\}$ as the m-th exterior power of e^{tR}. It moves quanta between modes and constituents, and never creates or annihilates one.

(ii) The binding-compatible elements that change N are the paired boosts, among them the copy boosts. Each of their terms creates or annihilates one pair, and a copy boost puts one quantum in each of the two constituents it joins.

(iii) Let $\hat{U}$ lift a compact η_n-isometry commuting with G_n, which acts on z as a unitary W. A change of frame by $\hat{U}$ takes the rate operator $\mathcal{N}(\tilde{K})$ to $\hat{U}\mathcal{N}(\tilde{K})\hat{U}^{-1} = \mathcal{N}(W\tilde{K}W^{-1})$. This operator commutes with N and has the same spectrum on every sector $\{N = m\}$, the sums of m rates. The numbers $N^{(k)}$ are not kept.

Proof. (i) The proof of Theorem 9.27(i) holds at every n, and $\mathcal{N}(R)$ is the derivation of the exterior algebra induced by R, so its exponential is $\Lambda(e^{tR})$. (ii) is Theorem 5.39(ii), with the pairing of Theorem 9.27(iv) at $n = 3$ and of Proposition 5.48(iii) at $n = 2$. (iii) $\mathrm{Ad}\,\hat{U}$ acts on vectors as the isometry, hence on the a_p^+ as W, which gives the formula. A compact isometry commuting with G_n commutes with $K_n = -\eta_n G_n$, so $\hat{U}$ commutes with N, and conjugation keeps spectra. This is the Fock form of Theorem 9.21(iii), and Theorem 9.27(v) exhibits a frame change that moves $N^{(1)}$ to $N^{(2)}$. □

The corollary is fixed. It answers the first question precisely. The gauge-admissible field between constituents transfers quanta and never creates them. It does not change a rate either (Theorem 9.21(iii)). Which constituent carries a quantum depends on the frame when the copy rotation is gauged, as Proposition 5.51 found at $n = 2$. How many quanta there are, and the multiset of their rates, does not. Pair creation across constituents needs a copy boost, which is a metric connection and not a gauge field (Proposition 9.22). Because H is one-body, its many-quantum lines carry no interaction energy. The field mixes quanta; it does not bind them. This is a statement about the number-preserving one-body class $\mathcal{N}(H)$ (the second quantisation $d\Gamma(H)$); it says nothing about extensions of the framework with two-body terms, which are not constructed here.

9.4.2 Time-odd invariants on n constituents

Theorem 9.23 found on three constituents one invariant that two do not carry, the sign of $\mathcal{J} = B_{12}B_{23}B_{31}$, odd under interior time reversal. For a cycle $\gamma = (k_1, \ldots, k_L)$ of distinct copy indices write $\Pi_\gamma B = B_{k_1k_2} B_{k_2k_3} \cdots B_{k_Lk_1}$; reversing the orientation multiplies it by $(-1)^L$.

Theorem 9.29 (The copy field on n constituents)**.** *Let $n \geq 3$, let the rates be distinct, and let $b = B \otimes \mathbf{1}_6$ with every $B_{kl} \neq 0$.*

(i) The endomorphisms of V_n commuting with G_n, the diagonal $\mathfrak{l}_G$ and $K \otimes G$ are $\sum_k (a_k + b_k G)$ on copy k, and the η_n-isometries among them are the 2^n copy reflections $\mathrm{diag}(s_1, \ldots, s_n) \otimes \mathbf{1}_6$, $s_k = \pm 1$, under which $B_{kl} \mapsto s_k s_l B_{kl}$.

(ii) Two fields with the same $|B_{kl}|$ are related by a copy reflection iff $\Pi_\gamma B$ has the same sign for both on every cycle γ. The signs of the $c_n = \frac{1}{2}(n-1)(n-2)$ triangles $B_{1k}B_{kl}B_{l1}$, $2 \leq k < l \leq n$, are independent and determine all the others. The invariants of the field are the $\frac{1}{2}n(n-1)$ magnitudes and these c_n signs.

(iii) Interior time reversal, $(K, B) \mapsto (K, -B)$ (Theorem 9.23(iv)), multiplies $\Pi_\gamma B$ by $(-1)^L$. It flips all c_n triangle signs, and no copy reflection undoes it.

(iv) The triangle invariants are spectral. With P_k the spectral projectors of K, polynomials in K, and $H = K - iB$ on the copy index, $\mathrm{tr}(P_k H P_l H P_m H) = i B_{kl} B_{lm} B_{mk}$ for every triangle (k, l, m).

Proof. (i) The commutant of G_n and the diagonal $\mathfrak{l}_G$ is $\{A \otimes \mathbf{1} + A' \otimes G\}$ with $A, A' \in \mathfrak{gl}(n)$ (Proposition 9.20). $[A \otimes \mathbf{1} + A' \otimes G, K \otimes G] = [A, K] \otimes G - [A', K] \otimes \mathbf{1}$, so A and A' commute with K and are diagonal. The isometry condition per copy is $a_k b_k = 0$, $a_k^2 - b_k^2 = 1$, as in the proof of Theorem 9.23(iii). (ii) Encode the signs of B_{kl}, $k < l$, as $\sigma \in \mathbb{Z}_2^E$, E the edges of the complete graph K_n. A reflection adds δs, $(\delta s)_{kl} = s_k + s_l$, so the orbits are the cosets of $\mathrm{im}\,\delta$. $\ker \delta$ is the constants because K_n is connected, so $\mathrm{rank}\,\delta = n - 1$ and there are $2^{|E|-n+1} = 2^{c_n}$ orbits. The functions of σ that are constant on cosets are the characters $\sigma \mapsto \sum_{e \in \gamma} \sigma_e$ with γ in the annihilator of $\mathrm{im}\,\delta$, the edge sets of even degree at every vertex, which form the cycle space [228]. Its dimension is $|E| - n + 1 = c_n$, and the fundamental cycles of the star at vertex 1, the triangles $(1, k, l)$, form a basis. A character pairs perfectly with the cosets, so the triangle signs separate the orbits. The oriented product $\Pi_\gamma B$ differs from the unoriented one by a sign fixed by γ. (iii) $-B$ adds the all-ones vector to σ, and its character on γ is $L \bmod 2$. A triangle is odd, so the all-ones vector is not in $\mathrm{im}\,\delta$. (iv) $P_k H P_l = H_{kl} E_{kl}$ and $H_{kl} = -i B_{kl}$ for $k \neq l$, and $P_k = \prod_{j \neq k} (K - \kappa_j)/(\kappa_k - \kappa_j)$ for distinct rates. □

The theorem is fixed. Part (i) is exact at $n = 4$, rates $(1, 2, 3, 5)$ (e4-2a), and at $n = 3$ it is Theorem 9.23(iii). Parts (ii) and (iii) are checked exhaustively on all sign patterns at $n = 3, 4, 5, 6$, with $c_n = 1, 3, 6, 10$ and $\mathrm{rank}\,\delta = n - 1$ over $\mathbb{Z}_2$ (e4-2b, e4-2c). Part (iv) is symbolic at $n = 4$, where the product of the four triangle invariants is a product of squares, because every edge of K_4 lies on two triangles (e4-2d). The count c_n is the cycle rank of K_n, inherited from graph theory. It is also the number of rephasing-invariant phases of n families of quarks [226, 227] (inherited). Here the phases are fixed at $\pm\pi/2$ because the field is real (Section 9.3.5), so they reduce to signs. On n constituents the copy field carries c_n time-odd signs, the holonomies of the constant copy connection around the triangles of the complete graph of constituents. These are discrete holonomies in copy space, not Wilson loops of a spacetime gauge field, and "time-odd" refers to the interior time reversal $(K, B) \mapsto (K, -B)$. The comparison with quark mixing is a count of phases only: no mixing matrix between mass and interaction states, no CP operation on a state space and no gauge connection on spacetime is specified, so the signs are not shown to be physical CP violation.

Remark 9.30 (Couplings that vanish)**.** For a general coupling graph Γ_B, with $|E'|$ nonzero couplings and $c(\Gamma_B)$ connected components, the same argument gives $2^{|E'|-n+c(\Gamma_B)}$ classes of sign patterns, so the number of independent cycle signs is the cycle rank $|E'| - n + c(\Gamma_B)$ (exhaustive over all 64 graphs at $n = 4$, rank over $\mathbb{Z}_2$ for all graphs at $n = 5, 6$; e4-2f). A chain of constituents carries no sign. A ring of n constituents carries one sign, odd under time reversal iff n is odd, in agreement with the bipartite criterion of Theorem 9.23(iv). This is fixed.

Proposition 9.31 (The Jarlskog-type determinant up the ladder)**.** *On the copy index* $[H,K] = iD$ *with* $D_{kl} = (\kappa_k - \kappa_l)B_{kl}$*, a real symmetric matrix with zero diagonal. Hence* $\det[H,K] = i^n \det D$*, and* $\det D$ *is the sum over derangements* π *of* $(-1)^{n-\#cycles(\pi)} \prod_k D_{k\pi(k)}$*. Under* $B \mapsto -B$ *it changes by* $(-1)^n$*.*

(i) $n = 3$*:* $\det[H,K] = 2iJ(\kappa_1 - \kappa_2)(\kappa_1 - \kappa_3)(\kappa_2 - \kappa_3)$*.*

(ii) $n = 4$*:* $\det[H,K] = D_{12}^2 D_{34}^2 + D_{13}^2 D_{24}^2 + D_{14}^2 D_{23}^2 - 2(D_{12}D_{23}D_{34}D_{41} + D_{12}D_{24}D_{43}D_{31} + D_{13}D_{32}D_{24}D_{41})$*. It is even under time reversal and contains no triangle.*

(iii) $n = 5$*:* $\det[H,K]$ *is odd under time reversal; every cycle cover contains a* 5*-cycle or a triangle.*

Proof. $[K - iB, K] = -i[B,K]$ and $[B,K]_{kl} = -(\kappa_k - \kappa_l)B_{kl}$. D is a product of two antisymmetric factors entrywise, hence symmetric. The Leibniz expansion of a matrix with zero diagonal runs over derangements, and $-D$ multiplies each term by $(-1)^n$. At $n = 4$ the derangements are the three double transpositions and the six 4-cycles. At $n = 5$ they are the 24 five-cycles and the 20 products of a transposition and a 3-cycle. □

The proposition is fixed. Part (i) restates Theorem 9.23(ii), and the identities are symbolic at $n = 3, 4, 5$, with the cycle-cover sum checked at $n = 4, 5$ (e4-2e). The determinant measures cycle covers. A 2-cycle contributes the magnitude $(\kappa_k - \kappa_l)^2 B_{kl}^2$, and a cycle of length $L \geq 3$ contributes twice the product of the rate differences around it times $\Pi_\gamma B$. At $n = 4$ the 4-cycle products are time-even: $B_{12}B_{23}B_{34}B_{41} = -B_{12}B_{23}B_{31} \cdot B_{13}B_{34}B_{41}/B_{13}^2$ is fixed by the product of two triangle signs. So $\det[H,K]$ cannot tell (K,B) from $(K,-B)$ at $n = 4$, although the field is time-odd. The time-odd invariants there are the triangle traces of Theorem 9.29(iv). Jarlskog's determinant [225] measures time reversal on three constituents and, among the rungs, only at odd n; the triangle traces measure it at every $n \geq 3$.

9.4.3 Copy isospin at four constituents

On two and three constituents the copy isospin is integral (Proposition 9.24), and Remark 9.26(d) notes that this does not extend to four (d7-6d). We determine the copy isospin content of S_4, of dimension 4096. Write

$$I_1^\pm = \tfrac{1}{2}(\hat{C}^{23} \pm \hat{C}^{14}), \qquad I_2^\pm = \tfrac{1}{2}(\hat{C}^{31} \pm \hat{C}^{24}), \qquad I_3^\pm = \tfrac{1}{2}(\hat{C}^{12} \pm \hat{C}^{34})$$

(the self-dual and anti-self-dual combinations; $J_a^\pm$ in the script), and let $j_\pm(j_\pm + 1)$ be the eigenvalues of the Casimirs $-\sum_a (I_a^\pm)^2$.

Theorem 9.32 (Copy isospin on four constituents)**.** *Let* $n = 4$*.*

(i) Each $\hat{C}^{kl}$ *equals* $\sum_i (a_{l,i}^+ a_{k,i}^- - a_{k,i}^+ a_{l,i}^-)$*, commutes with every* $N_{[i]}$*, and is a sum of six commuting terms of square* $-\frac{1}{4}$*, so* $e^{2\pi \hat{C}^{kl}} = +\mathbf{1}$*. The six close on* $\mathfrak{so}(4)$*, and the copy group acting on* S_4 *is* $SO(4)$*.*

(ii) $[I_a^+, I_b^-] = 0$ *and* $[I_1^\pm, I_2^\pm] = I_3^\pm$ *cyclically, on vectors and on* S_4*. On vectors* $4(I_a^\pm)^2 = -\mathbf{1}$*, so the copy index is the module* $(\frac{1}{2}, \frac{1}{2})$*. On* S_4*,* $e^{2\pi I_3^+} = e^{2\pi I_3^-} = (-1)^N$*. Hence* $I_3^+ = \frac{1}{2}(\hat{C}^{12} + \hat{C}^{34})$ *has eigenvalues* $i\mu$ *with* μ *half-integral exactly on the sectors of odd* N*. In particular* $e^{2\pi I_3^+} a_{1,1}^+ |0\rangle = -a_{1,1}^+ |0\rangle$*.*

(iii) As an $\mathfrak{so}(4)$*-module,* $S_4 \otimes \mathbb{C} \cong (\Lambda \mathbb{C}^4)^{\otimes 3}$ *with* $\Lambda \mathbb{C}^4 = 2(0,0) \oplus 2(\frac{1}{2}, \frac{1}{2}) \oplus (1,0) \oplus (0,1)$*. The modules* (j_+, j_-) *and their multiplicities are those of Table 9.2. Always* $j_+ - j_- \in \mathbb{Z}$*. The modules with half-integral* j_+ *fill exactly the odd-*N *sectors,* 2048 *dimensions. The one-quantum sector is* $(\frac{1}{2}, \frac{1}{2}) \otimes \mathbb{R}^3$*.*

even N: (j_+, j_-)	$(0,0)$	$(1,0)$	$(0,1)$	$(1,1)$	$(2,0)$	$(0,2)$	$(2,1)$	$(1,2)$	$(3,0)$	$(0,3)$
multiplicity	70	84	84	90	20	20	15	15	1	1
odd N: (j_+, j_-)	$(\frac{1}{2},\frac{1}{2})$	$(\frac{3}{2},\frac{1}{2})$	$(\frac{1}{2},\frac{3}{2})$	$(\frac{3}{2},\frac{3}{2})$	$(\frac{5}{2},\frac{1}{2})$	$(\frac{1}{2},\frac{5}{2})$				
multiplicity	140	64	64	20	6	6				

Table 9.2: The copy isospin content of $S_4 \otimes \mathbb{C}$ under $\mathfrak{so}(4) = \mathfrak{su}(2)_+ \oplus \mathfrak{su}(2)_-$. Each half has dimension 2048 (e4-3c).

Proof. (i) The one-body form is Theorem 9.27(i), and the six terms are those of Proposition 9.24(i) with $|D| = 6$. A module of $\mathfrak{so}(4)$ on which the 2π rotation in one plane acts as $\mathbf{1}$ integrates to $SO(4)$, not to its double cover. (ii) On vectors $2I_3^{\pm} = (c^{12} \pm c^{34})$ squares to $-\mathbf{1}$, so $e^{2\pi I_3^{\pm}} = -\mathbf{1}$ on V_4. Its lift is the product of the twelve commuting factors $e^{2\pi u}$, where u runs over the terms of $I_3^{\pm}$, $u^2 = -\frac{1}{16}$ and $e^{2\pi u} = 4u$. That product is a multiple of the volume element of $\mathrm{Cl}(12,12)$, and the computation gives $(-1)^N$. On a sector where $e^{2\pi I_3^{+}} = -\mathbf{1}$ the eigenvalues $i\mu$ of I_3^{+} satisfy $e^{2\pi i\mu} = -1$. (iii) S_4 is the exterior algebra on the modes, and the modes of direction i span a copy $\mathbb{C}_i^4$ of the copy index, so $S_4 \otimes \mathbb{C} = \Lambda(\mathbb{C}_1^4 \oplus \mathbb{C}_2^4 \oplus \mathbb{C}_3^4) = \bigotimes_i \Lambda\mathbb{C}_i^4$. By (i), $\mathfrak{so}(4)$ acts on each factor as the derivation induced by the vector module. $\Lambda^0 = \Lambda^4 = (0,0)$, $\Lambda^1 \cong \Lambda^3 = (\frac{1}{2},\frac{1}{2})$ and $\Lambda^2 = \Lambda^2_+ \oplus \Lambda^2_- = (1,0) \oplus (0,1)$ [237]. The Clebsch–Gordan rule gives the table. Odd degree in $\Lambda\mathbb{C}^4$ means half-integral (j_+, j_-), and the parity of a product is the parity of the total degree, $(-1)^N$. □

The theorem is fixed. (i) and (ii) are exact on S_4, with sparse integer matrices (e4-3a, e4-3b), and $e^{2\pi\hat{C}^{12}} = +\mathbf{1}$ also on S_5, of dimension 32768 (e4-3a). (iii) is checked on the joint spectrum of the two Casimirs, block-diagonalised by the direction numbers $N_{[i]}$, which commute with every copy rotation; the eigenvalues are integral to 10^{-8} and the multiplicities agree with the character (e4-3c). The decomposition rule is inherited. The same rule at $n = 3$, $\Lambda\mathbb{C}^3 = 2(j{=}0) \oplus 2(j{=}1)$, gives the Casimir spectrum $j = 0, 1, 2, 3$ on $40, 216, 200, 56$ dimensions of Proposition 9.24(ii), recomputed here (e4-3d, restated).

Corollary 9.33 (No spinor of the copy algebra at any rung)**.** *For every n, $e^{2\pi\hat{C}^{kl}} = +\mathbf{1}$ on S_n, and $S_n \otimes \mathbb{C} \cong (\Lambda\mathbb{C}^n)^{\otimes 3}$ as a module of the copy $\mathfrak{so}(n)$. The copy algebra acts through $SO(n)$, and for $n \geq 3$ no spinor module of $\mathfrak{so}(n)$ occurs. At $n = 4$ the half-integral isospin is that of the vector $(\frac{1}{2},\frac{1}{2})$ and of its odd exterior powers. At $n = 6$, where $\mathfrak{so}(6) \cong \mathfrak{su}(4)$, the fundamental $\mathbf{4}$ does not occur.*

Proof. The six terms of $\hat{C}^{kl}$ commute and square to $-\frac{1}{4}$ at every n, and the proof of Theorem 9.32(iii) holds with $\mathbb{C}^n$ in place of $\mathbb{C}^4$. Exterior powers of the vector module have integral weights in the orthonormal basis, and the spinor modules have half-integral ones [214]. □

The corollary is fixed, checked on S_4 and S_5 (e4-3a). It makes the doublets of Remark 9.26(d) precise. At $n = 4$ a state carries half-integral copy isospin under either $\mathfrak{su}(2)$ factor iff it has an odd number of quanta; the copy inversion $-\mathbf{1} \in SO(4)$, the rotation through π in the planes $(1,2)$ and $(3,4)$, acts as $(-1)^N$. This is possible without any double cover because $\mathfrak{so}(4)$ is not simple and its vector is a doublet of each factor. Up the ladder the copy algebra $\mathfrak{so}(n)$ is simple again ($n \geq 5$), so it has no $\mathfrak{su}(2)$ ideal. Its subalgebra $\mathfrak{so}(4)$ on any four constituents still has doublets: on S_5, $e^{2\pi I_3^{\pm}}$ built on the constituents $(1,2,3,4)$ or $(2,3,4,5)$ is $(-1)^{N'}$, with N' the number of quanta in those four constituents (e4-3e). Half-integral copy isospin therefore first appears at the fourth rung and persists above it, under the $\mathfrak{so}(4)$ of four constituents. The doublets are frames and their odd products, not spinors of a spin cover.

9.4.4 Rate patterns and the breaking of the copy symmetry

Proposition 9.34 (Rate patterns)**.** *Let the rates take r distinct values with multiplicities $n_1 + \cdots + n_r = n$.*

(i) The removable part of the copy $\mathfrak{so}(n)$, the part commuting with $K \otimes G$, is $\bigoplus_i \mathfrak{so}(n_i)$, block diagonal on the equal-rate constituents, of dimension $\sum_i \frac{1}{2} n_i(n_i - 1)$. The non-removable directions number $\frac{1}{2}n(n-1) - \sum_i \frac{1}{2} n_i(n_i-1) = \sum_{i<j} n_i n_j$. Within the compact binding-compatible $\mathfrak{so}(3n)$ the removable part is $\bigoplus_i \mathfrak{so}(3n_i)$, and the non-removable directions number $9 \sum_{i<j} n_i n_j$.

(ii) The quadratic form $B \mapsto \|[K, B]\|^2$ on $\mathfrak{so}(n)$ is diagonal in the basis c^{kl} (on the copy index), with eigenvalue $2(\kappa_k - \kappa_l)^2$.

Proof. (i) $[K, E_{lk} - E_{kl}] = (\kappa_l - \kappa_k)(E_{lk} + E_{kl})$, so $X \in \mathfrak{so}(n)$ commutes with K iff $X_{kl} = 0$ whenever $\kappa_k \neq \kappa_l$, and likewise for R and $\tilde{K}$ on $\mathbb{C}^{3n}$. Removability is commutation with $K \otimes G$ (Theorem 9.21(v)). The count uses $n^2 = \sum n_i^2 + 2\sum_{i<j} n_i n_j$. (ii) The matrices $E_{lk} + E_{kl}$, $k < l$, are orthogonal, of norm squared 2. □

The proposition is fixed, exact for every partition of $n = 2, \ldots, 5$ in $\mathfrak{so}(n)$ and of $n = 2, 3, 4$ in $\mathfrak{so}(3n)$ (e4-4a), and at rates $(1, 1, 2, 5)$, where the form is $\mathrm{diag}(0, 2, 32, 2, 32, 18)$ on $c^{12}, c^{13}, c^{14}, c^{23}, c^{24}, c^{34}$ (e4-4b). At $n = 3$ it restates the dimensions $36, 18, 9$ and $3, 1, 0$ of Theorem 9.21(v). With the factor 6 of $\mathbf{1}_6$, (ii) is the norm $6\sum_{k\neq l}(\kappa_k - \kappa_l)^2 B_{kl}^2$ of Remark 9.26(e).

Remark 9.35 (The breaking pattern)**.** The rate matrix is a symmetric tensor of the copy $\mathfrak{so}(n)$, and its stabiliser is $\bigoplus_i \mathfrak{so}(n_i)$; its stabiliser in $O(n)$ is $O(n_1) \times \cdots \times O(n_r)$, which for distinct rates is the group of copy reflections of Theorem 9.29(i). Read as in Remark 9.26(e), this is the pattern of a gauge symmetry broken by a symmetric-tensor vacuum value [229], with $\sum_{i<j} n_i n_j$ massive directions of masses proportional to $|\kappa_k - \kappa_l|$ [234, 235, 236]. Up the ladder, constituents with equal rates form a multiplet whose internal copy rotations are removable, so which of them carries a quantum is a gauge choice. Constituents with different rates are distinguishable, and the field between them is physical. The reading is interpretive, with the caveat of Section 9.3.7: b is a constant connection without kinetic term, and the rates are inputs.

9.4.5 A ladder of rates admits a non-abelian frame

Section 9.3.2 shows that a common binding action lets the rotor act by a non-central $A \in \mathfrak{u}(n)$. It also shows that the frame then admits equivariant brackets $[X, Y]_k = \sum_{i,j} \Gamma_{ij}^k \beta_0(X_i, Y_j)$, with $\beta_0(x, y) = z(x) \times z(y)$ the complex cross product and $\Gamma_{ij}^k = \Gamma_{ji}^k \in \mathbb{C}$, subject to $\Gamma_{ij}^k(\omega_k - \omega_i - \omega_j) = 0$, where A acts on copy k as $i\omega_k$. For $A = K \otimes G$ the ω_k are the rates. With a copy field, A acts on z as iH, and the condition applies to the eigenvalues of H, the lines.

Proposition 9.36 (Rotor-compatible frame brackets)**.** *(i) The real bilinear maps $V \times V \to V$ equivariant under $\mathfrak{l}_G$ form the two-dimensional space spanned by β_0 and $G\beta_0$, and all of them are antisymmetric. The equivariant antisymmetric brackets on V_n are the maps above, a space of real dimension $n^2(n+1) = 12, 36, 80$ for $n = 2, 3, 4$.*

(ii) $A = K \otimes G$ is a derivation of the bracket iff $\Gamma_{ij}^k = 0$ unless $\kappa_k = \kappa_i + \kappa_j$. The rotor-compatible brackets have real dimension 0, 0, 2 for rates $(1, 1)$, $(1, 3)$, $(1, 2)$; 0, 4, 0, 4 for $(1, 1, 1)$, $(1, 2, 3)$, $(1, 3, 5)$, $(1, 2, 4)$; and 8 for $(1, 2, 3, 4)$. These count the resonant coefficients; for $(1, 2, 3, 4)$ the Jacobi identity ($ad = bc$, item (v)) leaves a bracket variety of real dimension 6. For the harmonic ladder $\kappa_k = k$ the dimension is $2\sum_{k \leq n} \lfloor k/2 \rfloor = 2, 4, 8, 12, 18$ for $n = 2, \ldots, 6$.

(iii) Rates $(1,2,3)$. *The rotor-compatible brackets are* $[X,Y]_2 = a\,\beta_0(X_1,Y_1)$, $[X,Y]_3 = b\,(\beta_0(X_1,Y_2)+\beta_0(X_2,Y_1))$, $(a,b)\in\mathbb{C}^2$. *Every one of them satisfies the Jacobi identity. For* $ab\neq 0$ *the frame is a nilpotent Lie algebra* $V_1\oplus V_2\oplus V_3$, *graded by the rate and generated by* V_1, *with lower central series of real dimensions* $18,12,6,0$.

(iv) Rates $(1,2,4)$. *The rotor-compatible brackets are* $[X,Y]_2 = a\,\beta_0(X_1,Y_1)$, $[X,Y]_3 = b\,\beta_0(X_2,Y_2)$. *For* $X,Y\in V_1$ *and* $Z\in V_2$ *the Jacobiator is* $ab\,\beta_0(\beta_0(X,Y),Z)\in V_3$, *so the Jacobi identity holds iff* $ab=0$.

(v) Rates $(1,2,3,4)$, *with* $a=\Gamma^2_{11}$, $b=\Gamma^3_{12}$, $c=\Gamma^4_{13}$, $d=\Gamma^4_{22}$. *For* $X,Y\in V_1$ *and* $Z\in V_2$ *the Jacobiator is* $(ad-bc)\,\beta_0(\beta_0(X,Y),Z)\in V_4$, *and the Jacobi identity holds on all of* V_4 *iff* $ad=bc$.

Proof. (i) is an exact null-space computation on the 216 coefficients of a bilinear map (e4-5a); G acts as i, so $G\beta_0 = i\beta_0$, and the maps are complex bilinear. A bracket is a choice of Γ^k_{ij} for each k and each unordered pair $\{i,j\}$, $n\cdot\frac{1}{2}n(n+1)$ complex numbers. (ii) On z, A acts as $i\kappa_k$ on copy k, and the bracket is complex bilinear, so $A[X,Y]_k - [AX,Y]_k - [X,AY]_k = i(\kappa_k-\kappa_i-\kappa_j)\Gamma^k_{ij}\beta_0(X_i,Y_j)$. For $\kappa_k=k$ the pairs $i\le j$ with $i+j=k$ number $\lfloor k/2\rfloor$. (iii) The only composite is $[[X,Y]_2,Z]_3$, so the Jacobiator is $ab\sum_{\mathrm{cyc}}\beta_0(\beta_0(X_1,Y_1),Z_1)$, which vanishes by the Jacobi identity of the complex cross product, the bracket of $\mathfrak{so}(3,\mathbb{C})$. $\beta_0(V_1,V_1)=V_2$ and $\beta_0(V_1,V_2)=V_3$ for $ab\neq 0$. The Jacobiator is trilinear and alternating, and inputs of rates r_1,r_2,r_3 give an output of rate $r_1+r_2+r_3$, so each of its components is fixed by inputs whose rates add to the target rate; for $a\neq 0=b$ the series is $18,6,0$ (a12-3). (iv) The only composite is $[[X,Y]_2,Z]_3 = ab\,\beta_0(\beta_0(X_1,Y_1),Z_2)$, with inputs of rates $1,1,2$. For $X,Y\in V_1$ and $Z\in V_2$ the other two cyclic terms vanish, because $V_1\times V_2$ brackets to zero. For $ab=0$ there is no composite. (v) The V_3-component is as in (iii) and vanishes. For $X,Y\in V_1$ and $Z\in V_2$, $[[X,Y],Z]_4 = ad\,\beta_0(\beta_0(X,Y),Z)$, while $[[Y,Z],X]_4+[[Z,X],Y]_4 = bc\,(\beta_0(\beta_0(Y,Z),X)+\beta_0(\beta_0(Z,X),Y)) = -bc\,\beta_0(\beta_0(X,Y),Z)$ by the Jacobi identity of β_0. □

The proposition is fixed. (i) restates the space of Section 9.3.2 at $n=2$ (e4-5a). (ii) is exact, block by block, for the listed rates and for the harmonic ladder up to $n=4$; for $n=5,6$ the count is the number of resonant blocks (e4-5b). At $n=2$ it restates the bracket of $\mathrm{diag}(1,2)\otimes G$ of Section 9.3.2. (iii) to (v) are checked on random samples to 10^{-10} (e4-5c, e4-5d, e4-5e): antisymmetry, $\mathfrak{l}_G$-equivariance, the rotor as a derivation and the Jacobi identity; at $(1,2,4)$ and $(1,2,3,4)$ the samples off the stated conditions fail. That every rotor-compatible frame is nilpotent when all rates are positive is inherited: A is then a nonsingular derivation, and a Lie algebra with a nonsingular derivation is nilpotent [230].

The graded brackets of (iii)–(v) are of current type, $\mathfrak{g}\otimes A$ with $\mathfrak{g}$ the complex cross product and A a graded commutative coefficient algebra without unit, $t_it_j = \sum_k\Gamma^k_{ij}t_k$; Lie algebras extended over commutative algebras without unit are treated by Zusmanovich [311]. The truncated current algebra $\mathfrak{g}\otimes t\mathbb{C}[t]/(t^{n+1})$ is a particular case, not the general one: at rates $(1,2,3)$ it is the case $ab\neq 0$ of (iii), up to rescaling, while $a\neq 0$, $b=0$ gives a two-step nilpotent frame with V_3 central and decoupled, against the three-step lower central series of the truncation.

The reading is constructed. A harmonic ladder of rates admits a non-abelian frame, from $n=3$ on and already for $\mathrm{diag}(1,2)$ at $n=2$. Its bracket joins two frames whose rates add into the frame of the summed rate, so the frame algebra is graded by rate. With three constituents on the harmonic ladder the Lie frame brackets form the linear space $\mathbb{C}^2$. With four the Jacobi identity cuts out the cone $ad=bc$ in $\mathbb{C}^4$, and on a longer ladder the same condition recurs, because the V_4-component of the Jacobiator involves only the blocks with target rate at most 4. The resonant pattern $(1,2,4)$ admits a non-abelian frame only with one of its two blocks, $ab=0$. Nothing selects such a rotor. The paper's closure takes $A=G_n$ and an abelian frame (Theorem 4.22 and Proposition 4.33). The non-central rotor, its rate ladder and the bracket are admitted, not selected. The resonance condition $\omega_k=\omega_i+\omega_j$ is that of three-wave mixing and

sum-frequency generation [231], a parallel only (interpretive). Read with $mc^2 = \hbar\kappa$, the bracket composes frames whose rest energies add (interpretive).

9.4.6 Answer: what the field between constituents can and cannot do

Remark 9.37 (What the field does up the ladder)**.** (a) *It moves quanta and keeps their number* (fixed). A gauge-admissible field between the constituents acts as a one-body operator. It hops quanta between constituents and mixes them, and its N-quantum lines are sums of one-quantum lines (Theorem 9.27 and Corollary 9.28).

(b) *It does not take one to two* (fixed). It creates no quantum and changes no rate. Pair emergence across constituents needs a copy boost, which is not a gauge field (Corollary 9.28(ii); Proposition 9.22). Because it is one-body, it binds no quanta into composites; interactions beyond this one-body class are not examined. If two particles are to emerge from one, the process lies outside the compact field between constituents.

(c) *It carries time-odd invariants that grow with the rung* (fixed). From $n = 3$ on it carries $\frac{1}{2}(n-1)(n-2)$ independent triangle signs, each odd under interior time reversal. The triangle traces measure them at every n, Jarlskog's determinant only at odd n (Theorem 9.29 and Proposition 9.31).

(d) *It acts on doublets from the fourth rung on* (fixed). At $n = 4$ the copy isospin is half-integral exactly on odd N, and above it under the $\mathfrak{so}(4)$ of any four constituents, on an odd number of quanta in them. At every rung the copy algebra acts through $SO(n)$, so no spinor of it occurs (Theorem 9.32 and Corollary 9.33).

(e) *It is non-removable exactly between unequal rates* (fixed), with the prescribed binding tensor kept fixed. The non-removable directions number $\sum_{i<j} n_i n_j$; equal-rate constituents stay gauge-equivalent (Proposition 9.34). The reading as symmetry breaking by the rate matrix is interpretive (Remark 9.35).

(f) *It admits non-abelian frames on a harmonic ladder of rates* (constructed). The rotor must be non-central and its rates resonant, and nothing in the closure selects either (Proposition 9.36).

Up the ladder, then, the field between constituents distinguishes constituents by their rates, carries more time-odd structure, and from $n = 4$ on acts on doublets. It never changes the number of quanta that the rung carries. In the sense of Section 5.4.4, a rung carries N quanta, and the compact b-field keeps N fixed. It decides where the quanta are, not how many there are. This reading is interpretive, drawn from the fixed statements above.

9.5 The closure at two constituents

Section 4.4 closes the force law $\dot{p} = \kappa G^{\top} p$ into the Lie–Poisson system of $\mathfrak{s}_n = \mathbb{R} \ltimes_{G_n} V_n$, in which the rate is the angular velocity of a rotor. It treats one constituent, several constituents driven by one rotor (Proposition 4.33), and three constituents (Theorems 4.34 to 4.39): the closure symmetry, the absence of a common Lorentz algebra, the part of the Pati–Salam algebra that commutes with the binding root, what survives each projector, its chirality and anomalies, and the freedom left in the inertia. The case $n = 2$ is not treated there. Parts of it are in Chapter 5: the closure symmetry $\mathfrak{so}(6, \mathbb{C})$ (Proposition 5.27), the balance surface (Theorem 5.52), the balanced pairs (Proposition 5.55) and the pair invariant M_{pair} (Theorem 5.57). This section supplies the $n = 2$ analogue of every item, with the five Lorentz actions at $n = 2$ in place of the two projectors, which exist only at $n = 3$, and asks whether $n = 2$ is a closure rung in its own right. The answers, with their classes, are these. (1) The closure symmetry $\mathfrak{so}(6, \mathbb{C})$

n	copy algebra, group on S_n	non-removable directions, distinct rates	independent signs c_n	half-integral copy isospin	rotor brackets, rates $1,\dots,n$: resonant coefficients
2	$\mathfrak{so}(2)$, $SO(2)$	1	0	no; charges $i\{-3,\dots,3\}$	2
3	$\mathfrak{so}(3)$, $SO(3)$	3	1	no; $j=0,\dots,3$	4
4	$\mathfrak{su}(2)\oplus\mathfrak{su}(2)$, $SO(4)$	6	3	yes, exactly on odd N	8, Jacobi: $ad=bc$
5	$\mathfrak{so}(5)$, $SO(5)$	10	6	under the $\mathfrak{so}(4)$ of four constituents; no spinor of $\mathfrak{so}(5)$	12
6	$\mathfrak{so}(6)\cong\mathfrak{su}(4)$, $SO(6)$	15	10	as for $n=5$; no $\mathbf{4}$ of $\mathfrak{su}(4)$	18

Table 9.3: The field between constituents, rung by rung. The $n=2,3$ isospin entries restate Proposition 9.24. The others follow from Theorems 9.29 and 9.32, Corollary 9.33, and Propositions 9.34 and 9.36; the rotor column gives the real dimension of the space of resonant coefficients before the Jacobi identity is imposed (at $n=4$ the Jacobi identity cuts it to the variety $ad=bc$; at $n=5,6$ the Lie-bracket varieties are not determined here). Non-removable: with the prescribed binding tensor kept fixed.

is the complex orthogonal algebra of $q_{\mathrm{tot}}=z_1\cdot z_1+z_2\cdot z_2$. It mixes the two field strengths, keeps no quadratic invariant except q_{tot}, and does not preserve the pair invariant: on each of its orbits M_{pair} takes every value in $[\frac{1}{2}|q_{\mathrm{tot}}|,\infty)$ (fixed; Section 9.5.1). (2) There is no common Lorentz algebra, and under $\mathfrak{l}_G^{(1)}$ the spinor module carries no Weyl spinor (fixed, partly restated; Section 9.5.2). (3) None of the compact parts of the five candidate gauge algebras of the Lorentz actions at $n=2$ contains the Pati–Salam algebra (obstructed within that tested class; other constructions at $n=2$ are not excluded). The binding root leaves an $\mathfrak{su}(2)$ of the spectator gauge algebra, compact, non-abelian and kinetically admissible, which acts on the Weyl block with integer isospin and vector-like; of the diagonal connection algebra it leaves the copy rotation; of the mixed rung it leaves nothing (fixed; Section 9.5.3). (4) The inertia freedom is $(2,0)$ under the closure symmetry, as at $n=1$ and $n=3$, and it is $(14,10)$ or $(18,18)$ under the binding-compatible gauge parts (fixed, partly restated; Section 9.5.4). (5) The balance surface is connected and of codimension two. It is transversally neutral, since the imbalance is a constant of motion; the rate is stable, but the bounded frame orbits are not, and they need $h(0)\neq 0$. With equal inertias a massive balanced pair is never an exchange eigenstate of its momenta (fixed; Section 9.5.5). (6) Table 9.5 compares $n=1,2,3$, and Section 9.5.6 answers the question. Identifiers e5-n refer to the `PASS` lines of `e5_closure_two.py` (Table 9.6).

Notation. As in Section 5.6. $V_2=V\oplus V$ has the basis $e_a^{(1)}$, $e_a^{(2)}$, the metric $\eta_2=\eta\oplus\eta$ and the binding root $G_2=G\oplus G$. The temporal directions are $e_1^{(k)},e_2^{(k)},e_3^{(k)}$, each with its partner $e_{i+3}^{(k)}$, and the paired rotations and boosts $\mathrm{rot}(i,j)$, $\mathrm{bst}(i,j)$ of temporal $i<j$ are those of Section 4.4.6. A vector X of V_2 gives $Z=(z_1,z_2)\in\mathbb{C}^6$, $z_k=z(X^{(k)})$, on which G_2 acts as i; $q_k=z_k\cdot z_k$, $q_{\mathrm{tot}}=q_1+q_2$, $Q_{kl}=z_k\cdot z_l$ is the complex Gram matrix with singular values $\sigma_1\geq\sigma_2$, and $M_{\mathrm{pair}}=\frac{1}{2}(\sigma_1+\sigma_2)$ is the rest-energy density of the pair, a local invariant of the summed stress tensor and not an invariant mass (Theorem 5.57). $C_v=\varepsilon\otimes\mathbf{1}_6$ is the copy rotation and $\varepsilon\otimes G$ the copy boost, $\varepsilon=\left(\begin{smallmatrix}0&-1\\1&0\end{smallmatrix}\right)$. $\mathfrak{l}_G^{(k)}$ and $\mathfrak{l}_{\mathrm{int}}^{(k)}$ are the frame and interior Lorentz algebras of constituent k, and $\mathfrak{l}_G^{\Delta}$ is the diagonal $\mathfrak{l}_G$. On $S_2=\mathbb{R}^{64}$ the tower generators are $E_{(1,a)}=e_a\otimes\mathbf{1}$, $E_{(2,a)}=\omega\otimes e_a$, the spin lift of $M\in\mathfrak{so}(6,6)$ is $S(M)=\frac{1}{4}\sum_{a,b}(M\eta_2)_{ab}E_aE_b$, and $\Gamma_{12}=E_{(1,1)}\cdots E_{(2,6)}$. In the closure the per-constituent quantities are those of the momentum fields $\pi^{(k)}=\eta^{-1}p^{(k)}$, $w_k=w(p^{(k)})=q(z(\pi^{(k)}))$ (equation (4.15)); Lorentz-class inertias are $N_k=(a_k\eta^{-1}+b_kG\eta^{-1})/m_k$, $c_k=(a_k+ib_k)/m_k$, $F(p)=\sum_k c_kw_k$ and

statement	class	checks
Theorem 9.27(i): paired rotations are one-body operators $\mathcal{N}(R)$, conserve N	fixed (structure inherited, realisation fixed)	e4-1a, e4-1b
Theorem 9.27(ii): $\mathsf{H} = \mathcal{N}(H)$; $N = 1$ block is H; additive lines	fixed; external i constructed	e4-1e
Theorem 9.27(iii), (iv): copy rotations and copy boosts at $n = 3$ by constituent and direction	fixed	e4-1c, e4-1d
Theorem 9.27(v): quarter turn moves $N^{(1)}$ to $N^{(2)}$, keeps the rate spectrum	fixed	e4-1f
Corollary 9.28: transfer, not creation; rates gauge invariant	fixed (partly restated: Theorem 5.39; Theorem 9.21(iii))	e4-1b, e4-1d, e4-3a
Theorem 9.29(i): residual group is the copy reflections	fixed (restated at $n = 3$: Theorem 9.23(iii))	e4-2a
Theorem 9.29(ii), (iii): invariants are $\lvert B_{kl}\rvert$ and c_n triangle signs; all time-odd	fixed (cycle space inherited)	e4-2b, e4-2c
Theorem 9.29(iv): triangle traces	fixed	e4-2d
Remark 9.30: cycle rank of the coupling graph	fixed	e4-2f
Proposition 9.31: $\det[H, K] = i^n \det D$, parity $(-1)^n$, $n = 3, 4, 5$	fixed ($n = 3$ restated: Theorem 9.23(ii))	e4-2e, z1-4a
parallel with the $(n-1)(n-2)/2$ phases of n quark families	count inherited; physical CP reading not established	—
Theorem 9.32(i), (ii): $SO(4)$, $(\frac{1}{2}, \frac{1}{2})$, $e^{2\pi I_3^{\pm}} = (-1)^N$	fixed	e4-3a, e4-3b
Theorem 9.32(iii), Table 9.2: $(\Lambda\mathbb{C}^4)^{\otimes 3}$, doublets on odd N	fixed (decomposition rule inherited)	e4-3c
$n = 3$ Casimir spectrum $40, 216, 200, 56$	restated (Proposition 9.24(ii))	e4-3d
Corollary 9.33: no spinor of the copy algebra at any rung; doublets of the $\mathfrak{so}(4)$ of four constituents at $n = 5$	fixed	e4-3a, e4-3e
Proposition 9.34: removable $\bigoplus \mathfrak{so}(n_i)$, $\sum_{i<j} n_i n_j$ physical; mass form	fixed ($n = 3$ restated: Theorem 9.21(v))	e4-4a, e4-4b
Remark 9.35: breaking by the rate matrix	interpretive	—
Proposition 9.36(i), (ii): equivariant brackets, resonance condition, dimensions	fixed ($n = 2$ restated: Section 9.3.2)	e4-5a, e4-5b
Proposition 9.36(iii)–(v): Jacobi at $(1,2,3)$, $(1,2,4)$, $(1,2,3,4)$; nilpotency	fixed; nilpotency inherited	e4-5c, e4-5d, e4-5e
Section 9.4.5: a harmonic ladder admits a non-abelian frame	constructed; three-wave parallel interpretive	e4-5b, e4-5c
Remark 9.37, Table 9.3: what the field can and cannot do	(a)–(e) fixed, (f) constructed; summary interpretive	—

Table 9.4: Claims of Section 9.4, their classes and their checks, which are `PASS` lines of `e4_bfield_ladder.py`.

$R_{\text{eff}}e^{i\varphi_{\text{eff}}} = F(p(0))$. The *kinetic condition* is that of Proposition 2.29, used in Theorem 4.37 with the kernels compressed by the projector: $X^\top K^\mu + K^\mu X = 0$ for $\mu = 0, \dots, 3$, with the kernels $K^\mu = -\gamma^0\gamma^\mu$ of Proposition 2.21. At $\mu = 0$ it says that X is antisymmetric, and at $\mu = a$ that X commutes with the boosts $\gamma^0\gamma^a$.

9.5.1 The closure symmetry and the invariants of a pair

Proposition 9.38 (The closure symmetry at two constituents)**.** $G_2^\top \eta_2 G_2 = -\eta_2$ *and* $G_2^2 = -\mathbf{1}$. *The centraliser of* G_2 *in* $\mathfrak{so}(6,6)$ *has dimension* 30 *and is spanned by the* 15 *paired rotations and the* 15 *paired boosts. In the basis of the six temporal directions followed by their partners every element is* $X = \left(\begin{smallmatrix} R & S \\ -S & R \end{smallmatrix}\right)$ *with* R, S *antisymmetric, and* X *acts on* Z *as the complex antisymmetric matrix* $R - iS$. *The map* $X \mapsto R - iS$ *is a Lie isomorphism onto* $\mathfrak{so}(6,\mathbb{C})$, *the algebra of the complex orthogonal group of* q_{tot}, *and the compact part of the centraliser is* $\mathfrak{so}(6)$, *the paired rotations.*

This restates Proposition 5.27(i) (fixed there), the $n = 2$ member of Theorem 4.34; the bracket identity was checked on all 435 pairs of basis elements (e5-1a). By Proposition 5.27(ii), $\mathfrak{so}(6,\mathbb{C}) = \mathfrak{l}_G^{(1)} \oplus \mathfrak{l}_G^{(2)} \oplus (\mathbb{C}^3 \otimes \mathbb{C}^3)$, and the cross part maps z_1 into z_2: the closure symmetry does not keep the two field strengths apart.

Proposition 9.39 (Quadratic invariants of a pair)**.** *The holomorphic quadratic polynomials in* (z_1, z_2) *invariant under (a)* $\mathfrak{so}(6,\mathbb{C})$ *are the multiples of* q_{tot}*; (b)* $\mathfrak{l}_G^{(1)} \oplus \mathfrak{l}_G^{(2)}$ *form* $\operatorname{span}_{\mathbb{C}}\{q_1, q_2\}$*; (c)* $\mathfrak{l}_G^{\Delta}$ *form* $\operatorname{span}_{\mathbb{C}}\{q_1, q_2, z_1 \cdot z_2\}$. *In each case the invariant real quadratic forms on* V_2 *have real dimension* 2, 4, 6 *and are spanned by the real and imaginary parts of these.*

Proof. $\mathfrak{l}_G \cong \mathfrak{so}(3,\mathbb{C})$ acts on $\mathbb{C}^3$ by the vector representation. By the first fundamental theorem for the orthogonal group [161], the polynomial invariants of $SO(3,\mathbb{C})$ acting diagonally on (z_1, z_2) are generated by the inner products $z_k \cdot z_l$ (and $\det(z_1, z_2, \cdot)$, which needs three vectors), which gives (c). Under $\mathfrak{l}_G^{(1)} \oplus \mathfrak{l}_G^{(2)}$ a cross term $z_1 \cdot Az_2$ is invariant under $z_1 \mapsto \Lambda z_1$ with z_2 fixed only if $A = 0$, which gives (b). $\mathbb{C}^6$ is irreducible under $\mathfrak{so}(6,\mathbb{C})$, with q_{tot} its only invariant form, which gives (a). A real invariant form that is not the real part of a holomorphic one would be Hermitian, $h(Xu, v) + h(u, Xv) = 0$, and applied to X and iX this forces $h(Xu, v) = 0$ for all X, so $h = 0$. □

Inherited (classical invariant theory), with the real dimensions computed as exact null spaces (e5-1b).

Theorem 9.40 (The closure symmetry and the pair invariant)**.** *(i)* $q(z_1+z_2) = q_1+q_2+2\,z_1\cdot z_2$, $\det Q = q_1q_2 - (z_1 \cdot z_2)^2$, *and*

$$M_{\text{pair}}^2 = \tfrac{1}{4}(|q_1|^2 + |q_2|^2 + 2|z_1 \cdot z_2|^2) + \tfrac{1}{2}|q_1q_2 - (z_1 \cdot z_2)^2|.$$

So M_{pair}, $z_1 \cdot z_2$ *and* $q(z_1 + z_2)$ *are Lorentz invariants of the pair under* $\mathfrak{l}_G^{\Delta}$. *None of them is invariant under* $\mathfrak{l}_G^{(1)} \oplus \mathfrak{l}_G^{(2)}$*: a boost of constituent* 2 *alone fixes* q_1, q_2 *and changes* $z_1 \cdot z_2$ *and* M_{pair}. *Under the closure symmetry only* $q_{\text{tot}} = \operatorname{tr} Q$ *is invariant.*

(ii) M_{pair} *is invariant under* $\mathfrak{l}_G^{\Delta}$ *and the copy rotation* C_v, *and the subalgebra of* $\mathfrak{so}(6,\mathbb{C})$ *that leaves it invariant is* $\mathfrak{l}_G^{\Delta} \oplus \mathbb{R}C_v$, *of real dimension* 7.

(iii) The closure symmetry does not preserve M_{pair}. *The copy boost has the eigenvector* (e_1, ie_1) *with eigenvalue* 1, *and maps the balanced pair* (e_1, ie_1), *with* $q_{\text{tot}} = 0$ *and* $M_{\text{pair}} = 1$, *to* $e^t(e_1, ie_1)$, *with* $M_{\text{pair}} = e^{2t}$. *The compact rotation* $R_\varphi^{(1)} \in SO(6)$ *of Example 5.59 takes* $((1, i, 0), 0)$ *from* $M_{\text{pair}} = 0$ *to* $M_{\text{pair}} = |\sin\varphi|$.

(iv) For each $c \in \mathbb{C}$ the set $\{Z \neq 0 : \ q_{\mathrm{tot}}(Z) = c\}$ is a single orbit of $SO(6,\mathbb{C})$, and on it M_{pair} takes every value in $[\frac{1}{2}|c|, \infty)$. The bound $M_{\mathrm{pair}} \geq \frac{1}{2}|q_{\mathrm{tot}}|$ of Theorem 5.57(iii) is attained at $(z,0)$, and the copy boost carries $(z,0)$ to $(\cosh t\, z,\, i \sinh t\, z)$, with $q_{\mathrm{tot}} = q(z)$ and $M_{\mathrm{pair}} = \frac{1}{2}|q(z)| \cosh 2t$.

Proof. (i) The first two identities are expansions, and with them the formula is $M_{\mathrm{pair}}^2 = \frac{1}{4}\|Q\|_F^2 + \frac{1}{2}|\det Q|$, which follows from Theorem 5.57(ii). A diagonal Lorentz transformation Λ maps $Q = Z^\top Z$ to $Z^\top \Lambda^\top \Lambda Z = Q$; one acting on constituent 2 alone maps $z_1 \cdot z_2$ to $z_1 \cdot \Lambda z_2$. The last statement is Proposition 9.39(a). (ii) C_v maps Q to $RQR^\top$ with R real orthogonal, which keeps the singular values (Proposition 5.60(ii)). That nothing else in $\mathfrak{so}(6,\mathbb{C})$ keeps M_{pair} is computed exactly: at 32 pairs with Gaussian-integer entries and rational $|\det Q|$ the derivatives of M_{pair}^2 along the 30 real basis fields of $\mathfrak{so}(6,\mathbb{C})$ are rational numbers and form a matrix of exact rank 23, so the stabiliser has dimension at most 7; the seven directions of $\mathfrak{l}_G^{\Delta} \oplus \mathbb{R}C_v$ are independent and leave M_{pair} invariant identically (they act on Q trivially or by $Q \mapsto RQR^\top$), so it is exactly $\mathfrak{l}_G^{\Delta} \oplus \mathbb{R}C_v$ (z1-5b). (iii) The copy boost acts on the copy index as $\left(\begin{smallmatrix} 0 & -i \\ i & 0 \end{smallmatrix}\right)$ (Proposition 5.27(iii)), and M_{pair} is homogeneous of degree two. At (e_1, ie_1), $Q = vv^\top$ with $v = (1,i)$, so $\sigma_1 = 2$ and $\sigma_2 = 0$. The numbers for $R_\varphi^{(1)}$ restate Example 5.59. (iv) Transitivity on the level sets of a nondegenerate quadratic form is Witt's theorem [232]. At $(z,0)$, $Q = \mathrm{diag}(q(z),0)$. Since $(\varepsilon \otimes G)^2 = \mathbf{1}$, $e^{t\,\varepsilon\otimes G} = \cosh t + \sinh t\, \varepsilon \otimes G$, and the image has $Q = q(z)\, vv^\top$ with $v = (\cosh t, i \sinh t)$, so $\sigma_1 = |q(z)|\,|v|^2 = |q(z)| \cosh 2t$ and $\sigma_2 = 0$. The orbit is connected and M_{pair} is continuous, so it takes every value between its infimum and its unbounded values; for $c = 0$ use (iii). □

Items (i) and (iv) are fixed, with (i) partly restated from Theorem 5.57; (ii) is fixed, with the stabiliser certified by an exact rational rank; (iii) is fixed, with the second example restated. Checks: the identities symbolically, the formula against the singular values at 30 random pairs to 10^{-12} (e5-1e); invariance to 10^{-11}, and the rank 23 with a singular-value gap above 10^8 (e5-1c), now replaced by the exact rank certificate z1-5b; the orbits to 10^{-12} (e5-1d). So the closure symmetry mixes the two field strengths and does not preserve the pair invariant. The closure symmetry sees one field strength in $\mathbb{C}^6$ and its square q_{tot}. To split it into two field strengths that carry a pair invariant takes the seven-dimensional stabiliser $\mathfrak{l}_G^{\Delta} \oplus \mathbb{R}C_v$, a diagonal Lorentz action together with the copy rotation, and the closure selects neither. With equal inertias its dynamics agrees: the frame energy is $\frac{1}{2}\,\mathrm{Re}(c\, q_{\mathrm{tot}}(\pi))$ (Proposition 4.33; Proposition 9.47(i) with $c_1 = c_2 = c$), and its compact symmetry carries every balanced pair to one null constituent and an empty copy (Proposition 5.60(iii)). In that closure "two constituents" is a label that the dynamics does not see (interpretive).

9.5.2 No common Lorentz algebra

Proposition 9.41 (No common Lorentz algebra at two constituents)**.** *(i)* $\mathrm{Cent}(G_2) \cap \mathfrak{l}_{\mathrm{int}}^{(1)} = 0$. *The elements of $\mathfrak{so}(6,6)$ commuting with both G_2 and $\mathfrak{l}_{\mathrm{int}}^{(1)}$ form $\mathfrak{l}_G^{(2)} \cong \mathfrak{so}(3,\mathbb{C})$, of dimension 6. At rung n they form the $\mathfrak{so}(3n-3,\mathbb{C})$ of constituents $2,\dots,n$, of dimension $3(n-1)(3n-4)$: 0, 6 and 30 at $n = 1,2,3$.*

(ii) The bivectors of constituent 1 are $e_a e_b \otimes \mathbf{1}$, so $S_n \otimes \mathbb{C}$ carries 8^{n-1} copies of the content of $S \otimes \mathbb{C}$. Under $\mathfrak{l}_G^{(1)}$, $S_2 \otimes \mathbb{C} = 16\,(\frac{1}{2},\frac{1}{2})$: no Weyl spinor ($128\,(\frac{1}{2},\frac{1}{2})$ at $n = 3$). Under $\mathfrak{l}_{\mathrm{int}}^{(1)}$, $S_2 \otimes \mathbb{C} = 16\,(\frac{1}{2},0) \oplus 16\,(0,\frac{1}{2})$.

(iii) Under $\mathfrak{l}_G^{(2)}$, $S_2 \otimes \mathbb{C} = 16\,(\frac{1}{2},\frac{1}{2})$, and under $\mathfrak{l}_G^{\Delta}$, $S_2 \otimes \mathbb{C} = 4\big[(0,0) \oplus (1,0) \oplus (0,1) \oplus (1,1)\big]$: integer spin only.

Proof. (i) An element of $\mathfrak{l}_{\mathrm{int}}^{(1)}$ lives on constituent 1, where $\mathrm{Cent}(G) \cap \mathfrak{so}(3,3) = \mathfrak{l}_G \cong \mathfrak{so}(3,\mathbb{C})$ (Theorem 3.1). It fixes the surplus pair, and an element of $\mathfrak{so}(3,\mathbb{C})$ that kills $z = (0,1,0)$ and $z = (0,0,1)$ is an antisymmetric matrix with two zero columns, hence zero. Under $\mathfrak{l}_{\mathrm{int}}^{(1)}$, V_2

is the interior four-plane of constituent 1, irreducible and nontrivial, plus eight Lorentz-trivial directions: the surplus pair $F = \langle e_2^{(1)}, e_3^{(1)} \rangle$ and all of constituent 2. By Schur's lemma an equivariant $X \in \mathfrak{so}(6,6)$ vanishes on the four-plane and preserves the eight trivial directions. If X also commutes with G_2, then for $v \in F$, $G_2 v$ lies in the interior plane $\langle e_5^{(1)}, e_6^{(1)} \rangle$, so $G_2 X v = X G_2 v = 0$ and $Xv = 0$. Hence X vanishes on constituent 1 and is an element of $\mathfrak{so}(3,3)$ of constituent 2 commuting with G, that is of $\mathfrak{l}_G^{(2)}$ (Theorem 3.1). The same argument at rung n leaves $\mathrm{Cent}(G_{n-1})$ of constituents $2, \ldots, n$, which is $\mathfrak{so}(3n-3, \mathbb{C})$ by Proposition 5.27(i). (ii) $E_{(1,a)} E_{(1,b)} = e_a e_b \otimes \mathbf{1}$, and at one constituent $S \otimes \mathbb{C} = 2\,(\frac{1}{2}, \frac{1}{2})$ under $\mathfrak{l}_G$ and $2\,(\frac{1}{2}, 0) \oplus 2\,(0, \frac{1}{2})$ under $\mathfrak{l}_{\mathrm{int}}$ (Section 3.7). (iii) The bivectors of constituent 2 are $\mathbf{1} \otimes e_a e_b$, so $\mathfrak{l}_G^{\Lambda}$ acts on $(S \otimes \mathbb{C}) \otimes_{\mathbb{C}} (S \otimes \mathbb{C})$, and $(\frac{1}{2}, \frac{1}{2}) \otimes (\frac{1}{2}, \frac{1}{2}) = (0,0) \oplus (1,0) \oplus (0,1) \oplus (1,1)$. □

Part (i) restates Proposition 5.27(iv) and Proposition 9.19(iii) (fixed there), the dimension 30 restates Theorem 4.35, and the ladder formula is fixed (e5-2a). Parts (ii) and (iii) are fixed at $n = 2$, restated at $n = 3$, and the Clebsch–Gordan rule is inherited (Casimir spectra to 10^{-9}, computed on S_2 and on one constituent; e5-2b, e5-2c). No nonzero Lorentz generator leaves both the closure and the spinor structure of matter invariant, at $n = 2$ as at $n = 1$ and $n = 3$. What differs is what commutes with both: at $n = 2$ it is a whole second Lorentz algebra, the frame algebra of the spectator constituent. That is the material of the mixed rung, which pairs $\mathfrak{l}_{\mathrm{int}}^{(1)}$ with $\mathfrak{l}_G^{(2)}$ (Proposition 9.45).

9.5.3 Gauge algebras at two constituents and the binding root

Proposition 9.42 (No Pati–Salam algebra at two constituents)**.** *The compact parts of the candidate gauge algebras $\mathfrak{g}(k_I, k_B, k_S)$ of the five Lorentz actions at $n = 2$ (Theorem 5.15; Table 5.1) have dimensions* 13 *(spectator, $\mathfrak{so}(5) \oplus \mathfrak{so}(3)$),* 6 *(binding and spectator, $\mathfrak{so}(3) \oplus \mathfrak{so}(3)$),* 7 *(diagonal interior, $\mathfrak{so}(2) \oplus \mathfrak{so}(4)$),* 1 *(mixed) and* 1 *(binding on both). None contains the Pati–Salam algebra $\mathfrak{so}(4) \oplus \mathfrak{so}(6)$ [116], of dimension* 21*, or its colour factor $\mathfrak{su}(4)$, of dimension* 15*.*

Obstructed, by a dimension count; the five dimensions are re-verified (e5-3a, e5-3e, e5-3h, e5-3i). At $n = 3$ the compact spectator algebra $\mathfrak{so}(8) \oplus \mathfrak{so}(6)$, of dimension 43, already contains $\mathfrak{ps}_U$ (Section 5.2.7). The analogue at $n = 2$ of Theorems 4.36 to 4.38 is therefore the intersection of each compact candidate with $\mathrm{Cent}(G_2)$.

Proposition 9.43 (The spectator algebra and the binding root)**.** *Let the Lorentz algebra act by $\mathfrak{l}_{\mathrm{int}}^{(1)}$, with constituent* 2 *a spectator.*

(i) $\mathfrak{g}(1,0,1) = \mathrm{Cent}(\mathfrak{l}_{\mathrm{int}}^{(1)}) \cap \mathfrak{so}(6,6) \cong \mathfrak{so}(5,3)$ *has dimension* 28*. Its compact part is $\mathfrak{so}(5) \oplus \mathfrak{so}(3)$, with $\mathfrak{so}(5)$ on the positive Lorentz-trivial span $P = \langle e_2^{(1)}, e_3^{(1)}, e_1^{(2)}, e_2^{(2)}, e_3^{(2)} \rangle$ and $\mathfrak{so}(3)$ on $N = \langle e_4^{(2)}, e_5^{(2)}, e_6^{(2)} \rangle$.* $\mathfrak{g}(1,0,1) \cap \mathrm{Cent}(G_2) = \mathfrak{l}_G^{(2)}$*. Neither $\mathfrak{so}(5)$ nor $\mathfrak{so}(3)$ meets $\mathrm{Cent}(G_2)$, and with $E_2 = \langle e_1^{(2)}, e_2^{(2)}, e_3^{(2)} \rangle$,*

$$(\mathfrak{so}(5) \oplus \mathfrak{so}(3)) \cap \mathrm{Cent}(G_2) = \{ A + G_2 A G_2^{-1} : \ A \in \mathfrak{so}(E_2) \} \cong \mathfrak{su}(2),$$

the three paired rotations of constituent 2*: compact and non-abelian.*

(ii) Among the spin lifts of the 30 *elements of* $\mathrm{Cent}(G_2)$*, those satisfying the kinetic condition with the kernels of constituent* 1 *are exactly the paired rotations of constituent* 2*. The paired boosts of constituent* 2 *fail it.*

(iii) Let w be the product of the eight Lorentz-trivial generators. Then $w^2 = -\mathbf{1}$, w commutes with $\mathfrak{l}_{\mathrm{int}}^{(1)}$ and with the spin lift of $\mathfrak{so}(5,3)$, and $\Gamma = w\,\Omega^{(1)}$, with $\Omega^{(1)} = E_{(1,1)} E_{(1,4)} E_{(1,5)} E_{(1,6)}$ the interior volume element of constituent 1, is a chirality commuting with both. With

w as i, each chirality half is a Weyl block $(\frac{1}{2},0)\otimes(\mathbf{4},\mathbf{2})$*, respectively* $(0,\frac{1}{2})\otimes(\mathbf{4},\mathbf{2})$*, of* $\mathfrak{l}_{\rm int}^{(1)}\oplus\mathfrak{so}(5)\oplus\mathfrak{so}(3)$ *(Proposition 5.19). Under the binding-compatible* $\mathfrak{su}(2)$*,*

$$L=(\tfrac{1}{2},0)\otimes(2\cdot\mathbf{1}\oplus 2\cdot\mathbf{3}),\qquad R=(0,\tfrac{1}{2})\otimes(2\cdot\mathbf{1}\oplus 2\cdot\mathbf{3})\,:$$

integer isospin, and the same real representation on both halves, so the action is vector-like. With charges $h=-w\,S(X)$*,* $\mathcal{A}(h)=\mathcal{A}(h^3)=0$*, where* $\mathcal{A}(f)=\frac{1}{4}[\mathrm{Tr}(fP_L)-\mathrm{Tr}(fP_R)]$ *as in Theorem 4.38. There is no doublet, hence no global* $SU(2)$ *anomaly [169].*

Proof. (i) Let $A\in\mathfrak{so}(P)$ and $B\in\mathfrak{so}(N)$ with $[A+B,G_2]=0$, and follow the proof of Theorem 4.36. G_2 maps $F=\langle e_2^{(1)},e_3^{(1)}\rangle$ into the interior plane $\langle e_5^{(1)},e_6^{(1)}\rangle$, on which A and B vanish, so $G_2Av=(A+B)G_2v=0$ and $Av=0$ for $v\in F$. For $v\in E_2$, $Bv=0$ and $G_2v\in N$, so $G_2Av=BG_2v\in N$ and $Av\in G_2^{-1}N=E_2$. Hence $A\in\mathfrak{so}(E_2)$ and $B=G_2AG_2^{-1}$ on N; conversely every such pair commutes with G_2. Taking $B=0$ or $A=0$ shows that neither summand meets $\mathrm{Cent}(G_2)$. The rest is Proposition 9.41(i). (ii) At $\mu=0$ the condition is antisymmetry. The spin lift of a paired boost is a sum of boost bivectors, which are symmetric (Theorem 1.3). The spin lift of a paired rotation of constituent 2 is antisymmetric and commutes with every bivector of constituent 1, in particular with $\gamma^0\gamma^a$. A general element of $\mathrm{Cent}(G_2)$ was tested by an exact null space. (iii) $\mathfrak{so}(5)\supset\mathfrak{so}(E_2)\oplus\mathfrak{so}(F)$ restricts the spinor $\mathbf{4}$ to two isospin doublets, and $\mathfrak{so}(3)$ acts on $\mathbf{2}$ as a doublet, so the diagonal $\mathfrak{su}(2)$ sees $(\mathbf{2}\oplus\mathbf{2})\otimes\mathbf{2}=2\cdot(\mathbf{1}\oplus\mathbf{3})$. Every representation of $\mathfrak{su}(2)$ has weights symmetric under $m\mapsto-m$, so every odd trace of h vanishes on each half separately; integer isospin excludes doublets, and with them the global anomaly. □

Part (i) is fixed: it is the structure of Theorem 4.36 at $n=2$, with $U=P$ and $d=\dim E_2=3$. Part (ii) is fixed, and in (iii) the Weyl blocks restate Proposition 5.19, the branching is inherited, and the content and the anomaly traces are fixed. Checks: exact Lie algebra (e5-3a), exact null space of the kinetic condition (e5-3d), Casimirs to 10^{-9} and integer traces (e5-3b, e5-3c); the odd traces vanish as well for the $\mathfrak{so}(3)$ on N, which acts by doublets (e5-3j). The binding root therefore turns the isospin doublets of the spectator algebra, the "doublets, no colour, real" of Section 5.2.8, into isospin 0 and 1. No doublet survives at $n=2$, as for the copy isospin of Proposition 9.24. At $n=3$ the doublets survive in the $(\frac{1}{2},\frac{1}{2})_{\pm1}$ of $\mathfrak{so}(4)_\Delta$ on P_* (Theorem 4.38(ii)).

Proposition 9.44 (The diagonal connection algebra and the binding root)**.** *Let the Lorentz algebra act by the diagonal* $\mathfrak{l}_{\rm int}$*.*

(i) $\mathrm{Cent}(\textit{diagonal }\mathfrak{l}_{\rm int})\cap\mathfrak{so}(6,6)=\mathfrak{so}(2)\oplus\mathfrak{so}(4)$ *has dimension 7 and is compact. It meets* $\mathrm{Cent}(G_2)$ *in* $\mathbb{R}C_v$*, the copy rotation of all six directions: the copy rotation of the interior four-planes plus that of the surplus pairs. This algebra is abelian and compact, and* $S(C_v)=-C_b$ *with* $C_b=\frac{1}{2}\sum_a\eta_{aa}E_{(1,a)}E_{(2,a)}$*.*

(ii) There is no fermion under this action at $n=2$ *(Proposition 5.16) and no copy-invariant gamma matrix (Theorem 5.73), so the kinetic condition has no canonical kernel. With the summed kernels* $K_S^\mu=\sum_k\gamma_0^{(k)}\gamma_\mu^{(k)}$ *it holds for* C_b*. With the kernels* f_0f_μ *of the centre-of-mass generators* $f_a=(E_{(1,a)}+E_{(2,a)})/\sqrt{2}$ *(Proposition 5.22) it fails.*

(iii) $\Gamma_{12}^2=\mathbf{1}$*, and* Γ_{12} *commutes with the diagonal* $\mathfrak{l}_{\rm int}$ *and with* C_b*. With copy charge* q *defined by* $C_b=iq$*, the content of* $S_2\otimes\mathbb{C}$ *is*

$\Gamma_{12}=-1$	$(0,0)_{\pm3}\oplus(0,0)_{\pm1}\oplus2\,(\frac{1}{2},\frac{1}{2})_{\pm1}\oplus(1,0)_{\pm1}\oplus(0,1)_{\pm1}$
$\Gamma_{12}=+1$	$2\,(0,0)_{\pm2}\oplus(\frac{1}{2},\frac{1}{2})_{\pm2}\oplus2\,(\frac{1}{2},\frac{1}{2})_0\oplus2\,(1,0)_0\oplus2\,(0,1)_0$

Spin is integral, $\Gamma_{12}=-1$ *carries the odd and* $\Gamma_{12}=+1$ *the even copy charges, and each half is invariant under* $(q,(j_L,j_R))\mapsto(-q,(j_R,j_L))$*: self-conjugate, hence vector-like.* $\mathrm{Tr}(C_b^kP_\pm)=0$ *for every odd* k*.*

Proof. (i) Under the diagonal $\mathfrak{l}_{\text{int}}$, $V_2 = 2V \oplus 4 \cdot \mathbf{1}$, and Schur's lemma gives $\mathfrak{so}(2) \oplus \mathfrak{so}(4)$, the $n = 2$ case of Theorem 5.9. Within it, an element $aM + B$, with M the copy rotation of the interior four-planes and B in the surplus $\mathfrak{so}(4)$, commutes with G_2 exactly when B is a times the copy rotation of the surplus pairs, because G_2 maps each surplus pair into the interior plane of the same constituent. This is Proposition 9.19(ii), and the lift is Proposition 5.22(ii). (ii) C_b commutes with the diagonal boosts and is antisymmetric, while $f_0 f_\mu$ contains the cross terms $E_{(1,1)}E_{(2,\mu)}$ ($\gamma^0 = e_1$), with which it does not commute. (iii) Computed. □

Part (i) restates Proposition 9.19(ii) and Proposition 5.22 (fixed there; e5-3e), (ii) is fixed (exact; e5-3f), and (iii) is fixed (eigenvalues to 10^{-9}, traces exact; e5-3g). Under the diagonal action the binding root leaves an abelian, compact, Lorentz-invariant copy charge on matter that has no fermions; the row "$\mathfrak{so}(2) \oplus \mathfrak{so}(4)$, no fermions" of Section 5.2.8 becomes "$\mathfrak{so}(2)$, no fermions".

Proposition 9.45 (The mixed rung and the binding root)**.** *Let $\mathfrak{l}_{\text{mix}}$ pair $\mathfrak{l}^{(1)}_{\text{int}}$ with $\mathfrak{l}^{(2)}_{G}$ generator by generator (Proposition 5.74).*

(i) $\mathfrak{l}_{\text{mix}}$ satisfies the $\mathfrak{so}(1,3)$ brackets, its centraliser in $\mathfrak{so}(6,6)$ is $\mathbb{R}J_1$, $J_1 = e_2^{(1)}e_3^{(1)}$ acting on vectors (Proposition 5.75), and $\mathbb{R}J_1 \cap \mathrm{Cent}(G_2) = 0$. G_2 does not commute with $\mathfrak{l}_{\text{mix}}$.

(ii) The $\mathfrak{l}_{\text{mix}}$-equivariant endomorphisms of V_2 form a seven-dimensional algebra: the scalars on the four-vector, $\mathrm{span}\{\mathbf{1}, G\}$ on the field strength of constituent 2, and $\mathfrak{gl}(2)$ on the two Lorentz-trivial directions. Its Euclidean-antisymmetric part, the generators that rotate without dilating, is $\mathrm{span}\{G^{(2)}, J_1\}$, with $G^{(2)} = 0 \oplus G$, and $[J_1, G^{(2)}] = 0$.

Proof. (i) $\mathbb{R}^{6,6} = V \oplus \Lambda^2 V \oplus 2 \cdot \mathbf{1}$ under $\mathfrak{l}_{\text{mix}}$ (Proposition 5.74). J_1 rotates the surplus pair of constituent 1, and G_2 maps that pair into the interior four-plane, which J_1 fixes, so $[J_1, G_2] \neq 0$. No nonzero element of $\mathfrak{l}^{(1)}_{\text{int}}$ commutes with G_2 (Proposition 9.41(i)), so G_2 does not commute with $\mathfrak{l}_{\text{mix}}$. (ii) Schur's lemma on the three isotypic pieces, with $\mathrm{Hom}_{\mathfrak{l}_G}(V, V) = \mathrm{span}\{\mathbf{1}, G\}$ (Proposition 4.19). □

Part (i) restates Proposition 5.74 and Proposition 5.75 and is fixed in what it adds; (ii) is fixed (exact; e5-3h). The binding root of the closure is not covariant under the mixed action and leaves nothing of its algebra. The only covariant rotor generators that rotate without dilating are the combinations of $G^{(2)}$ and J_1. With the rotor $G^{(2)}$, which binds constituent 2 alone, all of $\mathbb{R}J_1$ survives (constructed: admitted, not selected). For completeness, under the two binding actions the binding root leaves $\mathfrak{l}^{(2)}_{G}$ of $\mathfrak{g}(0,1,1) = \mathfrak{so}(3,3)^{(2)}$, with compact part the $\mathfrak{su}(2)$ of Proposition 9.43, and all of $\mathfrak{g}(0,2,0) = \mathfrak{so}(2,\mathbb{C})$, with compact part $\mathbb{R}C_v$ (restating Proposition 5.27(iii) and Table 5.1; e5-3i). Neither action carries a Dirac operator, so the kinetic condition does not apply.

9.5.4 Inertia freedom at two constituents

Theorem 9.46 (Inertia freedom at two constituents)**.** *The real dimensions of the symmetric forms N on V_2^*, invariant under a symmetry ($rN + Nr^\top = 0$, Definition 4.25), in the Lorentz class $\{N, G_2\} = 0$ and in the duality class $[N, G_2] = 0$, are:*

symmetry	Lorentz class	duality class
$\mathrm{Cent}(G_2) = \mathfrak{so}(6,\mathbb{C})$	2	0
$\mathfrak{l}_G^{\Delta} \oplus \mathbb{R}C_v$ (the stabiliser of M_{pair})	2	0
$\mathfrak{l}_G^{(1)} \oplus \mathfrak{l}_G^{(2)}$	4	0
$\mathfrak{l}_G^{\Delta}$	6	0
$\mathfrak{l}_G^{(2)}$	14	9
$\mathfrak{su}(2)$, the binding-compatible spectator part (Proposition 9.43)	14	10
$\mathbb{R}C_v$, the binding-compatible diagonal part (Proposition 9.44)	18	18
none	42	36

Under the full closure symmetry $\mathfrak{so}(3n,\mathbb{C})$ the freedom is $(2,0)$ at $n = 1,2,3$, spanned by η_n^{-1} and $G_n\eta_n^{-1}$; with no symmetry it is $(3n(3n+1), 9n^2)$, that is $(12,9)$, $(42,36)$ and $(90,81)$.

Proof. Split N into blocks between constituents. Under $\mathfrak{l}_G^{(k)}$ the invariant symmetric forms of one constituent are $\mathrm{span}\{\eta^{-1}, G\eta^{-1}\}$, of Lorentz class (Proposition 4.19 and Theorem 4.26(ii),(iii)); an invariant cross block vanishes when only one constituent is acted on, and is a complex multiple of the invariant bilinear form under $\mathfrak{l}_G^{\Delta}$. This gives $2\cdot 2 = 4$ and $2\cdot 3 = 6$, the real and imaginary parts of the products $z_k \cdot z_l$ of Proposition 9.39, and $\mathfrak{so}(6,\mathbb{C})$ keeps only q_{tot}, which gives 2. A copy rotation acts on the copy index, and the copy-rotation-invariant complex symmetric 2×2 matrices are the multiples of $\mathbf{1}$, which gives 2 for the stabiliser. Under $\mathfrak{l}_G^{(2)}$ constituent 1 is free (12 and 9 by Theorem 4.26(i)), and its cross blocks vanish because $\mathfrak{so}(3)$ moves every vector of $\mathbb{R}^3$. Under the compact $\mathfrak{su}(2)$ alone the identity of constituent 2 is also invariant, which adds one duality form. Under $\mathbb{R}C_v$ alone, $N = \mathbf{1}\otimes A + \varepsilon\otimes B$ with A symmetric and B antisymmetric, which gives $12+6$ and $9+9$. With no symmetry the classes are the forms anticommuting and commuting with G_n. □

The first, third and fourth rows restate Section 5.6.2 ("four under $\mathfrak{l}_G^{(1)} \oplus \mathfrak{l}_G^{(2)}$, six under $\mathfrak{l}_G^{\Delta}$ alone and two under $\mathrm{Cent}(G_2)\cap\mathfrak{so}(6,6)$"); the other rows, and the ladder, are fixed (exact ranks; e5-4a, e5-4b). Two observations follow. The seven-dimensional stabiliser of the pair invariant fixes the inertia as much as the thirty-dimensional closure symmetry does: up to one complex scale. And, as at $n = 3$, imposing only what the binding root leaves of a gauge algebra enlarges the freedom to 14 or 18 real parameters in the Lorentz class, so the gauge parts do not select the inertia.

9.5.5 Two constituents and one rotor

The closure on $\mathfrak{s}_2 = \mathbb{R} \ltimes_{G_2} V_2$ with $H = h^2/(2I_0) + \sum_k \frac{1}{2} p^{(k)\top} N_k p^{(k)}$ has $p^{(k)} = e^{-\theta G} p^{(k)}(0)$ for both constituents, so $w_k = e^{2i\theta} w_k(0)$ and $I_0\ddot{\theta} = R_{\mathrm{eff}} \sin(2\theta + \varphi_{\mathrm{eff}})$. The rate is constant and nonzero iff $R_{\mathrm{eff}} = 0$ and $h(0) \neq 0$, and then every frame moves on a circle of diameter $|N_k p^{(k)}(0)|/|\kappa|$. The balance set $\mathcal{B} = \{F = 0\}$ has codimension two away from 0, its massive part is open and dense, and on it $|c_1|E_0(\pi^{(1)}) = |c_2|E_0(\pi^{(2)})$ (Theorem 5.52, restating Proposition 4.33). With equal inertias the pairs on the null subspaces $V_{\pm} = \{p^{(2)} = \mp G p^{(1)}\}$ are balanced (Proposition 5.55). The following items are new.

Proposition 9.47 (The rotor law with pair inertias)**.** *(i) $\frac{1}{2} p^{\top} N p = \mathrm{Re}\big((a+ib)w(p)\big)/(2m)$ for $N = (a\eta^{-1} + bG\eta^{-1})/m$, so block-diagonal Lorentz-class inertias give $U(\theta) = \frac{1}{2}\,\mathrm{Re}\big(e^{2i\theta}F(p(0))\big)$.*

(ii) The Lorentz-class forms on V_2^* invariant under $\mathfrak{l}_G^{\Delta}$ are exactly the real and imaginary parts of the three products $z_k\cdot z_l$ of the momentum fields. For such an inertia $U = \frac{1}{2}\,\mathrm{Re}\,\mathrm{tr}\big(C\,Q(\pi)\big)$ with C complex symmetric, $Q(\pi(\tau)) = e^{2i\theta}Q(\pi(0))$, and

$$I_0\ddot{\theta} = R\sin(2\theta+\varphi), \qquad Re^{i\varphi} = \mathrm{tr}\big(C\,Q(\pi(0))\big).$$

The balance condition is $\mathrm{tr}(CQ(\pi)) = 0$, and $C = \mathrm{diag}(c_1, c_2)$ recovers (i).

Proof. (i) is Theorem 4.26(ii) per constituent. (ii) The first statement is Theorem 9.46 (row $\mathfrak{l}_G^{\Delta}$) with Proposition 9.39(c). Every $z(\pi^{(k)})$ turns as $e^{i\theta}$, so $Q \mapsto e^{2i\theta}Q$, $U(\theta) = \frac{1}{2}R\cos(2\theta+\varphi)$, and $\dot{h} = -dU/d\theta$. □

Part (i) restates Theorem 4.26(ii) (symbolic). Part (ii) restates the remark in Section 5.6.2, which states the form and the balance condition; the rotor law with cross terms is fixed here. Checks: exact span, and the law along a flow with a random C to 10^{-9} (e5-5a).

Proposition 9.48 (The balance surface at two constituents)**.** *Let $c_1, c_2 \neq 0$.*

(i) In the data $(w_1, w_2) \in \mathbb{C}^2$ the balance condition $c_1w_1 + c_2w_2 = 0$ is a complex line, of real codimension two. In the momentum data $\mathbb{R}^{12}$, $\mathcal{B}$ has codimension two away from 0. If exactly one coefficient vanishes, say $c_2 = 0$, then $\mathcal{B} = \{w_1 = 0\} \times \mathbb{R}^6$ still has codimension two, but forces constituent 1 to be null; if both vanish, $\mathcal{B}$ is everything.

(ii) $\mathcal{B} \setminus 0$ is connected. For $c_1 = c_2$ it is the nonzero null cone $\{Z \cdot Z = 0,\ Z \neq 0\}$ of $Z = (z(\pi^{(1)}), z(\pi^{(2)})) \in \mathbb{C}^6$, a single orbit of $SO(6,\mathbb{C})$, diffeomorphic to $\mathbb{R}_{>0} \times V_2(\mathbb{R}^6)$ with $V_2(\mathbb{R}^6)$ the Stiefel manifold of orthonormal two-frames. For $c_1 \neq c_2$ it is the image of that cone under $(z_1, z_2) \mapsto (z_1, \rho^{-1}z_2)$, $\rho^2 = c_2/c_1$.

(iii) The all-null part of $\mathcal{B}$ has dimension 8. The massive part, on which both constituents are massive, has dimension 10, and w_1 maps it submersively onto $\mathbb{C}^*$, with fibre over s the product of the two complex spheres $\{w(p^{(1)}) = s\}$ and $\{w(p^{(2)}) = -c_1s/c_2\}$.

Proof. (i) is Theorem 5.52(iii); the degenerate cases are direct. (ii) $c_1q_1 + c_2q_2 = c_1(z_1, \rho z_2) \cdot (z_1, \rho z_2)$. A nonzero null vector of $\mathbb{C}^6$ is $u + iv$ with $u, v \in \mathbb{R}^6$, $|u| = |v| > 0$ and $u \perp v$; $SO(6,\mathbb{C})$ is transitive on these by Witt's theorem [232], and $V_2(\mathbb{R}^6)$ is connected. (iii) The null set $\{w = 0\} \subset \mathbb{R}^6$ has dimension four (Theorem 5.52(iii)). On the massive part $w_1 \neq 0$ and $w_2 = -c_1w_1/c_2 \neq 0$. □

Fixed, with (i) restated. Checks at 20 random points each: $\operatorname{rank} dF = 2$ for equal and unequal coefficients, orbit dimension 10, $\operatorname{rank} d(w_1, w_2) = 4$ on the all-null part, and $\operatorname{rank}(dF, dw_1) = 4$ on the massive part (e5-5f).

Theorem 9.49 (Stability of the balance surface)**.** *Let $c_1, c_2 \neq 0$ and let (θ, h, p, x) solve the two-constituent closure with $\theta(0) = 0$, $\kappa_0 = h(0)/I_0$ and $\psi_0 = \varphi_{\text{eff}}$.*

(i) R_{eff} is a constant of motion for every datum. The balance surface and every level set $\{R_{\text{eff}} = r\}$ are invariant, so the surface is neither attracting nor repelling. A perturbation δp of a balanced datum with nonzero momentum has imbalance $|dF(\delta p)| + O(|\delta p|^2)$, and since dF is onto there every small imbalance occurs (at $p = 0$, $dF = 0$).

(ii) Let $h(0) \neq 0$. The rate is stable. With $\varepsilon = R_{\text{eff}}/(I_0\kappa_0^2)$, $\kappa^2 = \kappa_0^2[1 + \varepsilon(\cos\psi_0 - \cos\psi)]$ and $|\kappa^2/\kappa_0^2 - 1| \leq 2\varepsilon$ for all τ, and the rotor turns for all time when $\varepsilon < \frac{1}{2}$.

(iii) Let $h(0) \neq 0$. The bounded frame orbits are not stable. In the rotating regime, during each period of the rate frame k is displaced by $\lambda\, e^{-\varphi_{\text{eff}}G} N_k p^{(k)}(0)$, with λ of Theorem 4.31, and its mean velocity is

$$\bar{v}_k = \tfrac{\varepsilon}{4}(1 - \varepsilon\cos\psi_0)\, e^{-\varphi_{\text{eff}}G} N_k p^{(k)}(0) + O(\varepsilon^3),$$

of first order in the imbalance. A perturbation off the surface therefore moves the frames by a distance that grows linearly in τ, at a rate linear in the perturbation. A perturbation along the surface, or of $h(0)$, keeps them on circles.

(iv) If $h(0) = 0$, a balanced datum is an equilibrium of the rotor, and each frame moves uniformly, $x_k = x_k(0) + \tau N_k p^{(k)}(0)$, without bound unless $N_k p^{(k)}(0) = 0$, in which case the frame is stationary. Off the surface, unless $\psi_0 \in \pi\mathbb{Z}$ (an equilibrium, at which the frames move uniformly), the rotor librates about $\psi = \pi$ with $\Omega^2 = 2R_{\text{eff}}/I_0$, amplitude $\xi_0 \in (0, \pi)$ given by $\cos\xi_0 = -\cos\psi_0$, and period $4\mathrm{K}(m)/\Omega$, $m = \sin^2(\xi_0/2)$, which diverges as $R_{\text{eff}} \to 0$; K and E are the complete elliptic integrals in the parameter m [36]. The frames move with the mean velocity $(1 - 2\mathrm{E}(m)/\mathrm{K}(m))e^{-\varphi_{\text{eff}}G}N_k p^{(k)}(0)$. They drift, except at the one amplitude with $2\mathrm{E}(m) = \mathrm{K}(m)$ ($m = 0.8261$, $\xi_0 = 130.7°$), at which every frame returns to its initial position after each period.

Proof. (i) $F(p(\tau)) = e^{2i\theta}F(p(0))$ because every w_k turns by $e^{2i\theta}$. (ii) Energy conservation, $H = \frac{1}{2}I_0\kappa^2 + \frac{1}{2}R_{\text{eff}}\cos\psi$, divided by $\frac{1}{2}I_0\kappa_0^2$; this is Corollary 4.30 with R_0/m replaced by R_{eff}. (iii) Lorentz-class N_k anticommute with G, so $\dot{x}_k = e^{2\theta G}N_k p^{(k)}(0)$ with the common angle θ, and the proof of Theorem 4.31 applies to each frame. Expanding $(1 + \varepsilon(\cos\psi_0 - \cos\psi))^{-1/2}$ to second order, $\lambda = \frac{\pi\varepsilon}{4|\kappa_0|}(1 - \frac{3}{2}\varepsilon\cos\psi_0) + O(\varepsilon^3)$ and the period is $\frac{\pi}{|\kappa_0|}(1 - \frac{1}{2}\varepsilon\cos\psi_0) + O(\varepsilon^2)$; their ratio gives $\bar{v}_k$. Along the surface $\varepsilon = 0$, and a change of $h(0)$ changes only the radius $|N_k p^{(k)}(0)|/(2|\kappa|)$ of the circles. (iv) At $h(0) = 0$ and $R_{\text{eff}} = 0$, $\ddot{\theta} = 0$ and $\theta \equiv 0$, so $\dot{x}_k = N_k p^{(k)}(0)$. Off the surface the rotor equation is the pendulum of Proposition 4.42 with R_0/m replaced by R_{eff}. With $\psi = \pi + \xi$, $\dot{x}_k = -e^{-\varphi_{\text{eff}}G}(\cos\xi + G\sin\xi)N_k p^{(k)}(0)$. The substitution $\sin(\xi/2) = \sin(\xi_0/2)\sin t$ of that proposition gives $d\tau = dt/(\Omega\sqrt{1 - m\sin^2 t})$, so over one period $\int \sin\xi\, d\tau = (2\sqrt{m}/\Omega)\int_0^{2\pi}\sin t\, dt = 0$ and $\int\cos\xi\, d\tau = (4/\Omega)\int_0^{\pi/2}(1 - 2m\sin^2 t)(1 - m\sin^2 t)^{-1/2}dt = (4/\Omega)\big(2\mathrm{E}(m) - \mathrm{K}(m)\big)$; divide by the period. □

Item (i) restates Theorem 5.52(i); (ii) restates Corollary 4.30 for two constituents; (iii) and (iv) are fixed. The second-order term in (iii) is new. Item (iv) sharpens the closing sentence of Proposition 4.33: on the balance surface the frames are bounded only if $h(0) \neq 0$, which is the qualifier "constant and nonzero" of Theorem 5.52(i). Checks (DOP853, relative and absolute tolerance 10^{-12}; e5-5b to e5-5e): with equal inertias $a + ib = 1 + 0.3i$, $m = I_0 = 1$, $h(0) = 2$, and $p^{(2)} = -Gp^{(1)}$ drawn at random ($E_0(\pi^{(k)}) = 1.2156$ each), the rate stays constant to 10^{-11} over $\tau \in [0, 60]$ and both frames lie on their circles (diameter 0.942) to 10^{-9}. A generic balanced pair with unequal inertias ($m = 1, 2.5$; $a + ib = 1 + 0.3i,\ 0.4 - 0.7i$) gives the same. Off the surface ($p^{(2)}$ perturbed by 0.05 along a fixed random direction, $R_{\text{eff}} = 0.391$, $\varepsilon = 0.098$), R_{eff} and $w_1\overline{w_2}$ stay constant to 10^{-10}, the rotor obeys the pendulum law to 10^{-9}, and the displacement per period is $\lambda e^{-\varphi_{\text{eff}}G}N_k p^{(k)}(0)$ to relative 10^{-8}. For perturbations 10^{-3}, 2×10^{-3}, 4×10^{-3} the imbalance, the rate excursion and the mean frame velocity double with the perturbation (to 2%). The mean velocity equals the quadrature value to 10^{-8}, the first-order law to within 1.5ε and the second-order law to within ε^2 (relative). With $h(0) = 0$ the rotor librates with the period of (iv) to 10^{-8}, and on the balance surface it stays at rest to 10^{-12} while the frames move uniformly. Off the surface with $h(0) = 0$ the displacement after one, two and three periods equals the mean-velocity law of (iv) to relative 10^{-9} at four amplitudes, among them $\xi_0 = 130.7°$, where it vanishes (e5-5i).

Proposition 9.50 (Exchange of the constituents)**.** *Let $\sigma(p^{(1)}, p^{(2)}) = (p^{(2)}, p^{(1)})$.*

(i) σ is an η_2-isometry commuting with G_2, so it preserves the Lie–Poisson bracket of $\mathfrak{s}_2$ and commutes with the rotor; it preserves the equal-inertia Hamiltonian; as an element of $O(6, \mathbb{C})$ it has determinant -1, so it lies outside the identity component of the closure symmetry group. It exchanges V_+ and V_-.

(ii) With equal coefficients $R_{\text{eff}} = 2|c|\,|w_s|$ with $w_s = \frac{1}{2}(w_1 + w_2)$, the exchange-even part of (w_1, w_2): a pair is balanced iff (w_1, w_2) is exchange-odd, $w_2 = -w_1$.

(iii) The exchange eigenstates of the momenta, $p^{(2)} = \pm p^{(1)}$, both have $w_2 = w_1$, so $F =$

$(c_1 + c_2)w_1$*: unless* $c_2 = -c_1$ *they are balanced only when null. With equal inertias a massive balanced pair is never an exchange eigenstate of its momenta.*

Proof. (i) σ permutes the copies; on Z it is the permutation of three pairs of coordinates, of determinant $(-1)^3$. If $p^{(2)} = -Gp^{(1)}$, then σp has second component $p^{(1)} = G(-Gp^{(1)})$. (ii) $R_{\text{eff}} = |c|\,|w_1 + w_2|$. (iii) w is quadratic, so $F = c_1 w_1 + c_2 w_1$. □

Fixed (e5-5g, e5-5j). Exchange parity of the duality invariants and exchange parity of the momenta are different things: the balanced pairs of $V_\pm$, a magnetic and an electric constituent a quarter duality turn apart (Proposition 5.55(iii)), are exchanged with each other by σ and are not exchange eigenstates. With $c_1 \neq c_2$, σ is not a symmetry of the closure.

Proposition 9.51 (What carries over from the rest closure)**.** *For one rotor on two constituents with Lorentz-class inertias:*

(i) (Theorem 4.40.) A function of p *is a Casimir iff it is invariant under* $p \mapsto e^{tG_2}p$*. At* $p \neq 0$ *the leaves are two-dimensional, so there are* 11 *local Casimirs, against* 5 *at* $n = 1$*, and the real and imaginary parts of* $\bar{z}_a z_b$*,* $a, b = 1, \dots, 6$*, with* $z \in \mathbb{C}^6$ *the pair of momentum fields, provide them. They include* E_6*,* E_0^2 *and* $\mathbf{S}$ *of each* $\pi^{(k)}$ *and the squared moduli* $|Q_{kl}|^2$*, which are smooth everywhere;* E_0 *and* $|Q_{kl}|$ *are conserved invariant functions, smooth where they do not vanish (as for* E_0 *in Theorem 4.40; a12-1). No Hamiltonian moves rest energy between the constituents (Corollary 5.54).*

(ii) (Theorem 4.28–4.8.) The pendulum, its small oscillations, the energy budget and the multiscale criterion hold with R_0/m *replaced by* R_{eff}*, which depends on the data only through* $F(p(0))$*. What changes is Corollary 4.43(i): a constant nonzero rate no longer needs a null momentum field.*

(iii) (Proposition 4.47 and Proposition 4.50.) $\Gamma_{12}^2 = \mathbf{1}$*, and* Γ_{12} *anticommutes with the twelve generators, so it implements* $e^{\pi G_2} = -\mathbf{1}$ *on* S_2*. Since* $(e^{\theta G_2})^\top \eta_2 e^{\theta G_2} = \cos 2\theta\, \eta_2 + \sin 2\theta\, \eta_2 G_2$*, the rotor lifts to* $\mathrm{Pin}(6,6)$ *only at* $\theta \in \pi\mathbb{Z}$*.*

(iv) (Section 4.5.5.) The rotor turns both momentum fields at one rate, while the graded rest Hamiltonian of the tower has the single frequency $(m_1^2 + m_2^2)^{1/2}$ *and the ungraded one the two frequencies* m_1*,* m_2 *(Proposition 5.34(v)). A closure with two rates exists, with the rotor acting by* $\mathrm{diag}(\kappa_1, \kappa_2) \otimes G$*,* $\kappa_1 \neq \kappa_2$ *(Proposition 9.20), and is admitted, not selected.*

(v) (Proposition 4.52.) Under Assumption 4.51, every leaf through the balance surface is flat, $U \equiv 0$*, so the conditional ladder* $\kappa_n = (n + \delta)\hbar/I_0$ *holds exactly with both constituents massive, where at* $n = 1$ *it needs null data.*

Proof. (i) The proof of Theorem 4.40 with G_2 in place of G; $zz^\dagger$ is a rank-one Hermitian matrix on $\mathbb{C}^6$, with $2 \cdot 6 - 1 = 11$ real parameters. (ii) The equations for (θ, h) are those of one constituent with $R_0 e^{i\varphi_0}/m$ replaced by $F(p(0))$. (iii) As in Proposition 4.50: a product of all twelve anticommuting generators anticommutes with each, and $G_2^\top \eta_2 = \eta_2 G_2$, $G_2^\top \eta_2 G_2 = -\eta_2$. (iv) and (v) are direct. □

Items (i)–(iii) restate the cited results for two constituents (fixed there) and are fixed in their $n = 2$ form (Jacobian rank 11 and 5, Lie–Poisson rank 2, exact Clifford identities; e5-5h). Item (iv) restates Proposition 5.34 and Proposition 9.20 and is constructed for the two-rate closure. Item (v) is constructed and conditional on the assumption. The copy rotor of Remark 5.61, a second rotor that exists because the uniqueness argument of Theorem 4.22 fails under $\mathfrak{l}_G^\Delta$, and the two-rate rotor of (iv) are the structures of the rest closure that exist only from $n = 2$ on.

	$n = 1$	$n = 2$	$n = 3$
closure symmetry $\mathrm{Cent}(G_n)$	$\mathfrak{l}_G \cong \mathfrak{so}(3,\mathbb{C})$, 6	$\mathfrak{so}(6,\mathbb{C})$, 30	$\mathfrak{so}(9,\mathbb{C})$, 72
$\mathrm{Cent}(G_n) \cap \mathfrak{l}_{\mathrm{int}}^{(1)}$	0	0	0
commuting with G_n and $\mathfrak{l}_{\mathrm{int}}^{(1)}$	0	$\mathfrak{l}_G^{(2)}$, 6	$\mathfrak{so}(6,\mathbb{C})$ of 2, 3, 30
$S_n \otimes \mathbb{C}$ under $\mathfrak{l}_G^{(1)}$	$2\,(\frac{1}{2},\frac{1}{2})$	$16\,(\frac{1}{2},\frac{1}{2})$	$128\,(\frac{1}{2},\frac{1}{2})$
Pati–Salam algebra	no	no	in $\mathfrak{so}(8) \oplus \mathfrak{so}(6)$
binding-compatible compact part of the spectator algebra	0	$\mathfrak{su}(2)$, 3; vector-like, isospin 0, 1	$\mathfrak{so}(6)$ of 2, 3, 15; contains $\mathfrak{so}(4)_\Delta$
the same for the diagonal connection algebra	0	$\mathbb{R}C_v$, 1; no fermions	copy rotations $\mathfrak{so}(3)$, 3
under the projectors	—	—	P_{fam}: $\mathfrak{u}(1)^2$; P_*: $\mathfrak{so}(4)_\Delta \oplus \mathbb{R}\,\mathrm{rot}(8,9)$, chiral
anomalies of what survives	—	none	P_{fam}: $\mathcal{A}(H^3) = -12\alpha\beta^2$; P_*: free on $\mathfrak{so}(4)_\Delta$
inertia, closure symmetry	$(2,0)$	$(2,0)$	$(2,0)$
inertia, per-constituent $\mathfrak{l}_G$	$(2,0)$	$(4,0)$	$(6,0)$
inertia, binding-compatible part	—	$(14,10)$ spectator; $(18,18)$ diagonal	$(18,13)$ P_{fam}; $(8,6)$ P_*
inertia, none	$(12,9)$	$(42,36)$	$(90,81)$
balance surface	$\{w = 0\}$: null data only, dim 4	codim 2, connected; massive part dense, dim 10	codim 2, dim 16

Table 9.5: The closure along the ladder. Dimensions are real; the inertia entries are (Lorentz class, duality class). The $n = 3$ column restates Theorems 4.34 to 4.39 and Section 5.2.7 and Theorem 5.52(iii), except the spectator and diagonal rows, which are computed here (e5-6a).

statement	class	checks
Proposition 9.38: $\mathrm{Cent}(G_2) = \mathfrak{so}(6,\mathbb{C})$, paired rotations and boosts, $R - iS$, compact part $\mathfrak{so}(6)$	restated (Proposition 5.27(i); Theorem 4.34 at $n = 2$)	e5-1a
Proposition 9.39: quadratic invariants q_{tot}; q_1, q_2; $q_1, q_2, z_1 \cdot z_2$	inherited (invariant theory); dimensions fixed	e5-1b
Theorem 9.40(i): M_{pair} through the Gram entries; diagonal against per-constituent invariance	fixed (partly restated, Theorem 5.57)	e5-1e
Theorem 9.40(ii): stabiliser of M_{pair} is $\mathfrak{l}_G^{\Delta} \oplus \mathbb{R}C_v$, dimension 7	fixed (exact rational rank 23)	z1-5b, e5-1c
Theorem 9.40(iii),(iv): $\mathfrak{so}(6,\mathbb{C})$ does not preserve M_{pair}; range $[\frac{1}{2}\lvert c\rvert, \infty)$ on each orbit	fixed (Example 5.59 restated; transitivity inherited)	e5-1d
Proposition 9.41: no common Lorentz algebra; $\mathfrak{l}_G^{(2)}$; $16(\frac{1}{2}, \frac{1}{2})$; integer spin under $\mathfrak{l}_G^{\Delta}$	fixed (restated: Proposition 5.27(iv), Proposition 9.19(iii); Theorem 4.35)	e5-2a to e5-2c
Proposition 9.42: no Pati–Salam algebra at $n = 2$	obstructed (dimension count)	e5-3a, 3e, 3h, 3i
Proposition 9.43: binding-compatible $\mathfrak{su}(2)$; kinetic condition; integer isospin, vector-like, anomaly free	fixed (Weyl block restated, Proposition 5.19; branching inherited)	e5-3a to e5-3d, 3j
Proposition 9.44: $\mathbb{R}C_v$; no canonical kinetic kernel; content, vector-like	fixed ((i) restated, Proposition 9.19(ii), Proposition 5.22)	e5-3e to e5-3g
Proposition 9.45: mixed rung, nothing survives G_2; covariant rotors $G^{(2)}, J_1$	fixed (restated: Proposition 5.74, Proposition 5.75); rotor $G^{(2)}$ constructed	e5-3h
binding actions: $\mathfrak{l}_G^{(2)}$ and $\mathfrak{so}(2,\mathbb{C})$ survive	restated (Proposition 5.27(iii), Table 5.1)	e5-3i
Theorem 9.46: inertia freedom at $n = 2$; $(2, 0)$ at $n = 1, 2, 3$	fixed (rows 1, 3, 4 restated, Section 5.6.2)	e5-4a, e5-4b
Proposition 9.47: rotor law with pair inertias	fixed ((i) restated, Theorem 4.26(ii))	e5-5a
Proposition 9.48: codimension two, connected, one orbit, parts	fixed ((i) restated, Theorem 5.52(iii))	e5-5f
Theorem 9.49: neutral surface, stable rate, secular frames, $h(0) \neq 0$ needed, libration drift	fixed ((i), (ii) restated)	e5-5b to e5-5e, 5i
Proposition 9.50: exchange, $\det \sigma = -1$, balanced iff w exchange-odd	fixed	e5-5g, 5j
Proposition 9.51: Casimirs, pendulum, half turn, rates, ladder at $n = 2$	(i)–(iii) restated (Section 4.5), fixed at $n = 2$; (iv), (v) constructed	e5-5h
Table 9.5: the ladder $n = 1, 2, 3$	fixed; $n = 3$ mostly restated (Section 4.4.6)	e5-6a, 6b, 2a, 4b
Section 9.5.6: answer	interpretive (drawn from fixed statements)	—

Table 9.6: Claims of Section 9.5, their classes and their checks, which are `PASS` lines of the script `e5_closure_two.py`.

9.5.6 The ladder: one, two and three constituents

The new entries of Table 9.5 at $n = 1$ and $n = 3$ are fixed (e5-6a). At $n = 1$, $\mathrm{Cent}(\mathfrak{l}_{\mathrm{int}}) \cap \mathfrak{so}(3,3) = \mathbb{R}J$ meets $\mathrm{Cent}(G)$ in 0, so the binding root leaves no gauge algebra on one constituent. At $n = 3$ the compact spectator algebra meets $\mathrm{Cent}(G_3)$ in the 15 paired rotations of constituents 2 and 3, which contain the $\mathfrak{so}(4)_\Delta$ of Theorem 4.36, and the diagonal connection algebra meets it in the three copy rotations. At every rung the binding-compatible parts are $\mathfrak{so}(3n-3)$, the paired rotations of constituents $2, \ldots, n$, and $\mathfrak{so}(n)$, the copy rotations: the proofs of Proposition 9.43(i) and Proposition 9.44(i) apply verbatim (fixed; exact at $n = 4$, e5-6b). The balance condition $F = 0$ has codimension two at every rung, because dF has rank two at every nonzero point. At $n = 1$ it is the null cone and carries no massive datum.

Answer. *Is $n = 2$ a closure rung in its own right?* Yes, in a precise sense (interpretive, drawn from the fixed statements cited here): every structure of the three-constituent analysis of Section 4.4.6 except the two projectors, which exist only at $n = 3$, has a computed $n = 2$ analogue, and four of them first appear at $n = 2$. The balance surface carries massive constituents at an exactly constant rate (Theorem 5.52). The closure symmetry mixes the constituents and does not see their pair invariant (Theorem 9.40). The closure symmetry compatible with the matter Lorentz algebra is nonzero, and is itself a Lorentz algebra (Proposition 9.41). And the binding root leaves a compact gauge algebra, non-abelian under the spectator action (Proposition 9.43). *What $n = 2$ adds to $n = 1$* is therefore a constant rate with mass, a gauge part that survives the binding root, a pair invariant and an exchange symmetry. *What it lacks*, and *what $n = 3$ adds*, is everything chiral. At $n = 2$ every binding-compatible gauge part acts vector-like, with zero anomaly coefficients and integer isospin, and no Pati–Salam algebra fits. At $n = 3$ the projectors P_{fam} and P_* exist, the surviving $\mathfrak{so}(4)_\Delta$ acts on inequivalent halves with doublets, the abelian charges of P_{fam} carry nonzero anomaly coefficients, and fermions exist under the diagonal action. The inertia question does not distinguish the rungs: the closure symmetry fixes the inertia up to one complex scale at $n = 1, 2, 3$ alike, and the gauge parts never do.

For the question of this section the closure gives a definite but modest answer (interpretive, drawn from the fixed statements above). The closure at $n = 2$ distinguishes two constituents only through structures it does not select. Equal inertias and the closure symmetry see one field strength in $\mathbb{C}^6$. A pair invariant needs a diagonal Lorentz action and the copy rotation. Per-constituent rest energies need the copies to be distinguished, and then nothing in the closure moves rest energy between them (Proposition 5.60 and Corollary 5.54). What two constituents can do that one cannot is to run at an exactly constant rate with both momentum fields massive. That holds on a set of codimension two which is invariant but whose frames are secularly unstable, which is closer to a balanced pair carried by the dynamics than to two particles produced by it.

9.6 The b-field on the temporal triple, c on the spatial triple

An earlier assignment places the arena of c (of c^2) on the negative triple $\langle e_4, e_5, e_6\rangle$, of signature $(-,-,-)$, and the arena of the b-field on the positive triple $\langle e_1, e_2, e_3\rangle$, of signature $(+,+,+)$, with the remark that the mass is unique in $T = e_1e_2e_3$. In the conventions of this manuscript the positive triple is the temporal triple V^+ and carries $\mathbf{B}$, the negative triple is the spatial triple V^- and carries $\mathbf{E}$ ($X = \sum_i (B_i e_i + E_i e_{i+3})$, Theorem 3.3), the clock is $\gamma^0 = e_1$, the spatial gamma matrices are $\gamma^a = e_{a+3}$, and c is carried by the conversion $C\colon V^+ \to V^-$, of which $C = c\,\mathbf{1}$ is the unique choice that commutes with the diagonal rotations (Theorem 2.2; Proposition 3.98). We therefore read the assignment as follows: the b-field takes values in the temporal rotations $\mathfrak{so}(3)_+ = \mathrm{span}\{e_2e_3, e_3e_1, e_1e_2\}$, and c^2 is a symmetric part of C. The trivector $T = \gamma^0 J$ ($J = e_2e_3$) is the Dirac mass form, the unique antisymmetric kernel invariant under all fifteen bivectors (Corollary 7.10), kept by $\mathrm{Spin}(3,3)$ at group level (Proposition 7.1).

The answers, with their classes, are these. (1) As a form T is invariant under all of $\mathfrak{so}(3,3)$; as an operator it commutes exactly with $\mathfrak{so}(3)_+ \oplus \mathfrak{so}(3)_-$, and its commutant is $M_4(\mathbb{C})$ (fixed; Section 9.6.1). (2) The temporal rotations keep the mass form and the rest flow, and the kinetic term breaks them to $\mathfrak{u}(1)_J$, in proportion to the speed of the clock. They and the Pauli–Gürsey $\mathfrak{su}(2)$ of the massless kinetic term are two diagonals of one $\mathfrak{so}(4)$ meeting in $\mathbb{R}J$ (fixed; Section 9.6.2). (3) A temporal b-field is half a field-strength rotation and half a counter-rotation. It keeps the rates, splits the lines into κ and $s \pm \omega/2$, $s = (\kappa^2 + \omega^2/4)^{1/2}$, cannot be removed along a worldline, and tears the binding (fixed; Section 9.6.3). (4) The binding root G_C preserves $\frac{1}{2}(c^2|\mathbf{B}|^2 + |\mathbf{E}|^2)$ iff $C = cR$, $R \in O(3)$. A temporal rotation keeps the symmetric factor $CC^\top$ on the spatial triple and moves only the orthogonal factor (fixed; the reading "c^2 on $(-,-,-)$, b on $(+,+,+)$" is interpretive; Section 9.6.4). (5) Under a temporal b-field the four-dimensional mass read by a fixed clock oscillates at fixed six-dimensional mass, unless the clock or the temporal energy is on the axis (fixed; reading interpretive; Section 9.6.5). (6) Between two constituents the temporal copy rotation rotates the graded masses at fixed $M = (m_1^2 + m_2^2)^{1/2}$, half of it creates quanta in pairs, and in the ungraded reading no continuous transfer at fixed spectrum exists, only the exchange of the two masses (fixed; Section 9.6.6). (7) At every rung the $\mathfrak{so}(3n,3n)$-invariant forms are spanned by the temporal and the spatial volume; their symmetry has period four in n, so the invariant Grassmann mass kernel is the temporal volume only for $n \equiv 1 \pmod 4$ (fixed; Section 9.6.7). Section 9.6.8 compares the assignment with the b-field of Sections 9.1 and 9.3. Identifiers e6-n refer to the `PASS` lines of `e6_bfield_triples.py` (Table 9.7).

Notation. $S = \mathbb{R}^8$ with the realisation of equation (1.1); $\omega_T = T = e_1e_2e_3$, $\omega_S = e_4e_5e_6 = \gamma^1\gamma^2\gamma^3$, $\Gamma = \omega_T\omega_S = e_1 \cdots e_6$, $J = e_2e_3$, $\Omega = e_1e_4e_5e_6$ (Section 2.3 and Theorem 7.7); $K^\mu = -\gamma^0\gamma^\mu$ are the kinetic kernels (Proposition 2.21) and $\mathfrak{g}_{\text{massless}} = \text{span}\{\Omega, J, e_2\Omega, e_3\Omega\} = \mathfrak{u}(1)_\Omega \oplus \mathfrak{su}(2)_{\text{PG}}$ is the massless kinetic algebra (Proposition 2.29). The temporal generators are symmetric and the spatial ones antisymmetric (Lemma 1.1, (1.4)). $\mathfrak{so}(3)_\pm$ are the rotations of $V^\pm$, the six compact bivectors, and the nine bivectors e_ie_{a+3} are the boosts (Theorem 1.3). A bivector Y acts on vectors by $v \mapsto Y(v) := \frac{1}{2}[Y, v]$. On $V = V^+ \oplus V^-$ we use the blocks $G = \left(\begin{smallmatrix} 0 & -\mathbf{1} \\ \mathbf{1} & 0 \end{smallmatrix}\right)$, $G_C = \left(\begin{smallmatrix} 0 & -C^{-1} \\ C & 0 \end{smallmatrix}\right)$ (Proposition 3.98), $N_C = \text{diag}(C^\top C, \mathbf{1})$ and $E_C(X) = \frac{1}{2}X^\top N_C X = \frac{1}{2}(|C\mathbf{B}|^2 + |\mathbf{E}|^2)$, with E_c for $C = c\,\mathbf{1}$; $\mathfrak{l}_G \oplus \mathfrak{p}_G$ is the splitting of $\mathfrak{so}(3,3)$ into the parts commuting and anticommuting with G (Proposition 9.2). The interior shift is $S = b + \kappa G$ with phase-space generator M (equation (9.2) and Lemma 9.3). Following Section 9.3, the *rates* are the spectrum of the η-self-adjoint part of S and the *lines* are the frequencies of S itself. On the tower $E_{(k,a)}$, N and the ungraded generators $u_a = \mathbf{1} \otimes e_a$ are as in Section 5.4; $\hat{C} = \sigma(\varepsilon \otimes \mathbf{1}_6) = -\frac{1}{2}\sum_a \eta_{aa} E_{(1,a)}E_{(2,a)} = -C_b$ is the spin lift of the copy rotation, as in Section 5.5; C_b has the sign fixed in Section 5.4. Units are $\hbar = c = 1$ unless c is shown.

9.6.1 T as a form and as an operator

Proposition 9.52 (The two invariances of the mass form)**.** *(i)* As a form. *For every $Y \in \mathfrak{so}(3,3)$, $Y^\top T + TY = 0$; hence $Y^\top TY = -TY^2$, and $(e^{sY})^\top T e^{sY} = T$ for every real s. (The ordering matters: $-Y^2T$ equals $-TY^2$ only when $[T, Y^2] = 0$, which holds for a blade but fails for $Y = e_1e_2 + e_3e_4$, whose square $2e_1e_2e_3e_4 - \mathbf{1} + \mathbf{1}$ anticommutes with T.) The same holds for ω_S. The forms on S invariant under $\mathfrak{so}(3,3)$ are $\text{span}\{T, \omega_S\}$, with T antisymmetric and ω_S symmetric.*

(ii) As an operator. *T commutes with the six compact bivectors and anticommutes with the nine boosts. Hence e^{sY} commutes with T for $Y \in \mathfrak{so}(3)_+ \oplus \mathfrak{so}(3)_-$, while $e^{sY}Te^{-sY} = e^{2sY}T$ for a boost.*

(iii) The commutant. *The commutant of T in $M_8(\mathbb{R})$ is spanned by the 32 blades with an even number of spatial indices. It is $\text{Cl}(V^+) \cdot \text{Cl}^0(V^-)$, a product of two commuting subalgebras*

with $\mathrm{Cl}(V^+) \cong \mathrm{Cl}(3,0) \cong M_2(\mathbb{C})$*, whose centre is* $\mathrm{span}\{\mathbf{1}, T\}$*, and* $\mathrm{Cl}^0(V^-) \cong \mathbb{H}$*. So it is* $M_2(\mathbb{C}) \otimes_{\mathbb{R}} \mathbb{H} \cong M_4(\mathbb{C})$*, of dimension* 32 *and with centre* $\mathrm{span}\{\mathbf{1}, T\}$*, and it meets* $\mathfrak{so}(3,3)$ *in* $\mathfrak{so}(3)_+ \oplus \mathfrak{so}(3)_- \cong \mathfrak{so}(4)$*. The* 32 *blades with an odd number of spatial indices anticommute with* T*.*

(iv) Ad_T *acts on* V *as* η*: on the frame it is parity.*

Proof. By the blade rule (Lemma 1.1) a blade e_I commutes with T iff $3|I| - |I \cap \{1,2,3\}|$ is even, that is iff $|I \cap \{4,5,6\}|$ is even, and anticommutes otherwise. Conjugation by T is diagonal in the blade basis, so its commutant is spanned by the blades it contains, as in Proposition 5.33(ii). This gives (ii) and the first sentence of (iii). A temporal generator commutes with a spatial bivector, so $\mathrm{Cl}(V^+)$ and $\mathrm{Cl}^0(V^-)$ commute, and their products are the 8×4 commuting blades. T is central in $\mathrm{Cl}(V^+)$; e_4e_5, e_5e_6 and their product $-e_4e_6$ square to $-\mathbf{1}$. The isomorphisms $\mathrm{Cl}(3,0) \cong M_2(\mathbb{C})$ and $M_2(\mathbb{C}) \otimes_{\mathbb{R}} \mathbb{H} \cong M_4(\mathbb{C})$ are inherited; equivalently, $T^2 = -\mathbf{1}$ makes $S \cong \mathbb{C}^4$, and the commutant of a complex structure is $\mathfrak{gl}(4,\mathbb{C})$. For (i): on the fifteen basis bivectors, a compact blade is antisymmetric and commutes with T, so $Y^\top T + TY = [T,Y] = 0$, and a boost blade is symmetric and anticommutes with T, so $Y^\top T + TY = \{T,Y\} = 0$. The condition is linear in Y, so it holds on all of $\mathfrak{so}(3,3)$. Then $Y^\top TY = -TYY$, and $\frac{d}{ds}\big[(e^{sY})^\top T e^{sY}\big] = (e^{sY})^\top (Y^\top T + TY) e^{sY} = 0$. ω_S has the same commutation pattern. The dimension two is Theorem 9.62(i) at $n = 1$. (iv) T commutes with e_1, e_2, e_3 and anticommutes with e_4, e_5, e_6. □

Part (i) restates Corollary 7.10 and Proposition 7.1(i) (fixed there), with the symmetric partner ω_S of Section 12.2. Part (iv) restates Proposition 7.5(1). Parts (ii) and (iii) are fixed (e6-1a to e6-1c); that ω_T and ω_S are fixed by all six compact rotations is noted after Theorem 7.7. The two invariances are one statement read twice. Because the compact bivectors are antisymmetric and the boosts symmetric, invariance of the form is commutation with the first and anticommutation with the second. As a form, T breaks nothing in $\mathfrak{so}(3,3)$. As an operator it singles out the maximal compact subalgebra, the rotations of the two triples, and among these it does not distinguish the temporal from the spatial ones. Two things distinguish T from ω_S: as a Grassmann kernel T is antisymmetric and ω_S symmetric, and as an operator $T^2 = -\mathbf{1}$ is a complex structure while $\omega_S^2 = \mathbf{1}$ gives a projector (Theorem 7.7).

9.6.2 The temporal rotations on the Dirac data

The Dirac data of one constituent are the kinetic kernels K^μ, the rest Hamiltonian $H(0) = m\gamma^0 = me_1$ of the real first-order form, and the rest flow $e^{-m\lambda T}$, with $JH(0) = mT$ (Section 2.3.4; Proposition 4.46 and Section 4.5.5).

Proposition 9.53 (What the temporal rotations do to the Dirac data)**.** *Let* $Y = \alpha J + \beta\, e_3e_1 + \gamma\, e_1e_2 \in \mathfrak{so}(3)_+$*.*

(i) Kinetic forms. $Y^\top K^0 + K^0 Y = 0$ *and* $Y^\top K^a + K^a Y = 2\,Y(\gamma^0)\,\gamma^a$*, where* $Y(\gamma^0) = \beta e_3 - \gamma e_2$ *is the velocity of the clock. Hence* $\sum_a \|Y^\top K^a + K^a Y\|^2 = 96(\beta^2 + \gamma^2)$ *(Frobenius norm), and* Y *maps the kinetic forms into their span,* $Y^\top K^\mu + K^\mu Y = \Lambda^\mu{}_\nu K^\nu$*, iff* $\beta = \gamma = 0$*:* $\mathfrak{so}(3)_+ \cap (\mathfrak{l}_{\mathrm{int}} \oplus \mathbb{R}J) = \mathbb{R}J$*.*

(ii) Rest Hamiltonian. $e^{\varphi e_1e_2} e_1 e^{-\varphi e_1e_2} = \cos 2\varphi\, e_1 - \sin 2\varphi\, e_2$*, and likewise with* e_1e_3 *and* e_3*. The stabiliser of* $H(0)$ *in* $\mathfrak{so}(3)_+$ *is* $\mathbb{R}J$*. A rotor of* $\mathrm{Spin}(V^+)$ *that carries* e_1 *to a clock* $\hat{n}$ *carries* J *to* $J(\hat{n}) = T\hat{n}$*,* K^a *to* $-\hat{n}\gamma^a$ *and* $H(0)$ *to* mn*, and it fixes* $K^0 = -\mathbf{1}$ *and* T*. Moreover* $\hat{n}J(\hat{n}) = J(\hat{n})\hat{n} = T$*.*

(iii) Rest flow. $e^{-m\lambda T}$ *commutes with* $\mathrm{Spin}(3)_+ \times \mathrm{Spin}(3)_-$*. The spin lift* $\frac{1}{2}e_je_k$ *of* $\mathfrak{so}(3)_+$ *has Casimir* $-\frac{3}{4}\mathbf{1}$ *on* S*, so* S *carries isospin* $\frac{1}{2}$*, and* $e^{\pi e_je_k} = -\mathbf{1}$*. The clock rotations* e_1e_2 *and* e_3e_1 *anticommute with* J *and commute with* T*. Their half turn* $\mathrm{Ad}_{e_1e_2} = \mathrm{Ad}_{e_1}\mathrm{Ad}_{e_2}$ *maps* (e_1, J, T, γ^a) *to* $(-e_1, -J, T, \gamma^a)$*.*

(iv) Field octants. *With the field-octant map $\Phi\colon V \to S$ of Theorem 3.44, $-T\Phi = \Phi G$, so $e^{-m\lambda T}\Phi = \Phi\, e^{m\lambda G}$: on the six field octants the Dirac rest flow is the duality rotation at angular frequency m.*

Proof. (i) Y is antisymmetric and commutes with each γ^a (a temporal bivector and a spatial generator), so $Y^\top K^a + K^a Y = [K^a, Y] = -[e_1, Y]\gamma^a = 2Y(e_1)\gamma^a$, and $K^0 = -\mathbf{1}$. With $v = Y(e_1)$ temporal, $(v\gamma^a)^\top(v\gamma^a) = |v|^2\mathbf{1}$, so each term has squared norm $4{\cdot}8|v|^2$. The covariance statement intersects the seven-dimensional solution space $\mathfrak{l}_{\text{int}} \oplus \mathbb{R}J$ of Proposition 7.1(ii) with $\mathfrak{so}(3)_+$; it is also solved directly (e6-2a). (ii) e_1 anticommutes with e_1e_2, so $e^{\varphi e_1e_2}e_1e^{-\varphi e_1e_2} = e^{2\varphi e_1e_2}e_1$. The stabiliser condition is $Y(e_1) = 0$. A rotor R of $\mathrm{Spin}(V^+)$ commutes with T and with every γ^a, and $J = Te_1$, so $\mathrm{Ad}_R J = T\hat{n}$ and $\mathrm{Ad}_R K^a = -\hat{n}\gamma^a$. Since $\hat{n}$ commutes with T and $\hat{n}^2 = \mathbf{1}$, $\hat{n}J(\hat{n}) = J(\hat{n})\hat{n} = T$. (iii) T commutes with the compact bivectors by Proposition 9.52(ii); the Casimir is $\frac{1}{4}\sum Y_j^2$ with $Y_j^2 = -\mathbf{1}$; the rest is the blade rule. (iv) Iterate $-T\Phi = \Phi G$. □

Part (i) is fixed (e6-2a); its covariance half restates Proposition 7.1(ii). Part (ii) is fixed (e6-2b, exact at $\cos\varphi = \frac{3}{5}$ and at the clock $\hat{n} = -\frac{7}{25}e_1 - \frac{24}{25}e_2$, where (i) holds with $Y(\hat{n})$ in place of $Y(\gamma^0)$). Its transport of the data restates Proposition 2.13 and Theorem 2.14, by which the Dirac mass form $\gamma^0 J(\hat{n}) = T$ is the same for every clock while $J(\hat{n})$ turns with $\hat{n}$. Part (iii) is fixed (e6-2c). Part (iv) restates Proposition 3.49(i), re-verified with Φ built from the Fock vacuum (e6-2c).

So the temporal $SU(2)$ is a symmetry of the mass form and of the rest flow that the kinetic term breaks to $\mathfrak{u}(1)_J$. That reverses the usual order, kinetic term symmetric and mass breaking, as Section 7.1.1 found for $\mathrm{Spin}(3,3)$. The broken directions are the two clock rotations: the kinetic term pairs one temporal direction with the spatial triple, and the breaking is the speed $|Y(\hat{n})|$ of the clock. They flip the charge, and their half turn acts as CP (Ad_{e_2} is the charge conjugation of Theorem 2.25, Ad_{e_1} the parity), reversing the sense of the rest rotation as in Proposition 7.3(iii). Interpretively, the temporal $SU(2)$ joins particle and antiparticle through the sphere of clocks at fixed mass form.

Proposition 9.54 (The mass symmetry and the kinetic symmetry)**.** *Let $P_\pm = \frac{1}{2}(\mathbf{1} \pm \omega_S)$.*

(i) $\mathfrak{so}(3)_+ \cap \mathfrak{g}_{\text{massless}} = \mathbb{R}J$.

(ii) $e_2\Omega = -(e_1e_2)\,\omega_S$ *and* $e_3\Omega = (e_3e_1)\,\omega_S$. ω_S *commutes with* $\mathfrak{so}(3)_+$, *and* $P_\pm$ *are orthogonal projectors of rank four.*

(iii) $\mathfrak{so}(3)_+ = \{YP_+ + YP_-\}$ *and* $\mathfrak{su}(2)_{\text{PG}} = \{YP_+ + (JYJ^{-1})P_-\}$, $Y \in \mathfrak{so}(3)_+$: *both are graphs of automorphisms of* $\mathfrak{so}(3)$ *over the two eigenspaces of* ω_S, *the identity and the half turn* Ad_J *about* J. *The Lie algebra they generate is* $\mathfrak{so}(3)_+P_+ \oplus \mathfrak{so}(3)_+P_- \cong \mathfrak{so}(4)$, *and they meet in the fixed algebra of* Ad_J, *which is* $\mathbb{R}J$.

(iv) $g = P_+ + JP_- = \exp(\frac{\pi}{2}JP_-)$ *is orthogonal and* $g\,\mathfrak{so}(3)_+\,g^{-1} = \mathfrak{su}(2)_{\text{PG}}$; $2g = \mathbf{1} + J + \omega_S - J\omega_S$ *has components of grades* $0, 2, 3, 5$, *so* $g \notin \mathrm{Pin}(3,3)$.

(v) $\mathfrak{so}(3)_+$ *preserves* T *and the kinetic term breaks it to* $\mathbb{R}J$; $\mathfrak{su}(2)_{\text{PG}}$ *preserves the kinetic forms and the mass breaks it to* $\mathbb{R}J$.

Proof. (ii) $e_2\Omega = e_2e_1e_4e_5e_6 = -e_1e_2\,\omega_S$, and similarly for e_3. ω_S is disjoint from each temporal bivector, and the grades 3 and 2 give a commuting sign. (iii) Substitute $\omega_S = P_+ - P_-$. J commutes with J and anticommutes with e_1e_2 and e_3e_1, so Ad_J is the half turn about J. A bracket of two graphs of automorphisms ϕ_1, ϕ_2 contains $[Y, Y']P_+ + [\phi_1 Y, \phi_2 Y']P_-$, and the generated algebra is computed (e6-2d). (iv) $(JP_-)^2 = -P_-$ and $gYg^{-1} = YP_+ + JYJ^{-1}P_-$, because $P_\pm$ commute with Y and J. (i) follows from (iii), and (v) is Proposition 9.53(i) together with Proposition 7.2(i)–(ii). □

This is fixed (e6-2d); (v) restates Proposition 2.29 and Proposition 7.2. The comparison has a sharp answer. The two $\mathfrak{su}(2)$'s are not the same, and neither contains the other. They are the untwisted and the Ad_J-twisted diagonals of $\mathfrak{so}(3)_+P_+ \oplus \mathfrak{so}(3)_+P_-$, and they share the Cartan direction J. One is the symmetry of the mass that the kinetic term breaks, the other the symmetry of the kinetic term that the mass breaks. Both leave the same $\mathfrak{u}(1)_J$ [117, 61]. The orthogonal map that exchanges them is not in the Clifford group.

9.6.3 A temporal b-field and the binding root

Proposition 9.55 (A b-field on the temporal triple)**.** *Let $b = (R, 0)$ on $V^+ \oplus V^-$, with $R = [\mathbf{r}]_\times \neq 0$, $\omega = |\mathbf{r}|$, $\kappa > 0$ and $S = b + \kappa G$.*

(i) $b = b_l + b_p$ with $b_l = \frac{1}{2}(R, R) \in \mathfrak{l}_G$, a field-strength rotation, and $b_p = \frac{1}{2}(R, -R) \in \mathfrak{p}_G$, a counter-rotation; they commute. In the notation of equation (9.4), $\mathbf{a} = \mathbf{c} = \mathbf{r}/2$, so $\mathbf{a} \cdot \mathbf{c} = \omega^2/4$.

(ii) $e^{\theta b} G e^{-\theta b} = G_{C(\theta)}$ with $C(\theta) = e^{-\theta R}$. $G_C^2 = -\mathbf{1}$ for every C, and G_C is η-self-adjoint iff C is orthogonal; so $\kappa G_{C(\theta)}$ stays in the self-adjoint summand and the rates, $\pm i\kappa$, are unchanged. $[b, G] = 2b_p G \neq 0$, so b is not removable along a worldline.

(iii) $\det(\lambda - S) = (\lambda^2 + \kappa^2)(\lambda^4 + (2\kappa^2 + \omega^2)\lambda^2 + \kappa^4)$. The lines are κ, on the plane $\langle \hat{\mathbf{r}}, G\hat{\mathbf{r}} \rangle$, and $s \pm \omega/2$, with $s = (\kappa^2 + \omega^2/4)^{1/2}$ and $(s + \omega/2)(s - \omega/2) = \kappa^2$; at $(\kappa, \omega) = (2, 3)$ they are $1, 2, 4$.

(iv) Not every solution is bounded. At $(\kappa, \omega) = (2, 3)$ and $(6, 5)$, M has Jordan blocks exactly at the frequencies $s \pm \omega/2$ and is semisimple at κ.

(v) On spinors $e_j e_k = \frac{1}{2}(e_j e_k - e_{j+3} e_{k+3}) + \frac{1}{2}(e_j e_k + e_{j+3} e_{k+3})$ for temporal j, k. The first summand, a paired rotation, commutes with N; the second changes N by ± 2 only.

Proof. (i) G commutes with $\mathrm{diag}(R, R)$ and anticommutes with $\mathrm{diag}(R, -R)$ in the $(\mathbf{B}, \mathbf{E})$ blocks. (ii) $e^{\theta b} = \mathrm{diag}(e^{\theta R}, \mathbf{1})$, and $\mathrm{diag}(Q, \mathbf{1})\, G\, \mathrm{diag}(Q^{-1}, \mathbf{1}) = G_{Q^{-1}}$. $G_C^\dagger = \eta G_C^\top \eta$ has blocks $-C^\top$ and $C^{-\top}$, which equal those of G_C iff $C^\top = C^{-1}$. The rates are the spectrum of the self-adjoint part, a tensor (Theorem 9.21(iii)). Removability fails by Proposition 9.8. (iii) b_l commutes with b_p and with G. By Larmor's theorem (Proposition 9.5) the substitution $X = e^{-b_l \tau} Y$ removes b_l and shifts the exponents by the eigenvalues $0, \pm i\omega/2$ of $-b_l$. On the support of b_p the remaining shift $b_p + \kappa G$ has the frequency $(\kappa^2 + \omega^2/4)^{1/2}$ of Theorem 9.7(ii), and on its kernel $\langle \hat{\mathbf{r}}, G\hat{\mathbf{r}} \rangle$ the frequency κ. The characteristic polynomial is also computed symbolically in $\mathbf{r}$. (iv) By Theorem 9.9(i), boundedness needs $\mathbf{a} \cdot \mathbf{c} = 0$. The Jordan structure is that of $b_p + \kappa G$ (Theorem 9.7(ii)), transported by the bounded frame $e^{-b_l \tau}$, and is computed exactly. (v) With $e = a^+ + a^-$ and $e' = a^+ - a^-$, the sum is $2(a_j^+ a_k^+ + a_j^- a_k^-)$ and the difference $2(a_j^+ a_k^- + a_j^- a_k^+)$. □

This is fixed (e6-3a to e6-3e); the statement on the rates restates Theorem 9.21(iii). A b-field on the temporal triple alone is therefore half binding-compatible and half binding-odd. Its paired half is the compact part of $\mathfrak{l}_G$, a Zeeman term removable by a rotating frame [83]. Its counter half lies in $\mathfrak{p}_G$. The latter gives the gauge-covariant remainder $D_\tau G = [b, G] = 2b_p G$, it moves the binding root to the root $G_{e^{-\theta R}}$ of a rotated conversion, and it tears the binding, with the screened frequency s and Jordan blocks. The rate $\kappa = c/\ell$ is never touched. In Proposition 9.55(iii) the two halves act independently, because $\mathbf{a} \parallel \mathbf{c}$: the lines are the screened frequency $\sqrt{\kappa^2 + |\mathbf{c}|^2}$ shifted by the Zeeman amounts $\pm|\mathbf{a}|$. Theorem 9.9 needed $\mathbf{a} \perp \mathbf{c}$ for screening; a field on one triple has the opposite geometry. On spinors the same split is the split of Proposition 5.38(ii): the paired half hops quanta within a constituent and the counter half creates or annihilates them in pairs, so the temporal rotations, J included, do not conserve N (Proposition 5.40(ii); Lemma 5.46).

9.6.4 c on the spatial triple: the polar form of the conversion

Proposition 9.56 (Conversions and the temporal rotations)**.** *Let $C\colon V^+ \to V^-$ be invertible, $z_C = C\mathbf{B} + i\mathbf{E}$ and $q_C = z_C \cdot z_C$.*

(i) $G_C^2 = -\mathbf{1}$, G_C acts on z_C as multiplication by i, and $e^{\theta G_C}$ preserves E_C, which depends on C only through $C^\top C$.

(ii) $G_C^\top N_c + N_c G_C = \left(\begin{smallmatrix} 0 & C^\top - c^2 C^{-1} \\ C - c^2 C^{-\top} & 0 \end{smallmatrix}\right)$, so G_C preserves $E_c = \frac{1}{2}(c^2|\mathbf{B}|^2 + |\mathbf{E}|^2)$ iff $C^\top C = c^2\mathbf{1}$, that is iff $C = cR$ with $R \in O(3)$.

(iii) For $C = cR$, the elements of $\mathfrak{so}(3) \oplus \mathfrak{so}(3)$ commuting with G_C are the twisted diagonal rotations (Q, RQR^{-1}). $G_{cR} = \mathrm{Ad}_{(R^{-1},\mathbf{1})} G_c = \mathrm{Ad}_{(\mathbf{1},R)} G_c$ for $R \in SO(3)$, and $\mathrm{Ad}_{(R_+,R_-)} G_c = G_{cR_-R_+^{-1}}$ for $(R_+, R_-) \in O(3)\times O(3)$. The Lie algebra preserving η and E_c is $\mathfrak{so}(3)\oplus\mathfrak{so}(3)$, and adding $[Y, G_c] = 0$ leaves the diagonal $\mathfrak{so}(3)$. Hence $\{G_C : C^\top C = c^2\mathbf{1}\}$ is one orbit of $O(3) \times O(3)$, of dimension three.

(iv) A temporal rotation $(R_+, \mathbf{1})$ carries G_C to $G_{CR_+^{-1}}$ and N_C to $N_{CR_+^{-1}}$. It conjugates $C^\top C$ to $R_+ C^\top C R_+^{-1}$ and fixes $CC^\top$; in the polar form $C = (CC^\top)^{1/2}U$ it moves only U. A spatial rotation $(\mathbf{1}, R_-)$ carries C to R_-C, fixes $C^\top C$ and moves only U' in $C = U'(C^\top C)^{1/2}$.

(v) For $C = cR$: E_c and $\mathrm{Re}\, q_C = c^2|\mathbf{B}|^2 - |\mathbf{E}|^2$ do not depend on R, while $\mathrm{Im}\, q_C = 2c\,(R\mathbf{B})\cdot\mathbf{E}$ does. At $c = 1$, $\mathbf{B} = e_1$, $\mathbf{E} = e_2$ and R the rotation through θ about e_3, $q_C = 2i\sin\theta$: the same field point is null for $\theta = 0$ and at rest for $\theta = \frac{\pi}{2}$, where $E_0 = E_6 = 1$.

Proof. (i) G_C^2 has blocks $-C^{-1}C$ and $-CC^{-1}$. On z_C, $G_C X$ has $C\mathbf{B}' = -\mathbf{E}$ and $\mathbf{E}' = C\mathbf{B}$, so $z_C(G_C X) = i z_C(X)$, and $|z_C|^2 = 2E_C$. (ii) is a block computation. Its vanishing means $C^\top C = c^2\mathbf{1}$, and then C/c is orthogonal. (iii) The lower-left block of $[G_C, \mathrm{diag}(Q, Q')]$ is $CQ - Q'C$. The conjugation formulas are block products. An η-isometry preserving the positive form N_c preserves the eigenspaces $V^\pm$ of $\eta^{-1}N_c = \mathrm{diag}(c^2, -1)$, and is orthogonal on each. The stabiliser dimensions are exact null spaces (e6-4c). (iv) $(CR_+^{-1})(CR_+^{-1})^\top = CC^\top$. (v) Expand $z_C \cdot z_C$; in the example $R\mathbf{B} = (\cos\theta, \sin\theta, 0)$. □

This is fixed (e6-4a to e6-4e). Parts (i)–(iii) extend Proposition 3.98 and Theorem 2.2 from $C = c\,\mathbf{1}$ to the orbit. They are elementary linear algebra: Schur's lemma applied to a twisted diagonal action gives $C = cR$. Part (v) is the passive form of Proposition 9.18(2), where the anti-diagonal part of a compact transport turns a null field into one at rest at fixed E_6.

Remark 9.57 (What "c^2 on $(-,-,-)$, b on $(+,+,+)$" fixes)**.** (a) *Fixed.* The complete invariant of a conversion under the temporal rotations is $CC^\top$, a symmetric form on the spatial triple: C and C' lie in one orbit of the orthogonal group of the temporal triple iff $C'C'^\top = CC^\top$ (Proposition 9.56(iv); e6-4f). A b-field on the temporal triple, a flow in $SO(3)$, can change anything about the conversion except this and the sign of $\det C$ (e6-4f). The paper's c is the isotropic case $CC^\top = c^2\mathbf{1}$, which is also $C^\top C = c^2\mathbf{1}$, and that is exactly the case in which the binding root preserves the weighted energy (Proposition 9.56(ii)). In this sense, which is the reading of (c), c^2 sits on the spatial triple and the temporal b-field cannot reach it. The energy E_C sees the other symmetric factor $C^\top C$, a form on the temporal triple that the temporal rotations conjugate (Proposition 9.56(i), (iv)). After the two triples are identified, the two factors coincide exactly when C is normal, $CC^\top = C^\top C$, for instance $C = \mathrm{diag}(2, 3, 4)$: isotropy is sufficient, not necessary. What the temporal b-field moves is the orthogonal factor R. The energy E_c does not see R; the binding root G_{cR} (which spatial direction is paired with which temporal one), the pseudoscalar $\mathrm{Im}\, q_C$ and the rest energy $\frac{1}{2}|q_C|$ do.

(b) *Fixed: a mirror.* With the triples exchanged the statement still holds: a spatial b-field keeps $C^\top C$ on the temporal triple and moves the other orthogonal factor. The conversion alone therefore does not decide which triple carries c^2 and which the b-field. The Dirac data decide it, through the clock in V^+ and the kinetic term (Proposition 9.53): in the interior reading $\mathfrak{so}(3)_- \subset \mathfrak{l}_{\mathrm{int}}$ is part of the Lorentz algebra, while $\mathfrak{so}(3)_+$ meets $\mathfrak{l}_{\mathrm{int}} \oplus \mathbb{R}J$ only in J. The frame reading does not decide it, since its Lorentz algebra $\mathfrak{l}_G$ contains only the diagonal rotations.

(c) *Relation to the clause "c^2 is the symmetric part of $E = mc^2$".* Section 3.9.6 gives the clause its exact forms: c is the unique isotropic conversion, its weight in E_6 a unit convention (Propositions 3.98 and 3.99), and the shift splits as $\mathrm{End}(V) = \mathfrak{so}(3,3) \oplus \{Y^\dagger = Y\}$ with κG in the self-adjoint summand, $(\kappa G)^2 = -c^2/\ell^2$, and the connections in $\mathfrak{so}(3,3)$ (Section 9.1). The polar decomposition of the conversion is a further, compatible form: its symmetric factor carries c^2 and is invariant under the temporal b-field, its orthogonal factor is moved by it, and for every orthogonal C, κG_C stays in the self-adjoint summand with the same spectrum (Proposition 9.55(ii)). It adds nothing else: it does not fix c, does not make c a symmetry (the stabiliser $O(3)\times O(3)$ of η and E_c is not a Lorentz group), and selects no rest energy. A constant R is a choice of identification of the triples (Theorem 2.2 selects $R = \mathbf{1}$ once a common $SO(3)$ is fixed); only a τ-dependent $R(\tau)$, a temporal b-field, has gauge-invariant content, its $D_\tau G \neq 0$. Reading the symmetric factor as "c^2" and the orthogonal factor as the place of the b-field is interpretive.

9.6.5 The six-momentum read by a clock

Let $P = \sum_i (\mathsf{E}_i/c) e_i + \sum_a p_a e_{a+3}$ with $\eta(P,P) = M_6^2 c^2$, so $M_6^2 c^4 = |\mathbf{E}|^2 - |\mathbf{p}|^2 c^2$, as in Proposition 3.101. For a clock $\hat{n}$ the energy read is $E(\hat{n}) = \hat{n} \cdot \mathbf{E}$ and $m_4(\hat{n})^2 c^4 = E(\hat{n})^2 - |\mathbf{p}|^2 c^2 = M_6^2 c^4 - |\hat{n} \times \mathbf{E}|^2$ (Proposition 3.31). A temporal b-field of constant rate ν about a unit axis $\hat{\mathbf{e}}$ acts on P by $\mathbf{E}(t) = R_{\hat{\mathbf{e}}}(\nu t)\mathbf{E}$, with $\mathbf{p}$ fixed.

Proposition 9.58 (The four-dimensional mass under a temporal b-field)**.** *(i) The flow keeps M_6 and $|\mathbf{E}|$ constant.*

(ii) For a fixed clock $\hat{n}$, $E(\hat{n},t) = A + B\cos(\nu t - \phi)$, where $A = (\hat{n}\cdot\hat{\mathbf{e}})(\hat{\mathbf{e}}\cdot\mathbf{E})$ and $B = |\hat{n}_\perp|\,|\mathbf{E}_\perp|$ with $\hat{n}_\perp$ and $\mathbf{E}_\perp$ the parts orthogonal to $\hat{\mathbf{e}}$. Hence $m_4(\hat{n})^2 c^4 = M_6^2 c^4 - |\mathbf{E}|^2 + E(\hat{n},t)^2$ oscillates. Its maximum is $M_6^2 c^4 - |\mathbf{E}|^2 + (|A|+B)^2 \le M_6^2 c^4$. Its minimum is $M_6^2 c^4 - |\mathbf{E}|^2 + (|A|-B)^2$ if $|A| \ge B$ and $M_6^2 c^4 - |\mathbf{E}|^2$ otherwise. It is constant iff $B = 0$, that is iff $\hat{n} \parallel \hat{\mathbf{e}}$ or $\mathbf{E} \parallel \hat{\mathbf{e}}$.

(iii) The rotation about the clock, whose spin lift is proportional to $J(\hat{n})$ (Proposition 2.13(iii)), keeps $E(\hat{n})$. On the reduced shell $\mathbf{E} \parallel \hat{n}$ of Theorem 12.12 (Proposition 3.28) the flow preserves the constraint iff $\hat{\mathbf{e}} \parallel \hat{n}$.

(iv) With $c = 1$, $\mathbf{E} = (2,3,6)$ and $\mathbf{p} = (0,0,3)$, so that $M_6^2 = 40$: the rotation about e_3 read by e_1 gives $E(\hat{n}) = 2\cos\nu t - 3\sin\nu t$ and $m_4^2 \in [-9,4]$; read by the symmetric clock $\widehat{W} = (e_1+e_2+e_3)/\sqrt{3}$ (Proposition 2.5) it gives $m_4^2 \in [\frac{35}{3} - 4\sqrt{26}, \frac{35}{3} + 4\sqrt{26}]$. The rotation about $\widehat{W}$ read by $\widehat{W}$ keeps $m_4^2 = \frac{94}{3}$ while the relative-time energies $\mathbf{E}_\perp$ turn.

Proof. Rodrigues' formula gives $\hat{n}\cdot\mathbf{E}(t) = A + (\hat{n}\cdot\mathbf{E} - A)\cos\nu t + \hat{n}\cdot(\hat{\mathbf{e}}\times\mathbf{E})\sin\nu t$, and the squared amplitude $(\hat{n}\cdot\mathbf{E} - A)^2 + (\hat{n}\cdot(\hat{\mathbf{e}}\times\mathbf{E}))^2$ equals $(|\hat{n}|^2 - (\hat{n}\cdot\hat{\mathbf{e}})^2)(|\mathbf{E}|^2 - (\hat{\mathbf{e}}\cdot\mathbf{E})^2)$ (Lagrange's identity in the plane orthogonal to $\hat{\mathbf{e}}$). $|A| + B \le |\mathbf{E}|$ by the Cauchy–Schwarz inequality. The rest is substitution; the reduced-shell statement says that a vector stays on the line $\mathbb{R}\hat{n}$ under all rotations about $\hat{\mathbf{e}}$ iff the line is the axis. □

This is fixed (e6-5a, symbolic for three rational axes; e6-5b, exact). The reading needs care. "The four-dimensional mass oscillates" holds only if the clock is held fixed while the temporal

b-field turns the temporal momentum. Equivalently, the clock turns against a fixed momentum along the circle of the clock sphere of Proposition 3.31 about the axis $\hat{\mathbf{e}}$ through $\hat{n}$, a great circle only when $\hat{n} \perp \hat{\mathbf{e}}$ (e6-5c). Read in the sense of Remark 3.32, the energy read by the clock is the projection of the undivided temporal energy, and the b-field moves energy periodically between the clock and the relative times. This is the dynamical form of the obstruction of Proposition 3.101: splitting the temporal energy lowers the mass. The six-dimensional mass is constant throughout. The rotation about the clock is the charge $J(\hat{n})$ and changes nothing; the two clock rotations are what make m_4 oscillate, and they are also what the kinetic term breaks (Proposition 9.53). On the reduced shell, the gauge-fixed reading of Section 12.4, a temporal b-field transverse to the clock is not admissible at fixed clock. Two clocks must be kept apart. For a *fixed* clock $\hat{n}$, $m_4(\hat{n})^2c^4 = M_6^2c^4 - |\hat{n}\times\mathbf{E}(t)|^2$ oscillates and is negative whenever $|\hat{n}\times\mathbf{E}| > M_6c^2$, as in (iv), where m_4^2 reaches -9; then the projected four-momentum is spacelike and m_4 is not a rest mass of anything. For a *co-rotating* clock, $\hat{n}(t) = R_{\hat{\mathbf{e}}}(\nu t)\hat{n}$, $|\hat{n}\times\mathbf{E}|$ is constant and so is m_4. Which clock is physical is the open clock-selection question; the oscillation is a statement about a fixed reading, not a derived variation of a particle mass.

9.6.6 Between constituents

Write $\hat{C}_+ = -\frac{1}{2}\sum_{i\le 3} E_{(1,i)}E_{(2,i)}$ and $\hat{C}_- = \frac{1}{2}\sum_{a\ge 4} E_{(1,a)}E_{(2,a)}$ for the temporal and the spatial parts of the copy rotation, $\hat{C} = \hat{C}_+ + \hat{C}_-$, and $C_\times = \hat{C}_+ - \hat{C}_-$. We write $C_\pm$ for $\hat{C}_\pm$.

Proposition 9.59 (Temporal and spatial copy rotations)**.** *(i) $C_\pm$ are antisymmetric and commute, and each is a sum of three commuting terms of square $-\frac{1}{4}\mathbf{1}$. Hence $e^{2\pi C_\pm} = -\mathbf{1}$ and $e^{2\pi\hat{C}} = \mathbf{1}$. The eigenvalues of $C_\pm$ are $\pm\frac{i}{2}$ (24 times each) and $\pm\frac{3i}{2}$ (8 times each).*

(ii) On vectors neither C_+ nor C_- commutes with G_2, $\hat{C}$ does, and $C_\times$ is compact and anticommutes with G_2.

(iii) $C_+ = \frac{1}{2}\hat{C} + \frac{1}{2}C_\times$. $\hat{C}$ conserves N and $C_\times = -\sum_i(a^+_{1,i}a^+_{2,i} + a^-_{1,i}a^-_{2,i})$ changes it by ± 2, one quantum in each constituent. From the Fock vacuum, $e^{\theta C_+}|0\rangle$ has weight $\binom{3}{k}\cos^{2(3-k)}\frac{\theta}{2}\sin^{2k}\frac{\theta}{2}$ in $N = 2k$, none in odd N, and $N^{(1)} = N^{(2)}$ throughout. So $\langle N\rangle = 3(1-\cos\theta)$, and $\theta = \pi$ fills all six modes.

(iv) As a constant field between two binding frames, $S = \kappa G_2 + \beta V_+$, where V_+ is the vector form of C_+, has $\det(\lambda - S) = \left(\lambda^4 + (2\kappa^2+\beta^2)\lambda^2 + \kappa^4\right)^3$. The rates are κ and the lines are $s \pm \beta/2$. At $(\kappa,\beta) = (2,3)$, M has Jordan blocks at both lines.

Proof. (i) $E_{(1,i)}E_{(2,i)}$ squares to $-\mathbf{1}$ for every i, and the terms for distinct i involve four distinct anticommuting generators. Then $e^{2\pi C_\pm} = \prod(\cos\pi \mp \sin\pi\, E_{(1,i)}E_{(2,i)}) = (-1)^3$. The multiplicities follow from the minimal polynomial $(\lambda^2+1)(\lambda^2+9)$ of $2C_\pm$ and $\mathrm{Tr}(2C_\pm)^2 = -192$. (ii) $V_\pm$ maps $e_a^{(1)}$ to $e_a^{(2)}$ on the temporal (respectively spatial) directions, and G_2 exchanges the two kinds; $V_+ - V_- = \varepsilon\otimes\eta$. (iii) As in Proposition 9.55(v): $E_{(1,i)}E_{(2,i)} \pm E_{(1,i+3)}E_{(2,i+3)}$ is a pair term or a hopping term. Since $[\hat{C}, C_\times] = 0$ and $\hat{C}|0\rangle = 0$, $e^{\theta C_+}|0\rangle = e^{\theta C_\times/2}|0\rangle$, a product of three rotations in the planes $\langle|0\rangle, a^+_{1,i}a^+_{2,i}|0\rangle\rangle$; this is checked exactly at $\cos\frac{\theta}{2} = \frac{3}{5}$. (iv) As in Proposition 9.55(iii), with $\varepsilon\otimes\mathbf{1}_3$ in place of $[\hat{\mathbf{r}}]_\times$. □

Part (i) restates Proposition 9.24(i); the rest is fixed (e6-6a, e6-6c, e6-6e). Part (iii) is the compact counterpart of the copy boost of Proposition 5.48. The pairs emerge unitarily, with probability $\binom{3}{k}\cos^{2(3-k)}\frac{\theta}{2}\sin^{2k}\frac{\theta}{2}$ in place of the hyperbolic weights there. As Section 5.5.4 says, compactness does not protect the number of quanta; commutation with the binding root does, and C_+ fails it. Part (iv) puts C_+ outside the binding-compatible class of Theorem 9.21: as a constant field it tears the binding of two manifolds as a temporal b-field tears that of one.

Proposition 9.60 (Mass transfer between constituents)**.** *Let* $H_g = m_1 E_{(1,1)} + m_2 E_{(2,1)}$, $M = (m_1^2 + m_2^2)^{1/2}$, $u_1 = e_1 \otimes \mathbf{1}$ *and* $u_2 = \mathbf{1} \otimes e_1$.

(i) $\mathrm{Ad}(e^{\theta C_+})E_{(1,1)} = \cos\theta\, E_{(1,1)} + \sin\theta\, E_{(2,1)}$ *and* $\mathrm{Ad}(e^{\theta C_+})E_{(2,1)} = \cos\theta\, E_{(2,1)} - \sin\theta\, E_{(1,1)}$. *Hence* $\mathrm{Ad}(e^{\theta C_+})H_g = m_1(\theta)E_{(1,1)} + m_2(\theta)E_{(2,1)}$, *where* $(m_1(\theta), m_2(\theta))$ *is* (m_1, m_2) *rotated through* θ. *The spectrum stays* $\{\pm M\}$, *and* $(M, 0)$ *goes to* $(M\cos\theta, M\sin\theta)$. C_- *commutes with* H_g.

(ii) The ungraded $H_u = m_1 u_1 + m_2 u_2$ *has spectrum* $\{\pm m_1 \pm m_2\}$, *each value* 16 *times, and this multiset determines* $\{|m_1|, |m_2|\}$. *Rotating the initial pair* $(M, 0)$ *inside the ungraded family changes the spectrum unless* $\sin 2\theta = 0$, *from* $\{\pm M\}$ *to* $\{\pm M(\cos\theta + \sin\theta), \pm M(\cos\theta - \sin\theta)\}$. *In general the spectrum is unchanged exactly when the unordered pair* $(|m_1|, |m_2|)$ *is:* $(2, 1)$ *rotated with* $\cos\theta = \frac{4}{5}$ *becomes* $(1, 2)$, *with the same spectrum, although* $\sin 2\theta = \frac{24}{25}$. $(\cos\theta\, u_1 + \sin\theta\, u_2)^2 = \mathbf{1} + \sin 2\theta\, u_1 u_2$, *so for* $\sin 2\theta \neq 0$ *no invertible matrix conjugates* u_1 *into* $\cos\theta\, u_1 + \sin\theta\, u_2$, *whereas* $(\cos\theta\, E_{(1,1)} + \sin\theta\, E_{(2,1)})^2 = \mathbf{1}$. *The product* $u_1 u_2$ *of the two ungraded clocks squares to* $\mathbf{1}$ *and commutes with both. The flip* $P(x \otimes y) = y \otimes x$ *is orthogonal with* $P u_1 P^{-1} = u_2$: *the exchange* $(m_1, m_2) \mapsto (m_2, m_1)$ *keeps the spectrum, it is the only change of* $(|m_1|, |m_2|)$ *that does, and no continuous family at fixed spectrum changes* $(|m_1|, |m_2|)$.

(iii) $\mathrm{Ad}(e^{\theta C_+})$ *keeps the spectrum of* H_u, *but it maps* u_1 *to* $\cos\theta\, u_1 + \sin\theta\, \omega \otimes e_1$. *This lies in* $\mathrm{Cl}(3,3) \otimes \mathbf{1} + \mathbf{1} \otimes \mathrm{Cl}(3,3)$ *only for* $\sin\theta = 0$.

Proof. (i) For $i = 2, 3$, $E_{(k,1)}$ anticommutes with both factors of $E_{(1,i)}E_{(2,i)}$, so these terms commute with $E_{(k,1)}$. The term $X = E_{(1,1)}E_{(2,1)}$ anticommutes with $E_{(1,1)}$ and $E_{(2,1)}$, so $e^{-\theta X/2}E_{(1,1)}e^{\theta X/2} = e^{-\theta X}E_{(1,1)} = \cos\theta\, E_{(1,1)} + \sin\theta\, E_{(2,1)}$. C_- commutes with $E_{(k,1)}$ by the same count. (ii) u_1 and u_2 are commuting involutions whose products have trace zero, so each joint sign occurs on 16 dimensions. The largest and smallest absolute values in the multiset are $|m_1| + |m_2|$ and $||m_1| - |m_2||$. Conjugation preserves the spectrum, and $\{\pm\cos\theta \pm \sin\theta\} \neq \{\pm 1\}$ for $\sin 2\theta \neq 0$. P exchanges the tensor factors, and since the spectrum fixes $\{|m_1|, |m_2|\}$, a continuous family at fixed spectrum keeps $(|m_1|, |m_2|)$. (iii) $\omega \otimes e_1$ is a tensor of two non-identity blades, independent of every $A \otimes \mathbf{1}$ and $\mathbf{1} \otimes B$. □

This is fixed (e6-6b, e6-6d, e6-6f). The checks are exact at $\cos\frac{\theta}{2} = \frac{3}{5}$, where the rotated ungraded spectrum is $\{\pm\frac{17}{25}, \pm\frac{31}{25}\}M$. Part (i) is Proposition 5.34(ii) realised by the temporal copy field. The single bivector $B = E_{(2,1)}E_{(1,1)}$ used there is one of the three terms of $2C_+$, and the other two leave the clocks alone.

Remark 9.61 (Does a second particle acquire mass from the first?)**.** The algebra is fixed. The temporal copy rotation turns the clock $E_{(1,1)}$ of the first constituent into the clock $E_{(2,1)}$ of the second, and from $(M, 0)$ it produces $(M\cos\theta, M\sin\theta)$ at fixed graded mass: a massless second copy acquires mass from the first. Reading this as "a second particle acquires mass from the first through the b-field" needs two further structures. One is a clock per copy, $E_{(k,1)}$ as the γ^0 of copy k. The other is the graded reading, and there Proposition 5.34 applies: the copies admit no simultaneous nonzero-mass Dirac constraints of the graded form, and the rest energy is the single value M. In that reading the process is a rotation of the mass direction of one particle between two internal clock directions, the masses adding in quadrature as in Proposition 5.62(iv). It is not the creation of a second massive particle. In the ungraded reading, where energies add and there are two particles (Proposition 5.34(iii)–(iv)), the continuous transfer is obstructed: no conjugation turns the clock u_1 into $\cos\theta\, u_1 + \sin\theta\, u_2$ with $\sin 2\theta \neq 0$, and the additive spectrum fixes the unordered pair of masses (Proposition 9.60(ii)). The only spectrum-preserving change is the exchange of the two masses by the flip of the tensor factors, a relabelling of the constituents rather than a transfer. The transfer is moreover not binding-compatible, and half of it creates quanta in pairs across the constituents (Proposition 9.59(iii)). So the mass transfer between

constituents is fixed algebra; the reading of it as the emergence of a second massive particle is interpretive in the one-particle reading and, as a continuous process, obstructed in the two-particle reading.

9.6.7 Up the ladder: the mass forms of a rung

Theorem 9.62 (Mass forms of a rung)**.** *At rung n let T_n be the product, in increasing order, of the $3n$ temporal generators $E_{(k,i)}$ ($i \le 3$), S_n that of the $3n$ spatial ones, and Γ_n the volume element of $\mathrm{Cl}(3n,3n)$ (here S_n is the spatial volume, not the spinor module $(\mathbb{R}^8)^{\otimes n}$ of equation (1.2)).*

(i) For every n, $Y^\top T_n + T_n Y = 0 = Y^\top S_n + S_n Y$ for every bivector Y, $T_n S_n = \pm\Gamma_n$, and the forms on $(\mathbb{R}^8)^{\otimes n}$ invariant under $\mathfrak{so}(3n,3n)$ are exactly $\mathrm{span}\{T_n, S_n\}$.

(ii) $T_n^\top = (-1)^{3n(3n-1)/2} T_n$ and $S_n^\top = (-1)^{3n+3n(3n-1)/2} S_n$. For $n \equiv 1,2,3,0 \pmod 4$ the pair (T_n, S_n) is (antisymmetric, symmetric), (antisymmetric, antisymmetric), (symmetric, antisymmetric), (symmetric, symmetric). Hence the Grassmann mass kernel invariant under $\mathfrak{so}(3n,3n)$ is unique iff n is odd. It is T_n for $n \equiv 1$ and S_n for $n \equiv 3 \pmod 4$. For $n \equiv 2$ the kernels form the plane $\mathrm{span}\{T_n, S_n\}$, and for $n \equiv 0$ there is none.

(iii) At $n = 2$, $T_2 = -\omega_S \otimes T$ and $S_2 = -T \otimes \omega_S$. The four products $M_1 \otimes M_2$, $M_k \in \{T, \omega_S\}$, span the forms invariant under $\mathfrak{so}(3,3)^{(1)} \oplus \mathfrak{so}(3,3)^{(2)}$; $T \otimes T$ and $\omega_S \otimes \omega_S$ are symmetric, and $\mathfrak{so}(6,6)$ keeps exactly the antisymmetric ones, T_2 and S_2. The per-copy $T \otimes \mathbf{1}$ is not invariant even under the boosts of copy 2.

(iv) The forms invariant under the temporal rotations $\mathfrak{so}(3n)_+$ are their commutant, of dimension 16, 128 and 1024 at $n = 1,2,3$, with (symmetric, antisymmetric) dimensions $(6,10)$, $(64,64)$ and $(528,496)$. They contain T_n and S_n.

(v) The forms invariant under the diagonal $\mathfrak{so}(3,3)$, the same bivector on every constituent, have dimension 2, 14 and 168 at $n = 1,2,3$, with (symmetric, antisymmetric) dimensions $(1,1)$, $(10,4)$ and $(84,84)$.

Proof. (i) At every rung the temporal generators are symmetric and the spatial ones antisymmetric, $E_A^\top = \eta_{AA} E_A$, since $\omega^\top = \omega$. By the blade rule in $\mathrm{Cl}(3n,3n)$, $T_n E_A T_n^{-1} = (-1)^{3n-1}\eta_{AA} E_A = (-1)^{3n-1} E_A^\top$ for every A, so for $Y = E_A E_B$, $T_n Y T_n^{-1} = E_A^\top E_B^\top = (E_B E_A)^\top = -Y^\top$, which is the invariance. The same holds for S_n with the sign $(-1)^{3n}$. Conversely, if M is invariant, then $Y^\top = -T_n Y T_n^{-1}$ gives $Y(T_n^{-1}M) = (T_n^{-1}M)Y$ for every bivector. The commutant of the bivectors is spanned by the blades it contains. A blade e_I commutes with $E_A E_B$ iff $|I \cap \{A,B\}|$ is even, and this holds for all $A \neq B$ iff $I = \emptyset$ or I is everything. So $M \in T_n\,\mathrm{span}\{\mathbf{1}, \Gamma_n\} = \mathrm{span}\{T_n, S_n\}$. (ii) The transpose of a product of m symmetric (respectively antisymmetric) generators is the reversed product times 1 (respectively $(-1)^m$), and reversal costs $(-1)^{m(m-1)/2}$. (iii) $E_{(2,1)}E_{(2,2)}E_{(2,3)} = \omega^3 \otimes T = \omega \otimes T$ and $T\omega = -\omega_S$. Within each copy the invariant forms are $\mathrm{span}\{T, \omega_S\}$, and blade conjugation is diagonal. (iv) A compact Y is antisymmetric, so invariance is commutation; the dimensions are exact orbit counts. (v) Let $c(\mu)$ be the weight multiplicities of a module for the split Cartan subalgebra spanned by the pole operators $e_i e_{i+3}$, which are diagonal. The multiplicity of the trivial module is $\sum_{w \in W(D_3)} \det(w)\, c(\rho - w\rho)$ [237]. It is evaluated on Sym^2 and Λ^2 of $(\mathbb{R}^8)^{\otimes n}$ with the exact weights; at $n = 2$ a direct exact solution on the 216 entries of opposite weight agrees. □

The theorem is fixed (e6-7a to e6-7e). At $n = 1$, (i) restates Corollary 7.10 and the pairing of Section 12.2, and the antisymmetric count 10 of (iv) restates the kernel space of $\mathfrak{so}(T)$ in Theorem 7.9. The pattern of (ii) is Chevalley's: on the spinors of $\mathbb{R}^{m,m}$ the two invariant forms have symmetries $(-1)^{m(m-1)/2}$ and $(-1)^{m(m+1)/2}$ [201, 214], here with $m = 3n$ (inherited).

What the arena adds is that the two forms are the temporal and the spatial volume of the rung, with the Klein factors of (iii). The uniqueness of the mass in $T = e_1e_2e_3$ is therefore a statement of the first rung. At the three-constituent rung the unique $\mathfrak{so}(9,9)$-invariant Grassmann kernel is the spatial volume S_3, and T_3 is a symmetric (bosonic) form. Under the diagonal action the mass is not unique from $n = 2$ on, and under the temporal rotations alone not even at $n = 1$. At the second rung the antisymmetric invariant kernels pair the temporal volume of one constituent with the spatial volume of the other (iii): no invariant mass kernel belongs to one constituent alone.

The invariant bilinear forms of spinor modules of $\mathrm{Spin}(p,q)$, with their symmetry types by signature, are classified by Alekseevsky and Cortés [312]; (i) and (ii) are that classification in the realisation of the tower.

9.6.8 The two assignments compared

Remark 9.63 (The temporal b-field and the b-field between constituents)**.** (a) *How they relate* (fixed). In Sections 9.1 and 9.3 the b-field is the η-antisymmetric part of the shift $S = \kappa G_n + b$, and κG_n is its η-self-adjoint part, $(\kappa G)^2 = -c^2/\ell^2$; κG maps V^+ to V^- and back, and that is where c enters. Section 9.3 studies the binding-compatible fields, $[b, G_n] = 0$. A b-field on the temporal triple is part of the connection, compact and gauge-admissible (Corollary 1.4), but it is not binding-compatible: half of it lies in $\mathfrak{p}_G$ (Proposition 9.55(i)). It belongs to the compact class of Section 9.1 with $\mathbf{a} = \mathbf{c}$, it transports κG to $\kappa G_{e^{-\theta R}}$ without changing the rates, and it tears the binding (Proposition 9.55). Between constituents the same holds for the temporal copy rotation (Proposition 9.59(iv)), which lies outside the Schrödinger class of Theorem 9.21.

(b) *A non-abelian Yang–Mills* $\mathfrak{su}(2)$ *with Cartan direction* J (fixed; the Cartan choice is the clock's). $\mathfrak{so}(3)_+ \cong \mathfrak{su}(2)$ is compact. It acts on V as isospin one on V^+ and trivially on V^-, and on spinors as isospin one-half (Proposition 9.53(iii)). The algebra fixes no Cartan direction, since all Cartan directions of $\mathfrak{so}(3)_+$ are conjugate (all its elements of a fixed nonzero norm are; a12-2). The kinetic term fixes it to $J = J(e_1)$, the rotation about the clock, which it keeps. The two off-diagonal directions are the clock rotations; they flip the charge, and their half turn acts as CP.

(c) *A mass term for the off-diagonal components* (identity fixed; reading interpretive; mechanism open). The breaking of the kinetic forms by $Y \in \mathfrak{so}(3)_+$ is $Y^\top K^a + K^a Y = 2Y(\hat{n})\gamma^a$, with total weight $96|Y(\hat{n})|^2$ (Proposition 9.53(i)). This is a quadratic form on $\mathfrak{so}(3)_+$ with kernel $\mathbb{R}J(\hat{n})$, invariant under the stabiliser of the clock: an invariant statement of the breaking. It has the form of the adjoint-Higgs mass term $g^2|[b, \langle\Phi\rangle]|^2$ of the Georgi–Glashow $SO(3)$ model [233], with the clock in the role of the Higgs vector [234, 235, 236]: the two charge-flipping components would be massive and $J(\hat{n})$ massless. That needs a clock field $\hat{n}(x)$ with a covariant kinetic term, and the arena supplies none: a dynamics that selects the clock is open (Section 3.4). Gauge invariance forbids a bare mass term (inherited), and the Dirac mass gives none, since $\mathfrak{so}(3)_+$ preserves T (Proposition 9.52). In Remark 9.26(e) the rate matrix played the role of the adjoint vacuum value; here the clock does. The pattern is that of an adjoint $SO(3) \to U(1)$ breaking, not of the electroweak $SU(2)_L \times U(1)_Y \to U(1)_{\mathrm{em}}$ by a doublet, and nothing here derives an electroweak Higgs mechanism.

(d) *What the assignment fixes, constructs, obstructs and leaves open.* Fixed: every finite-dimensional statement of Sections 9.6.1 to 9.6.7. Constructed: the clock, which fixes the Cartan direction and the kinetic term, and the temporal b-field itself. Obstructed: its removal along a worldline, bounded binding under it, continuous mass transfer at fixed

spectrum in the ungraded reading, and a unique Grassmann mass kernel at even rungs. Open: a dynamics for the clock, hence a mass for the off-diagonal components.

(e) *What it implies for the emergence of a second particle* (interpretive, drawn from the fixed statements). The temporal b-field gives the second rung a process that moves mass from one constituent to the other, but only as a rotation of one mass vector in the graded reading, in which the rung carries one particle. In the reading that has two particles it is obstructed as a continuous process; only the exchange of the two masses, a relabelling, keeps the spectrum. The uniqueness of the mass in T, which the assignment invokes, does not survive the ladder in a form that assigns a mass to each particle: the invariant kernels of the second rung pair the temporal volume of one constituent with the spatial volume of the other.

9.6.9 Answer

The assignment "c on $(-,-,-)$, b on $(+,+,+)$" has a precise and correct form. The temporal rotations cannot change the symmetric factor $CC^{\top}$ of the conversion, which lives on the spatial triple and is c^2 in the isotropic case; they keep the mass form T and the rest flow, and they move the orthogonal factor of the conversion, the clock and the binding root. The price is the binding and the kinetic term: a temporal b-field keeps the rates, splits the lines, cannot be gauged away and tears the binding, and the kinetic term breaks it to $\mathfrak{u}(1)_J$ in proportion to the speed of the clock. The mass is unique in $T = e_1e_2e_3$ on the first rung only; up the ladder the unique invariant Grassmann kernel is the temporal volume T_n for $n \equiv 1$ and the spatial volume S_n for $n \equiv 3 \pmod 4$, and at even rungs there is a plane of them or none. Between two constituents the temporal b-field rotates graded mass from one clock to the other at fixed total, a statement about one particle, not two.

9.7 The ladder: a table, and what changes along it

The two subsections below collect, rung by rung, what Chapter 5 and this chapter establish about the tower: the table of the ladder, and what changes, and what does not, in going from n constituents to $n+1$.

9.7.1 The ladder table

Table 9.8 collects the rung-by-rung data. The Lie algebra rows are fixed (e3-6a, exact over $\mathbb{Q}$ at $n \le 5$) and restate Theorem 5.1, Theorem 5.15 and Proposition 5.72; the row of copy-invariant four-vectors restates Theorem 5.73, with the case $n = 3$ and the absence at $n = 2$ re-verified (e3-6b). Read as a whole, the table says which structures single out which rungs.

No rung is singled out by the algebra. $\mathrm{Cl}(3n, 3n) = M_{8^n}(\mathbb{R})$, its module is S_n, and every Lie algebra row is a polynomial in n. The first structure with a period in n is the symmetry of the invariant bilinear forms on S_n, of period four (Theorem 9.62(ii)). It is Chevalley's pattern for spinors of $\mathbb{R}^{3n,3n}$, the dimension $6n$ running through the residues $6, 4, 2, 0$ modulo 8 with period four in n (inherited [201]). Through it the $\mathfrak{so}(3n, 3n)$-invariant Grassmann mass kernel is unique exactly at odd n. That is the only selection the algebra makes by itself, and it does not distinguish $n = 3$ from the other odd rungs.

Odd rungs are singled out by the interior Lorentz action. Under the diagonal $\mathfrak{l}_{\mathrm{int}}$, odd n is where the composite is a fermion (Proposition 5.66(i)), where an imaginary unit is selected (Proposition 5.36(iv)), where copy-invariant four-vectors and the Dirac block exist (Theorem 5.73 and Proposition 5.17). *Even rungs are singled out by the binding action*: only there does a Lorentz-invariant number of quanta exist under the diagonal $\mathfrak{l}_G$ (Theorem 5.50). *The third rung is singled out by added projectors and by the low-rank coincidences of the diagonal*

statement	class	checks
conventions; the triples, T, ω_S, Fock modes	restated ((1.1), (1.2), Theorem 7.7)	e6-0
Proposition 9.52(i): T and ω_S invariant forms under $\mathrm{Spin}(3,3)$	restated (Corollary 7.10, Section 12.2; Proposition 7.1)	e6-1a
Proposition 9.52(ii), (iii): commutes with $\mathfrak{so}(3)_+ \oplus \mathfrak{so}(3)_-$ only; commutant $M_4(\mathbb{C})$, 32 blades	fixed ($M_4(\mathbb{C})$ inherited)	e6-1b
Proposition 9.52(iv): $\mathrm{Ad}_T = \eta$; boosts conjugate T to $e^{2sY}T$	restated (Proposition 7.5); fixed	e6-1c
Proposition 9.53(i): kinetic breaking $2Y(\hat{n})\gamma^a$, weight $96\lvert Y(\hat{n})\rvert^2$; $\mathfrak{so}(3)_+ \cap (\mathfrak{l}_{\mathrm{int}} \oplus \mathbb{R}J) = \mathbb{R}J$	fixed (covariance restated, Proposition 7.1(ii))	e6-2a
Proposition 9.53(ii): clock rotation of $H(0)$; transport to the clock $\hat{n}$	fixed (transport restated, Proposition 2.13, Theorem 2.14)	e6-2b
Proposition 9.53(iii), (iv): rest flow kept; isospin $\frac{1}{2}$; CP half turn; octant duality	fixed; (iv) restated (Proposition 3.49)	e6-2c
Proposition 9.54: $\mathfrak{so}(3)_+$ and $\mathfrak{su}(2)_{\mathrm{PG}}$ two diagonals of $\mathfrak{so}(4)$, meeting in $\mathbb{R}J$	fixed ((v) restated, Proposition 2.29)	e6-2d
Proposition 9.55(i), (ii): half $\mathfrak{l}_G$, half $\mathfrak{p}_G$; $G \to G_{e^{-\theta R}}$; rates kept; not removable	fixed (rates restated, Theorem 9.21(iii))	e6-3a, e6-3b
Proposition 9.55(iii), (iv): lines κ, $s \pm \omega/2$; Jordan blocks	fixed	e6-3c, e6-3d
Proposition 9.55(v): paired half conserves N, counter half changes it by ± 2	fixed	e6-3e
Proposition 9.56: $C = cR$; twisted diagonal; orbit $O(3)$; $CC^\top$ invariant; q_C	fixed	e6-4a to e6-4e
Remark 9.57: "c^2 on $(-,-,-)$, b on $(+,+,+)$"	(a), (b) fixed; reading interpretive	e6-4d, e6-4f
Proposition 9.58: m_4 oscillates at fixed μ; constant iff on the axis	fixed; the reading interpretive	e6-5a to e6-5c
Proposition 9.59: $C_\pm$, $e^{2\pi C_\pm} = -\mathbf{1}$; pairs of quanta; tearing between frames	fixed ((i) restated, Proposition 9.24(i))	e6-6a, e6-6c, e6-6e
Proposition 9.60: graded mass transfer at fixed M; ungraded obstruction	fixed	e6-6b, e6-6d, e6-6f
Remark 9.61: "a second particle acquires mass"	interpretive (one-particle reading); obstructed as a continuous process (two-particle reading)	—
Theorem 9.62(i), (ii): invariant forms $\mathrm{span}\{T_n, S_n\}$; period four	fixed (Chevalley's pattern inherited)	e6-7a, e6-7b
Theorem 9.62(iii)–(v): Klein factors; temporal and diagonal counts	fixed	e6-7a, e6-7c to e6-7e
Remark 9.63: relation to RE; Yang–Mills $\mathfrak{su}(2)$; clock as adjoint Higgs	(a), (b) fixed; (c) identity fixed, reading interpretive, mechanism open; (e) interpretive	e6-2a

Table 9.7: Claims of Section 9.6, their classes and their checks, which are `PASS` lines of `e6_bfield_triples.py`. "Restated": fixed in an earlier part of the manuscript, re-verified here.

structure	$n=1$	2	3	4	5	source
$\dim V_n$, $\dim S_n$	6, 8	12, 64	18, 512	24, 8^4	30, 8^5	—
$\mathrm{Cl}(3n,3n)$		$M_{8^n}(\mathbb{R})$ at every rung				e3-1a ($n \le 4$)
Fock modes	3	6	9	12	15	e3-1c
spectator centraliser $\mathfrak{so}(3n-1,3n-3)$	1	28	91	190	325	Theorem 5.1; e3-6a
commutant of diagonal $\mathfrak{l}_{\mathrm{int}}$, $\mathfrak{so}(n)\oplus\mathfrak{so}(2n)$	1	7	18	34	55	Theorem 5.15; e3-6a
commutant of diagonal $\mathfrak{l}_G$, $\mathfrak{so}(n,\mathbb{C})$	0	2	6	12	20	Proposition 5.72; e3-6a
its compact part, $\mathfrak{so}(n)$	0	1	3	6	10	e3-6a
rotor algebra $\mathfrak{u}(n)$ (in $\mathfrak{so}(6n)$)	1	4	9	16	25	Proposition 5.72; e3-6a
copy algebra, commutant of diagonal $\mathfrak{so}(3,3)$	0	1	3	6	10	e3-4a
$C_{\mathrm{cop}}+C_{\mathrm{diag}}$	$\frac{15}{4}$	9	$\frac{63}{4}$	24	$\frac{135}{4}$	e3-4c
graded rest mass	m	$\sqrt{2}\,m$	$\sqrt{3}\,m$	$2m$	$\sqrt{5}\,m$	e3-2a
ungraded rest energies $\lvert m(n-2j)\rvert$	m	$0, 2m$	$m, 3m$	$0, 2m, 4m$	$m, 3m, 5m$	e3-2b
2π rotation, diagonal $\mathfrak{l}_{\mathrm{int}}$	$\mathbf{-1}$	$\mathbf{+1}$	$\mathbf{-1}$	$\mathbf{+1}$	$\mathbf{-1}$	e3-3a
2π rotation, diagonal $\mathfrak{l}_G$	$\mathbf{+1}$	$\mathbf{+1}$	$\mathbf{+1}$	$\mathbf{+1}$	$\mathbf{+1}$	e3-3b
copy-invariant four-vector	yes	no	yes	no	yes	Theorem 5.73; e3-6b
selected imaginary unit (diagonal $\mathfrak{l}_{\mathrm{int}}$)	yes	no	yes	no	yes	Proposition 5.36(iv)
invariant number of quanta (diagonal $\mathfrak{l}_G$)	no	yes	no	yes	no	Theorem 5.50
invariant Grassmann mass kernel	T_1	plane	S_3	none	T_5	Theorem 9.62(ii)
copy isospin	—	integral	integral	$j_\pm \in \frac{1}{2}\mathbb{Z}$	—	Proposition 9.24; e3-4d
projector P_*	—	—	yes	—	—	Theorem 6.10
$\Lambda^n_{\mathbb{C}}$ of the Dirac module on W_n	Dirac	$2(0,0)\oplus(\frac{1}{2},\frac{1}{2})$	Dirac	$(0,0)$	0	e3-5d
Koszul-antisymmetric part of S_n	8	32	88	192	360	e3-5e

Table 9.8: The ladder. Dimensions of Lie algebras are real. "Plane" and "none": at $n \equiv 2 \pmod 4$ the invariant Grassmann kernels form a plane, at $n \equiv 0$ there is none; T_n and S_n there are the temporal and spatial volumes of Theorem 9.62, not the spinor module. The copy isospin is that of the six-direction copy algebra.

algebra: the chiral Pati–Salam projector P_* is constructed at $n=3$ (Theorem 6.10), the diagonal connection algebra there is $\mathfrak{so}(3)\oplus\mathfrak{so}(6) \cong \mathfrak{su}(2)\oplus\mathfrak{su}(4)$ with colour $\mathfrak{su}(3)$ inside $\mathfrak{so}(6)$, and the copy isospin is integral (Proposition 5.17(iii); Proposition 9.24). At $n=5$ the Dirac block carries one Standard-Model family per copy multiplet (Proposition 5.17(iv)). Apart from the period of the forms, every one of these selections comes from a Lorentz action, a projector or a gauge-algebra identification added to the rung. None comes from the ladder.

9.7.2 What changes up the ladder

Rung by rung the algebra changes polynomially in n, and no rung is singled out by the algebra alone. The only periodic structure is the period four of the invariant spinor forms (Chevalley's pattern), through which the $\mathfrak{so}(3n,3n)$-invariant Grassmann mass kernel is unique exactly at odd n. Parity selections come from added structures: odd n from the diagonal interior action (spin parity $(-1)^n$, selected imaginary unit, copy-invariant four-vectors), even n from the common binding action (an invariant number of quanta), and $n=3$ from the projector P_* and the low-rank coincidences of the gauge algebra. Between rungs there is no process in the algebra: rung $n+1$ is not a state or an element of rung n, and the tensor embedding $\psi \mapsto \psi\otimes\phi_0$ needs a state

ϕ_0 that nothing selects.

claim	class	location	evidence
$\mathrm{Cl}(3n,3n) = M_{8^n}(\mathbb{R})$; step with multiplicity 8; commuting copy unique up to conjugation	proved algebra	Propositions 5.33 and 5.64	e1-1a–d, e3-1a–c, y1-1a–f
graded masses in quadrature, no simultaneous nonzero-mass graded Dirac constraints; ungraded spectra additive	proved algebra	Propositions 5.34 and 5.65	e1-2a–g, e3-2a–b, y1-2a–g, z1-1c
common complex structure only on W_n; selection by actions; none at even n	proved algebra; obstruction	Propositions 5.35 and 5.36 and Corollary 5.37	e1-3a–h, y1-3a–4f
N conserved exactly by $\mathfrak{gl}(3n,\mathbb{R})$; compact $\Leftrightarrow$ paired rotation (in $\mathfrak{so}(3n) \oplus \mathfrak{so}(3n)$); no odd change	proved algebra	Proposition 5.38, Theorem 5.39, and Lemma 5.46	e1-4a–g, e2-10–13, y2-11–14, z1-1a
compact fields outside $\mathrm{Cent}(G_n)$ (J, e_1e_2, S_k) create pairs of quanta	proved algebra	Propositions 5.40 and 5.47	e2-14–16, z1-1b
N not Lorentz invariant; invariant number iff n even (binding action)	proved algebra	Theorem 5.50	e2-19–22, y2-20–24
no equivariant vertex $S \to S_n$ for even n (interior action)	proved algebra (selection rule inherited)	Proposition 5.43	e2-2–6, y2-2–6
$C_{\mathrm{cop}} + C_{\mathrm{diag}} = \frac{3}{4}n(n+4)$; Howe duality; Koszul $\mathfrak{S}_n$	proved algebra (constant and duality inherited, Neergård, Howe)	Theorem 5.68 and Proposition 5.70	e3-4a–f, e3-5a–e, z1-3a–d
compact copy field = $d\Gamma(H)$: transfer, not creation; no binding in this class	proved algebra	Theorem 9.27 and Corollary 9.28	e4-1a–f
c_n triangle signs; $SO(4)$ doublets on odd N; no copy spinor	proved algebra; CP reading not established	Theorems 9.29 and 9.32 and Corollary 9.33	e4-2a–3e, z1-4a–b
pair invariant: stabiliser exactly $\mathfrak{l}_G^{\Delta} \oplus \mathbb{R}C_v$	proved (exact certificate)	Theorem 9.40	e5-1c, y5, z1-5b
mass form: $Y^{\top}T + TY = 0$ on $\mathfrak{so}(3,3)$, $Y^{\top}TY = -TY^2$	proved (corrected ordering)	Proposition 9.52	e6-1a–c, y6, z1-5a
W_n, H_u, plain exchange with antisymmetry; n fermions at one label, exclusion $n \leq 4$	constructed; not selected	Section 5.4.6 and Proposition 5.71	e3-5d, z1-3c
mixed-action vertex; Rabi transfer with external i; non-abelian frames on a harmonic ladder	constructed	Propositions 5.43, 5.51 and 9.36	e2-5, e2-23–25, e4-5a–e, z1-4c
no selected two-particle complex structure at $n = 2$; no PS in the five candidate algebras	obstruction (stated class)	Corollary 5.37 and Proposition 9.42	e1-3c–g, e5-3a–i
graded composite below free pair at equal labels; not a bound state	proved inequality; binding not supported	Proposition 5.44	e2-8, y2-8, z1-2a
uphill decays in $(3,3)$; none after reduction to one clock	inherited kinematics; conditional on the reduction	Proposition 10.6	e7-16, e7-17
"one particle" or "n particles"	interpretive	Sections 5.4.6 and 5.7.6	—
Poincaré state space, asymptotic states, interacting theory, dynamical selection, spin–statistics	missing	—	—

Table 9.9: Claims about particle multiplicity collected along the ladder, their classes and their evidence. A `PASS` line shows that the script's assertion holds; the class column records what the claim establishes. The checks are listed in Section 15.2.

Chapter 10

Classical and Astrophysical Consistency

The question of this chapter is what the classical readings of the arena say about observed physics. It reads the rest energy of a field point in the sky, the pointwise rest energy of isotropic radiation, pair creation, several particles in three times, the rate $\kappa = c/\ell$ and the sky, and time-odd invariants, clocks and the constancy of masses. Each comparison is kept within its scope: energy densities are compared in one environment, and the traceless stress tensor of radiation, not its density, decides whether it can stand in for matter; the distribution law of the rest energy holds in the classical Gaussian model; vacuum, two-photon and one-photon pair creation are kept apart, and the pair invariant is an identity of classical densities, not a threshold or a rate; the nuclear figure is a conditional sensitivity estimate; the birefringence bound on κ holds within a stated coupling model; the couplings of masses to clocks and to gravity are stated, and the clock tests constrain differential effects only; and the signed clock energy is not a mass. No observation singles out the arena; the bounds constrain readings, not the algebra.

10.1 Classical readings: astronomy and cosmology

This chapter reads the results of the preceding chapters in the sense of classical physics, astronomy and cosmology. Five classes are kept apart. A statement is *inherited* when it is ordinary physics, true in any notation, and cited from the literature. It is a *reading* (interpretive, in the classes of this manuscript) when it identifies a quantity of the arena with an observed one; we say what the identification needs. It is a *constraint* when, if a reading is adopted, an observation bounds a parameter. The constraint is then fixed arithmetic on cited inputs, conditional on the reading. It is *obstructed* when observation or inherited physics excludes the reading, and *open* otherwise. Mathematical statements carry the classes fixed and restated as before.

One fact governs the whole section. The arena is scale-free: the rate $\kappa = c/\ell$ of the interior flow has no fixed length ℓ, the identification $mc^2 = \hbar\kappa$ is added, and the multiscale ratio of Corollary 4.30 is an input (Proposition 4.49, Corollary 4.30 and Section 4.5.5). No number of astronomy or cosmology follows from the arena alone. Every number below is either inherited arithmetic on observed inputs or a bound on a parameter of a reading.

The answers, with their classes, are these. (1) The rest energy u_0 of a field point is the energy density in the frames where $\mathbf{E} \parallel \mathbf{B}_\mathrm{t}$ (inherited). Its stress is a tension u_0 along the common axis and a pressure u_0 across it, and every isotropic ensemble of field points gravitates as radiation, $w = \frac{1}{3}$, whatever its u_0. So the reading "field rest energy is dark matter" is obstructed (Section 10.1.1). (2) In the classical Gaussian model of isotropic thermal radiation the ratio E_0/E_6 at a point has density $2t$ on $[0, 1]$, independent of the energy, so $\langle E_0\rangle/\langle E_6\rangle = \frac{2}{3}$ (fixed within the model). Every plane wave is null, but the random superposition has $q \neq 0$ almost everywhere. "The CMB is locally massive" is a reading; for the quantised field it needs a

smearing, an operator ordering and a measurement model, and it says nothing about a photon mass (Section 10.1.2). (3) The pair invariant of two null fields (Theorem 5.57) vanishes on the same configurations as the Breit–Wheeler invariant. The thresholds of the gamma-ray horizon need m_e, which the arena does not supply. The vacuum pair rate depends on the canonical electric field ϵ_{can}, not on $E_0 = \frac{1}{2}(\epsilon_{\mathrm{can}}^2 + \beta_{\mathrm{can}}^2)$, so magnetar fields above B_Q create no pairs from the vacuum; the pairs of magnetised environments are made by photons converting in the field, and the photons supply the energy (inherited; Section 10.1.3). (4) Two six-momenta at mutual rest with temporal angle ϑ have $M^2 = m_1^2 + m_2^2 + 2m_1m_2\cos\vartheta$. In three times "uphill" decays are kinematically open, and one clock closes them (inherited, with an exact example); without amplitudes this motivates the reduction and bounds no lifetime. Under the reading, and under assumptions about the reaction that the arena does not derive, the $E = mc^2$ test of neutron capture on ^{28}Si corresponds to a temporal angle of order 10^{-4} rad between the neutron and the nucleus; this is a conditional sensitivity estimate, not a measured angle (Section 10.1.4). (5) A duality rotation of a null field is a rotation about its propagation axis (fixed). If the interior flow acted on free light as a duality rotation at a constant rate in cosmic time, a coupling the arena does not supply, CMB polarisation would bound that rate by $2.5\times10^{-20}\,\mathrm{s}^{-1}$, $10^{40.5}$ below $m_ec^2/\hbar$, on the unwrapped small-angle branch of that propagation hypothesis; the aliases $\beta + n\pi$ are not excluded by this division, and the bound concerns the model, not the arena (Section 10.1.5). (6) The three-manifold time-odd invariant (Theorem 9.23) parallels the Jarlskog invariant only as a reading. A temporal b-field acting on the clocks of matter is bounded by the constancy of m_p/m_e and by the equivalence principle, as conditional constraints on differential effects, under stated couplings of the masses to clocks and to gravity (Section 10.1.6). Section 10.1.7 collects the claims (Table 10.2). Identifiers e7-n refer to the `PASS` lines of `e7_classical.py`, and z5-n to those of the adjudication checks `z5_classical_adjudication.py`; physical constants are CODATA values from `scipy.constants`.

Notation. SI units. The frame variable is $\mathbf{B} = -\mathbf{B}_{\mathrm{t}}$, with $\mathbf{B}_{\mathrm{t}}$ the textbook field (Section I.1.4), $z_c = c\mathbf{B} + i\mathbf{E}$, $q_c = z_c \cdot z_c$, $u = \frac{\varepsilon_0}{2}|z_c|^2$, $u_0 = \frac{\varepsilon_0}{2}|q_c|$, $\mathbf{g} = \mathbf{E} \times \mathbf{B}_{\mathrm{t}}/(\mu_0 c^2)$ and $\rho_0 = u_0/c^2$. Then $u^2 = c^2|\mathbf{g}|^2 + u_0^2$ (Proposition 3.99; restated, e7-2). In Heaviside–Lorentz units with $c = 1$, $z = \mathbf{B} + i\mathbf{E}$, $E_6 = \frac{1}{2}|z|^2$, $E_0 = \frac{1}{2}|q|$, $q = z \cdot z = |\mathbf{B}_{\mathrm{t}}|^2 - |\mathbf{E}|^2 - 2i\,\mathbf{E}\cdot\mathbf{B}_{\mathrm{t}}$. The realisation is that of Appendix A.1, and G acts on z as i (e7-1).

10.1.1 The rest energy of a field point in the sky

The infimum of u over frames is u_0. For a non-null field ($q \neq 0$) it is an attained minimum, exactly in the frames where the Poynting vector vanishes, and there $\mathbf{E} \parallel \mathbf{B}_{\mathrm{t}}$ or one of them vanishes (Theorem 3.95). For a nonzero null field $u_0 = 0$ is approached but not attained: a boost along a plane wave gives $u(v) = u(0)(1-v)/(1+v) > 0$ for $|v| < 1$. This is inherited: the classification of a field at a point [148] and the problem of the frame with parallel fields in §25 of [82]. Let $\Theta^{00} = u$, $\Theta^{0i} = (\mathbf{E} \times \mathbf{B}_{\mathrm{t}})_i$, $\Theta^{ij} = -(E_iE_j + B_{\mathrm{t}i}B_{\mathrm{t}j}) + u\,\delta_{ij}$ be the stress–energy tensor of the field (Heaviside–Lorentz, $c = 1$).

Proposition 10.1 (The stress of a field point). *(i)* $\Theta^{\mu}{}_{\mu} = 0$ *and* $\Theta^{\mu}{}_{\lambda}\Theta^{\lambda}{}_{\nu} = E_0^2\,\delta^{\mu}{}_{\nu}$.

(ii) If $\mathbf{E} = \epsilon_{\mathrm{can}}\,\mathbf{n}$ *and* $\mathbf{B}_{\mathrm{t}} = \beta_{\mathrm{can}}\,\mathbf{n}$ *with* $|\mathbf{n}| = 1$, *then* $\Theta^{00} = u_0$, $\Theta^{0i} = 0$ *and* $\Theta^{ij} = u_0(\delta_{ij} - 2n_in_j)$: *a tension* u_0 *along* $\mathbf{n}$ *and a pressure* u_0 *in the two directions across it.*

(iii) In every ensemble of field points whose mean stress is isotropic in some frame, $\langle\Theta^{ij}\rangle = p\,\delta^{ij}$ *with* $p = \frac{1}{3}\langle\Theta^{00}\rangle$, *whatever the distribution of* u_0. *For field points at rest with isotropically distributed axes,* $p = \frac{1}{3}\langle u_0\rangle$.

(iv) In a Friedmann–Lemaître universe a component with $p = w\rho c^2$ *obeys* $\dot\rho = -3(1+w)\rho\,\dot a/a$. *So* $\rho \propto a^{-4}$ *for every such ensemble of free fields, against* a^{-3} *for dust.*

Proof. (i) $\sum_i \Theta^{ii} = -(|\mathbf{E}|^2 + |\mathbf{B}_\mathrm{t}|^2) + 3u = u$, so $\Theta^\mu{}_\mu = \Theta^{00} - \sum_i \Theta^{ii} = 0$; the square is the Rainich identity (Proposition 3.94). (ii) Substitute; $u = \frac{1}{2}(\epsilon_\mathrm{can}^2 + \beta_\mathrm{can}^2) = u_0$ since $|q| = \epsilon_\mathrm{can}^2 + \beta_\mathrm{can}^2$ when $\mathbf{E} \parallel \mathbf{B}_\mathrm{t}$, and $\mathbf{E} \times \mathbf{B}_\mathrm{t} = 0$. (iii) Tracelessness gives $\langle\Theta^{00}\rangle = \sum_i \langle\Theta^{ii}\rangle = 3p$. In the second case $\langle n_i n_j\rangle = \frac{1}{3}\delta_{ij}$, so $\langle\Theta^{ij}\rangle = u_0(1 - \frac{2}{3})\delta_{ij}$. (iv) Integrate the continuity equation. □

Inherited (§§33 and 35 of [82]; [129], as restated in Proposition 3.99; [238]), checked exactly (e7-3, e7-4). A bounded static configuration together with its sources has total mass U/c^2, with the field stresses balanced by those of the sources (von Laue's theorem [84]; Proposition 3.100). Field energy then gravitates as part of the mass of a matter system, and the matter fixes its share.

Corollary 10.2 (Field rest energy is not dark matter)**.** *Reading $\rho_0 = u_0/c^2$ as a rest-mass density does not make field rest energy behave as matter in cosmology. The reading "field rest energy is dark matter" is obstructed by inherited physics.*

Proof. What gravitates is $\Theta^{\mu\nu}$, and the reading leaves it unchanged. The dark matter of the standard model of cosmology is pressureless since before recombination, with $\Omega_c h^2 = 0.120 \pm 0.001$ [239]. By Proposition 10.1(iii)–(iv) every isotropic ensemble of free field points has $w = \frac{1}{3}$ and redshifts as a^{-4}, whatever u_0. A field point with $u_0 > 0$ is at rest only in the sense of Proposition 10.1(ii): its stress is not that of dust, and a free field propagates at c. The only configurations whose field energy weighs like rest mass are bound ones, by von Laue's theorem, and there the mass belongs to the matter system. The amounts are also small, when local densities are compared with local ones and cosmological means with cosmological means. Near the Sun the galactic magnetic energy density, 0.22 to 0.89 eV cm^{-3} for 3 to 6 μG, is between 4×10^{-10} and 4.5×10^{-9} of the local dark-matter density, 0.2 to 0.56 GeV cm^{-3} in the measurements collated by Read [266] (z5-1). In the cosmological mean, the rest energy of the CMB (Section 10.1.2), 0.17 eV cm^{-3}, is 1.4×10^{-4} of the mean cold-dark-matter energy density 1.26 keV cm^{-3} (e7-26, z5-1); the cosmological mean of the magnetic energy density is not known, and no comparison is made for it. The cause of the obstruction is the tracelessness, not the deficit. □

The fields of astronomy are, to a good approximation, magnetic in the frame of the conducting matter, where $\mathbf{E} \approx 0$, so $u = u_0 = B^2/(2\mu_0)$ and $\rho_0 = B^2/(2\mu_0 c^2)$ there. Table 10.1 gives the arithmetic (e7-5) for representative inputs: the geomagnetic field at the surface (about 50 μT; total intensity maps of [241]), a sunspot umbra (0.3 T [242]), a strongly magnetic white dwarf (isolated magnetic white dwarfs reach about 10^5 T [243]), the surface of a radio pulsar (about 10^8 T [244]), a magnetar (10^{10}–10^{11} T [245]) and the galactic field (a few microgauss, about 6 μG in total near the Sun [246]).

field	B (T)	$u = u_0$ (J m^{-3})	ρ_0 (kg m^{-3})
geomagnetic, surface	5×10^{-5}	9.9×10^{-4}	1.1×10^{-20}
sunspot umbra	0.3	3.6×10^{4}	4.0×10^{-13}
magnetic white dwarf	10^5	4.0×10^{15}	4.4×10^{-2}
radio pulsar, surface	10^8	4.0×10^{21}	4.4×10^{4}
magnetar	10^{10}–10^{11}	4.0×10^{25}–4.0×10^{27}	4.4×10^{8}–4.4×10^{10}
galactic, 3–6 μG	3–6 $\times 10^{-10}$	3.6–14 $\times 10^{-14}$	4.0–16 $\times 10^{-31}$
CMB, $T_\mathrm{CMB} = 2.72548$ K	—	4.17×10^{-14} (u)	3.10×10^{-31} ($\langle\rho_0\rangle$)

Table 10.1: Energy and rest-energy densities of astronomical fields in the frame where $\mathbf{E} \approx 0$ (e7-5), and of the CMB (e7-10). The last column is u_0/c^2; reading it as a mass density is the reading of Corollary 10.2.

The galactic field of 3 to 6 μG carries 0.22 to 0.89 eV cm^{-3}. The CMB carries $aT_\mathrm{CMB}^4 = 0.261$ eV cm^{-3}, with $T_\mathrm{CMB} = 2.72548$ K [240], and the cosmic rays of the local interstellar

medium about $1\,\mathrm{eV\,cm^{-3}}$ [247, 248]. The three agree to a factor of a few (e7-6). This rough equipartition of the magnetic, cosmic-ray and thermal energy densities is an inherited fact of the interstellar medium [247]. The arena neither explains it nor uses it.

10.1.2 The pointwise rest energy of isotropic radiation

Proposition 10.3 (Rest-energy fraction of a Gaussian field)**.** *Let $z = \mathbf{x} + i\mathbf{y}$ with $\mathbf{x}, \mathbf{y} \in \mathbb{R}^3$ independent and $N(0, \sigma^2\mathbf{1})$, and let $\lambda_1 \geq \lambda_2$ be the eigenvalues of the Gram matrix $\mathcal{G}_{xy}$ of $(\mathbf{x}, \mathbf{y})$. Then $|q| = \lambda_1 - \lambda_2$ and $|z|^2 = \lambda_1 + \lambda_2$. The ratio $t = E_0/E_6 = |z \cdot z|/|z|^2 \in [0, 1]$ has density $2t$ (distribution function t^2), is independent of $|z|^2$, and has mean $\frac{2}{3}$ and second moment $\frac{1}{2}$. Hence $\langle E_0\rangle/\langle E_6\rangle = \frac{2}{3}$, and $q \neq 0$ with probability one.*

Proof. $|q|^2 = (\mathbf{x}\cdot\mathbf{x} - \mathbf{y}\cdot\mathbf{y})^2 + 4(\mathbf{x}\cdot\mathbf{y})^2 = (\operatorname{tr}\mathcal{G}_{xy})^2 - 4\det\mathcal{G}_{xy} = (\lambda_1 - \lambda_2)^2$. $\mathcal{G}_{xy}$ is a real Wishart matrix $W_2(3, \sigma^2\mathbf{1})$, whose eigenvalues have the density $\propto (\lambda_1 - \lambda_2)e^{-(\lambda_1+\lambda_2)/2\sigma^2}$ on $\lambda_1 > \lambda_2 > 0$; the general factor $\prod_i \lambda_i^{(n-p-1)/2}$ is 1 at $n = 3$, $p = 2$ [250]. With $s = \lambda_1 + \lambda_2$, $t = (\lambda_1 - \lambda_2)/s$ and Jacobian $s/2$ the density becomes $\propto s^2 e^{-s/2\sigma^2} \cdot t$ on $s > 0$, $0 < t < 1$. It factorises, the first factor being the χ^2_6 law of $|z|^2/\sigma^2$. So t has density $2t$ and is independent of s, and $\langle ts\rangle/\langle s\rangle = \langle t\rangle = \frac{2}{3}$. □

The law is inherited: $f(t) = 2t$ is the eigenvalue law of a real Wishart matrix with $N = 2$, $K = 3$ [300, eq. (4.5)]; what the arena adds is the electromagnetic reading of t as E_0/E_6. Checked symbolically (e7-7); by Isserlis' theorem, $\langle|q|^2\rangle = 24\sigma^4$, $\langle|z|^4\rangle = 48\sigma^4$, $\langle|q|^4\rangle = 1920\sigma^8$ and $\langle|z|^8\rangle = 5760\sigma^8$, whose ratios are the moments $\frac{1}{2}$ and $\frac{1}{3}$ of the density $2t$ (e7-8); and by Monte Carlo with 2×10^6 samples, with mean 0.6665, Kolmogorov distance 7.5×10^{-4} to t^2 and correlation -1.1×10^{-3} between t and $|z|^2$ (e7-9).

The model. Thermal light is a circular complex Gaussian field, by the central limit theorem over many independent emitters (Sections 4.2 and 4.3 of [249]); this is inherited. For isotropic, unpolarised blackbody radiation at a point, isotropy gives $\langle E_i E_j\rangle = \langle B_{\mathrm{t}i} B_{\mathrm{t}j}\rangle = \frac{1}{3}u\,\delta_{ij}$ and $\langle E_i B_{\mathrm{t}j}\rangle = \frac{1}{3}\langle\mathbf{E}\cdot\mathbf{B}_{\mathrm{t}}\rangle\delta_{ij}$, and parity invariance gives $\langle\mathbf{E}\cdot\mathbf{B}_{\mathrm{t}}\rangle = 0$. The six components of z are therefore independent with a common variance, which is the hypothesis of Proposition 10.3, and $\langle E_6\rangle = u$, $\langle E_0\rangle = \frac{2}{3}u$. The distribution $2t$ is a statement of this classical Gaussian model, and only of it; it is not a prediction of the arena.

Scope of the model. For the quantised radiation field the statement needs a coarse-graining and a measurement model, and in general an operator ordering. A quantum field has no value at a point, so the fields must be smeared. If all six components are smeared with one spherically symmetric test function f at one time, they commute, since $[E_i(f), B_{\mathrm{t}j}(f)]$ is proportional to $\epsilon_{ijk}\int f\,\partial_k f\,d^3x = 0$ (z5-2). A joint measurement of them then has a distribution. In a thermal state, which is Gaussian [269], that distribution is Gaussian, and by isotropy, parity and duality invariance its six components are independent with a common variance. The law $2t$ then holds for the smeared ratio. Since it does not depend on the variance, it holds in the vacuum as well: it describes the shape of a Gaussian ensemble, not a property of the CMB. With different test functions, or at different times, the components need not commute (z5-2; concentric radial test functions can commute), $|q|$ is not a polynomial in them, and an ordering or a quasi-probability must be chosen, which is then not a distribution of measured ratios. The smeared distribution includes the vacuum fluctuations and is not by itself a statement about a renormalised local rest-energy observable. Nor is the law a statement about photons: every mode is null, and nothing here gives the photon a rest mass.

The reading. Every plane wave is null, $q = 0$ (Theorem 3.79). A random superposition has $q \neq 0$ almost everywhere, with mean rest-energy density $\frac{2}{3}u$ in the classical model. To say that the CMB is "locally massive" is a reading. Its content is $\langle E_0\rangle = \frac{2}{3}u$, under the assumptions just stated, and it says nothing about gravitation, which sees the traceless $\Theta^{\mu\nu}$ of Proposition 10.1. The whole behaves differently. Radiation in a box at rest in the CMB

frame has $\mathbf{P} = 0$ and $Mc^2 = U$, while $\int E_0 = \frac{2}{3}U$ on average; this is the strict inequality $M \geq \int E_0$ of Proposition 3.100(2), strict because the local energy–momenta are not proportional. Numbers (e7-10): today $u = aT_{\mathrm{CMB}}^4 = 4.17 \times 10^{-14}\,\mathrm{J\,m^{-3}}$, $\langle u_0 \rangle = 2.78 \times 10^{-14}\,\mathrm{J\,m^{-3}}$ and $\langle \rho_0 \rangle = 3.10 \times 10^{-31}\,\mathrm{kg\,m^{-3}} = 3.6 \times 10^{-5} \rho_{\mathrm{crit}}$, with $\rho_{\mathrm{crit}} = 3H_0^2/8\pi G_{\mathrm{N}}$ for $H_0 = 67.4\,\mathrm{km\,s^{-1}Mpc^{-1}}$ [239]. Like the whole CMB ($\Omega_\gamma = 5.4 \times 10^{-5}$), this density redshifts as a^{-4}.

10.1.3 Two from one in the sky: pair creation

Two photons. For null four-momenta k_1, k_2 with energies E_1, E_2 at angle ϑ, $s = (k_1 + k_2)^2 = 2E_1E_2(1 - \cos\vartheta)$, and $\gamma\gamma \to e^+e^-$ needs $s \geq 4m_e^2c^4$ [221]. Head-on, $E_1E_2 \geq (m_ec^2)^2$ (e7-11). On CMB photons of mean energy $\pi^4 kT_{\mathrm{CMB}}/30\zeta(3) = 6.3 \times 10^{-4}$ eV the threshold is 4.1×10^{14} eV, and on photons of energy kT_{CMB} it is 1.1×10^{15} eV. On optical and near-infrared extragalactic background light, 0.1–3 eV, it is 0.09–2.6 TeV. These thresholds enter the gamma-ray horizon [251, 252, 253], which is an optical-depth condition and needs in addition the background spectrum, the cross section, the distance and the cosmological evolution; Gould and Schréder predicted the cut-off of the cosmic photon spectrum above 10^{14} eV by the 3 K photons. The process has been observed with quasi-real, linearly polarised photons of ultraperipheral Au+Au collisions, with angular distributions in accord with QED [254]. All of this is inherited.

Relation to the arena. For two null fields $M_{\mathrm{pair}} = |z_1 \cdot z_2| = (E_6^{(1)}E_6^{(2)})^{1/2}(1 - \cos\vartheta_{12})$, with ϑ_{12} the angle between the Poynting vectors (Theorem 5.57(v); restated, e7-12). One null field does not become a massive pair under any covariant duplication, and no closure dynamics fills an empty constituent (Corollary 5.54; Corollary 5.45). The precise relation is this. M_{pair} and s vanish on the same configurations, the collinear ones, and are positive on all others: the selection rule "one photon makes no pair, two non-collinear photons can" is shared. The functions differ, and so do the objects: M_{pair} is an invariant of two classical field densities, an energy density, and $s^{1/2}$ the invariant energy of a two-photon state. M_{pair} is linear in $1 - \cos\vartheta$, because field densities are quadratic in the fields; s is linear in $1 - \cos\vartheta$, so the photon pair mass $s^{1/2}$ goes as its square root. The identity is not a threshold, a cross section or a rate. The threshold $2m_ec^2$ needs m_e, which the arena does not supply (Proposition 4.49). The gamma-ray horizon tests QED and the background light, not the arena.

Constant fields. If $q \neq 0$ there are frames with $\mathbf{E} = \epsilon_{\mathrm{can}}\,\mathbf{n}$, $\mathbf{B}_{\mathrm{t}} = \beta_{\mathrm{can}}\,\mathbf{n}$ (§25 of [82]).

Proposition 10.4 (Canonical fields)**.** *In such a frame, with $\epsilon_{\mathrm{can}} \geq 0$ and β_{can} real,*

$$\epsilon_{\mathrm{can}}^2 = \tfrac{1}{2}(|q| - \operatorname{Re} q), \qquad \beta_{\mathrm{can}}^2 = \tfrac{1}{2}(|q| + \operatorname{Re} q), \qquad E_0 = \tfrac{1}{2}|q| = \tfrac{1}{2}(\epsilon_{\mathrm{can}}^2 + \beta_{\mathrm{can}}^2).$$

The characteristic polynomial of $F^\mu{}_\nu$ is $\lambda^4 + (|\mathbf{B}_{\mathrm{t}}|^2 - |\mathbf{E}|^2)\lambda^2 - (\mathbf{E}\cdot\mathbf{B}_{\mathrm{t}})^2 = (\lambda^2 - \epsilon_{\mathrm{can}}^2)(\lambda^2 + \beta_{\mathrm{can}}^2)$, with eigenvalues $\pm\epsilon_{\mathrm{can}}$ and $\pm i\beta_{\mathrm{can}}$: $\epsilon_{\mathrm{can}} = \epsilon_E$ and $|\beta_{\mathrm{can}}| = \epsilon_M$ of Section 3.8.1. A purely magnetic F generates rotations and a purely electric one boosts.

Proof. q is invariant under the Lorentz group, which acts on z through $SO(3,\mathbb{C})$. In the canonical frame $\mathbf{E}\cdot\mathbf{B}_{\mathrm{t}} = \epsilon_{\mathrm{can}}\beta_{\mathrm{can}}$, so $q = \beta_{\mathrm{can}}^2 - \epsilon_{\mathrm{can}}^2 - 2i\epsilon_{\mathrm{can}}\beta_{\mathrm{can}} = (\beta_{\mathrm{can}} - i\epsilon_{\mathrm{can}})^2$, $|q| = \epsilon_{\mathrm{can}}^2 + \beta_{\mathrm{can}}^2$ and $\operatorname{Re} q = \beta_{\mathrm{can}}^2 - \epsilon_{\mathrm{can}}^2$. The coefficients of the characteristic polynomial are invariants and take these values. A real antisymmetric F exponentiates to rotations, a symmetric one to boosts. □

Inherited, checked exactly and on 200 random fields to 10^{-14} (e7-13). The vacuum decay rate in a constant field is

$$w = \frac{e^2\epsilon_{\mathrm{can}}\beta_{\mathrm{can}}}{4\pi^2\hbar^2 c} \sum_{n\geq 1} \frac{1}{n} \coth\frac{n\pi\beta_{\mathrm{can}}}{\epsilon_{\mathrm{can}}} \exp\Big(-\frac{n\pi E_S}{\epsilon_{\mathrm{can}}}\Big), \qquad E_S = \frac{m_e^2c^3}{e\hbar} = 1.32 \times 10^{18}\,\mathrm{V\,m^{-1}},$$

in Heaviside–Lorentz units [138, 66]. It is exponential in ϵ_{can} and vanishes for $\epsilon_{\mathrm{can}} = 0$, whatever β_{can}; Section 3.8.4 records the same fact for non-orthogonal fields. At fixed E_0 the first term varies by more than a factor 10 between the electric and the magnetic canonical field and

vanishes at the magnetic end; at fixed ϵ_{can} it grows with β_{can} (e7-27). The critical magnetic field is $B_Q = m_e^2c^2/(e\hbar) = 4.41 \times 10^9$ T. Radio pulsars sit at $0.02\,B_Q$; magnetars, at 10^{10}–10^{11} T, exceed it by factors 2–23 [245] (e7-14). Yet their fields are magnetic, $\epsilon_{\text{can}} \approx 0$, and create no pairs from the vacuum by this mechanism, although their E_0 is the largest in nature. Hence the reading "the rest energy of a field point materialises as pairs" is obstructed: the pair rate is not a function of E_0.

Pulsars. In a corotating, force-free magnetosphere $\mathbf{E} \cdot \mathbf{B}_{\text{t}} = 0$ [56]. Pair cascades need gaps where $\mathbf{E} \cdot \mathbf{B}_{\text{t}} \neq 0$, which accelerate the primary particles [133]. The curvature gamma-rays of the primaries convert into pairs in the strong magnetic field [255]. That one-photon conversion takes its energy from the photon; the magnetic field supplies momentum, not rest energy (inherited).

Three mechanisms kept apart. (a) Vacuum creation by an electric field, the Schwinger process: non-perturbative, with the rate above, and absent in a constant, purely magnetic field. (b) Photon–photon creation, the Breit–Wheeler process: a threshold $s \geq 4m_e^2c^4$ and a QED cross section. (c) One-photon creation in a magnetic field, $\gamma \to e^+e^-$, possible only with the field present: the photon supplies the energy and must have $\hbar\omega \sin\theta > 2m_ec^2$, with θ the angle between its direction and the field, and its conversion rate depends on the photon energy, the field strength and that angle [268]; this is the process of pulsar and magnetar magnetospheres. The arena supplies, for (b), the invariant M_{pair}, which has the zero set of s, and, for (a), the canonical fields of Proposition 10.4. It supplies no threshold, cross section or rate, since it contains neither m_e nor e. A pure magnetic background creates no pairs from the vacuum, and the pairs of magnetised environments are photon-assisted; neither fact concerns the rest energy E_0 of the field.

The analogy with the tower. A magnetic field acts on momenta as a rotation, which is compact; an electric field acts as a boost, which is non-compact (Proposition 10.4). In the tower the criterion is commutation with K_n, not compactness: elements that commute with K_n conserve the Fock number, and elements that anticommute with it create quanta in pairs from the Fock vacuum (Lemma 5.46(iii),(iv), Proposition 5.47). The two criteria differ: the compact charge $J = e_2e_3$ creates a pair from the Fock vacuum, while the pole operators ω_i are boosts that commute with N. The analogy is interpretive and rests on the commutator criterion. It does not identify the two questions. The Schwinger generator acts on the four-momentum of a charged particle in spacetime; the generators of Lemma 5.46 act on the interior Fock space, whose number N is not the number of electrons. The pairs of Proposition 5.47 appear at first order in the rapidity, with no threshold and no exponent, as in a change of reference vacuum, whereas the Schwinger rate is non-perturbative, with an exponent fixed by e and m_e, neither of which the arena contains.

10.1.4 Several particles in three times

The six-momentum $P = \sum_i (\mathsf{E}_i/c)\, e_i + \sum_a p_a\, e_{a+3}$ has three temporal components (Proposition 3.101). Set $c = 1$.

Proposition 10.5 (Two six-momenta at mutual rest)**.** *Let $u_1, u_2 \in V^+$ be unit vectors with $\eta(u_1, u_2) = \cos\vartheta$ and $P = m_1u_1 + m_2u_2$. Then*

$$M^2 := \eta(P, P) = m_1^2 + m_2^2 + 2m_1m_2\cos\vartheta,$$

so $M = m_1 + m_2$ at $\vartheta = 0$, $M = (m_1^2 + m_2^2)^{1/2}$ at $\vartheta = \pi/2$ and $M = |m_1 - m_2|$ at $\vartheta = \pi$. In $\mathrm{Cl}(3,3)$, $P^2 = M^2\mathbf{1}$. The graded rest Hamiltonian $H_g = m_1E_{(1,1)} + m_2E_{(2,1)}$ of the tower is the case of two orthogonal unit vectors of $\mathbb{R}^{6,6}$, $H_g^2 = (m_1^2 + m_2^2)\mathbf{1}$; the ungraded $H_u = m_1\, e_1 \otimes \mathbf{1} + m_2\, \mathbf{1} \otimes e_1$ has spectrum $\{\pm m_1 \pm m_2\}$ and is not a vector.

Proof. Expand $\eta(P, P)$; V^+ is positive definite. Vectors of a Clifford algebra square to their norm. The rest is Proposition 5.34(i),(iii). □

Fixed (e7-15: symbolic; P^2 exactly at $\cos\vartheta = \frac{3}{5}$; the tower exactly at $n = 2$). The reading is this: the graded composite of Proposition 5.34, with masses in quadrature (Proposition 5.62(iv)), is "two masses with orthogonal clocks", and the additive ungraded pair is "two masses with one clock". Its precise sense is limited. In $\mathbb{R}^{6,6}$ the temporal triples of the two copies are orthogonal, so $\vartheta = \pi/2$ is forced between vectors of different copies; inside one copy ϑ is free. The ungraded operator agrees with the case $\vartheta = 0$ only in its largest eigenvalue $m_1 + m_2$. Its additivity is that of the eigenvalues of commuting operators, one mass shell per particle, not that of a vector sum, which has two eigenvalues, not four. The identification is interpretive.

Proposition 10.6 (Uphill decays and one clock)**.** *(i) In signature $(1,3)$, future timelike P_1, P_2 satisfy $\eta(P_1, P_2) \geq m_1 m_2$, so $\eta(P_1 + P_2, P_1 + P_2) \geq (m_1 + m_2)^2$: a particle decays only into lighter ones in total.*

(ii) In signature $(3,3)$ this fails. $P = (10, 0, 0; 0, 0, 0)$ splits as $P_1 = (5, 5, 0; 1, 0, 0)$ plus $P_2 = (5, -5, 0; -1, 0, 0)$, with masses $10 \to 7 + 7 = 14$ and $\eta(P_1, P_2) = 1 < 49$, every clock component E_1 positive.

(iii) The clock e_1 reads $m_4^2 = \mathsf{E}_1^2 - |\mathbf{p}|^2 = 24$ for each product, and $2\sqrt{24} < 10$: read by the clock the decay is downhill.

(iv) If all temporal parts are parallel to one unit $u \in V^+$, the momenta lie in $\mathbb{R}u \oplus V^-$, of signature $(1,3)$, and (i) holds.

Proof. (i) $\eta(P_1, P_2) = E_1E_2 - \mathbf{p}_1 \cdot \mathbf{p}_2 \geq E_1E_2 - |\mathbf{p}_1||\mathbf{p}_2| \geq m_1 m_2$, the last step since $(E_1E_2 - |\mathbf{p}_1||\mathbf{p}_2|)^2 - m_1^2 m_2^2 = (E_1|\mathbf{p}_2| - E_2|\mathbf{p}_1|)^2 \geq 0$ with $E_1E_2 \geq |\mathbf{p}_1||\mathbf{p}_2|$. (ii)–(iii) Integer arithmetic. (iv) The Gram matrix of (u, e_4, e_5, e_6) is $\mathrm{diag}(1, -1, -1, -1)$. □

(i) is inherited; (ii) is the observation that with more than one time a particle can decay into heavier ones [256], recalled in [149], whose main argument is the loss of well-posed prediction; an ultrahyperbolic Cauchy problem can be made well posed by a nonlocal constraint [27], and Example 12.3 shows that compact surplus times give tachyons. All are checked exactly or, for (iv), with no violation among 20000 random pairs, against 12639 violations without a common clock (e7-16, e7-17). Item (iii) locates the paradox. "Uphill" is uphill in the six-dimensional mass M_6 and downhill in the mass m_4 read by a clock, since $m_4(\hat{n}) \leq M_6$ with equality iff the temporal momentum is parallel to the clock (Proposition 3.31; Proposition 3.101).

What survives the clock reduction. Under the reduction to one selected clock $\hat{n}$, the reduced shell $\mathbf{E} \parallel \hat{n}$ of Theorem 12.12, with the surplus temporal momenta removed by the constraints of Section 12.4 (Proposition 3.28), every momentum lies in $\mathbb{R}\hat{n} \oplus V^-$, of signature $(1,3)$. There $M_6 = m_4(\hat{n})$, the observable energy is the ordinary four-dimensional one, and by (iv) the inequality (i) holds: no uphill decay survives. Examples (ii)–(iii) use temporal momenta transverse to the clock, which the reduction removes; they are warnings about the unreduced six-dimensional kinematics and must not be read as predictions, and the ambient temporal momenta are not observable energies. The reduction itself is imposed, not derived (Assumption 2.16), and it also removes stable surplus-momentum modes, not only the unstable band.

A sensitivity estimate (conditional). Adopt the reading in which particles are labelled by M_6 and observed through one clock. Then a temporal angle between the constituents of ordinary matter would show as a mass defect. At mutual rest $M = m_1 + m_2 - \mu_{12}\vartheta^2/2 + O(\vartheta^4)$, with $\mu_{12} = m_1 m_2/(m_1 + m_2)$; the fractional defect is $(m_1 m_2/M_0^2)\vartheta^2/2$ with $M_0 = m_1 + m_2$, which is $\vartheta^2/8$ at equal masses, where $M = 2m\cos(\vartheta/2)$ exactly. Rainville *et al.* compared mass differences with γ-ray energies and found $1 - \Delta mc^2/E = (-1.4 \pm 4.4) \times 10^{-7}$ [257], a value that combines silicon and sulfur; silicon alone gives $(-9.7 \pm 8.0) \times 10^{-7}$. For neutron capture on ^{28}Si, $S_n = 8.4736$ MeV from the AME2020 masses [258]. One standard uncertainty is 3.7 eV, a fraction 1.4×10^{-10} of M_0, and the angle whose defect equals it is 9.1×10^{-5} rad between the neutron and the nucleus, or 3.3×10^{-5} rad for an equal-mass system at the same fractional precision

(e7-18). A defect Δ_{def} lowers the photon energy, $E = \Delta mc^2 - \Delta_{\mathrm{def}}$, so $1 - \Delta mc^2/E = -\Delta_{\mathrm{def}}/E$ and the central value lies on the side of a defect: the upper values of Δ_{def} at 1σ and 2σ are 4.9 and 8.6 eV, and the corresponding angles 1.0×10^{-4} and 1.4×10^{-4} rad (z5-3). These angles are sensitivity estimates, not bounds derived from a model and not silicon-specific bounds. They assume that the free neutron and the ^{28}Si nucleus meet at mutual rest with temporal angle ϑ, that the defect enters only that initial two-body system, that the trap masses and the γ energy are otherwise those of ordinary physics, and that ordinary nuclear binding is entirely accounted for by the measured masses. The arena supplies no model of the initial and final energies of the reaction that derives these assumptions, and the experimental residual measures an energy, not an internal angle. Proton stability enters differently. With off-clock final states accessible, the baryon-number-conserving channel $p \to n\, e^+ \nu_e$, closed in one time since $m_n > m_p$, opens kinematically by Proposition 10.6(ii), with a non-compact phase space. The proton mean life exceeds 10^{29} years in every mode searched and $\tau/B(p \to e^+\pi^0) > 2.4 \times 10^{34}$ years [118, 259]. Allowed weak decays with MeV energy release are typically fast, although lifetimes depend on matrix elements, selection rules and couplings. The argument is kinematic: it motivates, but does not derive, a selection, one clock for all matter, rather than a small angle. The arena does not compute the rate, since it has no amplitudes. This is the classical content of the clock selection (Proposition 2.5; the reduced shell of Theorem 12.12, Proposition 3.28). Classes: (i) and the kinematics of (ii) inherited; the angles are sensitivity estimates, conditional on the reading and on the assumptions about the reaction.

10.1.5 The rate $\kappa = c/\ell$ and the sky

The arena fixes no ℓ (Proposition 4.49). Three choices give (e7-19):

ℓ	ℓ (m)	$\kappa = c/\ell$ (s^{-1})	$\hbar\kappa$
reduced Compton wavelength $\hbar/(m_e c)$	3.86×10^{-13}	7.76×10^{20}	511.0 keV $= m_e c^2$
Planck length $(\hbar G_{\mathrm{N}}/c^3)^{1/2}$	1.62×10^{-35}	1.85×10^{43}	1.22×10^{19} GeV
Hubble radius c/H_0 [239]	1.37×10^{26}	2.18×10^{-18}	1.44×10^{-33} eV

The first is the identification $mc^2 = \hbar\kappa$ discussed in Section 4.5.5, for which $\ell = \hbar/(mc)$ is the only length (Proposition 4.49(ii)). The last puts $\hbar\kappa$ at the scale of ultralight fields. Each is a reading with no support from the arena.

Proposition 10.7 (Duality on a null field is a rotation about its axis)**.** *Let $z = \mathbf{B} + i\mathbf{E} \neq 0$ be null and $\hat{\mathbf{k}} = \mathbf{B} \times \mathbf{E}/|\mathbf{B} \times \mathbf{E}|$, which equals $\mathbf{E} \times \mathbf{B}_{\mathrm{t}}/|\mathbf{E} \times \mathbf{B}_{\mathrm{t}}|$, the direction of propagation. Then $\hat{\mathbf{k}} \times z = -iz$, $\hat{\mathbf{k}} \cdot z = 0$, and the duality rotation $e^{\vartheta G}$, $z \mapsto e^{i\vartheta} z$, acts as the rotation $R_{\hat{\mathbf{k}}}(-\vartheta)$ on $\mathbf{B}$ and on $\mathbf{E}$ separately. The plane wave $z = w\, f(\hat{\mathbf{k}} \cdot \mathbf{x} - t)$ with $\hat{\mathbf{k}} \times w = -iw$ solves $\partial_t z = -i\nabla \times z$ (Proposition 3.10). For $q \neq 0$ and $\vartheta \notin \pi\mathbb{Z}$ no rotation acts as $e^{\vartheta G}$.*

Proof. $\hat{\mathbf{k}} \times z = -iz$ is Theorem 3.97(ii), from $\mathbf{B} \cdot \mathbf{E} = 0$ and $|\mathbf{B}| = |\mathbf{E}|$; dotting with $\hat{\mathbf{k}}$ gives $\hat{\mathbf{k}} \cdot z = 0$. Then $e^{i\vartheta} z = \cos\vartheta\, z + \sin\vartheta\, iz = \cos\vartheta\, z - \sin\vartheta\, \hat{\mathbf{k}} \times z = R_{\hat{\mathbf{k}}}(-\vartheta) z$ by Rodrigues' formula, and $R_{\hat{\mathbf{k}}}(-\vartheta)$ is real. For the wave, $\nabla \times z = \hat{\mathbf{k}} \times w\, f' = -iw\, f'$ and $\partial_t z = -w\, f'$. Rotations fix q, while $e^{\vartheta G}$ sends q to $e^{2i\vartheta} q$. □

Fixed; checked on 50 null fields at three angles to 2×10^{-15}, the wave equation exactly (e7-20). It is the identity of helicity and duality on null fields [18, 121], Example 3.82, pulled back to the frame. Physically, a duality angle ϑ turns the plane of linear polarisation of a freely propagating wave about its line of sight by ϑ, the same for all frequencies, and only multiplies a circularly polarised wave by a phase.

A coupling model and its bound. The arena does not say that its interior flow acts on free light. Suppose, as a model, that it does: radiation in flight is carried by the flow $X(\tau) = e^{\mp\kappa G \tau} X_0$ (Section 4.5.5(ii)), with τ the cosmic time and κ constant. By Proposition 10.7 linearly polarised

light then arrives with its plane turned about the line of sight by κ times its propagation time, the same at all frequencies; the sign depends on orientation conventions that a bound on $|\kappa|$ does not need. For the CMB the propagation time is that since last scattering, $t_0 - t_{\rm LSS} \approx t_0 = 13.79$ Gyr [239]. If the rate varied, the bound below concerns its mean over the flight. The measured quantity is a uniform rotation β of the CMB polarisation. Minami and Komatsu find $\beta = 0.35^\circ \pm 0.14^\circ$ (68%) from the Planck 2018 data, determining the detector angles simultaneously from the Galactic foreground; their baseline assumes no intrinsic EB correlation of the CMB at last scattering or of the foreground, and a frequency-independent foreground EB of rotation form, an angle γ, would be read as $\beta - \gamma$ [260]. The joint WMAP and Planck analysis of Eskilt and Komatsu finds $\beta = 0.342^{\circ\,+0.094^\circ}_{\,-0.091^\circ}$, 3.6σ from zero, with no evidence of frequency dependence [267], and the ACT DR6 analysis of Diego-Palazuelos and Komatsu finds $\beta = 0.215^\circ \pm 0.074^\circ$, with its authors' caution that the systematics of those data are not yet understood [315]. Distant radio galaxies bound a uniform rotation over cosmological distances to a few degrees [22]. With $|\beta| < 0.63^\circ$ (|central value| + 2σ of [260]), $\kappa < 2.5 \times 10^{-20}\,\mathrm{s}^{-1} = 1.2 \times 10^{-2} H_0$, that is $\hbar\kappa < 1.7 \times 10^{-35}$ eV, $10^{40.5}$ below $m_e c^2/\hbar = 7.8 \times 10^{20}\,\mathrm{s}^{-1}$ (e7-21); with $|\beta| < 0.53^\circ$ from [267], $\kappa < 2.1 \times 10^{-20}\,\mathrm{s}^{-1}$ (z5-4). Dividing the angle by the age assumes the unwrapped, small-angle branch: rotations by β and by $\beta + n\pi$ leave linear polarisation unchanged, so for $\beta = 0.35^\circ$ the first alias is $\kappa = 7.2 \times 10^{-18}\,\mathrm{s}^{-1}$, about 290 times the quoted bound. The bound refers to that branch; excluding the others needs the visibility function, reionisation and the observation window. A time-dependent duality rotation is moreover not a symmetry of the free Maxwell equations, so the coupling hypothesis needs a stated propagation law. An oscillating rotation is, in addition, averaged over the visibility function, which washes it out [313]. Within the model and on the unwrapped branch, a rate tied to the electron mass by $mc^2 = \hbar\kappa$, or to any particle heavier than 1.7×10^{-35} eV, lies above the bound, and the Hubble rate does by a factor 86; a separate high-rate calculation (visibility, reionisation, observation window, propagation law) would be needed to exclude the aliased branches, and it is not done here. These are bounds on the coupling model, given a negligible foreground EB; they neither confirm nor exclude the arena, which does not couple its interior flow to light.

The converse reading. A cosmological duality angle of light is cosmic birefringence, as produced by a Chern–Simons or axion-like term [22] (Section 3.8.4). In the closure its potential vanishes on null fields. $U(\theta) = (R_0/2m)\cos\psi$ with $R_0 = 2|\nu|E_0(\pi_0) = 0$, so the duality angle of radiation is an exact flat direction (Theorem 4.28 and Corollary 4.43(i); restated, e7-22). The closure therefore neither forbids nor predicts a uniform rotation of polarisation. Its rate is a free datum, open, bounded as above if constant in time. A measured $\beta \neq 0$, at 2.4σ in [260] and 3.6σ in [267], would fix that datum and not test the arena; within the coupling model above it would correspond to $\kappa \approx 1.4 \times 10^{-20}\,\mathrm{s}^{-1}$ (z5-4).

10.1.6 Time-odd invariants, clocks and the constancy of masses

Three manifolds and three generations. For three manifolds with rates κ_k and couplings B_{kl} the plaquette is $H_{12}H_{23}H_{31} = i\mathcal{J}$ with $\mathcal{J} = B_{12}B_{23}B_{31}$, $\det[H, K] = 2i\mathcal{J}(\kappa_1 - \kappa_2)(\kappa_1 - \kappa_3)(\kappa_2 - \kappa_3)$, and interior time reversal sends $\mathcal{J}$ to $-\mathcal{J}$, which no copy reflection undoes at $n = 3$ (Theorem 9.23(ii),(iv); restated, e7-23). The Jarlskog invariant of quark mixing has the same shape: a rephasing-invariant plaquette of the mixing matrix, nonzero only for three generations, with $\det[M_u M_u^\dagger, M_d M_d^\dagger]$ equal to $2iJ$ times the two Vandermonde products of squared masses [225, 226]. The parallel is a reading. It says that in both cases a time-odd invariant of a Hermitian 3×3 structure needs three labels, and that the commutator determinant factorises through it. It does not establish CP violation in particle physics: within the three-copy interior model the invariant $\mathcal{J}$ is a property of the interior b-field between manifolds, with a free magnitude, not identified with the physical CKM invariant and not connected to a baryon-number-violating, out-of-equilibrium process (the matter and charge constructions of Chapters 6 and 7 do not supply such a process either). Sakharov's conditions for a baryon asymmetry, baryon-number

violation, C and CP violation and departure from equilibrium [261], are not touched. At most $\mathcal{J}$ is a candidate source of time-odd effects, and only under a reading that identifies interior time reversal with physical time reversal.

A temporal b-field on the clocks of matter. A temporal rotation turns the temporal momentum at fixed six-dimensional mass, and the mass read by a fixed clock oscillates (Proposition 9.58). At rest, for a libration $\phi = \theta_0 \sin \omega_{\text{lib}} t$ in the plane (e_1, e_2) read by e_1, $m_4 = M_6 |\cos \phi|$, which for $\theta_0 < \pi/2$ is $M_6 \cos \phi$, and $m_4/M_6 = 1 - \theta_0^2/4 + (\theta_0^2/4) \cos 2\omega_{\text{lib}} t + O(\theta_0^4)$. For a uniform rotation at rate ν about an axis at angle α from the clock, with the temporal energy initially on the clock, the energy read by the clock is $E(\hat{n})/M_6 = 1 - \sin^2 \alpha \, (1 - \cos \nu t)$ (fixed, e7-24), and the projected mass is its modulus, $m_4/M_6 = |1 - \sin^2 \alpha \, (1 - \cos \nu t)|$, since $m_4^2 = E(\hat{n})^2$ at rest. The signed expression is the clock energy, not a mass: it becomes negative near $\nu t = \pi$ when $\sin^2 \alpha > \frac{1}{2}$, and the modulus may be dropped exactly when $\sin^2 \alpha \leq \frac{1}{2}$, which includes the small-angle regime of the bounds below (z1-6a). In that regime both are $m_4/M_6 = 1 - A + A \cos \omega t$. A rotation common to all matter cancels in dimensionless mass ratios, so the bounds concern differential oscillations; all are conditional on the reading (e7-25).

Assumed couplings. The bounds below translate data into the parameters of the reading only under couplings that the arena does not derive. Atomic and molecular spectra are those of ordinary physics, with the masses m_4 read by a clock common to the apparatus, and their dependence on m_p/m_e is the one used in the cited analyses. For the equivalence principle, inertia is m_4 and gravity couples to M_6. A common rescaling of all masses, such as a temporal rotation shared by all matter, cancels in every dimensionless ratio and leaves $\eta_{\text{E}} = 0$, so these tests do not constrain it (z5-5); A below is the amplitude of a differential oscillation of m_p/m_e, one of the two masses taken as steady.

- *Quasar absorption.* Methanol lines at $z = 0.89$ give $|\Delta \mu_{pe}/\mu_{pe}| \lesssim 4 \times 10^{-7}$ (2σ) for $\mu_{pe} = m_p/m_e$, over a look-back time of about 7.5 Gyr [262]. For generic phases this gives $A < 4 \times 10^{-7}$, so $\theta_0 < 1.3 \times 10^{-3}$ rad for the libration and $\alpha < 6.3 \times 10^{-4}$ rad for the uniform rotation. One pair of epochs cannot exclude a coincidence of phases.

- *Atomic clocks.* Yb^+ against Cs gives $\dot{\mu}_{pe}/\mu_{pe} = (-0.5 \pm 1.6) \times 10^{-16} \, \text{yr}^{-1}$ [263], consistent with $(0.2 \pm 1.1) \times 10^{-16} \, \text{yr}^{-1}$ [264]. At 2σ, $A\omega < 3.7 \times 10^{-16} \, \text{yr}^{-1}$ for periods long compared with the years of comparison and a phase at which the drift is maximal: $A < 5.1 \times 10^{-6}$ for ω equal to the inverse age, and $A < 3.7 \times 10^{-16}$ for $\omega = 1 \, \text{yr}^{-1}$, a period of about 6.3 years, which is not long compared with the years of comparison.

- *Phases.* A bound on a drift constrains $A\omega$ only under an assumption on the phase: for $R = 1 - A + A \cos(\omega t + \varphi)$ the slope vanishes at an extremum, and for two epochs $R(t_2) - R(t_1) = -2A \sin(\frac{\omega}{2}(t_1 + t_2) + \varphi) \sin(\frac{\omega}{2}(t_2 - t_1))$, which vanishes whenever the separation is a whole number of periods. The figures above are order-of-magnitude sensitivities; a constraint on an oscillation needs a frequency-dependent analysis of a time series, as in the searches for oscillating constants with atomic clocks [314].

- *Equivalence principle.* Add the hypothesis that inertia is m_4 read by the clock while gravity couples to the undivided M_6. Then $m_g/m_i = 1/\cos \vartheta \approx 1 + \vartheta^2/2$, and the Eötvös parameter of two bodies is $\eta_{\text{E}} \approx \frac{1}{2}(\vartheta_A^2 - \vartheta_B^2)$. MICROSCOPE measured $\eta_{\text{E}}(\text{Ti}, \text{Pt}) = [-1.5 \pm 2.3(\text{stat}) \pm 1.5(\text{syst})] \times 10^{-15}$ [265], so $|\eta_{\text{E}}| < 7.0 \times 10^{-15}$ at 2σ, $|\vartheta_A^2 - \vartheta_B^2| < 1.4 \times 10^{-14}$, and $\vartheta < 1.2 \times 10^{-7}$ rad if one body is aligned. If inertia and gravity both couple to m_4, the test says nothing.

10.1.7 Summary and answer

Table 10.2: Claims of Section 10.1: the finding, its classical counterpart, its class, its check (`PASS` lines of `e7_classical.py`, e7-1 being the realisation) and what an observation could decide.

finding	classical counterpart	class	check	what an observation could decide
u_0 of a field point (Proposition 3.99)	energy density where $\mathbf{E} \parallel \mathbf{B}_\mathrm{t}$; infimum over frames, attained for non-null fields	restated; inherited	e7-2	nothing (identity)
traceless stress, tension and pressure u_0 (Proposition 10.1)	radiation, $w = \frac{1}{3}$; von Laue	inherited	e7-3, e7-4	nothing
ρ_0 as dark matter (Corollary 10.2)	pressureless dark matter; local with local, mean with mean	obstructed	e7-26, z5-1	already decided: dark matter has $w \approx 0$
u_0 of astronomical fields (Table 10.1)	magnetic energy densities; ISM equipartition	inherited arithmetic	e7-5, e7-6	nothing for the arena
E_0/E_6 has density $2t$ (Proposition 10.3)	Gaussian thermal field	fixed in the classical model; model inherited; quantum use needs a common smearing and a measurement model	e7-7 to e7-9, z5-2	joint statistics of commonly smeared $\mathbf{E}, \mathbf{B}_\mathrm{t}$; not a photon mass
CMB "locally massive", $\langle u_0 \rangle = \frac{2}{3}u$	photon gas, $Mc^2 = U$	reading	e7-10	nothing gravitational
M_pair of two null fields (Theorem 5.57)	Breit–Wheeler, gamma-ray horizon	restated; same zero set; an identity, not a threshold, cross section or rate; thresholds inherited	e7-11, e7-12	horizon tests QED and background light
canonical fields, $E_0 = \frac{1}{2}(\epsilon_\mathrm{can}^2 + \beta_\mathrm{can}^2)$ (Proposition 10.4)	Schwinger rate, E_S, B_Q	inherited	e7-13, e7-14	strong-field QED
rest energy materialises as pairs	magnetars above B_Q; vacuum, two-photon and one-photon creation kept apart	obstructed	e7-27	already decided (Schwinger; photon-assisted pairs in magnetospheres)
number conservation, $[X, N] = 0$, and pairs (Lemma 5.46)	magnetic against electric fields	interpretive (analogy); interior Fock number not identified with electron number	e7-13	nothing
$M^2 = m_1^2 + m_2^2 + 2m_1 m_2 \cos\vartheta$ (Proposition 10.5)	graded quadrature, additive copies	fixed; identification interpretive	e7-15	nothing
uphill decays (Proposition 10.6)	Dorling's instability	inherited; example fixed	e7-16, e7-17	off-clock channels kinematically open; motivates the imposed reduction (no lifetime bound derived)
one clock for matter	mass additivity, $E = mc^2$ test	sensitivity estimate, conditional on the reading and on a reaction model not derived	e7-18, z5-3	$\vartheta \sim 10^{-4}$ rad scale; a residual measures an energy, not an angle

finding	classical counterpart	class	check	what an observation could decide
$\kappa = c/\ell$, three lengths	Compton, Planck, Hubble	interpretive	e7-19	nothing until ℓ is fixed
duality = rotation about $\hat{\mathbf{k}}$ (Proposition 10.7)	helicity, polarisation rotation	fixed; inherited	e7-20	nothing
interior flow on light at $\kappa = m_e c^2/\hbar$	cosmic birefringence	bounded within a coupling model (flow on free light, cosmic time, constant rate) on the unwrapped small-angle branch; aliases not excluded; $\kappa < 2.5 \times 10^{-20}$ s^{-1} (2.1×10^{-20} with [267])	e7-21, z5-4	CMB polarisation, given a negligible foreground EB
flat duality angle of radiation (Corollary 4.43)	axion-like birefringence	open; restated	e7-22	β fixes a datum, tests nothing
three-manifold $\mathcal{J}$ (Theorem 9.23)	Jarlskog, Sakharov	reading	e7-23	nothing, unless manifolds become families
oscillating m_4 (Proposition 9.58)	constancy of m_p/m_e, equivalence principle	fixed; bounds conditional on stated couplings, differential only	e7-24, e7-25, z5-5	quasars, clocks, EP tests

Answer. Classical or astronomical content belongs to the inherited parts: the rest energy of a field point and its stress, the pointwise rest fraction $\frac{2}{3}$ of a Gaussian ensemble of fields, pair kinematics and the gamma-ray horizon, the canonical fields and the Schwinger rate, the kinematics of several times, and the identity of duality and helicity on light. There the arena agrees with ordinary physics and adds nothing measurable. Excluded as readings are field rest energy as dark matter and the materialisation of rest energy into pairs; bounded within an explicit coupling model, on its unwrapped small-angle branch, is the interior flow acting on free light at the rate $m_e c^2/\hbar$; unrestricted off-clock kinematics permits problematic channels and motivates the imposed common-clock reduction, but no proton-lifetime exclusion or uniqueness of that cure is derived. The numerical angles and rates are sensitivity estimates or bounds on those models, not measurements of quantities of the arena. Unconstrained, or open, are ℓ and with it κ, the origin of a cosmic birefringence angle, the magnitude and meaning of the three-manifold invariant, and the readings that identify the interior with particles. Two structures are missing before the arena can make an astronomical or cosmological prediction. One is a fixed length ℓ, equivalently a mass through $mc^2 = \hbar\kappa$. The other is a coupling of the interior to spacetime: on which fields the interior flow acts, how the temporal part of the six-momentum couples to clocks and to gravity, and which clock ordinary matter shares. With them, and with the dynamics and the states or boundary conditions that fix amplitudes, phases and clock orientations, the constraints above could become tests: the angles of Sections 10.1.4 and 10.1.6 and the rate of Section 10.1.5 would be predicted rather than bounded.

Chapter 11

Curvature, Geometry and Gravitational Obstructions

The question of this chapter is what survives when the metric of the arena is no longer flat. Three constructions are made. A curved arena of signature $(3,3)$ carries an almost para-Hermitian structure, and an explicit model with a normal $U(1)$ has a six-dimensional Einstein tensor in closed form (Section 11.1). In the field-strength reading the curvature operator of a four-metric lives on the frame, and the duality rotation of the curvature ends at the Einstein tensor (Section 11.2). An observation projector carried by the spin connection meets two competing curvatures (Section 11.3). Each of these is an identity of a *given* metric. None of them is a field equation for the metric, and Section 11.4 states, reading by reading, what a gravitational interpretation would still need: the dynamics of the metric or tetrad, an action, its constraints and physical degrees of freedom, the signs of the kinetic terms, the coupling to matter and a low-energy limit. The chapter relies on the clock and the interior four-plane of Chapter 2, the binding root and the field-strength reading of Chapter 3, and the paired-mode projector of Chapter 1; the causal analysis of Chapter 12 uses its normal-plane model.

Several letters carry more than one meaning in this chapter, each fixed by its context: B is the magnetic constant, the Stückelberg combination, a generic bivector and $\frac{1}{2}J_N$; G is the Hodge star, the shift of Chapter 4 and G_N; Ω is the para-Hermitian two-form and the spin connection; $\mathcal{V}$ is the space of algebraic curvature operators (not the volume element of Chapter 1); T is a tangent vector and the stress tensor; and Q is the Gauss–Bonnet functional.

11.1 A curved arena

Nothing in the algebra requires the ambient metric to be flat. Promoting the metric to a field of signature $(3,3)$, the generators to an orthonormal frame with a spin connection, and the pairing to a field of endomorphisms, one obtains a curved arena. This section describes its geometry, builds one explicit curved model against which statements can be tested, and computes the six-dimensional Einstein tensor of that model exactly. The two readings of $\mathbb{R}^{3,3}$ then each produce the Einstein tensor of a four-metric, in quite different ways, and neither determines the metric.

11.1.1 The pairing as a para-complex structure

Proposition 11.1 (The arena is almost para-Hermitian)**.** *Let (M,g) have signature $(3,3)$ and let $\mathcal{K}\in\mathrm{End}(TM)$ satisfy $\mathcal{K}^2=\mathbf{1}$ and $g(\mathcal{K}\cdot,\mathcal{K}\cdot)=-g$. Then $(M,g,\mathcal{K})$ is almost para-Hermitian: the eigendistributions $T^{\pm}M$ are isotropic of rank three, since $g(X,Y)=-g(\mathcal{K}X,\mathcal{K}Y)=-g(X,Y)$ on $T^{+}M$. It is para-Kähler, equivalently bi-Lagrangian, if and only if $\nabla^{\mathrm{LC}}\mathcal{K}=0$ for the Levi-Civita connection [28, 39]; then each eigendistribution is parallel, involutive and totally geodesic, and $\Omega(X,Y)=g(\mathcal{K}X,Y)$ is skew and closed.*

Executable check: g3-1 (exact symbolic or integer arithmetic; Section 15.2).

The para-complex structure is identified with the grading root of Proposition 2.3, $\mathcal{K}(v) = \frac{1}{2}[\varpi, v]$ with $\varpi = \sum_i e_i e_{i+3}$, once a pairing of the positive and negative directions is chosen; its eigenspaces are the two totally null three-planes spanned by $e_i \mp e_{i+3}$. The pairing is not canonical given the neutral metric alone, and a global construction needs the bundle reductions and spin assumptions to be stated. The null 3+3 splitting it induces is a different decomposition from the physical 4+2 splitting into the interior four-plane and the surplus plane, and conflating them is the first thing to avoid. A connection with torsion can preserve $\mathcal{K}$ without making the eigendistributions integrable or Ω closed, so a $\mathcal{K}$-preserving connection does not deliver the para-Kähler conclusion [92, 89]. Para-Kähler geometry is the world of $\mathcal{K}^2 = +\mathbf{1}$, while chirality is comfortable in the world of $J^2 = -\mathbf{1}$; this is one way of naming the difficulty that runs through the whole construction.

11.1.2 An explicit model with a normal $U(1)$

Split the tangent space as $TM = E \oplus N$ with $E = \mathrm{span}\{\widehat{W}, e_4, e_5, e_6\}$ of signature $(1,3)$, the symmetric clock of Proposition 2.5 playing the role of γ^0, and $N = \mathrm{span}\{u_1, u_2\}$ with

$$u_1 = \tfrac{1}{\sqrt{2}}(e_1 - e_2), \qquad u_2 = \tfrac{1}{\sqrt{6}}(e_1 + e_2 - 2e_3),$$

which complete the clock inside the positive triple: the relative times of Section 2.1.3. The metric does not select this split; it is extra data.

Lemma 11.2 (The normal plane carries a commuting rotation)**.** *$J_N := u_1 u_2 = \frac{1}{\sqrt{3}}(e_2e_3 + e_3e_1 + e_1e_2)$ satisfies $J_N^2 = -\mathbf{1}$ and commutes with $\widehat{W}$, e_4, e_5 and e_6.*

Proof. J_N is a sum of time–time bivectors and has no spatial index, so it commutes with e_4, e_5, e_6 by (1.3); it is the unit bivector of the plane orthogonal to $\widehat{W}$ inside the positive triple, so it commutes with $\widehat{W}$; and u_1, u_2 are orthonormal of square $+\mathbf{1}$, so $(u_1u_2)^2 = -u_1^2u_2^2 = -\mathbf{1}$. □

Executable check: g3-2 (exact symbolic or integer arithmetic; Section 15.2).

The normal plane therefore carries a compact $\mathfrak{u}(1)$ commuting with the physical generators. Give it a connection, $\nabla^\perp u_1 = -A u_2$, $\nabla^\perp u_2 = A u_1$, with spin connection $\Omega_N = \frac{1}{2} A J_N$. The model metric is

$$ds_6^2 = g^{(4)}_{\mu\nu}(x)\, dx^\mu dx^\nu + (du + vA)^2 + (dv - uA)^2, \qquad A = A_\mu(x)\, dx^\mu. \tag{11.1}$$

Proposition 11.3 (The model)**.** *The metric (11.1) has signature $(3,3)$; its restriction to $\Sigma = \{u = v = 0\}$ is $g^{(4)}$; the induced normal connection is A; and $R^u{}_{v\mu\nu}|_\Sigma = (dA)_{\mu\nu}$. For $g^{(4)}$ flat and $A = Bx\, dz$ one finds $R^u{}_{vxz}|_\Sigma = B$ while every component $R^\mu{}_{\nu\rho\sigma}|_\Sigma$ vanishes.*

Executable check: g2-1 (exact symbolic or integer arithmetic; Section 15.2).

This is the Kaluza–Klein construction for a rank-two real bundle with structure group $SO(2)$, and $R^u{}_{v\mu\nu} = F_{\mu\nu}$ is the standard formula for the curvature of a normal bundle [4, 111]. Two consequences are easy to misread in either direction: ambient curvature does not imply observable curvature, since the last example has a curved six-geometry with a flat induced four-metric; and a four-dimensional distribution is not a spacetime until an integral submanifold or a quotient is produced, which here Σ supplies.

Proposition 11.4 (The surplus constraints are first class)**.** *For a geodesic test particle in (11.1) put $\ell = v p_u - u p_v$, $k_\mu = p_\mu - A_\mu \ell$, so that $H = \frac{1}{2}(g^{\mu\nu}_{(4)} k_\mu k_\nu + p_u^2 + p_v^2)$. With $\phi_1 = p_u$, $\phi_2 = p_v$ and $C := A_\mu g^{\mu\nu}_{(4)} k_\nu$,*

$$\{\phi_1, \phi_2\} = 0, \qquad \{\phi_1, H\} = -C\phi_2, \qquad \{\phi_2, H\} = +C\phi_1,$$

so the constraints are first class for every $A_\mu(x)$ and every $g^{(4)}$. The reduced phase space has dimension $12 - 2 \times 2 = 8$, that of T^*M^4, and $H_{\mathrm{red}} = \frac{1}{2} g^{\mu\nu}_{(4)} p_\mu p_\nu$.

Executable check: c2-5 (exact symbolic or integer arithmetic; Section 15.2).

The potential has cancelled from the reduced Hamiltonian, and not by accident: the constraints generate translations in (u, v), so the normal $U(1)$ acts trivially on the reduced phase space and a spinless test particle in the reduced sector ($p_u = p_v = 0$) cannot see it; before the reduction, $H = \frac{1}{2} g^{\mu\nu}_{(4)}(p_\mu - A_\mu \ell)(p_\nu - A_\nu \ell) + \frac{1}{2}(p_u^2 + p_v^2)$ with $\ell = v p_u - u p_v$ couples to A whenever $\ell \neq 0$. Whatever makes the normal connection physical must come from the spinor sector, which Section 11.3 examines. This is a first instance of the constraint structure that Section 12.4 derives from a worldline gauge symmetry.

11.1.3 The six-dimensional Einstein tensor

In polar normal coordinates $u + iv = re^{i\phi}$ the model reads $ds_6^2 = h_{\mu\nu}dx^\mu dx^\nu + dr^2 + r^2(d\phi - A)^2$ with $h = g^{(4)}$. Write $W_{\mu\nu} = F_{\mu\rho}F_\nu{}^\rho - \frac{1}{4}h_{\mu\nu}F^2$. In signature $(+,-,-,-)$, $-W_{00} = \frac{1}{2}(|\mathbf{E}|^2 + |\mathbf{B}|^2)$, so a physical Maxwell field enters Einstein's equation as $\mathcal{G} = -2W$ in units where the coupling is absorbed.

Theorem 11.5 (The ambient Einstein tensor)**.** *For every $r > 0$, and not as a series in r,*

$$\mathcal{G}^{(6)}_{\mu\nu} = \mathcal{G}_{\mu\nu}[h] - \tfrac{r^2}{2}W_{\mu\nu}, \qquad \mathcal{G}^{(6)}_{rr} = -\tfrac{1}{2}R[h] + \tfrac{r^2}{8}F^2, \qquad \mathcal{G}^{(6)}_{\hat\phi\hat\phi} = -\tfrac{1}{2}R[h] + \tfrac{3r^2}{8}F^2,$$

$$\mathcal{G}^{(6)}_{\mu\hat\phi} = \tfrac{r}{2}\nabla^\rho F_{\rho\mu}, \qquad \mathcal{G}^{(6)}_{\mu r} = \mathcal{G}^{(6)}_{r\hat\phi} = 0, \qquad R^{(6)} = R[h] - \tfrac{r^2}{4}F^2,$$

with indices and ∇ referring to h. On the slice Σ the Einstein tensor is

$$\mathcal{G}^{(6)}\big|_\Sigma = \mathrm{diag}(\mathcal{G}_{\mu\nu}[h],\, -\tfrac{1}{2}R[h],\, -\tfrac{1}{2}R[h])\,:$$

its tangential block is the four-dimensional Einstein tensor exactly, with no stress term from F.

Proof. The projection onto (M, h) is a semi-Riemannian submersion with flat, totally geodesic fibres: for $U, V \in \{\partial_r, \partial_\phi\}$ and the horizontal lifts $E_\mu = \partial_\mu + A_\mu\partial_\phi$ one has $[U, E_\mu] = 0$ and $E_\mu\, g(U, V) = 0$, so the Koszul formula gives $g(\nabla_U V, E_\mu) = 0$. The integrability tensor is $\frac{1}{2}[E_\mu, E_\nu]^{\mathrm{vert}} = \frac{1}{2}F_{\mu\nu}\partial_\phi$ with $|\partial_\phi|^2 = r^2$. O'Neill's formulas, valid for semi-Riemannian submersions with nondegenerate fibres [110, 111, 12], then give $R^{(6)}_{\mu\nu} = R_{\mu\nu}[h] - \frac{r^2}{2}F_{\mu\rho}F_\nu{}^\rho$, $R^{(6)}_{\mu\hat\phi} = \frac{r}{2}\nabla^\rho F_{\rho\mu}$, $R^{(6)}_{\hat\phi\hat\phi} = \frac{r^2}{4}F^2$ and $R^{(6)}_{rr} = 0$, from which the Einstein components follow. They have also been checked symbolically on a Friedmann base $h = \mathrm{diag}(1, -t^2, -t^2, -t^2)$ with $A = z\,dt + \frac{1}{2}x^2 dy$, which carries both electric and magnetic fields. □

Executable check: g2-2 (exact symbolic or integer arithmetic; Section 15.2).

Corollary 11.6 (Sign, reduction and rigidity)**.** *(i) The tangential equation $\mathcal{G}[h] = +\frac{r^2}{2}W$ has the opposite sign to $\mathcal{G} = -2W$: relative to the ambient Einstein–Hilbert action and the stated connection-metric ansatz, the normal field strength gravitates with negative energy. The sign is forced: a Lorentzian four-plane in $\mathbb{R}^{3,3}$ uses one positive and three negative directions, so its normal plane has signature $(2, 0)$, the sign of time.*

(ii) Over a normal disc of radius r_c, $\iint \sqrt{|g_6|}\, R^{(6)}\, d\phi\, dr = \sqrt{|h|}(\pi r_c^2 R[h] - \frac{\pi r_c^4}{8}F^2)$. The Euler–Lagrange equation of $\int \sqrt{|h|}(a_1 R + a_2 F^2)$ is $a_1\mathcal{G} + 2a_2 W = 0$, so Einstein–Maxwell theory has $a_2/a_1 = +1$ while the reduced action has $a_2/a_1 = -r_c^2/8$: the wrong relative sign. Integrating over a finite normal disc is not a consistent compactification: the boundary terms and the discarded equations are not treated.

(iii) The vacuum is rigid: $\mathcal{G}^{(6)} = 0$ if and only if $F = 0$ and h is Ricci-flat. The internal equations differ by $\frac{r^2}{4}F^2$, which forces $F^2 = 0$ and then $R[h] = 0$; the tangential equation must hold at every r while $\mathcal{G}[h]$ and W do not depend on r, so $\mathcal{G}[h] = 0$ and $W = 0$ separately, and $W = 0$ forces $F = 0$ because $-W_{00}$ is positive definite.

Executable check: g2-3 (exact symbolic or integer arithmetic; Section 15.2).

Reduction on directions of timelike sign is known to give fields whose kinetic terms have the wrong sign [74], and the negative-energy vector found here is an instance. The two mechanisms of rigidity can be separated by changing the fibre. With a normal fibre of constant size, $ds_6^2 = h + \ell^2(d\psi - A)^2 + m^2 d\chi^2$, still of signature $(3,3)$, the internal equations force only $F^2 = 0$, and null fields survive.

Example 11.7 (Negative-energy null radiation)**.** Take $h = 2\,dU\,dV - H\,dU^2 - dX^2 - dY^2$ and $A = a(U)X\,dU$, so $F = a(U)\,dX \wedge dU$ is null. The only nonzero component of the six-dimensional Einstein tensor is $\mathcal{G}^{(6)}_{UU} = -\frac{1}{2}(H_{XX} + H_{YY} - \ell^2 a^2)$, so

$$H = \tfrac{1}{4}\ell^2 a(U)^2 (X^2 + Y^2)$$

solves the six-dimensional vacuum with $F \neq 0$. In four-dimensional terms $\mathcal{G}_{UU}[h] = -\frac{1}{2}\ell^2 a^2 < 0$, whereas a physical null Maxwell wave has $\mathcal{G}_{UU} = -2W_{UU} = +2a^2 > 0$: the wave is negative-energy radiation. With the fibre size the coordinate r, as in (11.1), the same field gives a tangential component graded in r that forces $a = 0$.

Executable check: g2-4, g2-5 (exact symbolic or integer arithmetic; Section 15.2).

More generally, with $ds_6^2 = h + e^{2\alpha(x)}(d\psi - A)^2 + e^{2\beta(x)} d\chi^2$,

$$\mathcal{G}^{(6)}_{\hat\psi\hat\psi} - \mathcal{G}^{(6)}_{\hat\chi\hat\chi} = \tfrac{1}{4} e^{2\alpha} F^2 - \Box(\alpha - \beta) - \nabla(\alpha - \beta)\cdot\nabla(\alpha + \beta).$$

For a breathing mode $\alpha = \beta$ the difference is a pure constraint and only null F survives; for a relative modulus it is a field equation for $\alpha - \beta$, and $F^2 \neq 0$ becomes admissible. Non-null F therefore needs the ratio of the two normal sizes to be a dynamical field. None of these escapes repairs the sign.

11.1.4 Off the slice

On Σ the F-couplings, which all carry r, switch off. Taking the four-geometry off the zero section requires care, because the level set $\{r = r_0\}$ is five-dimensional. Its quotient by the $U(1)$ is four-dimensional and carries h, and the ambient Einstein tensor in the horizontal frame is $\mathcal{G}[h] - \frac{r_0^2}{2} W$, with the sign above. A section $\phi = \chi(x)$ of $\{r = r_0\}$ is a four-dimensional submanifold with induced metric

$$h_\chi = h + r_0^2\, B \otimes B, \qquad B := d\chi - A,$$

where B is the Stückelberg combination, invariant under $A \mapsto A + d\lambda$, $\chi \mapsto \chi + \lambda$; $\det h_\chi = \det h\,(1 + r_0^2 |B|^2)$, so h_χ is Lorentzian only where $1 + r_0^2|B|^2 > 0$, and the section is totally geodesic on an open set only where $B = 0$, which requires $F = 0$.

Example 11.8 (A uniform magnetic field)**.** Take $h = \eta$ and $F = b\,dx \wedge dz$. On a static section the circulation of B around a circle of radius R_c in an x–z plane is $-\pi b R_c^2$, whatever χ, so $\sum_i B_i^2 \geq b^2 R_c^2/4$ somewhere on it, and the section metric has signature $(2,2)$ there as soon as $R_c > 2/(r_0|b|)$. In the gauge $A = bx\,dz$ with $\chi = 0$,

$$h_\chi = dt^2 - dx^2 - dy^2 - (1 - k^2x^2)\,dz^2, \qquad R[h_\chi] = -\frac{2k^2}{(1 - k^2x^2)^2}, \qquad k = r_0 b,$$

which degenerates at $|x| = 1/|k|$, where its curvature diverges, and has signature $(2,2)$ beyond. The Lorentzian region of a static section in a uniform field contains no circle of radius larger than $2/(r_0|b|)$.

Executable check: g2-6 (exact symbolic or integer arithmetic; Section 15.2).

11.1.5 Holonomy acting on the bound orbits

The interior dynamics of Section 4.3 can now be transported around a loop of the model. A circuit enclosing normal flux Φ rotates the frame by $h = \exp(\Phi R_N)$, with R_N the rotation of the normal plane N. R_N preserves η, but it rotates a temporal plane without its spatial partner, so it is not among the diagonal rotations of Theorem 3.1, and exactly

$$h = \mathbf{1} + \sin\Phi\, R_N + (1-\cos\Phi) R_N^2, \qquad hGh^{-1} - G = \sin\Phi\, A_1 + (1-\cos\Phi)\, A_2$$

with A_1, A_2 linearly independent, so $hGh^{-1} = G$ if and only if $\Phi \in 2\pi\mathbb{Z}$. The second-order equation $\ddot{X} = -\kappa^2 X$ does not see this, but the momentum does: a zero-momentum orbit $X(\tau) = e^{-\kappa G\tau} X_0$ carried once around becomes $hX(\tau)$, and

$$\alpha_+(hX) = \kappa\,(G - hGh^{-1})\, hX,$$

which vanishes for every X_0 only when $\Phi \in 2\pi\mathbb{Z}$. Normal curvature therefore converts bound orbits of zero momentum into orbits that carry momentum, through exactly the failure of the normal $U(1)$ to centralise the binding root. Executable check: g3-3 (exact symbolic arithmetic; Section 15.2).

11.1.6 Flows do not transfer

In DeTurck gauge Ricci flow reads $\partial_t g_{MN} = \Delta_g g_{MN} + Q(g, \partial g)$, and short-time existence requires the principal symbol $g^{PQ}\xi_P\xi_Q$ to be definite [64, 32]. In signature $(3,3)$ it has signature $(3,3)$: the operator is ultrahyperbolic, the gauge-fixed flow is not parabolic, and the standard Ricci–DeTurck theory does not apply. In the para-Kähler case the metric comes locally from a potential depending on coordinates along both foliations, $g_{i\bar{j}} = \partial^2\Phi/\partial x^i \partial y^j$ [141], but suppressing one set of variables degenerates the metric rather than producing a Hessian one; a real Monge–Ampère flow on a convex locus [20] would need its own domain, data and a proof that it preserves the arena's structures, and is a programme rather than a result.

11.2 Curvature in the field-strength reading

In the field-strength reading the six directions are $\Lambda^2\mathbb{R}^{1,3}$, η is the induced metric and G the Hodge star (Theorem 3.3). A tensor R_{abcd} antisymmetric in each pair is then an operator $\mathcal{R}$ on the frame, $(\mathcal{R}\omega)_{ab} = \frac{1}{2} R_{ab}{}^{cd}\omega_{cd}$ [143, 12], and the conditions that make it a Riemann tensor become conditions stated with η and G. Write $\mathrm{Ric}(\mathcal{R})_{bd} = R^a{}_{bad}$.

Proposition 11.9 (The curvature operator lives on the frame)**.** *$\mathcal{R}$ comes from a pair-symmetric tensor, $R_{abcd} = R_{cdab}$, if and only if $\mathcal{R}$ is η-self-adjoint, a 21-dimensional space $\mathcal{S}$; the η-antisymmetric operators give $R_{abcd} = -R_{cdab}$ instead. On $\mathcal{S}$, $\operatorname{Tr}\mathcal{R} = \frac{1}{2}R$ and*

$$R_{abcd} + R_{acdb} + R_{adbc} = -\tfrac{1}{2}\operatorname{Tr}(G\mathcal{R})\,\varepsilon_{abcd}, \qquad \varepsilon_{0123} = +1,$$

so the first Bianchi identity is the single linear condition $\operatorname{Tr}(G\mathcal{R}) = 0$, and the algebraic curvature operators form a 20-dimensional subspace $\mathcal{V}$, the dimension of the space of Riemann tensors in four dimensions.

Executable check: g1-1, g1-2 (exact symbolic or integer arithmetic; Section 15.2).

Pair symmetry is the same condition $G = G^\dagger$ that makes the interior shift factorise (Theorem 4.8), imposed on an operator of another type: the frame is shared, not the object. The binding root, which is η-self-adjoint, is itself in $\mathcal{S}$ but not in $\mathcal{V}$, since $\operatorname{Tr}(G\cdot G) = -6$; it is the one direction of $\mathcal{S}$ that violates the Bianchi identity.

11.2.1 A phase family that ends at the Einstein tensor

Theorem 11.10 (The phase family)**.** *For every $\mathcal{R} \in \mathcal{S}$, on or off the Bianchi shell, and every θ, the operator $\mathcal{R}_\theta := e^{\theta G}\mathcal{R}\, e^{\theta G}$ lies in $\mathcal{S}$ and*

$$\mathrm{Ric}(\mathcal{R}_\theta)_{ab} = R_{ab} - \tfrac{1}{2}\sin^2\theta\, R\, g_{ab} + \tfrac{1}{2}\sin 2\theta\ \mathrm{Tr}(G\mathcal{R})\, g_{ab}.$$

At $\theta = \pi/2$ the operator is the double dual $G\mathcal{R}G$ of Lanczos [81, 104], and its contraction is the Einstein tensor $R_{ab} - \frac{1}{2}Rg_{ab}$.

Proof. Two lemmas hold on all of $\mathcal{S}$: $\mathrm{Ric}(G\mathcal{R}G) = \mathrm{Ric}(\mathcal{R}) - \frac{1}{2}Rg$, and $\mathrm{Ric}(G\mathcal{R}) + \mathrm{Ric}(\mathcal{R}G) = \mathrm{Tr}(G\mathcal{R})\, g$. With $e^{\theta G} = \cos\theta + \sin\theta\, G$,

$$\mathcal{R}_\theta = \cos^2\theta\,\mathcal{R} + \sin\theta\cos\theta\,(G\mathcal{R} + \mathcal{R}G) + \sin^2\theta\, G\mathcal{R}G,$$

and the lemmas contract this to the formula. It has also been checked as a polynomial identity in the 21 entries, modulo $\cos^2\theta + \sin^2\theta = 1$. Since G is η-self-adjoint, so is $e^{\theta G}$, and $\mathcal{R}_\theta \in \mathcal{S}$. □

The map $\mathcal{R} \mapsto e^{\theta G}\mathcal{R}\, e^{\theta G}$ is a congruence, not a similarity, and it acts on algebraic curvature operators at a point: it is not a map between metrics or between solutions. Duality rotations of linearised gravity are a different structure [317].
Executable check: g1-3 (exact symbolic or integer arithmetic; Section 15.2).

Example 11.11 (A maximally symmetric curvature)**.** For de Sitter space of curvature k, $R_{abcd} = k(g_{ac}g_{bd} - g_{ad}g_{bc})$, the curvature operator is $\mathcal{R} = k\,\mathbf{1}$ on the frame up to the sign convention of the induced metric, so $\mathrm{Tr}\,\mathcal{R} = 6k = \frac{1}{2}R$ gives $R = 12k$ and $\mathrm{Tr}(G\mathcal{R}) = k\,\mathrm{Tr}\,G = 0$. Then $G\mathcal{R}G = kG^2 = -k\,\mathbf{1}$, and the formula gives $\mathrm{Ric}(\mathcal{R}_{\pi/2}) = R_{ab} - \frac{1}{2}Rg_{ab} = 3kg_{ab} - 6kg_{ab} = -3kg_{ab}$, the Einstein tensor of de Sitter space, which is $-\Lambda g_{ab}$ with $\Lambda = 3k$. The double dual of a constant-curvature operator is minus itself, and the trace reversal that turns Ricci into Einstein is the sign change $G^2 = -\mathbf{1}$.

Executable check: g1-4 (exact symbolic or integer arithmetic; Section 15.2).

Corollary 11.12 (What selects the endpoint)**.** *(i) For the curvature operator of a metric and a constant phase, the contracted Bianchi identity gives $\nabla^a H_{ab}(\theta) = \frac{1}{2}\cos^2\theta\,\nabla_b R$ for $H(\theta) := \mathrm{Ric}(\mathcal{R}_\theta)$, so divergence-freedom for every metric selects $\theta \equiv \pi/2 \pmod{\pi}$.*

(ii) $\mathrm{Tr}(G\mathcal{R}_\theta) = \cos 2\theta\ \mathrm{Tr}(G\mathcal{R}) - \frac{1}{2}\sin 2\theta\, R$, so on the Bianchi shell and for $R \neq 0$, $\mathcal{R}_\theta$ is itself a curvature operator only at $\sin 2\theta = 0$, compatible with (i) and narrowing nothing further.

(iii) $Q(\mathcal{R}) := \frac{1}{2}\mathrm{Tr}(\mathcal{R}G\mathcal{R}G) = -\frac{1}{8}(R^2 - 4R_{ab}R^{ab} + R_{abcd}R^{abcd})$ for every pair-symmetric $\mathcal{R}$, and $dQ[\mathcal{D}] = \mathrm{Tr}(G\mathcal{R}G\mathcal{D})$: the double dual is the algebraic gradient of the Gauss–Bonnet density with respect to the pairing $\mathrm{Tr}(AB)$, exactly so on $\mathcal{V}$, where $G\mathcal{R}G = \mathrm{grad}_{\mathcal{V}}\, Q$.

Executable check: g1-5 (exact symbolic or integer arithmetic; Section 15.2).

The gradient in (iii) does not lift to a field equation. The spacetime integral of the Gauss–Bonnet density has identically vanishing metric variation in four dimensions, which is Lanczos's identity [81, 90]: the Euler–Lagrange tensor $P_a{}^{cde}R_{bcde} - \frac{1}{2}g_{ab}\mathcal{E}_{\mathrm{GB}} - 2\nabla^c\nabla^d P_{acdb}$ with $P = -2\,{*R*}$ vanishes because the double dual is divergence-free and $({*R*})_a{}^{cde}R_{bcde} = -\frac{1}{4}g_{ab}\mathcal{E}_{\mathrm{GB}}$ in four dimensions. The two facts sit one Lovelock degree apart: the algebraic gradient of the quadratic density contracts to the metric gradient of the linear density $\mathrm{Tr}\,\mathcal{R} = \frac{1}{2}R$, and the quadratic density itself is inert. Which members of the family are variational at all is a sharper question, and it has a sharp answer.

Theorem 11.13 (Only the Einstein phase is variational)**.** *Consider the operators $g \mapsto H(\theta)[g]_{ab} = R_{ab} - \frac{1}{2}\sin^2\theta\, Rg_{ab}$. There is a Lagrangian $L(g, \partial g, \partial^2 g, \dots)$, covariant or not, defined on a neighbourhood of flat space, whose Euler–Lagrange expression is $H(\theta)$ if and only if $\cos\theta = 0$; then $R\sqrt{|g|}$ is one.*

Proof. An Euler–Lagrange operator has a formally self-adjoint Fréchet derivative at every point of its domain, the necessity half of the Helmholtz conditions [155, 109]. At $g = \eta$, $H(\theta)[\eta] = 0$, so the density factor does not contribute and the linearisation acts on plane waves $h_{ab}e^{ik\cdot x}$ with a symbol quadratic in k. Pairing it with a second perturbation q_{ab}, the antisymmetric part of the bilinear form is

$$q^{ab}\,\delta H_{ab}[h] - h^{ab}\,\delta H_{ab}[q] = \tfrac{1}{2}\cos^2\theta\,\big[(k\cdot q\cdot k)\ \mathrm{tr}\,h - (k\cdot h\cdot k)\ \mathrm{tr}\,q\big],$$

which vanishes identically only when $\cos\theta = 0$. There the operator is the Einstein tensor, and $\delta(R\sqrt{|g|}) = \sqrt{|g|}\,\mathcal{G}_{ab}\,\delta g^{ab}$ up to a divergence. □

The theorem concerns the displayed operator, with Euler–Lagrange expressions taken for variations of the metric at the density weight of $\sqrt{|g|}\,L$. It does not exclude other actions, constrained variations or equivalent formulations of the vacuum equations: since $\mathrm{tr}\,H(\theta) = \cos 2\theta\,R$, for $\cos 2\theta \neq 0$ the equations $H(\theta) = 0$ and $\mathrm{Ric} = 0$ have the same solutions, and at $\sin^2\theta = \frac{1}{2}$ the equation is the traceless one of the unimodular formulation [318].
Executable check: g1-6 (exact symbolic or integer arithmetic; Section 15.2).

For a diffeomorphism-invariant action Noether's second theorem makes the Euler–Lagrange tensor divergence-free, so Corollary 11.12(i) is the covariant necessary condition and Theorem 11.13 the full condition, obtained without assuming covariance. Both select the same phase.

11.2.2 The first-order form: the phase is a Barbero–Immirzi angle

Let e^a be a coframe, ω a Lorentz connection with curvature F^{ab}, and $\Sigma^{ab} = e^a \wedge e^b$. Both Σ and F are two-forms valued in $\Lambda^2\mathbb{R}^{1,3}$, which is the frame by Theorem 3.3. Pair them by $\langle \Sigma \wedge X\rangle = \frac{1}{2}\Sigma_{ab}\wedge X^{ab}$, orient by $\mathrm{vol} = e^0\wedge e^1\wedge e^2\wedge e^3$, and let $\mathcal{R}$ be the curvature operator of ω in the frame e.

Proposition 11.14 (Palatini and Holst on the frame)**.** *(i)* $\frac{1}{4}\varepsilon_{abcd}\,e^a\wedge e^b\wedge F^{cd} = \langle G\Sigma\wedge F\rangle = \mathrm{Tr}(\mathcal{R})\,\mathrm{vol}$*: the Palatini Lagrangian, which on the torsion-free locus is* $\frac{1}{2}R\,\mathrm{vol}$*.*

(ii) $e_a\wedge e_b\wedge F^{ab} = 2\langle\Sigma\wedge F\rangle = -2\,\mathrm{Tr}(G\mathcal{R})\,\mathrm{vol}$*: the Holst term is the first-Bianchi datum of Proposition 11.9.*

(iii) For the phase family, $\langle e^{\theta G}\Sigma\wedge e^{\theta G}F\rangle = \cos 2\theta\,\langle\Sigma\wedge F\rangle + \sin 2\theta\,\langle G\Sigma\wedge F\rangle = -\,\mathrm{Tr}(G\mathcal{R}_\theta)\,\mathrm{vol}$*. In the normalisation* $\frac{1}{2}\varepsilon_{abcd}e^ae^bF^{cd} - \gamma^{-1}e_ae_bF^{ab}$ *this is the Holst action [73] with*

$$\gamma = -\tan 2\theta,$$

the Barbero–Immirzi parameter.

(iv) The connection equation of $\int\langle e^{\theta G}\Sigma\wedge e^{\theta G}F\rangle$ *is* $e^{2\theta G}D\Sigma = 0$*, so the torsion vanishes, and the coframe equation is* $-2\sin 2\theta\,\mathcal{G}^{af} = 0$ *on the torsion-free locus. The vacuum equations are Einstein's for every* $\theta \notin \frac{\pi}{2}\mathbb{Z}$*. At the pure-Holst points* $\theta \in \frac{\pi}{2}\mathbb{Z}$ *no equation for the metric or coframe remains after the connection is eliminated, but the connection variation still gives* $D\Sigma = 0$ *and, for an invertible tetrad, vanishing torsion.*

Executable check: g1-7, g1-8 (exact symbolic or integer arithmetic; Section 15.2).

The roles of the phase differ between the two formalisms, as Figure 11.1 shows. In the second-order family the variational point is $\theta = \pi/2$; in the first-order family Einstein's equations hold for every $\theta \notin \frac{\pi}{2}\mathbb{Z}$, the Palatini point is $\theta = \pi/4$, and the endpoint $\theta = \pi/2$, where $\mathcal{R}_\theta = G\mathcal{R}G$, is exactly where the first-order Lagrangian degenerates to the pure Holst term. These are two uses of one rotation and must not be conflated. The self-dual value $\gamma = \pm i$ of Ashtekar's variables is not attained for any complex θ; it is approached as $\mathrm{Im}\,\theta \to \mp\infty$, where $e^{2\theta G}$ becomes

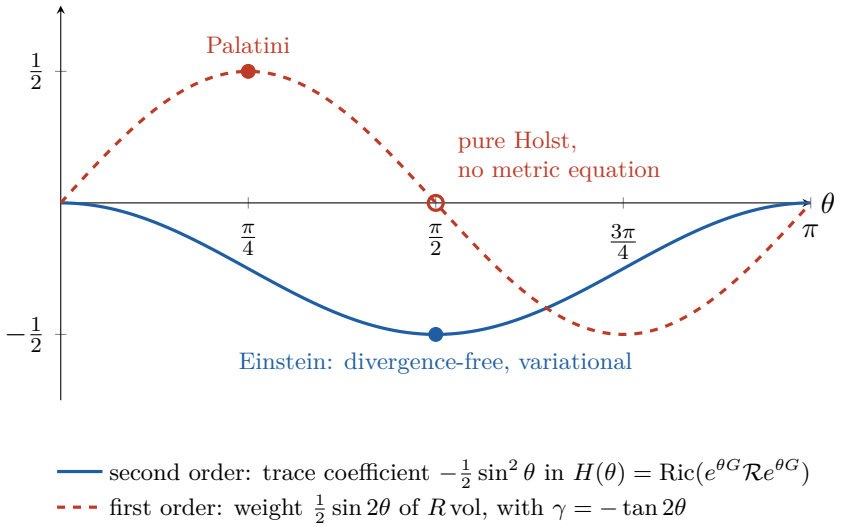


Figure 11.1: Two uses of one duality rotation of the curvature (Theorems 11.10 and 11.13 and Proposition 11.14). In the second-order family $H(\theta) = \mathrm{Ric}(e^{\theta G}\mathcal{R}e^{\theta G})$ the endpoint $\theta = \pi/2$ is the Einstein tensor, the only divergence-free and the only variational member. In the first-order family the same rotation weights the Einstein–Hilbert term by $\frac{1}{2}\sin 2\theta$: Einstein's equations follow for every $\theta \notin \frac{\pi}{2}\mathbb{Z}$, the Palatini point is $\pi/4$, and at $\theta = \pi/2$ the Lagrangian degenerates to the pure Holst term, with $\gamma = -\tan 2\theta$.

proportional to a projector onto an eigenspace of G. With torsion sources such as fermions the Holst term is no longer inert: for minimally coupled fermions the phase enters an effective four-fermion interaction [122, 42], while with a non-minimal coupling the Holst term can combine into the Nieh–Yan term and the phase can drop out [316]. Nothing here fixes either.

Both readings of $\mathbb{R}^{3,3}$ therefore construct the Einstein tensor of a given four-metric: the interior reading by restricting the six-dimensional Einstein tensor to the slice (Theorem 11.5), the field-strength reading as the contraction of $G\mathcal{R}G$ (Theorem 11.10). Neither determines the metric. No gravitational coupling, source or equation of motion for G follows, the selection criteria come from Riemannian geometry rather than from the Clifford structure, and the coefficient $\frac{1}{2}$ of the trace reversal comes from the algebra of the Hodge star.

11.3 The projector under transport

In a curved arena an observation projector becomes a field $\Pi(x) = \Pi(x)^2$ that is carried around by the spin connection, $D\Pi = d\Pi + [\Omega, \Pi]$. Two curvatures then compete on its image, and their competition decides whether the normal $U(1)$ of the model of Section 11.1.2 can be electromagnetism.

11.3.1 Two curvatures on the selected bundle

Differentiating $\Pi^2 = \Pi$ gives $\Pi(D\Pi)\Pi = 0$ and $(\mathbf{1}-\Pi)(D\Pi)(\mathbf{1}-\Pi) = 0$: the covariant derivative of a projector is purely off-diagonal, which is the configuration-space statement of Section 8.1 in covariant form. The induced connection $D^{\mathrm{obs}}s = \Pi Ds$ on the selected bundle has curvature

$$\mathcal{F}_{\mathrm{obs}} = \Pi\,\mathcal{R}\,\Pi + \Pi\,(D\Pi \wedge D\Pi)\,\Pi, \tag{11.2}$$

with $\mathcal{R}_{MN} = \frac{1}{4}R_{MNAB}\gamma^A\gamma^B$ the ambient spin curvature [85]. The two terms have distinct origins: ambient curvature, and the motion of the selected subspace relative to ambient transport. The flat case of Section 8.1 keeps only the second.

Proposition 11.15 (Transport obstruction)**.** *If ambient transport preserves the projector,* $D\Pi = 0$*, then* $[\mathcal{R}_{MN}, \Pi] = 0$*. For the paired-mode projector* $\Pi = \frac{1}{2}\mathbf{1} - \frac{1}{4}(\varpi + \omega)$ *of Example 1.6,* $\operatorname{rank}[J_N, \Pi] = 4$*, and in the model (11.1) with* $g^{(4)}$ *flat,* $\mathcal{R}_{\mu\nu} = \frac{1}{2}(F_N)_{\mu\nu} J_N$ *on* Σ*, so* $[\mathcal{R}_{\mu\nu}, \Pi] = \frac{1}{2}(F_N)_{\mu\nu}[J_N, \Pi] \neq 0$ *wherever the normal curvature* $F_N = dA$ *is nonzero.*

Proof. $D^2\Pi = [\mathcal{R}, \Pi]$ for any section of the endomorphism bundle, so $D\Pi = 0$ forces the condition. The rank is a direct computation in the 8×8 realisation. □

Executable check: g3-4 (exact symbolic or integer arithmetic; Section 15.2).

This statement is about the spinor projector and the spin curvature, and the corresponding tangent-level statement is false; the two must not be run together. At the tangent level J_N commutes with every generator of E (Lemma 11.2), so the rotation it generates fixes E pointwise, and in (11.1) the Christoffel symbols that mix E with N vanish on Σ: the splitting $TM = E \oplus N$ is Levi-Civita parallel there although $F_N \neq 0$. The symbols with two normal indices do not all vanish: on Σ, $\Gamma^u_{zv} = A_z$ and $\Gamma^v_{zu} = -A_z$ (for $A = bx\,dz$, $\Gamma^u_{zv} = bx$), the normal connection coefficients. At the spinor level the same rotation is not trivial, since $\operatorname{rank}[J_N, \Pi] = 4$. Both statements hold simultaneously, and together they are the content of the remark after Proposition 11.4: a spinless test particle in the zero-normal-momentum reduced sector cannot see the normal connection, and whatever makes it physical has to come from the spinor sector (the absence of that coupling does not establish a general spin requirement for Kaluza–Klein charge). The general principle that a normal-bundle curvature commutes with the projector is a statement about TM and does not transfer to End of the spinor bundle. Curvature cannot be chosen independently of the selection.

Proposition 11.16 (The adapted connection)**.** *Set* $\mathcal{C} := [\Pi, D\Pi]$ *and* $\widehat{D} := D + \mathcal{C}$*. Then* $\widehat{D}\Pi = 0$*. For the paired-mode projector, with the Levi-Civita spin connection,* $D\omega = 0$ *and* ω *commutes with bivectors, so* $\mathcal{C} = \frac{1}{16}[\varpi, D\varpi]$*, again a bivector: the correction stays inside* $\mathfrak{spin}(3,3)$*. The corresponding tangent connection is* $\widehat{\nabla}_X Y = \nabla_X Y + \frac{1}{2}\mathcal{K}(\nabla_X \mathcal{K})Y$*, metric-compatible and* $\mathcal{K}$*-preserving, hence generically with torsion.*

Proof. Write $\delta = D\Pi$, which is off-diagonal. Direct expansion gives $[[\Pi, \delta], \Pi] = -\delta$ for such δ, so $\widehat{D}\Pi = \delta + [\mathcal{C}, \Pi] = \delta + [[\Pi, \delta], \Pi] = 0$. The form of $\mathcal{C}$ follows from $D\Pi = -\frac{1}{4}D\varpi$ and $[\omega, D\varpi] = 0$. On tangent vectors $\mathcal{K}^2 = \mathbf{1}$ gives $\mathcal{K}(\nabla\mathcal{K})\mathcal{K} = -\nabla\mathcal{K}$, so the correction $\frac{1}{2}\mathcal{K}\nabla\mathcal{K}$ satisfies $\nabla\mathcal{K} + [\frac{1}{2}\mathcal{K}\nabla\mathcal{K}, \mathcal{K}] = 0$, and it is η-antisymmetric because $\mathcal{K}$ and $\nabla\mathcal{K}$ are η-antisymmetric and anticommute. The Levi-Civita connection is the unique torsion-free metric connection, so a metric connection preserving a structure that Levi-Civita transport does not preserve carries torsion. □

Executable check: g3-5, g3-6 (exact symbolic or integer arithmetic; Section 15.2).

The adapted generator is not an internal charge. In the model, with $\Omega = \frac{1}{2}AJ_+$ and $J_+ = J_N$, the adapted connection on Σ is $\widehat{\Omega} = \frac{A}{4}(J_+ - J_-)$ with $J_- = \frac{1}{\sqrt{3}}(e_5e_6 + e_6e_4 + e_4e_5)$, and indeed $[J_+ - J_-, \Pi] = 0$; but $[\frac{1}{2}(J_+ - J_-), e_4] = \frac{1}{\sqrt{3}}(e_6 - e_5) \neq 0$, and likewise for e_5 and e_6. Preserving the projector has bought commutation with Π at the cost of commutation with the physical generators, which is the first of the four conditions of Section 8.1.5. The identification of the normal $U(1)$ with electromagnetism is not available on this route.

Nor is the observed curvature the field-strength module of Proposition 3.7. $\mathcal{F}_{\text{obs}}$ takes values in $\operatorname{End}(\operatorname{im}\Pi)$, and the interior bivectors are not elements of that space: all six bivectors of the model's four-plane $E = \langle \widehat{W}, e_4, e_5, e_6 \rangle$ fail to commute with Π, for instance $[e_4e_5, \Pi] = \frac{1}{2}(e_2e_4 - e_1e_5)$ (g3-6). For the axial clock e_1 one of them, the boost e_1e_4, does commute with Π, being one of the pole operators whose sum is ϖ; the other five do not. Their compressions $\Pi B \Pi$ are endomorphisms of the image, but compression is not an algebra map and carries no Lie brackets. The model supplies a direct counterexample: with Π constant and $\Omega = \frac{1}{2}AJ_N$,

$\mathcal{F}_{\mathrm{obs}} = \frac{1}{2}\, dA\, \Pi J_N \Pi$, and exact expansion gives

$$4\sqrt{3}\, \Pi J_N \Pi = e_1e_2 - e_1e_3 + e_2e_3 - e_4e_5 + e_4e_6 - e_5e_6 \\ - (e_1e_2e_3e_4 + e_1e_2e_3e_5 + e_1e_2e_3e_6) + (e_1e_4e_5e_6 + e_2e_4e_5e_6 + e_3e_4e_5e_6),$$

with grade-four terms of the same size as the bivector terms. An electromagnetic two-form and a Lorentz-algebra-valued curvature two-form carry different kinds of index, and identifying them needs structure that is not supplied.

11.3.2 Projector dynamics is adapted transport

With the kinetic term $S_\Pi = \frac{f^2}{2} \int \sqrt{|g|}\, \mathrm{Tr}(D_\mu \Pi\, D^\mu \Pi)$ varied along the spin orbit, the Euler–Lagrange equation is (8.2) in curved form,

$$[\Pi, \Box_D \Pi] = 0 \qquad \Longleftrightarrow \qquad D_\mu \mathcal{C}^\mu = 0, \qquad \mathcal{C}_\mu = [\Pi, D_\mu \Pi],$$

so the conserved matrix current of the projector *is* the connection correction of Proposition 11.16: projector dynamics and adapted transport are the same data. Built with $\widehat{D}$ instead of D the functional vanishes identically, which is the same statement read backwards. By Theorem 1.13 and Example 8.3, the trace form on the spin orbit of the paired-mode projector has three positive and three negative directions, the compact and the boost generators, and on the manifold of orthogonal projectors it is positive definite.

Example 11.17 (The angle mode is a Stückelberg field)**.** The $U(1)$ direction is one of the healthy three: $T = [\frac{1}{2} J_N, \Pi]$ is symmetric with $\mathrm{Tr}(T^2) = +1$. But T is a tangent vector, not a generator, and conjugating by $e^{\theta T}$ with T symmetric leaves the orthogonal projectors; the correct one-angle ansatz conjugates by the compact generator itself,

$$\Pi(\theta) = e^{\theta J_N/2}\, \Pi_0\, e^{-\theta J_N/2}.$$

The one-angle geodesic condition $[\Pi, [B, [B, \Pi]]] = 0$ with $B = \frac{1}{2} J_N$ holds at Π_0 and hence along the orbit, so the ansatz is consistent with the projector equation, which reduces to an equation for $b_\mu = \partial_\mu \theta + A_\mu$. The action reduces to $\mathcal{L}_\theta = \frac{f^2}{2} b_\mu b^\mu$, invariant under $\theta \mapsto \theta - \lambda$, $A \mapsto A + d\lambda$: the projector angle is a Stückelberg field for the normal frame rotation [134]. Adjoining a Maxwell term by hand gives a Proca field with $m_A^2 = g_A^2 f^2$. Here $b = d\theta + A$ with $\theta \mapsto \theta - \lambda$, whereas the sections of the level set in the subsection "Off the slice" used $B = d\chi - A$ with $\chi \mapsto \chi + \lambda$; the two conventions are related by $\theta = -\chi$.

Executable check: g3-7 (exact symbolic or integer arithmetic; Section 15.2).

Read strictly, this is a constraint and not a prediction: neither the Maxwell term nor the scales g_A and f is derived, and a massive normal vector is not the photon. A Maxwell term added by hand does not cure the wrong-sign term obtained from the ambient Einstein–Hilbert action (Corollary 11.6), and the positive norm of the angle mode does not make the projector sector stable, since its trace form has signature $(3, 3)$. Together with the failure of the adapted generator to commute with the physical generators it leaves a dichotomy: either the normal $U(1)$ is not electromagnetic, or the paired-mode projector is not the observation projector. Which alternative holds is left open here; the torsion discussion below concerns the nature of the flat obstruction, not this choice.

11.3.3 Is the flat obstruction a torsion statement?

The factorisation condition $a = \frac{1}{2}(G - G^\top) = 0$ of Theorem 4.2 survives covariantisation, and requiring transport to preserve the observation projector adds to the Levi-Civita connection a frame-antisymmetric correction, which is contorsion. One may conjecture that these are the

same object: that in a first-order formulation with an independent connection [120] G is not a background but the connection's own shift. The types do not yet match, and that is part of the work: a is an endomorphism while contorsion is a one-form valued in metric-skew endomorphisms, so a relation such as $a = K_\mu \dot{x}^\mu$ needs a stated contraction and normalisation before it is even a proposition. And the flat exclusion cannot be borrowed for it: by Theorem 4.5 and Theorem 4.8, what repairs binding is the split form, not curvature or torsion, so the antisymmetric part a is not what binding was waiting for.

11.4 What the curvature results do and do not establish

An identity that contains the Einstein tensor is not a derivation of Einstein's equations. The results of this chapter construct the Einstein tensor of a four-metric that is given, in two different ways, and they find two sectors whose sign is wrong. They do not supply an equation that determines the metric. Table 11.1 sorts the readings by what each constructs and what it lacks.

The missing dynamics. Against the requirements that a gravitational interpretation has to meet, the chapter stands as follows.

(i) *Metric or tetrad dynamics and action.* No action for the six-dimensional metric, the pairing $\mathcal{K}$, the connection and the projector is derived. The only gravitational action that appears, the Palatini–Holst action of Proposition 11.14, is adopted; the arena contributes the reading of the Hodge star as G and of the Barbero–Immirzi parameter as a duality phase. If the six-dimensional Einstein–Hilbert action is adopted instead, its reduction over the normal disc has the wrong relative sign between curvature and field strength (Corollary 11.6(ii)), and its vacuum equations leave no Maxwell field (Corollary 11.6(iii)).

(ii) *Constraints and physical degrees of freedom.* No constraint analysis of a gravitational field is made, and no count of physical degrees of freedom follows from the arena. The constraints that do appear, $p_u = p_v = 0$ of Proposition 11.4, belong to a test worldline in a given metric. For the adopted Palatini–Holst action with an invertible tetrad and a nonzero Palatini coefficient, $\sin 2\theta \neq 0$, the count is the inherited one of general relativity, two in vacuum, and the Holst term does not change it. At the pure-Holst points $\theta \in \frac{\pi}{2}\mathbb{Z}$ the coframe equation on the torsion-free locus, $-2\sin(2\theta)G_{ab} = 0$, is empty (a torsion-free metric with $R_{ab} = 3Kg_{ab}$, $K \neq 0$, satisfies it although $G_{ab} \neq 0$), and the count of general relativity does not apply (Proposition 11.14).

(iii) *Kinetic signs.* Relative to the ambient Einstein–Hilbert action and the stated connection-metric ansatz, the vectors of the normal plane gravitate with negative energy (Corollary 11.6(i), Theorem 12.15). The remark below shows that this sign is not a convention.

(iv) *Matter coupling.* After the reduction a spinless test particle moves on the geodesics of h, $H_{\mathrm{red}} = \frac{1}{2}h^{\mu\nu}p_\mu p_\nu$ (Proposition 11.4), which is the minimal coupling of a test particle to a given metric. No stress tensor of the arena's matter is shown to source h. The Dirac operator of the chiral block on a curved base is not computed here; its spectrum and index are listed as open in Chapter 13.

(v) *Low-energy limit.* No Newtonian or post-Newtonian limit is derived, since h is given and no equation determines it.

Remark 11.18 (The wrong sign is not a convention)**.** Reversing the overall sign of the metric, $\eta \mapsto -\eta$, does not remove the negative energy of the normal-plane vectors. Under $g \mapsto -g$ the Ricci tensor and the Einstein tensor are unchanged, R and $W_{\mu\nu}$ change sign, and F^2 is unchanged. The physical Einstein–Maxwell equation reads $\mathcal{G} = -2W$ in signature $(+,-,-,-)$

and $\mathcal{G} = +2W$ in $(-,+,+,+)$, so the healthy ratio of the Lagrangian $a_1 R + a_2 F^2$ changes sign with the convention, $a_2/a_1 = +1$ in one and -1 in the other. The reduced ratio changes sign with it: $-r_c^2/8$ in the first convention and $+r_c^2/8$ in the second. The mismatch between the two is therefore the same in both conventions. The same conclusion follows without any action: a Lorentzian four-plane of $\mathbb{R}^{3,3}$ of either signature, $(1,3)$ or $(3,1)$, has a definite normal plane of the sign of its own time direction (Theorem 12.4(ii)), and the coefficient of F^2 in the six-dimensional scalar curvature is odd in that sign (Theorem 12.15). Choosing the normal plane of the sign of space would make the four-plane of signature $(2,2)$, which is not Lorentzian.

Executable check: g2-8 (exact symbolic or integer arithmetic; Section 15.2).

The results of this chapter are therefore of two kinds. The fixed ones are identities: the para-Hermitian structure, the closed-form Einstein tensor of the model, the curvature operator on the frame with the Bianchi identity as one trace, the phase family and its two selection criteria, and the first-order reading of the phase. The obstructed ones are the normal $U(1)$ as electromagnetism, on three counts, the healthy sign of every normal-plane vector, and the transfer of Ricci flow to split signature. A gravitational theory in the arena would need an action relating metric, pairing, connection and projector from which the metric follows rather than being given; that is the largest open problem of the curved arena, and it is listed with the others in Chapter 13. The binding, topological and quantum-stability qualifications of the interior dynamics are stated where it is developed, in Section 4.6.3.

reading	what is constructed	what is missing
ambient Einstein tensor on the slice (Theorem 11.5)	for the given model (11.1), the tangential block of $\mathcal{G}^{(6)}$ on Σ is $\mathcal{G}[h]$ exactly	an action for g_6; the six-dimensional vacuum equations force $F = 0$ and $\mathrm{Ric}[h] = 0$ (Corollary 11.6(iii)), and the reduced Einstein–Hilbert action has the wrong relative sign (Corollary 11.6(ii)); no source, no coupling constant, no field equation for h with matter
double dual in the field-strength reading (Theorems 11.10 and 11.13 and Corollary 11.12)	for every metric, $\mathrm{Ric}(G\mathcal{R}G)$ is the Einstein tensor; divergence-freedom and variationality select $\theta = \pi/2$	the selection criteria are those of Riemannian geometry and of the Helmholtz conditions, not of the Clifford structure; the variational principle $R\sqrt{\lvert g\rvert}$ is the classical one; no equation $\mathcal{G} = 8\pi G_{\mathrm{N}} T$ and no matter stress tensor
first-order form (Proposition 11.14)	the Palatini–Holst action, with the duality phase as Barbero–Immirzi parameter $\gamma = -\tan 2\theta$; vacuum equations are Einstein's for $\theta \notin \frac{\pi}{2}\mathbb{Z}$	the action is adopted, not derived, and it is ordinary general relativity; the phase is not fixed; with fermionic torsion it enters a four-fermion interaction under minimal coupling, and can drop out under a non-minimal one; nothing here determines it
normal-plane vectors (Corollary 11.6(i), Example 11.7 and Theorem 12.15)	a Kaluza–Klein vector from the normal plane	healthy sign: relative to the ambient Einstein–Hilbert action and the stated connection-metric ansatz, every such vector gravitates with negative energy, and spinless matter in the zero-normal-momentum reduced sector ($p_u = p_v = 0$, hence $\ell = 0$) does not couple to it; a spinless trajectory with $\ell \neq 0$ does (Proposition 11.4 and Theorem 12.12)
projector under transport (Propositions 11.15 and 11.16 and Example 11.17)	an adapted connection with torsion; the projector angle as a Stückelberg field	a gravitational role is not proposed; the vector it gives mass to is the normal one, not the photon and not a graviton

Table 11.1: The gravitational readings of this chapter: what each constructs for a given metric and what a gravitational theory would still need.

Chapter 12

Three Temporal Directions and Causal Reduction

The question of this chapter is whether the Lorentzian four-plane used in every earlier chapter, and with it the light cone and the distinction between past and future, can be obtained from the split geometry rather than postulated.

Motivation and scope. This investigation originated in the intuition that observable causal order might follow from the collective dynamics of the underlying system, rather than enter through an independently prescribed choice of a Lorentzian sector. The quantum motivation comes from the sum-over-histories formulation, in which complex probability amplitudes associated with admissible histories are combined before probabilities are extracted [322]. Applied to the Spacetime–Electromagnetic Framework, this suggests asking whether quantum dynamics and physical-state reduction associated with the split-signature geometry could determine an effective Lorentzian causal structure.

This possibility is a research conjecture. Summing amplitudes does not by itself derive a spacetime signature, select a physical four-plane, or establish microcausality. A complete realisation would require a specified space of histories, an action and integration prescription, a consistent treatment of constraints, and a positive physical state space. In particular, the Lorentzian cone whose emergence is sought cannot already be imposed through the admissibility conditions.

The present chapter investigates prerequisites and obstructions to this programme through explicit models. Its distinction between weak and strong emergence (Section 12.1) separates consistency within a selected Lorentzian sector from a dynamical explanation of that selection. The conditional scalar reduction of Corollary 12.14 achieves the former; the proposed emergence of causal order from the combined quantum dynamics remains open, and the obstructions below hold for the models and assumptions in which they are proved.

The chapter starts from the free field on $\mathbb{R}^{3,3}$, which for unrestricted data is ill-posed and has no time orientation (Section 12.1); identifies the choice of a four-plane with the choice of a complex structure and recovers one nappe of a light cone from twisted squares of spinors (Section 12.2); shows that no unconstrained $SO(3,3)$-invariant adjoint scalar order parameter selects that choice cleanly (Section 12.3); and derives the surplus-time constraints from a gauge symmetry of the worldline that is chosen, not derived (Section 12.4). Section 12.5 then analyses the clock reduction of Assumption 2.16, the two parameters called τ in this manuscript, and what the reduction keeps and removes; Section 12.6 states the universal-evolution hypothesis as an open hypothesis, with the ingredients a proposal would need. Section 12.7 collects the outcome. The chapter uses the clock and the interior four-plane of Chapter 2 and the normal-plane model of Chapter 11; the compatibility chapter (Chapter 13) uses its verdicts in the spacetime chain.

Several letters carry more than one meaning in this chapter, each fixed by its context: B is

the order parameter of Section 12.3, the Kaluza–Klein potentials B^u, B^v of Theorem 12.15 and, in bold, the magnetic field; C is the selector of Section 12.2 and the function $C = A_\mu h^{\mu\nu} k_\nu$ of Theorem 12.12; N is the normal plane and the lapse; n is the null rotation of Theorem 12.9 and a lattice vector in Example 12.3; and $K = \frac{1}{2} e_1 e_4$ in Theorem 12.9 is distinct from the grading root $\mathcal{K}$ and from the kernels $K_{\psi\chi}$. The letters $a, \dots, h$, κ, λ, λ_2 and m in Theorem 12.8 and its proof are coefficients local to that proof, and the degree-eight function there is written F.

12.1 Causal structure: the six-dimensional baseline

Every construction so far has used a Lorentzian four-plane chosen inside $\mathbb{R}^{3,3}$: the interior plane E of the Dirac structure, or the slice Σ of the curved model. Two levels must be distinguished. *Weak emergence* means that an already selected 3+1 sector carries a consistent quantum causal structure. *Strong emergence* means that the dynamics determine why that sector, with that causal structure, arises from the six-dimensional theory. The light cone must not be put into the admissibility conditions and then be said to emerge. We fix the failure conditions in advance: uncontrolled growing modes in the six-dimensional theory; a projector onto the four-dimensional sector imposed by hand; ghosts or negative spectral weights in the effective theory; propagation outside the derived cone; extra modes or imposed equations in the Maxwell sector.

12.1.1 The unrestricted scalar

Proposition 12.1 (A free field on $\mathbb{R}^{3,3}$)**.** *For the action $S_6 = \frac{1}{2}\int d^6X\,(\eta^{AB}\partial_A\phi\,\partial_B\phi - m^2\phi^2)$ in coordinates (t,u,v,x,y,z), with t,u,v temporal:*

(i) plane waves $e^{-i\omega t + i(q_1 u + q_2 v) + i\mathbf{k}\cdot\mathbf{x}}$ obey

$$\omega^2 = |\mathbf{k}|^2 + m^2 - q^2, \qquad q^2 = q_1^2 + q_2^2; \tag{12.1}$$

(ii) for $q^2 > |\mathbf{k}|^2 + m^2$ the growth rate is $\gamma = \sqrt{q^2 - |\mathbf{k}|^2 - m^2}$, and $(1+q^2)^{-s/2} e^{\gamma} \to \infty$ for every s. Plane waves are not of finite norm on the noncompact Cauchy surface, so the argument is applied to wave packets concentrated at large surplus momentum, or to plane waves in a periodic box: data small in any Sobolev norm produce arbitrarily large solutions at $t = 1$, so the Cauchy problem with respect to t is ill-posed in the sense of Hadamard [63];

(iii) the canonical energy density $\frac{1}{2}(\pi^2 - \phi_u^2 - \phi_v^2 + |\nabla_{\mathbf{x}}\phi|^2 + m^2\phi^2)$ averages to $\frac{1}{4}a^2(m^2 - q^2)$ on $\phi = a\sin qu$ in a periodic box in u, and approximately so on wave packets concentrated near such a mode: the energy is unbounded below.

Executable check: c2-1 (exact symbolic or integer arithmetic; Section 15.2).

Proposition 12.2 (No time orientation in signature $(3,3)$)**.** *The timelike set $\{\eta(X,X) > 0\}$ of $\mathbb{R}^{p,q}$ is connected for $p \geq 2$: the path $X(s) = (\cos s, \sin s, 0; 0, 0, 0)$ joins X to $-X$ with $\eta = 1$ throughout. Only for $p = 1$ does it split into a past and a future component.*

Executable check: c2-2 (exact symbolic or integer arithmetic; Section 15.2).

The split metric therefore supplies no canonical future cone invariant under the identity component $SO^+(3,3)$: a cone, or a causal order, can be introduced only by additional data. Anything called a light cone with a future has to come from such data, here a reduction to a single time. The arena already contains such a reduction, as an input. For the geodesic test particle of Proposition 11.4, $H = \frac{1}{2}(h^{\mu\nu}k_\mu k_\nu + p_u^2 + p_v^2)$ with $h = \mathrm{diag}(1,-1,-1,-1)$, the mass shell $2H = m^2$ is $p_0^2 = |\mathbf{p}|^2 + m^2 - p_u^2 - p_v^2$, the relation (12.1) with $q = (p_u, p_v)$. The first-class constraints $p_u = p_v = 0$ remove every mode with nonzero surplus momentum, as Figure 12.1 shows: the unstable band $q^2 > |\mathbf{p}|^2 + m^2$, and with it the stable modes $0 < q^2 \leq |\mathbf{p}|^2 + m^2$, keeping only $q = 0$. They are sufficient to remove the instability but not minimal; the minimal restriction, $q^2 \leq |\mathbf{p}|^2 + m^2$, depends on the four-momentum and is not of this local form.

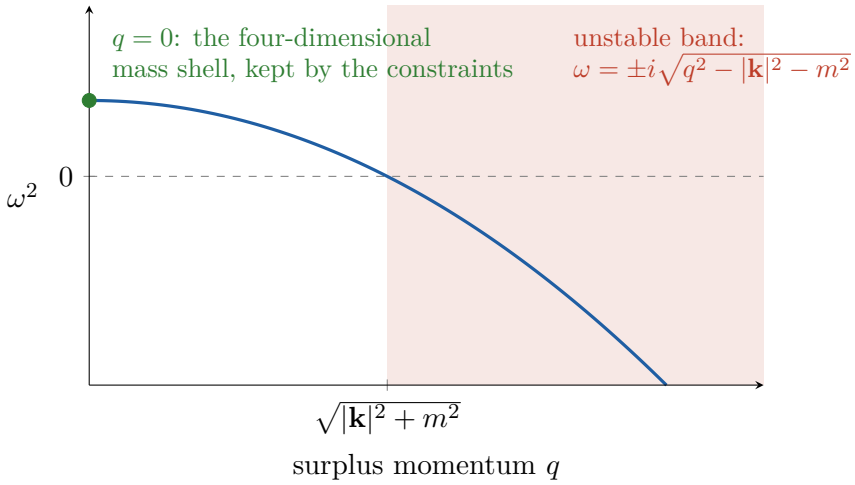


Figure 12.1: The dispersion relation of a free scalar on $\mathbb{R}^{3,3}$ at fixed spatial momentum (Proposition 12.1). Every surplus momentum beyond $\sqrt{|\mathbf{k}|^2 + m^2}$ gives an exponentially growing mode, with a rate that grows without bound. The constraints of Proposition 11.4 keep exactly the point $q = 0$.

Example 12.3 (Compact surplus times give tachyons)**.** If instead u and v are made periodic with period $2\pi R$, the surplus momenta become n/R with $n \in \mathbb{Z}^2$, and the four-dimensional masses are

$$m_n^2 = m^2 - \frac{|n|^2}{R^2}.$$

Every mode with $|n| > mR$ is a tachyon, and there are infinitely many: for $mR = 3$ there are 92, 412 and 1652 of them within $|n_i| \le 5$, 10 and 20. Compactification, the usual cure for extra space dimensions, is no cure for extra times.

Executable check: c2-3, c2-4 (exact symbolic or integer arithmetic; Section 15.2).

An ultrahyperbolic problem can be made well posed by a nonlocal constraint on the data [27], and the dimensionality of spacetime has been argued from such considerations [149]. The constraints of the arena, $\partial_u \phi = \partial_v \phi = 0$, are local and stronger: they keep only the zero modes, which is dimensional reduction. Either way the cure is a restriction, and at this stage it is an input. Section 12.4 asks where it can come from; the next two sections ask whether the four-plane it presupposes can be selected.

12.2 The kinematic selector

Choosing a Lorentzian four-plane inside $\mathbb{R}^{3,3}$ is equivalent to choosing its orthogonal complement, a two-plane, and a two-plane is described by its unit bivector. The question is which bivectors describe a Lorentzian complement, and what the arena does with them.

12.2.1 One constituent

Theorem 12.4 (The selector is a complex structure)**.** *Let $C = a \wedge b$ be a simple bivector of $\mathbb{R}^{3,3}$ and $N = \langle a, b \rangle$.*

(i) $C^2 = -\det \mathrm{Gram}(a, b)\, \mathbf{1}$.

(ii) For a nondegenerate N with C normalised: $C^2 = -\mathbf{1}$ exactly when N is definite, that is exactly when $N^\perp$ is Lorentzian, of signature $(1,3)$ when N is positive and $(3,1)$ when N is negative; and $C^2 = +\mathbf{1}$ exactly when $N^\perp$ has signature $(2,2)$. By (i), $C^2 = 0$ exactly when N is degenerate.

(iii) For a nondegenerate N, the centraliser of C in $\mathfrak{so}(3,3)$ is $\mathfrak{so}(N) \oplus \mathfrak{so}(N^\perp)$, of dimension seven, so the choices of N of a given nondegenerate type form an eight-dimensional homogeneous space. The degenerate strata differ: for the totally null plane $\langle e_1 + e_4, e_2 + e_5\rangle$, $C_0 = (e_1 + e_4)(e_2 + e_5)$ has $C_0^2 = 0$ and a nine-dimensional centraliser.

(iv) The surplus plane is such an N, with $C = J = e_2e_3$ and $N^\perp$ the interior four-plane.

Proof. (i) is the Clifford identity $(a \wedge b)^2 = (a \cdot b)^2 - a^2b^2$. (ii) For an orthonormal pair, $(ab)^2 = -a^2b^2$, which is -1 when $a^2 = b^2$ and $+1$ otherwise, and the signature of $N^\perp$ is the complement of that of N in $(3,3)$. (iii) holds for coordinate planes by (1.3), and every nondegenerate plane is an $SO(3,3)$-image of a coordinate plane of the same signature, by Witt's theorem; the centraliser of C_0 is computed directly. □

Executable check: c1-1, c1-12 (exact symbolic or integer arithmetic; Section 15.2).

For coordinate planes C is moreover antisymmetric on S when $C^2 = -\mathbf{1}$, an orthogonal complex structure; for tilted planes it is not. Orthogonality refers to the Euclidean structure of S, which singles out the temporal triple, while $C^2 = -\mathbf{1}$ is invariant. The distinction will matter.

Spinors give the light cone. The volume element $\omega_S = e_4e_5e_6$ satisfies $\omega_S e_A \omega_S^{-1} = -e_A^\top$ for every generator, so $\langle \psi, \varphi\rangle = \psi^\top \omega_S \varphi$ is a $\mathrm{Spin}(3,3)$-invariant pairing, and it pairs S^+ with S^-.

Theorem 12.5 (The light cone as twisted squares)**.** *Let ψ range over the chiral half S^+.*

(i) The forms $m_A = \omega_S e_A$ are antisymmetric and satisfy $X^\top m_A + m_A X = -\omega_S[X, e_A]$ for every bivector X; their restrictions to S^+ span $\Lambda^2(S^+)^*$, so $\psi \wedge \varphi \mapsto (\psi^\top \omega_S e_A \varphi)_A$ is an equivariant isomorphism $\Lambda^2 S^+ \cong \mathbb{R}^{3,3}$, the Klein correspondence of Theorem 3.84.

(ii) On S^+, $J^2 = -\mathbf{1}$, so $S^+ \cong \mathbb{C}^2$, and the interior $\mathfrak{so}(1,3)$ acts $\mathbb{C}$-linearly, as $\mathfrak{sl}(2,\mathbb{C})$.

(iii) For a unit simple bivector C with plane N define $v_C(\psi)_A = \psi^\top \omega_S e_A C \psi$. Then $v_C(\psi) \in N^\perp$ and $\eta(v_C, v_C) = 0$ identically, and the image has dimension three.

(iv) If $C^2 = -\mathbf{1}$, the component of v_C along the timelike direction of $N^\perp$ is definite: v_C fills one nappe of the light cone of $N^\perp$. If $C^2 = +\mathbf{1}$, no component is definite, and v_C fills the null vectors of a $(2,2)$ plane, which has no time orientation.

(v) $v_{-C} = -v_C$; on S^- the J-twisted square lies in the other nappe; and on S^+, $v_J = -j$, where $j_A = \psi^\top \gamma^0 e_A \psi$ is the Dirac current.

Executable check: c1-2, c1-3 (exact symbolic or integer arithmetic; Section 15.2).

The selector of the reduction is therefore not a new field. Choosing the spacetime $N^\perp$ and choosing the complex structure C are one choice, and in the arena that complex structure is the charge J, the same J that is the imaginary unit of the Dirac theory (Theorem 2.17). Null directions are the squares of Weyl spinors twisted by J, and the time orientation is fixed by the sign of J together with the chirality. The mathematics is the Klein–Penrose correspondence, in which a null vector is $\psi\bar{\psi}$ [319, 121, 157]; what is specific to the arena is that J already plays this role. All of this is kinematics: nothing yet says why J, rather than an indefinite or degenerate bivector, is chosen. For degenerate planes v_C is still null with a three-dimensional image, and for the totally null plane $\langle e_1 + e_4, e_2 + e_5\rangle$ it lies in the totally null three-plane $\langle e_i + e_{i+3}\rangle$, an eigenspace of the grading root $\mathcal{K}$. A six-dimensional Dirac model of signature $(3,3)$ with a locally gauged $SO(3,3)$, broken locally to the Lorentz group of a selected time direction, was studied by Dartora and Cabrera [288]. That general architecture is prior art; what is examined here is the identification of the selector with J and the obstructions to its dynamical selection (Section 14.2).

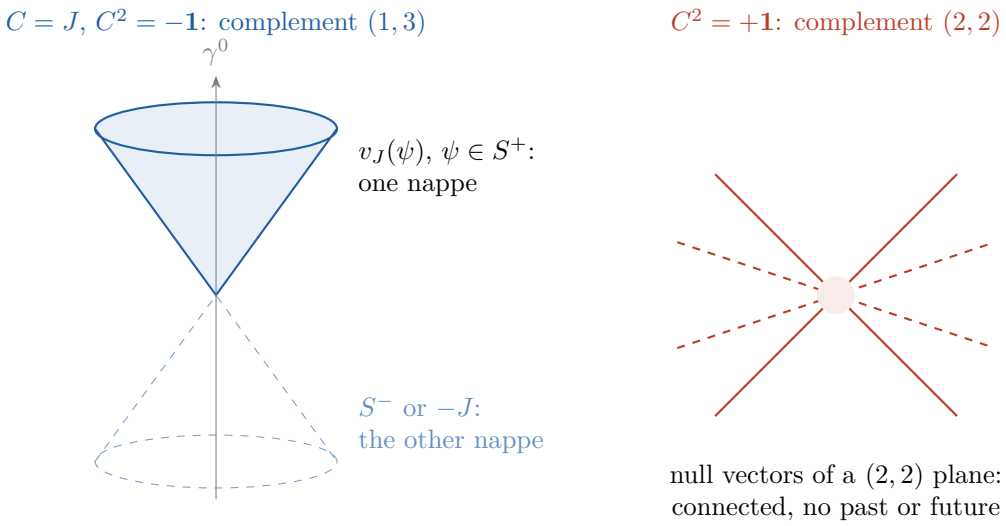


Figure 12.2: The selector at one constituent (Theorems 12.4 and 12.5). A unit bivector C with $C^2 = -\mathbf{1}$ selects a Lorentzian complement, and the C-twisted squares of chiral spinors fill one nappe of its light cone, the other nappe belonging to the other chirality or to $-C$. A bivector with $C^2 = +\mathbf{1}$ selects a $(2,2)$ complement, whose null set has no time orientation.

12.2.2 Three constituents

At three constituents J is the product of the fourteen generators of the Lorentz complement W_3, of signature $(8,6)$, and Ω is the volume element of E.

Theorem 12.6 (The selector at three constituents)**.** *(i)* $J^2 = \Omega^2 = -\mathbf{1}$, $[J,\Omega] = 0$ *and* $\Gamma = J\Omega$ *with* $\Gamma^2 = \mathbf{1}$*;* $\Omega = -J$ *on* $\Gamma = +1$*. Since* $\Gamma^2 = \mathbf{1}$ *at every rung, any coordinate four-plane* E' *with volume element* $\Omega_{E'}$ *and complementary volume element* $J_{E'}$ *has* $J_{E'}^2 = \Omega_{E'}^2 = \prod_{i\in E'} e_i^2$*: the volume element of the complement is a complex structure exactly when* E' *is Lorentzian, at every* n*.*

(ii) The centraliser of J *among the* 153 *bivectors has dimension* $97 = 6 + 91$*, so the choices form a space of dimension* 56*.*

(iii) With either $\mathrm{Spin}(9,9)$*-invariant pairing, the product* β_{odd} *of the nine negative generators, which is antisymmetric, or the product* β_{even} *of the nine positive ones, which is symmetric,* $\beta e_A J$ *is symmetric exactly for the fourteen directions of* W_3*: the* J*-twisted square lives on the complement, not on* E*. The untwisted* $\beta_{\mathrm{odd}} e_A$ *is symmetric for all eighteen directions, so the vector sits in* $\mathrm{Sym}^2 S^+$*, and the untwisted square is not null at random spinors of* S^+*.*

Executable check: c1-4, c1-5 (exact symbolic or integer arithmetic; Section 15.2).

The identification of the selector with J survives at three constituents, but the covariant construction of the cone from spinors is a feature of dimension six. At $n = 3$ the cone of E is recovered by the Dirac current $j_\mu = \psi^\top \gamma^0 \gamma_\mu \psi$, which is future causal, being a sum of one-constituent currents, and null on product states whose first factor is chiral (for the first factor $|000\rangle + |100\rangle$ it is $(2,0,0,0)$, timelike); but it uses the clock γ^0 and the transpose, that is, the positive inner product of the spinor module.

12.3 Type selection

Kinematics identifies the selector with a complex structure. Dynamics would have to explain why the field that plays this role takes a value of the complex-structure type at all, rather than

an indefinite or degenerate one. The natural test is a potential: take the order parameter to be an unconstrained bivector field $B \in \mathfrak{so}(3,3)$, an adjoint scalar, ask that the potential be invariant under $SO(3,3)$, so that nothing is put in by hand, and require that its minimum set be exactly the orbit of J, over an open set of couplings, with healthy fluctuations. Restriction to S^+ is the isomorphism $\mathfrak{so}(3,3) \cong \mathfrak{sl}(4,\mathbb{R})$, and the invariant polynomials are the polynomials in

$$t_k(B) = \operatorname{tr}\big((B|_{S^+})^k\big), \qquad k = 2,3,4,$$

the coefficients of the characteristic polynomial; a continuous invariant function is constant on orbit closures (for the real orbits and their closures see [320, 321]).

Proposition 12.7 (Invariants of the three types)**.** *Rotations of a positive plane and of a negative plane both have* $(t_2,t_3,t_4) = (-4,0,4)$ *and characteristic polynomial* $(y^2+1)^2$*; boosts of an indefinite plane have* $(4,0,4)$ *and* $(y-1)^2(y+1)^2$*. The positive and the negative planes induce opposite orientations of* $S^+ = \mathbb{R}^4$*: they are distinct* $SL(4,\mathbb{R})$*-orbits with identical invariants.*

Executable check: c1-6 (exact symbolic or integer arithmetic; Section 15.2).

So no invariant polynomial can prefer $(1,3)$ to $(3,1)$. Smooth invariant functions can separate the two orientations, since the orbits are distinct; the stronger obstruction is the orbit closure of Theorem 12.9(i). The two are exchanged by the anti-isometries $\mathcal{K}$ and G and correspond to the two sign conventions for a Lorentzian metric; physically this is harmless, but it means the selection is never unique. The type itself is harder.

Theorem 12.8 (No clean selection below degree eight)**.** *Let* V *be an* $SO(3,3)$*-invariant polynomial on* $\mathfrak{so}(3,3)$*, bounded below.*

(i) If $\deg V \le 4$*, then* $V = m\,t_2 + \lambda\,t_2^2 + \text{const}$.

(ii) If $\deg V \le 7$*, then* V *restricted to* $\{t_3 = 0\}$ *is a polynomial of degree at most two in* t_2 *alone.*

(iii) Hence if J *minimises* V*, so does* $B_w = \frac{3}{5}e_2e_3 + \frac{4}{5}e_5e_6$*, which has* $t_2 = -4$*,* $t_3 = 0$ *and* $t_4 = \frac{4804}{625} \neq 4$*, is not simple, and selects no four-plane.*

(iv) In degree eight, $F = (t_2+4)^2 + t_3^2 + (t_4-4)^2$ *vanishes exactly on the elements with characteristic polynomial* $(y^2+1)^2$*, and its Hessian at* J *has rank one.*

Proof. Use test elements of $\mathfrak{sl}(4,\mathbb{R})$ with rational entries. $A_0 = \operatorname{diag}(1,-1) \oplus \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ and $B_0 = \left(\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right) \oplus \left(\begin{smallmatrix} -1 & -1 \\ 1 & -1 \end{smallmatrix}\right)$ have $t_2 = t_3 = 0$ and $t_4 = 4, -16$; $C_0 = \operatorname{diag}(-1,-3) \oplus \left(\begin{smallmatrix} 2 & -3 \\ 3 & 2 \end{smallmatrix}\right)$ has $t_2 = 0$ and $t_3 = -120$; the complex and para-complex structures have $(t_2,t_4) = (\mp 4, 4)$; $D_0 = \operatorname{diag}(1,-1,2,-2)$ and two rotations of speeds $1,2$ have $(t_2,t_4) = (\pm 10, 34)$, all with $t_3 = 0$. (i) Along sA_0 and sB_0 a quartic term $\lambda_2 t_4$ would give $\lambda_2 t_4 s^4$ with both signs of t_4, so $\lambda_2 = 0$ for boundedness; along sC_0 a cubic term gives $-120\kappa s^3$, so $\kappa = 0$. (ii) The top homogeneous part must be non-negative, so an odd top degree vanishes, and in degree six the top part $e\,t_2^3 + f\,t_2^2 + g\,t_2t_4$ takes the values $\mp(64e+16g)$ on the complex and para-complex structures and $\pm(1000e+340g)$ on the last two elements; non-negativity forces $g = -4e$ and then $e = g = 0$, with $f \ge 0$ from C_0. The quartic coefficient of t_4 must again vanish, leaving $a\,t_2 + b\,t_3 + c\,t_2^2 + h\,t_2t_3 + f\,t_3^2$, which on $t_3 = 0$ is $a\,t_2 + c\,t_2^2$. (iii) follows. (iv) $F \ge 0$ with equality exactly when $(t_2,t_3,t_4) = (-4,0,4)$, which fixes the characteristic polynomial by Newton's identities; at J the gradients of t_3 vanish and those of t_2 and t_4 are proportional, so the Hessian has rank one. □

Executable check: c1-7, c1-8 (exact symbolic or integer arithmetic; Section 15.2).

Even where a potential does isolate the type, the minimum cannot be clean, and the reason holds at every degree.

Theorem 12.9 (Degeneracy and flat directions at every degree)**.** *Let* V *be any* $SO(3,3)$*-invariant function on* $\mathfrak{so}(3,3)$.

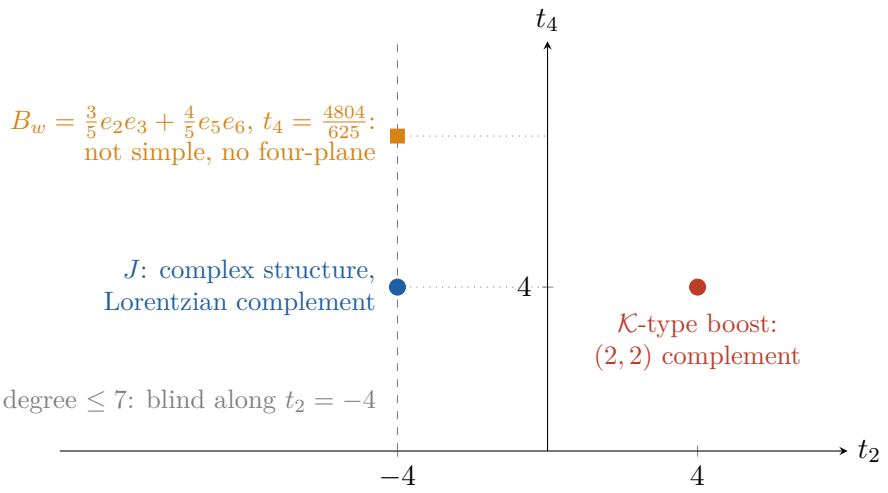


Figure 12.3: Type selection in the plane of the invariants (t_2, t_4) at $t_3 = 0$ (Theorem 12.8). A bounded-below invariant potential of degree at most seven depends on t_2 alone along $t_3 = 0$, so it cannot distinguish the complex structure J from the non-simple element B_w on the same vertical line. Degree eight is the first that can.

(i) Let $n = (e_1 + e_4)e_5$, a null rotation of the interior four-plane. Then $[J, n] = 0$ and $n^2 = 0$, so $J+n$ has the invariants of J, but its centraliser has dimension three and does not contain $\mathfrak{so}(1,3)$: it breaks Lorentz invariance. With $K = \frac{1}{2}e_1e_4$, $[K, n] = -n$ and $[K, J] = 0$, so conjugation by e^{tK} maps $J + n$ to $J + e^{-t}n$. Hence J lies in the closure of the orbit of $J + n$, and $V(J + n) = V(J)$ whenever V is continuous.

(ii) If V is smooth and has a local minimum at J, its Hessian vanishes on the six $\mathfrak{so}(1,3)$ directions. The $\mathrm{Stab}(J)$-invariant symmetric forms on $\mathfrak{so}(1,3)$ are the real and imaginary parts of the complex Killing form of $\mathfrak{sl}(2,\mathbb{C})$, and each nonzero one has signature $(3,3)$, so a positive semidefinite one is zero.

At best the fluctuations about J are one massive radial mode, eight Goldstone modes and six massless modes that are not Goldstone modes: non-Goldstone flat directions, whose vanishing masses are enforced by invariance and stability (ii), not by a broken symmetry.

Executable check: c1-9 (exact symbolic or integer arithmetic; Section 15.2).

Proposition 12.10 (Kinetic terms)**.** *Take B an unconstrained adjoint scalar with a factorised kinetic term, a spacetime contraction of ∂B with ∂B times an invariant symmetric form on $\mathfrak{so}(3,3)$. Since $\mathfrak{so}(3,3)$ is simple, that form is a multiple of the Killing form, which on the fifteen bivectors has signature $(9,6)$, boosts with one sign and rotations with the other. At J the radial direction is a rotation, the eight Goldstone directions are two rotations and six boosts, and the six $\mathfrak{so}(1,3)$ directions are three of each. Whatever the overall sign, at least six directions are ghosts, and the induced metric on the eight-dimensional space of choices has signature $(6,2)$.*

Executable check: c1-10 (exact symbolic or integer arithmetic; Section 15.2).

In covariant terms the Goldstone directions form a charged four-vector and the $\mathfrak{so}(1,3)$ directions a two-form, and healthy vectors and two-forms owe their positivity to a gauge symmetry that removes the wrong-sign components; a scalar order parameter has none. This is Theorem 1.13 again, in a new setting. There is a positive alternative, and it shows what is missing.

Proposition 12.11 (The positive alternative)**.** *On the fifteen bivectors the positive form $\langle X, Y\rangle = \mathrm{tr}(XY^{\top})/8$ is the identity, and its invariance algebra is the six compact rotations*

$\mathfrak{so}(3)_T \oplus \mathfrak{so}(3)_S$. *Split* $B = B_T + B_S + B_p$ *into these two parts and the nine boosts. The potential*

$$V_+ = \left(|B_T|^2 - 1\right)^2 + |B_S|^2 + |B_p|^2$$

has as minimum set the unit sphere of $\mathfrak{so}(3)_T$. *Every element there is a rotation of a plane* N *in the temporal triple, the clock is the axis of the rotation, the spacetime is the clock line plus the spatial triple, and* $J = \gamma^0 \omega_T$. *The Hessian at* J *has eigenvalues* 8, 0, 0 *and* 2 *(twelve times): two Goldstone modes and everything else massive with positive kinetic energy. But the boosts* e_1e_4, e_1e_5, e_1e_6 *of the selected spacetime are not symmetries of this sector.*

Executable check: c1-11 (exact symbolic or integer arithmetic; Section 15.2).

The requirement was a clean selection of the Lorentzian type over an open set of couplings, with every mode healthy and accounted for. An unconstrained $SO(3,3)$-invariant adjoint scalar order parameter cannot deliver it: below degree eight it does not isolate the type; at every degree the minimum is degenerate with Lorentz-violating configurations and has six non-Goldstone flat directions; and, with a factorised kinetic term, it has ghosts. The positive route selects cleanly only by giving up the boosts it was meant to produce.

12.4 Where the surplus constraints come from

The baseline showed that the constraints $p_u = p_v = 0$ remove every nonzero surplus momentum, the whole unstable band together with the stable surplus modes, and the curved model showed that they are first class. A first-class constraint generates a gauge symmetry, so the question is which symmetry, and whether it is a natural one.

Theorem 12.12 (A worldline gauge symmetry)**.** *For arbitrary* $A_\mu(x)$ *and flat* h, *with* $C = A_\mu h^{\mu\nu} k_\nu$, *the constraints satisfy* $\{p_u, H\} = -Cp_v$, $\{p_v, H\} = Cp_u$, $\{p_u, p_v\} = 0$ *and* $\{\ell, H\} = 0$. *The extended action*

$$\int d\tau \left(p \cdot \dot{x} - N(H - \tfrac{1}{2}m^2) - \lambda_1 p_u - \lambda_2 p_v\right)$$

is invariant under

$$\delta u = \varepsilon_1(\tau), \qquad \delta v = \varepsilon_2(\tau), \qquad \delta\lambda_1 = \dot{\varepsilon}_1 + NC\varepsilon_2, \qquad \delta\lambda_2 = \dot{\varepsilon}_2 - NC\varepsilon_1,$$

all other variables fixed: local translations along the normal plane, which are gauge symmetries although they are not isometries when $F \neq 0$, *the algebra closing with structure functions [69]. Constraint preservation is thereby shown for a test worldline in an arbitrary background* A_μ; *for interacting fields, with the normal-plane modes dynamical, it is not established here. In Dirac quantisation the physical states are the functions* $\Psi(x)$ *on spacetime, and on them* $\hat{H}\Psi = -\frac{1}{2}\Box_h \Psi$; *the potential* A *drops out.*

Executable check: c2-5, c2-6, c2-7 (exact symbolic or integer arithmetic; Section 15.2).

First-class constraints that remove surplus time directions by generating a worldline gauge symmetry are also the mechanism of two-time physics, where a local $Sp(2,\mathbb{R})$ acts on position and momentum in signature $(d,2)$ [284]; the comparison, and where it ends, is made in Section 14.2.

Proposition 12.13 (Gauging the isometry is not enough)**.** *In polar normal coordinates* $\ell = -p_\phi$ *and* $p_u^2 + p_v^2 = p_r^2 + p_\phi^2/r^2$. *Gauging only the rotation* ℓ, *the isometry of the normal plane, leaves* $p_0^2 = |\mathbf{p}|^2 + m^2 - p_r^2$: *the radial surplus momentum keeps the unstable band. For fields, the rotation-invariant modes* $J_0(qr)e^{-i\omega t + i\mathbf{k}\cdot\mathbf{x}}$ *obey* $\omega^2 = |\mathbf{k}|^2 + m^2 - q^2$.

Executable check: c2-9 (exact symbolic or integer arithmetic; Section 15.2).

The symmetry that removes the instability is therefore the full group of local translations along the normal plane, not its rotation subgroup.

Corollary 12.14 (Weak emergence for scalar matter)**.** *If the normal translations are gauged, the scalar sector reduces to the Klein–Gordon field on (M^4, h); the physical states do not depend on u, v, and they are not normalisable functions on $\mathbb{R}^{3,3}$, on which constants in u, v have infinite norm. For flat h (the setting of Proposition 12.13), a chosen time orientation and $m^2 \geq 0$, the one-particle space is that of the positive-frequency solutions with the Klein–Gordon inner product, and on its bosonic Fock space the energy is bounded below, the evolution is unitary, and $[\phi(x), \phi(y)] = 0$ at spacelike separation for the cone of h [160]. For a globally hyperbolic curved h the algebraic Klein–Gordon construction gives microcausality from the support of the advanced-minus-retarded propagator; a preferred positive-frequency ground state and a unitary implementation in a chosen Fock representation need further assumptions [326]. In the free flat product the zero modes ψ and the surplus modes χ do not mix, since translation invariance along the normal plane makes the mixed kernel $K_{\psi\chi}$ vanish, so the Schur complement is trivial and $K_{\text{eff}} = K_{\psi\psi}$; the rotating normal metric with a general A is not translation invariant, and there the reduced operator itself is needed; the content is not in the kernel but in whether χ is physical, and under the gauge principle it is not. The causal cone is that of h, the metric of the selected four-plane: it is inherited from the reduction, not derived from the six-dimensional sum.*

What the reduction leaves over is the geometry of the normal plane itself, and it is not healthy.

Theorem 12.15 (Every normal-plane vector is a ghost)**.** *For $ds^2 = h + \sigma[(du + B^u)^2 + (dv + B^v)^2]$ with flat h, $B^u = b_1 x^1 dx^2$ and $B^v = b_2 x^0 dx^3$, one finds $R_6 = -\frac{\sigma}{4} \sum F_{\mu\nu} F^{\mu\nu}$, and for the rotation ansatz $B^u = vA$, $B^v = -uA$, $R_6 = -\sigma\, r^2 F^2/4$, which at $\sigma = +1$ is the ambient scalar curvature of Theorem 11.5. The coefficient is odd in σ, so the healthy sign is that of a normal plane of the sign of space, $\sigma = -1$, the ordinary Kaluza–Klein case. By Theorem 12.4(ii) the normal plane of a Lorentzian four-plane in $\mathbb{R}^{3,3}$ always has the sign of time. Relative to the ambient Einstein–Hilbert action and these ansätze, every Kaluza–Klein vector from the normal plane, from its translations or its rotation, therefore gravitates with the wrong sign, and by Theorem 12.12 none of them couples to spinless matter in the zero-normal-momentum sector. The theorem concerns normal vectors obtained by this Einstein–Hilbert and Kaluza–Klein route.*

Executable check: g2-7 (exact symbolic or integer arithmetic; Section 15.2).

The failure condition "ghosts or negative spectral weights" is therefore met by the normal-plane vectors, and the photon cannot come from them. If the Maxwell field emerges, it must come from elsewhere: from the univectorial potential of the interior reading, or from the field-strength reading $\mathbb{R}^{3,3} = \Lambda^2 \mathbb{R}^{1,3}$, where the energy is the positive Hermitian form $|\mathbf{B}|^2 + |\mathbf{E}|^2$ (Section 3.7).

12.5 The clock reduction and the parameter τ

Every statement of this manuscript about a four-momentum, an observable four-plane or the absence of uphill decays rests on Assumption 2.16: physical states and observables are those of the reduced shell of one clock $\hat{n}$. This section says what the assumption does, how far its implementation reaches, and what the parameters along which things evolve are.

12.5.1 What the reduction is, and how far its implementation reaches

The assumption is implemented by the first-class constraints $p_u = p_v = 0$ of Proposition 11.4, gauged by the local normal translations of Theorem 12.12; for fields it is the restriction $\partial_u \phi = \partial_v \phi = 0$, which is dimensional reduction (Section 12.1). Four qualifications belong to it.

(i) It is *imposed*. The gauge principle is consistent but chosen: the multipliers λ_1, λ_2 are added to the worldline action by hand (Theorem 12.12), and Section 12.3 shows that no invariant potential for an unconstrained adjoint bivector order parameter has the orbit of J as

its minimum set, and that the factorised invariant kinetic term has ghosts; other representations, constrained order parameters, gauge or soldering constructions and derivative couplings are not excluded.

(ii) It is *sufficient, not minimal*. The constraints remove every nonzero surplus momentum, the stable modes $0 < q^2 \leq |\mathbf{p}|^2 + m^2$ together with the unstable band (Figure 12.1); the minimal restriction depends on the four-momentum and is not of this local form.

(iii) Its *preservation* is shown for a test worldline in an arbitrary background A_μ and a flat h (Theorem 12.12). For interacting fields with dynamical normal-plane modes it is not established.

(iv) It removes the surplus times and leaves the normal-plane *geometry*: relative to the ambient Einstein–Hilbert action, every Kaluza–Klein vector of the normal plane is a ghost (Theorem 12.15), and none couples to spinless matter in the zero-normal-momentum sector.

12.5.2 The worldline parameter

Proposition 12.16 (The worldline parameter is a gauge parameter)**.** *For the extended action* $S = \int d\tau\,(p\cdot\dot{x} - N(H - \frac{1}{2}m^2) - \lambda_1 p_u - \lambda_2 p_v)$ *of Theorem 12.12:*

(i) *S is invariant under $\tau \mapsto \tau' = f(\tau)$ with $f' > 0$, x and p transforming as scalars and N, λ_1, λ_2 as densities, $N' = N/f'$, $\lambda_i' = \lambda_i/f'$. Varying N gives the mass shell $2H = m^2$, a first-class constraint; on shell the reparametrisations are generated by the total first-class Hamiltonian $N(H - \frac{1}{2}m^2) + \lambda_1 p_u + \lambda_2 p_v$, multipliers included.*

(ii) *On the reduced shell $p_u = p_v = 0$, $h^{\mu\nu}p_\mu p_\nu = m^2 > 0$, the proper time of h along the worldline is $d\tau_h = |N|m\,d\tau$.*

Hence τ is not an observable. For $m > 0$, on a segment where the lapse does not vanish, the gauge $N = 1/m$ (if $N > 0$) or $N = -1/m$ (if $N < 0$; reparametrisations with $f' > 0$ preserve the sign of N) makes it the proper time of h; a lapse crossing zero needs separate treatment, and the time read by the clock is the proper time of the selected four-plane, not a seventh coordinate.

Proof. (i) Under the reparametrisation $p\cdot\dot{x}\,d\tau$ is invariant, and $N(H - \frac{1}{2}m^2)\,d\tau = N'(H - \frac{1}{2}m^2)\,d\tau'$, $\lambda_i p_i\,d\tau = \lambda_i' p_i\,d\tau'$. The variation of N gives $H - \frac{1}{2}m^2 = 0$; with $-NH$ alone it would give $H = 0$, the massless shell. The constraint algebra of Theorem 12.12 contains $\{p_u, H\} = -Cp_v$, $\{p_v, H\} = Cp_u$ and $\{p_u, p_v\} = 0$, so $H - \frac{1}{2}m^2$, p_u, p_v are first class. (ii) With $k_\mu = p_\mu - A_\mu\ell$ and $\ell = vp_u - up_v$, Hamilton's equation for x^μ is $\dot{x}^\mu = Nh^{\mu\nu}k_\nu$. On the reduced shell $\ell = 0$, so $k = p$ and $h_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = N^2h^{\mu\nu}p_\mu p_\nu = N^2m^2$, so $d\tau_h = |N|m\,d\tau$. □

Executable check: c2-8 (exact symbolic or integer arithmetic; Section 15.2).

This is the standard reparametrisation invariance of a relativistic worldline [69], written for the reduced arena. It says what τ is and what it is not: a label of points on a worldline, fixed only by a gauge choice, whose physical content is the proper time of h. It does not turn the six-dimensional manifold, or its tangent fibres, into a seven-dimensional spacetime.

12.5.3 The interior parameter

The interior dynamics of Chapter 4, the shifted operators $\partial_\tau \pm \kappa G$, the rotor and the closure, also evolves in a parameter written τ, with the dimension of a time. That parameter is not the worldline parameter of Proposition 12.16, and this manuscript does not relate it to the time read by a clock (Section 4.6.1). Two further steps would be needed to do so. The rate $\kappa = c/\ell$ of the binding would have to be identified with a rest frequency, $mc^2 = \hbar\kappa$, which is an added hypothesis (Section 4.5.6); and the rotor would have to be quantised, with its angle related to an external time. Both are open. Where the interior parameter is read as Newtonian time, in the orbital extension of Section 4.7, the reading is part of the identification and is stated there.

12.5.4 Local clock rotations

All unit clocks are related by rotations of V^+, and the rotor carrying e_1 to $\widehat{W}$ conjugates the free Dirac data (Remark 2.6). That is a global covariance of free data. A position-dependent rotor $R(x)$ adds the term $R^{-1}\partial_\mu R \in \mathfrak{so}(3)_+$ to the Dirac operator, so a local clock symmetry needs a connection with values in the rotations of V^+ and its transformation law. Since a rotation of the temporal triple alone changes the binding root G (Remark 2.6(b)), such a connection would also have to transport G, or be restricted to the matched rotations of both triples. It is not constructed here. Nor is it the gauge symmetry of Theorem 12.12: that symmetry translates along the normal plane of a fixed four-plane and does not rotate the clock. Global free equivalence therefore establishes neither a local clock gauge symmetry, nor a dynamical selection of the clock, nor a well-posed interacting theory with three times.

12.5.5 What survives the reduction

Under the reduction every momentum lies in $E_{\hat{n}} = \mathbb{R}\hat{n} \oplus V^-$, of signature $(1, 3)$, and the following hold.

(i) *Kinematics.* The six-dimensional mass equals the mass read by the clock, $M_6 = m_4(\hat{n})$, and the ordinary four-dimensional decay inequality holds: no uphill decay survives (Proposition 10.6(iv)). The uphill decays and the opened channels of Proposition 10.6(ii)–(iii) use temporal momenta transverse to the clock, which the reduction removes; they are warnings about the unreduced kinematics, not predictions, and the ambient temporal momenta are not observable energies (Chapter 10).

(ii) *Scalar matter.* With the normal translations gauged, the scalar sector is the Klein–Gordon field on (M^4, h), with energy bounded below, unitary evolution and microcausality for the cone of h for flat h; on a globally hyperbolic curved h, microcausality of the algebraic construction, with a preferred state and unitarity needing further assumptions (Corollary 12.14).

(iii) *Removed.* Every surplus-momentum mode, stable or not; with compact surplus times the tower of tachyons of Example 12.3 is removed with them.

(iv) *Not removed.* The ghost vectors of the normal plane (Theorem 12.15) and the type-selection obstructions of Section 12.3, which concern the four-plane the reduction presupposes.

12.6 A universal evolution parameter: an open hypothesis

A universal invariant evolution parameter is not new. In the parametrised relativistic dynamics of Horwitz and Piron [190], an invariant parameter evolves events in a fixed spacetime, and for a many-body system it is taken to be universal, as Newtonian time is [190, 289]. What may be specific to this framework is the connection of such a parameter with the clock reduction of Section 12.5: the parameter would act on the split six-dimensional arena, and the time read by a clock would follow from it through the reduction to the selected four-plane. The hypothesis is stated in a limited form, so that its requirements are explicit. It is not a result.

> *Universal-evolution hypothesis.* There is one parameter τ_U, external to $\mathbb{R}^{3,3}$, along which all fields of the arena evolve, and the time read by a clock is obtained from τ_U through the clock reduction.

As in the theory of Horwitz and Piron, the parameter is not a coordinate of spacetime. An external parameter does not turn $\mathbb{R}^{3,3}$, or its tangent fibres, into a seven-dimensional physical spacetime. A proposal would have to specify the following.

(1) *What evolves*: fields on $\mathbb{R}^{3,3}$, fields on the reduced four-plane, the interior variables of Chapter 4, or all of them, and in which configuration space.

(2) *The action and the constraint equations*, including the constraints that implement the clock reduction, and their preservation under the τ_U-evolution with interactions.

(3) *Reparametrisation properties.* If the action is invariant under $\tau_U \mapsto f(\tau_U)$, its generator is a constraint and τ_U is not observable, as for the worldline parameter of Proposition 12.16. If it is not, τ_U need not be a preferred time that breaks the Lorentz algebra of the selected four-plane or $SO(3,3)$: the action $\int d\tau_U \, (p \cdot dX/d\tau_U - \eta^{AB} p_A p_B/2M)$ is $SO(3,3)$-invariant with τ_U a scalar, as in the theory of Horwitz and Piron. Which symmetries are kept, and which broken, has to be stated.

(4) *The relation to measured time*: how the proper time of h, which is what a clock reads (Proposition 12.16(ii)), follows from τ_U.

(5) *Causal observables*: microcausality with respect to a stated cone. A self-adjoint generator gives unitary evolution in τ_U even when it is unbounded below, so unitarity and the existence of a ground state, which needs a generator bounded below, are separate requirements.

(6) *Compatibility with the obstructions of this chapter.* The type-selection results of Section 12.3 concern potentials and $SO(3,3)$-invariant kinetic terms, and they hold whatever the evolution parameter. The ghost vectors of Theorem 12.15 come from the six-dimensional curvature. The ill-posedness of Proposition 12.1 is a statement about evolution in a temporal coordinate. A first-order evolution in τ_U can pose a different Cauchy problem, but its generator then has to be bounded below for a ground state to exist, and an $SO(3,3)$-invariant second-order operator such as $\Box_\eta$ is not, since its symbol is indefinite.

Until a mechanism meeting these requirements is exhibited, the hypothesis remains open.

12.7 What the causal analysis establishes

Table 12.1 collects the outcome against the failure conditions fixed at the start of Section 12.1, together with the status of the clock reduction and of the evolution parameters.

Weak emergence holds for scalar matter, conditionally on a gauge principle, local translations along the normal plane, that is consistent but chosen rather than derived. Strong emergence through an unconstrained $SO(3,3)$-invariant adjoint scalar order parameter is excluded; whether the quantum dynamics of the whole system can select the Lorentzian sector, the motivation stated at the opening of the chapter, remains open. What survives is a sharper statement of the question. The selector is the complex structure J, and every construction here that produced a healthy Lorentzian sector also used one extra datum: a positive inner product, which is the transpose of the realisation, equivalently the temporal triple. Within these computations the totality of $SO(3,3)$-invariant dynamics fixes at most the type of the structure, never the instance, and it fixes even the type only imperfectly. The datum that singles out time, wherever things worked, is the positivity of an inner product; in quantum mechanics it is the positivity of the Hilbert-space norm. That this positivity fixes causal order is a conjecture; it indicates where a derivation of causal order from the split geometry would have to look.

Several routes remain open. A gauge mechanism for type selection, an $SO(3,3)$ analogue of the MacDowell–Mansouri and Stelle–West breaking [91, 145], $SO(3,3) \to SO(2) \times SO(1,3)$, might make the would-be ghosts gauge and solder the eight broken directions to the base; the six flat directions are not gauge directions at linear order, and removing them would need further constraints or a special action. A principle that makes the normal translations gauge, rather than adding the multipliers by hand, would turn weak emergence from a conditional statement

stage	result	verdict
baseline	growing modes and no ground state; the surplus constraints remove the band together with every stable surplus mode; compactification makes tachyons	obstruction confirmed, sufficient cure located
reduction	kinematics: selector $= J$, cone $=$ twisted squares at one constituent; dynamics: no mechanism from an unconstrained $SO(3,3)$-invariant adjoint scalar; the positive route breaks the boosts	kinematics exact; dynamical selection fails as posed
effective theory	scalar matter: no mixing in the flat product, $K_{\text{eff}} = K_{\psi\psi}$; normal-plane vectors are ghosts (Einstein–Hilbert route)	failure condition met by the vectors
causality	the scalar sector is microcausal for the cone of h (flat h; globally hyperbolic h for the algebraic construction)	conditional on gauged translations; inherited
Maxwell	normal-plane vectors excluded as photons	blocked on this route
clock reduction	Assumption 2.16 implemented by first-class constraints for a test worldline; removes stable surplus modes too	imposed; sufficient, not minimal; interacting case open
parameters	the worldline τ is a gauge parameter whose content is the proper time of h; the interior τ is not related to a clock	fixed (worldline); open (interior)
uphill decays	none after the reduction; the unreduced examples are warnings	conditional on the reduction
local clock symmetry	needs a connection in $\mathfrak{so}(3)_+$ that also transports G	not constructed
universal τ_U	prior art (Horwitz–Piron); no mechanism in this arena; requirements listed	open hypothesis

Table 12.1: The causal analysis against the failure conditions fixed in advance (Sections 12.1 to 12.6).

into a derived one. At three constituents, a $\mathrm{Spin}(9,9)$-covariant construction of the cone of E, for which pure spinors are the natural candidates, or a proof that none exists without the positive inner product, would settle whether the cone survives the tower.

Chapter 13

Compatibility of the Mathematical Constructions

The question of this chapter is how the results of Chapter 1–Chapter 12 fit together. The earlier chapters establish, one sector at a time, what the algebra fixes, what it admits, what it obstructs and what remains open. Here the sectors are held against one another. Section 13.1 sorts the results into the four classes of Section I.1.2 and lists the choices that the algebra leaves free. Section 13.2 follows five research chains, from Clifford data to causal evolution, to the physical photon, to interactions, to stable states and to gravitational dynamics, and marks every link as established, constructed, conditional, obstructed or open. Section 13.3 records for each sector its module, Lorentz action, complex structure, projector, bilinear form, action and status, and keeps the projectors apart. Section 13.4 names the readings that exclude one another and the bridges that are missing. The last two sections assess, once, the hypothesis on the rest energy that motivated part of the programme (Section 13.5) and its second question, duplication and the b-field between manifolds (Section 13.6).

The chains are research chains, not completed derivations. No common action for the sectors is derived or claimed. Where two sectors are realised together, the joint realisation is stated with its scope; everywhere else they are separate realisations on the same algebra.

13.1 What the algebra decides

We sort the results of this manuscript into the four classes announced in Section I.1.2, and then name the choices that remain. The test applied throughout is the deflationary one of Corollary 2.20: a statement about the arena carries information only if it could have come out otherwise, and compatible representation data is not a prediction.

13.1.1 Fixed by the algebra

At one constituent the algebra fixes more than one might expect. The signature rule separates the bivectors that generate a compact $U(1)$, the gauge-admissible ones, from those that generate $\mathbb{R}$: as operators on S the first are complex structures and the second have real ± 1 eigenspaces (conjugation by either kind is an involution of the algebra). The pole operators grade the algebra by $\mathbb{Z}_2^3$ through a maximally split Cartan subalgebra. The pairing of the two triples has exactly two square roots, the grading root $\mathcal{K}$ and the binding root G, related by the Cartan involution $\mathrm{Ad}_{e_1e_2e_3}$. Under permutation symmetry the clock is the barycentre of the Fisher–Rao octant, a description that is chosen; all unit clocks are related by rotations, and conversion along any single clock is invisible in the equations of motion, so no clock is selected dynamically. The imaginary unit is selected, not adjoined: inside the joint commutant of the interior Lorentz action and the internal bivectors the only complex structures commuting with the gamma matrices are $\pm J$.

Charge conjugation is the surplus plane, time reversal squares to $-\mathbf{1}$, the kinetic forms alone fix the internal symmetry algebra $\mathfrak{u}(1)_\Omega \oplus \mathfrak{su}(2)$ of the free massless action, which becomes a gauge algebra only once a connection is introduced, the mass form cuts it to $\mathfrak{u}(1)_J$, and that $\mathfrak{u}(1)$ is the whole internal centraliser. Along the tower the internal centraliser is $\mathfrak{so}(3n-1, 3n-3)$, nested and over one Lorentz frame, and the diagonal Lorentz action gives a compact internal algebra forming a reductive dual pair with the Lorentz algebra, with the exact identity $C_{\mathrm{cop}} + C_\Delta = \frac{1}{2}n(n+2)$; both the dual pair and the identity are inherited [224, 296].

In the field-strength reading the centraliser of G is a Lorentz algebra under which $\mathbb{R}^{3,3}$ *is* $\Lambda^2\mathbb{R}^{1,3}$, G is the Hodge star, the two Maxwell invariants are the real and imaginary parts of one holomorphic quadratic form, and covariance fixes the form of Maxwell's equations through a multiplicity count. In the interior dynamics, factorisation and binding with $G^2 = -\mathbf{1}$ coexist exactly in neutral signature, and, within the stated equivariant one-generator ansatz, the force law closes in one way only, into a rotor acting by G whose free motion is a pendulum in the duality angle. In the curved arena the curvature operator lives on the frame, the first Bianchi identity is the single trace $\mathrm{Tr}(G\mathcal{R}) = 0$, and the duality rotation of the curvature ends, where conservation and variationality both select it, at the double dual contracting to the Einstein tensor; in first-order form the same rotation is a Barbero–Immirzi angle. Kinematically, choosing a Lorentzian four-plane is choosing its definite orthogonal two-plane, represented by a normalised simple bivector whose two signs encode an orientation, which is a complex structure, and at one constituent the light cone is the set of chiral spinors squared with the help of J.

13.1.2 Constructed

The constructions are existence results; in each case the algebra admits the object and nothing selects it among alternatives. At three constituents the axial projector P_* of rank 64 carries the Pati–Salam algebra and one chiral family with a right-handed neutrino, and settles the question of colour with chirality. The diagonal Dirac block carries a gauge-invariant real $\mathrm{Cl}(1,3)$ built from a copy-singlet cubic. At the mixed second rung the univectorial potential, a Dirac block and the frame field can be coupled into two-flavour spinor electrodynamics. On the 272 mass kernels of the chiral block, with the complex structure $\widetilde{\mathcal{C}}$, the renormalizable potential of the $(\mathbf{10}, \mathbf{1}, \mathbf{3})$ component is classified, and sufficient conditions and separate necessary conditions under which the Standard-Model orbit is the global minimum of that restricted tensor potential are found (Theorems 7.25 and 7.26). In the full 272-dimensional leading-logarithm model (an $O(272)$-symmetric tree potential at the matching scale, one gauge and one Yukawa coupling) the radiative spectrum of all 271 angular modes is computed and the Standard-Model orbit is a local minimum below $x = 27$ (Theorem 7.32); its global minimality below x_* is a conjecture supported by numerics (Proposition 7.33), and Theorem 7.34 gives compatibility and spectra at the chosen vacuum, not full-space selection. For a given bidoublet vacuum value the electroweak stage of Proposition 7.35, the seesaw and the Georgi–Jarlskog factor follow. At the second rung a Lorentz-invariant, uncharged, non-rigid vacuum with a charged-scalar current exists on a compact S^4 of projectors, with a curved extension in which it is parallel. Both readings of $\mathbb{R}^{3,3}$ construct the Einstein tensor of a given four-metric.

13.1.3 Obstructed

Each obstruction holds under hypotheses stated where it was proved, and those hypotheses are part of the result. Hypercharge cannot be read off anomaly cancellation for charges built from three commuting involutions; $\mathfrak{su}(3)$ fits in the effective algebra of no axial projector of the five-projector family; the two central $\mathfrak{u}(1)$s of P_* cannot be gauged on its matter; the diagonal sector is vector-like on its complete blocks. The two readings of the same six directions exclude one another, and the obvious repair at the second rung costs the internal algebra, whose centraliser shrinks to the rotation of one surplus pair. The positive-form interior oscillates

only at zero momentum. No Lie–Poisson closure lives on the frame alone, and no inertia is both Lorentz invariant and rate-conserving. Non-orthogonality of $\mathbf{E}$ and $\mathbf{B}$ breaks P and T, is never spontaneous for the Maxwell energy, and generates no mass. The paired-mode projector sources only a massive vector, and at the first rung a vacuum that is both massless and Lorentz invariant is rigid. The normal $U(1)$ of the curved model fails as electromagnetism on three independent counts: a spinless particle in the zero-normal-momentum reduced sector cannot see it, its adapted generator does not commute with the physical generators, and it gravitates with the wrong sign relative to the ambient Einstein–Hilbert action. At the third rung the adapted-generator count is removed by a commuting vacuum, the sign count concerns a metric origin only, and a fourth count decides: on the chiral block its charge is half the fermion number, with a mixed anomaly with $\mathfrak{su}(2)_L$ (Remark 8.61). The electric charge is the photon candidate there, while the vacuum selected by the adopted potential is electroweak symmetric and the electroweak breaking mechanism is missing. The Ricci–DeTurck method does not transfer to split signature (other split-signature flows are not excluded). No vacuum value of a field built from the internal bivectors and J makes the matter chiral, Hermitian order parameters keep rank five, and vacuum alignment does not see chirality. In the causal analysis, the six-dimensional free field is ill-posed and has no time orientation, no $SO(3,3)$-invariant polynomial potential of an unconstrained adjoint scalar of degree below eight isolates the Lorentzian type, every such invariant potential is degenerate with a Lorentz-violating orbit and has six non-Goldstone flat directions, every factorised invariant kinetic term has ghosts, and, relative to the ambient Einstein–Hilbert action, every vector from the normal plane is a ghost.

13.1.4 Open, and the choices that remain

Table 13.1 lists the choices that the algebra leaves free, with the place where each enters. Most of the open problems are downstream of them.

The open problems that these choices leave, with the steps that could settle them, are collected in Section 16.3.

13.2 Five research chains

Each chain runs from algebraic data to a physical conclusion. A link is *established* when it is a theorem of the manuscript or an inherited result used within its hypotheses; *constructed* when the object exists and is exhibited but nothing selects it; *conditional* when it holds under an assumption, an adopted action or a chosen datum, stated where it enters; *obstructed* when it is impossible under stated hypotheses; and *open* otherwise. Where a link has two statuses, for instance existence established and selection conditional, Figure 13.1 draws the weaker one and Table 13.2 gives both. Figure 13.2 draws the same chains in detail: what is proved at each stage, with the place where it is proved, the choice or assumption on which each link rests, the no-go theorems that close routes along each chain, what remains open, and the results that join the chains.

Table 13.2: The links of the five research chains with their statuses and the places where each is proved or qualified.

link	status	evidence and limits
Spacetime		
Clifford data → Lorentz structure	established (existence); conditional (selection)	the interior $\mathfrak{so}(1,3)$ and $\mathfrak{l}_G$ exist (Chapters 1 to 3); choosing one needs a clock and stated hypotheses, and all unit clocks are equivalent under rotations (Remark 2.6); the imaginary unit is unique only under the hypotheses of Theorem 2.17

Table 13.2, continued

link	status	evidence and limits
Lorentz structure → observable four-plane	conditional	$E_{\hat{n}}$ is selected by a clock; the clock reduction is Assumption 2.16, imposed; no dynamics of an unconstrained $SO(3,3)$-invariant adjoint scalar selects the four-plane (Theorems 12.8 and 12.9 and Proposition 12.10)
four-plane → causal evolution	conditional (scalar matter); open (interacting)	weak emergence for scalar matter under the chosen gauge principle (Corollary 12.14); constraint preservation shown for a test worldline only (Theorem 12.12); no local clock symmetry is constructed (Section 12.5.4); a universal evolution parameter, prior art in the theory of Horwitz and Piron, is an open hypothesis in this arena (Section 12.6)
Electromagnetism		
field-strength reading → charge and gauge geometry	established (each part); conditional (the link); obstructed (normal charge as photon)	$\mathbb{R}^{3,3} = \Lambda^2\mathbb{R}^{1,3}$ under $\mathfrak{l}_G$ with G the Hodge star (Theorem 3.3), a representation theorem; the electric charge is identified among the compact generators at the aligned vacua (Theorem 8.37); no map or common action joins the two, so the link is conditional; the normal charge is half the fermion number on V_* and anomalous with $\mathfrak{su}(2)_L$ (Remarks 8.34 and 8.61), and the normal $U(1)$ of the curved model fails on three further counts (Section 11.3 and Theorem 12.15)
charge → Maxwell dynamics	conditional	covariance fixes the form of Maxwell's equations (Corollary 3.9); the Maxwell term is adopted, not derived
Maxwell dynamics → physical photon	open	the aligned vacuum is massless for the photon together with W and Z and is electroweak symmetric; the adopted potential opposes the neutral deformation (Proposition 8.49); the electroweak breaking mechanism is missing; no quantisation
Matter and interactions		
tower and module → chiral content	constructed	P_* is chosen among projectors; it carries one Pati–Salam family with ν_R (Theorem 6.10 and Corollary 6.12); chirality is imposed by the choice, not selected by dynamics (Proposition 8.5); the diagonal Lorentz action gives vector-like matter instead (Theorem 5.14)
chiral content → anomaly-free gauge action	conditional	the Standard-Model algebra is anomaly free on V_*; the two central $\mathfrak{u}(1)$s of P_* cannot be gauged on it (Proposition 6.14); the gauge action (7.4) is adopted
gauge action → interactions	open	Yukawa couplings and the scalar potential are adopted; no coupling or scale relation is derived
Energy and mass		
geometric operators → interior and field dynamics	constructed	the closure is unique given its premises and its free motion is a pendulum (Theorems 4.22 and 4.28); bounded orbits do not make the energy bounded below, which has signature $(6,6)$ (Section 4.6.3)
dynamics → energy and mass kernels	established (classification); conditional (link to dynamics)	the 272-dimensional kernel space and its complex structure $\widetilde{\mathcal{C}}$ (Theorem 7.17); a mass in a Dirac bilinear is an input, not generated; $E_6^2 = E_0^2 + \lvert\mathbf{S}\rvert^2$ is an identity, not a process (Section 13.5)
kernels → stable physical states	conditional (selection); open (stability)	the Standard-Model orbit is the minimum of the adopted potential on a stated open set of couplings (Theorems 7.25 and 7.34); global minimality on all of $\mathcal{M}$ is a conjecture supported by numerical searches (Proposition 7.33), within the model assumptions stated in Chapter 7; quantum stability is open
Gravity		
curved geometry → curvature identity	established	the ambient Einstein tensor of the model and the double dual of the field-strength reading (Theorems 11.5, 11.10 and 11.13)
curvature identity → variational field equations	open	no gravitational action is derived; the Palatini–Holst action is adopted (Proposition 11.14 and Table 11.1)

Table 13.2, continued

link	status	evidence and limits
field equations → healthy gravitational dynamics	obstructed (normal-plane route); open (otherwise)	relative to the ambient Einstein–Hilbert action and the stated ansatz, every normal-plane vector gravitates with negative energy, and the sign is not a convention (Theorem 12.15 and Remark 11.18); matter coupling, degrees of freedom and low-energy limit are not derived (Section 11.4)

No chain is complete. Two features recur in all five. Every chain passes through a choice that the algebra does not make, the clock, the projector, the Lorentz action, an action or a potential (Table 13.1); and every chain stops before dynamics that would be derived rather than adopted. The chains also do not share their data: the spacetime chain uses $\mathfrak{l}_{\mathrm{int}}$ after a clock, the electromagnetic chain uses $\mathfrak{l}_G$, and the two are not conjugate (Propositions 3.2 and 3.6). Section 13.4 lists the places where this matters.

13.3 Sectors, representations and projectors

Table 13.3 records for each sector the data that must agree before two sectors can be combined. It is the manuscript form of the compatibility map kept by the author.

Projectors kept apart. Six families of projectors occur, and no statement transfers from one to another without an explicit map.

(i) P_*, the chiral matter block of Section 6.2, a choice among the projectors that commute with the Lorentz action and J.

(ii) P_{fam}, the family block of Chapter 6: of the same rank, on a different orbit, with a different effective algebra.

(iii) The auxiliary projectors P_{ij}, $P_{L,R}$, P_ν on the module (Chapters 6 and 7).

(iv) The vacuum projectors Π of the compact families, with the aligned points $\Pi_{\pm A}$, $\Pi_{\pm W}$ (Chapter 8); their overlaps with P_L and P_{fam} enter the adopted potential, which encodes the target of the selection in its couplings.

(v) The observational projection $\mathrm{pr}_{\hat{n}}$ onto the four-plane of a clock (Chapters 2 and 12), which acts on the vector space $\mathbb{R}^6$.

(vi) The transported projector field $\Pi(x)$ of Section 11.3, which acts on the spinor module $\mathbb{R}^8$. It is not the curved counterpart of $\mathrm{pr}_{\hat{n}}$: Chapter 11 shows that the two behave differently under transport.

A transported projector keeps its rank and idempotence, but a projector fixed once, such as P_*, is not invariant under clock rotations unless it is transported with them (Remark 2.6(b)).

13.4 Incompatible readings and missing bridges

The sectors of Table 13.3 are realised on one algebra, but several of them are not realised together. The following incompatibilities are proved, and each is part of the result.

(1) *Displacement and field strength.* The two readings of the same six directions share η and the Clifford action, but their Lorentz algebras $\mathfrak{l}_{\mathrm{int}}$ and $\mathfrak{l}_G$ are not conjugate and no nonzero Lorentz-equivariant linear intertwiner joins the specified modules (Propositions 3.2 and 3.6 and Theorem 3.15). They meet only as base and fibre, which is a reading.

choice	status
the projector	a field with a computable configuration space; at rank 64 at least two admissible orbits, of dimensions 34 and 20; a spectrum-only potential is constant on them, so selection needs couplings that distinguish orbits
the Lorentz action	spectator and diagonal actions are not conjugate and trade against each other: a large internal algebra with ghost-signed directions and a chiral block, or a small compact algebra on vector-like matter
the hypercharge direction	a compatible embedding exists on P_* and is output by the vacuum of the kernel field, but the vacuum itself is selected only on an open set of couplings
the binding operator and its reading	G is adopted; conditional uniqueness once its Lorentz algebra is taken, not selection; the field-strength reading and the interior reading of the same six directions exclude one another
scales and inertias	no ratio among the rotor inertia, the frame inertia and the coupling scales is fixed; the inertia of the closure has 21 free directions satisfying neither requirement
the action	no parent action is derived; the gauge field, the metric and the kernel field are given dynamics by adopted actions whose equations follow
the four-plane and positivity	the selector is J; the type is not selected by the dynamics of an unconstrained invariant adjoint scalar, and every healthy construction used a positive inner product
families	the block carries one family; three copies of a module are a multiplicity index, not three constituents, and nothing reduces the copy multiplicity to three

Table 13.1: The choices the algebra leaves open.

(2) *Spin one-half and the octants.* $\rho_{\hat{n}}$ and ρ_{oct} are two Lorentz representations on one module; the shared operator T does not identify them.

(3) *Interior energy and field energy.* The interior energy has signature $(6,6)$ and the field energy E_6 is positive; no map makes one the other, and the two forms lie in the Lorentz class and the duality class of Theorem 4.26, which meet in 0.

(4) *Matter and the interior dynamics.* The chiral block is built on $\mathfrak{l}_{\mathrm{int}}$, the binding and the closure on $\mathfrak{l}_G$. At three constituents the centraliser of the binding root G_3 meets the interior Lorentz algebra of the first constituent in 0 (Theorem 4.35), and the part of the Pati–Salam algebra compatible with the binding is $\mathfrak{so}(4)_\Delta$ (Theorem 4.36).

(5) *Six-dimensional kinematics and the reduced shell.* Uphill decays exist without a common clock and disappear only because the reduction is imposed (Proposition 10.6).

(6) P_* *and* $\mathfrak{so}(4,6)$. The block V_* is not a module of the $\mathfrak{so}(4,6)$ of Krasnov's construction; only the compact part preserves it (Remark 6.24, z3-9). No intertwiner has been constructed.

(7) *The normal* $U(1)$ *and the photon.* The normal connection of the curved model is invisible to spinless matter in the zero-normal-momentum reduced sector, its adapted generator does not commute with the physical generators, relative to the ambient Einstein–Hilbert action it gravitates with the wrong sign, and on the chiral block its charge is anomalous (Section 11.3, Theorem 12.15, and Remark 8.61).

Joint realisations. One limited joint realisation is constructed. At the mixed second rung the univectorial potential, a Dirac block and the frame field are coupled in one gauge theory,

sector	algebra and module	Lorentz action	complex structure; projector	form; action	status
displacement reading (Chapter 3)	$V = \mathbb{R}^{3,3}$, vector module	$\mathfrak{l}_{\mathrm{int}}(\hat{n})$ after a clock	none; $\mathrm{pr}_{\hat{n}}$	η; free worldline, shifted interior Lagrangian	kinematics established; interior energy indefinite
field-strength reading (Chapter 3)	$V \cong \Lambda^2 E^*_{\hat{n}}$	$\mathfrak{l}_G$, $(1,0) \oplus (0,1)$	G acts as i on z; none	η, $q = z \cdot z$; Maxwell (inherited)	established; no nonzero Lorentz-equivariant linear intertwiner to the displacement reading (Theorem 3.15)
field octants (Chapter 3)	$S \cong \Lambda^*(\mathsf{L}) = \mathbb{R}^8$	ρ_{oct}, with odd Q_i	none chosen (the duality lift of Chapter 3 exists)	T, $T\Gamma$; Maxwell on the octants	fixed; representation-specific
Dirac field, one constituent (Chapter 2)	$S = \mathbb{R}^8$, Grassmann field	$\rho_{\hat{n}}$, clock-dependent	$J(\hat{n}) = T\hat{n}$; none	mass form T; real Dirac Lagrangian	equivalent to complex Dirac theory (Theorem 2.19); m an input
tower, $n \geq 2$ (Chapter 5)	$\mathrm{Cl}(3n, 3n)$ on $(\mathbb{R}^8)^{\otimes n}$	per constituent or diagonal	selected unit of the two-particle space; Fock-number projectors	graded or ungraded rest operators; no dynamics beyond free kernels	algebra proved; particle reading conditional
chiral block, $n = 3$ (Chapter 6)	$\mathrm{Cl}(9,9)$ on $\mathbb{R}^{512}$; $V_* = \mathrm{im}\, P_*$, rank 64	interior $\mathfrak{so}(1,3)$	J (14 internal generators); P_*	$K^\mu = P_* \gamma^0 \gamma^\mu P_*$; action (7.4) adopted	content proved (Theorem 6.10); preserved by compact Pati–Salam only, not by $\mathfrak{so}(4,6)$ (z3-9)
family block (Chapter 6)	$\mathrm{im}\, P_{\mathrm{fam}}$, rank 64	interior $\mathfrak{so}(1,3)$	P_{fam}, not P_*	as above	proved; a different orbit from P_*
mass kernels (Chapter 7)	$\mathcal{M}$, real dimension 272	invariant by definition	$\widetilde{\mathcal{C}}$ on $\mathcal{M}$; none	$\mathrm{Tr}\, M^\top N$; adopted potential, leading-log selection	classification proved (Theorem 7.17); selection conditional; global minimality conjectural
compact vacuum families, photon (Chapter 8)	projector fields on the family; aligned points $\Pi_{\pm A}, \Pi_{\pm W}$	–	–; Π, not P_*	trace form; adopted $V_{\mathfrak{h}}$ and Maxwell term	charge identified; normal charge anomalous; electroweak breaking mechanism missing
observational projection (Chapters 2 and 12)	$\mathrm{pr}_{\hat{n}}$ on V; reduced shell	$\mathfrak{l}_{\mathrm{int}}(\hat{n})$	–; $\mathrm{pr}_{\hat{n}}$	$\eta\|_{E_{\hat{n}}}$; surplus constraints	imposed (Assumption 2.16); worldline scope
interior dynamics (Chapter 4)	frame and rotor on $\mathfrak{so}(3,3)$; closure $\mathfrak{s}$	$\mathfrak{l}_G$	G; none	Lie–Poisson; inertia $\mathbb{I}$	bounded orbits proved; Hamiltonian not bounded below
b-field (Chapter 9)	connections commuting or anticommuting with G; copy rotations	per constituent	G per constituent; none	split virial identity	proved in the stated classes; rates not derived
curved arena (Chapter 11)	para-complex $\mathcal{K}$; normal $U(1)$ model	local $\mathfrak{l}_{\mathrm{int}}$	$\mathcal{K}$ (para-complex); $\Pi(x)$ transported	g_6; Einstein-tensor contractions	identities established; no gravitational action
causal structure (Chapter 12)	wave operator on $\mathbb{R}^{3,3}$; selector	–	selector $= J$; type selection	η; worldline gauge	conditional; worldline scope

Table 13.3: The sectors of the manuscript and the data that must agree before two of them can be combined.

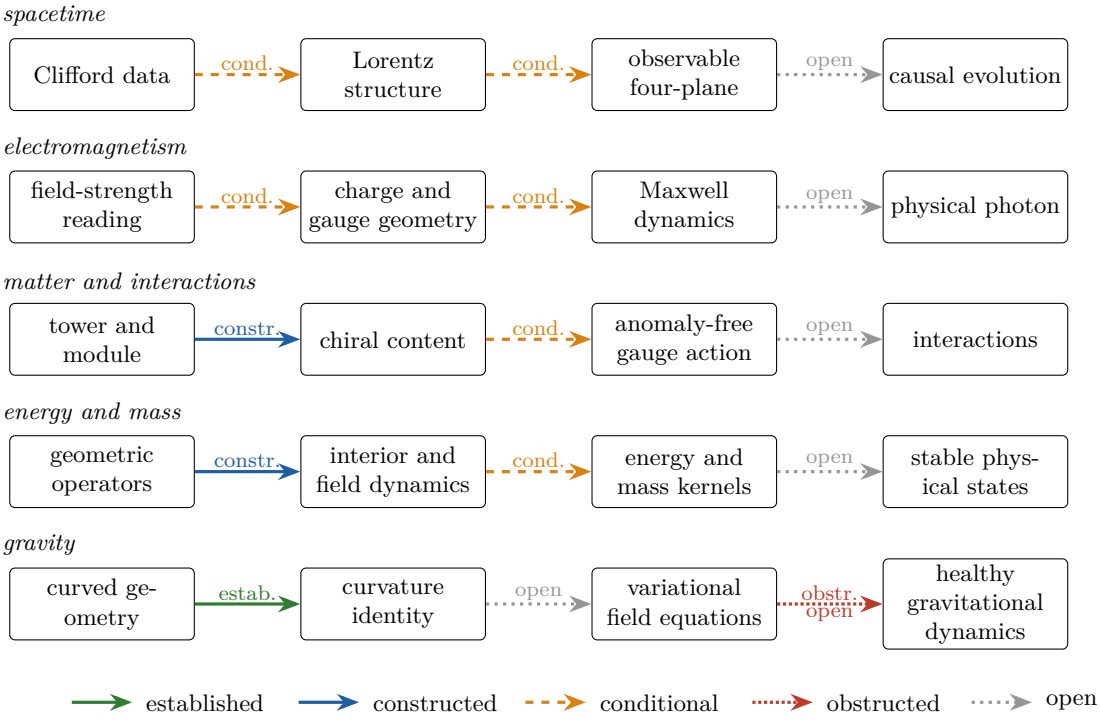


Figure 13.1: The five research chains of Section 13.2, each link drawn with its weakest status (Table 13.2 gives both statuses where there are two). The diagram is a summary of the statuses proved or stated in the cited sections; it is not a derivation, and no chain is complete.

two-flavour spinor electrodynamics (Theorems 5.76 and 5.77); the price is the internal algebra, whose centraliser shrinks to the rotation of one surplus pair (Proposition 5.75), and the block is not the chiral block of Section 6.2. No joint realisation of the chiral block, the field-strength reading, the interior dynamics and the curved arena exists in this manuscript.

Missing bridges. What would connect the separate realisations is, in each case, a construction that is not made: a common Lorentz action for the two readings, or a proof that none exists at an acceptable price beyond the five candidates tested; an explicit map between the octants and the spin module that intertwines a stated dynamics; a derivation of the clock reduction for interacting fields; an action relating metric, pairing, connection and projector; an origin of the adopted potentials; and an intertwiner, if one exists, between V_* and Krasnov's even forms. A common local action for a compatible subset of the sectors is a possible future milestone. It is not a result here, and it is not a prerequisite for the mathematical statements of the earlier chapters.

13.5 The rest-energy hypothesis, assessed

Part of the programme was motivated by a reading of $E = mc^2$ proposed by the author and stated in the introduction (Section I.2): the energy is one undivided, six-dimensional entity; the mass emerges through an internal differentiation of it; c^2 is the symmetric part of the relation; m is what breaks the symmetry; and m is generated from the b-field of Yang and Mills. The relation $E = mc^2$ itself is inherited. What can be assessed is how the arena realises its ingredients and which realisations are forced. The three clauses about the energy and c^2 are held against the results first; the two clauses about breaking and generation follow.

The results of Section 3.9 can be held against three clauses that are often attached to $E = mc^2$: that the energy is one undivided, six-dimensional quantity; that mass arises from an

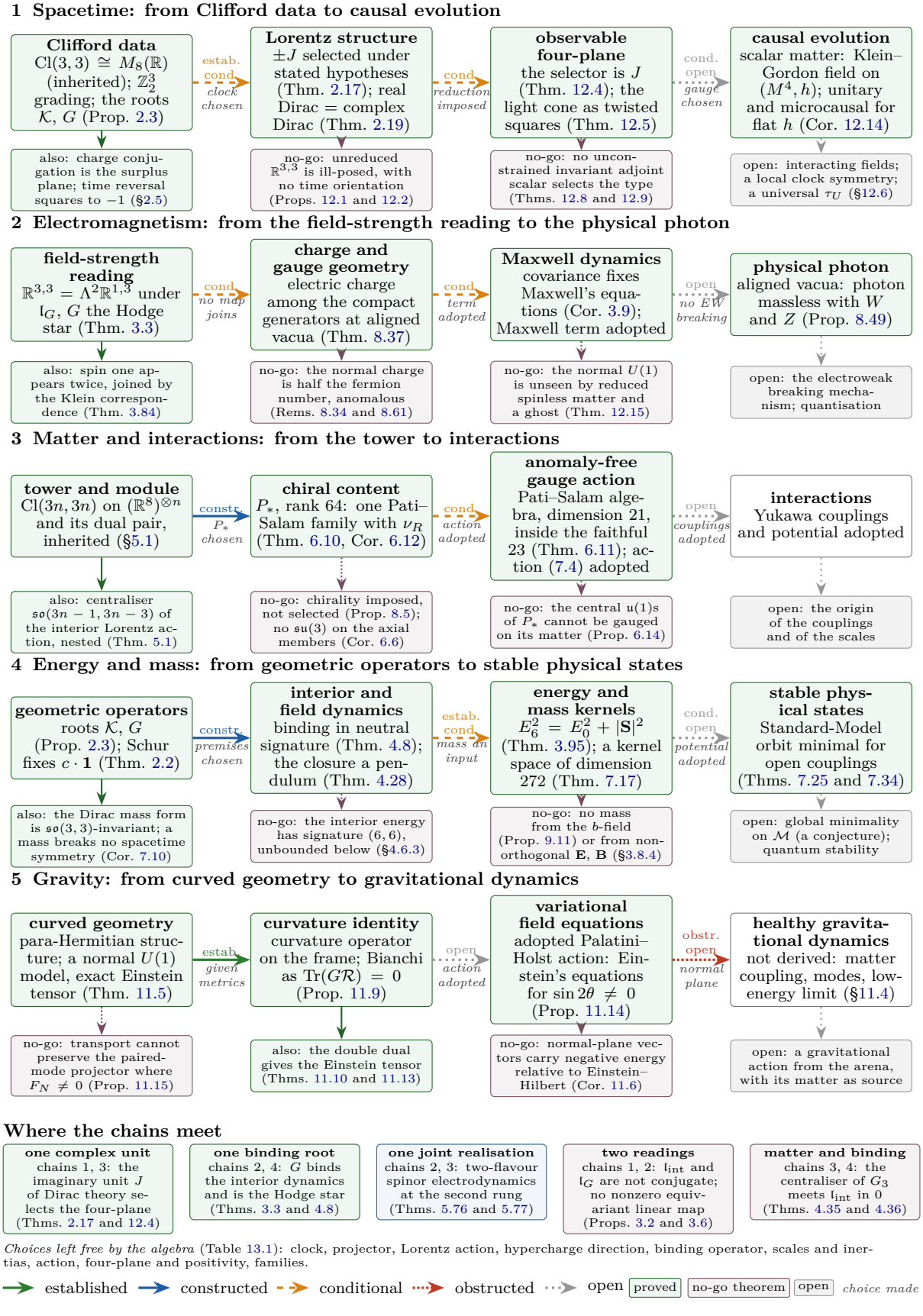


Figure 13.2: The research chains of Section 13.2 in detail. Green: results proved at each stage, under the hypotheses stated where they are proved, and in the bottom row results that join the chains; violet: no-go theorems, proved obstructions that close a route; grey: open. Each arrow carries its weakest status (both where Table 13.2 gives two) and, below it, the choice on which the link rests. A summary, not a derivation; no chain is complete.

internal differentiation of it; and that c^2 is the symmetric part of the relation.

The energy is one undivided, six-dimensional quantity. This holds in the field-strength reading. Among quadratic forms on the six frame components exactly one is invariant under the diagonal rotations and under duality, and it is positive: $E_6 = \frac{1}{2}(|\mathbf{B}|^2 + |\mathbf{E}|^2)$ (Theorem 3.90, inherited, restating Theorem 4.26(iii)). The transformations that preserve it and commute with G form $U(3)$; its full linear symmetry group is $O(6)$. The energy is unitary in this sense, but only as a pointwise quadratic form: $U(3)$ is not a symmetry of Maxwell's equations on the frame nor of the split-form interior (Proposition 3.91). In the closure the undivided energy of the momentum field, $E_6(\pi) = \frac{1}{2}|p|^2$, is a Casimir, conserved by every Hamiltonian together with $E_0(\pi)$ and $\mathbf{S}(\pi)$ (Remark 7.6, restating Proposition 4.23); the energies of the frame field are not conserved (Proposition 4.13 and Remark 7.6). The other reading of the clause, three temporal energies forming a vector in V^+, is obstructed: splitting that vector lowers the mass (Proposition 3.101).

Mass arises from an internal differentiation of that quantity. This holds as an identity, not as a process. With no interaction and no second field the undivided energy of a field point splits as $E_6^2 = E_0^2 + |\mathbf{S}|^2$, $E_0 = \frac{1}{2}|z \cdot z|$ (Proposition 3.93), in SI units $u^2 = c^2|\mathbf{g}|^2 + u_0^2$, the field-point form of $E^2 = (pc)^2 + (mc^2)^2$ (Proposition 3.99). The rest part is the complete invariant of Lorentz transformations and duality on nonzero fields, the zero field being a separate orbit (Theorem 3.92), the Lorentz scalar of the stress tensor whose 00 component is E_6 (Proposition 3.94), and the infimum of E_6 over boosts, attained exactly at vanishing Poynting vector for a non-null field and not attained for a nonzero null field, which has no rest frame (Theorem 3.95). All of this is inherited classical electrodynamics in the arena's variables. The arena adds which form plays which role: E_6 belongs to the duality class of Theorem 4.26 and E_0, the modulus of the two Lorentz-class invariants, is built from the Lorentz class, and reading $E = mc^2$ as the locus where the two coincide, the rest frame, is a reading. The differentiation does not happen in time: in the closure the split of the momentum field is fixed by the initial data, all three terms being Casimirs (Remark 7.6). Nor does a whole get its mass from interference: two disjoint counter-propagating null pulses have $q \equiv 0$ everywhere and $M = U > 0$ (Proposition 3.100).

c^2 is the symmetric part. This holds in one precise form. Schur's lemma fixes the isotropic form $C = c \cdot \mathbf{1}$ of the conversion between the temporal and the spatial triple, not the value of c (Theorem 2.2); c is a conversion, not a symmetry, and its weight in E_6 is a unit convention (Proposition 3.98). The form in which the clause is exact is the decomposition of the shift into the two summands of Theorem 4.18,

$$\mathrm{End}(V) = \mathfrak{so}(3,3) \oplus \{Y : Y^\dagger = Y\}, \qquad \kappa G \in \{Y : Y^\dagger = Y\}, \qquad (\kappa G)^2 = -\frac{c^2}{\ell^2}\,\mathbf{1}.$$

The η-self-adjoint summand is the one in which factorisation holds, the one that binds in neutral signature (Theorem 4.8), and the one that carries c, through $\kappa = c/\ell$; the antisymmetric summand holds the connections. The decomposition and the fact that only the first summand binds are fixed; the value $(\kappa G)^2 = -c^2/\ell^2$ is constructed, a definition with ℓ free; and reading the first summand as "c^2" and the second, which holds the connections, as the place of a gauge field is a reading; that a gauge generator differentiates the energy is not supported, since it selects no amount of rest energy (Proposition 9.18). Its limits are fixed results: duality is a symmetry of the positive form, while the interior Lagrangian is built on η, and the two forms lie in the duality and the Lorentz class of Theorem 4.26, which meet in 0; in the interior c^2 enters as $\kappa^2 = c^2/\ell^2$ and the energy is mc^2 times an indefinite, unselected amplitude (Proposition 4.13); and in the closure what $E_0(\pi)$ breaks is the constancy of the rotor rate, not anything carried by c (Remark 7.6).

Collected, the clauses become the following statement, which is a reading of fixed and inherited results. The undivided energy is $E_6 = \frac{1}{2}|z|^2$, the one rotation- and duality-invariant quadratic energy of a six-component frame vector, preserved by the $U(3)$ of transformations commuting with G. It differentiates internally, with no interaction, as $E_6^2 = E_0^2 + |\mathbf{S}|^2$, into a Lorentz-invariant rest energy $E_0 = \frac{1}{2}|z \cdot z|$ and a motion part $|\mathbf{S}|$. What differentiates is the arena's splitting into a temporal and a spatial triple, the real structure $\mathrm{Ad}_{e_1e_2e_3}$. Within $U(3)$ the real structure is preserved by $O(3)$ and E_0 by $U(1)\cdot O(3)$, symmetries of pointwise forms and not of any dynamics; nothing Lorentzian is broken, and the duality $U(1)_G$ is broken to $\mathbb{Z}_2$ only on the space of Lorentz orbits and only by a configuration. The constant c is a conversion, not a symmetry, and E_0/c^2 is a rest-mass density of a field point. The statement has three limits: a field point is not a particle (two boost factors, and the mass of a whole is not interference); in the closure only the momentum field's energies are conserved, while the frame field's oscillate or grow; and a compact gauge generator selects no amount of rest energy, its anti-diagonal part moving a field between null and rest at fixed E_6 and, at three constituents, $\mathfrak{so}(4)_\Delta$ changing the rest energies of the constituents at fixed $\frac{1}{2}|Z|^2$ and q_3. Whether some dynamics selects the rest frame of a field or the value of E_0 is open; nothing in this manuscript does.

m is what breaks the symmetry. This holds for internal symmetries and fails for spacetime ones (Section 7.1).

(i) A nonzero rest energy breaks duality: the duality $U(1)_G$ acts on the Lorentz invariant by $q \mapsto e^{2i\theta}q$, so for $E_0 \neq 0$ the stabiliser of q, or of the Lorentz orbit it labels, is $\{\pm\mathbf{1}\}$ (Proposition 3.96 and Theorem 3.97). Under duality alone a nonzero field point has trivial stabiliser, since $e^{\pi G}F = -F$. At a null field point duality joins the little group. This is inherited, pulled back to the frame.

(ii) For Dirac matter the mass breaks the chiral and Pauli–Gürsey symmetries, $\mathfrak{u}(1)_\Omega \oplus \mathfrak{su}(2) \to \mathfrak{u}(1)_J$, with three broken directions (Sections 7.1.1 and 7.1.4); for an explicit Dirac mass these are directions of the mass orbit, not Goldstone excitations.

(iii) The mass breaks no spacetime symmetry, not even the six-dimensional one: the Dirac mass form is the only Grassmann mass kernel invariant under all of $\mathfrak{so}(3,3)$ (Corollary 7.10). Within $\mathfrak{so}(3,3)$ it is the kinetic term that cuts the symmetry down. In this sense the clause is inverted.

(iv) In the closure the rest energy of the momentum field is the amplitude of the potential of the duality angle, so a null momentum field leaves the rate constant and a massive one makes the angle a pendulum (Theorem 4.28 and Corollary 4.29). Reading the angle as a pseudo-Goldstone boson of duality is interpretive: the breaking is explicit, carried by the Lorentz-class inertia.

m is generated from the b-field. Classically this fails, and the failure is sharp (Section 9.1).

(i) The binding rate is never a gauge potential: for every connection b the field equation factorises covariantly, κG transforming as a tensor and b as a connection, and without κ a constant b is pure gauge along the worldline (Theorem 9.1).

(ii) An admissible connection is compact, by the definition of gauge admissibility after Corollary 1.4. Its part commuting with G splits the rest frequency into charge-labelled lines and is removable along one worldline by a rotating frame (Proposition 9.5); the screening criterion of Theorem 9.9 decides when a mixed connection keeps every orbit bounded; and a connection alone binds nothing (Proposition 9.11).

clause	what the arena gives	verdict
energy undivided, six-dimensional	$E_6 = \frac{1}{2}(\lvert\mathbf{B}\rvert^2 + \lvert\mathbf{E}\rvert^2)$, the unique rotation- and duality-invariant quadratic energy; $U(3)$ preserves it and commutes with G (full group $O(6)$); a Casimir of the closure for the momentum field; the temporal-vector reading lowers the mass	holds in the field-strength reading (inherited); the temporal reading is obstructed
mass from internal differentiation	$E_6^2 = E_0^2 + \lvert\mathbf{S}\rvert^2$, $E_0 = \inf_{\text{boosts}} E_6$, attained at vanishing Poynting vector and not for a nonzero null field; in the closure the split is fixed by the initial data	holds as an identity, not as a process
c^2 the symmetric part	Schur's lemma fixes the isotropic form $c \cdot \mathbf{1}$ of the conversion, not the value of c (Theorem 2.2); κG the η-self-adjoint summand of the shift, the only one that binds	the decomposition is fixed; the reading is interpretive
m breaks the symmetry	breaks duality and the chiral algebra; breaks neither Lorentz nor Spin$(3,3)$	true for internal symmetries, false for spacetime ones
m generated from the b-field	the connection summand splits or tears the binding and never creates it; vector bosons acquire mass by eating Goldstones	fails classically; a scale by transmutation is conjectural

Table 13.4: The rest-energy hypothesis against the arena (Section 13.5).

(iii) At three constituents the part of the Pati–Salam algebra commuting with the binding root is $\mathfrak{so}(4)_\Delta \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$ (Theorem 4.36); it splits the rest frequency by weights and supplies no scale (Lemma 9.14).

(iv) In the selected vacuum the order is reversed: the Pati–Salam vector bosons acquire mass by eating Goldstone directions (Section 7.7 and Theorem 7.41) rather than generating the mass of anything.

The one role left for the b-field is the scale. The algebra fixes the isotropic form of the conversion map, while its normalisation and the dynamical length scale ℓ, hence $\kappa = c/\ell$, require additional input; the identification of $\hbar\kappa$ with a scale generated by dimensional transmutation is a conjecture (Conjecture 9.17), and reading the rate as a mass, $mc^2 = \hbar\kappa$, is a further hypothesis (Section 4.5.6).

Collected, the strongest statement of the hypothesis that the arena supports is the following (Table 13.4). The undivided energy is the duality-invariant energy $\frac{1}{2}(\lvert\mathbf{B}\rvert^2 + \lvert\mathbf{E}\rvert^2)$ of the six field components. The conversion between times and lengths is isotropic, $c \cdot \mathbf{1}$, by Schur's lemma, which does not fix the value of c, and it enters the dynamics through the η-self-adjoint part of the shift, the binding root. The mass is the Lorentz- and duality-invariant $\frac{1}{2}|q|$, the infimum of the energy over boosts, attained in the rest frame of a non-null field, and $E = mc^2$ is the rest frame, where the two coincide. What the mass breaks is duality and, for matter, chirality, never Lorentz invariance or Spin$(3,3)$. The b-field is the antisymmetric part of the same shift: it can split or destabilise the binding but cannot create it. Every part of this statement is fixed, inherited or interpretive as marked; none of it is a derivation of a mass.

13.6 Duplication and the b-field between manifolds

The second question of the motivating hypothesis (Section I.2) asked what process makes a unitary energy result in a duplication, from Cl$(3,3)$ to Cl$(6,6)$, and whether the b-field is a deal breaker between two manifolds or more. The answers of Chapters 5 and 9 are collected

statement	what the arena gives	verdict
the algebra is duplicated	$\mathrm{Cl}(6,6) = \mathrm{Cl}(3,3)\hat{\otimes}\mathrm{Cl}(3,3)$, Klein factor, Fock space (Section 5.3.1)	forced; inherited
the energy is duplicated	a beam-splitter S^3 of covariant embeddings, one η-respecting circle; $z \mapsto (z,z)$ doubles the energy, $(z,z)/\sqrt{2}$ divides it; cloning of unknown quantum states excluded (Theorem 5.24)	chosen; cloning obstructed (inherited)
the Lorentz action on the copies	one of five at $n = 2$; it decides the spin of the duplicate and what an inter-manifold field reaches (Table 5.1 and Propositions 5.29 and 9.19)	chosen
the duplicate adds structure	complex structure, copy charge, $\mathfrak{so}(2,\mathbb{C})$, S^4 vacuum, balance surface (Propositions 5.26, 5.27 and 5.30 and Theorem 5.52)	fixed (partly restated)
a pair from one null field	impossible covariantly; two null fields or a third body (Proposition 5.56 and Theorem 5.57)	obstructed; inherited kinematics
what performs the duplication	not the closure (Corollary 5.54); the copy rotation: a gauge turn at equal rates, Rabi mixing at unequal ones (Section 5.6.4); a copy rotor (Remark 5.61)	obstructed in the closure; mixing inherited; rotor constructed
the mass of one copy	not invariant under the copy rotation (Proposition 5.60)	fixed; a distinguishing structure open
a b-field between manifolds	$\mathfrak{so}(n,\mathbb{C})$, compact part $\mathfrak{so}(n)$; a spectator manifold decouples (Proposition 9.19)	fixed
it is a deal breaker	non-removable at a fixed rate matrix where it couples unequal rates (on an open interval $e^{-\tau b}$ removes it at the price of a time-dependent rate matrix): seesaw, confinement, resonance; a time-odd invariant at $n = 3$ (Theorems 9.21 and 9.23); removable in the common-rate closure (Remark 9.26)	fixed
it generates a rate	rates are constant in the fixed-rate model, not by gauge invariance (Theorem 9.21); closed worldlines need a holonomy statement	obstructed in the fixed-rate model
the rates differ	not in the common-rate closure; a non-central rotor in $\mathfrak{u}(n)$ admits them (Proposition 9.20)	constructed; origin open
isospin one-half between manifolds	copy isospin integral at $n = 2, 3$ under a common action; doublets from $n = 4$ (Propositions 9.24 and 9.25)	obstructed at $n = 2, 3$

Table 13.5: Verdicts on duplication and on the b-field between manifolds (Section 13.6).

in Table 13.5; as before, a rate becomes a rest energy only through the added identification $mc^2 = \hbar\kappa$.

How a unitary energy results in a duplication. The algebra step is forced and the energy step is chosen. Doubling the vector space forces the graded tensor product, whose spinor module is the Fock space of six fermionic modes. The undivided energy embeds isometrically in the duplicate, and the Lorentz-covariant embeddings form the three-sphere of a lossless beam splitter, of which one circle respects η. Read as a multiplication of the energy, the phrase fails: the well-defined classical map $z \mapsto (z,z)$ doubles the quadratic energy, $E_{12}(z,z) = 2E_6(z)$, while $z \mapsto (az, bz)$ with $|a|^2 + |b|^2 = 1$, for instance $(z,z)/\sqrt{2}$, preserves it and divides it between the copies; implementing an energy-increasing map physically requires an energy source. The no-cloning theorem, which concerns unknown quantum states, is a separate statement about copying a quantum. The algebra step creates no pair: in $\mathrm{Cl}(6,6)$ the graded masses add in quadrature, and the pair threshold $2m$ appears only for the ungraded operator (Proposition 5.62). What the duplicate can do that one copy cannot is precise: massive constituents of the closure can turn at an exactly constant rate on a codimension-two balance surface (Theorem 5.52), with no b-field.

The b-field between manifolds. The remark that the b-field is a deal breaker between manifolds is true for transmission and false for creation. A compact binding-compatible field coupling manifolds of unequal rates cannot be removed along a worldline while the rate matrix is kept fixed. On an open interval the gauge $e^{-\tau b}$ removes it at the price of a time-dependent rate matrix, with $\frac{d}{d\tau}K' = [b, K]$ at $\tau = 0$, and on a closed worldline a holonomy statement is needed. At a fixed rate matrix it shifts and spreads the lines, confines a frame without rate of its own by a seesaw, and from three manifolds on carries a time-odd invariant, the arena's analogue of Jarlskog's invariant (Theorems 9.21 and 9.23). But in the fixed-rate model the rates are the spectrum of the self-adjoint part of the shift, which transforms as a tensor, so no field creates or changes a rate; that the rates are constant follows from the model, not from gauge invariance; with equal rates every binding-compatible field is removable, $\mathfrak{so}(4)_\Delta$ included, and in the common-rate closure the deal breaker has no effect (Remark 9.26). The part of the copy field that survives every change of frame is proportional to the rate differences. Read with $mc^2 = \hbar\kappa$, the order of the hypothesis is reversed: the differentiation of the rates breaks the copy symmetry, and the b-field acts only because it is broken. Formally this has the structure of the Englert–Brout–Higgs–Kibble mechanism with the rate matrix as an adjoint vacuum value; that reading is interpretive, since here b has no kinetic term and the rates are inputs. Their origin is open: a non-central rotor generator, admitted but not selected, or a quantum scale by transmutation, which is conjectural.

Chapter 14

Comparison with Related Geometric Approaches

The question of this chapter is where the constructions of this manuscript stand among other attempts to read spacetime and particle physics off an algebraic or geometric structure. Comparisons that concern a single statement are made where the statement occurs; Section 14.5 lists them by chapter. The chapter first places the manuscript in the wider literature (Section 14.1) and among other work in signature $(3,3)$ or with several times (Section 14.2). It then treats in more detail the two programmes closest in method: Krasnov's realisation of one generation on the forms of a doubled split space, which shares the split signature and the 64-dimensional generation (Section 14.3), and the spectral programme of Chamseddine and Connes, which shares the strategy of reading the gauge symmetry off an algebra (Section 14.4). The last section sorts the statements of the manuscript by provenance, with the labels of Section I.1.2 (Section 14.6). The comparisons rest on the sources as read and recorded by the author. Equal dimensions are never taken as equivalence, and a shared signature is not taken as a shared theory.

14.1 Related work

Clifford algebras and real spinors. The classification of real Clifford algebras by signature, their periodicity and their gradings is classical [282, 85, 88, 125]. Floerchinger [283] reviews it for every signature in the language of relativistic fermions, with spinors as minimal ideals and the two Dirac adjoints. The identities on which this manuscript rests are instances of that classification and are inherited: $\mathrm{Cl}(3,3) \cong M_8(\mathbb{R})$, and the graded tensor decomposition $\mathrm{Cl}(3n,3n) \cong \widehat{\bigotimes}^n \mathrm{Cl}(3,3)$ behind the tower (Section 5.1), which goes back to [282]. The Kronecker realisation of Section 1.1 is a choice of basis for them. The use of Clifford algebras as a language for relativistic physics goes back to Hestenes' space-time algebra [70]. The one-equation form of Maxwell's equations used on the octants in Chapter 3 is Hestenes' form transferred to another module. The real form of Dirac theory is his [71], and is compared with Chapter 2 in the next paragraph.

Real spinor fields and the imaginary unit. In Hestenes' formulation [71] a Dirac spinor is an element ψ of the even subalgebra of the real space-time algebra $\mathrm{Cl}(1,3)$, a real eight-dimensional space. The gamma matrices act as $\psi \mapsto \gamma_\mu \psi \gamma_0$, and the imaginary unit of the Dirac equation is right multiplication by the bivector $\gamma_2\gamma_1$. The real operators that commute with this action are the right multiplications by the even elements commuting with γ_0. They form a quaternion algebra, and its square roots of -1 form a two-sphere. $\gamma_2\gamma_1$ is one point of that sphere, fixed by convention together with a spatial frame. Chapter 2 meets the same structure: the commutant of the four interior gamma matrices in $M_8(\mathbb{R})$ is again a quaternion algebra with a two-sphere of complex structures (Remark 2.18). What it adds is a rule that

removes this freedom under stated requirements. Commutation with the internal bivectors, or with the volume element of the temporal triple, leaves $\pm J$ (Theorem 2.17). Here J is an element of the ambient algebra, the rotation generator of the surplus plane, rather than a right action on an ideal. The equivalence with complex Dirac theory is inherited (Theorem 2.19 and Corollary 2.20). What the arena contributes is the selection rule and its conditions.

Algebraic approaches to Standard-Model structure. Several programmes extract Standard-Model structure from division and Clifford algebras [45, 58]. The representation-theoretic limits of embedding all of it in a single exceptional group are discussed in [34]. This manuscript shares the method of reading internal symmetry off centralisers of a Lorentz action in a Clifford algebra. It differs in the algebra, the split real algebras $\mathrm{Cl}(3n, 3n)$, and in its emphasis. It records which choices the algebra removes under stated requirements and which it leaves, and it treats a fit of Standard-Model content as a consistency result, not as evidence (Corollary 2.20).

Pati–Salam and $SO(10)$. The content of the chiral block is one Pati–Salam family with a right-handed neutrino (Theorem 6.10 and Corollary 6.12). It is that of the Pati–Salam model [116] and of the spinor **16** of $SO(10)$ [43], and its anomaly freedom is inherited [50]. The same content is realised by Krasnov's even forms (Section 14.3) and in the spectral Pati–Salam model (Section 14.4). It is shared prior art; what can distinguish this manuscript is how the content is realised, what selects it, and which dynamics it carries. On the family $(\mathbf{4}, \mathbf{2}, \mathbf{1}) \oplus (\bar{\mathbf{4}}, \mathbf{1}, \mathbf{2})$ the group that acts faithfully is $(SU(4) \times SU(2)_L \times SU(2)_R)/\mathbb{Z}_2$; "$SO(4) \times SO(6)$" below is shorthand at the level of Lie algebras. The 272 real dimensions of mass kernels are $2 \dim_{\mathbb{C}} \mathrm{Sym}^2(\mathbb{C}^{16}) = 2(10 + 126)$. With the complex structure $\widetilde{\mathcal{C}}$ of Theorem 7.17, the kernel space carries the Pati–Salam branching of $\mathbf{10} \oplus \overline{\mathbf{126}}$, the dual of $\mathrm{Sym}^2(\mathbf{16})$, as befits the coefficients of the mass bilinears of one family. These labels, and the seesaw and Georgi–Jarlskog structures of Chapter 7, are standard model building. What is specific here is that they occur inside the realisation $V_* = \mathrm{im}\, P_*$ and are computed in it. The radiative vacuum analysis of Chapter 7 is to be compared with that of the minimal non-supersymmetric $SO(10)$ models [302, 303].

Gravity and unification in split signature. Nesti and Percacci [191] and Maraner [192] proposed embedding one generation, together with the Lorentz group, in a single orthogonal group, $SO(3, 11)$ and $SO(7, 7)$ respectively. Krasnov's construction (Section 14.3) realises the $SO(7, 7)$ case on forms. The tower $\mathrm{Cl}(3n, 3n)$ at $n = 3$ has eighteen generators, not fourteen, and its matter block is selected by a projector rather than by a Weyl condition. None of these embeddings is reproduced here; the comparison at the level of the chiral block is made in Remark 6.24 and Section 14.3. Nor does this manuscript derive gravitational dynamics. Chapter 11 adopts a Palatini–Holst action and recovers the Einstein tensor of a prescribed four-metric (Section 11.4).

Forms as spinors. The realisation of the spinor module of a split space on the exterior algebra of a null half-space is Chevalley's [201]. Gualtieri [203, 295] develops it, together with the half-density factor that makes the pairing invariant. The reading of the operator $d - \delta$ on forms as a Dirac operator is the Dirac–Kähler construction [205, 202, 200, 209]. Both are inherited at one constituent (Proposition 1.11). At three constituents the manuscript uses no form realisation.

Electromagnetism. The comparisons for the field-strength reading are made in Chapter 3, each with its class:

- the Chevalley–Mukai pairing and the pure-spinor criterion [201, 203];
- Rainich's identity [129];

- the energy–inertia identities of a field point, due to Kaiser [298] (Section 3.9);
- duality rotations of one and of several Maxwell fields [46];
- the ModMax family [197, 210];
- the textbook Pauli and electric-dipole couplings [207], whose real forms and count at every clock are given in Theorem 3.64.

The duality between the copy algebra and the diagonal Lorentz algebra along the tower is an instance of Howe duality [224] (Corollary 5.12), and its Casimir relation is a special case of Neergård's [296, 297].

14.2 Signature $(3,3)$ and several times

Six-dimensional spacetimes of signature $(3,3)$, and physics with more than one time, have been studied for other reasons. The overlap with this manuscript lies more in the setting than in the results, and each comparison below says where it ends.

Two-time physics. Bars and collaborators work in signature $(d,2)$ [284, 285]. A worldline action with a local $Sp(2,\mathbb{R})$ symmetry, which treats position and momentum alike, imposes $X^2 = X \cdot P = P^2 = 0$ and removes one time and one space direction. Different gauge fixings of the same two-time system then give different one-time systems. The point of contact is Theorem 12.12: there too the selected worldline model removes the surplus directions through explicit first-class constraints that generate a worldline gauge symmetry; choosing that model and its gauge symmetry remains an input. The differences lie in each ingredient:

- here the signature is $(3,3)$, and two surplus directions of the sign of time are removed, not a $(1,1)$ pair;
- the gauge group is the abelian group of local translations along the normal plane, which closes with structure functions in a background field, not $Sp(2,\mathbb{R})$;
- the symmetry is chosen to remove the unstable band of Section 12.1, not derived from a duality between position and momentum;
- it is shown for a test worldline only.

No two-time relation between different one-time systems is claimed.

Three-dimensional time. Chen [287] posits three time dimensions and argues that quantum phenomena (non-locality, spin, gauge invariance) arise from motion in them. He also proposes a $3+3$ Kaluza–Klein unification of gravity, electromagnetism and the strong interaction. This manuscript takes the opposite position on the quantum side. The real Dirac theory on $\mathrm{Cl}(3,3)$ is complex Dirac theory written in real variables (Theorem 2.19), so no quantum phenomenon follows from the rewriting (Corollary 2.20). The temporal reading of the positive triple is an interpretation (Section 2.1), whose surplus directions must be removed or made harmless (Chapter 12). Relative to the ambient Einstein–Hilbert action, the Kaluza–Klein vectors of the normal plane are ghosts (Theorem 12.15). The two works share the signature and the temporal reading of three directions, and nothing else.

An $SO(3,3)$ gauge theory. Wesley, Singh and Isidro [286] formulate a BF-type gauge theory of $SO(3,3)$ on a six-dimensional spacetime of signature $(3,3)$ and break it by a MacDowell–Mansouri-type mechanism. They read two four-dimensional Lorentzian sectors of opposite signature, glued along a common $(1,1)$-plane, as gravity and the weak interaction. Their Clifford-algebra appendix uses $\mathrm{Cl}(3,3) \cong M_8(\mathbb{R})$ with $\eta = \mathrm{diag}(+,+,+,-,-,-)$, as here. The geometric difference is in the four-planes. Here a clock selects one Lorentzian four-plane $E_{\hat{n}} = \mathbb{R}\hat{n} \oplus V^-$, whose complement is a positive two-plane (Section 2.2), and no second sector is used. The weak interaction is given no geometric reading, the gauge algebras are centralisers in the tower (Chapters 5 and 6), and gravitational dynamics is not derived. The two constructions put different questions to the same algebra; whether their selections of four-planes can be related is not examined.

Dirac equations in six dimensions. Dartora and Cabrera [288] study a Dirac equation on a six-dimensional space of signature $(3,3)$ with a locally gauged $SO(3,3)$. The symmetry is broken locally to the Lorentz group of a selected time direction, and the result is a non-chiral, explicitly toy, "electroweak" theory. Their work is prior art for the general architecture of Chapters 2 and 12: a split-signature six-dimensional Dirac module from which a four-dimensional Lorentzian theory is obtained by selecting a time direction. What this manuscript adds there is:

- the selection rule for the imaginary unit (Theorem 2.17);
- the realification theorem and its corollary;
- the tower and its chiral block;
- the obstructions to causal selection within the invariant adjoint-scalar ansatz (Section 12.3).

Evolution parameters and ultrahyperbolic data. A universal invariant evolution parameter, distinct from coordinate time, is explicit in the work of Horwitz and Piron [190] and of Horwitz [289]. The parameter of Section 12.6 belongs to that line, and any originality concerns its implementation here and its connection with the clock reduction. That the unrestricted ultrahyperbolic problem is ill-posed, while suitably constrained evolution can be well posed, is due to Craig and Weinstein [27]. Tegmark's dimensional argument [149] is cited in Chapter 12 for the same baseline.

14.3 Krasnov's construction

Krasnov [211] realises the spinors of $SO(n,n)$ on the inhomogeneous forms of a null n-plane, in the setting of doubled geometry, where the Dirac operator acts by wedge in one half of the coordinates and by contraction in the other. The Lorentz group is selected by a second splitting, a choice that the paper does not make dynamical, and all fields depend on four of the $2n$ coordinates.

At $SO(3,3)$ the commutant of the Lorentz group is $SO(2)$, and the kinetic term reduces to that of one charged two-component Weyl fermion [211, §4]. At $SO(7,7)$ the commutant is $SO(4,6)$. Its compact part $SO(4) \times SO(6)$ is the Pati–Salam group at the level of Lie algebras, with $\mathrm{Spin}(4) \cong SU(2) \times SU(2)$ and $\mathrm{Spin}(6) \cong SU(4)$. One generation with ν_R is a real even form on $\mathbb{R}^7$, of real dimension 64. In two-component notation it consists of Weyl spinors ξ^{aI} of $SU(2)_L$ and $\bar{\xi}^{aI}$ of $SU(2)_R$, all of one Lorentz chirality, with the lepton number as a fourth colour. In the kinetic term [211, eq. (219)] the two sectors enter with opposite signs, and Krasnov states that the result coincides with the kinetic term of the Standard Model. Bosons and symmetry breaking are left open. Table 14.1 compares the two constructions along the axes that matter.

What is shared and what is not. Three statements are equivalent:

- the Chevalley realisation of the spinors of a split space on the forms of a null half-space, which both constructions inherit;
- the commutant $\mathfrak{so}(2)$ of a Lorentz subalgebra in $\mathfrak{so}(3,3)$;
- the representation content of the common algebra $\mathfrak{so}(1,3) \oplus \mathfrak{so}(4) \oplus \mathfrak{so}(6)$.

On the last point, both 64-dimensional modules are the underlying real representation (realification) of $(\mathbf{2},\mathbf{1}) \otimes [(\mathbf{4},\mathbf{2},\mathbf{1}) \oplus (\bar{\mathbf{4}},\mathbf{1},\mathbf{2})]$ (Corollary 6.12; [211, §6]). The right-handed block of V_*, written in the other Lorentz chirality, is the conjugate of the second summand, and a complex representation and its conjugate have the same underlying real representation. Finite-dimensional real representations of a semisimple Lie algebra are completely reducible, so the two modules are equivalent as representations of that algebra. This holds once its two embeddings are identified compatibly with the labels: the outer automorphisms that exchange $SU(2)_L$ with $SU(2)_R$, $\mathbf{4}$ with $\bar{\mathbf{4}}$, or the two Lorentz chiralities must be fixed in the same way on both sides. No explicit map is needed to know that one exists.

Three statements are analogous:

- Lorentz and Pati–Salam in one split-signature Clifford algebra with a 64-dimensional one-generation module;
- the non-dynamical selection of the Lorentz subgroup;
- the imposed reduction to four coordinates.

The equivalence concerns the common algebra only. The ambient actions differ: Krasnov's module carries $\mathfrak{so}(4,6)$, while V_* carries only its compact part, together with the two abelian generators of $\mathfrak{k}_{\text{eff}}(P_*)$ (Theorem 6.11).

What remains to be constructed. An intertwiner of the common algebra exists by the previous paragraph, but none is written down here. Constructing it does not require identifying a block of $\mathrm{Cl}(9,9)$ with $\mathrm{Cl}(7,7)$. $\mathrm{Cl}(7,7) \cong M_{128}(\mathbb{R})$ has no nonzero module of dimension below 128, and the 64-dimensional half-spin space carries only the even part $\mathrm{Cl}^0(7,7) \cong M_{64}(\mathbb{R}) \oplus M_{64}(\mathbb{R})$. Nor would removing from $\mathbb{R}^{9,9}$ the positive four-plane on which the normal charge Q_A lives leave $\mathbb{R}^{7,7}$: the complement has signature $(5,9)$, and a complement of signature $(7,7)$ requires removing a $(2,2)$-plane.

Three steps remain open:

(a) identify explicitly the generators of $\mathfrak{so}(1,3) \oplus \mathfrak{so}(4) \oplus \mathfrak{so}(6)$ on both sides, with their reality conventions and the outer automorphisms fixed;

(b) write the map;

(c) decide whether it carries the kinetic forms into one another.

Step (c) needs a common choice of complex structure, conjugation, Grassmann ordering and field variables. Without these, the opposite signs in Krasnov's eq. (219) and the single sign $K^0 = \mathbf{1}$ on V_* cannot be compared. The displayed signs alone establish neither a ghost on one side nor its absence on the other. The absence of the 24 non-compact generators of $\mathfrak{so}(4,6)$ on V_* (Remark 6.24) obstructs only the extension of such a map to the ambient $\mathfrak{so}(4,6)$ action, not the map itself.

14.4 The spectral programme of Chamseddine and Connes

Chamseddine and Connes obtain the Standard Model from a spectral triple. An algebra $\mathcal{A} = C^\infty(M) \otimes \mathcal{A}_F$ acts on a Hilbert space with a Dirac operator D, a real structure J_{C} and a grading γ. The classification of the irreducible finite geometries of KO-dimension 6 makes the number of fermions per generation a square, k^2, the full finite Hilbert space $\mathrm{Hom}_{\mathbb{C}}(V, W) \oplus \mathrm{Hom}_{\mathbb{C}}(W, V)$ having complex dimension $2k^2$, fermions and their charge conjugates [194, 195]. Two further inputs then fix $k = 4$, and with it 16 fermions per generation (32 for the finite space) [195, §3]: the hypothesis that the grading and the real form come from a quaternionic space, and the choice of its simplest non-trivially graded case, of quaternionic dimension two. The finite algebra $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ is then, up to an automorphism of the even part $\mathcal{A}^{\mathrm{ev}}$, the unique involutive subalgebra of maximal dimension that admits off-diagonal Dirac operators, those with $[D, Z(\mathcal{A})] \neq 0$ that satisfy the order-one condition [195, Thm. 4.1]. The condition on the centre is part of the selection: without it, $D = 0$ satisfies the order-one condition for every subalgebra.

Gauge and Higgs fields arise as inner fluctuations $D \mapsto D + A + J_{\mathrm{C}} A J_{\mathrm{C}}^{-1}$. In the Euclidean formulation [193, 25, 290], the spectral action $\mathrm{Tr}\, f(D/\Lambda)$ yields the bosonic Standard-Model action coupled to gravity. The gauge couplings are related at the cut-off scale, $g_3^2 = g_2^2 = \frac{5}{3} g_Y^2$. The relation comes from the relative coefficients of the gauge kinetic terms in the asymptotic expansion of the spectral action. It is therefore a prediction of the model at the cut-off scale, conditional on its hypotheses, before any low-energy consequence is computed [290]. The inputs are:

- the number of generations;
- the finite Dirac operator, with its Yukawa and Majorana data;
- the cut-off function and scale;
- the hypotheses and the minimal choice of the classification.

The spectral Pati–Salam model. Chamseddine, Connes and van Suijlekom drop the order-one condition [291, 292]. The inner fluctuations then acquire a part quadratic in the linear one, and the classification leads to a Pati–Salam model with gauge group $SU(2)_R \times SU(2)_L \times SU(4)$. It has 16 fermions per generation, in $(\mathbf{2}, \mathbf{1}, \mathbf{4}) \oplus (\mathbf{1}, \mathbf{2}, \mathbf{4})$ in their ordering of the factors. Its Higgs fields lie in representations fixed by the geometry, and they break the symmetry to that of the Standard Model at high energies. The running of its gauge couplings has been analysed at one loop [293].

The fermion content is the one realised here by V_*, so the Pati–Salam family is shared prior art in the spectral programme as well. The two programmes differ in three respects:

- the realisation: a spectral triple without the order-one condition there, a projected block of a split Clifford module here;
- what selects the content: a classification under stated hypotheses there, the choice of P_* here;
- the bosonic sector: constrained geometrically by the spectral action there, adopted here.

What the two programmes share. This manuscript shares with the spectral programme the strategy of reading the gauge symmetry off an algebraic structure, and none of its objects. It has no algebra subject to order-zero and order-one conditions, no real structure of KO-dimension 6 and no spectral action. Its gauge algebras are centralisers in a split Clifford algebra, its matter block is the image of a chosen projector, and its actions are adopted, not derived from a spectral

principle. Its inputs are the clock, the projector, the Lorentz action, the actions and potential, and the scales. It is therefore not a spectral-triple theory, and no number of the spectral-action programme transfers to it.

From a charge table to a mixing angle. Three steps lead from a charge table to a measured angle, and they must be kept apart.

(a) Normalising the hypercharge generator as $T_Y = \sqrt{3/5}\,Y$, so that $\operatorname{Tr} T_Y^2 = \operatorname{Tr}(T_L^3)^2$ over a family ($\operatorname{Tr} Y^2 = \frac{10}{3}$, $\operatorname{Tr}(T_L^3)^2 = 2$), is a convention. It defines $g_1 = \sqrt{5/3}\,g_Y$.

(b) The equality $g_1 = g_2$ at some scale is a physical boundary condition, which must be assumed or derived. With it, $\sin^2\theta_W = g_Y^2/(g_Y^2 + g_2^2) = \frac{3}{8}$ at that scale. With the same charges and $g_Y = g_2$ the value would be $\frac{1}{2}$: normalising the generators does not make the couplings equal.

(c) The spectrum, the thresholds and the running connect that boundary condition with couplings measured at other scales.

In the spectral action, step (b) is supplied by the action itself, as a conditional prediction at the cut-off. In this manuscript the charge traces fix only step (a). Step (b) comes from the single gauge coupling of the adopted action (7.4), at an unspecified scale (Theorem 7.34): it is a consequence of a chosen action, not of the algebra or of a spectral principle. Step (c) is not supplied, so no low-energy angle is predicted.

14.5 Comparisons made in the chapters

The comparisons below are made where the statement occurs; this section only lists them. Each entry gives the place, the source, and the relation: *inherited* (the result is the source's, used with its hypotheses), *prior art* (the source does the same thing in another setting), or *compared* (the source treats a related question, and the chapter says where the two differ).

- Chapter 1. Split spinors as forms and the half-density factor (Section 1.2): Gualtieri [203, 295], inherited. The $SO(3,3)$ case with its Lorentz embedding and extra $SO(2)$ (Proposition 1.7): Krasnov [211, §4], equivalent as a Lie-algebra statement. The algebras $\mathrm{Cl}(p,q)$: [282, 283], inherited.
- Chapter 2. Real spinor fields and the imaginary unit (Theorems 2.17 and 2.19 and Remark 2.18): Hestenes [71], inherited for the real form and compared for the selection (Section 14.1).
- Chapter 3. The rest-energy and inertia identities of a field point (Section 3.9): Kaiser [298], inherited.
- Chapter 4. A complex structure anti-isometric for the form forces neutral signature, and the complex-orthogonal centraliser (Theorems 4.8 and 4.34): the linear algebra of Norden (anti-Kähler) geometry [301], inherited.
- Chapter 5.
  - The copy–diagonal dual pair and its Casimir constants (Corollary 5.12 and Theorems 5.11 and 5.68): Howe [224] and Neergård [296, 297], inherited. The Clifford-algebra proof is an independent derivation in the real realisation.
  - The chirality strings: Jordan–Wigner [212], inherited.
  - Duality rotations of several Maxwell fields: Gaillard–Zumino [46], inherited.

  - The pair invariant (Theorem 5.57): compared with Kaiser [298] and with Thompson's results on complex symmetric matrices [299]; priority not established.

- Chapter 6. One chiral family with the Lorentz group in one orthogonal algebra (Theorems 6.10 and 6.11, Corollary 6.12, and Proposition 6.13): Maraner [192], Nesti–Percacci [191], Krasnov [211, §§6.4–6.7] and the spectral Pati–Salam model [292], prior art. With Krasnov's module, V_* is equivalent as a representation of $\mathfrak{so}(1,3)\oplus\mathfrak{so}(4)\oplus\mathfrak{so}(6)$ (Section 14.3). What the realisation adds is the projector P_* and the computations inside V_*. Remark 6.24 distinguishes the ambient actions only.

- Chapter 7. The radiative vacuum analysis (Theorems 7.30 and 7.32): Bertolini, Di Luzio and Malinský [302, 303], compared. The exclusion of an electroweak-scale axion (Theorem 7.42): Konaka et al. [304] and Peccei, Wu and Yanagida [305], inherited. The value $\sin^2\theta_W = \frac{3}{8}$ from the single coupling of the adopted action (Theorem 7.34): compared with the spectral action [290] (Section 14.4).

- Chapter 8. Sigma models on homogeneous spaces and their gauging (Sections 8.1 and 8.3): Coleman, Wess and Zumino and Callan, Coleman, Wess and Zumino [306, 307], inherited. Anomalies and massive anomalous gauge fields (Remark 8.34 and Theorem 8.52): Preskill [308], inherited.

- Chapter 9.

  - Roth's theorem (Theorem 9.4): the invariant proof of Flanders and Wimmer [309], inherited.
  - Quadratic fermion dynamics (Theorem 9.27): Bravyi [310], inherited.
  - The truncated graded brackets (Proposition 9.36): current Lie algebras [311], inherited.
  - Invariant bilinear forms on spinor modules (Theorem 9.62): Alekseevsky–Cortés [312], inherited.

- Chapter 10.

  - The rest-energy fraction of a Gaussian field (Proposition 10.3): the real Wishart eigenvalue law [300], inherited.
  - Oscillating birefringence and its washout (Section 10.1.5; the duality–helicity relation is Proposition 10.7): Fedderke, Graham and Rajendran [313], compared.
  - Oscillation searches with atomic clocks (Section 10.1.6): Van Tilburg et al. [314], compared.
  - The ACT DR6 angle $0.215^\circ \pm 0.074^\circ$: Diego-Palazuelos and Komatsu [315], quoted with its authors' caution on systematics.

- Chapter 11. Non-minimal fermion coupling in the Holst action (Proposition 11.14): Mercuri [316], compared. Duality-invariant linearised gravity (Theorem 11.10 and Corollary 11.12): Henneaux–Teitelboim [317], compared. The unimodular formulation: [318], compared.

- Chapter 12.

  - Spinor coordinates of the null cone (Theorem 12.5): Penrose [319], inherited.
  - Real orbits and invariants (Section 12.3): Richardson–Slodowy [320] and Jablonski [321], inherited.
  - Gauge constraints that remove surplus times (Theorem 12.12): Bars, Deliduman and Andreev [284], compared (Section 14.2).

  - A universal evolution parameter (Section 12.6): Horwitz–Piron [190] and Horwitz [289], prior art.
  - The general architecture: Dartora–Cabrera [288], prior art.
  - The sum-over-histories motivation of the causal programme: Feynman [322], motivation only.

- Chapter 16. The coupling relation at the cut-off (Section 16.4): Chamseddine, Connes and Marcolli [290], compared (Section 14.4). An index theorem on Lorentzian manifolds (Section 16.3): Bär and Strohmaier [294], inherited as a condition on any index statement.

14.6 What is inherited and what is new

The statements of this manuscript are sorted here with the labels of Section I.1.2. The labels record the logical status of a statement and where it comes from; they are not priority claims. *Constructed* means exhibited in this realisation, not new. A known theory written in the arena's variables, such as Dirac theory over the reals [71], stays inherited however it is realised. What may be specific to this manuscript lies in the additional requirements under which a structure is selected, and in the computations made inside the realisation.

(i) *Inherited: known results, restated in the arena's variables or used with their hypotheses.*

- Clifford algebras and spinors: the classification of real Clifford algebras, $\mathrm{Cl}(3,3) \cong M_8(\mathbb{R})$, and the graded tensor-product identity behind the tower; the Chevalley realisation and the Dirac–Kähler construction.
- Dirac theory: ordinary complex Dirac theory and its real form (Theorem 2.19).
- Electromagnetic field: the Lagrange and Rainich identities of the field energy; the energy–inertia identities of a field point; Maxwell's equations and their duality rotations; the Wishart law of Proposition 10.3.
- Linear algebra and dual pairs: the linear algebra of a complex structure anti-isometric for a neutral form; the reductive dual pair of the copy and diagonal algebras and the Casimir relation $C_{\mathrm{cop}} + C_{\Delta} = \frac{1}{2}n(n+2)$, with the Clifford-algebra proof as an independent derivation; Roth's theorem.
- Matter content: the Pati–Salam and $SO(10)$ content, its anomaly freedom, and the count $2\dim_{\mathbb{C}} \mathrm{Sym}^2(\mathbb{C}^{16}) = 272$ of mass kernels.
- Geometry and gravity: Kaluza–Klein reduction, O'Neill's formulas, Lanczos's identity and the Holst action.
- Worldlines: the reparametrisation invariance of a worldline and the invariant evolution parameter of Horwitz and Piron.

Centraliser computations are standard methods; their results along the tower are statements about this realisation and appear in (iii).

(ii) *Constructed: explicit realisations exhibited in the algebra, without anything selecting them among alternatives.*

- the tower $\mathrm{Cl}(3n, 3n)$ as a common setting for these structures;
- the chiral projector P_* built from three commuting involutions, with the normal charge on an extra positive four-plane;
- the field-strength reading, with G as the Hodge star and the duality phase of the curvature as a Barbero–Immirzi angle;

- the interior dynamics, its closure within the stated ansatz, and its rigid-body realisation;
- the worldline gauge symmetry of Theorem 12.12 for a test worldline.

(iii) *Fixed by the algebra, within this realisation: classifications that could have come out otherwise.*

- the selection of the imaginary unit under the stated commutation requirements (Theorem 2.17);
- the internal centralisers along the tower;
- the 272-dimensional kernel space with its complex structure (Theorem 7.17).

(iv) *Obstructed, under hypotheses stated with each result and each within its own model.*

- the absence of a nonzero Lorentz-equivariant linear intertwiner between the displacement and field-strength modules;
- hypercharge from anomaly cancellation in the class of three involutions;
- colour in the effective bivector algebras of the axial members of the five-projector family (Corollary 6.6);
- the normal $U(1)$ as the photon, each objection in its own model;
- causal type selection by an unconstrained adjoint scalar with an invariant bilinear kinetic term (Section 12.3).

The vacuum selections of Chapters 7 and 8 depend on the coefficients of the adopted potentials and are conditional results. In Chapter 7 global minimality on the full kernel space below x_* is a numerically supported conjecture; in Chapter 8 the global result on F_3 for the explicitly adopted potential is proved (Theorem 8.48), while the origin and robustness of that potential remain open; neither is a complete quantum effective potential.

(v) *Open.* A common action for a compatible set of sectors, a general interacting causal reduction, and an explicit intertwiner between V_* and Krasnov's module for the common Lorentz–Pati–Salam algebra, together with a comparison of the kinetic forms.

(vi) *Predictions of a specified model.* None is claimed.

Any claim of novelty for an item in (ii)–(iv) is relative to the sources read for this chapter. It rests on a targeted search, not on a survey of the literature, and it is not an independently established priority claim. Where priority is unresolved, as for the pair invariant of Theorem 5.57, the statement says so.

axis	Krasnov [211]	this manuscript	relation
ambient algebra	$\mathbb{R}^{7,7}$ doubled, $\mathrm{Cl}(7,7) \cong M_{128}(\mathbb{R})$	$\mathbb{R}^{3,3}$ and the tower $\mathrm{Cl}(3n,3n)$; at $n=3$, $\mathrm{Cl}(9,9)$ on $\mathbb{R}^{512}$ (Section 5.1)	analogous (split); different ambient (14 vs 18 generators)
module	$\Lambda^\bullet\mathbb{R}^7 = \mathbb{R}^{128}$; Weyl half $\Lambda^{\mathrm{even}}\mathbb{R}^7 = \mathbb{R}^{64}$, a module of the even part $\mathrm{Cl}^0(7,7)$ only	$V_* = \mathrm{im}\, P_*$, real dimension 64 (Section 6.2)	equivalent for the common algebra (row "one generation"); not as modules of the ambient algebras
forms or Clifford module	forms on a null half (Chevalley); D by wedge and contraction	Kronecker realisation; forms used at $n=1$ only (Proposition 1.11)	the same Chevalley construction at $n=1$
complex structure, chirality	fixed even parity; real Weyl spinor	J, the product of the 14 internal generators; $\Gamma = J\Omega$ (Theorem 12.6)	analogous; mechanisms differ
Lorentz embedding	a second splitting, not dynamical	the interior $\mathfrak{so}(1,3)$ after a clock, imposed (Assumption 2.16)	analogous: both are choices
lowest rung	$SO(3,3)$: commutant $SO(2)$, one charged Weyl fermion (§4)	$n=1$: internal centraliser $\mathfrak{u}(1) = \langle J \rangle$, J the imaginary unit of the Dirac module (Proposition 1.7 and Theorem 2.17)	equivalent as a Lie-algebra statement; field content differs
Pati–Salam rung	$\mathfrak{so}(4,6) \supset \mathfrak{so}(4) \oplus \mathfrak{so}(6)$ acts on the whole 64	P_* preserved by the 21 compact generators only; each of the 24 non-compact ones maps V_* to its complement (Remark 6.24, z3-9)	not the same action on $\mathfrak{so}(4,6)$; for the common algebra an intertwiner exists after compatible identification, but no explicit map is supplied
one generation	SM generation with ν_R: ξ^{aI}, $\bar{\xi}^{aI}$, one Lorentz chirality	one Pati–Salam family with ν_R (Theorem 6.10 and Corollary 6.12)	same content, the realification of $(\mathbf{2},\mathbf{1}) \otimes [(\mathbf{4},\mathbf{2},\mathbf{1}) \oplus (\bar{\mathbf{4}},\mathbf{1},\mathbf{2})]$: equivalent for $\mathfrak{so}(1,3) \oplus \mathfrak{so}(4) \oplus \mathfrak{so}(6)$; explicit map not constructed
projector	Weyl condition	$P_* = \frac{1}{8}(\mathbf{1}+\mathcal{X})(\mathbf{1}+\mathcal{Z})(\mathbf{1}+\mathcal{U})$, (6.1)	different mechanisms
kinetic term	$SU(2)_L$ and $SU(2)_R$ with opposite signs, eq. (219), stated to coincide with the SM kinetic term	$K^\mu = P_*\gamma^0\gamma^\mu P_*$, $K^0 = \mathbf{1}$ on V_*, one sign	kinetic equivalence not established; comparison needs a dictionary
mass bilinears	none for the Weyl Lagrangian	272-dimensional kernel space (Theorem 7.17)	treated here only
bosons and actions	proposed $D \to D + C$; not derived	adopted action (7.4) and potential	neither derives the bosonic action

Table 14.1: Krasnov's construction and the chiral block of this manuscript. "Equivalent" is used only where a statement is proved in both; equal dimensions are not equivalence.

Chapter 15

AI-Assisted Research and Verification Methodology

The question of this chapter is how the statements of this manuscript were produced and what kind of evidence stands behind each of them. Section 15.1 describes the AI-assisted workflow, the author's role and the limits of AI review. Section 15.2 describes the verification: the classes of evidence, the scripts and what a passing check shows, and the parts that have no executable evidence. The verification manifest, one row per check with its script, command, method and supporting label, is Appendix E.

15.1 The AI-assisted workflow

How the work was done. By the author's account, the programme was developed over roughly three years in an iterative process with AI systems. The questions, the starting hypothesis and the direction of the programme are the author's. Claude models assisted extensively: with candidate theorems, exploratory proofs, alternative constructions, code, calculations and drafting of the prose. ChatGPT assisted with discussion, methodology and referee-style criticism; the five reviews and the temporal audit consolidated in the review checklist that governs the present revision were AI-assisted reviews of this kind. The author examined the results throughout, raised objections and asked for corrections; several of the corrected statements are recorded in Section 15.2. The model versions changed over the period, and no particular version was in use throughout. The repository history of the programme begins in June 2026.

Roles in the present revision. In the revision that produced this manuscript, AI sessions under the author's direction:

(i) wrote derivation scripts together with the sections they check;

(ii) wrote referee scripts in separate sessions, instructed to refute the claims, not to import the derivation code, and to use a different method where possible (the surviving instructions are kept in the author's records);

(iii) adjudicated the items of the review checklist, each with a recorded outcome and evidence;

(iv) read and compared the literature of Chapter 14, recording what was read where;

(v) assembled and edited this manuscript.

The independence of a referee script is therefore an independence of code and method, not of a second human checker. None of the check scripts is known to have been written by a second human, and the manuscript does not claim that every result was checked by a human or by a second implementation.

What AI review is not. The reviews that shaped the revision were AI-assisted. They are not formal external peer review, they certify nothing, and agreement between models, or between runs of one model, is not a proof. A statement of this manuscript is supported by its own evidence, of one of the classes of Section 15.2, and not by the fact that an AI system produced or accepted it. The manuscript has not been peer reviewed.

Responsibility. The author is responsible for the manuscript, its verification and its conclusions. The disclosure above is meant to say how the work was done; it implies no endorsement, coauthorship or guarantee by any AI system or its provider.

15.2 How the statements were verified

15.2.1 Classes of evidence

Each statement belongs to one of the following classes, and the class is a property of the evidence, not of the confidence in it.

(i) *Written proof.* An argument given in the text, which a reader can check line by line. Many of these were first drafted with AI assistance; they stand or fall with the argument as written.

(ii) *Exact certificate.* An identity, rank, dimension, spectrum or decomposition computed in integer or rational arithmetic, or as a polynomial identity in symbolic variables. Every generator of the realisation of Appendix A.1 is a signed permutation matrix, so every blade at every rung is one, and products, commutators and traces involve no rounding. Kernel dimensions such as 272, centraliser and stabiliser dimensions and the orbit dimensions 34 and 20 are exact ranks.

(iii) *Numerical check.* A floating-point computation with a stated tolerance: integrations of the interior and orbital dynamics, eigenvalue computations, random tests at seeded inputs, and minimisations. It confirms an exact formula or supports a statement that is labelled numerical; it proves nothing beyond the sampled inputs.

(iv) *Independent re-check.* A referee script, written without importing the derivation script and where possible by a different method. The two share the definitions of the realisation and the conventions of the text, so a convention error common to both would not be caught; what they derive separately is the code and the method of computation.

(v) *Citation.* An established result of the literature, used with its hypotheses and not recomputed: anomaly cancellation of the Standard Model, the Vafa–Witten theorem, Michel's theorem, the Helmholtz conditions, Lanczos's identity, O'Neill's formulas and the others marked inherited.

(vi) *AI review.* A report of an AI-assisted reviewer. It produced objections, which were adjudicated with evidence of the classes above; it is not itself evidence.

15.2.2 The scripts

Each script prints one line `PASS <id>: <claim>` or `FAIL <id>: <claim>` per check and exits with status zero only if every check passes; in version 1.0.0 one script, `p2_alignment.py`, did not yet enforce this, and it does since version 1.0.1. They need Python 3 with NumPy, SciPy, SymPy and mpmath and run from any directory, for example

```
python3 verification/run_all.py --out reproducibility/selected-results/v1.0.1 -j 3
```

which runs all of them and writes one record per script. The scripts are kept by chapter in `verification/chNN_<topic>/`; by origin they are:

(i) *Two readings* (Chapter 3): derivation scripts `r1`–`r4` and referee scripts `s1`–`s4`. The Grassmann-valued Dirac field is handled by an exact Grassmann jet algebra in `r1`, and independently in `s1` by polarisation, which encodes a quadratic Grassmann expression as an antisymmetric bilinear in two commuting fields. Nine checks are floating-point: r2-10c, r2-10d and the random part of r2-18; s1-15, s1-31, s1-38; s2-10b, the random part of s2-14b, and s2-15.

(ii) *Rest energy* (Chapters 3 to 5 and 9): derivation scripts `d1`–`d7` and referee scripts `x1`–`x7`. Spectra and Jordan structure are checked with rational data, since floating-point inputs give false negatives in semisimplicity tests; dynamics is integrated with relative tolerances 10^{-11} to 10^{-12}, and elliptic and Mathieu quantities use `mpmath` and `scipy.special`. The referee findings of the second part, fifty in number, were each re-examined by a second referee session: forty confirmed, ten in part, none rejected, and no theorem refuted. This describes the internal review step of these scripts only; the external reviews of version 1.0.0 later found false statements in the same chapters (among them Proposition 4.9 and three statements of Chapter 9), which are corrected in this version.

(iii) *Emergence* (Chapters 5, 9 and 10): derivation scripts `e1`–`e7`; referee scripts `y1`, `y2`, `y5`, `y6` for four of the seven sections; the review adjudications `z1` and `z5`, which also hold an exact rank certificate in place of an earlier floating-point rank. Two sections were refereed by reading only. The classical section was corrected and reviewed in two passes; its observational inputs were checked against the primary papers where these were reachable, and the others were not rechecked.

(iv) *Photon selection* (Chapter 8): derivation scripts `p1`–`p3` and the adjudication script `z2`, which rebuilds the 512-dimensional realisation independently. No referee script exists for `p1` or `p2`. The global minimisations p2-16 and p3-22 are numerical.

(v) *Global manuscript* (Chapters 1, 6 and 7): the scripts `z3` and `z4` (`verification/ch06_chiral_matter`, `verification/ch07_mass_vacuum`), written to adjudicate the CORE review items; `z4` computes the Molien–Weyl counts of invariants on the kernel space in exact integer arithmetic.

(vi) *Rigid body* (Chapter 4): `verification/ch04_rigid_body/rb1`–`rb3`, including the numerical orbital examples with their inputs, solver and tolerances.

(vii) *Release audit* (Chapters 5, 7 and 8): `a6`–`a8`, written for version 1.0.0 for the statements that the audit of Appendix D had left without a recomputation. `a6` and `a7` are exact (blade parity, integer arithmetic on the 512-dimensional module, exact modular ranks, Molien–Weyl integrals); `a8` is numerical, on the kernel space of Chapter 7, and asserts a spectral gap at every rank it reads.

(viii) *Corrections audit* (Chapters 4 to 7, 9 and 12): written for version 1.0.1 in answer to the external reviews. `a9` and `a10` are exact certificates for statements whose decisive step had been asserted (Theorems 5.13, 6.5 and 6.11 and Corollary 6.12); `a11` is numerical, for the corrected complex structure of the kernel space and the strata of Theorem 7.42; the checks a3-9 and a3-10 became exact, c1-12 was added, c1-4 now samples S^+, and c2-8 derives the mass shell instead of inserting it. `a12` and the checks a3-11 and d7-4c record, exactly, the boundary cases and counterexamples found by a review of this version (Propositions 4.9, 9.22, 9.36 and 9.51), d6-8 now states $M_{\mathrm{pair}} = |\sin\varphi|$, and a11-3 adds the condensate stratum of Theorem 7.42. `a13` is exact: it builds the isomorphism ι of Proposition 7.22 and certifies the constant $\frac{3}{8}$ of Theorem 7.23(ii).

The independent review checks (`verification/ch15_ai_methodology`) add 22 checks, the vacuum checks (`verification/ch07_mass_vacuum`, v1–v3) 31, the gravity and causality checks (g1–g3, c1–c2) 44, the adversarial audit (a1–a5, Appendix D) 59, the release audit (a6–a8) 20, of which 9 are numerical, and the corrections audit (a9–a13) 13, of which 3 are numerical. The complete suite, 66 scripts, is run by `verification/run_all.py`; in the runs of `reproducibility/selected-results/v1.0.1/` the manifest of Appendix E lists 1405 checks, all passing, of which 171 are numerical, and every script with an earlier recorded run reproduces its identifiers and outcomes. The runs record the interpreter and library versions. Of the 1405 checks, 662 are cited in the manuscript by identifier; `verification/THEOREM_CODE.tsv` maps the others to the statements they test.

What a passing check shows. A `PASS` line shows that the coded assertion holds: exactly, for identities, ranks and symbolic derivations, and within the printed tolerance at the sampled inputs, for numerical checks. It validates a statement of the text only to the extent that the assertion is the statement. A pass does not change the class of a claim: an inherited statement that a script re-derives remains inherited, and an interpretive reading is not tested at all. The number of `PASS` lines is not a number of independent claims, since derivation and referee scripts check many statements twice.

15.2.3 What has no executable evidence

(i) *Statements without a cited check.* Of the 438 theorems, propositions, lemmas and corollaries of the manuscript, the claim-by-claim map `verification/COVERAGE.tsv` lists those that cite a current check, those with a written proof in the text, and those that rest on the argument given where they are stated, on the review adjudication scripts `z1`–`z5`, or on a check that the text does not yet cite. The algebraic identities and the explicit reductions of Chapters 11 and 12 are checked exactly by the scripts of `verification/ch11_curvature_gravity` and `verification/ch12_causality` (g1–g3, c1–c2); their gravitational and causal readings are not, since no gravitational action, interacting causal theory or stability analysis is constructed.

(ii) *The vacuum analysis.* Since the audit of this version, the second variation on the tensor model, the 271-dimensional angular Hessian, the random-start minimisations on $\mathcal{T}$ and on $\mathcal{M}$ and the crossing value x_* are reproduced by `verification/ch07_mass_vacuum/` (v1, v2, v3), with archived seeds and tolerances; the audit found and corrected one wrong entry of Theorem 7.27. The minimisations remain numerical experiments, and the global statements that rest on them remain conjectures (Proposition 7.33). The invariant-theory multiplicities quoted after Theorem 7.23 are certified exactly by z4-4.

(iii) *What the release audit leaves.* The electroweak statements of Chapter 7 (Propositions 7.22, 7.35, 7.38 and 7.43 and the structural part of Theorem 7.44) are recomputed numerically (a8), not certified exactly, except that the isomorphism ι of Proposition 7.22 and the spectral identity it carries are now certified exactly (a13-2), as is the constant $\frac{3}{8}$ of Theorem 7.23(ii) (a13-1); the one-loop formula of Proposition 7.21 is inherited; and Example 7.15 has no check.

(iv) *Literature.* Bibliographic data were confirmed on publisher, repository or citing pages; where only secondary confirmation was possible, the author's literature record says so. One attribution, the precise theorem for Howe's skew duality, could not be checked against the primary source.

15.2.4 Corrections found by the checks

The checks changed the mathematics, not only its presentation. Examples, each recorded with its evidence by the author: the surplus-time constraints remove every surplus mode, not only the unstable band; the ordering in the bilinear identity $Y^{\top}TY = -TY^{2}$; the binding gap is not a binding energy, and an earlier deficit formula was false; two false statements about which duplications preserve $|q_2|$; a proposed sign change of the copy-rotation convention was rejected; the factor between the vector-mass conventions; the counterexample to the vacuum-minimum hypotheses; and the local-stability exception in the photon family. A list of this kind is evidence that the procedure finds errors; it is not evidence that none remain. The external reviews of version 1.0.0 found further errors that the checks had not caught; they are listed in `docs/errata.md` and corrected in this version.

Chapter 16

Conclusions and Open Mathematical Problems

The question of the manuscript was which of the free choices of relativistic field theory the split Clifford algebra $\mathrm{Cl}(3,3)$ and its tower $\mathrm{Cl}(3n,3n)$ remove, and which they leave free. This chapter states the answer at the strength the proofs support. It names the strongest results (Section 16.1), keeps the mechanisms that failed as results (Section 16.2), lists the open problems with what would settle each (Section 16.3), and ends with the test that every statement here has been held to (Section 16.4).

16.1 What the manuscript establishes

The following are proved, with the hypotheses stated where they occur. Each is a statement about the algebra or about a specified construction in it; none is a prediction of a physical model.

(1) *The complex unit and Dirac theory.* Inside the joint commutant of the interior Lorentz action and the internal bivectors, the only complex structures commuting with the gamma matrices are $\pm J$ (Theorem 2.17), and the real Dirac theory on $\mathrm{Cl}(3,3)$ is ordinary complex Dirac theory, exactly (Theorem 2.19). The second statement is the reason for the deflationary test of Corollary 2.20.

(2) *Centralisers along the tower.* The bivectors commuting with the interior Lorentz action of $\mathrm{Cl}(3n,3n)$ form $\mathfrak{so}(3n-1,3n-3)$, nested along the tower; at one constituent this is the single generator J (Theorem 5.1 and Corollary 5.2). The diagonal action instead gives a compact algebra forming a reductive dual pair with the Lorentz algebra, with the Casimir identity $C_{\mathrm{cop}} + C_{\Delta} = \frac{1}{2}n(n+2)$; the dual pair and the identity are inherited [224, 296] and are realised here explicitly (Theorem 5.11 and Corollary 5.12).

(3) *Two readings of six directions.* Under the centraliser of G, $\mathbb{R}^{3,3}$ is $\Lambda^2\mathbb{R}^{1,3}$ with G the Hodge star (Theorem 3.3), and covariance fixes the form of Maxwell's equations (Corollary 3.9). The displacement and field-strength readings share the metric and the Clifford action, and no nonzero Lorentz-equivariant linear intertwiner joins the specified modules (Proposition 3.6 and Theorem 3.15).

(4) *Interior dynamics.* Factorisation and binding with $G^2 = -\mathbf{1}$ coexist exactly in neutral signature (Theorem 4.8); within the stated equivariant one-generator ansatz the force law closes in one way into a rotor, unique within that closure ansatz, whose free motion is a pendulum in the duality angle (Theorems 4.22 and 4.28); and the closure has a precise rigid-body form, whose equations, brackets and conserved quantities are compared with those of the rigid body (Theorem 4.53).

(5) *Chiral matter.* The projector P_* of rank 64 carries one Pati–Salam family with a right-handed neutrino (Theorem 6.10); its two central $\mathfrak{u}(1)$s cannot be gauged on it (Proposition 6.14).

(6) *Mass kernels.* The Lorentz-invariant mass kernels of the block form a 272-dimensional space with the complex structure $\widetilde{\mathcal{C}}$ and its Pati–Salam decomposition (Theorem 7.17); an adopted potential selects the Standard-Model orbit on a stated open set of couplings (Theorems 7.25 and 7.34).

(7) *Charge identification.* At the aligned vacua the electric charge is identified among the compact generators, and the normal charge is half the fermion number on the chiral block and anomalous (Theorem 8.37 and Remark 8.61).

(8) *Curvature.* The curvature operator of a four-metric lives on the frame, the first Bianchi identity is one trace, and the duality rotation of the curvature ends at the Einstein tensor, the only divergence-free and variational member of its family; in first-order form the phase is a Barbero–Immirzi angle (Theorems 11.10 and 11.13 and Proposition 11.14). These are identities of given metrics.

(9) *Causal kinematics.* Choosing a Lorentzian four-plane in $\mathbb{R}^{3,3}$ is choosing its definite orthogonal two-plane, represented by a normalised simple bivector whose two signs encode an orientation, a complex structure, and at one constituent the light cone is the set of chiral spinors squared with the help of J (Theorems 12.4 and 12.5).

(10) *Several constituents.* Massive constituents can turn at an exactly constant rate on a balance surface (Theorem 5.52), and a field between manifolds acts on frequencies and, in the fixed-rate model, never on rates (Theorem 9.21).

16.2 Mechanisms that fail

The following mechanisms were tried and fail under the hypotheses stated where each is proved. They are results of the manuscript, not omissions, and each removes a route that would otherwise look open.

(1) Hypercharge cannot be read off anomaly cancellation for charges built from three commuting involutions (Theorem 6.17).

(2) $\mathfrak{su}(3)$ fits in the effective algebra of no axial projector of the five-projector family (Corollary 6.6).

(3) A positive-form interior binds only at zero momentum (Theorem 4.5); no Lie–Poisson closure lives on the frame alone (Theorem 4.21).

(4) Non-orthogonality of $\mathbf{E}$ and $\mathbf{B}$ generates no mass (Section 3.8.4); momentum along the surplus times lowers the mass (Proposition 3.101); and, in the fixed-rate model, the b-field splits or tears the binding but never creates or changes a rate (Proposition 9.11 and Theorem 9.21).

(5) The normal $U(1)$ is not the photon, on four counts that concern different models (Section 13.4): in the curved model it is invisible to spinless matter in the zero-normal-momentum reduced sector and its adapted generator does not commute with the physical generators (Section 11.3); as a Kaluza–Klein vector, relative to the ambient Einstein–Hilbert action and the stated ansatz, it gravitates with negative energy, in either sign convention (Theorem 12.15 and Remark 11.18); and on the chiral block its charge is half the fermion number and anomalous (Remark 8.61).

(6) No unconstrained $SO(3,3)$-invariant adjoint scalar order parameter with an invariant bilinear kinetic term selects the Lorentzian type cleanly: below degree eight an invariant polynomial potential is blind, at every degree the minimum is degenerate with Lorentz-violating configurations, and the kinetic term has ghosts (Theorems 12.8 and 12.9 and Proposition 12.10).

(7) For unrestricted ultrahyperbolic data the six-dimensional free field is ill-posed, and it has no time orientation (Propositions 12.1 and 12.2); constrained evolution can be well posed [27]. Compact surplus times give infinitely many tachyons (Example 12.3). The Ricci–DeTurck method does not transfer to split signature (Section 11.1), which does not exclude every split-signature flow.

16.3 Open problems

The problems below are sharply posed by the results. For each, the step that would settle it is named.

(1) *Selection among the compact vacua.* A coupling that distinguishes the points of the compact family of vacua at $n \geq 3$, where colour survives, would turn the photon construction into a selection; the present potential encodes its target in its couplings.

(2) *The electroweak stage.* The selected vacuum is electroweak symmetric and the adopted potential opposes the neutral deformation (Proposition 8.49); a justified contribution that breaks it, with masses, mixing and the fate of the remaining scalar modes, is missing.

(3) *Global minimality.* The global structure of the potential on all 272 mass kernels, the choice between the Standard-Model vacuum and its mirror, the electroweak hierarchy, and the explicit breaking of the left-handed fermion number that the axion bound requires are open; the local statements are reproduced (Theorem 7.32), while the global ones rest on archived numerical searches (Proposition 7.33) and need a proof.

(4) *One theory for both readings.* A search over non-diagonal pairings of the two Lorentz bases at the second rung and above could settle whether the two readings of $\mathbb{R}^{3,3}$ admit a common algebraic realisation at an acceptable price. No finiteness of the relevant embeddings has been proved, so the search is not yet a decision procedure; and an algebraic answer would not by itself give a shared physical theory, which also needs compatible dynamics.

(5) *The Dirac operator of the blocks on a curved base.* An index is a signed count of zero modes, not a spectrum: on flat four-tori the index vanishes while the eigenvalues depend on the radii. A Lorentzian Dirac operator is not elliptic, so the base, the signature, the bundles, the domain and the boundary conditions must be specified before an index theorem applies, for example that of Bär and Strohmaier [294]; the spectrum is a separate problem. A nonvanishing index would not contradict vector-likeness.

(6) *The interior and time.* A mechanism placing the interior momenta on the null cone or on the balance surface, the only places where a constant rate and a bounded frame coexist; the quantisation of the rotor, whose angle would then have to be related to an external time; and an origin of the scale ℓ and of the rate differences between constituents.

(7) *Torsion.* Whether the antisymmetric part of the shift is the contorsion of an independent connection is a conjecture that is not yet type-correct (Section 11.3).

(8) *Gravity.* An action relating metric, pairing, connection and projector, from which the metric follows rather than being given, with its constraints, degrees of freedom, signs, matter coupling and low-energy limit (Section 11.4).

(9) *Causal order.* A gauge mechanism for type selection, an $SO(3,3)$ analogue of the MacDowell–Mansouri breaking; a principle that makes the normal translations gauge rather than adding them by hand; constraint preservation for interacting fields; a local clock symmetry with its connection; a construction of the cone at three constituents without the positive inner product, or a proof that none exists (Section 12.7); these are research directions. A universal evolution parameter, prior art in the theory of Horwitz and Piron, is open in this arena with the requirements of Section 12.6.

Realistic next steps. One of these is a computation that the existing tools can begin: the search over non-diagonal pairings, which becomes a decision procedure once, for a specified rung or class of embeddings, an effective exhaustive parametrisation or enumeration of them and decidable compatibility tests are available; finiteness alone does not supply one. Once the base, the bundles and the boundary conditions are fixed, the index computation is standard; the spectrum is a separate problem. The others are research problems, and a common local action for a compatible subset of the sectors would be a milestone, not a prerequisite for the mathematical results stated here.

16.4 The test, applied

The largest gap is the same in every sector, and it is one of selection rather than absence. The gauge algebra is derived and the gauge field is not; the arena is chosen and its consequences are derived, but its dynamics is not; the Einstein tensor of a given metric is constructed and the metric is not determined; the projector is a field whose equation follows once an action is adopted, but no action is derived and none of its admissible orbits is singled out; the vacuum of the kernel field is selected, but only on a stated open set of couplings and with the field content given; and the four-plane on which all of this is built rests on a clock reduction that is imposed. Recovering the hypercharge table or the anomaly conditions is a consistency result, not evidence, since any construction with that charge content gives them. The value $\sin^2\theta_W = \frac{3}{8}$ needs more than the charges: with the same charges $g_Y = g_2$ gives $\frac{1}{2}$. The traces $\operatorname{Tr} Y^2 = \frac{10}{3}$ and $\operatorname{Tr}(T_L^3)^2 = 2$ per family give $\frac{3}{8}$ only under a common kinetic normalisation, that is with equal normalised couplings $g_1 = g_2$, $g_Y^2 = \frac{3}{5}g_2^2$, at a stated scale; here that equality comes from the single coupling of the adopted action (7.4), which is a choice (Section 14.4).

Reducing the independent choices of Table 13.1 is the aim of the programme: a theorem that removes one of them without adding another would serve it, although new mathematics need not take that form. New physics would require a selected spectrum, a coupling or scale relation, a mass restriction, or a deviation that survives re-expression in the complex language of Theorem 2.19. None of these is claimed. What the manuscript does claim is a precise map of where the split-signature Clifford algebra decides and where it leaves the decision to the physicist: the fixed results, the constructions, the obstructions and the open problems, each with its hypotheses. On that map the decided territory is larger, and its borders sharper, than the size of the arena would suggest, and the undecided territory is named.

Appendix A

Explicit realisation, conventions and notation

A.1 The explicit realisation

The generators of every finite-dimensional realisation in this paper are integer matrices (signed permutation matrices), and this appendix collects the data needed to check by hand the statements that are integer computations with them. Parameter-dependent spectra, variational statements, irrational combinations of blades and numerical results are not integer computations (Section 15.2).

One constituent. Label the basis of $S = \mathbb{R}^8 = (\mathbb{R}^2)^{\otimes 3}$ by bit strings $b = b_1 b_2 b_3$, ordered as the binary numbers $0, \dots, 7$, and write δ_i for the string with a single 1 in position i. The generators (1.1) act by

$$e_i \,|b\rangle = (-1)^{b_1+\cdots+b_{i-1}} \,|b \oplus \delta_i\rangle, \qquad e_{i+3} \,|b\rangle = (-1)^{b_1+\cdots+b_i} \,|b \oplus \delta_i\rangle \qquad (i = 1, 2, 3) :$$

both e_i and e_{i+3} flip the i-th bit, and they differ only by the sign carried by that bit. As matrices, with $\bar{1} = -1$ and dots for zeros,

$$e_1 = \begin{pmatrix} \cdot & \cdot & \cdot & \cdot & 1 & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & 1 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 1 & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 1 \\ 1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & 1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & 1 & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & 1 & \cdot & \cdot & \cdot & \cdot \end{pmatrix}, \quad e_2 = \begin{pmatrix} \cdot & \cdot & 1 & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & 1 & \cdot & \cdot & \cdot & \cdot \\ 1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & 1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \bar{1} & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \bar{1} \\ \cdot & \cdot & \cdot & \cdot & \bar{1} & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \bar{1} & \cdot & \cdot \end{pmatrix}, \quad e_3 = \begin{pmatrix} \cdot & 1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \bar{1} & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \bar{1} & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \bar{1} & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \bar{1} & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 1 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 1 & \cdot \end{pmatrix},$$

$$e_4 = \begin{pmatrix} \cdot & \cdot & \cdot & \cdot & \bar{1} & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \bar{1} & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \bar{1} & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \bar{1} \\ 1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & 1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & 1 & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & 1 & \cdot & \cdot & \cdot & \cdot \end{pmatrix}, \quad e_5 = \begin{pmatrix} \cdot & \cdot & \bar{1} & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \bar{1} & \cdot & \cdot & \cdot & \cdot \\ 1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & 1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 1 & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 1 \\ \cdot & \cdot & \cdot & \cdot & \bar{1} & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \bar{1} & \cdot & \cdot \end{pmatrix}, \quad e_6 = \begin{pmatrix} \cdot & \bar{1} & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 1 & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & 1 & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \bar{1} & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & 1 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \bar{1} & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \bar{1} \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & 1 & \cdot \end{pmatrix}.$$

The first three are symmetric and the last three antisymmetric, as (1.4) requires, and $e_a e_b + e_b e_a = 2\eta_{ab}\mathbf{1}$ is checked entry by entry.

Derived operators at one constituent. The pole operators are diagonal, $\omega_i \,|b\rangle = (-1)^{b_i}|b\rangle$, so their joint eigenlines are the basis lines themselves, and the volume element is $\omega = e_1 \cdots e_6 = \Gamma$

with $\Gamma|b\rangle = -(-1)^{b_1+b_2+b_3}|b\rangle$; the half-spinors S^+ and S^- are spanned by the basis vectors with an odd, respectively even, number of ones. Further,

$$J = e_2e_3 :\ |b\rangle \mapsto (-1)^{b_2}\,|b \oplus \delta_2 \oplus \delta_3\rangle, \qquad \Omega = e_1e_4e_5e_6, \qquad T = \gamma^0 J = e_1e_2e_3, \qquad \Gamma = J\Omega.$$

On $V = \mathbb{R}^{3,3}$ the two square roots of the pairing are

$$\mathcal{K}e_i = -e_{i+3}, \quad \mathcal{K}e_{i+3} = -e_i, \qquad G\,e_i = e_{i+3}, \quad G\,e_{i+3} = -e_i,$$

with $\mathcal{K} = \frac{1}{2}\,\mathrm{ad}_{\varpi}$, $\varpi = \sum_i e_ie_{i+3}$, and $G = -\mathcal{K} \circ \mathrm{Ad}_{e_1e_2e_3}$. The interior four-plane is $E = \langle e_1, e_4, e_5, e_6\rangle$ with $\gamma^0 = e_1$ and $\gamma^a = e_{a+3}$; the surplus plane is $U = \langle e_2, e_3\rangle$; the paired-mode projector is $\Pi = \frac{1}{2}\mathbf{1} - \frac{1}{4}(\varpi + \mathcal{V})$, which in the basis above is the diagonal projector onto the strings with at least two ones.

The tower. At n constituents $S_n = (\mathbb{R}^8)^{\otimes n}$ and

$$E_{6(k-1)+a} = \omega \otimes \cdots \otimes \omega \otimes e_a \otimes \mathbf{1} \otimes \cdots \otimes \mathbf{1},$$

with $k-1$ factors ω. Since ω is diagonal and every e_a is a signed permutation, every E_A and every blade is a signed permutation matrix of size 8^n, and products, commutators, traces and ranks reduce to integer arithmetic. At three constituents the indices are grouped as follows:

temporal directions	$\{1,2,3,7,8,9,13,14,15\}$, partners $i+3$
interior four-plane E	$\{1,4,5,6\}$, $\gamma^0 = E_1$, $\gamma^a = E_{3+a}$
surplus pair of the first constituent	$\{2,3\}$, $q_A = E_2E_3$
weak directions W	$\{7,13,14,15\}$, $\chi_W = E_7E_{13}E_{14}E_{15}$
colour directions C	$\{10,11,12,16,17,18\}$, $\chi_C = \prod_{i\in C} E_i$
auxiliary directions	$\{2,3,8,9\}$

J is the ordered product of the fourteen generators outside E, $J^2 = -\mathbf{1}$, and $\Gamma = J\gamma^0\gamma^1\gamma^2\gamma^3$. The projectors used in the text are

$$P_* = \tfrac{1}{8}(\mathbf{1} + \Gamma\chi_W)(\mathbf{1} - J\chi_C)(\mathbf{1} - Jq_A), \qquad P_{\mathrm{fam}} = \tfrac{1}{8}(\mathbf{1} + X_1)(\mathbf{1} + X_2)(\mathbf{1} + X_3),$$

with $X_1 = J^{(1)}J^{(2)}$, $X_2 = \omega^{(3)}$ and $X_3 = \omega^{(1)}e_1^{(3)}e_4^{(3)}$, where $J^{(k)} = e_2^{(k)}e_3^{(k)}$ and $\omega^{(k)}$ is the volume element of constituent k. The colour Cartan element is $H = -(E_{10}E_{16} + E_{11}E_{17} + E_{12}E_{18})$, the weak generators are $t_i^{L,R} = \frac{1}{2}(\mathbf{1} \mp \chi_W)\tau_i$ with $\tau_i = -\frac{1}{2}w_jw_k$ for cyclic (i,j,k) of $(1,2,3)$, $w = (E_7, E_{13}, E_{14}, E_{15})$, and the Standard-Model line of mass kernels is

$$\Phi_0 = P_\nu\, P_*\, \gamma^0 E_3E_9E_{13}E_{14}E_{16}E_{17}E_{18}\, P_*\, P_\nu,$$

with P_ν the projector onto $\Gamma = +1$, $T_3^R = +\frac{1}{2}$, $B - L = -1$. The left–right involution is $O = E_1E_7E_8$, and the charge conjugation and CP blades are $C = E_2E_7E_{10}E_{11}E_{12}$ and $CP = E_1E_2E_7E_8E_{10}E_{11}E_{12}E_{14}$.

Second rung. At $n = 2$ the twelve generators are $e_a \otimes \mathbf{1}$ and $\omega \otimes e_a$ on $\mathbb{R}^{64}$. The mixed Lorentz action pairs the interior rotations $S_k = \frac{1}{2}\gamma^i\gamma^j$ and boosts $T_k = \frac{1}{2}\gamma^0\gamma^k$ of the first factor with the diagonal rotations L_k and boosts K_k of $\mathfrak{l}_G$ on the second, and the second-rung vacuum uses $a, b = e_2 \otimes \mathbf{1},\ e_3 \otimes \mathbf{1}$ and $c, d, e = \omega \otimes e_1,\ \omega \otimes e_2,\ \omega \otimes e_3$.

Appendix B

Units and normalisation dictionaries

B.1 Units of the two readings

The displacement reading of X is written in SI units with c explicit. The computations of the field-strength reading use Heaviside–Lorentz units with $c = 1$, and with $\hbar = c = 1$ for the Dirac field. Throughout, $\mathbf{B}$ is minus the textbook magnetic field, $\mathbf{B}_{\mathrm{t}} = -\mathbf{B}$, and $\mathbf{E}$ and $c\mathbf{B}$ share the unit $\mathrm{V\,m^{-1}}$. Table B.1 collects the objects that the readings compare; Table 3.1 adds their Lagrangians, canonical conjugates and Noether charges.

object	module	Lorentz action	SI unit	computed in
$Y = X_{\mathrm{cl}}$, displacement	$V = \mathbb{R}^{3,3}$: times ct^i on V^+, lengths on V^-	$\mathfrak{l}_{\mathrm{int}}(\hat{n})$ on $E_{\hat{n}} = \mathbb{R}\hat{n} \oplus V^-$ after a clock; the rest of V^+ scalar	m	SI, c explicit
$X_{\mathrm{EM}} = (c\mathbf{B}; \mathbf{E})$, field strength	V as the fibre $\Lambda^2 E^*_{\hat{n}}$	$\mathfrak{l}_G$ by the commutator, $(1,0) \oplus (0,1)$	$\mathrm{V\,m^{-1}}$	Heaviside–Lorentz, $c = 1$
field octants	$S \cong \Lambda^*(\mathsf{L}) \cong \mathbb{R}^8$; $\mathbf{E}$ on weight one, $\mathbf{B}$ on weight two	ρ_{oct}: L_i and the odd $Q_i = -TL_i$	$\mathrm{V\,m^{-1}}$	Heaviside–Lorentz, $c = 1$
ψ, Dirac field	real Grassmann field on $\mathbb{R}^{1,3}$ with values in S	$\rho_{\hat{n}}$, $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$	$\mathrm{m^{-3/2}}$	Heaviside–Lorentz, $\hbar = c = 1$
A_μ, potential	one-form on $E_{\hat{n}}$	$\mathfrak{l}_{\mathrm{int}}$, $(\frac{1}{2}, \frac{1}{2})$	$\mathrm{V\,s\,m^{-1}}$	Heaviside–Lorentz, $c = 1$
E_6, E_0, $\mathbf{S}$	quadratic in X_{EM}	$E_6 = \Theta^{00}$, $-\mathbf{S} = \Theta^{0a}$; E_0 invariant	$\mathrm{J\,m^{-3}}$	both
μ_{m}, d	Pauli and electric-dipole couplings	–	$\mathrm{J\,T^{-1}}$, $\mathrm{C\,m}$	mass dimension -1

Table B.1: Representations and units of the objects compared by the two readings.

B.2 Normalisation of gauge, scalar and mass terms

The action of the kernel field is written for gauge generators that are orthonormal for the module metric $\mathsf{G}_{ab} = \mathrm{Tr}(t_a^\top t_b)/512$, with unit gauge coupling; the coupling g_A is restored through the kinetic term $-\frac{1}{4g_A^2}\mathsf{G}_{ab}F^a_{\mu\nu}F^{b\,\mu\nu}$. Generators of the conventional normalisation $\mathrm{Tr}_{\mathrm{fund}}\,TT = \frac{1}{2}$ (such as t_i^L, t_i^R and the colour generator $\frac{1}{4}(E_{10}E_{16} - E_{11}E_{17})$) have module norm $\frac{1}{8}$, so the conventionally normalised coupling is $g = \sqrt{8}\,g_A$. With $v = |\Phi|/\sqrt{2}$ the vector masses at the

selected vacuum are $m^2 = 8g_A^2v^2$ and $40g_A^2v^2$, that is g^2v^2 and $5g^2v^2$ (Theorem 7.34; checks z3-5, z3-6). The photon analysis uses $-\operatorname{Tr}/128$ for the same generators, a factor 4 from $-\operatorname{Tr}/512$; its weak and hypercharge norms are $|t_3^L|^2 = \frac{1}{8}$ and $|t_Y|^2 = \frac{5}{24}$ in $-\operatorname{Tr}/512$.

B.3 Symbols of the matter, vacuum and photon chapters

Table B.2 lists the symbols that Chapters 6 to 8 fix, with their normalisation and the name they carry in the source notes and in the verification scripts, which were not renamed (Appendix E).

symbol	meaning and normalisation	where	in the scripts
$\mathcal{X}, \mathcal{Z}, \mathcal{U}$	the commuting involutions $\Gamma\chi_W$, $-J\chi_C$, $-Jq_A$; $P_* = \frac{1}{8}(\mathbf{1}+\mathcal{X})(\mathbf{1}+\mathcal{Z})(\mathbf{1}+\mathcal{U})$, rank 64	Section 6.2	X, Z, U
t_a	gauge generators acting on the module	Chapters 7 and 8	T_a
G_{ab}	module metric $\operatorname{Tr}(t_a^\top t_b)/512$; unit coupling g_A, conventional coupling $g = \sqrt{8}\, g_A$	(7.4), Appendix B.2	G_{ab}
$\langle X, Y\rangle$ on $\mathfrak{k}$	$-\operatorname{Tr}(XY)/128$ in the alignment analysis, in which the rotations $\frac{1}{2}e_ie_j$ are orthonormal; 4 times $-\operatorname{Tr}(XY)/512$	Section 8.3	same
Σ^{0a}	boosts $\frac{1}{2}\gamma^0\gamma^a$ of the interior Lorentz algebra	Theorem 7.17	K^a
$\mathcal{M}$	the real 272-dimensional space of Lorentz-invariant mass kernels; complex structure $\widetilde{\mathcal{C}}$	Theorem 7.17, Appendix C	K
v_R, v_u, v_d	amplitudes of $\Phi = v_R\Phi_0 + v_uK_{1u} + v_dK_{1d}$; $v = \lvert\Phi\rvert/\sqrt{2}$ in the vector masses	Example 7.36	v_R, κ_u, κ_d
ξ_1, ξ_2, ξ_3	weights, all ≥ 0, of the three terms $V_{\mathfrak{h}_{\mathrm{SM}}}$, $-\operatorname{Tr}(\Pi P_L)$ and $-\operatorname{Tr}(\Pi P_{\mathrm{fam}})$ of the selecting potential	Proposition 8.47	κ, λ, μ
Q	$T_3^L + T_3^R + \frac{1}{2}(B-L)$ on the block	Appendix C	same

Table B.2: Symbols of Chapters 6 to 8 and their names in the source notes. In Chapter 8 the letters T_a keep their meaning of tangent directions in the material on the projector as a field.

Appendix C

Branching and charge tables

C.1 One family of the chiral block

The block $V_* = \operatorname{im} P_*$ carries $(\mathbf{4},\mathbf{2},\mathbf{1})_L \oplus (\mathbf{4},\mathbf{1},\mathbf{2})_R$ of $\mathfrak{su}(4) \oplus \mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ (Theorem 6.10; proved, checks in Appendix E). Its decomposition under $\mathfrak{su}(3) \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_Y$, with $\mathbf{4} = \mathbf{3}_{1/3} \oplus \mathbf{1}_{-1}$ under $\mathfrak{su}(3) \oplus \mathfrak{u}(1)_{B-L}$ and $Q = T_3^L + T_3^R + \frac{1}{2}(B-L)$, is the standard one [116]; it is inherited, and the charge operators of the block reproduce it (Chapter 8).

Pati–Salam component	states	$\mathfrak{su}(3)$	T_3^L or T_3^R	$B-L$	Q
$(\mathbf{4},\mathbf{2},\mathbf{1})_L$	u_L, d_L	$\mathbf{3}$	$\pm\frac{1}{2}$	$\frac{1}{3}$	$\frac{2}{3}$, $-\frac{1}{3}$
	ν_L, e_L	$\mathbf{1}$	$\pm\frac{1}{2}$	-1	0, -1
$(\mathbf{4},\mathbf{1},\mathbf{2})_R$	u_R, d_R	$\mathbf{3}$	$\pm\frac{1}{2}$	$\frac{1}{3}$	$\frac{2}{3}$, $-\frac{1}{3}$
	ν_R, e_R	$\mathbf{1}$	$\pm\frac{1}{2}$	-1	0, -1

Table C.1: One family in the chiral block: 16 complex, 64 real dimensions with the Lorentz spinor components.

C.2 The space of mass kernels

With the complex structure $\widetilde{\mathcal{C}}$ of Theorem 7.17, the 272-dimensional real space $\mathcal{M}$ of Lorentz-invariant mass kernels is a complex space V of dimension 136, and $\mathcal{M} \otimes \mathbb{C} = V \oplus \bar{V}$. Table C.2 lists its components (proved: Theorem 7.17; the invariant counts built on them are checked exactly by z4, the conjugation of the tens by the cubic trace in a11-1). Replacing $\widetilde{\mathcal{C}}$ by $-\widetilde{\mathcal{C}}$ conjugates every label. Fermion number acts with charge 2 on the Majorana-type components and 0 on the Dirac-type ones (Proposition 7.20).

component of V	complex dimension	type	fermion number
$(\bar{\mathbf{10}},\mathbf{3},\mathbf{1})$	30	Majorana	2
$(\mathbf{10},\mathbf{1},\mathbf{3})$	30	Majorana	2
$(\mathbf{15},\mathbf{2},\mathbf{2})$	60	Dirac	0
$2\,(\mathbf{6},\mathbf{1},\mathbf{1})$	12	Majorana	2
$(\mathbf{1},\mathbf{2},\mathbf{2})$	4	Dirac	0
total	136		

Table C.2: The mass kernels of the chiral block. In the language of $\mathfrak{so}(10)$ the content is that of $\mathbf{10} \oplus \overline{\mathbf{126}}$, the dual of $\mathrm{Sym}^2(\mathbf{16}) = \mathbf{10} \oplus \mathbf{126}$: $\mathbf{10} = (\mathbf{6},\mathbf{1},\mathbf{1}) \oplus (\mathbf{1},\mathbf{2},\mathbf{2})$ and $\overline{\mathbf{126}} = (\mathbf{6},\mathbf{1},\mathbf{1}) \oplus (\bar{\mathbf{10}},\mathbf{3},\mathbf{1}) \oplus (\mathbf{10},\mathbf{1},\mathbf{3}) \oplus (\mathbf{15},\mathbf{2},\mathbf{2})$ (inherited), with $\mathbf{16} = (\mathbf{4},\mathbf{2},\mathbf{1}) \oplus (\bar{\mathbf{4}},\mathbf{1},\mathbf{2})$ and the complex structure $\widetilde{\mathcal{C}}$ of Theorem 7.17.

Appendix D

Verification and reproducibility

This appendix states what the executable checks of the manuscript compute, by which algorithms, under which conventions and with which limitations, and maps every labelled statement to the checks that test it. The mathematics of the manuscript does not depend on this appendix: every definition, theorem and proof is in the chapters or in Appendices A to C, and a check is evidence for a coded assertion, not a proof of the statement it tests. The scripts are supplementary material in the project repository, under `verification/chNN_<topic>/` (one directory per chapter, shared code in `verification/common/`) of the repository `https://github.com/driss-bencheikh/spacetime-electromagnetic-framework`, at the commit of version 1.0.0 recorded in `release/README.md`; this appendix gives what is needed to rerun and to judge them.

D.1 Status vocabulary

A statement is *analytically established* when its proof is in the text or it is cited, with its hypotheses, from published literature (*inherited*); *exactly verified* when a check establishes the coded assertion in exact arithmetic (integers, rationals, algebraic numbers, or symbolic expressions simplified to zero); *numerically supported* when the check uses floating-point arithmetic, a random sample or a numerical integration, with a stated tolerance; *conditional* when it rests on an explicit assumption; *obstructed* or *refuted* when a no-go or a counterexample is proved; *unresolved* otherwise. A `PASS` line certifies only the assertion printed with it: a check on one example verifies that example, and a check of an identity at a point does not verify the theory built on it.

D.2 Realisations and conventions

All checks use the integer realisation of Appendix A.1: $e_1 = \sigma_x \otimes \mathbf{1} \otimes \mathbf{1}$, $e_2 = \sigma_z \otimes \sigma_x \otimes \mathbf{1}$, $e_3 = \sigma_z \otimes \sigma_z \otimes \sigma_x$, $e_4 = \varepsilon \otimes \mathbf{1} \otimes \mathbf{1}$, $e_5 = \sigma_z \otimes \varepsilon \otimes \mathbf{1}$, $e_6 = \sigma_z \otimes \sigma_z \otimes \varepsilon$ with $\varepsilon = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, so $e_a^2 = \eta_{aa}\mathbf{1}$ with $\eta = \mathrm{diag}(1,1,1,-1,-1,-1)$, every blade is a signed permutation matrix and $e_a^\top = \eta_{aa} e_a$. At n constituents the generators are $E_{6k+a} = \omega^{\otimes k} \otimes e_a \otimes \mathbf{1}^{\otimes(n-1-k)}$ with $\omega = e_1 \cdots e_6$ (64×64 at $n = 2$, 512×512 at $n = 3$). The interior gamma matrices are $\gamma^\mu = (e_1, e_4, e_5, e_6)$ (or the clock four-plane $\langle \widehat{W}, e_4, e_5, e_6 \rangle$ where a statement says so), $\gamma_\mu = \eta_{\mu\nu}\gamma^\nu$ with $\eta = \mathrm{diag}(1,-1,-1,-1)$, $\Omega = \gamma^0\gamma^1\gamma^2\gamma^3$ and $I_4 = \gamma_0\gamma_1\gamma_2\gamma_3 = -\Omega$; $J = e_2 e_3$ at one constituent and the product of the internal generators at n; $\Gamma = J\Omega$; two-forms are identified with bivectors by $F \mapsto \frac{1}{2} F_{ab}\gamma^a\gamma^b$ and $\varepsilon_{0123} = +1$. The kinetic kernels are $K^\mu = -\gamma^0\gamma^\mu$ at one constituent (Proposition 2.21) and $P_* \gamma^0 \gamma^\mu P_*$ on the chiral block, an overall sign of the fermion Lagrangian that changes no field equation, gauge condition or spectrum (Proposition 2.24).

D.3 Algorithms

(A1) *Blade parity.* $e_I e_J = (-1)^{|I||J|-|I\cap J|} e_J e_I$ (Lemma 1.1) decides commutation of blades without matrices; counts over all 2^{18} blades of $\mathrm{Cl}(9,9)$ (Theorem 7.45) are exact enumerations.

(A2) *Exact linear solution spaces.* Commutants, intertwiners, invariant forms and gauge conditions ($X^\top K + KX = 0$, $[X, L] = 0$, $aN + Na^\top = 0$) are linear in the unknown matrix; the conditions are written on the blade basis (or on all matrix entries) and solved by exact rational nullspaces. No ansatz is made: the unknown ranges over the whole matrix algebra.

(A3) *Gram certificates on the large modules.* For operators A_k on $\mathbb{R}^{512}$ the integer Gram matrix $\mathrm{Tr}(A_k^\top A_l)$ has the rank of their span; commuting subspaces (the nullspace of $X \mapsto [X, P]$), kernels on a block and images are obtained from such Gram matrices in exact arithmetic (Theorem 6.5).

(A4) *Modular ranks with explicit solutions.* Where a dimension is asserted for a large integer system, the rank modulo a large prime bounds the rational rank from below; explicit exact solutions bound it from above, and the two are made to meet (e.g. Theorem 4.37). Where only ranks modulo two primes are computed (Theorem 4.39), the check says so.

(A5) *Symbolic geometry.* Curvature of the stated metrics by Christoffel symbols, Riemann, Ricci and Einstein tensors in exact symbolic arithmetic, with closed-form inverses for Kaluza–Klein coframes; Euler–Lagrange expressions and Poisson brackets by symbolic differentiation with general functions.

(A6) *Spectra of exact integer matrices.* Characteristic polynomials are computed exactly where possible (Theorem 2.31); joint spectra of commuting integer matrices on the 512-dimensional module are read by floating-point diagonalisation of a generic linear combination and accepted only within 10^{-8} of the stated rationals (numerical).

(A7) *Vacuum minimisation.* The radiative energy E_x on the 272-dimensional kernel space is minimised from random starts and its critical points are classified by the Hessian (Chapter 7); the stated minimum is numerically supported and its global minimality is a conjecture.

(A8) *Integrations.* Ordinary differential equations (closure dynamics, orbits) are integrated with stated tolerances (`DOP853` or `LSODA`, relative tolerance 10^{-12}) and compared with closed forms where these exist.

D.4 Exact certificates

The following identities are certified in exact arithmetic and carry the main structural statements: the Clifford relations of the realisation at $n = 1, 2, 3$; the trace-orthogonality $\mathrm{Tr}(\theta_S^\top \theta_R) = 16\,\delta_{SR}$ of the 256 monomials of the module of Theorem 2.30; the characteristic polynomials of Theorem 2.31; the solution spaces of Theorems 2.25 and 2.27 and Propositions 2.22 and 2.29 over all of $M_8(\mathbb{R})$; the commutants and invariant forms of Corollary 3.5, Proposition 4.19, and Theorem 4.26; the grade census of Proposition 3.76; $\mathrm{Tr}\, P_* = 64$ and the seven traceless products of Theorem 6.10; the projector-family table of Theorem 6.5; the $8192 = 4 \times 2048$ blades of Theorem 7.45; the centraliser dimensions 72, 30 and the Casimir $3\,\mathbf{1}$ of Theorems 4.34 and 4.35; the curvature identities of Chapter 11 for general operators and the stated metrics; the first-class bracket algebra of Proposition 11.4 for a general metric and potential. The release audit adds: the kernel spaces and internal algebras of Theorem 7.9 on the blades of $\mathrm{Cl}(3,3)$; the CP,

C and P images of Φ_0 in Theorem 7.46, with Φ_0 rebuilt in integer arithmetic; the counts of Theorem 7.47 graded by F and by the $J\Gamma$ charge; the centre and generators of the Lorentz commutant (Proposition 8.1); the basis of $P_{\mathrm{fam}}\mathcal{C}P_{\mathrm{fam}}$ by blade classes and the exact rank of the curvature values in Theorem 8.4 and Proposition 8.5; the second-rung family of Theorems 8.57 to 8.59 and Proposition 8.60; and the weights of the Weyl block at $n = 3$ in Proposition 5.19 (a6, a7).

D.5 Numerical checks and limitations

About one check in eight is numerical. They concern spectra on the large modules read in floating point (A6), random samples (random clocks, random spinors, random points of projector families), numerical integrations (A8), the vacuum minimisation (A7), loop-level coefficients, and the electroweak recomputations a8 on the kernel space, whose ranks are read with an asserted spectral gap. Each states its tolerance in its claim text. Their limitations are: a sample does not cover a family; a tolerance does not exclude a nearby degenerate case; the minimisation finds the lowest critical points reached from the starts used and does not prove global minimality; numerical integrations verify conserved quantities over finite time intervals only. Statements resting on such checks are classed as numerically supported, not as exactly verified.

D.6 Software and procedure

Python 3 with `numpy`, `scipy`, `sympy` and `mpmath`; no other dependency. Each script prints one `PASS` or `FAIL` line per check and exits with status zero exactly when all its checks pass. From the repository root:

- `python3 verification/run_all.py --out <directory> -j 3` runs every script and writes one output file per script, a summary and the software environment;
- `python3 verification/tools/make_manifest.py --runs <directory>` rebuilds `verification/MANIFEST.tsv` from the outputs; `verification/tools/coverage.py` maps statements to cited checks; `verification/tools/make_appendix_d.py` regenerates the theorem-to-code table `verification/THEOREM_CODE.tsv` and the list of scripts `verification/SCRIPTS.tsv`; and `verification/tools/make_reproducibility_index.py` writes `reproducibility/manifest.json` and the indexes of exact and numerical checks in `reproducibility/selected-results/`;
- the manuscript is built from `manuscript/` with `pdflatex`, `bibtex` and two further passes of `pdflatex` on `main.tex`.

The run records of this version are in `reproducibility/selected-results/v1.0.1/`; those of version 1.0.0 are kept in `reproducibility/selected-results/v1.0.0/`.

The theorem-to-code table is kept in the repository, not printed: `verification/THEOREM_CODE.tsv` lists every labelled statement with its environment, the checks that test it (cited in its text, added by the audit, or related without citation) and its evidence, which is *exact*, *numerical* or *exact+numerical* when checks exist and otherwise *proof in text*, *inherited*, *conjecture* or *argument in text* (argued in the surrounding text without a proof environment or a check). `verification/SCRIPTS.tsv` lists the scripts with their numbers of exact and numerical checks.

Appendix E

Verification manifest

The executable checks of the statements of this manuscript are listed in `verification/MANIFEST.tsv`: for every check its identifier, the script and command, the chapter and, where known, the manuscript label it supports, and whether it is exact or numerical. A passing check shows that the coded assertion holds; it does not change the class of a statement. The theorem-to-code table generated from it is `verification/THEOREM_CODE.tsv`; the algorithms, certificates and limitations of the checks are described in Appendix D.

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